

Nuclear Structure and Decay Data for A=34 Isobars*

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Abstract: This work presents a comprehensive and critical evaluation of experimental nuclear spectroscopic data from reactions and decays for all 11 nuclides with mass number 34 (Ne, Na, Mg, Al, Si, P, S, Cl, Ar, K, Ca). Recommended values are produced for level energies, spins and parities, half-lives, and radiation properties including energies, branching ratios, and multiplicities of γ rays, as well as characteristics of β decay, based on a rigorous assessment of all available experimental data. Discrepancies among existing results are carefully addressed. This work supersedes earlier full evaluations of A=34 published by [2012Ni10](#), [1990En08](#) (also [1998En04](#) update), and [1978En02](#).

^{34}Cl and ^{34}S remain the most extensively studied nuclides, investigated through more than 20 different reactions and decays, followed by ^{34}P , ^{34}Si , and ^{34}Ar . ^{34}Ne has been identified, but no nuclear-structure information is available. For ^{34}Na , no excited states have been confirmed, and no γ rays have been placed with certainty. No dataset exists for β^- decay of ^{34}Na to ^{34}Mg . ^{34}Ca has not yet been identified.

Cutoff Date: Literature available up to September 30, 2026 has been consulted, with the help of the primary bibliographic source, the NSR database ([2014Pr09](#)) available at the Brookhaven National Laboratory web page: www.nndc.bnl.gov/nsr/.

General Policies and Organization of Material: See the April 2026 issue of the *Nuclear Data Sheets* or <https://www.nndc.bnl.gov/nds/docs/NDSPolicies.pdf>.

General Comments: The statistical analysis of γ -ray data and deduced level schemes is carried out through a suite of current versions of computer codes, such as EvaluationToolkit, GLSC, Java-RULER, and ConsistencyCheck, available at the IAEA-NDS ENSDF codes web page (https://www-nds.iaea.org/public/ensdf_pgm). The direct feedings to excited states in β^- and ε decays have generally been computed from $I(\gamma+ce)$ intensity balances at each level; the associated $\log ft$ values are calculated using the BetaShape code. Theoretical total conversion coefficients are calculated using version 2.3e of the BrIcc code ([2008Ki07](#)) for the frozen-orbit option with an implicit uncertainty of 1.4% when not stated. Q values and particle separation energies have been adopted from [2021Wa16](#) (AME2020). Static magnetic dipole moments are taken from [2025StZZ](#) (long-lived states) and [2020StZV](#) (short-lived states) evaluations, and static electric quadrupole moments from [2021StZZ](#) evaluation, when available in these publications. Nuclear rms charge radii have been adopted from [2013An02](#) evaluation. In cases where weighted averaging has been used, the assigned uncertainty in the result is generally at least as large as the smallest uncertainty among the contributing data points. The evaluation workflow is aided by 'ENSDF-Agent'. Adopted Levels are placed with the assistance of the machine-learning tool 'Level-Matcher'.

Acknowledgements: Earlier evaluations ([2012Ni10](#), [1998En04](#) update, [1990En08](#), [1978En02](#), [1973EnVA](#), [1967En05](#), [1962En05](#), [1957En01](#), [1954En19](#)) and compilations available in the XUNDL database provided valuable resources for the present work. The evaluators are grateful to the reviewers at different stages of this work, and to the personnel at the National Nuclear Data Center (NNDC) at Brookhaven National Laboratory for facilitating this work.

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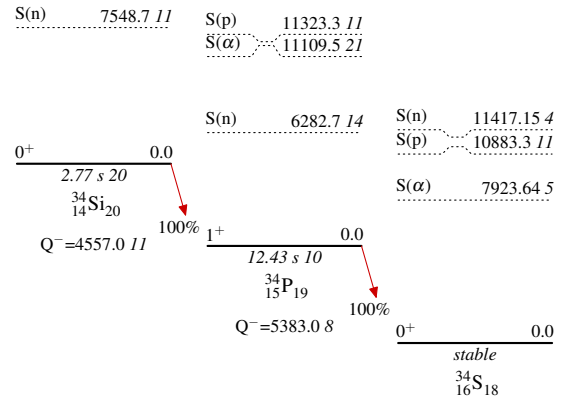
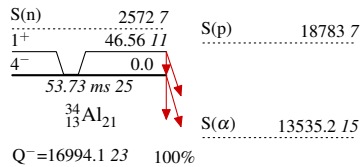
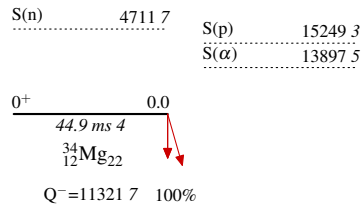
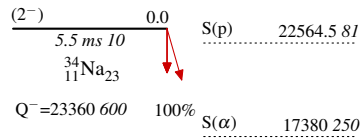
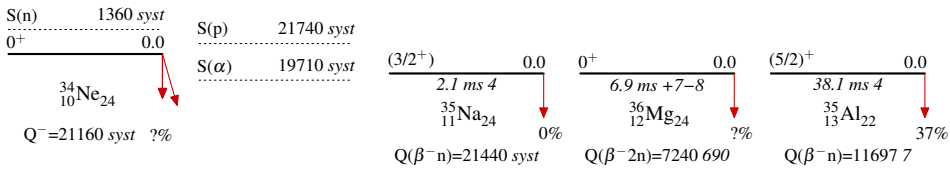
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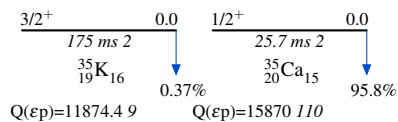
Nuclide	Data Type	Page	Nuclide	Data Type	Page
$^{34}_{10}\text{Ne}_{24}$	Skeleton Scheme for A=34	5	$^{34}_{16}\text{S}_{18}$	Adopted Levels, Gammas	131
$^{34}_{11}\text{Na}_{23}$	Adopted Levels	7	^{34}P	β^- decay (12.43 s)	180
$^{34}_{12}\text{Mg}_{22}$	Adopted Levels, Gammas	8	^{34}Cl	$\varepsilon+\beta^+$ decay (1.5266 s)	183
$^{34}_{13}\text{Al}_{21}$	C(^{36}Mg , $^{34}\text{Na}\gamma$)	10	^{34}Cl	$\varepsilon+\beta^+$ decay (31.99 min)	185
$^{34}_{14}\text{Si}_{20}$	Adopted Levels, Gammas	11	^3H (^{32}S , $p\gamma$)		188
$^{34}_{15}\text{P}_{19}$	^{35}Na β^- n decay (2.1 ms)	13	^3H (^{35}Cl , $\alpha\gamma$)		189
$^{34}_{16}\text{S}_{18}$	^1H (^{34}Mg , $^{34}\text{Mg}'\gamma$)	15	^4He (^{30}Si , α):resonances		191
$^{34}_{17}\text{Cl}_{17}$	^9Be (^{36}Si , $^{34}\text{Mg}\gamma$)	17	^{12}C (^{24}Mg , $^{34}\text{S}\gamma$)		194
$^{34}_{18}\text{Ar}_{16}$	C(^{37}Al , $^{34}\text{Mg}\gamma$)	19	^{24}Mg (^{16}O , $\alpha 2p\gamma$)		197
$^{34}_{19}\text{K}_{15}$	Coulomb excitation	21	^{27}Al (^{10}B , $2pn\gamma$)		203
$^{34}_{20}\text{Ca}_{14}$	Adopted Levels, Gammas	23	^{27}Al (^{12}C , $\alpha p\gamma$)		204
$^{34}_{21}\text{Sc}_{13}$	^{34}Mg β^- decay (44.9 ms)	28	^{28}Si (^{34}S , $^{34}\text{S}'\gamma$)		205
$^{34}_{22}\text{Ti}_{12}$	^{36}Mg $\beta^- 2n$ decay (6.9 ms)	32	^{30}Si (α, γ), (α, n):resonances		206
$^{34}_{23}\text{V}_{11}$	^2H (^{33}Al , $^{34}\text{Al}\gamma$), (^{34}Si , $^{34}\text{Al}\gamma$)	34	^{31}P ($\alpha, p\gamma$)		215
$^{34}_{24}\text{Cr}_{10}$	^4He (^{34}Al , $^{34}\text{Al}'\gamma$)	35	^{32}S (t, p)		226
$^{34}_{25}\text{Mn}_9$	^9Be (^{36}S , ^{34}Al)	37	^{32}S ($\alpha, ^2\text{He}$)		228
$^{34}_{26}\text{Fe}_8$	^9Be (^{40}Ar , ^{34}Al)	38	^{33}S (n, γ) E=thermal		229
$^{34}_{27}\text{Co}_7$	Pb (^{35}Al , $^{34}\text{Aln}\gamma$)	39	^{33}S (n, γ), (n, n):resonances		246
$^{34}_{28}\text{Ni}_6$	U(p , ^{34}Al)	41	^{33}S (d, p)		248
$^{34}_{29}\text{Cu}_5$	Coulomb excitation	42	^{34}S (e, e')		250
$^{34}_{30}\text{Zn}_4$	Adopted Levels, Gammas	43	^{34}S (γ, γ'), (pol γ, γ')		251
$^{34}_{31}\text{Ga}_3$	^{34}Al β^- decay (53.73 ms)	52	^{34}S (n, n), (n, n')		253
$^{34}_{32}\text{Ge}_2$	^{34}Al β^- decay (23.2 ms)	56	^{34}S ($\pi^+, \pi^{+'}$), ($\pi^-, \pi^{-'}$), ...		254
$^{34}_{33}\text{As}_1$	^{34}Al β^- decay:mixed	61	^{34}S (p, p), (p, p'), (pol p, p')		255
$^{34}_{34}\text{Se}_0$	^{35}Al β^- n decay (38.1 ms)	64	^{34}S ($p, p'\gamma$)		258
$^{34}_{35}\text{Br}_{-1}$	^2H (^{33}Si , $^{34}\text{Si}\gamma$), (^{35}Si , $^{34}\text{Si}\gamma$)	68	^{34}S (pol d, d), (pol d, d')		259
$^{34}_{36}\text{Kr}_{-2}$	^2H (^{34}Si , $^{34}\text{Si}'\gamma$)	70	^{34}S (α, α), (α, α'), ($\alpha, \alpha'\gamma$)		260
$^{34}_{37}\text{Rb}_{-3}$	^7Li (^{34}P , $^7\text{Be}\gamma$)	72	^{35}Cl (γ, p)		262
$^{34}_{38}\text{Sr}_{-4}$	^9Be (^{35}Si , $^{34}\text{Si}\gamma$)	74	^{35}Cl (n, d)		263
$^{34}_{39}\text{Y}_{-5}$	^{18}O (^{18}O , $2p\gamma$)	76	^{35}Cl ($d, ^3\text{He}$)		264
$^{34}_{40}\text{Zr}_{-6}$	Si (^{34}Si , $^{34}\text{Si}'\gamma$)	78	^{37}Cl ($p, \alpha\gamma$)		265
$^{34}_{41}\text{Nb}_{-7}$	^{36}S (^{11}B , ^{13}N)	79	^{206}Pb (^{34}S , $^{34}\text{S}'\gamma$)		266
$^{34}_{42}\text{Mo}_{-8}$	^{36}S (^{14}C , ^{16}O)	80	^{208}Pb (^{34}S , $^{34}\text{S}'$)		267
$^{34}_{43}\text{Tc}_{-9}$	^{160}Gd (^{34}S , $X\gamma$), (^{36}S , $X\gamma$)	81	^{208}Pb (^{36}S , $^{34}\text{S}\gamma$)		268
$^{34}_{44}\text{Ru}_{-10}$	^{208}Pb (^{36}S , $^{34}\text{Si}\gamma$)	82	$^{34}_{17}\text{Cl}_{17}$	Adopted Levels, Gammas	270
$^{34}_{45}\text{Rh}_{-11}$	Coulomb excitation	84	^{34}Ar	$\varepsilon+\beta^+$ decay (0.84646 s)	330
$^{34}_{46}\text{Pd}_{-12}$	Adopted Levels, Gammas	85	^{34}Cl	IT decay (31.99 min)	334
$^{34}_{47}\text{Ag}_{-13}$	^{34}Si β^- decay (2.77 s)	97	^{35}K	εp decay (175 ms)	336
$^{34}_{48}\text{Cd}_{-14}$	^{18}O (^{18}O , $p\gamma$) E=20 MeV	99	^9Be (^{37}Ca , $^{34}\text{Cl}\gamma$)		339
$^{34}_{49}\text{In}_{-15}$	^{18}O (^{18}O , $p\gamma$) E=24 MeV	101	^{24}Mg (^{12}C , $p\gamma$)		341
$^{34}_{50}\text{Sn}_{-16}$	^{18}O (^{18}O , $p\gamma$) E=25 MeV	107	^{27}Al (^{12}C , $\alpha n\gamma$)		344
$^{34}_{51}\text{Sb}_{-17}$	^{18}O (^{18}O , $p\gamma$) E=30 MeV	109	^{31}P ($\alpha, n\gamma$)		352
$^{34}_{52}\text{Te}_{-18}$	^{18}O (^{18}O , $p\gamma$) E=34 MeV	111	^{32}S ($^3\text{He}, p$)		355
$^{34}_{53}\text{I}_{-19}$	^{18}O (^{18}O , $p\gamma$) E=44 MeV	114	^{32}S ($^3\text{He}, p\gamma$)		356
$^{34}_{54}\text{Xe}_{-20}$	^{30}Si (^7Li , ^3He)	116	^{32}S (α, d)		363
$^{34}_{55}\text{Ba}_{-21}$	^{34}S (t , ^3He)	117	^{33}S (p, γ)		365
$^{34}_{56}\text{La}_{-22}$	^{34}S (^7Li , ^7Be), (^{11}B , ^{11}C)	118	^{33}S (p, p), ($p, p'\gamma$):resonances		425
$^{34}_{57}\text{Ce}_{-23}$	^{36}S (pol d, α), (POL $d, \alpha\gamma$)	119	^{33}S ($^3\text{He}, d$)		430
$^{34}_{58}\text{Pr}_{-24}$	^{115}In (^{34}S , $X\gamma$)	122	^{34}S (p, n)		432
$^{34}_{59}\text{Nd}_{-25}$	^{160}Gd (^{37}Cl , $X\gamma$)	124	^{34}S ($^3\text{He}, t$)		433
$^{34}_{60}\text{Pm}_{-26}$	^{176}Yb (^{36}S , $X\gamma$)	125	^{34}S (^6Li , ^6He)		435
$^{34}_{61}\text{Sm}_{-27}$	^{198}Pt (^{37}Cl , $X\gamma$)	127	^{35}Cl (γ, n)		436
$^{34}_{62}\text{Eu}_{-28}$	^{208}Pb (^{36}S , $^{34}\text{P}\gamma$)	128	^{35}Cl (p, d)		437
$^{34}_{63}\text{Gd}_{-29}$	Coulomb excitation	130	^{35}Cl (d, t)		438
$^{34}_{64}\text{Ter}_{-30}$			^{35}Cl ($^3\text{He}, \alpha$)		439
$^{34}_{65}\text{Er}_{-31}$			^{36}Ar (p , ^3He)		440
$^{34}_{66}\text{Yb}_{-32}$			^{36}Ar (d, α), (pol d, α)		441
$^{34}_{67}\text{Lu}_{-33}$			^{40}Ca (μ^- , $\nu\alpha 2n\gamma$)		442

$^{34}_{18}\text{Ar}$	444	$^{32}\text{S}(^3\text{He},\text{n})$	459
Adopted Levels, Gammas	444	$^{32}\text{S}(^3\text{He},\text{n}\gamma)$	460
^{35}Ca ep decay (25.7 ms)	449	$^{36}\text{Ar}(\text{p},\text{t})$	463
$^1\text{H}(^{35}\text{Cl},2\text{n})$	452	Coulomb excitation	465
$^1\text{H}(^{34}\text{Ar},\text{p}')$	453	$^{34}_{19}\text{K}_{15}$	466
$^3\text{He}(^{32}\text{S},\text{n}\gamma)$	454	Adopted Levels	466
$^4\text{He}(^{30}\text{S},\alpha)$:resonances	455	$^9\text{Be}(^{37}\text{Ca},\text{X})$	467
$^9\text{Be}(^{37}\text{Ca},^{34}\text{Ar}\gamma)$	456	$^{34}_{20}\text{Ca}_{14}$	468
$^{12}\text{C}(^{24}\text{Mg},^{34}\text{Ar}\gamma)$	457	Adopted Levels: not observed	468
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Skeleton Scheme for A=34

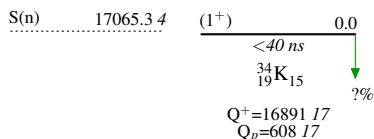


Skeleton Scheme for A=34 (continued)

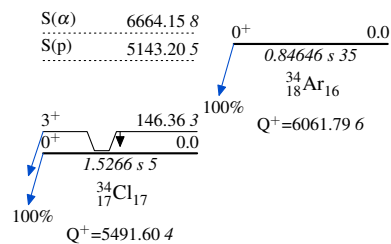


$S(n)$ 16830 syst

$S(\alpha)$ 8587 29



$S(\alpha)$ 6743.95 22
 $S(p)$ 4663.9 4
 $S(n)$ 11508.1 4



Ground-State and Isomeric-Level Properties

Nuclide	Level	$J\pi$	$T_{1/2}$	Decay Mode
^{34}Ne	0.0	0^+		$\% \beta^- = ?$; $\% \beta^- n = ?$; $\% \beta^- 2n = ? \dots$
^{34}Na	0.0	(2^-)	5.5 ms 10	$\% \beta^- = 100$; $\% \beta^- n = ?$; $\% \beta^- 2n > 0 \dots$
^{34}Mg	0.0	0^+	44.9 ms 4	$\% \beta^- = 100$; $\% \beta^- n = 21\ 7$; $\% \beta^- 2n < 0.1$
^{34}Al	0.0	4^-	53.73 ms 25	$\% \beta^- = 100$; $\% \beta^- n = 22\ 5$
^{34}Al	46.56	1^+	23.2 ms 14	$\% \beta^- = 100$; $\% \beta^- n = 11\ 4$
^{34}Si	0.0	0^+	2.77 s 20	$\% \beta^- = 100$
^{34}P	0.0	1^+	12.43 s 10	$\% \beta^- = 100$
^{34}S	0.0	0^+	stable	
^{34}Cl	0.0	0^+	1.5266 s 5	$\% \epsilon + \% \beta^+ = 100$
^{34}Cl	146.36	3^+	31.99 min 3	$\% \text{IT} = 46.0\ 4$; $\% \epsilon + \% \beta^+ = 54.0\ 4$
^{34}Ar	0.0	0^+	0.84646 s 35	$\% \epsilon + \% \beta^+ = 100$
^{34}K	0.0	(1^+)	$<40\text{ ns}$	$\% p = ?$
^{35}Na	0.0	$(3/2^+)$	2.1 ms 4	$\% \beta^- n > 0$
^{36}Mg	0.0	0^+	6.9 ms +7-8	$\% \beta^- 2n = ?$
^{35}Al	0.0	$(5/2^+)$	38.1 ms 4	$\% \beta^- n = 37\ 3$
^{35}K	0.0	$3/2^+$	175 ms 2	$\% \epsilon p = 0.37\ 15$
^{35}Ca	0.0	$1/2^+$	25.7 ms 2	$\% \epsilon p = 95.8\ 3$

Adopted Levels

$Q(\beta^-)=21160$ syst; $S(n)=1360$ syst [2021Wa16](#)

$\Delta Q(\beta^-)=790$, $\Delta S(n)=790$ (syst, [2021Wa16](#)).

$S(2n)=300$ 100, $Q(\beta^-n)=20990$ 680 (syst, [2021Wa16](#)).

Isotope discovery ([2012Th10](#)): $\text{Ta}(^{48}\text{Ca},X)$ at RIKEN ([2002No11](#)) and $\text{Ta}(^{48}\text{Ca},X)$ at GANIL ([2002Lu09](#)).

[2002No11](#): $\text{Ta}(^{48}\text{Ca},X)$ $E=64$ MeV/nucleon at RIKEN. Projectile fragments were separated and identified by the large-acceptance RIKEN Projectile-Fragment Separator (RIPS). Measured production yield and deduced production cross section of ^{34}Ne to be ≈ 0.1 pb. Also see annual report [2002NoZY](#).

[2002Lu09](#): $\text{Ta}(^{48}\text{Ca},X)$ $E=59.8$ MeV/nucleon at GANIL. Projectile fragments were separated and identified by the LISE-2000 fragment separator. Measured production yield and deduced production cross section of ^{34}Ne to be 0.17 pb 12. Also see [2002LuZT](#), [2002Lu19](#), [2003Lu11](#), and [2004Lu19](#).

[2019Ah07](#): $^9\text{Be}(^{48}\text{Ca},X)$ $E=345$ MeV/nucleon, 450-pnA ^{48}Ca beam at RIKEN. Projectile fragments were separated and identified using ΔE -ToF- $B\rho$ by the large-acceptance two-stage separator BigRIPS. Optimum settings of $B\rho$ were tuned to transmit ^{33}F for 14 hours and $^{36}\text{Ne}+^{39}\text{Na}$ for 7.8 hours. The Be target was irradiated with 1.4×10^{17} and 7.8×10^{16} ^{48}Ca ions, respectively. Measured Z vs A/Z particle-identification plot. 115 ^{34}Ne events were observed in the ^{33}F setting, and 4 ^{34}Ne events were observed in the $^{36}\text{Ne}+^{39}\text{Na}$ setting. No ^{35}Ne or ^{36}Ne events were observed in either setting.

[2022Ah02](#): same experimental setup as [2019Ah07](#) with 540-pnA ^{48}Ca beam. Optimum settings of $B\rho$ were tuned to transmit ^{39}Na for 46.1 hours and ^{36}Ne for 25.3 hours. Measured Z vs A/Z particle-identification plot. Some ^{34}Ne events were observed in the ^{36}Ne setting. No ^{35}Ne or ^{36}Ne events were observed in either setting.

[2025Ku15](#): review of two RIKEN experiments, [2019Ah07](#) and [2022Ah02](#). No events for ^{32}F , ^{33}F , ^{35}Ne , ^{36}Ne , and ^{38}Na were observed. The existence of bound ^{32}F , ^{33}F , ^{35}Ne , ^{36}Ne , and ^{38}Na can be excluded with high confidence, based on zero-event probabilities calculated from expected yields and the Poisson distribution. Authors conclude that the most neutron-rich and heaviest bound nuclei of fluorine and neon, corresponding to the dripline nuclei, are ^{31}F and ^{34}Ne , respectively.

 ^{34}Ne Levels

<u>E(level)</u>	<u>J^π</u>	<u>Comments</u>
0?	0 ⁺	$\% \beta^- = ?$; $\% \beta^- n = ?$; $\% \beta^- 2n = ?$; $\% \beta^- 3n = ?$ $T_{1/2}$: 2021Ko07 gives an estimate of 2 ms and cites $>1.5 \mu\text{s}$ from private communication newsletter of the French National Institute of Nuclear Physics and Particle Physics (IN2P3). 2025Ku15 introduced experimental plans for nuclear structure of ^{34}Ne and nearby nuclei at RIKEN RIBF. Experimental $T_{1/2}$ and $\% \beta^- n$ are unknown. Theoretical $T_{1/2}=4.7$ ms, $\% \beta^- 0n=9$, $\% \beta^- 1n=60$, $\% \beta^- 2n=23$, $\% \beta^- 3n=6$ (2021Mi17). Theoretical $T_{1/2}=3.7$ ms, $\% \beta^- 0n=4$, $\% \beta^- 1n=75$, $\% \beta^- 2n=15$, $\% \beta^- 3n=6$ (2019Mo01).

Adopted Levels, Gammas

$Q(\beta^-)=2.336\times 10^4$ 60; $S(n)=-1\times 10^1$ 60; $S(p)=21740$ syst; $Q(\alpha)=-19710$ syst 2021Wa16

$S(n)$: Deduced by evaluators using mass excesses of 23598.8 43 for ^{33}Na measured by 2025Ly01; 31680 600 for ^{34}Na from

2021Wa16. Value from 2021Wa16: $S(n)=170$ 750.

$\Delta S(p)=850$, $\Delta Q(\alpha)=780$ (syst, 2021Wa16).

$Q(\beta^-n)=18650$ 600 (2021Wa16).

Isotope discovery (2012Th10): Ir(p,X) at CERN (1983La12).

^{34}Na production: 1997Ha11: U(p,X) E=1 GeV at REX-ISOLDE CERN. Measured yield. Others: 1981To19, 1979De02, 1979Ra37.

^{34}Na decay measurements:

1983La12, 1983LaZU, 1984La03: Ir(p,X) E=10 GeV at CERN. Measured $T_{1/2}$ and β^- -delayed neutrons.

1984Gu19: Ir(p,X) E=10 GeV at CERN. Observed a 885-keV γ ray of ^{32}Mg in ^{34}Na β decay.

^{34}Na states may be populated by ^{34}Ne β^- decay. No decay data are available.

 ^{34}Na LevelsCross Reference (XREF) Flags

A C(^{36}Mg , ^{34}Na)

<u>E(level)</u>	<u>J^π</u>	<u>$T_{1/2}$</u>	<u>XREF</u>	<u>Comments</u>
0	(2 ⁻)	5.5 ms 10	A	$\% \beta^- = 100$; $\% \beta^- n = ?$; $\% \beta^- 2n > 0$; $\% \beta^- 3n = ?$ J^π : from shell-model calculations with the unmodified SDPF-M interaction (2014Do05). $T_{1/2}$: from decay curve of βn -coin (1983La12, 1984La03). ^{34}Na β^- -delayed neutrons have been observed by 1983La12 and 1984La03. An 885-keV γ ray of ^{32}Mg populated by ^{34}Na β^- -delayed two-neutron decay has been observed by 1984Gu19. 1984La03 reports $P_n=115$ 20, which is quoted as $P_n=113$ 22 by 1984Gu19. Theoretical $\% \beta^- 0n=5$, $\% \beta^- 1n=45$, $\% \beta^- 2n=48$, $\% \beta^- 3n=2$ (2021Mi17). Theoretical $\% \beta^- 0n=37$, $\% \beta^- 1n=43$, $\% \beta^- 2n=19$, $\% \beta^- 3n=1$ (2019Mo01). XREF: A(451?) The SDPF-M shell model predicts yrast states at 90 (4 ⁻), 106 (3 ⁻), 289 (0 ⁺), 297 (1 ⁻), 527 (5 ⁻), 583 (0 ⁻), 641 (2 ⁺), 777 (1 ⁺), 956 (3 ⁺), 1392 (4 ⁺), and 2110 (5 ⁺) keV. Due to the high level density, the interpretation of the observed γ ray at 451 keV 7 remains ambiguous (2014Do05).
451? 7			A	

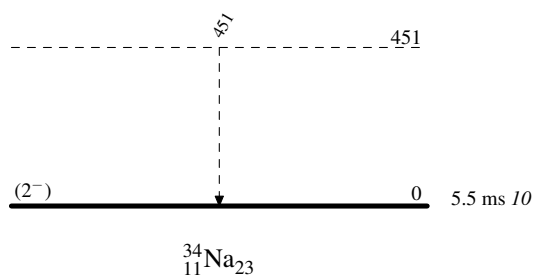
 $\gamma(^{34}\text{Na})$

<u>$E_i(\text{level})$</u>	<u>E_γ</u>	<u>E_f</u>	<u>J_f^π</u>
451?	451 [†] 7	0	(2 ⁻)

[†] Placement of transition in the level scheme is uncertain.

Adopted Levels, Gammas

Legend

Level Scheme-----► γ Decay (Uncertain)

$\text{C}(^{36}\text{Mg}, ^{34}\text{Na}\gamma)$ 2014Do05

2014Do05: a 345-MeV/nucleon ^{48}Ca primary beam was accelerated by the superconducting ring cyclotron (SRC) at RIKEN. A 236-MeV/nucleon ^{36}Mg secondary beam was produced by the fragmentation of ^{48}Ca on a Be target and separated by the BigRIPS separator using the B ρ - Δ E-B ρ method. The secondary reaction targets were 2.54-g/cm²-thick carbon and 2.13-g/cm²-thick CH₂ polyethylene. The secondary reaction products ^{33}Na , ^{34}Na , and ^{35}Na were identified by the magnetic spectrometer ZeroDegree using the B ρ - Δ E-ToF method. γ rays in coincidence with ^{34}Na were detected using the DALI2 array of 186 large NaI(Tl) detectors. Measured E_γ with Doppler correction, I_γ , and $(^{34}\text{Na})\gamma$ -coin. Deduced levels. Compared with shell-model calculations using the SDPF-M effective interaction.

 ^{34}Na Levels

<u>E(level)[†]</u>	<u>J^{π}</u>	<u>Comments</u>
0	(2 ⁻)	J ^{π} : from shell-model calculations with the unmodified SDPF-M interaction (2014Do05).
451? 7		The SDPF-M shell model predicts yrast states at 90 (4 ⁻), 106 (3 ⁻), 289 (0 ⁺), 297 (1 ⁻), 527 (5 ⁻), 583 (0 ⁻), 641 (2 ⁺), 777 (1 ⁺), 956 (3 ⁺), 1392 (4 ⁺), and 2110 (5 ⁺) keV. Due to the high level density, the interpretation of the observed γ ray at 451 keV 7 remains ambiguous (2014Do05).

[†] From E_γ data in 2014Do05.

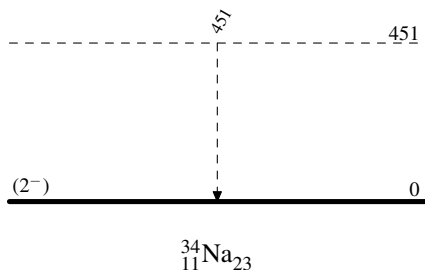
 $\gamma(^{34}\text{Na})$

<u>E_{γ}</u>	<u>E_i(level)</u>	<u>E_f</u>	<u>J^{π}_f</u>
451 [†] 7	451?	0	(2 ⁻)

[†] Placement of transition in the level scheme is uncertain.

 $\text{C}(^{36}\text{Mg}, ^{34}\text{Na}\gamma)$ 2014Do05

Legend

Level Scheme- - - - - ► γ Decay (Uncertain)

Adopted Levels, Gammas

$Q(\beta^-)=11321$ 7; $S(n)=4711$ 7; $S(p)=22564.5$ 81; $Q(\alpha)=-1.738\times 10^4$ 25 [2021Wa16](#)
 $S(p)$: Deduced by the evaluator using mass excesses of 23598.8 43 for ^{33}Na measured by [2025Ly01](#); 8323 7 for ^{34}Mg from [2021Wa16](#). Value from [2021Wa16](#): $S(p)=22750$ 450.
 $S(2n)=6990$ 8, $S(2p)=43250$ 500 (syst), $Q(\beta^-n)=8749$ 10 ([2021Wa16](#)).
 Isotope discovery ([2012Th10](#)): $^9\text{Be}(^{48}\text{Ca},X)$ $E=212$ MeV/nucleon at Berkeley ([1979We10](#)).
 ^{34}Mg production: [2015Mo17](#), [2012Kw02](#), [2007No13](#).
 ^{34}Mg decay measurements: [2019Li41](#), [2017Li03](#), [1984La03](#), [1984Gu19](#).
 ^{34}Mg mass measurements: [2025Ly01](#), [2019As04](#), [2013Ch49](#), [2007Ju03](#), [2001Sa72](#), [2001Sa21](#), [1991Or01](#), and [1991Zh24](#).
 ^{34}Mg radius measurements: [2011Ka01](#), [2006Kh08](#).
 Theoretical calculations: [2026Ji03](#), [2025Zh24](#), [2024Ha15](#), [2024Ab03](#), [2023Ra22](#), [2023Ba33](#), [2022Yo04](#), [2022Ot01](#), [2022Mo10](#), [2021Ka07](#), [2020Mi15](#).

 ^{34}Mg LevelsCross Reference (XREF) Flags

A	$^{35}\text{Na } \beta^-n$ decay (2.1 ms)	D	$\text{C}(^{37}\text{Al}, ^{34}\text{Mg}\gamma)$
B	$^1\text{H}(^{34}\text{Mg}, ^{34}\text{Mg}'\gamma)$	E	Coulomb excitation
C	$^9\text{Be}(^{36}\text{Si}, ^{34}\text{Mg}\gamma)$		

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	XREF	Comments
0	0^+	44.9 ms 4	ABCDE	$\% \beta^- = 100$; $\% \beta^-n = 21$ 7 (2017Li03); $\% \beta^- 2n < 0.1$ (2017Li03 , 2019Li41) $T_{1/2}$: from fit of β -delayed γ time distribution (2017Li03). Others: 20 ms 10 (1984La03). 63 ms 1 (2015RoZU) was discarded by 2017Li03 from the same group. Strong absorption $r_0^2 = 1.255$ fm ² 30 (2006Kh08). Interaction $\sigma = 1372$ mb 46 for ^{34}Mg on carbon and 568 mb 90 on hydrogen (2011Ka01). RMS matter radius = 3.23 fm 13 (2011Ka01). $B(E2)^\uparrow = 0.057$ 10 XREF: A(661?) $B(E2)^\uparrow$: weighted average of 0.054 10 (2005Ch66) and $B(E2) = 0.063$ 13 (2001Iw07). Other: ≤ 0.067 (1999Pr09). J^π : the only level strongly populated in Coulomb excitation, proton inelastic scattering, and fragmentation reactions. $T_{1/2}$: from $B(E2)^\uparrow = 0.057$ 10 in Coulomb excitation. XREF: B(2011)C(2120)E(2117?) J^π : Mg isotope systematics and shell-model calculations (2013Do22). XREF: B(3194)D(3131) J^π : 3130 γ to 0^+ g.s.
657 4	2^+	41 ps +9-6	ABCDE	
2043 19	(4^+)		BCDE	
3146 21	$(1, 2^+)$		B D	

[†] From a least-squares fit to γ -ray energies.

 $\gamma(^{34}\text{Mg})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	I_γ	E_f	J_f^π	Mult.	$\alpha^\ddagger$	Comments
657	2^+	657 4	100	0	0^+	[E2]	6.58×10^{-5} 15	$B(E2)(\text{W.u.}) = 17.2$ 30 E_γ : weighted average of 658 4 from $^1\text{H}(^{34}\text{Mg}, ^{34}\text{Mg}'\gamma)$, 660 10 from $^9\text{Be}(^{36}\text{Si}, ^{34}\text{Mg}\gamma)$, 652 6 from $\text{C}(^{37}\text{Al}, ^{34}\text{Mg}\gamma)$, and 657 7 from Coulomb excitation. E_γ : weighted average of 1353 28 from
2043	(4^+)	1386 18	100	657	2^+			

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) $\gamma(^{34}\text{Mg})$ (continued)

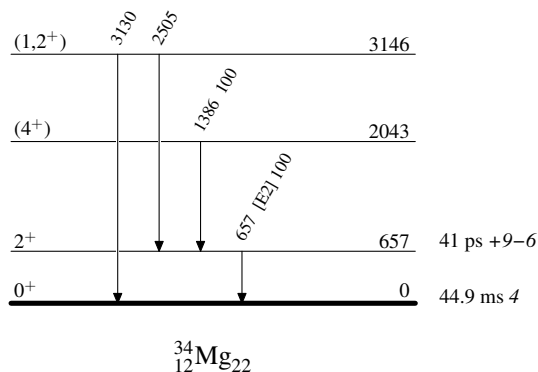
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^\dagger$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
					$^1\text{H}(^{34}\text{Mg}, ^{34}\text{Mg}'\gamma)$ and 1395 15 from $\text{C}(^{37}\text{Al}, ^{34}\text{Mg}\gamma)$. Other: 1460 20 from $^9\text{Be}(^{36}\text{Si}, ^{34}\text{Mg}\gamma)$. 2013Do22 explains that the E_γ discrepancy may originate from the low statistics and limited mass resolution in previous work 2001Yo03 in $^9\text{Be}(^{36}\text{Si}, ^{34}\text{Mg}\gamma)$.
3146	(1,2 ⁺)	2505 30	657	2 ⁺	E_γ : weighted average of 2536 34 from $^1\text{H}(^{34}\text{Mg}, ^{34}\text{Mg}'\gamma)$ and 2480 30 from $\text{C}(^{37}\text{Al}, ^{34}\text{Mg}\gamma)$.
		3130 30	0	0 ⁺	

[†] From $\text{C}(^{37}\text{Al}, ^{34}\text{Mg}\gamma)$, unless otherwise noted.

[‡] Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multiplicities, and mixing ratios, unless otherwise specified.

Adopted Levels, GammasLevel Scheme

Intensities: Relative photon branching from each level



^{35}Na β^- n decay (2.1 ms) 2013StZY,2022Cr03,1983La12

Parent: ^{35}Na : $E=0$; $J^\pi=(3/2^+)$; $T_{1/2}=2.1$ ms 4; $Q(\beta^-n)=21440$ syst; $\% \beta^-n$ decay >0

^{35}Na - J^π , $T_{1/2}$: From the Adopted Levels of ^{35}Na (2026Su18).

^{35}Na - $Q(\beta^-n)$: 21440 670 (syst, 2021Wa16).

^{35}Na - $\% \beta^-n$ decay: Experimental $\% \beta^-n$ is unknown.

2013StZY: ^{35}Na was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA $^{48}\text{Ca}^{20+}$ primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on a 15-mm-thick ^9Be target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the $B\rho$ - ΔE -ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beamline. To reduce background buildup, the scintillator barrel was rotated quickly and axially moved slowly in vertical direction, resulting in a helix-shaped motion. β particles were detected by the CAITEN and γ rays were detected using three HPGe clover detectors. Measured E_γ , $\beta\gamma$ -coin, and implant- β correlation, and deduced the $T_{1/2}$ of ^{35}Na . Comparisons with QRPA and shell-model calculations.

2022Cr03: ^{35}Na was produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA $^{48}\text{Ca}^{20+}$ primary beam from the FRIB linac impinging on an 8.89-mm-thick ^9Be target. The secondary cocktail beam centered around ^{42}Si was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing LaBr_3 detectors, and the VANDLE array of 88 neutron detectors. Two Si PIN detectors and a plastic scintillator were placed 1.5 m upstream in the last diagnostic detector box of the beamline. Ions were identified event by event by energy loss in the upstream PIN detector (Z determination) and time of flight between a plastic timing scintillator at the start of stage 3 of ARIS and the timing scintillator in the diagnostic detector box (A/Q determination) over a flight path of 33.5 m. Measured implant- β correlation and deduced $T_{1/2}$ of ^{35}Na .

Comparisons with QRPA and shell-model calculations.

1983La12, 1984La03: ^{35}Na was produced via fragmentation of a 30 g/cm² iridium target by 10 GeV protons from the CERN synchrotron. The recoiling nuclear fragments were thermalized in heated graphite. The alkali elements selectively diffused out of the target and were surface ionized and analyzed by the online mass spectrometer, then focused at the bottom of a well-shaped NE213 liquid scintillator neutron detector. β particles were detected using a well-shaped, thin plastic scintillator. Measured β and neutron counts. Deduced the $T_{1/2}$ of ^{35}Na . P_n values were deduced for other isotopes but not for ^{35}Na .

 ^{34}Mg Levels

<u>E(level)</u>	<u>$J^\pi^\dagger$</u>
0	0^+
661?	2^+

† From the Adopted Levels.

 $\gamma(^{34}\text{Mg})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
661 †	661?	2^+	0	0^+	2013StZY observed this γ ray in the background-subtracted γ -ray spectrum in a time window of 0-10 ms after the implantation of ^{35}Na but did not observe it in later time windows. 2013StZY proposed that it could be either the deexcitation of the first 2^+ level to 0^+ g.s. in ^{34}Mg or the deexcitation of an excited state in ^{35}Mg . Based on theoretical calculations (2019Mo01, 2021Mi17), the β^-1n should be the dominant branch, implying that this γ ray is more likely from ^{34}Mg .

† Placement of transition in the level scheme is uncertain.

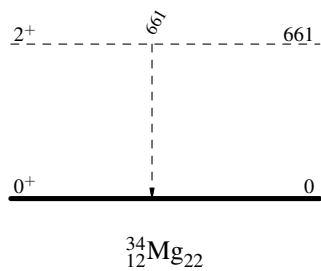
^{35}Na β^- -n decay (2.1 ms) 2013StZY,2022Cr03,1983La12

Legend

Decay Scheme

 -----► γ Decay (Uncertain)

$\% \beta^- n > 0$ $\nearrow$ $\begin{matrix} (3/2^+) & 0 \\ \hline Q=21440_{\text{syst}} \\ ^{35}_{11}\text{Na}_{24} \end{matrix}$ 2.1 ms 4



$^1\text{H}(^{34}\text{Mg}, ^{34}\text{Mg}'\gamma)$ 2014Mi09,2006EI03

- 2014Mi09:** a 63-MeV/nucleon ^{48}Ca primary beam was accelerated by the RIKEN RFQ-RILAC-CSM-RRC accelerator complex. Secondary beams containing ^{34}Mg were produced by fragmentation of the primary beam on ^{181}Ta and enriched ^{64}Ni targets, separated by the RIPS fragment separator, and bombarded the liquid hydrogen secondary target at 51.1 MeV/nucleon. The scattered particles were analyzed by the TOMBEE spectrometer and detected using a $\Delta\text{E-E}$ telescope of a silicon detector and a NaI(Tl) detector. The γ rays from the excited states populated by inelastic scattering were detected by the DALI2 array of 160 NaI(Tl) detectors. Measured E_γ with Doppler correction, γ yields, $(^{34}\text{Mg})\gamma$ -coin, and $\gamma\gamma$ -coin. Deduced levels, cross sections for population of excited states, deformation lengths, and deformation parameters β_2 . Compared with SDPF-M shell-model calculations.
- 2006EI03:** a 94-MeV/nucleon ^{40}Ar primary beam was transported to a ^{181}Ta primary target. The secondary beams of ^{30}Na , ^{31}Na , and ^{34}Mg were separated by the RIPS fragment separator and bombarded a liquid hydrogen target at a mean energy of 50 MeV/nucleon. The reaction products were detected and identified by a parallel-plate avalanche counter (PPAC) and a silicon detector telescope. The deexciting γ rays emitted by the inelastically scattered nuclei were detected by the DALI2 setup consisting of 146 NaI(Tl) detectors. Deduced mass deformation and neutron deformation parameters.

 ^{34}Mg Levels

$E(\text{level})^\dagger$	Comments
0	
658 4	Population $\sigma=63$ mb 5 (2014Mi09). Total population including feeding from higher state $\sigma=78$ mb 6 (2014Mi09) and $\sigma=111$ mb 37 (2006EI03). Deformation length $\delta=2.30$ fm $+9-10$ (stat) 16 (syst) (2014Mi09). $\beta_2=0.62$ 3 (stat) 5 (syst) (2014Mi09). Matter or mass deformation parameter $\beta_M=0.68$ 16 (2006EI03). Neutron deformation parameter $\beta_n=0.70$ 13 (2006EI03) using charge deformation parameter $\beta_C=0.58$ 6 from 2001Iw07.
2011 28	Population $\sigma=5$ mb 2 (2014Mi09).
3194 34	Population $\sigma=10$ mb 2 (2014Mi09).

† From a least-squares fit to γ -ray energies.

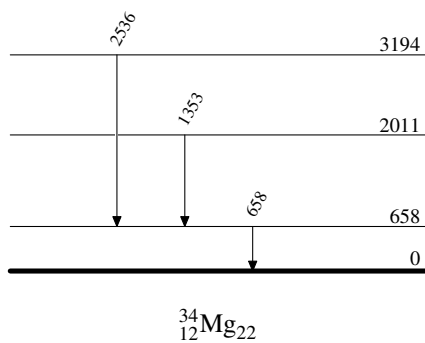
 $\gamma(^{34}\text{Mg})$

$E_\gamma^\dagger$	$E_i(\text{level})$	E_f	Comments
658 4	658	0	E_γ : other: 685 16 (2006EI03).
1353 28	2011	658	E_γ : 1353 $+29-27$ in 2014Mi09.
2536 34	3194	658	E_γ : 2536 $+34-33$ in 2014Mi09.

† From 2014Mi09.

$^1\text{H}(^{34}\text{Mg}, ^{34}\text{Mg}'\gamma)$ 2014Mi09,2006El03

Level Scheme



$^9\text{Be}(^{36}\text{Si}, ^{34}\text{Mg}\gamma)$ 2001Yo03

[2001Yo03](#), [2002Yo04](#): a 95-MeV/nucleon, 60-pnA primary beam of ^{40}Ar bombarded a 462.5-mg/cm² ^9Be primary target. A secondary beam of ^{36}Si was separated by the RIKEN projectile-fragment separator RIPS and transported to a secondary target of 385.5-mg/cm² ^9Be at a center-of-target energy of 38 MeV/nucleon. Reaction products from the $^{36}\text{Si}+^9\text{Be}$ fragmentation were detected and identified at forward angles with a parallel-plate avalanche counter (PPAC) and four sets of counter telescopes, each consisting of three silicon detectors and one Si(Li) detector. γ rays emitted from excited fragments were detected using 66 NaI(Tl) detectors. Measured E_γ with Doppler correction, I_γ , $(^{34}\text{Mg})\gamma$ -coin, and $\gamma\gamma$ -coin. Deduced levels. Also see review articles [2002Sa11](#) and [2002Mo35](#).

Advantages of using two-step fragmentations of $^9\text{Be}(^{40}\text{Ar}, X)$ and $^9\text{Be}(^{36}\text{Si}, X)$: when ^{40}Ar is the incident beam, the γ -ray yield for ^{34}Mg is low due to a small production cross-section, requiring high beam luminosity and would cause a heavy detection load. In contrast, using ^{36}Si , which is closer to ^{34}Mg , offers a larger production cross-section, allowing for more intense primary beams ([2002Mo35](#)).

 ^{34}Mg Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>
0	0^+
660 10	(2^+)
2120 23	(4^+)

[†] From E_γ data of [2001Yo03](#). The energies of the first 4^+ level and the first 2^+ level yield an $R_{4/2}=3.2$, which is very close to 10/3 expected for a rigid rotor and ^{34}Mg might be a nucleus with well-developed quadrupole deformation ([2001Yo03](#)).

[‡] Assigned based on fragmentation reaction systematics of magnesium isotopes ([2001Yo03](#)).

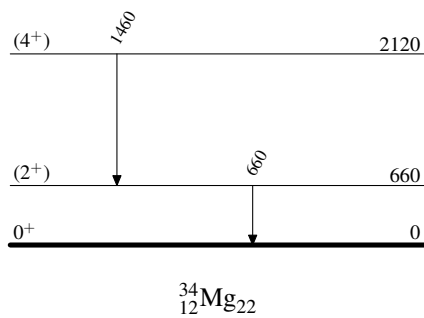
 $\gamma(^{34}\text{Mg})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
660 10	660	(2^+)	0	0^+
1460 20	2120	(4^+)	660	(2^+)

[†] From [2001Yo03](#).

$^9\text{Be}(^{36}\text{Si}, ^{34}\text{Mg}\gamma)$ 2001Yo03

Level Scheme



$^{34}_{12}\text{Mg}_{22}$

$\text{C}(^{37}\text{Al}, ^{34}\text{Mg}\gamma)$ 2013Do22

2013Do22: a 345-MeV/nucleon ^{48}Ca primary beam was accelerated by the superconducting ring cyclotron (SRC) at RIKEN. Secondary beams of $^{39}\text{Al}+^{40}\text{Si}$ and $^{37}\text{Al}+^{36}\text{Mg}$ were produced by the fragmentation of ^{48}Ca on a Be target and separated by the BigRIPS separator using the $\text{B}\rho\text{-}\Delta\text{E}\text{-B}\rho$ method. The secondary reaction target was 2.54-g/cm² thick carbon. The secondary reaction products ^{38}Mg and $^{34}\text{Mg}+^{36}\text{Mg}$ in another setting were identified by the magnetic spectrometer ZeroDegree using the $\text{B}\rho\text{-}\Delta\text{E}\text{-ToF}$ method. γ rays in coincidence with ^{34}Mg were detected using the DALI2 array of 186 large NaI(Tl) detectors. Measured E_γ with Doppler correction, I_γ , $(^{34}\text{Mg})\gamma\text{-coin}$, and $\gamma\gamma\text{-coin}$. Deduced levels. Compared with shell-model calculations using the SDPF-M effective interaction.

 ^{34}Mg Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$
0	0^+
652 6	(2^+)
2047 16	(4^+)
3131 22	

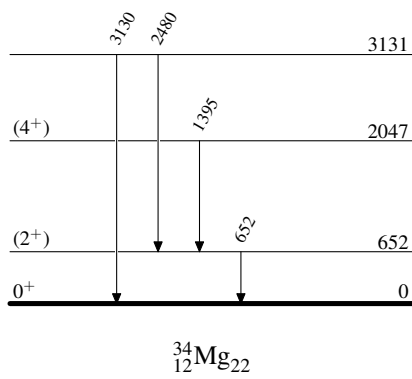
[†] From a least-squares fit to γ -ray energies. The energies of the first 4^+ level and the first 2^+ level yield an $R_{4/2}=3.14\ 5$, which demonstrates that ^{34}Mg is strongly deformed (2013Do22).

[‡] No direct evidence for definite spin assignments was obtained from the experiment but J^π can be assigned based on fragmentation reaction systematics of magnesium isotopes and shell-model predictions (2013Do22).

 $\gamma(^{34}\text{Mg})$

$E_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π
652 6	652	(2^+)	0	0^+
1395 15	2047	(4^+)	652	(2^+)
2480 30	3131		652	(2^+)
3130 30	3131		0	0^+

[†] From 2013Do22.

$\text{C}(^{37}\text{Al}, ^{34}\text{Mg}\gamma)$ 2013Do22Level Scheme

Coulomb excitation 2005Ch66,2001Iw07

- 2005Ch66:** $^{209}\text{Bi}(^{34}\text{Mg}, ^{34}\text{Mg}'\gamma)$. $E=76.4$ MeV/nucleon radioactive beam of ^{34}Mg was produced by fragmentation of a 110-MeV/nucleon ^{48}Ca primary beam on a ^9Be primary target and separated by the A1900 fragment separator at NSCL, MSU. The secondary target was 980 mg/cm² Bi. The γ rays were detected using an array of 24 position-sensitive trapezoidal NaI(Tl) detectors. Measured E_γ , I_γ , and scattered- $^{34}\text{Mg}\gamma$ -coin. Deduced cross sections for population of excited states, reduced electric quadrupole transition probabilities $B(E2)$, and deformation parameters β_2 .
- 2001Iw07:** $^{208}\text{Pb}(^{34}\text{Mg}, ^{34}\text{Mg}'\gamma)$. $E=44.9$ MeV/nucleon radioactive beam of ^{34}Mg was produced by fragmentation of a 95-MeV/nucleon ^{40}Ar primary beam on a ^9Be primary target and separated by the RIKEN RIPS fragment separator. The secondary target was 693 mg/cm² Pb. The γ rays were detected using a granular array of 66 NaI(Tl) detectors. Measured E_γ , I_γ , and scattered- $^{34}\text{Mg}\gamma$ -coin. Deduced cross sections for population of excited states and $B(E2)$. Also see [2002Mo35](#).
- 1999Pr09:** $^{197}\text{Au}(^{34}\text{Mg}, ^{34}\text{Mg}'\gamma)$. $E=50.6$ MeV/nucleon radioactive beam of ^{34}Mg was produced by fragmentation of an 80-MeV/nucleon ^{48}Ar primary beam on a ^9Be primary target and separated by the A1200 fragment separator at NSCL, MSU. The secondary target was 702-mg/cm² Au. Beams passing through the target were detected and stopped in a cylindrical fast-slow plastic phoswich detector located at zero degrees. The γ rays were detected using the NSCL NaI(Tl) array. Measured E_γ , I_γ , and (scattered- $^{34}\text{Mg}\gamma$)-coin. Deduced cross sections for population of excited states and $B(E2)$. Also see [2002Gl01](#) and [2000PrZX](#).

 ^{34}Mg Levels

$E(\text{level})^\dagger$	Comments
0	
657 7	$B(E2)\uparrow=0.057$ 10 $B(E2)\uparrow$: weighted average of 0.054 10 (2005Ch66) and 0.063 13 (2001Iw07). Other: ≤ 0.067 (1999Pr09). $B(E2)=0.054$ 10 (2005Ch66) from $\sigma=126$ mb 22 if no correction for feeding from higher-lying 2116 level is applied. $B(E2)>0.044$ 8 if possible correction for feeding from the 2116 level is considered. $\beta_2=0.58$ 6 (2001Iw07), 0.54 5 (2005Ch66).
2117?	

† From E_γ data.

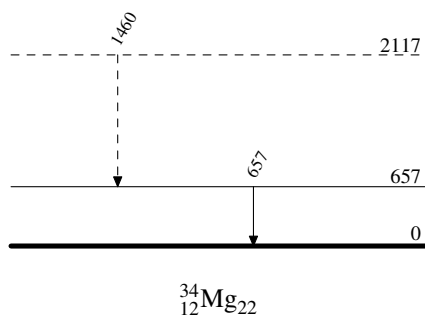
 $\gamma(^{34}\text{Mg})$

E_γ	$E_i(\text{level})$	E_f	Comments
657 7	657	0	E_γ : weighted average of 656 7 (2001Iw07) and 659 14 (2005Ch66). Other: 900-1400 assumed in 1999Pr09 .
1460 †	2117?	657	E_γ : quoted from 2001Yo03 . Not observed by any work in this dataset. An upper limit of intensity of this γ is estimated by 2005Ch66 to correct the possible feeding from the 2116 level.

† Placement of transition in the level scheme is uncertain.

Coulomb excitation 2005Ch66,2001Iw07

Legend

Level Scheme-----► γ Decay (Uncertain)

Adopted Levels, Gammas

$Q(\beta^-)=16994.1\ 23$; $S(n)=2572\ 7$; $S(p)=15249\ 3$; $Q(\alpha)=-13897\ 5$ [2021Wa16](#)

$S(2n)=8041\ 7$, $S(2p)=36220\ 40$, $Q(\beta^-n)=9445.4\ 22$ ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): $U(p,X)$ $E=800$ MeV at the Los Alamos Meson Physics Facility ([1977Bu11](#)).

^{34}Al production:

[2015Mo17](#): $^9\text{Be}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=95$ MeV/nucleon at RIKEN. Measured angular distributions and transverse momentum distributions of fragments. Deduced formulation for the width of transverse momentum distribution as a function of fragment velocity.

[2012Kw02](#): $E(^{40}\text{Ar})=140$ MeV/nucleon impinged on ^9Be , natural nickel, and ^{181}Ta targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

[2012Zh06](#): $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

[2007No13](#): $^9\text{Be}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=100$ MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

^{34}Al knockout-reaction measurements:

[2012No05](#): $^{33,34,35,36}\text{Al}$ produced by $^9\text{Be}(^{48}\text{Ca},X)$ fragmentation at $E(^{48}\text{Ca})=900$ MeV/nucleon at GSI. Measured $1n$ removal cross sections and longitudinal momentum distributions of the residues. Deduced single-particle occupancies in the ground states of $^{33,34,35}\text{Al}$. $\sigma(^{35}\text{Al}\rightarrow^{34}\text{Al})=75$ mb *4* and $\sigma(^{34}\text{Al}\rightarrow^{33}\text{Al})=81$ mb *4*.

[2010Ro23](#): ^{35}Al produced by $^9\text{Be}(^{40}\text{Ar},X)$ fragmentation at $E(^{40}\text{Ar})=700$ MeV/nucleon at GSI. Measured $1n$ knockout cross sections and longitudinal momentum distributions of the residues. $\sigma(^{35}\text{Al}\rightarrow^{34}\text{Al})=65$ mb *18* and $\sigma(^{34}\text{Al}\rightarrow^{33}\text{Al})=102$ mb *24*.

^{34}Al and ^{34m}Al decay measurements:

[2019Li41](#): $U(p,X)$ at CERN-ISOLDE. Measured $T_{1/2}$, β^- -delayed γ spectroscopy.

[2017Ha23](#), [2017Ha34](#): $^9\text{Be}(^{40}\text{Ar},^{34}\text{Al})$ at $E(^{40}\text{Ar})=95$ MeV/nucleon at HIRFL-RIBLL. Measured $T_{1/2}$, β^- -delayed γ spectroscopy.

[2012Ro25](#): $^9\text{Be}(^{36}\text{S},^{34}\text{Al})$ at $E(^{36}\text{S})=77.5$ MeV/nucleon at GANIL-LISE3. Measured $T_{1/2}$, β and γ spectroscopy.

[2001Nu01](#): $U(p,X)$ at $E=1$ GeV at CERN-ISOLDE. Measured $T_{1/2}$, β^- -delayed γ and neutron spectroscopy.

^{34}Al radius measurements:

[2006Kh08](#): ^{34}Al produced by $^{181}\text{Ta}(^{48}\text{Ca},X)$ fragmentation at $E(^{48}\text{Ca})=60.3$ MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

^{34}Al mass measurements:

[2019As04](#): ToF-ICR with ISOLTRAP, ISOLDE-CERN.

[2015Kw01](#): ToF-ICR with TITAN, TRIUMF.

Theoretical calculations:

[2025Zh57](#), [2024Sa36](#), [2022St03](#), [2013Li39](#).

 ^{34}Al LevelsCross Reference (XREF) Flags

A	^{34}Mg β^- decay (44.9 ms)	F	$\text{He}(^{34}\text{Al}, ^{34}\text{Al}'\gamma)$
B	^{36}Mg β^-2n decay (6.9 ms)	G	$\text{Pb}(^{35}\text{Al}, ^{34}\text{Al}n\gamma)$
C	$^2\text{H}(^{33}\text{Al}, ^{34}\text{Al}\gamma), (^{34}\text{Si}, ^{34}\text{Al}\gamma)$	H	$U(p, ^{34}\text{Al})$
D	$^9\text{Be}(^{36}\text{S}, ^{34}\text{Al})$	I	Coulomb excitation
E	$^9\text{Be}(^{40}\text{Ar}, ^{34}\text{Al})$		

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	XREF	Comments
0.0	4^-	53.73 ms <i>25</i>	ABCDEFGHI	$\% \beta^- = 100$; $\% \beta^-n = 22\ 5$ (2019Li41) $\mu = (+)2.157\ 16$ (2025StZZ , 2008Hi01) J^π : 4^- , 5^- from g factor measurement (2008Hi01); strong allowed β decay to a ^{34}Si spin=3, 4254.67 level. $T_{1/2}$: weighted average of 53.73 ms <i>13</i> in $U(p, ^{34}\text{Al})$ from 2019Li41 by fitting

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Adopted Levels, Gammas (continued) ^{34}Al Levels (continued)

E(level) [†]	J ^π	T _{1/2}	XREF	Comments
46.56 11	1 ⁺	23.2 ms 14	A DEFGH	time distribution gated on 929γ in ^{34}Si , 51.5 ms 9 in $^9\text{Be}(^{40}\text{Ar}, ^{34}\text{Al})$ from 2017Ha23 and 2017Ha34 by fitting implant-β-929γ decay curve, 54.4 ms 5 in $^9\text{Be}(^{36}\text{S}, ^{34}\text{Al})$ from 2012Ro25 by fitting β decay curve in coincidence with 926-keV β-delayed γ in ^{34}Si . Others: 56.3 ms 5 in U(p, ^{34}Al) from 2001Nu01 by β and γ counting rates, 50 ms 25 (1986Du07), 70 ms 25 (1988Mu08), and 42 ms 6 (1995ReZZ). μ: original (+)2.156 16 using β-NMR (2008Hi01). %β ⁻ n: from 2019Li41 . Others: 26 4 (2001Nu01), 27 5 (1989Ba50), 54 12 (1988Mu08), 12.5 25 (1995ReZZ). The existence two β-decaying states in ^{34}Al was proven by 2012Ro25 . Normal configuration of $\pi(d_{5/2})^{-1}\otimes\nu(f_{7/2})^1$ from shell-model calculations using the SDPF-U-MIX interaction (2017Li03). Strong absorption $r_0^2=1.187\text{ fm}^2$ 18 from Si(^{34}Al ,X) (2006Kh08). %β ⁻ =100; %β ⁻ n=11 4 (2019Li41) μ=+1.757 14 (2025StZZ , 2018Xu05) Q=0.038 5 (2021StZZ , 2018Xu05) μ: from β-detected nuclear magnetic resonance (2018Xu05). Q: from β-detected quadrupole resonance (2018Xu05). J ^π : spin=1 from nuclear magnetic resonance (β-NMR) and quadrupole resonance (β-NQR) by 2018Xu05 . β decay to a ^{34}Si 0 ⁺ , 2718.63 level with log ft=4.7 and the 0 ⁺ was determined by E0 transition via internal pair formation (IPF). T _{1/2} : unweighted average of 22.1 ms 2 in U(p, ^{34}Al) from 2019Li41 by fitting time distribution of E0 transition of 0 ⁺ , 2718 level to 0 ⁺ g.s. of ^{34}Si via the internal pair formation (IPF), 21.6 ms 15 in $^9\text{Be}(^{40}\text{Ar}, ^{34}\text{Al})$ from 2017Ha23 and 2017Ha34 by fitting implant-β-1193γ decay curve, and 26 ms 1 in $^9\text{Be}(^{36}\text{S}, ^{34}\text{Al})$ from 2012Ro25 by fitting β decay curve in coincidence with 511-keV γ ray arising from positron annihilation. Intruder configuration of $\pi(d_{5/2})^{-1}\otimes\nu(d_{3/2})^{-1}(f_{7/2})^2$ from shell-model calculations using the SDPF-U-MIX interaction (2017Li03).
383.7 12			C F	XREF: F(?) J ^π : (3 ⁻) from shell-model calculations (2007GeZX).
411.03 11			A	
471.85 12			A F	XREF: F(?)
487.77 11	(2 ⁻)		A G	J ^π : 441.20γ to 1 ⁺ , 46.56 level, and 487.67γ to 4 ⁻ g.s. There is no nearby levels with positive parity based on SDPF-U-MIX shell-model calculations (2017Li03).
597.52 10	(3 ⁻)		A	J ^π : 597.54γ to 4 ⁻ g.s. and no direct β-feeding from 0 ⁺ ^{34}Mg parent. There is no nearby levels with positive parity based on SDPF-U-MIX shell-model calculations (2017Li03).
657 9	(4 ⁻)		B I	B(E2)↑=0.0100 39 (2001Pr08) J ^π : 4 ⁻ from shell-model calculations in 2008Hi01 to reproduce the measured B(E2) in 2001Pr08 . T _{1/2} : 83 ps +63-27 ps +6-3 deduced from B(E2)↑ by evaluators, if J ^π (657)=4 ⁻ .
893.23 12	(0 ⁻ , 1, 2, 3 ⁺)		A	J ^π : 405.47γ to (2 ⁻), 487.77 level, 846.63γ to 1 ⁺ , 46.56 level.
969.98 12	(2 ⁻ , 3 ⁺)		A	J ^π : 923.69γ to 1 ⁺ , 46.56 level, 970.38γ to 4 ⁻ g.s.
1048.69 11	(1 ⁻ , 2, 3 ⁺)		A G	J ^π : 1002.02γ to 1 ⁺ , 46.56 level, 451.20γ to (3 ⁻), 597.52 level.
1179.04 13	(0, 1, 2)		A	J ^π : possible β ⁻ feeding from 0 ⁺ ^{34}Mg parent.
1231.33 13	(0, 1, 2)		A	J ^π : possible β ⁻ feeding from 0 ⁺ ^{34}Mg parent.
1299.7 16			C	J ^π : (3 ⁻) from shell-model calculations (2007GeZX).
1462.81 11	1 ⁺		A G	J ^π : allowed β ⁻ feeding from 0 ⁺ ^{34}Mg parent.
2004.07 12	(1 ⁺)		A G	J ^π : possible allowed β ⁻ feeding from 0 ⁺ ^{34}Mg parent.

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{34}Al Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>XREF</u>	<u>Comments</u>
2645.91 14	(1 ⁺)	A	J ^π : above S(n) of ^{34}Al and possible allowed β^- feeding from 0 ⁺ ^{34}Mg parent.

[†] From a least-squares fit to γ -ray energies.

 $\gamma(^{34}\text{Al})$

<u>E_i(level)</u>	<u>J_i^π</u>	<u>E_γ[†]</u>	<u>I_γ[†]</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.</u>	<u>α[‡]</u>	<u>Comments</u>
383.7		383.7 12	100	0.0	4 ⁻			E _γ : from He(^{33}Al , $^{34}\text{Al}\gamma$).
411.03		364.51 10	100	46.56	1 ⁺			
471.85		425.21 10	100	46.56	1 ⁺			E _γ : other: 425.3 11 from He(^{34}Al , $^{34}\text{Al}'\gamma$).
487.77	(2 ⁻)	76.72 12	27 4	411.03				
		441.20 10	100 14	46.56	1 ⁺			
		487.67 14	0.51 12	0.0	4 ⁻			
597.52	(3 ⁻)	597.54 11	100	0.0	4 ⁻			
657	(4 ⁻)	657 9	100	0.0	4 ⁻	[E2]	8.3×10 ⁻⁵ 4	E _γ , I _γ : from Coulomb excitation.
893.23	(0 ⁻ , 1, 2, 3 ⁺)	405.47 22	1.9 9	487.77 (2 ⁻)				
		421.30 14	7.1 9	471.85				
		846.63 10	100 14	46.56	1 ⁺			
969.98	(2 ⁻ , 3 ⁺)	482.19 20	15.6 22	487.77 (2 ⁻)				
		498.14 14	42 7	471.85				
		558.98 16	100 25	411.03				
		923.69 71	3.1 22	46.56	1 ⁺			
		970.38 41	5.1 22	0.0	4 ⁻			
1048.69	(1 ⁻ , 2, 3 ⁺)	451.20 11	2.7 5	597.52 (3 ⁻)				
		560.96 11	4.5 11	487.77 (2 ⁻)				
		576.96 12	1.32 27	471.85				
		637.64 12	5.0 8	411.03				
		1002.02 10	100 14	46.56	1 ⁺			
1179.04	(0, 1, 2)	285.85 26	84 14	893.23 (0 ⁻ , 1, 2, 3 ⁺)				
		767.96 11	47 7	411.03				
		1132.51 11	100 14	46.56	1 ⁺			
1231.33	(0, 1, 2)	338.15 11	47 8	893.23 (0 ⁻ , 1, 2, 3 ⁺)				
		759.18 18	11.3 19	471.85				
		820.33 10	100 13	411.03				
1299.7		916	100	383.7				
1462.81	1 ⁺	283.85 25	4.7 8	1179.04 (0, 1, 2)				
		414.16 10	50 7	1048.69 (1 ⁻ , 2, 3 ⁺)				
		492.88 10	9.0 13	969.98 (2 ⁻ , 3 ⁺)				
		569.63 10	20.9 29	893.23 (0 ⁻ , 1, 2, 3 ⁺)				
		974.96 10	100 14	487.77 (2 ⁻)				
		990.92 10	18.1 26	471.85				
		1051.70 10	94 13	411.03				
		1416.23 10	34 5	46.56	1 ⁺			
2004.07	(1 ⁺)	1110.45 14	1.8 7	893.23 (0 ⁻ , 1, 2, 3 ⁺)				
		1516.13 17	6.3 14	487.77 (2 ⁻)				
		1532.21 10	100 14	471.85				
		1593.14 19	5.6 14	411.03				
		1957.65 11	28 4	46.56	1 ⁺			
2645.91	(1 ⁺)	2158.08 14	92 25	487.77 (2 ⁻)				
		2173.89 14	100 25	471.85				
		2235.13 24	92 25	411.03				
		2598.9 5	50 8	46.56	1 ⁺			

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Adopted Levels, Gammas (continued)

 $\gamma(^{34}\text{Al})$ (continued)

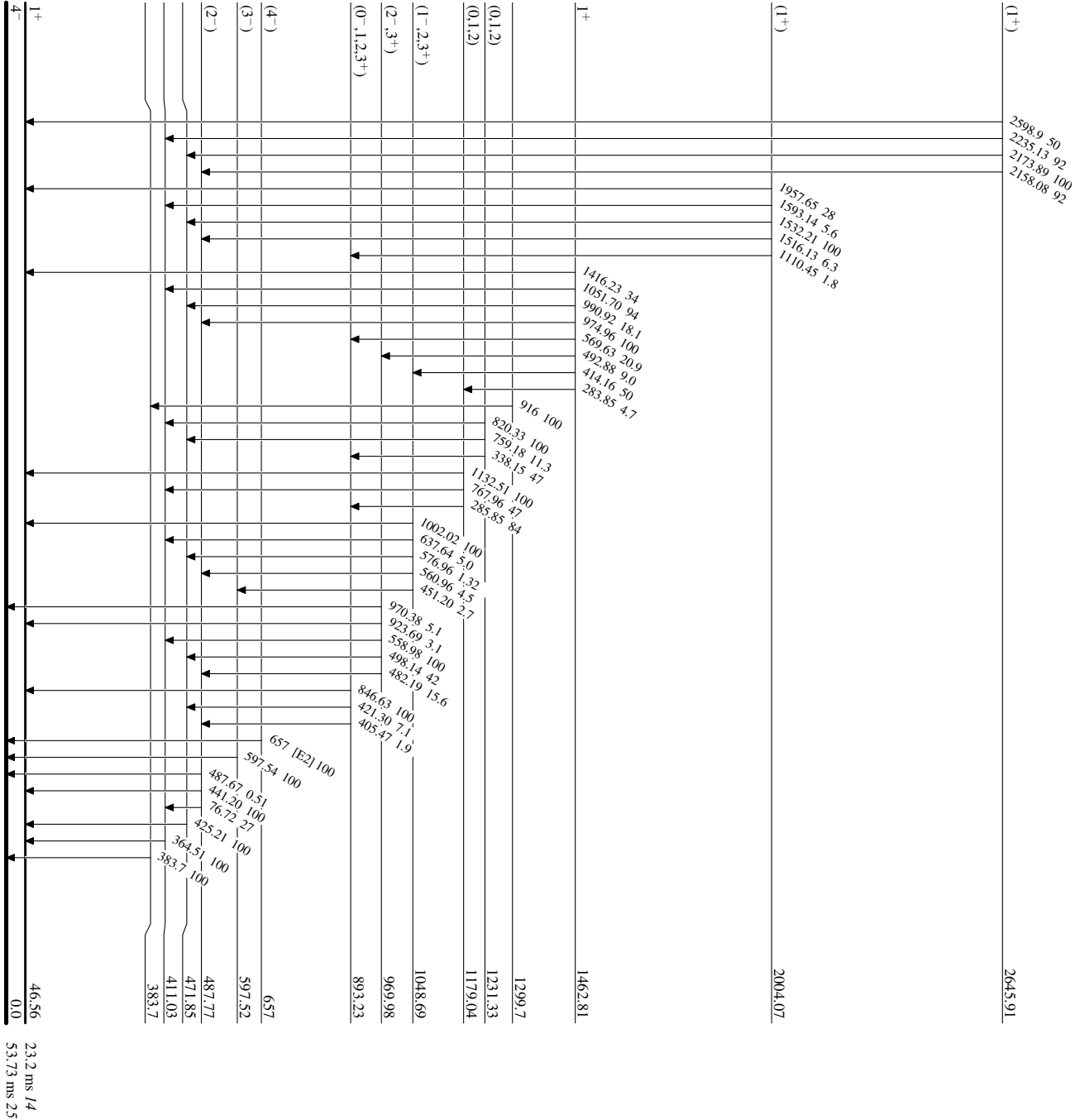
[†] From ^{34}Mg β^- decay (44.9 ms), unless otherwise noted.

[‡] Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

Adopted Levels, Gammas

Level Scheme

Intensities: Relative photon branching from each level



³⁴Al
₁₃²¹

^{34}Mg β^- decay (44.9 ms) 2017Li03

Parent: ^{34}Mg : $E=0$; $J^\pi=0^+$; $T_{1/2}=44.9$ ms 4; $Q(\beta^-)=11321$ 7; $\% \beta^-$ decay=100

^{34}Mg - $T_{1/2}$: From the Adopted Levels of ^{34}Mg , originally from fit of time distribution relative to the proton impact of four β -delayed γ rays in 2017Li03.

^{34}Mg - $Q(\beta^-)$: From 2021Wa16.

2017Li03: ^{34}Mg ions were produced by fragmentation between a 1.4-GeV pulsed proton beam from the Proton Synchrotron Booster and a UC_x target at ISOLDE-CERN. The Mg atoms that diffused out of the target matrix were selectively laser ionized using the resonance ionization laser ion source (RILIS) and accelerated to 40 keV. $A=34$ nuclei were selected by the ISOLDE General Purpose Separator and implanted onto a movable tape located at the center of the ISOLDE Decay station. The ^{34}Mg beam rate is 700 pps. I . β particles were detected using a 3-cm-thick NE102 plastic scintillator. γ rays were detected using five HPGe clover detectors. Measured E_γ , I_γ , $\beta\gamma$ -coin, and $\beta\gamma\gamma$ -coin. Deduced levels, J , π , decay branching ratios, $\log ft$, parent $T_{1/2}$ and $\% \beta^-$ n. Comparisons with large-scale shell-model calculations using the SDPF-U-MIX interaction.

1984La03: Sodium isotopes were produced via the fragmentation of a 30-g/cm² iridium target by 10-GeV protons from the CERN synchrotron. The recoiling nuclear fragments are thermalized in heated graphite. The alkali elements selectively diffuse out of the target, are surface ionized, and analyzed by the online mass spectrometer, then focused at the bottom of a well-shaped NE213 liquid scintillator neutron detector. β particles were detected using a well-shaped thin plastic scintillator. Measured β and neutron counts. Deduced the $T_{1/2}$ of ^{34}Mg . P_n values were reported for other Mg isotopes but not for ^{34}Mg .

1999YoZW (RIKEN Report): $^9\text{Be}(^{48}\text{Ca}, X)$ and $^{181}\text{Ta}(^{48}\text{Ca}, X)$. Measured $T_{1/2}$ and $\% \beta^-$ n.

2019As04: in-trap decay of ^{34}Mg was used to produce ^{34}Al isomer and ^{34}Si at ISOLDE-CERN. Measured masses of ^{34}Si , ^{33}Mg , ^{34}Mg , and ^{34}Al using the time-of-flight ion cyclotron resonance (ToF-ICR) at ISOLTRAP.

The decay scheme is considered incomplete due to a large gap of 8.7 MeV between the highest observed level at 2646 and $Q(\beta^-)$ value=11321 7 (2021Wa16). There may be missing transitions from unobserved levels in the gap.

 ^{34}Al Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\ddagger$	Comments
0.0	4^-	53.73 ms 25	
46.56 11	1^+	23.2 ms 14	
411.03 11			
471.85 12			
487.77 11	(2^-)		
597.52 10	(3^-)		
893.23 12	$(0^-, 1, 2, 3^+)$		
969.98 12	$(2^-, 3^+)$		
1048.69 11	$(1^-, 2, 3^+)$		
1179.04 13	$(0, 1, 2)$		
1231.33 13	$(0, 1, 2)$		
1462.81 11	1^+		
2004.07 12	(1^+)		J^π : 2017Li03 tentatively assigned (1^+) based on $\log ft=5.4$ 1.
2645.91 14	(1^+)		
2572+x			$E(\text{level})$: $x < 8749$ 10 from $Q(\beta^-)(^{34}\text{Mg}) - S(n)(^{34}\text{Al})$, where $Q(\beta^-)(^{34}\text{Mg})=11321$ 7 and $S(n)(^{34}\text{Al})=2572$ 7 from 2021Wa16. This represents a range of unidentified levels in ^{34}Al that subsequently decay to ^{33}Al via one-neutron emission.

† From a least-squares fit to γ -ray energies.

‡ From the Adopted Levels.

 β^- radiations

$E(\text{decay})$	$E(\text{level})$	$I\beta^-^\dagger^\ddagger$	$\log ft^\dagger$	Comments
$(4.4 \times 10^3 @ 44)$	2572+x	21 7		$I\beta^-$: $\% \beta^-$ n=21 7 derived from measurements in which the collection tape was not moved (2017Li03). $\% \beta^-$ 2n<0.1 inferred from the deduced intensities of the γ rays populating excited states in daughter nuclei (2017Li03).
(8675 7)	2645.91	0.400	6.0	

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^{34}Mg β^- decay (44.9 ms) **2017Li03** (continued) β^- radiations (continued)

E(decay)	E(level)	$I\beta^{-\ddagger}$	Log $ft^\dagger$	Comments
(9317 7)	2004.07	2.03	5.4	
(9858 7)	1462.81	28.4	4.4	$I\beta^-$, Log ft : Even if the β^- feeding intensity were to be reduced by an order of magnitude to account for possible missing feeding γ to this level due to the incomplete decay scheme, the log ft value would still remain within the range of allowed transitions.
(10090 7)	1231.33	0.840	6.0	
(10142 7)	1179.04	0.59	6.1	
(10272 7)	1048.69	<1.1	>5.9	
(10428 7)	893.23	<0.61	>6.2	
(10833 [#] 7)	487.77	<4.6	>7.8 ^{lu}	$I\beta^-$: 2 2 in 2017Li03 .
(10849 [#] 7)	471.85	<1.47	>5.9	$I\beta^-$: 0.6 6 in 2017Li03 .
(11274 7)	46.56	45 8	4.5 1	

[†] From $I(\gamma+\text{ce})$ intensity balance at each level. Quoted $I\beta^-$ values are considered upper limits due to the incomplete decay scheme, and the associated log ft values are considered lower limits. Note that possible unobserved transitions due to incomplete decay scheme might not have a significant impact on those very strong branches (e.g., $\%I\beta^-=45$ feeding to 46-keV level and $\%I\beta^-=28.4$ to 1462.8-keV level).

[‡] Absolute intensity per 100 decays.

[#] Existence of this branch is questionable.

[@] Estimated for a range of levels.

 $\gamma(^{34}\text{Al})$

I_γ normalization: 1.0 4 from $\Sigma\%I(\gamma+\text{ce to } 46)=34$ 11, deduced from $100-\%I\beta^-$ n- $\%I\beta^-(46)$, where $\%I\beta^-$ n=21 7 from the Adopted Levels of ^{34}Mg (originally from **2017Li03**) and $\%I\beta^-(46)=45$ 8 for the 46-keV isomer β decay from **2017Li03**. $\%I\beta^-(\text{g.s.})<0.3$ 1 from **2017Li03**. The deduced normalization factor may be considered an upper limit due to potential missing γ transitions from unobserved levels in the gap to the ground state. **2017Li03** deduced the number of implanted nuclei and reported absolute I_γ per ^{34}Mg decay.

$E_\gamma^\dagger$	$I_\gamma^{\ddagger}$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
76.72 12	2.33 38	487.77	(2 ⁻)	411.03		$\Delta E_\gamma(\text{stat})=0.06$ keV.
283.85 25	0.40 7	1462.81	1 ⁺	1179.04	(0,1,2)	$\Delta E_\gamma(\text{stat})=0.23$ keV.
285.85 26	0.36 6	1179.04	(0,1,2)	893.23	(0 ⁻ ,1,2,3 ⁺)	$\Delta E_\gamma(\text{stat})=0.24$ keV.
338.15 11	0.25 4	1231.33	(0,1,2)	893.23	(0 ⁻ ,1,2,3 ⁺)	$\Delta E_\gamma(\text{stat})=0.05$ keV.
364.51 10	11.2 15	411.03		46.56	1 ⁺	$\Delta E_\gamma(\text{stat})=0.01$ keV.
405.47 22	0.044 20	893.23	(0 ⁻ ,1,2,3 ⁺)	487.77	(2 ⁻)	$\Delta E_\gamma(\text{stat})=0.19$ keV.
414.16 10	4.29 59	1462.81	1 ⁺	1048.69	(1 ⁻ ,2,3 ⁺)	$\Delta E_\gamma(\text{stat})=0.01$ keV.
421.30 14	0.16 2	893.23	(0 ⁻ ,1,2,3 ⁺)	471.85		$\Delta E_\gamma(\text{stat})=0.09$ keV.
425.21 10	4.17 57	471.85		46.56	1 ⁺	$\Delta E_\gamma(\text{stat})=0.01$ keV.
441.20 10	8.8 12	487.77	(2 ⁻)	46.56	1 ⁺	$\Delta E_\gamma(\text{stat})=0.01$ keV.
451.20 11	0.10 2	1048.69	(1 ⁻ ,2,3 ⁺)	597.52	(3 ⁻)	$\Delta E_\gamma(\text{stat})=0.05$ keV.
482.19 20	0.07 1	969.98	(2 ⁻ ,3 ⁺)	487.77	(2 ⁻)	$\Delta E_\gamma(\text{stat})=0.17$ keV.
487.67 14	0.045 10	487.77	(2 ⁻)	0.0	4 ⁻	$\Delta E_\gamma(\text{stat})=0.10$ keV.
492.88 10	0.77 11	1462.81	1 ⁺	969.98	(2 ⁻ ,3 ⁺)	$\Delta E_\gamma(\text{stat})=0.02$ keV.
498.14 14	0.19 3	969.98	(2 ⁻ ,3 ⁺)	471.85		$\Delta E_\gamma(\text{stat})=0.09$ keV.
558.98 16	0.45 11	969.98	(2 ⁻ ,3 ⁺)	411.03		$\Delta E_\gamma(\text{stat})=0.12$ keV.
560.96 11	0.17 4	1048.69	(1 ⁻ ,2,3 ⁺)	487.77	(2 ⁻)	$\Delta E_\gamma(\text{stat})=0.05$ keV.
569.63 10	1.80 25	1462.81	1 ⁺	893.23	(0 ⁻ ,1,2,3 ⁺)	$\Delta E_\gamma(\text{stat})=0.01$ keV.
576.96 12	0.05 1	1048.69	(1 ⁻ ,2,3 ⁺)	471.85		$\Delta E_\gamma(\text{stat})=0.07$ keV.
597.54 11	0.077 10	597.52	(3 ⁻)	0.0	4 ⁻	$\Delta E_\gamma(\text{stat})=0.05$ keV.
637.64 12	0.19 3	1048.69	(1 ⁻ ,2,3 ⁺)	411.03		$\Delta E_\gamma(\text{stat})=0.07$ keV.

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$^{34}\text{Mg } \beta^-$ decay (44.9 ms) **2017Li03** (continued) $\gamma(^{34}\text{Al})$ (continued)

E_γ [†]	I_γ [‡]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
759.18 18	0.06 1	1231.33	(0,1,2)	471.85		$\Delta E_\gamma(\text{stat})=0.15$ keV.
767.96 11	0.20 3	1179.04	(0,1,2)	411.03		$\Delta E_\gamma(\text{stat})=0.05$ keV.
820.33 10	0.53 7	1231.33	(0,1,2)	411.03		$\Delta E_\gamma(\text{stat})=0.02$ keV.
846.63 10	2.27 31	893.23	(0 ⁻ ,1,2,3 ⁺)	46.56 1 ⁺		$\Delta E_\gamma(\text{stat})=0.01$ keV.
923.69 71	0.014 10	969.98	(2 ⁻ ,3 ⁺)	46.56 1 ⁺		$\Delta E_\gamma(\text{stat})=0.7$ keV.
						$\Delta I_\gamma=0.00$ is given in 2017LiZZ . Evaluators assign 0.01 based on minimum ΔI_γ in 2017LiZZ .
970.38 41	0.023 10	969.98	(2 ⁻ ,3 ⁺)	0.0 4 ⁻		$\Delta E_\gamma(\text{stat})=0.4$ keV.
974.96 10	8.6 12	1462.81	1 ⁺	487.77 (2 ⁻)		$\Delta E_\gamma(\text{stat})=0.01$ keV.
990.92 10	1.56 22	1462.81	1 ⁺	471.85		$\Delta E_\gamma(\text{stat})=0.021$ keV.
1002.02 10	3.78 52	1048.69	(1 ⁻ ,2,3 ⁺)	46.56 1 ⁺		$\Delta E_\gamma(\text{stat})=0.004$ keV.
1051.70 10	8.1 11	1462.81	1 ⁺	411.03		$\Delta E_\gamma(\text{stat})=0.004$ keV.
1110.45 14	0.025 10	2004.07	(1 ⁺)	893.23 (0 ⁻ ,1,2,3 ⁺)		$\Delta E_\gamma(\text{stat})=0.09$ keV.
1132.51 11	0.43 6	1179.04	(0,1,2)	46.56 1 ⁺		$\Delta E_\gamma(\text{stat})=0.032$ keV.
1416.23 10	2.92 40	1462.81	1 ⁺	46.56 1 ⁺		$\Delta E_\gamma(\text{stat})=0.013$ keV.
1516.13 17	0.09 2	2004.07	(1 ⁺)	487.77 (2 ⁻)		$\Delta E_\gamma(\text{stat})=0.14$ keV.
1532.21 10	1.43 20	2004.07	(1 ⁺)	471.85		$\Delta E_\gamma(\text{stat})=0.01$ keV.
1593.14 19	0.08 2	2004.07	(1 ⁺)	411.03		$\Delta E_\gamma(\text{stat})=0.16$ keV.
1957.65 11	0.40 6	2004.07	(1 ⁺)	46.56 1 ⁺		$\Delta E_\gamma(\text{stat})=0.04$ keV.
2158.08 14	0.11 3	2645.91	(1 ⁺)	487.77 (2 ⁻)		$\Delta E_\gamma(\text{stat})=0.10$ keV.
2173.89 14	0.12 3	2645.91	(1 ⁺)	471.85		$\Delta E_\gamma(\text{stat})=0.09$ keV.
2235.13 24	0.11 3	2645.91	(1 ⁺)	411.03		$\Delta E_\gamma(\text{stat})=0.22$ keV.
2598.9 5	0.06 1	2645.91	(1 ⁺)	46.56 1 ⁺		$\Delta E_\gamma(\text{stat})=0.5$ keV.

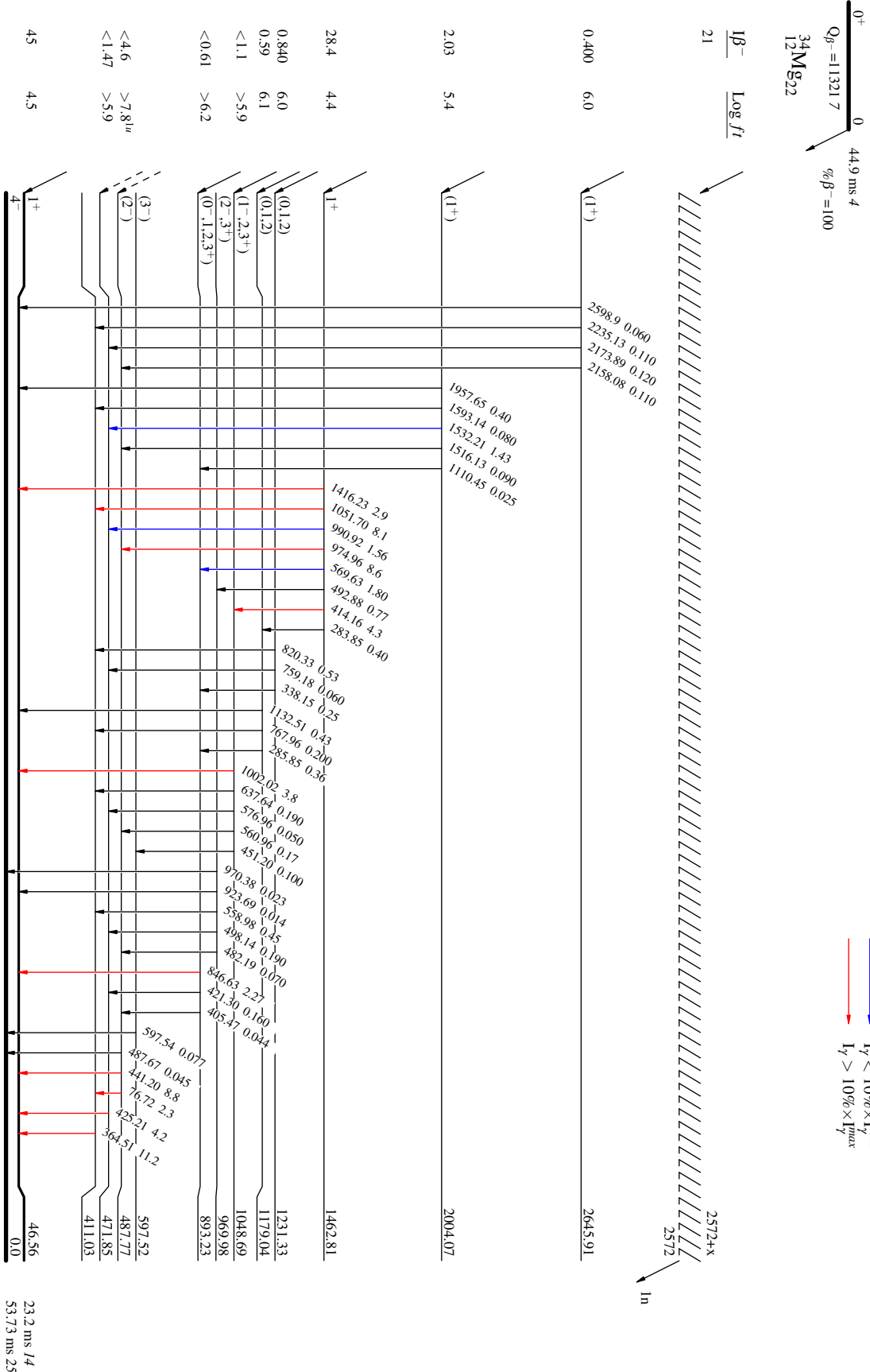
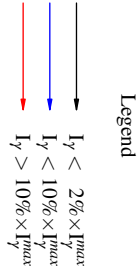
[†] From [2017Li03](#) and [2017LiZZ](#) (e-mail reply from A. Negret and R. Lica). E_γ systematic uncertainty of 0.1 keV from calibration added in quadrature to the statistical uncertainties provided in [2017LiZZ](#).

[‡] Absolute intensity per 100 decays.

³⁴Mg β⁻ decay (44.9 ms) 2017L103

Decay Scheme

Intensities: I_{γ+ce} per 100 parent decays



^{36}Mg β^-2n decay (6.9 ms) 2013StZY

Parent: ^{36}Mg : $E=0$; $J^\pi=0^+$; $T_{1/2}=6.9$ ms $+7-8$; $Q(\beta^-2n)=7.24\times 10^3$ 69; $\% \beta^-2n$ decay=?

^{36}Mg - $T_{1/2}$: Weighted average of 3.9 ms 13 (2004Gr20, 2003Gr22, implant- β correlation), 7.6 ms $+5-8$ (2013StZY, implant- β correlation), 7.2 ms 12 (2022Cr03, implant- β correlation), and 6.8 ms 10 (2023Lu07, implant- $\beta\gamma$ correlation). Other: ≈ 5 ms (1999YoZW, implant- β correlation, preliminary). 2013StZY reports original $T_{1/2}=7.6$ ms 1 (stat) $+5-8$ (syst). 2022Cr03 reports original $T_{1/2}=7.2$ ms 1 (stat) 12 (syst).

^{36}Mg - $Q(\beta^-2n)$: From 2021Wa16.

2013StZY: ^{36}Mg was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA $^{48}\text{Ca}^{20+}$ primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on a 15-mm-thick ^9Be target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the $B\rho$ - ΔE -ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beamline. To reduce background buildup, the scintillator barrel was rotated quickly and axially moved slowly in vertical direction, resulting in a helix-shaped motion. β particles were detected by the CAITEN and γ rays were detected using three HPGe clover detectors. Measured E_γ , I_γ , $\beta\gamma$ -coin, and implant- β correlation, and deduced the $T_{1/2}$ of ^{36}Mg . Comparisons with QRPA and shell-model calculations.

 ^{34}Al Levels

<u>E(level)</u>	<u>J^π[†]</u>	<u>$T_{1/2}$[†]</u>
0	4^-	53.73 ms 25
658	(4^-)	

[†] From the Adopted Levels.

 $\gamma(^{34}\text{Al})$

<u>E_γ</u>	<u>I_γ</u>	<u>$E_i(\text{level})$</u>	<u>J^π_i</u>	<u>E_f</u>	<u>J^π_f</u>	<u>Comments</u>
658 [†]	100 10	658	(4^-)	0	4^-	E_γ, I_γ: from 2013StZY. 2013StZY observed a 658γ that appears predominantly in the time 0-10 ms after an implantation of ^{36}Mg. 2013StZY states that the 658γ is a prompt line, so it must originate from deexcitation in one of the daughter nuclei $^{34,35,36}\text{Al}$, and it is very likely that the 658γ corresponds to the 657γ observed in 2001Pr08 via $^{197}\text{Au}(^{34}\text{Al}, ^{34}\text{Al}'\gamma)$ Coulomb excitation. 2023Lu07 observed a 657γ within the 100-ms window following the arrival of a ^{36}Mg implantation in both NSCL and FRIB experiments. 2023Lu07 states that if the 657γ belongs to ^{34}Al, the probability of two-neutron emission would need to be significantly larger than that of one-neutron emission based on the intensities of the relevant transitions in the γ-ray spectrum. They assign the 657γ to ^{36}Al, though some possible contributions from ^{34}Al cannot be ruled out.

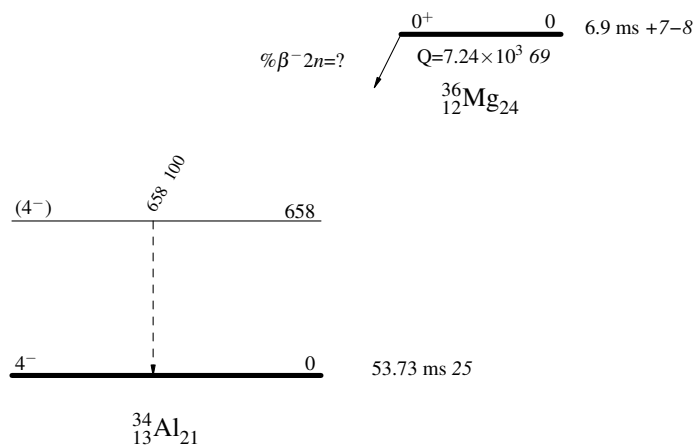
[†] Placement of transition in the level scheme is uncertain.

$^{36}\text{Mg} \beta^- 2n \text{ decay (6.9 ms)} \quad 2013\text{StZY}$

Legend

Decay Scheme

 Intensities: Relative I_γ

 - - - - - $\rightarrow$ γ Decay (Uncertain)


$^2\text{H}(^{33}\text{Al}, ^{34}\text{Al}\gamma), (^{34}\text{Si}, ^{34}\text{Al}\gamma)$ 2007GeZX

2007GeZX: 30-AMeV cocktail beam on 30-mg/cm² CD₂ secondary target at GANIL. Heavy ions produced in reactions were identified by the VAMOS spectrometer. γ rays were detected using the EXOGAM germanium clover array. Measured Doppler-corrected E_γ , I_γ , $\gamma\gamma$ -coin, and $(^{34}\text{Al})\gamma$ -coin. Deduced levels, J, π . Compared with shell-model calculations.

Translation of Sec. 4.6.4 in 2007GeZX: Many nuclei produce ^{34}Al : $^{33,34}\text{Al}$ and $^{34,35}\text{Si}$; the most favorable channels are those coming from ^{34}Si and ^{33}Al because the production of these nuclei is more significant.

 ^{34}Al Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>
0	
388	(3 ⁻)
1304	(3 ⁻)

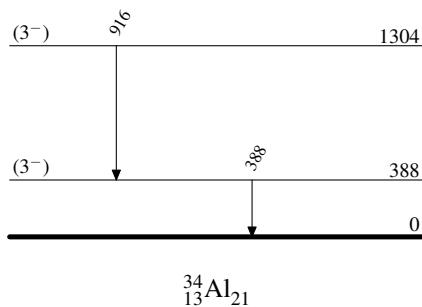
[†] From E_γ data in 2007GeZX.

[‡] Assigned in 2007GeZX by comparison with shell-model calculations.

 $\gamma(^{34}\text{Al})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
388	388	(3 ⁻)	0	
916	1304	(3 ⁻)	388	(3 ⁻)

[†] From 2007GeZX.

 $^2\text{H}(^{33}\text{Al}, ^{34}\text{Al}\gamma), (^{34}\text{Si}, ^{34}\text{Al}\gamma)$ 2007GeZXLevel Scheme

$\text{He}(^{34}\text{Al}, ^{34}\text{Al}'\gamma)$ 2006FuZX

2006FuZX: $^{28-31}\text{Na}$, $^{30-33}\text{Mg}$, $^{32-35}\text{Al}$, $^{34-36}\text{Si}$, and $^{36,37}\text{P}$ were produced via the projectile fragmentation of a 63-MeV/nucleon ^{40}Ar primary beam impinging on 1-mm-thick C and Be targets and selected by the RIKEN Projectile-Fragment Separator (RIPS). The secondary cocktail beam was directed onto a 150-mg/cm² liquid He target and the outgoing particles were identified by energy loss, kinetic energy, and time-of-flight measured using a Si detector, a NaI calorimeter, and a couple of plastic scintillators. The positions and directions of incident and outgoing particles were measured using PPACs. Only outgoing particles were identified so that the excited states could be populated by any allowed reactions. Deexcited γ rays emitted from outgoing particle moving at $v/c \approx 0.3$ were detected by the Gamma-Ray detector Array with Position and Energy sensitivity (GRAPE) consisting of segmented Ge detectors surrounding the liquid He target.

 ^{34}Al Levels

<u>$E(\text{level})^\dagger$</u>	<u>Comments</u>
0	
46.56 11	E(level): from the Adopted Levels.
383.7?	
471.8?	

[†] From E_γ data. The Evaluators place the γ rays observed in 2006FuZX based on the Adopted Levels and Gammas. No placement was proposed in 2006FuZX.

 $\gamma(^{34}\text{Al})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>
383.7 [‡] 12	383.7?	0
425.3 [‡] 11	471.8?	46.56

[†] From 2006FuZX.

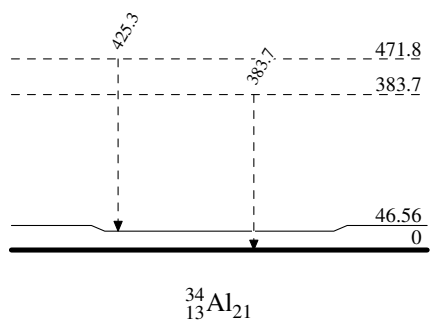
[‡] Placement of transition in the level scheme is uncertain.

$\text{He}(^{34}\text{Al}, ^{34}\text{Al}'\gamma)$ 2006FuZX

Legend

Level Scheme

-----► γ Decay (Uncertain)



$^9\text{Be}(^{36}\text{S}, ^{34}\text{Al})$ 2012Ro25

2012Ro25: a 600-pps, 93% pure ^{34}Al secondary beam was produced via the projectile fragmentation of a 77.5-MeV/nucleon ^{36}S primary beam on a 240-mg/cm² ^9Be target and selected by the LISE3 spectrometer at GANIL. ^{34}Al ions were implanted into a 50- μm Kapton foil in a beam-on (120 ms)/beam-off (300 ms) mode. Electrons and positrons were detected using four ΔE -E telescopes, each consisting of a 1-mm-thick Si detector and a 4.5-mm-thick Si(Li) detector. γ rays were detected using two Ge clover detectors of the EXOGAM array. Measured $\text{E}\beta$, Ee^+ , Ee^- , $\text{E}\gamma$, $\beta\gamma$ -coin, and $\beta\text{e}^+/\text{e}^-$ coin. Deduced ^{34}Al g.s. and isomer $\text{T}_{1/2}$, ^{34}Si levels, J, π , transition strengths, and configuration mixing. Comparisons with shell-model calculations.

2006Tu03: a 96% pure ^{34}Al secondary beam was produced via the projectile fragmentation of a 77.5-MeV/nucleon ^{36}S primary beam on a 194.78-mg/cm² ^9Be target and selected by the LISE3 spectrometer at GANIL. Measured spin-polarization of ^{34}Al induced by the single neutron pickup reaction $^9\text{Be}(^{36}\text{S}, ^{34}\text{Al})$, followed by the evaporation of three protons.

 ^{34}Al Levels

E(level)	J^π	$\text{T}_{1/2}$	Comments
0	4^-	54.4 ms 5	J^π : from the Adopted Levels. Dominated by neutron $1\text{p}0\text{h}$ configuration $(\text{f}_{7/2})^1$ with a proton hole in $\text{d}_{5/2}$ from shell-model calculations (2012Ro25). $\text{T}_{1/2}$: from fit of β decay curve in coincidence with 926-keV β -delayed γ ray in ^{34}Si (2012Ro25).
46.56 11	1^+	26 ms 1	$\text{E}(\text{level}), J^\pi$: from the Adopted Levels. $\text{E}(\text{level})=0.55$ MeV and $J^\pi=1^+$ with dominant neutron $2\text{p}1\text{h}$ configuration $(\text{d}_{3/2})^{-1}(\text{f}_{7/2})^2$ from shell-model calculations (2012Ro25). A fit of β decay curve in coincidence with 3326-keV β -delayed γ ray in ^{34}Si yields a shorter $\text{T}_{1/2}$ of 49.8 ms 2, which indicates that the 2^+ , 3326 level in ^{34}Si is also fed by the β decay of the isomer in ^{34}Al . Even without decay branching ratios, the fact that both the 0^+ and 2^+ levels in ^{34}Si are fed by the isomer suggests a 1^+ assignment for the isomer. $\text{T}_{1/2}$: from fit of β decay curve in coincidence with 511-keV γ ray arising from positron annihilation. Other than the 511 γ , no other γ rays are observed in coincidence with the electron-positron pair. The second 0^+ state at 2718 keV in ^{34}Si is observed to be populated by the β decay of the 1^+ isomer in ^{34}Al and to deexcite by $\text{E}0$ transition to 0^+ g.s. of ^{34}Si via the internal pair formation (IPF). The sum energy of the electron-positron pair is measured to be 1688 2 (2012Ro25).

$^9\text{Be}(^{40}\text{Ar}, ^{34}\text{Al})$ [2017Ha23](#), [2017Ha34](#)

[2017Ha23](#), [2017Ha34](#): a 69.2-MeV/nucleon ^{40}Ar primary beam was produced by the Sector Focusing Cyclotron and Separated Sector Cyclotron at the Heavy Ion Research Facility in Lanzhou (HIRFL). The secondary beam containing ^{34}Al was produced via the projectile fragmentation of ^{40}Ar impinging on a 182.6-mg/cm² ^9Be target and was selected using ΔE -tof- $B\rho$ by the Radioactive Ion Beam Line in Lanzhou (RIBLL). The secondary beam ions were transported into a target chamber filled with low-temperature nitrogen and implanted into a 1500- μm -thick DSSD sandwiched between two 3000- μm -thick plastic scintillators for beam diagnostics and veto, respectively. γ rays were detected using two Ge clover detectors. The fast timing signals of the γ rays were provided by a lanthanum bromide (LaBr_3) detector. Neutrons were detected using two plastic scintillator arrays. Measured implant- β correlation, $E\gamma$, $I\gamma$, $\beta\gamma$ -coin, $\gamma\gamma$ -coin, and βn -coin. Deduced levels, J , π , decay branching ratios, $\log ft$, parent ^{34}Al and ^{34}Al isomer $T_{1/2}$. Comparisons with shell-model calculations using Gogny D1S and SDPF-M interactions. Also see [2019Wa38](#) for detection setup.

 ^{34}Al Levels

<u>E(level)[†]</u>	<u>J^π[†]</u>	<u>T_{1/2}</u>	<u>Comments</u>
0	4 ⁻	51.5 ms 9	T _{1/2} : from fit of implant- β -929 γ decay curve (2017Ha23 , 2017Ha34).
46.56 11	1 ⁺	21.6 ms 15	T _{1/2} : from fit of implant- β -1193 γ decay curve (2017Ha23 , 2017Ha34). A total of 6.5×10^5 46.6-keV isomer decay events were observed in 2017Ha23 from fit of the inclusive β decay curve.

[†] From the Adopted Levels.

$\text{Pb}(^{35}\text{Al}, ^{34}\text{Al}n\gamma)$ 2017Ch36

Coulomb dissociation of $^{35}\text{Al} \rightarrow ^{34}\text{Al} + n$.

[2017Ch36](#), [2014ChZZ](#): ^{35}Al was produced via the projectile fragmentation of a 531-MeV/nucleon ^{40}Ar primary beam from the Heavy Ion Synchrotron (SIS18) at GSI. The secondary cocktail beam was separated by the FRS separator and impinged on a 2-g/cm² Pb target and a 0.93-g/cm² C target. Projectiles and reaction fragments were detected using 8 DSSDs, separated by a large-area dipole magnet (ALADIN) and tracked using two large scintillator fiber detectors (GFIs). Neutrons from the excited projectiles were detected using the high-efficiency Large Area Neutron Detector (LAND). γ rays from the deexciting projectile and projectile-like fragments were detected using a spherical 4π Crystal Ball detector array of 162 NaI(Tl) crystals. Measured $E(\text{fragment})$, E_n , E_γ , Coulomb dissociation cross sections. Deduced relative populations of ^{34}Al levels and ^{35}Al g.s. configuration. Comparisons with shell-model calculations with the SDPF-M interaction. The measured inclusive differential CD cross section (integrated up to 5.0 MeV relative energy) for $^{35}\text{Al} \rightarrow ^{34}\text{Al} + n$ using a Pb target is 78 mb *I3*.

[2021Bh12](#): A further analysis of the data from [2017Ch36](#). The $^{35}\text{Al}(\gamma, n)^{34}\text{Al}$ photoabsorption cross section was obtained from fitting the direct breakup model to the measured differential Coulomb dissociation cross section of ^{35}Al breaking up into ^{34}Al core excited states. The $^{34}\text{Al}(n, \gamma)^{35}\text{Al}$ neutron capture cross sections were obtained from the photoabsorption cross sections using the detailed balance theorem.

 ^{34}Al Levels

The total measured inclusive differential Coulomb dissociation cross section is 78 mb *I3*, integrated up to 5.0 MeV relative energy between the ^{34}Al core and the neutron.

$E(\text{level})^\dagger$	$\sigma \text{ (mb)}^\ddagger$	Comments
0	48 [#] <i>I3</i>	
46	48 [#] <i>I3</i>	
487		
1048	5 <i>I</i>	Relative population=6% <i>2</i> .
1460	4 <i>I</i>	Relative population=5% <i>2</i> .
2000	4 <i>2</i>	Relative population=5% <i>3</i> .
2550?	16 <i>6</i>	Relative population=21% <i>8</i> .

[†] From [2017Ch36](#).

[‡] $^{35}\text{Al} \rightarrow ^{34}\text{Al} + n$ Coulomb dissociation cross section.

[#] Sum σ for g.s.+ 46 isomer. Sum relative population=62% *20* for g.s.+ 46 isomer.

 $\gamma(^{34}\text{Al})$

$E_\gamma^\dagger$	$E_i(\text{level})$	E_f
441	487	46
1002 [‡]	1048	46
1090 [‡]	2550?	1460
1414	1460	46
1513	2000	487
1954	2000	46

[†] From level-energy difference in [2017Ch36](#).

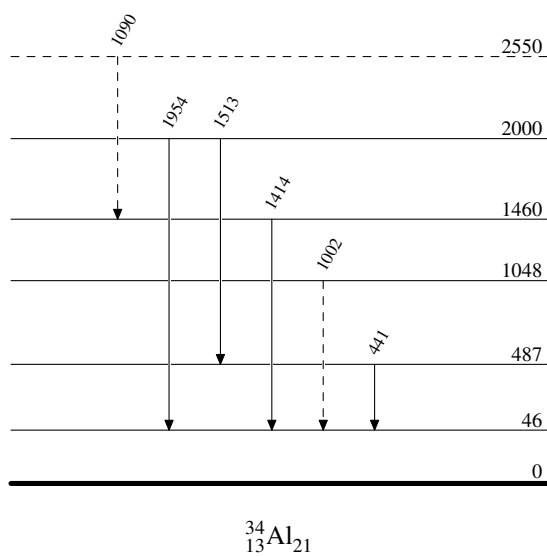
[‡] Placement of transition in the level scheme is uncertain.

Pb($^{35}\text{Al}, ^{34}\text{Aln}\gamma$) 2017Ch36

Legend

Level Scheme

-----► γ Decay (Uncertain)



U(p, ^{34}Al) **2019Li41**

2019Li41: ^{34}Mg and ^{34}Al ions were produced via fragmentation reactions induced by a 1.4-GeV pulsed proton beam from the Proton Synchrotron Booster on a UC_x target at ISOLDE-CERN. The atoms that diffused out of the target matrix were selectively laser ionized using the resonance ionization laser ion source (RILIS) and accelerated to 40 keV. $A=34$ nuclei were selected by the ISOLDE General Purpose Separator and implanted onto a movable tape located at the center of the ISOLDE Decay station. The ^{34}Mg beam rate is 700 pps *I* and the ^{34}Al beam rate is 800 pps *I*, both with purities of $>99\%$. β particles were detected using a 3-cm-thick NE102 plastic scintillator with an efficiency of 90% *5*. γ rays were detected using five HPGe clover detectors. Measured $E\gamma$, $I\gamma$, $\beta\gamma$ -coin, $\beta\gamma\gamma$ -coin, and $\beta e^+/e^-$ coin. Deduced levels, isomer, J , π , decay branching ratios, $\log ft$, $T_{1/2}$ and $\% \beta^- n$ of parent, and configuration mixings. Comparisons with large-scale shell-model calculations using the SDPF-U-MIX interaction.

2024Lu03: 480-MeV proton beam on uranium carbide target at ISAC-TRIUMF.

2001Nu01: 1-GeV proton beam on uranium carbide target at ISOLDE-CERN.

1989Ba50: 600-MeV proton beam on uranium carbide target at ISOLDE-CERN.

 ^{34}Al Levels

<u>E(level)[†]</u>	<u>J^π[†]</u>	<u>T_{1/2}</u>	<u>Comments</u>
0	4 ⁻	53.73 ms <i>13</i>	J ^π : from the Adopted Levels. T _{1/2} : from fit of time distribution gated on g.s.- β -delayed 929 γ , 3 ⁻ to 2 ⁺ transition in ^{34}Si (2019Li41). 89% 3 of ^{34}Al beam component (2019Li41). 86% 4 of ^{34}Al beam component (2024Lu03). T _{1/2} : from fit of time distribution of isomer- β -delayed E0 transition of 0 ⁺ , 2718 level to 0 ⁺ g.s. of ^{34}Si via the internal pair formation (IPF). 11% 3 of ^{34}Al beam component (2019Li41). 14% 4 of ^{34}Al beam component (2024Lu03).
46.56 <i>11</i>	1 ⁺	22.1 ms <i>2</i>	

[†] From the Adopted Levels.

Coulomb excitation 2001Pr08

2001Pr08: $^{197}\text{Au}(^{34}\text{Al}, ^{34}\text{Al}'\gamma)$. A 68.1-MeV/nucleon radioactive beam of ^{34}Al was produced by fragmentation of an 80-MeV/nucleon ^{48}Ar primary beam on a 376-mg/cm² ^9Be primary target, separated by the NSCL A1200 fragment separator, and bombarded on the 702 mg/cm² ^{197}Au secondary target at 59.70-MeV/nucleon middle of energy. Beams passing through the target were detected and stopped in a cylindrical fast-slow plastic phoswich detector located at zero degrees. The γ rays were detected using the NSCL NaI(Tl) array. Measured E_γ , I_γ , and (scattered- ^{34}Al) γ -coin. Deduced γ production cross sections and $B(E2)$. Also see **2000PrZX**.

 ^{34}Al Levels

2001Pr08 calculated $B(E2)$ with different J^π assumptions using shell- model and found that 4^- , g.s. to 3^- , 657 transition $B(E2)=0.00418$ is closest to their measured $B(E2)$.

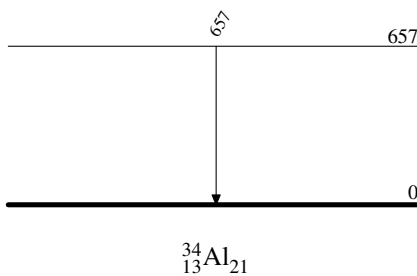
<u>E(level)[†]</u>	<u>Comments</u>
0	
657 9	$B(E2)^\dagger=0.0100$ 39 (2001Pr08)

[†] From E_γ data.

 $\gamma(^{34}\text{Al})$

<u>E_γ[†]</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>
657 9	657	0

[†] From **2001Pr08**.

Coulomb excitation 2001Pr08Level Scheme

Adopted Levels, Gammas

$Q(\beta^-)=4557.0$ 11; $S(n)=7548.7$ 11; $S(p)=18783$ 7; $Q(\alpha)=-13535.2$ 15 2021Wa16

$S(2n)=12056.6$ 9, $S(2p)=33741$ 3 (2021Wa16).

Isotope Discovery (2012Th10): $^{232}\text{Th}(^{40}\text{Ar},X)$ $E=290$ MeV at Dubna (1971Ar32).

^{34}Si production:

2015Mo17: $^9\text{Be}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=95$ MeV/nucleon at RIKEN.

2012Kw02: $E(^{40}\text{Ar})=140$ MeV/nucleon impinged on ^9Be , natural nickel, and ^{181}Ta targets at NSCL.

2012Zh06: $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=57$ MeV/nucleon at HIRFL, Lanzhou.

2008Wi09: $^{208}\text{Pb}(^{36}\text{S}, ^{34}\text{Si}\gamma)$ $E=230$ MeV. Measured $E\gamma$ using the Gammasphere Ge array at ANL. Known transitions in ^{34}Si at 3326, 930, 4255, 125, and 591 keV were observed. No additional information was extracted.

2007No13: $^9\text{Be}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=100$ MeV/nucleon at RIKEN.

^{34}Al radius measurements:

2006Kh08: ^{34}Al produced by $^{181}\text{Ta}(^{48}\text{Ca},X)$ fragmentation at $E(^{48}\text{Ca})=60.3$ MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

1999Ai02: $\text{Si}(^{34}\text{Si},X)$ at NSCL. Measured energy-integrated reaction cross sections at $E=38$ -80 MeV/nucleon. Deduced strong absorption radii.

Theoretical calculations: 2025Li35, 2023Ch17, 2022Lu03, 2020Mi15, 2020Fo04, 2020Ma25, 2019Sa23, 2019Sa02, 2018Va09, 2018Co03, 2017Wu11, 2017Ts01, 2017Ka11, 2017Du03, 2017De15, 2016Li27, 2016Eb02, 2015Xu06, 2015St17, 2015Pu01, 2015Gr15, 2014Wa03, 2014Ca21, 2013Su18, 2013Na17.

 ^{34}Si LevelsCross Reference (XREF) Flags

A	$^{34}\text{Al} \beta^-$ decay (53.73 ms)	F	$^2\text{H}(^{34}\text{Si}, ^{34}\text{Si}'\gamma)$	K	$^{36}\text{S}(^{14}\text{C}, ^{16}\text{O})$
B	$^{34}\text{Al} \beta^-$ decay (23.2 ms)	G	$^7\text{Li}(^{34}\text{P}, ^7\text{Be}\gamma)$	L	$^{160}\text{Gd}(^{34}\text{S}, X\gamma), (^{36}\text{S}, X\gamma)$
C	$^{34}\text{Al} \beta^-$ decay:mixed	H	$^9\text{Be}(^{35}\text{Si}, ^{34}\text{Si}\gamma)$	M	$^{208}\text{Pb}(^{36}\text{S}, ^{34}\text{Si}\gamma)$
D	$^{35}\text{Al} \beta^-n$ decay (38.1 ms)	I	$^{18}\text{O}(^{18}\text{O}, 2p\gamma)$	N	$\text{Si}(^{34}\text{Si}, ^{34}\text{Si}'\gamma)$
E	$^2\text{H}(^{33}\text{Si}, ^{34}\text{Si}\gamma), (^{35}\text{Si}, ^{34}\text{Si}\gamma)$	J	$^{36}\text{S}(^{11}\text{B}, ^{13}\text{N})$	O	Coulomb excitation

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	XREF	Comments
0.0	0^+	2.77 s 20	ABCDEFGHIJKLMNO	$\% \beta^- = 100$ $T_{1/2}$: from fit of β -delayed γ decay curve (1977Na05). Reduced strong absorption radius $r_0^2=1.229$ fm ² 44 (2006Kh08). $r_0^2=1.203$ fm ² 84 at a mean energy of 40.81 MeV/nucleon and $r_0^2=1.229$ fm ² 82 at a mean energy of 71.92 MeV/nucleon (1999Ai02).
2718.63 8	0^+	19.4 ns 5	ABC	J^π : observed $0^+ \rightarrow 0^+$ g.s. E0 transition via internal pair formation (IPF). $T_{1/2}$: from fit of time difference between β and electron-positron pair due to $0^+ \rightarrow 0^+$ E0 transition (2019Li41). Other: 19.4 ns 7 from time difference between β and electron-positron pair and 19.2 ns 8 from time difference between β and 511 annihilation of positron (2012Ro25).
3325.55 6	2^+	82 fs +49-51	ABCDEFGHI LMNO	XREF: O(3305) J^π : the only level strongly populated in Coulomb excitation and inelastic scattering. $T_{1/2}$: from $B(E2)\uparrow=0.0085$ 33 in Coulomb excitation.
3590 25 4254.67 7	3^-	<210 ns	ABCDEFGHI LMN K	XREF: N(4270) J^π : spin=3 from $\gamma\gamma(\theta)$ in $^{34}\text{Al} \beta^-$ decay:mixed. $\pi=-$ from allowed β -feeding from 4^- parent (^{34}Al g.s.). $T_{1/2}$: estimated from $\beta\gamma(t)$ (1989Ba50).
4378.68 11	4^-		A CDEFGHI LM	J^π : spin=4 from $\gamma\gamma(\theta)$ in $^{34}\text{Al} \beta^-$ decay:mixed. $\pi=-$ from allowed

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{34}Si Levels (continued)

E(level) [†]	J ^π	XREF	Comments
4518.99 6	(2 ⁺)	ABC EFGHI M	β -feeding from 4 ⁻ parent (^{34}Al g.s.). XREF: F(4519?)G(4519?)H(4520?) J ^π : spin=1,2 from $\gamma\gamma(\theta)$ in ^{34}Al β^- decay:mixed. Possibly allowed β -feeding from 1 ⁺ parent (^{34}Al 46.6-keV isomer). 2 ⁺ from shell-model calculations (2024Lu03).
4919.8 6		I	
4969.54 13	(5 ⁻)	A C EFGHI M	J ^π : spin=3,5 from $\gamma\gamma(\theta)$ in ^{34}Al β^- decay:mixed. 5 ⁻ from shell-model calculations (2024Lu03).
5041.44 9	(3 ⁻)	ABC EFGH M	XREF: F(5041?)G(5041?)H(5042?) J ^π : 1715.8 γ to 2 ⁺ , 3325.55 level. 662.7 γ to 4 ⁻ , 4378.68 level. 3 ⁻ from shell-model calculations (2024Lu03).
5243.21 15		C	
5348.69 8	(2 ⁺)	ABC G J M	XREF: G(5330)J(5330)M(5351) L=0 from 1 ⁺ in $^7\text{Li}(^3\text{P}, ^7\text{Be})$. 2 ⁺ from shell-model calculations (2010Ze03).
5637.18 26		C	
5773.45 8	(2 ⁺)	BC	J ^π : 5772.9 γ to 0 ⁺ , g.s.; 2 ⁺ from shell-model calculations (2024Lu03).
6022.13 8	(3 ⁻)	A C EFGH M	XREF: E(6001)F(6022?)G(6022?)H(6023?)M(6028) J ^π : possibly allowed β -feeding from 4 ⁻ parent (^{34}Al g.s.). 2696.5 γ to 2 ⁺ , 3325.55 level. 3 ⁻ from shell-model calculations (2024Lu03).
6201.71 10	(1,2 ⁺)	BC	J ^π : 6201.2 γ to 0 ⁺ , g.s.
6227.32 10	(4 ⁻)	A C	J ^π : 1972.7 γ to 3 ⁻ , 4254.67 level, and 1257.6 γ to (5 ⁻), 4969.54 level. Possible β -feeding from 4 ⁻ parent (^{34}Al g.s.). 4 ⁻ from shell-model calculations (2024Lu03).
6234.1 6		I	
6432.22 9		BC	
6676.20 11	(2 ⁺)	BC	J ^π : possibly allowed β -feeding from 1 ⁺ parent (^{34}Al 46.6-keV isomer). 2 ⁺ from shell-model calculations (2024Lu03).
6764.26 18	(2 ⁺)	BC	J ^π : 6763.3 γ to 0 ⁺ , g.s. and 3438.8 γ to 2 ⁺ , 3325.55 level. 2 ⁺ from shell-model calculations (2024Lu03).
6872.35 17	(2 ⁺)	C	J ^π : 6871.8 γ to 0 ⁺ , g.s. 2 ⁺ from shell-model calculations (2024Lu03).
6964.70 22		BC	
7107.09 8	(0 ⁺)	BC	J ^π : close to S(n)=7.5 MeV, possibly allowed β -feeding from 1 ⁺ parent (^{34}Al 46.6-keV isomer). 0 ⁺ from shell-model calculations (2024Lu03).
7476.27 7	0 ⁽⁺⁾	BC	J ^π : spin=0 from $\gamma\gamma(\theta)$ in ^{34}Al β^- decay:mixed. A high-lying level close to S(n)=7.5 MeV, probably allowed β -feeding from 1 ⁺ parent (^{34}Al 46.6-keV isomer). 0 ⁺ from shell-model calculations (2024Lu03).

[†] From a least-squares fit to γ -ray energies.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Si})$									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.	δ	$\alpha^\ddagger$	Comments
2718.63	0 ⁺	2718.4 1	100	0.0	0 ⁺	E0			E_γ : from ³⁴ Al β^- decay (23.2 ms). Mult.: decays via internal pair formation only.
3325.55	2 ⁺	606.8 1	0.103 47	2718.63	0 ⁺	[E2]		0.0001315 18	B(E2)(W.u.)=13 +16-7 E_γ : weighted average of 606.8 1 from ³⁴ Al β^- decay (53.73 ms) and 607.0 3 from ³⁴ Al β^- decay:mixed. I_γ : unweighted average of 0.056 6 from ³⁴ Al β^- decay (53.73 ms) and 0.150 20 from ³⁴ Al β^- decay:mixed.
		3325.45 10	100.0 15	0.0	0 ⁺	[E2]		0.000924 13	B(E2)(W.u.)=2.6 10 E_γ : weighted average of 3325.4 1 from ³⁴ Al β^- decay (53.73 ms), 3325.5 2 from ³⁴ Al β^- decay:mixed, 3326.96 70 from ³⁵ Al β^- n decay, 3326 1 from (¹⁸ O,2p γ), 3330 50 from Si(³⁴ Si, ³⁴ Si' γ), and 3305 55 from Coulomb excitation.
4254.67	3 ⁻	929.12 10	100.0 16	3325.55	2 ⁺	(E1+M2)	+0.02 1	1.645 $\times$ 10 ⁻⁵ 23	I_γ : other: 100.0 20 from ³⁴ Al β^- decay (53.73 ms). E_γ : weighted average of 929.1 1 from ³⁴ Al β^- decay (53.73 ms), 929.2 2 from ³⁴ Al β^- decay:mixed, and 929.12 40 from ³⁵ Al β^- n decay. Others: 929 1 from (¹⁸ O,2p γ) and 940 50 from Si(³⁴ Si, ³⁴ Si' γ). I_γ : others: 100.0 21 from ³⁴ Al β^- decay (53.73 ms), 100 4 from ³⁴ Al β^- decay (23.2 ms), 100.0 27 from ² H(³⁴ Si, ³⁴ Si' γ), and 100 12 from (³⁵ Si, ³⁴ Si γ). Mult., δ : D+Q from $\gamma\gamma(\theta)$ in ³⁴ Al β^- decay:mixed. $\Delta\pi$ =yes from level scheme.
		4254.4 1	26.9 9	0.0	0 ⁺	[E3]		0.000970 14	E_γ : weighted average of 4254.4 1 from ³⁴ Al β^- decay (53.73 ms) and 4254.6 2 from ³⁴ Al β^- decay:mixed. Other: 4255 1 from (¹⁸ O,2p γ). I_γ : weighted average of 29.8 11 from ³⁴ Al β^- decay (53.73 ms), 27 4 from ³⁴ Al β^- decay (23.2 ms), 26.5 5 from ³⁴ Al β^- decay:mixed, and 21.2 30 from (³⁵ Si, ³⁴ Si γ). Other: 53.4 31 from ² H(³⁴ Si, ³⁴ Si' γ).
4378.68	4 ⁻	123.95 10	100	4254.67	3 ⁻	(M1(+E2))	-0.012 17	0.00223 5	E_γ : weighted average of 123.9 1 from ³⁴ Al β^- decay (53.73 ms), 124.0 2 from ³⁴ Al β^- decay:mixed, 124.20 30 from ³⁵ Al β^- n decay, 124 1 from (¹⁸ O,2p γ), and 125 1 from (³⁶ S, ³⁴ Si γ).

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Si})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.	δ	Comments
4518.99	(2 ⁺)	1193.4 1	100.0 15	3325.55	2 ⁺	D+Q	+0.43 2	$I_\gamma, \text{Mult.}, \delta$: D(+Q) from $^{34}\text{Al } \beta^-$ decay:mixed. $\Delta\pi$ =no from level scheme. E_γ : weighted average of 1193.4 1 from $^{34}\text{Al } \beta^-$ decay (53.73 ms), 1193.5 2 from $^{34}\text{Al } \beta^-$ decay:mixed, 1191 1 from ($^{18}\text{O}, 2p\gamma$), and 1194 1 from ($^{36}\text{S}, ^{34}\text{Si}\gamma$). I_γ : others: 100 4 from $^{34}\text{Al } \beta^-$ decay (53.73 ms) and 100.0 22 from $^{34}\text{Al } \beta^-$ decay (23.2 ms). Mult., δ : from $^{34}\text{Al } \beta^-$ decay:mixed. E_γ : weighted average of 4518.6 1 from $^{34}\text{Al } \beta^-$ decay (53.73 ms) and 4518.8 4 from $^{34}\text{Al } \beta^-$ decay:mixed. I_γ : weighted average of 6.0 24 from $^{34}\text{Al } \beta^-$ decay (53.73 ms), 5.48 28 from $^{34}\text{Al } \beta^-$ decay (23.2 ms), and 4.46 30 from $^{34}\text{Al } \beta^-$ decay:mixed.
		4518.6 1	5.0 4	0.0	0 ⁺			
4919.8		665 1		4254.67	3 ⁻			
		1594 1		3325.55	2 ⁺			
4969.54	(5 ⁻)	590.8 1	100	4378.68	4 ⁻	D(+Q)	+0.01 4	E_γ : weighted average of 590.8 1 from $^{34}\text{Al } \beta^-$ decay (53.73 ms) and 591.0 2 from $^{34}\text{Al } \beta^-$ decay:mixed. Others: 590 1 from ($^{18}\text{O}, 2p\gamma$) and 591 1 from ($^{36}\text{S}, ^{34}\text{Si}\gamma$). Mult., δ : from $^{34}\text{Al } \beta^-$ decay:mixed.
5041.44	(3 ⁻)	662.7 2	5.4 6	4378.68	4 ⁻			
		1715.8 1	100.0 18	3325.55	2 ⁺			E_γ : weighted average of 1715.7 1 from $^{34}\text{Al } \beta^-$ decay (53.73 ms), 1715.8 1 from $^{34}\text{Al } \beta^-$ decay (23.2 ms), 1715.8 2 from $^{34}\text{Al } \beta^-$ decay:mixed, and 1715 1 from ($^{36}\text{S}, ^{34}\text{Si}\gamma$).
5243.21		988.5 2	100	4254.67	3 ⁻			
5348.69	(2) ⁺	2023.1 [#] 2	50.4 27	3325.55	2 ⁺			E_γ : weighted average of 2023.1 2 from $^{34}\text{Al } \beta^-$ decay (53.73 ms) and 2023.2 3 from $^{34}\text{Al } \beta^-$ decay:mixed. Other: 2025 2 from ($^{36}\text{S}, ^{34}\text{Si}\gamma$). I_γ : weighted average of 47 5 from $^{34}\text{Al } \beta^-$ decay (53.73 ms), 63 6 from $^{34}\text{Al } \beta^-$ decay (23.2 ms), 48.9 22 from $^{34}\text{Al } \beta^-$ decay:mixed, and 59 9 from ($^{34}\text{P}, ^7\text{Be}\gamma$). E_γ : weighted average of 5348.7 2 from $^{34}\text{Al } \beta^-$ decay (53.73 ms) and 5348.3 3 from $^{34}\text{Al } \beta^-$ decay:mixed. I_γ : others: 100 8 from $^{34}\text{Al } \beta^-$ decay (53.73 ms) and 100 6 from $^{34}\text{Al } \beta^-$ decay (23.2 ms).
		5348.6 2	100.0 22	0.0	0 ⁺			
5637.18		1117.9 3	50 17	4518.99	(2 ⁺)			
		2312.3 5	100 33	3325.55	2 ⁺			
5773.45	(2 ⁺)	2447.6 2	53 4	3325.55	2 ⁺			E_γ : weighted average of 2447.4 2 from $^{34}\text{Al } \beta^-$ decay (23.2 ms) and 2447.7 2 from $^{34}\text{Al } \beta^-$ decay:mixed. I_γ : weighted average of 55 4 from $^{34}\text{Al } \beta^-$ decay (23.2 ms) and 50 5 from $^{34}\text{Al } \beta^-$ decay:mixed. E_γ : weighted average of 5772.8 2 from $^{34}\text{Al } \beta^-$ decay (23.2 ms) and 5773.0 3 from $^{34}\text{Al } \beta^-$ decay:mixed. I_γ : other: 100 4 from $^{34}\text{Al } \beta^-$ decay (23.2 ms).
		5772.9 2	100.0 26	0.0	0 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Si})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Comments
6022.13	(3 ⁻)	673.4 1	17.00 35	5348.69	(2 ⁺)	E_γ : weighted average of 673.4 1 from ³⁴ Al β^- decay (53.73 ms) and 673.5 2 from ³⁴ Al β^- decay:mixed. I_γ : weighted average of 16.6 6 from ³⁴ Al β^- decay (53.73 ms) and 17.13 35 from ³⁴ Al β^- decay:mixed.
		778.9 2	4.55 35	5243.21		
		1503.1 1	25.3 4	4518.99	(2 ⁺)	E_γ : weighted average of 1503.1 1 from ³⁴ Al β^- decay (53.73 ms) and 1503.2 2 from ³⁴ Al β^- decay:mixed. I_γ : weighted average of 24.7 6 from ³⁴ Al β^- decay (53.73 ms) and 25.52 35 from ³⁴ Al β^- decay:mixed.
		1767.6 3	4.2 14	4254.67	3 ⁻	
		2696.5 1	100.0 18	3325.55	2 ⁺	E_γ : weighted average of 2696.5 1 from ³⁴ Al β^- decay (53.73 ms) and 2696.5 2 from ³⁴ Al β^- decay:mixed. Other: 2702 2 from (³⁶ S, ³⁴ Si γ). I_γ : other: 100.0 31 from ³⁴ Al β^- decay (53.73 ms).
6201.71	(1,2 ⁺)	6201.2 3	100	0.0	0 ⁺	E_γ : weighted average of 6201.2 3 from ³⁴ Al β^- decay (23.2 ms) and 6201.1 4 from ³⁴ Al β^- decay:mixed.
6227.32	(4 ⁻)	1185.8 1	32.2 29	5041.44	(3 ⁻)	E_γ : weighted average of 1185.8 1 from ³⁴ Al β^- decay (53.73 ms) and 1185.9 2 from ³⁴ Al β^- decay:mixed. I_γ : weighted average of 28.7 23 from ³⁴ Al β^- decay (53.73 ms) and 34.6 19 from ³⁴ Al β^- decay:mixed.
		1257.6 2	8.1 35	4969.54	(5 ⁻)	E_γ : weighted average of 1257.5 1 from ³⁴ Al β^- decay (53.73 ms) and 1257.9 2 from ³⁴ Al β^- decay:mixed. I_γ : unweighted average of 4.6 12 from ³⁴ Al β^- decay (53.73 ms) and 11.5 19 from ³⁴ Al β^- decay:mixed.
		1848.4 4	100 4	4378.68	4 ⁻	E_γ : unweighted average of 1848.0 1 from ³⁴ Al β^- decay (53.73 ms) and 1848.7 2 from ³⁴ Al β^- decay:mixed. I_γ : other: 100 6 from ³⁴ Al β^- decay (53.73 ms).
		1972.7 1	16.8 19	4254.67	3 ⁻	E_γ : weighted average of 1972.7 1 from ³⁴ Al β^- decay (53.73 ms) and 1972.5 2 from ³⁴ Al β^- decay:mixed. I_γ : weighted average of 16.1 23 from ³⁴ Al β^- decay (53.73 ms) and 17.3 19 from ³⁴ Al β^- decay:mixed.
6234.1		1264 1		4969.54	(5 ⁻)	
		1314 1		4919.8		
		1716 [#] 1		4518.99	(2 ⁺)	
6432.22		230.6 1	9.1 18	6201.71	(1,2 ⁺)	E_γ : weighted average of 230.6 1 from ³⁴ Al β^- decay (23.2 ms) and 230.7 3 from ³⁴ Al β^- decay:mixed. I_γ : weighted average of 10.3 15 from ³⁴ Al β^- decay (23.2 ms) and 6.5 22 from ³⁴ Al β^- decay:mixed.
		658.7 1	35 4	5773.45	(2 ⁺)	E_γ : weighted average of 658.7 1 from ³⁴ Al β^- decay (23.2 ms) and 658.8 2 from ³⁴ Al β^- decay:mixed.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Si})$ (continued)

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<u>E_i(level)</u>	<u>J_i^{π}</u>	<u>E_{γ}^{$\dagger$}</u>	<u>I_{γ}^{$\dagger$}</u>	<u>E_f</u>	<u>J_f^{π}</u>	<u>Comments</u>
6432.22		3106.5 1	100.0 30	3325.55	2 ⁺	I _{γ} : unweighted average of 30.9 15 from ³⁴ Al β^- decay (23.2 ms) and 39.1 22 from ³⁴ Al β^- decay:mixed. E _{γ} : weighted average of 3106.5 1 from ³⁴ Al β^- decay (23.2 ms) and 3106.6 2 from ³⁴ Al β^- decay:mixed.
6676.20	(2 ⁺)	2421.4 1	100.0 30	4254.67	3 ⁻	I _{γ} : from ³⁴ Al β^- decay (23.2 ms). Other: 100 5 from ³⁴ Al β^- decay:mixed. E _{γ} : weighted average of 2421.4 1 from ³⁴ Al β^- decay (23.2 ms) and 2421.3 2 from ³⁴ Al β^- decay:mixed.
		3350.6 2	40.6 30	3325.55	2 ⁺	I _{γ} : from ³⁴ Al β^- decay (23.2 ms). Other: 100 5 from ³⁴ Al β^- decay:mixed. E _{γ} : weighted average of 3350.5 2 from ³⁴ Al β^- decay (23.2 ms) and 3350.6 2 from ³⁴ Al β^- decay:mixed.
6764.26	(2 ⁺)	1722.8 2	50 13	5041.44	(3 ⁻)	I _{γ} : weighted average of 39.4 30 from ³⁴ Al β^- decay (23.2 ms) and 44 5 from ³⁴ Al β^- decay:mixed. E _{γ} : weighted average of 1722.9 2 from ³⁴ Al β^- decay (23.2 ms) and 1722.5 3 from ³⁴ Al β^- decay:mixed.
		3438.8 4	63 13	3325.55	2 ⁺	E _{γ} : weighted average of 3439.1 2 from ³⁴ Al β^- decay (23.2 ms) and 3438.2 3 from ³⁴ Al β^- decay:mixed.
6872.35	(2 ⁺)	6763.3 4	100 13	0.0	0 ⁺	
6964.70		6871.8 4	100	0.0	0 ⁺	
		1923.2 2	100	5041.44	(3 ⁻)	E _{γ} : weighted average of 1923.2 2 from ³⁴ Al β^- decay (23.2 ms) and 1923.1 6 from ³⁴ Al β^- decay:mixed.
7107.09	(0 ⁺)	235.1 3	1.3 4	6872.35	(2 ⁺)	
		905.3 1	12.9 9	6201.71	(1,2 ⁺)	E _{γ} : weighted average of 905.3 1 from ³⁴ Al β^- decay (23.2 ms) and 905.5 2 from ³⁴ Al β^- decay:mixed. I _{γ} : weighted average of 13.3 9 from ³⁴ Al β^- decay (23.2 ms) and 12.2 12 from ³⁴ Al β^- decay:mixed.
		1333.6 1	9.4 21	5773.45	(2 ⁺)	E _{γ} : weighted average of 1333.6 1 from ³⁴ Al β^- decay (23.2 ms) and 1333.9 8 from ³⁴ Al β^- decay:mixed.
		1758.4 1	14.8 16	5348.69	(2 ⁺)	I _{γ} : unweighted average of 11.5 9 from ³⁴ Al β^- decay (23.2 ms) and 7.3 12 from ³⁴ Al β^- decay:mixed. E _{γ} : weighted average of 1758.3 1 from ³⁴ Al β^- decay (23.2 ms) and 1758.6 2 from ³⁴ Al β^- decay:mixed.
		2588.0 1	100.0 25	4518.99	(2 ⁺)	I _{γ} : weighted average of 12.4 18 from ³⁴ Al β^- decay (23.2 ms) and 15.9 12 from ³⁴ Al β^- decay:mixed. E _{γ} : from ³⁴ Al β^- decay (23.2 ms). Other: 2588.0 2 from ³⁴ Al β^- decay:mixed. I _{γ} : other: 100 4 from ³⁴ Al β^- decay (23.2 ms).
7476.27	0 ⁽⁺⁾	603.8 2	7.4 11	6872.35	(2 ⁺)	
		1702.8 1	28.6 11	5773.45	(2 ⁺)	E _{γ} : weighted average of 1702.8 1 from ³⁴ Al β^- decay (23.2 ms) and 1703.0 2 from ³⁴ Al β^- decay:mixed.
		2127.6 1	7.0 7	5348.69	(2 ⁺)	I _{γ} : weighted average of 28.8 14 from ³⁴ Al β^- decay (23.2 ms) and 28.4 11 from ³⁴ Al β^- decay:mixed. E _{γ} : weighted average of 2127.6 1 from ³⁴ Al β^- decay (23.2 ms) and 2127.8 3 from ³⁴ Al β^- decay:mixed.
		2957.1 1	38.8 11	4518.99	(2 ⁺)	I _{γ} : weighted average of 6.9 7 from ³⁴ Al β^- decay (23.2 ms) and 7.4 21 from ³⁴ Al β^- decay:mixed. E _{γ} : weighted average of 2957.1 1 from ³⁴ Al β^- decay (23.2 ms) and 2957.2 2 from ³⁴ Al β^-

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Si})$ (continued)

<u>E_i(level)</u>	<u>E_{γ}[†]</u>	<u>I_{γ}[†]</u>	<u>E_f</u>	<u>J_f^{π}</u>	<u>Comments</u>
					decay:mixed.
7476.27	4150.4 1	100.0 34	3325.55	2 ⁺	I _{γ} : weighted average of 38.4 7 from ³⁴ Al β^- decay (23.2 ms) and 42.1 21 from ³⁴ Al β^- decay:mixed. E _{γ} : weighted average of 4150.4 1 from ³⁴ Al β^- decay (23.2 ms) and 4150.5 2 from ³⁴ Al β^- decay:mixed. I _{γ} : from ³⁴ Al β^- decay (23.2 ms). Other: 100 4 from ³⁴ Al β^- decay:mixed.

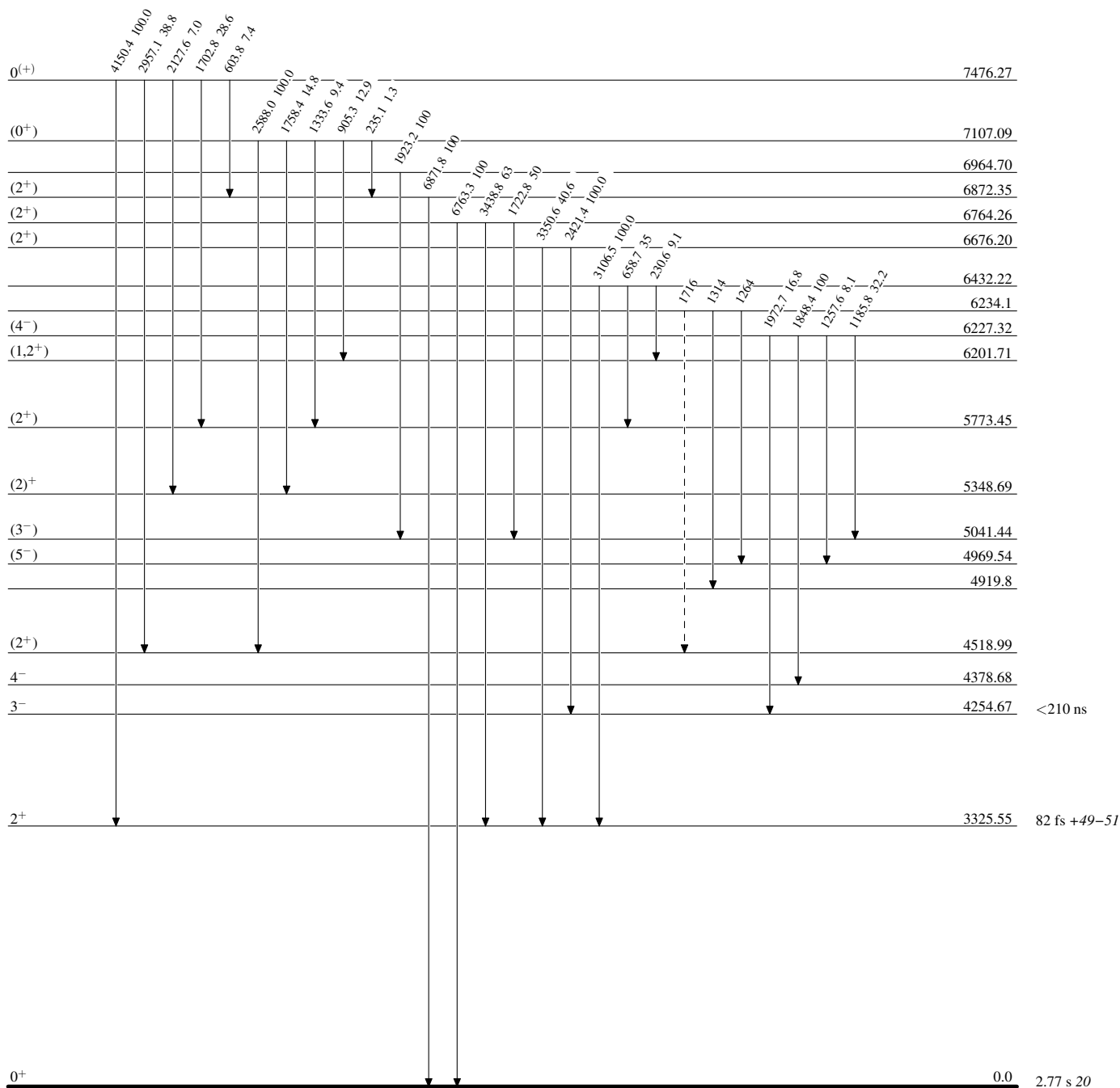
[†] From ³⁴Al β^- decay:mixed (2024Lu03), unless otherwise noted.
[‡] Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.
[#] Placement of transition in the level scheme is uncertain.

Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level

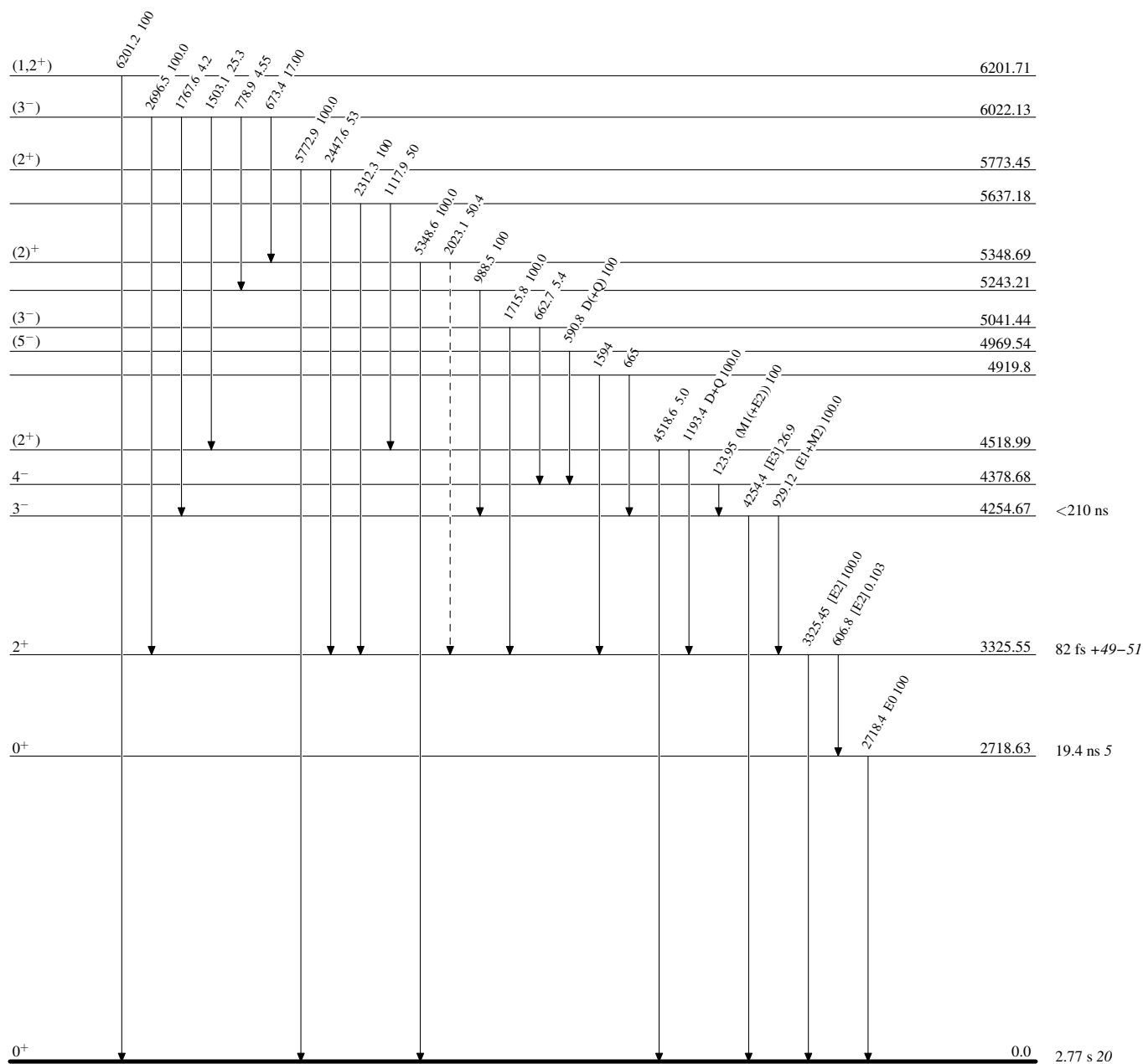
-----► γ Decay (Uncertain)

Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)

^{34}Al β^- decay (53.73 ms) 2019Li41,2017Ha23,2012Ro25

Parent: ^{34}Al : $E=0.0$; $J^\pi=4^-$; $T_{1/2}=53.73$ ms 25; $Q(\beta^-)=16994.1$ 23; $\% \beta^-$ decay=100

^{34}Al - J^π , $T_{1/2}$: From the Adopted Levels of ^{34}Al .

^{34}Al - $Q(\beta^-)$: From 2021Wa16.

2019Li41: ^{34}Mg and ^{34}Al ions were produced via fragmentation reactions induced by a 1.4-GeV pulsed proton beam from the Proton Synchrotron Booster on a UC_x target at CERN-ISOLDE. The atoms that diffused out of the target matrix were selectively laser ionized using the resonance ionization laser ion source (RILIS) and accelerated to 40 keV. $A=34$ nuclei were selected by the ISOLDE General Purpose Separator and implanted onto a movable tape located at the center of the ISOLDE Decay station. The ^{34}Mg beam rate is 700 pps 1 and the ^{34}Al beam rate is 800 pps 1, both with purities of $>99\%$. β particles were detected using a 3-cm-thick NE102 plastic scintillator with an efficiency of 90% 5. γ rays were detected using five HPGe clover detectors. Measured E_γ , I_γ , $\beta\gamma$ -coin, $\beta\gamma\gamma$ -coin, and $\beta e^+/e^-$ -coin. Deduced levels, isomer, J , π , decay branching ratios, $\log ft$, $T_{1/2}$ and $\% \beta^-$ n of parent, and configuration mixings. Comparisons with large-scale shell-model calculations using the SDPF-U-MIX interaction.

2017Ha23, 2017Ha34: a 69.2-MeV/nucleon ^{40}Ar primary beam was produced by the Sector Focusing Cyclotron and Separated Sector Cyclotron at the Heavy Ion Research Facility in Lanzhou (HIRFL). The secondary beam containing ^{34}Al was produced via the projectile fragmentation of ^{40}Ar impinging on a 182.6-mg/cm 2 ^9Be target and was selected using ΔE -tof-B ρ by the Radioactive Ion Beam Line in Lanzhou (RIBLL). The secondary beam ions were transported into a target chamber filled with low-temperature nitrogen and implanted into a 1500- μm -thick DSSD sandwiched between two 3000- μm -thick plastic scintillators for beam diagnostics and veto, respectively. γ rays were detected using two Ge clover detectors. The fast timing signals of the γ rays were provided by a lanthanum bromide (LaBr_3) detector. Neutrons were detected using two plastic scintillator arrays. Measured implant- β correlation, E_γ , I_γ , $\beta\gamma$ -coin, $\gamma\gamma$ -coin, and βn -coin. Deduced levels, J , π , decay branching ratios, $\log ft$, parent ^{34}Al and ^{34}Al isomer $T_{1/2}$. Comparisons with shell-model calculations using Gogny D1S and SDPF-M interactions. Also see 2019Wa38 for the same experiment with ^{35}Al β -delayed neutron data.

2012Ro25: a 600-pps, 93% pure ^{34}Al secondary beam was produced via the projectile fragmentation of a 77.5-MeV/nucleon ^{36}S primary beam on a 240-mg/cm 2 ^9Be target and selected by the LISE3 spectrometer at GANIL. ^{34}Al ions were implanted into a 50- μm Kapton foil in a beam-on (120 ms)/beam-off (300 ms) mode. Electrons and positrons were detected using four ΔE -E telescopes, each consisting of a 1-mm-thick Si detector and a 4.5-mm-thick Si(Li) detector. γ rays were detected using two Ge clover detectors of the EXOGAM array. Measured E_β , $E e^+$, $E e^-$, E_γ , $\beta\gamma$ -coin, and $\beta e^+/e^-$ coin. Deduced ^{34}Al g.s. and isomer $T_{1/2}$, ^{34}Si levels, J , π , transition strengths, and configuration mixing. Comparisons with shell-model calculations. Earlier measurements at GANIL: 1988Mu08, 1986Du07.

2001Nu01: a ^{34}Al secondary beam at 30 pps was produced via the fragmentation of a UC target with 1-GeV protons at ISOLDE, CERN with subsequent surface-ionization and mass separation. ^{34}Al ions were collected onto a moving tape. β particles were detected using a thin cylindrical plastic scintillator, γ rays were detected using two Ge detectors, and neutrons were detected using eight low-threshold plastic scintillators. Measured E_γ , I_γ , E_n , I_n , $\beta\gamma$ -coin, $\gamma\gamma$ -coin, βn -coin. Deduced the decay scheme consisting of ^{33}Si and ^{34}Si levels, ^{34}Al $T_{1/2}$, decay branching ratios, $\log ft$, and β -delayed neutron emission probability. Comparisons with shell-model calculations. Earlier measurement at ISOLDE: 1989Ba50.

The decay scheme is considered incomplete due to a large gap of about 10.8 MeV between the highest observed level at $E=6227$ and $Q(\beta^-)$ value=16994.1 23 (2021Wa16). There may be missing transitions from unobserved levels in the gap.

 ^{34}Si Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\ddagger$	Comments
0.0	0^+	2.77 s 20	
2718.77 12	0^+	19.4 ns 5	
3325.58 7	2^+	82 fs +49–51	
4254.68 8	3^-	<210 ns	
4378.85 10	4^-		
4518.99 7	(2^+)		
4969.65 12	(5^-)		
5041.34 10	(3^-)		
5348.84 11	(2^+)		
6022.19 9	(3^-)		
6227.17 10	(4^-)		
7549+x			

$E(\text{level})$: $x < 9445.4$ 22 from $Q(\beta^-)(^{34}\text{Al}) - S(n)(^{34}\text{Si})$, where $Q(\beta^-)(^{34}\text{Al})=16994.1$ 23

Continued on next page (footnotes at end of table)

^{34}Al β^- decay (53.73 ms) 2019Li41,2017Ha23,2012Ro25 (continued) ^{34}Si Levels (continued)

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>T_{1/2}[‡]</u>	<u>Comments</u>
and S(n)(^{34}Al)=7548.7 11 from 2021Wa16. This represents a range of unidentified levels in ^{34}Si that subsequently decay to ^{33}Si via one-neutron emission.			

[†] From a least-squares fit to γ -ray energies.

[‡] From the Adopted Levels.

 β^- radiations

<u>E(decay)</u>	<u>E(level)</u>	<u>Iβ^-^{†‡}</u>	<u>Log <i>ft</i>[†]</u>
(9.5×10 ³ @ 95)	7549+x	22 5	
(10766.9 25)	6227.17	0.79 7	6.22 5
(10971.9 25)	6022.19	2.76 21	5.72 4
(11645.3 25)	5348.84	<0.1	>9.8 ^{1u}
(11952.8 25)	5041.34	0.95 13	6.4 1
(12024.5 25)	4969.65	3.9 4	5.76 5
(12475.1 # 25)	4518.99	<0.1	>10.0 ^{1u}
(12615.3 25)	4378.85	28 3	5.00 6
(12739.4 25)	4254.68	41 7	4.9 1
(13668.5 25)	3325.58	<0.1	>10.3 ^{1u}

[†] β -feeding intensity is from γ -ray intensity balance at each level. Quoted I β^- values are considered upper limits due to the incomplete decay scheme, and the associated log *ft* values are considered lower limits.

[‡] Absolute intensity per 100 decays.

Existence of this branch is questionable.

@ Estimated for a range of levels.

³⁴Al β⁻ decay (53.73 ms) 2019Li41,2017Ha23,2012Ro25 (continued)

γ(³⁴Si)

I_γ normalization: From Σ%I(γ+ce to g.s.)=78.5, deduced from 100-%β⁻n, where %β⁻n=22.5 from the Adopted Levels of ³⁴Al (originally from 2019Li41). The deduced normalization factor may be considered an upper limit due to potential missing γ transitions from unobserved levels in the gap to the ground state. The original normalization factor with uncertainty is 0.61(6) in 2019Li41. 2019Li41 states that the numbers of implanted ³⁴Mg and ³⁴Al nuclei are derived from singles γ-ray spectra, using the transitions corresponding to the last populated daughters in the decay chains.

E _γ [†]	I _γ ^{†#}	E _i (level)	J _i ^π	E _f	J _f ^π	Mult. [‡]	δ [‡]	α [@]	I _(γ+ce) [#]	Comments
123.9 1	54 2	4378.85	4 ⁻	4254.68	3 ⁻	(M1(+E2))	-0.012 17	0.00223 5		%I _γ =33
590.8 1	6.5 2	4969.65	(5 ⁻)	4378.85	4 ⁻	D(+Q)	+0.01 4			%I _γ =4.0
606.8 1	0.056 6	3325.58	2 ⁺	2718.77	0 ⁺	[E2]		0.0001315 18		%I _γ =0.034
										I _γ : other: 0.073 +79-25 from I _γ (3326)/I _γ (607)=1380 717 (2012Ro25).
673.4 1	0.53 2	6022.19	(3 ⁻)	5348.84	(2) ⁺					%I _γ =0.32
929.1 1	94 2	4254.68	3 ⁻	3325.58	2 ⁺	(E1+M2)	+0.02 1	1.645×10 ⁻⁵ 23		%I _γ =57
1185.8 1	0.25 2	6227.17	(4 ⁻)	5041.34	(3 ⁻)					%I _γ =0.15
1193.4 1	0.83 3	4518.99	(2 ⁺)	3325.58	2 ⁺	D+Q	+0.43 2			%I _γ =0.51
1257.5 1	0.04 1	6227.17	(4 ⁻)	4969.65	(5 ⁻)					%I _γ =0.024
1503.1 1	0.79 2	6022.19	(3 ⁻)	4518.99	(2) ⁺					%I _γ =0.48
1715.7 1	1.8 1	5041.34	(3 ⁻)	3325.58	2 ⁺					%I _γ =1.1
1848.0 1	0.87 5	6227.17	(4 ⁻)	4378.85	4 ⁻					%I _γ =0.53
1972.7 1	0.14 2	6227.17	(4 ⁻)	4254.68	3 ⁻					%I _γ =0.085
2023.1 2	0.18 2	5348.84	(2) ⁺	3325.58	2 ⁺					%I _γ =0.11
2696.5 1	3.2 1	6022.19	(3 ⁻)	3325.58	2 ⁺					%I _γ =2.0
2718		2718.77	0 ⁺	0.0	0 ⁺	E0			<0.5	
3325.4 1	100 2	3325.58	2 ⁺	0.0	0 ⁺	[E2]		0.000923 13		%I _γ =61
4254.4 1	28 1	4254.68	3 ⁻	0.0	0 ⁺	[E3]		0.000970 14		%I _γ =17
4518.6 1	0.05 2	4518.99	(2 ⁺)	0.0	0 ⁺					%I _γ =0.031
5348.7 2	0.38 3	5348.84	(2) ⁺	0.0	0 ⁺					%I _γ =0.23

[†] From 2019Li41.

[‡] From the Adopted Gammas.

[#] For absolute intensity per 100 decays, multiply by 0.61.

[@] Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

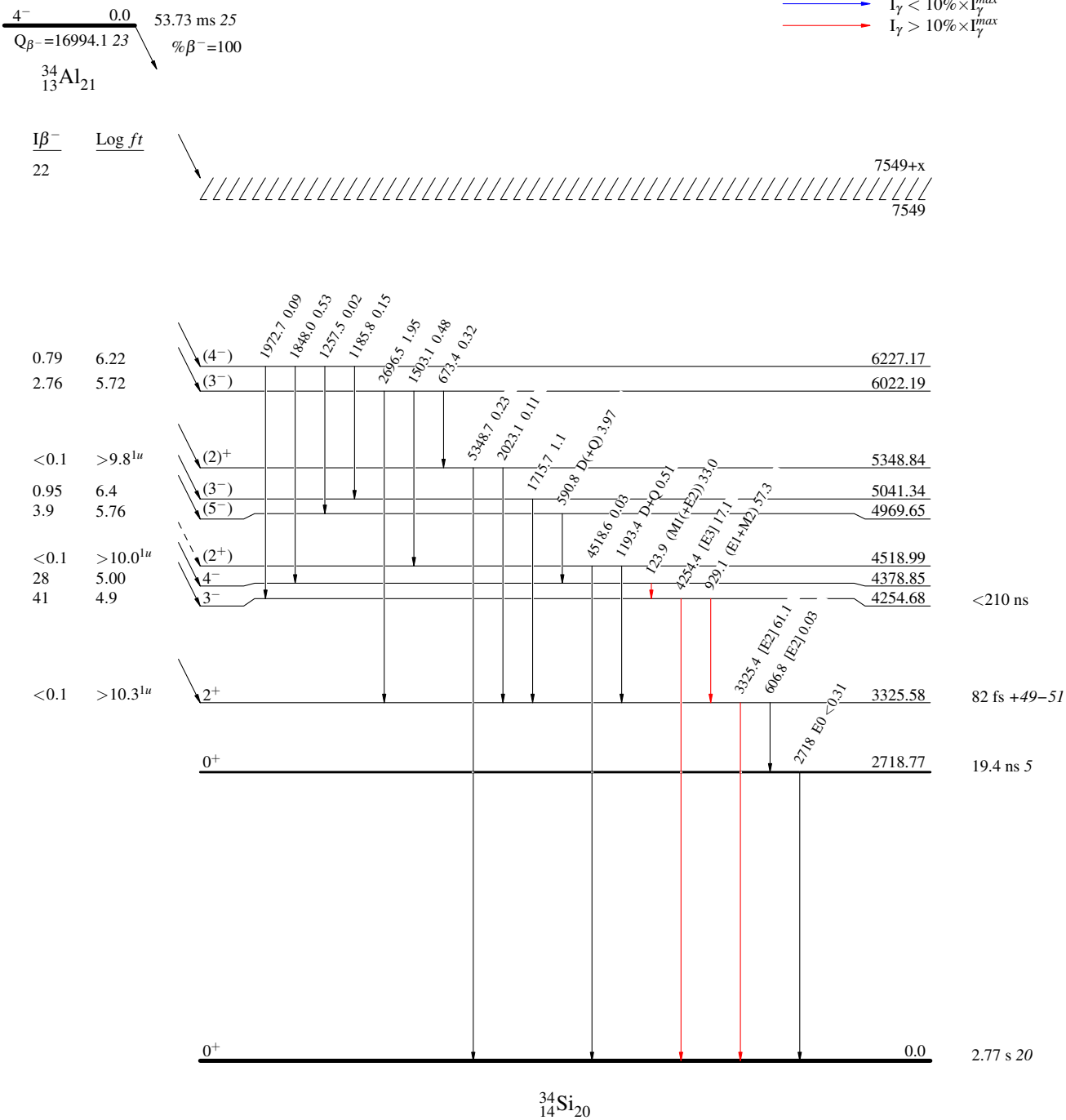
^{34}Al β^- decay (53.73 ms) 2019Li41,2017Ha23,2012Ro25

Decay Scheme

Intensities: $I_{(\gamma+ce)}$ per 100 parent decays

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$



^{34}Al β^- decay (23.2 ms) [2019Li41,2017Ha23,2012Ro25](#)

Parent: ^{34}Al : $E=46.56$ 11; $J^\pi=1^+$; $T_{1/2}=23.2$ ms 14; $Q(\beta^-)=16994.1$ 23; $\% \beta^-$ decay=100

^{34}Al -E, J^π , $T_{1/2}$: From the Adopted Levels of ^{34}Al .

^{34}Al - $Q(\beta^-)$: From [2021Wa16](#).

[2019Li41](#): ^{34}Mg and ^{34}Al ions were produced via fragmentation reactions induced by a 1.4-GeV pulsed proton beam from the Proton Synchrotron Booster on a UC_x target at CERN-ISOLDE. The atoms that diffused out of the target matrix were selectively laser ionized using the resonance ionization laser ion source (RILIS) and accelerated to 40 keV. $A=34$ nuclei were selected by the ISOLDE General Purpose Separator and implanted onto a movable tape located at the center of the ISOLDE Decay station. The ^{34}Mg beam rate is 700 pps 1 and the ^{34}Al beam rate is 800 pps 1, both with purities of $>99\%$. β particles were detected using a 3-cm-thick NE102 plastic scintillator with an efficiency of 90% 5. γ rays were detected using five HPGe clover detectors. Measured E_γ , I_γ , $\beta\gamma$ -coin, $\beta\gamma\gamma$ -coin, and $\beta e^+/e^-$ -coin. Deduced levels, isomer, J , π , decay branching ratios, $\log ft$, $T_{1/2}$ and $\% \beta^-$ n of parent, and configuration mixings. Comparisons with large-scale shell-model calculations using the SDPF-U-MIX interaction.

[2017Ha23](#), [2017Ha34](#): a 69.2-MeV/nucleon ^{40}Ar primary beam was produced by the Sector Focusing Cyclotron and Separated Sector Cyclotron at the Heavy Ion Research Facility in Lanzhou (HIRFL). The secondary beam containing ^{34}Al was produced via the projectile fragmentation of ^{40}Ar impinging on a 182.6-mg/cm 2 ^9Be target and was selected using ΔE -tof-B ρ by the Radioactive Ion Beam Line in Lanzhou (RIBLL). The secondary beam ions were transported into a target chamber filled with low-temperature nitrogen and implanted into a 1500- μm -thick DSSD sandwiched between two 3000- μm -thick plastic scintillators for beam diagnostics and veto, respectively. γ rays were detected using two Ge clover detectors. The fast timing signals of the γ rays were provided by a lanthanum bromide (LaBr_3) detector. Neutrons were detected using two plastic scintillator arrays. Measured implant- β correlation, E_γ , I_γ , $\beta\gamma$ -coin, $\gamma\gamma$ -coin, and βn -coin. Deduced levels, J , π , decay branching ratios, $\log ft$, parent ^{34}Al and ^{34}Al isomer $T_{1/2}$. Comparisons with shell-model calculations using Gogny D1S and SDPF-M interactions. Also see [2019Wa38](#) for the same experiment with ^{35}Al β -delayed neutron data.

[2012Ro25](#): a 600-pps, 93% pure ^{34}Al secondary beam was produced via the projectile fragmentation of a 77.5-MeV/nucleon ^{36}S primary beam on a 240-mg/cm 2 ^9Be target and selected by the LISE3 spectrometer at GANIL. ^{34}Al ions were implanted into a 50- μm Kapton foil in a beam-on (120 ms)/beam-off (300 ms) mode. Electrons and positrons were detected using four ΔE -E telescopes, each consisting of a 1-mm-thick Si detector and a 4.5-mm-thick Si(Li) detector. γ rays were detected using two Ge clover detectors of the EXOGAM array. Measured E_β , $E e^+$, $E e^-$, E_γ , $\beta\gamma$ -coin, and $\beta e^+/e^-$ coin. Deduced ^{34}Al g.s. and isomer $T_{1/2}$, ^{34}Si levels, J , π , transition strengths, and configuration mixing. Comparisons with shell-model calculations. Also see [2015RoZU](#). Earlier measurements at GANIL: [1988Mu08](#), [1986Du07](#).

The decay scheme is considered incomplete due to a large gap of about 10.8 MeV between the highest observed level at $E=6227$ and $Q(\beta^-)$ value=16994.1 23 ([2021Wa16](#)). There may be missing transitions from unobserved levels in the gap.

 ^{34}Si Levels

E(level) [†]	J^π [‡]	$T_{1/2}$ [‡]	Comments
0.0	0^+	2.77 s 20	
2718.52 10	0^+	19.4 ns 5	
3325.59 6	2^+	82 fs +49–51	
4254.70 8	3^-	<210 ns	
4518.98 7	(2^+)		
5041.49 11	(3^-)		
5348.72 9	(2^+)		
5773.42 8	(2^+)		
6201.69 10	($1,2^+$)		
6432.22 9			
6676.21 11	(2^+)		
6764.66 16	(2^+)		
6964.75 23			
7107.05 8	(0^+)		
7476.29 8	$0^{(+)}$		
7549+x			

E(level): $x < 9492.0$ 22 from $Q(\beta^-)(^{34}\text{Al}) + E(\text{isomer}) - S(n)(^{34}\text{Si})$, where $Q(\beta^-)(^{34}\text{Al})=16994.1$ 23 and $S(n)(^{34}\text{Al})=7548.7$ 11 from [2021Wa16](#). This represents a range of unidentified levels in ^{34}Si that subsequently decay to ^{33}Si via one-neutron emission.

Continued on next page (footnotes at end of table)

^{34}Al β^- decay (23.2 ms) 2019Li41,2017Ha23,2012Ro25 (continued) ^{34}Si Levels (continued)[†] From a least-squares fit to γ -ray energies.[‡] From the Adopted Levels. β^- radiations

E(decay)	E(level)	$I\beta^-$ [‡]	Log ft [†]	Comments
(9.5×10 ³ @ 95)	7549+x	11 4		
(9564.4 23)	7476.29	5.6 5	4.72 5	
(9933.6 23)	7107.05	3.4 4	5.02 6	
(10075.9 23)	6964.75	<0.1	>6.6	
(10276.0 23)	6764.66	0.26 4	6.20 7	
(10364.5 23)	6676.21	1.0 1	5.64 5	
(10608.4 23)	6432.22	2.1 2	5.36 5	
(10839.0 23)	6201.69	<0.1	>6.7	
(11267.2 23)	5773.42	<0.4	>6.2	
(11691.9 23)	5348.72	<0.1	>6.9	
(11999.2 # 23)	5041.49	<0.1	>6.9	
(12521.7 # 23)	4518.98	4.8 12	5.35 11	
(12786.0 23)	4254.70	<0.1	>7.1	
(13715.1 23)	3325.59	7 3	5.37 19	
(14322.1 23)	2718.52	37 6	4.74 7	
(17040.7 25)	0.0	27 10	5.24 16	$I\beta^-$: from 100-% β^- n-Σ% $I\beta^-$ (excited levels).

[†] β -feeding intensity is from γ -ray intensity balance at each level. Quoted $I\beta^-$ values are considered upper limits due to the incomplete decay scheme, and the associated log ft values are considered lower limits. But note that possible unobserved transitions due to incomplete decay scheme might not have a significant impact on those very strong branches (e.g., % $I\beta^-$ =37 feeding to 2718 level).

[‡] Absolute intensity per 100 decays.

Existence of this branch is questionable.

@ Estimated for a range of levels.

³⁴Al β⁻ decay (23.2 ms) 2019Li41,2017Ha23,2012Ro25 (continued)

γ(³⁴Si)

I_γ normalization: From %I_γ(3325γ)=22 3, which is determined from the total number of implants and this strongest γ ray. The deduced normalization factor may be considered an upper limit due to potential missing γ transitions from unobserved levels in the gap to the ground state. 2019Li41 states that the numbers of implanted ³⁴Mg and ³⁴Al nuclei are derived from singles γ-ray spectra, using the transitions corresponding to the last populated daughters in the decay chains.

E _γ [†]	I _γ ^{†#}	E _i (level)	J _i ^π	E _f	J _f ^π	Mult. [‡]	δ [‡]	α [@]	I _(γ+ce) [#]	Comments
230.6 1	0.7 1	6432.22		6201.69	(1,2 ⁺)					%I _γ =0.15
658.7 1	2.1 1	6432.22		5773.42	(2 ⁺)					%I _γ =0.46
905.3 1	1.5 1	7107.05	(0 ⁺)	6201.69	(1,2 ⁺)					%I _γ =0.33
929.1 1	2.6 1	4254.70	3 ⁻	3325.59	2 ⁺	(E1+M2)	+0.02 1	1.645×10 ⁻⁵ 23		%I _γ =0.57
1193.4 1	36.5 8	4518.98	(2 ⁺)	3325.59	2 ⁺	D+Q	+0.43 2			%I _γ =8.0
1333.6 1	1.3 1	7107.05	(0 ⁺)	5773.42	(2 ⁺)					%I _γ =0.29
1702.8 1	4.2 2	7476.29	0 ⁽⁺⁾	5773.42	(2 ⁺)					%I _γ =0.92
1715.8 1	0.7 1	5041.49	(3 ⁻)	3325.59	2 ⁺					%I _γ =0.15
1722.9 2	0.6 1	6764.66	(2 ⁺)	5041.49	(3 ⁻)					%I _γ =0.13
1758.3 1	1.4 2	7107.05	(0 ⁺)	5348.72	(2 ⁺)					%I _γ =0.31
1923.2 2	0.2 1	6964.75		5041.49	(3 ⁻)					%I _γ =0.04
2023.1 2	1.0 1	5348.72	(2 ⁺)	3325.59	2 ⁺					%I _γ =0.22
2127.6 1	1.0 1	7476.29	0 ⁽⁺⁾	5348.72	(2 ⁺)					%I _γ =0.22
2421.4 1	3.3 1	6676.21	(2 ⁺)	4254.70	3 ⁻					%I _γ =0.73
2447.4 2	2.9 2	5773.42	(2 ⁺)	3325.59	2 ⁺					%I _γ =0.64
2588.0 1	11.3 4	7107.05	(0 ⁺)	4518.98	(2 ⁺)					%I _γ =2.5
2718.4 1		2718.52	0 ⁺	0.0	0 ⁺	E0			168 3	
2957.1 1	5.6 1	7476.29	0 ⁽⁺⁾	4518.98	(2 ⁺)					%I _γ =1.2
3106.5 1	6.8 2	6432.22		3325.59	2 ⁺					%I _γ =1.5
3325.4 1	100 3	3325.59	2 ⁺	0.0	0 ⁺	[E2]		0.000923 13		%I _γ =22 3 I _γ : 2019Li41 measured %I _γ =22 3 based on the total number of implants and this strongest γ ray and derived a normalization factor of 0.22 3.
3350.5 2	1.3 1	6676.21	(2 ⁺)	3325.59	2 ⁺					%I _γ =0.29
3439.1 2	0.6 1	6764.66	(2 ⁺)	3325.59	2 ⁺					%I _γ =0.13
4150.4 1	14.6 5	7476.29	0 ⁽⁺⁾	3325.59	2 ⁺					%I _γ =3.2
4254.4 1	0.7 1	4254.70	3 ⁻	0.0	0 ⁺	[E3]		0.000970 14		%I _γ =0.15
4518.6 1	2.0 1	4518.98	(2 ⁺)	0.0	0 ⁺					%I _γ =0.44
5348.7 2	1.6 1	5348.72	(2 ⁺)	0.0	0 ⁺					%I _γ =0.35
5772.8 2	5.3 2	5773.42	(2 ⁺)	0.0	0 ⁺					%I _γ =1.2
6201.2 3	2.1 1	6201.69	(1,2 ⁺)	0.0	0 ⁺					%I _γ =0.46

[†] From 2019Li41.

³⁴Al β⁻ decay (23.2 ms) [2019Li41](#),[2017Ha23](#),[2012Ro25](#) (continued)

γ(³⁴Si) (continued)

‡ From the Adopted Gammas.
For absolute intensity per 100 decays, multiply by 0.22.
@ Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on γ-ray energies, assigned
multipolarities, and mixing ratios, unless otherwise specified.

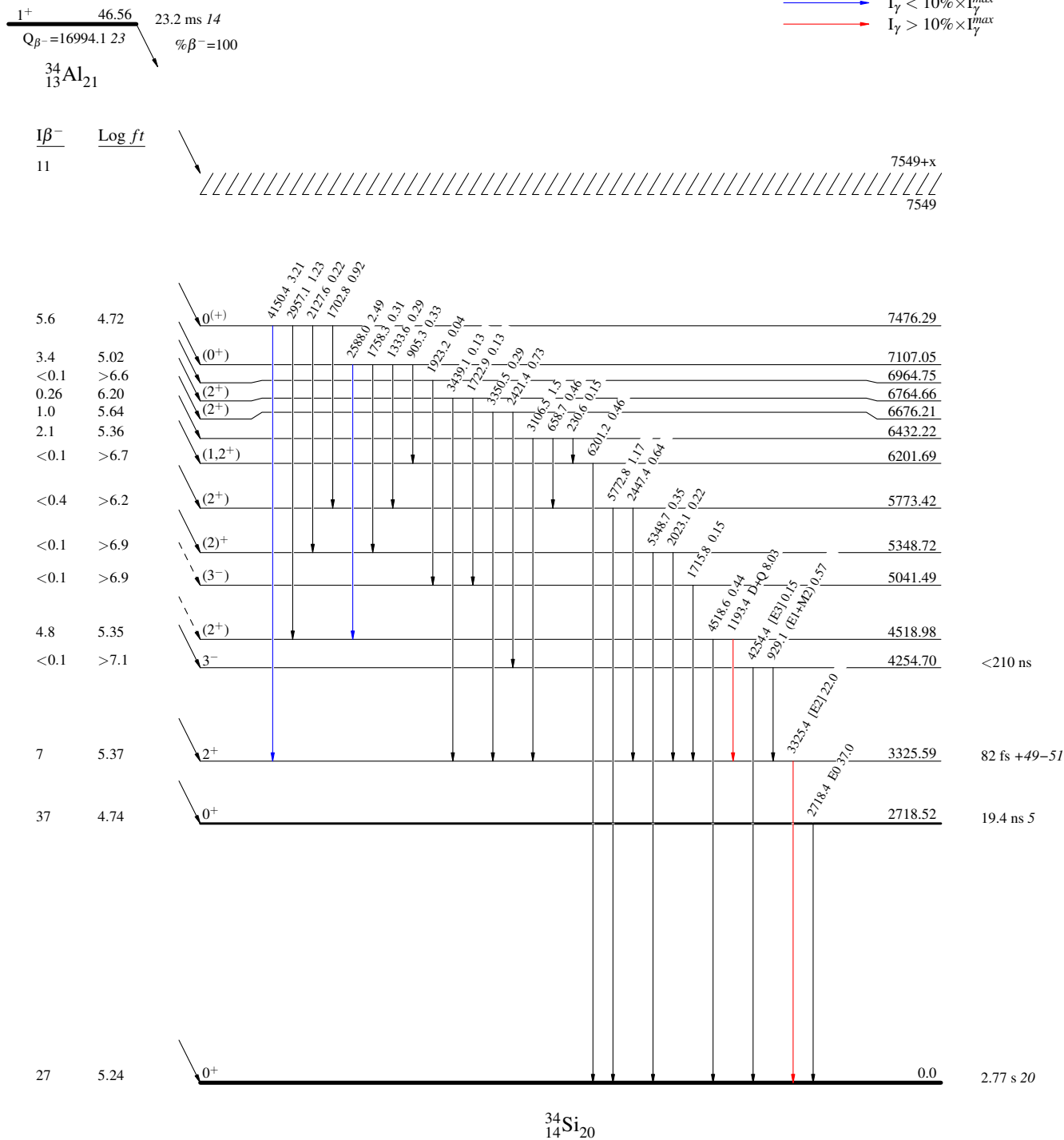
^{34}Al β^- decay (23.2 ms) 2019Li41,2017Ha23,2012Ro25

Decay Scheme

Intensities: $I_{(\gamma+ce)}$ per 100 parent decays

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$



^{34}Al β^- decay:mixed 2024Lu03

Parent: ^{34}Al : $E=46.56$ 11; $J^\pi=1^+$; $T_{1/2}=23.2$ ms 14; $Q(\beta^-)=16994.1$ 23; $\% \beta^-$ decay=100

Parent: ^{34}Al : $E=0$; $J^\pi=4^-$; $T_{1/2}=53.73$ ms 25; $Q(\beta^-)=16994.1$ 23; $\% \beta^-$ decay=100

$^{34}\text{Al}(46.56)-E, J^\pi, T_{1/2}$: From the Adopted Levels of ^{34}Al .

$^{34}\text{Al}(46.56)-Q(\beta^-)$: From 2021Wa16.

$^{34}\text{Al}(0)-J^\pi, T_{1/2}$: From the Adopted Levels of ^{34}Al .

$^{34}\text{Al}(0)-Q(\beta^-)$: From 2021Wa16.

2024Lu03: ^{34}Al ions were produced via reactions of a 480-MeV 14- μA proton beam impinging on a 9.89-g/cm² UC_x target at ISAC, TRIUMF. The ions were selectively ionized using the TRIUMF Resonant Ionization Laser Ion Source (TRILIS), accelerated to 40 keV, separated by the ISAC mass separator, and delivered to a movable Mylar tape. The deexciting γ rays were detected using the Gamma-Ray Infrastructure For Fundamental Investigations of Nuclei (GRIFFIN) Ge array, consisting of 16 Compton-suppressed HPGe clover detectors. β particles were detected using the Scintillating Electron Positron Tagging ARray (SCEPTAR), consisting of ten plastic scintillator paddles and a 1-mm-thin, fast plastic scintillator called the zero degree scintillator (ZDS). Measured E_γ , I_γ , $\gamma\gamma(\theta)$, $\beta\gamma$ -coin, and $\gamma\gamma$ -coin. Deduced J , π , decay branching ratios, $\log ft$. Comparisons with shell-model calculations using the FSU interaction Hamiltonian.

 ^{34}Si Levels

E(level)	J^π	$T_{1/2}$	E(level)	J^π	E(level)	J^π
0	0^+	2.77 s 20	5243.4 2		6676.4 2	(2 ⁺)
2718.6 3	0^+	19.4 ns 5	5348.8 2	(2 ⁺)	6764.0 2	(2 ⁺)
3325.7 1	2 ⁺	82 fs +49-51	5637.4 3		6872.6 2	(2 ⁺)
4254.9 1	3 ⁻	<210 ns	5773.5 2	(2 ⁺)	6964.7 6	
4378.8 2	4 ⁻		6022.3 2	(3 ⁻)	7107.4 2	(0 ⁺)
4519.2 2	(2 ⁺)		6201.8 2	(1,2 ⁺)	7476.5 2	0 ⁽⁺⁾
4969.7 2	(5 ⁻)		6227.5 2	(4 ⁻)		
5041.5 2	(3 ⁻)		6432.4 2			

 $\gamma(^{34}\text{Si})$

E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult.	δ
124.0 2	42.6 6	4378.8	4 ⁻	4254.9	3 ⁻	D(+Q)	-0.012 17
230.7 3	0.03 1	6432.4		6201.8	(1,2 ⁺)		
235.1 3	0.011 3	7107.4	(0 ⁺)	6872.6	(2 ⁺)		
591.0 2	6.4 1	4969.7	(5 ⁻)	4378.8	4 ⁻	D(+Q)	+0.01 4
603.8 2	0.07 1	7476.5	0 ⁽⁺⁾	6872.6	(2 ⁺)		
607.0 3	0.15 2	3325.7	2 ⁺	2718.6	0 ⁺		
658.8 2	0.18 1	6432.4		5773.5	(2 ⁺)		
662.7 2	0.09 1	5041.5	(3 ⁻)	4378.8	4 ⁻		
673.5 2	0.49 1	6022.3	(3 ⁻)	5348.8	(2 ⁺)		
778.9 2	0.13 1	6022.3	(3 ⁻)	5243.4			
905.5 2	0.10 1	7107.4	(0 ⁺)	6201.8	(1,2 ⁺)		
929.2 2	89.9 14	4254.9	3 ⁻	3325.7	2 ⁺	D+Q	+0.02 1
988.5 2	0.10 1	5243.4		4254.9	3 ⁻		
1117.9 3	0.03 1	5637.4		4519.2	(2 ⁺)		
1185.9 2	0.18 1	6227.5	(4 ⁻)	5041.5	(3 ⁻)		
1193.5 2	3.36 5	4519.2	(2 ⁺)	3325.7	2 ⁺	D+Q	+0.43 2
1257.9 2	0.06 1	6227.5	(4 ⁻)	4969.7	(5 ⁻)		
1333.9 8	0.06 1	7107.4	(0 ⁺)	5773.5	(2 ⁺)		
1503.2 2	0.73 1	6022.3	(3 ⁻)	4519.2	(2 ⁺)		
1703.0 2	0.27 1	7476.5	0 ⁽⁺⁾	5773.5	(2 ⁺)		
1715.8 2	1.68 3	5041.5	(3 ⁻)	3325.7	2 ⁺		
1722.5 3	0.04 1	6764.0	(2 ⁺)	5041.5	(3 ⁻)		
1758.6 2	0.13 1	7107.4	(0 ⁺)	5348.8	(2 ⁺)		
1767.6 3	0.12 4	6022.3	(3 ⁻)	4254.9	3 ⁻		
1848.7 2	0.52 2	6227.5	(4 ⁻)	4378.8	4 ⁻		

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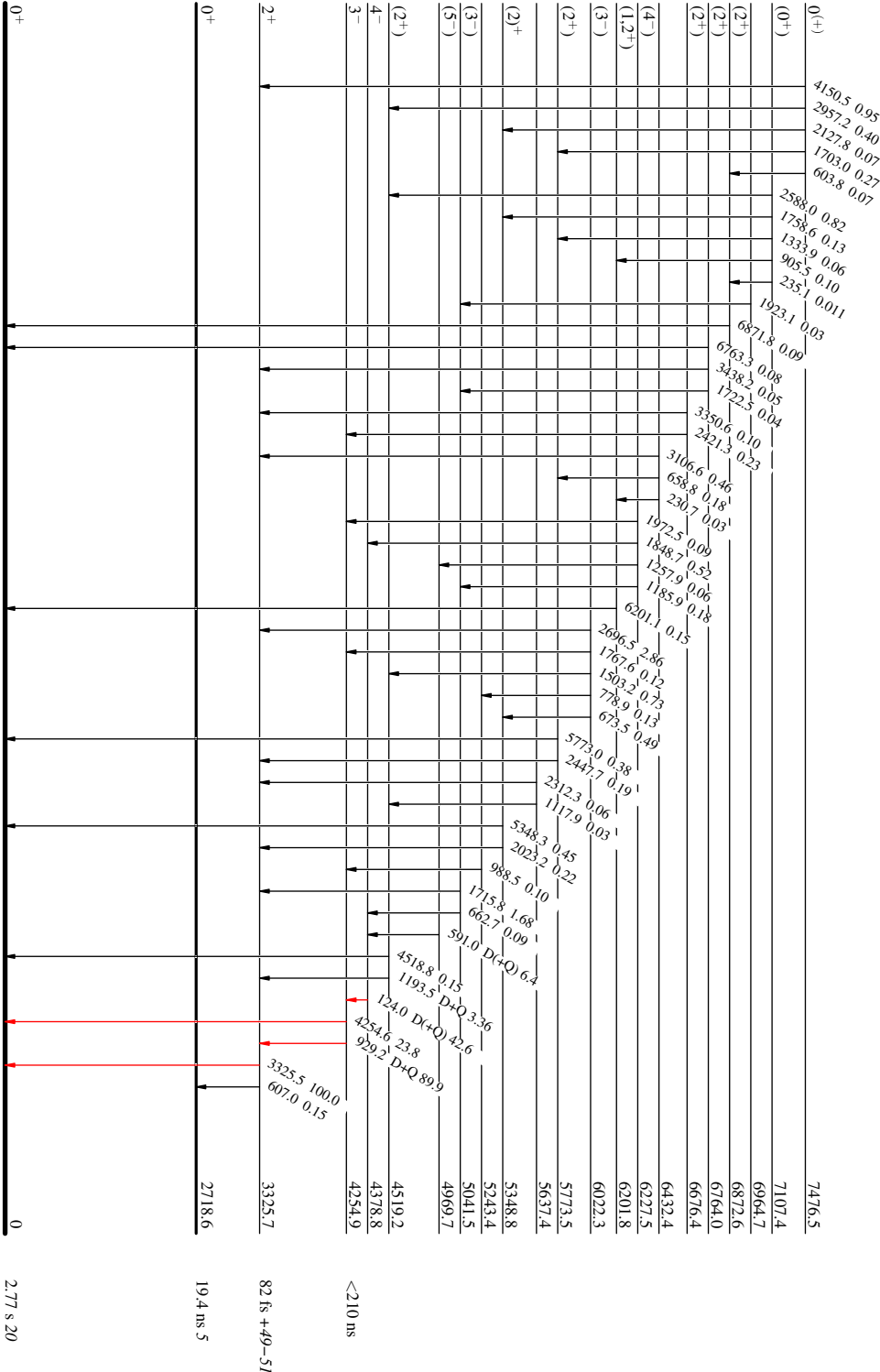
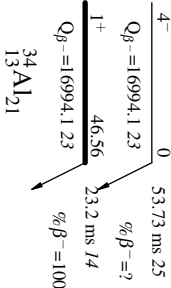
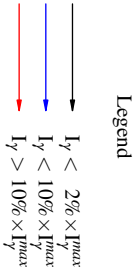
$^{34}\text{Al} \beta^-$ decay:mixed **2024Lu03** (continued) $\gamma(^{34}\text{Si})$ (continued)

E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π
1923.1 6	0.03 1	6964.7		5041.5 (3 ⁻)		3325.5 2	100.0 15	3325.7	2 ⁺	0	0 ⁺
1972.5 2	0.09 1	6227.5	(4 ⁻)	4254.9 3 ⁻		3350.6 2	0.10 1	6676.4	(2 ⁺)	3325.7	2 ⁺
2023.2 3	0.22 1	5348.8	(2) ⁺	3325.7 2 ⁺		3438.2 3	0.05 1	6764.0	(2 ⁺)	3325.7	2 ⁺
2127.8 3	0.07 2	7476.5	0 ⁽⁺⁾	5348.8 (2) ⁺		4150.5 2	0.95 4	7476.5	0 ⁽⁺⁾	3325.7	2 ⁺
2312.3 5	0.06 2	5637.4		3325.7 2 ⁺		4254.6 2	23.8 4	4254.9	3 ⁻	0	0 ⁺
2421.3 2	0.23 1	6676.4	(2 ⁺)	4254.9 3 ⁻		4518.8 4	0.15 1	4519.2	(2 ⁺)	0	0 ⁺
2447.7 2	0.19 2	5773.5	(2 ⁺)	3325.7 2 ⁺		5348.3 3	0.45 1	5348.8	(2) ⁺	0	0 ⁺
2588.0 2	0.82 2	7107.4	(0 ⁺)	4519.2 (2 ⁺)		5773.0 3	0.38 1	5773.5	(2 ⁺)	0	0 ⁺
2696.5 2	2.86 5	6022.3	(3 ⁻)	3325.7 2 ⁺		6201.1 4	0.15 1	6201.8	(1,2 ⁺)	0	0 ⁺
2957.2 2	0.40 2	7476.5	0 ⁽⁺⁾	4519.2 (2 ⁺)		6763.3 4	0.08 1	6764.0	(2 ⁺)	0	0 ⁺
3106.6 2	0.46 2	6432.4		3325.7 2 ⁺		6871.8 4	0.09 1	6872.6	(2 ⁺)	0	0 ⁺

³⁴Al β⁻ decay:mixed 2024I.n03

Decay Scheme

Intensities: Relative I_γ



^{35}Al β^- n decay (38.1 ms) 2019Wa38,2005Ti11,2001Nu01

Parent: ^{35}Al : $E=0$; $J^\pi=(5/2)^+$; $T_{1/2}=38.1$ ms 4; $Q(\beta^-n)=11697$ 7; $\% \beta^-n$ decay=37 3

^{35}Al - J^π , $T_{1/2}$: From the Adopted Levels of ^{35}Al (2026Su18).

^{35}Al - $Q(\beta^-n)$: From 2021Wa16.

^{35}Al - $\% \beta^-n$ decay: $\% \beta^-n=37$ 3: Weighted average of 40 2 (2005Ti11, 2006AnZW), 41 13 (2001Nu01, 2002Nu02), 26 4 (1995ReZZ, 2008ReZZ), and 40 10 (1989Le16, 1989MuZU). Others: 43 9 (1999YoZW, preliminary), 87 +37-25 (1988Mu08, 1988BaYZ) with a non-physical upper bound. Note that the original value reported in 2005Ti11 is 38 2 (2006AnZW report 38 3) and the evaluators have added the neutron feeding intensities of the 4379 (0.9% 2) and 4255 (1.1% 8) levels deduced from γ intensities in ^{34}Si that were not included in their reported $\% \beta^-n$ value, as the authors state that these levels are fed by either neutrons with energy below their detection threshold or by weak transitions that cannot be identified in the neutron spectrum.

2019Wa38: a 69.2-MeV/nucleon ^{40}Ar primary beam was produced by the Sector Focusing Cyclotron and Separated Sector Cyclotron at the Heavy Ion Research Facility in Lanzhou (HIRFL). The secondary beam containing ^{34}Al was produced via the projectile fragmentation of ^{40}Ar impinging on a 182.6-mg/cm² ^9Be target and was selected using ΔE -tof- $B\rho$ by the Radioactive Ion Beam Line in Lanzhou (RIBLL). The secondary beam ions were transported into a target chamber filled with low-temperature nitrogen and implanted into a 1500- μm -thick DSSD sandwiched between two 3000- μm -thick plastic scintillators for beam diagnostics and veto, respectively. γ rays were detected using two Ge clover detectors. The fast timing signals of the γ rays were provided by a lanthanum bromide (LaBr_3) detector. Neutrons were detected using two plastic scintillator arrays. Measured implant- β correlation, E_n , $\beta\gamma$ -coin, and βn -coin. Deduced parent ^{35}Al $T_{1/2}$. Also see 2017Ha23 and 2017Ha34 for the same experiment.

2005Ti11, 2006AnZW: a ^{35}Al secondary beam at ≈ 2 pps was produced via the fragmentation of a 78-MeV/nucleon ^{36}S primary beam and selected by the LISE3 spectrometer at GANIL. A total of 3.46×10^5 ^{35}Al ions were continuously implanted into an NE102A plastic scintillator also for detecting β . The implantation detector was sandwiched between two silicon detectors for monitoring beam and for veto, respectively. Neutrons were detected using the TONNERRE array consisting of 19 plastic scintillator modules. γ rays were detected using two EXOGAM clover modules and a LEPS detector. Measured E_γ , I_γ , E_n , I_n , $\beta\gamma$ -coin, βn -coin, and $\beta n\gamma$ -coin. Deduced the decay scheme consisting of ^{35}Si and ^{34}Si levels, ^{35}Al $T_{1/2}$, decay branching ratios, $\log ft$, $B(\text{GT})$, and β -delayed neutron emission probability. Comparisons with shell-model calculations.

2001Nu01, 2002Nu02: a ^{35}Al secondary beam at 8 pps was produced via the fragmentation of a UC target with 1.4-GeV protons at ISOLDE, CERN with subsequent surface-ionization and mass separation. ^{35}Al ions were collected onto a moving tape. β particles were detected using a thin cylindrical plastic scintillator, γ rays were detected using two Ge detectors, and neutrons were detected using eight low-threshold plastic scintillators. Measured E_γ , I_γ , E_n , I_n , $\beta\gamma$ -coin, $\gamma\gamma$ -coin, βn -coin. Deduced the decay scheme consisting of ^{35}Si and ^{34}Si levels, ^{35}Al $T_{1/2}$, decay branching ratios, $\log ft$, and β -delayed neutron emission probability. Comparisons with shell-model calculations. No evidence was found for any $A=33$ activity and $2n$ emission was thus found negligible.

Other experimental studies on ^{35}Al $T_{1/2}$ and β -delayed neutron emission probability: 1999YoZW, 1995ReZZ/2008ReZZ, 1989MuZU, 1989Le16, 1988MuZY, 1988Mu08, 1988DuZT, 1988BaYZ, 1987DuZU, 1987BaZI.

Theoretical studies involving ^{35}Al decay: 2018Yo06, 2013Li39.

 ^{34}Si Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>
0	0^+
3327.1 7	2^+
4256.3 8	3^-
4380.5 9	4^-

[†] From a least-squares fit to γ -ray energies.

[‡] From the Adopted Levels.

³⁵Al β⁻ n decay (38.1 ms) [2019Wa38](#),[2005Ti11](#),[2001Nu01](#) (continued)

γ(³⁴Si)

I_γ normalization: From %I_γ(3326.96)=5.5 *15* in [2001Nu01](#).

<u>E_γ[†]</u>	<u>I_γ^{‡@}</u>	<u>E_i(level)</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[#]</u>	<u>δ[#]</u>	<u>α^{&}</u>	<u>Comments</u>
124.20 <i>30</i>	16 <i>4</i>	4380.5	4 ⁻	4256.3	3 ⁻	(M1(+E2))	-0.012 <i>17</i>	0.00222 <i>5</i>	%I _γ =0.88 I _γ : weighted average of 16 <i>4</i> from 2005Ti11 and 14 <i>11</i> from 2001Nu01 .
929.12 <i>40</i>	33 <i>7</i>	4256.3	3 ⁻	3327.1	2 ⁺	(E1+M2)	+0.02 <i>1</i>	1.645×10 ⁻⁵ <i>23</i>	%I _γ =1.8 I _γ : weighted average of 36 <i>15</i> from 2005Ti11 and 32 <i>7</i> from 2001Nu01 .
3326.96 <i>70</i>	100 <i>20</i>	3327.1	2 ⁺	0	0 ⁺	[E2]		0.000924 <i>13</i>	%I _γ =5.5 I _γ : 100 <i>27</i> from 2005Ti11 and 100 <i>20</i> from 2001Nu01 .

[†] From [2001Nu01](#).

[‡] From [2005Ti11](#) and [2001Nu01](#), normalized so I_γ for the strongest 3327γ equals 100.

[#] From the Adopted Gammas.

[@] For absolute intensity per 100 decays, multiply by 0.055.

[&] Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

$^{35}\text{Al} \beta^- \text{n decay (38.1 ms)}$ **2019Wa38,2005Ti11,2001Nu01 (continued)**Delayed Neutrons (^{34}Si)

Neutron transitions in [2019Wa38](#) are placed by evaluators based on the similarity of the neutron energy spectra of Fig. 5 of [2019Wa38](#) to Fig. 1 of [2005Ti11](#).

$E(n)_{\text{lab}}^{\dagger}$	$E(^{34}\text{Si})$	$I(n)^{\ddagger @}$	$E(^{35}\text{Si})^{\#}$	Comments
	4256.3	1.02 8		
	4380.5	0.83 19		
510 10	3327.1	<6.57	6322	$E(n)_{\text{lab}}$: from 2019Wa38 short neutron wall. Other: 515 from 2005Ti11 . $I(n)$: 6.3 3 for sum of neutron branches to g.s. and the first excited state of ^{34}Si .
640 10	0	3.05 19	3129	$E(n)_{\text{lab}}$: from 2019Wa38 short neutron wall. Other: 647 from 2005Ti11 .
960 10	0	5.55 28	3458	$E(n)_{\text{lab}}$: weighted average of 950 10 from 2019Wa38 short neutron wall and 970 10 from 2019Wa38 long neutron wall. Other: 948 from 2005Ti11 , 980 from 2001Nu01 .
1323 20	0	2.96 19	3832	$E(n)_{\text{lab}}$: weighted average of 1320 20 from 2019Wa38 short neutron wall and 1340 50 from 2019Wa38 long neutron wall. Other: 1259 from 2005Ti11 .
1685 40	3327.1	2.41 19	7532	$E(n)_{\text{lab}}$: weighted average of 1690 40 from 2019Wa38 short neutron wall and 1670 70 from 2019Wa38 long neutron wall. Other: 1515 from 2005Ti11 .
2026 70	3327.1	2.50 19	7883	$E(n)_{\text{lab}}$: weighted average of 2030 70 from 2019Wa38 short neutron wall and 2020 90 from 2019Wa38 long neutron wall. Other: 1836 from 2005Ti11 .
2607 20	0	8.23 28	5154	$E(n)_{\text{lab}}$: weighted average of 2600 30 from 2019Wa38 short neutron wall and 2610 20 from 2019Wa38 long neutron wall. Other: 2638 from 2005Ti11 , 3000 from 2001Nu01 .
3287 50	0	4.16 19	5854	$E(n)_{\text{lab}}$: weighted average of 3340 90 from 2019Wa38 short neutron wall and 3270 50 from 2019Wa38 long neutron wall. Other: 3192 from 2005Ti11 .
3840 40	0	<6.57	6423	$E(n)_{\text{lab}}$: weighted average of 3850 240 from 2019Wa38 short neutron wall and 3840 40 from 2019Wa38 long neutron wall. Other: 3746 from 2005Ti11 . $I(n)$: 6.3 3 for sum of neutron branches to g.s. and the first excited state of ^{34}Si .

† Evaluators take a weighted average when multiple measurements of a $E(n)_{\text{lab}}$ are available. [2019Wa38](#) reported the $E(n)_{\text{lab}}$ measured by two neutron detectors. [2005Ti11](#) reported the $E(\text{level})$ of ^{35}Si neutron-emitting levels. Evaluators deduced neutron center-of-mass energies $E_{\text{c.m.}} = E(\text{level})(^{35}\text{Si}) - S(n)(^{35}\text{Si}) - E(\text{level})(^{34}\text{Si})$ using the original $S(n) = 2474$ 43 used by [2005Ti11](#). Evaluators then deduced each $E(n)_{\text{lab}} = E_{\text{c.m.}} \times m(^{34}\text{Si}) / [m(n) + m(^{34}\text{Si})]$ for [2005Ti11](#).

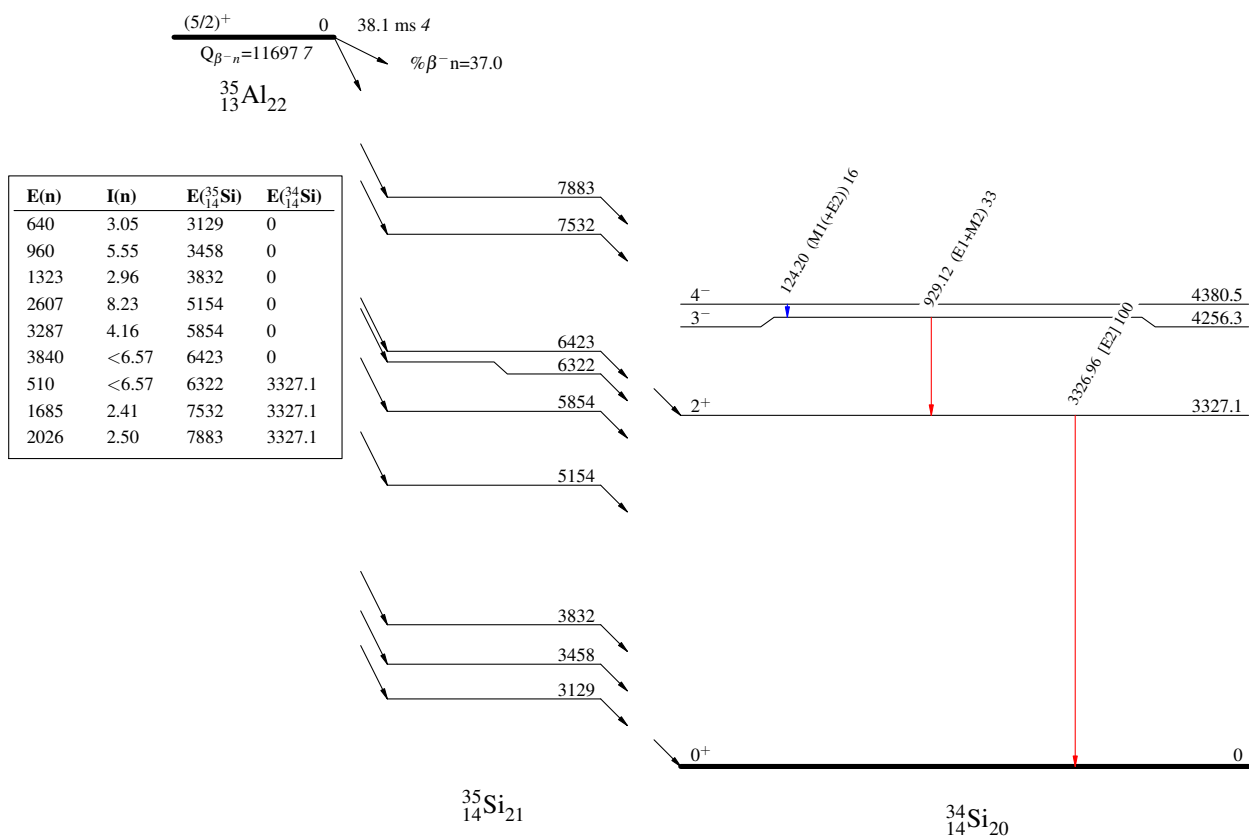
‡ From [2005Ti11](#). Considering the adopted $\% \beta^- n = 37$ 3 and $\% \beta^- n = 40$ 2 in [2005Ti11](#), evaluators scaled the original neutron intensities in [2005Ti11](#) by a factor of 0.925 to be consistent with the adopted $\% \beta^- n$.

$^{\#}$ $E(\text{level})(^{35}\text{Si}) = E(n)_{\text{lab}} \times [m(n) + m(^{34}\text{Si})] / [m(^{34}\text{Si}) + S(n)(^{35}\text{Si}) + E(\text{level})(^{34}\text{Si})]$, where $S(n)(^{35}\text{Si}) = 2470$ 40 ([2021Wa16](#)).

$^{\textcircled{a}}$ Absolute intensity per 100 decays.

^{35}Al β^- n decay (38.1 ms) 2019Wa38,2005Ti11,2001Nu01

Decay Scheme

 γ Intensities: Relative I_γ $I(n)$ Intensities: $I(n)$ per 100 parent decays

$^2\text{H}(^{33}\text{Si}, ^{34}\text{Si}\gamma), (^{35}\text{Si}, ^{34}\text{Si}\gamma)$ 2007GeZX

2007GeZX: 30-AMeV cocktail beam on 30-mg/cm² CD₂ secondary target at GANIL. Heavy ions produced in reactions were identified by the VAMOS spectrometer. γ rays were detected using the EXOGAM germanium clover array. Measured Doppler-corrected E_γ , I_γ , $\gamma\gamma$ -coin, $\gamma\gamma(\theta)$, and $(^{34}\text{Si})\gamma$ -coin. Deduced levels, J, π . Compared with shell-model calculations.

 ^{34}Si Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	Comments
0	0^+	
3326	(2^+)	
3996?	(0^+)	Not observed by any other experiment.
4255		
4379		
4519		
4969		
5041		
6001		

[†] From E_γ data in 2007GeZX.

[‡] Assigned in 2007GeZX from $\gamma\gamma(\theta)$.

 $\gamma(^{34}\text{Si})$

$E_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π
124	4379		4255	
590	4969		4379	
670 [‡]	3996?	(0^+)	3326	(2^+)
929	4255		3326	(2^+)
1191	4519		3326	(2^+)
1480	6001		4519	
1715	5041		3326	(2^+)
2676	6001		3326	(2^+)
3326	3326	(2^+)	0	0^+

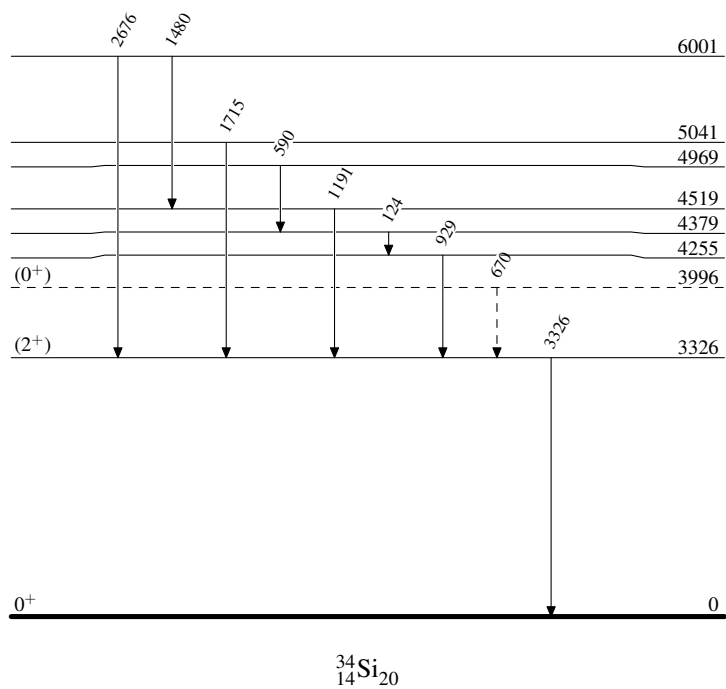
[†] From Fig. 4.26 of 2007GeZX.

[‡] Placement of transition in the level scheme is uncertain.

$^2\text{H}(^{33}\text{Si}, ^{34}\text{Si}\gamma), (^{35}\text{Si}, ^{34}\text{Si}\gamma)$ 2007GeZX

Legend

Level Scheme

 - - - - - $\rightarrow$ γ Decay (Uncertain)


$^2\text{H}(^{34}\text{Si}, ^{34}\text{Si}'\gamma)$ 2003Iw02

2003Iw02: A 38.4-MeV/nucleon, 2×10^4 -pps, 99% pure ^{34}Si secondary beam was produced via the fragmentation of a 95-MeV/nucleon ^{40}Ar primary beam on a 2-mm-thick ^9Be target separated by the RIKEN projectile-fragment separator (RIPS). The secondary target was liquid deuterium placed at the final achromatic focus F3 of RIPS. The reaction products were detected using four silicon counter telescopes each consisting of four 500- μm -thick ion-implanted silicon detectors. Deexcitation γ rays were detected in coincidence with reaction products by the DALI array of 68 NaI(Tl) scintillators. Measured E_γ with Doppler correction, I_γ , $\gamma\gamma$ -coin, and $(^{34}\text{Si})\gamma$ -coin.

 ^{34}Si Levels

<u>$E(\text{level})^\dagger$</u>	<u>J^π</u>
0.0	0^+
3326	2^+
4256	
4379	
4519?	
4970	
5041?	
6022?	

† From E_γ data in 2003Iw02.

 $\gamma(^{34}\text{Si})$

<u>$E_\gamma^\dagger$</u>	<u>I_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
591	4.1 5	4970		4379		
930	26.2 7	4256		3326	2^+	
1193 ‡	4.9 7	4519?		3326	2^+	
x 1480	2.0 8					E_γ : this γ was not seen in coin with 3326 γ . It may possibly be a candidate for populating excited 0^+ state from 3326, 2^+ level but sufficient evidence is lacking (2003Iw02).
1715 ‡	15.8 9	5041?		3326	2^+	
2696 ‡	14.7 13	6022?		3326	2^+	
3326	100.0	3326	2^+	0.0	0^+	
4255	14.0 8	4256		0.0	0^+	I_γ : contains contamination from ^{32}Si and ^{33}Si γ rays with similar energies (2003Iw02).

† From 2003Iw02.

‡ Observed in coincidence with 3326 γ . This indicates that the possible existence of the second 0^+ state at 2133 keV, as suggested by 2001Nu01 is unlikely.

x γ ray not placed in level scheme.

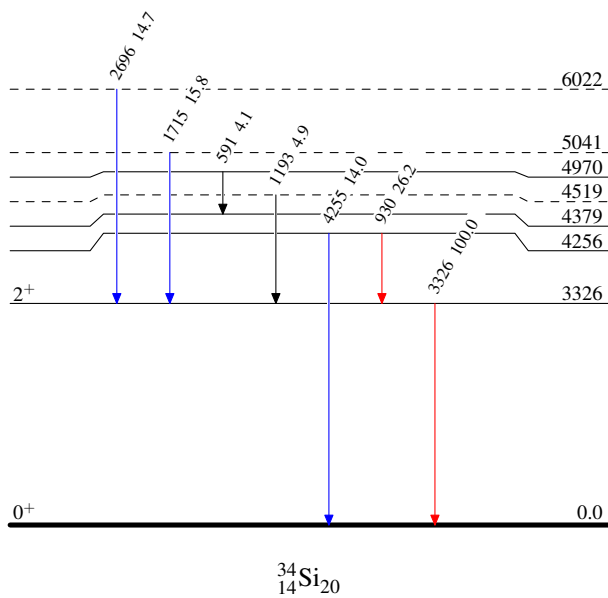
$^2\text{H}(^{34}\text{Si}, ^{34}\text{Si}'\gamma)$ 2003Iw02

Level Scheme

Intensities: Relative I_γ

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
- $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
- $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$



$^7\text{Li}(^{34}\text{P}, ^7\text{Be}\gamma)$ 2010Ze03

$J^\pi=1^+$ for ^{34}P ground state.

2010Ze03: a 100-MeV/nucleon ^{34}P secondary beam was produced via fragmentation of a ^{40}Ar primary beam from the NSCL CCF impinging on a 611-mg/cm² Be target, separated by the A1900 fragment separator, and impinged on a 2.6 mg/cm² natLi (92.4% ^7Li) target. γ rays were detected by SeGA array of fifteen 32-fold segmented Ge detectors. ^{34}Si nuclei were identified and momentum analyzed by the S800 spectrograph. Measured E_γ with Doppler correction, I_γ , (^{34}Si) γ -coin, and angular distributions of scattered ^{34}Si . Deduced Gamow-Teller strength distributions from ^{34}P to ^{34}Si . Comparisons with shell-model calculations.

 ^{34}Si Levels

<u>E(level)</u>	<u>J^π</u>	<u>$L^\ddagger$</u>	<u>Comments</u>
0 [†]		0	
3326			
4256			
4379			A summed 1053 γ from this level was also included to fit the γ -ray spectrum in 2010Ze03.
4519?			
4970			
5041?			
5330 [†]	(2) ⁺	0	J^π : 2 ⁺ from shell-model calculations. B(GT)=0.74 18 (stat) +00-14 (syst) from 2010Ze03.
6022?			

[†] ^{34}Si levels observed in coincidence with 429 γ from $^7\text{Li}\rightarrow^7\text{Be}$ 1/2⁻ level and deexcitation to the 3/2⁻ g.s., in which spin transfer $\Delta S=1$ is ensured.

[‡] From comparison of measured angular distribution with the DWBA calculation and taking into account experimental conditions in a Monte Carlo simulation.

 $\gamma(^{34}\text{Si})$

<u>$E_i(\text{level})$</u>	<u>J^π_i</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>E_f</u>
3326		3326		0
4256		929		3326
		4255		0
4379		125		4256
4519?		1193		3326
4970		591		4379
5041?		1715		3326
5330	(2) ⁺	2000 [#]	59 9	3326
		5330	100	0
6022?		2696		3326

[†] From literature and used by 2010Ze03 to fit their measured γ -ray spectrum, except for the transitions from the 5330 level.

[‡] From 2010Ze03.

[#] Placement of transition in the level scheme is uncertain.

^x γ ray not placed in level scheme.

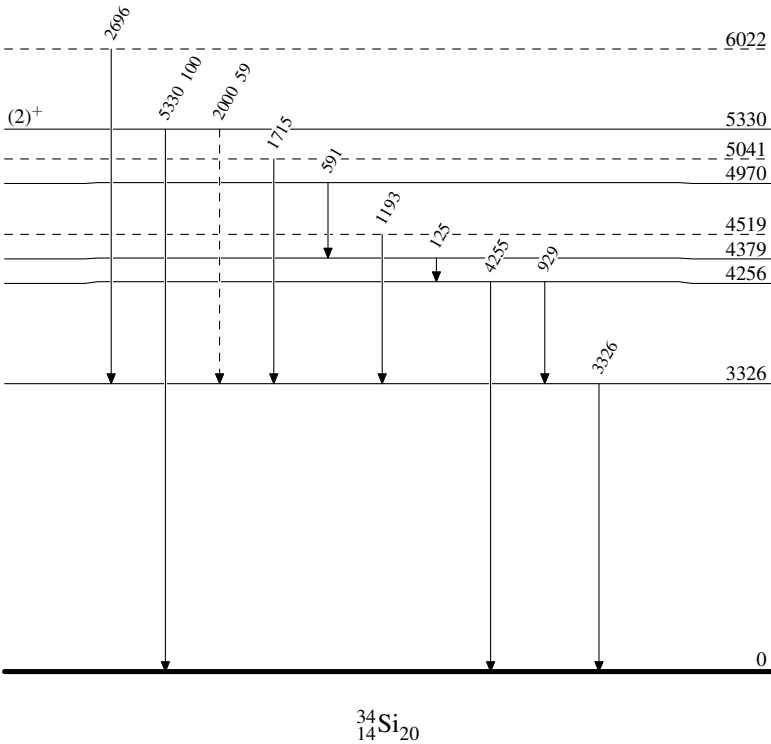
⁷Li(³⁴P,⁷Beγ) 2010Ze03

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)



³⁴Si₁₄²⁰⁻

$^9\text{Be}(^{35}\text{Si}, ^{34}\text{Si}\gamma)$ 2002En02

2002En02: a 73-MeV ^{35}Si secondary beam was produced via fragmentation of 100-MeV/nucleon ^{40}Ar primary beam on a ^9Be target, separated by the NSCL A1200 fragment separator, and impinged on a 93-mg/cm² ^9Be secondary target. The reaction residues were analyzed using the S800 spectrograph. γ rays in coincidence with scattered ^{34}Si were detected using an array of 38 NaI(Tl) detectors. Measured E_γ , I_γ , $\gamma\gamma$ -coin, and $(^{34}\text{Si})\gamma$ -coin. Comparisons with shell-model calculations.

2002En02 fits their γ spectrum using known ^{34}Al β -delayed γ rays from **2001Nu01**. **2002En02** states that an unambiguous placement of the γ rays from **2001Nu01** is not always possible, neither is there a one-to-one correspondence between the proposed excited states and the shell-model calculation. A disagreement between the overall fit and the measured data is clearly visible and indicates the possibility of an even more complex spectrum.

 ^{34}Si Levels $E(\text{level})^\dagger$

0
3327
4257
4380
4520?
4971
5042?
6023?

[†] From E_γ data in **2002En02**.

 $\gamma(^{34}\text{Si})$

<u>$E_\gamma^\dagger$</u>	<u>I_γ</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>
123		4380	4257
591	14 3	4971	4380
929	33 4	4257	3327
1053 [#]	4 2	4380	3327
1193 ^{‡#}	8 2	4520?	3327
1715 ^{‡#}	12 4	5042?	3327
2696 ^{‡#}	5 2	6023?	3327
3326	54 3	3327	0
4257	7 1	4257	0

[†] Quoted by **2002En02** from **2001Nu01**.

[‡] Placed by evaluators based on the Adopted Levels and Gammas.

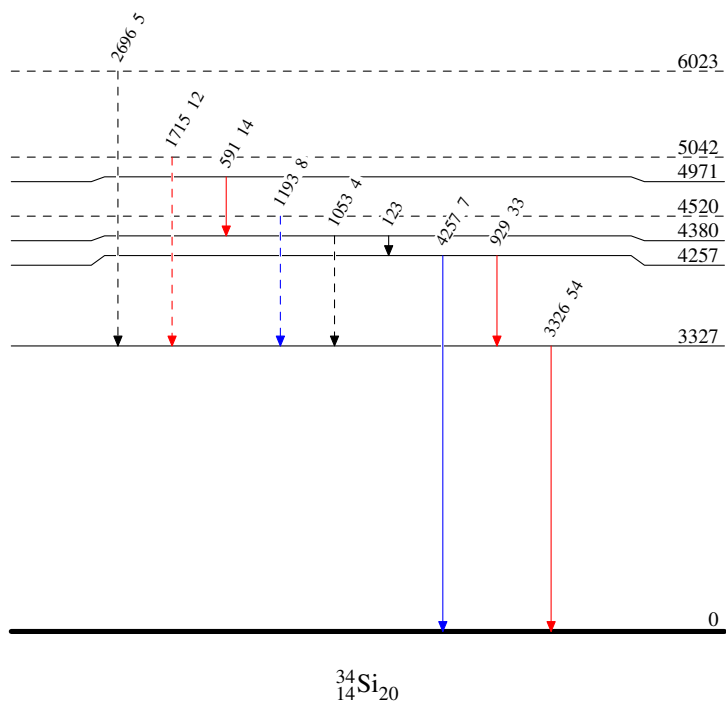
[#] Placement of transition in the level scheme is uncertain.

$^9\text{Be}(^{35}\text{Si}, ^{34}\text{Si}\gamma)$ 2002En02

Legend

Level Scheme
Intensities: Relative I_γ

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$
- - - - - → γ Decay (Uncertain)



$^{18}\text{O}(^{18}\text{O},2\text{p}\gamma)$ **2011Pa43**

2011Pa43: a 25-MeV ^{18}O beam was produced from the Argonne Tandem-Linac Accelerator System (ATLAS) and impinged on a $259\text{-}\mu\text{g}/\text{cm}^2$ ^{18}O target on a $22\text{-mg}/\text{cm}^2$ Ta backing. The fusion-evaporation products were stopped in the thick Ta backing. The evaporated protons were detected using the CsI(Tl) 4π -multidetector device Microball. The deexciting γ rays were detected using Gammasphere consisting of 101 high-purity Ge detectors. Measured E_γ , I_γ , $\gamma\gamma$ -coin, and $2\text{p}\gamma$ coin.

 ^{34}Si Levels $E(\text{level})^\dagger$

0
 3326.2 8
 4255.2 8
 4379.4 11
 4517.5 11
 4920.1 10
 4969.6 12
 6233.7 11

† From a least-squares fit to γ -ray energies.

 $\gamma(^{34}\text{Si})$

$E_\gamma^\dagger$	$E_i(\text{level})$	E_f	$E_\gamma^\dagger$	$E_i(\text{level})$	E_f	$E_\gamma^\dagger$	$E_i(\text{level})$	E_f
124 1	4379.4	4255.2	1191 1	4517.5	3326.2	1716 ‡ 1	6233.7	4517.5
590 1	4969.6	4379.4	1264 1	6233.7	4969.6	3326 1	3326.2	0
665 1	4920.1	4255.2	1314 1	6233.7	4920.1	4255 1	4255.2	0
929 1	4255.2	3326.2	1594 1	4920.1	3326.2			

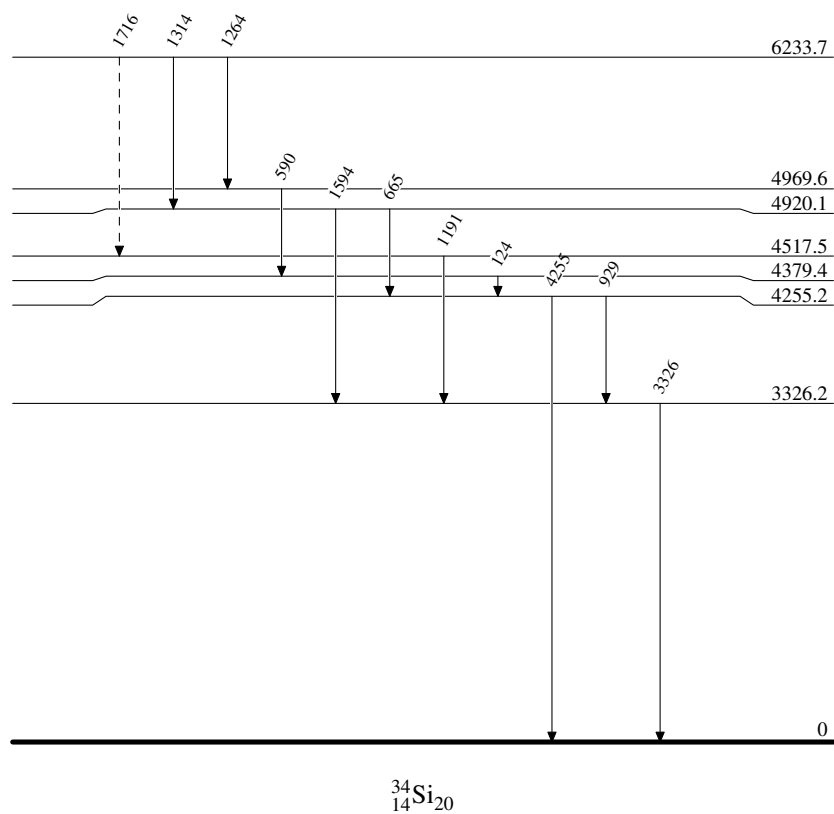
† From **2011Pa43**.

‡ Placement of transition in the level scheme is uncertain.

$^{18}\text{O}(^{18}\text{O},2\text{p}\gamma)$ 2011Pa43

Legend

Level Scheme

 - - - - - $\rightarrow$ γ Decay (Uncertain)

 $^{34}_{14}\text{Si}_{20}$

$\text{Si}(^{34}\text{Si}, ^{34}\text{Si}'\gamma)$ 2002Mi44

[2002Mi44](#), [2002Mi48](#): a 55-MeV/nucleon ^{34}Si secondary beam was produced via fragmentation of a ^{36}S primary beam at GANIL.

The beam was transported to the SPEG reaction chamber and was stopped in a telescope of ΔE and E silicon detectors. Isomeric internal conversion electrons were detected by a 3500- μm Si(Li) detector behind the Si telescope. γ rays were detected with a 4π array consisting of 72 BaF₂ detectors. Measured $E\gamma$, $I\gamma$, (particle)- γ -coin, and population cross sections.

No evidence for a low-lying shape isomeric 0^+ state below the first 2^+ state in ^{34}Si was found.

 ^{34}Si Levels

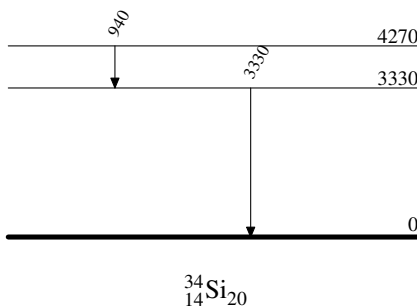
<u>$E(\text{level})^\dagger$</u>	<u>Comments</u>
0	
3330 50	Population $\sigma=9.5$ 2.
4270 71	Population $\sigma=4.3$ 15.

† From $E\gamma$ data in [2002Mi44](#).

 $\gamma(^{34}\text{Si})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>	<u>Comments</u>
940 50	4270	3330	E_γ : 950 50 is given in text of 2002Mi44 .
3330 50	3330	0	

† From [2002Mi44](#) Table 1.

 $\text{Si}(^{34}\text{Si}, ^{34}\text{Si}'\gamma)$ 2002Mi44Level Scheme

³⁶S(¹¹B,¹³N) **1985Fi03**

1985Fi03: a 83-MeV ¹¹B beam was produced from the Australian National University 14UD Pelletron accelerator. The target was 175-μg/cm² Ag₂S on a 16-μg/cm² carbon backing. Reaction products were momentum-analyzed by the Enge split-pole spectrometer with FWHM=230 keV and detected using a multi-element gas-filled detector. Measured cross sections. Deduced levels, Q value, and mass excesses.

³⁴Si Levels

<u>E(level)</u>	<u>J^π</u>	<u>Comments</u>
0	0 ⁺	
5330 50	(2 ⁺)	E(level): from 1985Fi03 . J ^π : proposed by 1985Fi03 based on agreement with excitation energy calculated by shell model and relative intensities calculated using DWBA with the shell-model spectroscopic amplitudes.

 $^{36}\text{S}(^{14}\text{C}, ^{16}\text{O})$ [1984Ma49](#)

[1984Ma49](#): a 71-MeV ^{14}C beam was produced by the Munich MP-Tandem accelerator. The target was Ag_2S with $14\text{-}\mu\text{g}/\text{cm}^2$ 80% enriched ^{36}S on a $3\text{-}\mu\text{g}/\text{cm}^2$ carbon backing. Reaction products were momentum-analyzed by the Munich Q3D spectrograph with FWHM=30-150 keV. Measured particle spectra. Deduced levels and mass excesses.

 ^{34}Si Levels

 $E(\text{level})$

0

3590 25

$^{160}\text{Gd}(^{34}\text{S},\text{X}\gamma),(^{36}\text{S},\text{X}\gamma)$ 1994Fo04

1994Fo04: 159-MeV ^{34}S and ^{36}S beams were produced from the Argonne Tandem Linac Accelerator System (ATLAS). Targets were 1 mg/cm² 98.1% enriched ^{160}Gd backed by 15 mg/cm² gold. γ rays were detected using the Argonne-Notre Dame BGO γ -ray facility consisting of 12 Compton-suppressed Ge detectors and a 50-element bismuth germanate (BGO) array. Measured E_γ , I_γ , $\gamma\gamma$ -coin. Deduced levels. $\gamma\gamma$ -coin gates were placed on known γ rays in specific $A \approx 160$ products to select individual reaction channels and identify coincident γ rays in light product partners. $\gamma\gamma$ -correlation matrices produced with the fold condition $4 \leq k \leq 9$ were found to include most of the events associated with deep inelastic processes.

 ^{34}Si Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>
0	0^+
3326	2^+
4255	3^-
4380	$(4,5)^-$

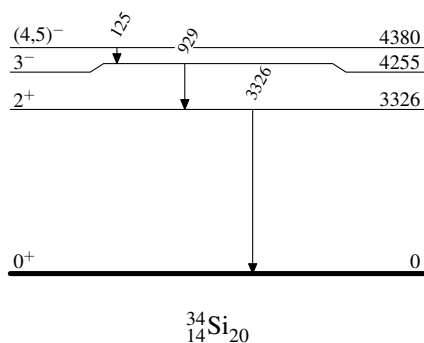
[†] From E_γ data in 1994Fo04.

[‡] Proposed based on yrast state depopulations.

 $\gamma(^{34}\text{Si})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
125	4380	$(4,5)^-$	4255	3^-
929	4255	3^-	3326	2^+
3326	3326	2^+	0	0^+

[†] From 1994Fo04.

 $^{160}\text{Gd}(^{34}\text{S},\text{X}\gamma),(^{36}\text{S},\text{X}\gamma)$ 1994Fo04Level Scheme

$^{208}\text{Pb}(^{36}\text{S}, ^{34}\text{Si}\gamma)$ 2010WaZT

Multi-nucleon transfer and deep-inelastic binary grazing reactions.

2010WaZT: a 215-MeV $^{36}\text{S}^{9+}$ beam was produced from the XTU-Tandem ALPI-superconducting linear accelerator complex at the INFN Legnaro National Laboratory, Italy. The target was $300\text{-}\mu\text{g}/\text{cm}^2$, 99.7% enriched ^{208}Pb on a $20\text{-}\mu\text{g}/\text{cm}^2$ carbon backing. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the CLARA array of 22 escape-suppressed Ge clover detectors covering the azimuthal angles from 98° to 180° . Measured E_γ with Doppler corrections, I_γ , and $(^{34}\text{Si})\gamma$ -coin. Deduced levels. Compared with shell-model calculations. Also see **2010Wa12** and **2010Wa20** from the same experiments.

 ^{34}Si Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>
0	0^+
3326 2	2^+
4255.2 23	(3^-)
4380.2 25	(4^-)
4520.2 23	
4971.2 27	(5^-)
5041.2 23	
5351 3	
6028 3	

† From a least-squares fit to γ -ray energies in **2010WaZT**.

‡ As given in **2010WaZT**, based on shell-model calculations.

 $\gamma(^{34}\text{Si})$

<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
125 1	29.7 19	4380.2	(4^-)	4255.2	(3^-)
591 1	32.2 27	4971.2	(5^-)	4380.2	(4^-)
929 1	50.0 33	4255.2	(3^-)	3326	2^+
1194 1	7.4 14	4520.2		3326	2^+
^x 1595 1	11.6 16				
1715 1	13.8 18	5041.2		3326	2^+
2025 2	9.9 14	5351		3326	2^+
2702 2	11.3 17	6028		3326	2^+
3326 2	100.0 45	3326	2^+	0	0^+

† From Table 6.7 of **2010WaZT**.

^x γ ray not placed in level scheme.

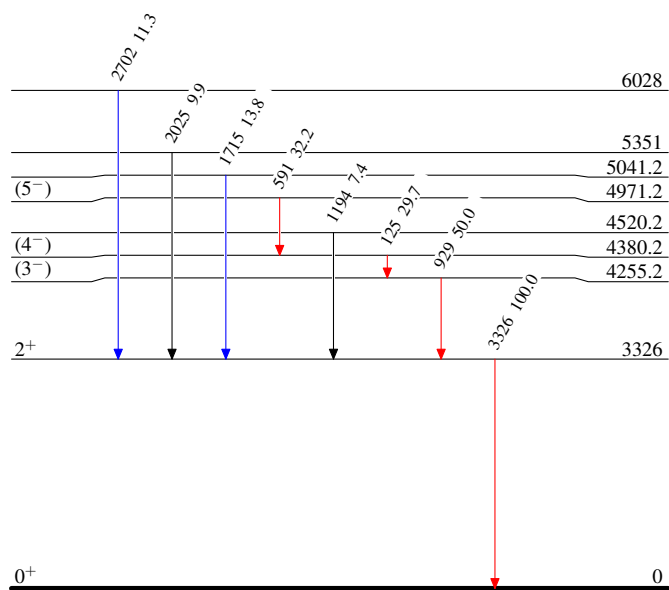
$^{208}\text{Pb}(^{36}\text{S}, ^{34}\text{Si}\gamma)$ 2010WaZT

Level Scheme

Intensities: Relative I_γ

Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$

 $^{34}_{14}\text{Si}_{20}$

Coulomb excitation 1998Ib01

1998Ib01: $^{197}\text{Au}(^{34}\text{Si}, ^{34}\text{Si}'\gamma)$. A 42.6-MeV/nucleon ^{34}Si secondary beam was produced by fragmentation of a 80-MeV/nucleon ^{40}Ar primary beam from the NSCL A1200 cyclotron on a 356-mg/cm² ^9Be production target at NSCL. The reaction target was a 184-mg/cm² ^{197}Au . Scattered beam particles were detected using a fast/slow plastic phoswich detector and γ rays were detected using an array of 39 cylindrical NaI(Tl) detectors. Measured E_γ , I_γ , and (scattered- ^{34}Si) γ -coin. Deduced excitation cross sections and $B(E2)$.

 ^{34}Si Levels

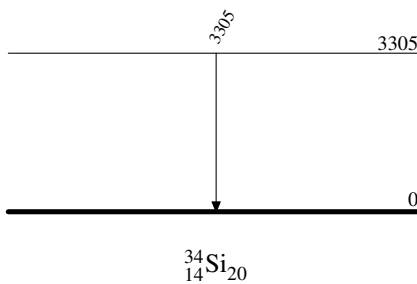
<u>$E(\text{level})^\dagger$</u>	<u>Comments</u>
0	
3305 55	$B(E2)^\dagger=0.0085$ 33 (1998Ib01) $\sigma=18.5$ mb 72 (1998Ib01).

† From E_γ data.

 $\gamma(^{34}\text{Si})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>
3305 55	3305	0

† From **1998Ib01**.

Coulomb excitation 1998Ib01Level Scheme

Adopted Levels, Gammas

$Q(\beta^-)=5383.0$ 8; $S(n)=6282.7$ 14; $S(p)=11323.3$ 11; $Q(\alpha)=-11109.5$ 21 [2021Wa16](#)

$S(2n)=16386.5$ 8, $S(2p)=28027$ 7 ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): $\text{Cl}(n,X)$ reaction at ETH Zurich ([1945Zu01](#), [1946Bl01](#)).

^{34}P Production:

[2012Zh06](#): $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=57$ MeV/nucleon at HIRFL, Lanzhou.

[2007No13](#): $^9\text{Be}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=100$ MeV/nucleon at RIKEN.

^{34}P radius measurement:

[1999Ai02](#): $\text{Si}(^{34}\text{P},X)$ at NSCL. Measured energy-integrated reaction cross sections at $E=38\text{--}80$ MeV/nucleon. Deduced strong absorption radii.

^{34}P mass measurement:

[2009Kw02](#): LEBIT Penning trap.

Theoretical calculations:

[2023Ch17](#): shell model with N3LO, JISP16, and DJ16A interactions. Calculated energy levels, J , π .

[2020Fo04](#): shell model, USD interaction. Uncertainty quantification of level energies, J , π , $B(E2)$, $B(M1)$.

[2017Tr02](#): shell model with SDPF-MU interaction. Calculated levels, J , π .

[2016Si02](#): ab initio with NN+3N interactions. Calculated $S(2n)$, $S(2p)$, energies of first 2^+ states.

[2012BoZT](#): shell model with PSDPF interaction, intruder states. Calculated levels, J , π , $T_{1/2}$.

[2012SrZZ](#): large-scale shell model. Calculated energy levels, J , π , $B(E2)$, $B(M1)$.

 ^{34}P LevelsCross Reference (XREF) Flags

A	^{34}Si β^- decay (2.77 s)	G	$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ $E=44$ MeV	M	$^{160}\text{Gd}(^{37}\text{Cl},X\gamma)$
B	$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ $E=20$ MeV	H	$^{30}\text{Si}(^7\text{Li},^3\text{He})$	N	$^{176}\text{Yb}(^{36}\text{S},X\gamma)$
C	$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ $E=24$ MeV	I	$^{34}\text{S}(t,^3\text{He})$	O	$^{198}\text{Pt}(^{37}\text{Cl},X\gamma)$
D	$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ $E=25$ MeV	J	$^{34}\text{S}(^7\text{Li},^7\text{Be}),(^{11}\text{B},^{11}\text{C})$	P	$^{208}\text{Pb}(^{36}\text{S},^{34}\text{P}\gamma)$
E	$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ $E=30$ MeV	K	$^{36}\text{S}(\text{pol } d,\alpha),(\text{POL } d,\alpha\gamma)$	Q	Coulomb excitation
F	$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ $E=34$ MeV	L	$^{115}\text{In}(^{34}\text{S},X\gamma)$		

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	XREF	Comments
0.0	1^+	12.43 s 10	ABCDEFGHIJKLMNO PQ	$\% \beta^- = 100$ J^π : allowed β^- -feeding to 0^+ ^{34}S ground state; unnatural parity from analyzing powers in $^{36}\text{S}(\text{pol } d,\alpha),(\text{pol } d,\alpha\gamma)$. $T_{1/2}$: weighted average of 12.40 s 12 (1946Bl01) and 12.45 s 10 (1973Go33). Other: 12.5 s (1958Sc31). Reduced strong absorption radius $r_0^2=1.144$ fm ² 76 (1999Ai02). XREF: H(441) J^π : spin=2 from $\alpha\gamma(\theta)$ in $^{36}\text{S}(\text{pol } d,\alpha),(\text{pol } d,\alpha\gamma)$. Spin=2 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ (2009Be26). Natural parity from analyzing powers in $^{36}\text{S}(\text{pol } d,\alpha),(\text{pol } d,\alpha\gamma)$. $T_{1/2}$: lifetime $\tau=1.6$ ps 5: weighted average of $\tau=1.9$ ps +9–5 in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ $E=24$ MeV from 2012Be11 with DSAM and $\tau=1.5$ ps 5 in $^{208}\text{Pb}(^{36}\text{S},^{34}\text{P}\gamma)$ from 2019Gr08 with DRDM. Other: $\tau<1.4$ ps in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ from 2009Be26 with DSAM.
429.07 13	2^+	1.11 ps 35	ABCDEFGHIJKLMNO PQ	J^π : spin=1 from $\alpha\gamma(\theta)$ in $^{36}\text{S}(\text{pol } d,\alpha),(\text{pol } d,\alpha\gamma)$ and from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ (2009Be26). Unnatural parity from analyzing powers in $^{36}\text{S}(\text{pol } d,\alpha),(\text{pol } d,\alpha\gamma)$. $T_{1/2}$: lifetime $\tau=1.1$ ps +4–3: weighted average of 1.3 ps 4 in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ from 2009Be26 with DSAM and 0.75 ps +65–20 in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ $E=24$ MeV from 2012Be11 with DSAM.
1607.73 19	1^+	0.76 ps +28–21	ABCDEFG IJKL PQ	XREF: I(2225)Q(2235?)
2228.53 34	2^-	>2 ps	C I K Q	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{34}P Levels (continued)

E(level) [†]	J ^π	T _{1/2}	XREF	Comments
2305.1 5	4 ⁻	2.0 ns 1	BCDEFG i j k l m n o p	J ^π : unnatural parity from analyzing powers in $^{36}\text{S}(\text{pol d}, \alpha)$, (pol d, $\alpha\gamma$). Spin=2 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV. T _{1/2} : lifetime $\tau > 3$ ps in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM. XREF: i(2309)j(2287) J ^π : spin=3,4 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV. 1875.9 γ , M2+E3, $\Delta J=2$ to 2 ⁺ , 429.07 level. Possible unnatural parity from analyzing powers in $^{36}\text{S}(\text{pol d}, \alpha)$, (pol d, $\alpha\gamma$). T _{1/2} : from $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ from 2012Ma21 with $\gamma\gamma(\text{t})$. Other: $0.3 < T_{1/2} < 2.5$ ns in $^{198}\text{Pt}(^{37}\text{Cl}, \text{X}\gamma)$ from 2002AsZY with fragment- $\gamma(\text{t})$. XREF: i(2309)
2320.5 5	3 ⁽⁻⁾	>6.9 ps	BCDEFG i L N P	J ^π : spin=3 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV. 3 ⁻ from shell model (2018Lu08 , 2012Be11). T _{1/2} : lifetime $\tau > 10$ ps in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM. Other: > 3 ps in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ from 2009Be26 with DSAM.
2372 20			K	
2628 12			K	
2675.0 7	(1 ⁺ , 2 ⁻ , 3 ⁺)		K P	XREF: P(2682.1?) J ^π : 2246 γ to 2 ⁺ , 429.07 level. 1067 γ to 1 ⁺ , 1607.73 level. Possible unnatural parity from analyzing powers in $^{36}\text{S}(\text{pol d}, \alpha)$, (pol d, $\alpha\gamma$).
3201 16			K	
3352.5 7	5 ⁽⁻⁾	0.28 ps +10-8	BCDEFG I L N P	XREF: I(3345?) J ^π : spin=5 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV. 5 ⁻ from shell model (2018Lu08 , 2012Be11). T _{1/2} : lifetime $\tau=0.41$ ps +15-12: weighted average of 0.30 ps 15 in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ from 2009Be26 with DSAM and 0.52 ps +18-12 in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM.
3482 8			K	
3752.4 7	(3,4)	0.26 ps 6	C F	J ^π : spin=(3,4) from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV. T _{1/2} : lifetime $\tau=0.37$ ps 8 in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM.
3806.5 26			C	
3911.5 11	(3,4)	0.14 ps 7	C	J ^π : spin=(3,4) from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV. T _{1/2} : lifetime $\tau=0.2$ ps 1 in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM.
3951.3 6	5 ⁻	0.118 ps 35	BCDEFG P	J ^π : spin=5 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV. 678.8 γ , M1+E2 from 6 ⁻ , 4630.1 level. T _{1/2} : lifetime $\tau=0.17$ ps 5: weighted average of 0.30 ps 15 in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ from 2009Be26 with DSAM and 0.16 ps 5 in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM.
4301.8 10	(1 ⁺ , 2 ⁻ , 3 ⁺)		K	J ^π : 2694 γ to 1 ⁺ , 1607.73 level. Possible unnatural parity from analyzing powers in $^{36}\text{S}(\text{pol d}, \alpha)$, (pol d, $\alpha\gamma$).
4424.2 27			C j	XREF: j(4470)
4454.6 11	(4 ⁻)	<0.10 ps	C jK	XREF: j(4470)k(4434) J ^π : spin=3,4 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV. 4 ⁻ from shell model (2018Lu08). T _{1/2} : lifetime $\tau < 0.14$ ps in $^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM.
4570.7 21			C	

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Adopted Levels, Gammas (continued) ^{34}P Levels (continued)

E(level) [†]	J ^π	T _{1/2}	XREF		Comments
4613.1 22			C		
4630.1 6	6 ⁻	0.31 ps 5	BCDEFG		J ^π : spin=6 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV. 2325.1 γ , E2(+M3), $\Delta J=2$ to 4 ⁻ , 2305.1 level. T _{1/2} : lifetime $\tau=0.45$ ps 7: weighted average of 0.5 ps 2 in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ from 2009Be26 with DSAM and 0.44 ps 7 in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM.
4710.1 16	(5,6 ⁻)		C	N	XREF: N(?) J ^π : spin=4,5,6 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV; 2405.5 γ to 4 ⁻ , 2305.1 level. 2018Lu08 adopts spin=(5,6) based on shell model and the tendency of fusion-evaporation reactions to favor higher-spin states.
4729.6 30			C	K	XREF: K(4725)
5012.6 21		<0.07 ps	C		T _{1/2} : lifetime $\tau<0.1$ ps in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM.
5344.5 29			C		
5394.1 7	6 ⁻	0.11 ps +8-5	C		J ^π : spin=6 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV; 3090 γ to 4 ⁻ , 2305.1 level cannot be M2 based on measured T _{1/2} and RUL. T _{1/2} : lifetime $\tau=0.16$ ps +12-7 in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM.
5594.5 26			C		
5726.4 13	4,6 ⁽⁻⁾		C		J ^π : spin=4,6 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV; 3421 γ to 4 ⁻ , 2305.1 level.
5788.7 29			C		
5975.7 30			C		
6118.6 31			C		
6174.6 29			C		
6193.6 11		<0.69 ps	BCDE G		T _{1/2} : lifetime $\tau<1$ ps in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ from 2009Be26 with DSAM.
6237.1 7	7 ⁽⁺⁾	>6.9 ps	BCDEFG	L N	J ^π : spin=7 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV. 2884.2 γ , M2(+E3), $\Delta J=2$ to 5 ⁽⁻⁾ , 3352.5 level. 7 ⁺ from shell model (2018Lu08 , 2012Be11). T _{1/2} : lifetime $\tau>10$ ps in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM. Other: >3 ps in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ from 2009Be26 with DSAM.
6294.8 22		<0.07 ps	C		T _{1/2} : lifetime $\tau<0.1$ ps in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM.
6317.0 31			C		
6357.2 9	(7) ⁻	<0.035 ps	C		J ^π : 2405.9 γ , E2 to 5 ⁻ , 3951.3 level. Shell-model suggests 7 ⁻ (2018Lu08 , 2012Be11). T _{1/2} : lifetime $\tau<0.05$ ps in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM.
6445.2 31			C		
6923.3 27	(6)		C		J ^π : spin=4,5,6 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$. 2018Lu08 adopts spin=(6) based on shell model and the tendency of fusion-evaporation reactions to favor higher-spin states.
6991.2 29			C		
7065.9 26			C		
7256.4 26			C		
7426.2 21		<0.07 ps	C	J	T _{1/2} : lifetime $\tau<0.1$ ps in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM.
7590.7 32			C		
7919.9 10		>0.35 ps	C		T _{1/2} : lifetime $\tau>0.5$ ps in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV from 2012Be11 with DSAM.

[†] From a least-squares fit to γ -ray energies.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{P})$									Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\delta^\ddagger$	$\alpha^\#$	
429.07	2 ⁺	429.07 13	100	0.0	1 ⁺	M1(+E2)	-0.07 7	0.000155 5	B(M1)(W.u.)=0.25 +12-6 E γ : from ^{34}Si β^- decay. Others: 429 1 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ (2009Be26), 429.1 4 from $^{18}\text{O},\text{pn}\gamma$ E=24 MeV, 429.4 10 from $^{18}\text{O},\text{pn}\gamma$ E=34 MeV, 429 1 from $(^{36}\text{S},^{34}\text{P}\gamma)$, and 422 7 from Coulomb excitation. Mult., δ : from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=34 MeV (2009Ch43). Other: D(+Q) with $\delta=+0.11$ +13-12 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV.
1607.73	1 ⁺	1178.62 16	100 5	429.07	2 ⁺	(M1+E2)	-0.93 26	2.87×10^{-5} 14	B(M1)(W.u.)=0.0060 +30-20; B(E2)(W.u.)=15 +7-6 E γ : weighted average of 1178.52 16 from ^{34}Si β^- decay, 1179 1 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ (2009Be26), 1179.1 4 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV, 1179.8 10 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=34 MeV, and 1178 1 from $(^{36}\text{S},^{34}\text{P}\gamma)$. I γ : from ^{34}Si β^- decay. Others: 100 22 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=20 MeV, 100 5 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV, 100 17 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=25 MeV, 100 33 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=30 MeV, 100 12 from (pol d, α), and 100 7 from $(^{36}\text{S},^{34}\text{P}\gamma)$. Mult., δ : D+Q from $\alpha\gamma(\theta)$ in $^{36}\text{S}(\text{pol d},\alpha\gamma)$. $\Delta\pi$ =no from level scheme. Other: D from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV and $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=34 MeV.
		1607.5 5	57 5	0.0	1 ⁺	(M1+E2)	+0.13 6	0.0001096 17	B(M1)(W.u.)=0.0025 +10-7; B(E2)(W.u.)=0.064 +78-45 E γ : weighted average of 1607.6 5 from ^{34}Si β^- decay, 1607 1 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ (2009Be26), 1608 1 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV, and 1607 1 from $(^{36}\text{S},^{34}\text{P}\gamma)$. I γ : weighted average of 56 8 from ^{34}Si β^- decay, 52 5 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV, 64 8 from (pol d, α), and 60 7 from $(^{36}\text{S},^{34}\text{P}\gamma)$. Mult., δ : D+Q from $\alpha\gamma(\theta)$ in $^{36}\text{S}(\text{pol d},\alpha\gamma)$. $\Delta\pi$ =no from level scheme.
2228.53	2 ⁻	620.6 4	68 14	1607.73	1 ⁺	(E1(+M2))	+0.07 16	5.0×10^{-5} 7	E γ : other: 627 9 from Coulomb excitation. Mult., δ : D(+Q) from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV. $\Delta\pi$ =yes from level scheme.
		1799.7 6	100 16	429.07	2 ⁺	[E1]		0.000504 7	
		2229 1	59 14	0.0	1 ⁺	[E1]		0.000803 11	
2305.1	4 ⁻	1875.9 6	100	429.07	2 ⁺	M2+E3	-0.65 38	0.000124 8	B(M2)(W.u.)=0.045 14; B(E3)(W.u.)=32 24

Adopted Levels, Gammas (continued)

<u>$\gamma(^{34}\text{P})$ (continued)</u>									Comments
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\dagger$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[‡]</u>	<u>$\delta^\ddagger$</u>	<u>$\alpha^\#$</u>	
2320.5	3 ⁽⁻⁾	1891.4 7	100	429.07	2 ⁺	(E1(+M2))	+0.07 14	0.000569 19	E_γ: weighted average of 1876 2 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ (2009Be26), 1875.8 6 from $(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV, 1876.1 10 from $(^{18}\text{O},\text{pn}\gamma)$ E=34 MeV, and 1876 1 from $(^{36}\text{S},^{34}\text{P}\gamma)$. Mult.,$\delta$: Q+O, $\Delta J=2$ from $\gamma\gamma(\theta)(\text{DCO})$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=34 MeV (2009Ch43). M3 ruled out by RUL. Others: Q(+O) with $\delta=0.00$ 12 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV. Q, $\Delta J=2$ from $\gamma\gamma(\theta)(\text{DCO})$ in $^{115}\text{In}(^{34}\text{S},\text{X}\gamma)$ (2006Kr07). Q from γ anisotropy in $^{198}\text{Pt}(^{37}\text{Cl},\text{X}\gamma)$.
2675.0	(1 ⁺ ,2 ⁻ ,3 ⁺)	1067 2246	46 18 100 26	1607.73 429.07	1 ⁺ 2 ⁺				E_γ, I_γ: from $^{36}\text{S}(\text{pol d},\alpha),(\text{pol d},\alpha\gamma)$. E_γ, I_γ: from $^{36}\text{S}(\text{pol d},\alpha),(\text{pol d},\alpha\gamma)$. Other: questionable 2253 1 with 100 10 and 2682 1 with 79 10 from $^{208}\text{Pb}(^{36}\text{S},^{34}\text{P}\gamma)$.
3352.5	5 ⁽⁻⁾	1031.6 9 1047.1 8	3.17 10 100.0 5	2320.5 2305.1	3 ⁽⁻⁾ 4 ⁻	[E2] (M1(+E2))	+0.02 9	2.53×10 ⁻⁵ 4	B(E2)(W.u.)=8.1 +33-21 B(M1)(W.u.)=0.066 +27-18 E_γ: weighted average of 1047 2 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ (2009Be26), 1047.3 8 from $(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV, 1047.8 10 from $(^{18}\text{O},\text{pn}\gamma)$ E=34 MeV, and 1046 1 from $(^{36}\text{S},^{34}\text{P}\gamma)$. Mult.,$\delta$: other: D, $\Delta J=1$ from $\gamma\gamma(\theta)(\text{DCO})$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=34 MeV (2009Ch43). $\Delta\pi=(\text{no})$ from level scheme.
3752.4	(3,4)	1431.8 6 1523.9 9	100.0 22 7.5 22	2320.5 2228.53	3 ⁽⁻⁾ 2 ⁻	D(+Q)			
3806.5		3377.2 26	100	429.07	2 ⁺				
3911.5	(3,4)	1591 1	100	2320.5	3 ⁽⁻⁾				

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{P})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\delta^\ddagger$	$\alpha^\#$	Comments
3951.3	5 ⁻	1631.5 11 1646.0 4	4.2 21 100.0 21	2320.5 2305.1	3 ⁽⁻⁾ 4 ⁻	[E2] (M1(+E2))	-0.03 18	0.0001562 22 0.0001216 24	B(E2)(W.u.)=2.6 +17-12 B(M1)(W.u.)=0.040 +18-11 E _γ : weighted average of 1645 2 from ¹⁸ O(¹⁸ O,pnγ) (2009Be26), 1646.1 4 from (¹⁸ O,pnγ) E=24 MeV, 1646.2 10 from (¹⁸ O,pnγ) E=34 MeV, and 1645 1 from (³⁶ S, ³⁴ Pγ). Mult.,δ: D, ΔJ=1 from γγ(θ)(DCO) and possible magnetic character from γγ(θ)(POL) in ¹⁸ O(¹⁸ O,pnγ) E=34 MeV (2009Ch43). D(+Q) with δ=-0.03 18 from γ(θ) in (¹⁸ O,pnγ) E=24 MeV.
4301.8	(1 ⁺ ,2 ⁻ ,3 ⁺)	2694	100	1607.73	1 ⁺	(M1+E2)		0.00035 4	Mult.: D+Q from γ(θ) in ¹⁸ O(¹⁸ O,pnγ) E=24 MeV. Δπ=(no) from level scheme. B(M1)(W.u.)>0.023 if M1, B(E2)(W.u.)>19 if E2.
4424.2		3994.9 27	100	429.07	2 ⁺				
4454.6		2134	100	2320.5	3 ⁽⁻⁾				
4570.7	6 ⁻	2250.1 20	100	2320.5	3 ⁽⁻⁾	M1+E2	+0.12 11	5.95×10 ⁻⁵ 22	B(M1)(W.u.)=0.091 +17-15; B(E2)(W.u.)=11 +29-10 E _γ : weighted average of 679 2 from ¹⁸ O(¹⁸ O,pnγ) (2009Be26), 678.7 4 from (¹⁸ O,pnγ) E=24 MeV, and 679.4 10 from (¹⁸ O,pnγ) E=34 MeV. I _γ : weighted average of 50 25 from (¹⁸ O,pnγ) E=20 MeV, 54 5 from (¹⁸ O,pnγ) E=24 MeV, 100 25 from (¹⁸ O,pnγ) E=25 MeV, 70 10 from (¹⁸ O,pnγ) E=30 MeV, 76 4 from (¹⁸ O,pnγ) E=34 MeV, and 82 18 from (¹⁸ O,pnγ) E=44 MeV. Mult.,δ: D, ΔJ=1 from γγ(θ)(DCO) and possible magnetic character from γγ(θ)(POL) in ¹⁸ O(¹⁸ O,pnγ) E=34 MeV (2009Ch43). D+Q with δ=+0.12 11 from γ(θ) in (¹⁸ O,pnγ) E=24 MeV (2012Be11). M2 ruled out by RUL.
4613.1		2292.5 21	100	2320.5	3 ⁽⁻⁾				
4630.1		678.8 4	68 5	3951.3	5 ⁻				
		2325.1 6	100 5	2305.1	4 ⁻	E2(+M3)	+0.02 +11-10	0.000480 8	B(E2)(W.u.)=2.4 +6-5

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{P})$ (continued)

<u>E_i(level)</u>	<u>J_i^{π}</u>	<u>E_{γ}^{$\dagger$}</u>	<u>I_{γ}^{$\dagger$}</u>	<u>E_f</u>	<u>J_f^{π}</u>	<u>Mult.^{$\ddagger$}</u>	<u>δ^{$\ddagger$}</u>	<u>α[#]</u>	<u>Comments</u>
									E_{γ}: others: 2325 2 from ¹⁸O(¹⁸O,pnγ) (2009Be26) and 2325.1 10 from (¹⁸O,pnγ) E=34 MeV. I_{γ}: others: 100 50 from (¹⁸O,pnγ) E=20 MeV, 100 25 from (¹⁸O,pnγ) E=25 MeV, 100 20 from (¹⁸O,pnγ) E=30 MeV, 100 7 from (¹⁸O,pnγ) E=34 MeV, and 100 18 from (¹⁸O,pnγ) E=44 MeV. Mult.,δ: Q, $\Delta J=2$ from $\gamma\gamma(\theta)$(DCO) and possible electric character from $\gamma\gamma(\theta)$(POL) in ¹⁸O(¹⁸O,pnγ) E=34 MeV (2009Ch43). M2 ruled out by RUL. Q(+O) with $\delta=+0.02$ +11-10 from $\gamma(\theta)$ in (¹⁸O,pnγ) E=24 MeV.
4710.1	(5,6 ⁻)	1356.9 23	33 11	3352.5	5 ⁽⁻⁾	D+Q			
		2405.5 20	100 5	2305.1	4 ⁻				
4729.6		4300.2 30	100	429.07	2 ⁺				
5012.6		2692 2	100	2320.5	3 ⁽⁻⁾				
5344.5		3039.3 28	100	2305.1	4 ⁻				
5394.1	6 ⁻	762.9 8	15 6	4630.1	6 ⁻	[M1+E2]			
		1442.6 7	100 11	3951.3	5 ⁻	(M1(+E2))	+0.02 5		
		2042 1	44 9	3352.5	5 ⁽⁻⁾	(M1(+E2))	+0.05 14		
		3090 2	28 9	2305.1	4 ⁻	[E2]			
5594.5		3289.2 25	100	2305.1	4 ⁻				
5726.4	4,6 ⁽⁻⁾	2373.9 15	100 8	3352.5	5 ⁽⁻⁾	D(+Q)			
		3421 2	28 8	2305.1	4 ⁻				
5788.7		3483.4 28	100	2305.1	4 ⁻				
5975.7		3670.4 29	100	2305.1	4 ⁻				
6118.6		3813.3 30	100	2305.1	4 ⁻				
6174.6		3869.3 28	100	2305.1	4 ⁻				
6193.6		1563.4 14	49 11	4630.1	6 ⁻				
		2840 2	100 11	3352.5	5 ⁽⁻⁾				
6237.1	7 ⁽⁺⁾	842.5 6	48 10	5394.1	6 ⁻	(E1(+M2))	+0.05 14		
		1607.7 5	100 13	4630.1	6 ⁻	(E1(+M2))	+0.05 9	0.000358 7	
									E_{γ}: weighted average of 2839 2 from ¹⁸O(¹⁸O,pnγ) (2009Be26) and 2841 2 from (¹⁸O,pnγ) E=24 MeV. B(E1)(W.u.)<4.0×10⁻⁵ Mult.,δ: D(+Q) and δ from $\gamma(\theta)$ in ¹⁸O(¹⁸O,pnγ) E=24 MeV; $\Delta\pi$=(yes) from level scheme. B(E1)(W.u.)<1.1×10⁻⁵; B(M2)(W.u.)<0.37 E_{γ}: weighted average of 1607 2 from ¹⁸O(¹⁸O,pnγ) (2009Be26), 1607.9 5 from (¹⁸O,pnγ) E=24 MeV, and 1607.1 10 from

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{P})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. ‡	$\delta^\ddagger$	$\alpha^\#$	Comments
									($^{18}\text{O},\text{pn}\gamma$) E=34 MeV. I_γ : others: 100 33 from ($^{18}\text{O},\text{pn}\gamma$) E=20 MeV, 100 16 from ($^{18}\text{O},\text{pn}\gamma$) E=25 MeV, 100 18 from ($^{18}\text{O},\text{pn}\gamma$) E=30 MeV, 100 12 from ($^{18}\text{O},\text{pn}\gamma$) E=34 MeV, and 100 16 from ($^{18}\text{O},\text{pn}\gamma$) E=44 MeV. Mult., δ : D, $\Delta J=1$ from $\gamma\gamma(\theta)(\text{DCO})$ and possible electric character from $\gamma\gamma(\theta)(\text{POL})$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=34 MeV (2009Ch43). D(+Q) with $\delta=+0.05$ 9 from $\gamma(\theta)$ in ($^{18}\text{O},\text{pn}\gamma$) E=24 MeV. B(M2)(W.u.)<0.95; B(E3)(W.u.)<53 E_γ : weighted average of 2883 2 from $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ (2009Be26), 2884.3 7 from ($^{18}\text{O},\text{pn}\gamma$) E=24 MeV, and 2884.3 10 from ($^{18}\text{O},\text{pn}\gamma$) E=34 MeV. I_γ : others: <17 from ($^{18}\text{O},\text{pn}\gamma$) E=20 MeV, 23 8 from ($^{18}\text{O},\text{pn}\gamma$) E=25 MeV, 46 9 from ($^{18}\text{O},\text{pn}\gamma$) E=30 MeV, 147 9 from ($^{18}\text{O},\text{pn}\gamma$) E=34 MeV, and 85 16 from ($^{18}\text{O},\text{pn}\gamma$) E=44 MeV. Mult., δ : M2 from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\theta)(\text{POL})$ in $^{115}\text{In}(^{34}\text{S},\text{X}\gamma)$ (2006Kr07). Q(+O) with $\delta=+0.11$ 18 from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV. Q, $\Delta J=2$ from $\gamma\gamma(\theta)(\text{DCO})$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=34 MeV (2009Ch43). B(E3)(W.u.)<16 E_γ : from ($^{18}\text{O},\text{pn}\gamma$) E=34 MeV. Other: 3932 3 from ($^{18}\text{O},\text{pn}\gamma$) E=24 MeV.
6237.1	$7^{(+)}$	2884.2 7	90 13	3352.5	$5^{(-)}$	M2(+E3)	+0.11 18	0.000425 8	
		3931.7 10	12.5 75	2305.1	4^-	[E3]		0.000870 12	
6294.8		2942.2 21	100	3352.5	$5^{(-)}$				
6317.0		4011.6 30	100	2305.1	4^-				
6357.2	$(7)^-$	2405.9 7	75 14	3951.3	5^-	E2		0.000519 7	B(E2)(W.u.)>11 Mult.: Q from $\gamma(\theta)$ in $^{18}\text{O}(^{18}\text{O},\text{pn}\gamma)$ E=24 MeV. M2 ruled out by RUL.
		3004 2	100 14	3352.5	$5^{(-)}$	[E2]		0.000791 11	B(E2)(W.u.)>4.9
6445.2		4139.8 30	100	2305.1	4^-				
6923.3	(6)	3570.6 26	100	3352.5	$5^{(-)}$	D+Q			
6991.2		3638.5 28	100	3352.5	$5^{(-)}$				
7065.9		3713.2 25	100	3352.5	$5^{(-)}$				
7256.4		3903.7 25	100	3352.5	$5^{(-)}$				
7426.2		1189 2	100	6237.1	$7^{(+)}$				
7590.7		4238.0 31	100	3352.5	$5^{(-)}$				
7919.9		1683 1	100 6	6237.1	$7^{(+)}$				
		1726 1	49 6	6193.6					

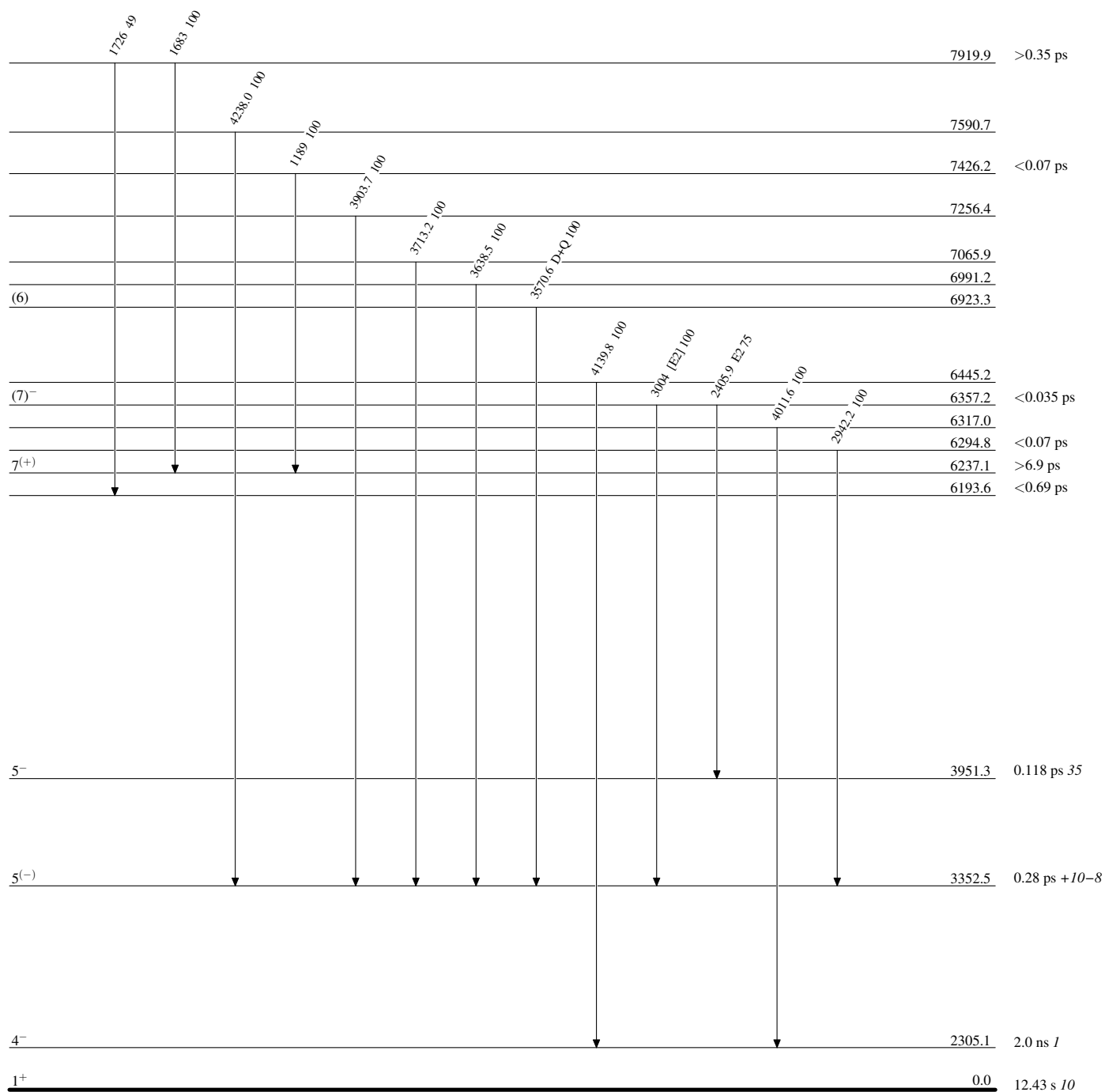
Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{P})$ (continued)

[†] From ¹⁸O(¹⁸O,pn γ) E=24 MeV, unless otherwise noted.
[‡] From $\gamma(\theta)$ in ¹⁸O(¹⁸O,pn γ) E=24 MeV, and RUL if applicable, unless otherwise noted.
[#] Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ -ray energies, assigned
multipolarities, and mixing ratios, unless otherwise specified.

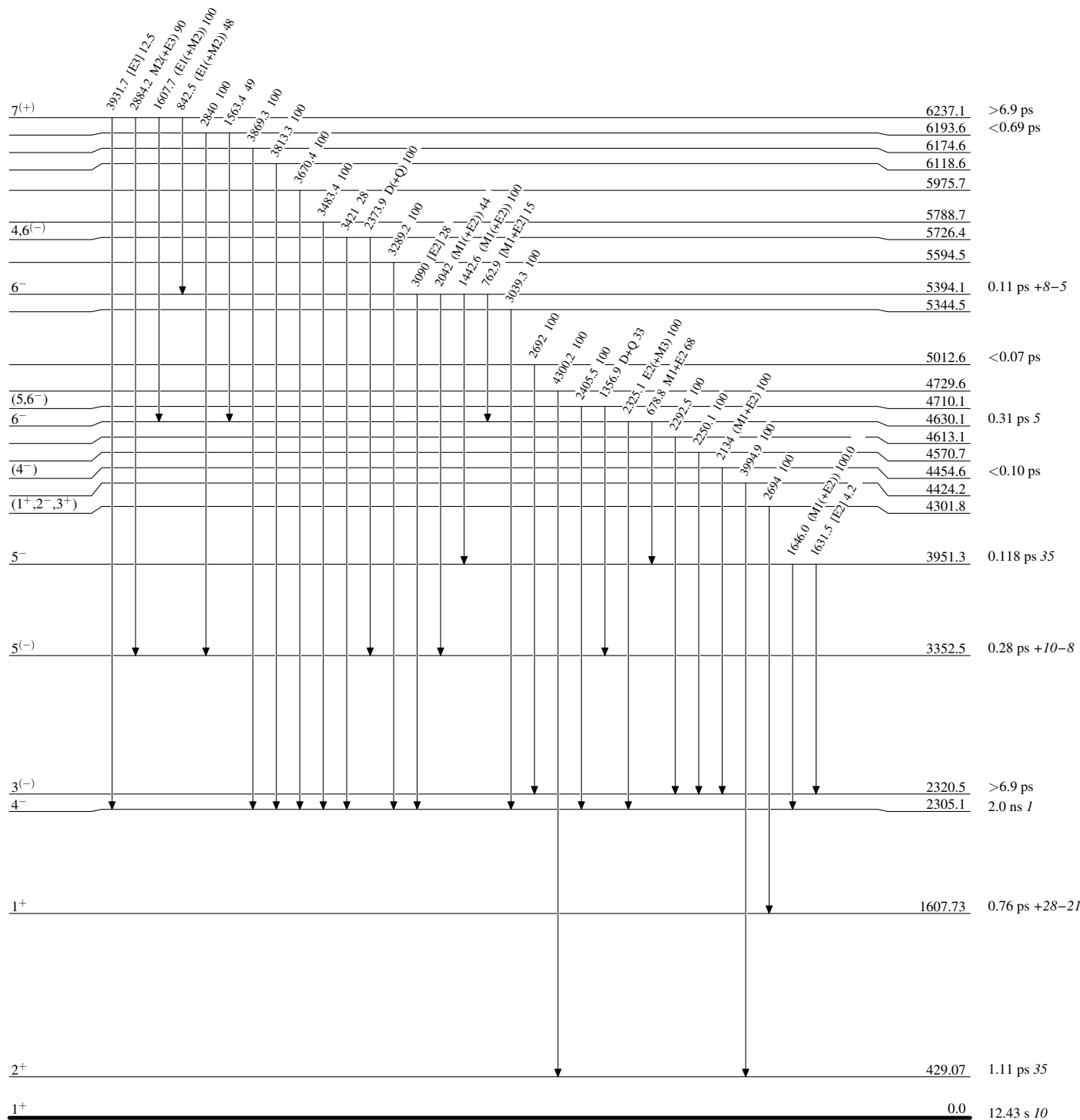
Adopted Levels, GammasLevel Scheme

Intensities: Relative photon branching from each level



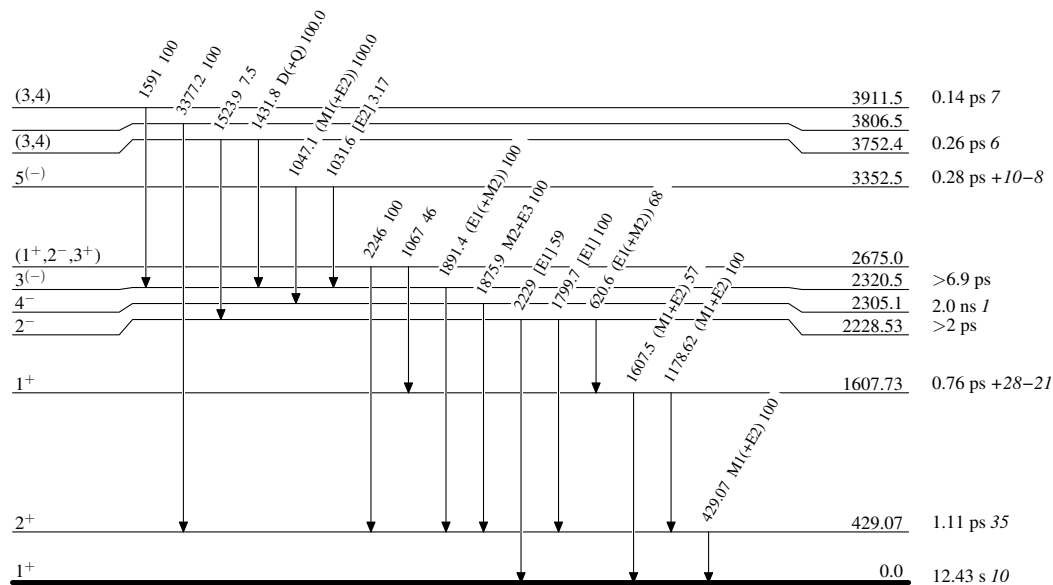
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{34}_{15}\text{P}_{19}$

$^{34}\text{Si} \beta^-$ decay (2.77 s) 1977Na05

Parent: ^{34}Si : $E=0$; $J^\pi=0^+$; $T_{1/2}=2.77$ s 20; $Q(\beta^-)=4557.0$ 11; $\% \beta^-$ decay=100

$^{34}\text{Si}-T_{1/2}$: From the Adopted Levels of ^{34}Si , originally from 1977Na05.

$^{34}\text{Si}-Q(\beta^-)$: From 2021Wa16.

1977Na05: ^{34}Si was produced via the $^{18}\text{O}(^{18}\text{O},2p)^{34}\text{Si}$ reaction with the 32-MeV $^{18}\text{O}^{5+}$ beam from one of the Brookhaven National Laboratory MP tandem Van de Graaffs. Targets were Ta_2O_5 enriched to 99% in ^{18}O . γ rays were detected using a coaxial Ge(Li) detector. β particles were detected using a NE102 plastic scintillator. Measured $E\beta$, $E\gamma$, $I\gamma$, and $\beta\gamma$ coin. Deduced levels, decay branching ratios, $\log ft$, parent ^{34}Si $T_{1/2}$ and masses.

 ^{34}P Levels

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>$T_{1/2}$[‡]</u>
0	1 ⁺	12.43 s 10
429.07 13	2 ⁺	1.11 ps 35
1607.62 19	1 ⁺	0.76 ps +28–21

[†] From $E\gamma$ data in 1977Na05.

[‡] From the Adopted Levels.

 β^- radiations

<u>E(decay)</u>	<u>E(level)</u>	<u>$I\beta^-$^{†‡}</u>	<u>$\log ft$</u>	<u>Comments</u>
(2949.4 15)	1607.62	≤ 100	≥ 3.3	$I\beta^-$: 1977Na05 assumes a β -feeding intensity of 100 to 1607.62, 1 ⁺ level. Evaluators adopt 100 as an upper limit.
(4127.9 [#] 15)	429.07			$I\beta^-$: 1977Na05 reports a β -feeding intensity <4 (2σ limit) to 429.07, 2 ⁺ level.
(4557.0 [#] 18)	0			$I\beta^-$: 1977Na05 assumes no β -feeding to the g.s. of ^{34}P . 52% is predicted by Millner (ORNL) using the two-body matrix elements from Chung and Wildenthal (private communications in 1977Na05). $<0.1\%$ is predicted by 1973La03 using a mixture of free plus modified-surface- δ -interaction matrix elements. Millner also predicts $T_{1/2}=2.62$ s, in good agreement with 1977Na05 measured $T_{1/2}$.

[†] From 1977Na05, which assumes a β -feeding intensity of 100 to 1607.62, 1⁺ level and no β feeding to the g.s. of ^{34}P and gave two shell-model predictions with the β feeding to g.s. 52% (1977Na05) or $<0.1\%$ (1973La03).

[‡] Absolute intensity per 100 decays.

[#] Existence of this branch is questionable.

 $\gamma(^{34}\text{P})$

$I\gamma$ normalization: 1977Na05 assumes a β -feeding intensity of 100 to 1607.62, 1⁺ level and the $\Sigma I\gamma=100$ from this level. Evaluators adopt 100 as an upper limit.

<u>E_γ[†]</u>	<u>I_γ^{†#}</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[‡]</u>	<u>δ[‡]</u>	<u>α[@]</u>	<u>Comments</u>
429.07 13	60 2	429.07	2 ⁺	0	1 ⁺	M1(+E2)	-0.07 7	0.000155 5	$\%I_\gamma \leq 62$
1178.52 16	64 3	1607.62	1 ⁺	429.07 2 ⁺		(M1+E2)	-0.93 26	2.87×10^{-5} 14	$\%I_\gamma \leq 67$
1607.6 5	36 5	1607.62	1 ⁺	0	1 ⁺	(M1+E2)	+0.13 6	0.0001096 17	$\%I_\gamma \leq 41$

[†] From 1977Na05.

[‡] From the Adopted Gammas.

[#] For absolute intensity per 100 decays, multiply by ≤ 1.0 .

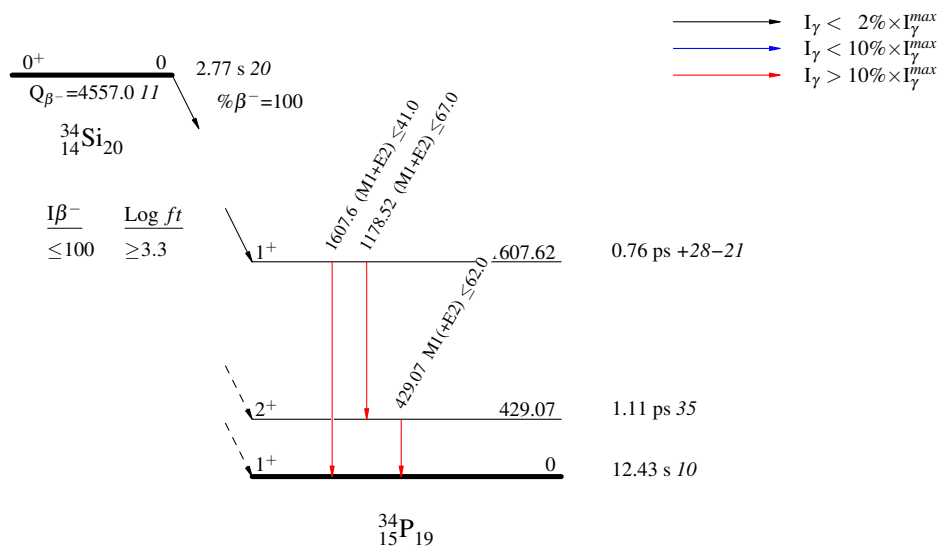
[@] Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

^{34}Si β^- decay (2.77 s) 1977Na05

Decay Scheme

Intensities: $I_{(\gamma+ce)}$ per 100 parent decays

Legend



$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) E=20\text{ MeV}$ **2009Be26**

2009Be26: 20, 24, 25, 30, and 44-MeV ^{18}O beams were used. The target was 200- $\mu\text{g}/\text{cm}^2$ 97%-enriched ^{18}O on a 12.7- μm Ta backing. Measured E_γ , I_γ , $\gamma\gamma$ -coin, $\text{p}\gamma$ -coin, and $\gamma(\theta)$. Deduced levels, J , π , multipolarities, and lifetimes using DSAM lineshape analysis. Comparisons to shell model calculations.

At FSU $E(^{18}\text{O})=24$ and 44 MeV. γ rays were detected by the Compton-suppressed high-purity germanium (HPGe) array, consisting of 6 single crystal HPGe detectors and 3 four-crystal clover detectors. Each Ge detector was surrounded by bismuth germanate (BGO) crystals used for Compton suppression. Charged particles were detected by a segmented ΔE -E Si telescope.

At LBNL $E(^{18}\text{O})=20, 25$, and 30 MeV. Charged particles were detected by the Silicon Telescope Array for Reaction Studies (STARS), consisting of a ΔE -E Si detector telescope. γ rays were detected by the Livermore Berkeley Array for Collaborative Experiments (LiBerACE) consisting of five HPGe clover detectors. Lifetimes were determined by Doppler-shifted line shapes using the LBNL data.

 ^{34}P Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0.0	1^+		J^π : from the Adopted Levels.
428.7 8	2	<0.97 ps	J^π : from 429 $\gamma(\theta)$ in 2009Be26 . $T_{1/2}$: $\tau < 1.4$ ps (2009Be26 , DSAM).
1607.4 8		0.90 ps 28	$T_{1/2}$: $\tau = 1.3$ ps 4 (2009Be26 , DSAM).
2304.7 22	(4)		J^π : spin=3,4 from 1876 $\gamma(\theta)$. 2009Be26 adopts spin=4.
2319.7 22	3	>2.1 ps	J^π : spin=3 from 1891 $\gamma(\theta)$ without showing $\gamma(\theta)$ plot. $T_{1/2}$: $\tau > 3$ ps (2009Be26 , DSAM).
3352.2 28	(5)	0.21 ps 10	J^π : spin=3,5 from 1047 $\gamma(\theta)$. 2009Be26 adopts spin=5 from tendency of populating yrast states in fusion-evaporation reactions. $T_{1/2}$: $\tau = 0.30$ ps 15 (2009Be26 , DSAM).
3950.0 27		0.21 ps 10	$T_{1/2}$: $\tau = 0.30$ ps 15 (2009Be26 , DSAM).
4629.2 26		0.35 ps 14	$T_{1/2}$: $\tau = 0.5$ ps 2 (2009Be26 , DSAM).
6191.3 34		<0.69 ps	$T_{1/2}$: $\tau < 1$ ps (2009Be26 , DSAM).
6235.8 29	(7)	>2.08 ps	J^π : spin=6,7 from 1607 $\gamma(\theta)$. 2009Be26 adopts spin=7 for a smaller δ value. $T_{1/2}$: $\tau > 3$ ps (2009Be26 , DSAM).

† From a least-squares fit to γ -ray energies.

‡ From $\gamma(\theta)$ measurements in **2009Be26**, unless otherwise noted.

 $\gamma(^{34}\text{P})$

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π
429 1	100	428.7	2	0.0	1^+	1645 2	11 2	3950.0		2304.7 (4)	
679 2	1.0 5	4629.2		3950.0		1876 2	37 4	2304.7 (4)		428.7 2	
1047 2	17 2	3352.2 (5)		2304.7 (4)		1891 2	17 2	2319.7	3	428.7 2	
1179 1	9 2	1607.4		428.7 2		2325 2	2 1	4629.2		2304.7 (4)	
1607 ‡ 1	6 ‡ 2	1607.4		0.0	1^+	2839 2	<1	6191.3		3352.2 (5)	
1607 ‡ 2	6 ‡ 2	6235.8 (7)		4629.2		2883 2	<1	6235.8 (7)		3352.2 (5)	

† From **2009Be26**. I_γ is given for $E(^{18}\text{O})=20$ MeV.

‡ Multiply placed with undivided intensity.

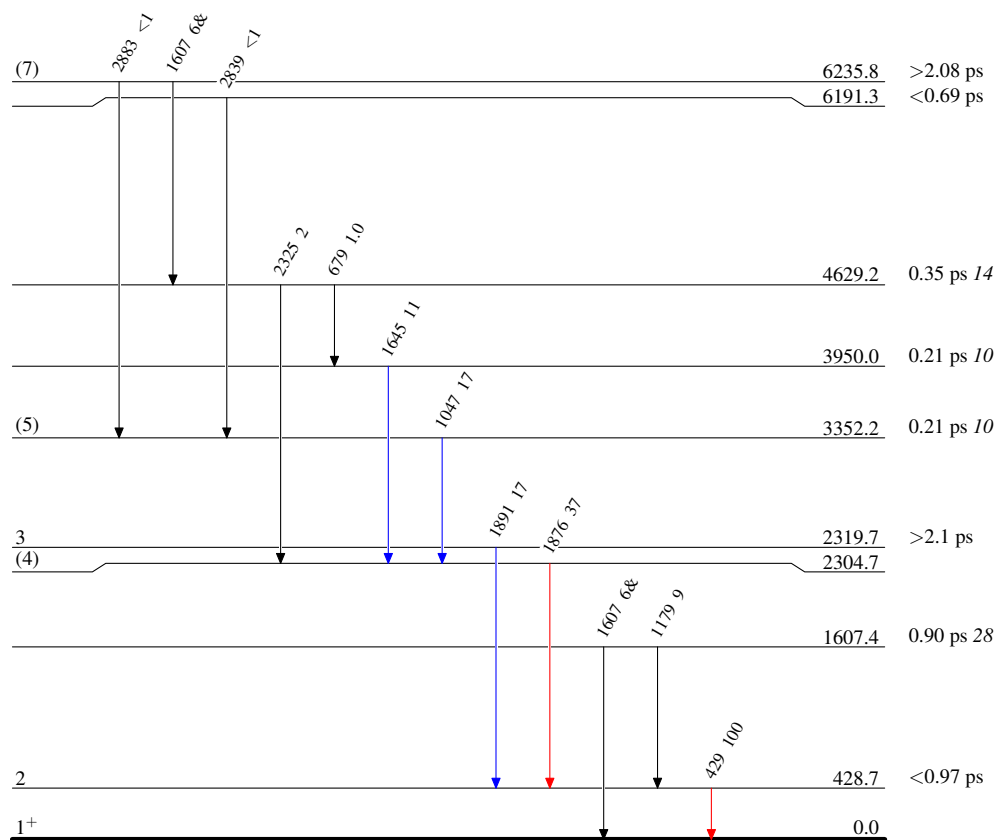
$^{18}\text{O}(^{18}\text{O,pn}\gamma) \text{E}=20 \text{ MeV}$ 2009Be26

Level Scheme

Intensities: Relative I_γ
& Multiply placed: undivided intensity given

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$

 $^{34}_{15}\text{P}_{19}$

$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) \text{E}=24 \text{ MeV}$ [2018Lu08](#),[2012Be11](#),[2009Be26](#)

[2018Lu08](#): a 24-MeV ^{18}O beam was provided by the ATLAS accelerator at Argonne National Laboratory. The target was $260\text{-}\mu\text{g}/\text{cm}^2$ 97%-enriched ^{18}O on a $12.7\text{-}\mu\text{m}$ Ta backing. The deexciting γ rays were detected by Gammasphere, consisting of 101 Compton-suppressed HPGe detectors. The evaporated protons and light charged particles were detected by the nearly 4π Microball detector, consisting of 95 CsI(Tl) scintillators. Measured E_γ with Doppler correction, I_γ , $\gamma\gamma$ -coin, $\text{p}\gamma$ -coin, and $\gamma(\theta)$. Deduced levels, J , π , and multiplicities. Comparisons to shell model calculations using the $(0+1)\hbar\omega$ PSDPF interaction.

[2018Lu08](#) is a reanalysis of the [2012Be11](#) data with proton emission channel selection enabled by the Microball and kinematic Doppler corrections. [2011BeZZ](#) is a draft of [2012Be11](#), providing more $\gamma(\theta)$ and DSAM fitting plots. The 6180 level and its 2828γ reported by [2012Be11](#) is removed by [2018Lu08](#). The 4447 level from [2012Be11](#) is revised to 4455. The 5281 level from [2012Be11](#) is revised to 6295. Two γ branches: 1446γ from 3911 level and 1606γ from 3911 level are only indicated by theories of [2012Be11](#), which are not included here.

 ^{34}P Levels

E(level) [†]	J^π [#]	$T_{1/2}$	Comments
0.0	1^+		J^π : from the Adopted Levels.
429.1 4	2	1.3 ps +6-3	$T_{1/2}$: $\tau=1.9$ ps +9-5 (2012Be11 , DSAM).
1608.2 $\frac{5}{2}^+$	1	0.52 ps +45-14	$T_{1/2}$: $\tau=0.75$ ps +65-20 (2012Be11 , DSAM).
2228.8 $\frac{5}{2}^+$	2	>2 ps	$T_{1/2}$: $\tau>3$ ps (2012Be11 , DSAM).
2305.1 6	4		
2320.6 6	3	>7 ps	$T_{1/2}$: $\tau>10$ ps (2012Be11 , DSAM).
3352.6 7	5	0.36 ps +12-8	$T_{1/2}$: $\tau=0.52$ ps +18-12 (2012Be11 , DSAM).
3752.6 $\frac{7}{2}^+$	(3,4)	0.26 ps 6	$T_{1/2}$: $\tau=0.37$ ps 8 (2012Be11 , DSAM).
3806.5 26			
3911.7 $\frac{7}{2}^+$	(3,4)	0.14 ps 7	$T_{1/2}$: $\tau=0.2$ ps 1 (2012Be11 , DSAM).
3951.4 $\frac{7}{2}^+$	5	0.111 ps 35	$T_{1/2}$: $\tau=0.16$ ps 5 (2012Be11 , DSAM).
4424.3 27			
4454.7 $\frac{7}{2}^+$	(4)	<0.10 ps	E(level): 4447 2 with a 2126γ in 2012Be11 . 2018Lu08 revised the energy with improved Doppler correction. J^π : spin=3,4 from $2134\gamma(\theta)$ in 2018Lu08 . 4^- from shell model calculations. $T_{1/2}$: $\tau<0.14$ ps (2012Be11 , DSAM).
4570.8 21			
4613.2 22			
4630.1 $\frac{7}{2}^+$	6	0.30 ps 5	$T_{1/2}$: $\tau=0.44$ ps 7 (2012Be11 , DSAM).
4710.2 16	(5,6)		J^π : spin=4,5,6 from $1357\gamma(\theta)$. Spin=(5,6) in Fig. 7 and spin=(6) in Table II of 2018Lu08 . Spin=5 suggested by shell model and spin=6 based on the tendency of fusion-evaporation reactions to favor higher-spin states.
4729.6 30			
5012.8 $\frac{21}{2}^+$	(2)	<0.07 ps	$T_{1/2}$: $\tau<0.1$ ps (2012Be11 , DSAM).
5344.5 29			
5394.2 $\frac{8}{2}^+$	6	0.11 ps +8-5	$T_{1/2}$: $\tau=0.16$ ps +12-7 (2012Be11 , DSAM).
5594.5 26			
5726.4 $\frac{14}{2}^+$	4,6		
5788.7 29			
5975.7 30			
6118.6 31			
6174.6 29			
6193.9 $\frac{11}{2}^+$			
6237.3 $\frac{8}{2}^+$	7	>6.9 ps	$T_{1/2}$: $\tau>10$ ps (2012Be11 , DSAM).
6294.9 22		<0.07 ps	$T_{1/2}$: $\tau<0.1$ ps (2012Be11 , DSAM) for a 5281 level with a 2960γ . 2018Lu08 revised the E_γ to be 2942 and the placement to be from a 6295 level.
6316.9 31			
6357.3 $\frac{9}{2}^+$	(7)	<0.035 ps	$T_{1/2}$: $\tau<0.05$ ps (2012Be11 , DSAM).
6445.2 31			
6923.4 27	(6)		J^π : spin=4,5,6 from $3571\gamma(\theta)$. 2018Lu08 adopts spin=(6) based on shell model and

Continued on next page (footnotes at end of table)

$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) E=24\text{ MeV}$ **2018Lu08,2012Be11,2009Be26 (continued)** ^{34}P Levels (continued)

E(level) [†]	$T_{1/2}$	Comments
6991.3 29		the tendency of fusion-evaporation reactions to favor higher-spin states.
7066.0 26		
7256.5 26		
7426.4 [‡] 21	<0.07 ps	
7590.8 32		$T_{1/2}$: $\tau < 0.1$ ps (2012Be11, DSAM).
7920.2 [‡] 11	>0.35 ps	$T_{1/2}$: $\tau > 0.5$ ps (2012Be11, DSAM).

[†] From a least-squares fit to γ -ray energies.[‡] Levels from 2012Be11 and also verified by 2018Lu08.[#] From 2012Be11 $\gamma(\theta)$ analysis, unless otherwise noted. $\gamma(^{34}\text{P})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
429.1	2	429.1 4	100	0.0	1 ⁺	D(+Q)	+0.11 +13-12	I_γ : 100.0 1 (2018Lu08). 100 (2009Be26). $A_2 = -0.09$ 3, $A_4 = +0.02$ 4 (2012Be11).
1608.2	1	1179.1 [‡] 4	100 [‡] 5	429.1	2	D		I_γ : 5 1 (2009Be26). $A_2 = -0.03$ 3, $A_4 = +0.01$ 4 (2012Be11).
		1608 [‡] 1	52 [‡] 5	0.0	1 ⁺			I_γ : 11 2 for doublet (2009Be26).
2228.8	2	620.6 [‡] 4	68 [‡] 14	1608.2	1	D(+Q)	+0.07 16	$A_2 = -0.12$ 8, $A_4 = +0.06$ 10 (2012Be11).
		1799.7 [‡] 6	100 [‡] 16	429.1	2			
		2229 [‡] 1	59 [‡] 14	0.0	1 ⁺			
2305.1	4	1875.8 6	100	429.1	2	Q(+O)	0.00 12	I_γ : 77.7 1 (2018Lu08). 76 6 (2009Be26). $A_2 = +0.32$ 6, $A_4 = +0.02$ 8 (2012Be11).
2320.6	3	1891.5 7	100	429.1	2	D(+Q)	+0.07 14	I_γ : 12.2 1 (2018Lu08). 10 2 (2009Be26). $A_2 = -0.15$ 7, $A_4 = +0.03$ 8 (2012Be11).
3352.6	5	1031.6 9	3.17 10	2320.6	3			I_γ : 0.64 2 (2018Lu08).
		1047.3 8	100.0 5	2305.1	4	D(+Q)	+0.02 9	I_γ : 20.2 1 (2018Lu08). 37 4 (2009Be26). $A_2 = -0.23$ 7, $A_4 = +0.04$ 8 (2012Be11).
3752.6	(3,4)	1431.8 [‡] 6	100.0 [‡] 22	2320.6	3	D(+Q)		$\delta(E2/M1) = +0.18 +15-14$ for 4 ⁻ or $\delta(E2/M1) = -0.36$ for 3 ⁻ (e-mail reply from the first author of 2012Be11, P.C. Bender, to B. Singh on Feb. 29, 2012).
		1523.9 [‡] 9	7.5 [‡] 22	2228.8	2			
3806.5		3377.2 26	100	429.1	2			I_γ : 1.2 2 (2018Lu08).
3911.7	(3,4)	1591 [‡] 1	100 [‡]	2320.6	3			
3951.4	5	1631.5 [‡] 11	4.2 [‡] 21	2320.6	3			
		1646.1 [‡] 4	100.0 [‡] 21	2305.1	4	D(+Q)	-0.03 18	I_γ : 19 2 (2009Be26). $A_2 = -0.31$ 7, $A_4 = +0.01$ 8 (2012Be11).
4424.3		3994.9 27	100	429.1	2			I_γ : 4.7 2 (2018Lu08).
4454.7	(4)	2134		2320.6	3	D+Q		E_γ : revised from 2126 2 in 2012Be11 by 2018Lu08.
4570.8		2250.1 20	100	2320.6	3			I_γ : 1.0 3 (2018Lu08).
4613.2		2292.5 21	100	2320.6	3			I_γ : 0.8 2 (2018Lu08).
4630.1	6	678.7 [‡] 4	54 [‡] 5	3951.4	5	D+Q	+0.12 11	I_γ : 7 1 (2009Be26). $A_2 = -0.05$ 8, $A_4 = +0.03$ 8 (2012Be11).
		2325.1 [‡] 6	100 [‡] 5	2305.1	4	Q(+O)	+0.02 +11-10	I_γ : 10 2 (2009Be26). $A_2 = +0.42$ 8, $A_4 = -0.05$ 8 (2012Be11).

Continued on next page (footnotes at end of table)

$^{18}\text{O}(^{18}\text{O}, \text{pn}\gamma) \text{E}=24 \text{ MeV}$ **2018Lu08,2012Be11,2009Be26 (continued)** $\gamma(^{34}\text{P})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
4710.2	(5,6)	1356.9 23	33 11	3352.6 5		D+Q		I_γ : 2.1 7 (2018Lu08).
		2405.5 20	100 5	2305.1 4				I_γ : 6.3 3 (2018Lu08).
4729.6		4300.2 30	100	429.1 2				I_γ : 0.5 2 (2018Lu08).
5012.8	(2)	2692 ‡ 2	100 ‡	2320.6 3				
5344.5		3039.3 28	100	2305.1 4				I_γ : 2.6 6 (2018Lu08).
5394.2	6	762.9 ‡ 8	15 ‡ 6	4630.1 6				
		1442.6 ‡ 7	100 ‡ 11	3951.4 5		D(+Q)	+0.02 5	$A_2=-0.24$ 9, $A_4=+0.12$ 10 (2012Be11).
		2042 ‡ 1	44 ‡ 9	3352.6 5		D(+Q)	+0.05 14	$A_2=-0.21$ 12, $A_4=+0.26$ 15 (2012Be11).
		3090 ‡ 2	28 ‡ 9	2305.1 4				
5594.5		3289.2 25	100	2305.1 4				I_γ : 1.1 5 (2018Lu08).
5726.4	4,6	2373.9 ‡ 15	100 ‡ 8	3352.6 5		D(+Q)		Mult.: from 2011BeZZ, not included in 2012Be11.
		3421 ‡ 2	28 ‡ 8	2305.1 4				
5788.7		3483.4 28	100	2305.1 4				I_γ : 1.1 8 (2018Lu08).
5975.7		3670.4 29	100	2305.1 4				I_γ : 1.5 5 (2018Lu08).
6118.6		3813.3 30	100	2305.1 4				I_γ : 0.8 6 (2018Lu08).
6174.6		3869.3 28	100	2305.1 4				I_γ : 1.3 3 (2018Lu08).
6193.9		1563.4 ‡ 14	49 ‡ 11	4630.1 6				
		2841 ‡ 2	100 ‡ 11	3352.6 5				I_γ : 6 1 (2009Be26).
6237.3	7	842.5 ‡ 6	48 ‡ 10	5394.2 6		D(+Q)	+0.05 14	$A_2=-0.19$ 12, $A_4=-0.02$ 12 (2012Be11).
		1607.9 ‡ 5	100 ‡ 13	4630.1 6		D(+Q)	+0.05 9	I_γ : 11 2 for doublet (2009Be26).
								$A_2=-0.17$ 10, $A_4=+0.11$ 11 (2012Be11).
		2884.3 ‡ 7	90 ‡ 13	3352.6 5		Q(+O)	+0.11 18	I_γ : 5 1 (2009Be26).
								$A_2=+0.52$ 10, $A_4=-0.05$ 11 (2012Be11).
		3932 ‡ 3	12.5 ‡ 75	2305.1 4				
6294.9		2942.2 21	100	3352.6 5				I_γ : 4.0 8 (2018Lu08).
6316.9		4011.6 30	100	2305.1 4				I_γ : 0.60 5 (2018Lu08).
6357.3	(7)	2405.9 ‡ 7	75 ‡ 14	3951.4 5		Q		$A_2=-0.53$ 12, $A_4=-0.57$ 13 (2012Be11).
		3004 ‡ 2	100 ‡ 14	3352.6 5				
6445.2		4139.8 30	100	2305.1 4				I_γ : 1.3 8 (2018Lu08).
6923.4	(6)	3570.6 26	100	3352.6 5		D+Q		I_γ : 0.60 9 (2018Lu08).
6991.3		3638.5 28	100	3352.6 5				I_γ : 0.8 3 (2018Lu08).
7066.0		3713.2 25	100	3352.6 5				I_γ : 0.20 5 (2018Lu08).
7256.5		3903.7 25	100	3352.6 5				I_γ : 0.30 5 (2018Lu08).
7426.4		1189 ‡ 2	100 ‡	6237.3 7				
7590.8		4238.0 31	100	3352.6 5				I_γ : 1.0 4 (2018Lu08).
7920.2		1683 ‡ 1	100 ‡ 6	6237.3 7				
		1726 ‡ 1	49 ‡ 6	6193.9				

[†] From 2018Lu08, unless otherwise noted. 2012Be11 reports γ branching ratios. 2018Lu08 and 2009Be26 report relative intensities with $I_\gamma(429)=100$. The original intensities in 2018Lu08 and 2009Be26 are given under comments.

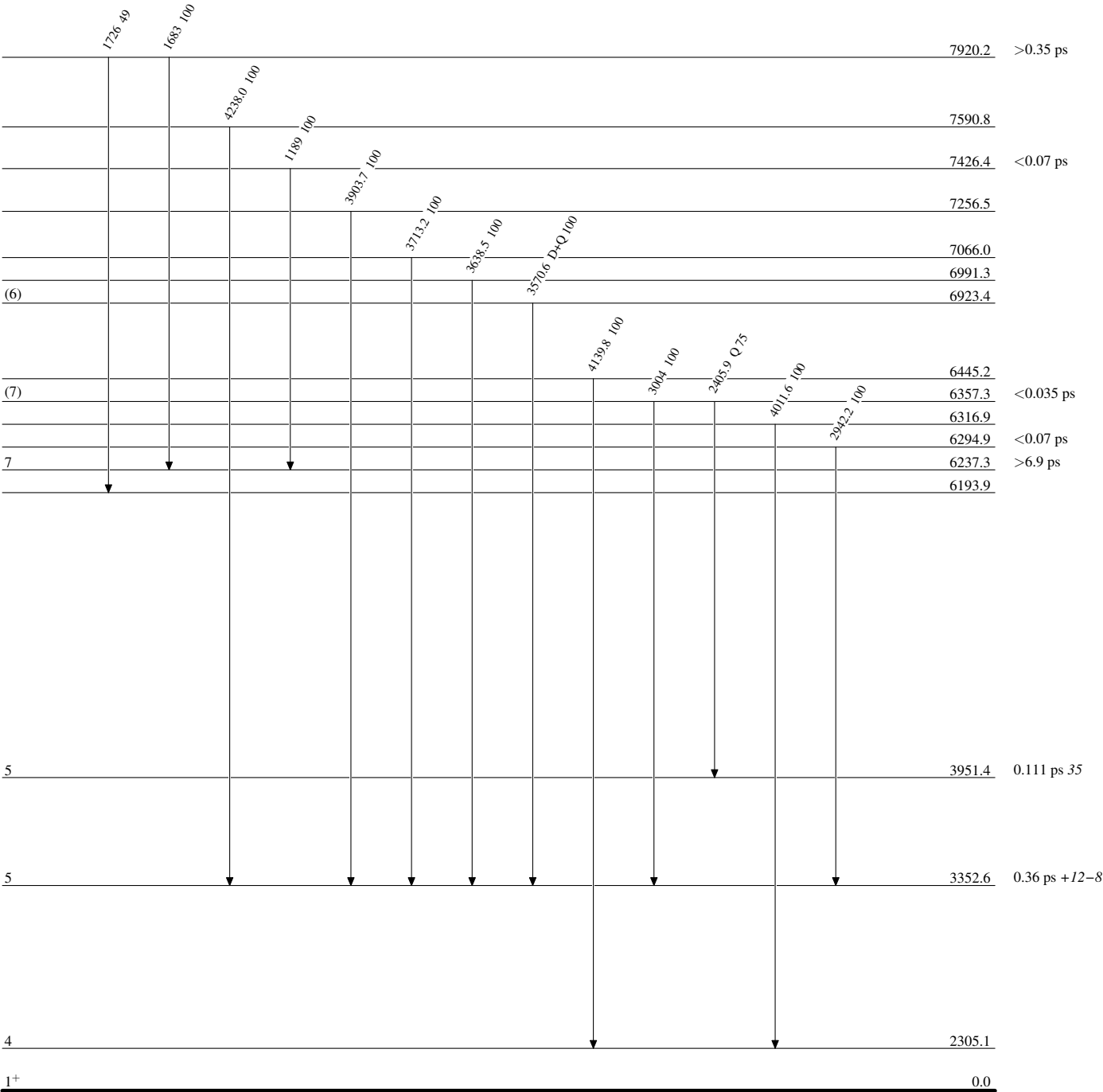
[‡] From 2012Be11 and also verified by 2018Lu08.

[#] From 2012Be11 $\gamma(\theta)$ analysis, unless otherwise noted.

¹⁸O(¹⁸O,pn γ) E=24 MeV 2018Lu08,2012Be11,2009Be26

Level Scheme

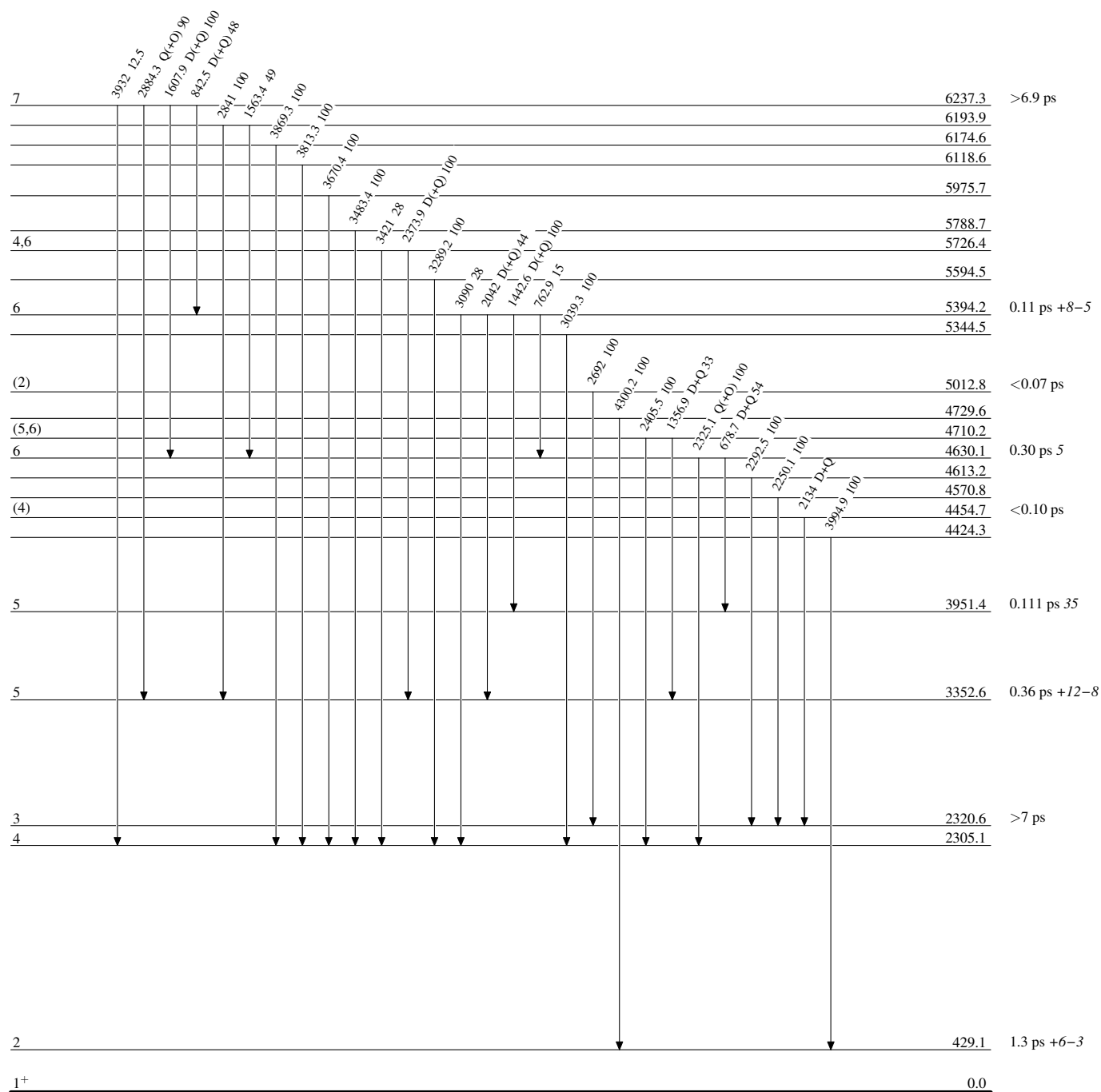
Intensities: Relative photon branching from each level



$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) E=24\text{ MeV}$ 2018Lu08,2012Be11,2009Be26

Level Scheme (continued)

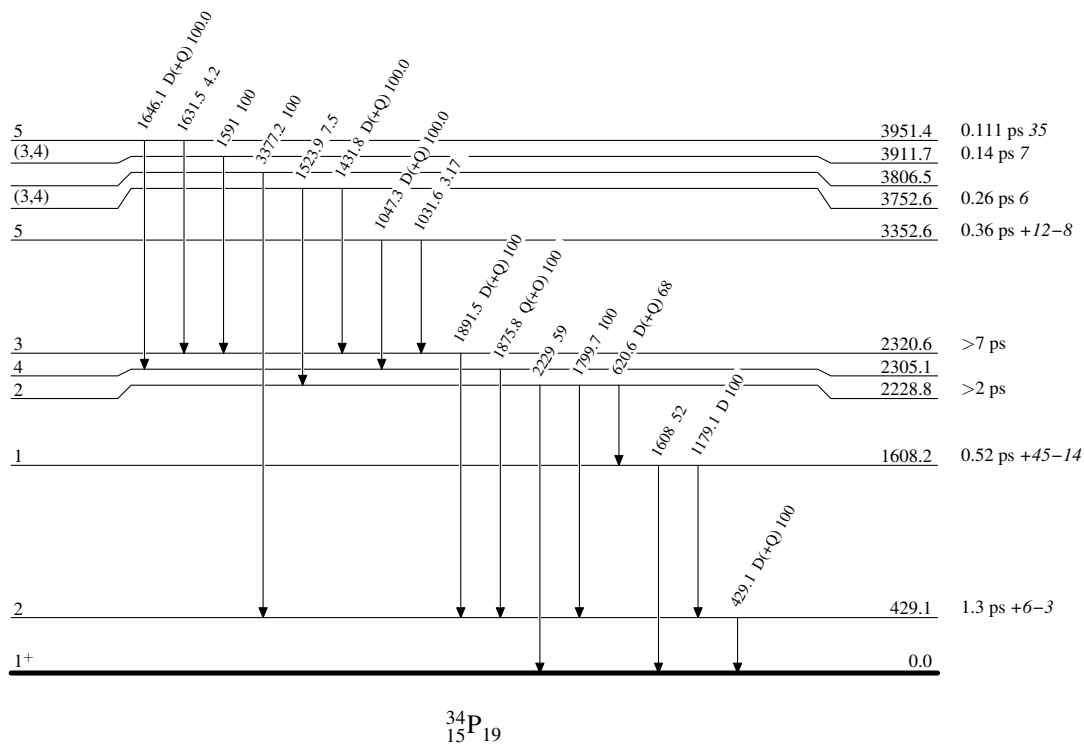
Intensities: Relative photon branching from each level



$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) E=24 \text{ MeV}$ 2018Lu08,2012Be11,2009Be26

Level Scheme (continued)

Intensities: Relative photon branching from each level



$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) E=25\text{ MeV}$ 2009Be26

2009Be26: 20, 24, 25, 30, and 44-MeV ^{18}O beams were used. The target was 200- $\mu\text{g}/\text{cm}^2$ 97%-enriched ^{18}O on a 12.7- μm Ta backing. Measured E_γ , I_γ , $\gamma\gamma$ -coin, $\text{p}\gamma$ -coin, and $\gamma(\theta)$. Deduced levels, J , π , multipolarities, and lifetimes using DSAM lineshape analysis. Comparisons to shell model calculations.

At FSU $E(^{18}\text{O})=24$ and 44 MeV. γ rays were detected by the Compton-suppressed high-purity germanium (HPGe) array, consisting of 6 single crystal HPGe detectors and 3 four-crystal clover detectors. Each Ge detector was surrounded by bismuth germanate (BGO) crystals used for Compton suppression. Charged particles were detected by a segmented ΔE -E Si telescope.

At LBNL $E(^{18}\text{O})=20, 25$, and 30 MeV. Charged particles were detected by the Silicon Telescope Array for Reaction Studies (STARS), consisting of a ΔE -E Si detector telescope. γ rays were detected by the Livermore Berkeley Array for Collaborative Experiments (LiBerACE) consisting of five HPGe clover detectors. Lifetimes were determined by Doppler-shifted line shapes using the LBNL data.

 ^{34}P Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0.0	1^+		J^π : from the Adopted Levels.
428.7 8	2	<0.97 ps	J^π : from 429 $\gamma(\theta)$ in 2009Be26. $T_{1/2}$: $\tau < 1.4$ ps (2009Be26, DSAM).
1607.4 8		0.90 ps 28	$T_{1/2}$: $\tau = 1.3$ ps 4 (2009Be26, DSAM).
2304.7 22	(4)		J^π : spin=3,4 from 1876 $\gamma(\theta)$. 2009Be26 adopts spin=4.
2319.7 22	3	>2.1 ps	J^π : spin=3 from 1891 $\gamma(\theta)$ without showing $\gamma(\theta)$ plot. $T_{1/2}$: $\tau > 3$ ps (2009Be26, DSAM).
3352.2 28	(5)	0.21 ps 10	J^π : spin=3,5 from 1047 $\gamma(\theta)$. 2009Be26 adopts spin=5 from tendency of populating yrast states in fusion-evaporation reactions. $T_{1/2}$: $\tau = 0.30$ ps 15 (2009Be26, DSAM).
3950.0 27		0.21 ps 10	$T_{1/2}$: $\tau = 0.30$ ps 15 (2009Be26, DSAM).
4629.2 26		0.35 ps 14	$T_{1/2}$: $\tau = 0.5$ ps 2 (2009Be26, DSAM).
6191.3 34		<0.69 ps	$T_{1/2}$: $\tau < 1$ ps (2009Be26, DSAM).
6235.8 29	(7)	>2.08 ps	J^π : spin=6,7 from 1607 $\gamma(\theta)$. 2009Be26 adopts spin=7 for a smaller δ value. $T_{1/2}$: $\tau > 3$ ps (2009Be26, DSAM).

† From a least-squares fit to γ -ray energies.

‡ From $\gamma(\theta)$ measurements in 2009Be26, unless otherwise noted.

 $\gamma(^{34}\text{P})$

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π
429 1	100	428.7	2	0.0	1^+	1645 2	31 2	3950.0		2304.7	(4)
679 2	4 1	4629.2		3950.0		1876 2	54 4	2304.7	(4)	428.7	2
1047 2	28 2	3352.2	(5)	2304.7	(4)	1891 2	15 2	2319.7	3	428.7	2
1179 1	6 1	1607.4		428.7	2	2325 2	4 1	4629.2		2304.7	(4)
1607 ‡ 1	13 ‡ 2	1607.4		0.0	1^+	2839 2	2 1	6191.3		3352.2	(5)
1607 ‡ 2	13 ‡ 2	6235.8	(7)	4629.2		2883 2	3 1	6235.8	(7)	3352.2	(5)

† From 2009Be26. I_γ is given for $E(^{18}\text{O})=25$ MeV.

‡ Multiply placed with undivided intensity.

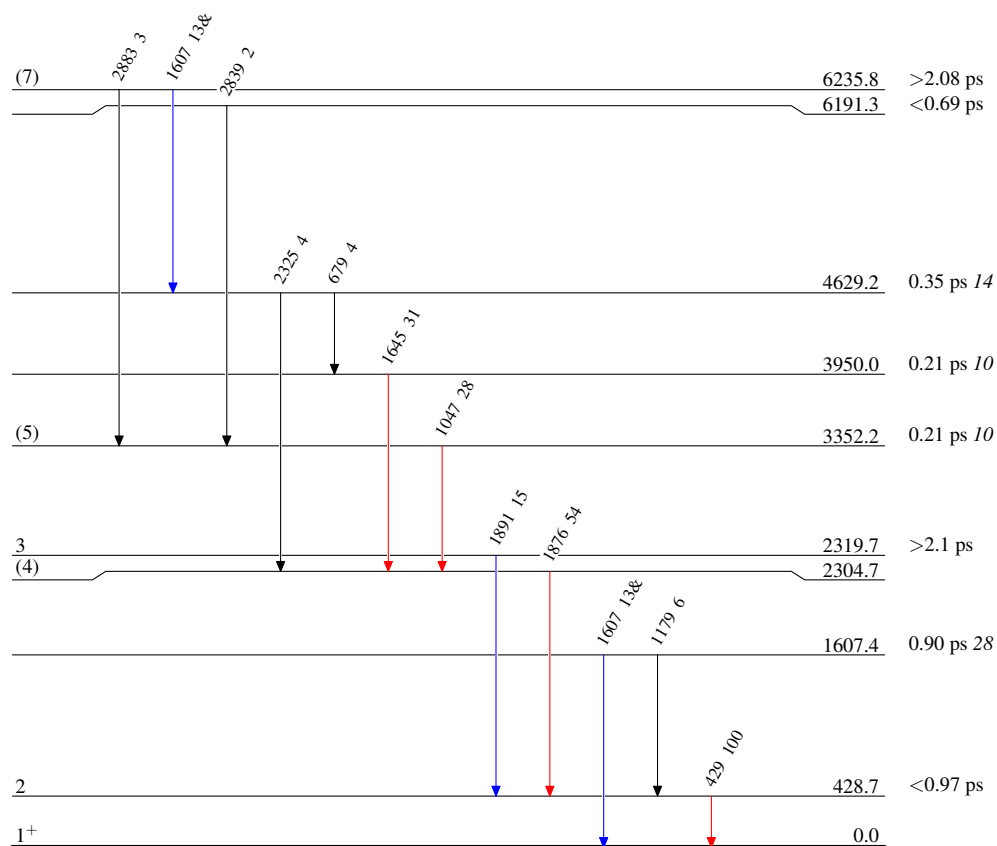
$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) \text{E}=25 \text{ MeV} \quad 2009\text{Be26}$

Level Scheme

Intensities: Relative I_γ
& Multiply placed: undivided intensity given

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\text{max}}$

 $^{34}_{15}\text{P}_{19}$

$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) E=30\text{ MeV}$ 2009Be26

2009Be26: 20, 24, 25, 30, and 44-MeV ^{18}O beams were used. The target was 200- $\mu\text{g}/\text{cm}^2$ 97%-enriched ^{18}O on a 12.7- μm Ta backing. Measured E_γ , I_γ , $\gamma\gamma$ -coin, $\text{p}\gamma$ -coin, and $\gamma(\theta)$. Deduced levels, J , π , multipolarities, and lifetimes using DSAM lineshape analysis. Comparisons to shell model calculations.

At FSU $E(^{18}\text{O})=24$ and 44 MeV. γ rays were detected by the Compton-suppressed high-purity germanium (HPGe) array, consisting of 6 single crystal HPGe detectors and 3 four-crystal clover detectors. Each Ge detector was surrounded by bismuth germanate (BGO) crystals used for Compton suppression. Charged particles were detected by a segmented ΔE -E Si telescope.

At LBNL $E(^{18}\text{O})=20, 25$, and 30 MeV. Charged particles were detected by the Silicon Telescope Array for Reaction Studies (STARS), consisting of a ΔE -E Si detector telescope. γ rays were detected by the Livermore Berkeley Array for Collaborative Experiments (LiBerACE) consisting of five HPGe clover detectors. Lifetimes were determined by Doppler-shifted line shapes using the LBNL data.

 ^{34}P Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0.0	1^+		J^π : from the Adopted Levels.
428.7 8	2	<0.97 ps	J^π : from 429 $\gamma(\theta)$ in 2009Be26. $T_{1/2}$: $\tau < 1.4$ ps (2009Be26, DSAM).
1607.4 8		0.90 ps 28	$T_{1/2}$: $\tau = 1.3$ ps 4 (2009Be26, DSAM).
2304.7 22	(4)		J^π : spin=3,4 from 1876 $\gamma(\theta)$. 2009Be26 adopts spin=4.
2319.7 22	3	>2.1 ps	J^π : spin=3 from 1891 $\gamma(\theta)$ without showing $\gamma(\theta)$ plot. $T_{1/2}$: $\tau > 3$ ps (2009Be26, DSAM).
3352.2 28	(5)	0.21 ps 10	J^π : spin=3,5 from 1047 $\gamma(\theta)$. 2009Be26 adopts spin=5 from tendency of populating yrast states in fusion-evaporation reactions. $T_{1/2}$: $\tau = 0.30$ ps 15 (2009Be26, DSAM).
3950.0 27		0.21 ps 10	$T_{1/2}$: $\tau = 0.30$ ps 15 (2009Be26, DSAM).
4629.2 26		0.35 ps 14	$T_{1/2}$: $\tau = 0.5$ ps 2 (2009Be26, DSAM).
6191.3 34		<0.69 ps	$T_{1/2}$: $\tau < 1$ ps (2009Be26, DSAM).
6235.8 29	(7)	>2.08 ps	J^π : spin=6,7 from 1607 $\gamma(\theta)$. 2009Be26 adopts spin=7 for a smaller δ value. $T_{1/2}$: $\tau > 3$ ps (2009Be26, DSAM).

† From a least-squares fit to γ -ray energies.

‡ From $\gamma(\theta)$ measurements in 2009Be26, unless otherwise noted.

 $\gamma(^{34}\text{P})$

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π
429 1	100	428.7	2	0.0	1^+	1645 2	46 4	3950.0		2304.7	(4)
679 2	7 1	4629.2		3950.0		1876 2	64 4	2304.7	(4)	428.7	2
1047 2	44 4	3352.2	(5)	2304.7	(4)	1891 2	9 1	2319.7	3	428.7	2
1179 1	3 1	1607.4		428.7	2	2325 2	10 2	4629.2		2304.7	(4)
1607 ‡ 1	11 ‡ 2	1607.4		0.0	1^+	2839 2	7 1	6191.3		3352.2	(5)
1607 ‡ 2	11 ‡ 2	6235.8	(7)	4629.2		2883 2	5 1	6235.8	(7)	3352.2	(5)

† From 2009Be26. I_γ is given for $E(^{18}\text{O})=30$ MeV.

‡ Multiply placed with undivided intensity.

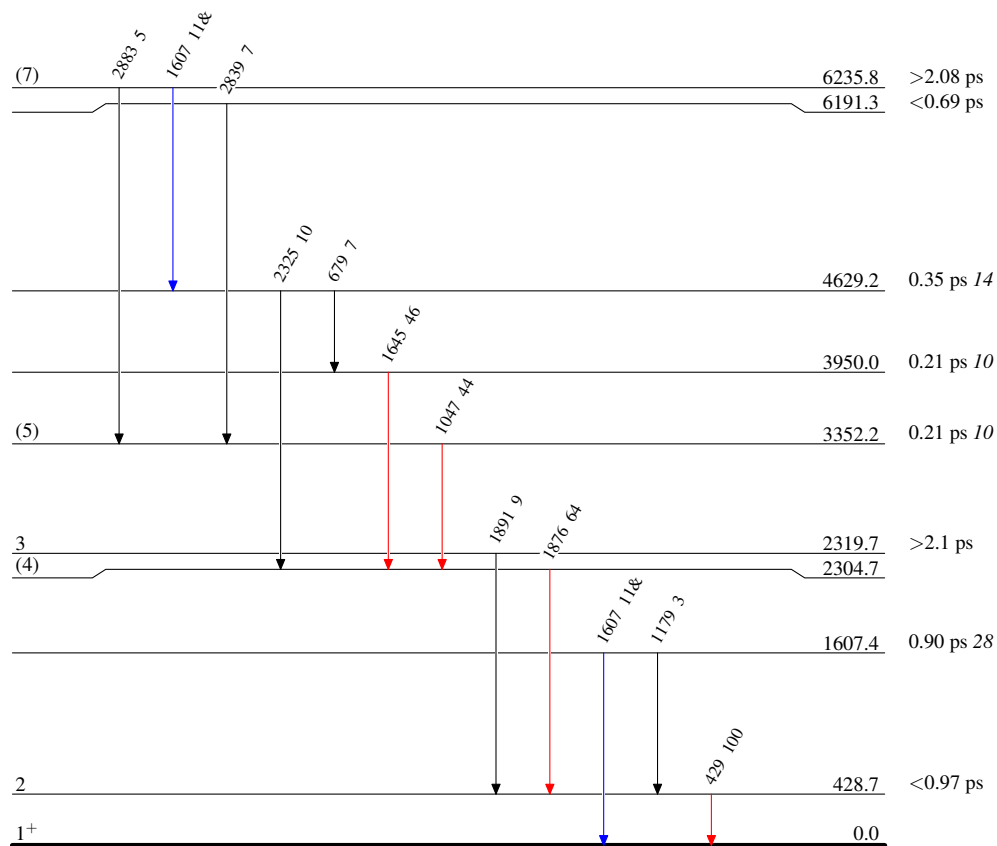
$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) \text{E}=30 \text{ MeV}$ 2009Be26

Level Scheme

Intensities: Relative I_γ
& Multiply placed: undivided intensity given

Legend

$\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\text{max}}$

 $^{34}_{15}\text{P}_{19}$

$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) E=34 \text{ MeV}$ [2009Ch43,2012Ma21](#)

[2009Ch43](#), [2010Gh02](#): a 34-MeV ^{18}O beam was provided from the 14 UD BARC-TIFR Pelletron facility at TIFR, Mumbai. The target was 1.6-mg/cm² ^{18}O on both sides of a 50-mg/cm²² Ta foil. The deexciting γ rays were detected by an array of seven Compton-suppressed Clover Ge detectors placed at 30°, 60°, 90°, 120°, and 150°. Measured E_γ , I_γ , $\gamma\gamma$, $\gamma\gamma(\theta)(\text{DCO})$, and $\gamma\gamma(\theta)(\text{POL})$. Deduced levels, J, π , and γ -ray multipolarities. Comparisons with shell-model calculations.

[2012Ma21](#): a 36-MeV ^{18}O beam was provided from the Tandem van de Graaff accelerator at the National Institute of Physics and Nuclear Engineering, Bucharest, Romania. The target was 1.6-mg/cm² ^{18}O on both sides of a 50-mg/cm²² Ta foil. γ rays were detected by an array of eight HPGe detectors and seven LaBr₃:Ce detectors. Measured E_γ , I_γ , and $\gamma\gamma$ -coin. Deduced levels, J, π , multipolarities, lifetimes, and transition strengths. Comparison with shell-model calculations. [2011MaZP](#) is a preprint.

 ^{34}P Levels

E(level) [†]	J π [‡]	T _{1/2}	Comments
0.0	1 ⁺		J π : from the Adopted Levels.
429.4 10	2 ⁺		
1609.2 14	(1 ⁺)		
2305.6 14	4 ⁽⁻⁾	2.0 ns 1	T _{1/2} : from $\gamma\gamma(t)$ in 2012Ma21 .
2321.1 14	(3 ⁻)		
3353.3 16	5 ⁽⁻⁾		
3749.3? 17			
3943.3? 17			
3951.6 16	5 ⁽⁻⁾		
4630.8 16	6 ⁽⁻⁾		
6237.7 16	7 ⁽⁺⁾		

[†] From a least-squares fit to the measured γ -ray energies.

[‡] As given in [2009Ch43](#). When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

 $\gamma(^{34}\text{P})$

E_γ [†]	I_γ [†]	$E_i(\text{level})$	J π_i	E_f	J π_f	Mult. [‡]	δ [‡]	Comments
429.4 10	100 3	429.4	2 ⁺	0.0	1 ⁺	M1(+E2)	-0.07 7	Mult.: DCO(Q)=0.57 10 gated on 1876 γ . Magnetic nature is assigned in 2009Ch43 as POL indicates.
679.4 10	3.17 16	4630.8	6 ⁽⁻⁾	3951.6	5 ⁽⁻⁾	(M1)		DCO(Q)=0.46 8 gated on 1876 γ . Magnetic nature is assigned in 2009Ch43 as POL qualitatively indicates.
1047.8 10	17.1 9	3353.3	5 ⁽⁻⁾	2305.6	4 ⁽⁻⁾	D		Mult.: DCO(Q)=0.28 5 gated on 1876 γ . (M1) is assigned in 2009Ch43 .
1179.8 10	0.74 10	1609.2	(1 ⁺)	429.4	2 ⁺	D		Mult.: DCO(D)=0.77 14 gated on 429 γ . (M1) is assigned in 2009Ch43 .
1443.7 [#] 10		3749.3?		2305.6	4 ⁽⁻⁾			
1607.1 10	4.5 5	6237.7	7 ⁽⁺⁾	4630.8	6 ⁽⁻⁾	(E1)		Mult.: DCO(Q)=0.43 8 gated on 1876 γ . Electric nature is assigned in 2009Ch43 as POL qualitatively indicates.
1637.7 [#] 10		3943.3?		2305.6	4 ⁽⁻⁾			
1646.2 10	3.6 4	3951.6	5 ⁽⁻⁾	2305.6	4 ⁽⁻⁾	(M1)		DCO(D)=0.88 15 gated on 1607 γ . Magnetic nature is assigned in 2009Ch43 as POL qualitatively indicates.
1876.1 10	88.6 11	2305.6	4 ⁽⁻⁾	429.4	2 ⁺	Q+O	-0.65 38	Mult., δ : DCO(D)=1.62 26 gated on 429 γ . (M2+E3) is assigned in 2009Ch43 based on RUL. $-1.03 < \delta < -0.27$ from analysis of DCO and POL data with $\sigma/J=0.4$. Other

Continued on next page (footnotes at end of table)

$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) \text{ E}=34 \text{ MeV}$ [2009Ch43,2012Ma21](#) (continued) $\gamma(^{34}\text{P})$ (continued)

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	Comments
1891.6 10	11.3 9	2321.1	(3 ⁻)	429.4	2 ⁺	D	possible $+1.2 < \delta < +4.0$ is unrealistic by RUL. Mult.: DCO(D)=1.07 18 gated on 429 γ . (E1+M2) is assigned in 2009Ch43 .
2325.1 10	4.2 3	4630.8	6 ⁽⁻⁾	2305.6	4 ⁽⁻⁾	(E2)	DCO(D)=1.8 3 gated on 1607 γ . Electric nature is assigned in 2009Ch43 as POL qualitatively indicates.
2884.3 10	6.6 4	6237.7	7 ⁽⁺⁾	3353.3	5 ⁽⁻⁾	Q	Mult.: DCO(Q)=1.35 22 gated on 1876 γ . (M2) is assigned in 2009Ch43 .
3931.7 10		6237.7	7 ⁽⁺⁾	2305.6	4 ⁽⁻⁾	[E3]	

[†] From [2009Ch43](#).

[‡] From measured $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL). Expected R_{DCO} values in a stretched quadrupole gate are $R_{\text{DCO}}(\text{Q})\approx 1$ for stretched quadrupole ($\Delta J=2$) or unstretched dipole ($\Delta J=0$) transitions; $R_{\text{DCO}}(\text{Q})\approx 0.5$ for stretched dipole ($\Delta J=1$) transitions. Expected R_{DCO} values in a stretched dipole gate are $R_{\text{DCO}}(\text{D})\approx 1$ for stretched dipole ($\Delta J=1$) transitions; $R_{\text{DCO}}(\text{D})\approx 2$ for stretched quadrupole ($\Delta J=2$) or unstretched dipole ($\Delta J=0$) transitions. A positive POL indicates an electric type transition, and a negative POL indicates a magnetic type transition. DCO and POL values are provided in an e-mail reply to B. Singh from the first author of [2009Ch43](#) on Sept 14, 2009.

Placement of transition in the level scheme is uncertain.

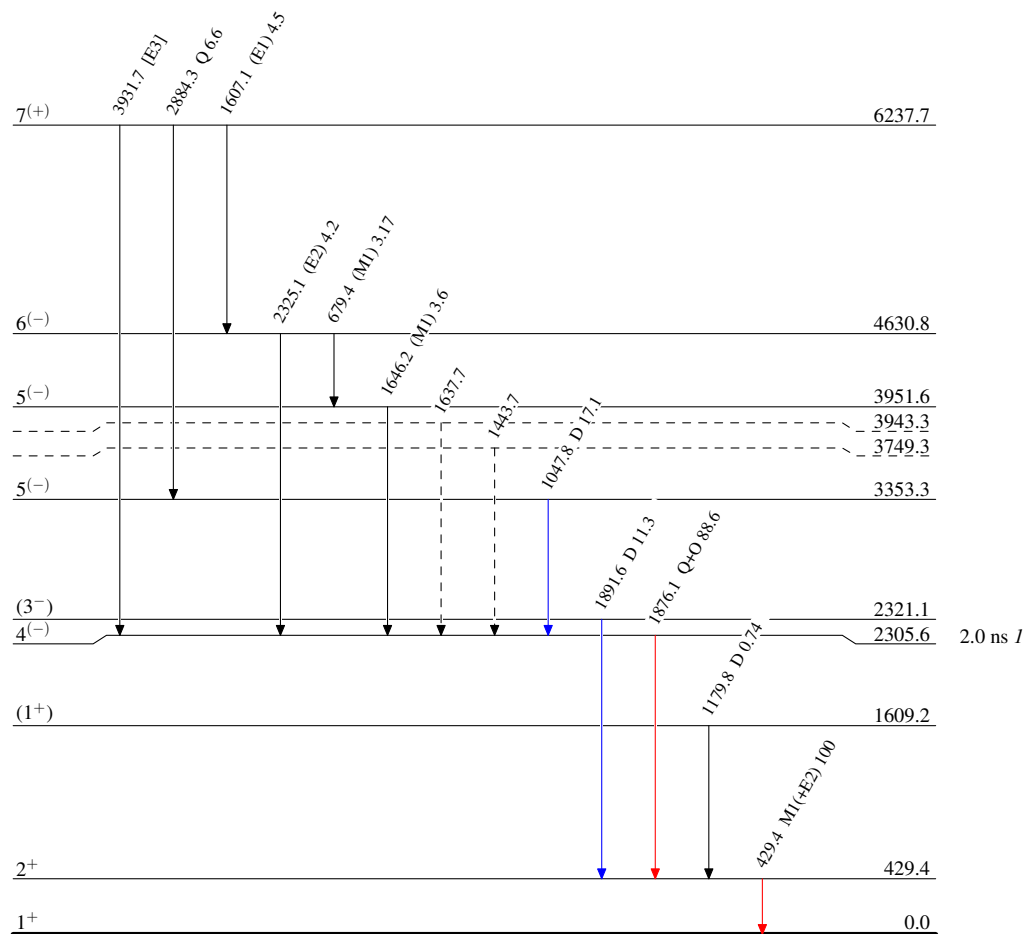
$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) E=34\text{ MeV}$ 2009Ch43,2012Ma21

Legend

Level Scheme

Intensities: Relative I_γ

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\text{max}}$
 $\cdots\cdots\cdots\longrightarrow$ γ Decay (Uncertain)

 $^{34}_{15}\text{P}_{19}$

$^{18}\text{O}(^{18}\text{O},\text{pn}\gamma) E=44\text{ MeV}$ 2009Be26

2009Be26: 20, 24, 25, 30, and 44-MeV ^{18}O beams were used. The target was 200- $\mu\text{g}/\text{cm}^2$ 97%-enriched ^{18}O on a 12.7- μm Ta backing. Measured E_γ , I_γ , $\gamma\gamma$ -coin, $\text{p}\gamma$ -coin, and $\gamma(\theta)$. Deduced levels, J , π , multipolarities, and lifetimes using DSAM lineshape analysis. Comparisons to shell model calculations.

At FSU $E(^{18}\text{O})=24$ and 44 MeV. γ rays were detected by the Compton-suppressed high-purity germanium (HPGe) array, consisting of 6 single crystal HPGe detectors and 3 four-crystal clover detectors. Each Ge detector was surrounded by bismuth germanate (BGO) crystals used for Compton suppression. Charged particles were detected by a segmented ΔE -E Si telescope.

At LBNL $E(^{18}\text{O})=20, 25$, and 30 MeV. Charged particles were detected by the Silicon Telescope Array for Reaction Studies (STARS), consisting of a ΔE -E Si detector telescope. γ rays were detected by the Livermore Berkeley Array for Collaborative Experiments (LiBerACE) consisting of five HPGe clover detectors. Lifetimes were determined by Doppler-shifted line shapes using the LBNL data.

 ^{34}P Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	Comments
0.0	1^+	J^π : from the Adopted Levels.
428.7 8	2	J^π : from 429 $\gamma(\theta)$ in 2009Be26.
1607.4 8		
2304.7 22	(4)	J^π : spin=3,4 from 1876 $\gamma(\theta)$. 2009Be26 adopts spin=4.
2319.7 22	3	J^π : spin=3 from 1891 $\gamma(\theta)$ without showing $\gamma(\theta)$ plot.
3352.2 28	(5)	J^π : spin=3,5 from 1047 $\gamma(\theta)$. 2009Be26 adopts spin=5 from tendency of populating yrast states in fusion-evaporation reactions.
3950.0 27		
4629.2 26		
6191.3 34		
6235.8 29	(7)	J^π : spin=6,7 from 1607 $\gamma(\theta)$. 2009Be26 adopts spin=7 for a smaller δ value.

† From a least-squares fit to γ -ray energies.

‡ From $\gamma(\theta)$ measurements in 2009Be26, unless otherwise noted.

 $\gamma(^{34}\text{P})$

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π
429 ‡ 1	100	428.7	2	0.0	1^+	1876 2	76 6	2304.7	(4)	428.7	2
679 2	9 2	4629.2		3950.0		1891 2	6 1	2319.7	3	428.7	2
1047 2	36 4	3352.2	(5)	2304.7	(4)	2325 2	11 2	4629.2		2304.7	(4)
1179 1	8 2	1607.4		428.7	2	2839 2	10 2	6191.3		3352.2	(5)
1607 2	13 2	6235.8	(7)	4629.2		2883 2	11 2	6235.8	(7)	3352.2	(5)
1645 2	20 3	3950.0		2304.7	(4)						

† From 2009Be26. I_γ is given for $E(^{18}\text{O})=44\text{ MeV}$. 2009Be26 states that the γ intensities at $E(^{18}\text{O})=44\text{ MeV}$ were measured in coincidence with the 429 γ gate, so the intensities of ground state transitions could not be measured. Intensities at $E(^{18}\text{O})=44\text{ MeV}$ are normalized to $I_\gamma(1876)=76\ 6$ that was determined at $E(^{18}\text{O})=24\text{ MeV}$.

‡ Placement of transition in the level scheme is uncertain.

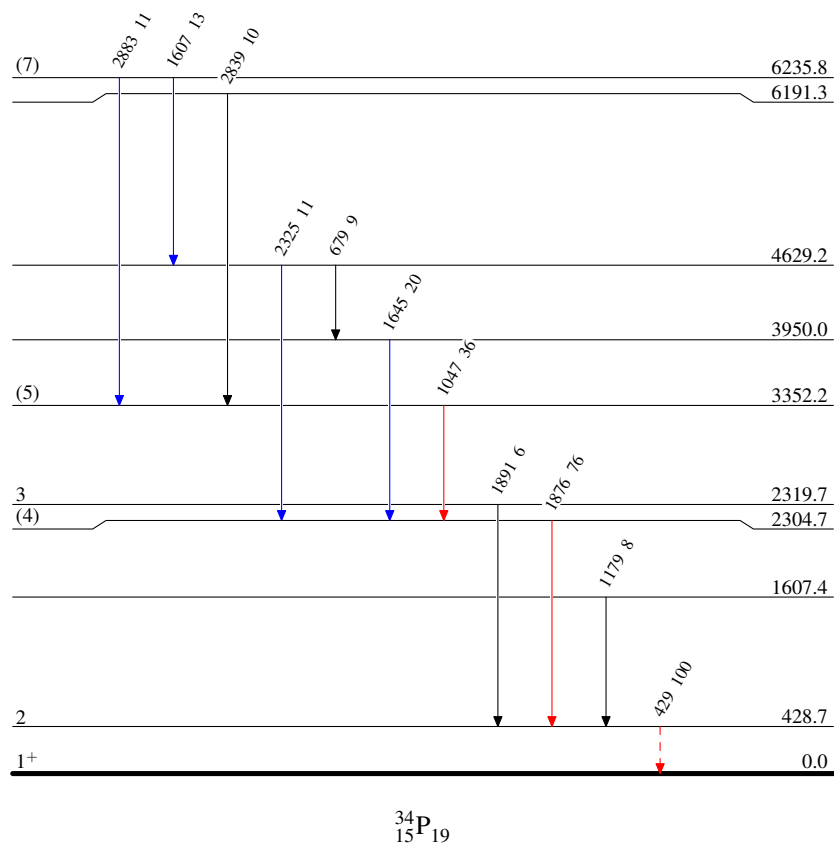
$^{18}\text{O}(^{18}\text{O,pn}\gamma) E=44 \text{ MeV}$ 2009Be26

Level Scheme

Intensities: Relative I_γ

Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$
- - - - - γ Decay (Uncertain)



 $^{30}\text{Si}(^7\text{Li}, ^3\text{He})$ [1977Pe17](#)

[1977Pe17](#): a 24-MeV ^7Li beam was provided from the EN Tandem accelerator in Oxford. The target was prepared by vacuum evaporation of $20\text{-}\mu\text{g}/\text{cm}^2$ silicon-dioxide enriched to 95.5% in ^{30}Si on a $10\text{ }\mu\text{g}/\text{cm}^2$ carbon backing. Measured ^3He spectra and $\sigma(E(^3\text{He}),\theta)$ using double-focusing magnetic spectrometer. Deduced levels and masses.

 ^{34}P Levels

<u>E(level)</u>	<u>Comments</u>
0	$d\sigma/d\Omega=0.21$ 4.
441 20	$d\sigma/d\Omega=0.26$ 5.
667? 40	E(level): 1977Pe17 states that this may be due to an impurity.
	$d\sigma/d\Omega=0.28$ 6.

³⁴S(t, ³He) 1977Aj01

1977Aj01: a 23-MeV triton beam was provided from the Los Alamos Scientific Laboratory three-stage Van de Graaff facility. The target was 140-μg/cm² Sb₂S₃, 85.6% enriched in ³⁴S, deposited on a 30-μg/cm² carbon foil. Measured ³He spectra and σ(θ) from 20-35° (lab) in 5° steps using the Q3D magnetic spectrometer and a 1-m-long helix focal plane detector. Deduced levels.

³⁴P Levels

E(level)	Comments
0	dσ/dΩ=4.2 13.
423 10	dσ/dΩ=15 5.
1605 10	dσ/dΩ=4.8 15.
2225 10	dσ/dΩ=22 7.
2309 10	dσ/dΩ=94 28.
3345?	dσ/dΩ=7 2. observed only at one angle.

 $^{34}\text{S}(^7\text{Li}, ^7\text{Be}), (^{11}\text{B}, ^{11}\text{C})$ **1985Dr06**

1985Dr06: a 52-MeV ^7Li beam and a 83-MeV ^{11}B beam were produced from the 14UD Pelletron accelerator at the Australian National University. The targets were 280- $\mu\text{g}/\text{cm}^2$ Ag_2S ^{34}S and ^{36}S . Measured ^7Be and ^{11}C spectra using Enge split-pole spectrometer at 10° (lab) for $(^7\text{Li}, ^7\text{Be})$ and at 8° for $(^{11}\text{B}, ^{11}\text{C})$ with FWHM=110 keV for ^7Be spectrum. Deduced mass excesses and levels.

 ^{34}P LevelsE(level)[†]

0
430
1600
2287 15
4470 70
7440 70

[†] From **1985Dr06**.

$^{36}\text{S}(\text{pol d},\alpha),(\text{POL d},\alpha\gamma)$ 1987TrZZ

1987TrZZ: $^{36}\text{S}(\text{pol d},\alpha),(\text{pol d},\alpha\gamma)$. $E_d=6\text{--}16$ MeV on 81.1%-enriched ^{36}S target at the McMaster University tandem accelerator.

The α particles were momentum analyzed by an Enge split-pole spectrograph and detected in a focal-plane position-sensitive delay-line counter. The γ rays were measured using a Ge detector. The $\alpha\gamma(\theta)$ measurements were performed using five large NaI(Tl) detectors. Measured E_α , E_γ , I_γ , $\alpha\gamma(\theta)$, and tensor analyzing powers (T_{20}). Deduced levels, J , π , multipolarities, and mixing ratios. Comparisons with shell-model calculations.

1978Ba30: $^{36}\text{S}(\text{d},\alpha)$. $E_d=3.05\text{--}3.26$ MeV on 81.1%-enriched ^{36}S target at the Radium Institute. Measured α spectra with a magnetic spectrograph. Deduced levels.

 ^{34}P Levels

$E(\text{level})^\dagger$	J^π	Comments
0	1^+	J^π : unnatural parity from analyzing powers (1987TrZZ); spin=1 from the Adopted Levels.
429.1 1	2^+	E(level): 429 7 from 1978Ba30 E_α data, 430 4 from 1987TrZZ E_α data.
1607.6 5	1^+	J^π : natural parity from analyzing powers (1987TrZZ) and spin=2 from $\alpha\gamma(\theta)$ (1987TrZZ). E(level): 1616 10 from 1978Ba30 E_α data, 1610 3 from 1987TrZZ E_α data.
2225 2		J^π : unnatural parity from analyzing powers (1987TrZZ) and spin=1 from $\alpha\gamma(\theta)$ (1987TrZZ). E(level): 2236 10 from 1978Ba30, 2232 10 from 1987TrZZ E_α data.
2299 3	$(1^+, 2^-, 3^+, 4^-)$	J^π : unnatural parity from analyzing powers (1987TrZZ). Spin=3 seems better than spin=1,2 from $\alpha\gamma(\theta)$ plot (1987TrZZ).
2372 20		E(level): 2302 20 from 1978Ba30 E_α data, 2307 7 from 1987TrZZ E_α data.
2628 12		E(level): from 1978Ba30 E_α data.
2675 3		E(level): from 1978Ba30 E_α data.
3086? 15	$(1^+, 2^-, 3^+)$	E(level): 2683 7 from 1987TrZZ E_α data.
3201 16		E(level): from 1987TrZZ E_α data.
3291? 23		E(level): from 1978Ba30 E_α data.
3482 8		E(level): from 1987TrZZ E_α data.
3546? 12	$(1^+, 2^-, 3^+)$	E(level): from 1987TrZZ E_α data.
4302 6		E(level): 4306 7 from 1987TrZZ E_α data.
4434 10		E(level): 4438 13 from 1987TrZZ E_α data.
4725 12		E(level): 4744 9 from 1987TrZZ E_α data.

† From E_γ data in 1987TrZZ. The $E(\text{level})$ values from E_α data are given under comments.

‡ From measured $\alpha\gamma(\theta)$ and T_{20} in 1987TrZZ. 1987TrZZ placed all J^π in parentheses for levels above 2 MeV.

 $\gamma(^{34}\text{P})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	$\delta^\#$	Comments
429.1	2^+	429.1		0	1^+	D+Q	+0.52 10	δ : or +1/(0.01 8) (1987TrZZ).
1607.6	1^+	1178.5	61 7	429.1	2^+	D+Q	-0.93 26	δ : or -0.96 64 (1987TrZZ).
		1607.6	39 5	0	1^+	D+Q	+0.13 6	δ : or +1/(0.13 6) (1987TrZZ).
2225		1796		429.1	2^+			
		2225 @		0	1^+			
2299	$(1^+, 2^-, 3^+, 4^-)$	1870 @		429.1	2^+			
2675	$(1^+, 2^-, 3^+)$	376 @	<9	2299	$(1^+, 2^-, 3^+, 4^-)$			
		1067	23 9	1607.6	1^+			
		2246	50 13	429.1	2^+			
		2675 @	18 10	0	1^+			
4302	$(1^+, 2^-, 3^+)$	1627 @		2675	$(1^+, 2^-, 3^+)$			
		2694		1607.6	1^+			
4434		2135 @		2299	$(1^+, 2^-, 3^+, 4^-)$			
4725	$(1^+, 2^-, 3^+, 4^-)$	4296		429.1	2^+			

Continued on next page (footnotes at end of table)

$^{36}\text{S}(\text{pol d},\alpha),(\text{POL d},\alpha\gamma)$ [1987TrZZ](#) (continued)

$\gamma(^{34}\text{P})$ (continued)

[†] From level-energy differences.

[‡] From [1987TrZZ](#).

From measured $\alpha\gamma(\theta)$ in [1987TrZZ](#). All D+Q assignments in this dataset are (M1+E2) from level scheme.

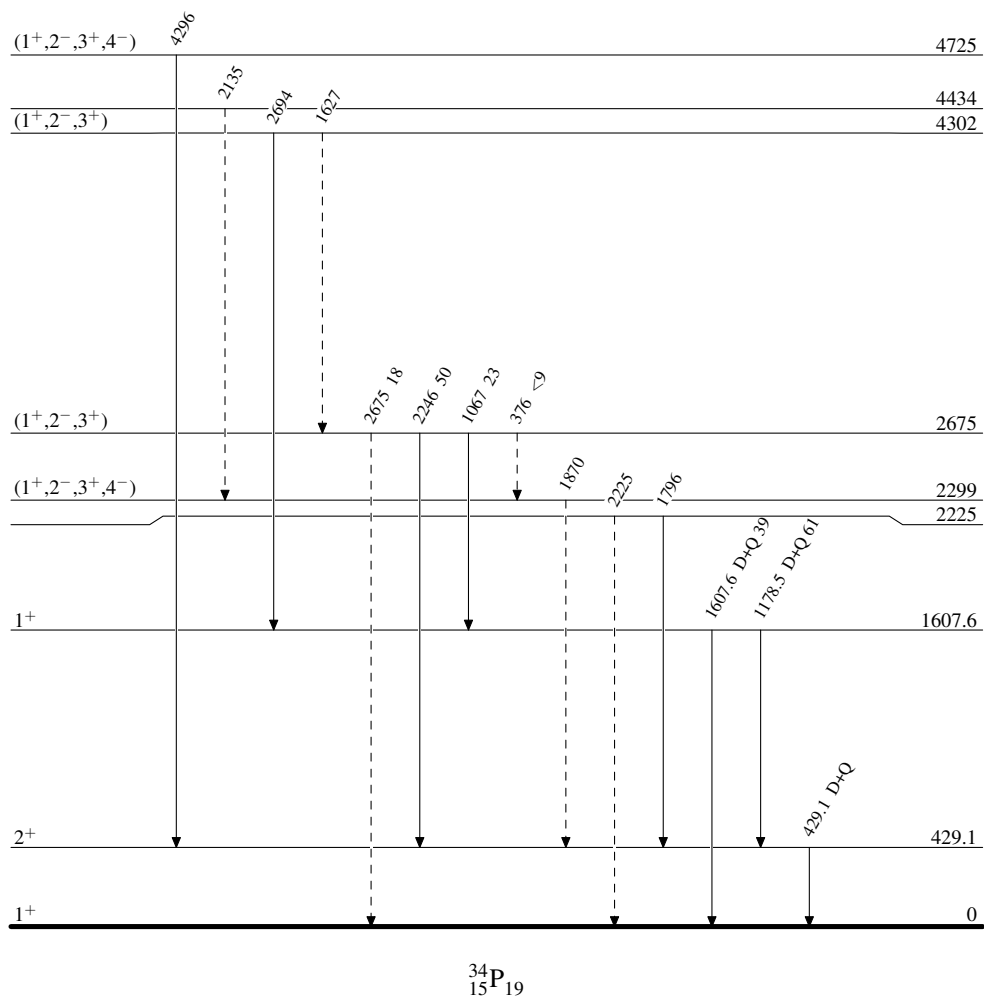
@ Placement of transition in the level scheme is uncertain.

$^{36}\text{S}(\text{pol d},\alpha),(\text{POL d},\alpha\gamma)$ 1987TrZZ

Legend

Level Scheme

Intensities: % photon branching from each level

 - - - - - $\rightarrow$ γ Decay (Uncertain)


$^{115}\text{In}(^{34}\text{S}, \text{X}\gamma)$ 2006Kr07

2006Kr07: a 140-MeV ^{34}S beam was produced from the 15-UD Pelletron accelerator at the Nuclear Science Centre, New Delhi, India. The target was 1.29-mg/cm² enriched ^{115}In . The deexciting γ rays were detected using the Indian National Gamma Array (INGA), consisting of eight Compton-suppressed HPGe Clover detectors. Measured E_γ , I_γ , $\gamma\gamma$ -coin, $\gamma\gamma(\theta)$ (DCO), and $\gamma\gamma(\theta)$ (POL). Deduced levels, J, π , and γ -ray multiplicities. Comparisons with shell-model calculations.

 ^{34}P Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	Comments
0	1 ⁺	J^π : from the Adopted Levels.
429.3 5	2 ⁺	
1608.4 7	(1 ⁺)	
2305.3 7	4	J^π : 4 ⁺ is assigned in 2006Kr07.
2320.6 7	3 ⁻	
3352.5 7	5 ⁽⁻⁾	$E(\text{level}), J^\pi$: from the Adopted Levels; final level for 2883 γ .
6237.1	7 ⁽⁺⁾	$E(\text{level})$: from the Adopted Levels. The 2883 γ was placed to be from a 5188 level in 2006Kr07.

[†] From a least-squares fit to the measured γ -ray energies, unless otherwise noted.

[‡] As given in 2006Kr07, unless otherwise noted. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

 $\gamma(^{34}\text{P})$

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	Comments
429.3	>155	429.3	2 ⁺	0	1 ⁺	M1	$R_{\text{DCO}}(\text{Q})=0.59$ 5, $\text{POL}=-0.16$ 13. (M1) is assigned in 2006Kr07.
1179.1	17.7 11	1608.4	(1 ⁺)	429.3	2 ⁺	Q	$R_{\text{DCO}}(\text{D})=1.68$ 14, $\text{POL}=+0.15$ 17. E2 is assigned in 2006Kr07.
1875.9	100	2305.3	4	429.3	2 ⁺	Q	
1891.2	38.5 18	2320.6	3 ⁻	429.3	2 ⁺	D	$R_{\text{DCO}}(\text{D})=0.89$ 10, $\text{POL}=+0.16$ 17. E1 is assigned in 2006Kr07.
2883.0	16.1 11	6237.1	7 ⁽⁺⁾	3352.5	5 ⁽⁻⁾	M2	$R_{\text{DCO}}(\text{Q})=0.88$ 9, $\text{POL}=-0.22$ 11.

[†] From 2006Kr07. $\Delta E_\gamma=1$ keV is assumed by evaluators based on typical uncertainties for the INGA array.

[‡] From measured $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL). Expected R_{DCO} values in a stretched quadrupole gate are $R_{\text{DCO}}(\text{Q})\approx 1.05$ for stretched quadrupole ($\Delta J=2$) or unstretched dipole ($\Delta J=0$) transitions; $R_{\text{DCO}}(\text{Q})\approx 0.55$ for stretched dipole ($\Delta J=1$) transitions. Expected R_{DCO} values in a stretched dipole gate are $R_{\text{DCO}}(\text{D})\approx 1.1$ for stretched dipole ($\Delta J=1$) transitions; $R_{\text{DCO}}(\text{D})\approx 1.8$ for stretched quadrupole ($\Delta J=2$) or unstretched dipole ($\Delta J=0$) transitions. A positive POL indicates an electric type transition, and a negative POL indicates a magnetic type transition.

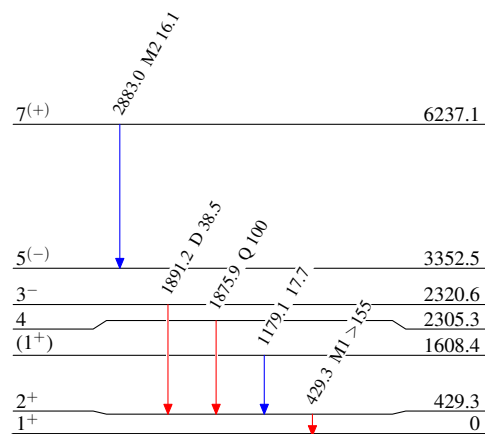
$^{115}\text{In}(^{34}\text{S},\text{X}\gamma)$ 2006Kr07

Level Scheme

Intensities: Relative I_γ

Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$

 $^{34}_{15}\text{P}_{19}$

$^{160}\text{Gd}(^{37}\text{Cl},\text{X}\gamma)$ 1994Fo04

1994Fo04: a 167-MeV ^{37}Cl beam was produced from the Argonne Tandem Linac Accelerator System (ATLAS). Targets were 1 mg/cm² 98.1% enriched ^{160}Gd backed by 15 mg/cm² gold. γ rays were detected using the Argonne-Notre Dame BGO γ -ray facility consisting of 12 Compton-suppressed Ge detectors and a 50-element bismuth germanate (BGO) array. Measured E_γ , I_γ , and $\gamma\gamma$ -coin. Deduced levels. $\gamma\gamma$ -coin gates were placed on known γ rays in specific $A\approx 160$ products to select individual reaction channels and identify coincident γ rays in light product partners. $\gamma\gamma$ -correlation matrices produced with the fold condition $4 \leq k \leq 9$ were found to include most of the events associated with deep inelastic processes.

 ^{34}P Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>	<u>Comments</u>
0	1^+	J^π : from the Adopted Levels.
429	2^+	J^π : proposed by 1994Fo04 from configuration $\pi s_{1/2} \nu d_{3/2}$.
2305	$(3^-, 4^-)$	J^π : proposed by 1994Fo04 from configuration $\pi s_{1/2} \nu f_{7/2}$.

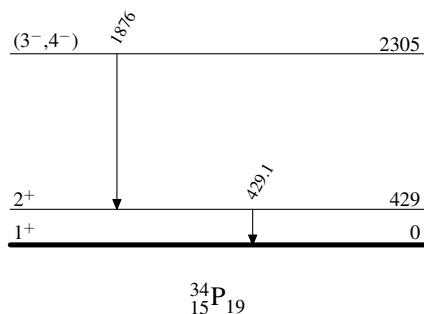
† From E_γ data in 1994Fo04.

‡ Proposed by 1994Fo04 based on yrast state depopulations, unless otherwise noted.

 $\gamma(^{34}\text{P})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
429.1	429	2^+	0	1^+
1876	2305	$(3^-, 4^-)$	429	2^+

† From 1994Fo04. Also observed in ^{34}S and ^{36}S induced reactions.

 $^{160}\text{Gd}(^{37}\text{Cl},\text{X}\gamma)$ 1994Fo04Level Scheme

$^{176}\text{Yb}(^{36}\text{S},\text{X}\gamma)$ 2005OI02

2005OI02, 2005OI04: A 230-MeV ^{36}S beam was produced using the combination of the XTU tandem Van de Graaff accelerator and the ALPI superconducting linear accelerator at the INFN Legnaro National Laboratory. The target was 14-mg/cm² 97.8%-enriched ^{176}Yb on a 35-mg/cm² ^{208}Pb backing. γ rays deexciting projectile-like and target-like fragments were detected using the GASP array of 40 HPGe Compton-suppressed detectors and an inner ball of 80 BGO crystals. Measured E_γ , I_γ , and $\gamma\gamma$ -coin. Deduced levels, J , π . Comparisons with shell-model calculations. 2005OI04 also reports γ spectra of deep-inelastic products populated by a 215-MeV ^{36}S beam on a 300 $\mu\text{g}/\text{cm}^2$ ^{208}Pb target. Only γ spectra are given and the three ^{34}P peaks in 2005OI04 are labelled with the same energy values as in 2005OI02.

 ^{34}P Levels

$E(\text{level})^\dagger$	$J\pi^\ddagger$
0	1^+
429	2^+
2305	4^-
2320	$3^{(-)}$
3351	$5^{(-)}$
4723?	$(5,6^-)$
6236	$7^{(+)}$

† From E_γ data in 2005OI02.

‡ From the Adopted Levels.

 $\gamma(^{34}\text{P})$

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π
429		429	2^+	0	1^+
1046	257 36	3351	$5^{(-)}$	2305	4^-
1876		2305	4^-	429	2^+
1891		2320	$3^{(-)}$	429	2^+
2403 ‡		4723?	$(5,6^-)$	2320	$3^{(-)}$
2885	72 30	6236	$7^{(+)}$	3351	$5^{(-)}$

† From 2005OI02.

‡ Placement of transition in the level scheme is uncertain.

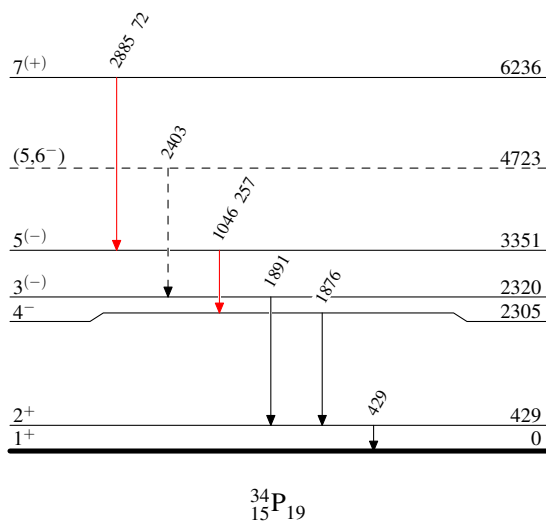
$^{176}\text{Yb}(^{36}\text{S},\text{X}\gamma)$ 2005OI02

Legend

Level Scheme

Intensities: Relative I_γ

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$
- - - - - → γ Decay (Uncertain)



$^{198}\text{Pt}(^{37}\text{Cl},\text{X}\gamma)$ 2002AsZY

2002AsZY: $^{198}\text{Pt}(^{37}\text{Cl},\text{X}\gamma)$ at 9 MeV/nucleon. Projectile-like fragments were detected using an annular Si detector. γ rays emitted from the stopped fragments were detected using four Ge detectors. Measured E_γ , I_γ , and (fragment) γ -coin. Deduced levels, J , π , multipolarities, and lifetimes.

 ^{34}P Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0	1^+		
429	2^+		
2305	4^-	1.4 ns <i>II</i>	$T_{1/2}$: $0.3 < T_{1/2} < 2.5$ ns from fragment- $\gamma(t)$ (2002AsZY).

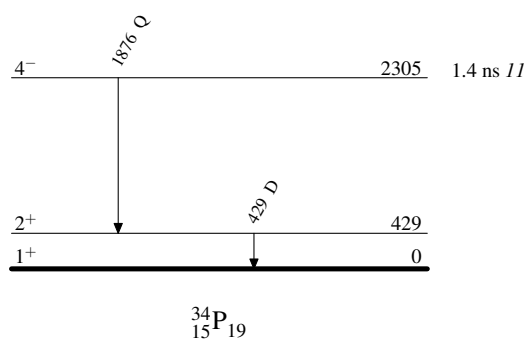
† From E_γ data in 2002AsZY.

‡ From the Adopted Levels.

 $\gamma(^{34}\text{P})$

$E_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult.	Comments
429	429	2^+	0	1^+	D	Mult.: stretched dipole from in-plane to out-of-plane γ anisotropy analysis and M1 is assigned by 2002AsZY.
1876	2305	4^-	429	2^+	Q	Mult.: stretched quadrupole from in-plane to out-of-plane γ anisotropy analysis and M2 is assigned by 2002AsZY.

† From 2002AsZY.

 $^{198}\text{Pt}(^{37}\text{Cl},\text{X}\gamma)$ 2002AsZYLevel Scheme

$^{208}\text{Pb}(^{36}\text{S}, ^{34}\text{P}\gamma)$ 2015Ch56,2019Gr08

2015Ch56: A 215-MeV ^{36}S beam was produced using the combination of XTU tandem Van de Graaff accelerator and ALPI superconducting linear accelerator at the INFN Legnaro National Laboratory. The target was $300\text{-}\mu\text{g}/\text{cm}^2$ 99.7% enriched ^{208}Pb on a $20\text{ }\mu\text{g}/\text{cm}^2$ carbon backing. Projectile-like fragments produced in multinucleon binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the CLARA array of 22 EUROBALL escape-suppressed HPGe clover detectors. Doppler corrections of γ -ray energies were performed event by event. Measured E_γ , I_γ , $(^{34}\text{P})\gamma$ -coin, and $\gamma\gamma$ -coin. Deduced levels, J , π . Comparisons with shell-model calculations. Preliminary results are given in [2005OI04](#).

2019Gr08: A 225-MeV ^{36}S beam was provided by Tandem-ALPI accelerator complex at the INFN Legnaro National Laboratory. The target was $1\text{ mg}/\text{cm}^2$ 99.7% enriched ^{208}Pb with $1\text{ mg}/\text{cm}^2$ Nb backing and mounted onto the Cologne differential plunger. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the AGATA demonstrator array of five triple cluster modules of 36-fold segmented Ge crystals covering backward angles from 135° to 175° . Doppler corrections of γ -ray energies were performed event by event. Measured E_γ , $(^{34}\text{P})\gamma$ -coin, and level lifetimes using the differential recoil-distance method (DRDM). Comparisons with shell-model calculations.

 ^{34}P Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0	1^+		
429.0 7	2^+	1.04 ps 35	$T_{1/2}$: $\tau=1.5$ ps 5 from differential recoil-distance method (DRDM) (2019Gr08).
1607.0 8	1^+		
2305.1 12	4^-		
2320.1 12	$3^{(-)}$		
2682.1? 8	$(1^+, 2^-, 3^+)$		
3351.1 16	$5^{(-)}$		
3950.1 16	5^-		

† From a least-squares fit to γ -ray energies in [2015Ch56](#).

‡ From the Adopted Levels.

 $\gamma(^{34}\text{P})$

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π
429 1	100.0 13	429.0	2^+	0	1^+
1046 1	9.6 7	3351.1	$5^{(-)}$	2305.1	4^-
1178 1	10.2 7	1607.0	1^+	429.0	2^+
1607 1	6.1 7	1607.0	1^+	0	1^+
1645 1	7.0 7	3950.1	5^-	2305.1	4^-
1876 1	24.4 11	2305.1	4^-	429.0	2^+
1891 1	7.1 6	2320.1	$3^{(-)}$	429.0	2^+
2253 ‡ 1	5.8 6	2682.1?	$(1^+, 2^-, 3^+)$	429.0	2^+
2682 ‡ 1	4.6 6	2682.1?	$(1^+, 2^-, 3^+)$	0	1^+

† From [2015Ch56](#).

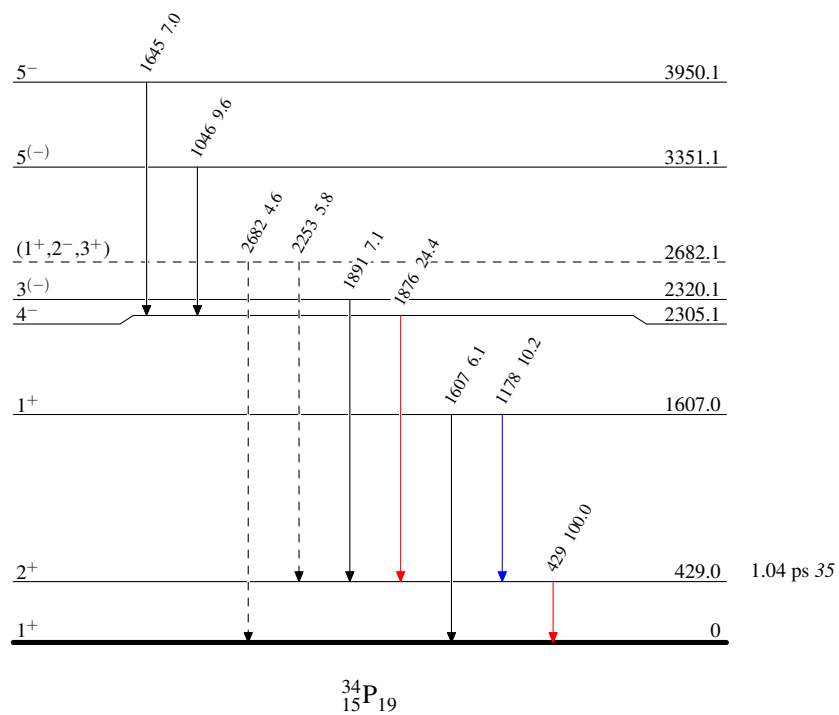
‡ Placement of transition in the level scheme is uncertain.

$^{208}\text{Pb}(^{36}\text{S}, ^{34}\text{P}\gamma)$ 2015Ch56,2019Gr08

Legend

Level Scheme
Intensities: Relative I_γ

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$
- - - - - γ Decay (Uncertain)



Coulomb excitation **2000Pr09**

2000Pr09: $^{197}\text{Au}(^{34}\text{P}, ^{34}\text{P}'\gamma)$. A 54.5-MeV/nucleon radioactive beam of ^{34}P was produced via fragmentation of a 90-MeV/nucleon ^{40}Ar primary beam on a ^9Be primary target at NSCL, MSU. The secondary target was ^{197}Au . The mid-target beam energy is 44.4 MeV/nucleon. γ rays were detected using the NSCL NaI(Tl) array. Measured E_γ , I_γ , and (scattered- ^{34}P)- γ coin. Deduced excitation cross sections and $B(E2)$. Also see [2002Gl01](#) and [2000PrZX](#).

 ^{34}P Levels

$E(\text{level})^\dagger$	Comments
0	
422 7	$B(E2)=0.0020$ 10 if the yield of 429γ is entirely from direct population of the 422 level in the Coulomb excitation reaction. 2000Pr09 adopts the $B(E2)^\dagger$ for the 2235 level and concludes that they have no evidence for direct E2 excitation of the 429 level. The yield of 429γ can be explained entirely by feeding from the decay of the 2235 level through the 1608 level.
1608	$E(\text{level})$: rounded value from the Adopted Levels. 1178γ and 1608γ are assumed to deexcite this level but are not observed in 2000Pr09 .
2235? 9	$B(E2)^\dagger=0.0026$ 12 (2000Pr09)

† From E_γ data in [2000Pr09](#).

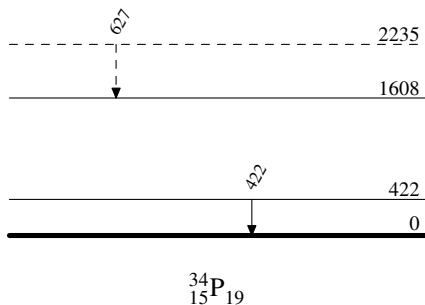
 $\gamma(^{34}\text{P})$

E_γ	$E_i(\text{level})$	E_f	Comments
422 7	422	0	$\sigma=5.2$ mb 24.
627 † 9	2235?	1608	$\sigma=6.8$ mb 30.

† Placement of transition in the level scheme is uncertain.

Coulomb excitation **2000Pr09**

Legend

Level Scheme-----► γ Decay (Uncertain)

Adopted Levels, Gammas

$Q(\beta^-) = -5491.60$ 4; $S(n) = 11417.15$ 4; $S(p) = 10883.3$ 11; $Q(\alpha) = -7923.64$ 5 [2021Wa16](#)

$S(2n) = 20058.789$ 40, $S(2p) = 20431.9$ 3 ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): identified using the mass spectrometer at the Cavendish Laboratory, UK ([1926As01](#)).

[2025Li67](#): calculated level energies, J, π , and rotational bands for ^{34}S using the variation after projection shell model (VAPSM).

[2023Ch34](#): calculated energy levels, J, π , electric quadrupole, and magnetic dipole moments for ^{34}S using microscopic effective sd-shell interactions with bare and effective values for the N3LO, JISP16, DJ16, DJ16A, and USDB interactions.

Adopted Levels, Gammas (continued)

³⁴S Levels

E(level), J^π , and width (Γ) for resonances (>11.4 MeV) are from ³³S(n, γ),(n,n):resonances, unless otherwise noted.

Cross Reference (XREF) Flags

A	³⁴ P β^- decay (12.43 s)	M	³¹ P(α , γ)	Y	³⁴ S(pol d,d),(pol d,d')
B	³⁴ Cl $\varepsilon+\beta^+$ decay (1.5266 s)	N	³² S(t,p)	Z	³⁴ S(α,α'),(α,α' '),($\alpha,\alpha'\gamma$)
C	³⁴ Cl $\varepsilon+\beta^+$ decay (31.99 min)	O	³² S(α , ² He)	Others:	
D	³ H(³² S, γ)	P	³³ S(n, γ) E=thermal	AA	³⁵ Cl(γ ,p)
E	³ H(³⁵ Cl, $\alpha\gamma$)	Q	³³ S(n, γ),(n,n):resonances	AB	³⁵ Cl(n,d)
F	⁴ He(³⁰ Si, α):resonances	R	³³ S(d,p)	AC	³⁵ Cl(d, ³ He)
G	¹² C(²⁴ Mg, ³⁴ S γ)	S	³⁴ S(γ,γ'),(pol γ,γ')	AD	³⁷ Cl(p, $\alpha\gamma$)
H	²⁴ Mg(¹⁶ O, α 2p γ)	T	³⁴ S(e,e')	AE	²⁰⁶ Pb(³⁴ S, ³⁴ S' γ)
I	²⁷ Al(¹⁰ B,2p γ)	U	³⁴ S($\pi^+,\pi^+'),(\pi^-, \pi^-'),(\pi^+,\pi^+),(\pi^-, \pi^-)$	AF	²⁰⁸ Pb(³⁴ S, ³⁴ S')
J	²⁷ Al(¹² C, α p γ)	V	³⁴ S(n,n),(n,n')	AG	²⁰⁸ Pb(³⁶ S, ³⁴ S' γ)
K	²⁸ Si(³⁴ S, ³⁴ S' γ)	W	³⁴ S(p,p),(p,p'),(pol p,p')		
L	³⁰ Si(α,γ),(α ,n):resonances	X	³⁴ S(p,p' γ)		

E(level) [†]	J^π &	T _{1/2}	XREF	Comments
0.0 ^a	0 ⁺	stable	ABCDE GHIJKLMNOP RSTUVWXYZ	XREF: Others: AA, AB, AC, AD, AE, AF, AG Evaluated rms nuclear charge radius R=3.2847 fm 21 (2013An02).
2127.561 ^a 15	2 ⁺	0.315 ps 12	A CDE GHIJKLMN P R TUVWXYZ	XREF: Others: AA, AB, AC, AD, AE, AF, AG $\mu=+1.0$ 2 (1979Za01,2020StZV) Q=+0.04 3 (1980Ba40,1981Sp07,2021StZZ) B(E2) $\uparrow$ =0.0197 7 J^π : spin=2 from p γ (θ) in ³¹ P(α , γ). $\pi=+$ from 2127.498 γ , E2 to 0 ⁺ , g.s. T _{1/2} : lifetime τ =454 fs 17: weighted average of 490 fs 30 in ³ H(³² S, γ) from 1977He12 with DSAM, 465 fs 50 in ²⁸ Si(³⁴ S, ³⁴ S' γ) from 1977Sc36 with DSAM, 350 fs 60 in ³¹ P(α , γ) from 1969Gr03 with DSAM, 460 fs 95 in ³¹ P(α , γ) from 1970Br18 with DSAM, 467 fs 90 in ³¹ P(α , γ) from 1970Cu02 with DSAM, 440 fs 50 in ³¹ P(α , γ) from 1970Gr11 with DSAM, 400 fs 32 in ³¹ P(α , γ) from 1970Ra17 with DSAM, 400 fs 40 in ³¹ P(α , γ) from 1974Gr06 with DSAM, and 476 fs 17 converted from the adopted B(E2) $\uparrow$ =0.0197 e ² b ² 7 by evaluators. μ : from 1979Za01 by perturbed angular correlation after ion implantation method. Q: +0.06 4 in 1980Ba40 by Coulomb excitation reorientation method, which was recalculated by 1981Sp07 to be +0.04 3. Also see 2021StZZ evaluation. B(E2) $\uparrow$: weighted average of B(E2) $\uparrow$ =0.0203 13 from ²⁸ Si(³⁴ S, ³⁴ S' γ) (1977Sc36), 0.01928 66 from ³⁴ S(e,e') (1985Wo06), 0.0263 41 from ²⁰⁶ Pb(³⁴ S, ³⁴ S' γ) (1974OI02,

Adopted Levels, Gammas (continued)

³⁴ S Levels (continued)							
E(level) [†]	J ^{π&}	T _{1/2}	XREF				Comments
3304.210 16	2 ⁺	122 fs 9	A CDE GHIJ LMN P R TUVW YZ	1980Ba40), and 0.0205 33 from ²⁰⁸Pb(³⁴S, ³⁴S') (1980Ba40). β₂(p,p')=0.28 1 (1985Al03); 0.24 2 (1999Ma63 by reanalysing 1985Al03 data with Becchetti-Greenless optical potential). XREF: Others: AA, AB, AC, AD, AG B(E2)↑=0.00256 29 J^π: spin=2 from pγ(θ) in ³¹P(α,pγ). π=+ from 3304.029γ, E2 to 0⁺, g.s. T_{1/2}: lifetime τ=176 fs 13: weighted average of 192 fs 13 in ³H(³²S,pγ) from 1977He12 with DSAM, 120 fs 30 in ³¹P(α,pγ) from 1969Gr03 with DSAM, 190 fs 40 in ³¹P(α,pγ) from 1970Br18 with DSAM, 144 fs 28 in ³¹P(α,pγ) from 1970Cu02 with DSAM, 218 fs 30 in ³¹P(α,pγ) from 1970Gr11 with DSAM, 175 fs 25 in ³¹P(α,pγ) from 1970Ra17 with DSAM, 145 fs 20 in ³¹P(α,pγ) from 1974Gr06 with DSAM, and 189 fs +24–20 converted from the adopted B(E2)↑=0.00256 e²b² 29 with the adopted γ branching ratios by evaluators. B(E2)↑: from ³⁴S(e,e') (1985Wo06). XREF: Others: AA XREF: B(3916?) J^π: L=0 from 0⁺ in ³²S(t,p). T_{1/2}: lifetime τ=1618 fs 130: weighted average of 1600 fs 130 in ³¹P(α,pγ) from 1970Gr11 with DSAM and 1890 fs 500 in ³¹P(α,pγ) from 1970Ra17 with DSAM. Other: >400 fs in ³¹P(α,pγ) from 1974Gr06 with DSAM.			
3916.40 4	0 ⁺	1.12 ps 9	AB G LMN P R W YZ	XREF: Others: AA, AC XREF: B(4075?)N(4085)ac(4090) J^π: spin=1 from pγ(θ) in ³¹P(α,pγ). π=+ from 1947.060γ, M1+E2 to 2⁺, 2127.561 level. T_{1/2}: lifetime τ<24 fs in ³¹P(α,pγ) from 1970Ra17 with DSAM. Others: <33 fs in ³¹P(α,pγ) from 1970Gr11 with DSAM and ≤50 fs in ³¹P(α,pγ) from 1974Gr06 with DSAM.			
4074.662 17	1 ⁺	<17 fs	ABC G LMN P R W Y	XREF: Others: AA, AC XREF: ac(4090) J^π: spin=2 from pγ(θ) in ³¹P(α,pγ). π=+ from 4114.52γ, E2 to 0⁺, g.s. T_{1/2}: lifetime τ=104 fs 10: weighted average of 89 fs 20 in ³¹P(α,pγ) from 1970Gr11 with DSAM, 110 fs 10 in ³¹P(α,pγ) from 1970Ra17 with DSAM, and 100 fs 15 in ³¹P(α,pγ) from 1974Gr06 with DSAM.			
4114.812 23	2 ⁺	72 fs 7	A C G MN P R VW YZ	XREF: Others: AA, AG J^π: spin=3 from pγ(θ) in ³¹P(α,pγ). L=3 from 0⁺ in ³²S(t,p). T_{1/2}: lifetime τ=122 fs 10: weighted average of 125 fs 20 in ³¹P(α,pγ) from 1970Gr11 with DSAM, 135 fs 17 in ³¹P(α,pγ) from 1970Ra17 with DSAM, 145 fs 50 in ³¹P(α,pγ) from 1971So01 with DSAM, and 115 fs 10 in ³¹P(α,pγ) from 1974Gr06 with DSAM.			
4624.404 ^b 18	3 ⁻	85 fs 7	A E GH J MNOP R VW Y	XREF: Others: AA, AC, AG XREF: N(4690)R(4688)V(4689)W(4689)Z(4690)AA(4690)AC(4710)			
4688.96 ^a 5	4 ⁺	88 fs 7	A C E GH J MN P R VW YZ				

Adopted Levels, Gammas (continued)

³⁴ S Levels (continued)									
E(level) [†]	J ^π &	T _{1/2}	XREF				Comments		
4876.842 25	3 ⁺	40 fs 15	A C	GH	M P R		J^π: spin=4 from $\text{py}(\theta)$ and $\gamma\gamma(\theta)$ in $^{31}\text{P}(\alpha,\text{py})$. $\pi=+$ from 2561.35γ, E2 to 2⁺, 2127.561 level. T_{1/2}: lifetime $\tau=127$ fs 10: weighted average of 132 fs 15 in $^{31}\text{P}(\alpha,\text{py})$ from 1970Gr11 with DSAM, 131 fs 13 in $^{31}\text{P}(\alpha,\text{py})$ from 1970Ra17 with DSAM, 110 fs 20 in $^{31}\text{P}(\alpha,\text{py})$ from 1971So01 with DSAM, 125 fs 10 in $^{31}\text{P}(\alpha,\text{py})$ from 1974Gr06 with DSAM, and 130 fs 20 in $^{31}\text{P}(\alpha,\text{py})$ from 1977GrZH with DSAM. XREF: Others: AA, AG XREF: R(4874)AA(4880)		
4889.755 23	2 ⁺	36 fs 10	A C	G	MN P R	W Z	J^π: spin=3 from $\text{py}(\theta)$ in $^{31}\text{P}(\alpha,\text{py})$. $\pi=+$ from 2749.24γ, M1+E2 to 2⁺, 2127.561 level. T_{1/2}: lifetime $\tau=57$ fs 22 in $^{31}\text{P}(\alpha,\text{py})$ from 1970Ra17 with DSAM. Others: <85 fs in $^{31}\text{P}(\alpha,\text{py})$ from 1970Gr11 with DSAM and ≤ 70 fs in $^{31}\text{P}(\alpha,\text{py})$ from 1974Gr06 with DSAM. XREF: Others: AA, AC XREF: W(4882)Z(4890)AC(4900)		
5228.170 25	0 ⁺		B	G	MN P R	W Z	J^π: L=0 from 0⁺ in $^{32}\text{S}(\text{t,p})$. Isotropic $\gamma\gamma(\theta)$ in $^{31}\text{P}(\alpha,\text{py})$. T_{1/2}: lifetime $\tau=52$ fs 14 in $^{31}\text{P}(\alpha,\text{py})$ from 1970Ra17 with DSAM. Other: <40 fs in $^{31}\text{P}(\alpha,\text{py})$ from 1970Gr11 with DSAM. XREF: B(5228?)		
5322.512 29	2 ⁺	17 fs 7		G	MN P R	W Z	XREF: Others: AC XREF: N(5320)W(5321)Z(5320)ac(5360) J^π: spin=2 from $\text{py}(\theta)$ in $^{31}\text{P}(\alpha,\text{py})$. $\pi=+$ from 3194.74γ, M1+E2 to 2⁺, 2127.561 level. Discrepancy: L=3 from 3/2⁺ for 5326 6 in $^{33}\text{S}(\text{d,p})$. T_{1/2}: lifetime $\tau=24$ fs 10 in $^{31}\text{P}(\alpha,\text{py})$ from 1970Gr11 with DSAM. Other: ≤ 40 fs in $^{31}\text{P}(\alpha,\text{py})$ from 1974Gr06 with DSAM.		
5380.989 32	1 ⁺	≤ 49 fs	B	G	MN P R	W	XREF: Others: AA, AC XREF: B(5381?)ac(5360) J^π: spin=1 from $\text{py}(\theta)$ in $^{31}\text{P}(\alpha,\text{py})$. $\pi=+$ from 3253.21γ, M1+E2 to 2⁺, 2127.561 level.		
5679.928 19	2 ⁺	263 fs 21		H	LM P R	W	T_{1/2}: lifetime $\tau \leq 70$ fs in $^{31}\text{P}(\alpha,\text{py})$ from 1974Gr06 with DSAM. XREF: R(5683)W(5684) J^π: spin=2 from $\text{py}(\theta)$ in $^{31}\text{P}(\alpha,\text{py})$. $\pi=+$ from 2375.657γ, M1+E2 to 2⁺, 3304.210 level. Discrepancy: L=1 from 3/2⁺ for 5683 7 level in $^{33}\text{S}(\text{d,p})$ and the DWBA fit of $\sigma(\theta)$ seems acceptable (1971Va21, 1972Cr08), which could be a different level; 3⁻ assigned by 2005Ma03 in $^{24}\text{Mg}(^{16}\text{O},\alpha 2\text{py})$.		
5690.62 ^b 27	5 ⁻	36.9 ps 17	E H J	MNO	R	W Z	T_{1/2}: lifetime $\tau=380$ fs 30 in $^{31}\text{P}(\alpha,\text{py})$ from 1972Jo10 with DSAM. XREF: Others: AG XREF: N(5679)O(5690)Z(5690) J^π: spin=5 from $\gamma(\theta)$ and $\gamma(\theta)(\text{POL})$ in $^{31}\text{P}(\alpha,\text{py})$. $\pi=-$ from 1001.5γ, E1 to 4⁺, 4688.96 level.		

Adopted Levels, Gammas (continued)

³⁴S Levels (continued)

E(level) [†]	J ^π &	T _{1/2}	XREF				Comments
							T _{1/2} : lifetime $\tau=53.3$ ps 24: weighted average of 55 ps 7 in ³ H(³⁵ Cl, $\alpha\gamma$) from 1976Co11 with RDM, 52.9 ps 24 in ²⁷ Al(¹² C, $\alpha\gamma$) from 1976Me03 with RDM, and 54 ps 5 in ³¹ P(α,γ) from 1972Gr15 with RDM. Other: >0.5 ps in ³¹ P(α,γ) from 1974Gr06 with DSAM.
5755.873 22	1 ⁻		MN	P	R	W	XREF: Others: AA J ^π : L=1 from 0 ⁺ in ³² S(t,p).
5847.521 30	0 ⁺		MN	P	R	W	XREF: N(5859)W(5848) J ^π : L=0 from 0 ⁺ in ³² S(t,p).
5998.10 8	2 ⁺	<6.9 fs	MN	P	R	W Z	XREF: N(6008) J ^π : 5997.30 γ , E2 to 0 ⁺ , g.s.
6121.56 10	2 ⁺	<52 fs	MN	P	R	W Z	T _{1/2} : lifetime $\tau<10$ fs in ³¹ P(α,γ) from 1972Jo10 with DSAM. XREF: N(6128)W(6118)Z(6130) J ^π : L=2 from 0 ⁺ in ³² S(t,p).
6168.860 29	3 ⁺	<9.0 fs	MN	P	R	W Z	T _{1/2} : lifetime $\tau<75$ fs in ³¹ P(α,γ) from 1972Jo10 with DSAM. XREF: M(6173)W(6170)Z(6130) J ^π : spin=3 from $\gamma(\theta)$ in ³¹ P(α,γ). $\pi=+$ from the strongest branch 2864.56 γ , M1+E2 to 2 ⁺ , 3304.210 level. Discrepancy: $\pi=-$ from L=1+3 from 3/2 ⁺ for 6174 8 level in ³³ S(d,p) and L=(3) from 0 ⁺ for 6179 14 level in ³² S(t,p), which could be a different level.
6251.73 ^c 7	(3,4)	0.42 ps +49–21	H	MN	P	R W	T _{1/2} : lifetime $\tau<13$ fs in ³¹ P(α,γ) from 1972Jo10 with DSAM. XREF: Others: AA , AC , AG XREF: M(6251)N(6256)R(6254)W(6248)AA(6250)AC(6220) J ^π : contradictory firm assignments based on different evidence: spin=4 from $\gamma(\theta)$ and $\pi=+$ is from 1374.43 γ M1+E2 to 3 ⁺ , 4876.842 level in ³¹ P(α,γ) (1972Jo10), whereas 4 ⁻ is in ²⁴ Mg(¹⁶ O, $\alpha 2\gamma$) based on 1539.7 γ E2, $\Delta J=2$ from 7790.8 level with J ^π =(6) ⁻ , which is determined from its deexciting 2099.6 γ M1+E2, $\Delta J=1$ to 5 ⁻ , 5690.62 level. $\pi=-$ is also supported by L=3 from 3/2 ⁺ for 6254 8 level in ³³ S(d,p), whereas $\pi=+$ seems to be supported by L=2 from 3/2 ⁺ for 6220 50 level in ³⁵ Cl(d, ³ He) and by expected natural parity for the 6256 14 level observed in ³² S(t,p). All of this evidence might indicate two different levels with similar energies, however, 2005Ma03 in ²⁴ Mg(¹⁶ O, $\alpha 2\gamma$) analyzed the 1539.7 γ -gated spectrum and confirms that only one level exists at 6251 and all four γ rays (572, 1374, 1562, 1375) are from this level. Also note that in ²⁴ Mg(¹⁶ O, $\alpha 2\gamma$), the 572.0 γ with D and $\Delta J=1$ feeds the 5679.9 level which is assumed to be 3 ⁻ but has been assigned and adopted to be 2 ⁺ based on $\gamma(\theta)$ and measured T _{1/2} with RUL in ³¹ P(α,γ). This would result in a spin=3 assignment for 6251.7 level, further complicating the situation here. In evaluators' view further investigation is warranted to resolve this puzzle. T _{1/2} : lifetime $\tau=390$ fs +70–40 in ³¹ P(α,γ) from 1972Jo10 with DSAM. Other: 600 fs +700–300 in ³¹ P(α,γ) from 1977GrZH with DSAM.

Adopted Levels, Gammas (continued)

³⁴S Levels (continued)

E(level) [†]	J ^{π&}	T _{1/2}	XREF	Comments
6342.52 10	1 ⁻	<25 fs	MN P R W	J ^π : L=1 from 0 ⁺ in ³² S(t,p). T _{1/2} : lifetime τ<36 fs in ³¹ P(α,pγ) from 1972Jo10 with DSAM.
6421.34 8	4 ⁻	<6.9 fs	Mn P R w	XREF: n(6423)w(6414) J ^π : spin=2,4 from pγ(θ) in ³¹ P(α,pγ). Spin=2 is ruled out because either E2 or M2 for 1732.39γ to 4 ⁺ , 4688.96 level would be ruled out by RUL. π=- from L=3 from 3/2 ⁺ in ³³ S(d,p). T _{1/2} : lifetime τ<10 fs in ³¹ P(α,pγ) from 1972Jo10 with DSAM.
6428.12 8	(2 ⁺ ,3 ⁺)		n P w	XREF: n(6423)w(6414) J ^π : 1739.32γ to 4 ⁺ , 4688.96 level; 2353.06γ to 1 ⁺ , 4074.662 level.
6478.768 23 6535 15	1 ⁻		MN P R W N R	J ^π : spin=1 from pγ(θ) in ³¹ P(α,pγ). L=1 from 3/2 ⁺ in ³³ S(d,p). XREF: R(6533?) E(level): from ³² S(t,p).
6639 1	1 ⁻ ,2 ⁽⁻⁾ ,3 ⁽⁻⁾	42 fs 10	MN R W Z	E(level): weighted average of 6639 1 from ³¹ P(α,pγ), 6639 15 from ³² S(t,p), 6640 9 from ³³ S(d,p), and 6635 14 from ³⁴ S(p,p),(p,p'),(pol p,p'). J ^π : spin=2,3,4 from pγ(θ) in ³¹ P(α,pγ); 959γ to 2 ⁺ , 5679.928 level cannot be M2 or E2 and 2015γ to 3 ⁻ , 4624.404 level cannot be M2, based on level T _{1/2} and RUL; L=(3) from 3/2 ⁺ for 6640 9 level in ³³ S(d,p) gives π=(-).
6685.330 30	(0 to 3) ⁻		N P R W	T _{1/2} : lifetime τ=60 fs 15 in ³¹ P(α,pγ) from 1977GrZH with DSAM. XREF: N(6690)R(6690)
6742 15	2 ⁽⁺⁾ ,4 ⁽⁺⁾	<6.9 fs	MN W	J ^π : 929.436γ to 1 ⁻ , 5755.873 level. π=- from L=1 from 3/2 ⁺ in ³³ S(d,p). E(level): weighted average of 6743 15 from ³² S(t,p) and 6738 25 from ³⁴ S(p,p). Other: 6731 from ³¹ P(α,pγ). J ^π : spin=2,4 from pγ(θ) in ³¹ P(α,pγ). π=(+) from 4614γ to 2 ⁺ , 2127.561 level and 2053γ to 4 ⁺ , 4688.96 level.
6828.82	2 ⁺	<46 fs	MN P R W	T _{1/2} : lifetime τ<10 fs in ³¹ P(α,pγ) from 1972Jo10 with DSAM. XREF: Others: AC XREF: N(6828)R(6832)ac(6830) J ^π : spin=2 from pγ(θ) in ³¹ P(α,pγ). L=2 from 0 ⁺ in ³² S(t,p).
6847.90 6	(1,2 ⁺)		P	T _{1/2} : lifetime τ<67 fs in ³¹ P(α,pγ) from 1972Jo10 with DSAM. XREF: Others: AC XREF: ac(6830)
6864 [‡] 1	2 ⁽⁺⁾ ,4 ⁽⁺⁾	28 fs 7	MN W	J ^π : 6846.37γ to 0 ⁺ , g.s. XREF: N(6869)W(6860) E(level): weighted average of 6864 1 from (α,pγ), 6869 15 from (t,p), and 6860 14 from ³⁴ S(p,p),(p,p'),(pol p,p'). J ^π : spin=2,4 from pγ(θ) in ³¹ P(α,pγ) (1972Jo10). π=(+) from 2175γ to 4 ⁺ , 4688.96 level and 4736γ to 2 ⁺ , 2127.561 level. T _{1/2} : lifetime τ=40 fs 10: weighted average of 70 fs 50 in ³¹ P(α,pγ) from 1972Jo10 with DSAM and 39 fs 10 in ³¹ P(α,pγ) from 1977GrZH with DSAM.

Adopted Levels, Gammas (continued)

³⁴S Levels (continued)

E(level) [†]	J ^{π&}	T _{1/2}	XREF		Comments
6890 <i>1</i>	(3,4) ⁺	<14 fs	MN	W	XREF: Others: AC XREF: AC(6910) E(level): weighted average of 6890 <i>1</i> from ³¹ P(α,py), 6898 <i>15</i> from ³² S(t,p), 6888 <i>14</i> from ³⁴ S(p,p), and 6910 <i>40</i> from ³⁵ Cl(d, ³ He). J ^π : spin=(3,4) from py(θ) in ³¹ P(α,py). π=+ from L=2 from 3/2 ⁺ in ³⁵ Cl(d, ³ He). T _{1/2} : lifetime τ<20 fs in ³¹ P(α,py) from 1977GrZH with DSAM. J ^π : spin=2 from py(θ) in ³¹ P(α,py). π=- from L=1 from 3/2 ⁺ in ³³ S(d,p). XREF: n(7112)R(7114)w(7110) J ^π : L=1+3 from 3/2 ⁺ for 7114 <i>10</i> level in ³³ S(d,p). Possible L=3 component of the L=3,(2) doublet from 0 ⁺ at 7112 <i>15</i> in ³² S(t,p).
6954.217 <i>31</i> 7110.450 <i>34</i>	2 ⁻ 3 ⁻		MN P R n P R	W w	XREF: Others: AC XREF: n(7112)w(7110) E(level): mean of 7110 from ³⁵ Cl(d, ³ He) and 7112 from ³¹ P(α,py). Neither reports uncertainties for E(level). Others: 7112 <i>15</i> doublet from ³² S(t,p) and 7110 <i>14</i> from ³⁴ S(p,p),(p,p'),(pol p,p'). J ^π : spin=2 from py(θ) in ³¹ P(α,py). π=+ from L=0 from 3/2 ⁺ in ³⁵ Cl(d, ³ He). Possible L=2 component of the L=3,(2) doublet from 0 ⁺ in ³² S(t,p). T _{1/2} : lifetime τ<10 fs in ³¹ P(α,py) from 1972Jo10 with DSAM. XREF: Others: AC XREF: ac(7190) J ^π : 3089.53γ to 1 ⁺ , 4074.662 level, 5036.4γ to 2 ⁺ , 2127.561 level. XREF: Others: AC XREF: ac(7190) J ^π : spin=2 from py(θ) in ³¹ P(α,py). π=+ from 7218.48γ, E2 to 0 ⁺ , g.s. T _{1/2} : lifetime τ<18 fs in ³¹ P(α,py) from 1972Jo10 with DSAM. Γ _{γ0} =1.09 eV 33 from ³⁴ S(γ,γ'),(pol γ,γ').
7111	2 ⁺	<6.9 fs	Mn	w	XREF: Others: AC XREF: n(7112)w(7110) E(level): mean of 7110 from ³⁵ Cl(d, ³ He) and 7112 from ³¹ P(α,py). Neither reports uncertainties for E(level). Others: 7112 <i>15</i> doublet from ³² S(t,p) and 7110 <i>14</i> from ³⁴ S(p,p),(p,p'),(pol p,p'). J ^π : spin=2 from py(θ) in ³¹ P(α,py). π=+ from L=0 from 3/2 ⁺ in ³⁵ Cl(d, ³ He). Possible L=2 component of the L=3,(2) doublet from 0 ⁺ in ³² S(t,p). T _{1/2} : lifetime τ<10 fs in ³¹ P(α,py) from 1972Jo10 with DSAM. XREF: Others: AC XREF: ac(7190) J ^π : 3089.53γ to 1 ⁺ , 4074.662 level, 5036.4γ to 2 ⁺ , 2127.561 level. XREF: Others: AC XREF: ac(7190) J ^π : spin=2 from py(θ) in ³¹ P(α,py). π=+ from 7218.48γ, E2 to 0 ⁺ , g.s. T _{1/2} : lifetime τ<18 fs in ³¹ P(α,py) from 1972Jo10 with DSAM. Γ _{γ0} =1.09 eV 33 from ³⁴ S(γ,γ'),(pol γ,γ').
7164.46 <i>17</i>	(0 ⁺ to 3 ⁺)		P		XREF: Others: AC XREF: ac(7190) J ^π : 3089.53γ to 1 ⁺ , 4074.662 level, 5036.4γ to 2 ⁺ , 2127.561 level. XREF: Others: AC XREF: ac(7190) J ^π : spin=2 from py(θ) in ³¹ P(α,py). π=+ from 7218.48γ, E2 to 0 ⁺ , g.s. T _{1/2} : lifetime τ<18 fs in ³¹ P(α,py) from 1972Jo10 with DSAM. Γ _{γ0} =1.09 eV 33 from ³⁴ S(γ,γ'),(pol γ,γ').
7219.28 <i>6</i>	2 ⁺	<12 fs	M P S		XREF: Others: AC XREF: ac(7190) J ^π : spin=2 from py(θ) in ³¹ P(α,py). π=+ from 7218.48γ, E2 to 0 ⁺ , g.s. T _{1/2} : lifetime τ<18 fs in ³¹ P(α,py) from 1972Jo10 with DSAM. Γ _{γ0} =1.09 eV 33 from ³⁴ S(γ,γ'),(pol γ,γ').
7248 [‡] <i>2</i>	(4)	14 fs 7	M o	w	XREF: o(7240)w(7248) J ^π : spin=(2,4) from py(θ) for 7248 <i>2</i> level in ³¹ P(α,py). The only deexciting 1557γ to 5 ⁻ , 5690.62 level in ³¹ P(α,py). T _{1/2} : lifetime τ=20 fs 10 in ³¹ P(α,py) from 1977GrZH with DSAM. XREF: Others: AC XREF: N(7245)o(7240)w(7248)ac(7190) J ^π : L=(2,3) from 0 ⁺ in ³² S(t,p). The only deexciting 2558.82γ to 4 ⁺ , 4688.96 level in ³³ S(n,γ) E=thermal. XREF: r(7398)w(7360?) J ^π : 2490.6γ to 3 ⁺ , 4876.842 level; 3451.5γ to 0 ⁺ , 3916.40 level in ³³ S(n,γ) E=thermal. XREF: r(7398)w(7389) E(level),J ^π : from ³² S(t,p); L(t,p)=3 from 0 ⁺ . XREF: r(7398)w(7389)
7248.04 <i>11</i>	(2 ⁺ ,3 ⁻)		NoP	w	XREF: Others: AC XREF: N(7245)o(7240)w(7248)ac(7190) J ^π : L=(2,3) from 0 ⁺ in ³² S(t,p). The only deexciting 2558.82γ to 4 ⁺ , 4688.96 level in ³³ S(n,γ) E=thermal. XREF: r(7398)w(7360?) J ^π : 2490.6γ to 3 ⁺ , 4876.842 level; 3451.5γ to 0 ⁺ , 3916.40 level in ³³ S(n,γ) E=thermal. XREF: r(7398)w(7389) E(level),J ^π : from ³² S(t,p); L(t,p)=3 from 0 ⁺ . XREF: r(7398)w(7389)
7367.42 <i>10</i>	(1 ⁺ ,2 ⁺)		P r	W	XREF: Others: AC XREF: N(7245)o(7240)w(7248)ac(7190) J ^π : L=(2,3) from 0 ⁺ in ³² S(t,p). The only deexciting 2558.82γ to 4 ⁺ , 4688.96 level in ³³ S(n,γ) E=thermal. XREF: r(7398)w(7360?) J ^π : 2490.6γ to 3 ⁺ , 4876.842 level; 3451.5γ to 0 ⁺ , 3916.40 level in ³³ S(n,γ) E=thermal. XREF: r(7398)w(7389) E(level),J ^π : from ³² S(t,p); L(t,p)=3 from 0 ⁺ . XREF: r(7398)w(7389)
7388 <i>15</i>	3 ⁻		N r	w	XREF: Others: AC XREF: N(7245)o(7240)w(7248)ac(7190) J ^π : L=(2,3) from 0 ⁺ in ³² S(t,p). The only deexciting 2558.82γ to 4 ⁺ , 4688.96 level in ³³ S(n,γ) E=thermal. XREF: r(7398)w(7360?) J ^π : 2490.6γ to 3 ⁺ , 4876.842 level; 3451.5γ to 0 ⁺ , 3916.40 level in ³³ S(n,γ) E=thermal. XREF: r(7398)w(7389) E(level),J ^π : from ³² S(t,p); L(t,p)=3 from 0 ⁺ . XREF: r(7398)w(7389)
7392 [‡] <i>1</i>	(4,5)	159 fs 35	M r	w	XREF: Others: AC XREF: N(7245)o(7240)w(7248)ac(7190) J ^π : L=(2,3) from 0 ⁺ in ³² S(t,p). The only deexciting 2558.82γ to 4 ⁺ , 4688.96 level in ³³ S(n,γ) E=thermal. XREF: r(7398)w(7360?) J ^π : 2490.6γ to 3 ⁺ , 4876.842 level; 3451.5γ to 0 ⁺ , 3916.40 level in ³³ S(n,γ) E=thermal. XREF: r(7398)w(7389) E(level),J ^π : from ³² S(t,p); L(t,p)=3 from 0 ⁺ . XREF: r(7398)w(7389)

Adopted Levels, Gammas (continued)

³⁴S Levels (continued)

E(level) [†]	J ^π &	T _{1/2}	XREF			Comments
7467.74 10	(0 ⁺ ,1,2)		N	P	W	J ^π : from py(θ) in ³¹ P(α,py). T _{1/2} : lifetime τ=230 fs 50 in ³¹ P(α,py) from 1977GrZH with DSAM. J ^π : 3392.86γ to 1 ⁺ , 4074.662 level; 1469.67γ to 2 ⁺ , 5998.10 level; 989.1γ to 1 ⁻ , 6478.768 level.
7552.70 8	(0 ⁺ to 3 ⁻)		N	P	R	J ^π : 4248.28γ to 2 ⁺ , 3304.210 level; 1210.04γ to 1 ⁻ , 6342.52 level.
7629.906 22	3 ⁻	14 fs 7	MN	P	R	XREF: N(7621)R(7632)W(7625) J ^π : L=3 from 0 ⁺ in ³² S(t,p). L=1+3 from 3/2 ⁺ in ³³ S(d,p). T _{1/2} : lifetime τ=20 fs 10 in ³¹ P(α,py) from 1977GrZH with DSAM.
7655 11	(1 ⁻ ,2 ⁻ ,3 ⁻ ,4 ⁻ ,5 ⁻)			R	W	E(level): weighted average of 7659 11 from ³³ S(d,p) and 7649 14 from ³⁴ S(p,p). J ^π : L=(3) from 3/2 ⁺ in ³³ S(d,p).
7730.79 15	(1 ⁻ ,2 ⁻ ,3 ⁻)		N	P	R	XREF: N(7714)W(7732) J ^π : L=(1+3) from 3/2 ⁺ in ³³ S(d,p). 5602.78γ to 2 ⁺ , 2127.561 level.
7739 16	2 ⁺		N			E(level): from ³² S(t,p). J ^π : L=2 from 0 ⁺ in ³² S(t,p).
7753 11	0 ⁻ ,1 ⁻ ,2 ⁻ ,3 ⁻			R		E(level): from ³³ S(d,p). J ^π : L=1 from 3/2 ⁺ in ³³ S(d,p).
7781.22 6	(1) ⁻		P	RS	W	XREF: W(7779) J ^π : 7780.22γ to 0 ⁺ , g.s. L=1 from 3/2 ⁺ for 7784 11 in ³³ S(d,p).
7790.8 ^C 4	(6) ⁻	97 fs 24	H	M		Γ _{γ0} =0.57 eV 9 from ³⁴ S(γ,γ'),(pol γ,γ'). XREF: Others: AG J ^π : π=- from 2099.6γ, M1+E2, ΔJ=0,1 to 5 ⁻ , 5690.62 level. 1539.7γ, E2, ΔJ=2 to (3,4), 6251.73 level. 6 ⁻ from shell-model calculation (2005Ma03) in ²⁴ Mg(¹⁶ O,α2py).
7804 5	2 ⁺		N		W	T _{1/2} : lifetime τ=140 fs 35: weighted average of 190 fs 50 in ²⁴ Mg(¹⁶ O,α2py) from 2005Ma03 with DSAM and 115 fs 35 in ³¹ P(α,py) from 1977GrZH with DSAM. XREF: Others: AC XREF: AC(7780) E(level): weighted average of 7801 16 from ³² S(t,p), 7805 5 from ³⁴ S(p,p), and 7780 30 from ³⁵ Cl(d, ³ He). J ^π : L=2 from 0 ⁺ in ³² S(t,p). L=2 from 3/2 ⁺ in ³⁵ Cl(d, ³ He).
7974.72 16	(1,2 ⁺)		N	P	W	J ^π : 7973.45γ to 0 ⁺ g.s.
8025 16	0 ⁺		N		w	XREF: w(8034) E(level): from ³² S(t,p). J ^π : L=0 from 0 ⁺ in ³² S(t,p).
8036.31 14	(1 ⁻ ,2 ⁺)			P	w	XREF: w(8034) J ^π : 8036.6γ to 0 ⁺ , g.s.; 925.79γ to 3 ⁻ , 7110.450 level.
8083 [‡] 1	5	44 fs 7	M			J ^π : from py(θ) in ³¹ P(α,py).
8138.08 7	(1) ⁻			P	R	T _{1/2} : lifetime τ=64 fs 10 in ³¹ P(α,py) from 1977GrZH with DSAM. XREF: R(8142)

Adopted Levels, Gammas (continued)

³⁴S Levels (continued)

E(level) [†]	J ^{π&}	T _{1/2}	XREF				Comments
8174.9 4	(1,2 ⁺)			P			J ^π : L=1 from 3/2 ⁺ in ³³ S(d,p). 8136.98γ to 0 ⁺ , g.s.
8185.45 12	1 ⁺			P	S		J ^π : 8173.8γ to 0 ⁺ , g.s.
8205.40 7	(1 ⁻ to 4 ⁺)			P			J ^π : spin=1 from γ(θ) in ³⁴ S(γ,γ'),(pol γ,γ'). π=+ from 8184.70γ, M1 to 0 ⁺ , g.s.
8255 16	2 ⁺			N			Γ _{γ0} =0.78 eV 20 from ³⁴ S(γ,γ'),(pol γ,γ').
8294.40 9	4 ⁽⁺⁾	<28 fs		MN	P R	W	J ^π : 6077.27γ to 2 ⁺ , 2127.561 level; 3581.2γ to 3 ⁻ , 4624.404 level.
							J ^π : L=2 from 0 ⁺ in ³² S(t,p).
							XREF: M(8293)N(8293)R(8299)W(8296)
							J ^π : spin=4 from pγ(θ) in ³¹ P(α,pγ). Strongest branch 6166.24γ to 2 ⁺ , 2127.561 level.
							T _{1/2} : lifetime τ<40 fs in ³¹ P(α,pγ) from 1977GrZH with DSAM.
8371.0 ^b 4	(7) ⁻	83 fs 14	H	M			XREF: Others: AG
							J ^π : spin=(5,7) from pγ(θ) in ³¹ P(α,pγ). 2680.5γ, E2, ΔJ=2 to 5 ⁻ , 5690.62 level.
							580.3γ, ΔJ=1 to (6) ⁻ , 7790.8 level. 7 ⁻ from 2005Ma03 shell model calculation and band assignments in ²⁴ Mg(¹⁶ O,α2pγ).
							T _{1/2} : lifetime τ=120 fs 20: weighted average of 122 fs 40 in ²⁴ Mg(¹⁶ O,α2pγ) from 2005Ma03 with DSAM and 120 fs 20 in ³¹ P(α,pγ) from 1977GrZH with DSAM.
8385.40 6	1 ⁻			N	P S	W	J ^π : L=1 from 0 ⁺ in ³² S(t,p).
8423 5	4 ⁺			NO		W	Γ _{γ0} =0.49 eV 15 from ³⁴ S(γ,γ'),(pol γ,γ').
							XREF: O(8450)
							E(level): weighted average of 8418 16 from ³² S(t,p) and 8423 5 from ³⁴ S(p,p).
8503.7 ^a 5	(6 ⁺)	28 fs 7	H	M			J ^π : L=4 from 0 ⁺ in ³² S(t,p).
							J ^π : 2812.7γ, D, ΔJ=1 to 5 ⁻ , 5690.62 level; 3813.6γ to 4 ⁺ , 4688.96 level. 6 ⁺ from band assignments in ²⁴ Mg(¹⁶ O,α2pγ).
							T _{1/2} : lifetime τ=40 fs 10 in ³¹ P(α,pγ) from 1977GrZH with DSAM.
8506.77 4	1 ⁻			N	P S	W	XREF: W(8511)
							J ^π : L=1 from 0 ⁺ in ³² S(t,p).
8580 5						W	Γ _{γ0} =0.52 eV 9 from ³⁴ S(γ,γ'),(pol γ,γ').
8615.74 4	(2 ⁻ ,3 ⁺)			P	R	W	E(level): from ³⁴ S(p,p'),(pol p,p').
							XREF: R(8622)
8656 5	1 ⁺				S	W	J ^π : 2363.97γ to (3,4), 6251.73 level; 4540.68γ to 1 ⁺ , 4074.662 level.
							E(level): weighted average of 8657 7 from ³⁴ S(γ,γ') and 8656 5 from ³⁴ S(p,p).
							J ^π : spin=1 from γ(θ) in ³⁴ S(γ,γ'),(pol γ,γ'). π=+ from 8655γ, M1 to 0 ⁺ , g.s.
							Γ _{γ0} =0.41 eV 19 from ³⁴ S(γ,γ'),(pol γ,γ').
8671 5						W	
8702.34 13	(0 ⁺ ,1,2)			P			J ^π : 6573.6γ to 2 ⁺ , 2127.561 level; 516.86γ to 1 ⁺ , 8185.45 level; 2945.8γ to 1 ⁻ , 5755.873 level.
8718 5						W	
8727.63 7	(1 ⁻ ,2 ⁺)			P			J ^π : 1617.00γ to 3 ⁻ , 7110.450 level; 8726.78γ to 0 ⁺ , g.s.
8734.8 6	(6 ⁻)		H			W	J ^π : 3044.1γ, D+Q, ΔJ=0,1 to 5 ⁻ , 5690.62 level. 1178γ from (7 ⁺), 9912.8 level. Spin

Adopted Levels, Gammas (continued)

³⁴S Levels (continued)

E(level) [†]	J ^π &	XREF		Comments
				increases with excitation energy in ²⁴ Mg(¹⁶ O,α2pγ). 6 ⁻ is given by 2005Ma03 in ²⁴ Mg(¹⁶ O,α2pγ).
8792 5			W	
8805.62 24	(1,2 ⁺)	P	W	J ^π : 8804.4γ to 0 ⁺ g.s.
8874.01 6	(1 ⁻ ,2,3 ⁺)	P		J ^π : 4799.11γ to 1 ⁺ , 4074.662 level. 1244.32γ to 3 ⁻ , 7629.906 level.
8953 5			W	
8970.6 5	(6)	H		J ^π : 3280.0γ to 5 ⁻ , 5690.62 level; 1180γ to (6) ⁻ , 7790.8 level; 942.3γ, D, ΔJ=1 from (7 ⁺), 9912.8 level. 6 ⁻ is given by 2005Ma03 in ²⁴ Mg(¹⁶ O,α2pγ).
8987 5			W	
9026.31 6	(1,2 ⁺)	P		J ^π : 9024.95γ to 0 ⁺ g.s.
9120 5			W	
9158.714 29	(1,2 ⁺)	P		J ^π : 3311.6γ to 0 ⁺ , 5847.521 level.
9171 5			W	
9208.03 5	(1,2 ⁺)	P	W	J ^π : 9206.65γ to 0 ⁺ g.s.
9226 6			W	
9257 10			W	
9413.8 5	(6 ⁻)	H		J ^π : 3722.6γ, D+Q, ΔJ=0,1 to 5 ⁻ , 5690.62 level; 1043.8γ to (7) ⁻ , 8371.0 level; 986.8γ from (8 ⁻), 10399.8 level. 6 ⁻ is given by 2005Ma03 in ²⁴ Mg(¹⁶ O,α2pγ).
9429 5			W	
9445 5			W	
9479 4	1 ⁺	ST	W	E(level): weighted average of 9478 4 from ³⁴ S(γ,γ'), 9480 10 from ³⁴ S(e,e'), and 9481 5 from ³⁴ S(p,p). J ^π : spin=1 from γ(θ) in ³⁴ S(γ,γ'),(pol γ,γ'). π=+ from 9478γ, M1 to 0 ⁺ , g.s. Γ _{γ0} =1.1 eV 3 from ³⁴ S(γ,γ'),(pol γ,γ').
9546.09 7	(1,2 ⁺)	P		J ^π : 9544.83γ to 0 ⁺ g.s.
9566 10			W	
9598.42 8		P	W	XREF: W(9601)
9641 4	(1,2 ⁺)	S		E(level): weighted average of 9598.42 8 from ³³ S(n,γ) E=thermal and 9601 5 from ³⁴ S(p,p). J ^π : from ³⁴ S(γ,γ'),(pol γ,γ'). Γ _{γ0} =3.6 eV 7 from ³⁴ S(γ,γ'),(pol γ,γ').
9665.742 34		P		
9707 6	(1,2 ⁺)	S	W	E(level): weighted average of 9711 5 from ³⁴ S(γ,γ') and 9700 6 from ³⁴ S(p,p). J ^π : 9706γ to 0 ⁺ , g.s. Γ _{γ0} =0.50 eV 14 from ³⁴ S(γ,γ'),(pol γ,γ').
9801.88 9	(1,2 ⁺)	P		J ^π : 5884.6γ to 0 ⁺ , 3916.40 level.
9836.70 6		P	W	XREF: W(9847)
9868 5	1 ⁺	ST	W	E(level): weighted average of 9860 7 from ³⁴ S(γ,γ'), 9870 10 from ³⁴ S(e,e'), and 9872 5 from ³⁴ S(p,p). J ^π : spin=1 from γ(θ) in ³⁴ S(γ,γ'),(pol γ,γ'). π=+ from 9867γ, M1 to 0 ⁺ , g.s. Γ _{γ0} =0.60 eV 12 from ³⁴ S(γ,γ'),(pol γ,γ').

Adopted Levels, Gammas (continued)

<u>^{34}S Levels (continued)</u>					
E(level) [†]	J ^π &	T _{1/2}	XREF		Comments
9912.8 4	(7 ⁺)	184 fs 38	H		W
					XREF: w(9925) J ^π : 1408.6γ, D, ΔJ=1 to (6 ⁺), 8503.7 level. 7 ⁺ from shell model calculation (2005Ma03) in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. T _{1/2} : lifetime τ=265 fs 55 in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$ from 2005Ma03 with DSAM. XREF: w(9925) E(level): weighted average of 9932 5 from $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances and 9933.36 12 from $^{33}\text{S}(n,\gamma)$ E=thermal. J ^π : spin=1 from γ(θ) and γγ(θ) in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances, and π=− from only resonances with natural parity can be populated. XREF: W(9969)
9933.36 12	1 [−]		L	P	W
9981 @ 5	1 [−] @		L		W
10000 # 10	1 ⁺ #				T W
10092.21 15			L	P	
10097 5			L		
10140 5			L		
10169 @ 5	1 [−] @		L		
10170 5	1 ⁺			S	
					J ^π : spin=1 from γ(θ) in $^{34}\text{S}(\gamma,\gamma'),(\text{pol } \gamma,\gamma')$. π=+ from 10168γ, M1 to 0 ⁺ , g.s. Γ _{γ0} =1.06 eV 20 from $^{34}\text{S}(\gamma,\gamma'),(\text{pol } \gamma,\gamma')$. XREF: T(10180)
10179.59 # 5	1 ⁺ #			P	T
10201 5			L		
10212.15 4				P	
10236 5			L		
10249 @ 5	1 [−] @		L		
10311.530 @ 27	2 ⁺ @		L	P	
10382 @ 4	1 [−] @		L		
10386 @ 4	3 [−] @		L		
10399.8 ^C 5	(8 [−])		H		
					J ^π : 2608.6γ, Q to (6) [−] , 7790.8 level; 2028.8γ to (7) [−] , 8371.0 level. 8 [−] from band assignments in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$.
10408 @ 5	2 ⁺ @		L		
10430 # 10	1 ⁺ #				T
10447 5			L		
10482 @ 4	0 ⁺ @		L		
10494 @ 5	1 [−] @		L		
10528 5			L		
10587 @ 5	1 [−] @		L		
10617 5			L		
10625 @ 5	1 [−] @		L		
10650.13 19				P	
					Γ _γ =0.84 eV from $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. Γ _γ >1.3 eV from $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. Γ _γ >0.7 eV from $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances.

Adopted Levels, Gammas (continued)

³⁴S Levels (continued)

E(level) [†]	J ^{π&}	T _{1/2}	XREF		Comments
10651.5 ^a 5	(8 ⁺)	35 fs 17	H		J ^π : 2147.2γ, E2, ΔJ=2 to (6 ⁺), 8503.7 level; 2280.4γ, D, ΔJ=1 to (7) ⁻ , 8371.0 level. 8 ⁺ from shell model calculation and band assignments (2005Ma03) in ²⁴ Mg(¹⁶ O,α2pγ). T _{1/2} : lifetime τ=50 fs 25 in ²⁴ Mg(¹⁶ O,α2pγ) from 2005Ma03 with DSAM.
10662 [#] 5	(1 ⁺ ,2 ⁻) [#]		L	T	E(level): weighted average of 10663 5 from ³⁰ Si(α,γ),(α,n):resonances and 10660 10 from ³⁴ S(e,e').
10670 [@] 5	1 ^{-@}		L		Γ _γ =0.73 eV from ³⁰ Si(α,γ),(α,n):resonances.
10705 5			L	O	XREF: O(10700)
10768 [@] 5	2 ⁺ @		L		
10790 5	1 ⁻		L	S	E(level): weighted average of 10791 5 from ³⁰ Si(α,γ),(α,n):resonances and 10786 13 from ³⁴ S(γ,γ'). J ^π : E1 ΔJ=1 γ to 0 ⁺ , g.s. Γ _{γ0} =0.75 eV 14 from ³⁴ S(γ,γ'),(pol γ,γ'). Γ _γ =3 eV from ³⁰ Si(α,γ),(α,n):resonances.
10802 [#] 6	1 ⁺ #			ST	E(level): weighted average of 10803 6 from ³⁴ S(γ,γ') and 10800 10 from ³⁴ S(e,e'). Γ _{γ0} =0.60 eV 11 from ³⁴ S(γ,γ'),(pol γ,γ'). XREF: L(10847)
10840.62 15	3 ⁻		L	P	J ^π : E1+M2 ΔJ=1 γ to 2 ⁺ , 2127.561 level; 6152.1γ to 4 ⁺ , 4688.96 level.
10869 5			L		
10895 5			L		
10917 5			L		
10931 [@] 5	1 ^{-@}		L		
10994 [@] 5	2 ⁺ @		L		
11015 [@] 5	2 ⁺ @		L		
11020 [#] 10	1 ⁺ #			T	
11024.95 [@] 11	1 ^{-@}		L	P	Γ _γ =1.7 eV from ³⁰ Si(α,γ),(α,n):resonances.
11047 5			L		
11088 [@] 5	2 ⁺ @		L		Γ _γ =0.2 eV from ³⁰ Si(α,γ),(α,n):resonances.
11108 [@] 5	3 ^{-@}		L		
11142 [@] 5	1 ^{-@}		L		Γ _γ =2.6 eV from ³⁰ Si(α,γ),(α,n):resonances.
11165 [@] 5	1 ^{-@}		L		Γ _γ =1.7 eV from ³⁰ Si(α,γ),(α,n):resonances.
11180 5			L		
11194 5			L		
11220 [@] 5	(2 ⁺) [@]		L		Γ _γ =0.2 eV from ³⁰ Si(α,γ),(α,n):resonances.
11233 [@] 5	1 ^{-@}		L		Γ _γ =2.8 eV from ³⁰ Si(α,γ),(α,n):resonances.
11272 [@] 5	2 ⁺ @		L		
11289 5			L		

Adopted Levels, Gammas (continued)

³⁴S Levels (continued)

E(level) [†]	J ^π &	T _{1/2}	XREF		Comments
11315 @ 5	2 ⁺ @		L		Γ _γ =0.08 eV from ³⁰ Si(α,γ),(α,n):resonances.
11323 @ 5	1 ⁻ @		L		Γ _γ =2.2 eV from ³⁰ Si(α,γ),(α,n):resonances.
11350 # 10	1 ⁺ #			T	
11358 @ 5	1 ⁻ @		L		Γ _γ =1.4 eV from ³⁰ Si(α,γ),(α,n):resonances.
11372 @ 5	3 ⁻ @		L		Γ _γ =1.5 eV from ³⁰ Si(α,γ),(α,n):resonances.
11374.2 6	(8 ⁺)		H		J ^π : 3002.8γ, D, ΔJ=1 to (7) ⁻ , 8371.0 level; 1461.7γ, D(+Q), ΔJ=1 to (7) ⁺ , 9912.8 level. 1966.8γ from (10 ⁺), 13341.6 level. 8 ⁺ is given by 2005Ma03 in ²⁴ Mg(¹⁶ O,α2pγ).
11381 @ 5	2 ⁺ @		L		Γ _γ =0.1 eV from ³⁰ Si(α,γ),(α,n):resonances.
11398 5			L		
11411.33	2 ⁺		L	Q	XREF: L(11405)
(11417.222 18)	2 ⁺			P	J ^π : s-wave neutron capture on 3/2 ⁺ g.s. of ³³ S gives 1 ⁺ , 2 ⁺ . 6727.5γ to 4 ⁺ , 4688.96 level and 3787.096γ to 3 ⁻ , 7629.906 level favor the 2 ⁺ assignment.
11420 @ 5	1 ⁻ @		L		Γ _γ =4.4 eV from ³⁰ Si(α,γ),(α,n):resonances.
11430.3	2 ⁺	0.116 keV 20		Q	Γ _n =75.0 eV 8, Γ _γ =0.21 eV 5, Γ _α =41 eV 5 from ³³ S(n,γ),(n,n):resonances.
11434.3	2 ⁻	0.049 keV 10	L	Q	XREF: L(11435)
11440.4	3 ⁻	0.0198 keV 10	1	Q	Γ _n =39.1 eV 8, Γ _γ =0.90 eV 5 from ³³ S(n,γ),(n,n):resonances. XREF: l(11444)
11448.1		<0.015 keV	1	Q	Γ _n =16.0 eV 9, Γ _γ =1.44 eV 10, Γ _α =2.5 eV 3 from ³³ S(n,γ),(n,n):resonances. XREF: l(11444)
11458 @ 5	3 ⁻ @		L		
11467.8	2 ⁺	0.368 keV 8		Q	
11469.2	3 ⁻	0.152 keV 15		Q	
11473 @ 5	1 ⁻ @		L		
11474.6	2 ⁻	0.45 keV 6		Q	
11490 @ 5	1 ⁻ @		L	Q	Γ _n =275 eV 5, Γ _γ =1.08 eV 7, Γ _α =0.17 keV 5 from ³³ S(n,γ),(n,n):resonances. E(level): other: 11486.0 from ³³ S(n,γ),(n,n):resonances. J ^π : 1 ⁻ in both ³⁰ Si(α,γ),(α,n):resonances and ³³ S(n,γ),(n,n):resonances. Γ _n =65 eV 10, Γ _α =0.11 keV 6 from ³³ S(n,γ),(n,n):resonances. Γ _γ =0.6 eV from ³⁰ Si(α,γ),(α,n):resonances.
11492.7	2 ⁻	0.51 keV 10		Q	Γ _n =507 eV 13, Γ _γ =2.11 eV 14 from ³³ S(n,γ),(n,n):resonances.
11496.2	2 ⁺	0.71 keV 3		Q	Γ _n =705 eV 19, Γ _γ =0.94 eV 6, Γ _α =4 eV 2 from ³³ S(n,γ),(n,n):resonances.
11499.6	1 ⁻			Q	Γ _n =1.33 keV 8, Γ _α =4.0 keV 6 from ³³ S(n,γ),(n,n):resonances.
11500 # 10	1 ⁺ #			T	
11502.3	1 ⁻	0.292 keV 25		Q	Γ _n =280 eV 20, Γ _γ =2.11 eV 14, Γ _α =10 eV 5 from ³³ S(n,γ),(n,n):resonances.
11506 @ 5	(1 ⁻) @	0.26 keV 5	L	Q	E(level): other: 11502.9 from ³³ S(n,γ),(n,n):resonances.
11515.3	2 ⁻	1.262 keV 25		Q	Γ _n =1.260 keV 25, Γ _γ =1.48 eV 13 from ³³ S(n,γ),(n,n):resonances.
11544		0.20 keV 4		Q	

Adopted Levels, Gammas (continued)

^{34}S Levels (continued)

E(level) [†]	J ^π &	T _{1/2}	S	XREF		Comments
11545 @ 5	1 ⁻ @	0.63 keV 7		L	Q	XREF: Q(11541) E(level): other: 11541 from $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances. J ^π : 1 ⁻ in both $^{30}\text{Si}(\alpha,\gamma),(\alpha,\text{n})$:resonances and $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances. Γ _n =0.36 keV 4, Γ _γ =1.4 eV 4, Γ _α =0.27 keV 6 from $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances. Γ _γ =1.0 eV from $^{30}\text{Si}(\alpha,\gamma),(\alpha,\text{n})$:resonances.
11546		0.23 keV 4			Q	
11551		0.15 keV 3			Q	
11564	≥1	0.134 keV 18			Q	
11575	(0 ⁻)	0.152 keV 32			Q	
11581	2 ⁻	3.42 keV 8			Q	Γ _n =3.42 keV 8, Γ _γ =2.6 eV 3 from $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances.
11590	2 ⁻	0.76 keV 4			Q	Γ _n =0.76 keV 4, Γ _γ =0.87 eV 11 from $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances.
11608	3 ⁻	0.62 keV 3			Q	Γ _n =0.61 keV 3, Γ _γ =1.33 eV 12 from $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances.
11610		0.70 keV 14			Q	
11614	3 ⁻	2.1 keV 8			Q	Γ _n =2.09 keV 8, Γ _γ =2.17 eV 20, Γ _α =14 eV 5 from $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances.
11622		0.31 keV 6		L	Q	XREF: L(11619)
11626		<0.12 keV			Q	
11632	2 ⁺	0.75 keV 7			Q	Γ _n =0.69 keV 7, Γ _γ =1.2 eV 4, Γ _α =55 eV 20 from $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances.
11634	0 ⁺	5.3 keV 10			Q	Γ _n =4.4 keV 9, Γ _α =0.9 keV 3 from $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances.
11637 5	3 ⁻	0.96 keV 6		L	Q	XREF: Q(11639) Γ _n =0.76 keV 5, Γ _γ =0.81 eV 13, Γ _α =0.20 keV 3 from $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances.
11643 @ 5	1 ⁻ @			L		Γ _γ =2.3 eV from $^{30}\text{Si}(\alpha,\gamma),(\alpha,\text{n})$:resonances.
11649	3 ⁻	0.61 keV 12			Q	Γ _n =0.46 keV 3, Γ _γ =1.82 eV 20 from $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances.
11669	2 ⁻	0.40 keV 8		1	Q	XREF: 1(11670) Γ _n =0.67 keV 6, Γ _γ =2.4 eV 2 from $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances.
11670	1 ⁺	0.55 keV 11		1	Q	XREF: 1(11670) Γ _n =0.23 keV 7, Γ _γ =2.1 eV 3 from $^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances.
11704		0.61 keV 12			Q	
11706		0.79 keV 16	298.2		Q	
11712 @ 5	1 ⁻ @			L		
11717		0.67 keV 14			Q	
11744		0.28 keV 6			Q	
11751 5				L		
11773		0.40 keV 8			Q	
11785		1.40 keV 25			Q	
11790 5				L		
11796		1.30 keV 25			Q	
11807.4 5	(8)			H		J ^π : 3436.1γ, D, ΔJ=1 to (7) ⁻ , 8371.0 level. Spin increases with excitation energy assumed in $^{24}\text{Mg}(^{16}\text{O},\alpha 2\text{p}\gamma)$. 8 ⁺ is given by 2005Ma03 in $^{24}\text{Mg}(^{16}\text{O},\alpha 2\text{p}\gamma)$.
11829		1.7 keV 3			Q	

Adopted Levels, Gammas (continued)

³⁴S Levels (continued)

E(level) [†]	J ^π &	T _{1/2}	XREF		Comments
11850 5			L		
11859 5			L		
11867 5		3.3 keV 5	L	Q	XREF: Q(11870)
11878 5			L		
11908 5			L		
11922 @ 5	(3 ⁻)@		L		
11932 @ 5	1 ⁻ @		L		
11949		2.3 keV 4		Q	
11957 @ 5	3 ⁻ @		L		
11979 5			L		
12034 @ 5	1 ⁻ @		F	L	XREF: F(12030)
12063 5			L		
12077 5			L		
12100 @ 5	1 ⁻ @		F	L	XREF: F(12100)
12120 # 10	1 ⁺ #				T
12137 5			L		
12141.3 5	(9 ⁺)	173 fs 35	H		J ^π : 2228.8γ, E2, ΔJ=2 to (7 ⁺), 9912.8 level; 1489.2γ to (8 ⁺), 10651.5 level; 1741.6γ to (8 ⁻), 10399.8 level. 9 ⁺ from shell-model calculation (2005Ma03) in ²⁴ Mg(¹⁶ O,α2pγ).
					T _{1/2} : lifetime τ=250 fs 50 in ²⁴ Mg(¹⁶ O,α2pγ) from 2005Ma03 with DSAM.
					XREF: F(12150)
12151 5			F	L	
12165 5				L	
12173 5				L	
12180 10	2 ⁻				T
12194 @ 5	1 ⁻ @			L	
12210		<50 keV	F		
12224 5				L	
12243 5				L	
12256 5				L	
12271 5				L	
12281 5				L	
12350		<20 keV	F		
12460 # 10	(1 ⁺ ,2 ⁻)#	<30 keV	F		T
12540		<30 keV	F		
12650		<50 keV	F		
12660 # 10	1 ⁺ #				T
12760		<50 keV	F		
12930 # 10	(1 ⁺ ,2 ⁻)#				T
12970		<50 keV	F		

Adopted Levels, Gammas (continued)

³⁴S Levels (continued)

E(level) [†]	J ^π &	T _{1/2}	XREF	Comments
12985.4 6	(9 ⁺)		H	J ^π : 2333.8γ to (8 ⁺), 10651.5 level; 1611.5γ to (8 ⁺), 11374.2 level; 356.3γ from (10 ⁺), 13341.6 level. Spin increases with excitation energy assumed in ²⁴ Mg(¹⁶ O,α2pγ). (9 ⁺) is given by 2005Ma03 in ²⁴ Mg(¹⁶ O,α2pγ).
13320.2 ^b 10	(9 ⁻)		H	J ^π : 4949.3γ to (7) ⁻ , 8371.0 level; 2920.1γ to (8 ⁻), 10399.8 level; 9 ⁻ from band assignments in ²⁴ Mg(¹⁶ O,α2pγ).
13340		<50 keV	F	
13341.6 6	(10 ⁺)	180 fs 28	H	J ^π : 1966.8γ, E2, ΔJ=2 to (8 ⁺), 11374.2 level. 10 ⁺ from shell model calculation (2005Ma03) in ²⁴ Mg(¹⁶ O,α2pγ). T _{1/2} : lifetime τ=260 fs 40 in ²⁴ Mg(¹⁶ O,α2pγ) from 2005Ma03 with DSAM.
13460		<40 keV	F	
13590 [#] 10	2 ⁻ [#]			T
13610		<50 keV	F	
13710		<50 keV	F	
13790 [#] 10	2 ⁻ [#]	<40 keV	F	T
13920		<50 keV	F	
13960.5 ^a 10	(10 ⁺)		H	J ^π : 3308.8γ to (8 ⁺), 10651.5 level; 10 ⁺ from band assignments in ²⁴ Mg(¹⁶ O,α2pγ).
13990 [#] 10	1 ⁺ [#]			T
14070		<50 keV	F	
14160		<50 keV	F	
14200 [#] 10	(1 ⁺ ,2 ⁻) [#]			T
14270		<100 keV	F	
14320 [#] 10	(1 ⁺ ,2 ⁻) [#]			T
14370		<50 keV	F	
14430 [#] 10	(1 ⁺ ,2 ⁻) [#]	<50 keV	F	T
14520		<50 keV	F	
14576.4 11	(8,9,10 ⁺)		H	J ^π : 2768.9γ to (8), 11807.4 level. Spin increases with excitation energy assumed in ²⁴ Mg(¹⁶ O,α2pγ). (10 ⁺) is given by 2005Ma03 in ²⁴ Mg(¹⁶ O,α2pγ).
14600		<50 keV	F	
14730		<100 keV	F	
14800 [#] 10	2 ⁻ [#]			T
14840		<50 keV	F	
14930		<100 keV	F	
15020		<100 keV	F	
15120		<50 keV	F	
15170		<50 keV	F	
15220		<50 keV	F	
15244.3 9	(10,11,12 ⁺)		H	J ^π : 1902.7γ to (10 ⁺), 13341.6 level. Spin increases with excitation energy assumed in ²⁴ Mg(¹⁶ O,α2pγ).
15281.0 ^c 17	(10 ⁻)		H	J ^π : 4880.8γ to (8 ⁻), 10399.8 level; 10 ⁻ from band assignments in ²⁴ Mg(¹⁶ O,α2pγ).

Adopted Levels, Gammas (continued)

<u>^{34}S Levels (continued)</u>				
E(level) [†]	J ^π &	T _{1/2}	XREF	Comments
15300		<100 keV	F	
15540		<100 keV	F	
15640		<50 keV	F	
15700		<50 keV	F	
15770		<50 keV	F	
15850		<50 keV	F	
15950		<50 keV	F	
16060		<50 keV	F	
16190		<100 keV	F	
16340		<50 keV	F	
16440		<50 keV	F	
16550		<50 keV	F	
16600		<50 keV	F	
16649.0 ^a 13	(12 ⁺)		H	J ^π : 2688.4γ to (10 ⁺), 13960.5 level; 12 ⁺ from band assignments in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$.
16650		<70 keV	F	
16730		<70 keV	F	
16820		<50 keV	F	
16880		<50 keV	F	
16940		<50 keV	F	
17010		<50 keV	F	
17110		<50 keV	F	
17180		<50 keV	F	
17250		<50 keV	F	
17330		<50 keV	F	
17460		<50 keV	F	
17520		<50 keV	F	
17590		<50 keV	F	
17620		<50 keV	F	
17660		<50 keV	F	
17690		<50 keV	F	
17740		<50 keV	F	
17790		<50 keV	F	
17840		<50 keV	F	
17860		<50 keV	F	
17900		<50 keV	F	
18030		<50 keV	F	
18110		<40 keV	F	
18210		<40 keV	F	
18300		<50 keV	F	
18350		<50 keV	F	
18430		<50 keV	F	
18540		<50 keV	F	
18600		<50 keV	F	

Adopted Levels, Gammas (continued)

³⁴S Levels (continued)

E(level) [†]	T _{1/2}	XREF	E(level) [†]	T _{1/2}	XREF
18670	<50 keV	F	21000	<80 keV	F
18730	<50 keV	F	21140	<50 keV	F
18860	<50 keV	F	21250	<50 keV	F
19080	<150 keV	F	21340	<40 keV	F
19200	<50 keV	F	21460	<50 keV	F
19280	<70 keV	F	21570	<50 keV	F
19370	<80 keV	F	21700	<50 keV	F
19550	<50 keV	F	21840	<80 keV	F
19650	<50 keV	F	22040	<50 keV	F
19720	<50 keV	F	22230	<80 keV	F
19820	<50 keV	F	22340	<50 keV	F
19900	<50 keV	F	22440	<40 keV	F
20080	<50 keV	F	22600	<100 keV	F
20170	<50 keV	F	22720	<50 keV	F
20300	<50 keV	F	22890	<40 keV	F
20390	<150 keV	F	22990	<40 keV	F
20520	<70 keV	F	23170	<50 keV	F
20680	<50 keV	F	23320	<30 keV	F
20770	<50 keV	F	23500	<100 keV	F
20840	<50 keV	F			

[†] From a least-squares fit to γ -ray energies, unless otherwise noted.

[‡] From ³¹P(α , γ) directly reported E(level) without E γ .

From ³⁴S(e,e'). J^π is assigned based on the multipolarities in ³⁴S(e,e'). The 180° inelastic electron scattering data establishes magnetic character and the multipolarity order is deduced from least-squares fits of the reduced transition probability versus momentum transfer (1994Pe08).

@ From ³⁰Si(α , γ),(α ,n):resonances. Spin is from $\gamma(\theta)$ and $\gamma\gamma(\theta)$ in ³⁰Si(α , γ),(α ,n):resonances, and $\pi=+/-$ from the ³⁴S resonances populated in ³⁰Si+ α can only have natural parity.

& From measured $p\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ in ³¹P(α , γ), unless otherwise noted.

^a Band(A): Ground-state band.

^b Band(B): Band based on 3⁻, 4624.404 level.

^c Band(C): Band based on (3,4), 6251.73 level.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. &	$\delta^\&$	α^b	Comments
2127.561	2 ⁺	2127.498 20	100	0.0	0 ⁺	E2		0.000387 5	B(E2)(W.u.)=6.30 +26-24 E _γ : weighted average of 2127.5 6 from ²⁴ Mg(¹⁶ O,α2pγ), 2126.7 8 from ³¹ P(α,pγ), 2127.499 20 from ³³ S(n,γ) E=thermal, 2127.45 20 from ³⁴ S(p,p'γ), and 2128 1 from ²⁰⁸ Pb(³⁶ S, ³⁴ Sγ). Mult.: Q from pγ(θ) and electric character from γ(θ)(POL) in ³¹ P(α,pγ). Q, ΔJ=2 from γγ(θ)(ADO) in ²⁴ Mg(¹⁶ O,α2pγ).
3304.210	2 ⁺	1176.650 26	100.0 9	2127.561	2 ⁺	M1+E2	-0.16 2	2.93×10 ⁻⁵ 4	B(M1)(W.u.)=0.0577 +47-40; B(E2)(W.u.)=4.2 +12-10 E _γ : weighted average of 1177.3 5 from ²⁴ Mg(¹⁶ O,α2pγ), 1175.2 8 from ³¹ P(α,pγ), 1176.650 20 from ³³ S(n,γ) E=thermal, and 1177 1 from ²⁰⁸ Pb(³⁶ S, ³⁴ Sγ). I _γ : from ³⁴ Cl ε+β ⁺ decay (31.99 min). Others: 100 10 from ²⁴ Mg(¹⁶ O,α2pγ), 100 10 from ³¹ P(α,pγ), 100 9 from ³³ S(n,γ) E=thermal, and 100 10 from ²⁰⁸ Pb(³⁶ S, ³⁴ Sγ). Mult.,δ: D+Q, ΔJ=0,1 from pγ(θ) in ³¹ P(α,pγ). M2 ruled out by RUL.
		3304.029 29	87.1 9	0.0	0 ⁺	E2		0.000916 13	B(E2)(W.u.)=0.84 +7-6 E _γ : weighted average of 3304.6 7 from ²⁴ Mg(¹⁶ O,α2pγ), 3303.1 4 from ³¹ P(α,pγ), 3304.031 20 from ³³ S(n,γ) E=thermal, and 3305 2 from ²⁰⁸ Pb(³⁶ S, ³⁴ Sγ). I _γ : weighted average of 87.2 9 from ³⁴ Cl ε+β ⁺ decay (31.99 min), 79 8 from ²⁴ Mg(¹⁶ O,α2pγ), 92 10 from ³¹ P(α,pγ), 84 8 from ³³ S(n,γ) E=thermal, and 81 13 from ²⁰⁸ Pb(³⁶ S, ³⁴ Sγ). Mult.: Q from pγ(θ) and electric character from γ(θ)(POL) in ³¹ P(α,pγ). Q, ΔJ=2 from A ₂ and A ₄ in ²⁴ Mg(¹⁶ O,α2pγ).
3916.40	0 ⁺	612.16 5 1787.3 8	0.33 4 100 10	3304.210 2127.561	2 ⁺ 2 ⁺	[E2] [E2]		0.0001878 26 0.0002259 32	B(E2)(W.u.)=3.0 +6-5 B(E2)(W.u.)=4.22 +37-32 E _γ : unweighted average of 1787 1 from ³⁴ P β ⁻ decay, 1786.0 8 from ³¹ P(α,pγ), and 1788.794 20 from ³³ S(n,γ) E=thermal.
4074.662	1 ⁺	770.428 20	8.5 9	3304.210	2 ⁺	(M1(+E2))	-1.3 15	8.1×10 ⁻⁵ 23	I _γ : weighted average of 6.2 22 from ³¹ P(α,pγ) and 8.8 8 from ³³ S(n,γ) E=thermal.

Adopted Levels, Gammas (continued)

$E_i(\text{level})$	J_i^π	$\gamma(^{34}\text{S})$ (continued)							Comments
		$E_\gamma^{\dagger}$	$I_\gamma^{\dagger}$	E_f	J_f^π	Mult. ^{&}	$\delta^{\&}$	α^b	
4074.662	1 ⁺	1947.060 20	95 8	2127.561	2 ⁺	M1+E2	+1.3 +32-9	0.000276 32	Mult.,δ: D(+Q) from $\text{p}\gamma(\theta)$ in $^{31}\text{P}(\alpha,\text{p}\gamma)$. $\Delta\pi=\text{no}$ from level scheme. E_γ: weighted average of 1947.1 15 from $^{34}\text{P}\beta^-$ decay, 1945.5 13 from $^{31}\text{P}(\alpha,\text{p}\gamma)$, and 1947.060 20 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. I_γ: weighted average of 61 22 from $^{34}\text{P}\beta^-$ decay, 102 8 from $^{31}\text{P}(\alpha,\text{p}\gamma)$, and 93 8 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. Mult.,δ: D+Q, $\Delta J=0,1$ from $\text{p}\gamma(\theta)$ in $^{31}\text{P}(\alpha,\text{p}\gamma)$. M2 ruled out by RUL.
		4074.42 4	100 9	0.0	0 ⁺	(M1)		1.06×10^{-3} 2	E_γ: weighted average of 4073.4 15 from $^{34}\text{P}\beta^-$ decay, 4070.9 13 from $^{31}\text{P}(\alpha,\text{p}\gamma)$, and 4074.418 20 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. I_γ: other: 100 13 from $^{34}\text{P}\beta^-$ decay, 100 10 from $^{31}\text{P}(\alpha,\text{p}\gamma)$.
4114.812	2 ⁺	1987.19 3	69.8 34	2127.561	2 ⁺	M1+E2	-0.40 5	0.000260 4	Mult.: D from $\text{p}\gamma(\theta)$ in $^{31}\text{P}(\alpha,\text{p}\gamma)$. $\Delta\pi=\text{no}$ from level scheme. B(M1)(W.u.)=0.0138 +16-14; B(E2)(W.u.)=2.2 +6-5 E_γ: weighted average of 1987.2 10 from $^{34}\text{P}\beta^-$ decay and 1987.19 3 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. I_γ: weighted average of 83 17 from $^{34}\text{P}\beta^-$ decay, 67.7 24 from $^{34}\text{Cl}\varepsilon+\beta^+$ decay (31.99 min), 92 10 from $^{31}\text{P}(\alpha,\text{p}\gamma)$, and 76 8 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal.
		4114.52 4	100.0 22	0.0	0 ⁺	E2		1.21×10^{-3} 2	Mult.,δ: D+Q, $\Delta J=0,1$ from $\text{p}\gamma(\theta)$ in $^{31}\text{P}(\alpha,\text{p}\gamma)$. M2 ruled out by RUL. B(E2)(W.u.)=0.60 +7-6 E_γ: weighted average of 4114.0 15 from $^{34}\text{P}\beta^-$ decay, 4115.0 20 from $^{34}\text{Cl}\varepsilon+\beta^+$ decay (31.99 min), 4113.7 9 from $^{31}\text{P}(\alpha,\text{p}\gamma)$, and 4114.52 4 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. I_γ: from $^{34}\text{Cl}\varepsilon+\beta^+$ decay (31.99 min). Others: 100 17 from $^{34}\text{P}\beta^-$ decay, 100 10 from $^{31}\text{P}(\alpha,\text{p}\gamma)$, and 100 11 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal.
4624.404	3 ⁻	1320.168 20	100 [#] 7	3304.210	2 ⁺	(E1(+M2))	-0.03 5	0.0001510 22	Mult.: Q from $\text{p}\gamma(\theta)$ in $^{31}\text{P}(\alpha,\text{p}\gamma)$. M2 ruled out by RUL. B(E1)(W.u.)=0.00233 +34-30 E_γ: weighted average of 1320.1 5 from

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)									Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.&	$\delta^\&$	α^b	
4624.404	3^-	2496.725 25	41 4	2127.561	2^+	(E1(+M2))	+0.02 4	0.000968 14	$^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$, 1318.8 10 from $^{31}\text{P}(\alpha,p\gamma)$, 1320.169 20 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal, and 1320 1 from $^{208}\text{Pb}(^{36}\text{S},^{34}\text{S}\gamma)$. I_γ : others: 100 11 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$ and 100 11 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. Mult., δ : D(+Q) from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha,p\gamma)$. D, $\Delta J=1$ from $\gamma\gamma(\theta)(\text{ADO})$ in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. $\Delta\pi$ =yes from level scheme. B(E1)(W.u.)= 1.41×10^{-4} +32-27 E_γ : weighted average of 2496.5 8 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$, 2495.5 7 from $^{31}\text{P}(\alpha,p\gamma)$, and 2496.726 20 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. I_γ : weighted average of 46 5 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$, 32 7 from $^{31}\text{P}(\alpha,p\gamma)$, and 41 4 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. Mult., δ : D(+Q) from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha,p\gamma)$. D, $\Delta J=1$ from $\gamma\gamma(\theta)(\text{ADO})$ in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. $\Delta\pi$ =yes from level scheme.
4688.96	4^+	4624.2 5 1384.5 ‡ 14	0.55 13 0.44 ‡ 30	0.0 3304.210	0^+ 2^+	[E3] [E2]		1.08×10^{-3} 2 7.32×10^{-5} 11	B(E3)(W.u.)=17.9 +47-44 B(E2)(W.u.)=0.85 +62-45 I_γ : other: <2 from $^{31}\text{P}(\alpha,p\gamma)$. B(E2)(W.u.)=8.9 +8-7 E_γ : weighted average of 2561.1 6 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$, 2560.5 4 from $^{31}\text{P}(\alpha,p\gamma)$, 2561.36 5 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal, and 2561 2 from $^{208}\text{Pb}(^{36}\text{S},^{34}\text{S}\gamma)$. I_γ : from ^{34}Cl $\varepsilon+\beta^+$ decay (31.99 min). Other: 100 10 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. Mult.: Q, $\Delta J=2$ from $\gamma\gamma(\theta)(\text{ADO})$ in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$ and $p\gamma(\theta)$ in $^{31}\text{P}(\alpha,p\gamma)$. M2 ruled out by RUL.
4876.842	3^+	1572.56 5	70 8	3304.210	2^+	M1+E2	-0.09 4	0.0001011 14	B(M1)(W.u.)=0.058 +35-17; B(E2)(W.u.)=0.7 +11-5 E_γ : weighted average of 1572.5 5 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$, 1571.9 5 from $^{31}\text{P}(\alpha,p\gamma)$, 1572.57 5 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal, and 1573 1 from $^{208}\text{Pb}(^{36}\text{S},^{34}\text{S}\gamma)$. I_γ : weighted average of 71 23 from ^{34}Cl $\varepsilon+\beta^+$

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. ^{&}	$\delta^{\&}$	α^b	Comments
4876.842	3 ⁺	2749.24 5	100 [#] 8	2127.561	2 ⁺	M1+E2	-0.11 3	0.000568 8	decay (31.99 min), 89 20 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2\gamma)$, 59 8 from $^{31}\text{P}(\alpha,\gamma)$, and 80 9 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. Mult., δ : D+Q, $\Delta J=0,1$ from $\gamma(\theta)$ in $^{31}\text{P}(\alpha,\gamma)$. M2 ruled out by RUL. B(M1)(W.u.)=0.015 +9-4; B(E2)(W.u.)=0.10 +10-5 E_γ : weighted average of 2749.6 6 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2\gamma)$, 2750.9 25 from $^{31}\text{P}(\alpha,\gamma)$, and 2749.24 5 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. I_γ : others: 100 14 from $^{34}\text{Cl } \varepsilon+\beta^+$ decay (31.99 min), 100 22 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2\gamma)$, and 100 10 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. Mult., δ : D+Q, $\Delta J=0,1$ from $\gamma(\theta)$ in $^{31}\text{P}(\alpha,\gamma)$. M2 ruled out by RUL.
4889.755	2 ⁺	1585.510 20	96 [#] 24	3304.210	2 ⁺	[M1,E2]		0.000122 18	I_γ : others: 94 9 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal, <100 from $^{34}\text{Cl } \varepsilon+\beta^+$ decay (31.99 min). B(M1)(W.u.)=0.064 +28-18 if M1, B(E2)(W.u.)=99 +44-28 if E2. B(E2)(W.u.)=99 +44-28 upper bound exceeds RUL=100 if E2.
		2762.10 8	<71 [#]	2127.561	2 ⁺	[M1,E2]		0.00063 6	I_γ : others: 112 11 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal, <61 from $^{34}\text{Cl } \varepsilon+\beta^+$ decay (31.99 min).
		4889.30 8	100 [#] 24	0.0	0 ⁺	E2		1.46 $\times 10^{-3}$ 2	B(E2)(W.u.)=0.37 +17-10 I_γ : others: 100 10 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal, 100 from $^{34}\text{Cl } \varepsilon+\beta^+$ decay (31.99 min). Mult.: Q from $\gamma(\theta)$ in $^{31}\text{P}(\alpha,\gamma)$. M2 ruled out by RUL.
5228.170	0 ⁺	1113.27 9	4.1 6	4114.812	2 ⁺				
		1153.492 20	100 9	4074.662	1 ⁺	D			Mult.: D from $\gamma(\theta)$ in $^{31}\text{P}(\alpha,\gamma)$.
5322.512	2 ⁺	698.18 13	1.37 19	4624.404	3 ⁻	[E1]		4.47 $\times 10^{-5}$ 6	B(E1)(W.u.)=0.0014 +10-5
		1247.92 6	8.0 10	4074.662	1 ⁺	[M1,E2]		4.0 $\times 10^{-5}$ 6	B(M1)(W.u.)=0.049 +35-16 if M1. B(E2)(W.u.)=1.2 $\times 10^2$ +9-4 exceeds RUL=100 if E2.
		3194.74 5	100 11	2127.561	2 ⁺	M1+E2	-0.17 6	0.000747 11	B(M1)(W.u.)=0.035 +23-11; B(E2)(W.u.)=0.39 +48-23 Mult., δ : D+Q with $\delta=-0.17$ 6 or $\delta=+1.8$ 10 from $\gamma(\theta)$ in $^{31}\text{P}(\alpha,\gamma)$. M2 ruled out by RUL.
5380.989	1 ⁺	1266.11 5	17.4 19	4114.812	2 ⁺	[M1,E2]		4.3 $\times 10^{-5}$ 6	
		2076.89 8	39 4	3304.210	2 ⁺	[M1,E2]		0.00032 4	
		3253.21 6	100 11	2127.561	2 ⁺	M1+E2	-1.1 10	0.00084 7	I_γ : other: 100 13 from $^{31}\text{P}(\alpha,\gamma)$. Mult., δ : D+Q with $-2.1<\delta\leq-0.1$ from $^{31}\text{P}(\alpha,\gamma)$. M2 ruled out by RUL.

Adopted Levels, Gammas (continued)

$E_i(\text{level})$	J_i^π	$\gamma(^{34}\text{S})$ (continued)							Comments
		$E_\gamma^{\dagger}$	$I_\gamma^{\dagger}$	E_f	J_f^π	Mult. ^{&}	$\delta^{\&}$	α^b	
5380.989	1 ⁺	5380.59 9	53 5	0.0	0 ⁺	(M1)		1.44×10^{-3} 2	I_γ : weighted average of 61 13 from $^{31}\text{P}(\alpha, \text{py})$ and 52 5 from $^{33}\text{S}(\text{n}, \gamma)$ E=thermal. Mult.: D from $\text{py}(\theta)$ in $^{31}\text{P}(\alpha, \text{py})$. $\Delta\pi$ =no from level scheme.
5679.928	2 ⁺	789.1 6	1.50 27	4889.755	2 ⁺	[M1,E2]		7.1×10^{-5} 18	B(M1)(W.u.)=0.00126 +27-25 if M1, B(E2)(W.u.)=7.9 +17-16 if E2.
		803.103 27	4.4 4	4876.842	3 ⁺	[M1,E2]		6.8×10^{-5} 17	B(M1)(W.u.)=0.00350 +48-42 if M1, B(E2)(W.u.)=21.3 +29-26 if E2.
		1055.491 20	26.9 27	4624.404	3 ⁻	[E1]		1.895×10^{-5} 27	B(E1)(W.u.)=2.77 $\times 10^{-4}$ 35 E_γ : weighted average of 1055.4 8 from $^{24}\text{Mg}(^{16}\text{O}, \alpha 2\text{py})$ and 1055.491 20 from $^{33}\text{S}(\text{n}, \gamma)$ E=thermal.
		1564.8 5	3.5 8	4114.812	2 ⁺	[M1,E2]		0.000115 17	I_γ : other: 49 25 from $^{24}\text{Mg}(^{16}\text{O}, \alpha 2\text{py})$. B(M1)(W.u.)=3.8 $\times 10^{-4}$ +10-9 if M1, B(E2)(W.u.)=0.60 15 if E2.
		2375.657 20	100 9	3304.210	2 ⁺	M1+E2	<-0.4	0.000417 9	B(E2)(W.u.)=0.3 +22-1 E_γ : weighted average of 2375.4 7 from $^{24}\text{Mg}(^{16}\text{O}, \alpha 2\text{py})$ and 2375.657 20 from $^{33}\text{S}(\text{n}, \gamma)$ E=thermal.
		3552.08 6	66.7 7	2127.561	2 ⁺	D+Q	-0.47 +7-11		I_γ : others: 100 49 from $^{24}\text{Mg}(^{16}\text{O}, \alpha 2\text{py})$, 100 16 from $^{31}\text{P}(\alpha, \text{py})$. Mult., δ : D+Q with $\delta < -0.4$ or $\delta > +2.4$ from $^{31}\text{P}(\alpha, \text{py})$. M2 ruled out by RUL.
									E_γ : weighted average of 3551.2 6 from $^{24}\text{Mg}(^{16}\text{O}, \alpha 2\text{py})$ and 3552.08 4 from $^{33}\text{S}(\text{n}, \gamma)$ E=thermal.
5690.62	5 ⁻	1001.5 5	100 [#] 4	4688.96	4 ⁺	E1		2.095×10^{-5} 29	I_γ : others: 149 75 from $^{24}\text{Mg}(^{16}\text{O}, \alpha 2\text{py})$, 54 16 from $^{31}\text{P}(\alpha, \text{py})$. Mult., δ : D+Q with $\delta = -0.47 +7-11$ from $\text{py}(\theta)$ in $^{31}\text{P}(\alpha, \text{py})$. B(E1)(W.u.)=9.1 $\times 10^{-6}$ +7-6 E_γ : weighted average of 1001.6 5 from $^{24}\text{Mg}(^{16}\text{O}, \alpha 2\text{py})$ and 1001 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{34}\text{S}\gamma)$. I_γ : others: 100 10 from $^{24}\text{Mg}(^{16}\text{O}, \alpha 2\text{py})$, 100 9 from $^{208}\text{Pb}(^{36}\text{S}, ^{34}\text{S}\gamma)$. Mult.: D from $\text{py}(\theta)$ and electric character from $\gamma(\theta)(\text{POL})$ in $^{31}\text{P}(\alpha, \text{py})$. D, $\Delta J=1$ from $\gamma\gamma(\theta)(\text{ADO})$ in $^{24}\text{Mg}(^{16}\text{O}, \alpha 2\text{py})$.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.&	$\delta^\&$	α^b	Comments
5690.62	5^-	1066.2 5	88 8	4624.404	3^-	E2		4.11×10^{-5} 6	B(E2)(W.u.)=0.78 6 E_γ : weighted average of 1066.2 5 from $^{24}\text{Mg}(^{16}\text{O}, \alpha 2p\gamma)$ and 1066 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{34}\text{S}\gamma)$. I_γ : unweighted average of 83 10 from $^{24}\text{Mg}(^{16}\text{O}, \alpha 2p\gamma)$, 104 4 from $^{31}\text{P}(\alpha, p\gamma)$, and 78 8 from $^{208}\text{Pb}(^{36}\text{S}, ^{34}\text{S}\gamma)$. Mult.: Q from $p\gamma(\theta)$ and electric character from $\gamma(\theta)(\text{POL})$ in $^{31}\text{P}(\alpha, p\gamma)$. Q, $\Delta J=2$ from $\gamma\gamma(\theta)(\text{ADO})$ in $^{24}\text{Mg}(^{16}\text{O}, \alpha 2p\gamma)$.
5755.873	1^-	3562.7 [‡] 6 1640.7 10 2451.557 20 3628.10 4 5755.5 5	2.9 [‡] 12 0.97 57 29.6 29 100 9 2.9 5	2127.561 4114.812 3304.210 2127.561 0.0	2^+ 2^+ 2^+ 2^+ 0^+	[E3]		0.000749 10	B(E3)(W.u.)=1.01 41
5847.521	0^+	1772.82 4 2543.13 ^c 10 3719.68 16	73 7 <550 ^c 100 11	4074.662 3304.210 2127.561	1^+ 2^+ 2^+				
5998.10	2^+	1121.33 9 1922.92 22 3870.51 31 5997.30 31	57 8 100 18 92 13 56 10	4876.842 4074.662 2127.561 0.0	3^+ 1^+ 2^+ 0^+	[M1+E2] [M1+E2] [M1+E2] E2	<0.33 <2.1	3.3×10^{-5} 5 0.000257 32 0.00106 7 1.74×10^{-3} 2	I_γ : other: 100 42 from $^{31}\text{P}(\alpha, p\gamma)$. I_γ : other: 128 42 from $^{31}\text{P}(\alpha, p\gamma)$. I_γ : other: 278 50 from $^{31}\text{P}(\alpha, p\gamma)$. Mult.: Q from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha, p\gamma)$. M2 ruled out by RUL.
6121.56	2^+	2817.76 25	100 [#] 9	3304.210	2^+	(M1+E2)	-0.09 4	0.000595 8	Mult., δ : D+Q, $\Delta J=0,1$ from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha, p\gamma)$. $\Delta\pi$ =no from level scheme.
6168.860	3^+	3994.8 8 846.1 13 1479.73 15 1544.41 ^c 10 2053.94 14 2864.56 4 4040.63 29	22 [#] 9 2.6 17 2.41 30 <25.9 ^c 5.4 8 100 10 5.0 7	2127.561 5322.512 4688.96 4624.404 4114.812 3304.210 2127.561	2^+ 2^+ 4^+ 3^- 2^+ 2^+ 2^+	[M1,E2] [M1,E2] D+Q [E1] [M1,E2] M1+E2 D+Q	+0.04 +6-3 -0.23 7 -0.43 16	0.00110 7 6.0×10^{-5} 14 0.000311 4 0.00031 4 0.000619 9	I_γ : other: 95 13 from $^{31}\text{P}(\alpha, p\gamma)$. Mult., δ : D+Q, $\Delta J=0,1$ from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha, p\gamma)$. I_γ : other: 100 13 from $^{31}\text{P}(\alpha, p\gamma)$. Mult., δ : D+Q, $\Delta J=0,1$ from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha, p\gamma)$. M2 ruled out by RUL. Other: $\delta=-1.9$ 6. I_γ : other: 53 13 from $^{31}\text{P}(\alpha, p\gamma)$. Mult., δ : D+Q with $\delta=-0.43$ 16 or $\delta=-1.0$ 3 from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha, p\gamma)$.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. $\&$	$\delta\&$	α^b	Comments
6251.73	(3,4)	572.0 † 1	28 † 14	5679.928	2 $^+$	D			B(E1)(W.u.)= $9\times 10^{-4} +10-6$; B(M1)(W.u.)=0.029 +34-18 E $_\gamma$: weighted average of 572.0 1 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$ and 571.7 6 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. Mult.: D, $\Delta J=1$ from $\gamma\gamma(\theta)$ (ADO) in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. B(M1)(W.u.)= $5\times 10^{-4} +6-3$; B(E2)(W.u.)=14 +14-7 E $_\gamma$: weighted average of 1375.0 5 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$ and 1374.34 20 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. Mult., δ : D+Q, $\Delta J=0,1$ from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha,p\gamma)$. M2 ruled out by RUL. E $_\gamma$: weighted average of 1562.5 5 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$, 1562.3 5 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal, and 1562 1 from $^{208}\text{Pb}(^{36}\text{S},^{34}\text{S}\gamma)$. E $_\gamma$: weighted average of 1626.7 5 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$ and 1627.2 10 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal.
		1374.43 † 23	97 † 21	4876.842	3 $^+$	M1+E2	-3.7 +7-26	6.96×10^{-5} 13	
		1562.4 † 5	100 † 21	4688.96	4 $^+$				
		1626.8 † 5	45 † 10	4624.404	3 $^-$				
6342.52	1 $^-$	3038.18 32	100 14	3304.210	2 $^+$	(E1(+M2))	-0.55 65	0.00108 28	Mult., δ : D(+Q) from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha,p\gamma)$. $\Delta\pi$ =yes from level scheme.
		6341.58 32	36 6	0.0	0 $^+$	(E1)			I $_\gamma$: weighted average of 43 14 from $^{31}\text{P}(\alpha,p\gamma)$ and 35 6 from $^{33}\text{S}(\text{n},\gamma)$ E=thermal. Mult.: D from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha,p\gamma)$. $\Delta\pi$ =yes from level scheme.
6421.34	4 $^-$	1544.41 10	100 $^\#$ 8	4876.842	3 $^+$	(E1)		0.000311 4	Mult., δ : 0.00 6 from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha,p\gamma)$. $\Delta\pi$ =yes from level scheme.
		1732.39 11	61 $^\#$ 8	4688.96	4 $^+$	(E1)		0.000456 6	δ : 0.00 +32-14 from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha,p\gamma)$. $\Delta\pi$ =yes from level scheme.
6428.12	(2 $^+$,3 $^+$)	306.63 16	19 4	6121.56	2 $^+$				
		1739.32 9	100 13	4688.96	4 $^+$				
		2353.06 21	48 8	4074.662	1 $^+$				
6478.768	1 $^-$	631.13 6	2.70 30	5847.521	0 $^+$				
		722.95 14	1.67 21	5755.873	1 $^-$				
		798.92 10	2.8 4	5679.928	2 $^+$				
		1156.39 7	15.0 17	5322.512	2 $^+$				
		1602.06 15	4.1 7	4876.842	3 $^+$				
		1854.28 4	12.2 13	4624.404	3 $^-$				

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.&	$\delta^\&$	α^b	Comments
6478.768	1 ⁻	2404.04 6 3174.37 5 4350.85 9	10.2 11 100 10 61 7	4074.662 1 ⁺ 3304.210 2 ⁺ 2127.561 2 ⁺					
						D+Q	-1.1 9		I_γ : weighted average of 70 17 from $^{31}\text{P}(\alpha, p\gamma)$ and 59 7 from $^{33}\text{S}(\text{n}, \gamma)$ E=thermal. Mult., δ : $-2.0 < \delta \leq -0.19$ from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha, p\gamma)$.
6639	1 ⁻ , 2 ⁽⁻⁾ , 3 ⁽⁻⁾	959 2015	28 [#] 13 100 [#] 13	5679.928 2 ⁺ 4624.404 3 ⁻	[E1] [M1, E2]			2.280×10 ⁻⁵ 32 0.000297 35	B(E1)(W.u.)=0.0038 +19-16 B(M1)(W.u.)=0.050 +17-11 if M1, B(E2)(W.u.)=48 +16-10 if E2.
6685.330	(0 to 3) ⁻	929.436 21	100	5755.873 1 ⁻					
6742	2 ⁽⁺⁾ , 4 ⁽⁺⁾	1865 2053 3438 4614	9.1 [#] 91 36 [#] 13 36 [#] 13 100 [#] 9	4876.842 3 ⁺ 4688.96 4 ⁺ 3304.210 2 ⁺ 2127.561 2 ⁺	[M1, E2] [E2] [M1, E2] [M1, E2]			0.000232 30 0.000351 5 0.00090 7 0.00130 8	
6828.82	2 ⁺	2204.34 2713.89 2753.3 13 6828		4624.404 3 ⁻ 4114.812 2 ⁺ 4074.662 1 ⁺ 0.0 0 ⁺	[E1] [M1, E2] [M1, E2] Q			0.000788 11 0.00061 6 0.00062 6	
6847.90	(1, 2) ⁺	1525.39 6 6846.37 32	100 10 50 6	5322.512 2 ⁺ 0.0 0 ⁺					Mult.: Q from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha, p\gamma)$.
6864	2 ⁽⁺⁾ , 4 ⁽⁺⁾	2175 4736		4688.96 4 ⁺ 2127.561 2 ⁺					
6954.217	2 ⁻	1274.30 4 1631.641 25 2064.40 2839.3 4 3649.88 12 4826.0 5	38 4 93 9 32 5 100 10 3.5 16	5679.928 2 ⁺ 5322.512 2 ⁺ 4889.755 2 ⁺ 4114.812 2 ⁺ 3304.210 2 ⁺ 2127.561 2 ⁺		D(+Q)	0.0 5		Mult., δ : D+Q with $\delta=+1.8$ 3 for $J_i=2$ or Q(+O) with $\delta=0.00$ 3 for $J_i=4$ from $p\gamma(\theta)$ in $^{31}\text{P}(\alpha, p\gamma)$.
7110.450	3 ⁻	281.34 24 941.59 6 989.1 ^c 3 2233.49 4 2995.8 6 4982.44 20	0.46 16 8.2 10 <2.1 ^c 100 10 7.4 20 26.2 28	6828.82 2 ⁺ 6168.860 3 ⁺ 6121.56 2 ⁺ 4876.842 3 ⁺ 4114.812 2 ⁺ 2127.561 2 ⁺					
7111	2 ⁺	3807	39 [#] 11	3304.210 2 ⁺	[M1, E2]			0.00104 7	

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.&	$\delta^\&$	α^b	Comments
7111	2 ⁺	4983	100 [#] 11	2127.561	2 ⁺	(M1+E2)	+0.27 +19-15	1.34×10 ⁻³ 3	Mult., δ : D+Q with $\delta=+0.27 +19-15$ or +1.2 +7-4 from ³¹ P(α ,p γ). $\Delta\pi$ =no from level scheme.
		7110	12.1 [#] 91	0.0	0 ⁺	(E2)			Mult.: Q from p γ (θ) in ³¹ P(α ,p γ). $\Delta\pi$ =no from level scheme.
7164.46	(0 ⁺ to 3 ⁺)	3089.53 26	100 20	4074.662	1 ⁺				
		5036.4 7	45 11	2127.561	2 ⁺				
7219.28	2 ⁺	2328.8 5	5.2 15	4889.755	2 ⁺	[M1,E2]		0.00044 5	
		2530.25 10	18.8 26	4688.96	4 ⁺	[E2]		0.000579 8	
		5091.31		2127.561	2 ⁺	[M1,E2]		0.00144 8	
		7218.48 13	100 10	0.0	0 ⁺	E2			Mult.: from $\gamma(\theta)$ and azimuthal asymmetry in ³⁴ S(γ,γ'),(pol γ,γ').
7248	(4)	1557	100 [#]	5690.62	5 ⁻				
7248.04	(2 ⁺ ,3 ⁻)	2558.82 13	100	4688.96	4 ⁺				
7367.42	(1 ⁺ ,2 ⁺)	2490.6 13	95 25	4876.842	3 ⁺				
		3451.5 9	54 16	3916.40	0 ⁺				
		5239.8 4	100 14	2127.561	2 ⁺				
7467.74	(0 ⁺ ,1,2)	989.1 ^c 3	<6.5 ^c	6478.768	1 ⁻				
		1469.67 24	14.7 26	5998.10	2 ⁺				
		3392.86 24	100 12	4074.662	1 ⁺				
7552.70	(0 ⁺ to 3 ⁻)	1210.04 13	10.2 14	6342.52	1 ⁻				
		2230.14 14	50 6	5322.512	2 ⁺				
		4248.28 21	100 11	3304.210	2 ⁺				
7629.906	3 ⁻	2940.42 31	8.3 12	4688.96	4 ⁺	[E1]		1.22×10 ⁻³ 2	B(E1)(W.u.)=8×10 ⁻⁵ +7-3
		3005.39 5	79 8	4624.404	3 ⁻	[M1,E2]		0.00073 6	B(M1)(W.u.)=0.023 +20-8 if M1, B(E2)(W.u.)=10 +9-4 if E2.
		3515.07 11	11.3 13	4114.812	2 ⁺	[E1]		1.49×10 ⁻³ 2	B(E1)(W.u.)=6×10 ⁻⁵ +5-2
		4325.397 30	100 10	3304.210	2 ⁺	[E1]		1.81×10 ⁻³ 3	B(E1)(W.u.)=2.9×10 ⁻⁴ +25-10
7730.79	(1 ⁻ ,2 ⁻ ,3 ⁻)	5602.78 15	100	2127.561	2 ⁺				
7781.22	(1 ⁻)	1353.46 16	10.0 13	6428.12	(2 ⁺ ,3 ⁺)				
		7780.22 10	100 13	0.0	0 ⁺				
7790.8	(6 ⁻)	1539.7 [‡] 5	19 [‡] 4	6251.73	(3,4)	E2		0.0001222 17	B(E2)(W.u.)=16 +7-5 E γ : weighted average of 1539.6 5 from ²⁴ Mg(¹⁶ O, α 2p γ) and 1540 1 from ²⁰⁸ Pb(³⁶ S, ³⁴ S γ). Mult.: Q, $\Delta J=2$ from A ₂ and A ₄ in ²⁴ Mg(¹⁶ O, α 2p γ). M2 ruled out by RUL.
		2099.6 [‡] 8	100 [‡] 11	5690.62	5 ⁻	M1+E2	-1.8 1	0.000355 5	B(M1)(W.u.)=0.0049 +17-10; B(E2)(W.u.)=14.0 +47-29

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.&	α^b	Comments
								Mult., δ : D+Q, $\Delta J=0,1$ from $\gamma\gamma(\theta)$ (ADO) in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. M2 ruled out by RUL.
7974.72	(1,2 ⁺)	4670.1 6	26 14	3304.210	2 ⁺			
		5847.4 5	60 14	2127.561	2 ⁺			
		7973.45 25	100 14	0.0	0 ⁺			
8036.31	(1 ⁻ ,2 ⁺)	925.79 14	95 12	7110.450	3 ⁻			
		8036.6 7	100 22	0.0	0 ⁺			
8138.08	(1) ⁻	1795.3 ^c 3	<18 ^c	6342.52	1 ⁻			
		2290.26 15	19 4	5847.521	0 ⁺			
		6010.3 3	36 6	2127.561	2 ⁺			
		8136.98 17	100 12	0.0	0 ⁺			
8174.9	(1,2 ⁺)	2945.8 ^c 10	<248 ^c	5228.170	0 ⁺			
		8173.8 9	100 20	0.0	0 ⁺			
8185.45	1 ⁺	8184.70 24	100	0.0	0 ⁺	M1		Mult.: from $\gamma(\theta)$ and azimuthal asymmetry in $^{34}\text{S}(\gamma,\gamma'),(\text{pol } \gamma,\gamma')$.
8205.40	(1 ⁻ to 4 ⁺)	3581.2 4	31 6	4624.404	3 ⁻			
		6077.27 12	100 11	2127.561	2 ⁺			
8294.40	4 ⁽⁺⁾	1951.77 19	34 7	6342.52	1 ⁻	[E3]	0.0001731 24	
		6166.24 13	100 11	2127.561	2 ⁺	[E2]		
8371.0	(7) ⁻	580.3 [‡] 6	2.0 [‡] 10	7790.8	(6) ⁻	(M1)	9.95×10 ⁻⁵ 14	B(M1)(W.u.)=0.027 +16-13 Mult.: D, $\Delta J=1$ from $\gamma\gamma(\theta)$ (ADO) in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. $\Delta\pi$ =no from level scheme.
		2680.5 [‡] 6	100 [‡] 10	5690.62	5 ⁻	E2	0.000648 9	B(E2)(W.u.)=7.4 +15-11 E _γ : weighted average of 2680.5 6 from $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$ and 2681 2 from $^{208}\text{Pb}(^{36}\text{S},^{34}\text{S}\gamma)$. Mult.: Q, $\Delta J=2$ from $\gamma\gamma(\theta)$ (ADO) in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. M2 ruled out by RUL.
8385.40	1 ⁻	8384.28 9	100	0.0	0 ⁺			
8503.7	(6 ⁺)	2812.7 [‡] 9	100 [‡] 18	5690.62	5 ⁻	(E1)	1.15×10 ⁻³ 2	B(E1)(W.u.)=6.8×10 ⁻⁴ +24-15 Mult.: D, $\Delta J=1$ from $\gamma\gamma(\theta)$ (ADO) in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. $\Delta\pi$ =(yes) from level scheme.
		3813.6 [‡] 7	51 [‡] 11	4688.96	4 ⁺	[E2]	1.11×10 ⁻³ 2	B(E2)(W.u.)=1.3 +5-4
		3183.9 7	2.6 17	5322.512	2 ⁺			
		4391.77 29	9.4 19	4114.812	2 ⁺			
		5202.06 6	64 6	3304.210	2 ⁺			
		8505.68 10	100 11	0.0	0 ⁺			
8615.74	(2 ⁻ ,3 ⁺)	2363.97 8	58 31	6251.73	(3,4)			
		3738.69 17	33 5	4876.842	3 ⁺			
		3990.7 7	8.1 20	4624.404	3 ⁻			
		4540.68 15	47 6	4074.662	1 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)							
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. $\&$	Comments
8615.74	(2 ⁻ ,3 ⁺)	5311.10 15	22.2 28	3304.210	2 ⁺		
		6487.48 6	100 11	2127.561	2 ⁺		
8656	1 ⁺	8655	100	0.0	0 ⁺	M1	Mult.: from $\gamma(\theta)$ and azimuthal asymmetry in $^{34}\text{S}(\gamma,\gamma'),(\text{pol } \gamma,\gamma')$.
8702.34	(0 ⁺ ,1,2)	516.86 12	29 5	8185.45	1 ⁺		
		2945.8 ^C 10	<36 ^C	5755.873	1 ⁻		
		3022.0 10	14.7 83	5679.928	2 ⁺		
		3812.0 5	23 6	4889.755	2 ⁺		
		6573.6 4	100 18	2127.561	2 ⁺		
8727.63	(1 ⁻ ,2 ⁺)	1617.00 12	100 13	7110.450	3 ⁻		
		3500.3 5	25 6	5228.170	0 ⁺		
		6600.1 7	11.9 26	2127.561	2 ⁺		
		8726.78 24	22.7 31	0.0	0 ⁺		
8734.8	(6 ⁻)	3044.1 [‡] 6	100 [‡]	5690.62	5 ⁻	D+Q	Mult.: D+Q, $\Delta J=0,1$ from $\gamma\gamma(\theta)(\text{ADO})$ in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$.
8805.62	(1,2 ⁺)	2326.2 ^C 10	<20 ^C	6478.768	1 ⁻		
		5501.4 5	100 20	3304.210	2 ⁺		
		8804.4 4	52 9	0.0	0 ⁺		
8874.01	(1 ⁻ ,2,3 ⁺)	1244.32 21	4.4 10	7629.906	3 ⁻		
		4758.79 27	17.0 30	4114.812	2 ⁺		
		4799.11 28	19.2 30	4074.662	1 ⁺		
		6745.64 16	100 11	2127.561	2 ⁺		
8970.6	(6)	1180 [‡] 1	5.7 [‡] 29	7790.8	(6) ⁻		
		3280.0 [‡] 6	100 [‡] 20	5690.62	5 ⁻		
9026.31	(1,2 ⁺)	3644.8 8	60 13	5380.989	1 ⁺		
		9024.95 17	100 11	0.0	0 ⁺		
9158.714	(1,2 ⁺)	3311.6 5	39 7	5847.521	0 ⁺		
		5043.3 4	100 17	4114.812	2 ⁺		
		5084.2 5	8.8 32	4074.662	1 ⁺		
9208.03	(1,2 ⁺)	334.21 15	4.8 12	8874.01	(1 ⁻ ,2,3 ⁺)		
		1840.52 12	64 10	7367.42	(1 ⁺ ,2 ⁺)		
		1959.67 17	100 13	7248.04	(2 ⁺ ,3 ⁻)		
		9206.65 26	40 6	0.0	0 ⁺		
9413.8	(6 ⁻)	1043.8 7	21 12	8371.0	(7) ⁻		
		3722.6 6	100 21	5690.62	5 ⁻	D+Q	Mult.: D+Q, $\Delta J=0,1$ from $\gamma\gamma(\theta)(\text{ADO})$ in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$.
9479	1 ⁺	9478	100	0.0	0 ⁺	M1	Mult.: from $\gamma(\theta)$ and azimuthal asymmetry in $^{34}\text{S}(\gamma,\gamma'),(\text{pol } \gamma,\gamma')$.
9546.09	(1,2 ⁺)	672.00 10	34 5	8874.01	(1 ⁻ ,2,3 ⁺)		
		2326.2 ^C 10	<20 ^C	7219.28	2 ⁺		
		6241.0 5	100 16	3304.210	2 ⁺		
		9544.83 28	84 11	0.0	0 ⁺		
9598.42		982.68 9	26 4	8615.74	(2 ⁻ ,3 ⁺)		
		3476.95 18	100 14	6121.56	2 ⁺		

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.&	α^b	Comments
9641	(1,2 ⁺)	9639.4	100	0.0	0 ⁺			
9665.742		2817.76 ^c 25	<221 ^c	6847.90	(1,2 ⁺)		0.000595 8	
		7536.2 7	100 23	2127.561	2 ⁺			
9707	(1,2 ⁺)	9706	100	0.0	0 ⁺			
9801.88	(1,2 ⁺)	5884.6 6	48 11	3916.40	0 ⁺			
		6496.62 23	100 13	3304.210	2 ⁺			
		7675.0 8	29 7	2127.561	2 ⁺			
9836.70		2989.9 7	41 21	6847.90	(1,2 ⁺)			
		7708.32 30	100 16	2127.561	2 ⁺			
9868	1 ⁺	9867	100	0.0	0 ⁺	M1		Mult.: from $\gamma(\theta)$ and azimuthal asymmetry in $^{34}\text{S}(\gamma,\gamma'),(\text{pol } \gamma,\gamma')$.
9912.8	(7 ⁺)	942.3 [‡] 5	29 [‡] 9	8970.6	(6)	D		Mult.: D, $\Delta J=1$ from A_2 and A_4 in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$.
		1178 [‡] 1	14.3 [‡] 72	8734.8	(6 ⁻)			
		1408.6 [‡] 9	30 [‡] 9	8503.7	(6 ⁺)	D		Mult.: D, $\Delta J=1$ from $\gamma\gamma(\theta)(\text{ADO})$ in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$.
		1541.5 [‡] 5	12.9 [‡] 72	8371.0	(7 ⁻)			
		2122.9 [‡] 6	100 [‡] 14	7790.8	(6 ⁻)			
9933.36	1 ⁻	725.25 22	68 11	9208.03	(1,2 ⁺)			
		1795.3 ^c 3	<142 ^c	8138.08	(1 ⁻)			
		2152.41 23	100 53	7781.22	(1 ⁻)			
		7804.84	14@	2127.561	2 ⁺			
		9932.1 6	48@ 11	0.0	0 ⁺	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi=\text{yes}$ from level scheme.
9981	1 ⁻	7853	100	2127.561	2 ⁺			
		9979	40@	0.0	0 ⁺	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi=\text{yes}$ from level scheme.
10092.21		1364.4 4	68 19	8727.63	(1 ⁻ ,2 ⁺)			
		3664.8 4	100 21	6428.12	(2 ⁺ ,3 ⁺)			
10097		7968	100@	2127.561	2 ⁺			
		10095	<10@	0.0	0 ⁺			
10169	1 ⁻	8040	100@	2127.561	2 ⁺			
		10167	30@	0.0	0 ⁺	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi=\text{yes}$ from level scheme.
10170	1 ⁺	10168.5	100	0.0	0 ⁺	M1		$E_\gamma, I_\gamma, \text{Mult.}$: from $\gamma(\theta)$ and azimuthal asymmetry in $^{34}\text{S}(\gamma,\gamma'),(\text{pol } \gamma,\gamma')$.
10179.59	1 ⁺	4499.7 10	89 27	5679.928	2 ⁺			
		8051.1 6	100 19	2127.561	2 ⁺			
10212.15		4532.6 7	49 15	5679.928	2 ⁺			
		8083.49 31	100 15	2127.561	2 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.&	α^b	Comments
10249	1 ⁻	8120	100 [@]	2127.561	2 ⁺			
		10247	20 [@]	0.0	0 ⁺	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
10311.530	2 ⁺	1925.94 17	44 13	8385.40	1 ⁻			
		2173.55 21	25 8	8138.08	(1) ⁻			
		2843.7 6	94 21	7467.74	(0 ⁺ ,1,2)			
		4988.6 4	100 14	5322.512	2 ⁺			
		6236.3 11	30 8	4074.662	1 ⁺			
		8182.912		2127.561	2 ⁺			
		10309.849		0.0	0 ⁺	(E2)		I _γ : 100 from $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. I _γ : 40 from $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances, relative to I _γ (8188)=100. Mult.: Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =no from level scheme.
10382	1 ⁻	6465	100 [@]	3916.40	0 ⁺			
10386	3 ⁻	4706	100 [@]	5679.928	2 ⁺			
10399.8	(8 ⁻)	986.8 [‡] 9	11.3 [‡] 56	9413.8	(6 ⁻)			
		2028.8 [‡] 6	47 [‡] 10	8371.0	(7) ⁻			
		2608.6 [‡] 6	100 [‡] 14	7790.8	(6) ⁻	Q		Mult.: Q, $\Delta J=2$ from $\gamma\gamma(\theta)(\text{ADO})$ in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$.
10408	2 ⁺	8279	100 [@]	2127.561	2 ⁺			
		10406	100 [@]	0.0	0 ⁺	(E2) ^a		Mult.: Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =no from level scheme.
10447		7142	100 [@]	3304.210	2 ⁺			
10482	0 ⁺	6407	100 [@]	4074.662	1 ⁺			
10494	1 ⁻	8365	<10 [@]	2127.561	2 ⁺			
		10492	100 [@]	0.0	0 ⁺	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
10587	1 ⁻	7282	100 [@]	3304.210	2 ⁺	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
		8458	60 [@]	2127.561	2 ⁺			
10625	1 ⁻	8496	100 [@]	2127.561	2 ⁺			
		10623	100 [@]	0.0	0 ⁺	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
10650.13		2919.7 5	100 26	7730.79	(1 ⁻ ,2 ⁻ ,3 ⁻)			
		5268.9 ^c 6	<79 ^c	5380.989	1 ⁺			
10651.5	(8 ⁺)	2147.2 [‡] 6	100 [‡] 21	8503.7	(6 ⁺)	E2	0.000396 6	B(E2)(W.u.)=27 +23-10 Mult.: Q, $\Delta J=2$ from $\gamma\gamma(\theta)(\text{ADO})$ in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. M2 ruled out by RUL.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. ^{&}	$\delta^\&$	α^b	Comments
10651.5	(8 ⁺)	2280.4 [‡] 10	100 [‡] 21	8371.0	(7) ⁻	(E1)		0.000837 12	B(E1)(W.u.)=8×10 ⁻⁴ +7-3 Mult.: D, $\Delta J=1$ from $\gamma\gamma(\theta)$ (ADO) in ²⁴ Mg(¹⁶ O, α 2p γ). $\Delta\pi$ =(yes) from level scheme.
10670	1 ⁻	7365	100 [@]	3304.210	2 ⁺	(E1) ^a			Mult.: D from $\gamma(\theta)$ in ³⁰ Si(α,γ),(α,n):resonances. $\Delta\pi$ =yes from level scheme.
		8541	30 [@]	2127.561	2 ⁺				
		10668	<10 [@]	0.0	0 ⁺				
10768	2 ⁺	8639	100 [@]	2127.561	2 ⁺	(M1+E2) ^a	+0.3		Mult., δ : D+Q from $\gamma(\theta)$ in ³⁰ Si(α,γ),(α,n):resonances. $\Delta\pi$ =no from level scheme.
		10766	<10 [@]	0.0	0 ⁺				
10790	1 ⁻	7485	5 [@]	3304.210	2 ⁺				
		8661	20 [@]	2127.561	2 ⁺				
		10788	100 [@]	0.0	0 ⁺	(E1) ^a			Mult.: D from $\gamma(\theta)$ in ³⁰ Si(α,γ),(α,n):resonances. $\Delta\pi$ =yes from level scheme.
10802	1 ⁺	10800	100	0.0	0 ⁺				
10840.62	3 ⁻	748.43 14	71 10	10092.21					
		6152.1 5	100 28	4688.96	4 ⁺				
		8711.86		2127.561	2 ⁺	(E1+M2)	-0.024 17		I $_\gamma$: 100 from ³⁰ Si(α,γ),(α,n):resonances. This is the only γ branch observed from this level in ³⁰ Si(α,γ),(α,n):resonances. Mult., δ : D+Q from $\gamma(\theta)$ in ³⁰ Si(α,γ),(α,n):resonances. $\Delta\pi$ =yes from level scheme. This γ .
10931	1 ⁻	8802	100 [@]	2127.561	2 ⁺	(E1+M2) ^a	+0.154 17		Mult., δ : D+Q from $\gamma(\theta)$ in ³⁰ Si(α,γ),(α,n):resonances. $\Delta\pi$ =yes from level scheme.
10994	2 ⁺	8865	100 [@]	2127.561	2 ⁺	(M1+E2) ^a	+0.078 32		Mult., δ : D+Q from $\gamma(\theta)$ in ³⁰ Si(α,γ),(α,n):resonances. $\Delta\pi$ =no from level scheme.
11015	2 ⁺	8886	100 [@]	2127.561	2 ⁺	(M1+E2) ^a	-0.52 22		Mult., δ : D+Q from $\gamma(\theta)$ in ³⁰ Si(α,γ),(α,n):resonances. $\Delta\pi$ =no from level scheme.
11024.95	1 ⁻	1998.3 4	50 18	9026.31	(1,2 ⁺)				
		4903.4 5	100 29	6121.56	2 ⁺				
		5268.9 6	96 25	5755.873	1 ⁻				
		7719.79		3304.210	2 ⁺				I $_\gamma$: 17 from ³⁰ Si(α,γ),(α,n):resonances, relative to I $_\gamma$ (11023)=100.
		8896.13		2127.561	2 ⁺				I $_\gamma$: 14 from ³⁰ Si(α,γ),(α,n):resonances, relative to I $_\gamma$ (11023)=100.
		11023.02		0.0	0 ⁺	(E1)			I $_\gamma$: 100 from ³⁰ Si(α,γ),(α,n):resonances.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
								Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11088	2 ⁺	7783	47@	3304.210	2 ⁺			
		8959	44@	2127.561	2 ⁺			
		11086	100@	0.0	0 ⁺	(E2) ^a		Mult.: Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =no from level scheme.
11108	3 ⁻	8979	100@	2127.561	2 ⁺	(E1+M2) ^a	+0.062 1	Mult., δ : D+Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11142	1 ⁻	7837	9@	3304.210	2 ⁺			
		9013	18@	2127.561	2 ⁺			
		11140	100@	0.0	0 ⁺	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11165	1 ⁻	7860	100@	3304.210	2 ⁺			
		9036	13@	2127.561	2 ⁺			
		11163	77@	0.0	0 ⁺	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11220	(2 ⁺)	7915	100@	3304.210	2 ⁺			
		9091	48@	2127.561	2 ⁺			
		11218	32@	0.0	0 ⁺	(E2) ^a		Mult.: Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =no from level scheme.
11233	1 ⁻	7928	100@	3304.210	2 ⁺			
		9104	3.7@	2127.561	2 ⁺			
		11231	24@	0.0	0 ⁺	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11272	2 ⁺	9143	100@	2127.561	2 ⁺	(M1+E2) ^a	+0.18 15	Mult., δ : D+Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =no from level scheme.
11315	2 ⁺	8010	67@	3304.210	2 ⁺			
		9186	38@	2127.561	2 ⁺			
		11313	100@	0.0	0 ⁺	(E2) ^a		Mult.: Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =no from level scheme.
11323	1 ⁻	8018	48@	3304.210	2 ⁺			
		9194	65@	2127.561	2 ⁺			
		11321	100@	0.0	0 ⁺	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11358	1 ⁻	8053	100@	3304.210	2 ⁺			
		9229	18@	2127.561	2 ⁺			
		11356	36@	0.0	0 ⁺	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11372	3 ⁻	8067	44@	3304.210	2 ⁺			
		9243	100@	2127.561	2 ⁺	(E1+M2) ^a	+0.022 6	Mult., δ : D+Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
		11370	5.9@	0.0	0 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.&	Comments
11374.2	(8 ⁺)	1461.7 9	90 20	9912.8	(7 ⁺)	D(+Q)	Mult.: D(+Q), $\Delta J=1$ from $\gamma\gamma(\theta)(\text{ADO})$ in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$.
		3002.8 6	100 20	8371.0	(7) ⁻	D	Mult.: D, $\Delta J=1$ from $\gamma\gamma(\theta)(\text{ADO})$ in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$.
11381	2 ⁺	8076	100@	3304.210	2 ⁺		
		9252	33@	2127.561	2 ⁺		
		11379	37@	0.0	0 ⁺	(E2) ^a	Mult.: Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =no from level scheme.
(11417.222)	2 ⁺	392.28 11	0.200 32	11024.95	1 ⁻		
		576.80 19	0.236 34	10840.62	3 ⁻		
		767.20 21	0.158 26	10650.13			
		1105.673 21	2.40 23	10311.530	2 ⁺		
		1205.05 4	0.98 10	10212.15			
		1237.61 5	0.84 10	10179.59	1 ⁺		
		1325.22 26	0.53 11	10092.21			
		1484.06 19	0.53 8	9933.36	1 ⁻		
		1580.50 6	1.07 11	9836.70			
		1615.24 10	3.6 5	9801.88	(1,2 ⁺)		
		1751.431 29	2.32 23	9665.742			
		1818.96 14	0.61 10	9598.42			
		1871.04 8	3.3 4	9546.09	(1,2 ⁺)		
		2209.10 6	1.39 15	9208.03	(1,2 ⁺)		
		2258.430 23	6.0 7	9158.714	(1,2 ⁺)		
		2390.82 6	2.15 23	9026.31	(1,2 ⁺)		
		2543.13 ^c 10	<17.0 ^c	8874.01	(1 ⁻ ,2,3 ⁺)		
		2611.7 4	1.9 5	8805.62	(1,2 ⁺)		
		2689.50 10	3.5 4	8727.63	(1 ⁻ ,2 ⁺)		
		2714.50 19	4.5 8	8702.34	(0 ⁺ ,1,2)		
		2801.33 5	16.3 16	8615.74	(2 ⁻ ,3 ⁺)		
		2910.28 5	16.1 16	8506.77	1 ⁻		
		3031.69 8	7.4 10	8385.40	1 ⁻		
		3122.65 15	4.4 7	8294.40	4 ⁽⁺⁾		
		3211.69 9	3.8 4	8205.40	(1 ⁻ to 4 ⁺)		
		3231.89 20	1.36 18	8185.45	1 ⁺		
		3241.9 5	0.58 11	8174.9	(1,2 ⁺)		
		3278.79 11	5.2 7	8138.08	(1) ⁻		
		3442.24 25	1.65 26	7974.72	(1,2 ⁺)		
		3635.83 8	8.4 10	7781.22	(1) ⁻		
		3787.096 20	43 4	7629.906	3 ⁻		
		3864.25 11	2.71 28	7552.70	(0 ⁺ to 3 ⁻)		
		3949.27 12	2.48 28	7467.74	(0 ⁺ ,1,2)		
		4049.68 15	1.89 21	7367.42	(1 ⁺ ,2 ⁺)		
		4197.69 9	4.8 7	7219.28	2 ⁺		
		4252.38 22	1.98 24	7164.46	(0 ⁺ to 3 ⁺)		

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
(11417.222)	2^+	4306.44 6	13.4 13	7110.450	3^-			
		4462.44 20	12.7 13	6954.217	2^-			
		4568.9 4	0.48 10	6847.90	(1,2 ⁺)			
		4588.37 26	0.95 16	6828.82	2^+			
		4731.37 10	2.55 26	6685.330	(0 to 3) ⁻			
		4938.06 3	35.8 34	6478.768	1^-			
		5074.79 25	0.68 13	6342.52	1^-			
		5247.94 4	19.0 18	6168.860	3^+			
		5294.94 24	0.68 13	6121.56	2^+			
		5569.30 5	9.0 10	5847.521	0^+			
		5660.78 6	29.7 29	5755.873	1^-			
		5736.76 4	69 7	5679.928	2^+			
		6035.68 7	7.1 8	5380.989	1^+			
		6094.4 4	0.34 8	5322.512	2^+			
		6188.45 6	14.0 15	5228.170	0^+			
		6526.84 6	8.9 10	4889.755	2^+			
		6539.66 16	1.60 20	4876.842	3^+			
		6727.5 9	0.11 7	4688.96	4^+			
		6792.10 3	39 4	4624.404	3^-			
		7302.2 8	0.45 8	4114.812	2^+			
		7341.67 6	59 6	4074.662	1^+			
		7499.90 5	100 10	3916.40	0^+			
		8111.99 9	9.8 11	3304.210	2^+			
		9288.28 16	1.77 20	2127.561	2^+			
		11415.17 11	11.5 11	0.0	0^+			
11420	1^-	8115	5@	3304.210	2^+			
		9291	19@	2127.561	2^+			
		11418	100@	0.0	0^+	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11458	3^-	9329	100@	2127.561	2^+	(E1+M2) ^a	+0.037 2	Mult., δ : D+Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11473	1^-	9344	100@	2127.561	2^+	(E1+M2) ^a	-0.13 7	Mult., δ : D+Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11490	1^-	8185	56@	3304.210	2^+			
		9361	46@	2127.561	2^+			
		11488	100@	0.0	0^+	(E1) ^a		Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11506	(1 ⁻)	9377	100@	2127.561	2^+	(E1+M2) ^a	-0.058 16	Mult., δ : D+Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =(yes) from level scheme.

Adopted Levels, Gammas (continued)

$E_i(\text{level})$	J_i^π	E_γ [†]	I_γ [†]	E_f	J_f^π	$\gamma(^{34}\text{S})$ (continued)			Comments
						Mult. ^{&}	δ ^{&}	α ^b	
11545	1 ⁻	8240	100 [@]	3304.210	2 ⁺	[E1]			B(E1)(W.u.)=0.00159 +66-37 B(E1)(W.u.): calculated using $\Gamma_\gamma=1.4$ eV 4 from $^{33}\text{S}(n,\gamma),(n,n)$:resonances.
		9416	60 [@]	2127.561	2 ⁺	[E1]			B(E1)(W.u.)= 6.4×10^{-4} +28-17 B(E1)(W.u.): calculated using $\Gamma_\gamma=1.4$ eV 4 from $^{33}\text{S}(n,\gamma),(n,n)$:resonances.
		11543	63 [@]	0.0	0 ⁺	(E1) ^a			B(E1)(W.u.)= 3.6×10^{-4} +16-10 Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme. B(E1)(W.u.): calculated using $\Gamma_\gamma=1.4$ eV 4 from $^{33}\text{S}(n,\gamma),(n,n)$:resonances.
11643	1 ⁻	8338	20 [@]	3304.210	2 ⁺				
		9514	25 [@]	2127.561	2 ⁺				
		11641	100 [@]	0.0	0 ⁺	(E1) ^a			Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11712	1 ⁻	9583	100 [@]	2127.561	2 ⁺	(E1(+M2)) ^a	-0.080	80	Mult., δ : D(+Q) from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11807.4	(8)	1894.6 [‡] 6	100 [‡] 20	9912.8	(7 ⁺)				
		3436.1 [‡] 6	100 [‡] 40	8371.0	(7 ⁻)	D			Mult.: D, $\Delta J=1$ from $\gamma\gamma(\theta)$ (ADO) in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$.
11922	(3 ⁻)	9793	100 [@]	2127.561	2 ⁺	(E1) ^a			Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =(yes) from level scheme.
11932	1 ⁻	11930	100 [@]	0.0	0 ⁺	(E1) ^a			Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
11957	3 ⁻	9828	100 [@]	2127.561	2 ⁺	(E1+M2) ^a	+0.031	4	Mult., δ : D+Q from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
12034	1 ⁻	12032	100 [@]	0.0	0 ⁺	(E1) ^a			Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
12100	1 ⁻	12098	100 [@]	0.0	0 ⁺	(E1) ^a			Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
12141.3	(9 ⁺)	1489.2 [‡] 6	7.4 [‡] 42	10651.5	(8 ⁺)	[M1,E2]		9.0×10^{-5} 13	B(M1)(W.u.)=0.0024 +16-11 if M1, B(E2)(W.u.)=4.2 +28-20 if E2.
		1741.6 [‡] 5	12.6 [‡] 32	10399.8	(8 ⁻)	[E1]		0.000462 6	B(E1)(W.u.)= 7.4×10^{-5} +29-21
		2228.8 [‡] 6	100 [‡] 12	9912.8	(7 ⁺)	E2		0.000435 6	B(E2)(W.u.)=7.6 +20-14 Mult.: Q, $\Delta J=2$ from $\gamma\gamma(\theta)$ (ADO) in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. M2 ruled out by RUL.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{S})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. &	$\delta^\&$	α^b	Comments
12194	1 ⁻	12192	100 [@]	0.0	0 ⁺	(E1) ^a			Mult.: D from $\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes from level scheme.
12985.4	(9 ⁺)	1178 1 1611.5 7 2333.8 7	42 25 50 25 100 40	11807.4 (8) 11374.2 (8 ⁺) 10651.5 (8 ⁺)					
13320.2	(9 ⁻)	2920.1 10 4949.3 18	26 16 100 21	10399.8 (8 ⁻) 8371.0 (7) ⁻					
13341.6	(10 ⁺)	356.3 [‡] 6	6.3 [‡] 31	12985.4 (9 ⁺)		[M1+E2]	<0.35	7	B(M1)(W.u.)=0.09 +6-4 δ : deduced by evaluators assuming RUL=100 for B(E2)(W.u.).
		1200.4 [‡] 7	100 [‡] 22	12141.3 (9 ⁺)		M1+E2		3.5×10 ⁻⁵ 5	Mult.: D+Q, ΔJ =0,1 from $\gamma\gamma(\theta)$ (ADO) in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. M2 ruled out by RUL. B(M1)(W.u.)=0.031 +9-7 if M1, B(E2)(W.u.)=83 +23-18 if E2.
		1966.8 [‡] 9	81 [‡] 19	11374.2 (8 ⁺)		E2		0.000309 4	B(E2)(W.u.)=5.7 +17-14 Mult.: Q, ΔJ =2 from $\gamma\gamma(\theta)$ (ADO) in $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$. M2 ruled out by RUL.
13960.5	(10 ⁺)	3308.8 [‡] 8	100 [‡]	10651.5 (8 ⁺)					
14576.4	(8,9,10 ⁺)	2768.9 [‡] 9	100 [‡]	11807.4 (8)					
15244.3	(10,11,12 ⁺)	1902.7 [‡] 6	100 [‡]	13341.6 (10 ⁺)					
15281.0	(10 ⁻)	4880.8 16	100	10399.8 (8 ⁻)					
16649.0	(12 ⁺)	2688.4 [‡] 8	100 [‡]	13960.5 (10 ⁺)					

[†] From $^{33}\text{S}(n,\gamma)$ E=thermal, unless otherwise noted. E_γ values without uncertainties are deduced from level-energy differences.

[‡] From $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$, unless otherwise noted. Multipolarities are also from $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$ if available.

From $^{31}\text{P}(\alpha,p\gamma)$.

@ From $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances.

& From measured $\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ in $^{31}\text{P}(\alpha,p\gamma)$, and RUL if applicable, unless otherwise noted.

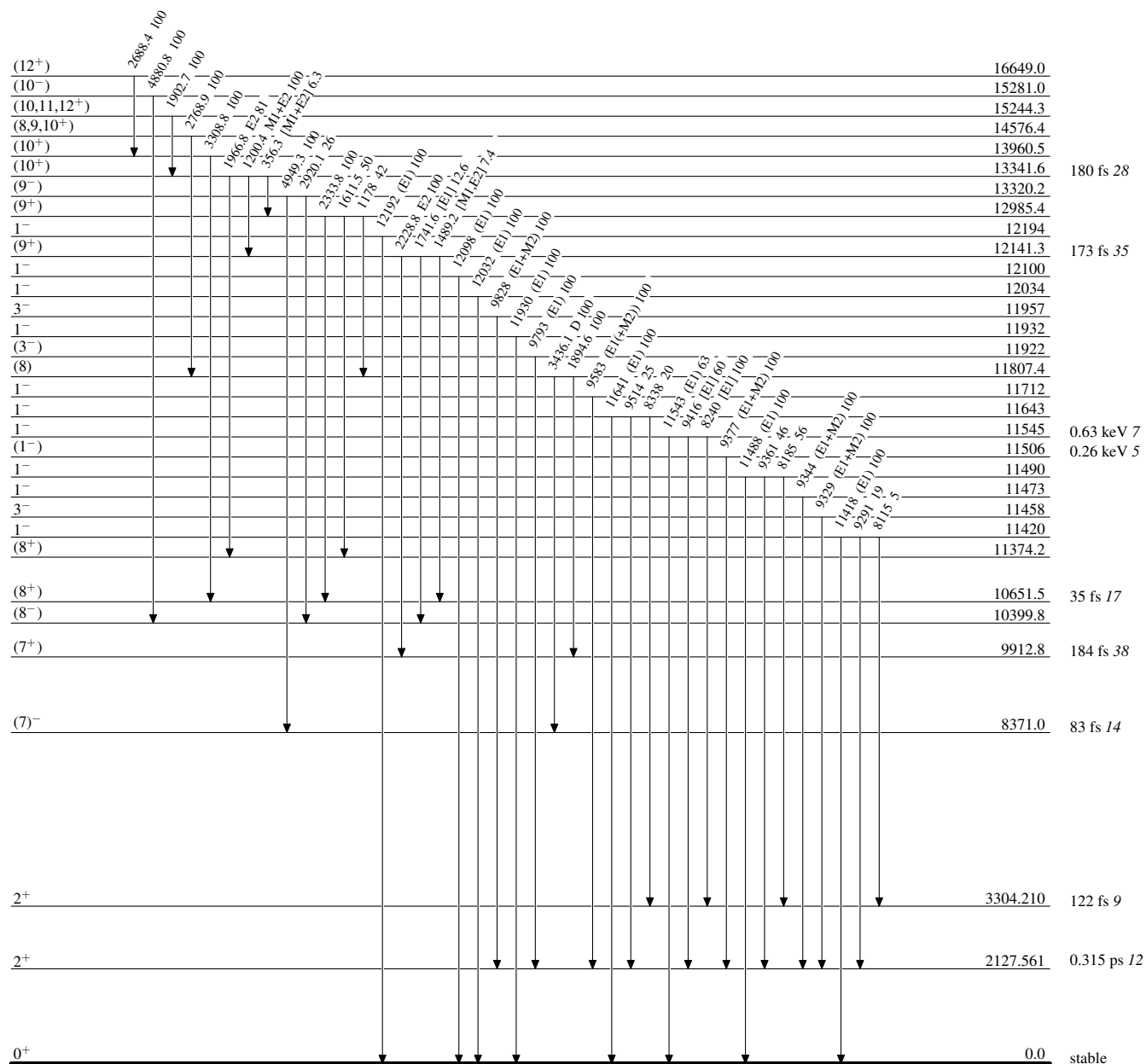
^a From $\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ in $^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances. $\Delta\pi$ =yes/no from level scheme.

^b Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

^c Multiply placed with undivided intensity.

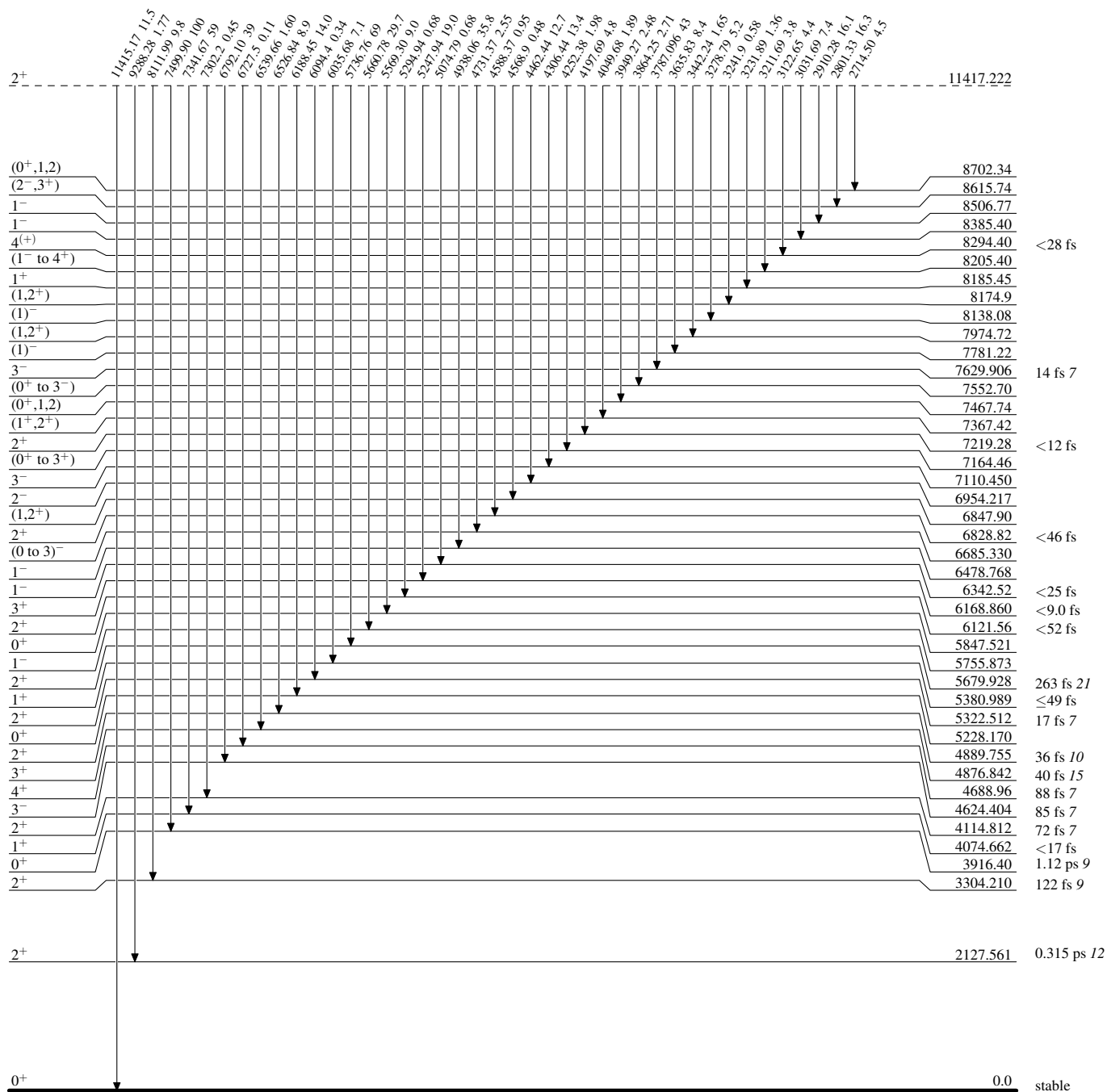
Adopted Levels, Gammas**Level Scheme**

Intensities: Relative photon branching from each level



Adopted Levels, Gammas**Level Scheme (continued)**

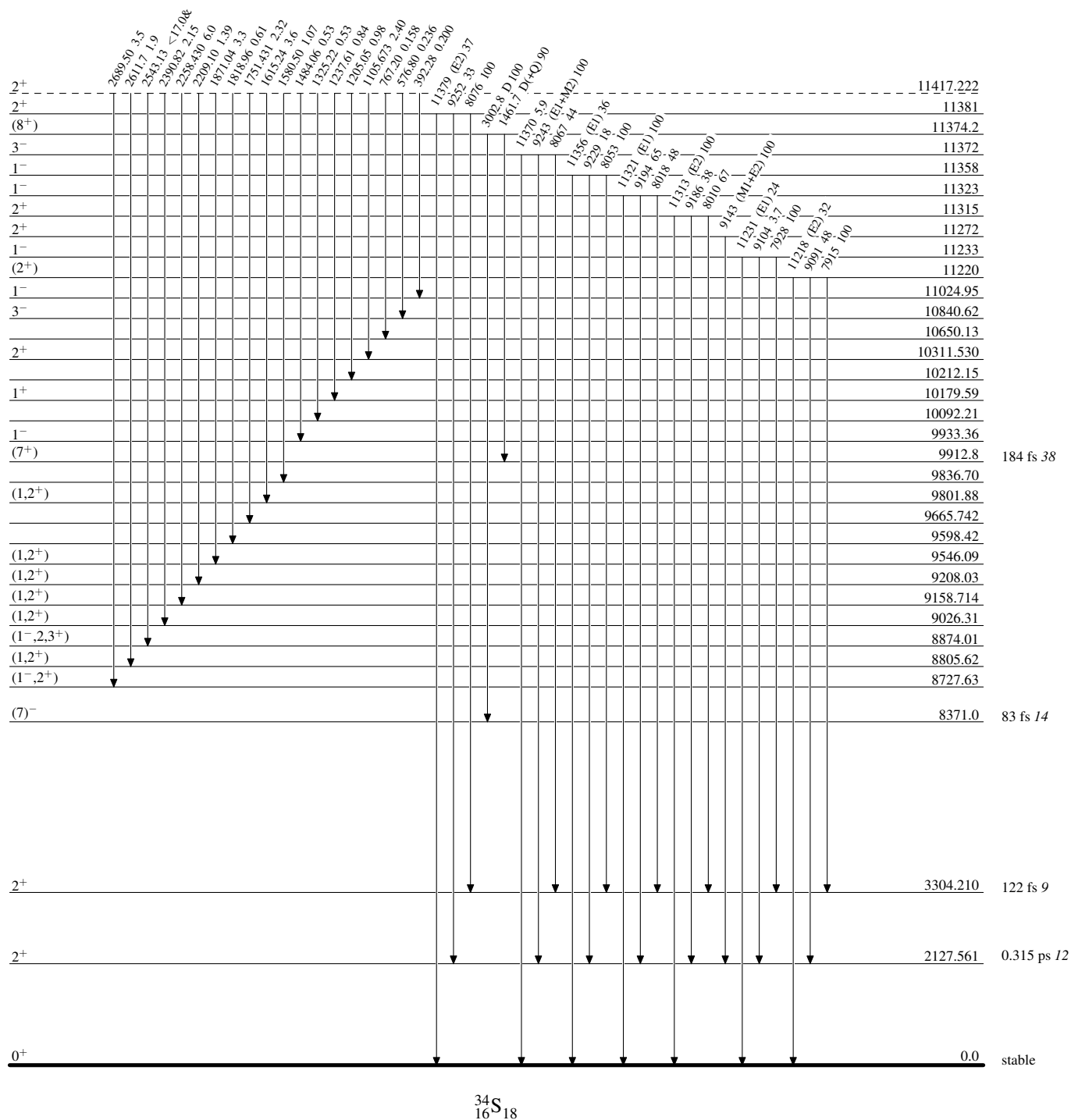
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

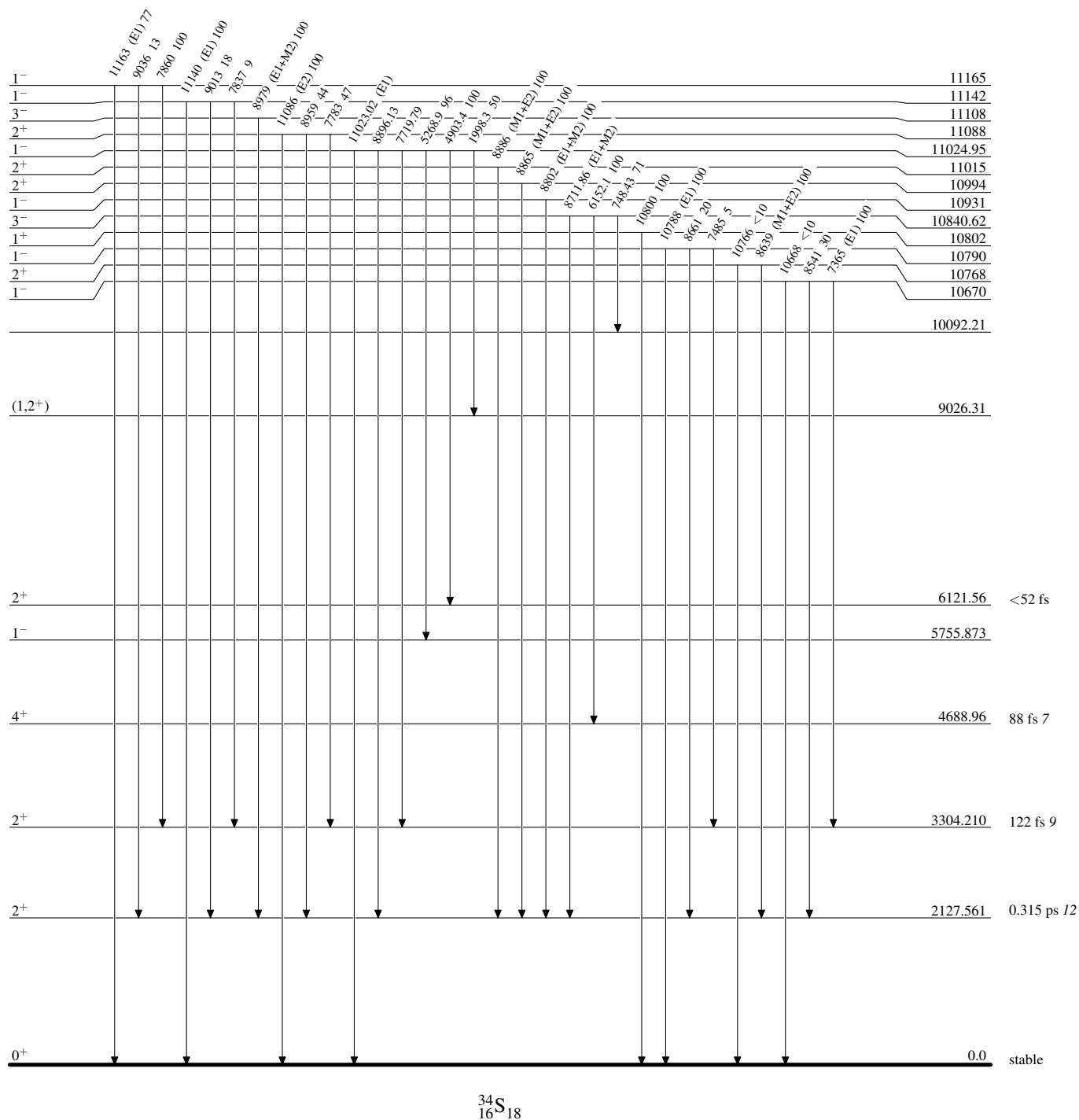
Level Scheme (continued)

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given



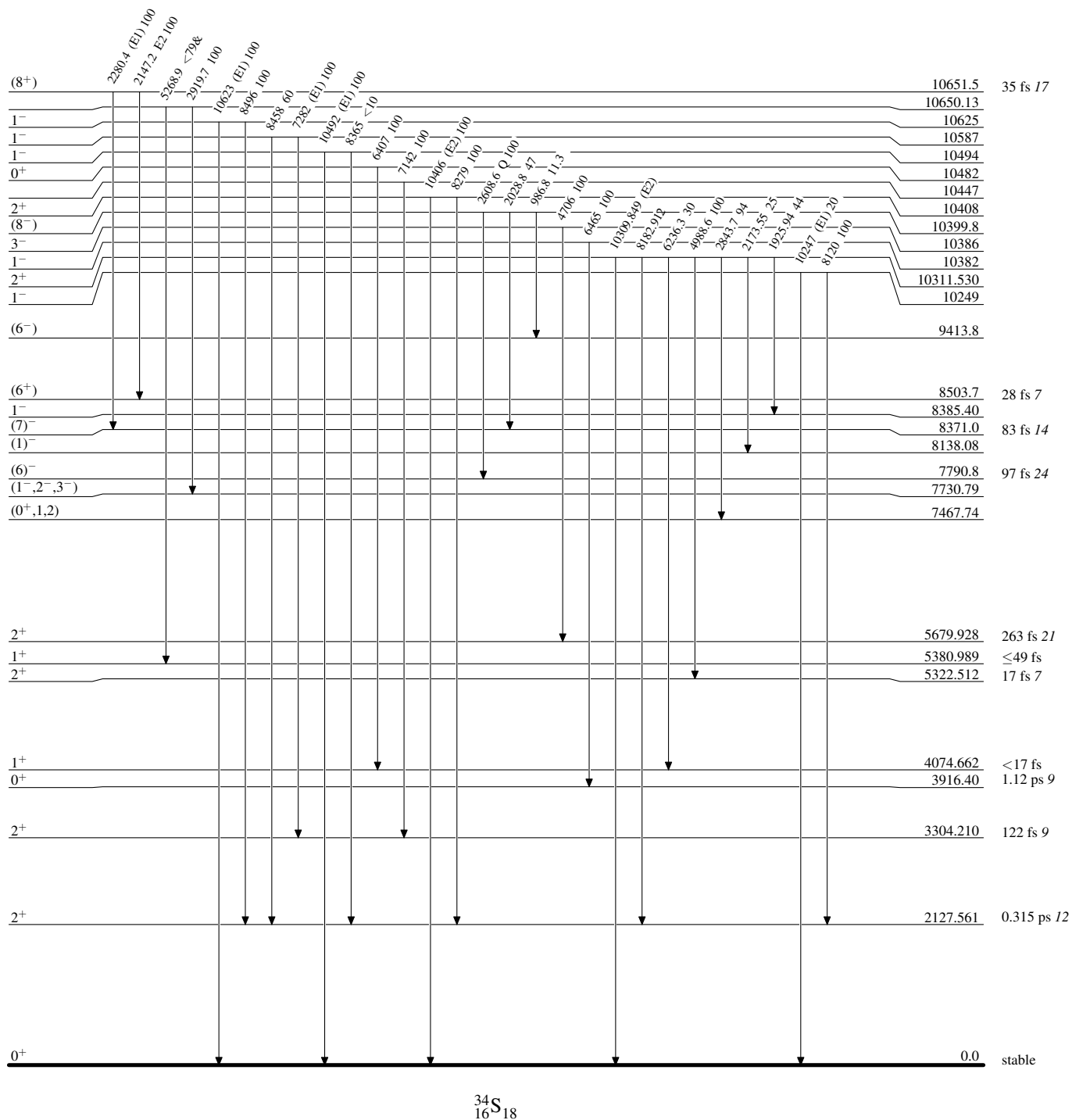
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given



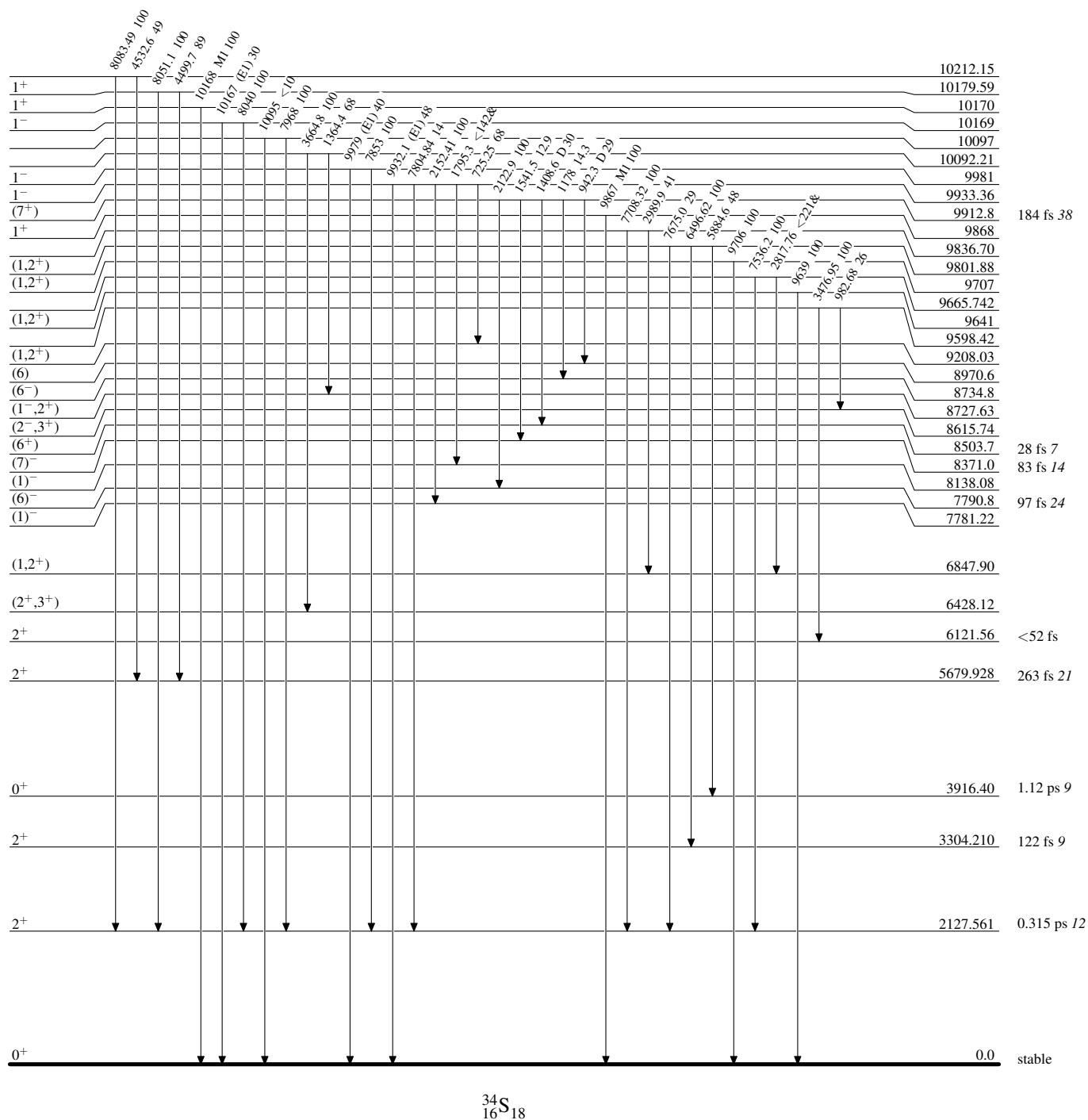
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given



Adopted Levels, Gammas**Level Scheme (continued)**

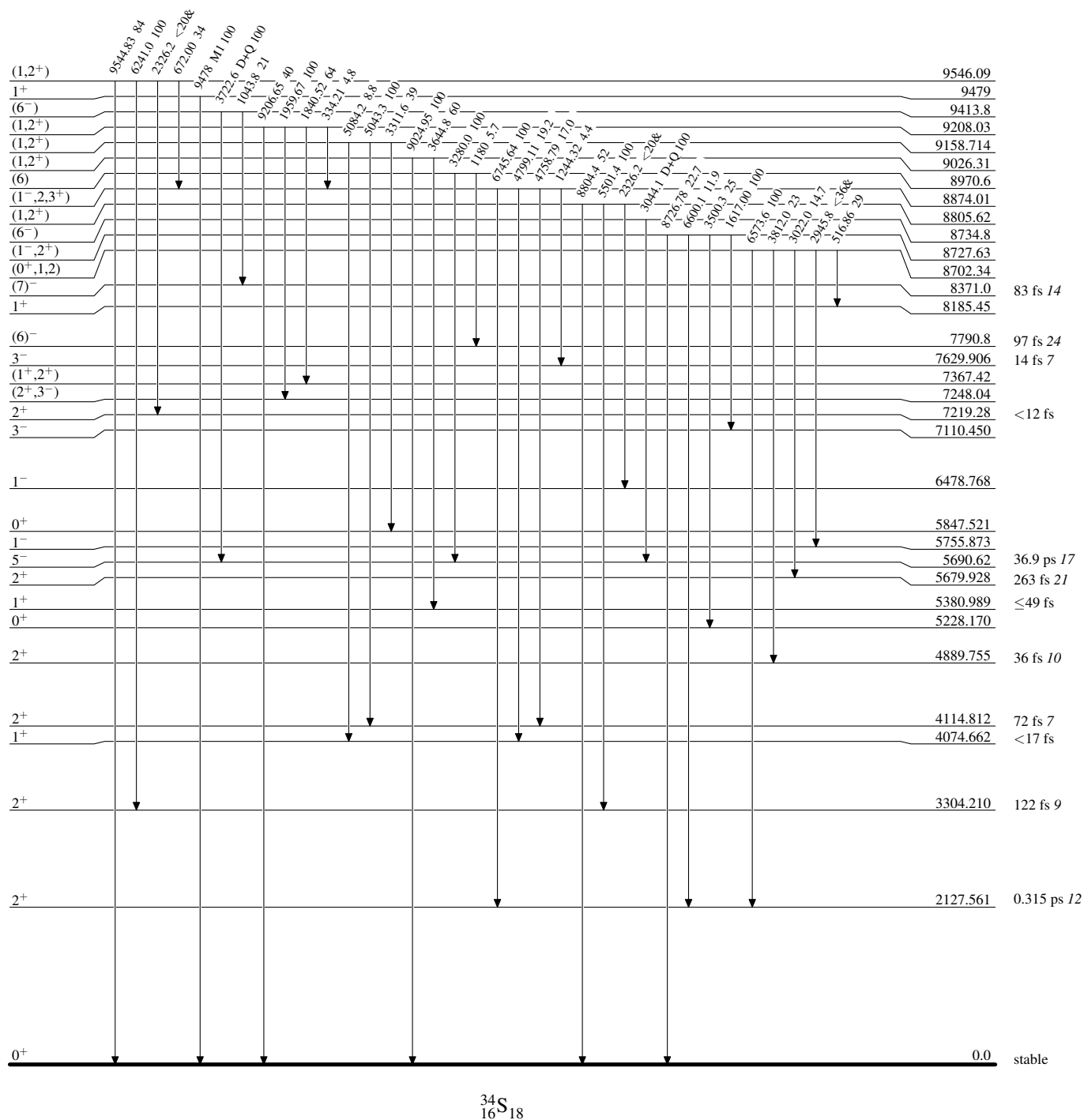
Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given



Adopted Levels, Gammas

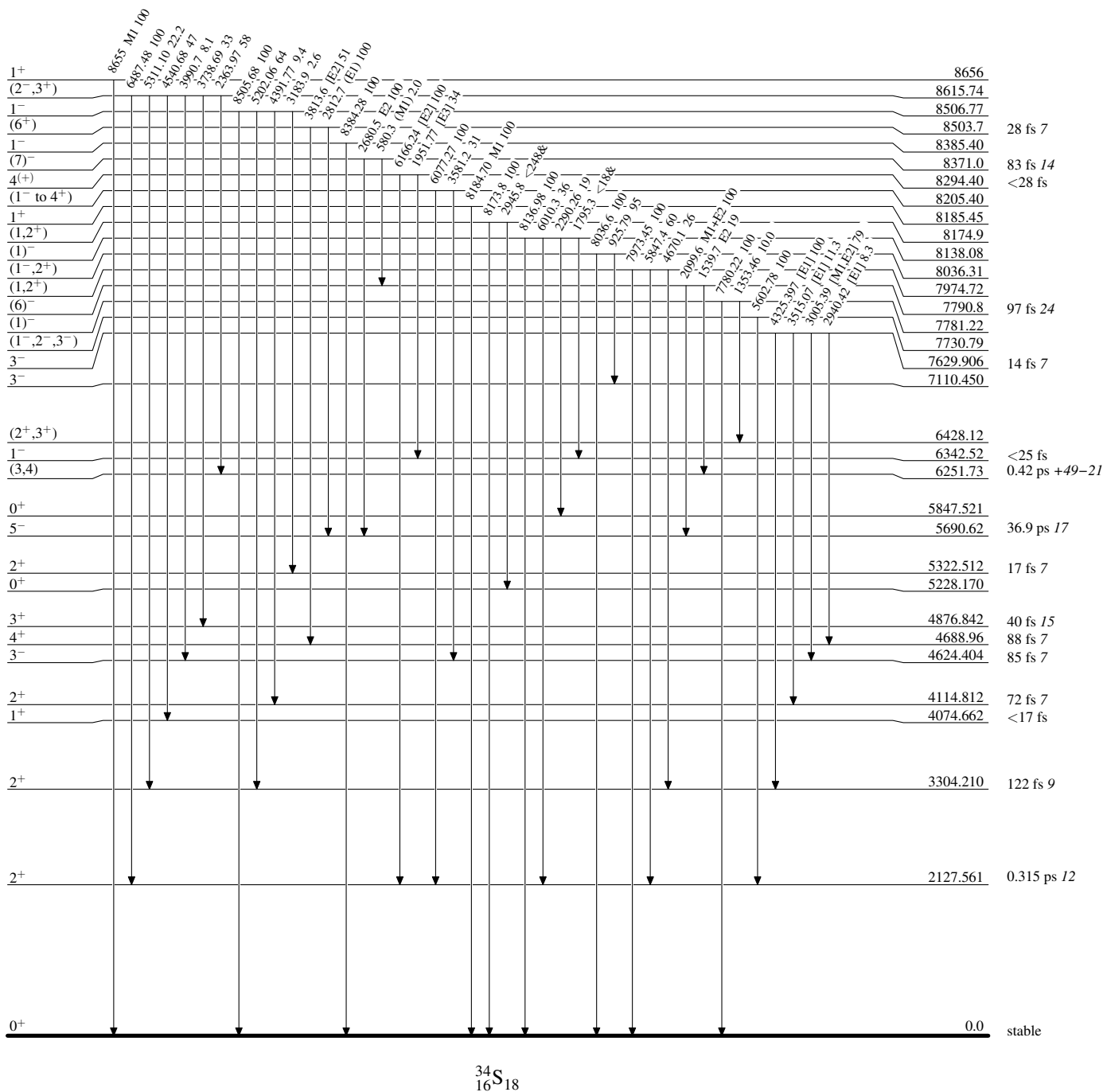
Level Scheme (continued)

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given



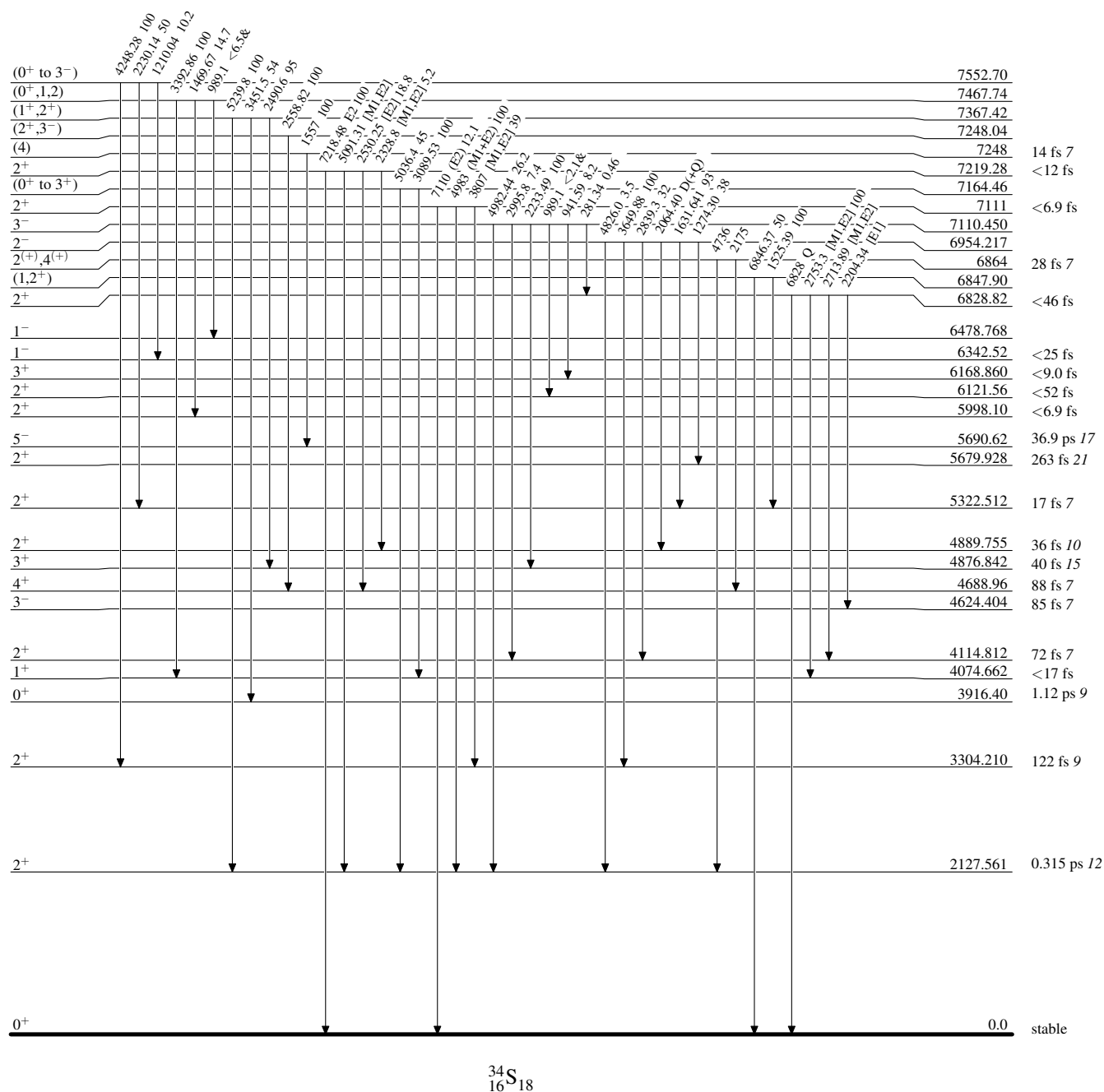
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given



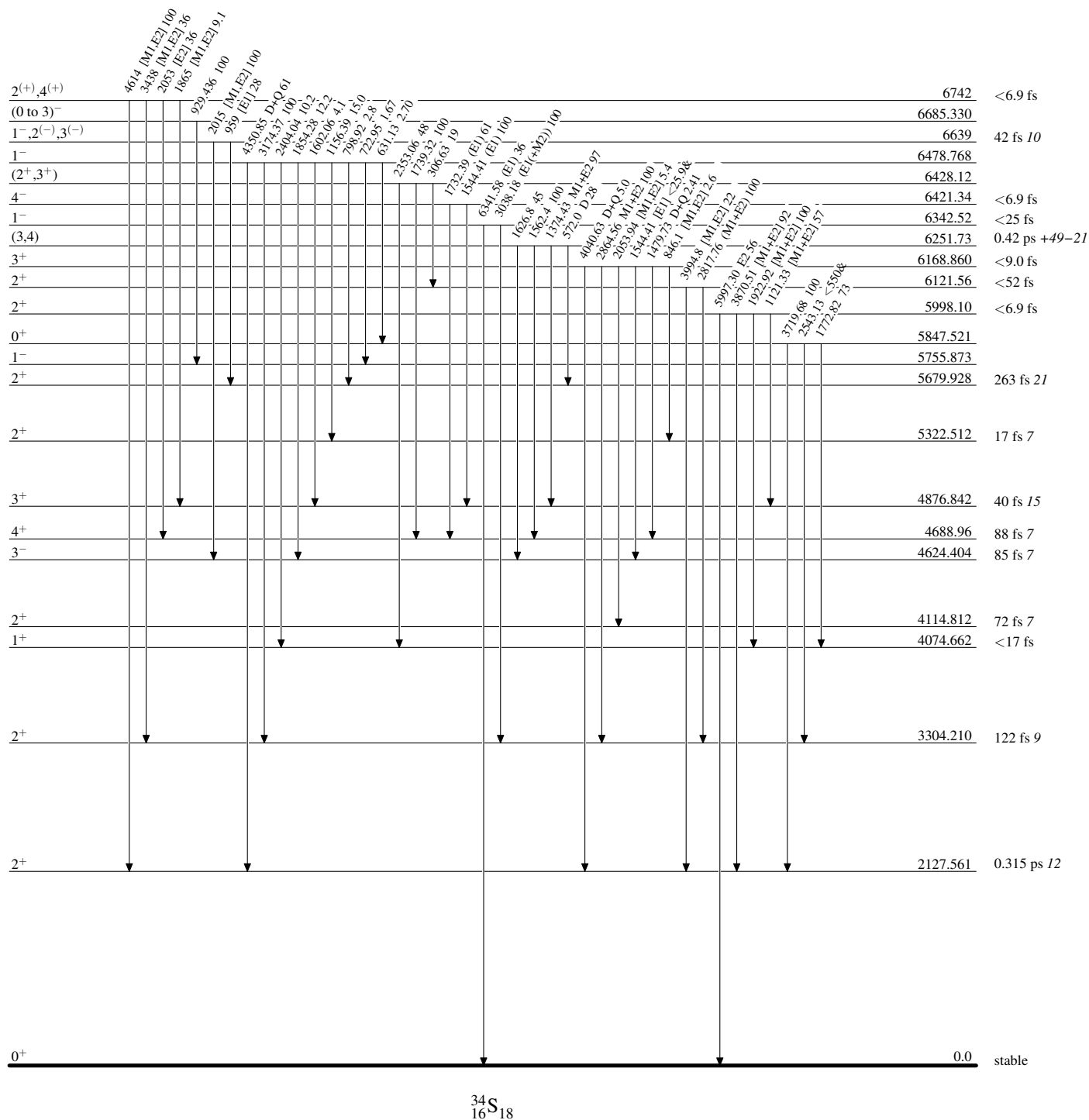
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given



Adopted Levels, Gammas

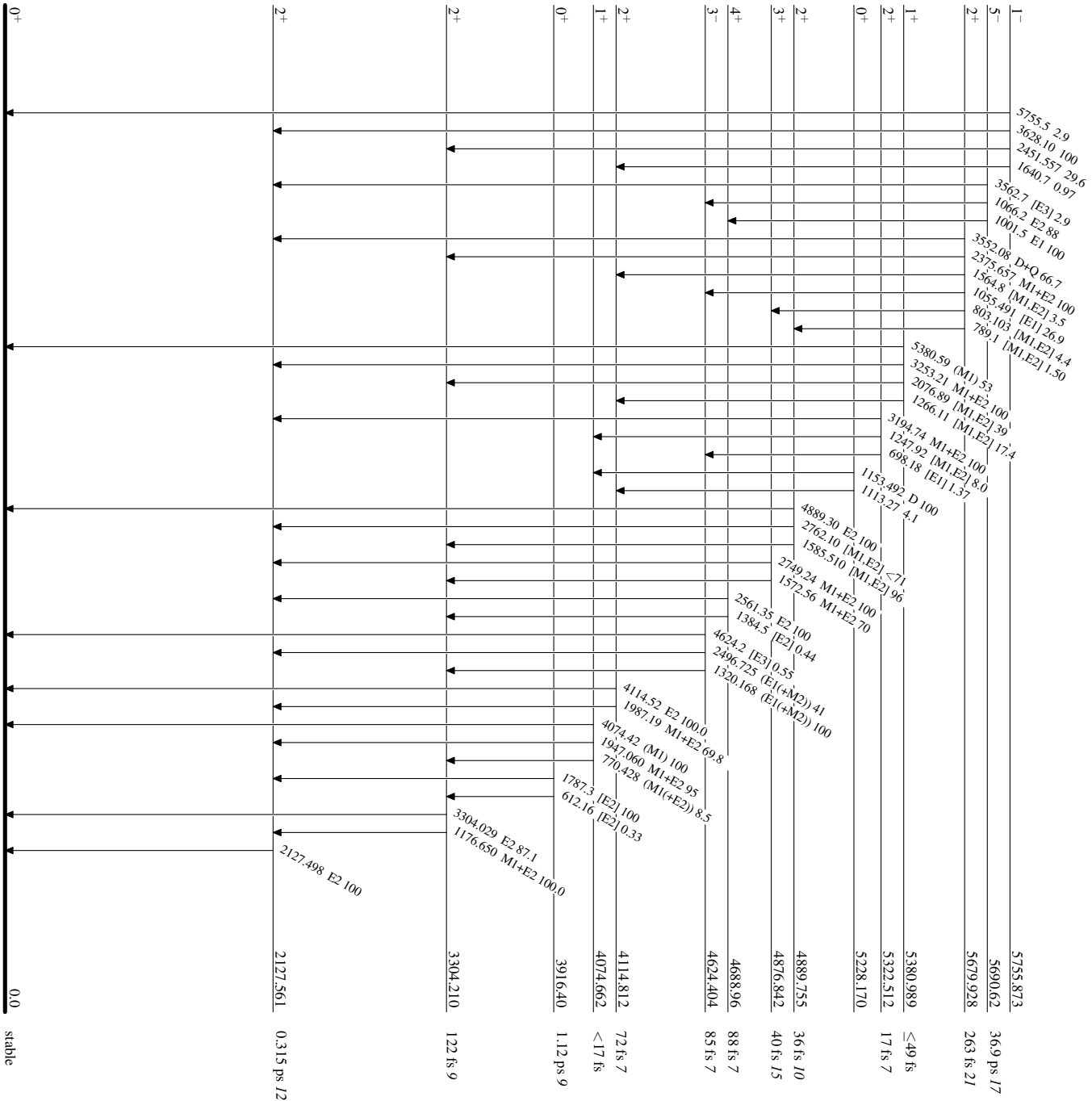
Level Scheme (continued)

 Intensities: Relative photon branching from each level
 & Multiply placed: undivided intensity given


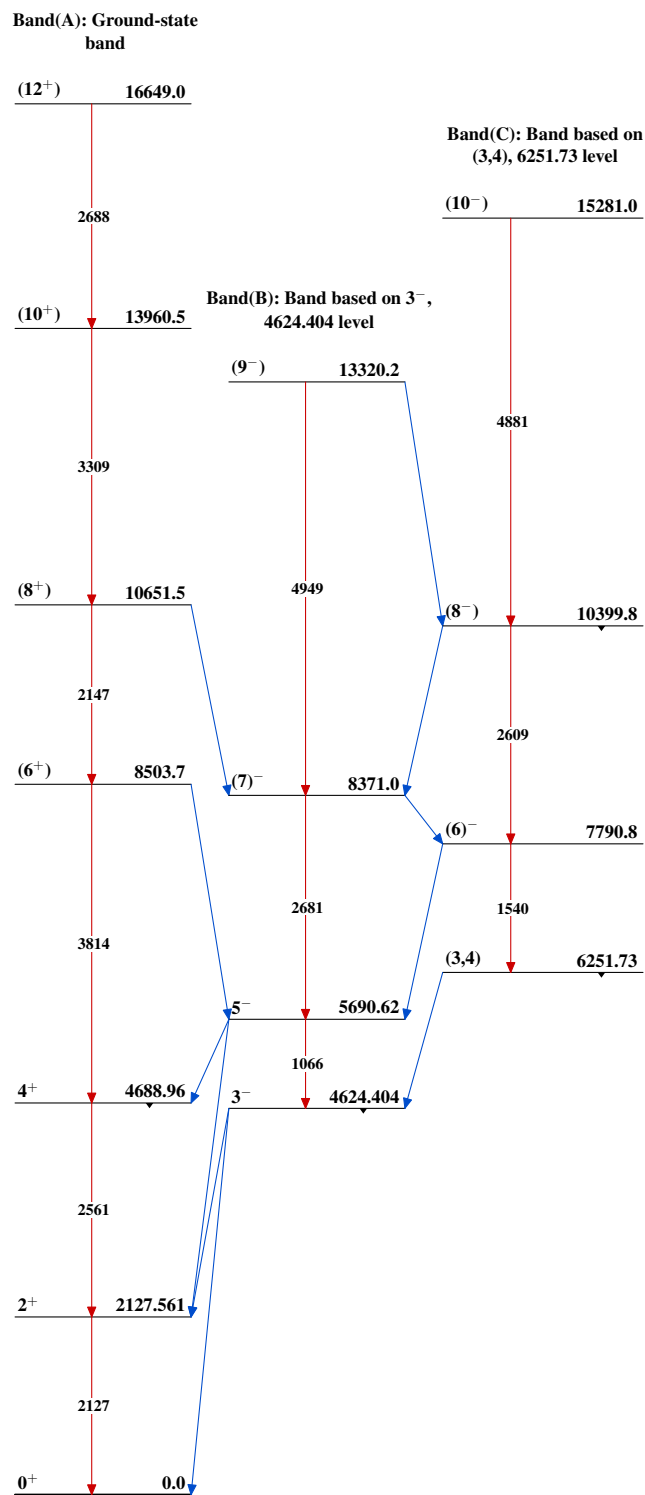
Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given



³⁴S
₁₆¹⁸

Adopted Levels, Gammas $^{34}_{16}\text{S}_{18}$

^{34}P β^- decay (12.43 s) 1973Go33

Parent: ^{34}P : $E=0.0$; $J^\pi=1^+$; $T_{1/2}=12.43$ s 10; $Q(\beta^-)=5383.0$ 8; $\% \beta^-$ decay=100

^{34}P - J^π , $T_{1/2}$: From the Adopted Levels of ^{34}P .

^{34}P - $Q(\beta^-)$: From 2021Wa16.

1973Go33: ^{34}P was produced via the $^{18}\text{O}(^{18}\text{O}, \text{pn})$ reaction using a 42-MeV ^{18}O beam bombarding a 3-mg/cm² Ta₂O₅ target, 99%-enriched in ^{18}O , at BNL. γ rays were detected using a 60 cm³ Ge(Li) detector. β particles were detected using a 7.6-cm-diameter by 5.1-cm-deep NE102 scintillator. Measured E_γ , I_γ , E_β , I_β , and $\beta\gamma$ -coin. Deduced levels, $T_{1/2}$, decay branching ratios, $\log ft$, Q values, and masses.

1971Wa04: ^{34}P was produced via the reaction $^{34}\text{S}(\text{n}, \text{p})$ using 14.8-MeV neutrons produced by the reaction $\text{T}(\text{D}, \text{n})^4\text{He}$ at the University of Arkansas 400-kV Cockcroft-Walton linear accelerator. γ rays were detected using an 8-cm³ Ge(Li) detector or a NaI(Tl) detector. β particles were detected using a 3.8-cm-diameter by 2.1-cm high cylindrical plastic scintillator. Measured E_γ , I_γ , E_β , I_β , and $\beta\gamma$ -coin. Deduced levels, $T_{1/2}$, decay branching ratios, $\log ft$.

1956Mo93: measured β -delayed γ from ^{34}P decay at Purdue University.

1946BI01: measured β from ^{34}P decay and isotope discovery of ^{34}P .

 ^{34}S Levels

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>$T_{1/2}$[‡]</u>
0.0	0 ⁺	stable
2127.3 5	2 ⁺	0.315 ps 12
3303.7	2 ⁺	122 fs 9
3914.4 11	0 ⁺	1.12 ps 9
4074.1 11	1 ⁺	<17 fs
4114.5 9	2 ⁺	72 fs 7
4622.2	3 ⁻	85 fs 7
4687.5	4 ⁺	88 fs 7
4875.2	3 ⁺	40 fs 15
4891	2 ⁺	36 fs 10

[†] From a least-squares fit to γ -ray energies measured by 1973Go33. Level energies for 3303, 4622, 4687, 4875, and 4891 were evaluated by 1973Go33 authors based on available literature data. Evaluators quote those energy values from 1973Go33 without uncertainties.

[‡] From the Adopted Levels.

 β^- radiations

<u>E(decay)</u>	<u>E(level)</u>	<u>$I\beta^-$^{†‡}</u>	<u>$\log ft$</u>	<u>Comments</u>
(1268.5 16)	4114.5	0.32 6	4.9 1	
(1308.9 17)	4074.1	0.109 23	5.4 1	
(1468.6 17)	3914.4	0.044 16	6.0 2	
(3255.7 14)	2127.3	14.5 19	5.0 1	$I\beta^-$: others: 15 2 from β and $\beta\gamma$ -coin spectra measured in 1971Wa04. 25 from β spectra measured in 1946BI01.
(5383.0 16)	0.0	85 2	5.17 2	$I\beta^-$: from β and $\beta\gamma$ -coin spectra measured in 1971Wa04. Other: 75 from β spectra measured in 1946BI01.

[†] β -feeding intensity is from γ +ce intensity balance at each level.

[‡] Absolute intensity per 100 decays.

^{34}P β^- decay (12.43 s) [1973Go33](#) (continued) $\gamma(^{34}\text{S})$

I_γ normalization: From $\Sigma I(\gamma + \text{ce to g.s.}) = 15.2$ and $I\beta^-(\text{g.s.}) = 85.2$ from [1971Wa04](#) based on measured β and $\beta\gamma$ -coin spectra.

$E_\gamma^\dagger$	$I_\gamma^{\dagger\#}$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. ‡	$\delta^\ddagger$	Comments
1318.5	<0.21	4622.2	3^-	3303.7	2^+	(E1(+M2))	-0.03 5	$I_\gamma < 0.0351$
1571.5	<1.0	4875.2	3^+	3303.7	2^+	M1+E2	-0.09 4	$I_\gamma < 0.167$
1787 1	0.3 1	3914.4	0^+	2127.3	2^+	[E2]		$I_\gamma = 0.044 +23-19$
1947.1 15	0.28 10	4074.1	1^+	2127.3	2^+	M1+E2	+1.3 +32-9	$I_\gamma = 0.041 +23-19$
1987.2 10	1.0 2	4114.5	2^+	2127.3	2^+	M1+E2	-0.40 5	$I_\gamma = 0.15 +6-5$
2127.3	100.0 3	2127.3	2^+	0.0	0^+	E2		$I_\gamma = 14.7 20$
2560.0	<0.11	4687.5	4^+	2127.3	2^+	E2		$I_\gamma < 0.0184$
3303.5	<0.12	3303.7	2^+	0.0	0^+	E2		$I_\gamma < 0.0200$
4073.4 15	0.46 6	4074.1	1^+	0.0	0^+	(M1)		$I_\gamma = 0.068 +19-17$
4114.0 15	1.2 2	4114.5	2^+	0.0	0^+	E2		$I_\gamma = 0.18 +6-5$
4891	<0.08	4891	2^+	0.0	0^+	E2		$I_\gamma < 0.0134$

† From [1973Go33](#). E_γ without uncertainty is deduced by [1973Go33](#) based on available literature $E(\text{level})$.

‡ From the Adopted Gammas.

$^\#$ For absolute intensity per 100 decays, multiply by 0.148 20.

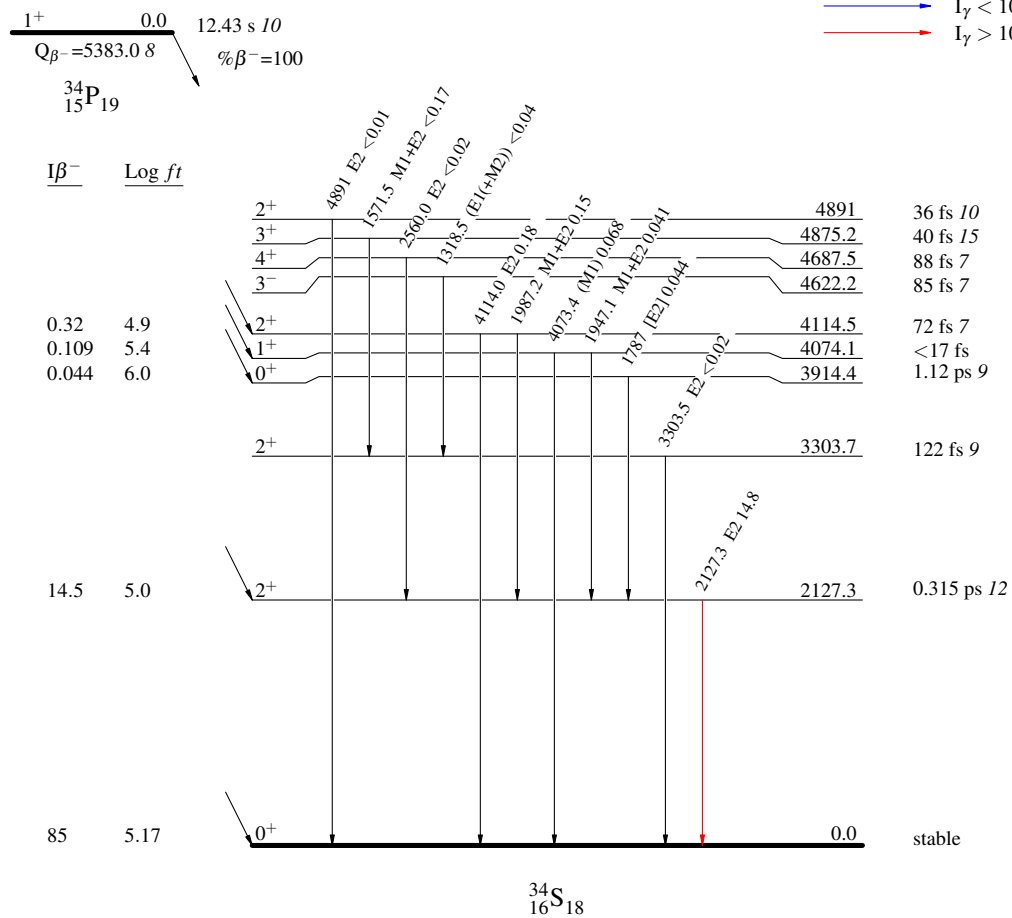
$^{34}\text{P} \beta^-$ decay (12.43 s) 1973Go33

Decay Scheme

Intensities: I_γ per 100 parent decays

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$



^{34}Cl $\varepsilon+\beta^+$ decay (1.5266 s) 1975Va02

Parent: ^{34}Cl : $E=0$; $J^\pi=0^+$; $T_{1/2}=1.5266$ s 5; $Q(\varepsilon+\beta^+)=5491.60$ 4; $\% \varepsilon+\beta^+$ decay=100

^{34}Cl - J^π , $T_{1/2}$: From the Adopted Levels of ^{34}Cl .

^{34}Cl - $Q(\varepsilon+\beta^+)$: From 2021Wa16.

^{34}Cl ground state electron capture and β^+ decay measurements:

1980Wi13: ^{34}Clm was produced via the $^{31}\text{P}(\alpha, n)$ reaction with 10 and 11-MeV α beams from the ONR-CIT tandem accelerator impinging on MnP powder pressed in Be, Zr, and Al foils. γ rays were detected using a 100-cm³ Ge(Li) detector. Measured E_γ , I_γ . Deduced ^{34}S levels, $T_{1/2}$, decay branching ratios, and $\log ft$ for $^{34}\text{Clm} \rightarrow ^{34}\text{S}$ decay. Also measured ^{34}Cl ground-state decay.

1975Va02: ^{34}Clm was produced via the $^{24}\text{Mg}(^{12}\text{C}, pn)$ reaction with a 35-MeV ^{12}C beam from the Utrecht EN tandem accelerator. After one hour of bombardment, the radioactivity was detected using a 100-cm² Ge(Li) detector and a 23×28-cm² NaI(Tl) detector for Compton suppression. Measured E_γ , I_γ . Deduced ^{34}S levels, $T_{1/2}$, decay branching ratios, and $\log ft$ for $^{34}\text{Clm} \rightarrow ^{34}\text{S}$ decay. Also measured ^{34}Cl ground-state decay.

1983Kr06: ^{34}Clm was produced via the $^{35}\text{Cl}(p, pn)$ reaction with 25-MeV protons from the isochronous cyclotron of the Karlsruhe Research Center impinging on a NaCl target on a thin gold backing. The reaction products were separated using a magnetic spectrometer and detected in its focal plane by a Si(Li) detector. Measured $E\beta$ using the iron-free magnetic double-lens spectrometer. Deduced end-point energies, β branching ratios, and Fermi and Gamow-Teller shape factors.

Half-life measurements: 2006Ia05, 1983Ko22, 1976Wi08, 1973Ry01, 1972Ha82, 1965Eb01, 1965Fr06, 1961Ja22, 1960Ja12, 1958Mi85, 1954KI36.

β^+ spectrum measurement: 1956Gr07, 1953St42, 1951Ru24.

 ^{34}S Levels

$E(\text{level})^\dagger$	$J^\pi^\dagger$	$T_{1/2}^\dagger$
0	0^+	stable
3916?	0^+	1.12 ps 9
4075?	1^+	<17 fs
5228?	0^+	
5381?	1^+	≤49 fs

[†] From the Adopted Levels. $E(\text{level})$ values are rounded to the nearest integer.

 ε, β^+ radiations

$E(\text{decay})$	$E(\text{level})$	$I\beta^+ \ddagger$	$I\varepsilon^\ddagger$	$\log ft$	$I(\varepsilon+\beta^+)^\ddagger\ddagger$	Comments
(110.6 [#] 14)	5381?		$<9.0 \times 10^{-4}$	>2.0	<0.0009	
(263.6 [#] 14)	5228?		<0.003	>2.3	<0.003	
(1416.6 [#] 14)	4075?	$<3.00 \times 10^{-4}$	$<4.80 \times 10^{-4}$	>4.5	<0.00078	$I(\varepsilon+\beta^+)$: from 1980Wi13. Other: <0.002 from 1975Va02. 1980Wi13 adjusted the calibration standards used by 1975Va02 and obtained $I(\varepsilon+\beta^+)(4075)<0.0022$ after the adjustment.
(1575.6 [#] 14)	3916?	<0.0040	<0.0020	>4.0	<0.006	
(5491.6 14)	0	99.9	0.1	3.4918 2	100	$I(\varepsilon+\beta^+)$: 1980Wi13 reported an $I(\varepsilon+\beta^+)(4075)<0.00078$. Evaluators obtain $I(\varepsilon+\beta^+)(\text{excited states})<0.01068$ by summing up the 4075 branching from 1980Wi13 and the other three branchings from 1975Va02. This corresponds to $I(\varepsilon+\beta^+)(\text{g.s.})>99.989$. The original $I(\varepsilon+\beta^+)(\text{excited states})<0.012$ reported by 1975Va02 was obtained by summing up all the upper limits of the four branchings from 1975Va02. This corresponds to $I(\varepsilon+\beta^+)(\text{g.s.})>99.988$.

Continued on next page (footnotes at end of table)

 ^{34}Cl $\varepsilon+\beta^+$ decay (1.5266 s) [1975Va02](#) (continued)

 ε, β^+ radiations (continued)

[†] From [1975Va02](#), unless otherwise noted. Upper limits for unobserved peaks were taken as $4(N \times \text{FWHM})^{1/2}$, where N is the number of counts in the peak region and FWHM is estimated from neighboring peaks. [1975Va02](#) states that the sum of the intensities of four allowed branches is less than 1.2×10^{-4} of ^{34}Cl ground-state decay.

[‡] Absolute intensity per 100 decays.

[#] Existence of this branch is questionable.

^{34}Cl $\varepsilon+\beta^+$ decay (31.99 min) 1980Wi13,1975Va02

Parent: ^{34}Cl : $E=146.36$ 3; $J^\pi=3^+$; $T_{1/2}=31.99$ min 3; $Q(\varepsilon+\beta^+)=5491.60$ 4; $\% \varepsilon+\beta^+$ decay=54.0 4

^{34}Cl -E, J^π , $T_{1/2}$: From the Adopted Levels of ^{34}Cl .

^{34}Cl - $Q(\varepsilon+\beta^+)$: From 2021Wa16.

^{34}Cl - $\% \varepsilon+\beta^+$ decay: From $\% \varepsilon+\beta^+$ decay=54.0 4 and $\% \text{IT}$ decay=46.0 4 deduced by evaluators using the $\Sigma I(\gamma \text{ to } ^{34}\text{S g.s.})=129.3$ 12 and $I(146\gamma \text{ to } ^{34}\text{Cl g.s.})=94.4$ 14 measured by 1980Wi13, taking into account the conversion coefficient $\alpha=0.1656$ 24 (2008Ki07).

^{34}Cl 146-keV isomeric state electron capture and β^+ decay measurements:

1980Wi13: $^{34}\text{Cl}^{\text{m}}$ was produced via the $^{31}\text{P}(\alpha,n)$ reaction with 10 and 11-MeV α beams from the ONR-CIT tandem accelerator impinging on MnP powder pressed in Be, Zr, and Al foils. γ rays were detected using a 100-cm³ Ge(Li) detector. Measured E_γ , I_γ . Deduced ^{34}S levels, $T_{1/2}$, decay branching ratios, and $\log ft$ for $^{34}\text{Cl}^{\text{m}} \rightarrow ^{34}\text{S}$ decay. Compared with shell-model calculations.

1975Va02: $^{34}\text{Cl}^{\text{m}}$ was produced via the $^{24}\text{Mg}(^{12}\text{C},pn)$ reaction with a 35-MeV ^{12}C beam from the Utrecht EN tandem accelerator. After one hour of bombardment, the radioactivity was detected using a 100-cm² Ge(Li) detector and a 23×28-cm² NaI(Tl) detector for Compton suppression. Measured E_γ , I_γ . Deduced ^{34}S levels, $T_{1/2}$, decay branching ratios, and $\log ft$ for $^{34}\text{Cl}^{\text{m}} \rightarrow ^{34}\text{S}$ decay. Also measured ^{34}Cl ground-state decay.

1983Kr06: $^{34}\text{Cl}^{\text{m}}$ was produced via the $^{35}\text{Cl}(p,pn)$ reaction with 25-MeV protons from the isochronous cyclotron of the Karlsruhe Research Center impinging on a NaCl target on a thin gold backing. Measured β spectra using the iron-free magnetic double-lens spectrometer. Deduced end-point energies, β -decay branching ratios, and Fermi and Gamow-Teller shape factors.

1970Mo09: ^{34}Cl decay activity was created by bombarding 2-mm of Zn_3P_2 by 10-MeV α particles from the Triangle Universities Nuclear Laboratory FN tandem accelerator. γ rays were detected using a 20-cm³ coaxial Ge(Li) detector. 1970Mo09 primarily measured $^{31}\text{P}(\alpha,p\gamma)^{34}\text{S}$.

1956Ha44: Measured $\gamma\gamma(\theta)$ of the $2 \rightarrow 2 \rightarrow 0$ cascade in ^{34}S using NaI(Tl) detectors at UCLA.

Others:

Measured γ , $\gamma\gamma$: 1971Av10.

Measured $E\beta$, $I\beta$: 1971Wa04, 1956Gr07, 1951Ru24.

Measured $\gamma\gamma(\text{pol})$: 1958Sh62.

 ^{34}S Levels

E(level) [†]	J^π [†]	$T_{1/2}$ [†]	Comments
0	0 ⁺	stable	
2128	2 ⁺	0.315 ps 12	
3304	2 ⁺	122 fs 9	
4075	1 ⁺	<17 fs	
4115.3 20	2 ⁺	72 fs 7	E(level): from E_γ data.
4689	4 ⁺	88 fs 7	
4877	3 ⁺	40 fs 15	
4890	2 ⁺	36 fs 10	

[†] From the Adopted Levels, unless otherwise noted. E(level) values are rounded to the nearest integer.

 ε, β^+ radiations

$I\beta$ normalization: From 100/ $\% \varepsilon+\beta^+$ decay.

E(decay)	E(level)	$I\beta^+$ [†]	$I\varepsilon$ [†]	Log ft	$I(\varepsilon+\beta^+)$ [†]	Comments
(748.0 10)	4890		<0.0127	>5.7	<0.0127	$I(\varepsilon+\beta^+)$: <0.0013 (1980Wi13); <0.0007 (1975Va02).
(761.0 10)	4877		0.037 6	5.2 1	0.037 6	$I(\varepsilon+\beta^+)$: 0.038 6 (1980Wi13); 0.032 3 (1975Va02).
(949.0 10)	4689		<0.037	>5.4	<0.037	$I(\varepsilon+\beta^+)$: 0.0336 29 (1980Wi13); 0.030 6 (1975Va02).
(1522.7 20)	4115.3	0.257 6	0.189 4	5.1 10	0.446 10	$I(\varepsilon+\beta^+)$: 0.458 10 (1980Wi13); 0.392 15 (1975Va02).
(2334.0 10)	3304	24.90 31	0.805 10	4.8 32	25.70 32	$I(\varepsilon+\beta^+)$: 27.8 8 (1983Kr06) measured β^+ branching; 26.4 3 (1980Wi13); 24.3 7 (1975Va02).

Continued on next page (footnotes at end of table)

^{34}Cl $\varepsilon+\beta^+$ decay (31.99 min) [1980Wi13,1975Va02](#) (continued) ε, β^+ radiations (continued)

<u>E(decay)</u>	<u>E(level)</u>	<u>$I\beta^+$</u> [†]	<u>$I\varepsilon$</u> [†]	<u>Log ft</u>	<u>$I(\varepsilon+\beta^+)$</u> [†]	Comments
(3510.0 10)	2128	27.65 34	0.1273 16	6.0 34	27.78 34	$I(\varepsilon+\beta^+)$: 27.7 6 (1983Kr06 measured β^+ branching); 28.5 5 (1980Wi13); 28.4 7 (1975Va02).

[†] Absolute intensity per 100 decays.

 $\gamma(^{34}\text{S})$

I_γ normalization: From $\Sigma I(\gamma \text{ to g.s.})=100$.

<u>E_γ</u> [†]	<u>I_γ</u> [‡] @	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.</u> [#]	<u>δ</u> [#]	Comments
1177	32.9 3	3304	2 ⁺	2128	2 ⁺	M1+E2	-0.16 2	% I_γ =13.74 20 E_γ : other: 1160 30 (1956Ha44). I_γ : other: 32.2 10 (1975Va02). Mult., δ : other: D+Q with $\delta=-0.133$ 24 from 1177-2127 $\gamma\gamma(\theta)$; $A_2=+0.66$ 10, $A_2=-0.088$ 99 (1956Ha44).
1385	<0.021	4689	4 ⁺	3304	2 ⁺	[E2]		% I_γ <0.0089
1573	0.037 12	4877	3 ⁺	3304	2 ⁺	M1+E2	-0.09 4	% I_γ =0.015 5 I_γ : other: 0.026 5 (1975Va02). % I_γ <0.0076
1586	<0.018	4890	2 ⁺	3304	2 ⁺	[M1,E2]		I_γ : other: <0.015 (1975Va02).
1987	0.431 15	4115.3	2 ⁺	2128	2 ⁺	M1+E2	-0.40 5	% I_γ =0.180 7 I_γ : other: 0.409 11 (1975Va02) quoted as 0.409 15 by 1980Wi13 .
2127	100.0 11	2128	2 ⁺	0	0 ⁺	E2		% I_γ =41.75 34 E_γ : other: 2100 30 (1956Ha44). I_γ : other: 100 (1975Va02). % I_γ =0.0326 30
2561	0.078 7	4689	4 ⁺	2128	2 ⁺	E2		I_γ : other: 0.072 14 (1975Va02).
2749	0.052 7	4877	3 ⁺	2128	2 ⁺	M1+E2	-0.11 3	% I_γ =0.0218 33 I_γ : other: 0.049 5 (1975Va02).
2762	<0.011	4890	2 ⁺	2128	2 ⁺	[M1,E2]		% I_γ <0.0047
3304	28.7 3	3304	2 ⁺	0	0 ⁺	E2		% I_γ =11.98 17 E_γ : other: 3220 30 (1956Ha44). I_γ : other: 25.4 8 (1975Va02) corrected to 27.3 9 by 1980Wi13 .
4074	<0.00081	4075	1 ⁺	0	0 ⁺	(M1)		% I_γ <3.4 $\times 10^{-4}$ I_γ : other: <0.0023 (1975Va02).
4115.0 20	0.637 14	4115.3	2 ⁺	0	0 ⁺	E2		% I_γ =0.266 7 E_γ : from 1970Mo09 . I_γ : other: 0.52 3 (1975Va02) corrected to 0.61 4 by 1980Wi13 .
4877 ^{&}	<0.00076	4877	3 ⁺	0	0 ⁺			% I_γ <3.2 $\times 10^{-4}$ I_γ : other: <0.0010 (1975Va02).
4889	<0.0015	4890	2 ⁺	0	0 ⁺	E2		% I_γ <6.4 $\times 10^{-4}$ I_γ : other: <0.0010 (1975Va02).

[†] From the Adopted Gammas. E_γ values are rounded to the nearest integer, unless otherwise noted. No measured E_γ with uncertainty is given in [1975Va02](#) or [1980Wi13](#). The 4115.0 20 γ ray from [1970Mo09](#) is the only precise E_γ in this dataset.

[‡] From [1980Wi13](#), unless otherwise noted.

Continued on next page (footnotes at end of table)

^{34}Cl $\varepsilon+\beta^+$ decay (31.99 min) [1980Wi13,1975Va02](#) (continued)

$\gamma(^{34}\text{S})$ (continued)

From the Adopted Gammas.

@ For absolute intensity per 100 decays, multiply by 0.417 5.

& Placement of transition in the level scheme is uncertain.

$^3\text{H}(^{32}\text{S},\text{p}\gamma)$ 1975He25,1977He12

1975He25: $^{32}\text{S}(\text{t},\text{p}\gamma)^{34}\text{S}$ in inverse kinematics. A 38-MeV ^{32}S beam was provided from the Utrecht EN tandem accelerator.

Targets were $200\text{ }\mu\text{g}/\text{cm}^2$ TiD and TiT on a 0.3-mm-thick Al or Cu backing, which were also slowing-down materials. Tritium targets have $45\text{-}\mu\text{g}/\text{cm}^2$ of Al or Au as a protective layer. Protons were detected using an annular Si counter at $165^\circ\text{--}172^\circ$. γ rays were detected using a Ge(Li) detector at 0° in coincidence with protons. Measured $E\pi$, $E\gamma$, and $\text{p}\gamma$ -coin. Deduced levels, and lifetimes from DSAM lineshape analysis.

1977He12: a series of lifetime measurements using (d,p) and (t,p) reactions in inverse kinematics at Utrecht, Netherlands.

 ^{34}S Levels

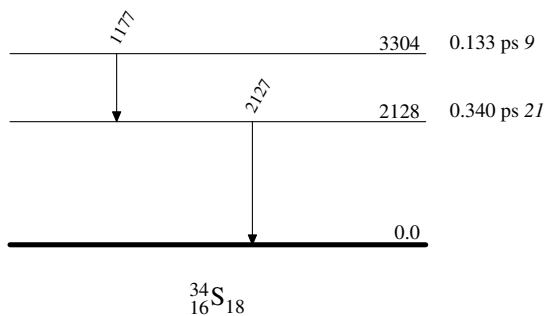
<u>$E(\text{level})^\dagger$</u>	<u>$T_{1/2}$</u>	<u>Comments</u>
0.0		
2128	0.340 ps 21	$T_{1/2}$: lifetime $\tau=490$ fs 30 from 1977He12 reanalysis of 1975He25 data using ^4He -scaled stopping power data of Northcliffe and Schilling. $\tau=470$ 20 from 1975He25 initial DSAM lineshape analysis.
3304	0.133 ps 9	$T_{1/2}$: lifetime $\tau=192$ fs 13 from 1977He12 reanalysis of 1975He25 data using ^4He -scaled stopping power data of Northcliffe and Schilling. $\tau=196$ 13 from 1975He25 initial DSAM lineshape analysis.

† From the Adopted Levels. $E(\text{level})$ values are rounded to the nearest integer.

 $\gamma(^{34}\text{S})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>
1177	3304	2128
2127	2128	0.0

† From the Adopted Gammas. E_γ values are rounded to the nearest integer.

 $^3\text{H}(^{32}\text{S},\text{p}\gamma)$ 1975He25,1977He12Level Scheme

$^3\text{H}(^{35}\text{Cl},\alpha\gamma)$ [1976Co11](#)

[1976Co11](#): $^{35}\text{Cl}(t,\alpha\gamma)^{34}\text{S}$ in inverse kinematics. A 55-MeV ^{35}Cl beam was provided from the Chalk River MP tandem accelerator and bombarded $300\text{-}\mu\text{g}/\text{cm}^2$ tritiated titanium hydride on a $1\text{-mg}/\text{cm}^2$ Cu backing. The residual nuclei were stopped in a $50\text{-mg}/\text{cm}^2$ Au foil placed at a variable distance downstream of the target. Particles were detected using a Si ΔE -E telescope consisting of a $50\text{-}\mu\text{m}$ surface barrier detector (ΔE) and a 5-mm Ge detector (E) at 0° . γ rays were detected using a Ge(Li) detector at 118° in coincidence with α particles. Measured $E\alpha$, $E\gamma$, and $\alpha\gamma$ coin. Deduced levels and lifetimes using the recoil-distance method.

 ^{34}S Levels

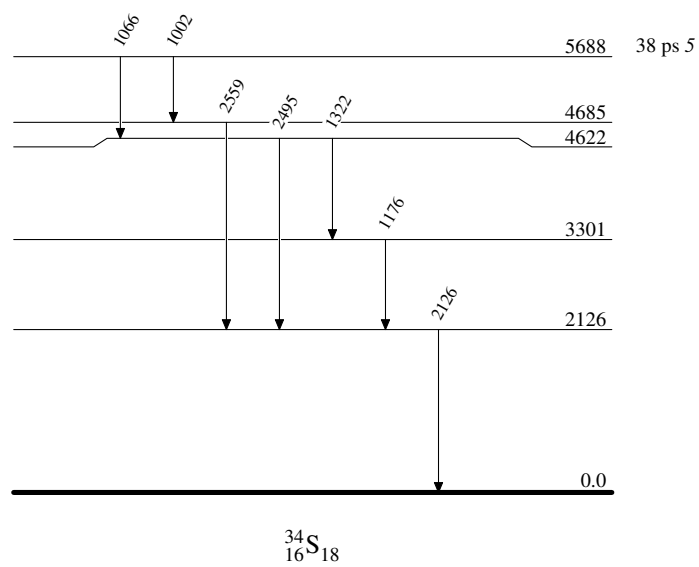
<u>$E(\text{level})^\dagger$</u>	<u>$T_{1/2}$</u>	<u>Comments</u>
0.0		
2126		
3301		
4622		
4685		
5688	38 ps 5	$T_{1/2}$: lifetime $\tau=55$ ps 7 from RDM (1976Co11).

† From $E\gamma$ data in [1976Co11](#).

 $\gamma(^{34}\text{S})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>
1002	5688	4685
1066	5688	4622
1176	3301	2126
1322	4622	3301
2126	2126	0.0
2495	4622	2126
2559	4685	2126

† From Fig. 8 of [1976Co11](#).

$^3\text{H}(^{35}\text{Cl}, \alpha\gamma)$ 1976Co11Level Scheme

$^4\text{He}(^{30}\text{Si},\alpha)$:resonances **2011No07**

$J^\pi=0^+$ for ^{30}Si ground state.

2011No07: $^{30}\text{Si}+\alpha$ elastic scattering was studied using the Thick Target Inverse Kinematics (TTIK) technique. A 150-MeV ^{30}Si beam was produced from the K-130 cyclotron at the University of Jyväskylä, Finland. The beam entered scattering chamber filled with helium gas through a Havar foil. The recoiling α particles were detected using an array of four silicon surface detectors. Measured E_α , I_α . Deduced excitation function, levels, J, widths (Γ), and partial widths (Γ_α/Γ) using a simplified R-matrix approach. Also see **2011No06**.

 ^{34}S Levels

E(level) [†]	WIDTH [‡]	L	Comments
12030	<50 keV	0	$\Gamma_\alpha/\Gamma=1.10$; $\gamma_g^2/\gamma_{sp}^2=96.6\%$.
12100	<50 keV	2	$\Gamma_\alpha/\Gamma=0.10$; $\gamma_g^2/\gamma_{sp}^2=22.4\%$.
12150	<50 keV	0	$\Gamma_\alpha/\Gamma=0.40$; $\gamma_g^2/\gamma_{sp}^2=27.3\%$.
12210	<50 keV	2	$\Gamma_\alpha/\Gamma=0.20$; $\gamma_g^2/\gamma_{sp}^2=33.7\%$.
12350	<20 keV	2	$\Gamma_\alpha/\Gamma=0.10$; $\gamma_g^2/\gamma_{sp}^2=5\%$.
12440	<30 keV	0	$\Gamma_\alpha/\Gamma=0.50$; $\gamma_g^2/\gamma_{sp}^2=11.2\%$.
12540	<30 keV	0	$\Gamma_\alpha/\Gamma=0.65$; $\gamma_g^2/\gamma_{sp}^2=12.6\%$.
12650	<50 keV	0	$\Gamma_\alpha/\Gamma=0.80$; $\gamma_g^2/\gamma_{sp}^2=20.9\%$.
12760	<50 keV	1	$\Gamma_\alpha/\Gamma=0.65$; $\gamma_g^2/\gamma_{sp}^2=19.5\%$.
12970	<50 keV	2	$\Gamma_\alpha/\Gamma=0.40$; $\gamma_g^2/\gamma_{sp}^2=15.9\%$.
13340	<50 keV	2	$\Gamma_\alpha/\Gamma=0.20$; $\gamma_g^2/\gamma_{sp}^2=4.6\%$.
13460	<40 keV	2	$\Gamma_\alpha/\Gamma=0.30$; $\gamma_g^2/\gamma_{sp}^2=4.9\%$.
13610	<50 keV	2	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=4.3\%$.
13710	<50 keV	2	$\Gamma_\alpha/\Gamma=0.15$; $\gamma_g^2/\gamma_{sp}^2=2.2\%$.
13790	<40 keV	2	$\Gamma_\alpha/\Gamma=0.30$; $\gamma_g^2/\gamma_{sp}^2=3.4\%$.
13920	<50 keV	3	$\Gamma_\alpha/\Gamma=0.05$; $\gamma_g^2/\gamma_{sp}^2=1.2\%$.
14070	<50 keV	2	$\Gamma_\alpha/\Gamma=0.80$; $\gamma_g^2/\gamma_{sp}^2=8.2\%$.
14160	<50 keV	3	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=4.8\%$.
14270	<100 keV	3	$\Gamma_\alpha/\Gamma=0.18$; $\gamma_g^2/\gamma_{sp}^2=6.1\%$.
14370	<50 keV	3	$\Gamma_\alpha/\Gamma=0.18$; $\gamma_g^2/\gamma_{sp}^2=2.8\%$.
14430	<50 keV	2	$\Gamma_\alpha/\Gamma=0.40$; $\gamma_g^2/\gamma_{sp}^2=3.1\%$.
14520	<50 keV	3	$\Gamma_\alpha/\Gamma=0.41$; $\gamma_g^2/\gamma_{sp}^2=5.4\%$.
14600	<50 keV	3	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=3.1\%$.
14730	<100 keV	3	$\Gamma_\alpha/\Gamma=0.43$; $\gamma_g^2/\gamma_{sp}^2=9.6\%$.
14840	<50 keV	3	$\Gamma_\alpha/\Gamma=0.26$; $\gamma_g^2/\gamma_{sp}^2=2.7\%$.
14930	<100 keV	3	$\Gamma_\alpha/\Gamma=0.51$; $\gamma_g^2/\gamma_{sp}^2=9.6\%$.
15020	<100 keV	3	$\Gamma_\alpha/\Gamma=0.30$; $\gamma_g^2/\gamma_{sp}^2=5.3\%$.
15120	<50 keV	3	$\Gamma_\alpha/\Gamma=0.96$; $\gamma_g^2/\gamma_{sp}^2=7.9\%$.
15170	<50 keV	3	$\Gamma_\alpha/\Gamma=0.48$; $\gamma_g^2/\gamma_{sp}^2=3.7\%$.
15220	<50 keV	3	$\Gamma_\alpha/\Gamma=0.93$; $\gamma_g^2/\gamma_{sp}^2=7.1\%$.
15300	<100 keV	4	$\Gamma_\alpha/\Gamma=0.10$; $\gamma_g^2/\gamma_{sp}^2=3\%$.
15540	<100 keV	4	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=6.3\%$.
15640	<50 keV	4	$\Gamma_\alpha/\Gamma=0.37$; $\gamma_g^2/\gamma_{sp}^2=4.1\%$.
15700	<50 keV	4	$\Gamma_\alpha/\Gamma=0.30$; $\gamma_g^2/\gamma_{sp}^2=3.4\%$.
15770	<50 keV	4	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=2.6\%$.
15850	<50 keV	4	$\Gamma_\alpha/\Gamma=0.45$; $\gamma_g^2/\gamma_{sp}^2=4.6\%$.
15950	<50 keV	4	$\Gamma_\alpha/\Gamma=0.45$; $\gamma_g^2/\gamma_{sp}^2=4.3\%$.
16060	<50 keV	4	$\Gamma_\alpha/\Gamma=0.33$; $\gamma_g^2/\gamma_{sp}^2=2.9\%$.
16190	<100 keV	4	$\Gamma_\alpha/\Gamma=0.35$; $\gamma_g^2/\gamma_{sp}^2=5.4\%$.
16340	<50 keV	4	$\Gamma_\alpha/\Gamma=0.36$; $\gamma_g^2/\gamma_{sp}^2=2.6\%$.
16440	<50 keV	4	$\Gamma_\alpha/\Gamma=0.52$; $\gamma_g^2/\gamma_{sp}^2=3.6\%$.
16550	<50 keV	4	$\Gamma_\alpha/\Gamma=0.42$; $\gamma_g^2/\gamma_{sp}^2=2.7\%$.

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$^4\text{He}(^{30}\text{Si},\alpha)$:resonances 2011No07 (continued) ^{34}S Levels (continued)

E(level) [†]	WIDTH [‡]	L	Comments
16600	<50 keV	4	$\Gamma_\alpha/\Gamma=0.18$; $\gamma_g^2/\gamma_{sp}^2=1.1\%$.
16650	<70 keV	4	$\Gamma_\alpha/\Gamma=0.31$; $\gamma_g^2/\gamma_{sp}^2=2.7\%$.
16730	<70 keV	4	$\Gamma_\alpha/\Gamma=0.30$; $\gamma_g^2/\gamma_{sp}^2=2.5\%$.
16820	<50 keV	4	$\Gamma_\alpha/\Gamma=0.08$; $\gamma_g^2/\gamma_{sp}^2=0.4\%$.
16880	<50 keV	4	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=1.3\%$.
16940	<50 keV	4	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=1.3\%$.
17010	<50 keV	4	$\Gamma_\alpha/\Gamma=0.20$; $\gamma_g^2/\gamma_{sp}^2=1\%$.
17110	<50 keV	4	$\Gamma_\alpha/\Gamma=0.20$; $\gamma_g^2/\gamma_{sp}^2=1\%$.
17180	<50 keV	4	$\Gamma_\alpha/\Gamma=0.31$; $\gamma_g^2/\gamma_{sp}^2=1.5\%$.
17250	<50 keV	4	$\Gamma_\alpha/\Gamma=0.24$; $\gamma_g^2/\gamma_{sp}^2=1.1\%$.
17330	<50 keV	4	$\Gamma_\alpha/\Gamma=0.22$; $\gamma_g^2/\gamma_{sp}^2=1\%$.
17460	<50 keV	4	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=1.1\%$.
17520	<50 keV	4	$\Gamma_\alpha/\Gamma=0.05$; $\gamma_g^2/\gamma_{sp}^2=0.2\%$.
17590	<50 keV	4	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=1\%$.
17620	<50 keV	5	$\Gamma_\alpha/\Gamma=0.05$; $\gamma_g^2/\gamma_{sp}^2=0.4\%$.
17660	<50 keV	4	$\Gamma_\alpha/\Gamma=0.12$; $\gamma_g^2/\gamma_{sp}^2=0.5\%$.
17690	<50 keV	5	$\Gamma_\alpha/\Gamma=0.11$; $\gamma_g^2/\gamma_{sp}^2=0.8\%$.
17740	<50 keV	4	$\Gamma_\alpha/\Gamma=0.03$; $\gamma_g^2/\gamma_{sp}^2=0.1\%$.
17790	<50 keV	5	$\Gamma_\alpha/\Gamma=0.18$; $\gamma_g^2/\gamma_{sp}^2=1.2\%$.
17840	<50 keV	6	$\Gamma_\alpha/\Gamma=0.17$; $\gamma_g^2/\gamma_{sp}^2=2.8\%$.
17860	<50 keV	5	$\Gamma_\alpha/\Gamma=0.21$; $\gamma_g^2/\gamma_{sp}^2=1.4\%$.
17900	<50 keV	6	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=3.9\%$.
18030	<50 keV	6	$\Gamma_\alpha/\Gamma=0.18$; $\gamma_g^2/\gamma_{sp}^2=2.6\%$.
18110	<40 keV	6	$\Gamma_\alpha/\Gamma=0.12$; $\gamma_g^2/\gamma_{sp}^2=1.3\%$.
18210	<40 keV	6	$\Gamma_\alpha/\Gamma=0.29$; $\gamma_g^2/\gamma_{sp}^2=3\%$.
18300	<50 keV	6	$\Gamma_\alpha/\Gamma=0.12$; $\gamma_g^2/\gamma_{sp}^2=1.5\%$.
18350	<50 keV	6	$\Gamma_\alpha/\Gamma=0.18$; $\gamma_g^2/\gamma_{sp}^2=2.2\%$.
18430	<50 keV	6	$\Gamma_\alpha/\Gamma=0.12$; $\gamma_g^2/\gamma_{sp}^2=13.4\%$.
18540	<50 keV	6	$\Gamma_\alpha/\Gamma=0.17$; $\gamma_g^2/\gamma_{sp}^2=1.7\%$.
18600	<50 keV	6	$\Gamma_\alpha/\Gamma=0.17$; $\gamma_g^2/\gamma_{sp}^2=1.7\%$.
18670	<50 keV	6	$\Gamma_\alpha/\Gamma=0.21$; $\gamma_g^2/\gamma_{sp}^2=2\%$.
18730	<50 keV	6	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=2.4\%$.
18860	<50 keV	6	$\Gamma_\alpha/\Gamma=0.26$; $\gamma_g^2/\gamma_{sp}^2=2.3\%$.
19080	<150 keV	6	$\Gamma_\alpha/\Gamma=0.20$; $\gamma_g^2/\gamma_{sp}^2=4.9\%$.
19200	<50 keV	6	$\Gamma_\alpha/\Gamma=0.09$; $\gamma_g^2/\gamma_{sp}^2=0.7\%$.
19280	<70 keV	6	$\Gamma_\alpha/\Gamma=0.15$; $\gamma_g^2/\gamma_{sp}^2=1.5\%$.
19370	<80 keV	6	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=2.7\%$.
19550	<50 keV	6	$\Gamma_\alpha/\Gamma=0.28$; $\gamma_g^2/\gamma_{sp}^2=1.8\%$.
19650	<50 keV	6	$\Gamma_\alpha/\Gamma=0.10$; $\gamma_g^2/\gamma_{sp}^2=0.6\%$.
19720	<50 keV	6	$\Gamma_\alpha/\Gamma=0.14$; $\gamma_g^2/\gamma_{sp}^2=0.8\%$.
19820	<50 keV	6	$\Gamma_\alpha/\Gamma=0.17$; $\gamma_g^2/\gamma_{sp}^2=1\%$.
19900	<50 keV	6	$\Gamma_\alpha/\Gamma=0.25$; $\gamma_g^2/\gamma_{sp}^2=1.4\%$.
20080	<50 keV	6	$\Gamma_\alpha/\Gamma=0.23$; $\gamma_g^2/\gamma_{sp}^2=1.2\%$.
20170	<50 keV	6	$\Gamma_\alpha/\Gamma=0.09$; $\gamma_g^2/\gamma_{sp}^2=0.4\%$.
20300	<50 keV	6	$\Gamma_\alpha/\Gamma=0.09$; $\gamma_g^2/\gamma_{sp}^2=0.4\%$.
20390	<150 keV	6	$\Gamma_\alpha/\Gamma=0.18$; $\gamma_g^2/\gamma_{sp}^2=2.5\%$.
20520	<70 keV	6	$\Gamma_\alpha/\Gamma=0.22$; $\gamma_g^2/\gamma_{sp}^2=1.4\%$.
20680	<50 keV	6	$\Gamma_\alpha/\Gamma=0.08$; $\gamma_g^2/\gamma_{sp}^2=0.3\%$.
20770	<50 keV	6	$\Gamma_\alpha/\Gamma=0.09$; $\gamma_g^2/\gamma_{sp}^2=0.4\%$.
20840	<50 keV	6	$\Gamma_\alpha/\Gamma=0.23$; $\gamma_g^2/\gamma_{sp}^2=0.9\%$.

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$^4\text{He}(^{30}\text{Si},\alpha)$:resonances **2011No07** (continued) ^{34}S Levels (continued)

E(level) [†]	WIDTH [‡]	L	Comments
21000	<80 keV	6	$\Gamma_\alpha/\Gamma=0.22$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=1.3\%$.
21140	<50 keV	6	$\Gamma_\alpha/\Gamma=0.15$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.5\%$.
21250	<50 keV	6	$\Gamma_\alpha/\Gamma=0.15$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.5\%$.
21340	<40 keV	6	$\Gamma_\alpha/\Gamma=0.27$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.7\%$.
21460	<50 keV	6	$\Gamma_\alpha/\Gamma=0.08$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.3\%$.
21570	<50 keV	6	$\Gamma_\alpha/\Gamma=0.17$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.6\%$.
21700	<50 keV	6	$\Gamma_\alpha/\Gamma=0.07$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.2\%$.
21840	<80 keV	6	$\Gamma_\alpha/\Gamma=0.16$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.8\%$.
22040	<50 keV	6	$\Gamma_\alpha/\Gamma=0.17$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.5\%$.
22230	<80 keV	6	$\Gamma_\alpha/\Gamma=0.13$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.6\%$.
22340	<50 keV	6	$\Gamma_\alpha/\Gamma=0.13$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.3\%$.
22440	<40 keV	6	$\Gamma_\alpha/\Gamma=0.05$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.1\%$.
22600	<100 keV	6	$\Gamma_\alpha/\Gamma=0.15$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.8\%$.
22720	<50 keV	6	$\Gamma_\alpha/\Gamma=0.11$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.3\%$.
22890	<40 keV	6	$\Gamma_\alpha/\Gamma=0.04$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.1\%$.
22990	<40 keV	6	$\Gamma_\alpha/\Gamma=0.05$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.1\%$.
23170	<50 keV	6	$\Gamma_\alpha/\Gamma=0.12$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.3\%$.
23320	<30 keV	6	$\Gamma_\alpha/\Gamma=0.06$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.1\%$.
23500	<100 keV	6	$\Gamma_\alpha/\Gamma=0.10$; $\gamma_\alpha^2/\gamma_{\text{sp}}^2=0.4\%$.

[†] **2011No07** states that the energy calibration is somewhat uncertain because of uncertainties in the cyclotron beam energy and the many stages of calculations when transforming the recorded spectra to the center-of-mass system. There is a small linear shift between the present data and the reference data, indicating an uncertainty of ≈ 100 keV in the absolute energy calibration.

[‡] Widths in Table 2 of **2011No07** are converted to upper limits by evaluators. Relative reduced widths= $\gamma_\alpha^2/\gamma_{\text{sp}}^2$, where γ_α^2 is the reduced α width and γ_{sp}^2 is the single-particle limit. **2011No07** states that the reported total widths represent only an upper limit, since most resonances are seen at about the experimental resolution of 50 keV. Γ_α/Γ is underestimated by 10% with 30% uncertainty. $\gamma_\alpha^2/\gamma_{\text{sp}}^2$ is overestimated by 30%. The errors are usually larger than the mean error at low energies, and smaller than the average at high energies.

$^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{S}\gamma)$ 2020Ke03,2021Ke04

$^{12}\text{C}+^{24}\text{Mg}\rightarrow^{36}\text{Ar}\rightarrow 2\text{p}+^{34}\text{S}$ fusion evaporation.

2020Ke03, 2021Ke04: a 95-MeV ^{24}Mg beam was produced from the ATLAS accelerator at ANL and bombarded a $\approx 200\text{-}\mu\text{g}/\text{cm}^2$

^{12}C target. Prompt γ rays were detected using the GRETTINA tracking array consisting of 12 modules. Fusion-evaporation residues were separated and identified by the Fragment Mass Analyzer (FMA) with a focal-plane position-sensitive parallel-grid avalanche counter (PGAC) and an ionization chamber. Measured E_γ , I_γ , $\gamma\gamma$ -coin, recoil- $\gamma(\theta)$. Deduced levels, J , π . Compared mirror nuclei ^{34}S - ^{34}Ar and shell model calculations. Calculated astrophysical $^{33}\text{Cl}(\text{p},\gamma)^{34}\text{Ar}$ reaction rate and discussed its implications for the identification of nova presolar grains from sulfur isotopic abundances.

 ^{34}S Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	Comments
0.0	0^+	
2128	2^+	
3304	2^+	
3916	0^+	
4075	1^+	
4115	2^+	
4625	3^-	
4689	4^+	
4877	3^+	
4890	2^+	
5228	0^+	
5323	2^+	J^π : from the Adopted Levels. 2^- was given in 2021Ke04 from 2012Ni10.
5381	1^+	

† From a least-squares fit to γ -ray energies.

‡ As given in 2020Ke03 and 2021Ke04 based on known assignments, unless otherwise noted. The main focus is to study the mirror nucleus ^{34}Ar .

 $\gamma(^{34}\text{S})$

$E_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. ‡	Comments
771	4075	1^+	3304	2^+		
1153	5228	0^+	4075	1^+		
1176	3304	2^+	2128	2^+		
1320	4625	3^-	3304	2^+	D	Mult.: $\Delta J=1$, dipole. $A_2=-0.23$ 3, $A_4=+0.07$ 4.
1573	4877	3^+	3304	2^+		
1586	4890	2^+	3304	2^+		
1788	3916	0^+	2128	2^+		
2077	5381	1^+	3304	2^+		
2128	2128	2^+	0.0	0^+		
2497	4625	3^-	2128	2^+		
2561	4689	4^+	2128	2^+	Q	Mult.: $\Delta J=2$, quadrupole. $A_2=+0.22$ 3, $A_4=-0.03$ 5.
2749	4877	3^+	2128	2^+		
2762	4890	2^+	2128	2^+		
3195	5323	2^+	2128	2^+		
3304	3304	2^+	0.0	0^+		
4075	4075	1^+	0.0	0^+		
4115	4115	2^+	0.0	0^+		
4890	4890	2^+	0.0	0^+		

† From Fig. 3 of 2020Ke03. No uncertainty is given. The primary focus of the study is the mirror nucleus ^{34}Ar .

Continued on next page (footnotes at end of table)

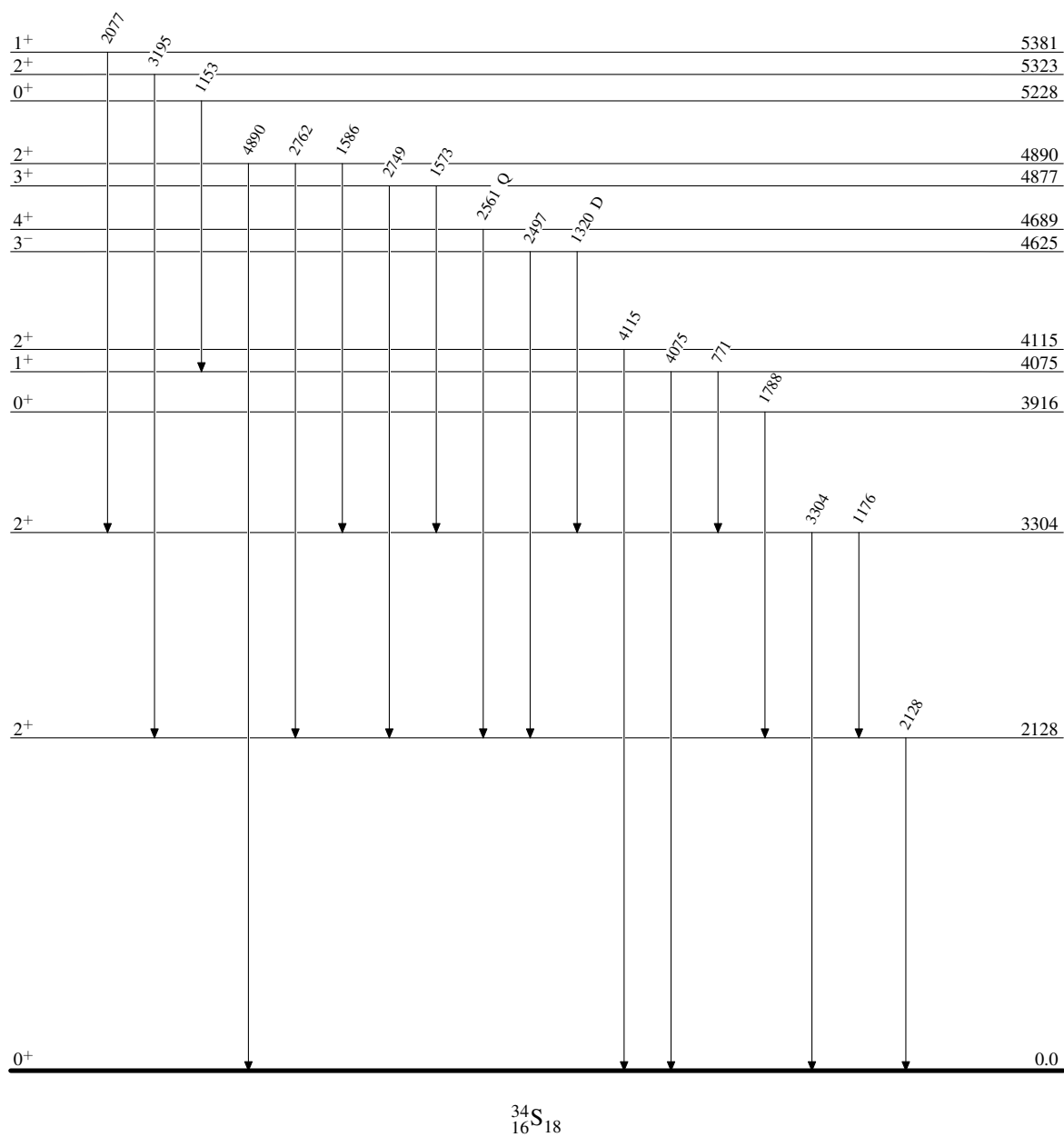
$^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{S}\gamma)$ [2020Ke03,2021Ke04](#) (continued)

$\gamma(^{34}\text{S})$ (continued)

[‡] Assigned based on Fig. 3 of [2021Ke04](#). The authors state that dipole transitions, $\Delta J=1$, have negative A_2 values while quadrupole transitions, $\Delta J=2$, have positive A_2 values.

$^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{S}\gamma)$ 2020Ke03,2021Ke04

Level Scheme

 $^{34}_{16}\text{S}_{18}$

$^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$ 2005Ma03

2005Ma03: $^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$ with a 70-MeV ^{16}O beam from the XTU Tandem accelerator at the Legnaro National Laboratory. Targets were 400- $\mu\text{g}/\text{cm}^2$ self-supporting ^{24}Mg and 750- $\mu\text{g}/\text{cm}^2$ ^{24}Mg on a 15- mg/cm^2 ^{197}Au backing. γ rays were detected using the GASP array consisting of 40 Compton-suppressed HPGe detectors and a multiplicity filter of 80 BGO scintillators. Charged particles were detected using the ISIS array consisting of 40 $\Delta\text{E-E}$ Si telescopes for channel selection and kinematical reconstruction. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, $\gamma(\theta)$, $\gamma\gamma(\theta)(\text{ADO})$, particle- γ -coin. Deduced levels, J , π , lifetimes using the Doppler-shift attenuation method (DSAM) lineshape analysis, band structures, multipolarities, and transition strengths.

 ^{34}S Levels

E(level) [†]	J^π [‡]	$T_{1/2}$	Comments
0.0 [#]	0 ⁺		
2127.7 [#] 5	2 ⁺		
3304.6 5	2 ⁺		
4624.6 [@] 6	3 ⁻		
4689.2 [#] 6	4 ⁺		
4877.0 6	3 ⁺		
5679.6 6	2 ⁺		J^π : from the Adopted Levels. 3 ⁻ was used in 2005Ma03.
5690.8 [@] 6	5 ⁻		
6251.6 ^{&} 5	4 ⁻		
7790.8 ^{&} 6	6 ⁻	132 fs 35	$T_{1/2}$: $\tau=190$ fs 50 (2005Ma03, DSAM).
8371.1 [@] 6	7 ⁻	85 fs 28	$T_{1/2}$: $\tau=122$ fs 40 (2005Ma03, DSAM).
8503.8 [#] 7	6 ⁺		
8735.0 8	6 ⁻		
8970.7 7	6 ⁻		J^π : from 2005Ma03.
9413.9 7	6 ⁻		J^π : 6 ⁻ in 2005Ma03.
9912.9 7	7 ⁺	184 fs 38	$T_{1/2}$: $\tau=265$ fs 55 (2005Ma03, DSAM).
10399.9 ^{&} 7	8 ⁻		
10651.6 [#] 7	8 ⁺	35 fs 17	$T_{1/2}$: $\tau=50$ fs 25 (2005Ma03, DSAM).
11374.3 8	8 ⁺		
11807.5 7	8 ⁺		
12141.4 7	9 ⁺	173 fs 35	$T_{1/2}$: $\tau=250$ fs 50 (2005Ma03, DSAM).
12985.5 8	(9 ⁺)		
13320.3 [@] 11	(9 ⁻)		
13341.7 8	10 ⁺	180 fs 28	$T_{1/2}$: $\tau=260$ fs 40 (2005Ma03, DSAM).
13960.6 [#] 11	(10 ⁺)		
14576.5 12	(10 ⁺)		
15244.4 10			
15281.0 ^{&} 18	(10)		
16649.1 [#] 14			

[†] From a least-squares fit to γ -ray energies in 2005Ma03.

[‡] As given in 2005Ma03, which assigns J^π based on the measured $\gamma\gamma(\theta)(\text{ADO})$ ratios. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

[#] Band(A): Ground-state band.

[@] Band(B): Band based on 3⁻, 4624.6 level.

[&] Band(C): Band based on 4⁻, 6251.6 level.

$^{24}\text{Mg}(^{16}\text{O}, \alpha 2p\gamma)$ 2005Ma03 (continued) $\gamma(^{34}\text{S})$

Expected R_{ADO} values are 0.75 for stretched dipole ($\Delta J=1$) and 1.25 for stretched quadrupole ($\Delta J=2$) transitions.

E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [†]	$\delta^\dagger$	Comments
356.3 6	0.2 1	13341.7	10 ⁺	12985.5	(9 ⁺)			
572.0 1	0.8 4	6251.6	4 ⁻	5679.6	2 ⁺			D from $R_{\text{ADO}}=0.72$ 13.
580.3 6	0.6 3	8371.1	7 ⁻	7790.8	6 ⁻	D		$R_{\text{ADO}}=0.88$ 12.
942.3 5	2.0 6	9912.9	7 ⁺	8970.7	6 ⁻	D		$A_2=-0.24$ 8, $A_4=-0.01$ 13.
986.8 9	0.8 4	10399.9	8 ⁻	9413.9	6 ⁻			
1001.6 5	42 4	5690.8	5 ⁻	4689.2	4 ⁺	D		$R_{\text{ADO}}=0.81$ 1, $A_2=-0.28$ 6, $A_4=+0.02$ 15.
1043.8 7	0.7 4	9413.9	6 ⁻	8371.1	7 ⁻			
1055.4 8	0.2 1	5679.6	2 ⁺	4624.6	3 ⁻			
1066.2 5	35 4	5690.8	5 ⁻	4624.6	3 ⁻	Q		$R_{\text{ADO}}=1.35$ 3, $A_2=+0.28$ 3, $A_4=-0.09$ 4.
1177.3 5	22.5 23	3304.6	2 ⁺	2127.7	2 ⁺			
1178 1	1.0 5	9912.9	7 ⁺	8735.0	6 ⁻			
1178 1	0.5 3	12985.5	(9 ⁺)	11807.5	8 ⁺			
1180 1	0.2 1	8970.7	6 ⁻	7790.8	6 ⁻			
1200.4 7	3.2 7	13341.7	10 ⁺	12141.4	9 ⁺	D+Q		Mult.: M1+E2 based on RUL. $R_{\text{ADO}}=0.93$ 9.
1320.1 5	27 3	4624.6	3 ⁻	3304.6	2 ⁺	D		$R_{\text{ADO}}=0.78$ 2, $A_2=-0.29$ 7, $A_4=-0.01$ 20.
1375.0 5	2.8 6	6251.6	4 ⁻	4877.0	3 ⁺			
1384.5 14	0.3 2	4689.2	4 ⁺	3304.6	2 ⁺			
1408.6 9	2.1 6	9912.9	7 ⁺	8503.8	6 ⁺	D		$R_{\text{ADO}}=0.8$ 1.
1461.7 9	3.7 8	11374.3	8 ⁺	9912.9	7 ⁺	D(+Q)		$R_{\text{ADO}}=0.87$ 7.
1489.2 6	0.7 4	12141.4	9 ⁺	10651.6	8 ⁺			
1539.6 5	5.0 10	7790.8	6 ⁻	6251.6	4 ⁻	Q		Mult.: E2 based on RUL. $A_2=+0.29$ 4, $A_4=-0.18$ 6.
1541.5 5	0.9 5	9912.9	7 ⁺	8371.1	7 ⁻			
1562.5 5	2.9 6	6251.6	4 ⁻	4689.2	4 ⁺			
1572.5 5	3.2 7	4877.0	3 ⁺	3304.6	2 ⁺			
1611.5 7	0.6 3	12985.5	(9 ⁺)	11374.3	8 ⁺			
1626.7 5	1.3 3	6251.6	4 ⁻	4624.6	3 ⁻			
1741.6 5	1.2 3	12141.4	9 ⁺	10399.9	8 ⁻			
1894.6 6	1.5 3	11807.5	8 ⁺	9912.9	7 ⁺			
1902.7 6	0.9 5	15244.4		13341.7	10 ⁺			
1966.8 9	2.6 6	13341.7	10 ⁺	11374.3	8 ⁺	Q		Mult.: E2 based on RUL. $R_{\text{ADO}}=1.2$ 1.
2028.8 6	3.3 7	10399.9	8 ⁻	8371.1	7 ⁻			
2099.6 8	27 3	7790.8	6 ⁻	5690.8	5 ⁻	D+Q	-1.8 1	Mult.: M1+E2 based on RUL. $R_{\text{ADO}}=0.37$ 1, $A_2=-0.93$ 27, $A_4=+0.38$ 69.
2122.9 6	7.0 10	9912.9	7 ⁺	7790.8	6 ⁻			
2127.5 6	100	2127.7	2 ⁺	0.0	0 ⁺	Q		$R_{\text{ADO}}=1.23$ 3, $A_2=+0.28$ 4, $A_4=-0.15$ 7.
2147.2 6	4.7 10	10651.6	8 ⁺	8503.8	6 ⁺	Q		Mult.: E2 based on RUL. $R_{\text{ADO}}=1.5$ 1, $A_2=+0.49$ 6, $A_4=-0.29$ 9.
2228.8 6	9.5 11	12141.4	9 ⁺	9912.9	7 ⁺	Q		Mult.: E2 based on RUL. $R_{\text{ADO}}=1.18$ 7.
2280.4 10	4.7 10	10651.6	8 ⁺	8371.1	7 ⁻	D		$R_{\text{ADO}}=0.74$ 4, $A_2=-0.11$ 8, $A_4=+0.02$ 14.
2333.8 7	1.2 5	12985.5	(9 ⁺)	10651.6	8 ⁺			
2375.4 7	0.4 2	5679.6	2 ⁺	3304.6	2 ⁺			
2496.5 8	12.3 12	4624.6	3 ⁻	2127.7	2 ⁺	D		$R_{\text{ADO}}=0.77$ 4, $A_2=-0.31$ 9, $A_4=-0.14$ 22.
2561.1 6	68 7	4689.2	4 ⁺	2127.7	2 ⁺	Q		$R_{\text{ADO}}=1.24$ 4, $A_2=+0.33$ 5, $A_4=-0.12$ 8.
2608.6 6	7.1 10	10399.9	8 ⁻	7790.8	6 ⁻	Q		$R_{\text{ADO}}=1.27$ 6, $A_2=+0.45$ 3, $A_4=-0.23$ 5.
2680.5 6	30 3	8371.1	7 ⁻	5690.8	5 ⁻	Q		Mult.: E2 based on RUL. $R_{\text{ADO}}=1.3$ 2, $A_2=+0.40$ 2, $A_4=-0.19$ 2.
2688.4 8	0.7 4	16649.1		13960.6	(10 ⁺)			
2749.6 6	3.6 8	4877.0	3 ⁺	2127.7	2 ⁺			
2768.9 9	0.7 4	14576.5	(10 ⁺)	11807.5	8 ⁺			
2812.7 9	7.7 14	8503.8	6 ⁺	5690.8	5 ⁻	D		$R_{\text{ADO}}=0.69$ 5, $A_2=-0.41$ 21, $A_4=-0.03$ 33.
2920.1 10	0.5 3	13320.3	(9 ⁻)	10399.9	8 ⁻			

Continued on next page (footnotes at end of table)

$^{24}\text{Mg}(^{16}\text{O},\alpha 2p\gamma)$ **2005Ma03** (continued) $\gamma(^{34}\text{S})$ (continued)

E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [†]	Comments
3002.8 6	4.1 8	11374.3	8 ⁺	8371.1 7 ⁻	7 ⁻	D	$R_{\text{ADO}}=0.65$ 5, $A_2=-0.36$ 24, $A_4=-0.03$ 33.
3044.1 6	3.1 7	8735.0	6 ⁻	5690.8 5 ⁻	5 ⁻	D+Q	$R_{\text{ADO}}=1.06$ 8.
3280.0 6	3.5 7	8970.7	6 ⁻	5690.8 5 ⁻	5 ⁻		
3304.6 7	17.8 18	3304.6	2 ⁺	0.0 0 ⁺	0 ⁺	Q	$A_2=+0.34$ 8, $A_4=-0.14$ 11.
3308.8 8	3.0 6	13960.6	(10 ⁺)	10651.6 8 ⁺	8 ⁺		
3436.1 6	1.5 6	11807.5	8 ⁺	8371.1 7 ⁻	7 ⁻	D	$R_{\text{ADO}}=0.67$ 8.
3551.2 6	0.6 3	5679.6	2 ⁺	2127.7 2 ⁺	2 ⁺		
3562.7 6	1.2 5	5690.8	5 ⁻	2127.7 2 ⁺	2 ⁺		
3722.6 6	3.3 7	9413.9	6 ⁻	5690.8 5 ⁻	5 ⁻	D+Q	$R_{\text{ADO}}=0.91$ 9.
3813.6 7	3.9 8	8503.8	6 ⁺	4689.2 4 ⁺	4 ⁺		
4880.8 16	1.8 5	15281.0	(10)	10399.9 8 ⁻	8 ⁻		
4949.3 18	1.9 4	13320.3	(9 ⁻)	8371.1 7 ⁻	7 ⁻		

[†] From **2005Ma03** measured $\gamma(\theta)$, $\gamma\gamma(\theta)(\text{ADO})$, and RUL if $T_{1/2}$ is available in this dataset.

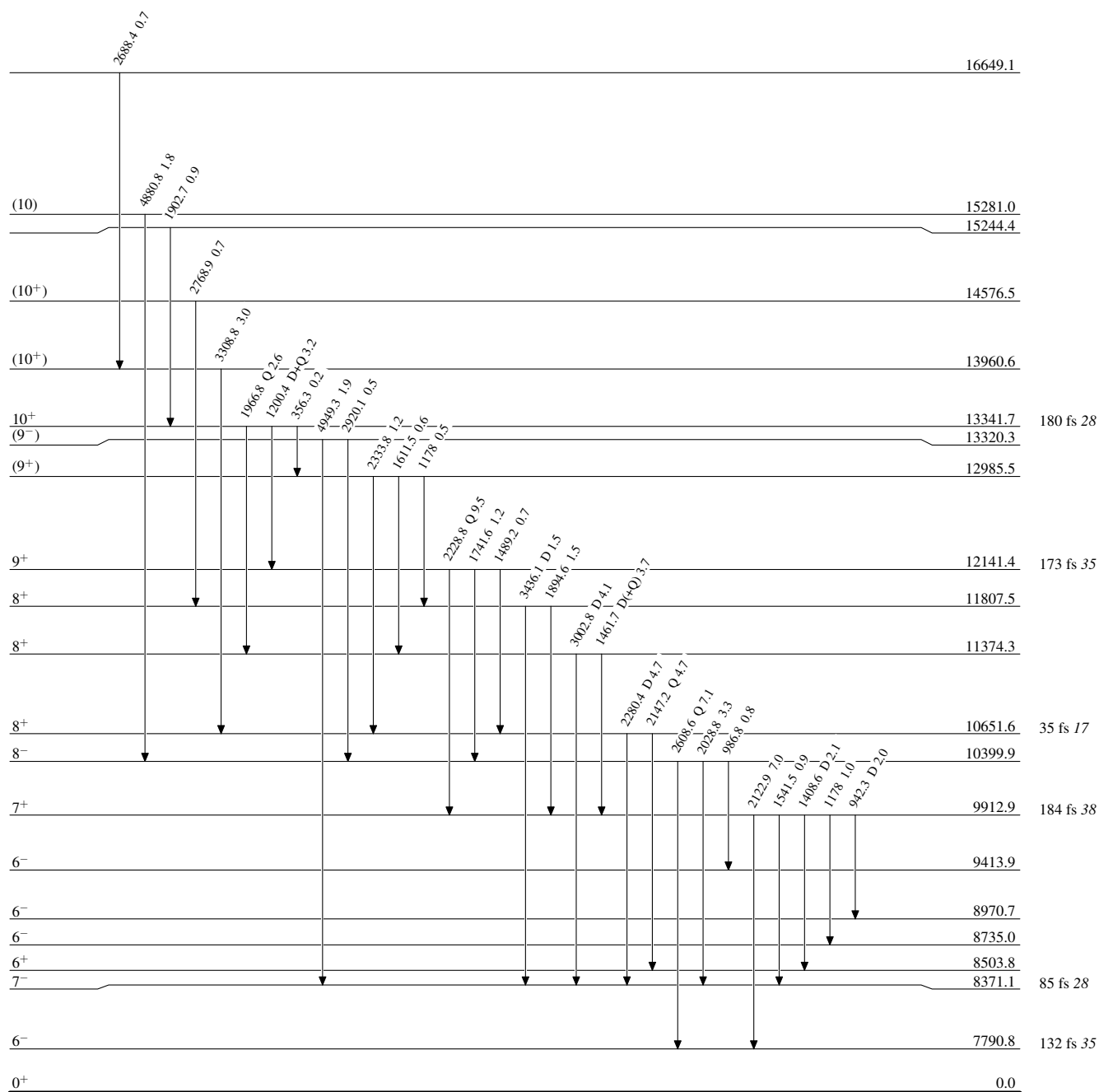
$^{24}\text{Mg}(^{16}\text{O}, \alpha 2p\gamma)$ 2005Ma03

Level Scheme

Intensities: Relative I_γ

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$

 $^{34}_{16}\text{S}_{18}$

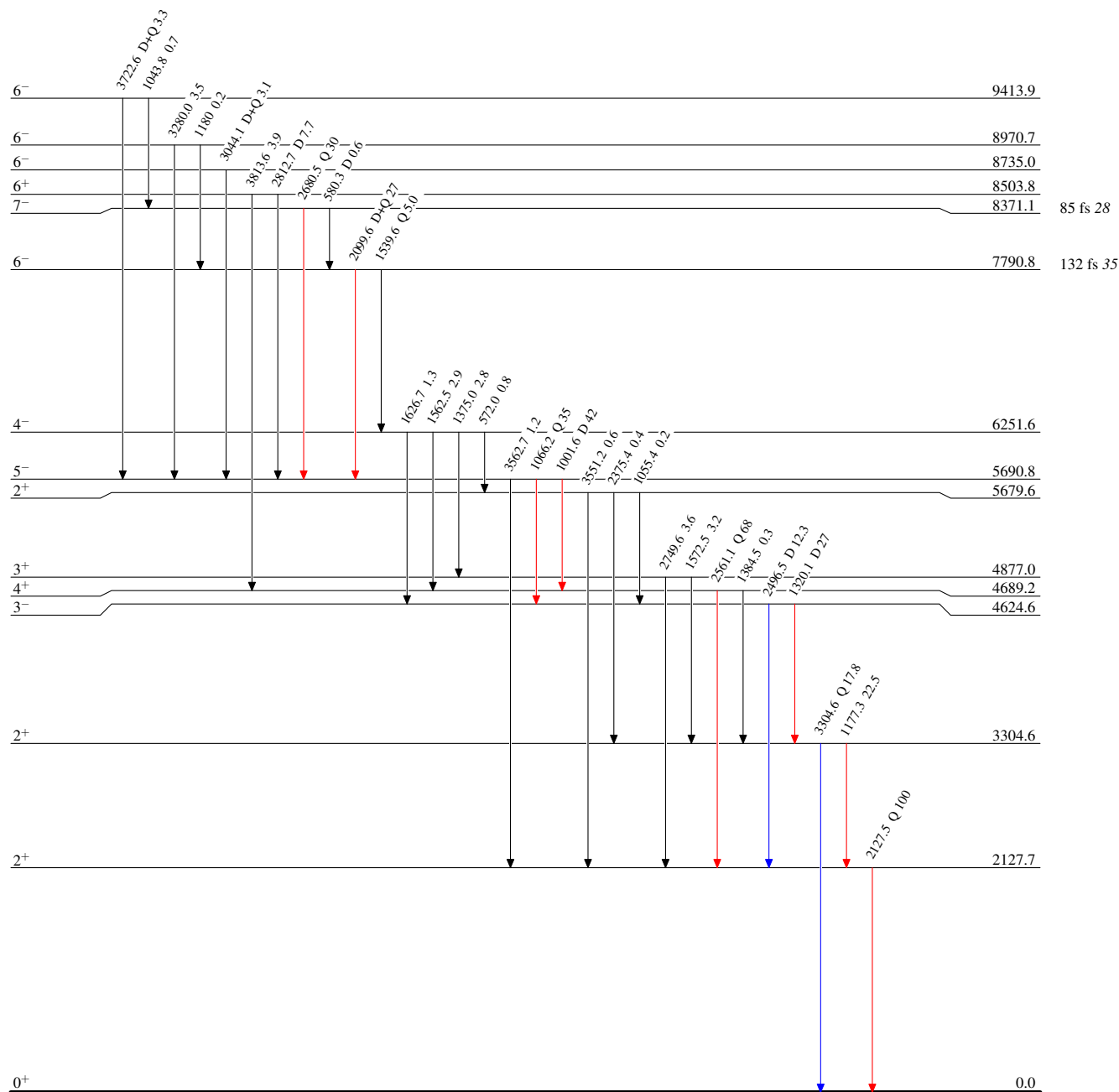
$^{24}\text{Mg}(^{16}\text{O}, \alpha 2p\gamma)$ 2005Ma03

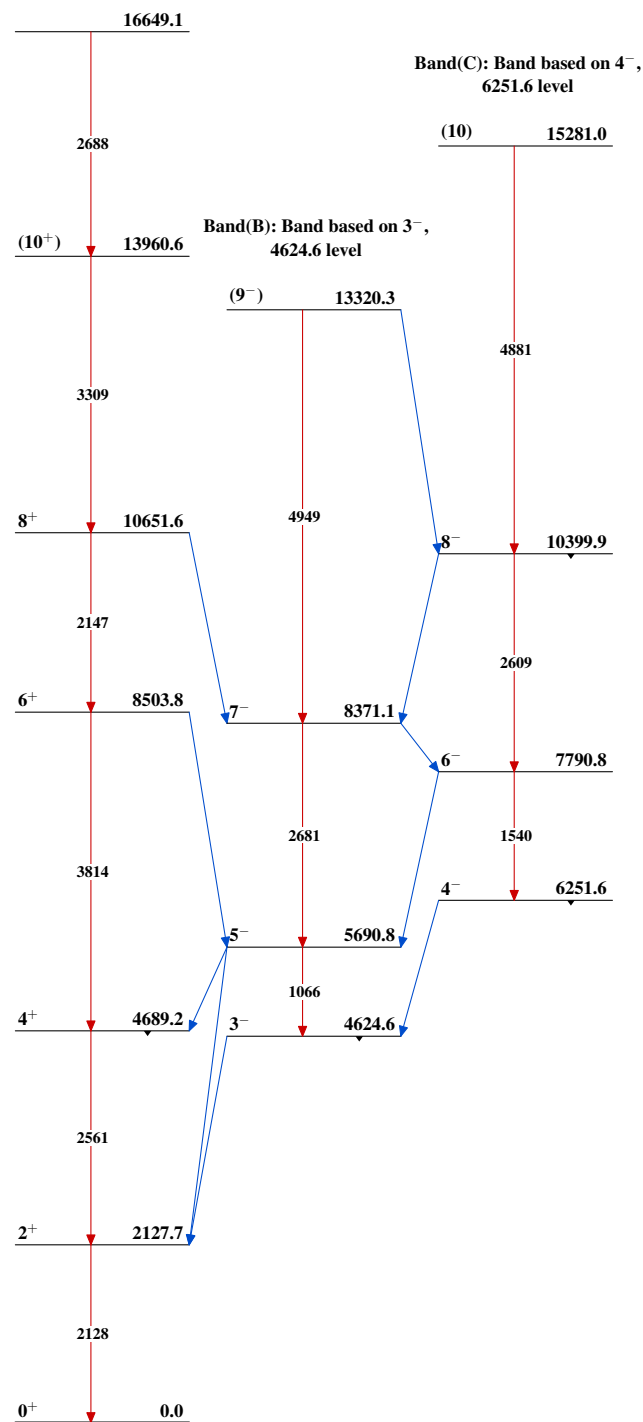
Level Scheme (continued)

Intensities: Relative I_γ

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$



$^{24}\text{Mg}(^{16}\text{O}, \alpha 2p\gamma)$ 2005Ma03Band(A): Ground-state
band $^{34}_{16}\text{S}_{18}$

$^{27}\text{Al}(^{10}\text{B},2\text{pn}\gamma)$ 2021Sh34

2021Sh34: a $^{10}\text{B}^{5+}$ beam was provided from the 6-MV tandem accelerator at the Instituto Nacional de Investigaciones Nucleares, Mexico and impinged on a $\approx 1.71 \text{ mg/cm}^2$ self-supporting ^{27}Al 99.9% isotopically pure target. Fusion-evaporation γ rays were detected using a HPGe detector at 125° . Measured E_γ , I_γ . Deduced production cross sections for evaporation residues.

 ^{34}S Levels

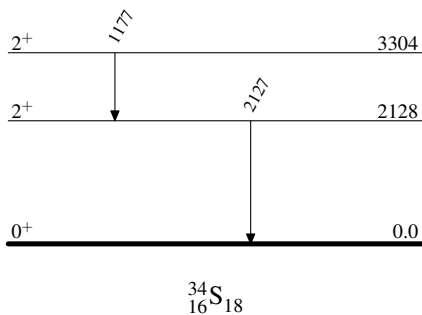
<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\dagger$</u>
0.0	0^+
2128	2^+
3304	2^+

† From the Adopted Levels. $E(\text{level})$ values are rounded to the nearest integer.

 $\gamma(^{34}\text{S})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
1177	3304	2^+	2128	2^+
2127	2128	2^+	0.0	0^+

† From the Adopted Gammas. E_γ values are rounded to the nearest integer. **2021Sh34** labels transitions with J^π on their measured γ spectrum from all reaction residues.

 $^{27}\text{Al}(^{10}\text{B},2\text{pn}\gamma)$ 2021Sh34Level Scheme

$^{27}\text{Al}(^{12}\text{C},\alpha\text{p}\gamma)$ 1976Me03

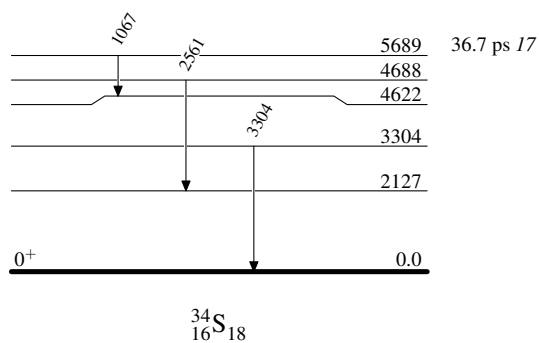
1976Me03: a 31-MeV ^{12}C beam was provided from the MP Tandem Van de Graaff accelerator at the Nuclear Research Center in Strasbourg, France. The target was a 100- $\mu\text{g}/\text{cm}^2$ self-supporting ^{27}Al target. A 7-mg/ cm^2 Au foil was used as the stopper for recoiling ions. The beam was stopped in a thick Ta foil behind the stopper. γ rays were detected using a Ge(Li) detector. Measured E_γ , I_γ . Deduced lifetimes using the recoil-distance method (RDM).

 ^{34}S Levels

<u>E(level)</u>	<u>J$^\pi$</u>	<u>T$_{1/2}$</u>	<u>Comments</u>
0.0	0 ⁺		
2127			
3304			
4622			
4688			
5689		36.7 ps 17	T $_{1/2}$: lifetime $\tau=52.9$ 24 from 1976Me03 by RDM.

 $\gamma(^{34}\text{S})$

<u>E$_\gamma$</u>	<u>E$_i$(level)</u>	<u>E$_f$</u>	<u>J$^\pi_f$</u>
1067	5689	4622	
2561	4688	2127	
3304	3304	0.0	0 ⁺

 $^{27}\text{Al}(^{12}\text{C},\alpha\text{p}\gamma)$ 1976Me03Level Scheme

$^{28}\text{Si}(^{34}\text{S}, ^{34}\text{S}'\gamma)$ 1977Sc36

1977Sc36: $^{28,29,30}\text{Si}(^{34}\text{S}, ^{34}\text{S}')$ Coulomb excitations. 49-52.9 MeV ^{34}S beams were produced from a single MP tandem Van de Graaff accelerator at the three-stage facility of Brookhaven National Laboratory. Deexcitation γ rays were detected using a 55-cm³ coaxial Ge(Li) detector at 0°. Measured E_γ , I_γ , γ -ray yields. Deduced levels, static quadrupole moments, $B(E2)$ from I_γ , and lifetimes from DSAM lineshape analysis.

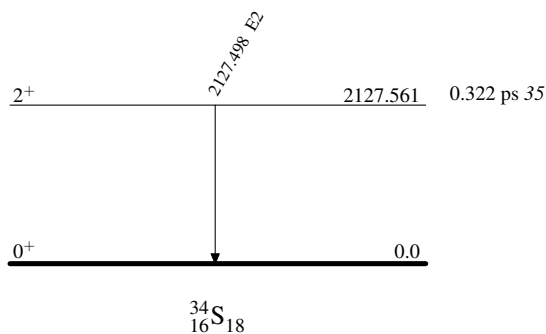
 ^{34}S Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\dagger$</u>	<u>$T_{1/2}$</u>	<u>Comments</u>
0.0	0 ⁺		
2127.561 15	2 ⁺	0.322 ps 35	$B(E2)\uparrow=0.0203$ 13 (1977Sc36) $T_{1/2}$: from $\tau=0.465$ ps 50 using DSAM lineshape analysis (Table 5 of 1977Sc36). Other: $\tau=0.469$ ps 31 from measured $B(E2)\uparrow$ (τ is given in Table 5 of 1977Sc36), but evaluators obtain $\tau=0.462$ ps 30 from the 1977Sc36 measured $B(E2)\uparrow=0.0203$ 13 (Table 9 of 1977Sc36). Static quadrupole moment $Q=+0.06$ 6 (1977Sc36).

[†] From the Adopted Levels.

 $\gamma(^{34}\text{S})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.</u>	<u>Comments</u>
2127.498 20	2127.561	2 ⁺	0.0	0 ⁺	E2	E_γ , Mult.: from the Adopted Gammas.

 $^{28}\text{Si}(^{34}\text{S}, ^{34}\text{S}'\gamma)$ 1977Sc36Level Scheme

$^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances **1967Wi01,1965Mc07,1964Va08**

The ground states of ^{30}Si , α , and ^{34}S all have $J^\pi=0^+$. The resonances in ^{34}S populated in $^{30}\text{Si}+\alpha$ have natural parity, and the γ transitions to the ^{34}S g.s. are most likely electric, such as E1 ($1^- \rightarrow 0^+$) and E2 ($2^+ \rightarrow 0^+$).

1967Wi01: $^{30}\text{Si}(\alpha,\gamma)$ and (α,n) with 3.25-4.95 MeV α beams from the 5.5 MeV Van de Graaff accelerator of the Southern Universities Nuclear Institute, South Africa. Targets were 15-40 $\mu\text{g}/\text{cm}^2$, 45%–95% enriched ^{30}Si targets on tantalum backings. γ rays were detected using NaI(Tl) detectors. Neutrons were detected using a Harwell-form neutron detector. Measured E_γ , I_γ , $\gamma(\theta)$, $\gamma\gamma(\theta)$, E_n , I_n , and yields. Deduced resonance levels, J , π , multipolarities, mixing ratios, and resonance strengths.

1965Mc07: $^{30}\text{Si}(\alpha,\gamma)$ with 3.25-4.25 MeV α beams. Work was done by the same group as **1967Wi01**.

1964Va08: $^{30}\text{Si}(\alpha,\gamma)$ with 2.1-3.25 MeV α beams. Work was done by the same group as **1967Wi01** but with the 3 MV Van de Graaff accelerator, Utrecht, Netherlands.

1975DeZS: $^{30}\text{Si}(\alpha,\gamma)$ with a 2.8-MeV α beam at Queen's University, Canada. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced 4 levels and resonance strengths.

1979Ku10: $^{30}\text{Si}(\alpha,\gamma)$ with 4-15 MeV α beams from the Stanford FN tandem Van de Graaff accelerator. Measured $\sigma(E_\gamma,\theta)$. Deduced E2 strength distributions.

 ^{34}S Levels

E(level) [†]	J^π ^b	E_α (lab) (keV) ^c	Comments
0.0 [‡]	0 ⁺ [‡]		
2127.561 [‡] 15	2 ⁺ [‡]		
3304.210 [‡] 16	2 ⁺ [‡]		
3916.40 [‡] 4	0 ⁺ [‡]		
4074.662 [‡] 19	1 ⁺ [‡]		
5679.928 [‡] 20	2 ⁺ [‡]		
9932 5	1 ⁻	2276 ^e 5	(2J+1) $\Gamma_\gamma\Gamma_\alpha/\Gamma=0.14$ eV (1964Va08).
9981 ^{&} 5	1 ⁻	2332 ^e 5	(2J+1) $\Gamma_\gamma\Gamma_\alpha/\Gamma=0.2$ eV (1964Va08).
10092 5		2457 ^e 5	
10097 ^{&} 5		2463 ^e 5	(2J+1) $\Gamma_\gamma\Gamma_\alpha/\Gamma=(0.2)$ eV (1964Va08).
10140 5		2512 ^e 5	
10169 5	1 ⁻	2545 ^e 5	(2J+1) $\Gamma_\gamma\Gamma_\alpha/\Gamma=0.7$ eV (1964Va08).
10201 5		2581 ^e 5	
10236 5		2621 ^e 5	
10249 5	1 ⁻	2635 ^e 5	(2J+1) $\Gamma_\gamma\Gamma_\alpha/\Gamma=0.9$ eV (1964Va08).
10317 5	2 ⁺	2712 ^e 5	(2J+1) $\Gamma_\gamma\Gamma_\alpha/\Gamma=0.4$ eV (1964Va08).
10382 ^a 4	1 ^{-a}		$\omega\gamma=0.05$ eV 1 (1975DeZS).
10386 ^a 4	3 ^{-a}		E(level): other: 10385 5, $E_\alpha=2790$ 5 from 1964Va08 . $\omega\gamma=0.20$ eV 4 (1975DeZS).
10408 5	2 ⁺	2815 ^e 5	(2J+1) $\Gamma_\gamma\Gamma_\alpha/\Gamma=0.2$ eV (1964Va08).
10447 5	3 ⁻	2860 ^e 5	E(level): other: 10443 4 from 1975DeZS . J^π : from 1975DeZS . $\omega\gamma=0.09$ eV 2 (1975DeZS).
10482 ^a 4	0 ⁺ ^a		$\omega\gamma=0.08$ eV 2 (1975DeZS).
10494 5	1 ⁻	2913 ^e 5	$\Gamma_\gamma=0.84$ eV (1964Va08).
			(2J+1) $\Gamma_\gamma\Gamma_\alpha/\Gamma=2$ eV (1964Va08).
10528 5		2952 ^e 5	
10587 ^{&} 5	1 ⁻	3018 ^e 5	$\Gamma_\gamma>1.3$ eV (1964Va08).
			(2J+1) $\Gamma_\gamma\Gamma_\alpha/\Gamma=4$ eV (1964Va08).
10617 5		3052 ^e 5	
10625 ^{&} 5	1 ⁻	3062 ^e 5	$\Gamma_\gamma>0.7$ eV (1964Va08).
			(2J+1) $\Gamma_\gamma\Gamma_\alpha/\Gamma=2$ eV (1964Va08).
10663 5		3104 ^e 5	
10670 5	1 ⁻	3113 ^e 5	$\Gamma_\gamma=0.73$ eV (1964Va08).
			(2J+1) $\Gamma_\gamma\Gamma_\alpha/\Gamma=2$ eV (1964Va08).
10705 5		3152 ^e 5	

Continued on next page (footnotes at end of table)

$^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances **1967Wi01,1965Mc07,1964Va08** (continued) ^{34}S Levels (continued)

E(level) [†]	J ^π ^b	E _α (lab) (keV) ^c	Comments
10768 5	2 ⁺	3223 ^e 5	(2J+1)Γ _γ Γ _α /Γ=0.7 eV (1964Va08).
10791 5	1 ⁻	3250 ^d 5	E _α (lab) (keV): other: 3249 5 (1964Va08). Γ _γ =3 eV (1965Mc07). (2J+1)Γ _γ Γ _α /Γ=9 eV (1964Va08). (2J+1)Γ _{γ0} Γ _α /Γ=7.2 eV (1965Mc07). (2J+1)Γ _{γ1} Γ _α /Γ=2.1 eV (1967Wi01).
10847 5	3 ⁻	3313 ^d 5	
10869 5		3338 ^d 5	
10895 5		3368 ^d 5	
10917 5		3392 ^d 5	
10931 5	1 ⁻	3408 ^d 5	(2J+1)Γ _{γ1} Γ _α /Γ=1.3 eV (1967Wi01).
10994 5	2 ⁺	3480 ^d 5	(2J+1)Γ _{γ1} Γ _α /Γ=1.5 eV (1967Wi01).
11015 5	2 ⁺	3503 ^d 5	(2J+1)Γ _{γ1} Γ _α /Γ=0.94 eV (1967Wi01).
11025 5	1 ⁻	3515 ^d 5	Γ _γ =1.7 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=3.9 eV (1965Mc07).
11047 5		3540 ^d 5	
11088 5	2 ⁺	3586 ^d 5	Γ _γ =0.2 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=0.5 eV (1965Mc07).
11108 5	3 ⁻	3609 ^d 5	(2J+1)Γ _{γ1} Γ _α /Γ=1.7 eV (1967Wi01).
11142 5	1 ⁻	3647 ^d 5	Γ _γ =2.6 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=6.2 eV (1965Mc07).
11165 5	1 ⁻	3674 ^d 5	Γ _γ =1.7 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=2.1 eV (1965Mc07).
11180 5		3690 ^d 5	
11194 5		3706 ^d 5	
11220 5	(2 ⁺)	3736 ^d 5	Γ _γ =0.2 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=0.2 eV (1965Mc07). 1965Mc07 states that γ intensities=419 from this resonance to ^{34}S levels higher than 3304 keV.
11233 5	1 ⁻	3751 ^d 5	Γ _γ =2.8 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=1.2 eV (1965Mc07). (2J+1)Γ _{γ1} Γ _α /Γ=1.6 eV (1967Wi01).
11272 5	2 ⁺	3795 ^d 5	
11289 5		3814 ^d 5	
11315 5	2 ⁺	3843 ^d 5	Γ _γ =0.08 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=0.2 eV (1965Mc07).
11323 5	1 ⁻	3853 ^d 5	Γ _γ =2.2 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=3.1 eV (1965Mc07).
11358 5	1 ⁻	3892 5	J ^π : from 1965Mc07. E _α (lab) (keV): other: 3894 5 (1965Mc07). Γ _γ =1.4 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=0.8 eV (1965Mc07).
11372 5	3 ⁻	3908 5	J ^π : other: 2 ⁺ (1965Mc07). E _α (lab) (keV): other: 3908 5 (1965Mc07). Γ _γ =1.5 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=0.1 eV (1965Mc07); (2J+1)Γ _{γ1} Γ _α /Γ=2.0 eV (1967Wi01).
11381 5	2 ⁺	3918 5	E _α (lab) (keV): other: 3918 5 (1965Mc07). Γ _γ =0.1 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=0.1 eV (1965Mc07). 1965Mc07 states that γ intensities=259 from this resonance to ^{34}S levels higher than 3304 keV.

Continued on next page (footnotes at end of table)

$^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances **1967Wi01,1965Mc07,1964Va08** (continued) ^{34}S Levels (continued)

E(level) [†]	J ^{πb}	E _α (lab) (keV) ^c	Comments
11398 5		3938 5	E _α (lab) (keV): other: 3940 5 (1965Mc07).
11405 5		3946 5	E _α (lab) (keV): other: 3948 5 (1965Mc07).
11420 5	1 ⁻	3962 5	E _α (lab) (keV): other: 3961 5 (1965Mc07). Γ _γ =4.4 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=10.7 eV (1965Mc07).
11435 [#] 5		3979 5	
11444 [@] 5		3990 5	E _α (lab) (keV): other: 3993 5 (1965Mc07).
11458 [@] 5	3 ⁻	4005 5	E _α (lab) (keV): other: 4007 5 (1965Mc07). (2J+1)Γ _{γ1} Γ _α /Γ=2.4 eV (1967Wi01).
11473 5	1 ⁻	4023 5	E _α (lab) (keV): other: 4024 5 (1965Mc07). (2J+1)Γ _{γ1} Γ _α /Γ=1.2 eV (1967Wi01).
11490 5	1 ⁻	4042 5	J ^π : from 1965Mc07. E _α (lab) (keV): other: 4043 5 (1965Mc07). Γ _γ =0.6 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=0.9 eV (1965Mc07). 1965Mc07 states that γ intensities=1300 from this resonance to ^{34}S levels higher than 3304 keV.
11506 5	(1 ⁻)	4060 5	E _α (lab) (keV): other: 4058 5 (1965Mc07). (2J+1)Γ _{γ1} Γ _α /Γ=0.74 eV (1967Wi01).
11545 5	1 ⁻	4104 5	J ^π : from 1965Mc07. E _α (lab) (keV): other: 4106 5 (1965Mc07). Γ _γ =1.0 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=0.9 eV (1965Mc07). 1965Mc07 states that γ intensities=1625 from this resonance to ^{34}S levels higher than 3304 keV.
11619 5		4188 5	E _α (lab) (keV): other: 4160 5 (1965Mc07).
11637 [#] 5		4208 5	
11643 5	1 ⁻	4215 5	J ^π : from 1965Mc07. E _α (lab) (keV): other: 4220 5 (1965Mc07). Γ _γ =2.3 eV (1965Mc07). (2J+1)Γ _{γ0} Γ _α /Γ=4.7 eV (1965Mc07). 1965Mc07 states that γ intensities=750 from this resonance to ^{34}S levels higher than 3304 keV.
11670 [#] 5		4246 5	
11712 5	1 ⁻	4293 5	(2J+1)Γ _{γ1} Γ _α /Γ=0.63 eV (1967Wi01).
11751 [#] 5		4338 5	
11790 5		4382 5	
11850 5		4450 5	
11859 5		4460 5	
11867 5		4469 5	
11878 5		4482 5	
11908 5		4516 5	
11922 [@] 5	(3 ⁻)	4531 5	(2J+1)Γ _{γ1} Γ _α /Γ=2.1 eV (1967Wi01).
11932 5	1 ⁻	4543 5	(2J+1)Γ _{γ0} Γ _α /Γ=5.2 eV (1967Wi01).
11957 5	3 ⁻	4571 5	(2J+1)Γ _{γ1} Γ _α /Γ=1.9 eV (1967Wi01).
11979 5		4596 5	
12034 5	1 ⁻	4658 5	(2J+1)Γ _{γ0} Γ _α /Γ=2.1 eV (1967Wi01).
12063 5		4691 5	
12077 5		4707 5	
12100 5	1 ⁻	4733 5	(2J+1)Γ _{γ0} Γ _α /Γ=1.5 eV (1967Wi01).
12137 [#] 5		4775 5	
12151 [#] 5		4791 5	

Continued on next page (footnotes at end of table)

$^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances **1967Wi01,1965Mc07,1964Va08** (continued) ^{34}S Levels (continued)

E(level) [†]	J^π ^b	$E_\alpha(\text{lab})$ (keV) ^c	Comments
12165 5		4807 5	
12173 5		4816 5	
12194 5	1 ⁻	4840 5	(2J+1) $\Gamma_{\gamma 0}\Gamma_\alpha/\Gamma=3.5$ eV (1967Wi01).
12224 5		4874 5	
12243 5		4895 5	
12256 5		4910 5	
12271 5		4927 5	
12281 5		4938 5	

[†] From $E_{\text{c.m.}}+S(\alpha)(^{34}\text{S})$, where $S(\alpha)(^{34}\text{S})=7923.64$ 5 (**2021Wa16**). $E_{\text{c.m.}}=E_\alpha(\text{lab})\times m(^{30}\text{Si})/[m_\alpha+m(^{30}\text{Si})]$.

[‡] From the Adopted Levels.

Resonances observed in (α,n) spectra only (**1967Wi01**).

@ Resonances observed in (α,γ) spectra only (**1967Wi01**).

& **1964Va08** states that there is an appreciable γ branch from this resonance to one of the levels higher than 3304 keV.

^a From **1975DeZS**. **1975DeZS** reports E(level) without giving $E_\alpha(\text{lab})$ or $S(\alpha)(^{34}\text{S})$ they used.

^b From **1964Va08** ($2276 < E_\alpha(\text{lab}) < 3249$), **1965Mc07** ($3250 < E_\alpha(\text{lab}) < 4220$), and **1967Wi01** ($3313 < E_\alpha(\text{lab}) < 4840$), unless otherwise noted. J^π are assigned by analyzing measured $\gamma(\theta)$ and $\gamma\gamma(\theta)$. In overlapping energy regions, the J^π assignments from the three experiments are in agreement.

^c From **1967Wi01**, unless otherwise noted.

^d From **1965Mc07**.

^e From **1964Va08**.

 $\gamma(^{34}\text{S})$

$E_i(\text{level})$	J_i^π	E_γ [†]	I_γ [‡]	E_f	J_f^π	Mult.&	Comments
9932	1 ⁻	7804	30 [#]	2127.561	2 ⁺		
		9930	100 [#]	0.0	0 ⁺	E1	
9981	1 ⁻	7853	100 [#]	2127.561	2 ⁺		
		9979	40 [#]	0.0	0 ⁺	E1	
10097		7968	100 [#]	2127.561	2 ⁺		
		10095	<10 [#]	0.0	0 ⁺		
10169	1 ⁻	8040	100 [#]	2127.561	2 ⁺		
		10167	30 [#]	0.0	0 ⁺	E1	
10249	1 ⁻	8120	100 [#]	2127.561	2 ⁺		
		10247	20 [#]	0.0	0 ⁺	E1	
10317	2 ⁺	8188	100 [#]	2127.561	2 ⁺		
		10315	40 [#]	0.0	0 ⁺	E2	
10382	1 ⁻	6465	100	3916.40	0 ⁺		I_γ : main γ -decay branch: 26% to 3914 level (1975DeZS).
10386	3 ⁻	4706	100	5679.928	2 ⁺		I_γ : main γ -decay branch: 40% to 5680 level (1975DeZS).
10408	2 ⁺	8279	100 [#]	2127.561	2 ⁺		
		10406	100 [#]	0.0	0 ⁺	E2	
10447	3 ⁻	7142	100	3304.210	2 ⁺		I_γ : main γ -decay branch: 51% to 3303 level (1975DeZS).
10482	0 ⁺	6407	100	4074.662	1 ⁺		I_γ : main γ -decay branch: 73% to 4073 level (1975DeZS).
10494	1 ⁻	8365	<10 [#]	2127.561	2 ⁺		
		10492	100 [#]	0.0	0 ⁺	E1	
10587	1 ⁻	7282	100 [#]	3304.210	2 ⁺	E1	

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$^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances [1967Wi01](#),[1965Mc07](#),[1964Va08](#) (continued) $\gamma(^{34}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. $\&$	$\delta^\&$	Comments
10587	1 ⁻	8458	60 [#]	2127.561	2 ⁺			
10625	1 ⁻	8496	100 [#]	2127.561	2 ⁺			
		10623	100 [#]	0.0	0 ⁺	E1		
10670	1 ⁻	7365	100 [#]	3304.210	2 ⁺	E1		
		8541	30 [#]	2127.561	2 ⁺			
		10668	<10 [#]	0.0	0 ⁺			
10768	2 ⁺	8639	100 [#]	2127.561	2 ⁺	M1+E2	+0.3	δ : from 1964Va08 .
		10766	<10 [#]	0.0	0 ⁺			
10791	1 ⁻	7486	5	3304.210	2 ⁺			
		8662	20	2127.561	2 ⁺			
		10789	100	0.0	0 ⁺	E1		
10847	3 ⁻	8718	100 [@]	2127.561	2 ⁺	E1+M2	-0.024 17	Mult.: from 1967Wi01 . $A_2=-0.289$, $A_4=-0.039$ (1967Wi01).
10931	1 ⁻	8802	100 [@]	2127.561	2 ⁺	E1+M2	+0.154 17	Mult.: from 1967Wi01 . $A_2=+0.099$, $A_4=-0.042$ (1967Wi01).
10994	2 ⁺	8865	100 [@]	2127.561	2 ⁺	M1+E2	+0.078 32	Mult.: from 1967Wi01 . $A_2=+0.339$, $A_4=+0.001$ (1967Wi01).
11015	2 ⁺	8886	100 [@]	2127.561	2 ⁺	M1+E2	-0.52 22	Mult.: from 1967Wi01 . $A_2=+0.645$, $A_4=-0.068$ (1967Wi01).
11025	1 ⁻	7720	17	3304.210	2 ⁺			
		8896	14	2127.561	2 ⁺			
		11023	100	0.0	0 ⁺	E1		
11088	2 ⁺	7783	47	3304.210	2 ⁺			
		8959	44	2127.561	2 ⁺			
		11086	100	0.0	0 ⁺	E2		
11108	3 ⁻	8979	100 [@]	2127.561	2 ⁺	E1+M2	+0.062 1	Mult.: from 1967Wi01 . $A_2=-0.457$, $A_4=-0.029$ (1967Wi01).
11142	1 ⁻	7837	9	3304.210	2 ⁺			
		9013	18	2127.561	2 ⁺			
		11140	100	0.0	0 ⁺	E1		
11165	1 ⁻	7860	100	3304.210	2 ⁺			
		9036	13.1	2127.561	2 ⁺			
		11163	76.9	0.0	0 ⁺	E1		
11220	(2 ⁺)	7915	100	3304.210	2 ⁺			
		9091	48.4	2127.561	2 ⁺			
		11218	32.3	0.0	0 ⁺	E2		
11233	1 ⁻	7928	100	3304.210	2 ⁺			
		9104	3.7	2127.561	2 ⁺			
		11231	24.4	0.0	0 ⁺	E1		
11272	2 ⁺	9143	100 [@]	2127.561	2 ⁺	M1+E2	+0.18 15	Mult.: from 1967Wi01 . $A_2=+0.188$, $A_4=+0.025$ (1967Wi01).
11315	2 ⁺	8010	67	3304.210	2 ⁺			
		9186	38	2127.561	2 ⁺			
		11313	100	0.0	0 ⁺	E2		
11323	1 ⁻	8018	48	3304.210	2 ⁺			
		9194	65	2127.561	2 ⁺			
		11321	100	0.0	0 ⁺	E1		
11358	1 ⁻	8053	100	3304.210	2 ⁺			
		9229	17.5	2127.561	2 ⁺			
		11356	35.7	0.0	0 ⁺	E1		Mult.: from 1965Mc07 .
11372	3 ⁻	8067	43.5	3304.210	2 ⁺			

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$^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances [1967Wi01](#),[1965Mc07](#),[1964Va08](#) (continued) $\gamma(^{34}\text{S})$ (continued)

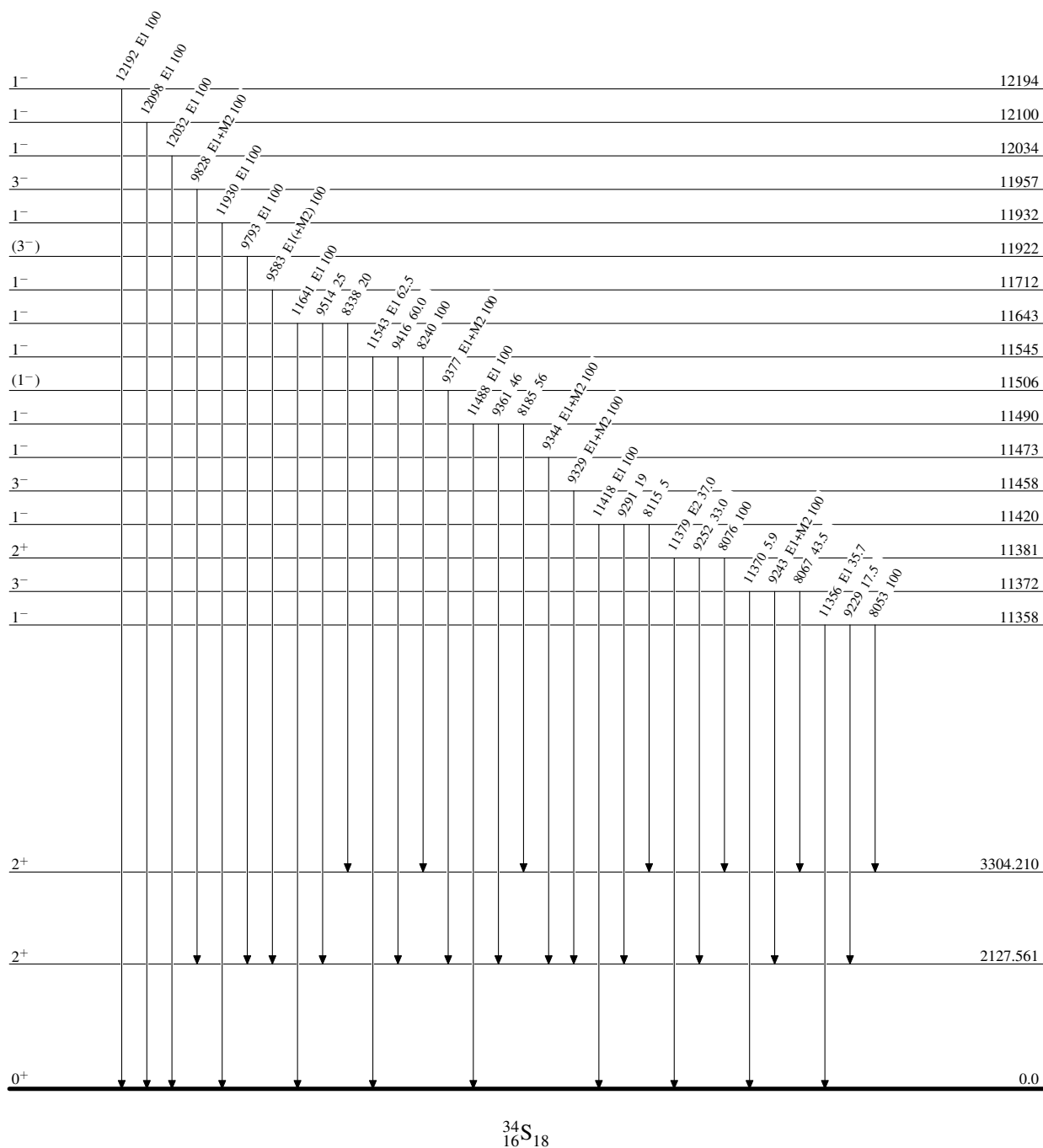
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
11372	3^-	9243	100	2127.561	2^+	E1+M2	+0.022 6	$A_2=-0.400$, $A_4=+0.007$ (1967Wi01).
		11370	5.9	0.0	0^+			
11381	2^+	8076	100	3304.210	2^+			
		9252	33.0	2127.561	2^+			
		11379	37.0	0.0	0^+	E2		
11420	1^-	8115	5	3304.210	2^+			
		9291	19	2127.561	2^+			
		11418	100	0.0	0^+	E1		
11458	3^-	9329	100@	2127.561	2^+	E1+M2	+0.037 2	$A_2=-0.405$, $A_4=-0.043$ (1967Wi01).
11473	1^-	9344	100@	2127.561	2^+	E1+M2	-0.13 7	$A_2=-0.242$, $A_4=-0.020$ (1967Wi01).
11490	1^-	8185	56	3304.210	2^+			
		9361	46	2127.561	2^+			
		11488	100	0.0	0^+	E1		
11506	(1^-)	9377	100@	2127.561	2^+	E1+M2	-0.058 16	$A_2=-0.127$, $A_4=-0.068$ (1967Wi01).
11545	1^-	8240	100	3304.210	2^+			
		9416	60.0	2127.561	2^+			
		11543	62.5	0.0	0^+	E1		
11643	1^-	8338	20	3304.210	2^+			
		9514	25	2127.561	2^+			
		11641	100	0.0	0^+	E1		
11712	1^-	9583	100@	2127.561	2^+	E1(+M2)	-0.080 80	$A_2=-0.195$, $A_4=-0.022$ (1967Wi01).
11922	(3^-)	9793	100@	2127.561	2^+	E1		δ : +0.005 21. $A_2=-0.321$, $A_4=-0.080$ (1967Wi01).
11932	1^-	11930	100@	0.0	0^+	E1		
11957	3^-	9828	100@	2127.561	2^+	E1+M2	+0.031 4	$A_2=-0.420$, $A_4=-0.009$ (1967Wi01).
12034	1^-	12032	100@	0.0	0^+	E1		
12100	1^-	12098	100@	0.0	0^+	E1		
12194	1^-	12192	100@	0.0	0^+	E1		

[†] Deduced by evaluators from level-energy difference.[‡] From [1965Mc07](#), unless otherwise noted.# From [1964Va08](#).@ From [1967Wi01](#).& From measured $\gamma(\theta)$ and $\gamma\gamma(\theta)$ in [1964Va08](#) ($2276 < E_\alpha(\text{lab}) < 3249$), [1965Mc07](#) ($3250 < E_\alpha(\text{lab}) < 4220$), and [1967Wi01](#) ($3313 < E_\alpha(\text{lab}) < 4840$). When considered in the Adopted Gammas, the firm electric multipolarities are put in parentheses if there are no other strong arguments to support the firm electric assignments.

$^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances 1967Wi01,1965Mc07,1964Va08

Level Scheme

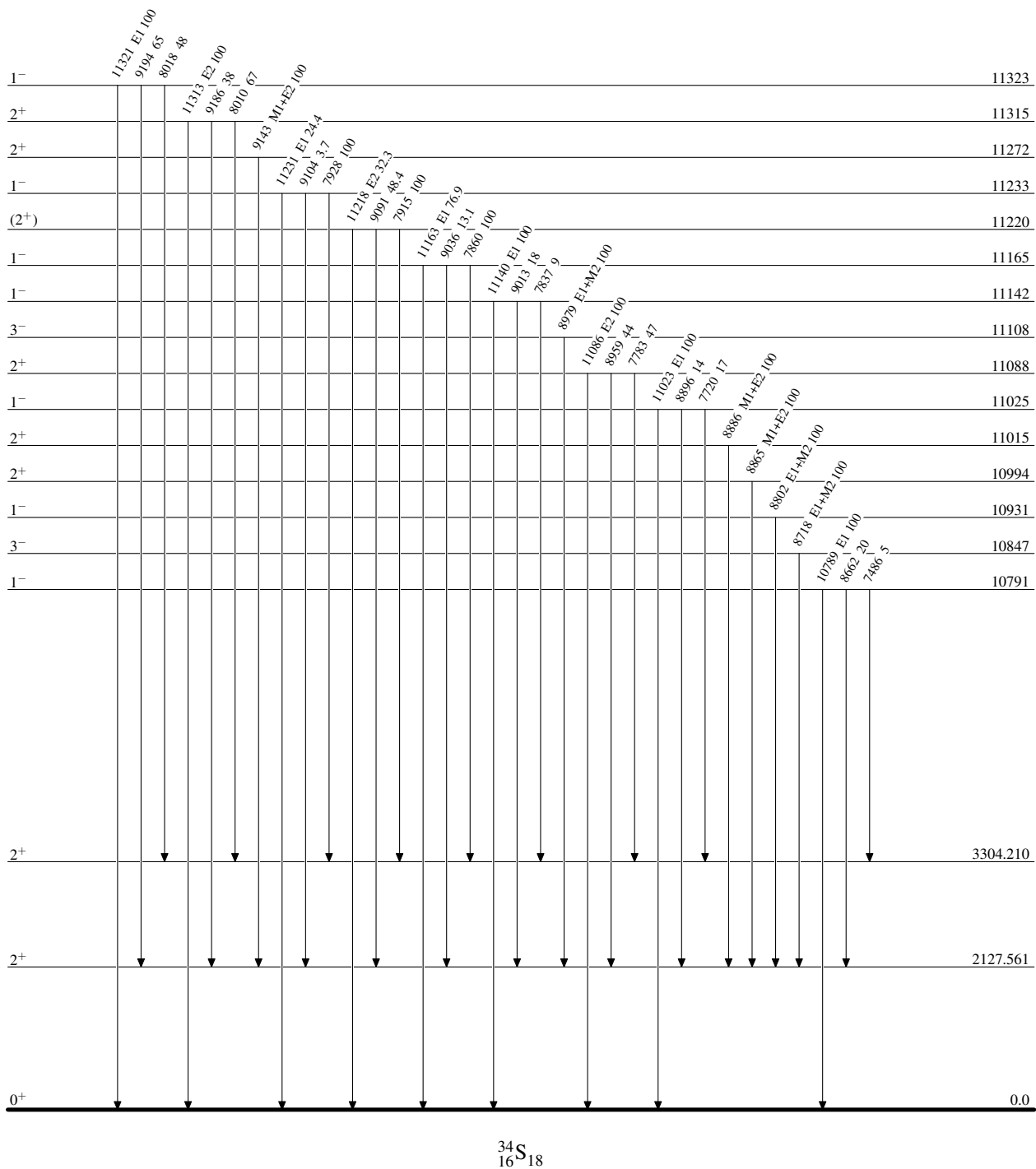
Intensities: Relative photon branching from each level



$^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances 1967Wi01,1965Mc07,1964Va08

Level Scheme (continued)

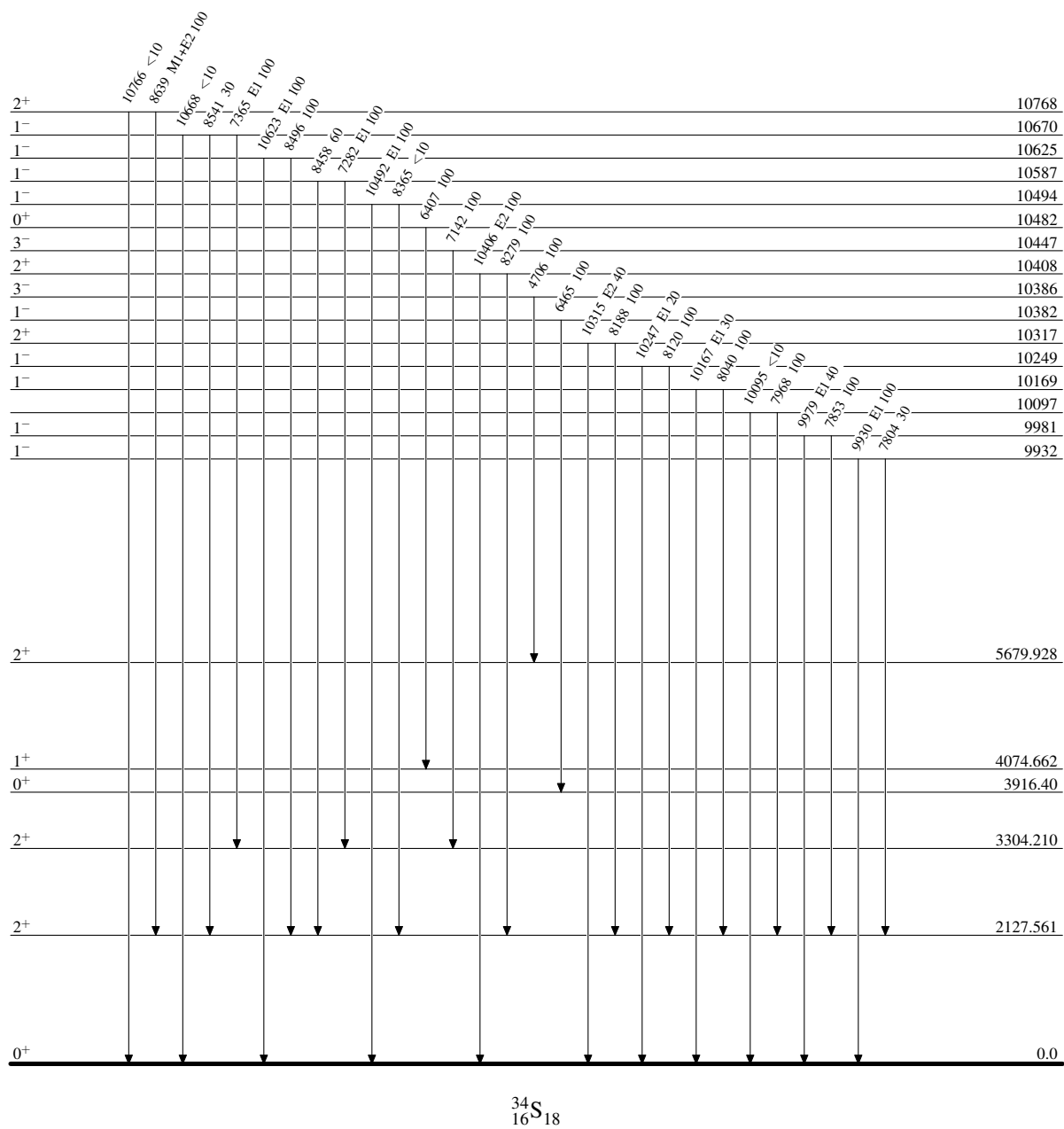
Intensities: Relative photon branching from each level



$^{30}\text{Si}(\alpha,\gamma),(\alpha,n)$:resonances 1967Wi01,1965Mc07,1964Va08

Level Scheme (continued)

Intensities: Relative photon branching from each level

 $^{34}_{16}\text{S}_{18}$

$^{31}\text{P}(\alpha, p\gamma)$ 1970Mo09, 1971Mu03, 1972Jo10

- 1970Mo09:** E=8.0–MeV α beam was provided from the Triangle Universities Nuclear Laboratory 4 MV Van de Graaff accelerator and bombarded a target of 200- $\mu\text{g}/\text{cm}^2$ Zn_3P_2 evaporated onto a 20- $\mu\text{g}/\text{cm}^2$ C backing. Protons were detected using a 100- μm annular silicon surface-barrier detector. γ rays were detected using a NaI(Tl) detector and 20- cm^3 coaxial Ge(Li) detector. Measured E_p , I_p , E_γ , I_γ , $p\gamma$ -coin, and $p\gamma(\theta)$. Deduced levels, J, π , multipolarities, and multipole mixing ratios. Also measured $^{31}\text{P}(\alpha, n\gamma)^{34}\text{Cl}$ and ^{34}Cl isomer decay.
- 1971Mu03:** E=11.0–MeV α beam was provided from the University of Liverpool tandem accelerator and bombarded a natural red phosphorus target on a C backing. Protons were detected using an annular silicon surface-barrier detector. γ rays were detected using an array of five NaI(Tl) detectors. Measured E_p , I_p , E_γ , I_γ , $p\gamma$ -coin, and $p\gamma(\theta)$. Deduced levels, J, π , γ branching ratios, multipolarities, and multipole mixing ratios.
- 1972Jo10:** E=9.3 and 11.0 MeV α beams were provided from the University of Liverpool tandem accelerator and bombarded a target of red phosphorus on a C backing. Protons were detected using an annular silicon surface-barrier detector. γ rays were detected using five NaI(Tl) detectors, one Ge(Li) detector, or five Ge(Li) detectors. Measured E_p , I_p , E_γ , I_γ , $p\gamma$ -coin, and $p\gamma(\theta)$. Deduced levels, J, π , γ branching ratios, multipolarities, and multipole mixing ratios. Reports lifetimes for higher-lying levels (>5.6 MeV) from private communications from the authors of **1970Gr11** using DSAM.
- 1970Gr11:** E=6–9 MeV α beams were provided from the University of Liverpool EN tandem accelerator and bombarded a natural red phosphorus target on an Au backing. γ rays were detected using a Ge(Li)-NaI(Tl) anti-Compton and pair escape spectrometer. Measured E_γ , I_γ . Deduced levels, γ branching ratios, and lifetimes using DSAM.
- 1970Ra17:** E=5.0–7.3 MeV α beams were provided from the Triangle Universities Nuclear Laboratory (TUNL) FN Van de Graaff accelerator and bombarded a target of gallium phosphide (GaP) on a Ni backing. γ rays were detected using a Ge(Li) coaxial detector. Measured E_γ , I_γ . Deduced lifetimes using DSAM.
- 1974Gr06:** E=7.2–8.0, 10.8, 11.0 MeV α beams were provided from the University of Freiburg 7 MV Van de Graaff accelerator and bombarded a target of Zn_3P_2 on a Ta or Ni backing. Protons were detected using an annular silicon surface-barrier detector. γ rays were detected using a Ge(Li) detector or two NaI(Tl) detectors. Measured E_p , I_p , E_γ , I_γ , $p\gamma$ -coin, $\gamma\gamma$ -coin, $\gamma\gamma(\theta)$, and $p\gamma(\theta)$. Deduced levels, J, π , γ branching ratios, multipolarities, multipole mixing ratios, and lifetimes from DSAM lineshape analysis. **1977GrZH:** an abstract by the same group of **1974Gr06** using $E_\alpha=13.9$ MeV at University of Freiburg. Measured $p\gamma(\theta)$ for higher-lying levels. **1969Gr03:** $E_\alpha=5.90$ MeV at University of Freiburg. Measured lifetimes of two ^{34}S levels from DSAM lineshape analysis.
- 1970Br18:** E=4.67–MeV α beam from the 5.5 MV Van de Graaff generator at the National Laboratory of Legnaro, Padova. Targets were red phosphorus on C foils. Deduced lifetimes of two ^{34}S levels using DSAM.
- 1970Cu02:** E=8.05, 8.14, 8.35 MeV α beams from the Atomic Energy Research Establishment (AERE), UK. Targets were red phosphorus on Ni and Ti backing foils. Protons were detected using an annular silicon surface-barrier detector. γ rays were detected using a NaI(Tl) detector and a Ge(Li) detector. Deduced lifetimes of two ^{34}S levels from DSAM lineshape analysis.
- 1971Gr26:** E=9.5 MeV α beam bombarded a target of red phosphorus on a Au backing. γ rays were detected using a single planar Ge(Li) polarimeter and a Ge(Li) detector. Measured E_γ , I_γ , $\gamma(\theta)$, and linear polarization. Deduced the level at 5.689 MeV has $J^\pi=5^-$ and its deexcitation γ multipolarities and multipole mixing ratios.
- 1972Gr15:** E=9.5 MeV α beam was provided from the Chalk River MP tandem accelerator and bombarded a target of InP on a stretched Au backing. γ rays were detected using a Ge(Li) detector. Deduced the lifetime of the 5.689-MeV level using the recoil-distance Doppler-shift method.
- 1979Ba54:** E=6.30 and 7.00 MeV α beams were provided from the Frankfurt 7.5 MV Van de Graaff accelerator. γ rays were detected using four cylindrical Ge(Li) detectors. Measured E_γ , I_γ , $\gamma(\theta)$, and polarization. Deduced multipolarities and multipole mixing ratios.
- 1971So01:** E=7 MeV α beam was provided from the Heidelberg EN Tandem Van de Graaff accelerator and bombarded a target of Zn_3P_2 on Au, Cu, and C backings. γ rays were detected using a Ge(Li) detector. Deduced lifetimes of two ^{34}S levels using DSAM lineshape analysis.
- 1988Mc05:** $^{31}\text{P}(\alpha, p)$ with E=1.7–5.5 MeV α beams from the 5.5-MV CN Van de Graaff accelerator at the University of Arizona. Measured proton yields.
- 1975So11:** $^{31}\text{P}(\alpha, p)$ with E=7.60 MeV α beam from the 5.5-MV Van de Graaff accelerator at Louis Pasteur University, Strasbourg, France. The target was red phosphorus on Au backings. Protons were detected using a 300- μm -thick silicon surface-barrier annular counter. Electron-positron pairs were detected using a ΔE -E telescope of plastic scintillators. Deduced $E0$ pair branching ratios.
- 1962Li07:** $^{31}\text{P}(\alpha, p)$ with E=42 MeV α beam from the University of Washington cyclotron. The target was red phosphorus on polystyrene backings. Protons were detected using a ΔE -E telescope of plastic and CsI(Tl) detectors. Measured $\sigma(E_p, \theta)$.

$^{31}\text{P}(\alpha, p\gamma)$ **1970Mo09, 1971Mu03, 1972Jo10** (continued) ^{34}S Levels

E(level) [†]	J ^π @	T _{1/2}	Comments
0.0	0 ⁺		$^{31}\text{P}(\alpha, p)$ $\sigma_{\text{total}}=17.5 +6-5 \mu\text{b}$ at E=42 MeV (1962Li07).
2127.0 5	2 ⁺	0.282 ps 14	E(level): others: 2126.8 8 (1970Gr11), 2127 1 (1974Gr06). J ^π : spin=2 from $\text{py}(\theta)$ in 1970Mo09 and 1971Mu03. $\pi=+$ from 2126.7 γ , E2 to 0 ⁺ , g.s. T _{1/2} : lifetime $\tau=407$ fs 20: weighted average of 350 fs 60 (1969Gr03, DSAM), 460 fs 95 (1970Br18, DSAM), 467 fs 90 (1970Cu02, DSAM), 440 fs 50 (1970Gr11, DSAM), 400 fs 32 (1970Ra17, DSAM), and 400 fs 40 (1974Gr06, DSAM).
3303.2 4	2 ⁺	112 fs 9	$^{31}\text{P}(\alpha, p)$ $\sigma_{\text{total}}=20 +8-6 \mu\text{b}$ at E=42 MeV (1962Li07). E(level): others: 3303.0 9 (1970Gr11), 3303 (1971Mu03), 3304 2 (1974Gr06), 3300 3 (1969Gr03). J ^π : spin=2 from $\text{py}(\theta)$ in 1970Mo09 and 1971Mu03. $\pi=+$ from 3303.1 γ , E2 to 0 ⁺ , g.s. T _{1/2} : lifetime $\tau=161$ fs 13: weighted average of 120 fs 30 (1969Gr03, DSAM), 190 fs 40 (1970Br18, DSAM), 144 fs 28 (1970Cu02, DSAM), 218 fs 30 (1970Gr11, DSAM), 175 fs 25 (1970Ra17, DSAM), and 145 fs 20 (1974Gr06, DSAM).
3913.0	(0,1,2)	1.12 ps 9	E(level): others: 3915.1 9 (1970Gr11), 3915 (1971Mu03), 3914 2 (1974Gr06). J ^π : spin=0,1,2,3,4 was used for $\text{py}(\theta)$ analysis in 1970Mo09, in which spin=3,4 has lower confidence levels. T _{1/2} : lifetime $\tau=1618$ fs 126: weighted average of 1600 fs 130 (1970Gr11, DSAM) and 1890 fs 500 (1970Ra17, DSAM). Other: ≥ 400 fs (1974Gr06, DSAM).
4071.8 10	1 ⁺	<17 fs	E(level): others: 4075 2 (1970Gr11), 4075 (1971Mu03), 4078 3 (1974Gr06). J ^π : spin=1 from $\text{py}(\theta)$ in 1970Mo09, 1971Mu03, and 1974Gr06. $\pi=+$ from 1945.5 γ , M1+E2 to 2 ⁺ , 2127.0 level. T _{1/2} : lifetime $\tau < 24$ fs (1970Ra17, DSAM). Others: <33 fs (1970Gr11, DSAM), ≤ 50 fs (1974Gr06, DSAM).
4114.0 9	2 ⁺	72 fs 6	E(level): others: 4116 2 (1970Gr11), 4116 (1971Mu03), 4119 3 (1974Gr06). J ^π : spin=2 from $\text{py}(\theta)$ in 1971Mu03 and 1974Gr06. $\pi=+$ from 4113.7 γ , E2 to 0 ⁺ , g.s. T _{1/2} : lifetime $\tau=104$ fs 8: weighted average of 89 fs 20 (1970Gr11, DSAM), 110 fs 10 (1970Ra17, DSAM), and 100 fs 15 (1974Gr06, DSAM).
4622.4	3 ⁻	84 fs 6	E(level): others: 4623 2 (1970Gr11), 4623 (1971Mu03), 4622 2 (1974Gr06). J ^π : spin=3 from $\text{py}(\theta)$ in 1971Mu03. Spin ≤ 4 from $\text{py}(\theta)$ in 1970Mo09. $\pi=-$ from 1066 γ , E2 from 5 ⁻ , 5688 level. T _{1/2} : lifetime $\tau=121$ fs 8: weighted average of 125 fs 20 (1970Gr11, DSAM), 135 fs 17 (1970Ra17, DSAM), 145 fs 50 (1971So01, DSAM), and 115 fs 10 (1974Gr06, DSAM).
4687.6	4 ⁺	88 fs 4	E(level): others: 4688 2 (1970Gr11), 4688 (1971Mu03), 4687 2 (1974Gr06), 4688 1 (1977GrZH). J ^π : spin=4 from $\text{py}(\theta)$ in 1971Mu03 and $\gamma\gamma(\theta)$ in 1974Gr06. $\pi=+$ from 2560.5 γ , E2 to 2 ⁺ , 2127.0 level. T _{1/2} : lifetime $\tau=127$ fs 6: weighted average of 132 fs 15 (1970Gr11, DSAM), 131 fs 13 (1970Ra17, DSAM), 110 fs 20 (1971So01, DSAM), 125 fs 10 (1974Gr06, DSAM), and 130 fs 20 (1977GrZH, DSAM).
4875.2	3 ⁺	40 fs 15	E(level): others: 4874 2 (1970Gr11), 4874 (1971Mu03), 4874 4 (1974Gr06). J ^π : spin=3 from $\text{py}(\theta)$ in 1971Mu03. $\pi=+$ from 2750.9 γ , M1+E2 to 2 ⁺ , 2127.0 level. T _{1/2} : lifetime $\tau=57$ fs 22 (1970Ra17, DSAM). Others: <85 fs (1970Gr11, DSAM), ≤ 70 fs (1974Gr06, DSAM).
4891.1 30	2 ⁺	36 fs 10	E(level): weighted average of 4889.6 40 (1970Mo09) and 4892 3 (1970Gr11). Other: 4892 (1971Mu03). J ^π : spin=2 from $\text{py}(\theta)$ in 1970Mo09 and 1971Mu03. $\pi=+$ from 4890.7 γ , E2 to 0 ⁺ , g.s. T _{1/2} : lifetime $\tau=52$ fs 14 (1970Ra17, DSAM). Other: <40 fs (1970Gr11, DSAM).
5225	0		E(level): from 1971Mu03. J ^π : spin=0 based on nonobservation of γ to 0 ⁺ , g.s. and $\gamma\gamma(\theta)$ is isotropic (1971Mu03).
5318.8 18	2 ⁺	17 fs 7	E(level): weighted average of 5318 2 (1970Gr11) and 5322 4 (1974Gr06). Other: 5318 (1971Mu03). J ^π : spin=2 from $\text{py}(\theta)$ in 1971Mu03. $\pi=+$ from 3191.6 γ , M1+E2 to 2 ⁺ , 2127.0 level. T _{1/2} : lifetime $\tau=24$ fs 10 (1970Gr11, DSAM). Other: ≤ 40 fs (1974Gr06, DSAM).
5382 4	1 ⁺	≤ 49 fs	E(level): from 1974Gr06. Other: 5380 (1971Mu03). J ^π : spin=1 from $\text{py}(\theta)$ in 1971Mu03. $\pi=+$ from 3255 γ , M1+E2 to 2 ⁺ , 2127.0 level.

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$^{31}\text{P}(\alpha, \text{p}\gamma)$ **1970Mo09, 1971Mu03, 1972Jo10 (continued)** ^{34}S Levels (continued)

E(level) [†]	J ^π @	T _{1/2}	Comments
5679 [‡] 1	2 ⁺	263 fs 21	T _{1/2} : lifetime $\tau \leq 70$ fs (1974Gr06, DSAM). E(level): other: 5680 (1972Jo10). J ^π : spin=2 from $\text{p}\gamma(\theta)$ in 1972Jo10. Other: 3 ⁻ , (2 ⁻) from 1977GrZH. $\pi=+$ from 2376 γ , M1+E2 to 2 ⁺ , 3303.2 level.
5688 4	5 ⁻	37.4 ps 35	T _{1/2} : lifetime $\tau=380$ fs 30 (1972Jo10, DSAM). E(level): from 1974Gr06. Other: 5689 (1972Jo10). J ^π : spin=5 from $\gamma(\theta)$ and $\gamma(\theta)$ (POL) in 1971Gr26. Spin=3,5 from $\text{p}\gamma(\theta)$ in 1972Jo10 and 1974Gr06. $\pi=-$ from 1000 γ , E1 to 4 ⁺ , 4687.6 level. T _{1/2} : lifetime $\tau=54$ ps 5 (1972Gr15, RDM). Others: ≥ 0.5 ps (1974Gr06, DSAM), 54 ps 6 from private communications between the authors of 1972Jo10 and 1972Gr15.
5753 [#]			
5848 [#]			
5993 [#]	2 ⁺	<6.9 fs	J ^π : spin=2 from $\text{p}\gamma(\theta)$ in 1972Jo10. $\pi=+$ from 5992 γ , E2 to 0 ⁺ , g.s. T _{1/2} : lifetime $\tau < 10$ fs (1972Jo10, DSAM).
6120 [#]	1,2	<52 fs	J ^π : spin=1,2 from $\text{p}\gamma(\theta)$ in 1972Jo10. T _{1/2} : lifetime $\tau < 75$ fs (1972Jo10, DSAM).
6173 [#]	3 ⁺	<9.0 fs	J ^π : spin=3 from $\text{p}\gamma(\theta)$ in 1972Jo10. $\pi=+$ from 2870 γ , M1+E2 to 2 ⁺ , 3303.2 level. T _{1/2} : lifetime $\tau < 13$ fs (1972Jo10, DSAM).
6251 [‡] 2	4 ⁺	0.42 ps +49-21	E(level): other: 6250 (1972Jo10). J ^π : spin=4 from $\text{p}\gamma(\theta)$ in 1972Jo10. Other: 4 ⁻ , (3 ⁻) from 1977GrZH. $\pi=+$ from 1376 γ , M1+E2 to 3 ⁺ , 4875.2 level. T _{1/2} : lifetime $\tau=390$ fs +70-40 (1972Jo10, DSAM). Other: $\tau=600$ fs +700-300 (1977GrZH, DSAM).
6347 [#]	1	<25 fs	J ^π : from $\text{p}\gamma(\theta)$ in 1972Jo10. T _{1/2} : lifetime $\tau < 36$ fs (1972Jo10, DSAM).
6415 [#]	4	<6.9 fs	J ^π : spin=2,4 from $\text{p}\gamma(\theta)$ in 1972Jo10. Spin=2 is ruled out because either E2 or M2 for 1727 γ to 4 ⁺ , 4687.6 level would be ruled out by RUL. T _{1/2} : lifetime $\tau < 10$ fs (1972Jo10, DSAM).
6480 [#]	1		J ^π : from $\text{p}\gamma(\theta)$ in 1972Jo10.
6639 [‡] 1	1 ⁻ , 2, 3	42 fs 10	E(level): other: 6640 (1972Jo10). J ^π : spin=2,3,4 from $\text{p}\gamma(\theta)$ 1972Jo10. 960 γ to 2 ⁺ , 5679 level cannot be M2 or E2 and 2017 γ to 3 ⁻ , 4622.4 level cannot be M2, based on level T _{1/2} and RUL. Other: 4 ⁻ from 1977GrZH with no supporting evidence. T _{1/2} : lifetime $\tau=60$ fs 15 (1977GrZH, DSAM).
6731 [#]	2 ⁽⁺⁾ , 4 ⁽⁺⁾	<6.9 fs	J ^π : spin=2,4 from $\text{p}\gamma(\theta)$ in 1972Jo10. $\pi=(+)$ from 4604 γ to 2 ⁺ , 2127.0 level and 2043 γ to 4 ⁺ , 4687.6 level. T _{1/2} : lifetime $\tau < 10$ fs (1972Jo10, DSAM).
6830 [#]	2	<46 fs	J ^π : spin=2 from $\text{p}\gamma(\theta)$ in 1972Jo10. T _{1/2} : lifetime $\tau < 67$ fs (1972Jo10, DSAM).
6864 [‡] 1	2 ⁽⁺⁾ , 4 ⁽⁺⁾	28 fs 7	E(level): other: 6864 (1972Jo10). J ^π : spin=2,4 from $\text{p}\gamma(\theta)$ in 1972Jo10. $\pi=(+)$ from 2176 γ to 4 ⁺ , 4687.6 level and 4737 γ to 2 ⁺ , 2127.0 level. 5 ⁻ from 1977GrZH is considered unlikely because of the presence of 4737 γ to 2 ⁺ , 2127.0 level. T _{1/2} : lifetime $\tau=40$ fs 10: weighted average of 39 fs 10 (1977GrZH, DSAM) and 70 fs 50 (1972Jo10, DSAM).
6890 [‡] 1	(3,4)	<14 fs	E(level): other: 6898 (1972Jo10). J ^π : (3,4) ⁺ from 1977GrZH. T _{1/2} : lifetime $\tau < 20$ fs (1977GrZH, DSAM).
6950 [#]	2		J ^π : spin=2 from $\text{p}\gamma(\theta)$ in 1972Jo10.
7112 [#]	2	<6.9 fs	J ^π : spin=2 from $\text{p}\gamma(\theta)$ in 1972Jo10.

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$^{31}\text{P}(\alpha, p\gamma)$ **1970Mo09, 1971Mu03, 1972Jo10** (continued) ^{34}S Levels (continued)

$E(\text{level})^\dagger$	$J^\pi@$	$T_{1/2}$	Comments
7220 [#]	2 ⁺	<13 fs	$T_{1/2}$: lifetime $\tau < 10$ fs (1972Jo10, DSAM). J^π : spin=2 from $p\gamma(\theta)$ in 1972Jo10. $\pi=+$ from 7219 γ , E2 to 0 ⁺ , g.s. $T_{1/2}$: lifetime $\tau < 18$ fs (1972Jo10, DSAM).
7248 [‡] 2	(4)	14 fs 7	E(level): other: 7245 (1972Jo10). J^π : (2,4) ⁺ from 1977GrZH; 1560 γ to 5 ⁻ , 5688 level (1972Jo10). $T_{1/2}$: lifetime $\tau=20$ fs 10 (1977GrZH, DSAM).
7392 [‡] 1	(4,5)	159 fs 35	J^π : 5 ⁻ , (4 ⁻ , 5 ⁺) from 1977GrZH. $T_{1/2}$: lifetime $\tau=230$ fs 50 (1977GrZH, DSAM).
7629 [‡] 1		14 fs 7	$T_{1/2}$: lifetime $\tau=20$ fs 10 (1977GrZH, DSAM).
7788 [‡] 1	6	80 fs 24	J^π : 6 ⁻ from 1977GrZH. $T_{1/2}$: lifetime $\tau=115$ fs 35 (1977GrZH, DSAM).
8083 [‡] 1	5	44 fs 7	J^π : 5 ⁻ from 1977GrZH. $T_{1/2}$: lifetime $\tau=64$ fs 10 (1977GrZH, DSAM).
8293 [‡] 2	4	<28 fs	J^π : 4 from 1977GrZH. $T_{1/2}$: lifetime $\tau < 40$ fs (1977GrZH, DSAM).
8369 [‡] 1	(5,7)	83 fs 14	J^π : 7 ⁻ , (5 ⁻) from 1977GrZH. $T_{1/2}$: lifetime $\tau=120$ fs 20 (1977GrZH, DSAM).
8502 [‡] 2	(4,6)	28 fs 7	J^π : (4,6 ⁺) from 1977GrZH. $T_{1/2}$: lifetime $\tau=40$ fs 10 (1977GrZH, DSAM).

[†] From a least-squares fit to γ -ray energies for levels connected with γ transitions, unless otherwise noted.

[‡] From 1977GrZH, unless otherwise noted.

[#] From 1972Jo10, unless otherwise noted.

@ From measured $p\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ in 1970Mo09, 1971Mu03, 1972Jo10, and/or 1974Gr06 given under comments. 1977GrZH is an abstract that reports spins and parities for higher-lying levels but lacks details. Evaluators adopt the reported spins considering that 1977GrZH measured $p\gamma(\theta)$.

 $\gamma(^{34}\text{S})$

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\ddagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. &	Comments
2127.0	2 ⁺	2126.7 8	100	0.0	0 ⁺	E2	E_γ : weighted average of 2126.8 8 (1970Mo09), 2126.7 8 (1970Gr11), 2127 1 (1974Gr06), and 2126 1 (1969Gr03). Other: 2127 (1971Mu03). I_γ : from 1970Mo09, 1970Gr11, 1971Mu03, and 1974Gr06. Mult.: Q from $p\gamma(\theta)$ in 1970Mo09 and 1971Mu03. Electric character from POL=+0.37 4 (1979Ba54, detector system 1); POL=+0.46 4 (1979Ba54, detector system 2). $A_2=+0.63$ 1, $A_4=-0.93$ 1 (1971Mu03). $A_2=+0.25$ 2, $A_4=-0.10$ 2 (1979Ba54, detector system 1). $A_2=+0.29$ 2, $A_4=-0.09$ 2 (1979Ba54, detector system 2). $A_2=+0.00$ 3, $A_4=-0.01$ 3 from 3913 level 1788-2127 $\gamma\gamma(\theta)$ (1971Mu03). $A_2=+0.03$ 2, $A_4=-0.02$ 2 from 4072 level 1948-2127 $\gamma\gamma(\theta)$ (1971Mu03). $A_2=-0.02$ 3, $A_4=+0.21$ 3 from 4114 level 1989-2127 $\gamma\gamma(\theta)$ (1971Mu03). $A_2=+0.45$ 2, $A_4=-0.28$ 2 from 4688 level 2561-2127 $\gamma\gamma(\theta)$ (1971Mu03). $A_2=+0.15$ 2, $A_4=-0.13$ 3 from 5319 level 3191-2127 $\gamma\gamma(\theta)$ (1971Mu03). $A_2=+0.14$ 5, $A_4=+0.07$ 6 from 5993 level 3866-2127 $\gamma\gamma(\theta)$ (1972Jo10). $A_2=-0.04$ 3, $A_4=+0.04$ 4 from 6480 level 3177-2127 $\gamma\gamma(\theta)$ (1972Jo10). $A_2=+0.15$ 3, $A_4=+0.09$ 4 from 6639 level 2017-2127 $\gamma\gamma(\theta)$ (1972Jo10).

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$^{31}\text{P}(\alpha, \text{p}\gamma)$ **1970Mo09, 1971Mu03, 1972Jo10 (continued)** $\gamma(^{34}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
3303.2	2 ⁺	1175.2 8	52 5	2127.0	2 ⁺	M1+E2	-0.16 2	I_γ: weighted average of 50 5 (1970Mo09), 57 5 (1970Gr11), 45 5 (1971Mu03), and 56 5 (1974Gr06). Mult.: from $\text{py}(\theta)$ in 1970Mo09, 1971Mu03, and 1974Gr06. M2 ruled out by RUL. δ: weighted average of -0.09 6 (1970Mo09) and -0.14 6 (1971Mu03), -0.17 2 (1974Gr06). $A_2=+0.28$ 3, $A_4=-0.04$ 3 (1971Mu03). $A_2=-0.08$ 6, $A_4=+0.05$ 7 from 5993 level 1918-1176 $\gamma\gamma(\theta)$ (1972Jo10). $A_2=-0.16$ 2, $A_4=+0.0$ 3 from 6480 level 3177-1176 $\gamma\gamma(\theta)$ (1972Jo10). $A_2=-0.30$ 5, $A_4=+0.19$ 7 from 6639 level 2017-1176 $\gamma\gamma(\theta)$ (1972Jo10). I_γ: weighted average of 50 5 (1970Mo09), 43 5 (1970Gr11), 55 5 (1971Mu03), and 44 5 (1974Gr06). Mult.: Q from $\text{py}(\theta)$ in 1970Mo09 and 1971Mu03. Electric character from POL=+0.53 4 (1979Ba54, detector system 1); POL=+0.65 4 (1979Ba54, detector system 2). $A_2=+0.65$ 2, $A_4=-1.22$ 2 (1971Mu03). $A_2=+0.31$ 2, $A_4=-0.06$ 2 (1979Ba54, detector system 1). $A_2=+0.37$ 2, $A_4=-0.06$ 2 (1979Ba54, detector system 2). $A_2=+0.53$ 6, $A_4=-0.32$ 8 from 4622 level 1320-3303 $\gamma\gamma(\theta)$. $A_2=+0.47$ 2, $A_4=-0.57$ 2 from 4875 level 1571-3303 $\gamma\gamma(\theta)$. $A_2=-0.02$ 9, $A_4=+0.11$ 11 from 6120 level 2817-3303 $\gamma\gamma(\theta)$ (1972Jo10). $A_2=+0.32$ 5, $A_4=-0.08$ 6 from 6173 level 2870-3303 $\gamma\gamma(\theta)$ (1972Jo10). $A_2=+0.62$ 13, $A_4=+0.08$ 14 from 6251 level 1376-3303 $\gamma\gamma(\theta)$ (1972Jo10). $A_2=-0.05$ 3, $A_4=-0.03$ 3 from 6347 level 3044-3303 $\gamma\gamma(\theta)$ (1972Jo10). $A_2=+0.36$ 10, $A_4=-0.29$ 13 from 6639 level 2017-3303 $\gamma\gamma(\theta)$ (1972Jo10).
3913.0	(0,1,2)	609.8 ^a 1786.0 8	<3 [@] 100	3303.2 2 ⁺ 2127.0 2 ⁺				I_γ: from 1970Mo09, 1970Gr11, and 1971Mu03. $A_2=+0.02$, $A_4=+0.03$ 3 (1971Mu03).
4071.8	1 ⁺	3913 ^a 159 ^a 768.6	<2 [@] <5 [@] 3 1	0.0 0 ⁺ 3913.0 (0,1,2) 3303.2 2 ⁺		D(+Q)	-1.3 15	I_γ: from 1970Gr11 and 1971Mu03. Other: 4 2 (1970Mo09). Mult., δ: from $\text{py}(\theta)$ in 1971Mu03. $A_2=+0.28$ 2, $A_4=+0.01$ 2 (1971Mu03). I_γ: weighted average of 46 15 (1970Mo09), 56 4 (1970Gr11), 43 4 (1971Mu03), and
		1945.5 13	49 4	2127.0 2 ⁺		M1+E2	+1.3 +32-9	

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$^{31}\text{P}(\alpha, p\gamma)$ **1970Mo09, 1971Mu03, 1972Jo10** (continued) $\gamma(^{34}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\ddagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
								51 7 (1974Gr06). Mult., δ : D+Q from $p\gamma(\theta)$ in 1970Mo09, 1971Mu03, and 1974Gr06. M2 ruled out by RUL. δ from 1974Gr06. Other: +0.1 +3-39 (1971Mu03). $A_2=+0.03$ 2, $A_4=-0.02$ 2 from 4072 level 1948-2127 $\gamma\gamma(\theta)$ (1971Mu03). I_γ : weighted average of 50 15 (1970Mo09), 41 5 (1970Gr11), 54 5 (1971Mu03), and 49 7 (1974Gr06). Mult.: D from $p\gamma(\theta)$ in 1970Mo09, 1971Mu03, and 1974Gr06. $A_2=-0.21$ 2, $A_4=-0.02$ 3 (1971Mu03).
4071.8	1 ⁺	4070.9 13	48 5	0.0 0 ⁺		D		
4114.0	2 ⁺	810.8 ^a 1986.9	<11 [@] 48 5	3303.2 2 ⁺ 2127.0 2 ⁺		M1+E2	-0.40 5	I_γ : weighted average of 20 20 (1970Mo09), 52 5 (1970Gr11), 45 5 (1971Mu03), and 50 7 (1974Gr06). Mult.: D+Q from $p\gamma(\theta)$ in 1971Mu03 and 1974Gr06. M2 ruled out by RUL. δ : weighted average of -0.57 12 (1971Mu03) and -0.37 5 (1974Gr06). $A_2=-0.02$ 3, $A_4=+0.21$ 3 from 4114 level 1989-2127 $\gamma\gamma(\theta)$ (1971Mu03). I_γ : weighted average of 80 20 (1970Mo09), 48 5 (1970Gr11), 55 5 (1971Mu03), and 50 7 (1974Gr06). Mult.: Q from $p\gamma(\theta)$ in 1971Mu03 and 1974Gr06. M2 ruled out by RUL. $A_2=+0.60$ 2, $A_4=-1.00$ 2 (1971Mu03).
		4113.7 9	52 5	0.0 0 ⁺		E2		
4622.4	3 ⁻	508.4 ^a 550.6 ^a 709 ^a 1318.8 10	<3 [@] <4 [@] <7 [@] 76 5	4114.0 2 ⁺ 4071.8 1 ⁺ 3913.0 (0,1,2) 3303.2 2 ⁺		D(+Q)	-0.03 5	I_γ : weighted average of 76 5 (1970Mo09), 75 5 (1970Gr11), 76 5 (1971Mu03), and 75 5 (1974Gr06). Mult., δ : from 1971Mu03. Other: +0.02 8 (1970Mo09). $A_2=-0.46$ 2, $A_4=+0.09$ 2 (1971Mu03). $A_2=-0.19$ 8, $A_4=-0.23$ 8 from 5688 level 1065-1320 $\gamma\gamma(\theta)$ (1972Jo10). I_γ : weighted average of 24 5 (1970Mo09), 25 5 (1970Gr11), 24 5 (1971Mu03), and 25 5 (1974Gr06). Mult., δ : from 1971Mu03. $A_2=-0.32$ 2, $A_4=-0.04$ 2 (1971Mu03). $A_2=-0.09$ 10, $A_4=+0.23$ 12 from 6639 level 2017-2496 $\gamma\gamma(\theta)$ (1972Jo10). I_γ : <1 (1970Mo09), ≤ 1 (1974Gr06).
		2495.5 7	24 5	2127.0 2 ⁺		D(+Q)	+0.02 4	
4687.6	4 ⁺	4622 ^a 573.6 ^a 615.8 ^a 775 ^a 1384.4 ^a 2560.5 4	<1 <3 [@] <4 [@] <7 [@] <2 [@] 100	0.0 0 ⁺ 4114.0 2 ⁺ 4071.8 1 ⁺ 3913.0 (0,1,2) 3303.2 2 ⁺ 2127.0 2 ⁺		E2		I_γ : from 1970Mo09, 1970Gr11, 1971Mu03, and

Continued on next page (footnotes at end of table)

$^{31}\text{P}(\alpha, \text{p}\gamma)$ **1970Mo09, 1971Mu03, 1972Jo10 (continued)** $\gamma(^{34}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\ddagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
								1974Gr06. Mult., δ : Q(+O) from $\text{p}\gamma(\theta)$ in 1971Mu03 and 1974Gr06. E2(+M3) with $\delta=0.00$ 1 is given in 1971Mu03. M2(+E3) ruled out by RUL. $A_2=+0.44$ 2, $A_4=-0.25$ 2 (1971Mu03). $A_2=+0.36$ 5, $A_4=-0.19$ 6 from 6864 level 2176-2561 $\gamma\gamma(\theta)$ (1972Jo10).
4687.6	4 ⁺	4687 ^a	<1 @	0.0	0 ⁺			
4875.2	3 ⁺	188 ^a	<14 @	4687.6	4 ⁺			
		253 ^a	<21 @	4622.4	3 ⁻			
		761.2 ^a	<6 @	4114.0	2 ⁺			
		803.4 ^a	<4 @	4071.8	1 ⁺			
		962 ^a	<8 @	3913.0	(0,1,2)			
		1571.9	5	3303.2	2 ⁺	M1+E2	-0.09 4	I_γ : weighted average of 32 7 (1970Mo09), 35 5 (1970Gr11), 40 5 (1971Mu03), and 40 10 (1974Gr06). Mult., δ : D+Q from 1971Mu03. M2 ruled out by RUL. $A_2=-0.57$ 2, $A_4=-0.05$ 2 (1971Mu03). I_γ : weighted average of 68 7 (1970Mo09), 65 5 (1970Gr11), 60 5 (1971Mu03), and 60 10 (1974Gr06). Mult., δ : D+Q from 1971Mu03. M2 ruled out by RUL. $A_2=-0.65$ 3, $A_4=+0.05$ 3 (1971Mu03).
		2750.9	25	2127.0	2 ⁺	M1+E2	-0.11 3	
		4875 ^a	<4 @	0.0	0 ⁺			
4891.1	2 ⁺	777.1 ^a	<13 @	4114.0	2 ⁺			
		819.3 ^a	<10 @	4071.8	1 ⁺			
		978 ^a	<22 @	3913.0	(0,1,2)			
		1587.9	49 12	3303.2	2 ⁺			I_γ : weighted average of 36 11 (1970Mo09) and 59 10 (1970Gr11).
		2764.0 ^a	<36 @	2127.0	2 ⁺			
		4890.7	51 12	0.0	0 ⁺	E2		I_γ : weighted average of 64 11 (1970Mo09) and 41 10 (1970Gr11). Mult.: Q from $\text{p}\gamma(\theta)$ in 1970Mo09 and 1971Mu03. M2 ruled out by RUL. $A_2=+0.51$ 4, $A_4=-0.74$ 5 (1971Mu03). $A_2=+0.21$ 5, $A_4=-0.45$ 6 from 6950 level 2058-4892 $\gamma\gamma(\theta)$ (1972Jo10).
5225	0	1153	100 @	4071.8	1 ⁺	D		Mult.: from 1971Mu03. $A_2=-0.01$ 3, $A_4=-0.02$ 4 (1971Mu03).
5318.8	2 ⁺	3191.6	100	2127.0	2 ⁺	M1+E2	-0.17 6	I_γ : from 1970Gr11, 1971Mu03, and 1974Gr06. Mult., δ : D+Q from 1971Mu03. M2 ruled out by RUL. Other: D+Q with $\delta=+1.8$ 10 (1974Gr06). $A_2=+0.12$ 2, $A_4=+0.05$ 2 (1971Mu03).
5382	1 ⁺	3255	62 8	2127.0	2 ⁺	M1+E2	-1.1 10	I_γ : from 1971Mu03. Mult., δ : D+Q with $-2.1 \leq \delta \leq -0.1$ (1971Mu03). M2 ruled out by RUL. $A_2=+0.04$ 3, $A_4=+0.02$ 3 (1971Mu03).
		5382	38 8	0.0	0 ⁺	D		I_γ , Mult.: from 1971Mu03. $A_2=-0.13$ 3, $A_4=+0.02$ 4 (1971Mu03).
5679	2 ⁺	2376	65 10	3303.2	2 ⁺	M1+E2	<-0.4	I_γ : from 1974Gr06.

Continued on next page (footnotes at end of table)

$^{31}\text{P}(\alpha, \text{p}\gamma)$ **1970Mo09, 1971Mu03, 1972Jo10 (continued)** $\gamma(^{34}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\ddagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
								Mult., δ : D+Q with $\delta < -0.4$ or $\delta > +2.4$ (1972Jo10). M2 ruled out by RUL with either δ . $A_2 = -0.13$ 12, $A_4 = -0.50$ 14 (1972Jo10). I_γ : from 1974Gr06. Mult., δ : from 1972Jo10. $A_2 = -0.14$ 2, $A_4 = -0.15$ 2 (1972Jo10). I_γ : weighted average of 51 2 (1972Jo10), 45 3 (1971Gr26), and 45 10 (1974Gr06). Mult., δ : D(+Q) with $\delta = 0.00$ 2 from $\gamma(\theta)$ and electric character from $\gamma(\theta)(\text{POL})$ in 1971Gr26. Other: D(+Q) with -0.05 9 (1972Jo10). $A_2 = -0.32$ 5, $A_4 = -0.22$ 5 (1972Jo10). I_γ : weighted average of 49 2 (1972Jo10), 55 3 (1971Gr26), and 55 10 (1974Gr06). Mult.: Q from $\gamma(\theta)$ and electric character from $\gamma(\theta)(\text{POL})$ in 1971Gr26. $A_2 = +0.29$ 7, $A_4 = -0.36$ 7 (1972Jo10).
5679	2 ⁺	3552	35 10	2127.0	2 ⁺	D+Q	-0.47 +7-11	
5688	5 ⁻	1000	49 2	4687.6	4 ⁺	E1		
		1066	51 2	4622.4	3 ⁻	E2		
5753		3626		2127.0	2 ⁺			
5848		3721		2127.0	2 ⁺			
5993	2 ⁺	1921	20 8	4071.8	1 ⁺			
		3866	25 8	2127.0	2 ⁺			
		5992	55 10	0.0	0 ⁺	E2		Mult.: Q from $\text{p}\gamma(\theta)$ in 1972Jo10. M2 ruled out by RUL. $A_2 = +0.40$ 2, $A_4 = -0.61$ 3. Mult., δ : -0.09 4 for $J_i = 2$ (1972Jo10). $A_2 = +0.38$ 5, $A_4 = +0.06$ 6 (1972Jo10). $A_2 = +0.05$ 5, $A_4 = -0.07$ 6 (1972Jo10).
6120	1,2	2817	82 7	3303.2	2 ⁺	D+Q		
6173	3 ⁺	3993	18 7	2127.0	2 ⁺			
		1282 ^a	6 5	4891.1	2 ⁺			
		1485	36 5	4687.6	4 ⁺	D+Q	+0.04 +6-3	Mult., δ : from 1972Jo10 by fixing $\delta(2870\gamma) = -0.23$. $A_2 = -0.34$ 3, $A_4 = +0.06$ 3 (1972Jo10). Mult., δ : D+Q from $\text{p}\gamma(\theta)$ in 1972Jo10. M2 ruled out by RUL. Other: $\delta = -1.9$ 6 (1972Jo10). $A_2 = -0.67$ 6, $A_4 = +0.01$ 7 (1972Jo10). Mult., δ : or -1.0 3 (1972Jo10). $A_2 = -0.94$ 4, $A_4 = +0.14$ 4 (1972Jo10).
		2870	38 5	3303.2	2 ⁺	M1+E2	-0.23 7	Mult., δ : D+Q from 1972Jo10. M2 ruled out by RUL. $A_2 = -0.44$ 6, $A_4 = +0.63$ 8 (1972Jo10).
		4046	20 5	2127.0	2 ⁺	D+Q	-0.43 16	
6251	4 ⁺	1376	57 10	4875.2	3 ⁺	M1+E2	-3.7 +7-26	
6347	1	1563	43 10	4687.6	4 ⁺			
		3044	70 10	3303.2	2 ⁺	D(+Q)	-0.55 65	Mult., δ : $-1.2 \leq \delta \leq -0.1$ from 1972Jo10. $A_2 = -0.05$ 3, $A_4 = -0.03$ 3 (1972Jo10). $A_2 = -0.20$ 5, $A_4 = +0.02$ 6 (1972Jo10). Mult., δ : 0.00 6 from 1972Jo10. $A_2 = -0.16$ 2, $A_4 = +0.0$ 3 (1972Jo10). Mult., δ : 0.00 +32-14 from 1972Jo10. $A_2 = +0.12$ 5, $A_4 = +0.05$ 6 (1972Jo10).
6415	4	6346	30 10	0.0	0 ⁺	D		
		1540	62 5	4875.2	3 ⁺	D		
		1727	38 5	4687.6	4 ⁺	D		
6480	1	3177	59 10	3303.2	2 ⁺			
		4353	41 10	2127.0	2 ⁺	D+Q	-1.1 9	Mult., δ : $-2.0 \leq \delta \leq -0.19$ from 1972Jo10. $A_2 = +0.12$ 5, $A_4 = +0.05$ 6 (1972Jo10). $A_2 = -0.46$ 8, $A_4 = +0.00$ 10 (1972Jo10).
6639	1 ⁻ , 2, 3	960	22 10	5679	2 ⁺			
		2017	78 10	4622.4	3 ⁻			

Continued on next page (footnotes at end of table)

$^{31}\text{P}(\alpha, p\gamma)$ **1970Mo09, 1971Mu03, 1972Jo10 (continued)** $\gamma(^{34}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
6731	$2^{(+)}, 4^{(+)}$	1856	5 5	4875.2	3^+			
		2043	20 7	4687.6	4^+			
		3428	20 7	3303.2	2^+			
		4604	55 5	2127.0	2^+			Mult., δ : D+Q with $\delta=+1.8$ 3 for $J_i=2$ or Q(+O) with $\delta=0.00$ 3 for $J_i=4$ (1972Jo10). $A_2=+0.44$ 2, $A_4=-0.23$ 2; $A_2=+0.50$ 2, $A_4=-0.27$ 2 (1972Jo10).
6830	2	2208		4622.4	3^-			
		2716		4114.0	2^+			
		6829		0.0	0^+	Q		Mult.: from $p\gamma(\theta)$ in 1972Jo10. $A_2=+0.49$ 9, $A_4=-0.39$ 12 (1972Jo10).
6864	$2^{(+)}, 4^{(+)}$	2176		4687.6	4^+			
		2242 ^a		4622.4	3^-			Mult.: E2, 5^- to 3^- , assigned by 1977GrZH, but 1972Jo10 did not observe this γ transition. Instead, 1972Jo10 observed 4737γ to 2^+ , 2127.0 level to be the strongest branch.
		4737		2127.0	2^+			Mult., δ : D+Q with $\delta=+2.2$ 5 or Q(+O) with $\delta=0.00$ 4 (1972Jo10). $A_2=+0.50$ 5, $A_4=-0.25$ 7 (1972Jo10).
6950	2	2059		4891.1	2^+	D(+Q)	0.0 5	Mult., δ : from $p\gamma(\theta)$ in 1972Jo10, which states that the 2058γ could not be resolved from the 2127γ by the NaI(Tl) detectors.
7112	2	3647		3303.2	2^+			
		3809	26 7	3303.2	2^+			
		4985	66 7	2127.0	2^+	D+Q	+0.27 +19-15	Mult., δ : or +1.2 +7-4 from $p\gamma(\theta)$ in 1972Jo10. $A_2=+0.64$ 3, $A_4=-0.04$ 3 (1972Jo10).
		7111	8 6	0.0	0^+	Q		Mult.: from $p\gamma(\theta)$ in 1972Jo10. $A_2=+0.70$ 12, $A_4=-1.03$ 17 (1972Jo10).
7220	2^+	2532		4687.6	4^+			
		5093		2127.0	2^+			
		7219		0.0	0^+	E2		Mult.: Q from $p\gamma(\theta)$ in 1972Jo10. M2 ruled out by RUL. $A_2=+0.46$ 4, $A_4=-0.73$ 5 (1972Jo10).
7248	(4)	1560		5688	5^-			
7788	6	2100		5688	5^-	D+Q		Mult.: M1+E2 is indicated in text of 1977GrZH.

[†] E_γ values without uncertainties are deduced from level-energy differences, unless otherwise noted.

[‡] E_γ values with uncertainties are from 1970Mo09, unless otherwise noted.

[#] From 1972Jo10, unless otherwise noted.

[@] From 1970Mo09, unless otherwise noted.

[&] From measured $p\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ given under comments, and RUL if applicable.

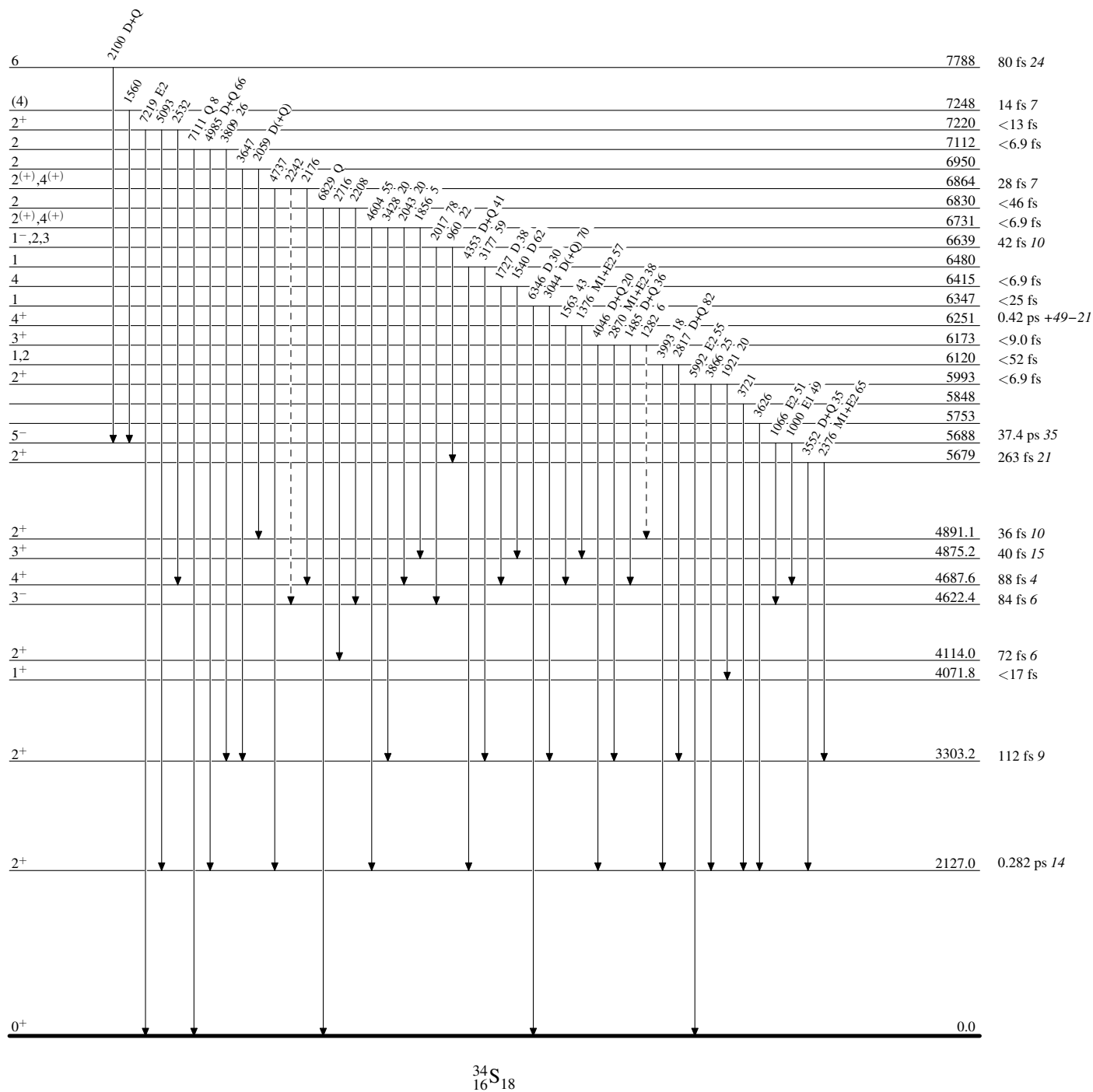
^a Placement of transition in the level scheme is uncertain.

$^{31}\text{P}(\alpha, p\gamma)$ 1970Mo09, 1971Mu03, 1972Jo10

Legend

Level Scheme

Intensities: % photon branching from each level

-----► γ Decay (Uncertain)

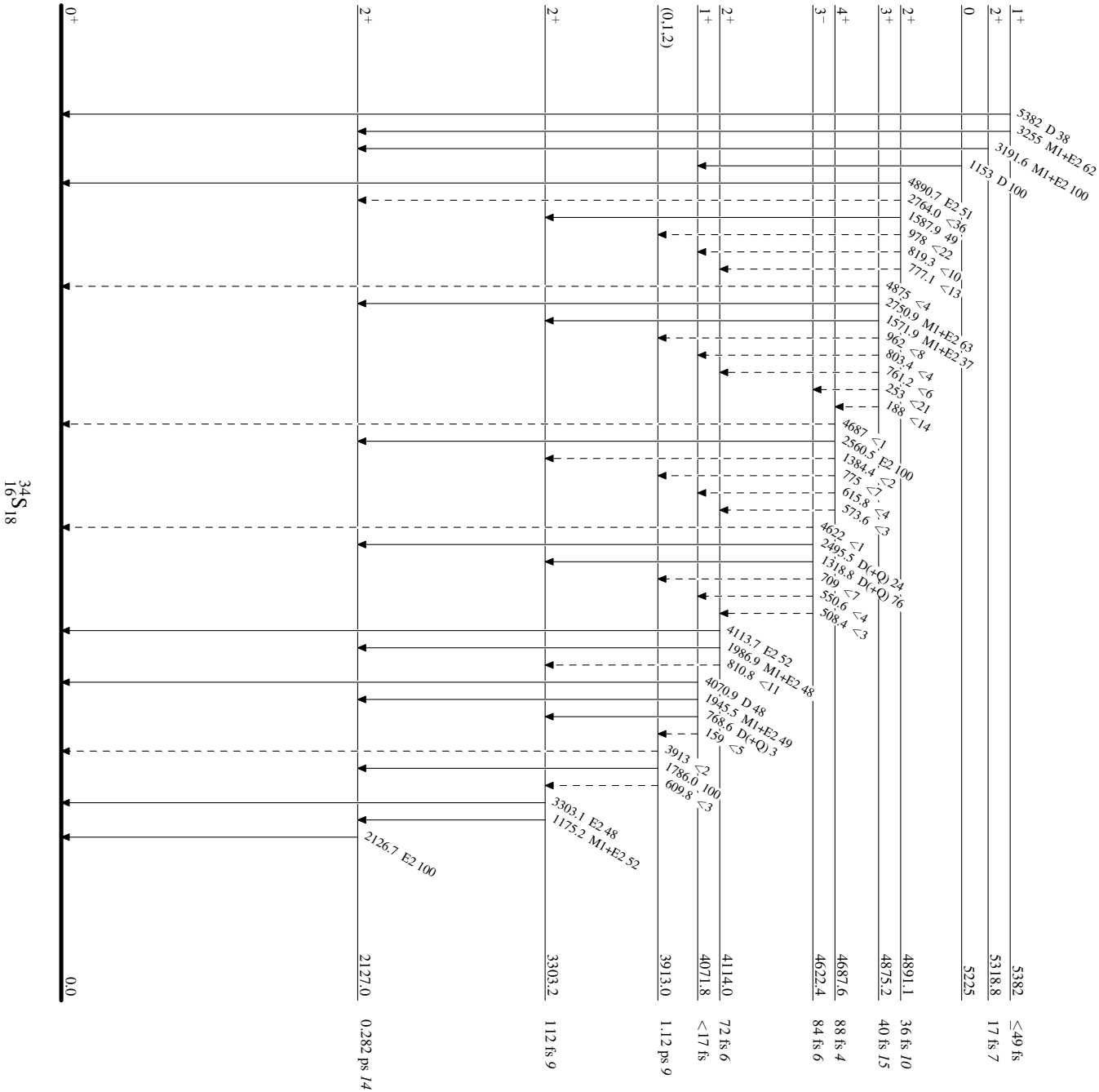
³¹P(α,pγ) 1970Mø09,1971Mø03,1972Jo10

Legend

Level Scheme (continued)

Intensities: % photon branching from each level

-----▶ γ Decay (Uncertain)



$^{32}\text{S}(\text{t,p})$ 1978Cr01

$J^\pi=0^+$ for ^{32}S ground state (2025Ch19).

1978Cr01: a 10-MeV triton beam was provided from the Aldermaston tandem accelerator, United Kingdom. The target was antimony trisulfide. The outgoing protons were analyzed in a multiangle magnetic spectrograph and recorded in nuclear emulsion plates. Measured $\sigma(E_p, \theta)$. Deduced levels, J , π , and L -transfers from DWUCK-DWBA analysis. Partial work on this measurement: 1973Cr04, 1975Cr07.

1979Fo17: theoretical discussion of the $^{32}\text{S}(\text{t,p})^{34}\text{S}$ reactions.

1978Cr01 states that since only relative cross sections were measured, and since the absolute normalizing factor for a (t,p) distorted-wave calculation is unknown, the normalization for the ground state has been chosen equal to 100. All configuration and normalization factors of the experimental to theoretical cross sections are given under comments.

 ^{34}S Levels

E(level) [†]	J^π [†]	L [†]	Comments
0	0^+	0	$(d_{3/2})^2$. N=100, 100. $(s_{1/2})^2$. N=13.8, 13.0.
2128 10	2^+	2	$(d_{3/2})^2$. N=11.8, 11.5. $d_{3/2}, s_{1/2}$. N=1.62, 1.52.
3308 11	(2^+)	(2)	$(d_{3/2})^2$. N=9.12, 8.79. $d_{3/2}, s_{1/2}$. N=1.18, 1.12.
3915 12	0^+	0	$(d_{3/2})^2$. N=3.62, 3.33. $(s_{1/2})^2$. N=0.48, 0.42. $(f_{7/2})^2$. N=1.75, 2.18.
4085 12			
4121 12	2^+	2	$(d_{3/2})^2$. N=4.50, 4.24. $d_{3/2}, s_{1/2}$. N=0.59, 0.53. $(f_{7/2})^2$. N=2.25, 2.61.
4623 13	3^-	3	$d_{3/2}, f_{7/2}$. N=33.8, 33.3. $d_{3/2}, p_{3/2}$. N=2.62, 2.67.
4690 13	(4^+)	(4)	$(f_{7/2})^2$. N=2.00, 1.88. $d_{3/2}, f_{7/2}$. N=0.48, 0.48.
4888 13	(2^+)	(2)	$(d_{3/2})^2$. N=3.50, 3.30. $(f_{7/2})^2$. N=1.88, 1.88.
			Doublet unresolved in 1978Cr01.
5225 13	0^+	0	$(d_{3/2})^2$. N=5.38, 4.85. $(f_{7/2})^2$. N=3.38, 3.64. $(p_{3/2})^2$. N=0.29, 0.28.
5320 13			
5380 13			
5679 14	5^-	5,(1)	$d_{3/2}, p_{3/2}$. N=3.25, 3.18. $d_{3/2}, f_{7/2}$. N=2.12, 2.18.
			Doublet unresolved in 1978Cr01.
5759 14	1^-	1	$d_{3/2}, p_{3/2}$. N=9.38, 8.18.
5859 14	0^+	0	$(f_{7/2})^2$. N=15.0.
6008 14	2^+	2	$(f_{7/2})^2$. N=33.75, 33.33. $(p_{3/2})^2$. N=3.75, 3.03.
6128 14	2^+	2	$(f_{7/2})^2$. N=7.75, 7.88.
6179 14	(3^-)	(3)	J^π : possible discrepancy: 3^+ assigned for 6168.860 level in the Adopted Levels.
			$d_{3/2}, p_{3/2}$. N=0.22, 0.21. $d_{3/2}, f_{7/2}$. N=3.12, 2.73.
6256 14			
6349 14	1^-	1	$d_{3/2}, p_{3/2}$. N=6.38, 5.76.
6423 14			
6488 14			
6535 15			
6639 15			
6690 15			
6743 15			
6828 15	2^+	2	$(f_{7/2})^2$. N=4.25, 4.55. $(p_{3/2})^2$. N=0.36.
6869 15			
6898 15			
6956 15			
7112 15	$3^-, (2^+)$	3,(2)	$d_{3/2}, f_{7/2}$. N=11.25, 10.00. $d_{3/2}, p_{3/2}$. N=0.78, 0.67. $(f_{7/2})^2$. N=6.50, 6.67. $(p_{3/2})^2$. N=0.62, 0.55.
			Possible doublet in 1978Cr01.
7245 15	$(2^+, 3^-)$	(2,3)	$d_{3/2}, f_{7/2}$. N=22.50, 20.61. $d_{3/2}, p_{3/2}$. N=1.62, 1.52.
7388 15	3^-	3	
7472 15			
7547 16			
7621 16	3^-	3	$d_{3/2}, f_{7/2}$. N=23.75, 23.03. $d_{3/2}, p_{3/2}$. N=1.88, 1.61.
7714 16			
7739 16	2^+	2	$(f_{7/2})^2$. N=4.75, 5.15. $(p_{3/2})^2$. N=0.48, 0.39.

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$^{32}\text{S}(\text{t,p})$ **1978Cr01** (continued) ^{34}S Levels (continued)

$E(\text{level})^\dagger$	$J^\pi^\dagger$	$L^\dagger$	Comments
7801 16	2 ⁺	2	(f _{7/2}) ² . N=20.62.
7971 16			
8025 16	0 ⁺	0	(p _{3/2}) ² . N=0.56.
8255 16	2 ⁺	2	(f _{7/2}) ² . N=5.12, 5.15. (p _{3/2}) ² . N=0.49, 0.36. d _{3/2} ,p _{3/2} . N=0.62, 0.52. d _{3/2} ,f _{7/2} . N=8.75, 4.24.
8293 16			(f _{7/2}) ² . N=4.75, 4.85. d _{3/2} ,f _{7/2} . N=1.12, 1.30.
8383 16	1 ⁻	1	d _{3/2} ,p _{3/2} . N=8.38.
8418 16	4 ⁺	4	(f _{7/2}) ² . N=11.25.
8496 16	1 ⁻	1	d _{3/2} ,p _{3/2} . N=12.25.

[†] From **1978Cr01**. The firm J^π assigned by **1978Cr01** is put in parentheses if their L-transfer is placed in parentheses.

 $^{32}\text{S}(\alpha, ^2\text{He})$ **1978Ja10**

1978Ja10: a 65-MeV α beam was provided from the Lawrence Berkeley Laboratory 88-Inch Cyclotron. The target was 750 $\mu\text{g}/\text{cm}^2$ Sb_2S_3 . Two protons from the breakup of ^2He were detected using two large solid-angle ΔE -E telescopes consisting of a 380- μm phosphorus diffused Si ΔE detector, a 5-mm Si(Li) E detector, and a 5-mm Si(Li) veto detector. Measured $\text{E}(^2\text{He})$, $\sigma(^2\text{He})$, and pp-coin. Deduced levels.

 ^{34}S Levels

E(level)

0
4620
5690
7240
8450
10700

$^{33}\text{S}(\text{n},\gamma)$ E=thermal 1985Ra15

$J^\pi=3/2^+$ for ^{33}S ground state (2025Ch07).

1985Ra15: 6×10^{11} n/cm²/s thermal neutrons at the target position were produced from the Los Alamos Omega West Reactor at a nominal 8-MW reactor power. γ rays following thermal neutron capture by a 1.10-g, 88.2% enriched ^{33}S target were detected using a 26 cm³ coaxial Ge(Li) detector inside a 20-cm-diam by 30-cm-long NaI(Tl) annulus with FWHM=2.3 keV at 1 MeV, 5.5 keV at 6 MeV and 8.8 keV at 11 MeV. Measured σ_γ , E_γ , and I_γ for ≈ 270 γ -ray transitions. Deduced 70 levels and γ -ray branching ratios.

Others:

1983Ra04: by the same group as 1985Ra15. Measured S(n) of ^{34}S and deduced (n, γ) reaction Q value based on 11 γ cascades.

All 22 E_γ values stated are the same as in 1985Ra15.

2009KiZW discussed method to identify nuclear levels based on neutron capture reactions.

2007ChZX: prompt γ activation analysis (PGAA database for elemental analysis), natural target. In the measurements at Budapest, 5 primary and 6 secondary γ rays were identified. The energies and relative intensities (deduced from measured elemental cross sections) are in good agreement with those from 1985Ra15. The normalization factor is also in agreement. The data in 1985Ra15 are much more complete and precise, thus adopted here.

 ^{34}S Levels

E(level) [†]	J^π [‡]	Comments
0.0	0 ⁺	
2127.564 13	2 ⁺	
3304.213 13	2 ⁺	
3916.408 21	0 ⁺	
4074.667 13	1 ⁺	
4114.81 2	2 ⁺	
4624.404 16	3 ⁻	
4688.97 4	4 ⁺	
4876.842 23	3 ⁺	
4889.76 2	2 ⁺	
5228.175 22	0 ⁺	
5322.514 28	2 ⁺	
5380.99 3	1 ⁺	
5679.928 17	2 ⁺	
5755.876 21	1 ⁻	
5847.524 29	0 ⁺	
5998.10 8	2 ⁺	
6121.56 10	2 ⁺	
6168.862 28	3 ⁺	
6251.68 9	(3,4)	1985Ra15 had two levels at 6251.23 and 6251.68, which are considered the same level and their γ ray transitions are combined. See the Adopted Levels for details.
6342.52 10	1 ⁻	
6421.34 8	4 ⁻	
6428.13 7	(2 ⁺ ,3 ⁺)	
6478.770 22	1 ⁻	
6685.333 29	(0 to 3) ⁻	
6828.83 18	2 ⁺	
6847.90 6	(1,2 ⁺)	
6954.218 30	2 ⁻	
7110.451 33	3 ⁻	
7164.47 17	(0 ⁺ to 3 ⁺)	
7219.28 6	2 ⁺	
7248.05 11	(2 ⁺ ,3 ⁻)	
7367.42 10	(1 ⁺ ,2 ⁺)	
7467.74 10	(0 ⁺ ,1,2)	
7552.70 8	(0 ⁺ to 3 ⁻)	
7629.907 21	3 ⁻	
7730.79 15	(1 ⁻ ,2 ⁻ ,3 ⁻)	

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15** (continued) ^{34}S Levels (continued)

E(level) [†]	J ^π [‡]	Comments
7781.22 6	(1) ⁻	
7974.72 16	(1,2) ⁺	
8036.31 14	(1 ⁻ ,2 ⁺)	
8138.08 7	(1) ⁻	
8174.9 4	(1,2) ⁺	
8185.46 12	1 ⁺	
8205.4 1	(1 ⁻ to 4 ⁺)	
8294.40 9	4 ⁽⁺⁾	
8385.41 6	1 ⁻	
8506.77 4	1 ⁻	
8615.74 4	(2 ⁻ ,3 ⁺)	
8702.35 12	(0 ⁺ ,1,2)	
8727.64 7	(1 ⁻ ,2 ⁺)	
8805.6 3	(1,2) ⁺	
8874.01 6	(1 ⁻ ,2,3 ⁺)	
9026.31 6	(1,2) ⁺	
9158.715 28	(1,2) ⁺	
9208.04 5	(1,2) ⁺	
9546.09 7	(1,2) ⁺	
9598.42 8		
9665.74 3		
9801.89 9	(1,2) ⁺	
9836.70 6		
9933.36 12	1 ⁻	
10092.21 15		
10179.59 5	1 ⁺	
10212.15 4		
10311.53 3	2 ⁺	
10650.13 19		
10840.62 14	3 ⁻	
11024.95 11	1 ⁻	
(11417.22 2)	2 ⁺	J ^π : s-wave capture in ^{33}S g.s., J ^π =3/2 ⁺ .

[†] From a least-squares fit to γ -ray energies.[‡] From the Adopted Levels. Supporting arguments from this dataset are given under comments where available. $\gamma(^{34}\text{S})$

I γ normalization: **1985Ra15** gives 0.220 without uncertainties. 0.220 15 deduced by evaluators from γ -ray intensity balance. **1985Ra15** states that the level scheme completeness is reasonable with $\Sigma I\gamma(\text{primary})/\Sigma I\gamma(\text{secondary to g.s.}) > 80\%$ for the $^{33}\text{S}(\text{n},\gamma)^{34}\text{S}$ reaction, which has a large Q value of 11417 keV and hence a greatly increased probability that many weak transitions go undetected.

E γ [†]	I γ ^{†#}	E _i (level)	J _i ^π	E _f	J _f ^π	E γ [†]	I γ ^{†#}	E _i (level)	J _i ^π	E _f	J _f ^π
^x 95.45 18	0.012 3					200.8 ^{‡&}	<0.02 [‡]	4889.76	2 ⁺	4688.97	4 ⁺
151.8 ^{‡&}	<0.02 [‡]	5380.99	1 ⁺	5228.175	0 ⁺	^x 229.71 16	0.054 10				
158.3 ^{‡&}	<0.06 [‡]	4074.667	1 ⁺	3916.408	0 ⁺	252.4 ^{‡&}	<0.03 [‡]	4876.842	3 ⁺	4624.404	3 ⁻
187.9 ^{‡&}	<0.03 [‡]	4876.842	3 ⁺	4688.97	4 ⁺	265.4 ^{‡&}	<0.02 [‡]	4889.76	2 ⁺	4624.404	3 ⁻
198.4 ^{‡&}	<0.03 [‡]	4114.81	2 ⁺	3916.408	0 ⁺	281.34 24	0.023 8	7110.451	3 ⁻	6828.83	2 ⁺

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15** (continued) $\gamma(^{34}\text{S})$ (continued)

E_γ †	I_γ ‡#	$E_i(\text{level})$	J_i^π	E_f	J_f^π
306.63 16	0.089 20	6428.13	(2 ⁺ ,3 ⁺)	6121.56	2 ⁺
334.21 15	0.042 10	9208.04	(1,2 ⁺)	8874.01	(1 ⁻ ,2,3 ⁺)
338.4 ‡&	<0.03 ‡	5228.175	0 ⁺	4889.76	2 ⁺
351.3 ‡&	<0.10 ‡	5228.175	0 ⁺	4876.842	3 ⁺
357.4 ‡&	<0.05 ‡	5679.928	2 ⁺	5322.514	2 ⁺
392.28 11	0.124 20	(11417.22)	2 ⁺	11024.95	1 ⁻
432.8 ‡&	<0.06 ‡	5322.514	2 ⁺	4889.76	2 ⁺
433.4 ‡&	<0.05 ‡	5755.876	1 ⁻	5322.514	2 ⁺
445.7 ‡&	<0.06 ‡	5322.514	2 ⁺	4876.842	3 ⁺
451.8 ‡&	<0.05 ‡	5679.928	2 ⁺	5228.175	0 ⁺
491.2 ‡&	<0.06 ‡	5380.99	1 ⁺	4889.76	2 ⁺
504.2 ‡&	<0.06 ‡	5380.99	1 ⁺	4876.842	3 ⁺
516.86 12	0.32 5	8702.35	(0 ⁺ ,1,2)	8185.46	1 ⁺
525.0 ‡&	<0.09 ‡	5847.524	0 ⁺	5322.514	2 ⁺
527.7 ‡&	<0.05 ‡	5755.876	1 ⁻	5228.175	0 ⁺
539.2 ‡&	<0.04 ‡	5228.175	0 ⁺	4688.97	4 ⁺
549.7 ‡&	<0.05 ‡	4624.404	3 ⁻	4074.667	1 ⁺
571.7 6	0.08 3	6251.68	(3,4)	5679.928	2 ⁺
576.80 19	0.146 21	(11417.22)	2 ⁺	10840.62	3 ⁻
603.8 ‡&	<0.04 ‡	5228.175	0 ⁺	4624.404	3 ⁻
612.16 5	0.263 28	3916.408	0 ⁺	3304.213	2 ⁺
619.4 ‡&	<0.09 ‡	5847.524	0 ⁺	5228.175	0 ⁺
631.13 6	0.283 31	6478.770	1 ⁻	5847.524	0 ⁺
633.5 ‡&	<0.11 ‡	5322.514	2 ⁺	4688.97	4 ⁺
672.00 10	0.152 20	9546.09	(1,2 ⁺)	8874.01	(1 ⁻ ,2,3 ⁺)
692.0 ‡&	<0.06 ‡	5380.99	1 ⁺	4688.97	4 ⁺
698.18 13	0.101 14	5322.514	2 ⁺	4624.404	3 ⁻
708.0 ‡&	<0.11 ‡	4624.404	3 ⁻	3916.408	0 ⁺
722.95 14	0.175 22	6478.770	1 ⁻	5755.876	1 ⁻
725.25 22	0.115 19	9933.36	1 ⁻	9208.04	(1,2 ⁺)
^x 743.50 20	0.098 15				
748.43 14	0.127 17	10840.62	3 ⁻	10092.21	
^x 752.30 8	0.222 26				
756.6 ‡&	<0.06 ‡	5380.99	1 ⁺	4624.404	3 ⁻
762.0 ‡&	<0.11 ‡	4876.842	3 ⁺	4114.81	2 ⁺
767.20 21	0.098 16	(11417.22)	2 ⁺	10650.13	
770.428 20	2.75 25	4074.667	1 ⁺	3304.213	2 ⁺
774.9 ‡&	<0.09 ‡	4889.76	2 ⁺	4114.81	2 ⁺
789.1 6	0.39 7	5679.928	2 ⁺	4889.76	2 ⁺
798.92 10	0.29 4	6478.770	1 ⁻	5679.928	2 ⁺
802.2 ‡&	<0.64 ‡	4876.842	3 ⁺	4074.667	1 ⁺
803.103 27	1.14 11	5679.928	2 ⁺	4876.842	3 ⁺
810.6 ‡&	<0.06 ‡	4114.81	2 ⁺	3304.213	2 ⁺
815.1 ‡&	<0.06 ‡	4889.76	2 ⁺	4074.667	1 ⁺
846.1 13	0.28 18	6168.862	3 ⁺	5322.514	2 ⁺
866.1 ‡&	<0.07 ‡	5755.876	1 ⁻	4889.76	2 ⁺
879.0 ‡&	<0.07 ‡	5755.876	1 ⁻	4876.842	3 ⁺
925.79 14	0.171 21	8036.31	(1 ⁻ ,2 ⁺)	7110.451	3 ⁻

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$^{33}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15** (continued) $\gamma(^{34}\text{S})$ (continued)

E_γ †	I_γ †#	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
929.436 21	1.07 10	6685.333	(0 to 3) ⁻	5755.876	1 ⁻	
940.7 ‡&	<0.29 ‡	6168.862	3 ⁺	5228.175	0 ⁺	
941.59 6	0.41 5	7110.451	3 ⁻	6168.862	3 ⁺	
957.8 ‡&	<0.14 ‡	5847.524	0 ⁺	4889.76	2 ⁺	
960.4 ‡&	<0.08 ‡	4876.842	3 ⁺	3916.408	0 ⁺	
970.7 ‡&	<0.14 ‡	5847.524	0 ⁺	4876.842	3 ⁺	
973.3 ‡&	<0.05 ‡	4889.76	2 ⁺	3916.408	0 ⁺	
982.68 9	0.187 27	9598.42		8615.74	(2 ⁻ , 3 ⁺)	
989.1 @ 3	0.079 @ 23	7110.451	3 ⁻	6121.56	2 ⁺	
989.1 @ 3	0.079 @ 23	7467.74	(0 ⁺ , 1, 2)	6478.770	1 ⁻	
990.9 ‡&	<0.11 ‡	5679.928	2 ⁺	4688.97	4 ⁺	
^x 1029.23 8	0.32 4					
^x 1035.82 17	0.108 31					
1055.491 20	7.0 7	5679.928	2 ⁺	4624.404	3 ⁻	
1066.9 ‡&	<0.09 ‡	5755.876	1 ⁻	4688.97	4 ⁺	
1105.673 21	1.49 14	(11417.22)	2 ⁺	10311.53	2 ⁺	
1113.27 9	0.41 6	5228.175	0 ⁺	4114.81	2 ⁺	
1121.33 9	0.35 5	5998.10	2 ⁺	4876.842	3 ⁺	
1131.5 ‡&	<0.09 ‡	5755.876	1 ⁻	4624.404	3 ⁻	
1153.492 20	10.0 9	5228.175	0 ⁺	4074.667	1 ⁺	
1156.39 7	1.57 18	6478.770	1 ⁻	5322.514	2 ⁺	
1158.6 ‡&	<0.26 ‡	5847.524	0 ⁺	4688.97	4 ⁺	
^x 1164.83 25	0.21 6					
1176.650 20	75 7	3304.213	2 ⁺	2127.564	2 ⁺	$E_\gamma=1176.61$ 7, $\sigma=82$ mb 5 (2007ChZX : PGAA database).
1205.05 4	0.61 6	(11417.22)	2 ⁺	10212.15		
1207.7 ‡&	<0.16 ‡	5322.514	2 ⁺	4114.81	2 ⁺	
1210.04 13	0.162 22	7552.70	(0 ⁺ to 3 ⁻)	6342.52	1 ⁻	
1223.1 ‡&	<0.20 ‡	5847.524	0 ⁺	4624.404	3 ⁻	
1237.61 5	0.52 6	(11417.22)	2 ⁺	10179.59	1 ⁺	
1244.32 21	0.120 26	8874.01	(1 ⁻ , 2, 3 ⁺)	7629.907	3 ⁻	
1247.92 6	0.59 7	5322.514	2 ⁺	4074.667	1 ⁺	
1250.6 ‡&	<0.22 ‡	6478.770	1 ⁻	5228.175	0 ⁺	
1266.11 5	0.66 7	5380.99	1 ⁺	4114.81	2 ⁺	
1274.30 4	1.17 11	6954.218	2 ⁻	5679.928	2 ⁺	
^x 1277.81 18	0.190 26					
1279.1 ‡&	<0.11 ‡	6168.862	3 ⁺	4889.76	2 ⁺	
1292.0 ‡&	<0.09 ‡	6168.862	3 ⁺	4876.842	3 ⁺	
1306.3 ‡&	<0.10 ‡	5380.99	1 ⁺	4074.667	1 ⁺	
1320.169 20	38 4	4624.404	3 ⁻	3304.213	2 ⁺	$E_\gamma=1320.07$ 13, $\sigma=34$ mb 4 (2007ChZX : PGAA database).
1325.22 26	0.33 7	(11417.22)	2 ⁺	10092.21		
1353.46 16	0.38 5	7781.22	(1) ⁻	6428.13	(2 ⁺ , 3 ⁺)	
1364.4 4	0.32 9	10092.21		8727.64	(1 ⁻ , 2 ⁺)	
1374.34 20	0.37 8	6251.68	(3, 4)	4876.842	3 ⁺	
1406.1 ‡&	<0.10 ‡	5322.514	2 ⁺	3916.408	0 ⁺	
^x 1435.00 11	0.30 5					
^x 1443.05 10	0.37 5					
1464.6 ‡&	<0.10 ‡	5380.99	1 ⁺	3916.408	0 ⁺	
1469.67 24	0.23 4	7467.74	(0 ⁺ , 1, 2)	5998.10	2 ⁺	

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$^{33}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15** (continued) $\gamma(^{34}\text{S})$ (continued)

E_γ †	I_γ ‡#	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
1479.73 15	0.263 32	6168.862	3 ⁺	4688.97	4 ⁺	
1484.06 19	0.33 5	(11417.22)	2 ⁺	9933.36	1 ⁻	
^x 1486.7 8	0.18 5					
1525.39 6	1.13 11	6847.90	(1,2 ⁺)	5322.514	2 ⁺	
1544.41 @ 10	2.58 @ 24	6168.862	3 ⁺	4624.404	3 ⁻	
1544.41 @ 10	2.58 @ 24	6421.34	4 ⁻	4876.842	3 ⁺	
1562.3 5	0.80 20	6251.68	(3,4)	4688.97	4 ⁺	
1564.8 5	0.91 20	5679.928	2 ⁺	4114.81	2 ⁺	
1572.57 5	5.6 6	4876.842	3 ⁺	3304.213	2 ⁺	
1580.50 6	0.66 7	(11417.22)	2 ⁺	9836.70		
1585.510 20	2.52 23	4889.76	2 ⁺	3304.213	2 ⁺	
1589.0 ‡&	<0.11 ‡	6478.770	1 ⁻	4889.76	2 ⁺	
1602.06 15	0.43 7	6478.770	1 ⁻	4876.842	3 ⁺	
1605.3 ‡&	<0.11 ‡	5679.928	2 ⁺	4074.667	1 ⁺	
1615.24 10	2.25 27	(11417.22)	2 ⁺	9801.89	(1,2 ⁺)	
1617.00 12	1.94 25	8727.64	(1 ⁻ ,2 ⁺)	7110.451	3 ⁻	
1627.2 10	0.19 7	6251.68	(3,4)	4624.404	3 ⁻	
1631.641 25	2.88 27	6954.218	2 ⁻	5322.514	2 ⁺	
1640.7 10	0.17 10	5755.876	1 ⁻	4114.81	2 ⁺	
1681.2 ‡&	<0.09 ‡	5755.876	1 ⁻	4074.667	1 ⁺	
1732.39 11	0.44 6	6421.34	4 ⁻	4688.97	4 ⁺	
1732.7 ‡&	<0.75 ‡	5847.524	0 ⁺	4114.81	2 ⁺	
1739.32 9	0.48 6	6428.13	(2 ⁺ ,3 ⁺)	4688.97	4 ⁺	
1751.431 29	1.44 14	(11417.22)	2 ⁺	9665.74		
1763.5 ‡&	<0.11 ‡	5679.928	2 ⁺	3916.408	0 ⁺	
1772.82 4	1.40 14	5847.524	0 ⁺	4074.667	1 ⁺	
1788.794 20	79 8	3916.408	0 ⁺	2127.564	2 ⁺	$E_\gamma=1788.86$ 12, $\sigma=67$ mb 7 (2007ChZX: PGAA database).
1795.3 @ 3	0.19 @ 5	8138.08	(1) ⁻	6342.52	1 ⁻	
1795.3 @ 3	0.19 @ 5	9933.36	1 ⁻	8138.08	(1) ⁻	
1818.96 14	0.38 6	(11417.22)	2 ⁺	9598.42		
1839.5 ‡&	<0.70 ‡	5755.876	1 ⁻	3916.408	0 ⁺	
1840.52 12	0.56 9	9208.04	(1,2 ⁺)	7367.42	(1 ⁺ ,2 ⁺)	
1854.28 4	1.28 13	6478.770	1 ⁻	4624.404	3 ⁻	
1871.04 8	2.04 22	(11417.22)	2 ⁺	9546.09	(1,2 ⁺)	
^x 1887.66 4	1.78 17					
1922.92 22	0.61 11	5998.10	2 ⁺	4074.667	1 ⁺	
1924.0 ‡&	<0.21 ‡	5228.175	0 ⁺	3304.213	2 ⁺	
1925.94 17	0.28 8	10311.53	2 ⁺	8385.41	1 ⁻	
1947.060 20	29.2 26	4074.667	1 ⁺	2127.564	2 ⁺	$E_\gamma=1947.0$ 5, $\sigma=13$ mb 8 (2007ChZX: PGAA database).
1951.77 19	0.53 11	8294.40	4 ⁽⁺⁾	6342.52	1 ⁻	
1959.67 17	0.88 11	9208.04	(1,2 ⁺)	7248.05	(2 ⁺ ,3 ⁻)	
^x 1980.15 12	0.64 9					
^x 1984.2 4	0.50 14					
1987.19 3	6.5 7	4114.81	2 ⁺	2127.564	2 ⁺	
1998.3 4	0.14 5	11024.95	1 ⁻	9026.31	(1,2 ⁺)	
2018.3 ‡&	<0.11 ‡	5322.514	2 ⁺	3304.213	2 ⁺	
^x 2046.29 5	1.78 18					
2053.94 14	0.59 9	6168.862	3 ⁺	4114.81	2 ⁺	
2076.89 8	1.48 16	5380.99	1 ⁺	3304.213	2 ⁺	

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$^{33}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15** (continued)

$\gamma(^{34}\text{S})$ (continued)						
E_γ †	I_γ ‡#	E_i (level)	J_i^π	E_f	J_f^π	Comments
2094.2 ‡&	<0.11 ‡	6168.862	3 ⁺	4074.667	1 ⁺	$E_\gamma=2127.47$ 6, $\sigma=324$ mb 15 (2007ChZX: PGAA database).
2127.499 20	318 29	2127.564	2 ⁺	0.0	0 ⁺	
2152.41 23	0.17 9	9933.36	1 ⁻	7781.22	(1) ⁻	
2173.55 21	0.16 5	10311.53	2 ⁺	8138.08	(1) ⁻	
2209.10 6	0.86 9	(11417.22)	2 ⁺	9208.04	(1,2 ⁺)	
2230.14 14	0.80 10	7552.70	(0 ⁺ to 3 ⁻)	5322.514	2 ⁺	
2233.49 4	5.0 5	7110.451	3 ⁻	4876.842	3 ⁺	
2252.5 ‡&	<0.11 ‡	6168.862	3 ⁺	3916.408	0 ⁺	
2258.430 23	3.7 4	(11417.22)	2 ⁺	9158.715	(1,2 ⁺)	
^x 2282.17 4	1.70 16					
2290.26 15	0.27 5	8138.08	(1) ⁻	5847.524	0 ⁺	
2307.4 ‡&	<0.13 ‡	7629.907	3 ⁻	5322.514	2 ⁺	
2326.2 @ 10	0.05 @ 4	8805.6	(1,2 ⁺)	6478.770	1 ⁻	
2326.2 @ 10	0.05 @ 4	9546.09	(1,2 ⁺)	7219.28	2 ⁺	
2328.8 5	0.14 4	7219.28	2 ⁺	4889.76	2 ⁺	
2353.06 21	0.23 4	6428.13	(2 ⁺ ,3 ⁺)	4074.667	1 ⁺	
2363.97 8	2.1 11	8615.74	(2 ⁻ ,3 ⁺)	6251.68	(3,4)	
2375.657 20	26.0 24	5679.928	2 ⁺	3304.213	2 ⁺	
2390.82 6	1.33 14	(11417.22)	2 ⁺	9026.31	(1,2 ⁺)	
2401.7 ‡&	<0.13 ‡	7629.907	3 ⁻	5228.175	0 ⁺	
2404.04 6	1.07 11	6478.770	1 ⁻	4074.667	1 ⁺	
^x 2441.31 4	1.75 17					
2451.557 20	5.2 5	5755.876	1 ⁻	3304.213	2 ⁺	
^x 2475.15 4	1.71 17					
2490.6 13	0.62 16	7367.42	(1 ⁺ ,2 ⁺)	4876.842	3 ⁺	
2496.726 20	15.4 14	4624.404	3 ⁻	2127.564	2 ⁺	
2530.25 10	0.51 7	7219.28	2 ⁺	4688.97	4 ⁺	
2543.13 @ 10	9.6 @ 9	5847.524	0 ⁺	3304.213	2 ⁺	
2543.13 @ 10	9.6 @ 9	(11417.22)	2 ⁺	8874.01	(1 ⁻ ,2,3 ⁺)	
2558.82 13	1.24 14	7248.05	(2 ⁺ ,3 ⁻)	4688.97	4 ⁺	
2561.36 5	3.6 4	4688.97	4 ⁺	2127.564	2 ⁺	
2611.7 4	1.2 3	(11417.22)	2 ⁺	8805.6	(1,2 ⁺)	
2689.50 10	2.16 24	(11417.22)	2 ⁺	8727.64	(1 ⁻ ,2 ⁺)	
2714.50 19	2.8 5	(11417.22)	2 ⁺	8702.35	(0 ⁺ ,1,2)	
2740.2 ‡&	<0.18 ‡	7629.907	3 ⁻	4889.76	2 ⁺	
2749.24 5	7.0 7	4876.842	3 ⁺	2127.564	2 ⁺	
2753.3 13	0.93 23	6828.83	2 ⁺	4074.667	1 ⁺	
2762.10 8	3.0 3	4889.76	2 ⁺	2127.564	2 ⁺	
2801.33 5	10.1 10	(11417.22)	2 ⁺	8615.74	(2 ⁻ ,3 ⁺)	
^x 2810.3 3	0.87 13					
2817.76 @ 25	0.84 @ 13	6121.56	2 ⁺	3304.213	2 ⁺	
2817.76 @ 25	0.84 @ 13	9665.74		6847.90	(1,2 ⁺)	
2839.3 4	1.00 16	6954.218	2 ⁻	4114.81	2 ⁺	
2843.7 6	0.59 13	10311.53	2 ⁺	7467.74	(0 ⁺ ,1,2)	
2864.56 4	10.9 11	6168.862	3 ⁺	3304.213	2 ⁺	
2910.28 5	10.0 10	(11417.22)	2 ⁺	8506.77	1 ⁻	
2919.7 5	0.43 11	10650.13		7730.79	(1 ⁻ ,2 ⁻ ,3 ⁻)	
2940.42 31	1.05 15	7629.907	3 ⁻	4688.97	4 ⁺	
2945.8 @ 10	0.30 @ 9	8174.9	(1,2 ⁺)	5228.175	0 ⁺	
2945.8 @ 10	0.30 @ 9	8702.35	(0 ⁺ ,1,2)	5755.876	1 ⁻	

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$^{33}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15** (continued) $\gamma(^{34}\text{S})$ (continued)

E_γ †	I_γ †#	E_i (level)	J_i^π	E_f	J_f^π	Comments
2989.9 7	0.18 9	9836.70		6847.90	(1,2 ⁺)	
2995.8 6	0.37 10	7110.451	3 ⁻	4114.81	2 ⁺	
3005.39 5	10.0 10	7629.907	3 ⁻	4624.404	3 ⁻	
3022.0 10	0.16 9	8702.35	(0 ⁺ ,1,2)	5679.928	2 ⁺	
3031.69 8	4.6 6	(11417.22)	2 ⁺	8385.41	1 ⁻	
3038.18 32	1.27 17	6342.52	1 ⁻	3304.213	2 ⁺	
^x 3051.83 26	0.64 12					
3089.53 26	0.56 11	7164.47	(0 ⁺ to 3 ⁺)	4074.667	1 ⁺	
3100.6‡&	<0.21‡	5228.175	0 ⁺	2127.564	2 ⁺	
3122.65 15	2.7 4	(11417.22)	2 ⁺	8294.40	4 ⁽⁺⁾	
^x 3149.29 15	0.89 12					
3174.37 5	10.5 10	6478.770	1 ⁻	3304.213	2 ⁺	
3183.9 7	0.12 8	8506.77	1 ⁻	5322.514	2 ⁺	
3194.74 5	7.4 8	5322.514	2 ⁺	2127.564	2 ⁺	
3211.69 9	2.36 23	(11417.22)	2 ⁺	8205.4	(1 ⁻ to 4 ⁺)	
3231.89 20	0.84 11	(11417.22)	2 ⁺	8185.46	1 ⁺	
3241.9 5	0.36 7	(11417.22)	2 ⁺	8174.9	(1,2 ⁺)	
3253.21 6	3.8 4	5380.99	1 ⁺	2127.564	2 ⁺	
3278.79 11	3.2 4	(11417.22)	2 ⁺	8138.08	(1) ⁻	
3304.031 20	63 6	3304.213	2 ⁺	0.0	0 ⁺	$E_\gamma=3304.19$ 9, $\sigma=50$ mb 11 (2007ChZX: PGAA database).
3311.6 5	0.62 11	9158.715	(1,2 ⁺)	5847.524	0 ⁺	
3392.86 24	1.57 19	7467.74	(0 ⁺ ,1,2)	4074.667	1 ⁺	
3442.24 25	1.02 16	(11417.22)	2 ⁺	7974.72	(1,2 ⁺)	
3451.5 9	0.35 10	7367.42	(1 ⁺ ,2 ⁺)	3916.408	0 ⁺	
3476.95 18	0.71 10	9598.42		6121.56	2 ⁺	
3500.3 5	0.48 11	8727.64	(1 ⁻ ,2 ⁺)	5228.175	0 ⁺	
3515.07 11	1.43 16	7629.907	3 ⁻	4114.81	2 ⁺	
3552.08 4	17.34 17	5679.928	2 ⁺	2127.564	2 ⁺	
3581.2 4	0.37 7	8205.4	(1 ⁻ to 4 ⁺)	4624.404	3 ⁻	
3628.10 4	17.6 16	5755.876	1 ⁻	2127.564	2 ⁺	
3635.83 8	5.2 6	(11417.22)	2 ⁺	7781.22	(1) ⁻	
3644.8 8	0.48 10	9026.31	(1,2 ⁺)	5380.99	1 ⁺	
3649.88 12	3.11 31	6954.218	2 ⁻	3304.213	2 ⁺	
3664.8 4	0.47 10	10092.21		6428.13	(2 ⁺ ,3 ⁺)	
3713.5‡&	<0.18‡	7629.907	3 ⁻	3916.408	0 ⁺	
3719.68 16	1.91 20	5847.524	0 ⁺	2127.564	2 ⁺	
3738.69 17	1.18 17	8615.74	(2 ⁻ ,3 ⁺)	4876.842	3 ⁺	
3787.096 20	26.5 25	(11417.22)	2 ⁺	7629.907	3 ⁻	$E_\gamma=3787.2$ 5, $\sigma=22$ mb 6 (2007ChZX: PGAA database).
3812.0 5	0.25 6	8702.35	(0 ⁺ ,1,2)	4889.76	2 ⁺	
3864.25 11	1.68 17	(11417.22)	2 ⁺	7552.70	(0 ⁺ to 3 ⁻)	
3870.51 31	0.56 8	5998.10	2 ⁺	2127.564	2 ⁺	
3949.27 12	1.54 17	(11417.22)	2 ⁺	7467.74	(0 ⁺ ,1,2)	
3990.7 7	0.29 7	8615.74	(2 ⁻ ,3 ⁺)	4624.404	3 ⁻	
3994.8 8	0.25 7	6121.56	2 ⁺	2127.564	2 ⁺	
4040.63 29	0.54 8	6168.862	3 ⁺	2127.564	2 ⁺	
4049.68 15	1.17 13	(11417.22)	2 ⁺	7367.42	(1 ⁺ ,2 ⁺)	
4074.418 20	31.3 29	4074.667	1 ⁺	0.0	0 ⁺	
4114.52 4	8.6 9	4114.81	2 ⁺	0.0	0 ⁺	
4197.69 9	3.0 4	(11417.22)	2 ⁺	7219.28	2 ⁺	
4248.28 21	1.59 18	7552.70	(0 ⁺ to 3 ⁻)	3304.213	2 ⁺	
4252.38 22	1.23 15	(11417.22)	2 ⁺	7164.47	(0 ⁺ to 3 ⁺)	
4306.44 6	8.3 8	(11417.22)	2 ⁺	7110.451	3 ⁻	
4325.397 30	12.7 12	7629.907	3 ⁻	3304.213	2 ⁺	

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15** (continued) $\gamma(^{34}\text{S})$ (continued)

E_γ †	I_γ †#	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult.	Comments
4350.85 9	6.2 7	6478.770	1 ⁻	2127.564	2 ⁺		
4391.77 29	0.44 9	8506.77	1 ⁻	4114.81	2 ⁺		
4462.44 20	7.9 8	(11417.22)	2 ⁺	6954.218	2 ⁻		
4499.7 10	0.23 7	10179.59	1 ⁺	5679.928	2 ⁺		
4532.6 7	0.23 7	10212.15		5679.928	2 ⁺		
4540.68 15	1.70 20	8615.74	(2 ⁻ ,3 ⁺)	4074.667	1 ⁺		
4568.9 4	0.30 6	(11417.22)	2 ⁺	6847.90	(1,2 ⁺)		
4588.37 26	0.59 10	(11417.22)	2 ⁺	6828.83	2 ⁺		
4624.2 5	0.21 5	4624.404	3 ⁻	0.0	0 ⁺	[E3]	
4670.1 6	0.11 6	7974.72	(1,2 ⁺)	3304.213	2 ⁺		
4731.37 10	1.58 16	(11417.22)	2 ⁺	6685.333	(0 to 3) ⁻		
4758.79 27	0.46 8	8874.01	(1 ⁻ ,2,3 ⁺)	4114.81	2 ⁺		
4799.11 28	0.52 8	8874.01	(1 ⁻ ,2,3 ⁺)	4074.667	1 ⁺		
4826.0 5	0.11 5	6954.218	2 ⁻	2127.564	2 ⁺		
4876.8 ‡&	<0.25 ‡	4876.842	3 ⁺	0.0	0 ⁺		
4889.30 8	2.68 26	4889.76	2 ⁺	0.0	0 ⁺		
4903.4 5	0.28 8	11024.95	1 ⁻	6121.56	2 ⁺		
4938.06 3	22.2 21	(11417.22)	2 ⁺	6478.770	1 ⁻		
4982.44 20	1.31 14	7110.451	3 ⁻	2127.564	2 ⁺		
4988.6 4	0.63 9	10311.53	2 ⁺	5322.514	2 ⁺		
5036.4 7	0.25 6	7164.47	(0 ⁺ to 3 ⁺)	2127.564	2 ⁺		
5043.3 4	1.59 26	9158.715	(1,2 ⁺)	4114.81	2 ⁺		
5074.79 25	0.42 8	(11417.22)	2 ⁺	6342.52	1 ⁻		
5084.2 5	0.14 5	9158.715	(1,2 ⁺)	4074.667	1 ⁺		
5202.06 6	3.00 29	8506.77	1 ⁻	3304.213	2 ⁺		
5239.8 4	0.65 9	7367.42	(1 ⁺ ,2 ⁺)	2127.564	2 ⁺		
5247.94 4	11.8 11	(11417.22)	2 ⁺	6168.862	3 ⁺		
5268.9 @ 6	0.27 @ 7	10650.13		5380.99	1 ⁺		
5268.9 6	0.27 7	11024.95	1 ⁻	5755.876	1 ⁻		
5294.94 24	0.42 8	(11417.22)	2 ⁺	6121.56	2 ⁺		
5311.10 15	0.80 10	8615.74	(2 ⁻ ,3 ⁺)	3304.213	2 ⁺		
5322.5 ‡&	<0.24 ‡	5322.514	2 ⁺	0.0	0 ⁺		
5380.59 9	1.97 20	5380.99	1 ⁺	0.0	0 ⁺		
5501.4 5	0.46 9	8805.6	(1,2 ⁺)	3304.213	2 ⁺		
5569.30 5	5.6 6	(11417.22)	2 ⁺	5847.524	0 ⁺		
5602.78 15	1.15 14	7730.79	(1 ⁻ ,2 ⁻ ,3 ⁻)	2127.564	2 ⁺		
5660.78 6	18.4 18	(11417.22)	2 ⁺	5755.876	1 ⁻		$E_\gamma=5660.0$ 4, $\sigma=42$ mb 8 (2007ChZX: PGAA database).
5679.9 ‡&	<0.53 ‡	5679.928	2 ⁺	0.0	0 ⁺		
5736.76 4	43 4	(11417.22)	2 ⁺	5679.928	2 ⁺		$E_\gamma=5736.75$ 23, $\sigma=63$ mb 8 (2007ChZX: PGAA database).
5755.5 5	0.51 8	5755.876	1 ⁻	0.0	0 ⁺		
5847.4 5	0.25 6	7974.72	(1,2 ⁺)	2127.564	2 ⁺		
5884.6 6	0.27 6	9801.89	(1,2 ⁺)	3916.408	0 ⁺		
5997.30 31	0.34 6	5998.10	2 ⁺	0.0	0 ⁺		
6010.3 3	0.50 8	8138.08	(1) ⁻	2127.564	2 ⁺		
6035.68 7	4.4 5	(11417.22)	2 ⁺	5380.99	1 ⁺		
6077.27 12	1.19 13	8205.4	(1 ⁻ to 4 ⁺)	2127.564	2 ⁺		
6094.4 4	0.21 5	(11417.22)	2 ⁺	5322.514	2 ⁺		
6152.1 5	0.18 5	10840.62	3 ⁻	4688.97	4 ⁺		
6166.24 13	1.55 17	8294.40	4 ⁽⁺⁾	2127.564	2 ⁺		
6188.45 6	8.7 9	(11417.22)	2 ⁺	5228.175	0 ⁺		
6236.3 11	0.19 5	10311.53	2 ⁺	4074.667	1 ⁺		
6241.0 5	0.45 7	9546.09	(1,2 ⁺)	3304.213	2 ⁺		

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15** (continued) $\gamma(^{34}\text{S})$ (continued)

$E_\gamma^\dagger$	$I_\gamma^{\ddagger\#}$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult.	Comments
6341.58 32	0.45 8	6342.52	1 ⁻	0.0	0 ⁺		
6478.8 ^{‡&}	<0.02 [‡]	6478.770	1 ⁻	0.0	0 ⁺		
6487.48 6	3.6 4	8615.74	(2 ⁻ ,3 ⁺)	2127.564	2 ⁺		
6496.62 23	0.56 7	9801.89	(1,2 ⁺)	3304.213	2 ⁺		
6526.84 6	5.5 6	(11417.22)	2 ⁺	4889.76	2 ⁺		
6539.66 16	0.99 12	(11417.22)	2 ⁺	4876.842	3 ⁺		
6573.6 4	1.09 19	8702.35	(0 ⁺ ,1,2)	2127.564	2 ⁺		
6600.1 7	0.23 5	8727.64	(1 ⁻ ,2 ⁺)	2127.564	2 ⁺		
6727.5 9	0.07 4	(11417.22)	2 ⁺	4688.97	4 ⁺	[E2]	
6745.64 16	2.71 30	8874.01	(1 ⁻ ,2,3 ⁺)	2127.564	2 ⁺		
6792.10 3	24.2 23	(11417.22)	2 ⁺	4624.404	3 ⁻		$E_\gamma=6792.1$ 6, $\sigma=22$ mb 5 (2007ChZX : PGAA database).
6846.37 32	0.56 7	6847.90	(1,2 ⁺)	0.0	0 ⁺		
7218.48 13	2.71 28	7219.28	2 ⁺	0.0	0 ⁺		
7302.2 8	0.28 5	(11417.22)	2 ⁺	4114.81	2 ⁺		
7341.67 6	36.5 34	(11417.22)	2 ⁺	4074.667	1 ⁺		$E_\gamma=7341.4$ 5, $\sigma=33$ mb 8 (2007ChZX : PGAA database).
7499.90 5	62 6	(11417.22)	2 ⁺	3916.408	0 ⁺		
7536.2 7	0.44 10	9665.74		2127.564	2 ⁺		
7629.9 ^{‡&}	<0.33 [‡]	7629.907	3 ⁻	0.0	0 ⁺		
7675.0 8	0.16 4	9801.89	(1,2 ⁺)	2127.564	2 ⁺		
7708.32 30	0.44 7	9836.70		2127.564	2 ⁺		
7780.22 10	3.8 5	7781.22	(1) ⁻	0.0	0 ⁺		
7973.45 25	0.42 6	7974.72	(1,2 ⁺)	0.0	0 ⁺		
8036.6 7	0.18 4	8036.31	(1 ⁻ ,2 ⁺)	0.0	0 ⁺		
8051.1 6	0.26 5	10179.59	1 ⁺	2127.564	2 ⁺		
8083.49 31	0.47 7	10212.15		2127.564	2 ⁺		
8111.99 9	6.1 7	(11417.22)	2 ⁺	3304.213	2 ⁺		
8136.98 17	1.40 16	8138.08	(1) ⁻	0.0	0 ⁺		
8173.8 9	0.157 31	8174.9	(1,2 ⁺)	0.0	0 ⁺		
8184.70 24	0.64 7	8185.46	1 ⁺	0.0	0 ⁺		
8384.28 9	3.43 33	8385.41	1 ⁻	0.0	0 ⁺		
8505.68 10	4.7 5	8506.77	1 ⁻	0.0	0 ⁺		
8726.78 24	0.44 6	8727.64	(1 ⁻ ,2 ⁺)	0.0	0 ⁺		
8804.4 4	0.24 4	8805.6	(1,2 ⁺)	0.0	0 ⁺		
9024.95 17	0.80 9	9026.31	(1,2 ⁺)	0.0	0 ⁺		
9206.65 26	0.35 5	9208.04	(1,2 ⁺)	0.0	0 ⁺		
9288.28 16	1.10 12	(11417.22)	2 ⁺	2127.564	2 ⁺		
9544.83 28	0.38 5	9546.09	(1,2 ⁺)	0.0	0 ⁺		
9932.1 6	0.082 19	9933.36	1 ⁻	0.0	0 ⁺		
11415.17 11	7.1 7	(11417.22)	2 ⁺	0.0	0 ⁺		

[†] From **1985Ra15**, unless otherwise noted. γ -ray cross sections in mb are reported in **1985Ra15**, multiplying these by 0.220 yields the number of photons per 100 thermal neutron captures.

[‡] Extra possible γ transitions are reported in Table IX of **1985Ra15** that are not included in Table V of **1985Ra15**. Table IX provides γ -ray branching ratios. E_γ values can be deduced from level-energy differences. Intensities are all upper limits.

[#] For intensity per 100 neutron captures, multiply by 0.220 15.

@ Multiply placed with undivided intensity.

& Placement of transition in the level scheme is uncertain.

^x γ ray not placed in level scheme.

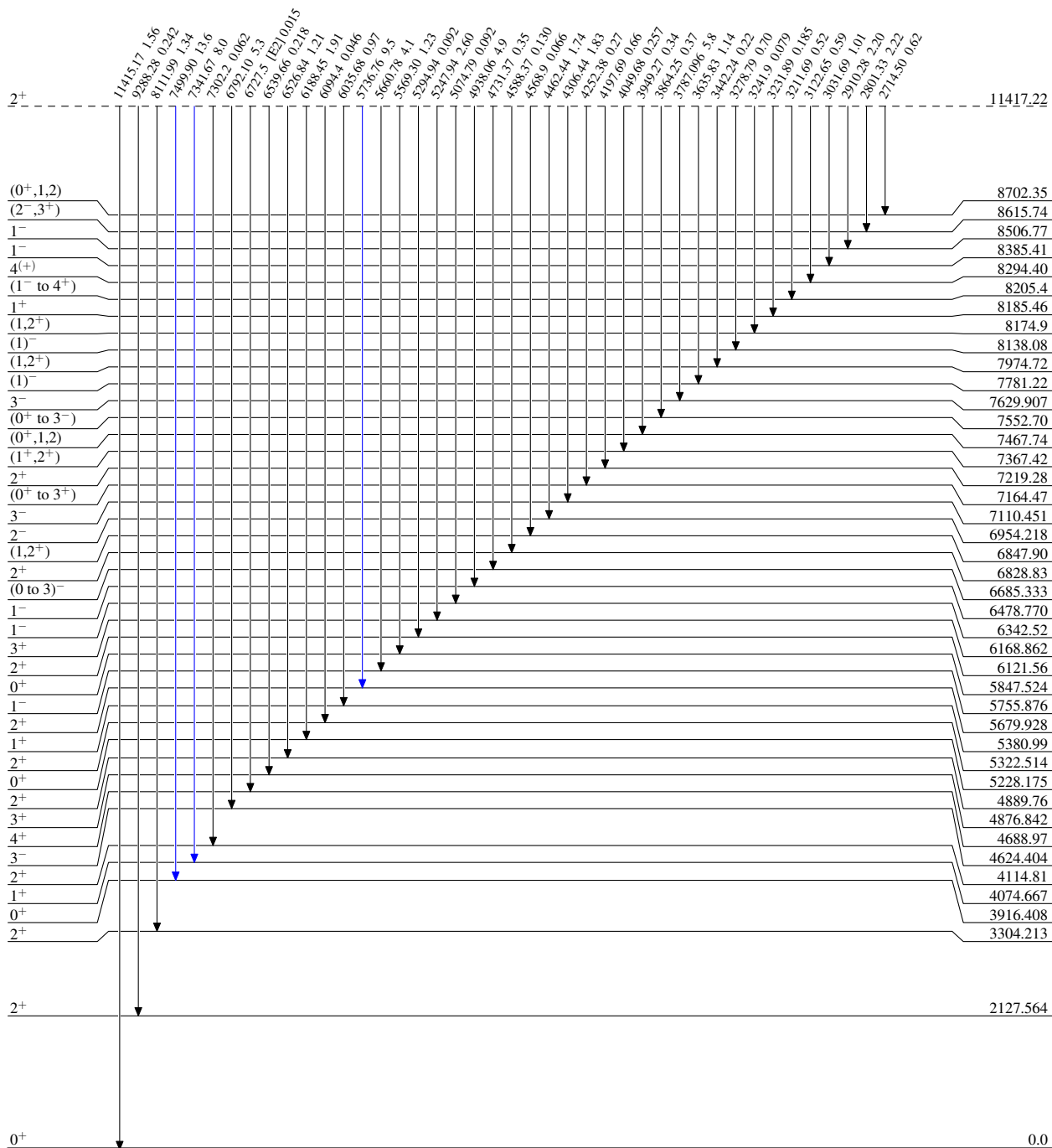
$^{33}\text{S}(n,\gamma)$ E=thermal 1985Ra15

Legend

Level Scheme

Intensities: I_γ per 100 neutron captures

$\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$



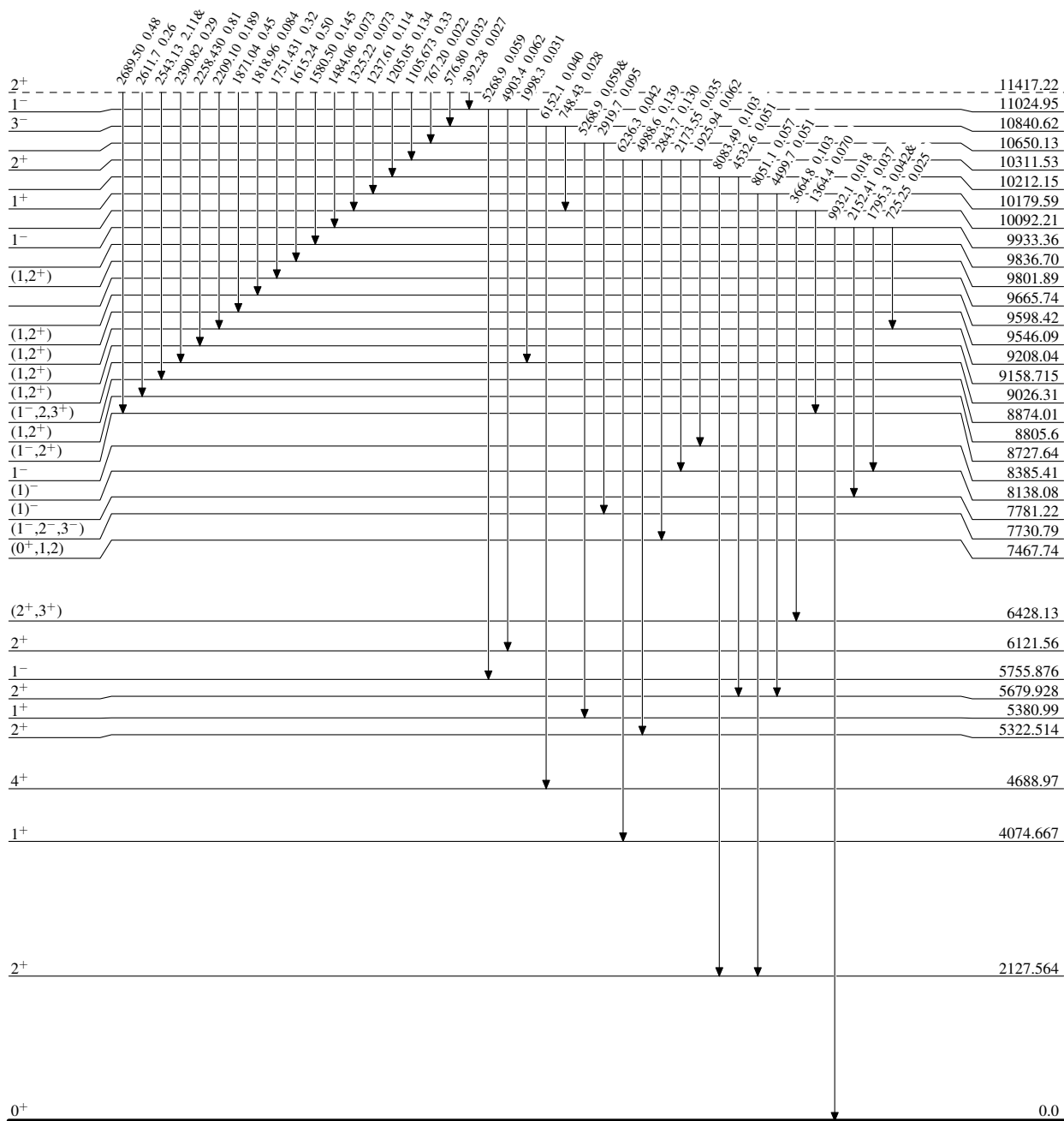
$^{33}\text{S}(\text{n},\gamma)$ E=thermal 1985Ra15

Level Scheme (continued)

Legend

Intensities: I_γ per 100 neutron captures
& Multiplied by: undivided intensity given

— $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 — $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 — $I_\gamma > 10\% \times I_\gamma^{\text{max}}$



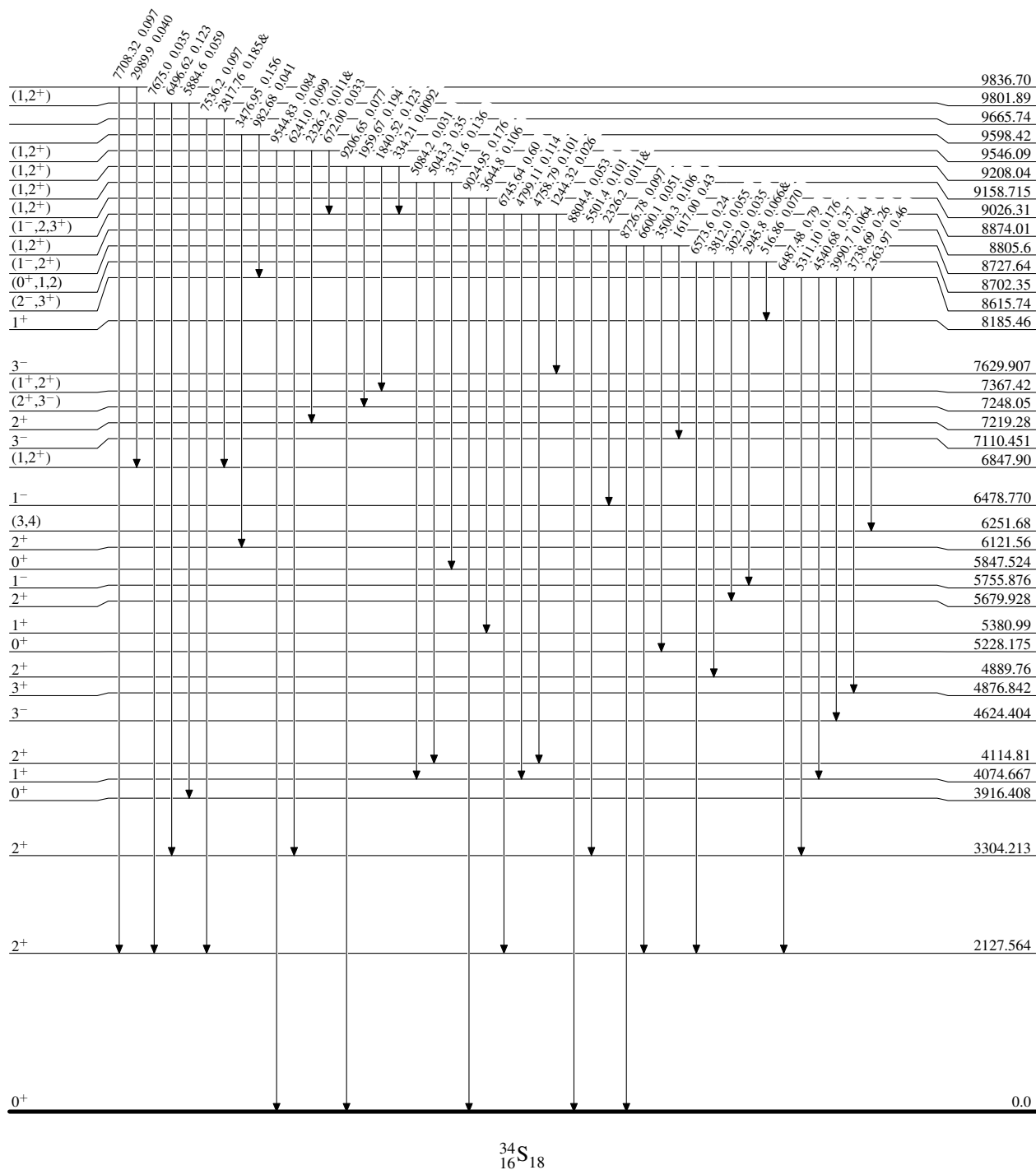
$^{33}\text{S}(\text{n},\gamma) \text{E=thermal}$ 1985Ra15

Level Scheme (continued)

Legend

Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given

→ $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
→ $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
→ $I_\gamma > 10\% \times I_\gamma^{\text{max}}$



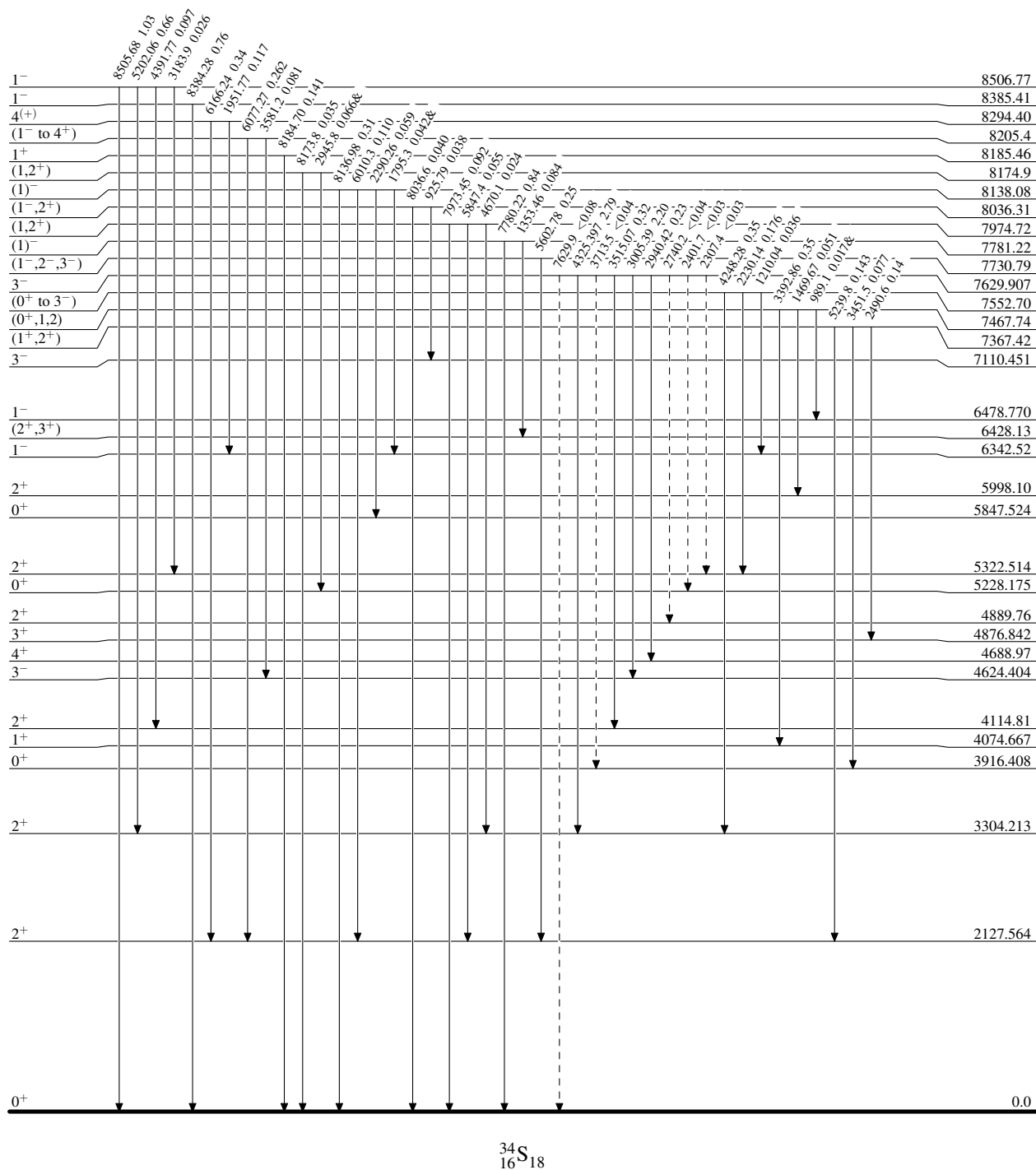
$^{33}\text{S}(\text{n},\gamma) \text{E=thermal}$ 1985Ra15

Legend

Level Scheme (continued)

Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given

→ $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 → $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 → $I_\gamma > 10\% \times I_\gamma^{\text{max}}$
 - - - - - γ Decay (Uncertain)

 $^{34}\text{S}_{18}$

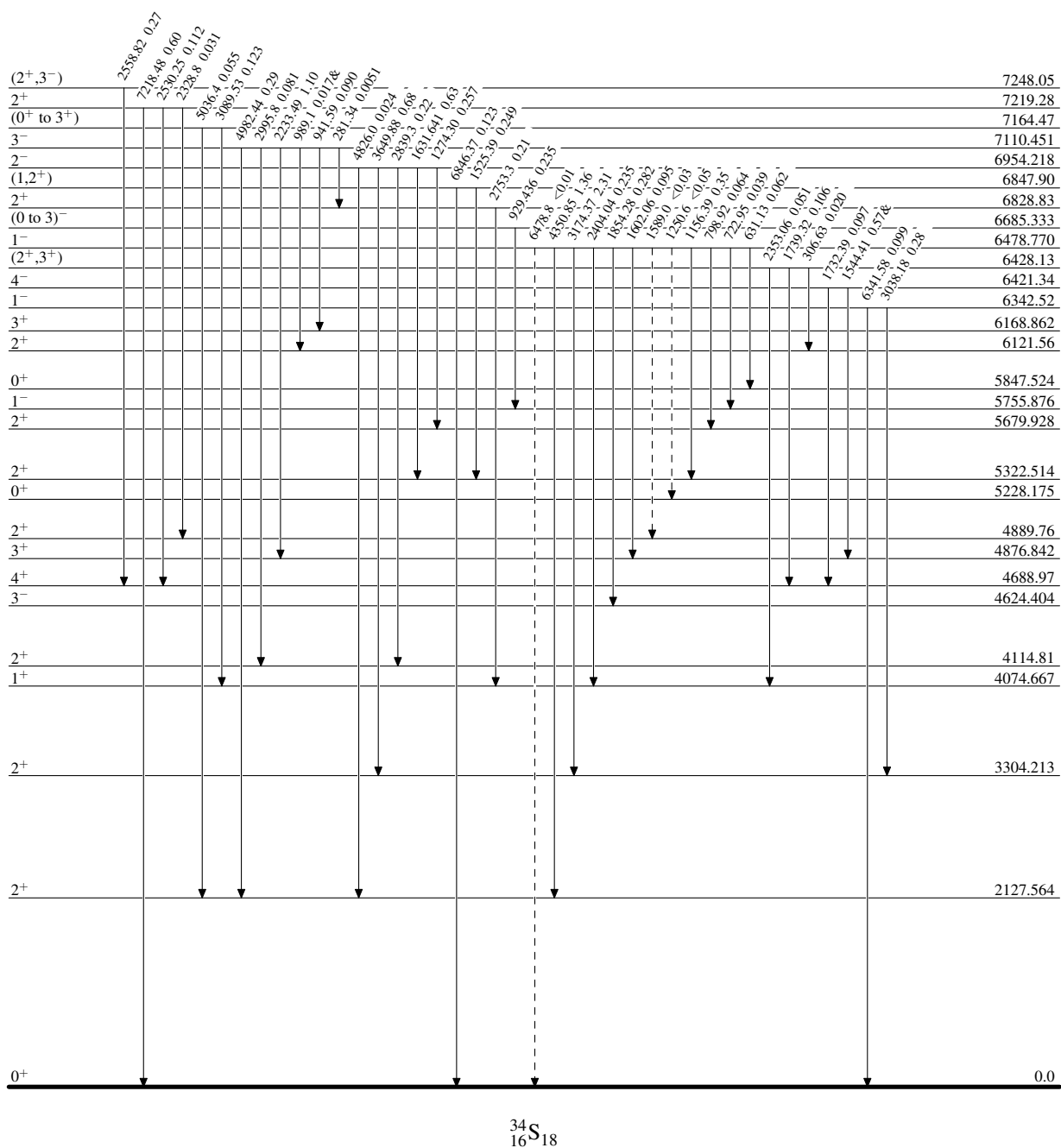
$^{33}\text{S}(\text{n},\gamma)$ E=thermal 1985Ra15

Level Scheme (continued)

Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given

Legend

-  $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 $I_\gamma > 10\% \times I_\gamma^{\text{max}}$
 γ Decay (Uncertain)



³³S(n,γ) E=thermal 1985Ra15

Level Scheme (continued)

Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given

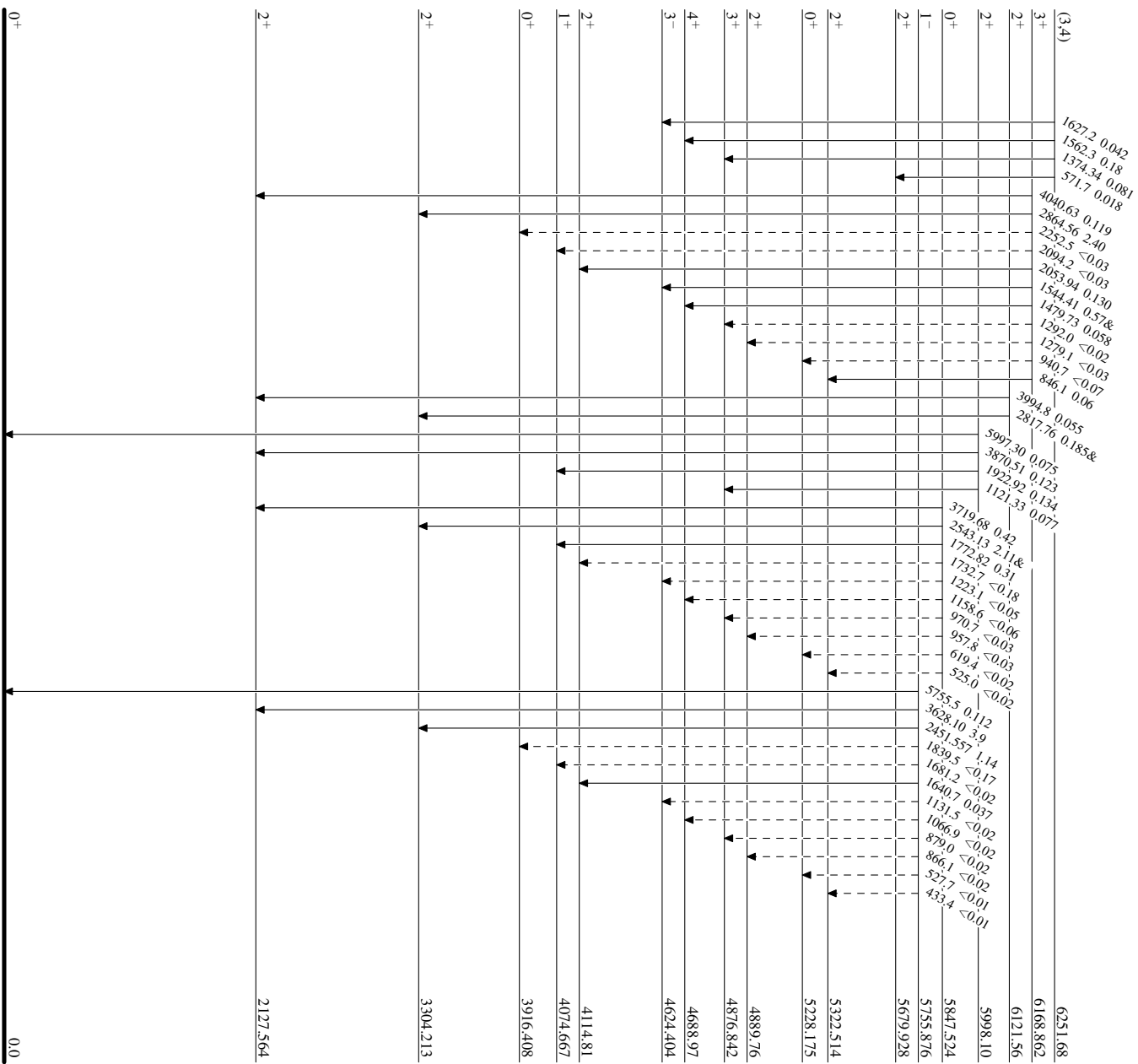
Legend

$I_\gamma < 2\% \times I_\gamma^{max}$

$I_\gamma < 10\% \times I_\gamma^{max}$

$I_\gamma > 10\% \times I_\gamma^{max}$

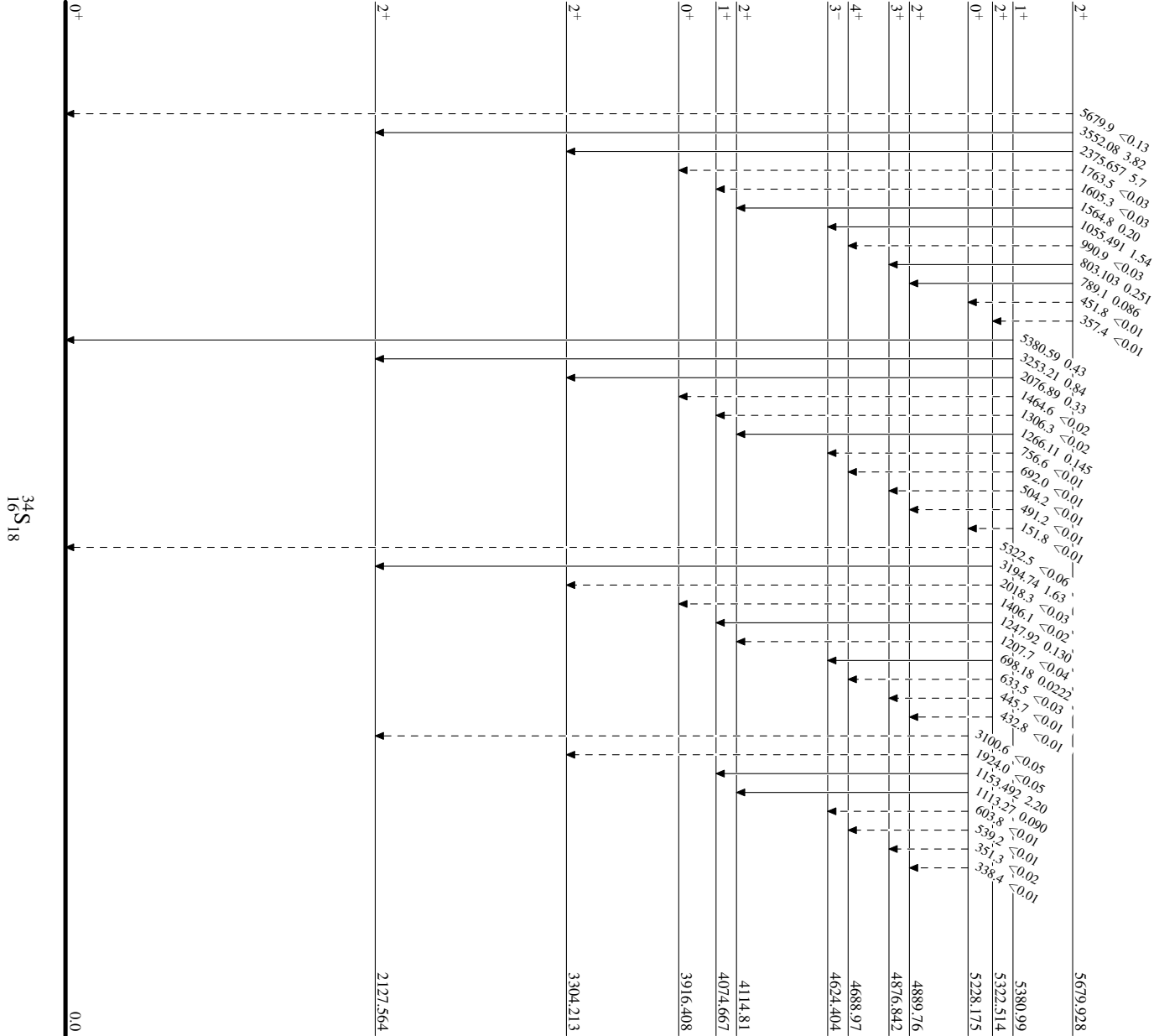
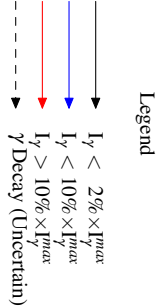
γ Decay (Uncertain)



³³S(n,γ) E=thermal 1985Ra15

Level Scheme (continued)

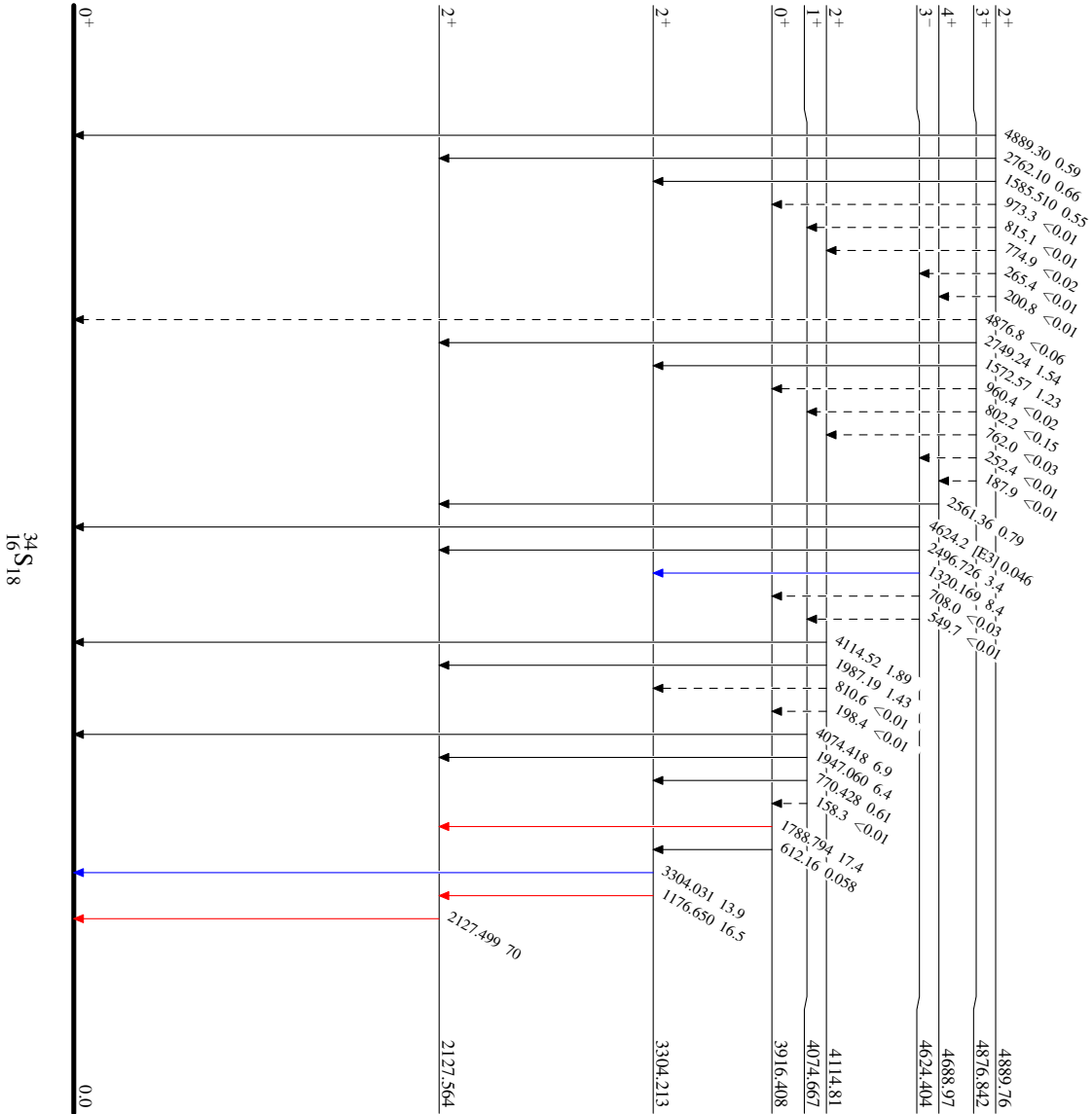
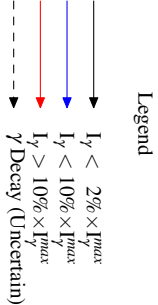
Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given



³³S(n, γ) E=thermal 1985Ra15

Level Scheme (continued)

Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given



$^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n})$:resonances **2018MuZY**

$J^\pi=3/2^+$ for ^{33}S ground state (2025Ch07).

1987Wa20: $^{33}\text{S}(\text{n},\alpha)$ with $E_n=0.010$ -1 MeV at the Geel Electron Linear Accelerator (GELINA). α particles were detected using a large surface-barrier detector. Measured integrated (n,α) cross sections. Deduced resonance parameters.

1987Co23: $^{33}\text{S}(\text{n},\text{n})$ with $E=0.010$ -2 MeV at the Oak Ridge Electron Linear Accelerator (ORELA). Neutrons were detected using a NE-110 scintillator. Measured total $^{33}\text{S}(\text{n},\text{n})$ cross sections. Deduced resonance parameters.

1975Au06: $^{33}\text{S}(\text{n},\alpha_0)$ with $E_n=10$ -700 keV at the Los Alamos Scientific Laboratory (LASL) vertical Van de Graaff accelerator. α particles were detected using silicon surface-barrier detectors. Measured $^{33}\text{S}(\text{n},\alpha_0)$ cross sections. $^{33}\text{S}(\text{n},\gamma)$ with $E_n=2.5$ -850 keV at the Oak Ridge Electron Linear Accelerator (ORELA). γ rays were detected using two NE-226 fluorocarbon liquid scintillators. Measured $^{33}\text{S}(\text{n},\gamma)$ cross sections. Deduced resonance parameters.

2018MuZY: Atlas of Neutron Resonances. Evaluation of neutron resonance energies, spins, width parameters, and resonance strengths for nuclei of $Z=1$ -60.

 ^{34}S Levels

J^π , L, and resonance parameters E_n , Γ_n , Γ_γ , Γ_α , and $g\Gamma_n\Gamma_\gamma/\Gamma$ are all from 2018MuZY. Statistical factor $g=(2J+1)/[(2J_n+1)(2J_T+1)]$, where J is the resonance spin, $J_n=1/2$ is the neutron spin, and $J_T=3/2$ is the ^{33}S ground state spin.

E(level) [†]	J^π	Γ	L	$E_n(\text{lab})$	Comments
11411.33?	2^+		0	-5.987	E(level): fictitious level from analysis of resonance spectrum below the S(n) threshold.
11430.3	2^+	0.116 keV 20	0	13.45	$\Gamma_\gamma=1.5$ eV. $g\Gamma_n\Gamma_\gamma/\Gamma=0.086$ 6.
11434.3	2^-	0.049 keV 10	1	17.63	$\Gamma_n=75.0$ eV 8. $\Gamma_\gamma=0.21$ eV 5. $\Gamma_\alpha=41$ eV 5. $g\Gamma_n\Gamma_\gamma/\Gamma=0.55$ 3.
11440.4	3^-	0.0198 keV 10	1	23.95	$\Gamma_n=39.1$ eV 8. $\Gamma_\gamma=0.90$ eV 5. $g\Gamma_n\Gamma_\gamma/\Gamma=1.02$ 5.
11448.1		<0.015 keV		31.79	$\Gamma_n=16.0$ eV 9. $\Gamma_\gamma=1.44$ eV 10. $\Gamma_\alpha=2.5$ eV 3.
11467.8	2^+	0.368 keV 8	0	52.11	$g\Gamma_n\Gamma_\gamma/\Gamma=0.062$ 4. $g\Gamma_n\Gamma_\gamma/\Gamma=0.309$ 20.
11469.2	3^-	0.152 keV 15	1	53.58	$\Gamma_n=349$ eV 6. $\Gamma_\gamma=0.52$ eV 4. $\Gamma_\alpha=18$ eV 2. $g\Gamma_n\Gamma_\gamma/\Gamma=0.54$ 3.
11474.6	2^-	0.45 keV 6	1	59.15	$\Gamma_n=68$ eV 3. $\Gamma_\gamma=1.4$ eV 3. $\Gamma_\alpha=83$ eV 13. $g\Gamma_n\Gamma_\gamma/\Gamma=0.66$ 4.
11486.0	1^-		1	70.88	$\Gamma_n=275$ eV 5. $\Gamma_\gamma=1.08$ eV 7. $\Gamma_\alpha=0.17$ keV 5.
11492.7	2^-	0.51 keV 10	1	77.83	$\Gamma_n=65$ eV 10. $\Gamma_\alpha=0.107$ keV 63. $g\Gamma_n\Gamma_\gamma/\Gamma=1.31$ 8.
11496.2	2^+	0.71 keV 3	0	81.36	$\Gamma_n=507$ eV 13. $\Gamma_\gamma=2.11$ eV 14. $g\Gamma_n\Gamma_\gamma/\Gamma=0.58$ 4.
11499.6	1^-		1	84.88	$\Gamma_n=705$ eV 19. $\Gamma_\gamma=0.94$ eV 6. $\Gamma_\alpha=4$ eV 2.
11502.3	1^-	0.292 keV 25	1	87.63	$\Gamma_n=1.33$ keV 8. $\Gamma_\alpha=4.0$ keV 6. $g\Gamma_n\Gamma_\gamma/\Gamma=0.76$ 49.
11502.9		0.26 keV 5		88.32	$\Gamma_n=280$ eV 20. $\Gamma_\gamma=2.11$ eV 14. $\Gamma_\alpha=10$ eV 5. $g\Gamma_n\Gamma_\gamma/\Gamma=0.14$ 10.
11515.3	2^-	1.262 keV 25	1	101.09	$g\Gamma_n\Gamma_\gamma/\Gamma=0.93$ 8. $\Gamma_n=1.260$ keV 25. $\Gamma_\gamma=1.48$ eV 13.
11541	1^-	0.63 keV 7	1	127.76	$g\Gamma_n\Gamma_\gamma/\Gamma=0.30$ 4. $\Gamma_n=0.36$ keV 4. $\Gamma_\gamma=1.4$ eV 4. $\Gamma_\alpha=0.27$ keV 6.
11544		0.20 keV 4		130.6	$g\Gamma_n\Gamma_\gamma/\Gamma=0.34$ 4.
11546		0.23 keV 4		133.1	$g\Gamma_n\Gamma_\gamma/\Gamma=0.41$ 3.
11551		0.15 keV 3		138.2	$g\Gamma_n\Gamma_\gamma/\Gamma=0.91$ 7.
11564	≥ 1	0.134 keV 18		151.57	$g\Gamma_n\Gamma_\gamma/\Gamma=0.85$ 7.
11575	(0^-)	0.152 keV 32	(1)	162.34	
11581	2^-	3.42 keV 8	1	168.51	$g\Gamma_n\Gamma_\gamma/\Gamma=1.65$ 16. $\Gamma_n=3.42$ keV 8. $\Gamma_\gamma=2.6$ eV 3.

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{n},\gamma),(\text{n},\text{n}):$ resonances **2018MuZY (continued)** ^{34}S Levels (continued)

$E(\text{level})^\dagger$	J^π	Γ	L	$E_n(\text{lab})$	Comments
11590	2^-	0.76 keV 4	1	178.29	$g\Gamma_n\Gamma_\gamma/\Gamma=0.54$ 7. $\Gamma_n=0.76$ keV 4. $\Gamma_\gamma=0.87$ eV 11.
11608	3^-	0.62 keV 3	1	196.6	$g\Gamma_n\Gamma_\gamma/\Gamma=1.16$ 9. $\Gamma_n=0.61$ keV 3. $\Gamma_\gamma=1.33$ eV 12.
11610		0.70 keV 14		199.1	$g\Gamma_n\Gamma_\gamma/\Gamma=1.07$ 11.
11614	3^-	2.1 keV 8	1	203.17	$g\Gamma_n\Gamma_\gamma/\Gamma=1.89$ 15. $\Gamma_n=2.09$ keV 8. $\Gamma_\gamma=2.17$ eV 20. $\Gamma_\alpha=14$ eV 5.
11622		0.31 keV 6		210.8	$g\Gamma_n\Gamma_\gamma/\Gamma=0.30$ 5.
11626		<0.12 keV		215.6	$g\Gamma_n\Gamma_\gamma/\Gamma=0.32$ 5.
11632	2^+	0.75 keV 7	0	221.2	$g\Gamma_n\Gamma_\gamma/\Gamma=0.72$ 11. $\Gamma_n=0.69$ keV 7. $\Gamma_\gamma=1.24$ eV 40. $\Gamma_\alpha=55$ eV 20.
11634	0^+	5.3 keV 10	2	223.17	$\Gamma_n=4.4$ keV 9. $\Gamma_\alpha=0.9$ keV 3.
11639	3^-	0.96 keV 6	1	228.6	$g\Gamma_n\Gamma_\gamma/\Gamma=0.56$ 8. $\Gamma_n=0.76$ keV 5. $\Gamma_\gamma=0.81$ eV 13. $\Gamma_\alpha=0.20$ keV 3.
11649	3^-	0.61 keV 12	1	238.6	$g\Gamma_n\Gamma_\gamma/\Gamma=1.58$ 14. $\Gamma_n=0.46$ keV 3. $\Gamma_\gamma=1.82$ eV 20.
11669	2^-	0.40 keV 8	1	259.51	$g\Gamma_n\Gamma_\gamma/\Gamma=1.48$ 12. $\Gamma_n=0.67$ keV 6. $\Gamma_\gamma=2.4$ eV 2.
11670	1^+	0.55 keV 11	0	260.92	$g\Gamma_n\Gamma_\gamma/\Gamma=0.80$ 9. $\Gamma_n=0.225$ keV 70. $\Gamma_\gamma=2.1$ eV 3.
11704		0.61 keV 12		295.4	$g\Gamma_n\Gamma_\gamma/\Gamma=2.02$ 16.
11706		0.79 keV 16		298.2	$g\Gamma_n\Gamma_\gamma/\Gamma=1.89$ 16.
11717		0.67 keV 14		308.7	$g\Gamma_n\Gamma_\gamma/\Gamma=1.18$ 22.
11744		0.28 keV 6		335.9	$g\Gamma_n\Gamma_\gamma/\Gamma=0.69$ 13.
11773		0.40 keV 8		367.4	$g\Gamma_n\Gamma_\gamma/\Gamma=1.24$ 18.
11785		1.40 keV 25		377.9	$g\Gamma_n\Gamma_\gamma/\Gamma=2.7$ 5.
11796		1.30 keV 25		391.3	$g\Gamma_n\Gamma_\gamma/\Gamma=3.0$ 4.
11829		1.7 keV 3		425.3	$g\Gamma_n\Gamma_\gamma/\Gamma=1.41$ 22.
11870		3.3 keV 5		465.4	$g\Gamma_n\Gamma_\gamma/\Gamma=3.1$ 3.
11949		2.3 keV 4		548.4	$g\Gamma_n\Gamma_\gamma/\Gamma=2.7$ 4.

† From $E_{\text{c.m.}}+S(\text{n})(^{34}\text{S})$ where $S(\text{n})(^{34}\text{S})=11417.15$ 4 (2021Wa16) and $E_{\text{c.m.}}=E_n(\text{lab})\times m(^{33}\text{S})/[m_n+m(^{33}\text{S})]$.

$^{33}\text{S}(\text{d,p})$ 1972Cr08,1971Va21,1963Br05

$J^\pi=3/2^+$ for ^{33}S ground state (2025Ch07).

1972Cr08: $^{33}\text{S}(\text{d,p})$ with a 12-MeV deuteron beam from the Argonne FN tandem Van de Graaff accelerator. The target was 83%-enriched ^{33}S evaporated onto a $30\text{-}\mu\text{g}/\text{cm}^2$ C foil and sandwiched beneath a second C foil. Reaction protons were analyzed by an Enge split-pole magnetic spectrograph and detected using nuclear track emulsions placed in the focal plane. Measured $\sigma(E_p, \theta)$ from 9° to 59° with FWHM=16 keV. Deduced levels, J, π , L-transfers, and spectroscopic factors from DWUCK-DWBA analysis.

1971Va21: $^{33}\text{S}(\text{d,p})$ with a 12-MeV deuteron beam from the Utrecht 6 MV tandem accelerator. The target was 77%-enriched ^{33}S on an aluminum backing. Reaction protons were detected using position-sensitive detectors in the focal plane of a split-pole magnetic spectrograph. Measured $\sigma(E_p, \theta)$ from 5° to 95° in steps of 5° . Deduced levels, J, π , L-transfers, and spectroscopic factors from finite-range DWUCK-DWBA analysis.

1963Br05: $^{33}\text{S}(\text{d,p})$ with a 6-MeV deuteron beam from the MIT-ONR electrostatic accelerator. Targets were natural sulfur and 25% ^{33}S -enriched sulfur sublimated and layered between silver thin films. Reaction protons were detected using nuclear emulsion plates inside a broad-range magnetic spectrograph. Measured $\sigma(E_p)$. Deduced 23 levels.

1949Da16: $^{33}\text{S}(\text{d,p})$ with 3.22-MeV deuterons. Measured $\sigma(E_p)$. Deduced Q values and a 0.82-MeV level in ^{34}S .

 ^{34}S Levels

E(level)	L @	(2J+1)C ² S @&	Comments
0.0	2	1.84	E(level): observed by 1972Cr08, 1971Va21, and 1963Br05. (2J+1)C ² S: other: 1.9 (1971Va21).
2122 [†] 10	0+2	0.52+5.28	E(level): other: 2127.5 (quoted energy value, 1972Cr08), 2127 (1971Va21). (2J+1)C ² S: other: 0.15+5.2 (1971Va21).
3304.9 20	0+2	0.64+1.44	E(level): weighted average of 3305 2 (1972Cr08) and 3302 10 (1963Br05). (2J+1)C ² S: other: 0.22+1.7 (1971Va21).
3915 [#]			
4072 [#]	0+2	0.033+0.06	L,(2J+1)C ² S: from 1971Va21.
4118 4	0+2	1.64+3.20	E(level): weighted average of 4118 4 (1972Cr08) and 4120 10 (1963Br05). (2J+1)C ² S: other: 0.46+2.4 (1971Va21).
4627 5	1+3	0.84+1.84	E(level): weighted average of 4627 5 (1972Cr08) and 4629 10 (1963Br05). (2J+1)C ² S: other: 0.35+1.4 (1971Va21).
4688 [#]			
4702 [†] 14			
4874 [#]	(1+3)	(0.01+0.3)	L,(2J+1)C ² S: from 1971Va21.
4888 [†] 10	(0)	(0.08)	E(level): other: 4891 (1971Va21). L,(2J+1)C ² S: from 1971Va21.
5225 [#]			
5326 [‡] 6	3	4.60	E(level): other: 5318 (1971Va21). J^π : possible discrepancy: 2^+ is assigned for 5322.512 level in the Adopted Levels. L: $(1^+)_3$ (1971Va21). (2J+1)C ² S: (0.029), 2.7 (1971Va21).
5384 6	0	0.92	E(level): weighted average of 5384 6 (1972Cr08) and 5384 10 (1963Br05). (2J+1)C ² S: other: 0.20+0.36 for L=0+2 (1971Va21).
5683 [‡] 7	1	2.60	E(level): other: 5680 (1971Va21). J^π : possible discrepancy: 2^+ is assigned for the 5679.928 level in the Adopted Levels. L: other: $1(+3)$ (1971Va21).
5694 [‡] 7	3	12.48	E(level): other: 5687 (1971Va21). (2J+1)C ² S: other: $(1.2^+)_7.9$ for L= $(1^+)_3$ (1971Va21).
5758 [‡] 7	1	1.68	E(level): other: 5759 (1971Va21). (2J+1)C ² S: other: 0.64(+0.62) for L= $1(+3)$ (1971Va21).
5859 [#]			
6008 [#]			

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{d,p})$ **1972Cr08,1971Va21,1963Br05** (continued) ^{34}S Levels (continued)

E(level)	L [@]	(2J+1)C ² S ^{@&}	Comments
6128 [‡]			E(level): weak peak from 1972Cr08 and 1971Va21 .
6174 8	1+3	1.00+2.84	E(level): weighted average of 6174 8 (1972Cr08) and 6173 10 (1963Br05).
6254 8	3	2.28	E(level): weighted average of 6256 8 (1972Cr08) and 6251 10 (1963Br05).
6345 8	1	0.80	E(level): weighted average of 6346 8 (1972Cr08) and 6344 10 (1963Br05).
6422 [‡] 8	3	0.88	E(level): other: 6423 (1971Va21).
6482 8	1	3.64	E(level): weighted average of 6483 8 (1972Cr08) and 6480 14 (1963Br05). (2J+1)C ² S: other: 1.20(+0.89) for L=1(+3) (1971Va21).
6533? [#]			
6640 9	(3)	(6.56)	E(level): weighted average of 6644 9 (1972Cr08) and 6634 10 (1963Br05). (2J+1)C ² S: other: (0.041),2.2 for L=(1 ⁺)3 (1971Va21).
6690 9	1	1.28	E(level): weighted average of 6690 9 (1972Cr08) and 6690 10 (1963Br05). L: other: (1 ⁺)3 (1971Va21).
6832 [‡] 9	0	0.56	
6959 10	1	0.84	E(level): weighted average of 6959 10 (1972Cr08) and 6959 10 (1963Br05).
7114 10	1+3	0.52+1.76	E(level): weighted average of 7115 10 (1972Cr08) and 7112 10 (1963Br05).
7398 [†] 14			E(level): other: 7388 (1972Cr08).
7547 [‡]			
7632 10	1+3	3.92,12	E(level): weighted average of 7633 11 (1972Cr08) and 7631 10 (1963Br05).
7659 [‡] 11	(3)	(2.28)	
7732 [‡] 11	(1+3)	(0.12,0.6)	
7753 11	1	0.32	E(level): weighted average of 7755 11 (1972Cr08) and 7750 14 (1963Br05).
7784 11	1	1.08	E(level): weighted average of 7785 11 (1972Cr08) and 7783 14 (1963Br05).
8142 [‡] 12	1	1.04	
8299 [†] 14			
8622? [†] 14			

[†] From **1963Br05**.[‡] From **1972Cr08**.[#] From **1971Va21**.[@] From **1972Cr08**, unless otherwise noted.[&] Scaled from the spectroscopic strengths in **1972Cr08**. **1972Cr08** reports spectroscopic strengths $G=(2J_i+1)/(2J_f+1) \times C^2S$. The spectroscopic factors $C^2S=(2J_f+1)/(2J_i+1) \sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$, where the normalization factor $N=1.53$. J is the spin of the populated ^{34}S level and $J_i=3/2$ is the spin for the initial target ^{33}S g.s.

$^{34}\text{S}(\text{e},\text{e}')$ **1985Wo06,1994Pe08**

1985Wo06: $^{34}\text{S}(\text{e},\text{e}')$ with 120, 240, 320-MeV incident electrons at the Mainz electron-scattering setup. The targets were varied between 80 and 120-mg/cm² enriched ^{34}S . Measured $\sigma(\text{E}(\text{e}'),\theta)$. Deduced form factors, transition charge densities, transition radii, and transition strengths from the measured cross sections.

1994Pe08: $^{34}\text{S}(\text{e},\text{e}')$ with 28-64-MeV electron beams from MUSL-2 (microtron using a superconducting linac 2) at the Nuclear Physics Laboratory of the University of Illinois at Urbana-Champaign. The target was CaS powder with 95%-enriched ^{34}S . Scattered electrons were momentum-analyzed and detected in a focal plane vertical-drift wire chamber. Measured $\sigma(\text{E}(\text{e}'),\theta)$. Deduced transition radii and transition strengths from the measured cross sections.

 ^{34}S Levels

$\text{E}(\text{level})^\dagger$	J^π	$T_{1/2}$	Comments
0.0	0^+		
2127	2^+	0.337 ps <i>12</i>	$T_{1/2}$: lifetime $\tau=0.486$ ps <i>17</i> from $\text{B}(\text{E}2)\uparrow$ in 1985Wo06 . $\text{B}(\text{E}2)\uparrow=0.01928$ <i>66</i> , transition radius $R_{\text{tr}}=4.54$ fm <i>7</i> , proton transition matrix elements $M_{\text{p}}=13.88$ fm ² <i>24</i> (1985Wo06).
3304	2^+		$\text{B}(\text{E}2)\uparrow=0.00256$ <i>29</i> , $R_{\text{tr}}=4.71$ fm <i>19</i> , $M_{\text{p}}=5.05$ fm ² <i>29</i> (1985Wo06).
9480 <i>10</i>	1^+		$\text{B}(\text{M}1)\uparrow=0.58$ <i>4</i> , $R_{\text{tr}}=3.40$ fm (1994Pe08).
9870 <i>10</i>	1^+		$\text{B}(\text{M}1)\uparrow=0.25$ <i>7</i> , $R_{\text{tr}}=3.40$ fm (1994Pe08).
10000 <i>10</i>	1^+		$\text{B}(\text{M}1)\uparrow=0.19$ <i>4</i> , $R_{\text{tr}}=3.40$ fm (1994Pe08).
10180 <i>10</i>	1^+		$\text{B}(\text{M}1)\uparrow=0.66$ <i>12</i> , $R_{\text{tr}}=4.33$ fm (1994Pe08).
10430 <i>10</i>	1^+		$\text{B}(\text{M}1)\uparrow=0.21$ <i>4</i> , $R_{\text{tr}}=3.40$ fm (1994Pe08).
10660 <i>10</i>	$(1^+,2^-)$		$\text{B}(\text{M}1)\uparrow=0.16$ <i>5</i> , $R_{\text{tr}}=3.40$ fm (1994Pe08).
10800 <i>10</i>	1^+		$\text{B}(\text{M}1)\uparrow=0.26$ <i>9</i> , $R_{\text{tr}}=3.48$ fm (1994Pe08).
11020 <i>10</i>	1^+		$\text{B}(\text{M}1)\uparrow=0.11$ <i>6</i> , $R_{\text{tr}}=2.69$ fm (1994Pe08).
11350 <i>10</i>	1^+		$\text{B}(\text{M}1)\uparrow=0.20$ <i>4</i> , $R_{\text{tr}}=3.40$ fm (1994Pe08).
11500 <i>10</i>	1^+		$\text{B}(\text{M}1)\uparrow=0.23$ <i>4</i> , $R_{\text{tr}}=3.40$ fm (1994Pe08).
12120 <i>10</i>	1^+		$\text{B}(\text{M}1)\uparrow=0.25$ <i>12</i> , $R_{\text{tr}}=3.10$ fm (1994Pe08).
12180 <i>10</i>	2^-		$\text{B}(\text{M}2)\uparrow=46$ <i>5</i> , $R_{\text{tr}}=4.00$ fm (1994Pe08).
12460 <i>10</i>	$(1^+,2^-)$		$\text{B}(\text{M}1)\uparrow=0.16$ <i>5</i> , $R_{\text{tr}}=3.40$ fm (1994Pe08).
12660 <i>10</i>	1^+		$\text{B}(\text{M}1)\uparrow=0.39$ <i>11</i> , $R_{\text{tr}}=3.21$ fm (1994Pe08).
12930 <i>10</i>	$(1^+,2^-)$		$\text{B}(\text{M}2)\uparrow=16$ <i>5</i> , $R_{\text{tr}}=4.00$ fm (1994Pe08).
13590 <i>10</i>	2^-		$\text{B}(\text{M}2)\uparrow=31$ <i>22</i> , $R_{\text{tr}}=3.70$ fm (1994Pe08).
13790 <i>10</i>	2^-		$\text{B}(\text{M}2)\uparrow=18$ <i>5</i> , $R_{\text{tr}}=4.00$ fm (1994Pe08).
13990 <i>10</i>	1^+		$\text{B}(\text{M}1)\uparrow=0.13$ <i>10</i> , $R_{\text{tr}}=3.42$ fm (1994Pe08).
14200 <i>10</i>	$(1^+,2^-)$		$\text{B}(\text{M}1)\uparrow=0.12$ <i>5</i> , $R_{\text{tr}}=3.40$ fm (1994Pe08).
14320 <i>10</i>	$(1^+,2^-)$		$\text{B}(\text{M}2)\uparrow=32$ <i>6</i> , $R_{\text{tr}}=4.00$ fm (1994Pe08).
14430 <i>10</i>	$(1^+,2^-)$		$\text{B}(\text{M}1)\uparrow=0.27$ <i>5</i> , $R_{\text{tr}}=3.40$ fm (1994Pe08).
14800 <i>10</i>	2^-		$\text{B}(\text{M}2)\uparrow=90$ <i>7</i> , $R_{\text{tr}}=4.00$ fm (1994Pe08).

[†] $\text{E}(\text{level}) > 9$ MeV is from **1994Pe08**. J^π are assigned by evaluators based on the multipolarities assigned in **1994Pe08**. Magnetic (M) character is established because **1994Pe08** observed that longitudinal electric contributions vanish at 180° back-scattering, confirmed by agreement of forward-angle measurements at matched momentum transfer (q) with the 180° data. M1 versus M2 assignments from least-squares fits of the reduced transition probability $\text{B}(\text{M}\Lambda, \text{q})$ versus q^2 for $\Lambda=1$ and 2, selecting the multipolarity that produced both a linear low-q dependence and a physically reasonable transition radius R_{tr} near the ground-state rms charge radius.

$^{34}\text{S}(\gamma, \gamma'), (\text{pol } \gamma, \gamma')$ **1984Be26, 1978Be46**

Nuclear resonance fluorescence (NRF) measurements:

1984Be26: $^{34}\text{S}(\text{pol } \gamma, \gamma')$ with 18-MeV unpolarized and linearly polarized bremsstrahlung beams from the 65-MeV electron linac at the University of Giessen. Targets were mixtures of H_3BO_3 , C, MgO, Si, and S. γ rays were detected using four Ge(Li) detectors at azimuthal angles of 0° , 90° , 180° , and 270° relative to the polarization plane of the linearly polarized beam. Measured E_γ , I_γ , and $\gamma(\theta)$. Deduced levels, J, decay widths to the ground state (Γ_0), and electric/magnetic characters from azimuthal asymmetry of scattered γ rays.

1978Be46: $^{34}\text{S}(\gamma, \gamma')$ with an 18-MeV electron bremsstrahlung beam from the 65-MeV electron linac at the University of Giessen on a 1-g self-supporting 77.9%-enriched ^{34}S target. γ rays were detected using a Ge(Li) detector at 125° relative to the bremsstrahlung beam. Measured E_γ , I_γ , and integrated (γ, γ') scattering cross sections. Deduced levels, J, and decay widths to the ground state (Γ_0). Also see **1977BeYD**.

 ^{34}S Levels

$E(\text{level})^\dagger$	$J^\pi^\dagger$	$\Gamma_0^\dagger$	Comments
0.0	0^+		
7220 2	$(1, 2^+)$	1.09 eV 33	Γ : assuming $J^\pi=2^+$. Other: $\Gamma_{\gamma 0}=0.92$ eV 29 assuming $J^\pi=1^+$ (1978Be46).
7781 2	$(1, 2^+)$	0.57 eV 9	
8185 3	$1^+ \ddagger$	0.78 3 eV 20	Γ : weighted average of $\Gamma_{\gamma 0}=0.7$ eV 3 (1984Be26) and 0.86 eV 28 (1978Be46).
8385 3	$(1, 2^+)$	0.49 eV 15	
8511 3	$(1, 2^+)$	0.52 eV 9	
8657 7	$1^+ \ddagger$	0.41 3 eV 19	Γ : weighted average of $\Gamma_{\gamma 0}=0.4$ eV 3 (1984Be26) and 0.41 eV 25 (1978Be46).
9478 4	$1^+ \ddagger$	1.1 3 eV 3	
9640 4	$(1, 2^+)$	3.6 eV 7	Γ : weighted average of $\Gamma_{\gamma 0}=1.0$ eV 5 (1984Be26) and 1.13 eV 49 (1978Be46).
9711 5	$(1, 2^+)$	0.50 eV 14	
9860 7	$1^+ \ddagger$	0.60 3 eV 12	Γ : weighted average of $\Gamma_{\gamma 0}=0.54$ eV 19 (1984Be26) and 0.63 eV 15 (1978Be46).
10170 5	$1^+ \ddagger$	1.06 3 eV 20	
10786 13	$(1, 2^+)$	0.75 eV 14	Γ : weighted average of $\Gamma_{\gamma 0}=1.0$ eV 3 (1984Be26) and 1.11 eV 28 (1978Be46).
10803 6	$(1, 2^+) \ddagger$	0.60 3 eV 11	

† From **1978Be46**, unless otherwise noted. Due to the low momentum transfer of photons, primary dipole and to a lesser extent quadrupole, transitions are induced. For 0^+ g.s. target, most likely $J=1$ states are populated. The decay widths to the ground state (Γ_0) are deduced by authors assuming $J=1$ and 100% decay to g.s.

‡ From **1984Be26** based on γ multiplicities deduced from $\gamma(\theta)$ and azimuthal asymmetry of scattered γ rays.

 $\gamma(^{34}\text{S})$

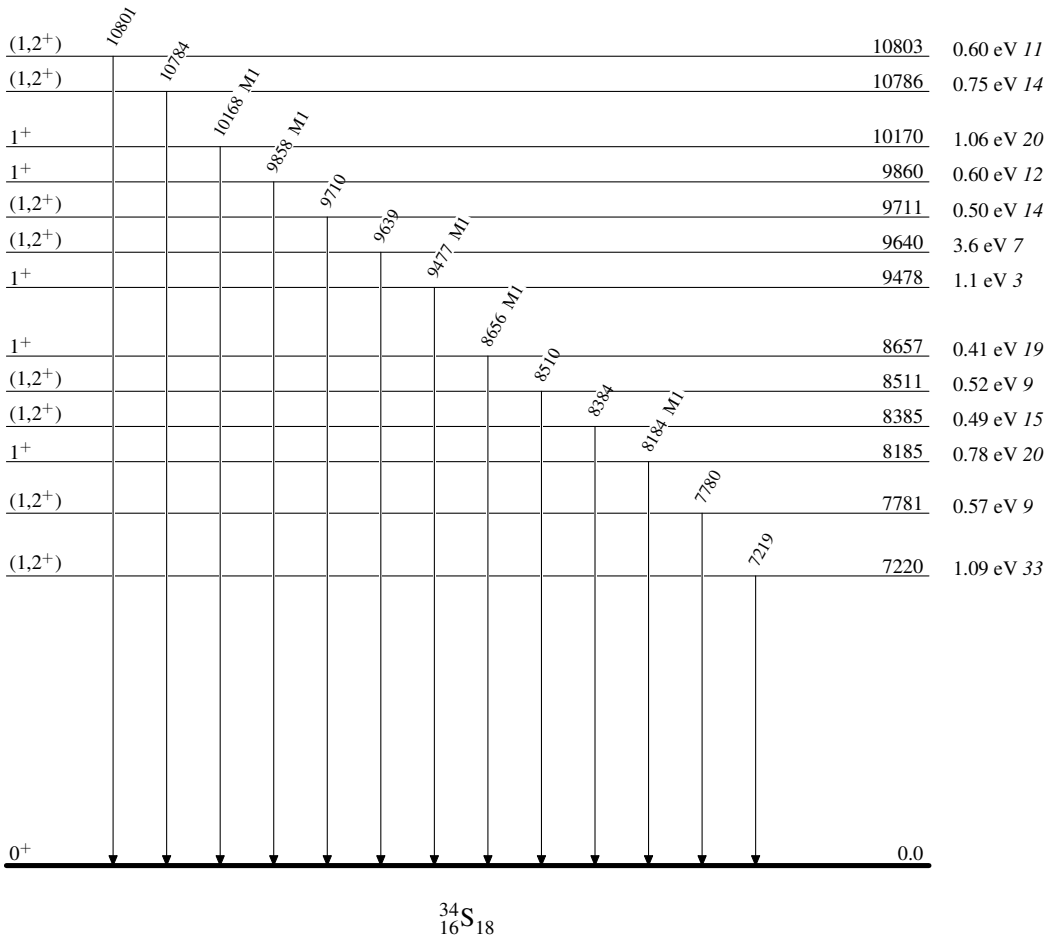
$E_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. ‡	$E_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. ‡
7219	7220	$(1, 2^+)$	0.0	0^+		9639	9640	$(1, 2^+)$	0.0	0^+	
7780	7781	$(1, 2^+)$	0.0	0^+		9710	9711	$(1, 2^+)$	0.0	0^+	
8184	8185	1^+	0.0	0^+	M1	9858	9860	1^+	0.0	0^+	M1
8384	8385	$(1, 2^+)$	0.0	0^+		10168	10170	1^+	0.0	0^+	M1
8510	8511	$(1, 2^+)$	0.0	0^+		10784	10786	$(1, 2^+)$	0.0	0^+	
8656	8657	1^+	0.0	0^+	M1	10801	10803	$(1, 2^+)$	0.0	0^+	
9477	9478	1^+	0.0	0^+	M1						

† From level energy differences.

‡ From measured $\gamma(\theta)$ and azimuthal asymmetry of scattered γ rays.

³⁴S(γ,γ'),(pol γ,γ') 1984Be26,1978Be46

Level Scheme



$^{34}\text{S}(\text{n},\text{n}),(\text{n},\text{n}')$ [1985AI03,1986AI21](#)

[1985AI03](#), [1986AI21](#): $^{34}\text{S}(\text{n},\text{n}),(\text{n},\text{n}')$ with 21.7 and 25.5-MeV monoenergetic neutrons produced via the $^3\text{H}(\text{d},\text{n})^4\text{He}$ reaction using pulsed and bunched deuteron beam at the Ohio University Beam Swinger time-of-flight facility. The targets were 93.6% and 94.5%-enriched ^{34}S powder. Scattered neutrons were detected utilizing a seven-fold array of NE213 liquid scintillation detectors. Measured $\sigma(E_{\text{n}},\theta)$. [1985AI03](#) deduced proton (M_{p}) and neutron (M_{n}) multipole matrix elements for E2 transitions to the first three 2^+ states in ^{34}S using coupled-channels (CC) analysis. [1986AI21](#) deduced isoscalar E2, E3, and E4 reduced transition matrix elements using the anharmonic vibrational (AHV) collective model within a coupled-channels framework.

 ^{34}S Levels

<u>E(level)[†]</u>	<u>Jπ[‡]</u>
0.0	0 ⁺
2128	2 ⁺
3304	2 ⁺
4115	2 ⁺
4624	3 ⁻
4689	4 ⁺

[†] From the Adopted Levels. E(level) values are rounded to the nearest integer.

[‡] From [1986AI21](#) coupled-channels analysis of measured $\sigma(\theta)$.

$^{34}\text{S}(\pi^+, \pi^{+'}), (\pi^-, \pi^{-'}), (\pi^+, \pi^+), (\pi^-, \pi^-)$ 1991Kr06, 1984So13

- 1991Kr06:** $^{34}\text{S}(\pi^+, \pi^+), (\pi^+, \pi^{+'}), (\pi^-, \pi^-), (\pi^-, \pi^{-'})$ with 50-MeV pions provided from the πE3 channel beamline at the Paul Scherrer Institute (PSI). Targets were self-supporting wafers of crystallized cast sulfur with 93.4%-enriched ^{34}S . Scattered pions were analyzed by the Low-Energy Pion Spectrometer (LEPS). Measured elastic and inelastic scattering $\sigma(\theta)$. Deduced difference in the neutron and proton r.m.s. radii Δr_{np} and ratio of neutron to proton matrix elements M_n/M_p from Distorted Wave Impulse Approximation (DWIA) analysis of the measured $\sigma(\theta)$.
- 1984So13:** $^{34}\text{S}(\pi^+, \pi^+), (\pi^+, \pi^{+'}), (\pi^-, \pi^-), (\pi^-, \pi^{-'})$ with 50-MeV pions provided from the M13 beamline at TRIUMF. Measured elastic scattering $\sigma(\theta)$ with the Quadrupole-Quadrupole-Dipole (QQD) Spectrometer. Deduced deformation parameters β_p and β_n , and neutron/proton matrix elements M_n, M_p from distorted-wave DWPI analysis. **1991Bi07** reports pion-induced double charge exchange reactions $\sigma(\theta)$ on ^{34}S and ^{56}Fe targets at the same facility as **1991Kr06**.
- 1987Mo07:** $^{34}\text{S}(\pi^+, \pi^{+'}), (\pi^-, \pi^{-'})$ with 180-MeV pions provided from the Clinton P. Anderson Meson Physics Facility at LANL. Measured inelastic scattering $\sigma(\theta)$ with the Energetic Pion Channel and Spectrometer. Deduced M_n and M_p from DWIA analysis.
- 1987Zu03:** $^{34}\text{S}(\pi^+, \pi^-)$ with 292-MeV pions provided from the Clinton P. Anderson Meson Physics Facility at LANL. Targets were 94.3%-enriched ^{34}S . Measured double-charge exchange differential cross sections at 5° with the Energetic Pion Channel and Spectrometer.
- 1985Gy02:** $^{34}\text{S}(\pi^-, \pi^-)$ with 48.4-MeV pions provided from the M13 beamline at TRIUMF. Targets were 290-mg/cm² 92.29%-enriched ^{34}S . Measured elastic scattering $\sigma(\theta)$ with the Quadrupole-Quadrupole-Dipole (QQD) Spectrometer. Deduced neutron radius differences.

 ^{34}S Levels

E(level) [†]	Comments
0.0	The difference in neutron r.m.s. radii $r_n(^{34}\text{S})-r_n(^{32}\text{S})=0.06$ fm <i>l</i> and neutron skin $\Delta r_{np}=0.03$ fm <i>l</i> (1991Kr06). The difference in neutron radii $r_n(^{34}\text{S})-r_n(^{32}\text{S})=0.103$ fm <i>32</i> (Fermi fit) and 0.061 fm <i>35</i> (Fourier-Bessel fit) (1985Gy02).
2128	$M_n/M_p=+1.17$ <i>9</i> (1991Kr06). $M_n=17.1$ <i>11</i> , $M_p=15.5$ <i>10</i> (1987Mo07). $M_n/M_p=0.88$ <i>6</i> (1984So13). $\beta_p=0.283$ <i>32</i> , $\beta_n=0.287$ <i>32</i> (1984So13).
3304	$M_n/M_p=-0.4$ <i>2</i> (1991Kr06).

[†] From the Adopted Levels, rounded to the nearest integer.

$^{34}\text{S}(\text{p,p}), (\text{p,p}'), (\text{pol p,p}')$ 1978BeXU, 1968Mo22

- 1978BeXU: $^{34}\text{S}(\text{p,p}')$ with a 40-MeV proton beam on an $80\text{-}\mu\text{g}/\text{cm}^2$ thick sandwich-type ^{34}S target. Scattered protons were momentum-analyzed by an Enge split-pole spectrometer and detected using a 50-cm-long position-sensitive proportional counter backed by a plastic scintillator. Measured $\sigma(E_p, \theta)$. Deduced levels, J, and π .
- 1968Mo22: $^{34}\text{S}(\text{p,p}')$ with a 12.011-MeV ^{17}O proton beam from the ONR-CIT tandem accelerator on a CdS target 85%-enriched in ^{34}S . The scattered protons were detected using an array of 16 Au-Si surface barrier detectors with FWHM=25 keV at the focal plane of a double focusing magnetic spectrometer. Measured $\sigma(E_p)$. Deduced levels.
- 1999ShZX: $^{34}\text{S}(\text{p,p}')$ with 160-MeV protons. Measured proton spectra at 0° .
- 1999Ma63: systematic discussion of (p,p) and (p,p') on sulfur isotopes.
- 1991Kh06: $^{34}\text{S}(\text{pol p,p}')$ with a 318-MeV polarized proton beam from the Los Alamos Meson Physics Facility on a 67.92%-enriched ^{34}S target. Scattered protons were analyzed by the high-resolution spectrometer and detected using a standard focal-plane detector at 5° – 39° . Measured $\sigma(E_p, \theta)$ and analyzing powers. Deduced neutron and proton transition densities and multipole matrix elements. 1991Ke09 reports ^{32}S results and 1991Kh06 reports ^{34}S results.
- 1985AI03: $^{34}\text{S}(\text{p,p}')$ with a 29.8-MeV proton beam from the Princeton University AVF cyclotron on 94.3%-enriched ^{34}S targets on C backings. Scattered protons were analyzed by a QDDD spectrometer detected using a position-sensitive gas proportional counter with FWHM=20 keV. Measured $\sigma(E_p, \theta)$. Deduced levels, J, and π . Also see 1984AI2V.
- 1985SaZS: $^{34}\text{S}(\text{pol p,p}')$ with a 65-MeV polarized proton beam. Scattered protons were analyzed by a high-resolution magnetic spectrograph. Measured $\sigma(E_p, \theta)$ and analyzing powers. Deduced the relative signs of neutron and proton transition matrix elements to the 2^+ state at 3.30 MeV.
- 1983Iv01: $^{34}\text{S}(\text{p,p}')$ with a 6-MeV proton beam. Measured $\sigma(E_p, \theta)$. Also see 1982IvZX.
- 1982Be32, 1982MiZQ: $^{34}\text{S}(\text{p,p}')$ with 650-MeV protons. Deduced relative signs of neutron and proton transition matrix elements to the 2^+ state at 3.30 MeV. See 1983Mi25 for distorted wave impulse approximation calculations.
- 1980PrZT: $^{34}\text{S}(\text{p,p}')$ with 6.9-MeV protons. Deduced spin-flip probability in the inelastic scattering to the 2^+ state at 2.13 MeV.
- 1980Fa07: systematic study of (p,p) and (p,p') with optical model and coupled-channel analysis.
- 1975AI08, 1975AI28: $^{34}\text{S}(\text{p,p})$ with a 1-GeV proton beam on a target 98.0%-enriched in ^{34}S . Scattered protons were detected using a magnetic spectrometer with FWHM=1.6 MeV. Measured $\sigma(E_p, \theta)$.
- 1974Gu26: $^{34}\text{S}(\text{p,p})$ with 5.4-6.2-MeV proton beams on a 97.97%-enriched ^{34}S target. Measured excitation functions and elastic $\sigma(E_p, \theta)$.
- 1972PI06, 1973Go42: $^{34}\text{S}(\text{pol p,p}')$ with a 24.5-MeV polarized proton beam from the Saclay variable energy cyclotron on Indium sulfide (InS) targets. Scattered protons were detected using cooled Si(Li) detectors. Measured $\sigma(E_p, \theta)$ and asymmetry(E_p, θ). Deduced levels, J, π , and deformation parameters.
- 1969An35: $^{34}\text{S}(\text{p,p}')$ with a 6.03-MeV proton beam on an InS target 57.17%-enriched in ^{34}S . Scattered protons were detected using a Si(Li) detector. Measured $\sigma(E_p, \theta)$. Deduced levels, J, and π .
- Others: 2005Ko28, 2001Kh17, 1990SeZR, 1988Za04, 1988AI14, 1983Ch50.

 ^{34}S Levels

E(level) [†]	J π	Comments
0.0	0 ⁺	$\beta_2(\text{p,p}')=0.28$ 1 (1985AI03); 0.24 2 (1999Ma63 by reanalysing 1985AI03 data with Becchetti-Greenless optical potential).
2129 [‡] 18	2 ⁺	J π : from (p,p') $\sigma(\theta)$ analysis in 1991Kh06 (pol p,p'), 1973Go42 (pol p,p'), 1985SaZS (pol p,p'), 1985AI03, 1983Iv01, and 1975AI28.
3306 [‡] 14	2 ⁺	J π : from (p,p') $\sigma(\theta)$ analysis in 1991Kh06 (pol p,p'), 1985SaZS (pol p,p'), 1985AI03, and 1983Iv01.
3917 [‡] 14	0	E(level): other: 3914 (1983Iv01). J π : from (p,p') $\sigma(\theta)$ analysis in 1983Iv01.
4077 [‡] 15		
4115 [‡] 14	2 ⁺	J π : from (p,p') $\sigma(\theta)$ analysis in 1991Kh06 (pol p,p'), 1985SaZS (pol p,p'), and 1985AI03.
4625 [‡] 14	3 ⁻	J π : from (pol p,p') $\sigma(\theta)$ analysis in 1973Go42.
4689 [‡] 25	4 ⁺	J π : from (p,p') $\sigma(\theta)$ analysis in 1991Kh06 (pol p,p').
4882 [‡] 14	2 ⁺	J π : from (pol p,p') $\sigma(\theta)$ analysis in 1991Kh06.
5228 [‡] 14		
5321 [‡] 14		

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p,p}),(\text{p,p}'),(\text{pol p,p}') \quad \text{1978BeXU}, \text{1968Mo22} \text{ (continued)}$ ^{34}S Levels (continued)

E(level) [†]	Comments
5381 $\frac{1}{2}^+$ 14	
5684 $\frac{1}{2}^+$ 14	
5689	
5755 $\frac{1}{2}^+$ 14	
5848 $\frac{1}{2}^+$ 18	
5995 $\frac{1}{2}^+$ 18	
6118 $\frac{1}{2}^+$ 14	
6170 $\frac{1}{2}^+$ 14	
6248 $\frac{1}{2}^+$ 18	
6340 $\frac{1}{2}^+$ 14	
6414 $\frac{1}{2}^+$ 14	
6479 $\frac{1}{2}^+$ 18	
6635 $\frac{1}{2}^+$ 14	
6680 $\frac{1}{2}^+$ 14	
6738 $\frac{1}{2}^+$ 25	
6819 $\frac{1}{2}^+$ 26	
6860 $\frac{1}{2}^+$ 14	
6888 $\frac{1}{2}^+$ 14	
6950 $\frac{1}{2}^+$ 14	
7110 $\frac{1}{2}^+$ 14	
7248 $\frac{1}{2}^+$ 18	
7264 $\frac{1}{2}^+$ 18	
7360 $\frac{1}{2}^+$ 18	
7389 $\frac{1}{2}^+$ 18	
7479 5	E(level): other: 7479 14 (1968Mo22).
7556 5	E(level): weighted average of 7557 5 (1978BeXU) and 7549 14 (1968Mo22).
7625 5	E(level): other: 7621 14 (1968Mo22).
7649 $\frac{1}{2}^+$ 14	
7732 5	
7779 5	
7805 5	
7976 5	
8034 5	
8296 5	
8385 5	
8423 5	
8511 5	
8580 5	
8623 5	
8656 5	
8671 5	
8718 5	
8740 10	
8792 5	
8809 5	
8953 5	
8987 5	
9120 5	
9171 5	

Continued on next page (footnotes at end of table)

³⁴S(p,p),(p,p'),(pol p,p') **1978BeXU,1968Mo22** (continued)

³⁴S Levels (continued)

<u>E(level)[†]</u>	<u>E(level)[†]</u>	<u>E(level)[†]</u>
9205 7	9445 5	9700 6
9226 6	9481 5	9847 10
9257 10	9566 10	9872 5
9429 5	9601 5	9925 5
		9969 10

[†] From **1978BeXU**, unless otherwise noted.
[‡] From **1968Mo22**.

$^{34}\text{S}(\text{p},\text{p}'\gamma)$ 1972Bo02

1972Bo02: $^{34}\text{S}(\text{p},\text{p}'\gamma)$ inelastic proton scattering using a 3.6-MeV proton beam from the Brookhaven National Laboratory 3.5-MeV Van de Graaff accelerator. The target was $400\ \mu\text{g}/\text{cm}^2$ Sb_2S_3 on Mo backing. γ rays were detected using a Ge(Li) detector. Measured E_γ .

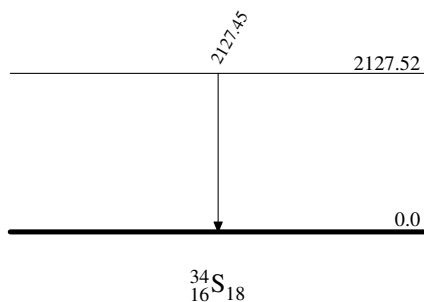
1985Ki07: $^{34}\text{S}(\text{p},\text{p}'\gamma)^{34}\text{S}$ with $E_p=2.4\text{--}4.2$ MeV from the 5 MV Van de Graaff accelerator of the Institute of Nuclear Research, Hungary. γ rays were detected using a 25-cm^3 Ge(Li) detector. Measured γ -ray yields for $^{37}\text{Cl}(\text{p},\alpha\gamma)^{34}\text{S}$ and $^{34}\text{S}(\text{p},\text{p}'\gamma)^{34}\text{S}$ reactions, which are part of a series of proton-induced prompt γ -ray emission (PIGE) measurements.

 ^{34}S Levels

<u>E(level)</u>	<u>Comments</u>
0.0	
2127.52 20	E(level): from E_γ data.

 $\gamma(^{34}\text{S})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>	<u>Comments</u>
2127.45 20	2127.52	0.0	E_γ : from 1972Bo02. Other: 2127 from 1985Ki07.

 $^{34}\text{S}(\text{p},\text{p}'\gamma)$ 1972Bo02Level Scheme

$^{34}\text{S}(\text{pol d,d}),(\text{pol d,d}')$ 1978Ab02,1987Nu01

1978Ab02: elastic and inelastic scattering of vector polarized deuterons on ^{34}S . An 11.8-MeV polarized deuteron beam was provided from the University of Wisconsin Lamb-shift polarized ion source and tandem accelerator. The target was 610- $\mu\text{g}/\text{cm}^2$ 90%-enriched ^{34}S on a 100- $\mu\text{g}/\text{cm}^2$ Au backing. Scattered deuterons were detected using four Si $\Delta\text{E-E}$ telescopes. Measured angular distributions of cross sections $\sigma(E_d, \theta)$ and angular distributions of vector analyzing powers $iT_{11}(E_d, \theta)$. Deduced levels, J, π , optical model parameters, and vibrational deformation parameters β_2 from DWUCK-DWBA and CHUCK coupled-channels analysis.

1987Nu01: elastic and inelastic scattering of vector polarized deuterons on a series of nuclei including ^{34}S . A 52-MeV polarized deuteron beam was provided from the Lamb-shift polarized ion source and the Karlsruhe isochronous cyclotron. The target was 89.8%-enriched $^{34}\text{SH}_2$. Scattered deuterons were detected using six $\Delta\text{E-E}$ solid-state counter telescopes. Measured angular distributions of cross sections $\sigma(E_d, \theta)$ and angular distributions of vector analyzing powers $iT_{11}(E_d, \theta)$. Deduced levels, J, π , optical model parameters, and vibrational deformation parameters β_2 from DWBA and coupled-channels analysis.

Other: $^{34}\text{S}(\text{pol d,d})$ elastic scattering (**1975Ba29**).

 ^{34}S Levels

<u>E(level)[†]</u>	<u>Jπ[‡]</u>	<u>Comments</u>
0.0	0 ⁺	
2127	2 ⁺	J π : from 1987Nu01 . J π =(2 ⁺) from 1978Ab02 . $\beta_2=0.24$ (DWBA); $\beta_2=0.21$ (coupled-channels) from 1978Ab02 . $\beta_2=+0.20$ and $\beta_4=-0.20$ (coupled-channels); $\beta_2=-0.24$ and $\beta_4=-0.20$ (coupled-channels); $\beta_{02}=0.20$ (coupled-channels) from 1987Nu01 .
3304	(2 ⁺)	$\beta_2=0.06$ (DWBA); $\beta_2=-0.04$ for one phonon (coupled-channels); $\beta_2=0.10$ for two phonons (coupled-channels) from 1978Ab02 .
3914	(0 ⁺)	$\beta_2=0.14$ for two phonons (coupled-channels) from 1978Ab02 .
4072		
4115		4115+4072 doublet from Fig. 1 of 1978Ab02 .
4622		
4688		
4696		4622+4688+4696 triplet from Fig. 1 of 1978Ab02 . Numbers are illegible.

[†] From **1978Ab02**, unless otherwise noted.

[‡] From DWBA and coupled-channels analysis of the measured $\sigma(\theta)$ in **1978Ab02**, unless otherwise noted.

$^{34}\text{S}(\alpha, \alpha'), (\alpha, \alpha'), (\alpha, \alpha')\gamma$ **1974Gr15, 1980Ba40, 1984Sa11**

1974Gr15: $^{34}\text{S}(\alpha, \alpha')$ with 16-18-MeV ^4He beams from the McMaster University FN tandem accelerator. Targets were natural and ^{34}S -enriched CdS evaporated on C backings. Scattered α particles were detected using an annular Si surface-barrier detector at 176.8 - 178.7° . Measured E_α , I_α . Primary measurements were $^{31}\text{P}(p, \gamma)$ and $^{32}\text{S}(\alpha, \alpha')$.

1980Ba40: $^4\text{He}(^{34}\text{S}, \alpha)$ with a 70-MeV ^{34}S beam at Chalk River Nuclear Laboratories. The targets were 25- μm Cu foils implanted with 6×10^{17} atoms/ cm^2 of ^4He . Inelastically scattered α particles were detected using a ΔE -E telescope of surface-barrier detectors at 0° and γ rays were detected using a 100- cm^3 Ge(Li) detector at 0° . Measured $\alpha\gamma$ -coin. Deduced lifetime using DSAM lineshape analysis and $B(E2)\uparrow$ for the first 2^+ state in ^{34}S . Also measured $^{208}\text{Pb}(^{28}\text{Si}, ^{28}\text{Si}')$, $^{208}\text{Pb}(^{32}\text{S}, ^{32}\text{S}')$, and **1979DaZV** is superseded by **1980Ba40**. $^{208}\text{Pb}(^{34}\text{S}, ^{34}\text{S}')$ for $B(E2)\uparrow$ and quadrupole moments.

1984Sa11: $^{34}\text{S}(\alpha, \alpha'), (\alpha, \alpha')$ with a 120-MeV α beam from the Indiana University Cyclotron Facility. The target was 94.3%-enriched ^{34}S on a ^{12}C backing. Scattered α particles were detected using two scintillators at the focal plane of a quadrupole-dipole-dipole-multipole spectrometer. Measured $\sigma(E_\alpha, \theta)$ for g.s. and first two 2^+ states. Deduced the neutron and proton transition matrix elements from DWBA and CHUCK3-CCBA analyses of the measured $\sigma(\theta)$.

Others:

1994Br19: $^{34}\text{S}(\alpha, \alpha)$ $E=20$ MeV, $\sigma(\theta)$.

1979Za01: $^{34}\text{S}(\alpha, \alpha')\gamma$ $E=8.90$ MeV, measured g-factor for the first 2^+ state.

1973An24: $^{34}\text{S}(\alpha, \alpha'), (\alpha, \alpha')$ $E=24.2$ MeV.

 ^{34}S Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>	<u>$T_{1/2}$</u>	<u>Comments</u>
0.0	0^+		
2128	2^+	306 fs 18	$E(\text{level})$: rounded from the Adopted Levels. $T_{1/2}$: lifetime $\tau=0.442$ ps 26 (1980Ba40 , DSAM lineshape analysis). g-factor=+0.50 8 (1979Za01).
3300	2^+		
3920			
4120			
4690			
4890			
5230			
5320			
5690			
6010			
6130			
6640			

† From **1974Gr15**, unless otherwise noted.

‡ g.s. and the first two 2^+ states angular distributions are well fit with DWBA and CCBA calculations. **1974Gr15** states that the results for ^{34}S are consistent with the assignments 2^- for 5320 level, 5^- for 5690 level, and 4^- for 6640 level, but lack details to justify these assignments.

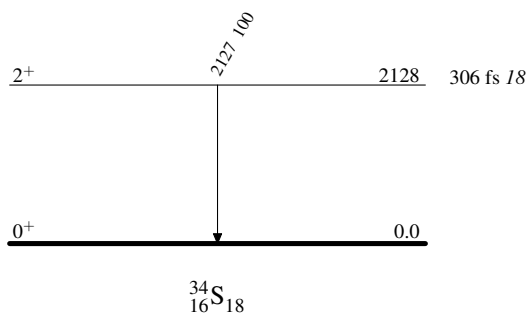
 $\gamma(^{34}\text{S})$

<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_γ</u>	<u>I_γ</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
2128	2^+	2127	100	0.0	0^+	E_γ, I_γ : rounded from the Adopted Gammas.

$^{34}\text{S}(\alpha,\alpha),(\alpha,\alpha'),(\alpha,\alpha'\gamma)$ **1974Gr15,1980Ba40,1984Sa11**

Level Scheme

Intensities: Relative photon branching from each level



$^{35}\text{Cl}(\gamma, \text{p})$ 1997Is02

1997Is02: a $E_{\text{max}}=32\text{--MeV}$ bremsstrahlung photon beam was produced from the Moscow State University betatron. The target was 380-g LiCl with natural isotopic composition. γ rays were detected using a 100-cm^3 Ge(Li) detector. Measured E_γ , I_γ . Deduced levels, integrated cross sections (σ_{int}), and Giant Dipole Resonance (GDR) features.

 ^{34}S Levels

<u>E(level)</u>	<u>Comments</u>
0.0	
2130	$\sigma_{\text{int}}=27.7\text{--}85.5\text{ MeV}\times\text{mb.}$
3300	$\sigma_{\text{int}}<76.0\text{ MeV}\times\text{mb.}$
3920	$\sigma_{\text{int}}=11.7\text{--}15.3\text{ MeV}\times\text{mb.}$
4070	$\sigma_{\text{int}}=13.8\text{ MeV}\times\text{mb}$ for 4070+4110.
4110	$\sigma_{\text{int}}=13.8\text{ MeV}\times\text{mb}$ for 4070+4110.
4620	$\sigma_{\text{int}}<15.5\text{ MeV}\times\text{mb.}$
4690	$\sigma_{\text{int}}<23.1\text{ MeV}\times\text{mb.}$
4880	$\sigma_{\text{int}}=29.2\text{ MeV}\times\text{mb.}$
4890	
5380	
5760	$\sigma_{\text{int}}=3.4\text{--}3.7\text{ MeV}\times\text{mb.}$
6250	

$^{35}\text{Cl}(\text{n,d})$ 1977Pa29,1968Mi02

$J^\pi=3/2^+$ for ^{35}Cl ground state (2026Su18).

1977Pa29: 14.1-MeV neutrons were produced via the $^3\text{H}(\text{d,n})^4\text{He}$ reaction at an average rate of 10^9 neutrons per second into 4π sr from the Brown University neutron generator and impinged on a 6.71-mg/cm², 99.32%-enriched ^{35}Cl target. Deuterons were detected using a time-of-flight counter telescope consisting of two proportional counters and a NaI(Tl) crystal. Measured $\sigma(E_{\text{d}},\theta)$. Deduced levels, L-transfers, spectroscopic factors from finite-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

1968Mi02: $^{35}\text{Cl}(\text{n,d})$ using 14.4-MeV neutrons produced at the Ruder Boskovic Institute in Zagreb. The target was 6.86-mg/cm², natural PbCl_2 deposited on a gold backing. Deuterons were detected using a counter telescope system consisting of three CO_2 gas proportional counters and a CsI(Tl) scintillator. Measured $\sigma(E_{\text{d}},\theta)$. Deduced levels, L-transfers, spectroscopic factors from zero-range DWBA and Plane Wave Born Approximation (PWBA) analyses of the measured $\sigma(\theta)$.

1967Fa10: $^{35}\text{Cl}(\text{n,d})$ using 14.1-MeV neutrons produced at the Institute of Physics of the University of Milan. The target was 3.77-mg/cm² natural C_6Cl_6 on a platinum backing. Deuterons were detected using a $\Delta\text{E-E}$ counter telescope. Measured $\sigma(E_{\text{d}},\theta)$. Deduced levels, L-transfers, spectroscopic factors from DWBA analysis of the measured $\sigma(\theta)$.

1982QaZX: discussions of (n,d), (n,t), (n, α) reactions.

 ^{34}S Levels

E(level)	L	Comments
0.0	2	L: from 1977Pa29, 1968Mi02, and 1967Fa10. S: 1.06 14 and 1.16 15 from DWBA using two sets of optical parameters (1977Pa29); S=1.58 from DWBA (1968Mi02); S=1.05 from PWBA (1968Mi02); S=1.074 (1967Fa10).
2130	0	E(level),L: from 1977Pa29 and 1968Mi02. Other: L=0+2 from 1967Fa10. S: 0.34 5 and 0.45 6 from DWBA using two sets of optical parameters (1977Pa29); S=0.29 from DWBA with no cut-off (1968Mi02); S=0.36 from DWBA with a cut-off radius of 4.1 fm (1968Mi02); S=0.19 from PWBA (1968Mi02); S=0.345 37 (1967Fa10).
3310	0	E(level),L: from 1968Mi02. Other: L=0+2 from 1967Fa10. S: 0.87 from DWBA (1968Mi02); S=0.30 from PWBA (1968Mi02); S=0.92 98 (1967Fa10).

$^{35}\text{Cl}(\text{d}, ^3\text{He})$ **1969Pu03, 1968Wi20**

$J^\pi=3/2^+$ for ^{35}Cl ground state (2026Su18).

1969Pu03: $^{35}\text{Cl}(\text{d}, ^3\text{He})$ with a 23.35-MeV deuteron beam from the Argonne cyclotron. The target was enriched ^{35}Cl lead chloride on thin C backing. Reaction products were detected using a ΔE -E counter covering the angular domain from 12° to 30° . Measured $\sigma(\text{E}(^3\text{He}), \theta)$. Deduced 11 levels, J, π , L-transfers, spectroscopic factors from the zero-range JULIE-DWBA analysis of the angular distributions.

1968Wi20: $^{35}\text{Cl}(\text{d}, ^3\text{He})$ with a 34.5-MeV deuteron beam from the Oak Ridge isochronous cyclotron. The target was NaCl evaporated onto carbon backing. The reaction products were detected using a solid-state ΔE -E telescope. Measured $\sigma(\text{E}(^3\text{He}), \theta)$. Deduced 9 levels, J, π , L-transfers, spectroscopic factors from the zero-range DWBA analysis of the angular distributions.

1962Cu07: $^{35}\text{Cl}(\text{d}, ^3\text{He})$ with a 15 MeV deuteron beam. Targets were $\text{C}_4\text{H}_5\text{Cl}_2$ with evaporated Au. The reaction products were analyzed by a magnetic spectrometer and detected in its focal plane by a CsI scintillation crystal. Measured $\sigma(\text{E}(^3\text{He}), \theta)$. Deduced two levels, J, π , L-transfers, spectroscopic factors from the zero-range SALLY-DWBA analysis of the angular distributions.

1974Cl09: theoretical discussion on the $^{35}\text{Cl}(\text{d}, ^3\text{He})^{34}\text{S}$ reaction.

 ^{34}S Levels

E(level) [†]	L [†]	C ² S [†]	Comments
0.0	2	1.00	C ² S: other: 1.29 (1968Wi20); 1.26 (1962Cu07). Pickup proton: $\text{d}_{3/2}$ (1969Pu03).
2120 10	0+2	0.25+0.37	C ² S: other: 0.39-0.32, (<0.39) (1968Wi20); none, <0.93 (1962Cu07). Pickup proton: $\text{s}_{1/2}, \text{d}_{3/2}$ (1969Pu03).
3310 15	0	0.98	C ² S: other: 1.11-0.99, (<0.78) (1968Wi20, second value for L=(2)). Pickup proton: $\text{s}_{1/2}$ (1969Pu03).
4090 15	0	0.82	C ² S: other: 0.93-0.86, (<0.42) (1968Wi20, second value for L=(2)). Pickup proton: $\text{s}_{1/2}$ (1969Pu03).
4710 20	2	0.5	C ² S: other: 0.62 (1968Wi20). Pickup proton: $\text{d}_{5/2}$ (1969Pu03).
4900 20	2	0.8	C ² S: other: 1.19 (1968Wi20). Pickup proton: $\text{d}_{5/2}$ (1969Pu03).
5360 40	2	0.2	Pickup proton: $\text{d}_{5/2}$ (1969Pu03).
6220 50	2	0.24	L, C ² S: from 1968Wi20. Other: L=(2), S=(0.5) (1969Pu03). Pickup proton: $\text{d}_{5/2}$ (1969Pu03).
6830	0	0.08	L, C ² S: from 1968Wi20. E(level) is quoted by 1968Wi20 from 1967En05. Pickup proton: $\text{d}_{5/2}$ (1968Wi20).
6910 40	2	0.5	Pickup proton: $\text{d}_{5/2}$ (1969Pu03).
7110	0	0.19	L, C ² S: from 1968Wi20. E(level) is quoted by 1968Wi20 from 1967En05. Pickup proton: $\text{d}_{5/2}$ (1968Wi20).
7190 40	2	0.6	Pickup proton: $\text{d}_{5/2}$ (1969Pu03).
7780 30	2	1.4	Pickup proton: $\text{d}_{5/2}$ (1969Pu03).

[†] From 1969Pu03, unless otherwise noted. Spectroscopic factor: $\text{C}^2\text{S}=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$, where $N=2.95$ is a normalization factor adopted by 1969Pu03 from 1966Ba54.

$^{37}\text{Cl}(\text{p},\alpha\gamma)$ 1985We07,1966Vo03

1985We07: $^{37}\text{Cl}(\text{p},\alpha\gamma)^{34}\text{S}$ with $E_{\text{p}}=0.65\text{-}2.15$ MeV from the University of Melbourne 5U Pelletron accelerator. The targets were BaCl_2 , 98.2% enriched in ^{37}Cl , on gold and aluminum backings. γ rays were detected by a $\text{Ge}(\text{Li})$ detector and neutrons were detected by two BF_3 proportional counters. Measured E_{γ} , I_{γ} , γ yields for $^{37}\text{Cl}(\text{p},\gamma)^{38}\text{Ar}$ and $^{37}\text{Cl}(\text{p},\alpha\gamma)^{34}\text{S}$ reactions, and neutron yields for the $^{37}\text{Cl}(\text{p},\text{n})^{37}\text{Ar}$ reaction. Deduced cross sections and thermonuclear reaction rates.

1985Ki07: $^{37}\text{Cl}(\text{p},\alpha\gamma)^{34}\text{S}$ with $E_{\text{p}}=2.4\text{-}4.2$ MeV from the 5 MV Van de Graaff accelerator of the Institute of Nuclear Research, Hungary. γ rays were detected using a 25-cm^3 $\text{Ge}(\text{Li})$ detector. Measured γ -ray yields for $^{37}\text{Cl}(\text{p},\alpha\gamma)^{34}\text{S}$ and $^{34}\text{S}(\text{p},\text{p}'\gamma)^{34}\text{S}$ reactions, which are part of a series of proton-induced prompt γ -ray emission (PIGE) measurements.

1975De16: $^{37}\text{Cl}(\text{p},\alpha\gamma)^{34}\text{S}$ with $E_{\text{p}}=1\text{-}3$ MeV.

1974Al05: $^{37}\text{Cl}(\text{p},\alpha_0)^{34}\text{S}$ with $E_{\text{p}}=1.17\text{-}1.70$ MeV from Utrecht 3-MV Van de Graaff accelerator. Measured yields.

1970An22: $^{37}\text{Cl}(\text{p},\alpha_0)^{34}\text{S}$, $^{37}\text{Cl}(\text{p},\alpha_1)^{34}\text{S}$ with $E_{\text{p}}=5.84, 5.93$, and 6.03 MeV. Measured $\sigma(E_{\alpha},\theta)$. Deduced ^{38}Ar level widths and densities.

1966Vo03: $^{37}\text{Cl}(\text{p},\alpha_0)^{34}\text{S}$, $^{37}\text{Cl}(\text{p},\alpha_1)^{34}\text{S}$, and $^{37}\text{Cl}(\text{p},\alpha_2)^{34}\text{S}$ with $E_{\text{p}}=11.0\text{-}11.952$ MeV from the Heidelberg Tandem Van de Graaff accelerator. Measured $\sigma(E_{\text{p}},E_{\alpha},\theta)$. Deduced ^{38}Ar level widths and densities.

 ^{34}S Levels

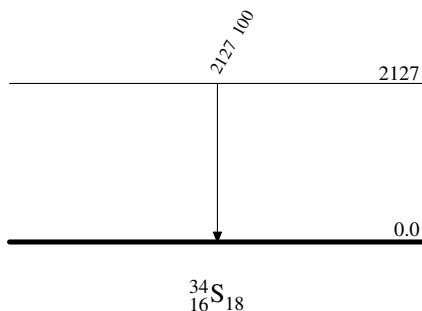
<u>$E(\text{level})$</u>	<u>Comments</u>
0.0	
2127	Observed by 1985We07 , 1985Ki07 , 1970An22 , and 1966Vo03 .
3304	Observed by 1966Vo03 .

 $\gamma(^{34}\text{S})$

<u>$E_i(\text{level})$</u>	<u>E_{γ}</u>	<u>I_{γ}</u>	<u>E_f</u>	<u>Comments</u>
2127	2127	100	0.0	E_{γ}, I_{γ} : from 1985We07 and 1985Ki07 . 2128 from 1975De16 .

 $^{37}\text{Cl}(\text{p},\alpha\gamma)$ 1985We07,1966Vo03Level Scheme

Intensities: Relative photon branching from each level



$^{206}\text{Pb}(^{34}\text{S}, ^{34}\text{S}'\gamma)$ 1974OI02

1974OI02: a 122-MeV ^{34}S beam was produced from the Chalk River Tandem. The target was 1-mg/cm² 99%-enriched ^{206}Pb .

Scattered particles were detected using an annular particle counter and γ rays were detected using NaI(Tl) detectors. Measured (particle) γ -coin, yield. Deduced static quadrupole moment (Q), B(E2) $\uparrow$, and g-factor.

 ^{34}S Levels

E(level) [†]	J ^{π} [†]	T _{1/2}	Comments
0.0	0 ⁺	stable	
2127.561 15	2 ⁺	0.25 ps +5-3	B(E2) $\uparrow$ =0.0263 41 (1974OI02,1980Ba40) T _{1/2} : lifetime τ =0.36 ps +7-5 converted by evaluators from B(E2) $\uparrow$ measured by 1974OI02 and corrected by 1980Ba40. 1980Ba40 also from Chalk River Lab reported a correction related to the magnitude of the giant dipole resonance correction factor. The correction changes B(E2) $\uparrow$ from the 1974OI02 original 0.025 e ² b ² 4 to 0.0263 e ² b ² 41 and Q from +0.026 eb 23 to +0.045 eb 25.

[†] From the Adopted Levels.

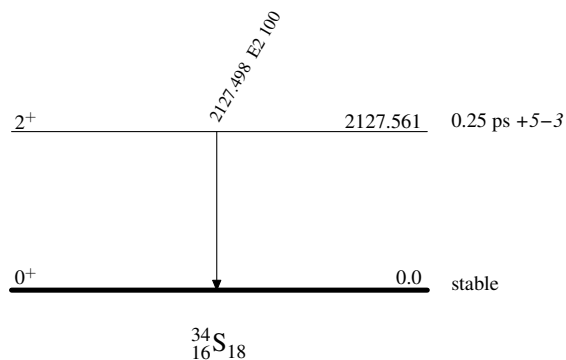
 $\gamma(^{34}\text{S})$

E _i (level)	J ^{π} _i	E _{γ} [†]	I _{γ}	E _f	J ^{π} _f	Mult. [†]
2127.561	2 ⁺	2127.498 20	100	0.0	0 ⁺	E2

[†] From the Adopted Gammas.

 $^{206}\text{Pb}(^{34}\text{S}, ^{34}\text{S}'\gamma)$ 1974OI02Level Scheme

Intensities: Relative photon branching from each level



$^{208}\text{Pb}(^{34}\text{S}, ^{34}\text{S}')$ **1980Ba40**

1980Ba40: a 122-MeV ^{34}S beam was produced from the Chalk River Tandem. Targets were 99.14%-enriched ^{208}Pb and ^{208}PbO .

Elastically and inelastically scattered particles were analyzed with the Chalk River QD3 type II spectrometer and detected using a 50-cm-long position sensitive heavy-ion gas counter. Measured $\sigma(E(^{34}\text{S}), \theta)$. Deduced Coulomb excitation probabilities, static electric quadrupole moments (Q_{2+}), and $B(E2)\uparrow$. Also measured $^4\text{He}(^{32}\text{S}, \alpha)$ and $^4\text{He}(^{34}\text{S}, \alpha)$ with 70-MeV ^{32}S and ^{34}S beams at Chalk River for DSAM lineshape analysis.

 ^{34}S Levels

<u>E(level)[†]</u>	<u>J^π[†]</u>	<u>Comments</u>
0	0 ⁺	
2127.561 15	2 ⁺	B(E2) $\uparrow$ =0.0205 33 (1980Ba40) B(E2) $\uparrow$ =0.0205 e ² b ² 33 and Q=+0.060 eb 8/ are from fit of measured Coulomb excitation probabilities in 1980Ba40 by allowing both Q and B(E2) $\uparrow$ to vary. Q=+0.063 eb 43 with a statistical uncertainty of 25 is from fit of measured Coulomb excitation probabilities in 1980Ba40 by fixing B(E2) $\uparrow$ =0.0203 e ² b ² 6 and allowing only Q to vary.

[†] From the Adopted Levels.

$^{208}\text{Pb}(^{36}\text{S}, ^{34}\text{S}\gamma)$ 2010WaZT

Multi-nucleon transfer and deep-inelastic binary grazing reactions.

2010WaZT: a 215-MeV $^{36}\text{S}^{9+}$ beam was produced from the XTU-Tandem ALPI-superconducting linear accelerator complex at the INFN Legnaro National Laboratory, Italy. The target was $300\text{-}\mu\text{g}/\text{cm}^2$, 99.7% enriched ^{208}Pb on a $20\text{-}\mu\text{g}/\text{cm}^2$ carbon backing. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the CLARA array of 22 escape-suppressed Ge clover detectors covering the azimuthal angles from 98° to 180° . Measured E_γ with Doppler corrections, I_γ , and $(^{34}\text{S})\gamma$ -coin. Deduced levels. Compared with shell-model calculations. Also see **2010Wa12** and **2010Wa20** from the same experiments.

 ^{34}S Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>
0.0	0^+
2128.1 9	(2^+)
3305.0 11	(2^+)
4624.9 14	(3^-)
4689.7 16	(4^+)
4878.1 15	(3^+)
5690.8 16	(5^-)
6251.7 19	(4^-)
7791.8 22	(6^-)
8372.0 26	(7^-)

† From a least-squares fit to γ -ray energies in **2010WaZT**.

‡ From **2010WaZT** based on shell-model calculations and $^{24}\text{Mg}(^{16}\text{O}, \alpha 2p\gamma)$ data from **2005Ma03**. The parentheses are added by evaluators.

 $\gamma(^{34}\text{S})$

<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
1001 1	26.8 25	5690.8	(5^-)	4689.7	(4^+)	1573 1	35.5 33	4878.1	(3^+)	3305.0	(2^+)
1066 1	21.0 21	5690.8	(5^-)	4624.9	(3^-)	2128 1	100.0 52	2128.1	(2^+)	0.0	0^+
1177 1	26.3 25	3305.0	(2^+)	2128.1	(2^+)	2561 2	39.6 36	4689.7	(4^+)	2128.1	(2^+)
1320 1	26.5 26	4624.9	(3^-)	3305.0	(2^+)	2681 2	15.8 27	8372.0	(7^-)	5690.8	(5^-)
1540 1	23.3 25	7791.8	(6^-)	6251.7	(4^-)	3305 2	21.2 35	3305.0	(2^+)	0.0	0^+
1562 1	18.6 23	6251.7	(4^-)	4689.7	(4^+)						

† From Table 6.11 of **2010WaZT**.

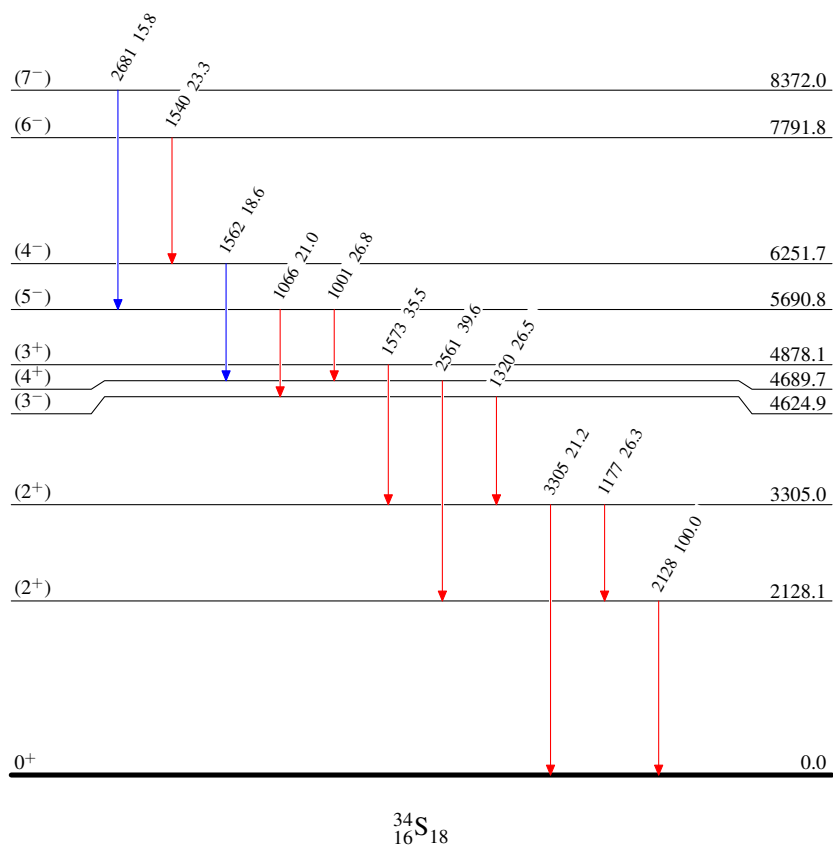
$^{208}\text{Pb}(^{36}\text{S}, ^{34}\text{S}\gamma)$ 2010WaZT

Level Scheme

Intensities: Relative I_γ

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$



Adopted Levels, Gammas

$Q(\beta^-) = -6061.79$ 6; $S(n) = 11508.1$ 4; $S(p) = 5143.20$ 5; $Q(\alpha) = -6664.15$ 8 2021Wa16

$S(2n) = 27248.0$ 6, $S(2p) = 14713.15$ 6, $Q(\varepsilon) = 5491.60$ 4 (2021Wa16).

Isotope discovery (2012Th10): $^{31}\text{P}(\alpha, n)^{34}\text{Cl}$ (1934Fr01), measured $T_{1/2}$ of the isomer at 146 keV. 1953St42 measured $T_{1/2}$ of the ground state.

^{34}Cl production:

2022Sh22: $^{36}\text{Ar}(\text{Be}, \text{X})$ at $E(^{36}\text{Ar}) = 149.3$ MeV/nucleon at NSCL, MSU. Measured the population of ^{34}Cl beam in the ground and isomeric states.

2016Ka36: $^{181}\text{Ta}(^{11}\text{B}, \text{X})$ and $^{209}\text{Bi}(^{11}\text{B}, \text{X})$ at $E(^{11}\text{B}) = 24$ MeV/nucleon. Measured production cross sections using γ -activation method and offline γ -spectroscopy at the U-400M accelerator and ACCULINA setup of JINR, Dubna.

2013SeZV: $^9\text{Be}(^{40}\text{Ar}, \text{X})$ at $E(^{40}\text{Ar}) = 35$ MeV/nucleon at the COMBAS fragment separator at the Flerov Laboratory of Nuclear Reactions, JINR. Measured production cross sections.

2013De37: $^{237}\text{Np}, ^{238}\text{U}(\text{p}, \text{F})$ at $E(\text{p}) = 660$ MeV at the LNR Phasotron U-400M Cyclotron at JINR, Dubna. Measured E_γ , I_γ , and production cross sections of fragments.

2012Zh06: $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar}, \text{X})$ at $E(^{40}\text{Ar}) = 57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

2011Ti04: $^{\text{nat}}\text{Ni}(\text{p}, \text{X})$ at $E(\text{p}) = 46\text{--}2605$ MeV using the U-10 ITEP proton synchrotron. Measured production cross sections.

2011Ti03: $^{\text{nat}}\text{Cr}(\text{p}, \text{X})$ and $^{56}\text{Fe}(\text{p}, \text{X})$ at $E(\text{p}) = 42\text{--}2605$ MeV using the U-10 ITEP proton synchrotron. Measured production cross sections.

2008Ti05: $^{56}\text{Fe}(\text{p}, \text{X})$ at $E(\text{p}) = 300\text{--}2600$ MeV using the U-10 ITEP proton synchrotron. Measured production cross sections.

2008Na03: $^{\text{nat}}\text{S}(\alpha, \text{x})^{34\text{m}}\text{Cl}$ at $E(\alpha) = 14.2\text{--}69.5$ MeV using the NIRS isochronous cyclotron (AVF930). Measured thick target yields.

Other ^{34}Cl cross-section measurements: 2007Na31, 2004De41, 2003Ya20, 2002DeZU, 2001Pi10.

^{34}Cl mass measurement: 2009Er07 by Penning trap.

 ^{34}Cl LevelsCross Reference (XREF) Flags

A	$^{34}\text{Ar} \varepsilon + \beta^+$ decay (0.84646 s)	I	$^{32}\text{S}(^3\text{He}, \text{p}\gamma)$	Q	$^{35}\text{Cl}(\gamma, \text{n})$
B	^{34}Cl IT decay (31.99 min)	J	$^{32}\text{S}(\alpha, \text{d})$	R	$^{35}\text{Cl}(\text{p}, \text{d})$
C	$^{35}\text{K} \varepsilon \text{p}$ decay (175 ms)	K	$^{33}\text{S}(\text{p}, \gamma)$	S	$^{35}\text{Cl}(\text{d}, \text{t})$
D	$^9\text{Be}(^{37}\text{Ca}, ^{34}\text{Cl}\gamma)$	L	$^{33}\text{S}(\text{p}, \text{p}), (\text{p}, \text{p}'\gamma)$: resonances	T	$^{35}\text{Cl}(^3\text{He}, \alpha)$
E	$^{24}\text{Mg}(^{12}\text{C}, \text{p}\text{n}\gamma)$	M	$^{33}\text{S}(^3\text{He}, \text{d})$	U	$^{36}\text{Ar}(\text{p}, ^3\text{He})$
F	$^{27}\text{Al}(^{12}\text{C}, \alpha \text{n}\gamma)$	N	$^{34}\text{S}(\text{p}, \text{n})$	V	$^{36}\text{Ar}(\text{d}, \alpha), (\text{pol d}, \alpha)$
G	$^{31}\text{P}(\alpha, \text{n}\gamma)$	O	$^{34}\text{S}(^3\text{He}, \text{t})$	W	$^{40}\text{Ca}(\mu^-, \nu \alpha 2 \text{n}\gamma)$
H	$^{32}\text{S}(^3\text{He}, \text{p})$	P	$^{34}\text{S}(^6\text{Li}, ^6\text{He})$		

$E(\text{level})^\dagger$	J^π	$T_{1/2}^\#$	XREF	Comments
0.0	0^+	1.5266 s 5	ABCDEFGHIJ K MNOPQRSTUVWXYZ	$\% \varepsilon + \% \beta^+ = 100$ XREF: V(?) J^π : superallowed $\varepsilon + \beta^+$ feeding from 0^+ parent ^{34}Ar g.s. with $\log ft = 3.486$. $L=2$ from $3/2^+$ in $^{33}\text{S}(^3\text{He}, \text{d})$. $T_{1/2}$: weighted average of 1.5268 s 5 (2006Ia05), 1.5277 s 22 (1983Ko22), 1.5252 s 11 (1976Wi08), and 1.526 s 2 (1973Ry01). Others: 1.530 s 4 (1975Ha24), 1.534 s 3 (1972Ha28), 1.565 s 7 (1965Eb01), 1.565 s 7 (1965Fr06), 1.560 s 14 (1961Ja22), 1.58 s 1 (1960Ja12), 1.61 s 1 (1958Mi85), and 1.53 s 2 (1954KI36). Isobaric analog state of ^{34}S g.s. $\% \text{IT} = 46.0$ 4; $\% \varepsilon + \% \beta^+ = 54.0$ 4 XREF: C(?)J(165) $E(\text{level})$: weighted average of 146.36 3 from $^{27}\text{Al}(^{12}\text{C}, \alpha \text{n}\gamma)$, 146.36 3 from $^{32}\text{S}(^3\text{He}, \text{p}\gamma)$, 146.4 1 from $^{33}\text{S}(\text{p}, \gamma)$, and 146.3
146.36 3	3^+	31.99 min 3	ABC EFGHIJK M QRST VW	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2}[#]</u>	<u>XREF</u>	<u>Comments</u>
				3 from $^{31}\text{P}(\alpha, n\gamma)$. Others: 145.3 16 from $^{32}\text{S}(^3\text{He}, p)$, 165 40 from $^{32}\text{S}(\alpha, d)$, 140 15 from $^{35}\text{Cl}(p, d)$, and 147 20 from $^{36}\text{Ar}(d, \alpha)$, (pol d, α). J^π: L=4 from 0⁺ in $^{36}\text{Ar}(d, \alpha)$, (pol d, α) and L-1 transfer from analyzing power. T_{1/2}: weighted average of 31.98 min 5 (1982Gr07), 31.93 min 9 (1980Wi13), 32.06 min 8 (1975Va02), and 31.99 min 5 (1965Eb01). Others: 32.40 min 4 (1956Gr07), 40 min 5 (1934Fr01). %IT and %ε+%β⁺ decay of the isomeric state are deduced by evaluators using the ΣI(γ to ^{34}S g.s.)=129.3 12 and I(146γ to ^{34}Cl g.s.)=94.4 14 measured by 1980Wi13, taking into account the conversion coefficient α=0.1656 24 (2008Ki07). Others: %IT=44.5 7 and %I(ε+β⁺)=55.5 7 (1980Wi13) deduced from relative I_γ and α=0.100 9). %IT=46.9 10 and %I(ε+β⁺)=53.1 10 (1975Va02) deduced from relative I_γ and α=0.100 9). %IT=44.1 6 and %I(ε+β⁺)=55.5 10 (1983Kr06) deduced from I_β).
461.01 4	1 ⁺	5.1 ps 5	A CDEFGHIJK MNOPQRSTUVWXYZ	XREF: C(?)J(479)R(450)S(450) E(level): weighted average of 460.98 26 from $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$, 461.00 4 from $^{32}\text{S}(^3\text{He}, p\gamma)$, 461.2 2 from $^{33}\text{S}(p, \gamma)$, and 461.5 4 from $^{31}\text{P}(\alpha, n\gamma)$. Others: 460.8 10 from ^{34}Ar ε+β⁺ decay, 461.8 34 from $^{32}\text{S}(^3\text{He}, p)$, 479 30 from $^{32}\text{S}(\alpha, d)$, 461.0 12 from $^{33}\text{S}(^3\text{He}, d)$, 458 4 from $^{34}\text{S}(^3\text{He}, t)$, 450 15 from $^{35}\text{Cl}(p, d)$, and 459 20 from $^{36}\text{Ar}(d, \alpha)$, (pol d, α). J^π: 461.00γ M1 to 0⁺, g.s. T_{1/2}: weighted average of 4.0 ps +16–12 in $^9\text{Be}(^{37}\text{Ca}, ^{34}\text{Cl}\gamma)$ from 2010Do03 with DSAM lineshape analysis with Cluster detectors, 6.2 ps 28 in $^{31}\text{P}(\alpha, n\gamma)$ from 1970Br10 with DSAM, 5.6 ps 6 in $^{31}\text{P}(\alpha, n\gamma)$ from 1971Ba98 with RDM, and 4.9 ps 5 in $^{32}\text{S}(^3\text{He}, p\gamma)$ from 1971Sn01 with DSAM. Others: in $^9\text{Be}(^{37}\text{Ca}, ^{34}\text{Cl}\gamma)$, 2.2 ps 20 (stat) 36 (syst) from lineshape analysis with MINIBALL detectors (2010Do03), 5.0 ps +20–15 from peak-shift analysis with Cluster detectors (2007DoZV), and 6 ps 2 from peak-shift analysis with MINIBALL detectors (2007DoZV).
665.57 5	1 ⁺	9.2 ps 7	A CDEFGHI K MNOPQRST VW	XREF: P(680) E(level): weighted average of 665.8 10 from ^{34}Ar ε+β⁺ decay, 666.2 5 from $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$, 664.7 54 from $^{32}\text{S}(^3\text{He}, p)$, 665.58 5 from $^{32}\text{S}(^3\text{He}, p\gamma)$, 665.3 2 from $^{33}\text{S}(p, \gamma)$, 665.1 14 from $^{33}\text{S}(^3\text{He}, d)$, 666 4 from $^{34}\text{S}(^3\text{He}, t)$, 670 15 from $^{35}\text{Cl}(p, d)$, 665 20 from $^{36}\text{Ar}(d, \alpha)$, (pol d, α), and 663.7 12 from $^{31}\text{P}(\alpha, n\gamma)$. J^π: 665.57γ M1 to 0⁺, g.s. T_{1/2}: lifetime τ=13.2 ps 10: weighted average of 14 ps +11–6 in $^{31}\text{P}(\alpha, n\gamma)$ from 1970Br10 with DSAM, 14.2 ps 15 in $^{31}\text{P}(\alpha, n\gamma)$ from 1971Ba98 with RDM, and 12.8 ps 10 in $^{32}\text{S}(^3\text{He}, p\gamma)$ from 1971Sn01 with DSAM. Others: in $^9\text{Be}(^{37}\text{Ca}, ^{34}\text{Cl}\gamma)$, 2007DoZV reports two preliminary values: T_{1/2}=13 ps 5 from peak-shift analysis with the MINIBALL detectors and T_{1/2}=32 ps +10–14 with the Cluster detectors.
1230.28 6	2 ⁺	13.7 ps 10	A D FGH IJK MN QRSTUVW	XREF: S(1210) E(level): weighted average of 1230.2 5 from $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$,

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2}[#]</u>	<u>XREF</u>	<u>Comments</u>
1887.29 8	2 ⁺	0.76 ps 28	A FGHIJK MN QRSTUV	1230.4 73 from $^{32}\text{S}(^3\text{He},\text{p})$, 1230.26 6 from $^{32}\text{S}(^3\text{He},\text{p}\gamma)$, 1230 40 from $^{32}\text{S}(\alpha,\text{d})$, 1230.33 8 from $^{33}\text{S}(\text{p},\gamma)$, 1229.4 16 from $^{33}\text{S}(^3\text{He},\text{d})$, 1230 15 from $^{35}\text{Cl}(\text{p},\text{d})$, and 1229 20 from $^{36}\text{Ar}(\text{d},\alpha)$, (pol d, α). J ^π : 564.67γ M1+E2 to 1 ⁺ , 665.57 level, 1083.9γ M1+E2 to 3 ⁺ , 146.36 level. T _{1/2} : lifetime τ=19.7 ps 14: weighted average of 22 ps 4 in $^{31}\text{P}(\alpha,\text{n}\gamma)$ from 1971Ba98 with RDM and 19.4 ps 14 in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$ from 1971Sn01 with DSAM. Other: >14 ps in $^{31}\text{P}(\alpha,\text{n}\gamma)$ from 1970Br10 with DSAM. XREF: j(1896)S(1840) E(level): weighted average of 1887.2 6 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$, 1880.0 53 from $^{32}\text{S}(^3\text{He},\text{p})$, 1887.29 8 from $^{33}\text{S}(\text{p},\gamma)$, 1885 3 from $^{33}\text{S}(^3\text{He},\text{d})$, 1880 15 from $^{35}\text{Cl}(\text{p},\text{d})$, and 1890 20 from $^{36}\text{Ar}(\text{d},\alpha)$, (pol d, α). J ^π : L=2 from 0 ⁺ in $^{36}\text{Ar}(\text{d},\alpha)$, (pol d, α) with J=L transfer from analyzing power. T _{1/2} : lifetime τ=1.1 ps 4: weighted average of 0.94 ps 40 in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$ from 1970Br11 with DSAM and 2.4 ps 13 in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$ from 1973Ca15 with DSAM.
1923.3 17	0 ⁺ , 1 ⁺ , 2 ⁺ , 3 ⁺ , 4 ⁺		j M	XREF: j(1896) J ^π : L=2 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He},\text{d})$.
2157.9 1	2 ⁺	33.3 fs 35	A EFGHI K M QR Tu	XREF: u(2162) E(level): weighted average of 2158.2 5 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$ and 2157.9 1 from $^{33}\text{S}(\text{p},\gamma)$. Others: 2163.0 87 from $^{32}\text{S}(^3\text{He},\text{p})$, 2157.5 15 from $^{33}\text{S}(^3\text{He},\text{d})$, and 2160 15 from $^{35}\text{Cl}(\text{p},\text{d})$. J ^π : L=0 from 3/2 ⁺ in $^{35}\text{Cl}(\text{p},\text{d})$ gives 1 ⁺ , 2 ⁺ . 1697.5γ, M1+E2, to 1 ⁺ , 461.01 level; 2012γ, M1+E2, to 3 ⁺ , 146.36 level. T _{1/2} : lifetime τ=48 fs 5: weighted average of 46 fs 15 from 1970Br11, 40 fs 20 from 1972AlZT, 100 fs 30 from 1973Ca15 in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$; 36 fs 15 from 1977Da02, 46 fs 10 from 1983Wa27, 47 fs 5 from 1984Ke01, and 49 fs 5 from 1985La16 in $^{33}\text{S}(\text{p},\gamma)$; all with DSAM.
2181.09 8	3 ⁺	0.35 ps 7	A EF IJK Q TuV	XREF: u(2162) E(level): weighted average of 2180.4 5 from $^{24}\text{Mg}(^{12}\text{C},\text{p}\text{n}\gamma)$, 2181.3 5 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$, and 2181.10 8 from $^{33}\text{S}(\text{p},\gamma)$. Others: 2175 7 from $^{32}\text{S}(\alpha,\text{d})$ and 2191 20 from $^{36}\text{Ar}(\text{d},\alpha)$, (pol d, α). J ^π : spin=3 from γ(θ) in $^{33}\text{S}(\text{p},\gamma)$; π=+ from 1720γ E2 to 1 ⁺ , 461.01 level. T _{1/2} : lifetime τ=504 fs 100: weighted average of 400 fs 100 from 1970Br11, 600 fs 150 from 1973Ca15 in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$; 600 fs 350 from 1977Da02, 630 fs 190 from 1983Wa27, and 800 fs 400 from 1985La16 in $^{33}\text{S}(\text{p},\gamma)$; all with DSAM.

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Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

E(level) [†]	J ^π	T _{1/2} [#]	XREF			Comments
2375.67 14	4 ⁺	0.150 ps 35	EFGHI	K	Q V	XREF: Q(2380) E(level): weighted average of 2375.64 14 from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$, 2375.8 5 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$, and 2375.7 2 from $^{33}\text{S}(\text{p},\gamma)$. Others: 2376.8 75 from $^{32}\text{S}(^3\text{He},\text{p})$, 2382 20 from $^{36}\text{Ar}(\text{d},\alpha)$, (pol d, α). J ^π : spin=4 from $\text{p}\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$ and $^{32}\text{S}(^3\text{He},\text{p}\gamma)$; $\pi=+$ from 2230.0 γ , M1+E2, to 3 ⁺ , 146.36 level. L=0+2 from 0 ⁺ in $^{36}\text{Ar}(\text{d},\alpha)$, (pol d, α) gives 1 ⁺ for 2382 20 seems to be a discrepancy. T _{1/2} : lifetime $\tau=216$ fs 50: weighted average of 260 fs 75 in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$ from 1970Br11 with DSAM, 275 fs 70 in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$ from 1973Ca15 with DSAM, 150 fs 50 in $^{33}\text{S}(\text{p},\gamma)$ from 1973An13 with DSAM, 210 fs 80 in $^{33}\text{S}(\text{p},\gamma)$ from 1977Da02 with DSAM, 200 fs 50 in $^{33}\text{S}(\text{p},\gamma)$ from 1983Wa27 with DSAM, and 250 fs 50 in $^{33}\text{S}(\text{p},\gamma)$ from 1985La16 with DSAM.
2580.3 2	1 ⁺	15 fs 9	A	HI K MNOP	r TuV	XREF: P(2560)r(2600)u(2600) E(level): weighted average of 2579.5 14 from $^{34}\text{Ar } \varepsilon+\beta^+$ decay, 2583.0 41 from $^{32}\text{S}(^3\text{He},\text{p})$, 2580.3 2 from $^{33}\text{S}(\text{p},\gamma)$, 2579.1 16 from $^{33}\text{S}(^3\text{He},\text{d})$, 2579 4 from $^{34}\text{S}(^3\text{He},\text{t})$, and 2589 20 from $^{36}\text{Ar}(\text{d},\alpha)$, (pol d, α). J ^π : spin=1 from $\text{p}\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$; $\pi=+$ from L=0 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He},\text{d})$. T _{1/2} : lifetime $\tau=22$ fs 13: unweighted average of 15 fs 6 in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$ from 1972AIZT with DSAM, 47 fs 12 in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$ from 1973Ca15 with DSAM, and 3.0 fs 25 in $^{33}\text{S}(\text{p},\gamma)$ from 1977Da02 with DSAM. Others: <5 fs in $^{33}\text{S}(\text{p},\gamma)$ from 1983Wa27 with DSAM and <7 fs in $^{33}\text{S}(\text{p},\gamma)$ from 1985La16 with DSAM.
2611.04 11	3 ⁺	0.170 ps 50	FGHI	K M	Qr TuV	XREF: r(2600)u(2600) E(level): weighted average of 2611.3 5 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$ and 2611.03 11 from $^{33}\text{S}(\text{p},\gamma)$. Others: 2614.2 38 from $^{32}\text{S}(^3\text{He},\text{p})$, 2608.5 13 from $^{33}\text{S}(^3\text{He},\text{d})$, and 2620 20 from $^{36}\text{Ar}(\text{d},\alpha)$, (pol d, α). J ^π : spin=3 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$; $\pi=+$ from L=2 from 3/2 ⁺ in $^{35}\text{Cl}(^3\text{He},\alpha)$ and $^{33}\text{S}(^3\text{He},\text{d})$. T _{1/2} : lifetime $\tau=245$ fs 72: weighted average of 820 fs 350 in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$ from 1973Ca15 with DSAM, 210 fs 60 in $^{33}\text{S}(\text{p},\gamma)$ from 1983Wa27 with DSAM, and 340 fs 140 in $^{33}\text{S}(\text{p},\gamma)$ from 1985La16 with DSAM. Other: >450 fs in $^{33}\text{S}(\text{p},\gamma)$ from 1977Da02 with DSAM.
2718 10	0 ⁺ ,1 ⁺ ,2 ⁺ ,3 ⁺ ,4 ⁺				o T	XREF: o(2718) J ^π : L=2 from 3/2 ⁺ in $^{35}\text{Cl}(^3\text{He},\alpha)$.
2721.09 20	2 ⁻	>1.4 ps	EFGHIJK	M o	V	XREF: J(2710)o(2718)V(2725)

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Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2}[#]</u>	<u>XREF</u>	<u>Comments</u>
3129.11 12	1 ⁺	<3 fs	A HI K MNOPQ TUV	E(level): weighted average of 2720.78 24 from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$, 2721.13 34 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$, and 2721.3 2 from $^{33}\text{S}(\text{p},\gamma)$. Others: 2720.5 39 from $^{32}\text{S}(^3\text{He},\text{p})$, 2720.7 25 from $^{33}\text{S}(^3\text{He},\text{d})$. J^π: 2721γ, M2, ΔJ=2 to 0⁺, g.s. T_{1/2}: lifetime τ>2 ps in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$ from 1973Ca15 with DSAM. Others: >1.0 ps in $^{33}\text{S}(\text{p},\gamma)$ from 1973An13 with DSAM, >400 fs in $^{33}\text{S}(\text{p},\gamma)$ from 1977Da02 with DSAM, and >1500 fs in $^{33}\text{S}(\text{p},\gamma)$ from 1985La16 with DSAM. XREF: V(3146)
3334.0 2	(1,2,3) ⁺	48 fs 19	HI K M O Qr TuV	E(level): weighted average of 3129.0 10 from $^{34}\text{Ar } \varepsilon+\beta^+$ decay, 3127.0 27 from $^{32}\text{S}(^3\text{He},\text{p})$, 3129.12 12 from $^{33}\text{S}(\text{p},\gamma)$, 3128.0 22 from $^{33}\text{S}(^3\text{He},\text{d})$, 3129 4 from $^{34}\text{S}(^3\text{He},\text{t})$, 3130 30 from $^{36}\text{Ar}(\text{p},^3\text{He})$, and 3146 20 from $^{36}\text{Ar}(\text{d},\alpha)$, (pol d,α). J^π: L=0+2 from 0⁺ in $^{36}\text{Ar}(\text{d},\alpha)$, (pol d,α). T_{1/2}: lifetime τ<4 fs in $^{33}\text{S}(\text{p},\gamma)$ from 1985La16 with DSAM. Others: <10 fs in $^{33}\text{S}(\text{p},\gamma)$ from 1983Wa27 with DSAM and 40 fs 12 in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$ from 1973Ca15 with DSAM. XREF: Q(3330)r(3360)u(3350)
3383.3 2	1 ⁺ ,2 ⁺	5.3 fs 12	HI K MN Qr TuV	E(level): weighted average of 3336.4 31 from $^{32}\text{S}(^3\text{He},\text{p})$, 3334.0 2 from $^{33}\text{S}(\text{p},\gamma)$, 3333.0 24 from $^{33}\text{S}(^3\text{He},\text{d})$, and 3330 4 from $^{34}\text{S}(^3\text{He},\text{t})$. J^π: L=2 from 0⁺ in $^{32}\text{S}(^3\text{He},\text{p})$; L=2 from 3/2⁺ in $^{33}\text{S}(^3\text{He},\text{d})$ and $^{35}\text{Cl}(^3\text{He},\alpha)$ gives 0⁺,1⁺,2⁺,3⁺,4⁺. L=3 from 0⁺ in $^{36}\text{Ar}(\text{d},\alpha)$, (pol d,α) gives 2⁻,3⁻,4⁻ for 3334 20 seems to be a discrepancy, probably indicating a different level. T_{1/2}: lifetime τ=68 fs 27: weighted average of 115 fs 35 in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$ from 1973Ca15 with DSAM and 53 fs 20 in $^{33}\text{S}(\text{p},\gamma)$ from 1985La16 with DSAM. XREF: r(3360)u(3350)
3545.08 15	3 ⁻	0.101 ps 21	HI K M Q T	E(level): weighted average of 3383.2 43 from $^{32}\text{S}(^3\text{He},\text{p})$, 3383.3 2 from $^{33}\text{S}(\text{p},\gamma)$, 3382.7 20 from $^{33}\text{S}(^3\text{He},\text{d})$, and 3377 20 from $^{36}\text{Ar}(\text{d},\alpha)$, (pol d,α). J^π: L=0+2 from 3/2⁺ in $^{35}\text{Cl}(^3\text{He},\alpha)$. L=2 from 0⁺ in $^{32}\text{S}(^3\text{He},\text{p})$. T_{1/2}: lifetime τ=7.7 fs 17: weighted average of 7.5 fs 17 in $^{33}\text{S}(\text{p},\gamma)$ from 1984Ke01 with DSAM and 9 fs 4 in $^{33}\text{S}(\text{p},\gamma)$ from 1985La16 with DSAM. Others: <50 fs in $^{32}\text{S}(^3\text{He},\text{p}\gamma)$ from 1973Ca15 with DSAM, <3 fs in $^{33}\text{S}(\text{p},\gamma)$ from 1977Da02 with DSAM, and <10 fs in $^{33}\text{S}(\text{p},\gamma)$ from 1983Wa27 with DSAM.
3545.08 15	3 ⁻	0.101 ps 21	HI K M Q T	E(level): weighted average of 3548.4 52 from $^{32}\text{S}(^3\text{He},\text{p})$, 3545.08 15 from $^{33}\text{S}(\text{p},\gamma)$, and 3544.9 25 from $^{33}\text{S}(^3\text{He},\text{d})$. J^π: spin=3 from γ(θ) in $^{33}\text{S}(\text{p},\gamma)$. π=- from L=1 from 3/2⁺ in $^{33}\text{S}(^3\text{He},\text{d})$ and $^{35}\text{Cl}(^3\text{He},\alpha)$. T_{1/2}: lifetime τ=145 fs 30: weighted average of 140 fs 40

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Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2}[#]</u>	<u>XREF</u>	<u>Comments</u>
3600.14 11	4 ⁻	15.9 ps 35	EFGHIJK	in $^{32}\text{S}(^3\text{He},p\gamma)$ from 1973Ca15 with DSAM, 140 fs 45 in $^{33}\text{S}(p,\gamma)$ from 1983Wa27 with DSAM, and 150 fs 30 in $^{33}\text{S}(p,\gamma)$ from 1985La16 with DSAM. Other: >500 fs in $^{33}\text{S}(p,\gamma)$ from 1973An13 with DSAM. E(level): weighted average of 3599.90 13 from $^{24}\text{Mg}(^{12}\text{C},pn\gamma)$, 3600.46 32 from $^{27}\text{Al}(^{12}\text{C},\alpha n\gamma)$, 3601.1 49 from $^{32}\text{S}(^3\text{He},p)$, 3603 14 from $^{32}\text{S}(\alpha,d)$, and 3600.28 11 from $^{33}\text{S}(p,\gamma)$. J ^π : spin=4 from $\gamma(\theta)$ in $^{33}\text{S}(p,\gamma)$. L=3+5 from 0 ⁺ in $^{32}\text{S}(\alpha,d)$. T _{1/2} : lifetime $\tau=23$ ps 5 in $^{31}\text{P}(\alpha,n\gamma)$ from 1978Ba61 with RDM. Others: >0.7 ps in $^{33}\text{S}(p,\gamma)$ from 1973An13 with DSAM and >2 ps in $^{32}\text{S}(^3\text{He},p\gamma)$ from 1973Ca15 with DSAM.
3631.40 12	5 ⁻	0.194 ns 42	EFGHI K M Q T	XREF: Q(3630) E(level): weighted average of 3631.33 12 from $^{24}\text{Mg}(^{12}\text{C},pn\gamma)$, 3631.7 4 from $^{27}\text{Al}(^{12}\text{C},\alpha n\gamma)$, and 3631.7 3 from $^{33}\text{S}(p,\gamma)$. Others: 3635.2 48 from $^{32}\text{S}(^3\text{He},p)$ and 3631.4 25 from $^{33}\text{S}(^3\text{He},d)$. J ^π : spin=5 from $^{33}\text{S}(p,\gamma)$. L=3 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He},d)$. T _{1/2} : lifetime $\tau=280$ ps 60 in $^{31}\text{P}(\alpha,n\gamma)$ from 1978Ba61 with RDM. Others: >0.7 ps in $^{33}\text{S}(p,\gamma)$ from 1977Da02 with DSAM, >1.1 ps in $^{33}\text{S}(p,\gamma)$ from 1985La16 with DSAM, and >3 ps in $^{32}\text{S}(^3\text{He},p\gamma)$ from 1973Ca15 with DSAM.
3646.3 2	5 ⁺	0.152 ps 62	F K	E(level): weighted average of 3646.5 7 from $^{27}\text{Al}(^{12}\text{C},\alpha n\gamma)$ and 3646.3 2 from $^{33}\text{S}(p,\gamma)$. J ^π : 3500 γ , E2, $\Delta J=2$ to 3 ⁺ , 146.36 level. 725 γ , M1+E2, from (4) ⁺ , 4371.5 level. T _{1/2} : lifetime $\tau=220$ fs 90 in $^{33}\text{S}(p,\gamma)$ from 1985La16 with DSAM.
3660.0 3	(0 ⁺ to 4 ⁺)	<3.5 fs	K	J ^π : 1502.1 γ to 2 ⁺ , 2157.9 level. T _{1/2} : lifetime $\tau<5$ fs in $^{33}\text{S}(p,\gamma)$ from 1985La16 with DSAM.
3773.7 4	1 ⁻	55 fs 14	hI K M	XREF: h(3804) E(level): weighted average of 3773.72 41 from $^{33}\text{S}(p,\gamma)$ and 3773.0 24 from $^{33}\text{S}(^3\text{He},d)$. J ^π : spin=1 from $\gamma(\theta)$ in $^{33}\text{S}(p,\gamma)$. L=1 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He},d)$. T _{1/2} : lifetime $\tau=80$ fs 20: weighted average of 115 fs 30 in $^{32}\text{S}(^3\text{He},p\gamma)$ from 1973Ca15 with DSAM, 55 fs 15 in $^{33}\text{S}(p,\gamma)$ from 1983Wa27 with DSAM, and 110 fs 20 in $^{33}\text{S}(p,\gamma)$ from 1985La16 with DSAM.
3791.7 2	(0 to 3)		h K	XREF: h(3804) J ^π : 3330.5 γ to 1 ⁺ , 461.01 level.
3847 10				T
3940.1 3	(1) ⁺	<4 fs	K	TU XREF: T(3937)U(3940) E(level): from $^{33}\text{S}(p,\gamma)$. Other: 3937 10 from $^{35}\text{Cl}(^3\text{He},\alpha)$. J ^π : L=0+2 from 3/2 ⁺ in $^{35}\text{Cl}(^3\text{He},\alpha)$ gives 1 ⁺ , 2 ⁺ ;

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Adopted Levels, Gammas (continued)

³⁴ Cl Levels (continued)					
E(level) [†]	J ^π	T _{1/2} [#]	XREF		Comments
3964.1 3	(2,3,4) ⁺	90 fs 42	K	Q T	L=(0) from 0 ⁺ in ³⁶ Ar(p, ³ He) gives (0 ⁺ ,1 ⁺). T _{1/2} : lifetime τ<6 fs in ³³ S(p,γ) from 1985La16 with DSAM. XREF: Q(3960) J ^π : γ to 2 ⁺ , 2157.9 level, 3 ⁺ , 146.36 level, and 4 ⁺ , 2375.67 level; π=+ from L=2 from 3/2 ⁺ in ³⁵ Cl(³ He,α). T _{1/2} : lifetime τ=130 fs 60: weighted average of 160 fs 60 in ³³ S(p,γ) from 1977Da02 with DSAM and 90 fs 70 in ³³ S(p,γ) from 1985La16 with DSAM. XREF: Q(3980)
3983.0 3	3 ⁻	0.131 ps 28	H K M	Q T	E(level): weighted average of 3983.0 3 from ³³ S(p,γ) and 3983.4 25 from ³³ S(³ He,d). Other: 3982.6 72 doublet from ³² S(³ He,p). J ^π : spin=3 from γ(θ) in ³³ S(p,γ). L=3 from 3/2 ⁺ in ³³ S(³ He,d) and ³⁵ Cl(³ He,α). T _{1/2} : lifetime τ=189 fs 40: weighted average of 165 fs 60 in ³³ S(p,γ) from 1983Wa27 with DSAM and 200 fs 40 in ³³ S(p,γ) from 1985La16 with DSAM. Other: >100 fs in ³³ S(p,γ) from 1973An13 with DSAM. XREF: q(4075)
4075 10	1 ⁺ ,2 ⁺			q T	J ^π : L=0+2 from 3/2 ⁺ in ³⁵ Cl(³ He,α). XREF: J(4060)q(4075)
4076.2 2	4 ⁻	0.9 ps +12-4	HIJK M	q	E(level): weighted average of 4076.5 58 from ³² S(³ He,p), 4060 18 from ³² S(α,d), 4076.2 2 from ³³ S(p,γ), and 4076 3 from ³³ S(³ He,d). J ^π : spin=4 from γ(θ) in ³³ S(p,γ). L=3+5 from 0 ⁺ in ³² S(α,d) gives 4 ⁻ . T _{1/2} : lifetime τ=1.3 ps +17-6 in ³² S(³ He,pγ) from 1973Ca15 with DSAM. Others: >0.25 ps in ³³ S(p,γ) from 1973An13 with DSAM and >0.5 ps in ³³ S(p,γ) from 1977Da02 with DSAM.
4136.6	1 ⁺ ,2 ⁺			T	J ^π : L=0+2 from 3/2 ⁺ in ³⁵ Cl(³ He,α) gives 1 ⁺ ,2 ⁺ . XREF: M(4146)
4139.7 3	(2 ⁻)	94 fs 28	HI K M		E(level): weighted average of 4141.1 79 from ³² S(³ He,p), 4139.7 2 from ³³ S(p,γ), and 4146 3 from ³³ S(³ He,d). J ^π : spin=2,3 from γ(θ) in ³³ S(p,γ); L=1+3 from 0 ⁺ in ³² S(³ He,p) gives (2 ⁻); L=(1+3) from 3/2 ⁺ in ³³ S(³ He,d) gives (1 ⁻ ,2 ⁻ ,3 ⁻). T _{1/2} : lifetime τ=136 fs 40: weighted average of 150 fs 50 in ³² S(³ He,pγ) from 1973Ca15 with DSAM, 120 fs 40 in ³³ S(p,γ) from 1983Wa27 with DSAM, and 150 fs 60 in ³³ S(p,γ) from 1985La16 with DSAM.
4147.8 2	1 ⁺ ,2 ⁺		K	R	XREF: R(4160) E(level): from ³³ S(p,γ). Other: 4160 20 from ³⁵ Cl(p,d). J ^π : L=0 from 3/2 ⁺ in ³⁵ Cl(p,d).
4211 4	1 ⁺ ,2 ⁺			O T	E(level): weighted average of 4212 4 from ³⁴ S(³ He,t) and 4206 10 from ³⁵ Cl(³ He,α). J ^π : L=0+2 from 3/2 ⁺ in ³⁵ Cl(³ He,α).
4325.89 15	1 ⁺ ,2 ⁺	5.1 fs 21	K	T	E(level): weighted average of 4325.89 15 from ³³ S(p,γ)

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Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2}[#]</u>	<u>XREF</u>	<u>Comments</u>
4354.2 2	(1) ⁻	32 fs 6	HI K M	and 4321 10 from $^{35}\text{Cl}(^3\text{He},\alpha)$. J ^π : L=0+2 from 3/2 ⁺ in $^{35}\text{Cl}(^3\text{He},\alpha)$. T _{1/2} : lifetime $\tau=7.4$ fs 30: weighted average of 11 fs 9 in $^{33}\text{S}(p,\gamma)$ from 1977Da02 with DSAM and 7 fs 3 in $^{33}\text{S}(p,\gamma)$ from 1985La16 with DSAM. E(level): weighted average of 4354.5 20 from $^{32}\text{S}(^3\text{He},p)$, 4354.2 2 from $^{33}\text{S}(p,\gamma)$, and 4353 3 from $^{33}\text{S}(^3\text{He},d)$. J ^π : spin=(1) from $\gamma(\theta)$ in $^{33}\text{S}(p,\gamma)$. L=1 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He},d)$ gives 0 ⁻ , 1 ⁻ , 2 ⁻ , 3 ⁻ . L=1 from 0 ⁺ in $^{32}\text{S}(^3\text{He},p)$ gives (0 ⁻ , 1 ⁻ , 2 ⁻). T _{1/2} : lifetime $\tau=46$ fs 8: weighted average of 43 fs 8 in $^{33}\text{S}(p,\gamma)$ from 1985La16 with DSAM and 65 fs 20 in $^{32}\text{S}(^3\text{He},p\gamma)$ from 1973Ca15 with DSAM. Other: <10 fs in $^{33}\text{S}(p,\gamma)$ from 1983Wa27 with DSAM.
4371.5 13	(4) ⁺	>2.08 ps	F	J ^π : 725 γ , M1+E2, $\Delta J=1$ to 5 ⁺ , 3646.3 level. T _{1/2} : lifetime $\tau>3$ ps in $^{27}\text{Al}(^{12}\text{C},\alpha n\gamma)$ from 1978Ba39 with RDM.
4417.3 2	(2) ⁻	24 fs 8	H JK M	E(level): weighted average of 4412.1 31 from $^{32}\text{S}(^3\text{He},p)$, 4417.3 2 from $^{33}\text{S}(p,\gamma)$, and 4418 3 from $^{33}\text{S}(^3\text{He},d)$. Other: 4417 17 from $^{32}\text{S}(\alpha,d)$. J ^π : L=1, 1+3 from 0 ⁺ in $^{32}\text{S}(\alpha,d)$: L=1 gives 0 ⁻ , 1 ⁻ , 2 ⁻ ; L=1+3 gives 2 ⁻ . L=1 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He},d)$ gives 0 ⁻ , 1 ⁻ , 2 ⁻ , 3 ⁻ ; L=3 from 0 ⁺ in $^{32}\text{S}(^3\text{He},p)$ gives (2 ⁻ , 3 ⁻ , 4 ⁻). T _{1/2} : lifetime $\tau=35$ fs 11 in $^{33}\text{S}(p,\gamma)$ from 1985La16 with DSAM.
4446.6 2	4 ⁻	0.116 ps 24	K Q	XREF: Q(4450) J ^π : 4300.0 γ , D(+Q), to 3 ⁺ , 146.36 level. Primary transition 2424.5 γ , M1+E2, from 5 ⁻ , 6871.18 level. T _{1/2} : lifetime $\tau=167$ fs 35: weighted average of 160 fs 60 in $^{33}\text{S}(p,\gamma)$ from 1977Da02 with DSAM, 180 fs 60 in $^{33}\text{S}(p,\gamma)$ from 1983Wa27 with DSAM, and 165 fs 35 in $^{33}\text{S}(p,\gamma)$ from 1985La16 with DSAM.
4461.4 3	(2,3) ⁻	0.145 ps 42	HI K M Q	E(level): weighted average of 4461.4 3 from $^{33}\text{S}(p,\gamma)$ and 4461 3 from $^{33}\text{S}(^3\text{He},d)$ and 4455.8 40 from $^{32}\text{S}(^3\text{He},p)$. J ^π : L=1 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He},d)$ gives 0 ⁻ , 1 ⁻ , 2 ⁻ , 3 ⁻ . L=3 from 0 ⁺ in $^{32}\text{S}(^3\text{He},p)$ gives (2 ⁻ , 3 ⁻ , 4 ⁻). T _{1/2} : lifetime $\tau=209$ fs 60: weighted average of 200 fs 60 in $^{32}\text{S}(^3\text{He},p\gamma)$ from 1973Ca15 with DSAM and 370 fs 250 in $^{33}\text{S}(p,\gamma)$ from 1985La16 with DSAM.
4515.7 2	2 ⁻	10.4 fs 35	H K M	E(level): weighted average of 4514.5 32 from $^{32}\text{S}(^3\text{He},p)$, 4515.7 2 from $^{33}\text{S}(p,\gamma)$, and 4516.3 22 from $^{33}\text{S}(^3\text{He},d)$. J ^π : L=1 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He},d)$ gives 0 ⁻ , 1 ⁻ , 2 ⁻ , 3 ⁻ ; L=1+3 from 0 ⁺ in $^{32}\text{S}(^3\text{He},p)$ gives (2 ⁻). T _{1/2} : lifetime $\tau=15$ fs 5: weighted average of 15 fs 7 in $^{33}\text{S}(p,\gamma)$ from 1983Wa27 with DSAM and 15 fs 5 in $^{33}\text{S}(p,\gamma)$ from 1985La16 with DSAM.
4605.8 10	(-)	0.24 ps 17	h K m Q t	XREF: Q(4606)t(4608) E(level), T _{1/2} : from $^{33}\text{S}(p,\gamma)$. J ^π : L=1+3 from 3/2 ⁺ in $^{35}\text{Cl}(^3\text{He},\alpha)$ gives 1 ⁻ , 2 ⁻ , 3 ⁻ for 4608 level. L=1 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He},d)$ gives 0 ⁻ , 1 ⁻ , 2 ⁻ , 3 ⁻ for

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

E(level) [†]	J ^π	T _{1/2} [#]	XREF	Comments
				4607.9 21 level. L=1+3 from 0 ⁺ in $^{32}\text{S}(^3\text{He},\text{p})$ gives (2 ⁻) for 4611.9 70 level. T _{1/2} : lifetime $\tau=350$ fs 240 in $^{33}\text{S}(\text{p},\gamma)$ from 1985La16 with DSAM. XREF: q(4610)t(4608) J ^π : same arguments as 4605.8 level also apply to this level.
4609.7 15	(-)		h K m q t	XREF: q(4610) E(level): from $^{34}\text{S}(^3\text{He},\text{t})$. J ^π : L=0 from 0 ⁺ in $^{34}\text{S}(^3\text{He},\text{t})$. XREF: j(4665)u(4670)
4611 4	0 ⁺ ,1 ⁺		0 q	E(level): weighted average of 4638.9 4 from $^{33}\text{S}(\text{p},\gamma)$ and 4638.7 21 from $^{33}\text{S}(^3\text{He},\text{d})$. J ^π : L=1+3 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He},\text{d})$ gives 1 ⁻ ,2 ⁻ ,3 ⁻ ; 4492.2γ to 3 ⁺ , 146.36 level and 4177.6γ to 1 ⁺ , 461.01 level cannot be M2 based on T _{1/2} and RUL. T _{1/2} : lifetime $\tau=45$ fs 14 in $^{33}\text{S}(\text{p},\gamma)$ from 1985La16 with DSAM.
4638.9 4	2 ⁻	31 fs 10	jK M u	XREF: j(4665)u(4670) E(level): weighted average of 4693.0 34 from $^{32}\text{S}(^3\text{He},\text{p})$, 4695.7 2 from $^{33}\text{S}(\text{p},\gamma)$, and 4695 2 from $^{33}\text{S}(^3\text{He},\text{d})$. J ^π : L=2,1+3 from 0 ⁺ in $^{32}\text{S}(^3\text{He},\text{p})$ gives (1 ⁺ ,2,3 ⁺). T _{1/2} : lifetime $\tau=9$ fs 4 in $^{33}\text{S}(\text{p},\gamma)$ from 1985La16 with DSAM. Other: <10 fs in $^{33}\text{S}(\text{p},\gamma)$ from 1983Wa27 with DSAM.
4695.7 2	(1 ⁺ ,2,3 ⁺)	6.2 fs 28	H jK M u	E(level): weighted average of 4717.4 6 from $^{33}\text{S}(\text{p},\gamma)$ and 4718 2 from $^{33}\text{S}(^3\text{He},\text{d})$. Other: 4715 10 from $^{35}\text{Cl}(^3\text{He},\alpha)$. J ^π : L=2 from 3/2 ⁺ in $^{35}\text{Cl}(^3\text{He},\alpha)$ gives 0 ⁺ ,1 ⁺ ,2 ⁺ ,3 ⁺ ,4 ⁺ . 2341.7γ to 4 ⁺ , 2375.67 level. T _{1/2} : lifetime $\tau<13$ fs in $^{33}\text{S}(\text{p},\gamma)$ from 1985La16 with DSAM.
4717.5 6	2 ⁺ ,3 ⁺ ,4 ⁺	<9 fs	K M Q T	E(level): weighted average of 4743.08 13 from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$ and 4743.59 34 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$. J ^π : 1143.0γ, E2, ΔJ=2 to 4 ⁻ , 3600.14 level; 1111.77γ, M1+E2, to 5 ⁻ , 3631.40 level. T _{1/2} : lifetime $\tau=7$ ps 3 in $^{31}\text{P}(\alpha,\text{n}\gamma)$ from 1978Ba61 with RDM. Other: >3 ps in $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$ from 1978Ba39 with RDM.
4743.15 17	6 ⁻	4.9 ps 21	EFG	J ^π : L=1+3,3,4 from 0 ⁺ in $^{32}\text{S}(\alpha,\text{d})$: L=1+3 gives 2 ⁻ ; L=3 gives 2 ⁻ ,3 ⁻ ,4 ⁻ ; L=4 gives 3 ⁺ ,4 ⁺ ,5 ⁺ . E(level): weighted average of 4824.03 12 from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$, 4824.52 34 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$, 4824.8 24 from $^{32}\text{S}(^3\text{He},\text{p})$, and 4824.41 18 from $^{33}\text{S}(\text{p},\gamma)$. J ^π : 1224.1γ, E1(+M2), ΔJ=1 to 4 ⁻ , 3600.14 level; 4677.4γ, E2, ΔJ=2 to 3 ⁺ , 146.36 level. T _{1/2} : lifetime $\tau=464$ fs 200: weighted average of 450 fs 200 in $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$ from 1982Va02 with DSAM and 480 fs 220 in $^{33}\text{S}(\text{p},\gamma)$ from 1983Wa27 with DSAM. Other: >500 fs in $^{33}\text{S}(\text{p},\gamma)$ from 1985La16 with DSAM.
4786 10	2 ⁻ to 5 ⁺		J	J ^π : L=2,1+3 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He},\text{d})$: L=2 gives
4824.18 12	5 ⁺	0.32 ps 14	EFGH K	
4857 2	0 ⁺ to 4 ⁺		M	

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Adopted Levels, Gammas (continued)

³⁴ Cl Levels (continued)						
E(level) [†]	J ^π	T _{1/2} [#]	XREF			Comments
						0 ⁺ , 1 ⁺ , 2 ⁺ , 3 ⁺ , 4 ⁺ ; L=1+3 from 3/2 ⁺ in ³³ S(³ He,d) gives 1 ⁻ , 2 ⁻ , 3 ⁻ .
4862.4 8	5 ⁻		F			J ^π : 2681γ, M2, ΔJ=2 to 3 ⁺ , 2181.09 level. 453γ from 7 ⁺ , 5314.95 level.
4925 4	(⁺)		j	0	uv	XREF: j(4930)u(4970)v(4970) E(level): from (³ He,t). J ^π : L=0+2,2 from 0 ⁺ in ³² S(α,d) for 4930 20 level: L=0+2 gives 1 ⁺ ; L=2 gives 1 ⁺ , 2 ⁺ , 3 ⁺ .
4941.9 4	1 ⁺	<5.5 fs	jK M O		Tuv	XREF: j(4930)T(4939)u(4970)v(4970) E(level): weighted average of 4941.9 4 from ³³ S(p,γ), 4942 4 from ³⁴ S(³ He,t), and 4939 11 from ³⁵ Cl(³ He,α). J ^π : L=0 from 0 ⁺ in ³⁴ S(³ He,t) gives 0 ⁺ , 1 ⁺ ; L=2 from 3/2 ⁺ in ³⁵ Cl(³ He,α) gives 0 ⁺ , 1 ⁺ , 2 ⁺ , 3 ⁺ , 4 ⁺ ; 2330.8γ to 3 ⁺ , 2611.04 level.
						T _{1/2} : lifetime τ<8 fs in ³³ S(p,γ) from 1985La16 with DSAM.
4957.5 11	1 ⁺ , 2 ⁺ , 3 ⁺	<11.8 fs	H K M O		Tuv	XREF: u(4970)v(4970) E(level): weighted average of 4965.7 71 from ³² S(³ He,p), 4957.3 11 from ³³ S(p,γ), and 4957 4 from ³⁴ S(³ He,t). Other: 4958 11 from ³⁵ Cl(³ He,α). J ^π : L=2 from 3/2 ⁺ in ³³ S(³ He,d) and ³⁵ Cl(³ He,α) gives 0 ⁺ , 1 ⁺ , 2 ⁺ , 3 ⁺ , 4 ⁺ ; L=(1,2) from 0 ⁺ in ³² S(³ He,p) gives (0 ⁻ , 1, 2, 3 ⁺). 4810.8γ to 3 ⁺ , 146.36 level.
						T _{1/2} : lifetime τ<17 fs in ³³ S(p,γ) from 1985La16 with DSAM.
4979 2	1 ⁺ , 2 ⁺		MnO qr		Tuv	XREF: n(4985)q(4990)r(5000)u(4970)v(4970) E(level): weighted average of 4979 2 from ³³ S(³ He,d), 4981 4 from ³⁴ S(³ He,t), and 4971 11 from ³⁵ Cl(³ He,α). J ^π : L=1,0 from 3/2 ⁺ in ³³ S(³ He,d): L=1 gives 0 ⁻ , 1 ⁻ , 2 ⁻ , 3 ⁻ ; L=0 gives 1 ⁺ , 2 ⁺ ; L=0+2 from 3/2 ⁺ in ³⁵ Cl(³ He,α) gives 1 ⁺ , 2 ⁺ .
4995.6 3	1 ⁺ , 2 ⁺	4.9 fs 28	K MnO qr		Tuv	XREF: n(4985)q(4990)r(5000)u(4970)v(4970) E(level): weighted average of 4995.6 3 from ³³ S(p,γ) and 4994 2 from ³³ S(³ He,d). Others: 4994 4 from ³⁴ S(³ He,t) and 4998 12 from ³⁵ Cl(³ He,α). J ^π : L=0+2 from 3/2 ⁺ in ³⁵ Cl(³ He,α) gives 1 ⁺ , 2 ⁺ . L=2 from 3/2 ⁺ in ³⁵ Cl(p,d) gives 0 ⁺ , 1 ⁺ , 2 ⁺ , 3 ⁺ , 4 ⁺ for 5000 30 level.
						T _{1/2} : lifetime τ=7 fs 4 in ³³ S(p,γ) from 1985La16 with DSAM. Other: <5 fs in ³³ S(p,γ) from 1983Wa27 with DSAM.
5009 2	0 ⁺ , 1 ⁺ , 2 ⁺ , 3 ⁺ , 4 ⁺		M O Qr		Tuv	XREF: r(5000)u(4970)v(4970) E(level): weighted average of 5009 2 from ³³ S(³ He,d) and 5010 4 from ³⁴ S(³ He,t). Other: 5010 13 from ³⁵ Cl(³ He,α). J ^π : L=2 from 3/2 ⁺ in ³⁵ Cl(³ He,α). L=2,3 from 3/2 ⁺ in ³³ S(³ He,d): L=2 gives 0 ⁺ , 1 ⁺ , 2 ⁺ , 3 ⁺ , 4 ⁺ ; L=3 gives 1 ⁻ , 2 ⁻ , 3 ⁻ , 4 ⁻ , 5 ⁻ .

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Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

E(level) [†]	J ^π	T _{1/2} [#]	XREF	Comments
5028 2	0 ⁻ , 1 ⁻ , 2 ⁻ , 3 ⁻		K M r	XREF: K(5020?)r(5000) J ^π : L=1 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$ gives 0 ⁻ , 1 ⁻ , 2 ⁻ , 3 ⁻ .
5061 4			0	
5093 4			0	
5154 4			0	
5171.6 3	4	51 fs 2/	K M O	E(level): weighted average of 5171.6 3 from $^{33}\text{S}(\text{p}, \gamma)$ and 5173 4 from $^{34}\text{S}(^3\text{He}, \text{t})$. J ^π : spin=4 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p}, \gamma)$. L=3, 2 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$: L=3 gives 1 ⁻ , 2 ⁻ , 3 ⁻ , 4 ⁻ , 5 ⁻ ; L=2 gives 0 ⁺ , 1 ⁺ , 2 ⁺ , 3 ⁺ , 4 ⁺ . T _{1/2} : lifetime $\tau=73$ fs 30: weighted average of 90 fs 35 in $^{33}\text{S}(\text{p}, \gamma)$ from 1985La16 with DSAM and 60 fs 30 in $^{33}\text{S}(\text{p}, \gamma)$ from 1977Da02 with DSAM.
5233 4			0	
5246 2			M	J ^π : L=3 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$ gives 1 ⁻ , 2 ⁻ , 3 ⁻ , 4 ⁻ , 5 ⁻ .
5263 4			0	
5278 7	5 ⁺ , 6 ⁺ , 7 ⁺		J	J ^π : L=6 from 0 ⁺ in $^{32}\text{S}(\alpha, \text{d})$ gives 5 ⁺ , 6 ⁺ , 7 ⁺ .
5292			H	
5314.95 20	7 ⁺	65 ps 8	EFG	E(level): weighted average of 5314.86 18 from $^{24}\text{Mg}(^{12}\text{C}, \text{pn}\gamma)$ and 5315.4 4 from $^{27}\text{Al}(^{12}\text{C}, \alpha\gamma)$. J ^π : spin=7 from $\gamma(\theta)$ in $^{31}\text{P}(\alpha, \text{n}\gamma)$ and $^{24}\text{Mg}(^{12}\text{C}, \text{pn}\gamma)$; 490.8 γ , E2, $\Delta J=2$ to 5 ⁺ , 4824.18 level; 571.8 γ , E1+M2, $\Delta J=1$ to 6 ⁻ , 4743.15 level. T _{1/2} : lifetime $\tau=94$ ps 12 in $^{27}\text{Al}(^{12}\text{C}, \alpha\gamma)$ from 1978Ba39 with RDM.
5326 4			K 0	E(level): from $^{34}\text{S}(^3\text{He}, \text{t})$.
5338 2			M	J ^π : L=1+3, 2 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$: L=1+3 gives 0 ⁻ , 1 ⁻ , 2 ⁻ , 3 ⁻ , 4 ⁻ , 5 ⁻ ; L=2 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$ gives 0 ⁺ , 1 ⁺ , 2 ⁺ , 3 ⁺ , 4 ⁺ .
5357 4			K 0	E(level): from $^{34}\text{S}(^3\text{He}, \text{t})$.
5386.8 15	(2, 3, 4, 5) ⁻		K M O	XREF: O(5388?) J ^π : L=3 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$ gives 1 ⁻ , 2 ⁻ , 3 ⁻ , 4 ⁻ , 5 ⁻ ; primary transition 1786.6 γ to 4 ⁻ , 3600.14 level gives (2 ⁻ , 3, 4, 5, 6 ⁻).
5424 4			K 0	E(level): from $^{34}\text{S}(^3\text{He}, \text{t})$.
5435 2	1 ⁺ , 2 ⁺		K M	XREF: K(?) E(level): from $^{33}\text{S}(^3\text{He}, \text{d})$. Both 2014Pa44 in $^{33}\text{S}(^3\text{He}, \text{d})$ and 2013Fa10 in $^{33}\text{S}(\text{p}, \gamma)$ assign this as a new level different from adjacent known levels. J ^π : L=0 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$ gives 1 ⁺ , 2 ⁺ .
5444 4			0	
5453 2	1 ⁺ , 2 ⁺		K M	XREF: K(5454?) J ^π : L=0 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$ gives 1 ⁺ , 2 ⁺ .
5485 4			K 0	E(level): from $^{34}\text{S}(^3\text{He}, \text{t})$.
5541.1 11	(2 ⁻ to 5 ⁻)	>0.7 ps	K 0	E(level): weighted average of 5540.8 11 from $^{33}\text{S}(\text{p}, \gamma)$ and 5545 4 from $^{34}\text{S}(^3\text{He}, \text{t})$. J ^π : 1940.9 γ to 4 ⁻ , 3600.14 level gives (2 ⁻ , 3, 4, 5, 6 ⁻); 1996.0 γ to 3 ⁻ , 3545.08 level gives (1 ⁻ , 2, 3, 4, 5 ⁻). T _{1/2} : lifetime $\tau>1000$ fs in $^{33}\text{S}(\text{p}, \gamma)$ from 1977Da02 with DSAM.
5577.2 10	(2) ⁺		K M O uv	XREF: u(5600)v(5600) E(level): weighted average of 5577.6 10 from $^{33}\text{S}(\text{p}, \gamma)$,

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Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

E(level) [†]	J ^π	XREF	Comments
5608 2	1 ⁻ , 2 ⁻ , 3 ⁻ , 4 ⁻ , 5 ⁻	M O uv	5577 2 from $^{33}\text{S}(^3\text{He}, \text{d})$, and 5572 4 from $^{34}\text{S}(^3\text{He}, \text{t})$. J ^π : L=0 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$ gives 1 ⁺ , 2 ⁺ . Primary transitions with I _γ >10: 2032.1γ to 3 ⁻ , 3545.08 level, 5430.4γ to 3 ⁺ , 146.36 level, 1977.0γ to 4 ⁻ , 3600.14 level, 2856.0γ to 2 ⁻ , 2721.09 level, 4346.6γ to 2 ⁺ , 1230.28 level. XREF: u(5600)v(5600) E(level): weighted average of 5608 2 from $^{33}\text{S}(^3\text{He}, \text{d})$ and 5606 4 from $^{34}\text{S}(^3\text{He}, \text{t})$.
5635.75 20	(1, 2 ⁺)	K M O uv	J ^π : L=3 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$. XREF: u(5600)v(5600) E(level): weighted average of 5635.75 20 from $^{33}\text{S}(\text{p}, \gamma)$ and 5635 4 from $^{34}\text{S}(^3\text{He}, \text{t})$. J ^π : L=1+3, 2 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$; L=1+3 gives 1 ⁻ , 2 ⁻ , 3 ⁻ ; L=2 gives 0 ⁺ , 1 ⁺ , 2 ⁺ , 3 ⁺ , 4 ⁺ . Primary transitions: 5635.25γ to 0 ⁺ , g.s., 4969.79γ to 1 ⁺ , 665.57 level, and 3055.3γ to 1 ⁺ , 2580.3 level. XREF: O(5671?) J ^π : L=0 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$ gives 1 ⁺ , 2 ⁺ . Primary transition 5672.5γ to 0 ⁺ , g.s. gives (1, 2 ⁺).
5673.0 10	1 ⁺ , 2 ⁺	K M O	XREF: O(5761?) J ^π : primary transitions: 5762.8γ to 0 ⁺ , g.s. and 5616.4γ to 3 ⁺ , 146.36 level.
5705 5		O	
5763.3 10	(1 ⁺ , 2 ⁺)	jK O	E(level): weighted average of 5785.6 10 from $^{33}\text{S}(\text{p}, \gamma)$, 5784 2 from $^{33}\text{S}(^3\text{He}, \text{d})$, and 5782 4 from $^{34}\text{S}(^3\text{He}, \text{t})$. J ^π : L=1 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$ gives 0 ⁻ , 1 ⁻ , 2 ⁻ , 3 ⁻ . Primary transition 5638.2γ to 3 ⁺ , 146.36 level gives (1 ⁺ , 2, 3, 4, 5 ⁺). XREF: O(5806) J ^π : L=1+3 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$ gives 1 ⁻ , 2 ⁻ , 3 ⁻ . Primary transitions: 5805.4γ to 0 ⁺ , g.s., 5344.4γ to 1 ⁺ , 461.01 level, 5659.0γ to 3 ⁺ , 146.36 level, and 3430.0γ to 4 ⁺ , 2375.67 level give (2 ⁺).
5785.1 10	(2, 3) ⁻	jK M O	E(level): weighted average of 5852.91 30 from $^{33}\text{S}(\text{p}, \gamma)$ and 5852 4 from $^{34}\text{S}(^3\text{He}, \text{t})$. J ^π : L=1+3 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$ gives 1 ⁻ , 2 ⁻ , 3 ⁻ . Primary transitions: 5706.04γ to 3 ⁺ , 146.36 level, 2307.75γ to 3 ⁻ , 3545.08 level, and 2252.69γ to 4 ⁻ , 3600.14 level.
5805.9 10	(2)	jK M O	
5852.91 30	(2, 3) ⁻	jK M O	E(level): weighted average of 5897.2 10 from $^{33}\text{S}(\text{p}, \gamma)$ and 5898 4 from $^{34}\text{S}(^3\text{He}, \text{t})$. J ^π : L=1 from 3/2 ⁺ in $^{33}\text{S}(^3\text{He}, \text{d})$ gives 0 ⁻ , 1 ⁻ , 2 ⁻ , 3 ⁻ . Primary transitions: 5435.8γ to 1 ⁺ , 461.01 level and 5750.4γ to 3 ⁺ , 146.36 level, and 5896.8γ (I _γ =5) to 0 ⁺ , g.s.
5868 4		jK O	
5897.3 10	(2)	jK M O	E(level): weighted average of 5940.8 10 from $^{33}\text{S}(\text{p}, \gamma)$ and 5941 4 from $^{34}\text{S}(^3\text{He}, \text{t})$. J ^π : spin=1:3 from γ(θ) in $^{33}\text{S}(\text{p}, \gamma)$. Primary transitions: 5479.3γ to 1 ⁺ , 461.01 level and 5793.9γ to 3 ⁺ , 146.36 level.
5917 4		j O	
5940.8 10	1 ⁺ , 2, 3 ⁺	jK O	J ^π : primary transitions: 6029.5γ to 0 ⁺ , g.s. and 5883.2γ to 3 ⁺ , 146.36 level.
5980 4		O	
6030.1 20	(1 ⁺ , 2 ⁺)	K	J ^π : spin=1, 2 from γ(θ) in $^{33}\text{S}(\text{p}, \gamma)$. Primary transition 6088.36γ to 0 ⁺ , g.s.
6088.94 6	1, 2 ⁽⁺⁾	K	XREF: u(6160)v(6160)
6136.2 11	1 ⁽⁺⁾ , 2, 3 ⁽⁺⁾	K uv	

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Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2}[#]</u>	<u>XREF</u>	<u>Comments</u>
6141.8 11	1,2 ⁽⁺⁾		K	uv J ^π : spin=1:3 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. Primary transitions: 5470.2 γ to 1 ⁺ , 665.57 level and 5989.3 γ to 3 ⁺ , 146.36 level. XREF: u(6160)v(6160)
6169.4 [‡] 11	3 ^{-‡}	25 eV 4	KL	v J ^π : spin=1,2 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. Primary transition 6141.2 γ to 0 ⁺ , g.s. XREF: v(6160)
6181.27 20	1 ⁺ ,2		K	v E(level): from $^{33}\text{S}(\text{p},\gamma)$. J ^π : spin=3 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. γ D(+Q) to 2 ⁺ , 3 ⁺ , 3 ⁻ , and 4 ⁻ levels. XREF: v(6160)
6208.2 5	4 ⁽⁻⁾	<1.4 fs	K	J ^π : spin=1,2 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. Primary transitions: 5515.22 γ to 1 ⁺ , 665.57 level and 6034.34 γ to 3 ⁺ , 146.36 level. J ^π : spin=4 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. Primary transitions: 2576.7 γ to 5 ⁻ , 3631.40 level, 2608.0 γ , to 4 ⁻ , 3600.14 level, and 2663.0 γ to 3 ⁻ , 3545.08 level. T _{1/2} : lifetime $\tau < 0.002$ ps in $^{33}\text{S}(\text{p},\gamma)$ from 1973An13 with DSAM.
6219.3 12	(3 ⁺ ,4 ⁺)		K	J ^π : primary transitions: 4038.0 γ to 3 ⁺ , 2181.09 level, 4988.6 γ to 2 ⁺ , 1230.28 level, 2572.9 γ to 5 ⁺ , 3646.3 level, and 3843.4 γ to 4 ⁺ , 2375.67 level.
6229.5 5	1		K	J ^π : spin=1 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. Primary transitions: 3508.2 γ , D(+Q), to 2 ⁻ , 2721.09 level, 4341.9 γ to 2 ⁺ , 1887.29 level, and 2455.7 γ , D(+Q), to 1 ⁻ , 3773.7 level.
6266.6 13	(1 ⁺ ,2,3 ⁺)		jK	XREF: j(6300) J ^π : primary transitions: 6119.7 γ to 3 ⁺ , 146.36 level and 5600.5 γ to 1 ⁺ , 665.57 level.
6273.3 13	(2 ⁺ ,3,4 ⁺)		jK	XREF: j(6300) J ^π : spin=3 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. Primary transitions: 3897.4 γ , D+Q, to 4 ⁺ , 2375.67 level, 4092.0 γ to 3 ⁺ , 2181.09 level, and 4385.7 γ to 2 ⁺ , 1887.29 level.
6321.7 [‡] 13	2 ^{-‡}	1.60 keV 16	H jKL	XREF: j(6300) E(level): weighted average of 6322.0 13 from $^{33}\text{S}(\text{p},\gamma)$ and 6321.2 15 from $^{33}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$: resonances.
6361.3 13	(1 ⁺ ,2,3 ⁺)		K	J ^π : spin=1,2 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. J ^π : primary transitions: 6214.3 γ to 3 ⁺ , 146.36 level and 5899.7 γ to 1 ⁺ , 461.01 level.
6370.2 [‡] 2	2 ^{-‡}	1.60 keV 16	KL	E(level): weighted average of 6370.2 2 from $^{33}\text{S}(\text{p},\gamma)$ and 6369.0 15 from $^{33}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$: resonances.
6382.4 14	(3 ⁺)		K	J ^π : spin=2 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. J ^π : primary transitions: 6235.4 γ to 3 ⁺ , 146.36 level, 2782.1 γ to 4 ⁻ , 3600.14 level, 2837.2 γ to 3 ⁻ , 3545.08 level, 4006.5 γ to 4 ⁺ , 2375.67 level, and 5716.3 γ to 1 ⁺ , 665.57 level.
6399.8 14	(3,4 ⁺)		K	J ^π : primary transitions: 6252.8 γ to 3 ⁺ , 146.36 level, 5169.1 γ to 2 ⁺ , 1230.28 level, 4023.9 γ to 4 ⁺ , 2375.67 level, and 2799.5 γ to 4 ⁻ , 3600.14 level.
6441.2 14	(1 ⁺ ,2,3 ⁺)		K	J ^π : primary transitions: 4283.0 γ to 2 ⁺ , 2157.9 level,

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Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

E(level) [†]	J ^π	T _{1/2} [#]	XREF	Comments
				6294.2γ to 3 ⁺ , 146.36 level, and 5979.6γ to 1 ⁺ , 461.01 level.
6450.54 20	3 ⁽⁻⁾ , 4		K	J ^π : spin=3,4 from γ(θ) in $^{33}\text{S}(\text{p},\gamma)$. Primary transitions: 6303.55γ to 3 ⁺ , 146.36 level, 2850.27γ to 4 ⁻ , 3600.14 level, 2905.33γ to 3 ⁻ , 3545.08 level, and 2819.02γ to 5 ⁻ , 3631.40 level.
6477.5 14	(0,1,2,3 ⁺)		K	J ^π : primary transition 6015.9γ to 1 ⁺ , 461.01 level.
6486.9 14	(1,2 ⁺)		K	J ^π : spin=1:4 from γ(θ) in $^{33}\text{S}(\text{p},\gamma)$. Primary transitions: 4328.7γ to 2 ⁺ , 2157.9 level, 6486.2γ to 0 ⁺ g.s., and 6025.3γ to 1 ⁺ , 461.01 level.
6527.6 15	(2 ⁻ ,3 ⁺)		jK	XREF: j(6540) J ^π : primary transitions: 2927.3γ to 4 ⁻ , 3600.14 level, 6380.6γ to 3 ⁺ , 146.36 level, 5861.5γ to 1 ⁺ , 665.57 level, and 4369.4γ to 2 ⁺ , 2157.9 level.
6545.9 [‡] 15	2 ^{-[‡]}	2.0 keV 2	jKL	XREF: j(6540) E(level): weighted average of 6546.6 15 from $^{33}\text{S}(\text{p},\gamma)$ and 6545.2 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
6574.7 15	(2 ⁺ ,3,4 ⁺)		jK	XREF: j(6540) J ^π : primary transitions: 4416.5γ to 2 ⁺ , 2157.9 level and 6427.7γ to 3 ⁺ , 146.36 level.
6583.7 15	(1 ⁻ ,2 ⁺)		K	J ^π : primary transitions: 6122.1γ to 1 ⁺ , 461.01 level, 3038.5γ to 3 ⁻ , 3545.08 level, 4425.5γ to 2 ⁺ , 2157.9 level, and 6583.0γ to 0 ⁺ g.s.
6605? 2			K	XREF: K(?)
6624.7 15	(3 ⁻)		K	J ^π : primary transitions: 3024.4γ to 4 ⁻ , 3600.14 level, 6163.1γ to 1 ⁺ , 461.01 level, 6477.7γ to 3 ⁺ , 146.36 level, 4466.5γ to 2 ⁺ , 2157.9 level, 3079.5γ to 3 ⁻ , 3545.08 level, and 2993.2γ to 5 ⁻ , 3631.40 level.
6632.7 [‡] 20	[‡]		L	
6640.92 [‡] 7	4 ^{-[‡]}	40 eV 6	KL	J ^π : spin=4 from γ(θ) in $^{33}\text{S}(\text{p},\gamma)$.
6695.1 16	(3 ⁺ ,4,5 ⁺)		K	J ^π : primary transitions: 4083.8γ to 3 ⁺ , 2611.04 level and 3048.7γ to 5 ⁺ , 3646.3 level.
6702.8 16	(3,4)		h K	XREF: h(6724) J ^π : primary transitions: 6555.8γ to 3 ⁺ , 146.36 level and 4326.8γ to 4 ⁺ , 2375.67 level. Weak 3102.5γ to 4 ⁻ , 3600.14 level and weak 3157.6γ to 3 ⁻ , 3545.08 level.
6719.8 [‡] 15	3 ^{-[‡]}	3.0 keV 3	h KL	XREF: h(6724) E(level): weighted average of 6720.2 16 from $^{33}\text{S}(\text{p},\gamma)$ and 6719.5 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
6723.7 16	(3 ⁺ ,4 ⁺)		h K	XREF: h(6724) J ^π : primary transitions: 1899.5γ to 5 ⁺ , 4824.18 level, 4347.7γ to 4 ⁺ , 2375.67 level, 4112.4γ to 3 ⁺ , 2611.04 level, and 4565.5γ to 2 ⁺ , 2157.9 level.
6737.3 [‡] 15	2 ^{-[‡]}	2.0 keV 2	h KL	XREF: h(6724) E(level): weighted average of 6737.0 16 from $^{33}\text{S}(\text{p},\gamma)$ and 6737.5 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
6748.9 [‡] 15	2 ^{-[‡]}	0.80 keV 8	L	
6778 [‡] 2	[‡]		L	
6790.8 16	(1,2 ⁺)		K	J ^π : primary transition 6790.1γ to 0 ⁺ g.s.

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Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

E(level) [†]	J ^π	T _{1/2} [#]	XREF	Comments
6797.4 16	(1 ⁺ to 5 ⁺)		K	J ^π : spin=1;5 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. Primary transition 6650.3 γ to 3 ⁺ , 146.36 level.
6807.1 [‡] 15	1 ^{-‡}	1.0 keV 1	KL	E(level): weighted average of 6807.2 16 from $^{33}\text{S}(\text{p},\gamma)$ and 6807.0 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
6828.6 14	(2,3 ⁺)		K	J ^π : primary transitions: 4670.4 γ to 2 ⁺ , 2157.9 level, 6681.5 γ to 3 ⁺ , 146.36 level, 6367.0 γ to 1 ⁺ , 461.01 level, and 3283.4 γ to 3 ⁻ , 3545.08 level.
6842.8 [‡] 14	2 ^{-‡}	2.0 keV 2	KL	E(level): weighted average of 6842.8 14 from $^{33}\text{S}(\text{p},\gamma)$ and 6842.8 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
6852.4 14	2 ⁺	0.80 keV 8	KL	E(level): weighted average of 6853.0 14 from $^{33}\text{S}(\text{p},\gamma)$ and 6851.8 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
6871.18 [‡] 30	5 ^{-‡}	26 eV 7	KL	J ^π : spin=5 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$.
6887.97 30	2		K	J ^π : spin=2 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$.
6890.78 [‡] 30	1 ^{-‡}	0.85 keV 9	L	
6901.75 30	(1 ⁺ ,2 ⁺)		K	J ^π : primary transitions: 4743.5 γ to 2 ⁺ , 2157.9 level and 6901.00 γ to 0 ⁺ g.s.
6918.0 17	3,4		jK	XREF: j(6950) J ^π : spin=3,4 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$.
6931.5 [‡] 17	1 ^{-‡}	5.0 keV 5	jKL	XREF: j(6950) E(level): weighted average of 6931.5 17 from $^{33}\text{S}(\text{p},\gamma)$ and 6934.84 30 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
6977.4 17	(2 ⁻ ,3,4 ⁺)		jK	XREF: j(6950) J ^π : primary transitions: 4819.1 γ to 2 ⁺ , 2157.9 level and 6830.3 γ to 3 ⁺ , 146.36 level. No γ to 0 ⁺ g.s. suggests 1,2 ⁺ less likely.
6984.9 10			jK	XREF: j(6950)
6986.85 30	1 ⁻	0.60 keV 6	jKL	XREF: j(6950) E(level): weighted average of 6986.9 20 from $^{33}\text{S}(\text{p},\gamma)$ and 6986.85 30 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
6991.41 30	1 ⁺	0.220 keV 22	KL	E(level): weighted average of 6992.3 9 from $^{33}\text{S}(\text{p},\gamma)$ and 6991.31 30 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7018.9 10			K	
7037.79 [‡] 30	2 ^{-‡}	8.0 keV 8	L	XREF: u(7070)v(7070)
7051.6 18	(1 ⁻ ,2,3 ⁺)		K	XREF: u(7070)v(7070) J ^π : primary transitions: 6589.9 γ to 1 ⁺ , 461.01 level and 3506.3 γ to 3 ⁻ , 3545.08 level.
7059.01 30	1,2		KL	XREF: u(7070)v(7070) E(level): weighted average of 7059.04 30 from $^{33}\text{S}(\text{p},\gamma)$ and 7057.7 20 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7061.6 10			K	J ^π : spin=1,2 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$.
7078.90 9	3 ⁻	0.89 keV 11	KL	XREF: u(7070)v(7070) E(level): weighted average of 7078.89 9 from

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Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

E(level) [†]	J ^π	T _{1/2} [#]	XREF	Comments
				$^{33}\text{S}(\text{p},\gamma)$ and 7080.4 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances. J ^π : spin=3 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$; 2 ⁻ ,3 ⁻ from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances. T _{1/2} : unweighted average of 0.787 keV 30 (1977Da03) and 1.00 keV 10 (1989Va15) in $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7111.6 [‡] 15	2 ^{-‡}	1.20 keV 12	L	
7115.9 10			K	
7123.2 [‡] 10	2 ^{-‡}	4.0 keV 4	KL	E(level): weighted average of 7122.7 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7124.3 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7126.6 10			K	
7130.5 [‡] 15	2 ^{-‡}	15.0 keV 30	L	
7145.0 10			K	
7165.4 10			K	
7175.1 10			K	
7183.8 10			K	
7196.2 [‡] 15	2 ^{-‡}	1.60 keV 16	L	
7203.2 10			K	
7213.7 [‡] 15	1 ^{-‡}	2.0 keV 2	L	
7218.8 10			K	
7239.1 [‡] 14	1 ^{+‡}	0.55 keV 6	KL	E(level): weighted average of 7238.2 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7241.2 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7249.8 10		0.14 ps 5	G K	T _{1/2} : lifetime $\tau=200$ fs 70 in $^{31}\text{P}(\alpha,\text{n}\gamma)$ from 1982Va02 with DSAM.
7250.0 5	(9) ⁺	0.159 ps 28	EF	E(level): weighted average of 7249.9 5 from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$, 7250.5 11 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$. J ^π : 1935.0 γ , E2, $\Delta J=2$ to 7 ⁺ , 5314.95 level in $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$. T _{1/2} : lifetime $\tau=0.23$ ps 4 in $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$ from 2014Bi03 with DSAM.
7271.2 10			K	
7284.7 10			K	
7295.4 10			K	
7317.1 [‡] 15	2 ^{-‡}	3.5 keV 4	L	
7321.6 10			KL	E(level): weighted average of 7321.6 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7321.6 20 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7336 2			j L	XREF: j(7350)
7346.8 10			jK	XREF: j(7350) E(level): from $^{33}\text{S}(\text{p},\gamma)$. J ^π : L=3,6 from 0 ⁺ in $^{32}\text{S}(\alpha,\text{d})$ gives 2 ⁻ ,3 ⁻ ,4 ⁻ ,5 ⁺ ,6 ⁺ ,7 ⁺ for 7350 20.
7358.5 [‡] 15	2 ^{-‡}	20 keV 4	j L	XREF: j(7350)
7368.4 [‡] 15	1 ⁻ ,2 ⁻ ,3 ^{-‡}		j L	XREF: j(7350)
7376.0 10			K	
7382.7 10			K	
7386.6 [‡] 15	2 ^{-‡}	0.70 keV 7	L	
7406.4 10			KL	E(level): weighted average of 7406.0 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7408.0 20 from

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Adopted Levels, Gammas (continued)

^{34}Cl Levels (continued)				
E(level) [†]	J ^π	T _{1/2} [#]	XREF	Comments
				$^{33}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
7412.0 [‡] 15	1 ⁻ ,2 ⁻ [‡]		L	
7424.5 10			K	
7441.0 10			K	
7450.1 10			KL	E(level): weighted average of 7449.7 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7451.6 20 from $^{33}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
7457.6 [‡] 15	1 ⁻ [‡]	4.0 keV 4	L	
7462.3 10			K	
7470.1 10			K	
7475.3 [‡] 15	2 ⁺ [‡]	0.170 keV 17	L	
7486.6 10			K	
7491.3 [‡] 15	2 ⁻ [‡]	0.10 keV 1	L	
7495.6 [‡] 11	2 ⁻ [‡]	1.40 keV 14	KL	E(level): weighted average of 7496.3 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7493.9 15 from $^{33}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
7506.0 10			K	
7511.3 [‡] 10	1 ⁺ [‡]	0.30 keV 3	KL	E(level): weighted average of 7511.8 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7510.2 15 from $^{33}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
7517.6 10			K	
7525.7 [‡] 15	1 ⁺ ,2 ⁺ [‡]		L	
7527.2 [‡] 10	1 ⁺ [‡]	3.0 keV 3	KL	E(level): weighted average of 7527.3 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7527.1 15 from $^{33}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
7531.2 [‡] 15	3 ⁻ [‡]	40 keV 8	L	
7533.9 [‡] 15	1 ⁻ ,2 ⁻ ,3 ⁻ ,4 ⁻ ,5 ⁻ [‡]		L	
7536.4 [‡] 15	2 ⁻ [‡]	3.6 keV 4	L	
7539.9 10			K	
7544.5 11	1 ⁺	0.150 keV 15	KL	E(level): weighted average of 7543.8 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7546.1 15 from $^{33}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
7553.3 [‡] 15	2 ⁺ [‡]	2.50 keV 25	L	
7562.3 10			K	
7577.8 10			K	
7582.6 10			K	
7585.8 [‡] 15	1 ⁺ [‡]	1.40 keV 14	L	
7588.5 10			K	
7595.4 [‡] 15	2 ⁺ [‡]	0.70 keV 7	L	
7609.8 10			K	
7613.7 [‡] 15	2 ⁺ [‡]	0.30 keV 3	L	
7618.1 [‡] 15	(1) ⁻ [‡]	0.50 keV 5	L	
7624.4 10			K	
7630.2 10			K	
7643.8 10			K	
7649.3 [‡] 10	1 ⁻ [‡]	32 keV 7	KL	E(level): weighted average of 7649.6 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7648.6 15 from $^{33}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
7652.1 [‡] 15	2 ⁻ [‡]	0.70 keV 7	L	

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Adopted Levels, Gammas (continued)

^{34}Cl Levels (continued)				
E(level) [†]	J ^π	T _{1/2} [#]	XREF	Comments
7666.1 10			K	
7673.6 [‡] 15	1 ⁻ [‡]	2.0 keV 2	L	
7675.3 10	2 ⁺ , 3 ⁺		KL	E(level): weighted average of 7675.8 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7674.3 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances. J ^π : spin=2,3 from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$; 1 ⁺ , 2 ⁺ , 3 ⁺ , 4 ⁺ from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7677.9 [‡] 15	3 ⁻ [‡]	2.0 keV 2	L	
7684.7 [‡] 15	2 ⁺ , 3 ⁺ [‡]		L	
7690.4 10	1 ⁺	0.30 keV 3	KL	E(level): weighted average of 7690.3 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7690.5 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7696.8 [‡] 15	1 ⁻ [‡]	1.30 keV 13	L	
7699.5 11	(7) ⁺	<0.34 ps	F	J ^π : 2384γ, M1, ΔJ=0 to 7 ⁺ , 5314.95 level. T _{1/2} : lifetime τ<0.49 ps in $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$ from 2014Bi03 with DSAM.
7704.6 [‡] 15	1 ⁻ , 2 ⁻ , 3 ⁻ , 4 ⁻ , 5 ⁻ [‡]		L	
7708.8 10			K	
7720.1 [‡] 10	1 ⁻ [‡]	6.0 keV 6	KL	E(level): weighted average of 7720.4 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7719.3 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7730.1 10			K	
7753.4 10			K	
7761.6 [‡] 15	0 ⁺ , 1 ⁺ , 2 ⁺ [‡]		L	
7768.5 [‡] 15	1 ⁻ , 2 ⁻ , 3 ⁻ [‡]		L	
7784.5 10			K	
7787.6 [‡] 15	1 ⁺ , 2 ⁺ [‡]		L	
7799.9 [‡] 15	3 ⁺ , 4 ⁺ [‡]		L	
7801.5 10	7 ⁺ , 8 ⁺ , 9 ⁺	<83.0 fs	EF	E(level): weighted average of 7802.4 10 from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$ and 7800.5 11 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$. J ^π : 2486.2γ, M1+E2, to 7 ⁺ , 5314.95 level. T _{1/2} : lifetime τ<0.12 ps in $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$ from 2014Bi03 with DSAM.
7803.9 10		69 fs 49	G K	XREF: G(7800) T _{1/2} : lifetime τ=100 fs 70 in $^{31}\text{P}(\alpha,\text{n}\gamma)$ from 1982Va02 with DSAM.
7809.7 10			K	
7825.3 [‡] 15	3 ⁺ [‡]	0.040 keV 4	L	
7827.3 [‡] 15	2 ⁻ [‡]	10.0 keV 20	j L	XREF: j(7840)
7831.0 10			jK	XREF: j(7840)
7848.8 [‡] 10	3 ⁺ , 4 ⁺ [‡]		KL	E(level): weighted average of 7848.5 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7849.6 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7853.4 10			jK	XREF: j(7840)
7859.2 29			j L	XREF: j(7840)
7878.6 10			K	
7886.1 10			KL	E(level): weighted average of 7886.4 10 from

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Adopted Levels, Gammas (continued)

^{34}Cl Levels (continued)				
E(level) [†]	J ^π	T _{1/2} [#]	XREF	Comments
7898.8 10			KL	$^{33}\text{S}(\text{p},\gamma)$ and 7883.4 29 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances. E(level): weighted average of 7899.0 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7897.0 29 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7904.8 10			K	
7909.5 15			L	
7909.6 [‡] 10	2 ^{-‡}	8.0 keV 8	KL	E(level): weighted average of 7909.6 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7909.5 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7929.1 [‡] 15	2 ^{+‡}	1.40 keV 14	L	
7934.7 [‡] 10	3 ^{+‡}	0.50 keV 5	KL	E(level): weighted average of 7934.9 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7934.2 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7943.1 [‡] 10	1 ⁺ ,2 ⁺ ,3 ⁺ ,4 ^{+‡}		KL	E(level): weighted average of 7942.6 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7944.2 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7953.3 [‡] 29	[‡]		L	
7967.2 [‡] 10	1 ^{+‡}	0.20 keV 2	KL	E(level): weighted average of 7966.9 10 from $^{33}\text{S}(\text{p},\gamma)$ and 7967.8 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
7975.1 [‡] 15	2 ^{+‡}	3.0 keV 3	L	
7977.5 [‡] 15	1 ^{+‡}	2.70 keV 27	kL	
7980.3 [‡] 15	3 ^{+‡}	0.30 keV 3	kL	
7985.3 [‡] 15	2 ^{-‡}	19 keV 4	L	
7990.2 10			K	
7997.0 10			K	
8002.6 [‡] 15	1 ⁺ ,2 ^{+‡}		L	
8012.5 [‡] 15	1 ⁺ ,2 ⁺ ,3 ⁺ ,4 ^{+‡}		L	
8020.8 [‡] 15	2 ^{-‡}	0.60 keV 6	L	
8029.9 15	1 ⁻	8.0 keV 8	KL	E(level): weighted average of 8030.9 10 from $^{33}\text{S}(\text{p},\gamma)$ and 8027.6 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
8035.8 10			K	
8043.2 [‡] 15	2 ^{-‡}	4.5 keV 5	L	
8048.3 [‡] 10	2 ^{+‡}	0.20 keV 2	KL	E(level): weighted average of 8048.4 10 from $^{33}\text{S}(\text{p},\gamma)$ and 8048.1 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
8052.5 [‡] 15	1 ^{-‡}	0.30 keV 3	L	
8059.3 [‡] 15	3 ^{+‡}	1.20 keV 12	L	J ^π : other: (2) ⁺ , (4) ⁺ also possible $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances (1989Va15).
8070.5 [‡] 10	1 ⁻ ,2 ⁻ ,3 ⁻ ,4 ⁻ ,5 ^{-‡}		KL	E(level): weighted average of 8070.7 10 from $^{33}\text{S}(\text{p},\gamma)$ and 8069.9 15 from $^{33}\text{S}(\text{p},\text{p}),(\text{p},\text{p}'\gamma)$:resonances.
8084.6 [‡] 15	2 ^{+‡}	0.70 keV 7	L	
8087.5 [‡] 15	1 ^{-‡}	0.50 keV 5	L	
8106.4 [‡] 15	(2) ^{-‡}	5.0 keV 5	L	

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Adopted Levels, Gammas (continued)

^{34}Cl Levels (continued)				
E(level) [†]	J ^π	T _{1/2} [#]	XREF	Comments
8118.7 [±] ₁₅	2 ⁺ _±	1.10 keV 11	L	
8127.4 [±] ₁₅	0 ⁺ _±	1.0 keV 1	L	
8132.8 10			K	
8141.6 10			K	
8149.2 [±] ₁₅	2 ⁻ _±	26 keV 5	L	
8154.6 [±] ₁₀	1 ⁻ _±	1.20 keV 12	KL	E(level): weighted average of 8154.2 10 from ³³ S(p,γ) and 8155.6 15 from ³³ S(p,p),(p,p'γ):resonances.
8155.5 11	(6) ⁺	<56 fs	F	J ^π : 2840γ, M1+E2, to 7 ⁺ , 5314.95 level. T _{1/2} : lifetime τ<0.08 ps in ²⁷ Al(¹² C,αnγ) from 2014Bi03 with DSAM.
8159.0 [±] ₁₅	1 ⁺ _±	0.90 keV 9	L	
8164.6 [±] ₁₅	1 ⁻ , 2 ⁻ _±		L	
8186.38 [±] ₃₀	3 ⁺ , 4 ⁺ _±		L	
8192.5 [±] ₅	2 ⁺ _±	0.50 keV 5	KL	E(level): weighted average of 8191.0 10 from ³³ S(p,γ) and 8192.59 30 from ³³ S(p,p),(p,p'γ):resonances.
8195.9 10			K	
8203.75 [±] ₃₀	1 ⁻ _±	15.0 keV 30	L	
8210.54 [±] ₃₀	(3) ⁺ _±	0.20 keV 2	L	
8214.86 [±] ₃₀	3 ⁻ _±	0.30 keV 3	KL	E(level): weighted average of 8214.3 10 from ³³ S(p,γ) and 8214.91 30 from ³³ S(p,p),(p,p'γ):resonances.
8231.99 [±] ₃₀	2 ⁻ _±	10.0 keV 20	L	
8236.30 [±] ₃₀	2 ⁺ _±	0.95 keV 10	KL	E(level): weighted average of 8235.7 10 from ³³ S(p,γ) and 8236.35 30 from ³³ S(p,p),(p,p'γ):resonances.
8248.3 10			K	
8251.39 [±] ₃₀	(2 ⁺) _±	0.10 keV 1	L	
8254.30 [±] ₃₀	2 ⁻ , 3 ⁻ _±	0.70 keV 7	L	
8260.12 [±] ₃₄	4 ⁺ _±	0.80 keV 8	KL	E(level): weighted average of 8259.0 10 from ³³ S(p,γ) and 8260.22 30 from ³³ S(p,p),(p,p'γ):resonances.
8265.56 [±] ₃₀	2 ⁺ , 3 ⁺ _±		L	
8271.58 [±] ₃₀	0 ⁻ _±	0.30 keV 3	L	XREF: L(?)
8276.72 [±] ₃₀	2 ⁻ _±	2.20 keV 22	L	
8280.3 10			K	
8305.5 10	2, 3		K	J ^π : spin=2,3 from γ(θ) in ³³ S(p,γ).
8316.60 [±] ₃₀	4 ⁺ _±	2.0 keV 2	L	
8317.7 [±] ₄	1 ⁺ , 2 ⁺ _±		KL	E(level): weighted average of 8319.1 10 from ³³ S(p,γ) and 8317.57 30 from ³³ S(p,p),(p,p'γ):resonances.
8322.91 [±] ₃₀	1 ⁻ , 2 ⁻ , 3 ⁻ , 4 ⁻ , 5 ⁻ _±		L	
8330.19 [±] ₃₀	1 ⁻ , 2 ⁻ , 3 ⁻ , 4 ⁻ , 5 ⁻ _±		L	
8337.6 10			K	
8358.42 [±] ₃₀	2 ⁺ _±	1.50 keV 15	L	
8361.8 10			K	

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Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

E(level) [†]	J ^π	T _{1/2} [#]	XREF	Comments
8372.5 10			jK	XREF: j(8390)
8376.18 [‡] 30	2+ [‡]	0.80 keV 8	j L	XREF: j(8390)
8381.3 [‡] 15	(2)- [‡]	0.30 keV 3	j L	XREF: j(8390)
				J ^π : other: L=3,5,6 from 0 ⁺ in $^{32}\text{S}(\alpha, d)$ gives 2 ⁻ , 3 ⁻ , 4 ⁻ , 5, 6, 7 ⁺ for 8390 20.
8386.5 [‡] 15	3 ⁺ , 4+ [‡]		j L	XREF: j(8390)
8390.0 10			jK	XREF: j(8390)
8460.4 [‡] 15	1+ [‡]	0.80 keV 8	L	
8468.2 [‡] 15	2- [‡]	2.80 keV 28	L	
8477.2 [‡] 15	(2)+ [‡]	1.90 keV 19	L	
8485.9 [‡] 15	1- [‡]	1.40 keV 14	L	
8494.2 [‡] 15	2+ [‡]	1.50 keV 15	L	
8494.7 [‡] 15	2- [‡]	0.45 keV 5	L	
8509.3 [‡] 15	2+ [‡]	25 keV 5	L	
8537.6 [‡] 15	3- [‡]	2.0 keV 2	L	
8543.4 [‡] 15	3- [‡]	2.0 keV 2	L	
8548.1 [‡] 15	1+ [‡]	25 keV 5	L	
8549.2 [‡] 15	2- [‡]	3.0 keV 3	L	
8563.4 [‡] 15	4+ [‡]	1.50 keV 15	L	
8579.4 [‡] 15	1+ [‡]	3.10 keV 31	L	
8587.1 [‡] 15	4- [‡]	0.40 keV 4	L	
8597.9 [‡] 15	1+ [‡]	3.0 keV 3	L	
8601.8 [‡] 15	2+ [‡]	3.0 keV 3	L	
8603.2 [‡] 15	3- [‡]	0.20 keV 2	L	
8610.5 [‡] 15	3- [‡]	6.0 keV 6	L	
8613.4 [‡] 15	4-, 5- [‡]		L	
8643.6 [‡] 15	1- [‡]	0.60 keV 6	L	
8647.5 [‡] 15	2+ [‡]	10.0 keV 20	L	
8651.9 [‡] 15	(1)+ [‡]	1.0 keV 1	L	
8655.7 [‡] 15	(1)+ [‡]	1.80 keV 18	L	
8665.1 [‡] 15	4-, 5- [‡]		L	
8693.3 [‡] 15	1+ [‡]	0.50 keV 5	L	
8697.2 [‡] 15	2+ [‡]	1.50 keV 15	L	
8700.1 [‡] 15	1- [‡]	1.50 keV 15	L	
8706.4 [‡] 15	1+ [‡]	0.80 keV 8	L	
8714.6 [‡] 15	3+ [‡]	4.0 keV 4	L	
8733.2 [‡] 15	3- [‡]	35 keV 7	L	
8754.6 [‡] 15	1- [‡]	0.70 keV 7	L	
8756.6 [‡] 15	1- [‡]	4.0 keV 4	L	
8761.4 [‡] 15	3+ [‡]	1.60 keV 16	L	
8779.4 [‡] 15	3+ [‡]	0.80 keV 8	L	
8790.6 [‡] 15	2- [‡]	6.0 keV 6	L	
8794.0 [‡] 15	1- [‡]	0.20 keV 2	L	
9392.7 11	(7)+	<6.9 fs	F	J ^π : 4077γ, M1(+E2), ΔJ=0 to 7 ⁺ , 5314.95 level in

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{34}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2}[#]</u>	<u>XREF</u>	<u>Comments</u>
10631.7 15	(11) ⁺	<28 fs	F	²⁷ Al(¹² C,αnγ). T _{1/2} : lifetime τ<0.01 ps in ²⁷ Al(¹² C,αnγ) from 2014Bi03 with DSAM. J ^π : 3381γ, E2, ΔJ=2 to (9) ⁺ , 7250.0 level in ²⁷ Al(¹² C,αnγ). T _{1/2} : lifetime τ<0.04 ps in ²⁷ Al(¹² C,αnγ) from 2014Bi03 with DSAM.

[†] From ³³S(p,γ), unless otherwise noted.

[‡] From ³³S(p,p),(p,p'γ):resonances, unless otherwise noted. J^π is primarily from multilevel, multichannel R-matrix analysis in ³³S(p,p),(p,p'γ):resonances ([1989Va15](#)).

[#] Widths are from multilevel, multichannel R-matrix analysis in ³³S(p,p),(p,p'γ):resonances ([1989Va15](#)), unless otherwise noted. Half-lives are from sources given in each comment.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$								Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. @	$\delta^@$	
146.36	3 ⁺	146.36 3	100	0.0	0 ⁺	[M3]		B(M3)(W.u.)=0.08639 +79-81 E _γ : weighted average of 146.36 3 from ³⁴ Cl IT decay and 146.36 3 from ³² S(³ He,pγ).
461.01	1 ⁺	461.00 4	100.0 26	0.0	0 ⁺	M1		B(M1)(W.u.)=0.0441 +48-40 E _γ : weighted average of 460.8 10 from ³⁴ Ar ε+β ⁺ decay, 460.9 3 from ²⁷ Al(¹² C,αnγ), and 461.00 4 from ³² S(³ He,pγ). I _γ : from ³⁴ Ar ε+β ⁺ decay; others: 100 from ³² S(³ He,pγ) and ³³ S(p,γ). Mult.: M1, ΔJ=1, from γγ(θ)(DCO) and γγ(θ)(POL) in ²⁷ Al(¹² C,αnγ). B(E2)(W.u.)=3.59 +59-55
665.57	1 ⁺	519.21	1.46 19	146.36	3 ⁺	[E2]		B(E2)(W.u.)=3.59 +59-55 I _γ : others: 2.9 10 from ³⁴ Ar ε+β ⁺ decay, 8.3 48 from ³² S(³ He,pγ). B(M1)(W.u.)=0.00800 +65-57 E _γ : weighted average of 665.8 10 from ³⁴ Ar ε+β ⁺ decay, 666 1 from ²⁷ Al(¹² C,αnγ), and 665.57 5 from ³² S(³ He,pγ). I _γ : from ³² S(³ He,pγ). Others: 100 from ³⁴ Ar ε+β ⁺ decay, 100 from ³³ S(p,γ), and 100 from ²⁷ Al(¹² C,αnγ). Mult.: M1, ΔJ=1, from γγ(θ)(DCO) and γγ(θ)(POL) in ²⁷ Al(¹² C,αnγ). B(M1)(W.u.)=0.00230 +40-42; B(E2)(W.u.)=7.1 +46-42
		665.57 5	100 6	0.0	0 ⁺	M1		E _γ : weighted average of 564 1 from ²⁷ Al(¹² C,αnγ) and 564.67 6 from ³² S(³ He,pγ). I _γ : unweighted average of 74 5 from ³² S(³ He,pγ), 98.6 14 from ³³ S(p,γ). Mult.,δ: D+Q from pγ(θ) in ³² S(³ He,pγ). M2 ruled out by RUL. B(M1)(W.u.)=0.00125 +11-9; B(E2)(W.u.)=0.013 +29-12 E _γ : weighted average of 769 1 from ²⁷ Al(¹² C,αnγ) and 769.25 7 from ³² S(³ He,pγ). I _γ : others: 100 5 from ²⁷ Al(¹² C,αnγ), 100 8 from ³² S(³ He,pγ). Mult.: D+Q, ΔJ=1, from γγ(θ)(DCO) in ²⁷ Al(¹² C,αnγ) and from pγ(θ) in ³² S(³ He,pγ). Δπ=no from level scheme. δ: from ²⁷ Al(¹² C,αnγ); δ=+3.3 +18-12 from ³² S(³ He,pγ). B(M1)(W.u.)=1.6×10 ⁻⁴ +16-12; B(E2)(W.u.)=0.76 +38-52
1230.28	2 ⁺	564.67 6	91 9	665.57	1 ⁺	M1+E2	+0.5 2	E _γ : weighted average of 1084 1 from ²⁷ Al(¹² C,αnγ) and 1083.9 3 from ³² S(³ He,pγ). I _γ : weighted average of 84 5 from ³² S(³ He,pγ) and 87.1 14 from ³³ S(p,γ). Mult.,δ: D+Q and δ from pγ(θ) in ³² S(³ He,pγ); E1+M2 ruled out by RUL. B(E2)(W.u.)=0.042 20 I _γ : from ³² S(³ He,pγ). Other: <1.4 from ³³ S(p,γ). B(M1)(W.u.)=0.0057 +33-16; B(E2)(W.u.)=0.13 +19-10
		769.25 7	100.0 14	461.01	1 ⁺	(M1+E2)	+0.04 3	E _γ : weighted average of 1426 1 from ²⁷ Al(¹² C,αnγ) and 1426.7 7 from
		1083.9 3	86.9 14	146.36	3 ⁺	M1+E2	+1.2 +23-7	
		1230.26	5.3 26	0.0	0 ⁺	[E2]		
1887.29	2 ⁺	1426.5 7	100.0 12	461.01	1 ⁺	(M1+E2)	-0.11 5	

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)							
$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. @	δ @
							³² S(³ He,p γ). I_γ : others: 100 5 from ³² S(³ He,p γ), 100 6 from ²⁷ Al(¹² C, α n γ). Mult.: D+Q from $\gamma\gamma(\theta)$ (DCO) in ²⁷ Al(¹² C, α n γ) and from p $\gamma(\theta)$ in ³² S(³ He,p γ). $\Delta\pi$ =no from level scheme. δ : $\delta=-2.1^{8-4}$ from ³² S(³ He,p γ). B(M1)(W.u.)=0.0014 +12-10; B(E2)(W.u.)=1.2 +15-11 E_γ : weighted average of 1741 1 from ²⁷ Al(¹² C, α n γ) and 1740.2 16 from ³² S(³ He,p γ).
1887.29	2 ⁺	1740.8 10	72.8 16	146.36	3 ⁺	M1+E2	-0.8 +6-15
2157.9	2 ⁺	928 1	8.61 31	1230.28	2 ⁺	(M1+(E2))	+0.1 +6-3
		1697.5 4	100.0 5	461.01	1 ⁺	M1+E2	-0.09 2
		2012 1	15 5	146.36	3 ⁺	M1+E2	+0.22 10
		2158 1	23.5 5	0.0	0 ⁺	[E2]	
2181.09	3 ⁺	1515 1	25.2 16	665.57	1 ⁺	E2	
		1720 1	80 7	461.01	1 ⁺	E2	
		2035.1 7	100.0 25	146.36	3 ⁺	M1+E2	-0.4 1
							I_γ : weighted average of 66 5 from ³² S(³ He,p γ) and 73.2 12 from ³³ S(p, γ). Mult., δ : D+Q from p $\gamma(\theta)$ in ³² S(³ He,p γ). M2 ruled out by RUL. B(M1)(W.u.)=0.048 +11-21; B(E2)(W.u.)<88 E_γ : from ³² S(³ He,p γ). I_γ : weighted average of 10.2 29 from ³² S(³ He,p γ) and 8.59 31 from ³³ S(p, γ). Mult., δ : D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =no from level scheme. B(M1)(W.u.)=0.091 +11-9; B(E2)(W.u.)=1.01 +53-40 E_γ : weighted average of 1697 1 from ²⁷ Al(¹² C, α n γ) and 1697.6 4 from ³² S(³ He,p γ). I_γ : from ³² S(³ He,p γ). Others: 100 5 from ²⁷ Al(¹² C, α n γ) and 100.0 6 from ³³ S(p, γ). Mult.: D+Q from $\gamma\gamma(\theta)$ (DCO) in ²⁷ Al(¹² C, α n γ) and $\gamma(\theta)$ in ³³ S(p, γ). M2 ruled out by RUL. δ : weighted average of -0.10 2 from ²⁷ Al(¹² C, α n γ) and -0.05 3 from ³³ S(p, γ). B(M1)(W.u.)=0.0079 +24-26; B(E2)(W.u.)=0.37 +39-27 E_γ : from ³² S(³ He,p γ). I_γ : unweighted average of 10.3 29 from ³² S(³ He,p γ) and 20.2 5 from ³³ S(p, γ). Mult., δ : D+Q from $\gamma(\theta)$ in ³³ S(p, γ). M2 ruled out by RUL. B(E2)(W.u.)=8.9 +11-9 I_γ : weighted average of 24.9 29 from ³² S(³ He,p γ) and 23.5 5 from ³³ S(p, γ). B(E2)(W.u.)=3.8 +10-7 I_γ : weighted average of 39 12 from ³² S(³ He,p γ) and 25.0 14 from ³³ S(p, γ). Mult.: Q, $\Delta J=2$, from $\gamma\gamma(\theta)$ (DCO) in ²⁷ Al(¹² C, α n γ). M2 ruled out by RUL. B(E2)(W.u.)=6.4 +17-11 I_γ : weighted average of 89 9 from ³² S(³ He,p γ) and 74 7 from ³³ S(p, γ). Mult.: E2, $\Delta J=2$, from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, α n γ). B(M1)(W.u.)=0.00314 +82-57; B(E2)(W.u.)=0.48 +27-20 E_γ : weighted average of 2035 1 from ²⁷ Al(¹² C, α n γ) and 2035.1 7 from ³² S(³ He,p γ).

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J_i^{π}</u>	<u>E_{γ}^{$\dagger\dagger$}</u>	<u>I_{γ}[#]</u>	<u>E_f</u>	<u>J_f^{π}</u>	<u>Mult.[@]</u>	<u>δ[@]</u>	<u>Comments</u>
								I _{γ} : other: 100 9 from ³² S(³ He,p γ). Mult., δ : D+Q from $\gamma\gamma(\theta)$ (DCO) in ²⁷ Al(¹² C, $\alpha\gamma$) and p $\gamma(\theta)$ in ³² S(³ He,p γ). M2 ruled out by RUL. B(E2)(W.u.)=2.44 +98-71
2375.67	4 ⁺	1145.37	0.84 20	1230.28	2 ⁺	[E2]		I _{γ} : other: 11.1 78 from ³² S(³ He,p γ). B(M1)(W.u.)=0.0113 +35-23; B(E2)(W.u.)=1.43 +83-61
		2230.0 4	100.00 20	146.36	3 ⁺	M1+E2	+0.4 1	E _{γ} : weighted average of 2229 1 from ²⁷ Al(¹² C, $\alpha\gamma$) and 2230.1 4 from ³² S(³ He,p γ). I _{γ} : other: 100 8 from ³² S(³ He,p γ). Mult., δ : magnetic character dominant from $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, $\alpha\gamma$). Others: D+Q with δ =+5.7 25 from ³² S(³ He,p γ). D+Q with δ =+6.3 +32-20 from ³³ S(p, γ). B(M1)(W.u.)=0.086 +89-34
2580.3	1 ⁺	2579.4 14	100.0 24	0.0	0 ⁺	(M1)		I _{γ} : ffrom ³⁴ Ar $\varepsilon+\beta^+$ decay (0.84646 s). Other: 100 from ³³ S(p, γ). Mult., δ : D with δ =0 from p $\gamma(\theta)$ in ³² S(³ He,p γ). $\Delta\pi$ =no from level scheme. B(M1)(W.u.)=0.31 +13-8; B(E2)(W.u.)=7 $\times$ 10 ¹ +11-6
2611.04	3 ⁺	453 1	61 5	2157.9	2 ⁺	M1+E2	+0.11 6	Mult., δ : D+Q from $\gamma\gamma(\theta)$ (DCO) in ²⁷ Al(¹² C, $\alpha\gamma$). M2 ruled out by RUL. B(E2)(W.u.)=72 +197-61 upper bound exceeds RUL=100. B(M1)(W.u.)=0.0123 +49-29 if M1, B(E2)(W.u.)=25 +10-6 if E2. B(E2)(W.u.)=2.9 +12-8 B(M1)(W.u.)=0.0032 +13-7 if M1, B(E2)(W.u.)=2.06 +82-47 if E2.
		1381 1	68.1 33	1230.28	2 ⁺	[M1,E2]		B(E1)(W.u.)<1.8 $\times$ 10 ⁻⁴
		1945 1	43 7	665.57	1 ⁺	[E2]		I _{γ} : weighted average of 14.1 8 from ²⁷ Al(¹² C, $\alpha\gamma$) and 14.9 11 from ³³ S(p, γ). Other: 39 5 from ³² S(³ He,p γ). Mult., δ : D+Q from $\gamma\gamma(\theta)$ (DCO) in ²⁷ Al(¹² C, $\alpha\gamma$). $\Delta\pi$ =yes from level scheme. B(E1)(W.u.)<2.0 $\times$ 10 ⁻⁵
		2465 1	100.0 29	146.36	3 ⁺	[M1,E2]		B(E1)(W.u.)<3.6 $\times$ 10 ⁻⁶
2721.09	2 ⁻	563 1	14.4 8	2157.9	2 ⁺	(E1+M2)	-0.38 6	I _{γ} : other: 12 5 from ³² S(³ He,p γ). B(E1)(W.u.)<4.4 $\times$ 10 ⁻⁶ ; B(M2)(W.u.)<0.030
		834 1	4.4 4	1887.29	2 ⁺	[E1]		I _{γ} : weighted average of 15.3 8 from ²⁷ Al(¹² C, $\alpha\gamma$) and 15.5 11 from ³³ S(p, γ). Other: 29 5 from ³² S(³ He,p γ). Mult., δ : D+Q from $\gamma\gamma(\theta)$ (DCO) in ²⁷ Al(¹² C, $\alpha\gamma$). Other: D(+Q) with δ =-0.1 2 from ³² S(³ He,p γ). $\Delta\pi$ =yes from level scheme. B(E1)(W.u.)<2.0 $\times$ 10 ⁻⁵ ; B(M2)(W.u.)<0.089
		1491 1	4.6 4	1230.28	2 ⁺	[E1]		E _{γ} : weighted average of 2260 1 from ²⁷ Al(¹² C, $\alpha\gamma$) and 2260.7 5 from ³² S(³ He,p γ). I _{γ} : others: 100 6 from ²⁷ Al(¹² C, $\alpha\gamma$), 100 7 from ³² S(³ He,p γ).
		2055 1	15.4 8	665.57	1 ⁺	(E1+M2)	-0.05 3	
		2260.6 5	100.0 15	461.01	1 ⁺	E1+M2	+0.05 2	

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\ddagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. @	$\delta^@$	Comments
								Mult., δ : E1+M2, $\Delta J=1$, from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\theta)(\text{POL})$ in $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$. Others: D(+Q) with $\delta=-0.1$ 3 from $^{32}\text{S}(^3\text{He}, p\gamma)$. D(+Q) with $\delta=-0.03$ 5 from $^{33}\text{S}(p, \gamma)$.
2721.09	2 ⁻	2575 1	37.8 15	146.36	3 ⁺	E1+M2	-0.3 1	B(E1)(W.u.) $<5.1\times 10^{-6}$; B(M2)(W.u.) <0.51 I_γ : weighted average of 34.3 18 from $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$, 42 5 from $^{32}\text{S}(^3\text{He}, p\gamma)$, and 38.9 11 from $^{33}\text{S}(p, \gamma)$.
		2721 1	27.1 32	0.0	0 ⁺	M2		Mult., δ : E1+M2, $\Delta J=1$, from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\theta)(\text{POL})$ in $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$. Other: D(+Q) with $\delta=0.0$ 3 from $^{32}\text{S}(^3\text{He}, p\gamma)$. B(M2)(W.u.) <2.1 I_γ : unweighted average of 26.4 14 from $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$, 22 5 from $^{32}\text{S}(^3\text{He}, p\gamma)$, and 32.8 11 from $^{33}\text{S}(p, \gamma)$.
3129.11	1 ⁺	3128.8 10	100.0 20	0.0	0 ⁺	[M1]		Mult.: M2, $\Delta J=2$, from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\theta)(\text{POL})$ in $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$. B(M1)(W.u.) >0.24 E_γ : from $^{34}\text{Ar } \varepsilon+\beta^+$ decay (0.84646 s). I_γ : from $^{34}\text{Ar } \varepsilon+\beta^+$ decay. Others: 100 from $^{33}\text{S}(p, \gamma)$ and $^{32}\text{S}(^3\text{He}, p\gamma)$.
3334.0	(1,2,3) ⁺	1176.1	100 15	2157.9	2 ⁺	[M1+E2]	<1.1 &	B(M1)(W.u.)=0.089 +140-58 B(M1)(W.u.)=0.0089 +59-27 if M1, B(E2)(W.u.)=4.9 +33-15 if E2.
		2668.3	90 12	665.57	1 ⁺	[M1,E2]		B(M1)(W.u.)=0.0043 +29-14 if M1, B(E2)(W.u.)=2.0 +14-7 if E2.
		2872.9	54 12	461.01	1 ⁺	[M1,E2]		B(M1)(W.u.)=0.069 +21-14 if M1, B(E2)(W.u.)=59 +18-11 if E2.
3383.3	1 ⁺ , 2 ⁺	2153.0	52.0 25	1230.28	2 ⁺	[M1,E2]		B(M1)(W.u.)=0.057 +17-11 if M1, B(E2)(W.u.)=30.3 +90-58 if E2.
		2717.6	86 4	665.57	1 ⁺	[M1,E2]		B(M1)(W.u.)=0.040 +12-8 if M1, B(E2)(W.u.)=18.4 +54-36 if E2.
		2922.2	75 5	461.01	1 ⁺	[M1,E2]		B(E2)(W.u.)=14.7 +43-28
		3236.8	100.0 34	146.36	3 ⁺	[E2]		B(E1)(W.u.)=1.79 $\times 10^{-5}$ +57-41
3545.08	3 ⁻	2314.72	3.6 6	1230.28	2 ⁺	[E1]		B(E1)(W.u.)=1.56 $\times 10^{-4}$ +41-27; B(M2)(W.u.)=0.50 +58-35
		3398.54	100.0 6	146.36	3 ⁺	(E1+M2)	-0.09 4	Mult., δ : D+Q from $\gamma(\theta)$ in $^{33}\text{S}(p, \gamma)$. $\Delta\pi$ =yes from level scheme. B(E2)(W.u.)=4.6 +13-9
3600.14	4 ⁻	879.2 2	94.1 11	2721.09	2 ⁻	E2		E_γ : weighted average of 879.1 2 from $^{24}\text{Mg}(^{12}\text{C}, p n\gamma)$ and 879.4 4 from $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$. I_γ : others: 122 5 from $^{24}\text{Mg}(^{12}\text{C}, p n\gamma)$, 115 6 from $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$.
		1224.1 2	17.7 9	2375.67	4 ⁺	(E1(+M2))	+0.3 3	Mult.: E2, $\Delta J=2$, from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\theta)(\text{POL})$ in $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$. B(E1)(W.u.)=1.69 $\times 10^{-6}$ +81-65; B(M2)(W.u.) <2.0 E_γ : weighted average of 1224.1 2 from $^{24}\text{Mg}(^{12}\text{C}, p n\gamma)$ and 1224.3 6 from $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$. I_γ : weighted average of 17 5 from $^{24}\text{Mg}(^{12}\text{C}, p n\gamma)$, 18.2 9 from $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$, and 15.4 19 from $^{33}\text{S}(p, \gamma)$.
		3453.4 2	100.0 11	146.36	3 ⁺	(E1+M2)	+0.06 3	Mult., δ : D(+Q) consistent with $\Delta J=0$ from $\gamma\gamma(\theta)(\text{DCO})$ in $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$. $\Delta\pi$ =yes from level scheme. B(E1)(W.u.)=4.6 $\times 10^{-7}$ +13-9; B(M2)(W.u.)=6.4 $\times 10^{-4}$ +86-48 E_γ : weighted average of 3453.3 2 from $^{24}\text{Mg}(^{12}\text{C}, p n\gamma)$ and 3453.8 6 from $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)$.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)							
$E_i(\text{level})$	J_i^π	$E_\gamma^{\ddagger\ddagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. [@]	$\delta^\text{@}$
I_γ: others: 100 5 from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$, 100 5 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$. Mult.,$\delta$: D+Q from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. $\Delta\pi$=yes from level scheme. Others: D(+Q) with $\delta=+0.05$ 5 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$. D(+Q) with $\delta=+0.07$ 17 from $^{31}\text{P}(\alpha,\text{n}\gamma)$. B(E1)(W.u.)=$9.8\times 10^{-7} +32-21$; B(M2)(W.u.)$<0.0094$ E_γ: weighted average of 1255.7 1 from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$ and 1256.0 8 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$. I_γ: others: 100.0 33 from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$, 100 5 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$. Mult.,$\delta$: E1(+M2), $\Delta J=1$, from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\theta)(\text{POL})$ in $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$. Others: D(+Q) with $\delta=0.00$ 2 from $^{33}\text{S}(\text{p},\gamma)$. D(+Q) with $\delta=0.00$ 2 from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$. B(M2)(W.u.)=$0.0118 +32-23$; B(E3)(W.u.)=$0.31 +32-21$ E_γ: weighted average of 3484.7 2 from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$ and 3485.1 6 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$. I_γ: unweighted average of 66.7 33 from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$, 67.0 35 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$, and 83.4 13 from $^{33}\text{S}(\text{p},\gamma)$. Mult.: M2+E3, $\Delta J=2$, from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\theta)(\text{POL})$ in $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$. Other: Q+O from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$. δ: from $^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$. Other: -0.23 12 from $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$. B(E2)(W.u.)=$1.08 +71-32$ I_γ: from (p,γ). Mult.: E2, $\Delta J=2$, from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\theta)(\text{POL})$ in $^{27}\text{Al}(^{12}\text{C},\alpha\text{n}\gamma)$.							
3631.40	5 ⁻	1255.7 1	100.0 13	2375.67	4 ⁺	E1(+M2)	-0.01 4
		3484.7 2	72 6	146.36	3 ⁺	M2+E3	-0.23 10
3646.3	5 ⁺	3500 1	100 7	146.36	3 ⁺	E2	
3660.0	(0 ⁺ to 4 ⁺)	1502.1	100	2157.9	2 ⁺		
3773.7	1 ⁻	3773.5	100	0.0	0 ⁺	[E1]	B(E1)(W.u.)= $2.18\times 10^{-4} +74-46$
3791.7	(0 to 3)	3330.5	100 7	461.01	1 ⁺		
3940.1	(1) ⁺	3274.4	100 9	665.57	1 ⁺	[M1,E2]	B(M1)(W.u.) >0.083 if M1, B(E2)(W.u.) >30 if E2.
		3478.9	72 9	461.01	1 ⁺	[M1,E2]	B(M1)(W.u.) >0.048 if M1, B(E2)(W.u.) >16 if E2.
3964.1	(2,3,4) ⁺	1588.4	11 5	2375.67	4 ⁺	[E2]	B(E2)(W.u.)= $5.9 +59-29$
		1806.2	100 16	2157.9	2 ⁺	[M1,E2]	B(M1)(W.u.)= $0.023 +19-8$ if M1, B(E2)(W.u.)= $28 +23-10$ if E2.
		3817.5	67 25	146.36	3 ⁺	[M1,E2]	B(M1)(W.u.)= $0.0017 +13-7$ if M1, B(E2)(W.u.)= $0.45 +36-18$ if E2.
3983.0	3 ⁻	1261.9	13.5 11	2721.09	2 ⁻	M1+E2	B(M1)(W.u.)= $0.0044 +20-14$; B(E2)(W.u.)= 6.9 35
		1825.1	43.2 17	2157.9	2 ⁺	(E1(+M2))	Mult., δ : D+Q from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. M2 ruled out by RUL. B(E1)(W.u.)= $2.23\times 10^{-4} +77-50$; B(M2)(W.u.) <1.5
		3836.4	100.0 31	146.36	3 ⁺	(E1(+M2))	Mult., δ : D(+Q) from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. $\Delta\pi$ =yes from level scheme. B(E1)(W.u.)= $5.6\times 10^{-5} +17-11$; B(M2)(W.u.) <0.081
4076.2	4 ⁻	1355.1	9.1 11	2721.09	2 ⁻	[E2]	Mult., δ : D(+Q) from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. $\Delta\pi$ =yes from level scheme. B(E2)(W.u.)= $1.2 +10-6$
		1700.5	35.5 13	2375.67	4 ⁺	(E1+M2)	B(E1)(W.u.)= $2.9\times 10^{-5} +25-14$

Adopted Levels, Gammas (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\ddagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. [@]	$\delta^@$	Comments
4076.2	4 ⁻	1895.1	22.0 13	2181.09	3 ⁺	(E1+M2)	+0.05 +3-1	Mult., δ : D+Q from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. $\Delta\pi=\text{yes}$ from level scheme. B(M2)(W.u.)=3.3 +29-18 exceeds RUL=3.
		3929.6	100.0 25	146.36	3 ⁺	(E1(+M2))	0.00 +1-2	B(E1)(W.u.)=1.4 $\times 10^{-5}$ +12-7; B(M2)(W.u.)=0.044 +75-24 Mult., δ : D+Q from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. $\Delta\pi=\text{yes}$ from level scheme. B(E1)(W.u.)=7.1 $\times 10^{-6}$ +62-35; B(M2)(W.u.)<8.7 $\times 10^{-4}$
4139.7	(2 ⁻)	1559.4	37 6	2580.3	1 ⁺	[E1]		Mult., δ : D(+Q) from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. $\Delta\pi=\text{yes}$ from level scheme. B(E1)(W.u.)=3.5 $\times 10^{-4}$ +15-9
		1981.7	43 4	2157.9	2 ⁺	[E1]		B(E1)(W.u.)=1.95 $\times 10^{-4}$ +86-48
		2909.3	14 4	1230.28	2 ⁺	[E1]		B(E1)(W.u.)=2.0 $\times 10^{-5}$ +11-7
		3993.1	100 6	146.36	3 ⁺	[E1]		B(E1)(W.u.)=5.6 $\times 10^{-5}$ +24-13
4147.8	1 ⁺ ,2 ⁺	1966.7	7.0 10	2181.09	3 ⁺			
		1989.8	6.1 13	2157.9	2 ⁺			
		2917.4	100.0 23	1230.28	2 ⁺			
		3482.0	66.2 19	665.57	1 ⁺			
		4147.5	9.8 15	0.0	0 ⁺			
4325.89	1 ⁺ ,2 ⁺	2167.9	100 6	2157.9	2 ⁺	[M1+E2]	<0.8 ^{&}	B(M1)(W.u.)=0.37 +36-18
4354.2	(1) ⁻	2196.2	33.2 28	2157.9	2 ⁺	[E1]		B(E1)(W.u.)=4.4 $\times 10^{-4}$ +11-8
		3893.0	11.5 28	461.01	1 ⁺	[E1]		B(E1)(W.u.)=2.71 $\times 10^{-5}$ +89-72
		4353.9	100.0 28	0.0	0 ⁺	[E1]		B(E1)(W.u.)=1.69 $\times 10^{-4}$ +39-27
4371.5	(4) ⁺	725 1	100	3646.3	5 ⁺	M1+E2		Mult.: M1+E2, $\Delta J=1$, from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\theta)(\text{POL})$ in $^{27}\text{Al}(^{12}\text{C},\alpha n\gamma)$.
								B(M1)(W.u.)<0.028 if M1.
4417.3	(2) ⁻	3751.5	58 5	665.57	1 ⁺	[E1]		B(E1)(W.u.)=1.87 $\times 10^{-4}$ +93-47
		3956.0	100 5	461.01	1 ⁺	[E1]		B(E1)(W.u.)=2.8 $\times 10^{-4}$ +14-7
4446.6	4 ⁻	4300.0	100	146.36	3 ⁺	(E1(+M2))	-0.02 +2-1	B(E1)(W.u.)=7.0 $\times 10^{-5}$ +18-12; B(M2)(W.u.)<0.020
								Mult., δ : D(+Q) from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. $\Delta\pi=\text{yes}$ from level scheme.
4461.4	(2,3) ⁻	4314.8	100	146.36	3 ⁺	[E1]		B(E1)(W.u.)=5.5 $\times 10^{-5}$ +23-13
4515.7	2 ⁻	3849.9	100.0 25	665.57	1 ⁺	[E1]		B(E1)(W.u.)=8.7 $\times 10^{-4}$ +42-22
		4369.0	25.0 25	146.36	3 ⁺	[E1]		B(E1)(W.u.)=1.49 $\times 10^{-4}$ +72-40
4605.8	(⁻)	1884.7	100	2721.09	2 ⁻	[M1,E2]		B(M1)(W.u.)=0.0059 +69-26 if M1, B(E2)(W.u.)=6.5 +76-28 if E2.
		2447.8	44	2157.9	2 ⁺	[E1]		B(E1)(W.u.)=3.5 $\times 10^{-5}$ +40-16
		2718.4	35	1887.29	2 ⁺	[E1]		B(E1)(W.u.)=2.0 $\times 10^{-5}$ +24-9
		3375.3	54	1230.28	2 ⁺	[E1]		B(E1)(W.u.)=1.6 $\times 10^{-5}$ +19-8
4609.7	(⁻)	1888.6	71	2721.09	2 ⁻			
		3379.2	100	1230.28	2 ⁺			
		4463.0	73	146.36	3 ⁺			
4638.9	2 ⁻	4177.6	100 8	461.01	1 ⁺	[E1]		B(E1)(W.u.)=1.81 $\times 10^{-4}$ +88-45
		4492.2	58 10	146.36	3 ⁺	[E1]		B(E1)(W.u.)=8.4 $\times 10^{-5}$ +40-23
4695.7	(1 ⁺ ,2,3 ⁺)	2537.7	100	2157.9	2 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. @	$\delta^@$	Comments
4717.5	2 ⁺ , 3 ⁺ , 4 ⁺	2106.4	16 5	2611.04	3 ⁺	[M1,E2]		B(M1)(W.u.)>0.015 if M1, B(E2)(W.u.)>14 if E2.
		2341.7	11.1 32	2375.67	4 ⁺	[E2]		B(E2)(W.u.)>5.7
		2536.3	35 5	2181.09	3 ⁺	[M1,E2]		B(M1)(W.u.)>0.024 if M1, B(E2)(W.u.)>15 if E2.
		2559.5	11 5	2157.9	2 ⁺	[M1,E2]		B(M1)(W.u.)>0.0047 if M1, B(E2)(W.u.)>2.8 if E2.
		4570.8	100 5	146.36	3 ⁺	[M1,E2]		B(M1)(W.u.)>0.013 if M1, B(E2)(W.u.)>2.5 if E2.
4743.15	6 ⁻	1111.77 10	83 4	3631.40	5 ⁻	M1+E2	+0.36 6	B(M1)(W.u.)=8.6×10 ⁻⁴ +60-27; B(E2)(W.u.)=0.36 +28-14 E _γ : weighted average of 1111.77 10 from ²⁴ Mg(¹² C,pn γ) and 1111.8 6 from ²⁷ Al(¹² C, α n γ). I _γ : weighted average of 84 6 from ²⁴ Mg(¹² C,pn γ) and 86 4 from ²⁷ Al(¹² C, α n γ). Mult., δ : D+Q from $\gamma\gamma(\theta)$ (DCO) in ²⁷ Al(¹² C, α n γ). Other: D+Q with δ =+4.7 6 from $\gamma(\theta)$ in ²⁴ Mg(¹² C,pn γ). M2 ruled out by RUL. B(E2)(W.u.)=3.2 +23-10 E _γ : weighted average of 1143.0 2 from ²⁴ Mg(¹² C,pn γ) and 1143.0 6 from ²⁷ Al(¹² C, α n γ). I _γ : from ²⁷ Al(¹² C, α n γ). Other: 100 6 from ²⁴ Mg(¹² C,pn γ). Mult.: E2, ΔJ =2, from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, α n γ). Other: Q(+O) with δ =+0.02 7 from ²⁴ Mg(¹² C,pn γ).
		1143.0 2	100 5	3600.14	4 ⁻	E2		
		2562 1		2181.09	3 ⁺			
		4596.6 3	96 9	146.36	3 ⁺	E3		B(E3)(W.u.)=29 +20-9 E _γ : weighted average of 4596.5 3 from ²⁴ Mg(¹² C,pn γ) and 4597.1 7 from ²⁷ Al(¹² C, α n γ). I _γ : unweighted average of 87 6 from ²⁴ Mg(¹² C,pn γ) and 105 5 from ²⁷ Al(¹² C, α n γ). Mult.: electric character from $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, α n γ). O(+H) with δ =+0.07 9 from $\gamma(\theta)$ in ²⁴ Mg(¹² C,pn γ). M4 ruled out by RUL. B(E1)(W.u.)=4.5×10 ⁻⁵ +44-20 E _γ : weighted average of 1192.67 5 from ²⁴ Mg(¹² C,pn γ) and 1193 1 from ²⁷ Al(¹² C, α n γ). I _γ : weighted average of 5.7 14 from ²⁴ Mg(¹² C,pn γ), 5.15 29 from ²⁷ Al(¹² C, α n γ), and 6.5 16 from ³³ S(p, γ). Mult., δ : consistent with ΔJ =0 from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, α n γ). Other: D(+Q) with δ =+0.3 6 from ²⁴ Mg(¹² C,pn γ). B(E1)(W.u.)=1.7×10 ⁻⁴ +17-7; B(M2)(W.u.)<1.6 E _γ : weighted average of 1224.1 2 from ²⁴ Mg(¹² C,pn γ) and 1223.9 6 from ²⁷ Al(¹² C, α n γ). I _γ : weighted average of 27.1 29 from ²⁴ Mg(¹² C,pn γ), 20.1 11 from ²⁷ Al(¹² C, α n γ), and 37 13 from ³³ S(p, γ).
4824.18	5 ⁺	1192.67 5	5.21 29	3631.40	5 ⁻	E1(+M2)	-0.1 3	
		1224.1 2	21.1 19	3600.14	4 ⁻	E1(+M2)	-0.02 2	

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J^{π}_i</u>	<u>E_{γ}^{$\dagger \ddagger$}</u>	<u>I_{γ}[#]</u>	<u>E_f</u>	<u>J^{π}_f</u>	<u>Mult.[@]</u>	<u>$\delta^@$</u>	<u>Comments</u>
4824.18	5 ⁺	2213 <i>I</i> 2449 <i>I</i> 2643.4 <i>5</i>	1.21 <i>I2</i> 8.4 <i>8</i>	2611.04 3 ⁺ 2375.67 4 ⁺ 2181.09 3 ⁺		[E2] E2		Mult., δ : E1(+M2), $\Delta J=1$, from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, $\alpha\gamma$). B(E2)(W.u.)=0.045 +34-15 B(E2)(W.u.)=0.129 +96-41 E _{γ} : weighted average of 2643.5 <i>5</i> from ²⁴ Mg(¹² C,pn γ) and 2643 <i>I</i> from ²⁷ Al(¹² C, $\alpha\gamma$). I _{γ} : weighted average of 10.0 <i>I4</i> from ²⁴ Mg(¹² C,pn γ), 8.5 <i>5</i> from ²⁷ Al(¹² C, $\alpha\gamma$), and 4.8 <i>I6</i> from ³³ S(p, γ). Mult., δ : Q, $\Delta J=2$, from $\gamma\gamma(\theta)$ (DCO) in ²⁷ Al(¹² C, $\alpha\gamma$). M2 ruled out by RUL. Other: Q(+O) with $\delta=-0.03$ <i>6</i> from ²⁴ Mg(¹² C,pn γ). B(E2)(W.u.)=0.089 +65-28 E _{γ} : weighted average of 4677.4 <i>2</i> from ²⁴ Mg(¹² C,pn γ) and 4677.9 <i>7</i> from ²⁷ Al(¹² C, $\alpha\gamma$). I _{γ} : from ²⁴ Mg(¹² C,pn γ). Others: 100.0 <i>32</i> from ³³ S(p, γ), 100 <i>5</i> from ²⁷ Al(¹² C, $\alpha\gamma$). Mult., δ : E2, $\Delta J=2$, from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, $\alpha\gamma$). Other: Q(+O) with $\delta=0.00$ <i>6</i> from ²⁴ Mg(¹² C,pn γ). Mult.: M2, $\Delta J=2$, from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, $\alpha\gamma$). B(E2)(W.u.)>24 B(M1)(W.u.)>0.035 if M1, B(E2)(W.u.)>24 if E2. B(M1)(W.u.)>0.011 B(M1)(W.u.)>0.017 if M1, B(E2)(W.u.)>4.7 if E2. B(E2)(W.u.)>1.2 B(M1)(W.u.)=0.20 +19-8 if M1, B(E2)(W.u.)=96 +91-36 if E2. B(E2)(W.u.)=96 +91-36 upper bound exceeds RUL=100 if E2, indicating M1 component.
4862.4	5 ⁻	2681 <i>I</i>	100	2181.09 3 ⁺		M2		
4941.9	1 ⁺	2330.8 2361.5 4941.5	43 <i>I9</i> 45 <i>I9</i> 1.0×10 ² <i>4</i>	2611.04 3 ⁺ 2580.3 1 ⁺ 0.0 0 ⁺		[E2] [M1,E2] [M1]		
4957.5	1 ⁺ ,2 ⁺ ,3 ⁺	3727.0 4810.8	100 <i>I0</i> 94 <i>I0</i>	1230.28 2 ⁺ 146.36 3 ⁺		[M1,E2] [E2]		
4995.6	1 ⁺ ,2 ⁺	2837.6	100	2157.9 2 ⁺		[M1,E2]		
5028	0 ⁻ ,1 ⁻ ,2 ⁻ ,3 ⁻	2870 ^{<i>a</i>}	100	2157.9 2 ⁺				
5171.6	4	1571.4 5024.8	100 100	3600.14 4 ⁻ 146.36 3 ⁺				
5314.95	7 ⁺	453 <i>I</i> 490.8 <i>2</i>		4862.4 5 ⁻ 4824.18 5 ⁺				B(E2)(W.u.)=36.7 +62-52 E _{γ} : weighted average of 490.8 <i>2</i> from ²⁴ Mg(¹² C,pn γ) and 490.9 <i>3</i> from ²⁷ Al(¹² C, $\alpha\gamma$). I _{γ} : from ²⁴ Mg(¹² C,pn γ). Other: 100 <i>5</i> from ²⁷ Al(¹² C, $\alpha\gamma$). Mult., δ : E2, $\Delta J=2$, from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, $\alpha\gamma$). Other: Q(+O) with $\delta=+0.02$ <i>2</i> from ²⁴ Mg(¹² C,pn γ). B(E1)(W.u.)=9.1×10 ⁻⁶ +44-43; B(M2)(W.u.)=0.32 +36-24 E _{γ} : weighted average of 571.8 <i>2</i> from ²⁴ Mg(¹² C,pn γ) and 571.8 <i>3</i> from
		571.8 <i>2</i>	22 <i>I3</i>	4743.15 6 ⁻		E1+M2	-0.05 <i>2</i>	

Adopted Levels, Gammas (continued)

<u>$\gamma(^{34}\text{Cl})$ (continued)</u>							Comments
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_γ ^{†‡}</u>	<u>I_γ [#]</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult. [@]</u>	
²⁷ Al(¹² C, α n γ). I_γ : unweighted average of 35.1 14 from ²⁴ Mg(¹² C,pn γ) and 8.6 5 from ²⁷ Al(¹² C, α n γ). Mult., δ : E1+M2, $\Delta J=1$, from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\theta)(\text{POL})$ in ²⁷ Al(¹² C, α n γ). Other: D+Q with $\delta=-0.05$ 2 from ²⁴ Mg(¹² C,pn γ). D(+Q) with $\delta=-0.02$ 7 from ³¹ P(α ,n γ). B(E2)(W.u.)=0.00440 +80-67 Mult.: Q, $\Delta J=2$, from $\gamma\gamma(\theta)(\text{DCO})$ in ²⁷ Al(¹² C, α n γ). $\Delta\pi=\text{no}$ from level scheme.							
5314.95	7 ⁺	1669 1	5.46 32	3646.3	5 ⁺	(E2)	
5386.8	(2,3,4,5) ⁻	1786.6	100	3600.14	4 ⁻		
5541.1	(2 ⁻ to 5 ⁻)	1940.9	100 8	3600.14	4 ⁻		
		1996.0	56 8	3545.08	3 ⁻		
5577.2	(2) ⁺	1061.5	9.0 30	4515.7	2 ⁻		
		1977.0	29 7	3600.14	4 ⁻		
		2032.1	100	3545.08	3 ⁻		
		2856.0	22.6 23	2721.09	2 ⁻		
		3201.4	3.0 12	2375.67	4 ⁺		
		3689.7	5.5 10	1887.29	2 ⁺		
		4346.6	10.3 14	1230.28	2 ⁺		
		5430.4	62 6	146.36	3 ⁺		
5635.75	(1,2 ⁺)	1844.0	3.7 7	3791.7	(0 to 3)		
		2506.54	3.9 4	3129.11	1 ⁺		
		2914.53	1.00 30	2721.09	2 ⁻		
		3055.3	12.2 7	2580.3	1 ⁺		
		3477.7	0.60 30	2157.9	2 ⁺		
		3748.24	1.80 30	1887.29	2 ⁺		
		4405.16	0.60 40	1230.28	2 ⁺		
		4969.79	15.4 10	665.57	1 ⁺		
		5174.32	2.2 7	461.01	1 ⁺		
		5635.25	100	0.0	0 ⁺		
5673.0	1 ⁺ ,2 ⁺	3092.6	11.6 26	2580.3	1 ⁺		
		3514.9	8.8 24	2157.9	2 ⁺		
		5007.0	6.4 34	665.57	1 ⁺		
		5672.5	100	0.0	0 ⁺		
5763.3	(1 ⁺ ,2 ⁺)	1823.2	2.2	3940.1	(1) ⁺		
		2379.9	17	3383.3	1 ⁺ ,2 ⁺		
		2634.1	1.5	3129.11	1 ⁺		
		3605.2	100	2157.9	2 ⁺		
		3875.8	1.3	1887.29	2 ⁺		
		5301.9	3	461.01	1 ⁺		
		5616.4	16	146.36	3 ⁺		
		5762.8	44	0.0	0 ⁺		

Adopted Levels, Gammas (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π	$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π
5785.1	(2,3) ⁻	2011.3	0.8	3773.7	1 ⁻	5897.3	(2)	3176.1	5	2721.09	2 ⁻
		2401.7	52	3383.3	1 ⁺ ,2 ⁺			3316.8	2.4	2580.3	1 ⁺
		2451.0	1.2	3334.0	(1,2,3) ⁺			4009.8	11	1887.29	2 ⁺
		3063.9	0.6	2721.09	2 ⁻			4666.7	41	1230.28	2 ⁺
		3173.9	2.8	2611.04	3 ⁺			5231.3	32	665.57	1 ⁺
		3603.8	3	2181.09	3 ⁺			5435.8	100	461.01	1 ⁺
		3627.0	32	2157.9	2 ⁺			5750.4	85	146.36	3 ⁺
		3897.6	1.6	1887.29	2 ⁺			5896.8	5	0.0	0 ⁺
		4554.5	6	1230.28	2 ⁺	5940.8	1 ⁺ ,2,3 ⁺	983.3 ^a	9.0 30	4957.5	1 ⁺ ,2 ⁺ ,3 ⁺
		5638.2	100	146.36	3 ⁺			1245.1	0.85	4695.7	(1 ⁺ ,2,3 ⁺)
5805.9	(2)	1200.1	9.4	4605.8	(⁻)			2557.4	6.3	3383.3	1 ⁺ ,2 ⁺
		1658.1	3.2	4147.8	1 ⁺ ,2 ⁺			3329.6	0.56	2611.04	3 ⁺
		1822.9	1.9	3983.0	3 ⁻			3360.3	1.6	2580.3	1 ⁺
		2205.7	1.6	3600.14	4 ⁻			3564.9	0.56	2375.67	4 ⁺
		2260.7	1.3	3545.08	3 ⁻			3782.7	100	2157.9	2 ⁺
		3225.4	58	2580.3	1 ⁺			4053.3	1.1	1887.29	2 ⁺
		3430.0	3.9	2375.67	4 ⁺			5274.8	0.99	665.57	1 ⁺
		3918.4	1.3	1887.29	2 ⁺			5479.3	21	461.01	1 ⁺
		4575.3	13	1230.28	2 ⁺			5793.9	6.8	146.36	3 ⁺
		5139.9	58	665.57	1 ⁺			5940.2	0.99	0.0	0 ⁺
		5344.4	100	461.01	1 ⁺	6030.1	(1 ⁺ ,2 ⁺)	2646.7	4.7	3383.3	1 ⁺ ,2 ⁺
		5659.0	65	146.36	3 ⁺			5883.2	6.4	146.36	3 ⁺
		5805.4	6.1	0.0	0 ⁺			6029.5	100	0.0	0 ⁺
5852.91	(2,3) ⁻	1705.1	18	4147.8	1 ⁺ ,2 ⁺	6088.94	1,2 ⁽⁺⁾	2297.2	0.33	3791.7	(0 to 3)
		1869.9	2.8	3983.0	3 ⁻			2315.2	0.56	3773.7	1 ⁻
		2079.1	0.4	3773.7	1 ⁻			3367.67	4.7	2721.09	2 ⁻
		2252.69	8.4	3600.14	4 ⁻			3508.5	0.33	2580.3	1 ⁺
		2307.75	13	3545.08	3 ⁻			4201.37	2.6	1887.29	2 ⁺
		2469.5	3.6	3383.3	1 ⁺ ,2 ⁺			5627.43	2.7	461.01	1 ⁺
		3131.7	7.6	2721.09	2 ⁻			6088.36	100	0.0	0 ⁺
		3241.70	0.6	2611.04	3 ⁺	6136.2	1 ⁽⁺⁾ ,2,3 ⁽⁺⁾	2752.8	1.82 91	3383.3	1 ⁺ ,2 ⁺
		3671.61	2.4	2181.09	3 ⁺			3414.9	1.82 91	2721.09	2 ⁻
		3694.8	8.4	2157.9	2 ⁺			3525.0	3.6 18	2611.04	3 ⁺
		3965.37	6.2	1887.29	2 ⁺			3555.7	1.82 91	2580.3	1 ⁺
		4622.29	28	1230.28	2 ⁺			3954.9	1.82 91	2181.09	3 ⁺
		5706.04	100	146.36	3 ⁺			3978.1	10.9 11	2157.9	2 ⁺
5897.3	(2)	1291.5	5	4605.8	(⁻)			4905.5	40 4	1230.28	2 ⁺
		1749.5 ^a	24 8	4147.8	1 ⁺ ,2 ⁺			5470.2	100 10	665.57	1 ⁺
		2352.1	5	3545.08	3 ⁻			5989.3	20.0 20	146.36	3 ⁺
		2513.9	1.2	3383.3	1 ⁺ ,2 ⁺	6141.8	1,2 ⁽⁺⁾	1626.1	1.16 58	4515.7	2 ⁻
		2563.2	0.88	3334.0	(1,2,3) ⁺			1724.5	0.58 29	4417.3	(2) ⁻

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\ddagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. @	$\delta^@$	Comments
6141.8	$1,2^{(+)}$	1787.6	1.74 93	4354.2	$(1)^-$			
		2368.0	2.3 12	3773.7	1^-			
		2596.6	0.58 29	3545.08	3^-			
		3420.5	6.4 7	2721.09	2^-			
		3561.3	1.16 58	2580.3	1^+			
		3983.7	0.58	2157.9	2^+			
		4911.1	2.3 12	1230.28	2^+			
		6141.2	100 10	0.0	0^+			
		6169.4	3 ⁻					
		1653.7	4.4 24	4515.7	2^-			
6169.4	3^-	2093.1	16 6	4076.2	4^-	(M1(+E2))	+0.03 5	Mult., δ : D(+Q) from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. $\Delta\pi$ =no from level scheme.
		2186.3	100 10	3983.0	3^-	(M1(+E2))	-0.03 +5-4	Mult., δ : D(+Q) from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. $\Delta\pi$ =no from level scheme.
		2569.2	23.6 24	3600.14	4^-			
		2624.2	17 6	3545.08	3^-	(M1(+E2))	-0.05 +19-14	Mult., δ : D(+Q) from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. $\Delta\pi$ =no from level scheme.
		3448.1	20.7 21	2721.09	2^-			
		3558.2	7.3 30	2611.04	3^+			
		3988.1	16 6	2181.09	3^+			
		4011.3	1.47 74	2157.9	2^+			
		4281.8	7.1 31	1887.29	2^+			
		4938.7	10.7 31	1230.28	2^+	(E1(+M2))	0.00 1	Mult., δ : D(+Q) from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. $\Delta\pi$ =yes from level scheme.
		6022.5	80 8	146.36	3^+	(E1(+M2))	-0.02 3	Mult., δ : D(+Q) from $\gamma(\theta)$ in $^{33}\text{S}(\text{p},\gamma)$. $\Delta\pi$ =yes from level scheme.
		6181.27	1 ⁺ ,2					
		1855.33	11.4 57	4325.89	$1^+,2^+$			
		2217.1	2.3 12	3964.1	$(2,3,4)^+$			
		2797.9	4.6 23	3383.3	$1^+,2^+$			
6181.27	$1^+,2$	2847.1	6.8 34	3334.0	$(1,2,3)^+$			
		3052.01	1.14 57	3129.11	1^+			
		3459.99	1.14 57	2721.09	2^-			
		3570.03	9.1 46	2611.04	3^+			
		3600.8	18.0 18	2580.3	1^+			
		3999.93	1.4	2181.09	3^+			
		4023.1	6.8 38	2157.9	2^+			
		4293.69	13 4	1887.29	2^+			
		4950.60	25.0 25	1230.28	2^+			

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\ddagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. [@]	$\delta^@$	Comments
6181.27	1 ⁺ ,2	5515.22	100 10	665.57	1 ⁺	(D+Q)		
		5719.74	11 4	461.01	1 ⁺			
		6034.34	20.1 21	146.36	3 ⁺			
		6180.67	1.9	0.0	0 ⁺			
6208.2	4 ⁽⁻⁾	1384.0	1.04 52	4824.18	5 ⁺	[E1]		B(E1)(W.u.)>3.7×10 ⁻⁴
		2131.9	18.6 19	4076.2	4 ⁻	[M1,E2]		B(M1)(W.u.)>0.11 if M1, B(E2)(W.u.)>98 if E2.
		2225.1	10 4	3983.0	3 ⁻	[M1,E2]		B(M1)(W.u.)>0.037 if M1, B(E2)(W.u.)>29 if E2.
		2561.8	1.04 52	3646.3	5 ⁺	[E1]		B(E1)(W.u.)>5.9×10 ⁻⁵
		2576.7	20.8 21	3631.40	5 ⁻	[M1,E2]		B(M1)(W.u.)>0.072 if M1, B(E2)(W.u.)>43 if E2.
		2608.0	100 10	3600.14	4 ⁻	(M1(+E2))	-0.03 5	B(M1)(W.u.)>0.36
								Mult., δ : D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =(no) from level scheme.
		2663.0	49 5	3545.08	3 ⁻	(M1(+E2))	0.00 2	B(M1)(W.u.)>0.16
								Mult., δ : D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =(no) from level scheme.
		3832.3	4.2 21	2375.67	4 ⁺	[E1]		B(E1)(W.u.)>7.2×10 ⁻⁵
		4026.9	7.6 21	2181.09	3 ⁺	[E1]		B(E1)(W.u.)>1.6×10 ⁻⁴
		6061.3	1.04 52	146.36	3 ⁺	[E1]		B(E1)(W.u.)>4.5×10 ⁻⁶
6219.3	(3 ⁺ ,4 ⁺)	1501.8	100 10	4717.5	2 ⁺ ,3 ⁺ ,4 ⁺			
		1893.4	3.9 21	4325.89	1 ⁺ ,2 ⁺			
		2572.9	18.0 18	3646.3	5 ⁺			
		2587.8	2.6 13	3631.40	5 ⁻			
		2885.2	7.7 39	3334.0	(1,2,3) ⁺			
		3608.1	5.1 26	2611.04	3 ⁺			
		3843.4	11.5 59	2375.67	4 ⁺			
		4038.0	46 5	2181.09	3 ⁺			
		4988.6	28.2 28	1230.28	2 ⁺			
		6072.4	33.3 33	146.36	3 ⁺			
6229.5	1	1590.6	4.5 18	4638.9	2 ⁻			
		1713.8	24.0 25	4515.7	2 ⁻			
		1812.2	6.2 20	4417.3	(2) ⁻			
		1875.2	20.0 20	4354.2	(1) ⁻			
		1903.6	2.2 11	4325.89	1 ⁺ ,2 ⁺			
		2089.7	13.2 13	4139.7	(2) ⁻			
		2455.7	31.6 33	3773.7	1 ⁻	D(+Q)	+0.03 5	
		3100.2	2.4	3129.11	1 ⁺			
		3508.2	100 10	2721.09	2 ⁻	D(+Q)	-0.05 5	

Adopted Levels, Gammas (continued) $\gamma(^{34}\text{Cl})$ (continued)

<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^{\dagger\ddagger}$</u>	<u>$I_\gamma^\#$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult. @</u>	<u>$\delta^@$</u>
6229.5	1	4071.3	10 4	2157.9	2 ⁺		
		4341.9	39	1887.29	2 ⁺		
		5768.0	2.4 10	461.01	1 ⁺		
		6228.9	1.9 8	0.0	0 ⁺		
6266.6	(1 ⁺ ,2,3 ⁺)	3545.3	5.8 29	2721.09	2 ⁻		
		4085.3	5.8	2181.09	3 ⁺		
		4108.4	100 10	2157.9	2 ⁺		
		4379.0	11.6 12	1887.29	2 ⁺		
		5035.9	8.6	1230.28	2 ⁺		
		5600.5	27.5 28	665.57	1 ⁺		
		6119.7	72	146.36	3 ⁺		
		1826.7	0.31	4446.6	4 ⁻		
6273.3	(2 ⁺ ,3,4 ⁺)	2197.0	1.11 64	4076.2	4 ⁻		
		2309.1	1.27 64	3964.1	(2,3,4) ⁺		
		2939.2	0.77	3334.0	(1,2,3) ⁺		
		3662.1	12.7 13	2611.04	3 ⁺		
		3897.4	100 10	2375.67	4 ⁺	D+Q	-0.08 5
		4092.0	25.4 26	2181.09	3 ⁺		
		4115.1	1.8 8	2157.9	2 ⁺		
		4385.7	11.1 11	1887.29	2 ⁺		
		5042.6	1.2 12	1230.28	2 ⁺		
		6126.4	4.8 24	146.36	3 ⁺		
		1715.9	4.6 23	4605.8	(⁻)		
		1967.4	2.3 12	4354.2	(1) ⁻		
		2245.4	6.8 34	4076.2	4 ⁻		
		2690.2	2.3 12	3631.40	5 ⁻		
6321.7	2 ⁻	2776.5	9.1 46	3545.08	3 ⁻		
		2938.3	1.14 57	3383.3	1 ⁺ ,2 ⁺		
		3600.4	12.5 14	2721.09	2 ⁻		
		3710.4	4.6 23	2611.04	3 ⁺		
		3741.2	20.4 21	2580.3	1 ⁺		
		4140.3	2.3 12	2181.09	3 ⁺		
		4434.1	2.3 12	1887.29	2 ⁺		
		5091.0	4.6 23	1230.28	2 ⁺		
		5655.6	34 4	665.57	1 ⁺		
		5860.2	16.5 18	461.01	1 ⁺		
		6174.7	100 10	146.36	3 ⁺		

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. @	$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π
6361.3	(1 ⁺ ,2,3 ⁺)	2977.9	71 7	3383.3	1 ⁺ ,2 ⁺		6382.4	(3 ⁺)	6235.4	100 10	146.36	3 ⁺
		4203.1	100 10	2157.9	2 ⁺		6399.8	(3,4 ⁺)	2799.5	5.5 27	3600.14	4 ⁻
		5130.6	25.0 25	1230.28	2 ⁺				3016.4	58 6	3383.3	1 ⁺ ,2 ⁺
		5899.7	14.8 16	461.01	1 ⁺				3678.5	3.6 18	2721.09	2 ⁻
		6214.3	17.1 18	146.36	3 ⁺				4023.9	5.5 27	2375.67	4 ⁺
6370.2	2 ⁻	1412.7	3.5 17	4957.5	1 ⁺ ,2 ⁺ ,3 ⁺				5169.1	9.1 46	1230.28	2 ⁺
		1731.3	1.72 86	4638.9	2 ⁻				6252.8	100 10	146.36	3 ⁺
		1854.5	3.5 17	4515.7	2 ⁻		6441.2	(1 ⁺ ,2,3 ⁺)	2364.9	2.7 14	4076.2	4 ⁻
		2015.9	8.6 45	4354.2	(1) ⁻				3057.8	9.6 10	3383.3	1 ⁺ ,2 ⁺
		2387.1	1.72 86	3983.0	3 ⁻				4283.0	100 10	2157.9	2 ⁺
		2406.0	1.72 86	3964.1	(2,3,4) ⁺				5775.1	5.5 28	665.57	1 ⁺
		2825.0	6.9 35	3545.08	3 ⁻				5979.6	8.2 8	461.01	1 ⁺
		2986.8	1.72 86	3383.3	1 ⁺ ,2 ⁺				6294.2	11.0 11	146.36	3 ⁺
		3036.1	2.1	3334.0	(1,2,3) ⁺		6450.54	3 ⁽⁻⁾ ,4	1844.7	8.3 42	4605.8	(⁻)
		3240.9	3.5 17	3129.11	1 ⁺				1989.1	6.9	4461.4	(2,3) ⁻
		3648.9	20.7 21	2721.09	2 ⁻				2003.9	7.3 38	4446.6	4 ⁻
		3758.9	3.5 17	2611.04	3 ⁺				2374.3	4.2 21	4076.2	4 ⁻
		3789.7	23.5 24	2580.3	1 ⁺				2819.02	10.4 52	3631.40	5 ⁻
		4188.8	10.3 52	2181.09	3 ⁺				2850.27	38 4	3600.14	4 ⁻
		4212.0	8.6 45	2157.9	2 ⁺				2905.33	7.3 38	3545.08	3 ⁻
		4482.6	3.5 17	1887.29	2 ⁺				3116.4	1	3334.0	(1,2,3) ⁺
		5139.5	11.2 23	1230.28	2 ⁺				3839.27	29.2 29	2611.04	3 ⁺
		5704.1	100 10	665.57	1 ⁺	D+Q			4269.16	4.2 21	2181.09	3 ⁺
		5908.6	25.6 23	461.01	1 ⁺				6303.55	100 10	146.36	3 ⁺
		6223.2	37 5	146.36	3 ⁺		6477.5	(0,1,2,3 ⁺)	2123.2	26.2 26	4354.2	(1) ⁻
		6369.6	46 5	0.0	0 ⁺				3897.0	9.2 9	2580.3	1 ⁺
6382.4	(3 ⁺)	1772.7	49 5	4609.7	(⁻)				5811.4	18.5 19	665.57	1 ⁺
		1776.6	36 4	4605.8	(⁻)				6015.9	100 10	461.01	1 ⁺
		2306.1	6.4 33	4076.2	4 ⁻		6486.9	(1,2 ⁺)	2546.7	16.7 17	3940.1	(1) ⁺
		2399.3	5.1 26	3983.0	3 ⁻				3357.6	1.5	3129.11	1 ⁺
		2782.1	20.5 21	3600.14	4 ⁻				4328.7	100 10	2157.9	2 ⁺
		2837.2	9.0 46	3545.08	3 ⁻				5820.8	3.3 17	665.57	1 ⁺
		3048.3	6.4 33	3334.0	(1,2,3) ⁺				6025.3	13.3 13	461.01	1 ⁺
		3661.1	2.6 13	2721.09	2 ⁻				6486.2	33.3 33	0.0	0 ⁺
		3771.1	10.3 51	2611.04	3 ⁺		6527.6	(2 ⁻ ,3 ⁺)	2927.3	21.0 21	3600.14	4 ⁻
		4006.5	6.4 33	2375.67	4 ⁺				3144.1	100 10	3383.3	1 ⁺ ,2 ⁺
		5716.3	5.1 26	665.57	1 ⁺				4369.4	9.7 10	2157.9	2 ⁺

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. [@]	$\delta^@$	Comments
6527.6	(2 ⁻ ,3 ⁺)	4640.0	6.5 32	1887.29	2 ⁺			
		5861.5	11.3 11	665.57	1 ⁺			
		6380.6	12.9 13	146.36	3 ⁺			
6545.9	2 ⁻	1374.3	1.6	5171.6	4			
		3162.4	0.95	3383.3	1 ⁺ ,2 ⁺			
		3965.4	7.7 8	2580.3	1 ⁺			
		5315.2	20.5 21	1230.28	2 ⁺			
		6398.9	100 10	146.36	3 ⁺			
6574.7	(2 ⁺ ,3,4 ⁺)	4198.8	1.9	2375.67	4 ⁺			
		4416.5	100 10	2157.9	2 ⁺			
		6427.7	5.3 26	146.36	3 ⁺			
6583.7	(1 ⁻ ,2 ⁺)	3038.5	53 5	3545.08	3 ⁻			
		4425.5	21.3 21	2157.9	2 ⁺			
		5917.6	21.3 21	665.57	1 ⁺			
		6122.1	100 10	461.01	1 ⁺			
		6583.0	17.0 17	0.0	0 ⁺			
6624.7	(3 ⁻)	1237.9	30	5386.8	(2,3,4,5) ⁻			
		1907.1	32	4717.5	2 ⁺ ,3 ⁺ ,4 ⁺			
		2484.9	19	4139.7	(2 ⁻)			
		2548.4	75	4076.2	4 ⁻			
		2993.2	28	3631.40	5 ⁻			
		3024.4	100	3600.14	4 ⁻			
		3079.5	52	3545.08	3 ⁻			
		3290.5	52	3334.0	(1,2,3) ⁺			
		4013.4	23	2611.04	3 ⁺			
		4443.3	69	2181.09	3 ⁺			
		4466.5	77	2157.9	2 ⁺			
		5394.0	21	1230.28	2 ⁺			
		6163.1	100	461.01	1 ⁺			
		6477.7	92	146.36	3 ⁺			
6640.92	4 ⁻	2314.95	1.32 66	4325.89	1 ⁺ ,2 ⁺			
		2564.6	100 10	4076.2	4 ⁻	M1+E2	-0.14 3	Mult., δ : D+Q from $\gamma(\theta)$ in ³³ S(p, γ). M2 ruled out by RUL.
		2657.8	7.9 40	3983.0	3 ⁻			
		2994.5	1.32 66	3646.3	5 ⁺			
		3009.38	1.32 66	3631.40	5 ⁻			
		3040.63	13.2 66	3600.14	4 ⁻	M1+E2	-0.12 3	Mult., δ : D+Q from $\gamma(\theta)$ in ³³ S(p, γ). M2 ruled out by RUL.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. @	$\delta^@$	Comments
6640.92	4 ⁻	3095.69	47 5	3545.08	3 ⁻	M1+E2	+0.04 2	Mult., δ : D+Q from $\gamma(\theta)$ in ³³ S(p, γ). M2 ruled out by RUL. Mult., δ : D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =yes from level scheme. Mult., δ : D+Q from $\gamma(\theta)$ in ³³ S(p, γ). M2 ruled out by RUL and $\Delta\pi$ =yes from level scheme seem to be a discrepancy.
		4029.62	15.8 16	2611.04	3 ⁺	(E1(+M2))	+0.02 +1-2	
		4264.96	15.8 16	2375.67	4 ⁺	D+Q	-0.14 4	
		4459.52	1.32 66	2181.09	3 ⁺			
		6493.89	58 6	146.36	3 ⁺	(E1(+M2))	+0.02 +1-2	Mult., δ : D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =yes from level scheme.
6695.1	(3 ⁺ ,4,5 ⁺)	1870.9	50 5	4824.18	5 ⁺			
		1999.3	13 6	4695.7	(1 ⁺ ,2,3 ⁺)			
		3048.7	31.3 31	3646.3	5 ⁺			
		3094.8	47 5	3600.14	4 ⁻			
		4083.8	72 7	2611.04	3 ⁺			
		4319.1	100 10	2375.67	4 ⁺			
6702.8	(3,4)	2626.5	2.4 12	4076.2	4 ⁻			
		3102.5	4.8 24	3600.14	4 ⁻			
		3157.6	3.6 18	3545.08	3 ⁻			
		3368.6	2.4 12	3334.0	(1,2,3) ⁺			
		4326.8	6.0 30	2375.67	4 ⁺			
		6555.8	100 10	146.36	3 ⁺			
6719.8	3 ⁻	2110.0	31.3 31	4609.7	(⁻)			
		2113.9	22.9 23	4605.8	(⁻)			
		2736.7	8.3 42	3983.0	3 ⁻			
		3336.3	9.4 48	3383.3	1 ⁺ ,2 ⁺			
		4832.1	11.5 13	1887.29	2 ⁺			
		5489.0	25.0 25	1230.28	2 ⁺			
		6572.8	100 10	146.36	3 ⁺			
6723.7	(3 ⁺ ,4 ⁺)	1899.5	100	4824.18	5 ⁺			
		2027.9	20	4695.7	(1 ⁺ ,2,3 ⁺)			
		2759.5	4.8	3964.1	(2,3,4) ⁺			
		3077.3	33	3646.3	5 ⁺			
		3340.2	1.6	3383.3	1 ⁺ ,2 ⁺			
		4112.4	72	2611.04	3 ⁺			
		4347.7	92	2375.67	4 ⁺			
		4542.3	5.6	2181.09	3 ⁺			

Adopted Levels, Gammas (continued) $\gamma(^{34}\text{Cl})$ (continued)

<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^{\dagger\ddagger}$</u>	<u>$I_\gamma^\#$</u>	<u>E_f</u>	<u>J_f^π</u>
6723.7	(3 ⁺ ,4 ⁺)	4565.5	7.6	2157.9	2 ⁺
		6576.7	64	146.36	3 ⁺
6737.3	2 ⁻	3608.0	5.1 26	3129.11	1 ⁺
		4579.1	100 10	2157.9	2 ⁺
		4849.6	3.6	1887.29	2 ⁺
		5506.5	12.8 64	1230.28	2 ⁺
		6071.2	36 4	665.57	1 ⁺
		6275.7	28.2 28	461.01	1 ⁺
		6590.3	74 8	146.36	3 ⁺
6790.8	(1,2 ⁺)	4903.1	5.3 26	1887.29	2 ⁺
		6790.1	100 10	0.0	0 ⁺
6797.4	(1 ⁺ to 5 ⁺)	2833.2	1.10 55	3964.1	(2,3,4) ⁺
		3413.9	0.55 28	3383.3	1 ⁺ ,2 ⁺
		4186.1	2.2 11	2611.04	3 ⁺
		4421.4	1.10 55	2375.67	4 ⁺
		4639.2	0.55 28	2157.9	2 ⁺
		4909.7	4.4 22	1887.29	2 ⁺
		6650.3	100 10	146.36	3 ⁺
6807.1	1 ⁻	3423.6	12.7 13	3383.3	1 ⁺ ,2 ⁺
		4226.5	16.9 17	2580.3	1 ⁺
		4648.9	11.3 11	2157.9	2 ⁺
		6140.9	100 10	665.57	1 ⁺
6828.6	(2,3 ⁺)	3283.4	7.3	3545.08	3 ⁻
		3445.1	15.5 16	3383.3	1 ⁺ ,2 ⁺
		4670.4	100 10	2157.9	2 ⁺
		6162.4	5.2 26	665.57	1 ⁺
		6367.0	19.0 19	461.01	1 ⁺
		6681.5	32.8 33	146.36	3 ⁺
6842.8	2 ⁻	2236.9	5.0 25	4605.8	(⁻)
		2327.0	25 13	4515.7	2 ⁻
		2488.5	45 5	4354.2	(1) ⁻
		2694.9	34	4147.8	1 ⁺ ,2 ⁺
		2703.0	16	4139.7	(2 ⁻)
		2859.7	48 5	3983.0	3 ⁻
		3069.0	6.3	3773.7	1 ⁻
		3459.3	23 12	3383.3	1 ⁺ ,2 ⁺
		3508.6	10.0 50	3334.0	(1,2,3) ⁺

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J_i^{π}</u>	<u>E_{γ}^{$\dagger \ddagger$}</u>	<u>I_{γ}[#]</u>	<u>E_f</u>	<u>J_f^{π}</u>	<u>Mult.[@]</u>	<u>$\delta^{\textcircled{a}}$</u>	<u>Comments</u>
6842.8	2 ⁻	3713.5	9.4	3129.11	1 ⁺			
		4231.5	12.5 65	2611.04	3 ⁺			
		4262.2	45 5	2580.3	1 ⁺			
		4661.4	6.9	2181.09	3 ⁺			
		4684.6	30.0 30	2157.9	2 ⁺			
		4955.1	35.0 35	1887.29	2 ⁺			
		5612.0	65 7	1230.28	2 ⁺			
		6176.6	35.0 35	665.57	1 ⁺			
		6381.2	100 10	461.01	1 ⁺			
		6695.7	12.5 65	146.36	3 ⁺			
		1856.8	2.5 13	4995.6	1 ⁺ ,2 ⁺			
		2156.6	2.5 13	4695.7	(1 ⁺ ,2,3 ⁺)			
		2498.1	2.5 13	4354.2	(1) ⁻			
		3307.2	5.1	3545.08	3 ⁻			
6852.4	2 ⁺	3723.1	7.5 38	3129.11	1 ⁺			
		4271.8	7.5 38	2580.3	1 ⁺			
		4964.7	58 6	1887.29	2 ⁺			
		5621.6	32.5 33	1230.28	2 ⁺			
		6186.2	15.0 15	665.57	1 ⁺			
		6390.8	100 10	461.01	1 ⁺			
		6705.3	10.0 50	146.36	3 ⁺			
		6851.7	13 6	0.0	0 ⁺			
		1330.1	13.6 14	5541.1	(2 ⁻ to 5 ⁻)			
		1484.4	6.8 34	5386.8	(2,3,4,5) ⁻			
		1699.5	25.0 25	5171.6	4	D(+Q)	+0.02 3	
		2424.5	64 7	4446.6	4 ⁻	M1+E2	+0.05 +1-3	Mult., δ : D+Q from $\gamma(\theta)$ in ³³ S(p, γ). M2 ruled out by RUL.
		2794.9	5.7	4076.2	4 ⁻			
		3239.61	100 10	3631.40	5 ⁻	(M1(+E2))	+0.04 7	Mult., δ : D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =no from level scheme.
6887.97	2	4495.19	18.2 18	2375.67	4 ⁺	(E1(+M2))	+0.04 +3-5	Mult., δ : D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =yes from level scheme.
		1930.4	1.16 58	4957.5	1 ⁺ ,2 ⁺ ,3 ⁺			
		2170.4	2.3 12	4717.5	2 ⁺ ,3 ⁺ ,4 ⁺			
		2278.2	1.16 58	4609.7	(⁻)			
		2282.1	1.16 58	4605.8	(⁻)			
		3504.5	2.3 12	3383.3	1 ⁺ ,2 ⁺			

Adopted Levels, Gammas (continued) $\gamma(^{34}\text{Cl})$ (continued)

<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^{\dagger\dagger}$</u>	<u>$I_\gamma^\#$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[@]</u>	<u>$\delta^@$</u>
6887.97	2	4706.53	1.16 58	2181.09	3 ⁺		
		4729.7	100 10	2157.9	2 ⁺	D+Q	-0.52 +3-2
		5000.29	1.16 58	1887.29	2 ⁺		
		6740.89	5.8 29	146.36	3 ⁺		
6901.75	(1 ⁺ ,2 ⁺)	3518.3	21.1 21	3383.3	1 ⁺ ,2 ⁺		
		4743.5	100 10	2157.9	2 ⁺		
		5670.96	0.81	1230.28	2 ⁺		
		6440.09	1.2	461.01	1 ⁺		
		6754.67	2.6 13	146.36	3 ⁺		
		6901.00	7.9 8	0.0	0 ⁺		
		1376.9	0.48	5541.1	(2 ⁻ to 5 ⁻)		
6918.0	3,4	3271.5	2.4 13	3646.3	5 ⁺		
		3286.4	3.2 16	3631.40	5 ⁻		
		3372.7	0.81	3545.08	3 ⁻		
		3583.8	2.4 13	3334.0	(1,2,3) ⁺		
		4306.7	4.0 21	2611.04	3 ⁺		
		4542.0	40 4	2375.67	4 ⁺		
		4736.6	8.9 10	2181.09	3 ⁺		
		6770.9	100 10	146.36	3 ⁺		
		2213.9	24	4717.5	2 ⁺ ,3 ⁺ ,4 ⁺		
		2321.7	100	4609.7	(⁻)		
6931.5	1 ⁻	2325.6	64	4605.8	(⁻)		
		2415.7	8.2	4515.7	2 ⁻		
		2470.0	9.1	4461.4	(2,3) ⁻		
		2577.2	9.1	4354.2	(1) ⁻		
		2948.4	15	3983.0	3 ⁻		
		3331.2	4.6	3600.14	4 ⁻		
		3386.2	21	3545.08	3 ⁻		
		3548.0	55	3383.3	1 ⁺ ,2 ⁺		
		4210.1	13	2721.09	2 ⁻		
		4350.9	19	2580.3	1 ⁺		
		4773.2	19	2157.9	2 ⁺		
		5043.8	10.0 50	1887.29	2 ⁺		
		6265.3	46	665.57	1 ⁺		
		6784.4	50	146.36	3 ⁺		
		3593.9	11.2 12	3383.3	1 ⁺ ,2 ⁺		
6977.4	(2 ⁻ ,3,4 ⁺)	4819.1	100 10	2157.9	2 ⁺		

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\dagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. [@]	$\delta^@$	Comments
6977.4	(2 ⁻ ,3,4 ⁺)	5746.6	8.2 9	1230.28	2 ⁺			
		6830.3	29.9 30	146.36	3 ⁺			
6986.85	1 ⁻	3003.7	18.3 20	3983.0	3 ⁻			
		3386.53	18.3 20	3600.14	4 ⁻			
		3441.58	19.5 20	3545.08	3 ⁻			
		4265.5	11.0 56	2721.09	2 ⁻			
		4828.6	100 10	2157.9	2 ⁺			
		5756.05	59 6	1230.28	2 ⁺			
		6839.75	18.3 20	146.36	3 ⁺			
7051.6	(1 ⁻ ,2,3 ⁺)	2094.0	39 4	4957.5	1 ⁺ ,2 ⁺ ,3 ⁺			
		3506.3	17.9 89	3545.08	3 ⁻			
		3668.1	36 4	3383.3	1 ⁺ ,2 ⁺			
		4893.3	93 9	2157.9	2 ⁺			
		6385.4	71 7	665.57	1 ⁺			
		6589.9	100 10	461.01	1 ⁺			
7059.01	1,2	2063.3	2.6 14	4995.6	1 ⁺ ,2 ⁺			
		2117.0	1.75 88	4941.9	1 ⁺			
		2363.2	0.35	4695.7	(1 ⁺ ,2,3 ⁺)			
		2449.2	0.86	4609.7	(⁻)			
		2733.00	0.88 44	4325.89	1 ⁺ ,2 ⁺			
		3398.8	1.75 88	3660.0	(0 ⁺ to 4 ⁺)			
		3929.66	4.4 23	3129.11	1 ⁺			
		4337.6	1.75 88	2721.09	2 ⁻			
		4447.66	1.2	2611.04	3 ⁺			
		4478.4	0.88 44	2580.3	1 ⁺			
		4877.54	2.6 14	2181.09	3 ⁺			
		4900.7	0.88 44	2157.9	2 ⁺			
		5171.30	8.8 44	1887.29	2 ⁺			
		5828.19	5.3 26	1230.28	2 ⁺			
		6392.80	44 5	665.57	1 ⁺			
		6597.31	100 10	461.01	1 ⁺			
7078.90	3 ⁻	2473.0	6.3	4605.8	(⁻)			
		2752.89	2.2	4325.89	1 ⁺ ,2 ⁺			
		3002.6	3.6 18	4076.2	4 ⁻			
		3095.8	57 6	3983.0	3 ⁻	(M1(+E2))	0.00 +6-4	Mult., δ : D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =no from level scheme.
		3478.57	61 6	3600.14	4 ⁻	(M1(+E2))	0.00 3	Mult., δ : D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =no from level scheme.

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^{\dagger\ddagger}$	$I_\gamma^\#$	E_f	J_f^π	Mult. [@]	$\delta^@$	Comments
7078.90	3 ⁻	3533.62	54 6	3545.08	3 ⁻	(M1(+E2))	-0.07 +7-4	Mult., δ : D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =no from level scheme. Mult.: D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =no from level scheme. δ : or -3.5 5, both from ³³ S(p, γ).
		4357.51	39 4	2721.09	2 ⁻	(M1(+E2))	-0.03 +2-3	
		4897.43	3.6 18	2181.09	3 ⁺			Mult.: D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =yes from level scheme. Mult.: D(+Q) from $\gamma(\theta)$ in ³³ S(p, γ). $\Delta\pi$ =yes from level scheme.
		5191.19	10.7 54	1887.29	2 ⁺			
		5848.08	100 10	1230.28	2 ⁺	(E1(+M2))	-0.04 +4-2	
		6931.78	28.6 29	146.36	3 ⁺	(E1(+M2))	-0.05 7	
7249.8		1934.8	100	5314.95	7 ⁺			B(E2)(W.u.)=20.1 +44-30 E γ : weighted average of 1935.0 5 from ²⁴ Mg(¹² C,pn γ) and 1935 1 from ²⁷ Al(¹² C, α n γ). Mult.: E2, $\Delta J=2$, from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, α n γ).
7250.0	(9) ⁺	1935.0 5	100	5314.95	7 ⁺	E2		
7675.3	2 ⁺ ,3 ⁺	5516.9		2157.9	2 ⁺	D+Q	0.12 8	B(M1)(W.u.)>0.0048 Mult.: consistent with $\Delta J=0$ from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, α n γ).
		7528.1		146.36	3 ⁺	D+Q	0.32 25	
7699.5	(7) ⁺	2384 1	100	5314.95	7 ⁺	M1		B(M1)(W.u.)>0.016; B(E2)(W.u.)>0.24 E γ : weighted average of 2487.4 10 from ²⁴ Mg(¹² C,pn γ) and 2485 1 from ²⁷ Al(¹² C, α n γ). Mult., δ : D+Q from $\gamma\gamma(\theta)$ (DCO) in ²⁷ Al(¹² C, α n γ). M2 ruled out by RUL.
7801.5	7 ⁺ ,8 ⁺ ,9 ⁺	2486.2 12	100	5314.95	7 ⁺	M1+E2	+0.22 7	B(M1)(W.u.)>0.015; B(E2)(W.u.)>0.32 Mult., δ : M1+E2 from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, α n γ).
8155.5	(6) ⁺	2840 1	100	5314.95	7 ⁺	M1+E2	-0.3 1	B(M1)(W.u.)>0.032 Mult., δ : M1(+E2) consistent with $\Delta J=0$ from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, α n γ).
8305.5	2,3	8158.1	100	146.36	3 ⁺	D+Q	0.27 18	B(M1)(W.u.)>0.032 Mult., δ : M1(+E2) consistent with $\Delta J=0$ from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, α n γ).
9392.7	(7) ⁺	4077 1	100	5314.95	7 ⁺	M1(+E2)	+0.2 5	
10631.7	(11) ⁺	3381 1	100	7250.0	(9) ⁺	E2		B(E2)(W.u.)>7.0 Mult.: E2, $\Delta J=2$, from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\theta)$ (POL) in ²⁷ Al(¹² C, α n γ).

[†] E γ values without uncertainties are deduced from level-energy differences, unless otherwise noted.

[‡] E γ values with uncertainties are from ²⁷Al(¹²C, α n γ), unless otherwise noted.

[#] From ³³S(p, γ), unless otherwise noted.

[@] From $\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ in ³³S(p, γ), and RUL if applicable, unless otherwise noted.

Adopted **Levels, Gammas** (continued)

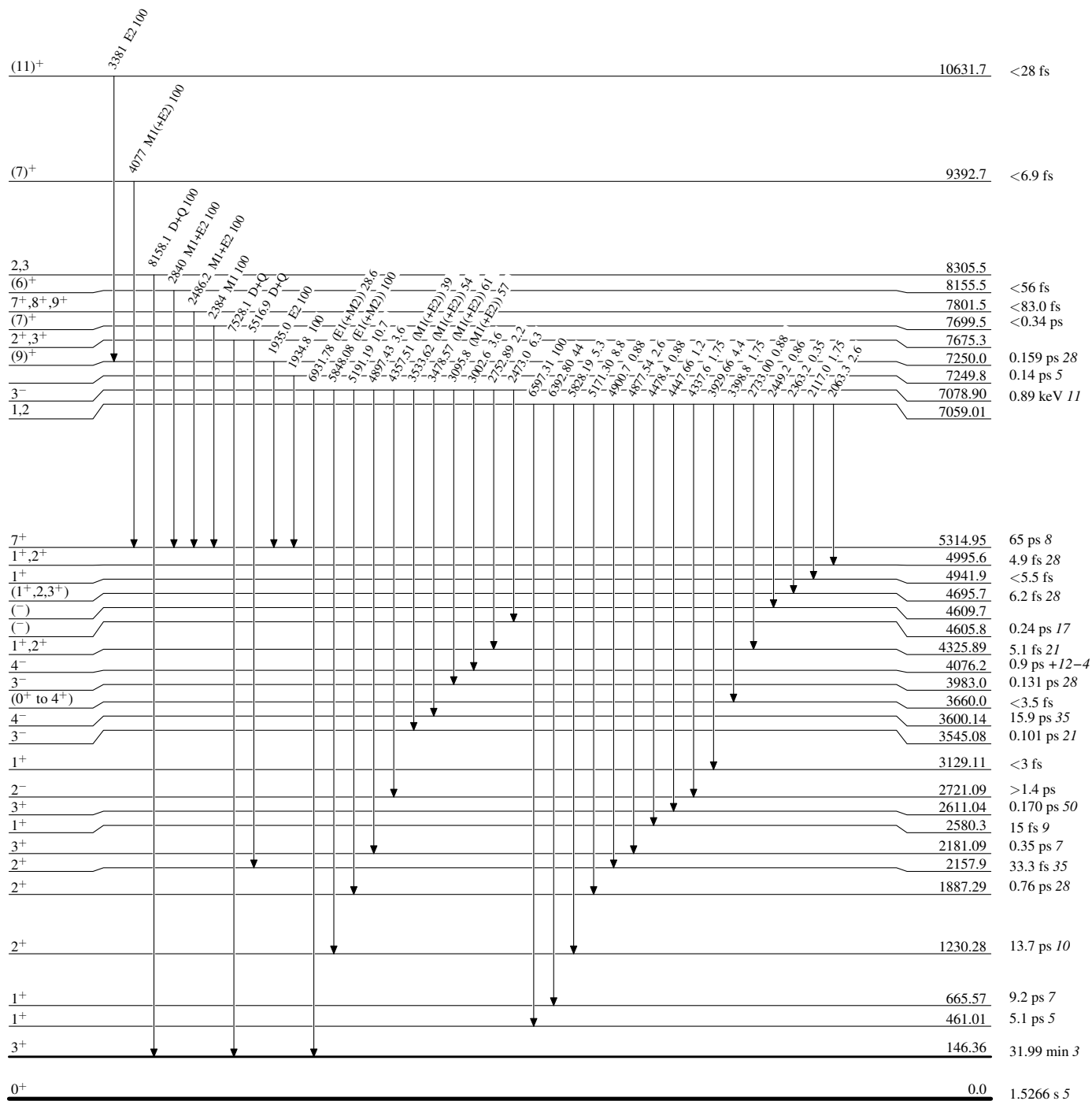
$\gamma(^{34}\text{Cl})$ (continued)

& Deduced by the evaluators from adopted level $T_{1/2}$ and $RUL < 100$ for B(E2)(W.u.).

^a Placement of transition in the level scheme is uncertain.

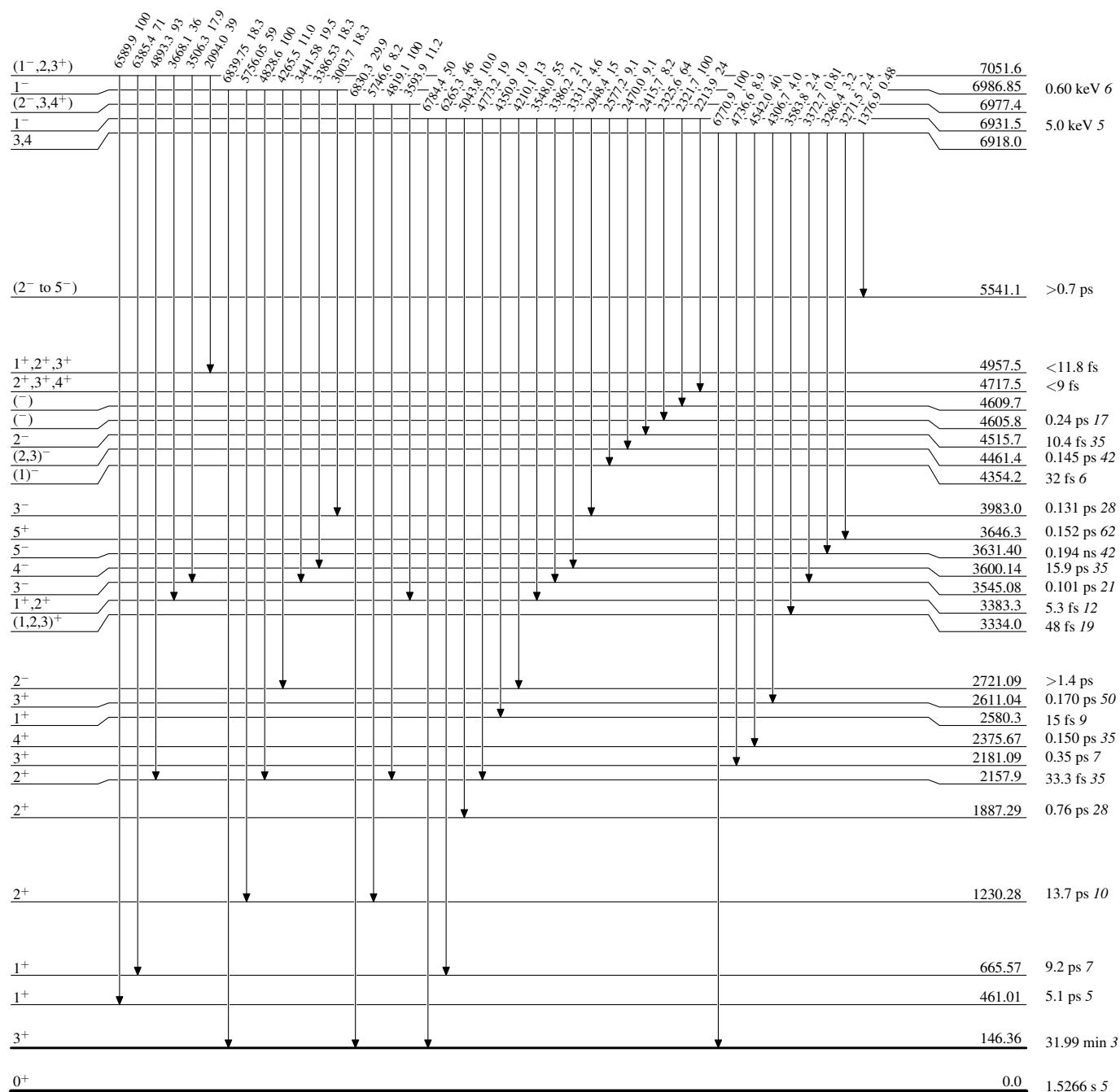
Adopted Levels, Gammas**Level Scheme**

Intensities: Relative photon branching from each level



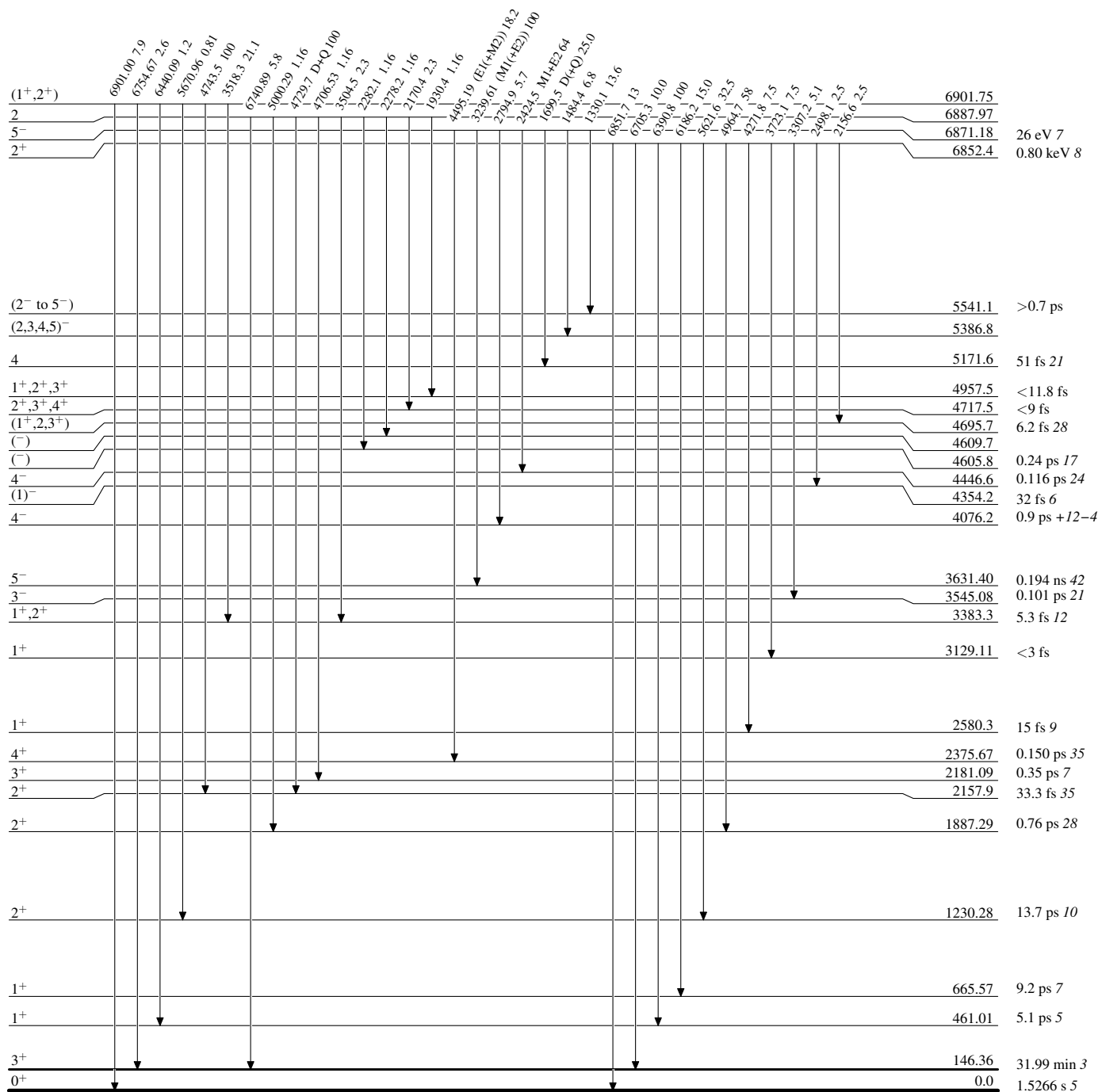
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

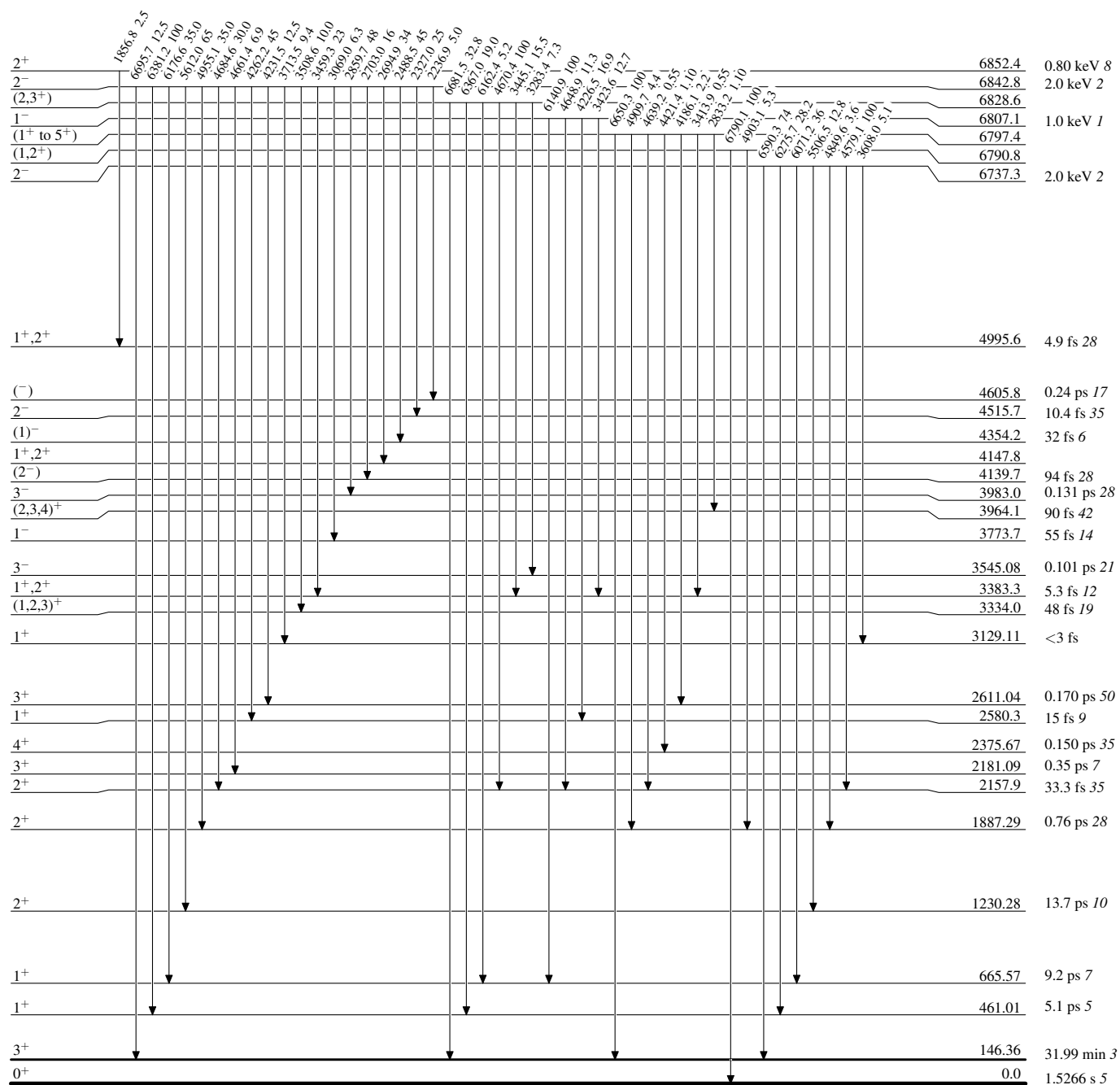
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

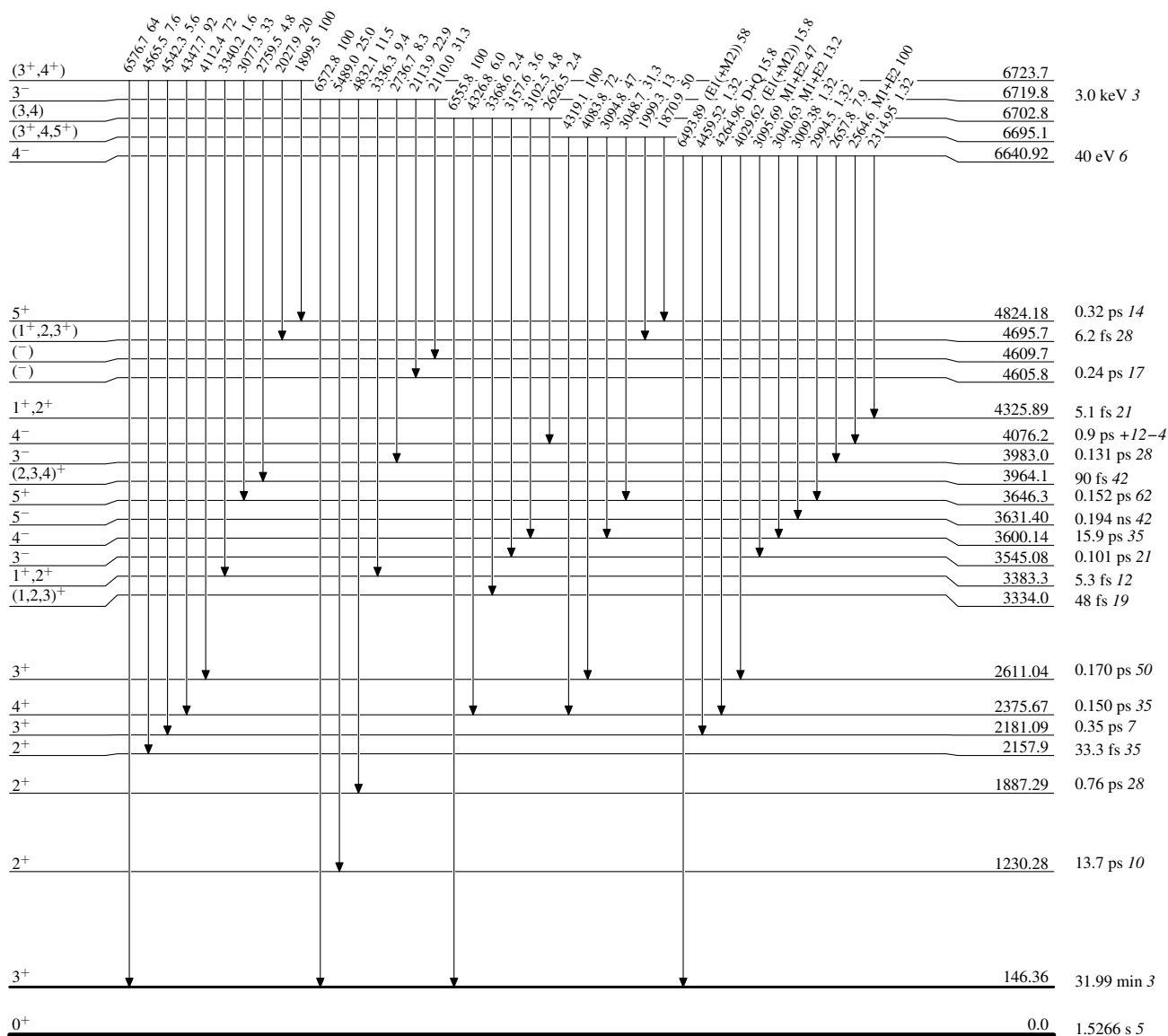
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

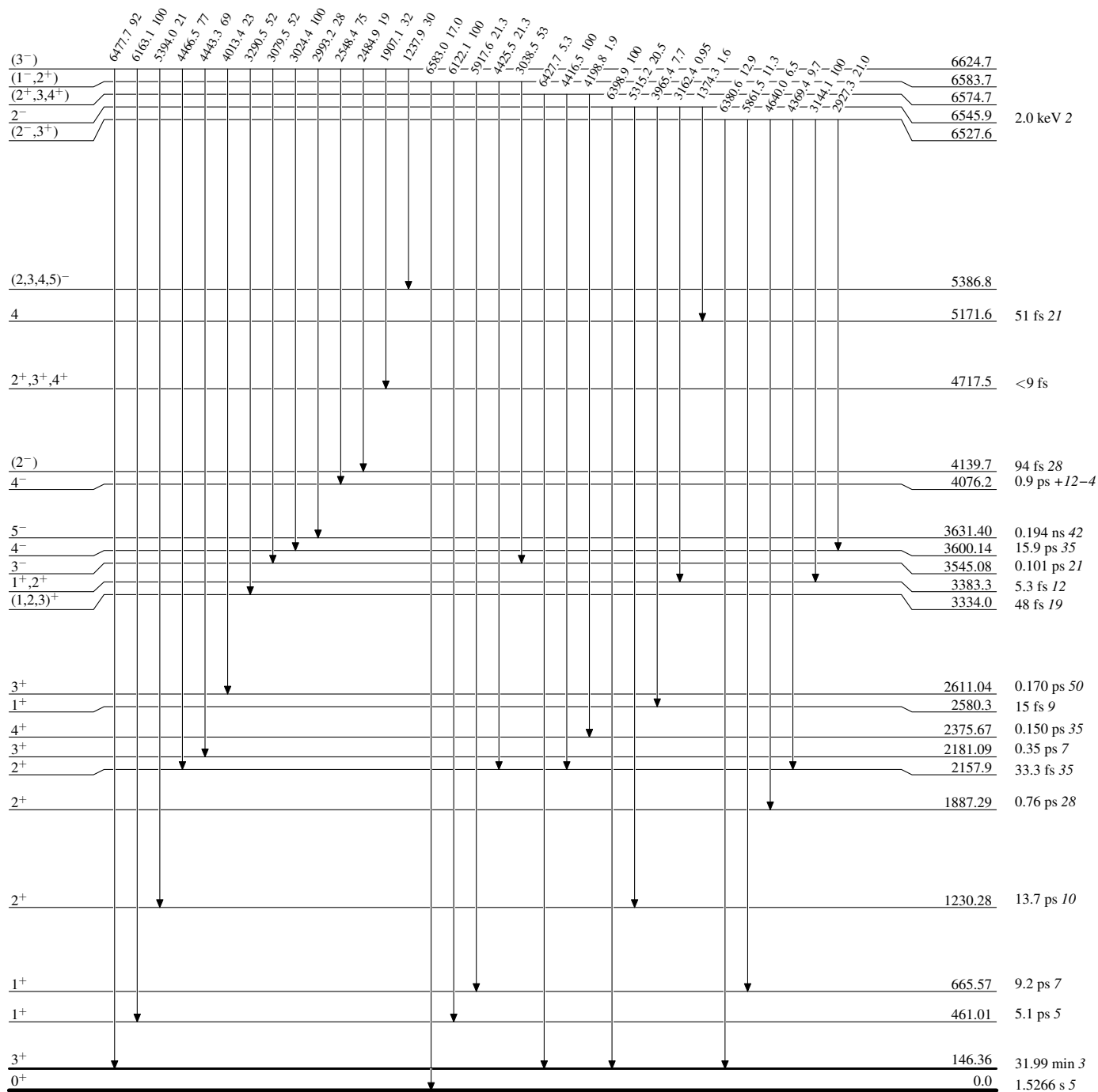
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



Adopted Levels, Gammas**Level Scheme (continued)**

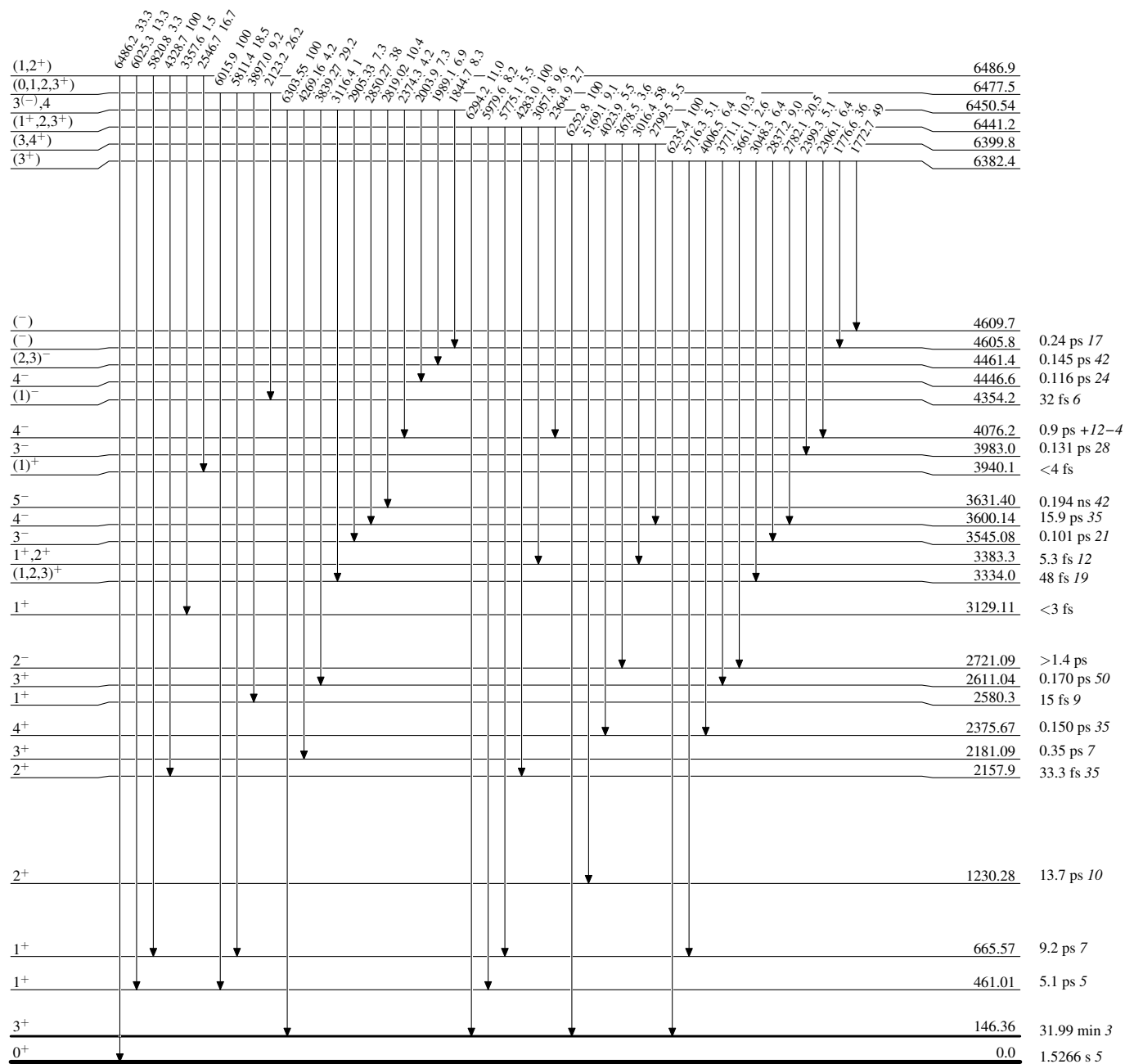
Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

Adopted Levels, Gammas

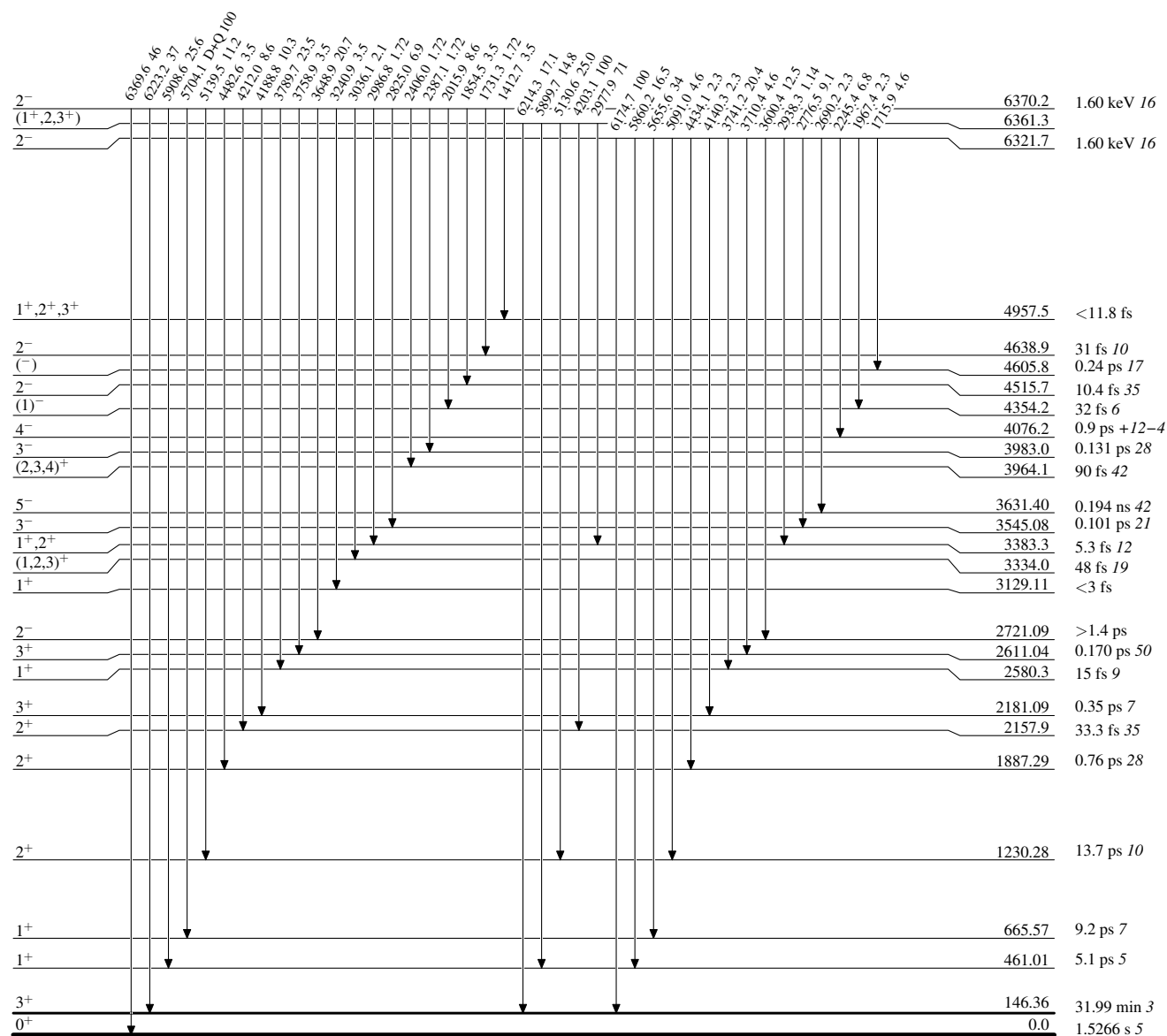
Level Scheme (continued)

Intensities: Relative photon branching from each level



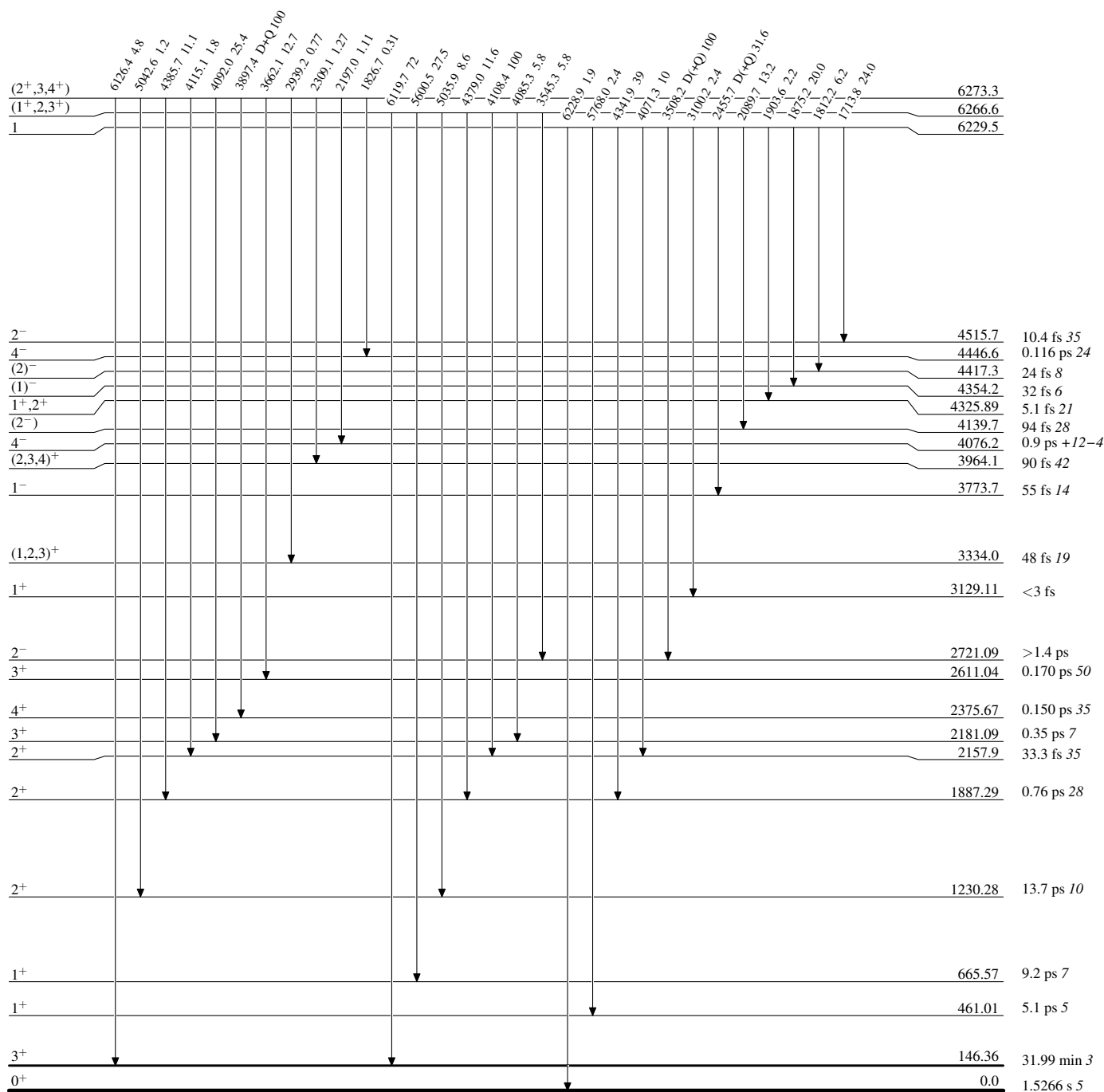
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

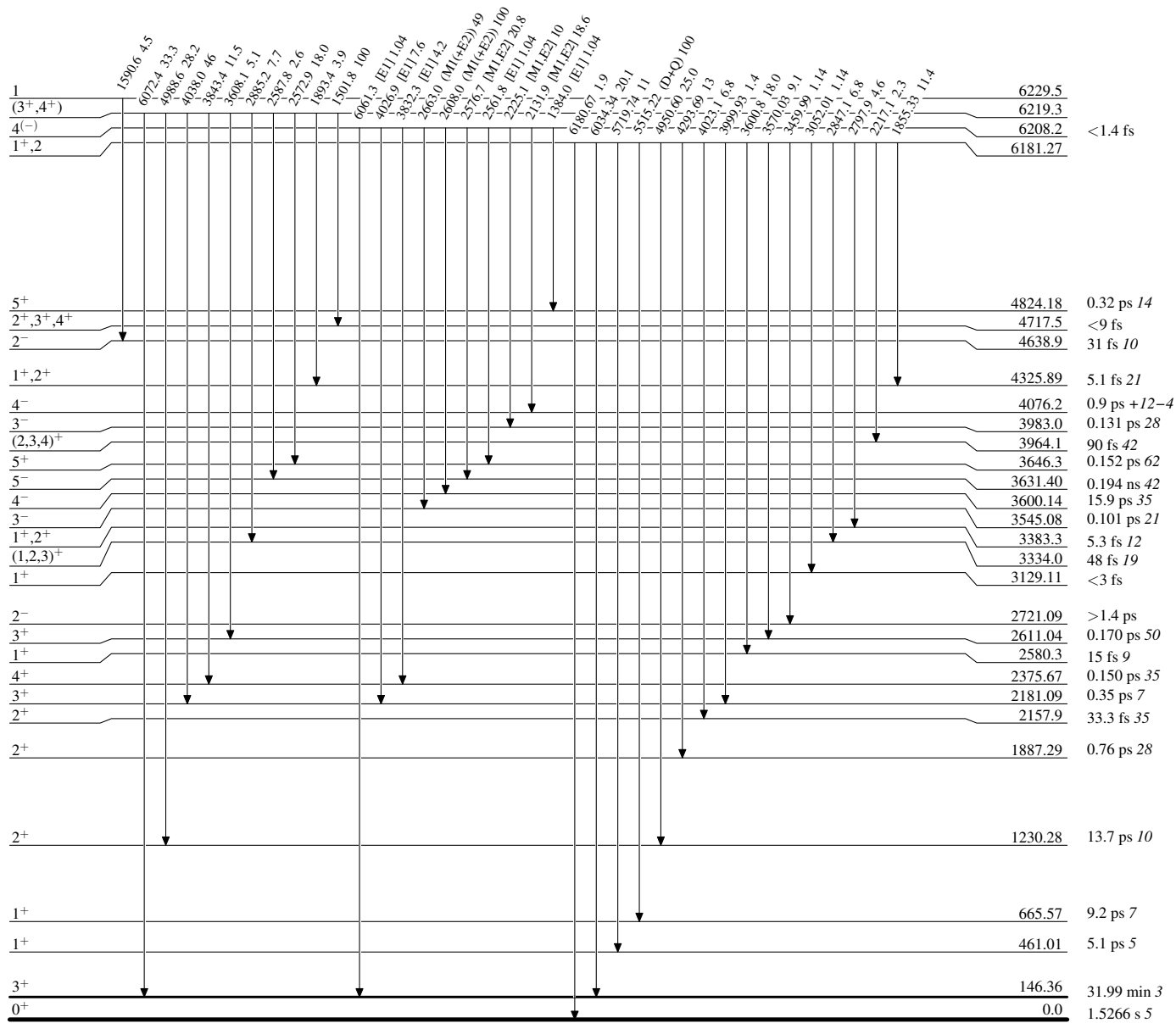
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

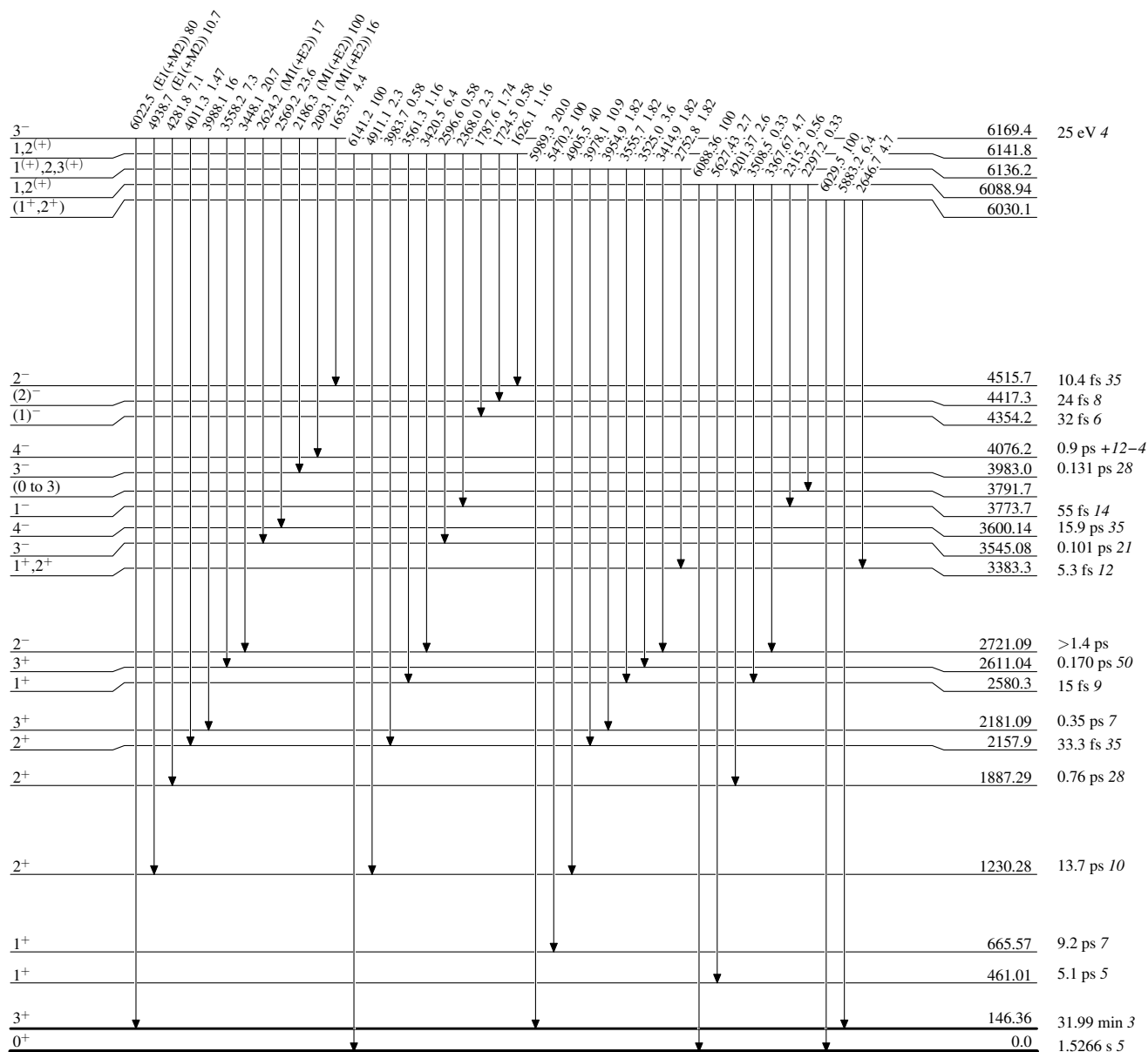
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

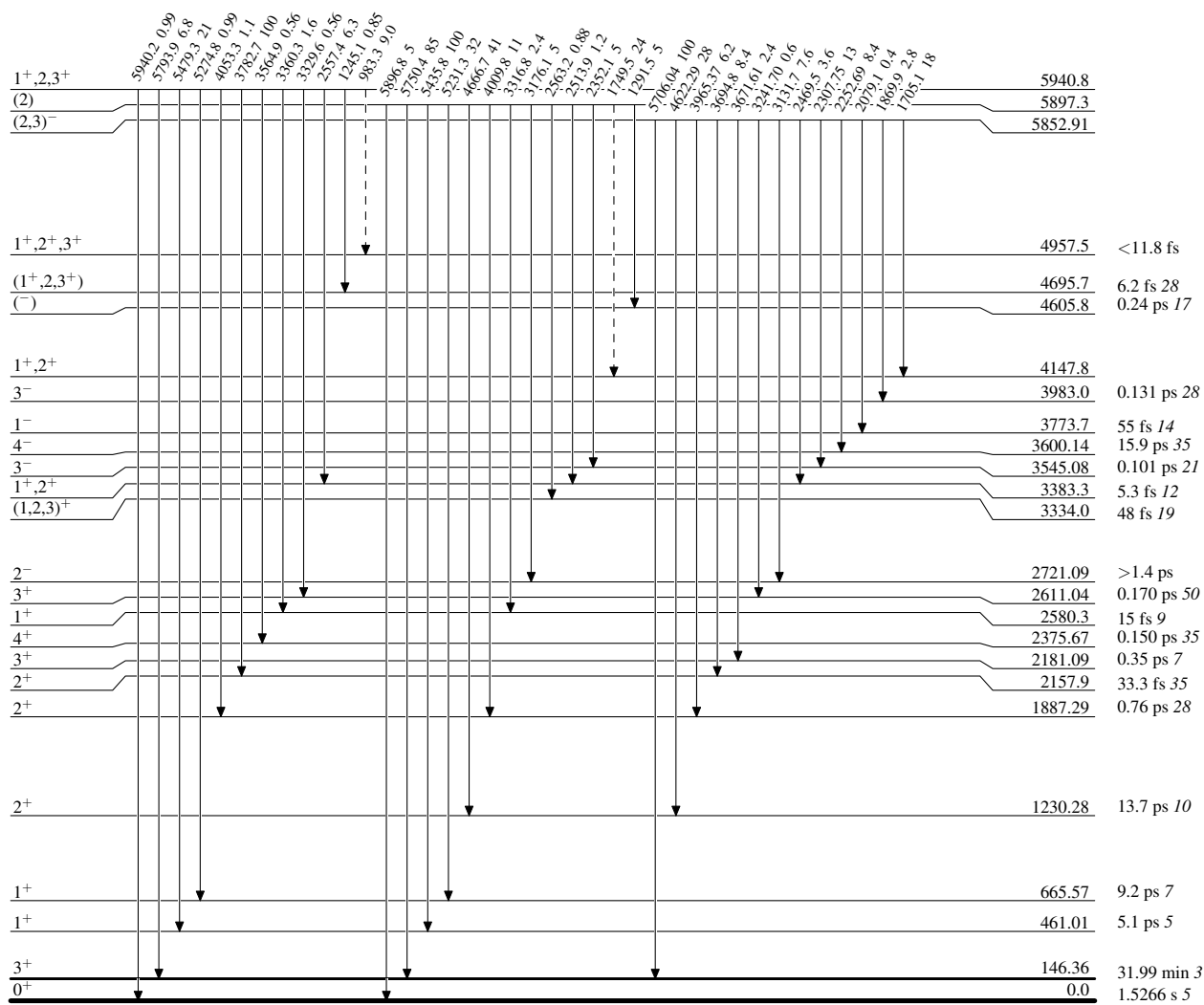
 $^{34}_{17}\text{Cl}_{17}$

Adopted Levels, Gammas

Legend

Level Scheme (continued)

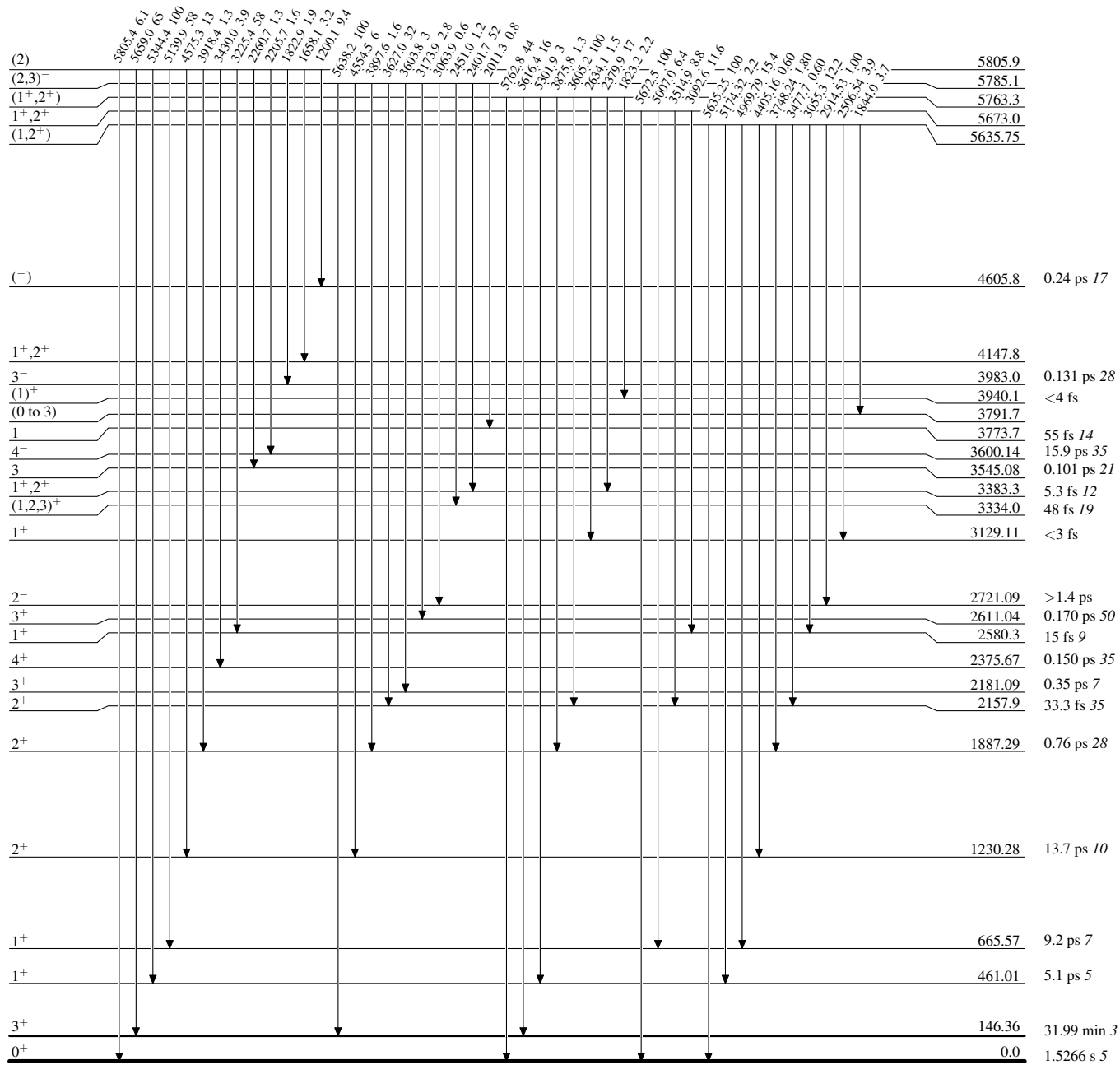
Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

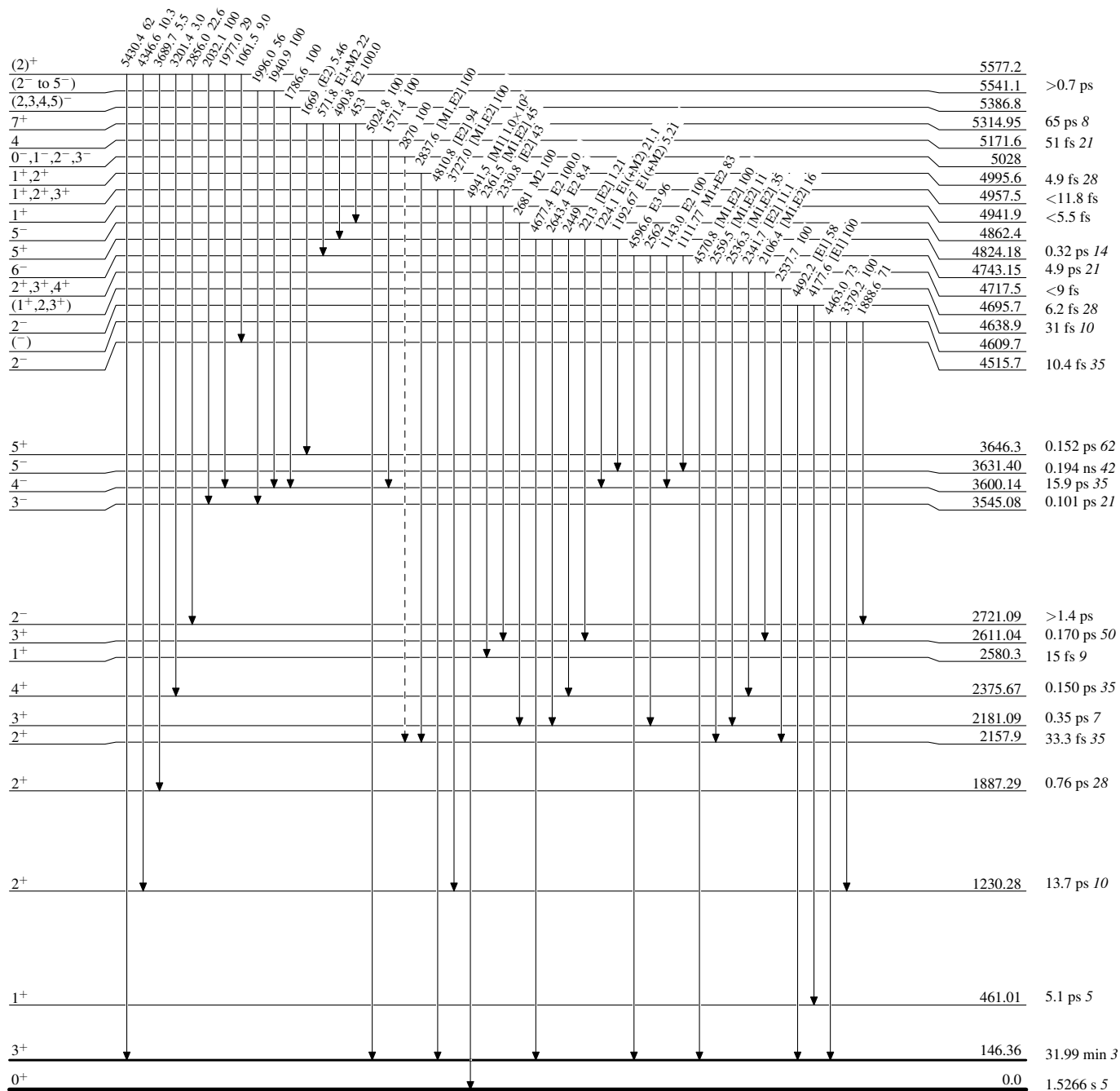


Adopted Levels, Gammas

Legend

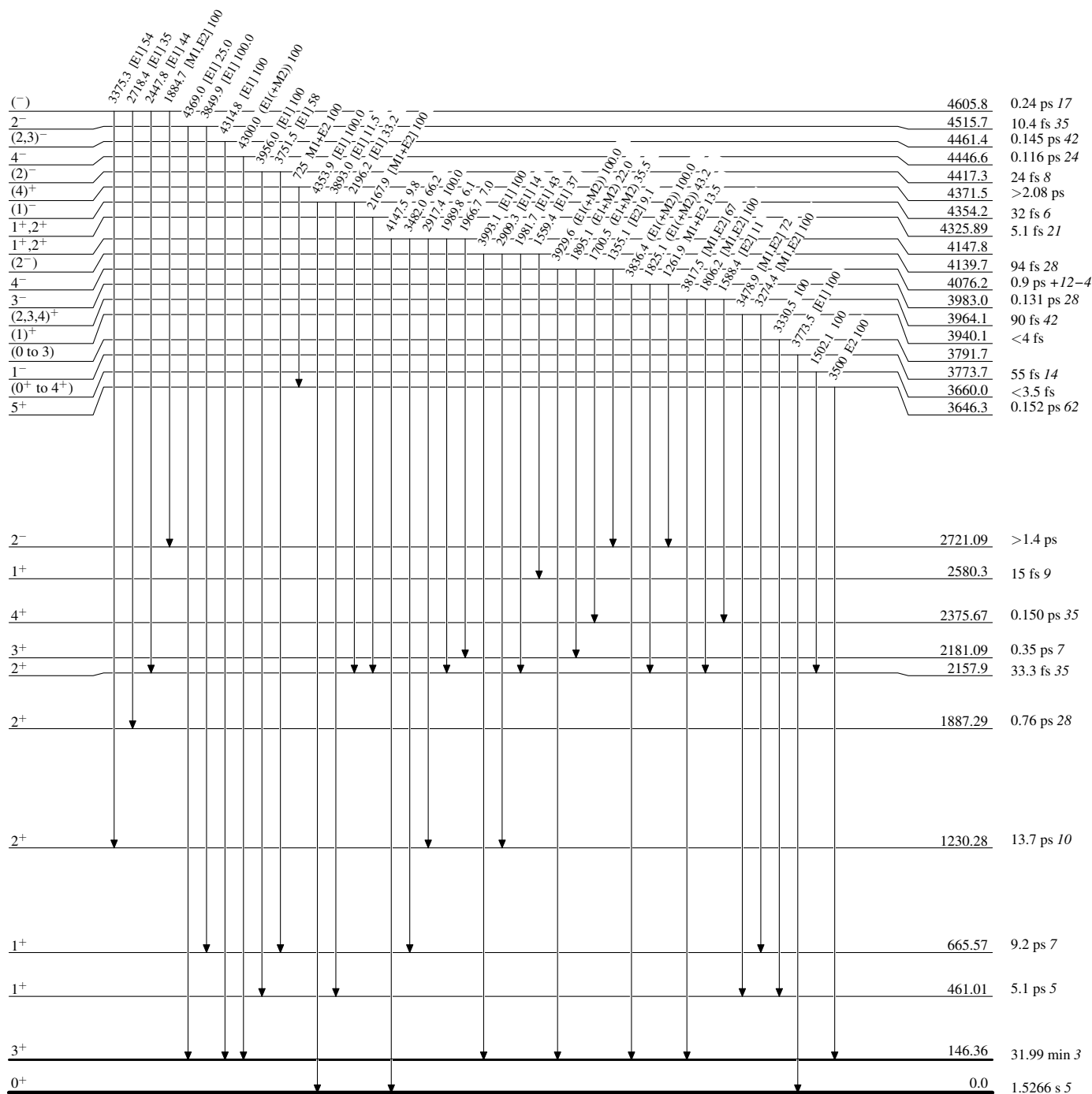
Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)


Adopted Levels, Gammas**Level Scheme (continued)**

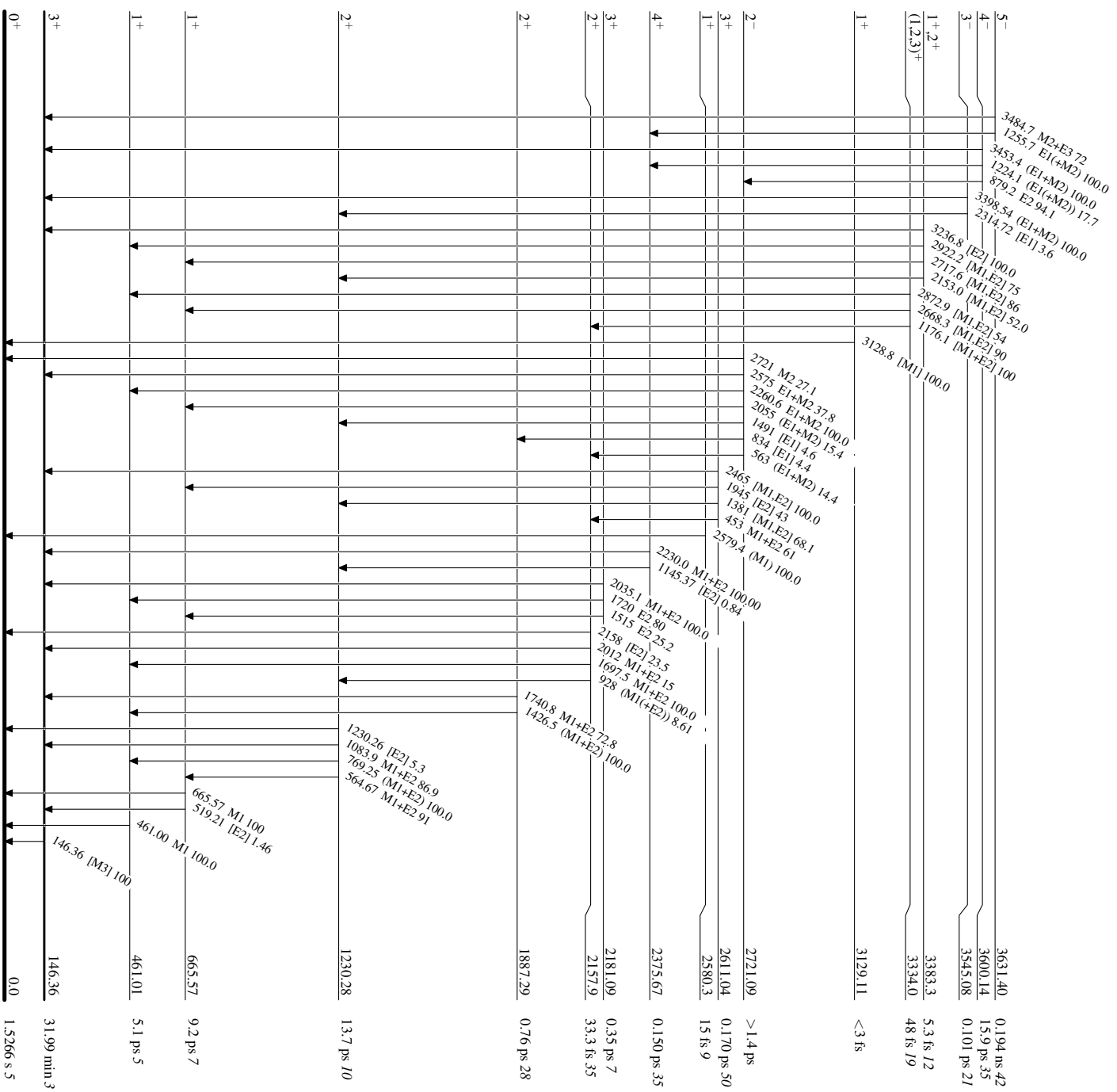
Intensities: Relative photon branching from each level

 $^{34}_{17}\text{Cl}_{17}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level



^{34}Ar $\varepsilon+\beta^+$ decay (0.84646 s) 2020Ia01,2020Ia02

Parent: ^{34}Ar : $E=0.0$; $J^\pi=0^+$; $T_{1/2}=0.84646$ s 35; $Q(\varepsilon+\beta^+)=6061.79$ 6; % $\varepsilon+\beta^+$ decay=100

^{34}Ar - J^π , $T_{1/2}$: From the Adopted Levels of ^{34}Ar . The Adopted $T_{1/2}$ of ^{34}Ar is determined by 2020Ia01 in this dataset.

^{34}Ar - $Q(\varepsilon+\beta^+)$: From 2021Wa16.

2020Ia01: a 30-MeV/nucleon ^{35}Cl beam was produced from the K500 cyclotron at Texas A&M University and impinged on a liquid-nitrogen-cooled hydrogen gas target. The fully stripped reaction products recoiled into the Momentum Achromat Recoil Spectrometer (MARS), where they were spatially separated according to their charge-to-mass ratio. A 99.5% pure ^{34}Ar beam was implanted into aluminized Mylar tape of a fast tape-transport system and then moved to a well-shielded location 90 cm away. Decay positrons were detected using a 4π proportional gas counter. Nearly 12000 collect-move-detect cycles were repeated and a total of 1.7×10^8 decay positrons were recorded. Measured $E\beta$, $\beta(t)$. Deduced $T_{1/2}$ of ^{34}Ar .

2020Ia02: similar setup as 2020Ia01. Nearly 70000 collect-move-detect cycles were repeated. A total of 2.6710×10^8 20 decay positrons were recorded. Decay positrons were detected using a 1-mm-thick BC-404 scintillator and γ rays were detected using a HPGe detector. Measured $E\beta$, $E\gamma$, $\beta\gamma$ -coin. Deduced levels, decay branching ratios, $\log ft$, Gamow-Teller strengths. Comparisons with shell-model calculations.

2006Ia05: similar setup as 2020Ia01. Measured $T_{1/2}$ of ^{34}Ar .

1974Ha26: ^{34}Ar was produced via the $^{32}\text{S}(^3\text{He},n)$ reaction with a 12-MeV ^3He beam from the Chalk River MP tandem accelerator, fast gas-transport system, Ge(Li) detector. Measured $I\gamma$, $E\gamma$, $T_{1/2}$, and absolute $\varepsilon+\beta^+$ branching ratios.

Others: 1972Ha58 ($T_{1/2}$), 1971Mo27 ($I\gamma$, $E\gamma$, $T_{1/2}$).

The decay scheme is considered fairly complete. Shell-model calculations estimate the possible unobserved β -feeding intensities to higher excited states in ^{34}Cl to be <0.005%, more than an order-of-magnitude less than the quoted uncertainty (2020Ia02).

 ^{34}Cl Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\ddagger$
0	0^+	1.5266 s 5
146.36	3^+	31.99 min 3
460.8 10	1^+	5.1 ps 5
665.8 10	1^+	9.2 ps 7
1230	2^+	13.7 ps 10
1887	2^+	0.76 ps 28
2158	2^+	33.3 fs 35
2181	3^+	0.35 ps 7
2579.5 14	1^+	15 fs 9
3129.0 10	1^+	<3 fs

† Level energies with uncertainties are from 1974Ha26 based on measured $E\gamma$. Level energies without uncertainties are rounded from the Adopted Levels.

‡ From the Adopted Levels.

 ε, β^+ radiations

$E(\text{decay})$	$E(\text{level})$	$I\beta^+^\ddagger$	$I\varepsilon^\ddagger$	$\log ft$	$I(\varepsilon+\beta^+)^\dagger^\ddagger$	Comments
(2932.8 14)	3129.0	1.26 4	0.0159 5	3.48 2	1.28 4	$I(\varepsilon+\beta^+)$: original value with asymmetric uncertainties=1.28 +4-2 (2020Ia02).
(3482.3 17)	2579.5	0.85 3	0.0050 2	4.13 2	0.85 3	$I(\varepsilon+\beta^+)$: original value with asymmetric uncertainties=0.85 +3-2 (2020Ia02).
(5396.0 14)	665.8	2.52 4	0.00268 4	4.78 1	2.52 4	$I(\varepsilon+\beta^+)$: original value with asymmetric uncertainties=2.52 +4-3 (2020Ia02) is obtained by multiplying the γ/β normalization factor of 0.02480 28 to the relative β^+ +ec feeding of 1.015 +10-5 to the 665.8 level, which was deduced by taking into account the possible feeding and deexcitation γ branchings of this level. The correction for the electron-capture-to-positron ratio (ec/ β^+) is negligible.
(5601.0 14)	460.8	0.88 4	8.2×10^{-4} 4	5.33 2	0.88 4	$I(\varepsilon+\beta^+)$: original value with asymmetric

Continued on next page (footnotes at end of table)

^{34}Ar $\varepsilon+\beta^+$ decay (0.84646 s) **2020Ia01,2020Ia02** (continued) ε, β^+ radiations (continued)

E(decay)	E(level)	$I\beta^+$ ‡	$I\varepsilon$ ‡	Log ft	$I(\varepsilon+\beta^+)$ ‡‡	Comments
(6061.8 14)	0	94.41 8	0.06647 6	3.4931 6	94.48 8	uncertainties=0.88 +2-4 (2020Ia02).

‡ Deduced from γ -ray intensity balance, relative to $I\gamma(666)=100$, normalized to absolute $\%I\gamma(666\gamma)=0.0248\ 28$, which was deduced from the measured counts of γ rays and positrons.

‡‡ Absolute intensity per 100 decays.

 $\gamma(^{34}\text{Cl})$

$I\gamma$ normalization: The number of β -coincident 666 γ relative to the total number of positrons emitted from the decay of ^{34}Ar was deduced based on the measured numbers of $\beta\gamma$ coincidences and β singles. See 2020Ia02 for meticulous corrections on detector efficiency, impurity, pile-up, summing, dead time, random coincidence, and background.

E_{γ} ‡	I_{γ} $^{\ddagger\textcircled{a}}$	$E_i(\text{level})$	J_i^{π}	E_f	J_f^{π}	Mult. $^{\#}$	α $^{\&}$	Comments
205 a	<1.0	665.8	1 $^{+}$	460.8	1 $^{+}$	[M1,E2]	0.006 5	$\%I\gamma<0.0251$ I_{γ} : <1.3 (2020Ia02).
315 a	<0.18	460.8	1 $^{+}$	146.36	3 $^{+}$	[E2]	2.14×10^{-3} 3	$\%I\gamma<0.00451$ I_{γ} : <0.86 (2020Ia02).
460.8 10	35.4 9	460.8	1 $^{+}$	0	0 $^{+}$	M1	0.0001974 29	$\%I\gamma=0.878\ 25$ I_{γ} : 35.3 9 (2020Ia02), 36.5 36 (1974Ha26).
519	1.46 19	665.8	1 $^{+}$	146.36	3 $^{+}$	[E2]	0.000376 5	$\%I\gamma=0.036\ 5$ I_{γ} : 2.9 10 (2020Ia02).
549 a	<0.45	3129.0	1 $^{+}$	2579.5	1 $^{+}$	[M1,E2]	2.2×10^{-4} 9	$\%I\gamma<0.0113$
665.8 10	100	665.8	1 $^{+}$	0	0 $^{+}$	M1	9.09×10^{-5} 13	$\%I\gamma=2.5$
693 a	<0.26	2579.5	1 $^{+}$	1887	2 $^{+}$	[M1,E2]	1.19×10^{-4} 35	$\%I\gamma<0.00652$
948 a	<0.33	3129.0	1 $^{+}$	2181	3 $^{+}$	[E2]	6.55×10^{-5} 9	$\%I\gamma<0.00828$
971 a	<0.34	3129.0	1 $^{+}$	2158	2 $^{+}$	[M1,E2]	5.2×10^{-5} 9	$\%I\gamma<0.00853$
1242 a	<0.56	3129.0	1 $^{+}$	1887	2 $^{+}$	[M1,E2]	4.5×10^{-5} 6	$\%I\gamma<0.0140$
1350 a	<0.34	2579.5	1 $^{+}$	1230	2 $^{+}$	[M1,E2]	6.1×10^{-5} 9	$\%I\gamma<0.0103$ I_{γ} : <0.41 (2020Ia02).
1899 a	<0.55	3129.0	1 $^{+}$	1230	2 $^{+}$	[M1,E2]	0.000249 31	$\%I\gamma<0.0138$
1915 a	<0.32	2579.5	1 $^{+}$	665.8	1 $^{+}$	[M1,E2]	0.000256 31	$\%I\gamma<0.00803$
2119 a	<0.35	2579.5	1 $^{+}$	460.8	1 $^{+}$	[M1,E2]	0.00035 4	$\%I\gamma<0.00878$
2434 a	<0.26	2579.5	1 $^{+}$	146.36	3 $^{+}$	[E2]	0.000535 7	$\%I\gamma<0.00652$
2464 a	<0.27	3129.0	1 $^{+}$	665.8	1 $^{+}$	[M1,E2]	0.00050 5	$\%I\gamma<0.00677$
2579.4 14	34.1 6	2579.5	1 $^{+}$	0	0 $^{+}$	(M1)	0.000499 7	$\%I\gamma=0.846\ 18$ I_{γ} : 33.8 8 (2020Ia02), 34.5 10 (1974Ha26).
2668 a	<0.29	3129.0	1 $^{+}$	460.8	1 $^{+}$	[M1,E2]	0.00059 5	$\%I\gamma<0.00727$
2983	<0.22	3129.0	1 $^{+}$	146.36	3 $^{+}$	[E2]	0.000784 11	$\%I\gamma<0.00552$
3128.8 10	51.5 8	3129.0	1 $^{+}$	0	0 $^{+}$	[M1]	0.000720 10	$\%I\gamma=1.277\ 25$ I_{γ} : 51.1 10 (2020Ia02), 52.1 12 (1974Ha26).

‡ No E_{γ} with uncertainty is given by any paper in this dataset. If a γ transition is to the g.s. and also the strongest γ branch, the E_{γ} is deduced from the level energy. Other E_{γ} values are deduced from the level energy differences.

$^{\ddagger\textcircled{a}}$ I_{γ} are from 2020Ia02, which takes into account their own data, 1974Ha26, 1990En08, and an unpublished (p, γ) I_{γ} data by one of the authors of 2020Ia02.

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^{34}Ar $\varepsilon+\beta^+$ decay (0.84646 s) [2020Ia01,2020Ia02](#) (continued)

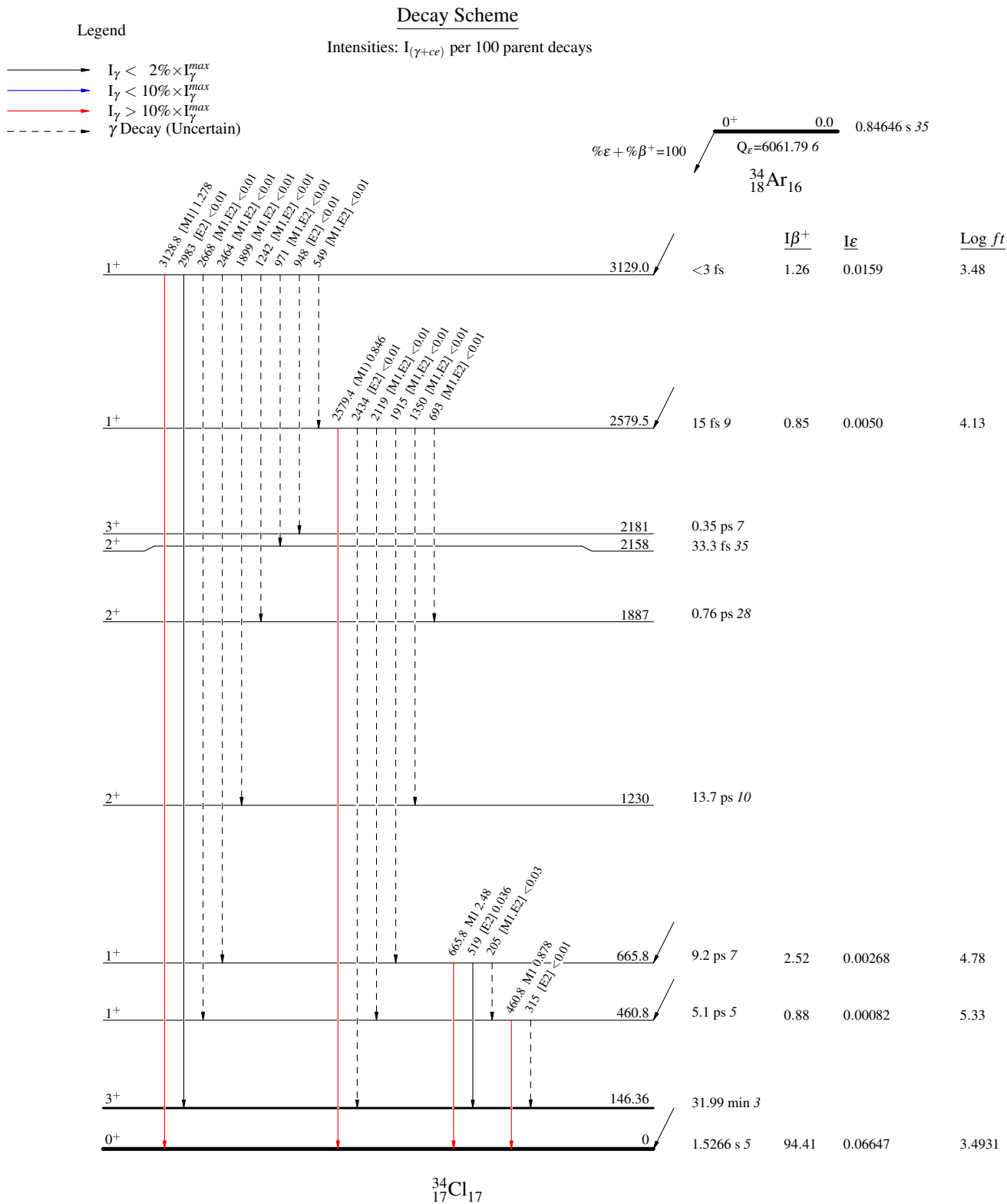
$\gamma(^{34}\text{Cl})$ (continued)

From the Adopted Gammas.

@ For absolute intensity per 100 decays, multiply by 0.02480 28.

& Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

^a Placement of transition in the level scheme is uncertain.

^{34}Ar $\varepsilon + \beta^+$ decay (0.84646 s) 2020Ia01,2020Ia02

^{34}Cl IT decay (31.99 min) 1980Wi13,1975Va02

Parent: ^{34}Cl : $E=146.36$ 3; $J^\pi=3^+$; $T_{1/2}=31.99$ min 3; %IT decay=46.0 4

^{34}Cl -E, J^π , $T_{1/2}$: From the Adopted Levels.

^{34}Cl -%IT decay: Relative γ intensities in $^{34}\text{Cl}^m$ decay: $I(\gamma \text{ to } ^{34}\text{S g.s.})=129.3$ 12 and $I(146\gamma \text{ to } ^{34}\text{Cl g.s.})=94.4$ 14 measured by 1980Wi13. $I(\varepsilon+\beta^+)$ feeding from 3^+ , $^{34}\text{Cl}^m$ to 0^+ , $^{34}\text{S g.s.}$ is zero. The $^{34}\text{Cl}^m$ decay scheme is considered fairly complete. $I(146\text{ce to } ^{34}\text{Cl g.s.})=15.6$ 3 calculated by evaluators using $I(146\gamma \text{ to } ^{34}\text{Cl g.s.})=94.4$ (1980Wi13) and conversion coefficient $\alpha=0.1656$ 24 (2008Ki07). Based on $I(146\gamma+\text{ce to } ^{34}\text{Cl g.s.})=110.0$ 17 and $I(\gamma \text{ to } ^{34}\text{S g.s.})=129.3$ 12, evaluators deduce the decay branching ratios: %IT=46.0 4 and % $I(\varepsilon+\beta^+)=54.0$ 4.

The isomeric transition of the ^{34}Cl 146-keV metastable state.

1980Wi13: $^{34}\text{Cl}^m$ was produced via the $^{31}\text{P}(\alpha,n)$ reaction with 10 and 11-MeV α beams from the ONR-CIT tandem accelerator impinging on MnP powder pressed in Be, Zr, and Al foils. γ rays were detected using a 100-cm³ Ge(Li) detector. Measured E_γ , I_γ . Deduced ^{34}S levels, $T_{1/2}$, decay branching ratios, and log ft for $^{34}\text{Cl}^m \rightarrow ^{34}\text{S}$ decay. Compared with shell-model calculations.

1975Va02: $^{34}\text{Cl}^m$ was produced via the $^{24}\text{Mg}(^{12}\text{C},pn)$ reaction with a 35-MeV ^{12}C beam from the Utrecht EN tandem accelerator. After one hour of bombardment, the radioactivity was detected using a 100-cm² Ge(Li) detector and a 23×28-cm² NaI(Tl) detector for Compton suppression. Measured E_γ , I_γ . Deduced levels, $T_{1/2}$, decay branching ratios, and log ft for $^{34}\text{Cl}^m \rightarrow ^{34}\text{S}$ decay. Also briefly reported ^{34}Cl ground-state decay.

1983Kr06: $^{34}\text{Cl}^m$ was produced via the $^{35}\text{Cl}(p,pn)$ reaction with 25-MeV protons from the isochronous cyclotron of the Karlsruhe Research Center impinging on a NaCl target on a thin gold backing. Measured β spectra using the iron-free magnetic double-lens spectrometer. Deduced end-point energies, β -decay branching ratios, and Fermi and Gamow-Teller shape factors.

2006Ia05: Measured $T_{1/2}$ at Cyclotron Institute, Texas A&M University.

1983Ko22: $^{12}\text{C}(^{28}\text{Si},3p3n)$ $E=115$ MeV at Chalk River. Tape transport system. 4 π proportional counter. Mass-separated sources. Measured $T_{1/2}$.

1982Gr07: $^{35}\text{Cl}(p,d)$ $E=72$ MeV at Schweizerisches Institut für Nuklearforschung (SIN) in Villigen, Switzerland. RbCl target. Ge(Li) detector. Measured $T_{1/2}$.

1973Ry01: $^{31}\text{P}(\alpha,n\gamma)$ $E=6.5, 6.65$ MeV. Plastic scintillator, multi-scaling decay spectra. Measured $T_{1/2}$, $E\beta$, Q value.

1972Ha82: $^{32}\text{S}(^3\text{He},n)$ $E=3.2$ MeV. NE102 plastic scintillator. Measured $T_{1/2}$, $E\beta$.

1971Sn01: $^{32}\text{S}(^3\text{He},p\gamma)$ $E=6.8$ MeV. Measured E_γ , I_γ , DSAM.

1971Wa04: $^{35}\text{Cl}(n,2n\gamma)^{34}\text{Cl}$ $E=14.8$ MeV. Ge(Li), NaI(Tl), and plastic detectors. Measured $T_{1/2}$, E_γ , I_γ , $E\beta$, $I\beta$.

1965Eb01: $^{35}\text{Cl}(n,2n\gamma)^{34}\text{Cl}$ $E=14.7$ MeV, natural target, NaI(Tl) detectors. Measured $T_{1/2}$.

 ^{34}Cl Levels

$E(\text{level})^\dagger$	$J^\pi^\dagger$	$T_{1/2}^\dagger$	Comments
0.0	0^+	1.5266 s 5	
146.36 3	3^+	31.99 min 3	%IT=46.0 4; % ε +% β^+ =54.0 4 1983Kr06 measured the $^{34}\text{Cl}^m$ and $^{34}\text{Cl g.s.}$ β^+ branchings and deduced %IT=44.1 6 based on the $^{34}\text{Cl g.s.}$ β^+ branching and % $I(\varepsilon+\beta^+)=55.5$ 10 based on two $^{34}\text{Cl}^m$ β^+ branchings. 1980Wi13 measured decay branching ratios %IT=44.5 7 and % $I(\varepsilon+\beta^+)=55.5$ 7 based on their relative γ intensities in $^{34}\text{Cl}^m$ decay: $I(\gamma \text{ to } ^{34}\text{S g.s.})=129.3$ 12 and $I(146\gamma \text{ to } ^{34}\text{Cl g.s.})=94.4$ 14. 1975Va02 measured decay branching ratios %IT=46.9 10 and % $I(\varepsilon+\beta^+)=53.1$ 10 based on their relative γ intensities in $^{34}\text{Cl}^m$ decay: $I(\gamma \text{ to } ^{34}\text{S g.s.})=125.9$ 12 and $I(146\gamma \text{ to } ^{34}\text{Cl g.s.})=101$ 4. Both 1980Wi13 and 1975Va02 used $\alpha=0.100$ 9 calculated by 1971Wa04.

† From the Adopted Levels.

³⁴Cl IT decay (31.99 min) 1980Wi13,1975Va02 (continued)

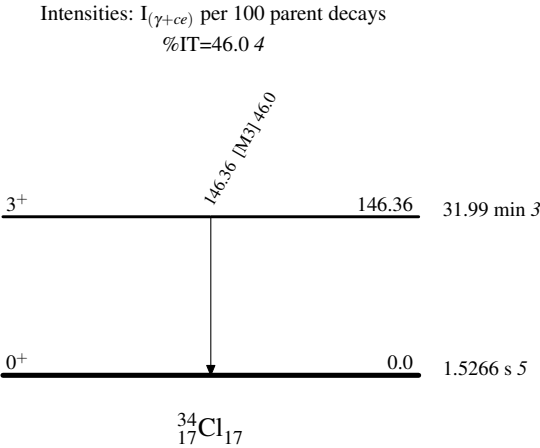
γ(³⁴Cl)

<u>E_γ[†]</u>	<u>I_γ[‡]</u>	<u>E_i(level)</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[†]</u>	<u>α[#]</u>	<u>I_(γ+ce)[‡]</u>	<u>Comments</u>
146.36 3	85.79	146.36	3 ⁺	0.0	0 ⁺	[M3]	0.1656 24	100	I _γ : deduced by evaluators from I(γ+ce)=100 and α=0.1656 24.

[†] From the Adopted Gammas.
[‡] For absolute intensity per 100 decays, multiply by 0.460 4.
[#] Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ-ray energies, assigned multiplicities, and mixing ratios, unless otherwise specified.

³⁴Cl IT decay (31.99 min) 1980Wi13,1975Va02

Decay Scheme



^{35}K εp decay (175 ms) 1980Ew02,2019ChZU

Parent: ^{35}K : $E=0$; $J^\pi=3/2^+$; $T_{1/2}=175$ ms 2; $Q(\varepsilon\text{p})=11874.4$ 9; $\% \varepsilon\text{p}$ decay=0.37 15

^{35}K - J^π , $T_{1/2}$: From the Adopted Levels of ^{35}K (2026Su18). Adopted $T_{1/2}$ from weighted average of 175 ms 2 (2018Sa54), 178 ms 8 (1998Sc19), and 190 ms 30 (1980Ew02) in this dataset.

^{35}K - $Q(\varepsilon\text{p})$: From 2021Wa16.

^{35}K - $\% \varepsilon\text{p}$ decay: 1980Ew02 reports $\% \text{SI}(\text{p})=0.37$ 15 for $E(\text{p})>0.9$ MeV. Other proton branches have also been observed in 2018Sa54 and 2019ChZU, but only one weak $\% \text{I}(\text{p})=2.5 \times 10^{-3}$ is reported in 2019ChZU. The evaluators consider the $\% \text{SI}(\text{p})=0.37$ 15 from 1980Ew02 a lower limit for the total proton intensity.

1980Ew02, 1979Ca15: A 600-MeV proton beam was produced from the synchrocyclotron at CERN-ISOLDE and bombarded a ScC_2 target. The $^{45}\text{Sc}(\text{p}, 8\text{n}3\text{p})$ spallation reaction products diffused out of the target and reached a tungsten surface ionization source where potassium isotopes were selectively ionized. The beam was extracted from the ion source, separated by the ISOLDE analyzing magnet, and collected by a mylar foil for γ -ray measurements and then a carbon foil for proton measurements. γ rays were detected using a Ge(Li) detector. Time for positron activities were determined using a 700- μm -thick silicon detector. Protons were detected using a 20- μm -700- μm -thick ΔE -E telescope of silicon surface barrier detectors with FWHM=50 keV. Measured $E_\gamma(<5$ MeV), I_γ , $E_{\text{p}}(>0.9$ MeV), I_{p} . Deduced levels, J , π , decay branching ratios, $\log ft$, parent ^{35}K $T_{1/2}$, and coefficients of the isobaric multiplet mass equation for $A=35$, $T=3/2$ quartets. Comparisons with shell-model calculations and the mirror nucleus ^{35}Cl . Also see abstracts 1978HaYH, 1979HaZY, 1979HaZT, and 1979AnZZ.

2018Sa54: A 36-MeV/nucleon ^{36}Ar primary beam was produced from the K500 cyclotron at Texas A&M University. The secondary ^{35}K beam was produced via the $^1\text{H}(^{36}\text{Ar}, ^{35}\text{K})2\text{n}$ reaction of ^{36}Ar bombarding a LN_2 -cooled hydrogen gas target, separated by MARS, and implanted into a 45- μm DSSD sandwiched between a 140- μm SSSD and a 1-mm Si-pad detector in a pulsed-beam mode. $\varepsilon+\beta^+$ -delayed protons were detected by the implantation detector. γ rays were detected by two HPGe detectors. Measured $E_{\text{p}}(>300$ keV), I_{p} , E_γ , I_γ , $\text{p}\gamma$ -coin, $\gamma\gamma$ -coin. Deduced parent ^{35}K $T_{1/2}$.

2019ChZU: Same beam production as 2018Sa54. ^{35}K was implanted into the AstroBox2 detector filled with 800-Torr P5 gas. $\varepsilon+\beta^+$ -delayed protons were detected by the implantation detector. γ rays were detected by 4 Clover Ge detectors. Measured $E_{\text{p}}(>100$ keV), I_{p} , E_γ , I_γ , $\text{p}\gamma$ -coin, $\gamma\gamma$ -coin.

Others: 1998Sc19, 2006Me04 (^{35}K β -NMR studies).

Theoretical studies involving ^{35}K decay: shell model (1985Br29, 2003Sm02).

 ^{34}Cl Levels

$E(\text{level})^\dagger$	$J^\pi^\dagger$	$T_{1/2}^\dagger$
0	0^+	1.5266 s 5
146.36? 3	3^+	31.99 min 3
461.01? 4	1^+	
665.57 4	1^+	

† From the Adopted Levels.

Delayed Protons (^{34}Cl)

$E(\text{p})^\dagger$	$E(^{34}\text{Cl})$	$I(\text{p})^{#b}$	$E(^{35}\text{Ar})^\textcircled{a}$	$E(\text{p})^\dagger$	$E(^{34}\text{Cl})$	$I(\text{p})^{#b}$	$E(^{35}\text{Ar})^\textcircled{a}$
1282 20		0.017 7		937 ‡ 11	665.57		7518 $\&$
1705 20		0.045 18		1124 ‡ 11	0		7053
1875 20		0.025 10		1347 ‡ 11	0		7283
2575 20		0.011 4		1425 c 20	146.36?	0.11 4	7518 $\&$
2807 20		0.0060 24		1555 c 20	0	0.044 17	7518 $\&$
439 ‡ 11	0	2.5×10^{-3} 5	6348	1980 c 20	461.01?	0.018 7	8393 a
673 ‡ 11	665.57		7255	2282 c 20	146.36?	0.024 10	8393 a
844 ‡ 11	665.57		7431	2425 c 20	0	0.020 8	8393 a

Continued on next page (footnotes at end of table)

^{35}K ε p decay (175 ms) 1980Ew02,2019ChZU (continued)Delayed Protons (^{34}Cl) (continued)

[†] $E(p)_{\text{lab}}$ from 1980Ew02, unless otherwise noted.

[‡] $E(p)_{\text{lab}}$ deduced by evaluators from the level energies reported in Figure 5.1 of 2019ChZU.

1980Ew02 reports 10 relative I_p in Table 2 and reports $\% \Sigma I(p) = 0.37\ 15$, which contains 10 proton branches and unidentified proton branches in the continuum of the proton spectrum. 1980Ew02 reports $\% I(\varepsilon + \beta^+) = 0.15\ 6$ and $0.062\ 26$ for two proton-emitting states in ^{35}Ar corresponding to 5 tentatively placed proton branches in Table 3. This provides the only quantitative information to normalize the relative I_p from 1980Ew02 Table 2. Based on the ratio of $I_p(\text{abs})/I_p(\text{rel}) = 0.00106$, evaluators deduce $I_p(\text{abs})$ for each proton branch. After normalization, the 10 proton branches identified in 1980Ew02 account for $\% \Sigma I(p) = 0.32\ 13$.

@ For $E(p)_{\text{lab}}$ from 1980Ew02, $E(\text{level})(^{35}\text{Ar}) = E(p)_{\text{lab}} \times [m(p) + m(^{34}\text{Cl})] / m(^{34}\text{Cl}) + S(p)(^{35}\text{Ar}) + E(\text{level})(^{34}\text{Cl})$, with $S(p)(^{35}\text{Ar}) = 5896.2\ 7$ from 2021Wa16, $E(\text{level})(^{34}\text{Cl})$ from the Adopted Levels of ^{34}Cl . For $E(p)_{\text{lab}}$ from 2019ChZU, $E(\text{level})(^{35}\text{Ar})$ is directly from Figure 5.1 of 2019ChZU.

& Weighted average of $E(\text{level})(^{35}\text{Ar})$ of 7497 20, 7510 20, and 7527 11. The former two $E(\text{level})$ values are deduced from $E(p0)_{\text{lab}} = 1555\ 20$ (1980Ew02) and $E(p1)_{\text{lab}} = 1425\ 20$ (1980Ew02), respectively. The third $E(\text{level})$ value, 7527 11, is directly from Figure 5.1 of 2019ChZU, which decays by emitting $E(p3)_{\text{lab}} = 937\ 11$.

^a Weighted average of $E(\text{level})(^{35}\text{Ar})$ of 8392 20, 8392 20, and 8395 20, deduced from $E(p0)_{\text{lab}} = 2425\ 20$ (1980Ew02), $E(p1)_{\text{lab}} = 2282\ 20$ (1980Ew02), and $E(p2)_{\text{lab}} = 1980\ 20$ (1980Ew02), respectively.

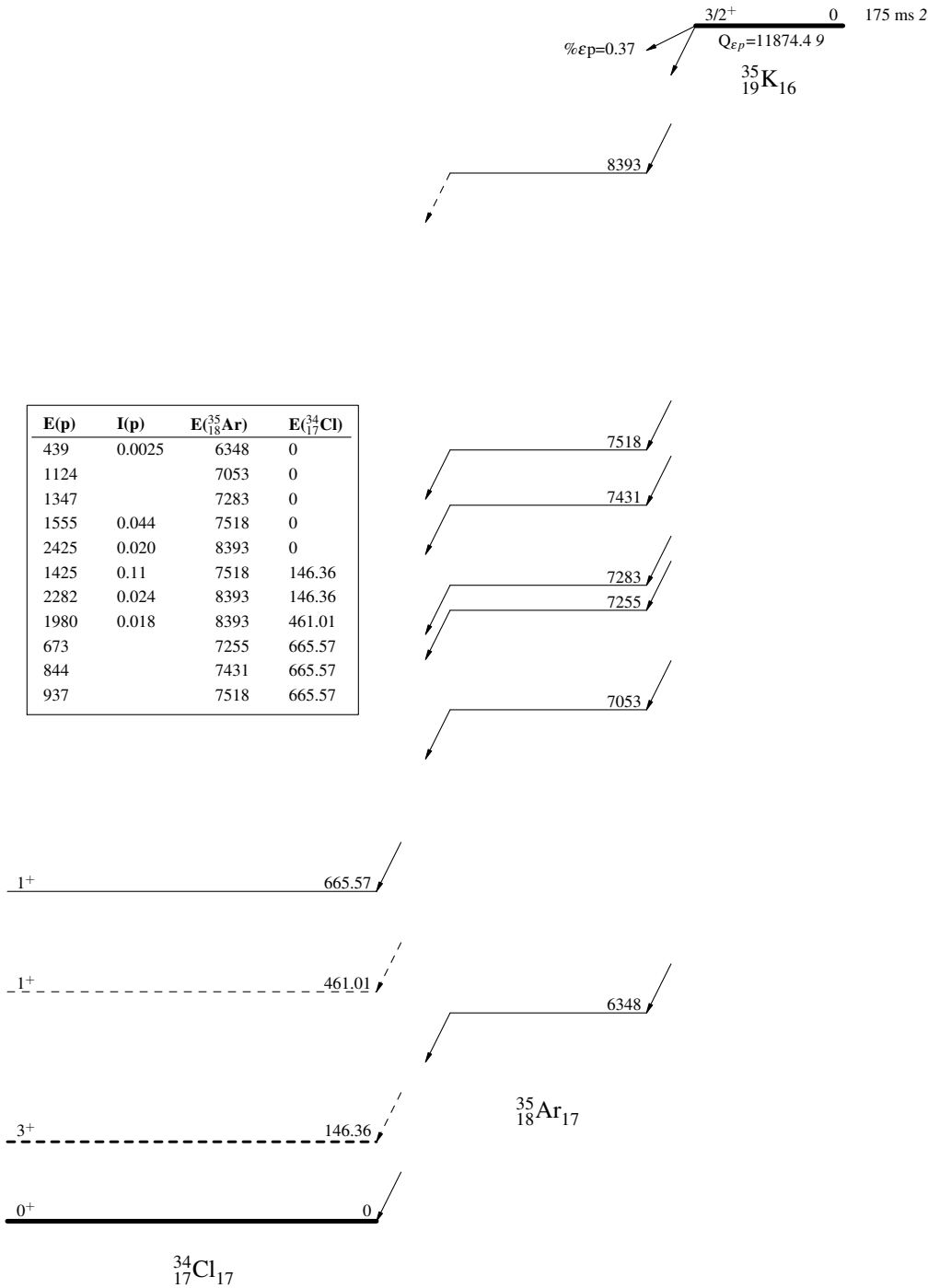
^b Absolute intensity per 100 decays.

^c Placement of transition in the level scheme is uncertain.

³⁵K εp decay (175 ms) 1980Ew02,2019ChZU

Decay Scheme

I(p) Intensities: I(p) per 100 parent decays



$^9\text{Be}(^{37}\text{Ca}, ^{34}\text{Cl})$ **2010Do03**

$^{37}\text{Ca} \rightarrow 3\text{p} + ^{34}\text{Cl}$ on ^9Be target.

2010Do03: a 420-MeV/nucleon ^{40}Ca primary beam was provided from the heavy ion synchrotron SIS at GSI and impinged on a 4-mg/cm² ^9Be target. ^{37}Ca was selected by FRS and was incident on a 700-mg/cm² secondary ^9Be target at 195.7 MeV/u. ^{34}Cl was produced via a secondary three proton removal reaction. γ rays were detected using the RISING setup consisting of 15 Cluster HPGe detectors, 8 MINIBALL HPGe detectors, and 8 Hector BaF₂ detectors. Fragments were detected using the calorimeter telescope CATE consisting of position sensitive Si ΔE detectors and CsI(Tl) E_{res} detectors. Measured E_γ , I_γ . Deduced lifetime using Doppler Shift Attenuation lineshape analysis (**2010Do03**) and peak-shift analysis (**2007DoZV**).

 ^{34}Cl Levels

<u>$E(\text{level})^\dagger$</u>	<u>$T_{1/2}$</u>	<u>Comments</u>
0		
461	4.0 ps +16–12	$T_{1/2}$: from original $T_{1/2}=4.0$ ps 9 (stat) +13–8 (syst) from lineshape analysis with Cluster detectors (2010Do03). Others: $T_{1/2}=2.2$ ps 20 (stat) 36 (syst) from lineshape analysis with MINIBALL detectors (2010Do03). $T_{1/2}=5.0$ ps +20–15 from peak-shift analysis with Cluster detectors (2007DoZV). $T_{1/2}=6$ ps 2 from peak-shift analysis with MINIBALL detectors (2007DoZV).
666	13 ps 5	$T_{1/2}$: from peak-shift analysis with the MINIBALL detectors (2007DoZV). Other: 32 ps +10–14 from peak shift analysis with the Cluster detectors (2007DoZV).
1230		

[†] Rounded from the Adopted Levels.

 $\gamma(^{34}\text{Cl})$

<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>
461	100	461	0
666		666	0
769	12 6	1230	461

[†] Rounded from the Adopted Gammas.

[‡] From **2010Do03**.

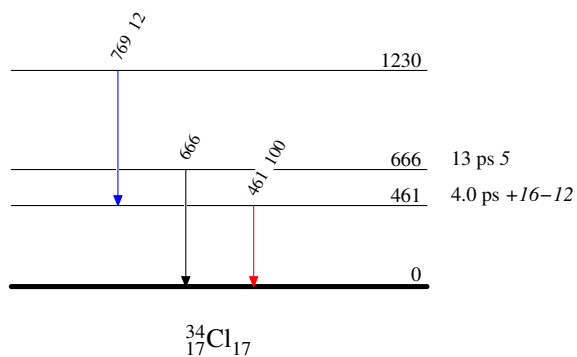
$^9\text{Be}(^{37}\text{Ca}, ^{34}\text{Cl}\gamma)$ 2010Do03

Level Scheme

Intensities: Relative I_γ

Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$



$^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$ **1982Va02**

1982Va02: 32-35-MeV ^{12}C beams from the Utrecht Van de Graaff accelerator impinged on a target of 99.92%-enriched $380\text{-}\mu\text{g}/\text{cm}^2$ ^{24}Mg on Au backing. γ rays were detected using a large solid angle Compton-suppression spectrometer (LACSS) and a HPGe central detector in combination with four gating Ge(Li) detectors. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, and $\gamma\gamma(\theta)$. Deduced levels, J, γ -ray branching ratios, and mixing ratios. Comparisons with shell-model calculations. Also measured the $^{31}\text{P}(\alpha,n\gamma)^{34}\text{Cl}$ reaction for lifetimes using DSAM.

 ^{34}Cl Levels

E(level) [†]	J ^π [‡]	Comments
0.0	0 ⁺ [#]	
146.36	3 ⁺ [#]	
461.00		
665.56		
2158.05		
2180.4 5	3 ⁺ [#]	E(level): 2181.4 2 given in Table 5 of 1982Va02 . No deexciting γ rays are reported in 1982Va02 .
2375.64 14	4 ⁺ [#]	E(level): 2375.7 2 given in Table 5 of 1982Va02 .
2720.78 24	2 ⁻ [#]	E(level): 2721.2 2 given in Table 5 of 1982Va02 .
3599.90 13	4 ⁻ [#]	
3631.33 12	5 ⁻ [#]	
4743.08 13	6	
4824.03 12	5	
5314.86 18	7	
7249.9 5	(9)	
7802.4 10	(8)	

[†] From a least-squares fit to γ -ray energies in [1982Va02](#), except for the four low-lying levels at 146, 461, 665, and 2158 keV, for which [1982Va02](#) did not report data and the evaluators take the E(level) values from the Adopted Levels.

[‡] Spin from $\gamma\gamma(\theta)$ in [1982Va02](#), unless otherwise noted.

[#] From the Adopted Levels.

 $\gamma(^{34}\text{Cl})$

$E_i(\text{level})$	J_i^{π}	$E_{\gamma}^{\dagger}$	$I_{\gamma}^{\dagger}$	E_f	J_f^{π}	Mult. [†]	$\delta^{\dagger}$	Comments
461.00		461		0.0	0 ⁺			$A_2 = -0.231$ 7, A_4 set to zero.
665.56		666		0.0	0 ⁺			$A_2 = -0.134$ 25, A_4 set to zero.
2375.64	4 ⁺	2229		146.36	3 ⁺			
2720.78	2 ⁻	564		2158.05				
		2056		665.56				
		2260		461.00				
		2575		146.36	3 ⁺			
		2722		0.0	0 ⁺			
3599.90	4 ⁻	879.1 2	51 2	2720.78	2 ⁻	Q(+O)	-0.05 6	$A_2 = +0.305$ 18, $A_4 = -0.129$ 19.
		1224.1 [‡] 2	7 [‡] 2	2375.64	4 ⁺			
		3453.3 2	42 2	146.36	3 ⁺			$A_2 = -0.18$ 3, A_4 set to zero.
3631.33	5 ⁻	1255.7 1	60 2	2375.64	4 ⁺	D(+Q)	0.00 2	$A_2 = -0.50$ 5, A_4 set to zero.
		3484.7 2	40 2	146.36	3 ⁺	Q+O	-0.23 10	$A_2 = +0.13$ 3, $A_4 = -0.07$ 4.
4743.08	6	1111.77 10	31 2	3631.33	5 ⁻	D+Q	+4.7 6	$A_2 = +0.30$ 4, $A_4 = +0.25$ 4.
		1143.0 2	37 2	3599.90	4 ⁻	Q(+O)	+0.02 7	$A_2 = +0.48$ 12, $A_4 = -0.18$ 11.
		4596.5 3	32 2	146.36	3 ⁺	O		$A_2 = +0.64$ 6 A_4 set to zero.
4824.03	5	1192.67 5	4 1	3631.33	5 ⁻	D(+Q)	+0.3 6	$A_2 = +0.24$ 12, A_4 set to zero.
		1224.1 [‡] 2	19 [‡] 2	3599.90	4 ⁻			$A_2 = -0.33$ 5, A_4 set to zero.
		2643.5 5	7 1	2180.4	3 ⁺	Q(+O)	-0.03 6	$A_2 = +0.34$ 9, $A_4 = -0.24$ 10.
		4677.4 2	70 2	146.36	3 ⁺	Q(+O)	0.00 5	$A_2 = +0.324$ 5, $A_4 = -0.168$ 5.

Continued on next page (footnotes at end of table)

²⁴ Mg(¹² C,pnγ) 1982Va02 (continued)								
<u>γ(³⁴Cl) (continued)</u>								
E _i (level)	J _i ^π	E _γ [†]	I _γ [†]	E _f	J _f ^π	Mult. [†]	δ [†]	Comments
5314.86	7	490.8 2	74 1	4824.03	5	Q(+O)	+0.02 2	A ₂ =+0.327 2, A ₄ =-0.140 2.
		571.8 2	26 1	4743.08	6	D+Q	-0.05 2	A ₂ =-0.310 7, A ₄ set to zero.
7249.9	(9)	1935.0 5	100	5314.86	7			
7802.4	(8)	2487.4 10	100	5314.86	7			

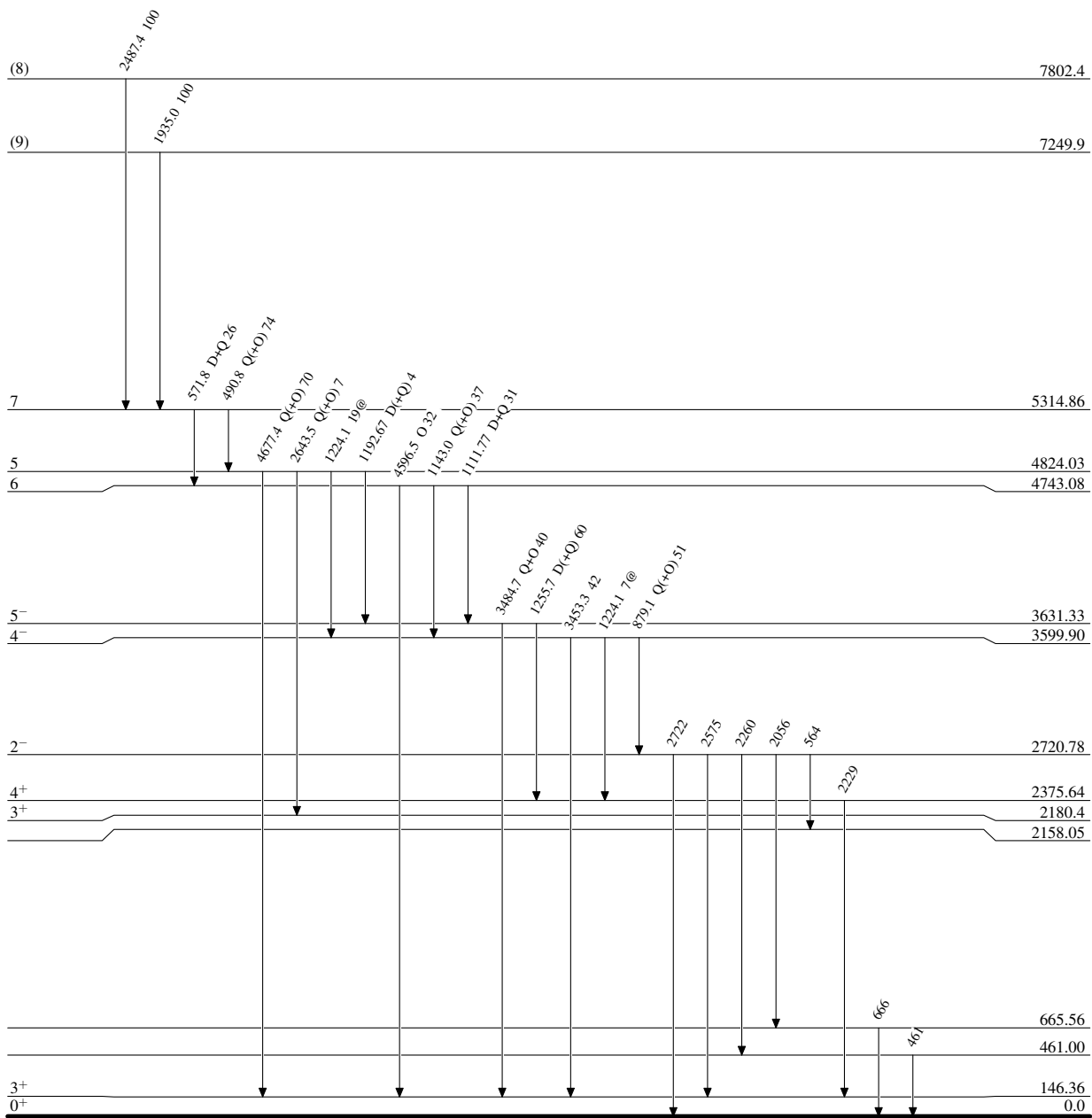
[†] From **1982Va02**, unless otherwise noted.
[‡] Multiply placed with intensity suitably divided.

$^{24}\text{Mg}(^{12}\text{C},\text{pn}\gamma)$ 1982Va02

Level Scheme

Intensities: % photon branching from each level

@ Multiply placed: intensity suitably divided

 $^{34}_{17}\text{Cl}_{17}$

$^{27}\text{Al}(^{12}\text{C},\alpha n\gamma)$ **2014Bi03**

2014Bi03: a 40-MeV ^{12}C beam was provided from the 14-UD Pelletron accelerator at the Tata Institute of Fundamental Research, Mumbai. The target was 0.50-mg/cm² ^{27}Al on 10-mg/cm² Au backing. States of ^{34}Cl were populated via the $^{27}\text{Al}(^{12}\text{C},\alpha n)^{34}\text{Cl}$ heavy-ion reaction and the deexciting γ rays were detected using the Indian National Gamma Array, consisting of 15 Compton-suppressed clover detectors. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, $\gamma\gamma(\theta)$ (DCO), and integrated polarization asymmetry (IPDCO). Deduced levels, J, π , γ -ray multipolarities, lifetimes from DSAM lineshape analysis, and transition strengths. Comparisons with large-basis shell-model calculations.

1978Ba39: 27-45-MeV ^{12}C beams from the Strasbourg tandem accelerator, France, impinged on 350- $\mu\text{g}/\text{cm}^2$ Al targets on Ta backings. γ rays were detected using two Ge(Li) detectors. Measured $\gamma\gamma$ -coin and $\gamma\gamma(\theta)$. Deduced levels, J, π , γ -ray multipolarities, and lifetimes using the Recoil Distance Method (RDM). Also measured the $^{31}\text{P}(\alpha,n\gamma)^{34}\text{Cl}$ reaction.

 ^{34}Cl Levels

E(level) [†]	J π [‡]	T _{1/2}	Comments
0.0	0 ⁺ [#]		
146.36 3	3 ⁺ [#]		
460.98 26	1 ⁺		
666.2 5	1 ⁺		
1230.2 5	2 ⁺		
1887.2 6	2 ⁺		
2158.2 5	2 ⁺		
2181.3 5	3 ⁺		
2375.8 5	4 ⁺		
2611.3 5	3 ⁺		
2721.13 34	2 ⁻		
3600.46 32	4 ⁻		
3631.7 4	5 ⁻		
3646.5 7	5 ⁺		
4371.5 13	4 ⁺	>2.08 ps	T _{1/2} : $\tau > 3$ ps (1978Ba39, RDM).
4743.59 34	6 ⁻		
4824.52 34	5 ⁺		
4862.4 8	5 ⁻		
5315.4 4	7 ⁺	65 ps 8	T _{1/2} : $\tau = 94$ ps 12 (1978Ba39, RDM).
7250.5 11	9 ⁺	0.159 ps 28	T _{1/2} : $\tau = 0.23$ 4 ps (2014Bi03, DSAM).
7699.5 11	(7 ⁺)	<0.34 ps	T _{1/2} : $\tau < 0.49$ ps (2014Bi03, DSAM).
7800.5 11	7 ⁺ , 8 ⁺ , 9 ⁺ [#]	<83.0 fs	J π : 8 ⁻ is given in 2014Bi03. T _{1/2} : $\tau < 0.12$ ps (2014Bi03, DSAM).
8155.5 11	6 ⁺	<56 fs	T _{1/2} : $\tau < 0.08$ ps (2014Bi03, DSAM).
9392.7 11	7 ⁺	<6.9 fs	J π : 9 ⁻ is given in 2014Bi03. Evaluators adopt 7 ⁺ based on 4077 γ , M1(+E2), consistent with $\Delta J = 0$ to 7 ⁺ , 5315.4 level. T _{1/2} : $\tau < 0.01$ ps (2014Bi03, DSAM).
10631.7 15	11 ⁺	<28 fs	T _{1/2} : $\tau < 0.04$ ps (2014Bi03, DSAM).

[†] From a least-squares fit to γ -ray energies.

[‡] From 2014Bi03 based on the measured γ -ray multipolarities, unless otherwise noted.

[#] From the Adopted Levels.

$^{27}\text{Al}(^{12}\text{C},\alpha n\gamma)$ **2014Bi03 (continued)**

$\gamma(^{34}\text{Cl})$									
$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. ‡	δ	γ branching ratio $^\#$	Comments
146.36 3		146.36	3 ⁺	0.0	0 ⁺	[M3]			E_γ : from the Adopted Levels.
453 1	7.9 4	2611.3	3 ⁺	2158.2	2 ⁺	D+Q	+0.11 6	19 13	$R_{\text{DCO}}(\text{D})=1.30$ 17, 3 ⁺ to 2 ⁺ from 2014Bi03 level scheme.
453 1		5315.4	7 ⁺	4862.4	5 ⁻			2.8 2	
460.9 3	100 7	460.98	1 ⁺	0.0	0 ⁺	M1			E_γ : weighted average of 460.9 3 (1978Ba39) and 461 1 (2014Bi03).
									γ branching ratio: other: 100 (1978Ba39).
									$R_{\text{DCO}}(\text{Q})=0.49$ 2, POL=-0.08 1, 1 ⁺ to 0 ⁺ from 2014Bi03 level scheme.
									$A_2=-0.25$ 4 (1978Ba39).
490.9 3	253 13	5315.4	7 ⁺	4824.52	5 ⁺	E2		72 4	E_γ : weighted average of 490.9 3 (1978Ba39) and 491 1 (2014Bi03).
									δ : 0.00 5 (1978Ba39).
									γ branching ratio: other: 66 2 (1978Ba39).
									$R_{\text{DCO}}(\text{Q})=0.99$ 3, POL=+0.14 2, 7 ⁺ to 5 ⁺ from 2014Bi03 level scheme.
									$A_2=0.14$ 2, $A_4=-0.08$ 2 (1978Ba39).
563 1	12.4 7	2721.13	2 ⁻	2158.2	2 ⁺	D+Q	-0.38 6	9.8 11	$R_{\text{DCO}}(\text{D})=1.13$ 10, 2 ⁻ to 2 ⁺ from 2014Bi03 level scheme.
564 1		1230.2	2 ⁺	666.2	1 ⁺			36 5	
571.8 3	21.8 11	5315.4	7 ⁺	4743.59	6 ⁻	E1+M2	-0.04 2	21.6 11	E_γ : weighted average of 571.8 3 (1978Ba39) and 572 1 (2014Bi03).
									δ : other: -0.03 12 (1978Ba39).
									γ branching ratio: other: 34 2 (1978Ba39).
									$R_{\text{DCO}}(\text{Q})=0.44$ 3, POL=+0.15 10, 7 ⁺ to 6 ⁻ from 2014Bi03 level scheme.
									$A_2=-0.08$ 4 (1978Ba39).
									Note that the branching ratio of 21.6 2 in Table II of 2014Bi03 differs from 7.35 7 deduced from the relative intensity in Table I of 2014Bi03.
666 1	45.0 23	666.2	1 ⁺	0.0	0 ⁺	M1			$R_{\text{DCO}}(\text{Q})=0.48$ 4, POL=-0.10 3, 1 ⁺ to 0 ⁺ from 2014Bi03 level scheme.
725 1	5.0 3	4371.5	4 ⁺	3646.5	5 ⁺	M1+E2			$R_{\text{DCO}}(\text{Q})=0.38$ 7, POL=-0.17 12, 4 ⁺ to 5 ⁺ from 2014Bi03 level scheme.
769 1	19.4 10	1230.2	2 ⁺	460.98	1 ⁺	D+Q	+0.04 3	32 5	$R_{\text{DCO}}(\text{D})=1.13$ 9, 2 ⁺ to 1 ⁺ from 2014Bi03 level scheme.
834 1		2721.13	2 ⁻	1887.2	2 ⁺			2.0 2	
879.4 4	98 5	3600.46	4 ⁻	2721.13	2 ⁻	E2		49.2 31	E_γ : weighted average of 879.5 5 (1978Ba39) and 879 1 (2014Bi03).
									$R_{\text{DCO}}(\text{Q})=0.95$ 5, POL=+0.12 2, 4 ⁻ to 2 ⁻ from 2014Bi03 level scheme.
									$A_2=0.19$ 8 (1978Ba39).
928 1		2158.2	2 ⁺	1230.2	2 ⁺			8.6 11	
1084 1		1230.2	2 ⁺	146.36	3 ⁺			31 4	
1111.8 6	29.5 15	4743.59	6 ⁻	3631.7	5 ⁻	D+Q	+0.36 6	29.2 32	E_γ : weighted average of 1111.6 8 (1978Ba39) and 1112 1

$^{27}\text{Al}(^{12}\text{C}, \alpha n \gamma)$ **2014Bi03 (continued)**

$\gamma(^{34}\text{Cl})$ (continued)									
$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. ‡	δ	γ branching ratio $^\#$	Comments
									(2014Bi03). γ branching ratio: other: 30 2 (1978Ba39). $R_{\text{DCO}}(\text{D})=1.92$ 17 and $\text{POL}=-0.03$ 1 indicate M2+E3. 6 $^-$ to 5 $^-$ is given in 2014Bi03 level scheme. This is a discrepancy. Evaluators adopt a conservative assignment of D+Q. $A_2=-0.08$ 20 (1978Ba39). E_γ : weighted average of 1143.0 8 (1978Ba39) and 1143 1 (2014Bi03). Mult., δ : other: Q(+O) with $\delta=0.09$ 53 (1978Ba39). γ branching ratio: other: 43 2 (1978Ba39). $R_{\text{DCO}}(\text{D})=2.08$ 18, $\text{POL}=+0.07$ 2, 6 $^-$ to 4 $^-$ from 2014Bi03 level scheme. $A_2=0.16$ 9, $A_4=-0.16$ 9 (1978Ba39). $R_{\text{DCO}}(\text{D})=1.94$ 36, consistent with $\Delta J=0$, $\text{POL}=+0.08$ 2, 5 $^+$ to 5 $^-$ from 2014Bi03 level scheme. E_γ : weighted average of 1223.8 8 (1978Ba39) and 1224 1 (2014Bi03). γ branching ratio: other: 33 5 (1978Ba39). $R_{\text{DCO}}(\text{Q})=0.42$ 2, $\text{POL}=+0.04$ 1, 5 $^+$ to 4 $^-$ from 2014Bi03 level scheme. $A_2=-0.50$ 6 (1978Ba39). E_γ : weighted average of 1223.8 8 (1978Ba39) and 1225 1 (2014Bi03). $R_{\text{DCO}}(\text{Q})=1.29$ 14, consistent with $\Delta J=0$, 4 $^-$ to 4 $^+$ from 2014Bi03 level scheme. E_γ : weighted average of 1256.0 15 (1978Ba39) and 1256 1 (2014Bi03). γ branching ratio: other: 67 2 (1978Ba39). $R_{\text{DCO}}(\text{Q})=0.39$ 5, $\text{POL}=+0.010$ 2, 5 $^-$ to 4 $^+$ from 2014Bi03 level scheme. $A_2=-0.29$ 7 (1978Ba39). $R_{\text{DCO}}(\text{D})=0.71$ 11, 2 $^+$ to 1 $^+$ from 2014Bi03 level scheme. $R_{\text{DCO}}(\text{D})=2.24$ 20, no POL data but E2 is assigned by 2014Bi03, 3 $^+$ to 1 $^+$ from 2014Bi03 level scheme. $R_{\text{DCO}}(\text{Q})=1.03$ 38 from 65 $^\circ$ –90 $^\circ$ matrix. 2014Bi03 assigns E2. 7 $^+$ to 5 $^+$ from 2014Bi03 level scheme. $R_{\text{DCO}}(\text{D})=0.74$ 6, 2 $^+$ to 1 $^+$ from 2014Bi03 level scheme. $R_{\text{DCO}}(\text{Q})=1.10$ 16, $\text{POL}=+0.15$ 6, 3 $^+$ to 1 $^+$ from 2014Bi03 level scheme.
1143.0 6	34.4 18	4743.59	6 $^-$	3600.46	4 $^-$	E2		34 4	
1193 1	8.9 5	4824.52	5 $^+$	3631.7	5 $^-$	E1(+M2)	-0.1 3	5.1 3	
1223.9 6	34.7 18	4824.52	5 $^+$	3600.46	4 $^-$	E1(+M2)	-0.02 2	14.8 8	
1224.3 6	15.5 8	3600.46	4 $^-$	2375.8	4 $^+$	D(+Q)	+0.3 3	7.8 6	
1256.0 8	23.0 12	3631.7	5 $^-$	2375.8	4 $^+$	E1(+M2)	-0.01 4	52.5 29	
1381 1		2611.3	3 $^+$	1230.2	2 $^+$			28.0 25	
1426 1	8.5 5	1887.2	2 $^+$	460.98	1 $^+$	D+Q	-0.11 5	48 4	
1491 1		2721.13	2 $^-$	1230.2	2 $^+$			2.1 2	
1515 1	14.5 7	2181.3	3 $^+$	666.2	1 $^+$	Q		14.5 19	
1669 1	13.8 8	5315.4	7 $^+$	3646.5	5 $^+$	Q		3.5 3	
1697 1	22.3 11	2158.2	2 $^+$	460.98	1 $^+$	D+Q	-0.10 2	62 4	
1720 1	41.3 21	2181.3	3 $^+$	460.98	1 $^+$	E2		41 5	
1741 1		1887.2	2 $^+$	146.36	3 $^+$			52 5	

²⁷Al(¹²C,αnγ) **2014Bi03** (continued)

γ(³⁴Cl) (continued)

E_γ †	I_γ †	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult.‡	δ	γ branching ratio#	Comments
1935 1	232 12	7250.5	9 ⁺	5315.4	7 ⁺	E2			R _{DCO} (Q)=1.10 3, POL=+0.08 1, 9 ⁺ to 7 ⁺ from 2014Bi03 level scheme.
1945 1		2611.3	3 ⁺	666.2	1 ⁺			17.0 16	
2012 1		2158.2	2 ⁺	146.36	3 ⁺			10.2 11	
2035 1	2.9 2	2181.3	3 ⁺	146.36	3 ⁺	D+Q	-0.4 1	44 6	I _γ : 2.9 2 Note that the branching ratio of 44.1 46 in Table II of 2014Bi03 differs from 4.9 2 deduced from the relative intensity in Table I of 2014Bi03 . R _{DCO} (Q)=0.51 8 and POL=+0.04 1 indicate E1+M2. 3 ⁺ to 3 ⁺ from 2014Bi03 level scheme and B(M1) is calculated in 2014Bi03 Table V. This is a discrepancy. Evaluators adopt a conservative assignment of D+Q. R _{DCO} (D)=0.87 6, 2 ⁻ to 1 ⁺ from 2014Bi03 level scheme.
2055 1	13.5 7	2721.13	2 ⁻	666.2	1 ⁺	D+Q	-0.05 3	7.5 9	
2158 1		2158.2	2 ⁺	0.0	0 ⁺			19.1 22	
2213 1	2.1 2	4824.52	5 ⁺	2611.3	3 ⁺			1.6 1	R _{DCO} (Q)=0.69 15. No multipolarity or δ is given in 2014Bi03 . 5 ⁺ to 3 ⁺ from 2014Bi03 level scheme.
2229 1	40.9 21	2375.8	4 ⁺	146.36	3 ⁺	D+Q	+0.4 1		R _{DCO} (Q)=0.90 9 and POL=-0.09 3 indicate M2+E3. 4 ⁺ to 3 ⁺ from 2014Bi03 level scheme. This is a discrepancy. Evaluators adopt a conservative assignment of D+Q.
2260 1	88 5	2721.13	2 ⁻	460.98	1 ⁺	E1+M2	+0.05 2	49 6	R _{DCO} (D)=1.15 5, POL=+0.04 1, 2 ⁻ to 1 ⁺ from 2014Bi03 level scheme.
2384 1	18.2 9	7699.5	(7 ⁺)	5315.4	7 ⁺	M1			R _{DCO} (Q)=0.89 25 from 65°-90° matrix, consistent with $\Delta J=0$, POL=-0.17 11. No multipolarity or δ is given in 2014Bi03 . B(M1) is calculated in 2014Bi03 Table V. (7 ⁺) to 7 ⁺ from 2014Bi03 level scheme.
2449 1		4824.52	5 ⁺	2375.8	4 ⁺			1.5 1	
2465 1		2611.3	3 ⁺	146.36	3 ⁺			36.0 31	
2485 1	89 5	7800.5	7 ⁺ ,8 ⁺ ,9 ⁺	5315.4	7 ⁺	D+Q	+0.22 7		R _{DCO} (Q)=0.74 8 and POL=+0.016 2 indicate E1+M2. 8 ⁻ to 7 ⁺ from 2014Bi03 level scheme. B(M2)(W.u.)>61.1 exceeds RUL=3. Evaluators adopt a conservative assignment of D+Q.
2562 1		4743.59	6 ⁻	2181.3	3 ⁺	[E3]		1.0 1	
2575 1	30.2 16	2721.13	2 ⁻	146.36	3 ⁺	E1+M2	-0.3 1	16.8 19	R _{DCO} (Q)=0.51 12, POL=+0.03 1, 2 ⁻ to 3 ⁺ from 2014Bi03 level scheme.
2643 1	14.7 8	4824.52	5 ⁺	2181.3	3 ⁺	Q		4.8 7	R _{DCO} (Q)=1.13 13. 2014Bi03 assigns E2. 5 ⁺ to 3 ⁺ from 2014Bi03 level scheme.
2681 1	3.3 2	4862.4	5 ⁻	2181.3	3 ⁺	M2			R _{DCO} (D)=1.88 26, POL=-0.1 1. No multipolarity or δ is given in 2014Bi03 . 5 ⁻ to 3 ⁺ from 2014Bi03 level scheme.
2721 1	23.2 12	2721.13	2 ⁻	0.0	0 ⁺	M2		12.9 15	R _{DCO} (Q)=0.98 29, POL=-0.06 3, 2 ⁻ to 0 ⁺ from 2014Bi03 level scheme.
2840 1	47.3 24	8155.5	6 ⁺	5315.4	7 ⁺	M1+E2	-0.3 1		R _{DCO} (Q)=0.83 12, POL=-0.02 1, 6 ⁺ to 7 ⁺ from 2014Bi03 level scheme.
3381 1	74 4	10631.7	11 ⁺	7250.5	9 ⁺	E2			R _{DCO} (Q)=1.00 10, POL=+0.03 2, 11 ⁺ to 9 ⁺ from 2014Bi03 level scheme.

²⁷Al(¹²C, α n γ) **2014Bi03** (continued)

$\gamma(^{34}\text{Cl})$ (continued)

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	δ	γ branching ratio [#]	Comments
3453.8 6	85 4	3600.46	4 ⁻	146.36	3 ⁺	D(+Q)	+0.05 5	43.0 28	E_γ : weighted average of 3453.6 8 (1978Ba39) and 3454 1 (2014Bi03). $R_{\text{DCO}}(\text{Q})=0.68$ 5 and $\text{POL}=-0.008$ 1 indicate M1(+E2). 4 ⁻ to 3 ⁺ from 2014Bi03 level scheme. This is a discrepancy. Evaluators adopt a conservative assignment of D(+Q). $A_2=-0.22$ 8, $A_4=-0.11$ 8 (1978Ba39).
3485.1 6	15.4 8	3631.7	5 ⁻	146.36	3 ⁺	M2+E3	-0.23 12	47.5 27	E_γ : weighted average of 3485.2 8 (1978Ba39) and 3485 1 (2014Bi03). γ branching ratio: other: 33 2 (1978Ba39). $R_{\text{DCO}}(\text{Q})=1.74$ 47, $\text{POL}=-0.10$ 4, 5 ⁻ to 3 ⁺ from 2014Bi03 level scheme. $A_2=0.04$ 14 (1978Ba39).
3500 1	37.2 19	3646.5	5 ⁺	146.36	3 ⁺	E2			$R_{\text{DCO}}(\text{Q})=0.88$ 11, $\text{POL}=+0.003$ 3, 5 ⁺ to 3 ⁺ from 2014Bi03 level scheme.
4077 1	17.9 10	9392.7	7 ⁺	5315.4	7 ⁺	M1(+E2)	+0.2 5		$R_{\text{DCO}}(\text{Q})=1.20$ 43, $\text{POL}=-0.02$ 1, 9 ⁻ to 7 ⁺ from 2014Bi03 level scheme indicates M2(+E3), which makes $B(\text{M2})(\text{W.u.}) > 1208$ exceed $\text{RUL}=3$ and $B(\text{E3})(\text{W.u.}) < 7.57 \times 10^{11}$ upper limit exceed $\text{RUL}=100$. Evaluators consider this assignment unlikely and reassign M1(+E2) consistent with $\Delta J=0$.
4597.1 7	36.2 19	4743.59	6 ⁻	146.36	3 ⁺	(E3)		36 5	E_γ : weighted average of 4597.1 10 (1978Ba39) and 4597 1 (2014Bi03). γ branching ratio: other: 27 3 (1978Ba39). $R_{\text{DCO}}(\text{D})=3.45$ 26, $\text{POL}=+0.010$ 2, 6 ⁻ to 3 ⁺ from 2014Bi03 level scheme.
4677.9 7	173 9	4824.52	5 ⁺	146.36	3 ⁺	E2		72 4	E_γ : weighted average of 4677.7 10 (1978Ba39) and 4678 1 (2014Bi03). δ : -0.03 10 (1978Ba39). γ branching ratio: other: 67 5 (1978Ba39). $R_{\text{DCO}}(\text{Q})=1.00$ 2, $\text{POL}=+0.010$ 1, 5 ⁺ to 3 ⁺ from 2014Bi03 level scheme. $A_2=0.19$ 4, $A_4=-0.09$ 4 (1978Ba39).

[†] From 2014Bi03, unless otherwise noted. No uncertainties for E_γ are provided in 2014Bi03. Based on the typical uncertainties reported in other work by the authors using the same INGA detector array, the evaluators assume a $\Delta E_\gamma=1$ keV for averaging and least-squares fitting. 2014Bi03 Table I provides only statistical uncertainties for I_γ . The systematic uncertainties are $\approx 5\%$, arising from the efficiency curve and fits, as communicated by M. Saha Sarkar in an email reply to B. Singh on February 17, 2014. The evaluators have combined the statistical and systematic uncertainties in quadrature.

[‡] Deduced by evaluators based on measured $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\theta)(\text{POL})$ in 2014Bi03. $R_{\text{DCO}}(\text{D})$ is from gating on a known stretched dipole ($\Delta J=1$) transition and $R_{\text{DCO}}(\text{Q})$ is from gating on a known stretched quadrupole ($\Delta J=2$) transition. For stretched transition with same multipolarity as the gating transition, R_{DCO} value should be very close to unity. For different multipolarities of the gating and projected transitions, the R_{DCO} value depends on the angle between the detectors and the

²⁷Al(¹²C,αnγ) **2014Bi03 (continued)**

γ(³⁴Cl) (continued)

amount of mixing present in the mixed multipolarity transition. However, the authors did not specify the expected R_{DCO} values. Based on Fig. 5 in **2014Bi03**, the expected R_{DCO}(Q)≈1.0 and R_{DCO}(D)≈2.0 for stretched quadrupole (ΔJ=2) or unstretched dipole (ΔJ=0) transitions; R_{DCO}(Q)≈0.5 and R_{DCO}(D)≈1.0 for stretched dipole (ΔJ=1) transitions. A positive POL indicates an electric character, and a negative POL indicates a magnetic character. Twelve multipolarities are assigned in **1978Ba39**, which are consistent with the **2014Bi03** assignments.

2014Bi03 Table II provides only statistical uncertainties for the γ branching ratios. According to the email reply to B. Singh from the corresponding author M. Saha Sarkar on February 17, 2014, the γ branching ratios listed in Table II of **2014Bi03** were measured separately by gating on transitions above the level. The systematic uncertainties for these measurements are ≈5%, which arise from the efficiency curve and fits. The evaluators have combined the statistical and systematic uncertainties in quadrature. In some cases, the branching ratios differ significantly from those derived from the I_γ data in Table I of **2014Bi03**. The I_γ data in Table I should be considered more reliable, as the authors suggested in their email reply.

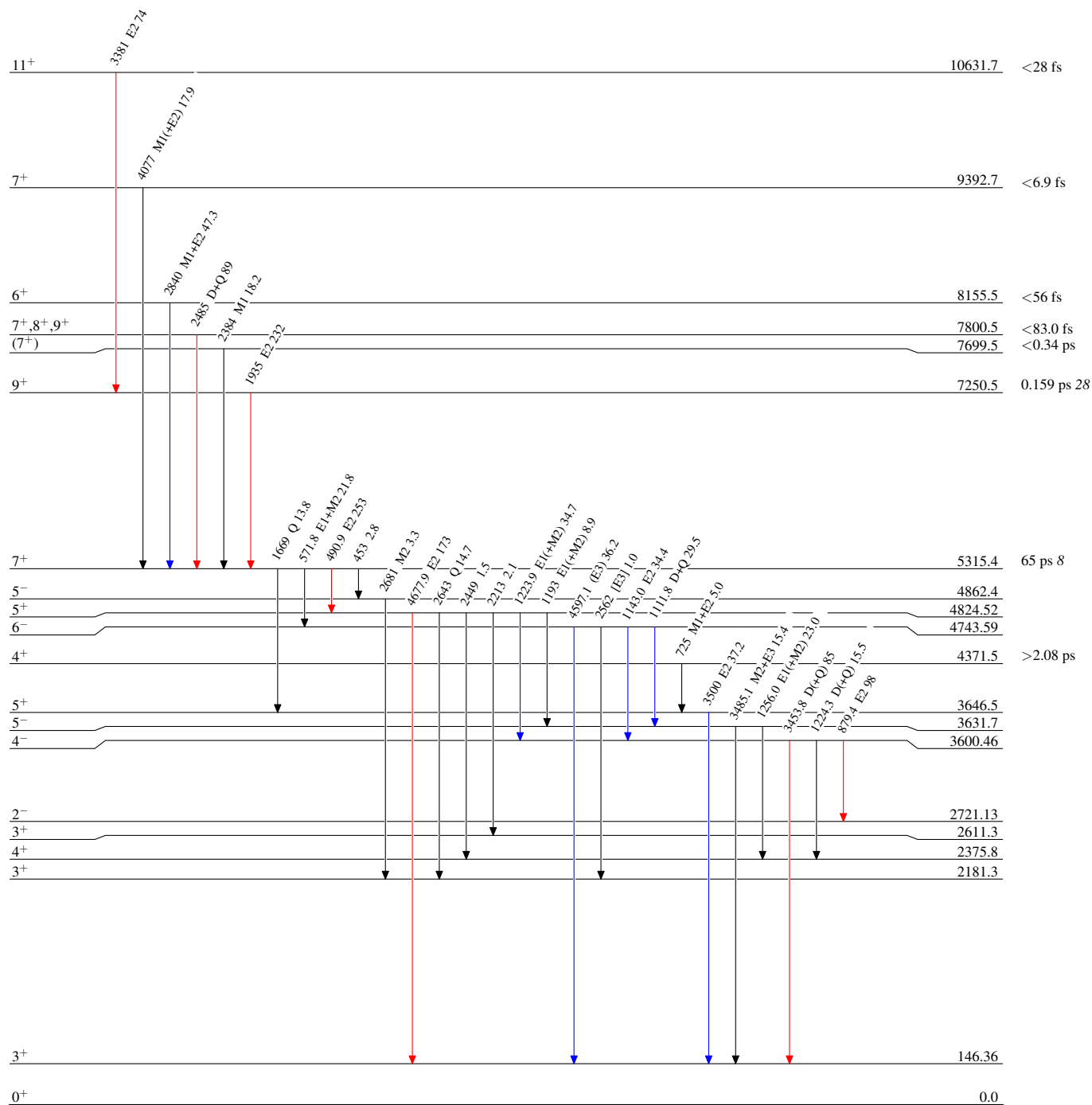
$^{27}\text{Al}(^{12}\text{C}, \alpha n \gamma)$ 2014Bi03

Level Scheme

Intensities: Relative I_γ

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$

 $^{34}_{17}\text{Cl}_{17}$

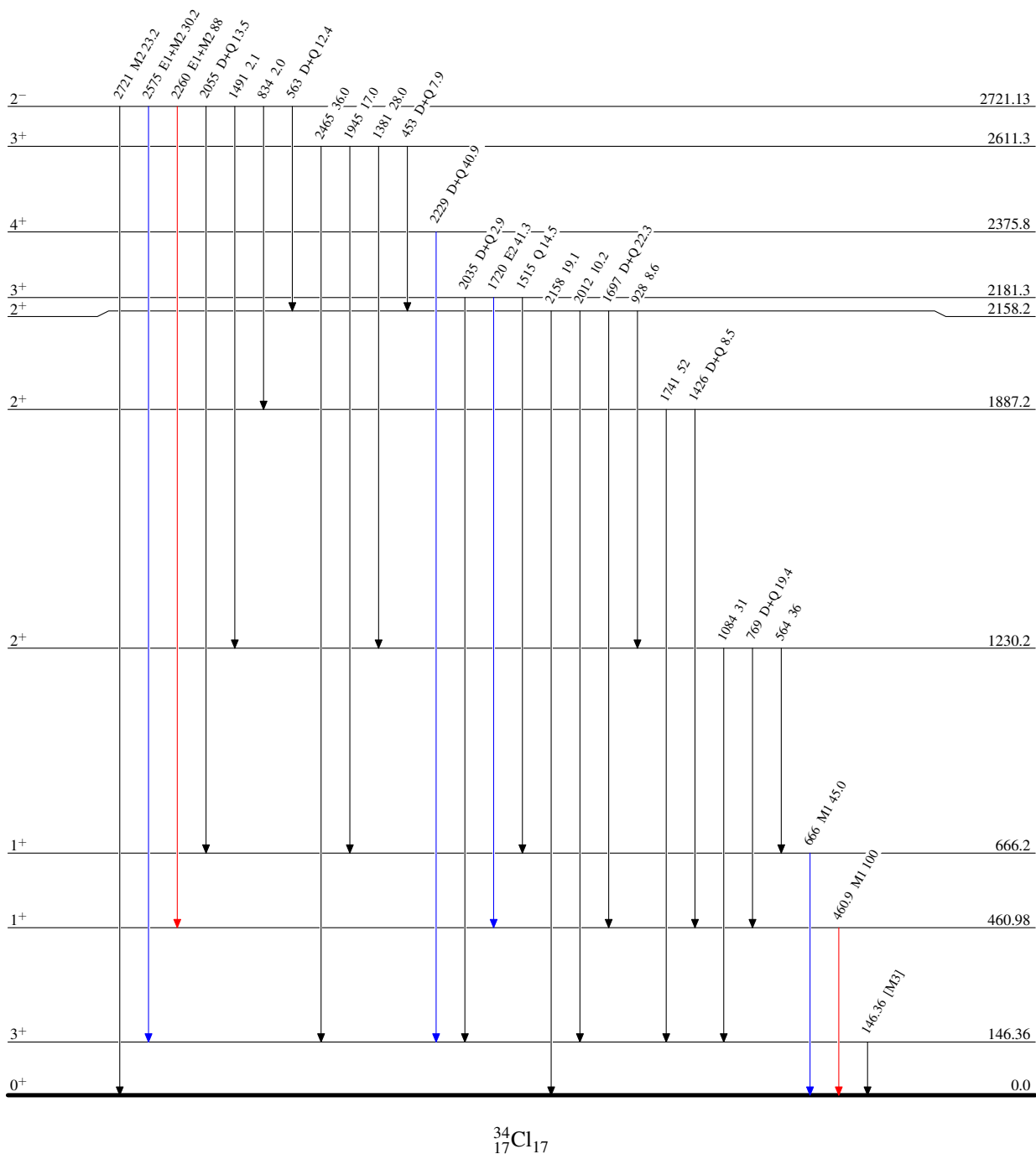
$^{27}\text{Al}(^{12}\text{C}, \alpha n \gamma)$ 2014Bi03

Level Scheme (continued)

Intensities: Relative I_γ

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$

 $^{34}_{17}\text{Cl}_{17}$

$^{31}\text{P}(\alpha, n\gamma)$ [1978Ba61](#), [1971Ba98](#), [1970Br10](#)

- [1982Va02](#): 11.7 and 16.3-MeV α beams from the Utrecht 6-MV Van de Graaff accelerator impinged on a target of 100- $\mu\text{g}/\text{cm}^2$ P on thick Au backing. γ rays were detected using a large solid angle Compton-suppression spectrometer (LACSS) and a HPGe central detector in combination with four gating Ge(Li) detectors. Measured $E\gamma$, $I\gamma$. Deduced lifetimes from Doppler pattern (DSAM). Also measured the $^{24}\text{Mg}(^{12}\text{C}, p n\gamma)^{34}\text{Cl}$ reaction.
- [1978Ba61](#), [1978Ba77](#): 13 and 14.6-MeV α beams from the 7.5-MV Van de Graaff accelerator at the Strasbourg Nuclear Research Center, France, impinged on a 200- $\mu\text{g}/\text{cm}^2$ red phosphorus target. γ rays were detected using two Ge(Li) detectors. Measured $\gamma(\theta)$. Deduced mixing ratios, lifetimes using the Recoil Distance Method (RDM).
- [1978Ba39](#): 14.4-MeV α beams from the 7.5-MV Van de Graaff accelerator at the Strasbourg Nuclear Research Center, France, impinged on a 200- $\mu\text{g}/\text{cm}^2$ red phosphorus target. Neutrons were detected using a NE213 liquid scintillator. γ rays were detected using a Ge(Li) detector. Measured $n\gamma$ -coin. [1978Ba39](#) is primarily a $^{27}\text{Al}(^{12}\text{C}, \alpha n\gamma)^{34}\text{Cl}$ reaction measurement.
- [1971Ba98](#): 0.3-0.7 MeV > threshold α beams was from the Padova 5.5 MV Van de Graaff generator. Targets were 100- $\mu\text{g}/\text{cm}^3$ evaporated Zn_3P_2 . γ rays were detected using a 55- cm^3 Ge(Li) detector. Deduced lifetimes using RDM.
- [1970Br10](#): 7.50-8.75-MeV α beams impinged on a 600- $\mu\text{g}/\text{cm}^2$ red phosphorus target. γ rays were detected using a Ge(Li) detector. Deduced lifetimes using DSAM. Also measured γ -ray angular distributions in $^{32}\text{S}(^3\text{He}, p\gamma)^{34}\text{Cl}$ reaction.
- [1970Mo09](#): an 8.0-MeV α beam was provided from the Triangle Universities Nuclear Laboratory 4 MV Van de Graaff accelerator and bombarded a target of 200- $\mu\text{g}/\text{cm}^2$ Zn_3P_2 evaporated onto a 20- $\mu\text{g}/\text{cm}^2$ C backing. γ rays were detected using a 20- cm^3 coaxial Ge(Li) detector. [1970Mo09](#) primarily measured $^{31}\text{P}(\alpha, p\gamma)^{34}\text{S}$.
- Others: $^{31}\text{P}(\alpha, n)^{34}\text{Cl}$ Q-value, yield, excitation function measurements: [2021VI03](#), [2015VI01](#), [1974ClZK](#), [1973Ry01](#), [1970Um01](#), [1969UmZZ](#), [1956Qu04](#).

 ^{34}Cl Levels

Except for the three lowest levels from [1970Mo09](#), no other literature in this dataset reports any $E\gamma$ or E(level) values with uncertainties. All E(level) values from 1 to 7 MeV presented here are taken from Fig. 3 of [1978Ba39](#). E(level) values above 7 MeV are taken from Fig. 10 of [1982Va02](#). All $E\gamma$ values are deduced from level-energy differences.

E(level)	J^π [†]	$T_{1/2}$	Comments
0	0		
146.3 3	3		
461.5 4		6.17 ps 28	$T_{1/2}$: lifetime $\tau=8.9$ ps 4: weighted average of 9 ps 4 (1970Br10 , DSAM) and 8.1 ps 9 (1971Ba98 , RDM).
663.7 12		9.8 ps 6	$T_{1/2}$: lifetime $\tau=14.1$ ps 9: weighted average of 14 ps +11-6 (1970Br10 , DSAM) and 14.2 ps 15 (1971Ba98 , RDM).
1230		15.3 ps 28	$T_{1/2}$: lifetime $\tau=22$ ps 4 (1971Ba98 , RDM). Other: >14 ps (1970Br10 , DSAM).
1887			
2158			
2376			
2611			
2722	2		
3600	4	15.9 ps 35	$T_{1/2}$: lifetime $\tau=23$ ps 5 (1978Ba61 , RDM).
3632		1.9×10^2 ps 4	$T_{1/2}$: lifetime $\tau=280$ ps 60 (1978Ba61 , RDM).
4743	6	4.9 ps 21	$T_{1/2}$: lifetime $\tau=7$ ps 3 (1978Ba61 , RDM).
4824		0.31 ps 14	$T_{1/2}$: lifetime $\tau=450$ fs 200 (1982Va02 , DSAM).
5315	7		
7250		0.14 ps 5	$T_{1/2}$: lifetime $\tau=200$ fs 70 (1982Va02 , DSAM).
7800		69 fs 49	$T_{1/2}$: lifetime $\tau=100$ fs 70 (1982Va02 , DSAM).

[†] Spin from $\gamma(\theta)$ in [1978Ba61](#), except for the ground state, 146, 2722, 3600 are known and used by [1978Ba61](#).

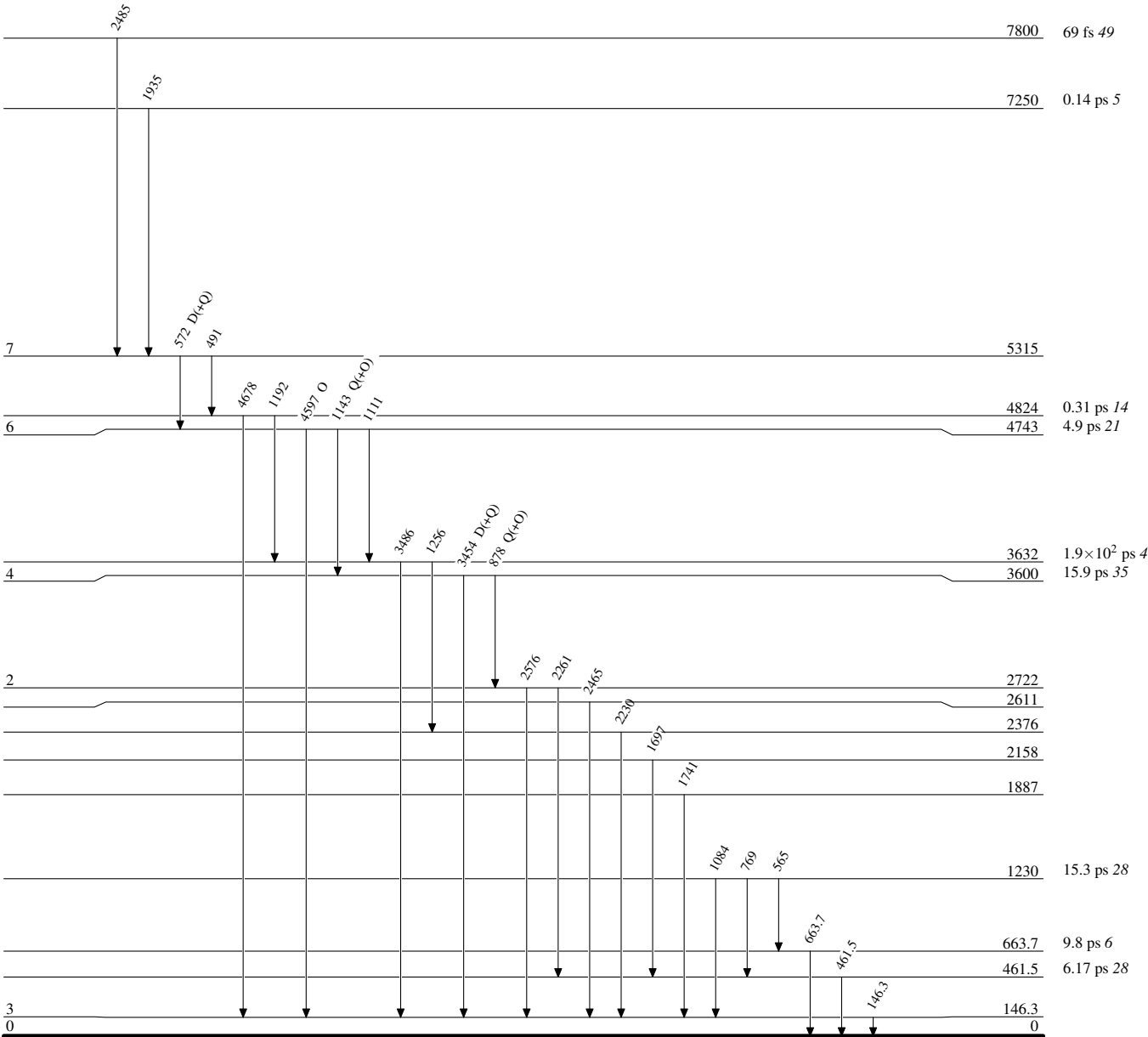
$^{31}\text{P}(\alpha, n\gamma)$ [1978Ba61](#), [1971Ba98](#), [1970Br10](#) (continued)

$\gamma(^{34}\text{Cl})$							
E_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [†]	$\delta^\dagger$	Comments
146.3 3	146.3	3	0	0			
461.5 4	461.5		0	0			
491	5315	7	4824				
565	1230		663.7				
572	5315	7	4743	6	D(+Q)	-0.02 7	$A_2=-0.33$ 5 (1978Ba61).
663.7 12	663.7		0	0			
769	1230		461.5				
878	3600	4	2722	2	Q(+O)	-0.02 3	$A_2=0.40$ 2, $A_4=-0.14$ 3 (1978Ba61).
1084	1230		146.3	3			
1111	4743	6	3632				
1143	4743	6	3600	4	Q(+O)	-0.03 9	$A_2=0.31$ 5, $A_4=-0.17$ 4 (1978Ba61).
1192	4824		3632				
1256	3632		2376				
1697	2158		461.5				
1741	1887		146.3	3			
1935	7250		5315	7			
2230	2376		146.3	3			
2261	2722	2	461.5				
2465	2611		146.3	3			
2485	7800		5315	7			
2576	2722	2	146.3	3			
3454	3600	4	146.3	3	D(+Q)	+0.07 17	$A_2=-0.16$ 3 (1978Ba61).
3486	3632		146.3	3			
4597	4743	6	146.3	3	O		Mult., δ : O(+H) with $\delta=+0.07$ 9 (1978Ba61). $A_2=0.52$ 6, $A_4=0.20$ 7, $A_6=0.13$ 8 (1978Ba61).
4678	4824		146.3	3			

[†] From $\gamma(\theta)$ in [1978Ba61](#). As for the three lowest levels, evaluators also adopt the uncertainties of the level energies reported in [1970Mo09](#) as the uncertainties of the γ -ray energies since the ground-state γ ray is the dominant branch for each of the three levels.

³¹P(α ,n γ) 1978Ba61,1971Ba98,1970Br10

Level Scheme



³⁴Cl₁₇

$^{32}\text{S}(^3\text{He},\text{p})$ 1977Bi04

$^{32}\text{S}+1\text{p}+1\text{n}\rightarrow^{34}\text{Cl}$ from $J^\pi=0^+$ ^{32}S ground state.

1977Bi04, 1976Bi14: an 18-MeV ^3He beam was provided from the University of Pennsylvania tandem accelerator. The target was gaseous H_2S . Protons were detected on nuclear emulsions in a multi-angle broad range magnetic spectrograph with an energy resolution of 22-25 keV. Measured $\sigma(E_{\text{p}},\theta)$. Deduced levels, L-transfers, J, π , and spectroscopic factors from DWUCK-DWBA analysis.

1976De24: a 24.5-MeV ^3He beam was produced from the Princeton azimuthally varying field cyclotron. The target was CdS with natural ^{32}S . Reaction products were detected using a freon-cooled $\Delta\text{E-E}$ Si detector telescope with FWHM=40 keV. Measured $\sigma(E_{\text{p}},\theta)$. Deduced levels, L-transfers, J, and π from DWUCK-DWBA analysis. Also measured $^{32}\text{S}(\alpha,\text{d})^{34}\text{Cl}$.

1972Na23: a 24-MeV ^3He beam was produced from the MP tandem accelerator of the Max Planck Institut fur Kernphysik in Heidelberg. Targets were PbS on carbon backings. The reaction protons were momentum-analyzed by a multigap magnetic spectrograph and detected using 50- μm Ilford L4 nuclear emulsions with a resolution of 40 keV. Measured $\sigma(E_{\text{p}},\theta)$. Deduced levels, L-transfers, J, π , and spectroscopic factors from zero-range DWUCK-DWBA analysis.

Others: 1991Ru05 measured cross sections of $^{32}\text{S}(^3\text{He},\text{p})^{34\text{m}}\text{Cl}$ at E=4.0-12.0 MeV. 2017Ay03 measured $^{32}\text{S}(^3\text{He},\text{p})^{34}\text{Cl}$ as a byproduct of $^{24}\text{Mg}(^3\text{He},\text{p})^{26}\text{Al}$.

 ^{34}Cl Levels

E(level) [†]	L [†]	Comments
0.0 25	0	L: 0 also from 1972Na23.
145.3 16	2+4	
461.8 34	2	L: other: L=2(+0) from 1972Na23.
664.7 54	0	L: other: L=0(+2) from 1972Na23.
1230.4 73	(2)	L: other: L=2 from 1972Na23.
1880.0 53	2	
2163.0 87	2	E(level): doublet.
2376.8 75	(4)	
2583.0 41	0	L: 0 also from 1972Na23.
2614.2 38	(2)+(4)	
2720.5 39	1+3	
3127.0 27	0	L: other: L=0+2 in 1972Na23.
3336.4 31	2	L: 2 also from 1972Na23.
3383.2 43	2	
3548.4 52	3	
3601.1 49	(3,4)	
3635.2 48	5	
3804? 29		
3982.6 72	(0)+3	E(level): doublet.
4076.5 58	5(+3)	
4141.1 79	1+3	
4354.5 20	1	
4412.1 31	3	
4455.8 40	3	
4514.5 32	1+3	
4611.9 70	1+3	E(level): doublet.
4693.0 34	2,1+3	E(level): other: 4674 (1976De24).
4824.8 24	3,4	E(level): other: 4790 (1976De24).
4965.7 71	(1,2)	E(level): doublet.
5292		E(level): from 1976De24.
6319		E(level): from 1976De24.
6724		E(level): from 1976De24.

[†] From 1977Bi04, unless otherwise noted.

$^{32}\text{S}(^3\text{He}, p\gamma)$ 1973Ca15, 1971De27, 1971Sn01

$^{32}\text{S}+1p+1n \rightarrow ^{34}\text{Cl}$ from $J^\pi=0^+$ ^{32}S ground state.

1973Ca15: 9.0-MeV ^3He beam from the Utrecht 6 MV tandem Van de Graaff generator impinging on a natural Sb_2S_3 target.

Protons were detected using two Si surface-barrier detectors and γ rays were detected using a Ge(Li) detector. Measured $p\gamma$ -coin. Deduced lifetimes using DSAM.

1971De27: 8.0-9.0-MeV ^3He beams from the Notre Dame model FN tandem Van de Graaff accelerator impinging on natural CdS targets. Protons were detected using an annular Si surface-barrier detector and γ rays were detected using NaI(Tl) detectors and coaxial Ge(Li) detectors. Measured E_p , E_γ , I_γ , and $p\gamma(\theta)$ -coin. Deduced levels, J , γ -ray branching ratios, and mixing ratios.

1971Sn01: 6.8-MeV ^3He beam from the Brookhaven National Laboratory 3.5-MeV Van de Graaff accelerator impinging on a 95% ^{32}S target. γ rays were detected using a Ge(Li) detector. Measured E_γ , I_γ . Deduced levels and lifetimes using DSAM. Also measured $^{31}\text{P}(\alpha, n\gamma)^{34}\text{Cl}$ reaction at the Stony Brook tandem Van de Graaff accelerator and deduced lifetimes using RDM.

1970Br11: 7.5-MeV ^3He beam from the Utrecht 6-MV tandem Van de Graaff generator impinging on a natural ZnS target. Protons were detected using two Si surface-barrier detectors and γ rays were detected using a Ge(Li) detector. Measured $p\gamma$ -coin. Deduced levels, J , γ -ray branching ratios, and lifetimes using DSAM.

1970Br10: 5.2-MeV ^3He beam from the Padova 5.5-MV Van de Graaff generator impinging on a natural CdS target. Protons were detected using an annular Si surface-barrier detector and γ rays were detected using a Ge(Li) detector and a NaI(Tl) detector. Measured E_p , E_γ , I_γ , and $p\gamma(\theta)$ -coin. Deduced levels, J , γ -ray branching ratios, and mixing ratios. Also measured $^{31}\text{P}(\alpha, n\gamma)^{34}\text{Cl}$ reaction using a Ge(Li) detector and deduced lifetimes using DSAM.

1970Sy01: 5.75-MeV ^3He beam from the University of Groningen Van de Graaff accelerator impinging on natural ZnS target. Protons were detected using a Si annular detector and γ rays were detected using a NaI(Tl) detector. Measured E_p , E_γ , I_γ , and $p\gamma(\theta)$ -coin. Deduced levels, J , γ -ray branching ratios, and mixing ratios.

1970Th10: 9-, 9.5-, 10-MeV ^3He beams impinging on natural ZnS target at Nuclear Research Institute, Strasbourg, France. Protons were detected using a Si surface-barrier detector and γ rays were detected using a NaI detector. Measured $\gamma p(\theta)$. Deduced levels and J .

1972AIZT, 1971AIZB: DSAM with 8.5-MeV ^3He beam at University of Notre Dame.

 ^{34}Cl Levels

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	Comments
0.0	0^+		J^π : from the Adopted Levels.
146.36 3	3^+		J^π : from the Adopted Levels.
461.00 4	1	4.9 ps 5	J^π : spin=1 from $p\gamma(\theta)$ in 1970Sy01, 1970Br10, 1970Th10, 1971De27. $T_{1/2}$: lifetime $\tau=7.1$ ps 7 (1971Sn01, DSAM).
665.58 5	1	8.9 ps 7	J^π : spin=1 from $p\gamma(\theta)$ in 1970Sy01, 1970Br10, 1970Th10, 1971De27. $T_{1/2}$: lifetime $\tau=12.8$ ps 10 (1971Sn01, DSAM).
1230.26 6	2	13 ps 1	J^π : spin=2 from $p\gamma(\theta)$ in 1970Br10, 1971De27. Spin=1,2,(3) from $p\gamma(\theta)$ in 1970Sy01. $T_{1/2}$: lifetime $\tau=19.4$ ps 14 (1971Sn01, DSAM).
1887.6	2	0.8 ps 3	J^π : spin=2 from $p\gamma(\theta)$ in 1970Br10, 1971De27. Spin=1,2,(0) from $p\gamma(\theta)$ in 1970Sy01. $T_{1/2}$: lifetime $\tau=1.1$ ps 4: weighted average of 2.4 ps 13 (1973Ca15, DSAM) and 0.94 ps 40 (1970Br11, DSAM).
2158.7		36 fs 10	$T_{1/2}$: lifetime $\tau=52$ fs 15: weighted average of 100 fs 30 (1973Ca15, DSAM), 46 fs 15 (1970Br11, DSAM), and 40 fs 20 (1972AIZT, DSAM).
2181.5	1,2,3	0.32 ps 6	J^π : spin=1,2,3 from $p\gamma(\theta)$ in 1971De27. $T_{1/2}$: lifetime $\tau=462$ fs 100: weighted average of 600 fs 150 (1973Ca15, DSAM) and 400 fs 100 (1970Br11, DSAM).
2376.5	4	186 fs 35	J^π : spin=4 from $p\gamma(\theta)$ in 1971De27. $T_{1/2}$: lifetime $\tau=268$ fs 70: weighted average of 275 fs 70 (1973Ca15, DSAM) and 260 fs 75 (1970Br11, DSAM).
2581	1	21 fs 11	$E(\text{level})$: from 1971De27, 2587 in 1970Sy01, 2.58 MeV in 1973Ca15. J^π : spin=1 from $p\gamma(\theta)$ in 1971De27. $T_{1/2}$: lifetime $\tau=31$ fs 16: unweighted average of 47 fs 12 (1973Ca15, DSAM) and 15 fs 6 (1972AIZT, DSAM).
2610		0.57 ps 24	$E(\text{level})$: from 1973Ca15. $T_{1/2}$: lifetime $\tau=820$ fs 350 (1973Ca15, DSAM).
2721.8	2	>1.4 ps	J^π : spin=2 from $p\gamma(\theta)$ in 1971De27. Spin=2,(1) from $p\gamma(\theta)$ in 1970Sy01. $T_{1/2}$: lifetime $\tau>2$ ps (1973Ca15, DSAM).

Continued on next page (footnotes at end of table)

$^{32}\text{S}(^3\text{He}, p\gamma)$ **1973Ca15, 1971De27, 1971Sn01 (continued)** ^{34}Cl Levels (continued)

E(level) [†]	T _{1/2}	Comments
3126	28 fs 8	E(level): from 1971De27. Other: 3.13 MeV in 1973Ca15. J ^π : spin=1 from pγ(θ) in 1971De27. T _{1/2} : lifetime τ=40 fs 12 (1973Ca15, DSAM).
3330	80 fs 24	T _{1/2} : lifetime τ=115 fs 35 (1973Ca15, DSAM).
3380	<35 fs	T _{1/2} : lifetime τ<50 fs (1973Ca15, DSAM).
3550	97 fs 28	T _{1/2} : lifetime τ=140 fs 40 (1973Ca15, DSAM).
3600	>1.4 ps	T _{1/2} : lifetime τ>2 ps (1973Ca15, DSAM).
3630	>2 ps	T _{1/2} : lifetime τ>3 ps (1973Ca15, DSAM).
3770	80 fs 21	T _{1/2} : lifetime τ=115 fs 30 (1973Ca15, DSAM).
4080	0.9 ps +12-4	T _{1/2} : lifetime τ=1.3 ps +17-6 (1973Ca15, DSAM).
4140	104 fs 35	T _{1/2} : lifetime τ=150 fs 50 (1973Ca15, DSAM).
4350	45 fs 14	T _{1/2} : lifetime τ=65 fs 20 (1973Ca15, DSAM).
4460	139 fs 42	T _{1/2} : lifetime τ=200 fs 60 (1973Ca15, DSAM).

[†] From a least-squares fit to γ-ray energies with uncertainties for levels connected with γ transitions. The E(level) of 2581, 2610, and all levels above 3 MeV, including all their connected γ rays, are reported without any energy uncertainties.

 $\gamma(^{34}\text{Cl})$

E _i (level)	J _i ^π	E _γ [†]	I _γ	E _f	J _f ^π	Mult.	δ	Comments
146.36	3 ⁺	146.36 3		0.0	0 ⁺			E _γ : weighted average of 146.36 3 (1971Sn01), 146.4 3 (1971De27).
461.00	1	314.64 [#]	<0.5	146.36	3 ⁺			E _γ : other: 314.64 (1971Sn01).
		461.00 4	100	0.0	0 ⁺	D		I _γ : from 1971Sn01. Others: <3 (1970Br10), <5 (1970Sy01). E _γ : weighted average of 461.00 4 (1971Sn01), 461.0 1 (1971De27). I _γ : 100 (1971Sn01), 100 (1971De27), 100 (1970Sy01), 100 (1970Br10). Mult., δ: D with δ=0 from pγ(θ) in 1970Sy01. A ₂ =-0.51 2, A ₄ set to 0 (1970Br10); A ₂ =+0.34 6 in coincidence with proton feeding the 1887 level (1970Br10). A ₂ =-0.47 4, A ₄ =+0.03 7 (1970Sy01); A ₂ =-0.30 7, A ₄ =-0.05 14 at proton feeding the 2720 level (1970Sy01).
665.58	1	204.58 [#]	9 4	461.00	1			E _γ : other: 204.55 (1971Sn01). I _γ : from 1971De27. Others: <3 (1971Sn01), <5 (1970Sy01), <3 (1970Br10).
		519.22	7 4	146.36	3 ⁺			E _γ : other: 519.19 (1971Sn01). I _γ : from 1971De27. Others: <1 (1971Sn01), 7 5 (1970Sy01), <3 (1970Br10).
		665.57 5	84 5	0.0	0 ⁺	D		E _γ : weighted average of 665.56 5 (1971Sn01), 665.6 1 (1971De27). I _γ : from 1971De27. Others: 100 (1971Sn01), 93 5 (1970Sy01), 100 (1970Br10). Mult., δ: D with δ=0 from pγ(θ) in 1970Sy01. A ₂ =-0.30 3, A ₄ =0 (1970Br10). A ₂ =-0.42 7, A ₄ =+0.10 14 (1970Sy01). A ₂ =-0.12 8, A ₄ =-0.02 15 at proton feeding the 1230 level (1970Sy01).
1230.26	2	564.67 6	28 2	665.58	1	D+Q	+0.5 2	E _γ : weighted average of 564.68 6 (1971Sn01), 564.6 2 (1971De27). I _γ : from 1971De27. Others: 40.7 20 (1971Sn01), 37 3

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$^{32}\text{S}(^3\text{He}, p\gamma)$ **1973Ca15, 1971De27, 1971Sn01 (continued)** $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	I_γ	E_f	J_f^π	Mult.	δ	Comments
1230.26	2	769.25 7	38 3	461.00	1	D+Q	+3.3 +18-12	(1970Sy01), 31 4 (1970Br10). Mult., δ : from $p\gamma(\theta)$ in 1971De27. Other: D+Q with $\delta=+0.37$ 9 (1970Br10). $A_2=+0.37$ 10, $A_4=+0.10$ 10 (1970Br10). $A_2=+0.36$ 8, $A_4=-0.05$ 14 (1970Sy01). E_γ : weighted average of 769.25 7 (1971Sn01), 769.3 3 (1971De27). I_γ : from 1971De27. Others: 31.8 20 (1971Sn01), 35 3 (1970Sy01), 37 5 (1970Br10). Mult., δ : from $p\gamma(\theta)$ in 1971De27. Other: D+Q with $+0.7<\delta<+1.7$ (1970Br10). $A_2=+0.94$ 9, $A_4=+0.41$ 10 (1970Br10). $A_2=+0.63$ 9, $A_4=-0.08$ 16 (1970Sy01). E_γ : from 1971De27. Other: 1083.88 (1971Sn01). I_γ : from 1971De27. Others: 27.5 20 (1971Sn01), 28 3 (1970Sy01), 32 6 (1970Br10). Mult., δ : from $p\gamma(\theta)$ in 1971De27. Other: D+Q with $+0.4<\delta<+4.0$ (1970Br10). $A_2=-0.80$ 8, $A_4=+0.11$ 9 (1970Br10). $A_2=-0.57$ 9, $A_4=-0.15$ 18 (1970Sy01). E_γ : other: 1230.24 (1971Sn01). I_γ : from 1971De27. Others: <4 (1971Sn01), <3 (1970Sy01), <2 (1970Br10). 1887.6 2 657.3 [#] 2 2 1230.26 2 E_γ : other: 656.86 (1971Sn01). I_γ : from 1970Sy01. Others: <8 (1971Sn01), <3 (1970Br10). 1222.0 [#] 2 2 665.58 1 E_γ : other: 1221.55 (1971Sn01). I_γ : from 1970Sy01. Others: <3 (1970Br10). 1426.7 7 58 3 461.00 1 D+Q -2.1 +8-4 E_γ : from 1971De27. Other: 1426.10 (1971Sn01). I_γ : from 1970Sy01. Others: 67.2 20 (1971Sn01), 57 2 (1971De27), 61 4 (1970Br10). Mult., δ : from 1971De27. Others: D+Q with $\delta=-1.8$ 2 (1970Br10); D+Q with $-20<\delta<-+0.36$ (1970Sy01). $A_2=-0.66$ 6, $A_4=+0.39$ 8 (1970Br10). $A_2=-0.37$ 14, $A_4=-0.19$ 27 (1970Sy01). E_γ : from 1971De27. Other: 1740.74 (1971Sn01). I_γ : from 1970Sy01. Others: 32.8 20 (1971Sn01), 43 2 (1971De27), 39 4 (1970Br10). Mult., δ : from 1971De27. Others: D+Q with $-2.0<\delta<-0.3$ (1970Br10); D+Q with $-1.7<\delta<+0.8$ (1970Sy01). $A_2=+0.28$ 8, $A_4=+0.05$ 9 (1970Br10). $A_2=+0.10$ 23, $A_4=-0.39$ 39 (1970Sy01). E_γ : other: 1887.10 (1971Sn01). I_γ : from 1970Sy01. Others: <14 (1971Sn01), <1 (1970Br10). 2158.7 271 [#] <1 1887.6 2 E_γ : other: 270.90 (1971Sn01). I_γ : from 1971Sn01. 928.4 7.0 20 1230.26 2 E_γ : other: 927.76 (1971Sn01). I_γ : from 1971Sn01. Other: 10 (tentative, 1970Sy01). 1493.1 [#] <1 665.58 1 E_γ : other: 1492.45 (1971Sn01). I_γ : from 1971Sn01. Other: <10 (1970Br11). 1697.6 4 68.8 3 461.00 1 E_γ : from 1971De27. Other: 1697.00 (1971Sn01).

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$^{32}\text{S}(^3\text{He},p\gamma)$ **1973Ca15,1971De27,1971Sn01 (continued)** $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	I_γ	E_f	J_f^π	Mult.	δ	Comments
2158.7		2012.3	7.1 20	146.36	3 ⁺			I_γ : from 1971Sn01. Others: 70 (tentative, 1970Sy01), 73 8 (1970Br11). E_γ : other: 2011.64 (1971Sn01). I_γ : from 1971Sn01. Other: <8 (1970Br11). E_γ : other: 2158.00 (1971Sn01). I_γ : from 1971Sn01. Others: 20 (tentative, 1970Sy01), 27 8 (1970Br11).
		2159	17.1 20	0.0	0 ⁺			I_γ : from 1970Br11. Other: 10 (tentative, 1970Sy01). I_γ : from 1970Br11. Other: 50 (tentative, 1970Sy01). E_γ : from 1971De27.
2181.5	1,2,3	1515.9	17 5	665.58	1			I_γ : from 1970Br11. Other: 40 (tentative, 1970Sy01). Mult., δ : D+Q with $\delta=0.0$ +3-2 or $\delta=\pm 6$ +3-24 for $J_i=2$ (1971De27); D+Q with $\delta=-0.5$ 1 or $\delta<-10$ or $\delta>+3.3$ for $J_i=3$ (1971De27).
		1720.5	39 4	461.00	1			I_γ : from 1970Br11. Other: 40 (tentative, 1970Sy01).
		2035.1 7	44 4	146.36	3 ⁺			E_γ : from 1971De27.
2376.5	4	2181 [#]	<1	0.0	0 ⁺			I_γ : from 1970Br11.
		1146.2	10 7	1230.26	2			I_γ : from 1970Sy01.
		1710.9 [#]	<40	665.58	1			I_γ : from 1970Sy01. Other: <4 (1970Br11).
		1915.4 [#]	<40	461.00	1			I_γ : from 1970Sy01. Other: <2 (1970Br11).
		2230.1 4	90 7	146.36	3 ⁺	D+Q	+5.7 25	E_γ : from 1971De27. I_γ : from 1970Sy01. Others: 100 (1970Br11), 100 (1971De27). Mult., δ : from $p\gamma(\theta)$ in 1971De27. $A_2=+0.4$, $A_4=+0.4$ 1 (1971De27).
2581	1	2376 [#]	<10	0.0	0 ⁺			I_γ : from 1970Sy01. Other: <3 (1970Br11).
		2581	100	0.0	0 ⁺	D		I_γ : tentative 100 from 1970Sy01 and 1971De27. Mult., δ : D and $\delta=0$ from $p\gamma(\theta)$ in 1970Sy01. $A_2=+0.37$ 1, $A_4=+0.01$ 2 at $E_p=8.30$ MeV (1971De27). $A_2=+0.15$ 2, $A_4=+0.02$ 4 at $E_p=8.40$ MeV (1971De27). $A_2=+0.19$ 2, $A_4=+0.08$ 2 at $E_p=8.50$ MeV (1971De27). $A_2=+0.16$ 4, $A_4=+0.01$ 6 at $E_p=8.54$ MeV (1971De27). $A_2=-0.06$ 5, $A_4=-0.12$ 9 at $E_p=8.70$ MeV (1971De27). $A_2=+0.21$ 4, $A_4=+0.04$ 7 at $E_p=8.76$ MeV (1971De27). $A_2=+0.08$ 8, $A_4=-0.11$ 15 (1970Sy01).
2610		1380 [#]	20	1230.26	2			I_γ : tentative from 1970Sy01.
		1944 [#]	20	665.58	1			I_γ : tentative from 1970Sy01.
		2149 [#]	20	461.00	1			I_γ : tentative from 1970Sy01.
		2464 [#]	40	146.36	3 ⁺			I_γ : tentative from 1970Sy01.
2721.8	2	540 [#]	16 ⁺ 2	2181.5	1,2,3			I_γ : from 1971De27 for both 540 γ and 564 γ . Other: 7 4 (1970Sy01) for both 540 γ and 564 γ . 563 γ is also observed in other reactions, but 540 γ is not, and therefore, the 540 γ branch is considered tentative.
		563 ⁺	16 ⁺ 2	2158.7				I_γ : from 1971De27 for both 540 γ and 564 γ . Other: 7 4 (1970Sy01) for both 540 γ and 564 γ .

Continued on next page (footnotes at end of table)

$^{32}\text{S}(^3\text{He}, p\gamma)$ **1973Ca15, 1971De27, 1971Sn01 (continued)** $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	I_γ	E_f	J_f^π	Mult.	δ	Comments
2721.8	1491.5	5 2	1230.26	2			I_γ : from 1971De27. Other: 12 5 (1970Sy01).
	2056.2	12 2	665.58	1	D(+Q)	-0.1 2	I_γ : from 1971De27. Other: 10 6 (1970Sy01).
							Mult., δ : from $p\gamma(\theta)$ in 1971De27. Other: $\delta < -2.5$ (1971De27).
	2260.7 5	41 3	461.00	1	D(+Q)	-0.1 3	E_γ : from 1971De27.
							I_γ : from 1971De27. Other: 54 9 (1970Sy01).
							Mult., δ : from $p\gamma(\theta)$ in 1970Sy01. Others: $\delta = -3.7 + 13-28$ (1970Sy01); $\delta = -2.9 + 10$ (1971De27).
							$A_2 = -0.21$ 14, $A_4 = +0.52$ 25 (1970Sy01).
	2575.3	17 2	146.36	3 ⁺	D(+Q)	0.0 3	I_γ : from 1971De27. Other: 11 6 (1970Sy01).
							Mult., δ : from $p\gamma(\theta)$ in 1971De27. Other: $\delta < -2.5$ (1971De27).
	2722	9 2	0.0	0 ⁺			I_γ : from 1971De27. Other: 6 5 (1970Sy01).
3126	3126	100	0.0	0 ⁺			I_γ : from 1971De27.
							$A_2 = -0.13$ 17, $A_4 = +0.26$ 26 at $E_p = 8.40$ MeV (1971De27).
							$A_2 = -0.31$ 4, $A_4 = +0.04$ 6 at $E_p = 8.60$ MeV (1971De27).
							$A_2 = -0.62$ 5, $A_4 = -0.09$ 8 at $E_p = 8.70$ MeV (1971De27).
3330	1171		2158.7				
3380	2150		1230.26	2			
	3234		146.36	3 ⁺			
3550	3404		146.36	3 ⁺			
3600	878		2721.8	2			
	3454		146.36	3 ⁺			
3630	1254		2376.5	4			
3770	3770		0.0	0 ⁺			
4080	3933		146.36	3 ⁺			
4140	3993		146.36	3 ⁺			
4350	4350		0.0	0 ⁺			
4460	4313		146.36	3 ⁺			

[†] E_γ values without uncertainties are deduced from level-energy differences by evaluators, unless otherwise noted. γ rays with I_γ values reported only as upper limits are considered questionable if there are no supporting observations in the literature.

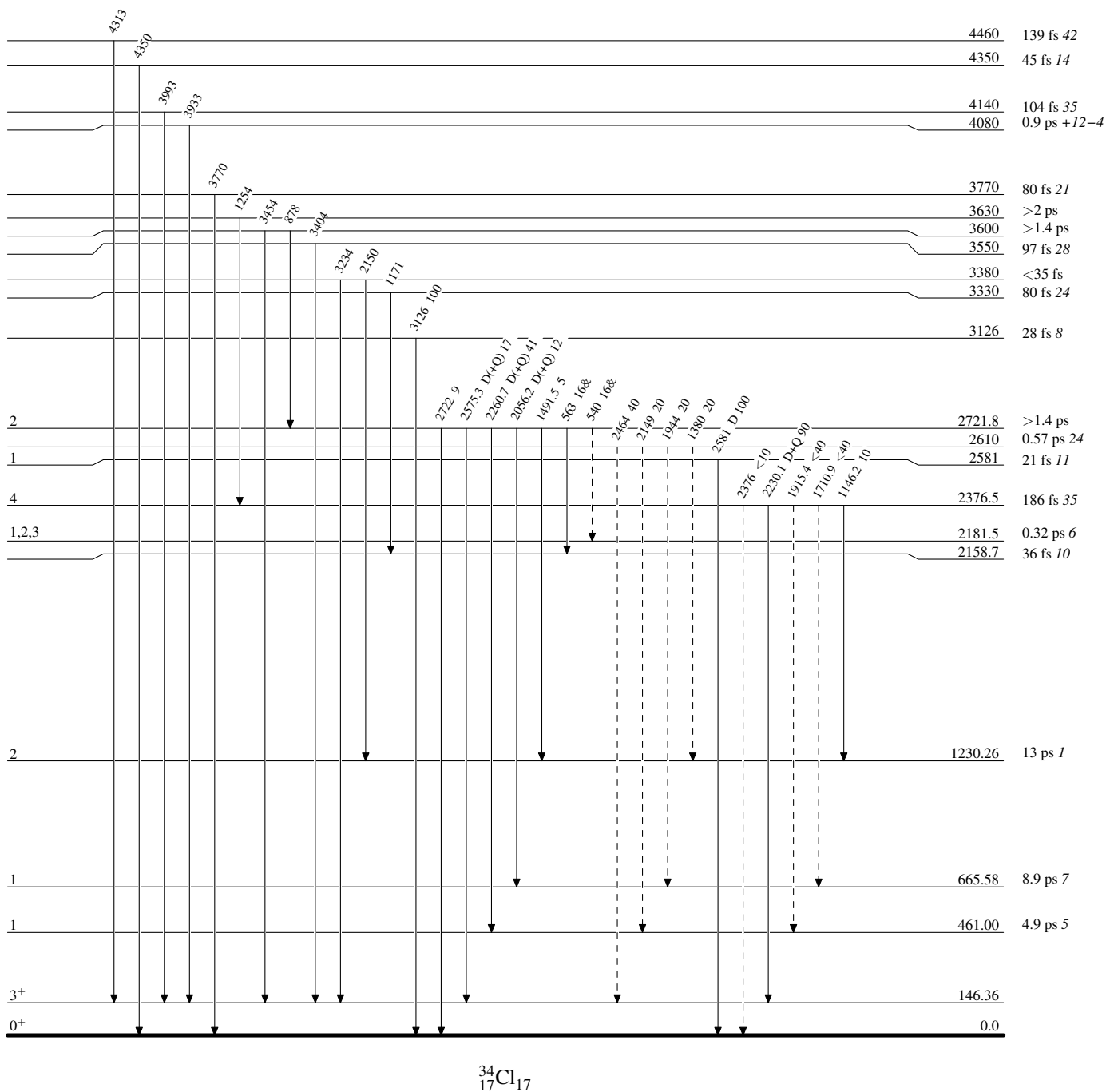
[‡] Multiply placed with undivided intensity.

Placement of transition in the level scheme is uncertain.

$^{32}\text{S}(^3\text{He},\text{p}\gamma)$ 1973Ca15,1971De27,1971Sn01

Legend

Level Scheme

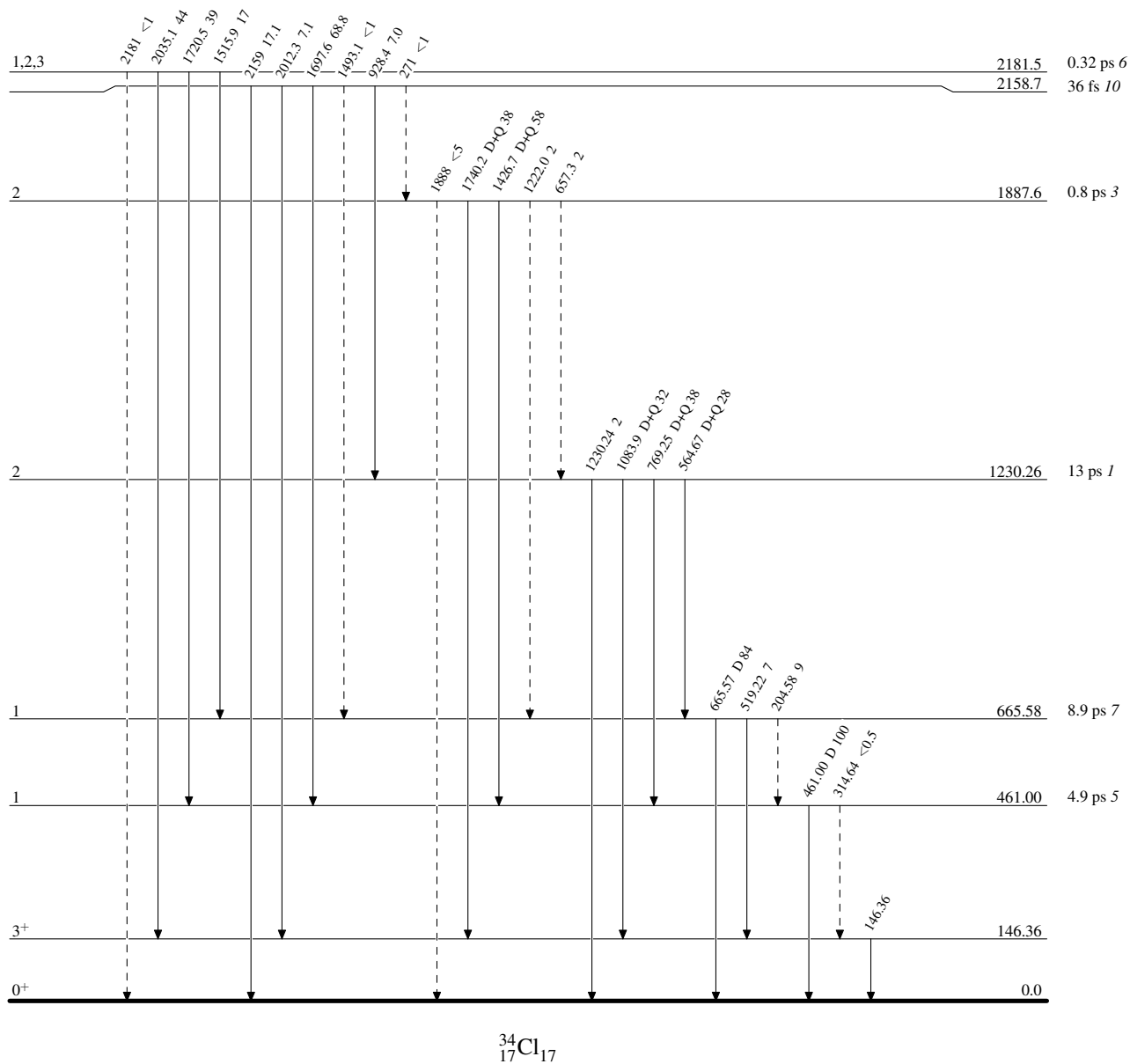
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given-----► γ Decay (Uncertain)

$^{32}\text{S}(^3\text{He},\text{p}\gamma)$ 1973Ca15,1971De27,1971Sn01

Legend

Level Scheme (continued)

Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

-----► γ Decay (Uncertain)

³²S(α,d) **1982De29,1977Na10,1976De24**

³²S+1p+1n→³⁴Cl from $J^\pi=0^+$ ³²S ground state.

1982De29, 1981De37: a 50-MeV α beam was produced from the KVI cyclotron. Targets were solid S evaporated onto C foil and covered with a C layer and gaseous H₂S. Reaction products were detected using a Si ΔE-E telescope. Measured σ(E_d,θ). Deduced levels, L-transfers, J, π, and spectroscopic factors from zero-range DWUCK-DWBA analysis.

1977Na10: a 40-MeV α beam was produced from the Michigan State University cyclotron. Targets were 99.9%-enriched ³²S of 110 μg/cm² sandwiched between layers of Formvar and carbon foils. Reaction products were detected in the focal plane of an Enge split-pole magnetic spectrograph with a proportional-counter plastic-scintillator combination with FWHM=40-60 keV. Measured σ(E_d,θ). Deduced levels, J, π, and L-transfers.

1976De24: a 34.8-MeV α beam was produced from the Princeton azimuthally varying field cyclotron. The target was CdS with natural ³²S. Reaction products were detected using a freon-cooled ΔE-E Si detector telescope with FWHM=50 keV. Measured σ(E_d,θ). Deduced levels, L-transfers, J, and π from DWUCK-DWBA analysis. Also measured ³²S(³He,p)³⁴Cl.

1966Ki10: a 29.0-MeV α beam was produced from the 105-cm Kyoto University cyclotron. Targets were 0.78-mg/cm²-thick sulfur covered by thick gold foils. Reaction products were detected using a 50-μm ΔE surface barrier and 5-mm E lithium drifted detectors. Deduced levels, L-transfers, J, and π from DWBA analysis.

³⁴Cl Levels

The experimental cross section is described as a sum over contributions from different angular momentum transfers, incorporating the spectroscopic factors A_L^2 : $(d\sigma/d\Omega)_{\text{exp}} = \alpha C \sum 1/(2J+1)[A_L^2(d\sigma/d\Omega)_{\text{macro}}^L + A_{L+2}^2(d\sigma/d\Omega)_{\text{macro}}^{L+2}]$, where $\alpha=1/3$ for (d,α) reactions and $\alpha=2J+1$ for (α,d) reactions on $J_i=0$ nuclei (**1982De29**).

E(level) [†]	L [#]	Comments
165 40	2+4	E(level): weighted average of 150 40 (1977Na10) and 180 40 (1982De29). L: other: L=(2) from 1976De24 . $A_{L=2}^2=0.71$ 16 for 3 ⁺ , $A_{L=4}^2<0.5$ for 3 ⁺ (1982De29).
479 30	0+2	E(level): weighted average of 460 40 (1977Na10) and 490 30 (1982De29). L: other: L=2 from 1976De24 . $A_{L=0}^2=1.2$ 5 for 1 ⁺ , $A_{L=2}^2=0.3$ 3 (1982De29).
1230 40	2	E(level): other: 1230 (1966Ki10). $A_{L=2}^2=0.19$ 3 for 2 ⁺ (1982De29).
1896 40		E(level): weighted average of 1880 40 (1977Na10) and 1920 50 (1966Ki10).
2175 7	2+4	E(level): weighted average of 2180 40 (1977Na10) and 2175 7 (1982De29). Other: 2210 (1966Ki10). $A_{L=2}^2=0.26$ 3 for 3 ⁺ , $A_{L=4}^2$ could not be determined (1982De29).
2710 20	1+3	E(level): weighted average of 2730 70 (1966Ki10) and 2708 20 (1982De29). L: L=(3) from 1976De24 . $A_{L=1}^2=0.4$ 2 for 2 ⁻ , $A_{L=3}^2=0.24$ 15 for 2 ⁻ (1982De29).
3603 14	3+5	E(level): weighted average of 3630 40 (1977Na10) and 3600 14 (1982De29). L: others: L=5 from 1976De24 and 1977Na10 . $A_{L=3}^2=0.88$ 8 for 4 ⁻ , $A_{L=5}^2$ could not be determined (1982De29).
4060 18	3+5	E(level): weighted average of 4080 40 (1977Na10) and 4056 18 (1982De29). Other: 4060 100 (1966Ki10). L: others: L=(5) from 1976De24 and L=5 from 1977Na10 . $A_{L=3}^2=0.39$ 4 for 4 ⁻ , $A_{L=5}^2$ could not be determined (1982De29).
4417 17	1,1+3	$A_{L=1}^2=2.6$ 5 for 0 ⁻ , $A_{L=1}^2=0.95$ 16 for 1 ⁻ , $A_{L=1}^2=0.4$ 3 for 2 ⁻ , $A_{L=3}^2<0.3$ (1982De29).
4665 40	1+3	E(level): weighted average of 4680 40 (1977Na10) and 4650 40 (1982De29). $A_{L=1}^2<0.8$ for 2 ⁻ , $A_{L=3}^2=0.6$ 3 for 2 ⁻ (1982De29).
4786 10	1+3,3,4	E(level): weighted average of 4790 40 (1977Na10) and 4786 10 (1982De29). Other: 4790 (1966Ki10). L: others: L=4 from 1977Na10 , L=6 from 1976De24 . $A_{L=3}^2=2.3$ 3 for 2 ⁻ , $A_{L=3}^2=1.48$ 15 for 3 ⁻ , $A_{L=4}^2=0.56$ 6 for 4 ⁺ (1982De29).
4930 20	0+2,2	L: 2 is slightly preferred based on the DWBA fit (1982De29). $A_{L=0}^2<2.14$ for 1 ⁺ , $A_{L=2}^2=1.4$ 10 for 1 ⁺ , $A_{L=2}^2=1.11$ 11 for 2 ⁺ (1982De29).
5277 10	6	E(level): weighted average of 5280 40 (1977Na10) and 5277 10 (1982De29). Others: 5230 (1966Ki10), 5283 (1976De24). L: 6 from 1976De24 , 1977Na10 , and 1982De29 . $A_{L=6}^2=1.0$ for 7 ⁺ (1982De29).

Continued on next page (footnotes at end of table)

$^{32}\text{S}(\alpha, \text{d})$ [1982De29](#), [1977Na10](#), [1976De24](#) (continued) ^{34}Cl Levels (continued)

E(level) [†]	L [#]	Comments
5.85×10^3 $\ddagger$ 15		
6300 $\ddagger$		
6540 $\ddagger$		
6950 $\ddagger$		
7350 20	3,6	E(level): other: 7370 (1966Ki10).
		$A_{L=3}^2=1.47$ 13 for 3^- , $A_{L=6}^2=0.60$ 5 for 6^+ , $A_{L=6}^2=0.34$ 3 for 7^+ (1982De29).
7840 30	3,5,6	$A_{L=3}^2=1.08$ 11 for 3^- , $A_{L=5}^2=0.65$ 6 for 5^- , $A_{L=6}^2=0.41$ 4 for 6^+ , $A_{L=6}^2=0.23$ 3 for 7^+ (1982De29).
8390 20	3,5,6	$A_{L=3}^2=1.86$ 17 for 3^- , $A_{L=5}^2=1.05$ 9 for 5^- , $A_{L=6}^2=0.64$ 6 for 6^+ , $A_{L=6}^2=0.36$ 4 for 7^+ (1982De29).

[†] From [1982De29](#), unless otherwise noted. $\ddagger$ From [1966Ki10](#).[#] From DWBA analysis of the measured $\sigma(\theta)$ in [1982De29](#), unless otherwise noted.

 $^{33}\text{S}(\text{p},\gamma)$ [1983Wa27](#), [1977Da02](#), [1992Ka39](#)

$^{33}\text{S} + \text{p} \rightarrow ^{34}\text{Cl}$ from $J^\pi = 3/2^+$ ^{33}S ground state.

- [1983Wa27](#): 0.4-2.0-MeV proton beams were provided from the 2.5-MV Van de Graaff accelerator at Potchefstroom University, South Africa. The targets were Ag_2S (92%-enriched in ^{33}S) on Ta backings. γ rays were detected using a NaI detector and two Ge(Li) detectors. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced reaction Q values, J, π , 41 bound levels, 41 resonance strengths, and lifetimes for 17 bound levels using DSAM.
- [1977Da02](#): 1.0-2.0-MeV proton beams were provided from the 4-MV Van de Graaff accelerator at CENBG, France. The targets were Ag_2S (92%-enriched in ^{33}S) on Au backings. γ rays were detected using a NaI detector and two Ge(Li) detectors. Measured E_γ , I_γ , and radiative capture excitation functions. Deduced reaction Q values, levels, 45 resonance strengths, γ -ray branching ratios, and lifetimes using DSAM.
- [1977Da03](#): similar setup as [1977Da02](#). Protons were detected using four Si(Li) surface barrier detectors. Measured excitation functions for (p,p) elastic scattering. Deduced γ multipole mixing ratios, resonance energies, J, and π from an R-matrix fit on (p,p) excitation functions.
- [1992Ka39](#): 1.8-3.5-MeV proton beams impinged on a Ag_2S target. γ rays were detected using Ge(Li) and NaI(Tl) detectors. Measured E_γ , I_γ , and $\gamma(\theta)$. Deduced J, π , and γ multipolarities and multipole mixing ratios.
- [2020Pa40](#): a 1.072-MeV proton beam was produced from the 5U 5-MV single-ended vertical Pelletron at the Nuclear Science Laboratory of University of Notre Dame. The target was prepared by implanting 70-keV negative ^{33}S ions into a tantalum foil. γ rays were detected with the compact detector array GEORGINA, consisting of five Ge detectors. Measured E_γ , I_γ , $\gamma\gamma$ -coin. Deduced precise γ -ray branching ratios for the 666-keV level.
- [2013Fa10](#): Direct measurement of the $^{33}\text{S}(\text{p},\gamma)^{34}\text{Cl}$ reaction in inverse kinematics using 194-514 keV/nucleon ^{33}S beams produced from the Isotope Separator and Accelerator (ISAC) radioactive beam facility at TRIUMF. The beams impinged on a windowless gas target of H_2 gas on the DRAGON (Detector of Recoils And Gammas Of Nuclear reactions) recoil separator. γ rays from radiative captures were detected by 30 BGO detectors surrounding the target. Recoil products were separated by the DRAGON separator and detected in a double-sided silicon strip detector (DSSSD) at the end. Measured E_γ , reaction yields. Discovered two new resonances. Deduced resonance energies (E_r) and resonance strengths ($\omega\gamma$). Summarized in Conference proceedings [2010PaZT](#).
- [2011Fr04](#): Direct measurement of the $^{33}\text{S}(\text{p},\gamma)^{34}\text{Cl}$ reaction in normal kinematics using 200-710 keV proton beams provided from a tandem Van de Graaff accelerator at the University of Washington. Targets were prepared by implanting 45-keV 10^{16} ^{33}S ions into oxygen-free high-conductivity copper substrates. γ rays were detected using two HPGe detectors at $\pm 55^\circ$, surrounded by plastic scintillators operating in anticoincidence mode. Measured E_γ and I_γ for three resonances at 432, 492, and 529 keV. Deduced precise γ -ray branching ratios for the three resonances and the percentage of decays producing ^{34}Cl g.s. and $^{34\text{m}}\text{Cl}$ at 147 keV. Summarized in Conference proceedings [2010PaZT](#).
- [1994Li20](#): 0.973-1.997-MeV proton beams were provided from the AURA2 accelerator at the University of Auckland, New Zealand and impinged on Ag_2S targets 98%-enriched in ^{33}S . γ rays were detected by a 22% Ge(Li) at 90° relative to the beam direction and a NaI detector. Positrons from ^{34}Cl decay were detected by a plastic scintillator ΔE -E telescope. A separate measurement was performed using the proton beams from the 3 MeV Van de Graaff laboratory at the University of Florence, Italy. γ rays were detected using an 80% intrinsic Ge detector. Measured E_γ , I_γ , and positron yields. Deduced $^{33}\text{S}(\text{p},\gamma)^{34}\text{Cl}$ reaction Q values, S(p) of ^{34}Cl , and precise excitation energies for three resonances.
- [1985La16](#): 1.05-1.98-MeV proton beams were provided from the 2.5-MV Van de Graaff accelerator at the University of Helsinki and impinged on ^{33}S targets on Ta backings. γ rays were detected using a PGT 110- cm^3 Ge(Li) detector. Deduced lifetimes for 35 bound levels using DSAM.
- [1984Ke01](#): 0.82- and 1.43-MeV proton beams were provided from the 2.5-MV Van de Graaff accelerator at the University of Helsinki and impinged on ^{33}S targets on Ta backings. γ rays were detected using a PGT Ge(Li) detector. Deduced lifetimes for two bound levels using DSAM.
- [1983Ra04](#): 0.9-1.4-MeV proton beams impinged on an 88.2%-enriched ^{33}S target (CdS on carbon backing). γ rays were detected using a NaI detector at 90° relative to the beam direction and protons were detected using a Si surface barrier detector at 135° relative to the beam direction. Measured E_γ , I_γ , reaction Q, and S(p).
- [1973An13](#): 1.3-1.8-MeV proton beams were provided from the 2.5-MV van de Graaff accelerator at the University of Helsinki and impinged on ^{33}S targets on Ta backings. γ rays were detected using a 40- cm^3 Ge(Li) detector. Measured E_γ , I_γ . Deduced levels and lifetimes using DSAM.
- [1971Hy02](#): 1058, 1098, 1121-keV proton beams were provided from the Aerospace Research Laboratories 2-MeV Van de Graaff accelerator and impinged on an 84%-enriched ^{33}S target (Ag backing soldered on brass disk). γ rays were detected using a Ge(Li) detector and a NaI detector. Measured E_γ , I_γ , and $\text{py}(\theta)$. Deduced levels, J, π , resonance strengths for three resonances, γ multipolarities, and multipole mixing ratios.

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

1969Gr29: 1.0-1.3-MeV proton beams impinged on a 25%-enriched ^{33}S target. γ rays were detected using a Ge(Li) detector at 0° and 115° or 120° relative to the beam direction. Measured E_γ , I_γ , $\text{p}\gamma(\theta)$, and $\text{p}\gamma(\theta)$. Deduced levels, J , π , γ multipolarities, multipole mixing ratios, and lifetimes using DSAM.

1964Gl04: 0.3-1.3-MeV proton beams impinged on a 22%-enriched ^{33}S target (CdS on Cu or Ta backings). γ rays were detected using NaI detectors at $+85^\circ$ and -85° relative to the beam direction. Measured E_γ , I_γ , $\gamma\gamma$ -coin for nine resonances ($E_{\text{p}}(\text{lab})=447$, 507, 546, 639, 662, 683, 731, 777, 822 keV). Deduced resonance strengths.

1959Ku79: protons were accelerated with the Utrecht 850-kV Cockcroft-Walton generator and impinged on a CdS target enriched in ^{33}S . Measured γ -ray yield curves. Deduced two resonances.

Standard definition of resonance strength $\omega\gamma=(2J_{\text{r}}+1)/[(2J_{\text{p}}+1)(2J_{\text{T}}+1)]\times\Gamma_{\gamma}\Gamma_{\text{p}}/\Gamma$, where J_{r} is the spin of the resonance, $J_{\text{p}}=1/2$ is the spin of the proton, and $J_{\text{T}}=3/2$ is the spin of the target ^{33}S ground state. The resonance strengths S from the original papers in this dataset are mainly obtained from resonance γ -ray yields and are defined without the $(2J_{\text{p}}+1)(2J_{\text{T}}+1)$ factor. Therefore, to obtain the standard resonance strength, all the resonance strengths provided in the comments below should be divided by 8, except for $\omega\gamma$ values from **2013Fa10**, which are already in the standard definition.

 ^{34}Cl Levels

$E(\text{level})^{\dagger\ddagger}$	$J^{\pi\#}$	$T_{1/2}$	Comments
0	$0^+ @$		
146.4 1	$3^+ @$		E(level): weighted average of 146.5 5 (1977Da02), 146.4 1 (1973An13), and 146.8 10 (1969Gr29).
461.2 2	$1^+ @$		E(level): weighted average of 461.2 5 (1977Da02), 461.0 2 (1973An13), and 461.5 3 (1969Gr29).
665.3 2	1		E(level): weighted average of 665.7 5 (1977Da02), 665.6 2 (1973An13), and 664.6 3 (1969Gr29).
1230.33 8	2^+		J^π : from $\gamma(\theta)$ γ cascade at $E_{\text{p}}(\text{lab})=1266$ in 1969Gr29 . E(level): weighted average of 1230.5 3 (1977Da02), 1230.33 8 (1983Wa27), 1230.5 5 (1973An13), and 1228.8 7 (1969Gr29).
1887.29 8	$2^+ @$		E(level): weighted average of 1887.2 3 (1977Da02), 1887.31 8 (1983Wa27), 1887.0 5 (1973An13), and 1885.9 16 (1969Gr29).
2157.9 1	$2^+ @$	33 fs 4	E(level): weighted average of 2158.1 3 (1977Da02), 2157.90 8 (1983Wa27), 2157.9 1 (1973An13), and 2158.4 12 (1969Gr29).
2181.10 8	3	0.45 ps 13	$T_{1/2}$: lifetime $\tau=47$ fs 5: weighted average of 36 fs 15 (1977Da02 , DSAM), 46 fs 10 (1983Wa27 , DSAM), 47 fs 5 (1984Ke01 , DSAM), and 49 fs 5 (1985La16 , DSAM). E(level): weighted average of 2181.2 5 (1977Da02), 2181.10 8 (1983Wa27), and 2179.9 14 (1973An13).
2375.7 2	4	139 fs 35	J^π : from $\gamma(\theta)$ γ cascade at $E_{\text{p}}(\text{lab})=1545$ in 1977Da03 . Other: spin=(3) from $\gamma(\theta)$ γ cascade at $E_{\text{p}}(\text{lab})=1098$ in 1971Hy02 . $T_{1/2}$: lifetime $\tau=649$ fs 190: weighted average of 600 fs 350 (1977Da02 , DSAM), 630 fs 190 (1983Wa27 , DSAM), and 800 fs 400 (1985La16 , DSAM). E(level): weighted average of 2375.6 5 (1977Da02), 2375.7 2 (1983Wa27), 2368 4 (1973An13), 2377.3 12 (1971Hy02), and 2375.6 7 (1969Gr29).
2580.3 2		2.1 fs 17	J^π : from $\gamma(\theta)$ γ cascade at $E_{\text{p}}(\text{lab})=1165$ in 1969Gr29 . $T_{1/2}$: lifetime $\tau=201$ fs 50: weighted average of 150 fs 50 (1973An13 , DSAM), 210 fs 80 (1977Da02 , DSAM), 200 fs 50 (1983Wa27 , DSAM), and 250 fs 50 (1985La16 , DSAM). E(level): weighted average of 2579.9 5 (1977Da02), 2580.4 2 (1983Wa27), 2580.4 5 (1973An13), and 2580.6 13 (1969Gr29).
2611.03 11	3	159 fs 42	$T_{1/2}$: lifetime $\tau=3.0$ fs 25 (1977Da02 , DSAM). Others: <5 fs (1983Wa27 , DSAM) and <7 fs (1985La16 , DSAM). E(level): weighted average of 2610.6 5 (1977Da02), 2611.05 11 (1983Wa27), 2611.1 5 (1973An13), and 2611.2 13 (1971Hy02).
2721.3 2	2	>1.0 ps	J^π : from $\gamma(\theta)$ γ cascade at $E_{\text{p}}(\text{lab})=1057$ in 1977Da03 . $T_{1/2}$: lifetime $\tau=230$ fs 60: weighted average of 210 fs 60 (1983Wa27 , DSAM) and 340 fs 140 (1985La16 , DSAM). Other: >450 fs (1977Da02 , DSAM). E(level): weighted average of 2721.4 7 (1977Da02), 2721.1 2 (1983Wa27), 2721 1 (1973An13), 2721.7 3 (1971Hy02), and 2720.4 16 (1969Gr29).
			J^π : from $\gamma(\theta)$ γ cascade at $E_{\text{p}}(\text{lab})=1098$ in 1971Hy02 .

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π #	T _{1/2}	Comments
			T _{1/2} : lifetime $\tau > 1500$ fs (1985La16 , DSAM). Others: > 1000 fs (1973An13 , DSAM) and > 400 fs (1977Da02 , DSAM).
3129.12 12		<3 fs	E(level): weighted average of 3128.9 5 (1977Da02), 3129.13 12 (1983Wa27), and 3129 2 (1973An13).
			T _{1/2} : lifetime $\tau < 4$ fs (1985La16 , DSAM). Other: < 10 fs (1983Wa27 , DSAM).
3334.0 2		37 fs 14	E(level): weighted average of 3333.9 10 (1977Da02) and 3334.0 2 (1983Wa27).
			T _{1/2} : lifetime $\tau = 53$ fs 20 (1985La16 , DSAM).
3383.3 2		5.3 fs 12	E(level): weighted average of 3383.6 6 (1977Da02) and 3383.3 2 (1983Wa27).
			T _{1/2} : lifetime $\tau = 7.7$ fs 17: weighted average of 7.5 fs 17 (1984Ke01 , DSAM) and 9 fs 4 (1985La16 , DSAM). Others: < 3 fs (1977Da02 , DSAM) and < 10 fs (1983Wa27 , DSAM).
3545.08 15	3	102 fs 21	E(level): weighted average of 3545.2 8 (1977Da02), 3545.07 15 (1983Wa27), 3545 1 (1973An13), 3545.3 6 (1971Hy02), and 3544.6 13 (1969Gr29).
			J ^π : from $\gamma(\theta)$ γ cascade at E _p (lab)=1057, 1545, and 1997 in 1977Da03 and from $\gamma(\theta)$ γ cascade at E _p (lab)=1098 in 1971Hy02 .
			T _{1/2} : lifetime $\tau = 147$ fs 30: weighted average of 140 fs 45 (1983Wa27 , DSAM) and 150 fs 30 (1985La16 , DSAM). Other: > 500 fs (1973An13 , DSAM).
3600.28 11	4	>0.5 ps	E(level): weighted average of 3600.3 8 (1977Da02), 3600.27 11 (1983Wa27), 3600.2 3 (1973An13), 3600.7 4 (1971Hy02), and 3598.4 12 (1969Gr29).
			J ^π : from $\gamma(\theta)$ γ cascade at E _p (lab)=1545 and 1997 in 1977Da03 and $\gamma(\theta)$ γ cascade at E _p (lab)=1098 in 1971Hy02 .
			T _{1/2} : lifetime $\tau > 0.7$ ps (1973An13 , DSAM).
3631.7 3	5	>0.76 ps	E(level): weighted average of 3631.1 8 (1977Da02), 3631.8 3 (1983Wa27), 3631.6 3 (1973An13), and 3632.5 14 (1971Hy02).
			J ^π : from $\gamma(\theta)$ γ cascade at E _p (lab)=1782 in 1977Da03 .
			T _{1/2} : lifetime $\tau > 1100$ fs (1985La16 , DSAM). Other: > 700 fs (1977Da02 , DSAM).
3646.3 2		0.15 ps 6	E(level): weighted average of 3645.6 15 (1977Da02) and 3646.3 2 (1983Wa27).
			T _{1/2} : lifetime $\tau = 220$ fs 90 (1985La16 , DSAM).
			Another I γ =33 decays to unknown final levels (1983Wa27).
3660.0 3		<3 fs	E(level): weighted average of 3659.4 15 (1977Da02) and 3660.0 3 (1983Wa27).
			T _{1/2} : lifetime $\tau < 5$ fs (1985La16 , DSAM).
3773.72 41	1	57 fs 19	E(level): weighted average of 3773.9 10 (1977Da02) and 3773.84 15 (1983Wa27), and 3771.0 7 (1971Hy02).
			J ^π : from $\gamma(\theta)$ γ cascade at E _p (lab)=1121 in 1971Hy02 .
			T _{1/2} : lifetime $\tau = 82$ fs 28: unweighted average of 55 fs 15 (1983Wa27 , DSAM) and 110 fs 20 (1985La16 , DSAM).
			Another I γ =30 decays to unknown final levels (1983Wa27).
3791.7 2			E(level): from 1983Wa27 .
			Another I γ =18 decays to unknown final levels (1983Wa27).
3940.1 3		<4 fs	E(level): weighted average of 3941.2 15 (1977Da02) and 3940.1 3 (1983Wa27).
			T _{1/2} : lifetime $\tau < 6$ fs (1985La16 , DSAM).
3964.1 3		90 fs 42	E(level): weighted average of 3964.2 15 (1977Da02) and 3964.1 3 (1983Wa27).
			T _{1/2} : lifetime $\tau = 130$ fs 60: weighted average of 160 fs 60 (1977Da02 , DSAM) and 90 fs 70 (1985La16 , DSAM).
3983.0 3	3	131 fs 28	E(level): weighted average of 3982.2 5 (1977Da02), 3983.5 2 (1983Wa27), 3983 1 (1973An13), 3982.1 3 (1971Hy02), and 3982.0 11 (1969Gr29).
			J ^π : from $\gamma(\theta)$ γ cascade at E _p (lab)=1058 and 1098 in 1971Hy02 and from $\gamma(\theta)$ γ cascade at E _p (lab)=1057 and 1997 in 1977Da03 .
			T _{1/2} : lifetime $\tau = 189$ fs 40: weighted average of 165 fs 60 (1983Wa27 , DSAM) and 200 fs 40 (1985La16 , DSAM). Other: > 100 fs (1973An13 , DSAM).
4076.2 2	4	>0.3 ps	E(level): weighted average of 4075.9 10 (1977Da02), 4076.3 2 (1983Wa27), 4074 12 (1973An13), and 4075.4 9 (1971Hy02).
			J ^π : from $\gamma(\theta)$ γ cascade at E _p (lab)=1058 and 1098 in 1971Hy02 and from $\gamma(\theta)$ γ cascade at E _p (lab)=1545 in 1977Da03 .
			T _{1/2} : lifetime $\tau > 0.5$ ps (1977Da02 , DSAM). Other: > 0.25 ps (1973An13 , DSAM).
4139.7 2	2	89 fs 28	E(level): weighted average of 4139.9 15 (1977Da02) and 4139.8 2 (1983Wa27), and 4136.6

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π #	T _{1/2}	Comments
			12 (1971Hy02). J ^π : spin=2,3 from $\gamma(\theta)$ γ cascade at E _p (lab)=1121 in 1971Hy02 and the authors ruled out J=3 by RUL. T _{1/2} : lifetime τ =129 fs 40: weighted average of 120 fs 40 (1983Wa27, DSAM) and 150 fs 60 (1985La16, DSAM).
4147.8 2 4325.89 15		5.1 fs 21	E(level): weighted average of 4325.6 5 (1977Da02) and 4325.91 15 (1983Wa27). T _{1/2} : lifetime τ =7.4 fs 30: weighted average of 11 fs 9 (1977Da02, DSAM) and 7 fs 3 (1985La16, DSAM). Another I γ =25 decays to unknown final levels (1983Wa27).
4354.2 2	(1)	30 fs 6	E(level): weighted average of 4352.9 10 (1977Da02) and 4354.3 2 (1983Wa27), and 4352.7 9 (1971Hy02). J ^π : from $\gamma(\theta)$ γ cascade at E _p (lab)=1121 in 1971Hy02. T _{1/2} : lifetime τ =43 fs 8 (1985La16, DSAM). Other: <10 fs (1983Wa27, DSAM).
4417.3 2		24 fs 8	E(level): weighted average of 4416.4 5 (1977Da02), 4417.4 2 (1983Wa27), and 4415.8 23 (1971Hy02). T _{1/2} : lifetime τ =35 fs 11 (1985La16, DSAM).
4446.6 2	4	116 fs 24	E(level): weighted average of 4444.7 15 (1977Da02) and 4446.6 2 (1983Wa27). J ^π : from $\gamma(\theta)$ γ cascade at E _p (lab)=1782 in 1977Da03. T _{1/2} : lifetime τ =167 fs 35: weighted average of 160 fs 60 (1977Da02, DSAM), 180 fs 60 (1983Wa27, DSAM), and 165 fs 35 (1985La16, DSAM).
4461.4 3		0.26 ps 17	E(level): weighted average of 4462.2 15 (1977Da02) and 4461.4 3 (1983Wa27). T _{1/2} : lifetime τ =370 fs 250 (1985La16, DSAM).
4515.7 2		10 fs 4	E(level): weighted average of 4515.3 5 (1977Da02), 4515.8 2 (1983Wa27), and 4514.5 7 (1971Hy02). T _{1/2} : lifetime τ =15 fs 5: weighted average of 15 fs 7 (1983Wa27, DSAM) and 15 fs 5 (1985La16, DSAM).
4605.8 10		0.24 ps 17	E(level): from 1977Da02. T _{1/2} : lifetime τ =350 fs 240 (1985La16, DSAM).
4609.7 15 4638.9 4	1,2	31 fs 10	E(level): from 1977Da02. E(level): weighted average of 4638.3 15 (1977Da02) and 4638.9 4 (1983Wa27), and 4639.0 23 (1971Hy02). J ^π : from $\gamma(\theta)$ γ cascade at E _p (lab)=1121 in 1971Hy02. T _{1/2} : lifetime τ =45 fs 14 (1985La16, DSAM).
4695.7 2		6.2 fs 28	E(level): weighted average of 4695 2 (1977Da02) and 4695.7 2 (1983Wa27). T _{1/2} : lifetime τ =9 fs 4 (1985La16, DSAM). Other: <10 fs (1983Wa27, DSAM).
4717.4 6		<9 fs	E(level): weighted average of 4717.5 10 (1977Da02) and 4717.4 6 (1983Wa27). T _{1/2} : lifetime τ <13 fs (1985La16, DSAM).
4824.41 18		0.33 ps 15	E(level): weighted average of 4823.4 10 (1977Da02), 4824.5 2 (1983Wa27), and 4824 10 (1973An13). T _{1/2} : lifetime τ =480 fs 220 (1983Wa27, DSAM). Other: >500 fs (1985La16, DSAM).
4941.9 4		<6 fs	E(level): weighted average of 4941.4 15 (1977Da02) and 4941.9 4 (1983Wa27). T _{1/2} : lifetime τ <8 fs (1985La16, DSAM).
4957.3 11		<12 fs	E(level): weighted average of 4959 2 (1977Da02) and 4957.3 11 (1983Wa27). T _{1/2} : lifetime τ <17 fs (1985La16, DSAM).
4995.6 3		5 fs 3	E(level): weighted average of 4993 2 (1977Da02) and 4995.6 3 (1983Wa27). T _{1/2} : lifetime τ =7 fs 4 (1985La16, DSAM). Other: <5 fs (1983Wa27, DSAM). Another I γ =43 decays to unknown final levels (1983Wa27).
5020? 5171.6 3	4	51 fs 21	E(level): from 1977Da02. E(level): weighted average of 5170 2 (1977Da02) and 5171.6 3 (1983Wa27). J ^π : from $\gamma(\theta)$ γ cascade at E _p (lab)=1782 in 1977Da03. T _{1/2} : lifetime τ =73 fs 30: weighted average of 60 fs 30 (1977Da02, DSAM) and 90 fs 35 (1985La16, DSAM).
5326			Observed in 2013Fa10 and E _r =183 is quoted. $\omega\gamma$ <1.42 μeV (2013Fa10).
5357			Observed in 2013Fa10 and E _r =214 is quoted.

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$^{33}\text{S}(\text{p},\gamma)$ [1983Wa27](#),[1977Da02](#),[1992Ka39](#) (continued) ^{34}Cl Levels (continued)

E(level) $\dagger\dagger$	J^π #	$T_{1/2}$	$E_{\text{p}}(\text{lab})$ (keV) $\&$	Comments
5386.8 15				$\omega\gamma < 10.5 \mu\text{eV}$ (2013Fa10). E(level): weighted average of 5383 2 (1977Da02) and 5386.8 15 (1983Wa27). Also observed in 2013Fa10 and $E_{\text{r}}=243.6$ is quoted.
5424				$\omega\gamma < 21.8 \mu\text{eV}$ (2013Fa10). Observed in 2013Fa10 and $E_{\text{r}}=281$ is quoted.
5436? 2				$\omega\gamma < 10.1 \mu\text{eV}$ (2013Fa10). $E_{\text{r}}=293 \text{ keV}$ 2 (2013Fa10).
5454? 2				$\omega\gamma=82 \mu\text{eV} +18-21$ (2013Fa10). $E_{\text{r}}=311 \text{ keV}$ 2 (2013Fa10).
5485				$\omega\gamma=28 \mu\text{eV} +10-8$ (2013Fa10). Observed in 2013Fa10 and $E_{\text{r}}=342$ is quoted.
5540.8 11		$>0.7 \text{ ps}$		$\omega\gamma < 14.2 \mu\text{eV}$ (2013Fa10). E(level): weighted average of 5540 2 (1977Da02) and 5540.8 11 (1983Wa27). $T_{1/2}$: lifetime $\tau > 1000 \text{ fs}$ (1977Da02 , DSAM). Observed in 2013Fa10 and $E_{\text{r}}=399$ is quoted.
5577.6 10			447.0 10	$\omega\gamma < 3.99 \mu\text{eV}$ (2013Fa10). E(level): from $E_{\text{c.m.}}=434.4 10$: weighted average of $E_{\text{c.m.}}=433.7 10$ (1959Ku79 , 1964Gl04) and $E_{\text{c.m.}}=435 1$ (2013Fa10). $E_{\text{p}}(\text{lab})$ (keV): weighted average of 446.5 35 (1959Ku79) and 447 1 (1964Gl04), corresponding to $E_{\text{c.m.}}=433.7 10$. $\omega\gamma=50 \text{ meV} +10-7$ (2013Fa10). $\omega\gamma=0.4 \text{ eV}$ 1 (1983Wa27). A γ ray at 1700 40 with $I_{\gamma}=7 3$ is reported by 1964Gl04 . The authors assigned a 3.87 MeV level as the final level, which has not been observed by any other work.
5635.75 20	1,2		507.6 2	E(level): from $E_{\text{c.m.}}=492.51 20$: weighted average of 492.1 10 (1959Ku79), 492.0 10 (1964Gl04), 492.55 20 (1983Wa27), and 491 keV 5 (2013Fa10). J^π : from $\gamma(\theta)$ in 1983Wa27 . $E_{\text{p}}(\text{lab})$ (keV): weighted average of 507.1 10 (1959Ku79), 507 1 (1964Gl04), and 507.6 2 (1983Wa27). $\omega\gamma=0.16 \text{ eV}$ 8 (1964Gl04). $\omega\gamma=0.7 \text{ eV}$ 2 (1983Wa27).
5673.0 10			546 1	$\omega\gamma=61 \text{ meV} +11-8$ (2013Fa10). $E_{\text{p}}(\text{lab})$ (keV): from 1964Gl04 . $\omega\gamma=0.040 \text{ eV}$ 20 (1964Gl04).
5763.3 10			639 1	$\omega\gamma=0.7 \text{ eV}$ 3 (1983Wa27). $E_{\text{p}}(\text{lab})$ (keV): from 1964Gl04 . $\omega\gamma=0.011 \text{ eV}$ 6 (1964Gl04).
5785.6 10			662 1	$\omega\gamma=0.06 \text{ eV}$ 3 (1983Wa27). $E_{\text{p}}(\text{lab})$ (keV): from 1964Gl04 . $\omega\gamma=0.034 \text{ eV}$ 17 (1964Gl04).
5805.9 10			683 1	$\omega\gamma=0.4 \text{ eV}$ 2 (1983Wa27). $E_{\text{p}}(\text{lab})$ (keV): from 1964Gl04 . $\omega\gamma=0.05 \text{ eV}$ 3 (1964Gl04).
5852.91 30			731.4 3	$\omega\gamma=0.4 \text{ eV}$ 2 (1983Wa27). $E_{\text{p}}(\text{lab})$ (keV): weighted average of 731 1 (1964Gl04) and 731.4 3 (1983Wa27). $\omega\gamma=0.05 \text{ eV}$ 3 (1964Gl04).
5868.0 10			747 1	$\omega\gamma=0.5 \text{ eV}$ 2 (1983Wa27). $E_{\text{p}}(\text{lab})$ (keV): from 1964Gl04 . $\omega\gamma=0.010 \text{ eV}$ 5 (1964Gl04).
5897.2 10			777 1	$E_{\text{p}}(\text{lab})$ (keV): from 1964Gl04 . $\omega\gamma=0.05 \text{ eV}$ 3 (1964Gl04). $\omega\gamma=0.5 \text{ eV}$ 2 (1983Wa27).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π [#]	T _{1/2}	E _p (lab) (keV) ^{&}	Comments
5940.8 10	1 to 3		822 1	J ^π : from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): from 1964Gl04. $\omega\gamma=0.12$ eV 6 (1964Gl04). $\omega\gamma=0.8$ eV 2 (1983Wa27).
6030.1 20			914 2	E _p (lab) (keV): from 1964Gl04. $\omega\gamma=0.050$ eV 25 (1964Gl04). $\omega\gamma=0.4$ eV 2 (1983Wa27).
6088.94 6	1,2		974.62 4	E(level): weighted average of E(level)=6090.3 20 deduced at E _p (lab)=976 2 (1964Gl04), E(level)=6089.12 16 deduced at E _p (lab)=974.83 16 measured by 1983Ra04, E(level)=6088.90 6 deduced from E _p (lab)=974.61 4 measured by 1994Li20, and E(level)=6088.97 8 also by 1994Li20 based on γ data without reporting E _γ . J ^π : from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 976 2 (1964Gl04), 974.83 16 (1983Ra04), and 974.61 4 (1994Li20). $\omega\gamma=1.0$ eV 3 (1983Wa27). $\omega\gamma=0.13$ eV 7 (1964Gl04).
6136.2 11	1 to 3		1023.3 11	J ^π : from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1023 2 (1964Gl04) and 1023.4 11 (1977Da02). $\omega\gamma=0.02$ eV 1 (1964Gl04). $\omega\gamma=0.7$ eV 2 (1983Wa27). $\omega\gamma=0.6$ eV 2 (1977Da02).
6141.8 11	1,2		1029.1 11	J ^π : from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1029 2 (1964Gl04) and 1029.1 11 (1977Da02). $\omega\gamma=0.15$ eV 8 (1964Gl04). $\omega\gamma=1.1$ eV 3 (1983Wa27). $\omega\gamma=0.9$ eV 5 (1977Da02).
6169.4 11	3		1057.6 5	J ^π : from $\gamma(\theta)$ in 1971Hy02 and 1977Da03. Other: spin=2:5 from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1057 2 (1964Gl04), 1057.3 11 (1977Da02), and 1057.7 5 (1971Hy02). $\omega\gamma=0.45$ eV 23 (1964Gl04). $\omega\gamma=1.8$ eV 5 (1983Wa27). $\omega\gamma=1.5$ eV 7 (1977Da02). $\omega\gamma=3.6$ eV 8 (1969Gr29). $\omega\gamma=2.8$ eV 9 (1971Hy02).
6181.27 20	1,2		1069.8 2	J ^π : from $\gamma(\theta)$ of E _p (lab)=1071→666 and E _p (lab)=1071→146 (1969Gr29). E _p (lab) (keV): weighted average of 1071 2 (1964Gl04), 1071.1 11 (1977Da02), and 1069.7 2 (1983Wa27). $\omega\gamma=0.60$ eV 30 (1964Gl04). $\omega\gamma=3.6$ eV 5 (1977Da02), used for absolute strength calibration by 1983Wa27.
6208.2 5	4	<1.4 fs	1097.5 5	$\omega\gamma=3.5$ eV 6 (1969Gr29). J ^π : from $\gamma(\theta)$ in 1971Hy02. T _{1/2} : lifetime $\tau < 0.002$ ps (1973An13, DSAM). E _p (lab) (keV): weighted average of 1096 2 (1964Gl04), 1096.5 12 (1977Da02), and 1097.7 5 (1971Hy02). $\omega\gamma=0.29$ eV 15 (1964Gl04). $\omega\gamma=1.4$ eV 3 (1983Wa27). $\omega\gamma=2.2$ eV 10 (1977Da02). $\omega\gamma=5.3$ eV 11 (1969Gr29). $\omega\gamma=3.1$ eV 9 (1971Hy02).
6219.3 12			1109.0 12	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.10$ eV 6 (1977Da02).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π [#]	E _p (lab) (keV) ^{&}	Comments
6229.5 5	1	1119.5 5	J ^π : from $\gamma(\theta)$ in 1971Hy02. E _p (lab) (keV): unweighted average of 1119 2 (1964Gl04), 1119.6 12 (1977Da02), 1118.5 3 (1983Wa27), and 1120.8 5 (1971Hy02). $\omega\gamma=1.2$ eV 3 (1983Wa27). $\omega\gamma=0.21$ eV 11 (1964Gl04). $\omega\gamma=0.9$ eV 3 (1977Da02). $\omega\gamma=1.0$ eV 3 (1969Gr29). $\omega\gamma=1.9$ eV 6 (1971Hy02).
6266.6 13		1157.7 13	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.4$ eV 2 (1983Wa27). $\omega\gamma=0.2$ eV 1 (1977Da02).
6273.3 13	3	1164.6 13	J ^π : from $\gamma(\theta)$ E _p (lab)=1165→2376→146 γ cascade (1969Gr29). Other: spin=2,3 from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1165 2 (1964Gl04) and 1164.5 13 (1977Da02). $\omega\gamma=0.76$ eV 38 (1964Gl04). $\omega\gamma=3.3$ eV 7 (1983Wa27). $\omega\gamma=2.7$ eV 10 (1977Da02). $\omega\gamma=3.7$ eV 8 (1969Gr29).
6322.0 13	1,2	1214.8 13	J ^π : spin=1,2 from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1214 2 (1964Gl04) and 1215.2 13 (1977Da02). $\omega\gamma=0.19$ eV 10 (1964Gl04). $\omega\gamma=2.2$ eV 9 (1983Wa27). $\omega\gamma=1.9$ eV 7 (1977Da02). $\omega\gamma=2.3$ eV 5 (1969Gr29).
6361.3 13		1255.3 13	E _p (lab) (keV): weighted average of 1255 2 (1964Gl04) and 1255.4 13 (1977Da02). $\omega\gamma=0.05$ eV 3 (1964Gl04). $\omega\gamma=0.4$ eV 2 (1977Da02).
6370.2 2	2	1264.5 2	J ^π : from $\gamma(\theta)$ E _p (lab)=1266→666→g.s. γ cascade (1969Gr29). E _p (lab) (keV): weighted average of 1266 2 (1964Gl04), 1266.4 13 (1977Da02), and 1264.4 2 (1983Wa27). $\omega\gamma=2.7$ eV 6 (1983Wa27). $\omega\gamma=0.23$ eV 12 (1964Gl04). $\omega\gamma=2.5$ eV 8 (1977Da02). $\omega\gamma=4.2$ eV 9 (1969Gr29).
6382.4 14		1277.1 14	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.3$ eV 1 (1977Da02).
6399.8 14		1295.0 14	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.2$ eV 1 (1977Da02).
6441.2 14		1337.7 14	E _p (lab) (keV): weighted average of 1336.8 15 (1973An13) and 1338.4 14 (1977Da02). $\omega\gamma=0.7$ eV (1973An13). $\omega\gamma=0.2$ eV 1 (1977Da02).
6450.54 20	3,4	1347.3 2	J ^π : spin=3,4 from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1346.5 15 (1973An13), 1348.9 14 (1977Da02), and 1347.3 2 (1983Wa27). $\omega\gamma=0.9$ eV 3 (1983Wa27). $\omega\gamma=1.5$ eV (1973An13). $\omega\gamma=0.9$ eV 3 (1977Da02).
6477.5 14		1375.1 14	E _p (lab) (keV): weighted average of 1373.1 15 (1973An13) and 1376.9 14 (1977Da02). $\omega\gamma=0.7$ eV (1973An13). $\omega\gamma=0.2$ eV 1 (1977Da02).
6486.9 14	1 to 4	1384.8 14	J ^π : spin=1:4 from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1383.0 15 (1973An13) and 1386.3 14 (1977Da02). $\omega\gamma=0.7$ eV (1973An13). $\omega\gamma=0.6$ eV 3 (1983Wa27). $\omega\gamma=0.8$ eV 3 (1977Da02).
6527.6 15		1426.7 15	E _p (lab) (keV): from 1977Da02.

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π [#]	E _p (lab) (keV) ^{&}	Comments
6546.6 15		1446.3 15	$\omega\gamma=0.2$ eV 1 (1977Da02). E _p (lab) (keV): weighted average of 1445.0 15 (1973An13) and 1447.6 15 (1977Da02). $\omega\gamma=1.6$ eV (1973An13). $\omega\gamma=1.4$ eV 4 (1983Wa27). $\omega\gamma=1.6$ eV 5 (1977Da02).
6574.7 15		1475.3 15	E _p (lab) (keV): weighted average of 1473.8 15 (1973An13) and 1476.8 15 (1977Da02). $\omega\gamma=0.9$ eV (1973An13). $\omega\gamma=0.7$ eV 3 (1983Wa27). $\omega\gamma=0.6$ eV 2 (1977Da02).
6583.7 15		1484.5 15	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.4$ eV 2 (1977Da02).
6605? 2		1506.7 20	E _p (lab) (keV): from 1973An13. $\omega\gamma=1.1$ eV (1973An13).
6624.7 15		1526.8 15	E(level): the presence of the primary transitions from this resonance to 1 ⁺ and 5 ⁻ levels suggests there may be a doublet with very closely spaced levels. A careful examination of the shape of the resonance peak showed no sign of a possible doublet (1983Wa27). E _p (lab) (keV): weighted average of 1524.0 20 (1973An13) and 1528.4 15 (1977Da02). $\omega\gamma=0.5$ eV (1973An13). $\omega\gamma=0.4$ eV 1 (1983Wa27). $\omega\gamma=0.2$ eV 1 (1977Da02).
6640.92 7	4	1543.50 5	E(level): unweighted average of 6640.85 9 (at E _p (lab)=1543.49 5 in 1994Li20) and 6640.97 3 reported by 1994Li20 based on γ data without reporting E γ . J ^π : from $\gamma(\theta)$ in 1977Da03. Other: spin=3,4 from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1542.0 20 (1973An13), 1545.4 15 (1977Da02), 1543.6 2 (1983Wa27), and 1543.49 5 (1994Li20). Other: 1542 (1975Ke11). $\omega\gamma=3.8$ eV 6 (1977Da02), used for absolute strength calibration by 1983Wa27. $\omega\gamma=6.4$ eV (1973An13). $\omega\gamma=1.4$ eV 2 (1975Ke11).
6695.1 16		1599.3 16	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.4$ eV 2 (1977Da02).
6702.8 16		1607.3 16	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.3$ eV 2 (1977Da02).
6720.2 16		1625.2 16	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.3$ eV 1 (1977Da02).
6723.7 16		1628.8 16	E _p (lab) (keV): weighted average of 1626.5 20 (1973An13) and 1630.3 16 (1977Da02). $\omega\gamma=1.0$ eV 4 (1983Wa27). $\omega\gamma=1.6$ eV (1973An13). $\omega\gamma=0.9$ eV 4 (1977Da02).
6737.0 16		1642.5 16	E _p (lab) (keV): weighted average of 1640.2 20 (1973An13) and 1644.0 16 (1977Da02). $\omega\gamma=1.2$ eV (1973An13). $\omega\gamma=0.7$ eV 3 (1983Wa27). $\omega\gamma=0.5$ eV 2 (1977Da02).
6790.8 16		1698.0 16	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.2$ eV 1 (1983Wa27). $\omega\gamma=0.6$ eV 3 (1977Da02).
6797.4 16	1 to 5	1704.8 16	J ^π : spin=1:5 from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1703.2 20 (1973An13) and 1705.9 16 (1977Da02). $\omega\gamma=4.5$ eV (1973An13). $\omega\gamma=4.8$ eV 10 (1983Wa27). $\omega\gamma=4.0$ eV 25 (1977Da02).
6807.2 16		1714.9 16	E _p (lab) (keV): weighted average of 1713.8 20 (1973An13) and 1715.6 16 (1977Da02). $\omega\gamma=1.0$ eV (1973An13). $\omega\gamma=0.10$ eV 7 (1977Da02).
6828.6 14		1736.9 14	E _p (lab) (keV): weighted average of 1734.3 20 (1973An13) and 1738.2 14 (1977Da02). $\omega\gamma=1.2$ eV (1973An13).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π #	E _p (lab) (keV)&	Comments
6842.8 14		1751.5 14	$\omega\gamma=0.4$ eV 1 (1983Wa27). $\omega\gamma=0.6$ eV 3 (1977Da02). E _p (lab) (keV): from 1977Da02. $\omega\gamma=4.7$ eV 20 (1983Wa27); much larger than $\omega\gamma=1.8$ 10 (1977Da02), and 1983Wa27 states that a new, clean and thin target was used and the resonance was nearly totally resolved from the nearby $^{13}\text{C}(\text{p},\gamma)$ resonance.
6853.0 14		1762.1 14	E _p (lab) (keV): from 1977Da02. $\omega\gamma=2.1$ eV 5 (1983Wa27). $\omega\gamma=2.3$ eV 7 (1977Da02).
6871.18 30	5	1780.8 3	J ^π : spin=5 from $\gamma(\theta)$ in 1977Da03. E _p (lab) (keV): weighted average of 1782.2 16 (1977Da02) and 1780.7 3 (1983Wa27). $\omega\gamma=1.6$ eV 8 (1977Da02). $\omega\gamma=0.4$ eV 2 (1983Wa27).
6887.97 30	2	1798.1 3	J ^π : spin=2 from $\gamma(\theta)$ in 1977Da03. E _p (lab) (keV): weighted average of 1796.4 20 (1973An13), 1799.3 16 (1977Da02), and 1798.1 3 (1983Wa27). $\omega\gamma=2.9$ eV 10 (1983Wa27). $\omega\gamma=2.8$ eV (1973An13). $\omega\gamma=2.0$ eV 6 (1977Da02).
6901.75 30		1812.3 3	E _p (lab) (keV): weighted average of 1809.5 20 (1973An13), 1813.4 16 (1977Da02), and 1812.3 3 (1983Wa27). $\omega\gamma=2.4$ eV 6 (1983Wa27). $\omega\gamma=2.7$ eV (1973An13). $\omega\gamma=2.8$ eV 9 (1977Da02).
6918.0 17	3,4	1829.0 17	J ^π : from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): from 1977Da02. $\omega\gamma=11$ eV 2 (1977Da02), used for absolute strength calibration by 1983Wa27.
6931.5 17		1843.0 17	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.8$ eV 3 (1983Wa27). $\omega\gamma=0.8$ eV 5 (1977Da02).
6977.4 17		1890.3 17	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.5$ eV 2 (1977Da02).
6984.9 10		1898 1	E _p (lab) (keV): other: 1898 2 component of a triplet (1977Da02). $\omega\gamma=0.7$ eV 3 (1992Ka39), possibly part of the 1898+1900+1904 triplet reported by 1977Da02.
6986.9 20		1900 2	E _p (lab) (keV): component of a triplet (1977Da02). $\omega\gamma=1.7$ eV 10 for the 1898+1900+1904 triplet (1977Da02).
6992.3 9		1906 1	E _p (lab) (keV): weighted average of 1904 2 1977Da02 and 1906 1 (1992Ka39). $\omega\gamma=2.5$ eV 6 (1992Ka39), possibly part of the 1898+1900+1904 triplet reported by 1977Da02.
7018.9 10		1933 1	$\omega\gamma=0.7$ eV 3 (1992Ka39).
7051.6 18		1966.7 18	E _p (lab) (keV): from 1977Da02. $\omega\gamma=0.3$ eV 2 (1977Da02).
7059.04 30	1,2	1974.4 3	J ^π : spin=1,2 from $\gamma(\theta)$ in 1983Wa27. E _p (lab) (keV): weighted average of 1975.3 18 (1977Da02), 1974.4 3 (1983Wa27), and 1974 1 (1992Ka39). $\omega\gamma=8$ eV 2 (1977Da02), used for absolute strength calibration by 1983Wa27 and 1992Ka39.
7061.6 10		1977 1	$\omega\gamma=5$ eV 2 (1992Ka39).
7078.89 9	3	1994.86 7	E(level): unweighted average of 7078.83 10 (at E _p (lab)=1994.86 7 in 1994Li20) and 7079.00 6 reported by 1994Li20 based on γ data without reporting E _γ . J ^π : from $\gamma(\theta)$ in 1977Da03. E _p (lab) (keV): weighted average of 1997.2 18 (1977Da02) and 1994.86 7 (1994Li20). $\omega\gamma=1.7$ eV 4 (1983Wa27). $\omega\gamma=4.4$ eV 20 (1977Da02).
7115.9 10		2033 1	$\omega\gamma=0.7$ eV 3 (1992Ka39).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π [#]	E _p (lab) (keV) ^{&}	Comments
7122.7 10		2040 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7126.6 10		2044 1	$\omega\gamma=0.7$ eV 3 (1992Ka39).
7145.0 10		2063 1	$\omega\gamma=2.5$ eV 6 (1992Ka39).
7165.4 10		2084 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7175.1 10		2094 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7183.8 10		2103 1	
7203.2 10		2123 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7218.8 10		2139 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7238.2 10		2159 1	$\omega\gamma=3$ eV 1 (1992Ka39).
7249.8 10		2171 1	$\omega\gamma=4$ eV 1 (1992Ka39).
7271.2 10		2193 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7284.7 10		2207 1	
7295.4 10		2218 1	$\omega\gamma=2.5$ eV 6 (1992Ka39).
7321.6 10		2245 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7346.8 10		2271 1	$\omega\gamma=1.7$ eV 5 (1992Ka39).
7376.0 10		2301 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7382.7 10		2308 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7406.0 10		2332 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7424.5 10		2351 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7441.0 10		2368 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7449.7 10		2377 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7462.3 10		2390 1	$\omega\gamma=0.7$ eV 3 (1992Ka39).
7470.1 10		2398 1	$\omega\gamma=0.7$ eV 3 (1992Ka39).
7486.6 10		2415 1	$\omega\gamma=3$ eV 1 (1992Ka39).
7496.3 10		2425 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7506.0 10		2435 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7511.8 10		2441 1	
7517.6 10		2447 1	$\omega\gamma=4$ eV 1 (1992Ka39).
7527.3 10		2457 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7539.9 10		2470 1	
7543.8 10		2474 1	$\omega\gamma=0.5$ eV 3 (1992Ka39).
7562.3 10		2493 1	$\omega\gamma=0.7$ eV 2 (1992Ka39).
7577.8 10		2509 1	
7582.6 10		2514 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7588.5 10		2520 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7609.8 10		2542 1	$\omega\gamma=4.5$ eV 10 (1992Ka39).
7624.4 10		2557 1	$\omega\gamma=0.7$ eV 2 (1992Ka39).
7630.2 10		2563 1	
7643.8 10		2577 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7649.6 10		2583 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7666.1 10		2600 1	$\omega\gamma=4$ eV 1 (1992Ka39).
7675.8 10	2,3	2610 1	J ^π : spin from $\gamma(\theta)$ in 1992Ka39.
7690.3 10		2625 1	$\omega\gamma=12$ eV 3 (1992Ka39).
7708.8 10		2644 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7720.4 10		2656 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7730.1 10		2666 1	$\omega\gamma=0.7$ eV 2 (1992Ka39).
7753.4 10		2690 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7784.5 10		2722 1	$\omega\gamma=2.5$ eV 6 (1992Ka39).
7803.9 10		2742 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7809.7 10		2742 1	$\omega\gamma=5$ eV 2 (1992Ka39).
7831.0 10		2748 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7831.0 10		2770 1	$\omega\gamma=3$ eV 1 (1992Ka39).
7848.5 10		2788 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7853.4 10		2793 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7878.6 10		2819 1	
7886.4 10		2827 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7899.0 10		2840 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) ^{34}Cl Levels (continued)

E(level) ^{†‡}	J ^π [#]	E _p (lab) (keV) ^{&}	Comments
7904.8 10		2846 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7909.6 10		2851 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7934.9 10		2877 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7942.6 10		2885 1	$\omega\gamma=4.5$ eV 10 (1992Ka39).
7966.9 10		2910 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
7979.5 10		2923 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
7990.2 10		2934 1	$\omega\gamma=1.0$ eV 3 (1992Ka39).
7997.0 10		2941 1	$\omega\gamma=3.5$ eV 10 (1992Ka39).
8030.9 10		2976 1	$\omega\gamma=2.0$ eV 6 (1992Ka39).
8035.8 10		2981 1	$\omega\gamma=4$ eV 1 (1992Ka39).
8048.4 10		2994 1	$\omega\gamma=1.5$ eV 5 (1992Ka39).
8070.7 10		3017 1	$\omega\gamma=3$ eV 1 (1992Ka39).
8132.8 10		3081 1	
8141.6 10		3090 1	$\omega\gamma=7$ eV 3 (1992Ka39).
8154.2 10		3103 1	
8191.0 10		3141 1	$\omega\gamma=4$ eV 1 (1992Ka39).
8195.9 10		3146 1	
8214.3 10		3165 1	$\omega\gamma=9$ eV 3 (1992Ka39).
8235.7 10		3187 1	$\omega\gamma=4$ eV 2 (1992Ka39).
8248.3 10		3200 1	$\omega\gamma=6$ eV 2 (1992Ka39).
8259.0 10		3211 1	
8280.3 10		3233 1	$\omega\gamma=2$ eV 1 (1992Ka39).
8305.5 10	2,3	3259 1	J ^π : spin from $\gamma(\theta)$ in 1992Ka39. $\omega\gamma=27$ eV 5 (1992Ka39). Another I _γ =33 decays to unknown levels (1992Ka39).
8319.1 10		3273 1	$\omega\gamma=7$ eV 2 (1992Ka39).
8337.6 10		3292 1	$\omega\gamma=12$ eV 3 (1992Ka39).
8361.8 10		3317 1	$\omega\gamma=5$ eV 3 (1992Ka39).
8372.5 10		3328 1	$\omega\gamma=9$ eV 4 (1992Ka39).
8390.0 10		3346 1	$\omega\gamma=4$ eV 2 (1992Ka39).

[†] For resonances above S(p)(^{34}Cl)=5143.20 5 (2021Wa16), E(level)=E_{c.m.}+S(p)(^{34}Cl), where E_{c.m.}=E_p(lab)×m(^{33}S)/[m_p+m(^{33}S)], unless otherwise noted. For levels below 7.08 MeV (E_p(lab)<2 MeV), E_p(lab) is mainly from 1964Gl04, 1977Da02, and 1983Wa27. The average is taken when multiple E_p(lab) values are available. For levels above 7.08 MeV (E_p(lab)>2 MeV), E_p(lab) is mainly from 1992Ka39.

[‡] Most papers in this dataset report precise E(level) values based on E_γ data without reporting their measured E_γ values. The only exception is 1964Gl04, which reports E_γ measured by a NaI detector with uncertainties ranging from 10 to 50 keV. In addition to the large uncertainties, 1964Gl04 also used a much lower S(p)(^{34}Cl)=5.12 MeV in their analysis, which contributes to generally lower E(level) values. A least-squares fit to available E_γ values with uncertainties would not yield E(level) values that are more precise or complete than those already reported in the original papers. Therefore, the evaluators have adopted the precise E(level) values reported in papers and the average is taken when multiple E(level) values are available.

[#] Spins are from γ -ray angular distribution analysis, unless otherwise noted.

[@] For low-lying levels with J and π that are both given, J^π is from the Adopted Levels.

[&] From 1992Ka39, unless otherwise noted.

 $\gamma(^{34}\text{Cl})$

E _i (level)	J _i ^π	E _γ [†]	I _γ [‡]	E _f	J _f ^π	Comments
146.4	3 ⁺	146.4		0	0 ⁺	
461.2	1 ⁺	315 ^a	<1	146.4	3 ⁺	I _γ : other: <5 (1969Gr29).
		461.2	100	0	0 ⁺	E _γ : other: 460 20, γ cascade at E _p (lab)=507, 546, 731, 822 (1964Gl04). I _γ : other: 100 (1969Gr29).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
665.3	1	204 ^a 518.9	<0.3 1.46 19	461.2 146.4	1 ⁺ 3 ⁺			I_γ : from 2020Pa40. Other: <5 (1969Gr29). E_γ : other: 530 10, γ cascade at $E_p(\text{lab})=507$ (1964Gl04). I_γ : from 2020Pa40. Others: <1 (1983Wa27), <5 (1969Gr29). E_γ : others: 670 20, γ cascade at $E_p(\text{lab})=447, 507, 546, 662, 683, 731, 777, 822$ (1964Gl04). I_γ : from 2020Pa40, 1983Wa27, 1977Da02, and 1969Gr29. $A_2=+0.28$ 4, A_4 set to zero; $A_2=+0.31$ 5, $A_4=-0.07$ 7 (1969Gr29). E_γ : other: 560 20, γ cascade at $E_p(\text{lab})=731$ (1964Gl04). I_γ : others: 102.8 28 (1977Da02), 167 35 (1969Gr29). E_γ : others: 770 30, γ cascade at $E_p(\text{lab})=662, 822; 770$ 20, γ cascade at $E_p(\text{lab})=731$ (1964Gl04). I_γ : others: 100.0 28 (1977Da02), 100 18 (1969Gr29). E_γ : others: 1080 50, γ cascade at $E_p(\text{lab})=662; 1090$ 20, γ cascade at $E_p(\text{lab})=731; 1100$ 40, γ cascade at $E_p(\text{lab})=546$ (1964Gl04). I_γ : others: 75.0 28 (1977Da02), 80 25 (1969Gr29). I_γ : others: <5.7 (1983Wa27), <7.0 (1969Gr29).
1230.33	2 ⁺	565.0	98.6 14	665.3	1			I_γ : other: <16 (1969Gr29). E_γ : others: 1420 40, γ cascade at $E_p(\text{lab})=731; 1350$ 60, γ cascade at $E_p(\text{lab})=546$ (1964Gl04). I_γ : others: 100.0 33 (1977Da02), 100 36 (1969Gr29), 100 7 (1971Hy02). E_γ : other: 1750 30, γ cascade at $E_p(\text{lab})=546, 683, 731$ (1964Gl04). I_γ : weighted average of 66.7 33 (1977Da02), 74.2 12 (1983Wa27), 100 40 (1969Gr29), and 67 7 (1971Hy02).
		769.1	100.0 14	461.2	1 ⁺			I_γ : others: <1.7 (1983Wa27), <16 (1969Gr29). E_γ : others: 930 30, γ cascade at $E_p(\text{lab})=822; 950$ 40, γ cascade at $E_p(\text{lab})=662$ (1964Gl04). I_γ : weighted average of 9.1 15 (1977Da02), 8.54 31 (1983Wa27), 19 6 (1969Gr29), and 10.5 60 (1971Hy02). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=+0.14$ 10, $A_4=0.00$ 10 (1977Da03). I_γ : others: <3.1 (1983Wa27), <3.1 (1969Gr29), <1.5 (1971Hy02). E_γ : others: 1700 30, γ cascade at $E_p(\text{lab})=546, 731; 1700$ 40, γ cascade at $E_p(\text{lab})=662; 1710$ 30, γ cascade at $E_p(\text{lab})=639$ (1964Gl04). I_γ : others: 100.0 30 (1977Da02), 100 17 (1969Gr29), 100 9 (1971Hy02). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=-0.12$ 1, $A_4=-0.01$ 1 (1977Da03). $A_2=-0.20$ 3 (1971Hy02, γ cascade at $E_p(\text{lab})=1058; A_2=+0.03$ 4 (1971Hy02, γ cascade at $E_p(\text{lab})=1119$).
		1083.9	87.1 14	146.4	3 ⁺			
1887.29	2 ⁺	1230 ^a 657 ^a 1222 ^a 1426.1	<1.4 [#] <3.5 <0.83 100.0 12	0 1230.33 665.3 461.2	0 ⁺ 2 ⁺ 1 1 ⁺			
		1740.8	73.2 12	146.4	3 ⁺			
2157.9	2 ⁺	1887 ^a 927.6	<0.83 [#] 8.59 31	0 1230.33	0 ⁺ 2 ⁺	D(+Q)	+0.1 +6-3	
		1493 ^a	<0.76 [#]	665.3	1			
		1696.7	100.0 6	461.2	1 ⁺	D+Q	-0.05 3	

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
2157.9	2^+	2011.4	20.2 5	146.4	3^+	D+Q	+0.22 10	E_γ : others: 2010 40, γ cascade at $E_p(\text{lab})=822$; 2020 50, γ cascade at $E_p(\text{lab})=639$, 662; 2030 30, γ cascade at $E_p(\text{lab})=546$ (1964Gl04). I_γ : weighted average of 21.2 15 (1977Da02), 20.1 5 (1983Wa27), and 19 9 (1971Hy02). Other: <9.4 (1969Gr29). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=-0.11$ 7, $A_4=+0.02$ 7 (1977Da03). E_γ : others: 2160 30, γ cascade at $E_p(\text{lab})=639$, 822; 2160 50, γ cascade at $E_p(\text{lab})=662$; 2170 40, γ cascade at $E_p(\text{lab})=546$ (1964Gl04). I_γ : weighted average of 21.2 15 (1977Da02), 23.8 5 (1983Wa27), 38 14 (1969Gr29), and 19 5 (1971Hy02).
		2157.8	23.5 5	0	0^+			E_γ : others: 2160 30, γ cascade at $E_p(\text{lab})=639$, 822; 2160 50, γ cascade at $E_p(\text{lab})=662$; 2170 40, γ cascade at $E_p(\text{lab})=546$ (1964Gl04). I_γ : weighted average of 21.2 15 (1977Da02), 23.8 5 (1983Wa27), 38 14 (1969Gr29), and 19 5 (1971Hy02).
2181.10	3	951 ^a	<2.1 [#]	1230.33	2^+			I_γ : other: <3.9 (1983Wa27). I_γ : weighted average of 25.5 21 (1977Da02), 24.7 14 (1983Wa27), and 26 12 (1971Hy02). I_γ : weighted average of 87 4 (1977Da02), 69.8 27 (1983Wa27), and 49 9 (1971Hy02). $A_2=+0.03$ 12 (1971Hy02). I_γ : others: 100 4 (1977Da02), 100 14 (1971Hy02). $A_2=-0.02$ 6 (1971Hy02). I_γ : other: <3.9 (1983Wa27). I_γ : weighted average of 1.01 51 (1977Da02) and 0.81 20 (1983Wa27) Other: <2 (1969Gr29). I_γ : other: <4 (1969Gr29). I_γ : other: <2 (1969Gr29). I_γ : others: 100.0 5 (1977Da02), 100 (1969Gr29), 100 (1971Hy02). Mult., δ : from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1165$ in 1969Gr29. $A_2=+0.11$ 3, $A_4=-0.29$ 4 (1969Gr29). I_γ : other: <3 (1969Gr29). I_γ : from 1969Gr29. I_γ : other: <6 (1969Gr29). I_γ : other: <6 (1969Gr29). I_γ : other: <15 (1969Gr29). I_γ : other: <8 (1969Gr29). E_γ : others: 2570 30, γ cascade at $E_p(\text{lab})=683$; 2580 30, γ cascade at $E_p(\text{lab})=507$; 2590 30, γ cascade at $E_p(\text{lab})=546$ (1964Gl04). I_γ : others: 100 (1977Da02), 100 (1969Gr29). I_γ : unweighted average of 65.7 29 (1977Da02) and 55.3 33 (1983Wa27).
		1515.8	25.0 14	665.3	1			
		1719.9	73.7 70	461.2	1^+			
		2034.6	100.0 25	146.4	3^+			
		2181 ^a	<2.1 [#]	0	0^+			
2375.7	4	1145.4	0.84 20	1230.33	2^+			
		1710 ^a	<2.0	665.3	1			
		1914 ^a	<4	461.2	1^+			
		2229.2	100.00 20	146.4	3^+	D+Q	+6.3 +32-20	
		2376 ^a	<2.0	0	0^+			
2580.3		693 ^a	<8	1887.29	2^+			
		1350 ^a	<1	1230.33	2^+			
		1915 ^a	<2	665.3	1			
		2119 ^a	<2	461.2	1^+			
		2434 ^a	<2	146.4	3^+			
		2580.2	100	0	0^+			
2611.03	3	453.1	61 5	2157.9	2^+			
		724 ^a	<2.5	1887.29	2^+			
		1380.67	68.1 33	1230.33	2^+			
		1945.7	42.6 73	665.3	1			
		2150 ^a	<5.1	461.2	1^+			

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39 (continued)** $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
2611.03	3	2464.5	100.0 [#] 29	146.4	3 ⁺			I_γ : others: 100.0 33 (1983Wa27), 90 18 (1971Hy02).
2721.3	2	563.4	14.9 11	2157.9	2 ⁺			I_γ : weighted average of 16.3 21 (1977Da02), 14.5 11 (1983Wa27), and 17 7 (1971Hy02).
		834.0	4.4 4	1887.29	2 ⁺			I_γ : weighted average of 6.1 21 (1977Da02), 4.3 4 (1983Wa27), and 4.3 21 (1971Hy02).
		1490.9	4.6 4	1230.33	2 ⁺			$A_2=+0.1$ 3 (1971Hy02).
		2055.9	15.5 11	665.3	1			I_γ : weighted average of 4.1 21 (1977Da02), 4.5 4 (1983Wa27), and 6.4 21 (1971Hy02).
								$A_2=-0.07$ 21 (1971Hy02).
								I_γ : weighted average of 12.2 21 (1977Da02), 16.4 11 (1983Wa27), and 17 7 (1971Hy02).
								$A_2=-0.20$ 4 (1971Hy02, γ cascade at $E_p(\text{lab})=1058$); $A_2=-0.04$ 11 (1971Hy02, γ cascade at $E_p(\text{lab})=1119$).
		2260.0	100.0 15	461.2	1 ⁺	D(+Q)	-0.03 5	I_γ : others: 100 4 (1977Da02), 100 9 (1971Hy02).
								Mult., δ : or $\delta=-2.3$ 5 from $\gamma(\theta)$ in 1971Hy02.
								$A_2=-0.17$ 4 (1971Hy02, γ cascade at $E_p(\text{lab})=1097$); $A_2=+0.11$ 3 (1971Hy02, γ cascade at $E_p(\text{lab})=1119$).
		2574.8	38.9 11	146.4	3 ⁺			I_γ : weighted average of 36.7 21 (1977Da02), 39.5 11 (1983Wa27), and 40 7 (1971Hy02).
								$A_2=+0.07$ 5 (1971Hy02).
		2721.2	32.8 11	0	0 ⁺			I_γ : weighted average of 28.6 21 (1977Da02), 34.1 11 (1983Wa27), and 28 7 (1971Hy02).
								$A_2=+0.26$ 13 (1971Hy02, γ cascade at $E_p(\text{lab})=1058$); $A_2=+0.21$ 9 (1971Hy02, γ cascade at $E_p(\text{lab})=1097$); $A_2=-0.20$ 6 (1971Hy02, γ cascade at $E_p(\text{lab})=1119$).
3129.12		1899 ^a	<2	1230.33	2 ⁺			
		2668 ^a	<2	461.2	1 ⁺			
		2983 ^a	<2	146.4	3 ⁺			
		3128.97	100	0	0 ⁺			E_γ : other: 3150 40, γ cascade at $E_p(\text{lab})=546$ (1964Gl04).
3334.0		1176.1	100 15	2157.9	2 ⁺			I_γ : other: 100 (1977Da02).
		1447 ^a	<4.9	1887.29	2 ⁺			I_γ : other: 100 18 (1977Da02).
		2104 ^a	<4.9	1230.33	2 ⁺			
		2668.6	90 12	665.3	1			I_γ : other: 33 9 (1977Da02). 1983Wa27 pointed out that it seems likely that 1977Da02 was unaware of a $^{34\text{m}}\text{Cl}$ contamination and therefore obtained too large $I_\gamma(1176)$.
		2872.7	54 12	461.2	1 ⁺			I_γ : other: 16.4 90 (1977Da02). Due to a contamination issue pointed out by 1983Wa27, $I_\gamma(1176)$ from 1977Da02 is too large, which makes other I_γ too small.
		3187 ^a	<4.9	146.4	3 ⁺			
		3334 ^a	<7.3	0	0 ⁺			
3383.3		2152.9	52.0 25	1230.33	2 ⁺			I_γ : weighted average of 52 10 (1977Da02) and 52.0 25 (1983Wa27).
		2717.9	86 4	665.3	1			E_γ : other: 2710 50, γ cascade at $E_p(\text{lab})=662$ (1964Gl04).
								I_γ : weighted average of 87 7 (1977Da02) and 85 4 (1983Wa27).
		2922.0	75 5	461.2	1 ⁺			E_γ : other: 2920 30, γ cascade at $E_p(\text{lab})=662$

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
								(1964Gl04).
3383.3		3236.7	100.0 34	146.4	3 ⁺			I_γ : weighted average of 84 7 (1977Da02) and 72 4 (1983Wa27).
								E_γ : other: 3240 30, γ cascade at $E_p(\text{lab})=662$ (1964Gl04).
								I_γ : other: 100 7 (1977Da02).
3545.08	3	3383 ^a	<3.1	0	0 ⁺			I_γ : from 1969Gr29.
		965 ^a	<3	2580.3				I_γ : from 1969Gr29.
		1169 ^a	<15	2375.7	4			I_γ : from 1969Gr29.
		1387 ^a	<3	2157.9	2 ⁺			I_γ : from 1969Gr29.
		1658 ^a	<5	1887.29	2 ⁺			I_γ : from 1969Gr29.
		2314.67	3.6 6	1230.33	2 ⁺			I_γ : weighted average of 4.2 21 (1977Da02) and 3.5 6 (1983Wa27) Other: <5 (1969Gr29).
		2880 ^a	<12	665.3	1			I_γ : from 1969Gr29.
		3084 ^a	<3.1	461.2	1 ⁺			I_γ : other: <8 (1969Gr29).
		3398.5	100.0 6	146.4	3 ⁺	D+Q	-0.09 4	E_γ : other: 3400 50, γ cascade at $E_p(\text{lab})=731$ (1964Gl04).
								I_γ : others: 100.0 21 (1977Da02), 100 (1969Gr29), 100 (1971Hy02).
								Mult.: from $\gamma(\theta)$ in 1971Hy02 and 1977Da03.
								δ : weighted average of -0.06 5 (1971Hy02) and -0.11 4 (1977Da03). 1977Da03 reports $\delta=-0.11$ 4 based on $\delta=+1.5$ +10-7 or -0.1 +3-2 from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1057$; $\delta=-0.14$ 4 from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1545$; $\delta=-0.07$ +5-1 or +1.5 +3-2 from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1997$.
								$A_2=+0.19$ 3, $A_4=+0.022$ 21 (1977Da03).
								$A_2=+0.18$ 9, $A_4=-0.02$ 9 (1977Da03, γ cascade at $E_p(\text{lab})=1057$); $A_2=+0.25$ 3, $A_4=+0.02$ 3 (1977Da03, γ cascade at $E_p(\text{lab})=1545$); $A_2=+0.17$ 2, $A_4=+0.03$ 3 (1977Da03, γ cascade at $E_p(\text{lab})=1997$); $A_2=+0.23$ 8 (1971Hy02, γ cascade at $E_p(\text{lab})=1058$); $A_2=+0.25$ 2 (1971Hy02, γ cascade at $E_p(\text{lab})=1097$).
3600.28	4	3545 ^a	<3.1	0	0 ⁺			I_γ : other: <4 (1969Gr29).
		879.0	94.1 11	2721.3	2	Q+O	+0.08 5	I_γ : weighted average of 98 4 (1977Da02), 93.9 11 (1983Wa27), and 85 13 (1969Gr29), and 92 11 (1971Hy02).
								Mult.: from $\gamma(\theta)$ in 1977Da03 and 1971Hy02.
								δ : or +14 +43-5 (1977Da03). Other: $\delta=+0.01$ 4 (1971Hy02).
								$A_2=+0.30$ 3, $A_4=+0.02$ 3, $A_6=+0.01$ 4 (1977Da03).
								$A_2=+0.30$ 3, $A_4=+0.02$ 3, $A_6=+0.01$ 4 (1977Da03); $A_2=+0.24$ 6 (1971Hy02, γ cascade at $E_p(\text{lab})=1058$); $A_2=+0.25$ 1 (1971Hy02, γ cascade at $E_p(\text{lab})=1097$).
		1020 ^a	<3.7	2580.3				I_γ : from 1969Gr29.
		1224.6	15.4 19	2375.7	4			I_γ : weighted average of 14.9 21 (1977Da02), 15.7 19 (1983Wa27), and 17 8 (1971Hy02). Others: <7.4 (1969Gr29).
								$A_2=-0.14$ 7 (1971Hy02).
		1419 ^a	<4.3	2181.10	3			
		1442 ^a	<5.6	2157.9	2 ⁺			I_γ : from 1969Gr29.
		1713 ^a	<7.4	1887.29	2 ⁺			I_γ : from 1969Gr29.

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
3600.28	4	2370 ^a	<7.4	1230.33	2 ⁺			I_γ : from 1969Gr29.
		2935 ^a	<4.2	665.3	1			I_γ : other: <3.7 (1969Gr29).
		3139 ^a	<4.2	461.2	1 ⁺			I_γ : other: <7.4 (1969Gr29).
		3453.7	100.0 11	146.4	3 ⁺	D+Q	+0.06 3	I_γ : others: 100 4 (1977Da02), 100 22 (1969Gr29), 100 11 (1971Hy02).
								Mult.: from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1997$ in 1977Da03.
								δ : weighted average of $\delta=+0.07$ 4 (1971Hy02) and $\delta=+0.05$ 3 (1977Da03). Other: $\delta=-7$ 2 (1977Da03).
								$A_2=-0.12$ 3, $A_4=-0.01$ 3 (1977Da03).
								$A_2=-0.12$ 3, $A_4=-0.01$ 3 (1977Da03); $A_2=-0.13$ 2 (1971Hy02).
3631.7	5	3600 ^a	<4.2	0	0 ⁺			I_γ : other: <15 (1969Gr29).
		1256.0	100.0 13	2375.7	4	D(+Q)	0.00 2	I_γ : others: 100 4 (1977Da02), 100 14 (1971Hy02).
								Mult., δ : $\delta=0.00$ 2 in Table 5 and $\delta=0.00 +2-1$ in Table 3 (1977Da03).
								$A_2=-0.22$ 3, $A_4=-0.04$ 3 (1977Da03).
								$A_2=-0.22$ 3, $A_4=-0.04$ 3 (1977Da03); $A_2=-0.24$ 5 (1971Hy02).
		2401 ^a	<3.7	1230.33	2 ⁺			
		2966 ^a	<3.7	665.3	1			
		3170 ^a	<5.5	461.2	1 ⁺			
		3485.1	83.4 13	146.4	3 ⁺			I_γ : weighted average of 82 4 (1977Da02), 83.5 13 (1983Wa27), and 92 14 (1971Hy02).
								$A_2=+0.21$ 7 (1971Hy02).
3646.3		3632 ^a	<5.5	0	0 ⁺			
		2416 ^a	<2.7	1230.33	2 ⁺			
		3185 ^a	<4	461.2	1 ⁺			
		3499.7	100 7	146.4	3 ⁺			I_γ : other: 100 (1977Da02).
		3646 ^a	<4	0	0 ⁺			
3660.0		1502.1	100	2157.9	2 ⁺			I_γ : other: 100 (1977Da02).
		1773 ^a	<2	1887.29	2 ⁺			
		2430 ^a	<2	1230.33	2 ⁺			
		2995 ^a	<2	665.3	1			
		3199 ^a	<2	461.2	1 ⁺			
		3513 ^a	<5	146.4	3 ⁺			
		3660 ^a	<3	0	0 ⁺			
3773.72	1	3773.5	100	0	0 ⁺			I_γ : other: 100 (1971Hy02).
								$A_2=-0.20$ 5 (1971Hy02).
3791.7		2561 ^a	<3.5	1230.33	2 ⁺			
		3330.3	100 7	461.2	1 ⁺			
		3645 ^a	<8.2	146.4	3 ⁺			
		3792 ^a	<7.1	0	0 ⁺			
3940.1		2710 ^a	<3.5	1230.33	2 ⁺			
		3274.6	100 9	665.3	1			I_γ : other: 100 15 (1977Da02).
		3478.7	72 9	461.2	1 ⁺			I_γ : other: 133 15 (1977Da02).
		3940 ^a	<5.2	0	0 ⁺			
3964.1		1588 ^a	11 [#] 5	2375.7	4			I_γ : 1983Wa27 states that the existence of this transition observed by 1977Da02 is considered doubtful.
		1806.2	100 [#] 16	2157.9	2 ⁺			I_γ : other: 100 25 (1983Wa27).
		3817.5	67 25	146.4	3 ⁺			I_γ : other: 23 11 (1977Da02).
3983.0	3	1261.7	13.5 11	2721.3	2	D+Q	-0.8 3	I_γ : weighted average of 12.7 16 (1977Da02), 13.9 11 (1983Wa27), 12.3 62 (1971Hy02).

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39 (continued)** $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
3983.0	3							Mult., δ : from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1057$ in 1977Da03 .
								$A_2=-0.59$ 7, $A_4=+0.05$ 7 (1977Da03).
		1403 ^a	<12	2580.3				I_γ : from 1969Gr29 .
		1607 ^a	<12	2375.7	4			I_γ : from 1969Gr29 .
		1825.1	43.2 17	2157.9	2 ⁺	D(+Q)	+0.02 4	I_γ : weighted average of 46.0 32 (1977Da02), 42.3 17 (1983Wa27), and 72 17 (1969Gr29), and 40 8 (1971Hy02).
								Mult.: from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1057$ and 1997 in 1977Da03 .
								δ : 1977Da03 reports +0.02 4 based on $\delta=+0.05$ 3 or $\delta=-5.1$ +6-12 from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1057$; $\delta=-0.02$ +5-4 or $\delta=-3.7$ +4-10 from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1997$.
								$A_2=-0.11$ 3, $A_4=0.00$ 4 (1977Da03 , γ cascade at $E_p(\text{lab})=1057$); $A_2=-0.17$ 4, $A_4=+0.04$ 4 (1977Da03 , γ cascade at $E_p(\text{lab})=1997$); $A_2=-0.20$ 3 (1971Hy02).
		2096 ^a	<26	1887.29	2 ⁺			I_γ : from 1969Gr29 .
		2753 ^a	<6.4 [#]	1230.33	2 ⁺			I_γ : others: <7.8 (1983Wa27), <12 (1969Gr29).
4076.2	4	3318 ^a	<12	665.3	1			I_γ : from 1969Gr29 .
		3522 ^a	<1.6	461.2	1 ⁺			I_γ : other: <10 (1969Gr29).
		3836.4	100.0 31	146.4	3 ⁺	D(+Q)	-0.02 4	I_γ : others: 100.0 32 (1977Da02), 100 19 (1969Gr29), 100 9 (1971Hy02).
								Mult.: from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1057$ and 1997 in 1977Da03 .
								δ : 1977Da03 reports -0.02 4 based on $\delta=0.0$ 1 or +1.3 3 from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1057$; $\delta=-0.04$ 10 or $\delta=+1.4$ 3 from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1997$.
								$A_2=+0.20$ 4, $A_4=0.00$ 4 (1977Da03 , γ cascade at $E_p(\text{lab})=1057$); $A_2=+0.19$ 4, $A_4=+0.05$ 5 (1977Da03 , γ cascade at $E_p(\text{lab})=1997$); $A_2=+0.21$ 3 (1971Hy02 , γ cascade at $E_p(\text{lab})=1058$); $A_2=+0.00$ 5 (1971Hy02 , γ cascade at $E_p(\text{lab})=1097$).
		3983 ^a	<5.2	0	0 ⁺			I_γ : from 1969Gr29 .
		1354.9	9.1 11	2721.3	2			I_γ : weighted average of 13.6 34 (1977Da02) and 8.8 8 (1983Wa27).
		1700.5	35.5 13	2375.7	4	D+Q	-0.27 +4-3	I_γ : weighted average of 35.6 34 (1977Da02) and 35.5 13 (1983Wa27) Other: <15 (1971Hy02).
								Mult., δ : from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1545$ in 1977Da03 .
		1895.0	22.0 13	2181.10	3	D+Q	+0.05 +3-1	$A_2=+0.18$ 2, $A_4=0.00$ 2 (1977Da03).
								I_γ : weighted average of 20.3 34 (1977Da02) and 22.3 13 (1983Wa27).
								Mult., δ : from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1545$ in 1977Da03 .
								$A_2=-0.16$ 3, $A_4=+0.02$ 3 (1977Da03).
		2846 ^a	<5	1230.33	2 ⁺			
		3615 ^a	<3.3	461.2	1 ⁺			
		3929.6	100.0 25	146.4	3 ⁺	D(+Q)	0.00 +1-2	I_γ : others: 100.0 34 (1977Da02), 100 (1971Hy02).
								Mult., δ : from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1545$ in 1977Da03 .

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39 (continued)** $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
								$A_2=-0.27\ 2$, $A_4=+0.01\ 2$ (1977Da03). $A_2=-0.27\ 2$, $A_4=+0.01\ 2$ (1977Da03); $A_2=-0.07\ 5$ (1971Hy02).
4076.2	4	4076 ^a	<3.3	0	0 ⁺			
4139.7	2	1559.4	37 6	2580.3				
		1981.7	43 4	2157.9	2 ⁺			I_γ : weighted average of 36 9 (1977Da02), 45 4 (1983Wa27), and 39 14 (1971Hy02).
		2909.2	14 4	1230.33	2 ⁺			
		3474 ^a	<9.8	665.3	1			
		3678 ^a	<3.9	461.2	1 ⁺			
		3993.1	100 6	146.4	3 ⁺			I_γ : others: 100 9 (1977Da02), 100 14 (1971Hy02). $A_2=+0.20\ 20$ (1971Hy02).
		4139 ^a	<3.9	0	0 ⁺			
4147.8		1966.6	7.0 10	2181.10	3			
		1989.8	6.1 13	2157.9	2 ⁺			
		2917.3	100.0 23	1230.33	2 ⁺			
		3482.3	66.2 19	665.3	1			
		4001 ^a	<3.8	146.4	3 ⁺			
		4147.5	9.8 15	0	0 ⁺			E_γ : other: 4130 50, γ cascade at $E_p(\text{lab})=777$ (1964Gl04). I_γ : other: 100 (1977Da02).
4325.89		2167.9	100 6	2157.9	2 ⁺			
		2439 ^a	<1.3	1887.29	2 ⁺			
		3660 ^a	<2.5	665.3	1			
		3865 ^a	<3.8	461.2	1 ⁺			
		4179 ^a	<2.5	146.4	3 ⁺			
		4326 ^a	<5	0	0 ⁺			
4354.2	(1)	2196.2	33.2 28	2157.9	2 ⁺			I_γ : weighted average of 35.8 30 (1977Da02) and 31.0 28 (1983Wa27).
		3689 ^a	<4.2	665.3	1			
		3892.8	11.5 28	461.2	1 ⁺			I_γ : weighted average of 13.4 30 (1977Da02) and 9.9 28 (1983Wa27).
		4208 ^a	<2.8	146.4	3 ⁺			
		4353.9	100.0 28	0	0 ⁺			I_γ : others: 100.0 30 (1977Da02), 100 (1971Hy02). $A_2=-0.16\ 8$ (1971Hy02).
4417.3		3187 ^a	<3.1	1230.33	2 ⁺			
		3751.8	58 5	665.3	1			I_γ : weighted average of 64 8 (1977Da02) and 56 5 (1983Wa27). I_γ : other: 100 8 (1977Da02).
		3955.9	100 5	461.2	1 ⁺			
		4271 ^a	<6.3	146.4	3 ⁺			
		4417 ^a	<7.8	0	0 ⁺			I_γ : other: 100 (1971Hy02).
4446.6	4	4299.9	100	146.4	3 ⁺	D(+Q)	-0.02 +2-1	Mult., δ : from $\gamma(\theta)$ γ cascade at $E_p(\text{lab})=1782$ in 1977Da03. $A_2=-0.29\ 3$, $A_4=0.00\ 4$ (1977Da03).
		4446 ^a	<3	0	0 ⁺			
4461.4		4314.7	100	146.4	3 ⁺			
4515.7		3285 ^a	<3.8	1230.33	2 ⁺			
		3850.2	100.0 25	665.3	1			I_γ : others: 100 4 (1977Da02), 100 12 (1971Hy02). $A_2=-0.00\ 8$ (1971Hy02).
		4054 ^a	<2.5	461.2	1 ⁺			
		4369.0	25.0 25	146.4	3 ⁺			E_γ : other: 4370 50, γ cascade at $E_p(\text{lab})=447$ (1964Gl04). I_γ : weighted average of 27 4 (1977Da02), 25.0 25 (1983Wa27), and 19 12 (1971Hy02). $A_2=-0.10\ 20$ (1971Hy02).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
4515.7		4515 ^a	<5	0	0 ⁺	
4605.8		1884.4	100 [#]	2721.3	2	
		2447.8	44 [#]	2157.9	2 ⁺	
		2718.4	35 [#]	1887.29	2 ⁺	
		3375.3	54 [#]	1230.33	2 ⁺	
		4459 ^a	<12	146.4	3 ⁺	
4609.7		1888.3	71 [#]	2721.3	2	
		3379.2	100 [#]	1230.33	2 ⁺	
		4463.0	73 [#]	146.4	3 ⁺	
4638.9	1,2	4177.4	100 [#] 8	461.2	1 ⁺	I_γ : other: 100 13 (1983Wa27).
		4492.2	58 10	146.4	3 ⁺	I_γ : weighted average of 64 8 (1977Da02) and 41 13 (1983Wa27).
4695.7		2537.7	100	2157.9	2 ⁺	I_γ : other: 100 (1977Da02).
		2808 ^a	<4	1887.29	2 ⁺	
		3465 ^a	<4	1230.33	2 ⁺	
		4030 ^a	<5	665.3	1	
		4234 ^a	<4	461.2	1 ⁺	
		4695 ^a	<2	0	0 ⁺	
4717.4		2106.3	16 5	2611.03	3	
		2341.6	11.1 32	2375.7	4	
		2536.2	35 5	2181.10	3	I_γ : weighted average of 40 5 (1977Da02) and 30 5 (1983Wa27).
		2559.4	11 5	2157.9	2 ⁺	I_γ : weighted average of 7.9 32 (1977Da02) and 18 5 (1983Wa27).
		2830 ^a	<3.3	1887.29	2 ⁺	
		3487 ^a	<4.9	1230.33	2 ⁺	
		4052 ^a	<4.9	665.3	1	
		4256 ^a	<4.9	461.2	1 ⁺	
		4570.7	100 5	146.4	3 ⁺	I_γ : other: 100 5 (1977Da02).
4824.41		1192.7	6.5 16	3631.7	5	
		1224.11	37 13	3600.28	4	I_γ : unweighted average of 24 5 (1977Da02) and 50.0 32 (1983Wa27).
		2643.20	4.8 16	2181.10	3	
		3594 ^a	<11	1230.33	2 ⁺	
		4363 ^a	<3.2	461.2	1 ⁺	
		4677.7	100.0 32	146.4	3 ⁺	I_γ : other: 100 5 (1977Da02).
		4824 ^a	<3.2	0	0 ⁺	
4941.9		2330.8	43 [#] 19	2611.03	3	
		2361.5	45 [#] 19	2580.3		
		4941.5	100 [#] 38	0	0 ⁺	I_γ : other: 100 and another $I_\gamma=43$ decays to unknown final levels (1983Wa27).
4957.3		3070 ^a	<3.9	1887.29	2 ⁺	
		3726.8	100 10	1230.33	2 ⁺	I_γ : other: 100 10 (1977Da02).
		4292 ^a	<5.9	665.3	1	
		4496 ^a	<5.9	461.2	1 ⁺	
		4810.5	94 10	146.4	3 ⁺	I_γ : weighted average of 92 10 (1977Da02) and 96 10 (1983Wa27).
		4957 ^a	<3.9	0	0 ⁺	
4995.6		2837.6	100	2157.9	2 ⁺	
5020?		2862 ^a	100 [#]	2157.9	2 ⁺	
5171.6	4	1571.3	100	3600.28	4	
		5024.8	100 [#]	146.4	3 ⁺	
5386.8		1786.5	100	3600.28	4	
		1842 ^a		3545.08	3	The only γ branch from 1977Da02 but is not present in the more complete data from 1983Wa27.

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
5386.8		4156 ^a	<2	1230.33	2 ⁺	
		4721 ^a	<2	665.3	1	
		4925 ^a	<4	461.2	1 ⁺	
		5240 ^a	<3	146.4	3 ⁺	
		5386 ^a	<3	0	0 ⁺	
5540.8		1940.5	100 8	3600.28	4	I_γ : other: 100 (1977Da02).
		1995.7	56 8	3545.08	3	
		4875 ^a	<9.4	665.3	1	
		5079 ^a	<4.7	461.2	1 ⁺	
		5394 ^a	<7.8	146.4	3 ⁺	
		5540 ^a	<6.3	0	0 ⁺	
5577.6		1061.9	9 3	4515.7		E_γ, I_γ : 1050 30 from 1964Gl04.
		1977.3	28.7 @ 66	3600.28	4	I_γ : other: 38 (1983Wa27).
		2032.5	100 @	3545.08	3	E_γ : other: 2020 30 (1964Gl04). I_γ : others: 100 (1983Wa27), 100 14 (1964Gl04).
		2448 ^a	<3.9 @	3129.12		
		2856.2	22.6 @ 23	2721.3	2	I_γ : other: 21 (1983Wa27).
		2966 ^a	0.71	2611.03	3	Observed in 1983Wa27 but not in 2011Fr04.
		2997 ^a	<1.2 @	2580.3		
		3201.7	3.0 @ 12	2375.7	4	I_γ : other: 0.95 (1983Wa27).
		3690.1	5.5 @ 10	1887.29	2 ⁺	I_γ : other: 4.8 (1983Wa27).
		4347.0	10.3 @ 14	1230.33	2 ⁺	I_γ : other: 10 (1983Wa27).
		4912 ^a	<0.71	665.3	1	Observed in 1983Wa27 but not in 2011Fr04.
		5116 ^a	<1.4	461.2	1 ⁺	Observed in 1983Wa27 but not in 2011Fr04.
		5430.7	61.9 @ 55	146.4	3 ⁺	I_γ : others: 62 (1983Wa27), 51 7 (1964Gl04).
		5577 ^a	<0.24	0	0 ⁺	I_γ : other: <8 (1964Gl04); not observed in 2011Fr04.
5635.75	1,2	997 ^a	0.44	4638.9	1,2	Observed in 1983Wa27 but not in 2011Fr04.
		1030 ^a	4 1	4605.8		E_γ, I_γ : 1040 30 from 1964Gl04, but not observed in the more complete data from 1983Wa27 or 2011Fr04.
		1218 ^a	0.44	4417.3		Observed in 1983Wa27 but not in 2011Fr04.
		1653 ^a	3 1	3983.0	3	E_γ, I_γ : 1650 30 from 1964Gl04, but not observed in the more complete data from 1983Wa27 or 2011Fr04.
		1844.0	3.7 @ 7	3791.7		I_γ : other: 2.5 (1983Wa27).
		1862 ^a	1.0	3773.72	1	Observed in 1983Wa27 but not in 2011Fr04.
		1976 ^a	<0.3 @	3660.0		
		1989 ^a	<0.4 @	3646.3		
		2004 ^a	<0.3 @	3631.7	5	
		2506.53	3.9 @ 4	3129.12		I_γ : other: 4.1 (1983Wa27).
		2914.3	1.0 @ 3	2721.3	2	I_γ : other: 0.73 (1983Wa27).
		3025 ^a	<0.4 @	2611.03	3	
		3055.3	12.2 @ 7	2580.3		I_γ : others: 15 (1983Wa27), 17 3 (1964Gl04). $A_2 = -0.13$ 6, $A_4 = +0.05$ 7 (1983Wa27).
		3260 ^a	<0.3 @	2375.7	4	
		3454 ^a	<0.5 @	2181.10	3	
		3477.7	0.6 @ 3	2157.9	2 ⁺	
		3748.24	1.8 @ 3	1887.29	2 ⁺	I_γ : other: 1.9 (1983Wa27).
		4405.11	0.6 @ 4	1230.33	2 ⁺	I_γ : other: 0.58 (1983Wa27).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
5635.75	4970.1	15.4 [@] 10	665.3	1	E_γ : other: 4960 40 (1964Gl04). I_γ : others: 16 (1983Wa27), 11 3 (1964Gl04).
	5174.1	2.2 [@] 7	461.2	1 ⁺	I_γ : others: 2.9 (1983Wa27), 2 1 (1964Gl04).
	5489 ^a	<0.65 [@]	146.4	3 ⁺	I_γ : others: <0.44 (1983Wa27).
	5635.25	100 [@]	0	0 ⁺	I_γ : others: 100 (1983Wa27), 100 13 (1964Gl04). $A_2=+0.27$ 3, $A_4=+0.05$ 4 (1983Wa27).
5673.0	1256 ^a	0.41	4417.3		Observed in 1983Wa27 but not in 2011Fr04.
	1319 ^a	0.81	4354.2	(1)	Observed in 1983Wa27 but not in 2011Fr04.
	1733 ^a	2.0	3940.1		Observed in 1983Wa27 but not in 2011Fr04.
	1881 ^a	<6.7 [@]	3791.7		I_γ : other: 1.9 (1983Wa27).
	1899 ^a	<4.6 [@]	3773.72	1	I_γ : other: 0.54 (1983Wa27).
	2013 ^a	<3.4 [@]	3660.0		
	2027 ^a	<3.3 [@]	3646.3		
	2041 ^a	<4.0 [@]	3631.7	5	
	2073 ^a	<5.5 [@]	3600.28	4	
	2290 ^a	<1.3 [@]	3383.3		
	2544 ^a	2.8	3129.12		E_γ : other: 2520 30 (1964Gl04). I_γ : other: 4 2 (1964Gl04). Observed in 1983Wa27 but not in 2011Fr04.
	2952 ^a	<3.8 [@]	2721.3	2	
	3062 ^a	<1.5 [@]	2611.03	3	
	3092.6	11.6 [@] 26	2580.3		E_γ : other: 3070 30 (1964Gl04). I_γ : others: 6.6 (1983Wa27), 8 2 (1964Gl04).
	3297 ^a	<3.0 [@]	2375.7	4	
	3492 ^a	<3.2 [@]	2181.10	3	
	3514.9	8.8 [@] 24	2157.9	2 ⁺	E_γ : other: 3500 30 (1964Gl04). I_γ : others: 10 (1983Wa27), 14 3 (1964Gl04).
	3786 ^a	<4.4 [@]	1887.29	2 ⁺	E_γ : others: 3760 40 (1964Gl04). I_γ : others: 1.8 (1983Wa27), 3 2 (1964Gl04).
	4442 ^a	3 2	1230.33	2 ⁺	E_γ, I_γ : 4440 60 from 1964Gl04, but not observed in the more complete data from 1983Wa27 or 2011Fr04.
	5007.3	6.4 [@] 34	665.3	1	E_γ : others: 4980 60 (1964Gl04). I_γ : others: 8.2 (1983Wa27), 8 3 (1964Gl04).
	5211 ^a	<3.8 [@]	461.2	1 ⁺	I_γ : others: <0.54 (1983Wa27).
	5526 ^a	<1.4 [@]	146.4	3 ⁺	I_γ : others: <0.41 (1983Wa27).
	5672.5	100 [@]	0	0 ⁺	I_γ : others: 100 (1983Wa27), 100 20 (1964Gl04).
5763.3	1823.2	2.2	3940.1		
	2379.9	17	3383.3		
	2634.1	1.5	3129.12		
	3605.2	100	2157.9	2 ⁺	E_γ : other: 3600 40 (1964Gl04). I_γ : other: 100 18 (1964Gl04).
	3875.8	1.3	1887.29	2 ⁺	
	5098 ^a	<2.4	665.3	1	
	5301.7	3.0	461.2	1 ⁺	
	5616.4	16	146.4	3 ⁺	
	5762.8	44	0	0 ⁺	I_γ : other: 56 11 (1964Gl04).
5785.6	2011.8	0.8	3773.72	1	
	2402.2	52	3383.3		E_γ : other: 2390 30 (1964Gl04). I_γ : other: 44 10 (1964Gl04).

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
5785.6	2451.5	1.2	3334.0		
	3064.2	0.6	2721.3	2	
	3174.4	2.8	2611.03	3	
	3604.3	3	2181.10	3	
	3627.5	32	2157.9	2 ⁺	E_γ : other: 3620 30 (1964Gl04). I_γ : other: 40 8 (1964Gl04).
	3898.1	1.6	1887.29	2 ⁺	
	4554.9	6	1230.33	2 ⁺	E_γ : other: 4550 50 (1964Gl04). I_γ : other: 26 8 (1964Gl04).
	5120 ^a	<0.6	665.3	1	I_γ : other: <1 (1964Gl04).
	5324 ^a	<1.2	461.2	1 ⁺	
	5638.7	100	146.4	3 ⁺	I_γ : other: 100 16 (1964Gl04).
	5785 ^a	<0.6	0	0 ⁺	I_γ : other: <20 (1964Gl04).
5805.9	1200.1	9.4	4605.8		
	1658.1	3.2	4147.8		
	1822.9	1.9	3983.0	3	
	2205.5	1.6	3600.28	4	
	2260.7	1.3	3545.08	3	
	3225.4	58	2580.3		I_γ : other: 58 12 (1964Gl04).
	3430.0	3.9	2375.7	4	
	3918.4	1.3	1887.29	2 ⁺	E_γ : other: 3920 40 (1964Gl04). I_γ : other: 16 5 (1964Gl04).
	4575.2	13	1230.33	2 ⁺	
	5140.2	58	665.3	1	E_γ : other: 5120 50 (1964Gl04). I_γ : other: 38 10 (1964Gl04).
	5344.3	100	461.2	1 ⁺	E_γ : other: 5330 40 (1964Gl04). I_γ : other: 100 21 (1964Gl04).
	5659.0	65	146.4	3 ⁺	I_γ : other: 72 16 (1964Gl04).
	5805.4	6.1	0	0 ⁺	I_γ : other: <30 (1964Gl04).
5852.91	1705.1	18	4147.8		
	1869.9	2.8	3983.0	3	
	2079.1	0.4	3773.72	1	
	2252.55	8.4	3600.28	4	
	2307.75	13	3545.08	3	E_γ : other: 2300 50 (1964Gl04). I_γ : other: 19 4 (1964Gl04).
	2469.5	3.6	3383.3		
	3131.5	7.6	2721.3	2	
	3241.71	0.6	2611.03	3	
	3671.60	2.4	2181.10	3	E_γ : other: 3680 50 (1964Gl04). I_γ : other: 16 7 (1964Gl04).
	3694.8	8.4	2157.9	2 ⁺	
	3965.37	6.2	1887.29	2 ⁺	E_γ : other: 3950 100 (1964Gl04). I_γ : other: 24 7 (1964Gl04).
	4622.24	28	1230.33	2 ⁺	E_γ : other: 4650 50 (1964Gl04). I_γ : other: 36 11 (1964Gl04).
	5187 ^a	<1	665.3	1	I_γ : other: 7 3 (1964Gl04).
	5391 ^a	<1	461.2	1 ⁺	
5897.2	5706.0	100	146.4	3 ⁺	I_γ : other: 100 20 (1964Gl04).
	5852 ^a	<0.4	0	0 ⁺	I_γ : other: <15 (1964Gl04).
	1291.4	5	4605.8		
	1749 ^a	24 8	4147.8		E_γ, I_γ : 1720 30 from 1964Gl04, but not observed in the more complete data from 1983Wa27.
	2352.0	5	3545.08	3	
	2513.8	1.2	3383.3		
	2563.1	0.88	3334.0		

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments		
5897.2		3175.7	5	2721.3	2			
		3316.7	2.4	2580.3				
		4009.7	11	1887.29	2 ⁺			
		4666.5	41	1230.33	2 ⁺			
		5231.5	32	665.3	1	E_γ : other: 5200 50 (1964GI04).		
						I_γ : other: 43 18 (1964GI04).		
		5435.5	100	461.2	1 ⁺	E_γ : other: 5400 50 (1964GI04).		
						I_γ : other: 100 30 (1964GI04).		
		5750.3	85	146.4	3 ⁺	E_γ : other: 5700 50 (1964GI04).		
						I_γ : other: 73 20 (1964GI04).		
5940.8	1 to 3	5896.7	5	0	0 ⁺	E_γ : other: 5850 60 (1964GI04).		
							I_γ : other: 28 10 (1964GI04).	
		984 ^a	9 3	4957.3			E_γ, I_γ : 980 30 from 1964GI04, but not observed in the more complete data from 1983Wa27.	
		1245.1	0.85	4695.7				
		2557.4	6.3	3383.3				
		3329.6	0.56	2611.03	3			
		3360.3	1.6	2580.3				
		3564.9	0.56	2375.7	4			
		3782.7	100	2157.9	2 ⁺		I_γ : other: 100 18 (1964GI04).	
							$A_2=+0.05$ 2, $A_4=-0.05$ 3 (1983Wa27).	
		4053.3	1.1	1887.29	2 ⁺			
		5275.1	0.99	665.3	1			
		5479.1	21	461.2	1 ⁺		E_γ : other: 5470 50 (1964GI04).	
							I_γ : other: 12 5 (1964GI04).	
		5793.9	6.8	146.4	3 ⁺		I_γ : other: <5 (1964GI04).	
		5940.2	0.99	0	0 ⁺		I_γ : other: <5 (1964GI04).	
		6030.1		2646.7	4.7	3383.3		
				3872 ^a	<0.33	2157.9	2 ⁺	
4143 ^a	<0.67			1887.29	2 ⁺			
4799 ^a	<0.56			1230.33	2 ⁺			
5364 ^a	<0.56			665.3	1			
5568 ^a	<0.89			461.2	1 ⁺			
5883.2	6.4			146.4	3 ⁺			
6029.5	100			0	0 ⁺		I_γ : other: 100 (1964GI04).	
6088.94	1,2			2297.2	0.33	3791.7		
				2315.1	0.56	3773.72	1	
		3367.5	4.7	2721.3	2			
		3508.5	0.33	2580.3				
		3931 ^a	<0.33	2157.9	2 ⁺			
		4201.37	2.6	1887.29	2 ⁺		I_γ : other: 13 (1964GI04).	
		4858 ^a	<0.67	1230.33	2 ⁺			
		5423 ^a	<1.1	665.3	1		I_γ : other: 4.7 (1964GI04).	
		5627.2	2.7	461.2	1 ⁺			
		5942 ^a	<0.33	146.4	3 ⁺			
6088.36	100	0	0 ⁺		I_γ : other: 100 (1964GI04).			
					$A_2=+0.53$ 7, $A_4=-0.00$ 8 (1983Wa27).			
6136.2	1 to 3	1719 ^a	<1.8 [#]	4417.3				
		2153 ^a	<1.8 [#]	3983.0	3			
		2752.8	1.82 91	3383.3			I_γ : other: 1.4 (1983Wa27).	
		3414.7	1.82 91	2721.3	2		I_γ : other: 1.9 (1983Wa27).	
		3525.0	3.6 18	2611.03	3		I_γ : other: 3.8 (1983Wa27).	
		3555.7	1.82 91	2580.3			I_γ : other: 1.2 (1983Wa27).	
		3954.9	1.82 91	2181.10	3		I_γ : other: 1.7 (1983Wa27).	

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
6136.2	1 to 3	3978.1	10.9 11	2157.9	2 ⁺			I_γ : other: 11 (1983Wa27).
		4905.5	40 4	1230.33	2 ⁺			I_γ : other: 35 (1983Wa27), 48 (1964GI04). $A_2=-0.32$ 4, $A_4=+0.02$ 5 (1983Wa27).
		5470.4	100 10	665.3	1			I_γ : other: 100 (1983Wa27), 100 (1964GI04). $A_2=+0.28$ 5, $A_4=-0.01$ 6 (1983Wa27).
6141.8	1,2	5675 ^a	<0.52	461.2	1 ⁺			I_γ : other: 21 (1964GI04).
		5989.2	20 2	146.4	3 ⁺			I_γ : other: 17 (1983Wa27), 23 (1964GI04).
		1626.1	1.16 [#] 58	4515.7				I_γ : other: 1.3 (1983Wa27).
		1724.5	0.58 [#] 29	4417.3				I_γ : other: 1.5 (1983Wa27).
		1787.6	1.74 [#] 93	4354.2	(1)			I_γ : other: 1.7 (1983Wa27).
		2368.0	2.3 [#] 12	3773.72	1			I_γ : other: 2.6 (1983Wa27).
		2596.6	0.58 [#] 29	3545.08	3			
		2758 ^a	<0.58 [#]	3383.3				
		3420.3	6.4 [#] 7	2721.3	2			I_γ : other: 6.6 (1983Wa27).
		3561.3	1.16 [#] 58	2580.3				I_γ : other: 1.4 (1983Wa27).
		3983.7	0.58	2157.9	2 ⁺			
		4911.1	2.3 [#] 12	1230.33	2 ⁺			
		5476 ^a	<0.23	665.3	1			
		5680 ^a	<0.12	461.2	1 ⁺			
		5995 ^a	<0.35	146.4	3 ⁺			
		6141.2	100 [#] 10	0	0 ⁺			I_γ : other: 100 (1983Wa27), 100 (1964GI04). $A_2=-0.14$ 3, $A_4=+0.08$ 3 (1983Wa27).
6169.4	3	1653.7	4.4 [#] 24	4515.7				I_γ : other: 20 (1983Wa27).
		2030 ^a	<2.9 [#]	4139.7	2			I_γ : other: 10 (1983Wa27).
		2093.1	15.5 60	4076.2	4	D(+Q)	+0.03 5	I_γ : weighted average of 14.7 74 (1977Da02) and 16 6 (1971Hy02). Other: 15 (1983Wa27).
								Mult., δ : or $\delta<-7$ from $\gamma(\theta)$ in 1971Hy02.
								$A_2=-0.12$ 5 (1971Hy02).
		2186.3	100 [#] 10	3983.0	3	D(+Q)	-0.03 +5-4	I_γ : other: 100 (1983Wa27), 100 (1964GI04), 100 25 (1969Gr29), 100 19 (1971Hy02). Mult., δ : weighted average of -0.05 6 (1971Hy02) and -0.02 +5-4 (1977Da03). Other: $\delta=+1.3$ 2 (1971Hy02).
								$A_2=+0.27$ 3, $A_4=+0.01$ 3 (weighted average of values from 1983Wa27 and 1977Da03).
								$A_2=+0.25$ 6, $A_4=+0.07$ 7 (1983Wa27); $A_2=+0.27$ 3, $A_4=0.00$ 3 (1977Da03); $A_2=+0.25$ 3 (1971Hy02).
		2569.0	23.6 24	3600.28	4			I_γ : weighted average of 23.5 24 (1977Da02), 39 22 (1969Gr29), and 22 13 (1971Hy02). Other: 23 (1983Wa27).
		2624.2	16.5 60	3545.08	3	D(+Q)	-0.05 +19-14	I_γ : weighted average of 14.7 74 (1977Da02), 22 11 (1969Gr29), and 16 6 (1971Hy02). Other: 15 (1983Wa27), 28 (1964GI04). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=+0.25$ 5, $A_4=+0.01$ 5 (1977Da03).

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
6169.4	3							$A_2=+0.25\ 5$, $A_4=+0.01\ 5$ (1977Da03); $A_2=+0.28\ 5$ (1971Hy02).
		3447.9	20.7 21	2721.3	2			I_γ : weighted average of 20.6 21 (1977Da02) and 25 13 (1971Hy02). Other: 25 (1983Wa27).
		3558.2	7.3 30	2611.03	3			I_γ : weighted average of 5.9 30 (1977Da02) and 13 6 (1971Hy02). Other: 8.8 (1983Wa27).
								$A_2=+0.10\ 10$ (1971Hy02).
		3589 ^a	<31	2580.3				I_γ : from 1969Gr29.
		3794 ^a	<19	2375.7	4			I_γ : from 1969Gr29.
		3988.1	15.5 60	2181.10	3			I_γ : weighted average of 14.7 74 (1977Da02) and 16 6 (1971Hy02). Other: 15 (1983Wa27).
								$A_2=+0.28\ 7$ (1971Hy02).
		4011.3	1.47 [#] 74	2157.9	2 ⁺			I_γ : other: 2.1 (1983Wa27), <5.6 (1969Gr29).
		4281.8	7.1 31	1887.29	2 ⁺			I_γ : weighted average of 8.8 44 (1977Da02) and 6.3 31 (1971Hy02). Other: 6.4 (1983Wa27), <8.3 (1969Gr29).
								$A_2=-0.21\ 12$ (1971Hy02).
		4938.7	10.7 31	1230.33	2 ⁺	D(+Q)	0.00 1	I_γ : weighted average of 11.8 59 (1977Da02), 17 8 (1969Gr29), and 9.4 31 (1971Hy02). Other: 9.4 (1983Wa27).
								Mult., δ : from $\gamma(\theta)$ in 1971Hy02.
								$A_2=-0.24\ 10$ (1971Hy02).
6181.27	1,2	5504 ^a	<0.61	665.3	1			I_γ : other: <5.6 (1969Gr29).
		5708 ^a	<0.91	461.2	1 ⁺			I_γ : other: 35 (1964Gl04), <5.6 (1969Gr29).
		6022.4	80 8	146.4	3 ⁺	D(+Q)	-0.02 3	I_γ : weighted average of 74 8 (1977Da02), 100 17 (1969Gr29), and 91 22 (1971Hy02). Other: 85 (1983Wa27), 70 (1964Gl04).
								Mult.: from $\gamma(\theta)$ in 1971Hy02 and 1977Da03.
								δ : from 1971Hy02. Other: $\delta=-0.03\ 5$ (1977Da03).
								$A_2=+0.264\ 24$, $A_4=-0.01\ 3$ (weighted average of values from 1983Wa27 and 1977Da03).
								$A_2=+0.27\ 4$, $A_4=-0.03\ 5$ (1983Wa27); $A_2=+0.26\ 3$, $A_4=0.00\ 3$ (1977Da03); $A_2=+0.28\ 3$ (1971Hy02).
		6169 ^a	<0.91	0	0 ⁺			I_γ : other: <8.3 (1969Gr29).
		1855.33	11.4 [#] 57	4325.89				I_γ : other: 12 (1983Wa27).
		2217.1	2.3 [#] 12	3964.1				I_γ : other: 3.1 (1983Wa27).
		2797.9	4.6 [#] 23	3383.3				I_γ : other: 7.4 (1983Wa27).
		2847.1	6.8 [#] 34	3334.0				I_γ : other: 5.5 (1983Wa27).
		3052.00	1.14 [#] 57	3129.12				I_γ : other: 0.71 (1983Wa27).
		3459.8	1.14 [#] 57	2721.3	2			I_γ : other: 0.95 (1983Wa27).
		3570.04	9.1 [#] 46	2611.03	3			I_γ : other: 8.6 (1983Wa27).
		3600.8	18.0 18	2580.3				I_γ : weighted average of 18.2 18 (1977Da02) and 17 4 (1969Gr29). Other: 19 (1983Wa27), 22 (1964Gl04).
								I_γ : from 1969Gr29.
		3805 ^a	<0.94	2375.7	4			
		3999.92	1.4	2181.10	3			
		4023.1	6.8 38	2157.9	2 ⁺			I_γ : weighted average of 8.0 41 (1977Da02) and 5.7 38 (1969Gr29). Other: 10 (1983Wa27), 22 (1964Gl04).
								I_γ : weighted average of 9.1 46 (1977Da02) and 15 4 (1969Gr29). Other: 9.8 (1983Wa27), 14 (1964Gl04).
		4293.69	12.5 40	1887.29	2 ⁺			

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
6181.27	1,2	4950.55	25.0 25	1230.33	2 ⁺			I_γ : weighted average of 25.0 25 (1977Da02) and 25 4 (1969Gr29). Other: 23 (1983Wa27), 18 (1964Gl04). $A_2=-0.07$ 4, $A_4=-0.05$ 5 (1983Wa27); $A_2=-0.06$ 13 with $A_4=0$; $A_2=+0.09$ 19, $A_4=-0.23$ 24 (1969Gr29). I_γ : other: 100 (1983Wa27), 100 (1964Gl04), 100 8 (1969Gr29). Mult.: 1969Gr29 states that $\gamma(\theta)$ of $E_p(\text{lab})=1071$, 6181 level to 666, 1 ⁺ can be fit with $J(6181)=1,2$ for a variety of mixing ratios. $A_2=+0.00$ 3 $A_4=+0.07$ 4 (1983Wa27); $A_2=-0.06$ 5 with $A_4=0$; $A_2=-0.04$ 7, $A_4=-0.03$ 9 (1969Gr29). I_γ : weighted average of 10.2 52 (1977Da02) and 11 4 (1969Gr29). Other: 11 (1983Wa27). I_γ : weighted average of 20.5 21 (1977Da02) and 15.1 76 (1969Gr29). Other: 24 (1983Wa27), 28 (1964Gl04). Mult.: 1969Gr29 states that $\gamma(\theta)$ of $E_p(\text{lab})=1071$, 6181 level to 146, 3 ⁺ can be fit with $J(6181)=1,2$ for a variety of mixing ratios. $A_2=-0.26$ 13 with $A_4=0$; $A_2=-0.24$ 23, $A_4=-0.02$ 25 (1969Gr29). I_γ : other: <0.94 (1969Gr29). I_γ : other: 1.9 (1983Wa27). I_γ : weighted average of 18.8 19 (1977Da02) and 14.3 82 (1971Hy02). Other: 18 (1983Wa27). I_γ : weighted average of 7.3 38 (1977Da02), 26 15 (1969Gr29), and 12 4 (1971Hy02). Other: 13 (1983Wa27), 19 (1964Gl04). I_γ : weighted average of 20.8 21 (1977Da02) and 20 10 (1971Hy02). Other: 19 (1983Wa27). I_γ : other: 100 (1983Wa27), 100 20 (1969Gr29), 100 21 (1971Hy02). Mult., δ : from $\gamma(\theta)$ in 1971Hy02. $A_2=+0.31$ 1 (1971Hy02). I_γ : weighted average of 48 5 (1977Da02), 56 17 (1969Gr29), and 49 12 (1971Hy02). Other: 49 (1983Wa27), 100 (1964Gl04). Mult., δ : from $\gamma(\theta)$ in 1971Hy02. $A_2=-0.22$ 2 (1971Hy02). I_γ : from 1969Gr29. I_γ : other: 4.5 (1983Wa27), <7.3 (1969Gr29). I_γ : weighted average of 6.3 31 (1977Da02) and 8.2 21 (1971Hy02). Other: 5.7 (1983Wa27). $A_2=-0.24$ 7 (1971Hy02). I_γ : from 1969Gr29. I_γ : other: 6 (1964Gl04), <3.6 (1969Gr29). I_γ : from 1969Gr29. I_γ : other: <1.8 (1969Gr29). I_γ : other: <5.5 (1969Gr29). I_γ : other: 1.7 (1983Wa27), <3.6 (1969Gr29). I_γ : other: <1.8 (1969Gr29).
		5515.5	100 [#] 10	665.3	1	(D+Q)		
		5719.6	10.7 40	461.2	1 ⁺			
		6034.3	20.1 21	146.4	3 ⁺			
6208.2	4	6180.67	1.9	0	0 ⁺			
		1383.8	1.04 [#] 52	4824.41				
		2131.9	18.6 19	4076.2	4			
		2225.1	10.1 38	3983.0	3			
		2561.8	1.04 [#] 52	3646.3				
		2576.4	20.8 21	3631.7	5			
		2607.8	100 [#] 10	3600.28	4	D(+Q)	-0.03 5	
		2663.0	49 5	3545.08	3	D(+Q)	0.00 2	
		3628 ^a	<7.3	2580.3				
		3832.3	4.2 [#] 21	2375.7	4			
		4026.8	7.6 21	2181.10	3			
		4050 ^a	<7.3	2157.9	2 ⁺			
		4320.6 ^a	0.43	1887.29	2 ⁺			
		4978 ^a	<1.8	1230.33	2 ⁺			
		5542 ^a	<0.21	665.3	1			
		5747 ^a	<0.21	461.2	1 ⁺			
		6061.2	1.04 [#] 52	146.4	3 ⁺			
		6208 ^a	<0.21	0	0 ⁺			

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
6219.3		1501.9	100 [#] 10	4717.4				
		1893.4	3.9 [#] 21	4325.89				
		2572.9	18.0 [#] 18	3646.3				
		2587.5	2.6 [#] 13	3631.7	5			
		2885.2	7.7 [#] 39	3334.0				
		3608.1	5.1 [#] 26	2611.03	3			
		3843.4	11.5 [#] 59	2375.7	4			
		4037.9	46 [#] 5	2181.10	3			
		4988.6	28.2 [#] 28	1230.33	2 ⁺			
		6072.3	33.3 [#] 33	146.4	3 ⁺			
6229.5	1	1590.6	4.5 18	4638.9	1,2			I_γ : weighted average of 3.3 18 (1977Da02) and 6.0 20 (1971Hy02). Other: 3.9 (1983Wa27). $A_2=-0.18$ 15 (1971Hy02).
		1713.8	24.0 25	4515.7				I_γ : weighted average of 24.4 25 (1977Da02) and 22 6 (1971Hy02). Other: 29 (1983Wa27). $A_2=+0.05$ 3 (1971Hy02).
		1812.2	6.2 20	4417.3				I_γ : weighted average of 6.7 33 (1977Da02) and 6.0 20 (1971Hy02). Other: 6.3 (1983Wa27). $A_2=+0.32$ 16 (1971Hy02).
		1875.2	20 2	4354.2	(1)			I_γ : weighted average of 20 2 (1977Da02) and 16 6 (1971Hy02). Other: 23 (1983Wa27). $A_2=-0.24$ 5 (1971Hy02).
		1903.6	2.2 [#] 11	4325.89				I_γ : other: 2.2 (1983Wa27).
		2089.7	13.2 13	4139.7	2			I_γ : weighted average of 13.3 13 (1977Da02) and 12 4 (1971Hy02). Other: 12 (1983Wa27). $A_2=+0.01$ 9 (1971Hy02).
		2455.7	31.6 33	3773.72	1	D(+Q)	+0.03 5	I_γ : weighted average of 33.3 33 (1977Da02) and 26 6 (1971Hy02). Other: 44 (1983Wa27), 71 (1964Gl04). Mult., δ : from $\gamma(\theta)$ in 1971Hy02. $A_2=-0.23$ 3 (1971Hy02).
		3100.2	2.4	3129.12				
		3508.0	100 [#] 10	2721.3	2	D(+Q)	-0.05 5	I_γ : other: 100 (1983Wa27), 100 (1964Gl04), 100 12 (1971Hy02). Mult., δ : from $\gamma(\theta)$ in 1971Hy02. $A_2=+0.01$ 5 (1971Hy02).
		4071.3	10.4 40	2157.9	2 ⁺			I_γ : weighted average of 11.1 56 (1977Da02) and 10 4 (1971Hy02). Other: 9.5 (1983Wa27), 29 (1964Gl04). $A_2=-0.13$ 10 (1971Hy02).
		4341.9	39	1887.29	2 ⁺			I_γ : from 1964Gl04, but not observed in the more complete data from 1983Wa27 or 1977Da02.
		5564 ^a	<0.49	665.3	1			I_γ : other: 24 (1964Gl04).
		5767.8	2.4 10	461.2	1 ⁺			I_γ : weighted average of 4.4 22 (1977Da02) and 2.0 10 (1971Hy02). Other: 5.1 (1983Wa27).
		6083 ^a	<0.49	146.4	3 ⁺			
		6228.9	1.9 8	0	0 ⁺			I_γ : weighted average of 3.3 18 (1977Da02) and 1.6 8 (1971Hy02). Other: 5.6 (1983Wa27).
6266.6		3545.1	5.8 [#] 29	2721.3	2			I_γ : other: 4.9 (1983Wa27).
		4085.2	5.8	2181.10	3			
		4108.4	100 [#] 10	2157.9	2 ⁺			I_γ : other: 100 (1983Wa27).

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
6266.6		4379.0	11.6 [#] 12	1887.29	2 ⁺			I_γ : other: 11 (1983Wa27).
		5035.9	8.6	1230.33	2 ⁺			
		5600.8	27.5 [#] 28	665.3	1			I_γ : other: 30 (1983Wa27).
		5805 ^a	<2.3	461.2	1 ⁺			
		6119.6	72	146.4	3 ⁺			I_γ : other: <33 (1977Da02).
6273.3	3	6266 ^a	<2.6	0	0 ⁺			
		1826.7	0.31	4446.6	4			
		2197.0	1.11 [#] 64	4076.2	4			I_γ : other: 0.62 (1983Wa27).
		2309.1	1.27 [#] 64	3964.1				I_γ : other: 1.4 (1983Wa27).
		2939.2	0.77	3334.0				
		3662.1	12.7 [#] 13	2611.03	3			I_γ : other: 10 (1983Wa27).
		3693 ^a	<0.99	2580.3				I_γ : from 1969Gr29.
		3897.4	100 [#] 10	2375.7	4	D+Q	-0.08 5	I_γ : other: 100 (1983Wa27), 100 (1964Gl04), 100 5 (1969Gr29). Mult., δ : from $\gamma(\theta)$ $E_p(\text{lab})=1165\rightarrow 2376\rightarrow 146$ γ cascade (1969Gr29). $A_2=0.00$ 2 $A_4=-0.04$ 2 (1983Wa27); $A_2=-0.01$ 3 with $A_4=0$; $A_2=+0.02$ 3, $A_4=-0.06$ 4 (1969Gr29).
		4091.9	25.4 [#] 26	2181.10	3			I_γ : other: 22 (1983Wa27). $A_2=+0.18$ 4, $A_4=-0.05$ 5 (1983Wa27).
		4115.1	1.80 80	2157.9	2 ⁺			I_γ : weighted average of 1.59 80 (1977Da02) and 6.2 37 (1969Gr29). Other: 1.4 (1983Wa27), 23 (1964Gl04).
		4385.7	11.1 11	1887.29	2 ⁺			I_γ : weighted average of 11.1 11 (1977Da02) and 11.1 13 (1969Gr29). Other: 9.9 (1983Wa27), 19 (1964Gl04). $A_2=0.00$ 7, $A_4=+0.05$ 9 (1983Wa27).
		5042.6	1.2 12	1230.33	2 ⁺			I_γ : weighted average of 0.79 40 (1977Da02) and 4.9 13 (1969Gr29). I_γ : other: 8 (1964Gl04), 1.24 62 (1969Gr29). I_γ : other: 6 (1964Gl04), <0.99 (1969Gr29).
		5608 ^a	<0.15	665.3	1			I_γ : other: 7.7 (1983Wa27), <4.9 (1969Gr29).
		5812 ^a	<0.15	461.2	1 ⁺			I_γ : other: <0.74 (1969Gr29).
		6126.3	4.8 [#] 24	146.4	3 ⁺			
		6273 ^a	<0.15	0	0 ⁺			
6322.0	1,2	1716.2	4.6 [#] 23	4605.8				I_γ : other: 1.6 (1983Wa27).
		1967.7	2.3 [#] 12	4354.2	(1)			I_γ : other: 2.0 (1983Wa27).
		2245.7	6.8 [#] 34	4076.2	4			
		2690.2	2.3 [#] 12	3631.7	5			
		2776.8	9.1 [#] 46	3545.08	3			I_γ : other: 8.4 (1983Wa27).
		2938.6	1.14 [#] 57	3383.3				
		3600.5	12.5 [#] 14	2721.3	2			I_γ : other: 10 (1983Wa27).
		3710.8	4.6 [#] 23	2611.03	3			I_γ : other: 0.61 (1983Wa27).
		3741.5	20.4 21	2580.3				I_γ : weighted average of 20.5 21 (1977Da02) and 19 7 (1969Gr29). Other: 17 (1983Wa27). $A_2=-0.23$ 5, $A_4=-0.07$ 6 (1983Wa27).
		3946 ^a	<4.8	2375.7	4			I_γ : from 1969Gr29.
		4140.6	2.3 [#] 12	2181.10	3			I_γ : other: 1.6 (1983Wa27), <6.5 (1969Gr29).
		4434.4	2.3 [#] 12	1887.29	2 ⁺			I_γ : other: 1.4 (1983Wa27), <8.1 (1969Gr29).
		5091.3	4.6 [#] 23	1230.33	2 ⁺			I_γ : other: 3.5 (1983Wa27), <4.8 (1969Gr29).
		5656.2	34 4	665.3	1			I_γ : weighted average of 36 4 (1977Da02) and 31 5 (1969Gr29). Other: 41 (1983Wa27), 28 (1964Gl04).

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	Comments
6322.0	1,2	5860.3	16.5 18	461.2	1 ⁺		I_γ : weighted average of 18.2 18 (1977Da02) and 11.3 32 (1969Gr29). Other: 16 (1983Wa27).
		6175.0	100# 10	146.4	3 ⁺		I_γ : other: 100 (1983Wa27), 100 (1964Gl04), 100 11 (1969Gr29). $A_2=-0.04$ 3, $A_4=+0.01$ 3 (1983Wa27). I_γ : other: <4.8 (1969Gr29).
		6321 ^a	<0.61	0	0 ⁺		
6361.3		2977.9	71# 7	3383.3			
		4203.1	100# 10	2157.9	2 ⁺		I_γ : other: 100 (1964Gl04).
		5130.6	25.0# 25	1230.33	2 ⁺		
		5899.6	14.8# 16	461.2	1 ⁺		
		6214.3	17.1# 18	146.4	3 ⁺		
6370.2	2	1412.9	3.5# 17	4957.3			
		1731.3	1.72# 86	4638.9	1,2		I_γ : other: 3.3 (1983Wa27).
		1854.5	3.5# 17	4515.7			I_γ : other: 3.9 (1983Wa27).
		2015.9	8.6# 45	4354.2	(1)		I_γ : other: 6.1 (1983Wa27).
		2387.1	1.72# 86	3983.0	3		I_γ : other: 1.5 (1983Wa27).
		2406.0	1.72# 86	3964.1			I_γ : other: 1.5 (1983Wa27).
		2825.0	6.9# 35	3545.08	3		I_γ : other: 6.7 (1983Wa27).
		2986.8	1.72# 86	3383.3			I_γ : other: 2.1 (1983Wa27).
		3036.1	2.1	3334.0			
		3240.9	3.5# 17	3129.12			I_γ : other: 1.5 (1983Wa27).
		3648.7	20.7# 21	2721.3	2		I_γ : other: 18 (1983Wa27).
		3759.0	3.5# 17	2611.03	3		I_γ : other: 2.1 (1983Wa27).
		3789.7	23.5 24	2580.3			I_γ : weighted average of 24.1 24 (1977Da02) and 21 5 (1969Gr29). Other: 21 (1983Wa27), 40 (1964Gl04).
		3994 ^a	<2.3	2375.7	4		I_γ : from 1969Gr29.
		4188.8	10.3# 52	2181.10	3		I_γ : other: 7.3 (1983Wa27).
		4212.0	8.6# 45	2157.9	2 ⁺		I_γ : other: 6.7 (1983Wa27), <4.6 (1969Gr29).
		4482.6	3.5# 17	1887.29	2 ⁺		I_γ : other: 3.0 (1983Wa27), <2.3 (1969Gr29).
		5139.5	11.2 23	1230.33	2 ⁺		I_γ : weighted average of 10.3 52 (1977Da02) and 11.4 23 (1969Gr29). Other: 9.1 (1983Wa27).
		5704.4	100# 10	665.3	1	D+Q	I_γ : other: 100 (1983Wa27), 100 (1964Gl04), 100 5 (1969Gr29). Mult.: from $\gamma(\theta)$ $E_p(\text{lab})=1266 \rightarrow 666 \rightarrow \text{g.s.}$ γ cascade (1969Gr29). $A_2=-0.04$ 5 with $A_4=0$; $A_2=-0.04$ 6, $A_4=0.00$ 8 (1969Gr29).
		5908.5	25.6 23	461.2	1 ⁺		I_γ : weighted average of 31.0 31 (1977Da02) and 22.7 23 (1969Gr29). Other: 22 (1983Wa27). $A_2=-0.07$ 14 with $A_4=0$; $A_2=-0.11$ 16, $A_4=+0.09$ 21 (1969Gr29).
		6223.2	36.5 53	146.4	3 ⁺		I_γ : weighted average of 48 5 (1977Da02) and 34.1 23 (1969Gr29). Other: 42 (1983Wa27), 60 (1964Gl04). $A_2=+0.11$ 12 with $A_4=0$; $A_2=+0.11$ 13, $A_4=0.00$ 17 (1969Gr29).
		6369.6	46 5	0	0 ⁺		I_γ : weighted average of 52 5 (1977Da02) and 39 5 (1969Gr29). Other: 42 (1983Wa27). $A_2=-0.18$ 10 with $A_4=0$; $A_2=-0.21$ 13, $A_4=-0.03$ 15 (1969Gr29).
6382.4		1772.7	49# 5	4609.7			
		1776.6	36# 4	4605.8			

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
6382.4		2306.1	6.4 [#] 33	4076.2	4	
		2399.3	5.1 [#] 26	3983.0	3	
		2782.0	20.5 [#] 21	3600.28	4	
		2837.2	9.0 [#] 46	3545.08	3	
		3048.3	6.4 [#] 33	3334.0		
		3660.9	2.6 [#] 13	2721.3	2	
		3771.2	10.3 [#] 51	2611.03	3	
		4006.5	6.4 [#] 33	2375.7	4	
		5716.6	5.1 [#] 26	665.3	1	
		6235.4	100 [#] 10	146.4	3 ⁺	
6399.8		2799.4	5.5 [#] 27	3600.28	4	
		3016.4	58 [#] 6	3383.3		
		3678.3	3.6 [#] 18	2721.3	2	
		4023.8	5.5 [#] 27	2375.7	4	
		5169.1	9.1 [#] 46	1230.33	2 ⁺	
		6252.8	100 [#] 10	146.4	3 ⁺	
6441.2		2364.9	2.7 [#] 14	4076.2	4	
		3057.8	9.6 [#] 10	3383.3		
		4283.0	100 [#] 10	2157.9	2 ⁺	
		5775.4	5.5 [#] 28	665.3	1	
		5979.4	8.2 [#] 8	461.2	1 ⁺	
		6294.2	11.0 [#] 11	146.4	3 ⁺	
6450.54	3,4	1841 ^a	<4.2 [#]	4609.7		
		1844.7	8.3 [#] 42	4605.8		I_γ : other: 8.1 (1983Wa27).
		1989.1	6.9	4461.4		
		2003.9	7.3 [#] 38	4446.6	4	I_γ : other: 7.1 (1983Wa27).
		2374.3	4.2 [#] 21	4076.2	4	I_γ : other: 2.7 (1983Wa27).
		2818.7	10.4 [#] 52	3631.7	5	I_γ : other: 8.8 (1983Wa27).
		2850.13	38 [#] 4	3600.28	4	I_γ : other: 38 (1983Wa27). $A_2=+0.46$ 10, $A_4=-0.18$ 12 (1983Wa27).
		2905.33	7.3 [#] 38	3545.08	3	I_γ : other: 8.8 (1983Wa27).
		3116.4	1.0	3334.0		
		3839.28	29.2 [#] 29	2611.03	3	I_γ : other: 25 (1983Wa27).
		4269.15	4.2 [#] 21	2181.10	3	I_γ : other: 2.5 (1983Wa27).
		4292 ^a	<0.42	2157.9	2 ⁺	
		5785 ^a	<1.0	665.3	1	
		5989 ^a	<0.83	461.2	1 ⁺	
		6303.5	100 [#] 10	146.4	3 ⁺	I_γ : other: 100 (1983Wa27). $A_2=+0.03$ 7, $A_4=-0.06$ 9 (1983Wa27).
		6450 ^a	<1.7	0	0 ⁺	
6477.5		2123.2	26.2 [#] 26	4354.2	(1)	
		2704 ^a	<3.1 [#]	3773.72	1	
		3897.0	9.2 [#] 9	2580.3		
		5811.7	18.5 [#] 19	665.3	1	

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
6477.5		6015.7	100 [#] 10	461.2	1 ⁺	
6486.9	1 to 4	2546.7	16.7 [#] 17	3940.1		I_γ : other: 17 (1983Wa27).
		3357.6	1.5	3129.12		
		4328.7	100 [#] 10	2157.9	2 ⁺	I_γ : other: 100 (1983Wa27). $A_2=+0.15$ 7, $A_4=+0.07$ 8 (1983Wa27).
		5821.1	3.3 [#] 17	665.3	1	I_γ : other: 2.0 (1983Wa27).
		6025.1	13.3 [#] 13	461.2	1 ⁺	I_γ : other: 13 (1983Wa27).
		6340 ^a	<0.85	146.4	3 ⁺	
		6486.2	33.3 [#] 33	0	0 ⁺	I_γ : other: 36 (1983Wa27).
6527.6		2927.2	21.0 [#] 21	3600.28	4	
		3144.1	100 [#] 10	3383.3		
		4369.4	9.7 [#] 10	2157.9	2 ⁺	
		4640.0	6.5 [#] 32	1887.29	2 ⁺	
		5861.8	11.3 [#] 11	665.3	1	
		6380.6	12.9 [#] 13	146.4	3 ⁺	
6546.6		1375.0	1.6	5171.6	4	
		3163.1	0.95	3383.3		
		3966.1	7.7 [#] 8	2580.3		I_γ : other: 5.5 (1983Wa27).
		5315.8	20.5 [#] 21	1230.33	2 ⁺	I_γ : other: 11 (1983Wa27).
		5881 ^a	<0.36	665.3	1	
		6085 ^a	<0.48	461.2	1 ⁺	
		6399.6	100 [#] 10	146.4	3 ⁺	I_γ : other: 100 (1983Wa27).
		6546 ^a	<0.48	0	0 ⁺	
6574.7		4198.7	1.9	2375.7	4	
		4416.5	100 [#] 10	2157.9	2 ⁺	I_γ : other: 100 (1983Wa27).
		5909 ^a	<0.65	665.3	1	
		6427.7	5.3 [#] 26	146.4	3 ⁺	I_γ : other: 5.6 (1983Wa27).
		6574 ^a	<0.65	0	0 ⁺	
6583.7		3038.5	53 [#] 5	3545.08	3	
		4425.5	21.3 [#] 21	2157.9	2 ⁺	
		5917.9	21.3 [#] 21	665.3	1	
		6121.9	100 [#] 10	461.2	1 ⁺	
		6583.0	17.0 [#] 17	0	0 ⁺	
6624.7		1237.9	30	5386.8		
		1907.2	32	4717.4		
		2484.9	19	4139.7	2	
		2548.4	75	4076.2	4	I_γ : other: 34.3 34 (1977Da02).
		2992.9	28	3631.7	5	
		3024.3	100	3600.28	4	I_γ : other: 100 10 (1977Da02).
		3079.5	52	3545.08	3	I_γ : other: 34.3 34 (1977Da02).
		3290.5	52	3334.0		
		4013.4	23	2611.03	3	
		4443.3	69	2181.10	3	
		4466.5	77	2157.9	2 ⁺	
		5393.9	21	1230.33	2 ⁺	
		5959 ^a	<3.9	665.3	1	
		6162.9	100	461.2	1 ⁺	I_γ : other: 74 8 (1977Da02).
		6477.6	92	146.4	3 ⁺	I_γ : other: 43 4 (1977Da02).
6640.92	4	2314.95	1.32 [#] 66	4325.89		

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.&	$\delta^\&$	Comments
6640.92	2564.6	100 [#] 10	4076.2	4	D+Q	-0.14 3	I_γ : other: 100 (1983Wa27). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=+0.31$ 2, $A_4=+0.03$ 2 (1977Da03). $A_2=+0.44$ 5, $A_4=+0.01$ 6 (1983Wa27); $A_2=+0.31$ 2, $A_4=+0.03$ 2 (1977Da03).
	2657.8	7.9 [#] 40	3983.0	3			I_γ : other: 4.1 (1983Wa27).
	2994.5	1.32 [#] 66	3646.3				
	3009.1	1.32 [#] 66	3631.7	5			
	3040.49	13.2 [#] 66	3600.28	4	D+Q	-0.12 3	I_γ : other: 6.4 (1983Wa27). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=+0.32$ 3, $A_4=+0.08$ 3 (1977Da03).
	3095.69	47 [#] 5	3545.08	3	D+Q	+0.04 2	I_γ : other: 36 (1983Wa27). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=-0.25$ 2, $A_4=0.00$ 2 (1977Da03). $A_2=-0.27$ 9, $A_4=0.00$ 11 (1983Wa27); $A_2=-0.25$ 2, $A_4=0.00$ 2 (1977Da03).
	4029.63	15.8 [#] 16	2611.03	3	D(+Q)	+0.02 +1-2	I_γ : other: 8.6 (1983Wa27). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=-0.27$ 3, $A_4=-0.01$ 3 (1977Da03).
	4264.9	15.8 [#] 16	2375.7	4	D+Q	-0.14 4	I_γ : other: 11 (1983Wa27). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=+0.30$ 3, $A_4=+0.06$ 3 (1977Da03).
	4459.51	1.32 [#] 66	2181.10	3			I_γ : other: 1.4 (1983Wa27).
	5975 ^a	<0.68	665.3	1			
	6179 ^a	<0.46	461.2	1 ⁺			
	6493.9	58 [#] 6	146.4	3 ⁺	D(+Q)	+0.02 +1-2	I_γ : other: 59 (1983Wa27). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=-0.28$ 2, $A_4=+0.02$ 2 (1977Da03). $A_2=-0.27$ 9, $A_4=-0.04$ 9 (1983Wa27); $A_2=-0.28$ 2, $A_4=+0.02$ 2 (1977Da03).
	6640 ^a	<0.91	0	0 ⁺			
6695.1	1870.6	50 [#] 5	4824.41				
	1999.3	13 [#] 6	4695.7				
	3048.7	31.3 [#] 31	3646.3				
	3094.7	47 [#] 5	3600.28	4			
	4083.8	72 [#] 7	2611.03	3			
	4319.1	100 [#] 10	2375.7	4			
6702.8	2626.5	2.4 [#] 12	4076.2	4			
	3102.4	4.8 [#] 24	3600.28	4			
	3157.6	3.6 [#] 18	3545.08	3			
	3368.6	2.4 [#] 12	3334.0				
	4326.8	6 [#] 3	2375.7	4			
	6555.7	100 [#] 10	146.4	3 ⁺			
6720.2	2110.4	31.3 [#] 31	4609.7				
	2114.3	22.9 [#] 23	4605.8				
	2737.1	8.3 [#] 42	3983.0	3			
	3336.7	9.4 [#] 48	3383.3				

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$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)						
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
6720.2		4832.5	11.5 [#] 13	1887.29	2 ⁺	
		5489.4	25.0 [#] 25	1230.33	2 ⁺	
		6573.1	100 [#] 10	146.4	3 ⁺	
6723.7		1899.2	100	4824.41		I_γ : other: 100 11 (1977Da02).
		2027.9	20	4695.7		I_γ : other: 18 9 (1977Da02).
		2759.5	4.8	3964.1		
		3077.3	33	3646.3		I_γ : other: 31.6 32 (1977Da02).
		3340.2	1.6	3383.3		
		4112.4	72	2611.03	3	I_γ : other: 91 9 (1977Da02).
		4347.7	92	2375.7	4	I_γ : other: 118 12 (1977Da02).
		4542.3	5.6	2181.10	3	I_γ : other: 9 5 (1977Da02).
		4565.5	7.6	2157.9	2 ⁺	I_γ : other: 14 7 (1977Da02).
		6058 ^a	<1.2	665.3	1	
		6262 ^a	<2	461.2	1 ⁺	
		6576.6	64	146.4	3 ⁺	I_γ : other: 73 7 (1977Da02).
		6723 ^a	<0.8	0	0 ⁺	
6737.0		3607.7	5.1 [#] 26	3129.12		I_γ : other: 6.7 (1983Wa27).
		4578.8	100 [#] 10	2157.9	2 ⁺	I_γ : other: 100 (1983Wa27).
		4849.3	3.6	1887.29	2 ⁺	
		5506.2	12.8 [#] 64	1230.33	2 ⁺	I_γ : other: 12 (1983Wa27).
		6071.1	36 [#] 4	665.3	1	I_γ : other: 42 (1983Wa27).
		6275.2	28.2 [#] 28	461.2	1 ⁺	I_γ : other: 31 (1983Wa27).
		6589.9	74 [#] 8	146.4	3 ⁺	I_γ : other: 83 (1983Wa27).
		6736 ^a	<1.7	0	0 ⁺	
6790.8		4903.1	5.3 [#] 26	1887.29	2 ⁺	I_γ : other: <3 (1983Wa27).
		6125 ^a	<3	665.3	1	
		6329 ^a	<4	461.2	1 ⁺	
		6644 ^a	<6	146.4	3 ⁺	
6797.4	1 to 5	6790.1	100 [#] 10	0	0 ⁺	I_γ : other: 100 (1983Wa27).
		2833.2	1.10 [#] 55	3964.1		I_γ : other: 1.1 (1983Wa27).
		3413.9	0.55 [#] 28	3383.3		
		4186.1	2.2 [#] 11	2611.03	3	I_γ : other: 2.2 (1983Wa27).
		4421.4	1.10 [#] 55	2375.7	4	I_γ : other: 1.1 (1983Wa27).
		4639.2	0.55 [#] 28	2157.9	2 ⁺	I_γ : other: 0.44 (1983Wa27).
		4909.7	4.4 [#] 22	1887.29	2 ⁺	I_γ : other: 3.9 (1983Wa27).
		6132 ^a	<0.22	665.3	1	
		6336 ^a	<0.33	461.2	1 ⁺	
		6650.3	100 [#] 10	146.4	3 ⁺	I_γ : other: 100 (1983Wa27).
						$A_2=+0.16$ 5, $A_4=-0.04$ 7 (1983Wa27).
		6797 ^a	<0.44	0	0 ⁺	
6807.2		3423.7	12.7 [#] 13	3383.3		
		4226.6	16.9 [#] 17	2580.3		
		4649.0	11.3 [#] 11	2157.9	2 ⁺	
6828.6		6141.3	100 [#] 10	665.3	1	
		3283.4	7.3	3545.08	3	
		3445.1	15.5 [#] 16	3383.3		I_γ : other: 14 (1983Wa27).
		4670.4	100 [#] 10	2157.9	2 ⁺	I_γ : other: 100 (1983Wa27).
		5598 ^a	<1.2	1230.33	2 ⁺	

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
6828.6		6162.7	5.2 [#] 26	665.3	1			I_γ : other: 12 (1983Wa27).
		6366.8	19.0 [#] 19	461.2	1 ⁺			I_γ : other: 33 (1983Wa27).
		6681.5	32.8 [#] 33	146.4	3 ⁺			I_γ : other: 27 (1983Wa27).
		6828 ^a	<1.4	0	0 ⁺			
6842.8		2233 ^a	<10 [#]	4609.7				I_γ : other: 6.9 (1983Wa27).
		2236.9	5.0 [#] 25	4605.8				I_γ : other: 18 (1983Wa27).
		2327.0	25 [#] 13	4515.7				I_γ : other: 35 (1983Wa27).
		2488.5	45 [#] 5	4354.2	(1)			I_γ : other: 49 (1983Wa27).
		2694.9	34	4147.8				
		2703.0	16	4139.7	2			
		2859.7	48 [#] 5	3983.0	3			I_γ : other: 54 (1983Wa27).
		3068.9	6.3	3773.72	1			
		3459.3	23 [#] 12	3383.3				I_γ : other: 18 (1983Wa27).
		3508.6	10 [#] 5	3334.0				I_γ : other: 11 (1983Wa27).
		3713.5	9.4	3129.12				
		4231.5	12.5 [#] 65	2611.03	3			I_γ : other: 16 (1983Wa27).
		4262.2	45 [#] 5	2580.3				I_γ : other: 46 (1983Wa27).
		4661.4	6.9	2181.10	3			
		4684.6	30 [#] 3	2157.9	2 ⁺			I_γ : other: 29 (1983Wa27).
		4955.1	35.0 [#] 35	1887.29	2 ⁺			I_γ : other: 38 (1983Wa27).
		5612.0	65 [#] 7	1230.33	2 ⁺			I_γ : other: 75 (1983Wa27).
		6176.9	35.0 [#] 35	665.3	1			I_γ : other: 44 (1983Wa27).
		6381.0	100 [#] 10	461.2	1 ⁺			I_γ : other: 100 (1983Wa27).
		6695.7	12.5 [#] 65	146.4	3 ⁺			I_γ : other: 13 (1983Wa27).
		6842 ^a	<1.9	0	0 ⁺			
6853.0		1857.4	2.5 [#] 13	4995.6				I_γ : other: 2.3 (1983Wa27).
		2157.2	2.5 [#] 13	4695.7				I_γ : other: 2.3 (1983Wa27).
		2498.7	2.5 [#] 13	4354.2	(1)			
		3307.8	5.1	3545.08	3			
		3723.7	7.5 [#] 38	3129.12				I_γ : other: 8.2 (1983Wa27).
		4272.4	7.5 [#] 38	2580.3				I_γ : other: 9 (1983Wa27).
		4965.3	58 [#] 6	1887.29	2 ⁺			I_γ : other: 56 (1983Wa27).
		5622.2	32.5 [#] 33	1230.33	2 ⁺			I_γ : other: 33 (1983Wa27).
		6187.1	15.0 [#] 15	665.3	1			I_γ : other: 18 (1983Wa27).
		6391.2	100 [#] 10	461.2	1 ⁺			I_γ : other: 100 (1983Wa27).
		6705.9	10 [#] 5	146.4	3 ⁺			I_γ : other: 10 (1983Wa27).
		6852.3	13 [#] 6	0	0 ⁺			I_γ : other: 12 (1983Wa27).
6871.18	5	1330.4	13.6 [#] 14	5540.8				I_γ : other: 10 (1983Wa27). $A_2=+0.49$ 9, $A_4=-0.17$ 9 (1977Da03).
		1484.4	6.8 [#] 34	5386.8				I_γ : other: 2.8 (1983Wa27).
		1699.5	25.0 [#] 25	5171.6	4	D(+Q)	+0.02 3	I_γ : other: 24 (1983Wa27). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=-0.24$ 4, $A_4=+0.02$ 4 (1977Da03).
		2424.5	64 [#] 7	4446.6	4	D+Q	+0.05 +I-3	I_γ : other: 61 (1983Wa27). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=-0.19$ 3, $A_4=0.00$ 3 (1977Da03).
		2794.9	5.7	4076.2	4			

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. &	$\delta^\&$	Comments
6871.18	5	3239.3	100 [#] 10	3631.7	5	D(+Q)	+0.04 7	I_γ : other: 100 (1983Wa27). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=+0.41$ 3, $A_4=+0.05$ 4 (1977Da03).
		4495.2	18.2 [#] 18	2375.7	4	D(+Q)	+0.04 +3-5	I_γ : other: 14 (1983Wa27). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=-0.21$ 7, $A_4=-0.06$ 7 (1977Da03).
		4713 ^a	<0.87	2157.9	2 ⁺			
		6205 ^a	<1.3	665.3	1			
		6409 ^a	<1.5	461.2	1 ⁺			
		6724 ^a	<1.3	146.4	3 ⁺			
		6870 ^a	<1.3	0	0 ⁺			
6887.97	2	1930.6	1.16 [#] 58	4957.3				I_γ : other: 1.2 (1983Wa27).
		2170.5	2.3 [#] 12	4717.4				I_γ : other: 4.3 (1983Wa27).
		2278.2	1.16 [#] 58	4609.7				
		2282.1	1.16 [#] 58	4605.8				I_γ : other: 1.3 (1983Wa27).
		3504.5	2.3 [#] 12	3383.3				I_γ : other: 1.7 (1983Wa27).
		4706.52	1.16 [#] 58	2181.10	3			I_γ : other: 3.5 (1983Wa27).
		4729.7	100 [#] 10	2157.9	2 ⁺	D+Q	-0.52 +3-2	I_γ : other: 100 (1983Wa27). Mult., δ : from $\gamma(\theta)$ in 1977Da03. $A_2=-0.13$ 2, $A_4=+0.01$ 3 (1977Da03).
		5000.29	1.16 [#] 58	1887.29	2 ⁺			I_γ : other: 1.7 (1983Wa27).
		6222 ^a	<0.37	665.3	1			
		6426 ^a	<0.37	461.2	1 ⁺			
		6740.9	5.8 [#] 29	146.4	3 ⁺			I_γ : other: 8.2 (1983Wa27).
		6887 ^a	<0.49	0	0 ⁺			
6901.75		3518.3	21.1 [#] 21	3383.3				I_γ : other: 20 (1983Wa27).
		4743.5	100 [#] 10	2157.9	2 ⁺			I_γ : other: 100 (1983Wa27).
		5670.91	0.81	1230.33	2 ⁺			
		6236 ^a	<0.27	665.3	1			
		6439.9	1.2	461.2	1 ⁺			
		6754.6	2.6 [#] 13	146.4	3 ⁺			I_γ : other: 2.8 (1983Wa27).
		6901.00	7.9 [#] 8	0	0 ⁺			I_γ : other: 10 (1983Wa27).
6918.0	3,4	1377.2	0.48	5540.8				
		3271.5	2.4 [#] 13	3646.3				I_γ : other: 1.9 (1983Wa27).
		3286.1	3.2 [#] 16	3631.7	5			I_γ : other: 3.6 (1983Wa27).
		3372.7	0.81	3545.08	3			
		3583.8	2.4 [#] 13	3334.0				I_γ : other: 2.3 (1983Wa27).
		4306.7	4.0 [#] 21	2611.03	3			I_γ : other: 3.9 (1983Wa27).
		4542.0	40 [#] 4	2375.7	4			I_γ : other: 40 (1983Wa27). $A_2=+0.50$ 4, $A_4=+0.04$ 5 (1983Wa27).
		4736.6	8.9 [#] 10	2181.10	3			I_γ : other: 8.1 (1983Wa27).
		6252 ^a	<0.16	665.3	1			
		6456 ^a	<0.32	461.2	1 ⁺			
		6770.9	100 [#] 10	146.4	3 ⁺			I_γ : other: 100 (1983Wa27). $A_2=+0.06$ 2, $A_4=-0.04$ 2 (1983Wa27).
		6917 ^a	<0.65	0	0 ⁺			
6931.5		2214.0	24	4717.4				I_γ : other: 14 7 (1977Da02).
		2321.7	100	4609.7				I_γ : other: 100 10 (1977Da02).
		2325.6	64	4605.8				I_γ : other: 110 11 (1977Da02).

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued) $\gamma(^{34}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
6931.5		2415.7	8.2	4515.7		I_γ : other: 4.8 24 (1977Da02).
		2470.0	9.1	4461.4		I_γ : other: <9.6 (1977Da02).
		2577.2	9.1	4354.2	(1)	I_γ : other: 10 5 (1977Da02).
		2948.4	15	3983.0	3	I_γ : other: 19 10 (1977Da02).
		3331.0	4.6	3600.28	4	I_γ : other: 10 5 (1977Da02).
		3386.2	21	3545.08	3	I_γ : other: 14 7 (1977Da02).
		3548.0	55	3383.3		I_γ : other: 67 7 (1977Da02).
		4209.9	13	2721.3	2	I_γ : other: 14 7 (1977Da02).
		4350.9	19	2580.3		I_γ : other: 10 5 (1977Da02).
		4773.2	19	2157.9	2 ⁺	I_γ : other: 10 5 (1977Da02).
		5043.8	10 [#] 5	1887.29	2 ⁺	
		6265.6	46	665.3	1	I_γ : other: 19 10 (1977Da02).
		6470 ^a	<9.1	461.2	1 ⁺	
		6784.4	50	146.4	3 ⁺	I_γ : other: 67 7 (1977Da02).
		6931 ^a	<6.8	0	0 ⁺	
6977.4		3593.9	11.2 [#] 12	3383.3		
		4819.1	100 [#] 10	2157.9	2 ⁺	
		5746.6	8.2 [#] 9	1230.33	2 ⁺	
		6830.3	29.9 [#] 30	146.4	3 ⁺	
6986.9		2381 ^a	<4.9 [#]	4605.8		
		3003.8	18.3 [#] 20	3983.0	3	
		3386.4	18.3 [#] 20	3600.28	4	
		3441.6	19.5 [#] 20	3545.08	3	
		4265.3	11.0 [#] 56	2721.3	2	
		4828.6	100 [#] 10	2157.9	2 ⁺	
		5756.1	59 [#] 6	1230.33	2 ⁺	
		6839.8	18.3 [#] 20	146.4	3 ⁺	
7051.6		2094.2	39 [#] 4	4957.3		
		3506.3	17.9 [#] 89	3545.08	3	
		3668.1	36 [#] 4	3383.3		
		4893.3	93 [#] 9	2157.9	2 ⁺	
		6385.7	71 [#] 7	665.3	1	
		6589.7	100 [#] 10	461.2	1 ⁺	
7059.04	1,2	2063.4	2.6 [#] 14	4995.6		I_γ : other: 2.2 (1983Wa27).
		2117.1	1.75 [#] 88	4941.9		I_γ : other: 2.1 (1983Wa27).
		2363.3	0.35	4695.7		
		2449.3	0.86	4609.7		
		2733.03	0.88 [#] 44	4325.89		I_γ : other: 0.69 (1983Wa27).
		3285 ^a	<0.88 [#]	3773.72	1	I_γ : other: 0.17 (1983Wa27).
		3398.9	1.75 [#] 88	3660.0		I_γ : other: 1.4 (1983Wa27).
		3929.68	4.4 [#] 23	3129.12		I_γ : other: 5.3 (1983Wa27).
		4337.4	1.75 [#] 88	2721.3	2	I_γ : other: 1.2 (1983Wa27).
		4447.70	1.2	2611.03	3	
		4478.4	0.88 [#] 44	2580.3		I_γ : other: 1.4 (1983Wa27).
		4877.56	2.6 [#] 14	2181.10	3	I_γ : other: 1.4 (1983Wa27).
		4900.8	0.88 [#] 44	2157.9	2 ⁺	I_γ : other: 0.86 (1983Wa27).
		5171.33	8.8 [#] 44	1887.29	2 ⁺	I_γ : other: 7.1 (1983Wa27).

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p},\gamma)$ **1983Wa27,1977Da02,1992Ka39** (continued)

$\gamma(^{34}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. ^{&}	$\delta^\&$	Comments
7059.04	1,2	5828.17	5.3 [#] 26	1230.33	2 ⁺			I_γ : other: 5.7 (1983Wa27).
		6393.1	44 [#] 5	665.3	1			I_γ : other: 40 (1983Wa27).
								$A_2=+0.03$ 6, $A_4=-0.13$ 7 (1983Wa27).
		6597.2	100 [#] 10	461.2	1 ⁺			I_γ : other: 100 (1983Wa27).
								$A_2=+0.04$ 2, $A_4=-0.03$ 3 (1983Wa27).
7078.89	3	6912 ^a	<1.8 [#]	146.4	3 ⁺			I_γ : other: 0.86 (1983Wa27).
		7058 ^a	<0.35	0	0 ⁺			
		2473.0	6.3	4605.8				
		2752.88	2.2	4325.89				
		3002.6	3.6 [#] 18	4076.2	4			
		3095.7	57 [#] 6	3983.0	3	D(+Q)	0.00 +6-4	I_γ : other: 59 (1983Wa27).
								Mult., δ : from $\gamma(\theta)$ in 1977Da03.
								$A_2=+0.28$ 3, $A_4=+0.01$ 3 (1977Da03).
		3478.42	61 [#] 6	3600.28	4	D(+Q)	0.00 3	I_γ : other: 56 (1983Wa27).
								Mult., δ : from $\gamma(\theta)$ in 1977Da03.
								$A_2=-0.09$ 3, $A_4=+0.01$ 3 (1977Da03).
		3533.61	54 [#] 6	3545.08	3	D(+Q)	-0.07 +7-4	I_γ : other: 56 (1983Wa27).
								Mult., δ : from $\gamma(\theta)$ in 1977Da03.
								$A_2=+0.24$ 3, $A_4=+0.04$ 4 (1977Da03).
		3745 ^a	<3.6 [#]	3334.0				
		4357.3	39 [#] 4	2721.3	2	D(+Q)	-0.03 +2-3	I_γ : other: 41 (1983Wa27).
								Mult., δ : or $\delta=-3.5$ 5 from $\gamma(\theta)$ in 1977Da03.
								$A_2=-0.26$ 3, $A_4=0.00$ 3 (1977Da03).
		4897.41	3.6 [#] 18	2181.10	3			I_γ : other: 8.5 (1983Wa27).
		5191.18	10.7 [#] 54	1887.29	2 ⁺			I_γ : other: 8.9 (1983Wa27).
		5848.02	100 [#] 10	1230.33	2 ⁺	D(+Q)	-0.04 +4-2	I_γ : other: 100 (1983Wa27).
								Mult., δ : from $\gamma(\theta)$ in 1977Da03.
								$A_2=-0.25$ 3, $A_4=+0.02$ 3 (1977Da03).
		6413 ^a	<2.2	665.3	1			
		6617 ^a	<2.2	461.2	1 ⁺			
		6931.7	28.6 [#] 29	146.4	3 ⁺	D(+Q)	-0.05 7	I_γ : other: 33 (1983Wa27).
								Mult., δ : from $\gamma(\theta)$ in 1977Da03.
								$A_2=+0.24$ 4, $A_4=0.00$ 5 (1977Da03).
7675.8	2,3	7078 ^a	<3.0	0	0 ⁺			
		5517.4		2157.9	2 ⁺	D+Q	0.12 8	Mult., δ : from $\gamma(\theta)$ in 1992Ka39 (sign of δ is not given).
								$A_2=-0.14$ 9, $A_4=-0.18$ 12, $A_6=-0.005$ 12 (1992Ka39).
		7528.5		146.4	3 ⁺	D+Q	0.32 25	Mult., δ : from $\gamma(\theta)$ in 1992Ka39 (sign of δ is not given).
								$A_2=-0.51$ 7, $A_4=+0.23$ 7, $A_6=-0.08$ 8 (1992Ka39).
8305.5	2,3	8158.1	100	146.4	3 ⁺	D+Q	0.27 18	I_γ : from 1992Ka39.
								Mult., δ : from $\gamma(\theta)$ in 1992Ka39 (sign of δ is not given).
								$A_2=-0.41$ 5, $A_4=-0.12$ 6, $A_6=+0.05$ 6 (1992Ka39).

[†] Deduced by the evaluators from level-energy differences. E_γ values are not explicitly listed in the references in this dataset. γ rays with I_γ values reported only as upper limits in the literature are considered questionable if there are no supporting

$^{33}\text{S}(\text{p},\gamma)$ [1983Wa27](#),[1977Da02](#),[1992Ka39](#) (continued)

$\gamma(^{34}\text{Cl})$ (continued)

observations. Their E_γ values are rounded to the nearest keV.

‡ From [1983Wa27](#), unless otherwise noted.

From [1977Da02](#).

@ From [2011Fr04](#).

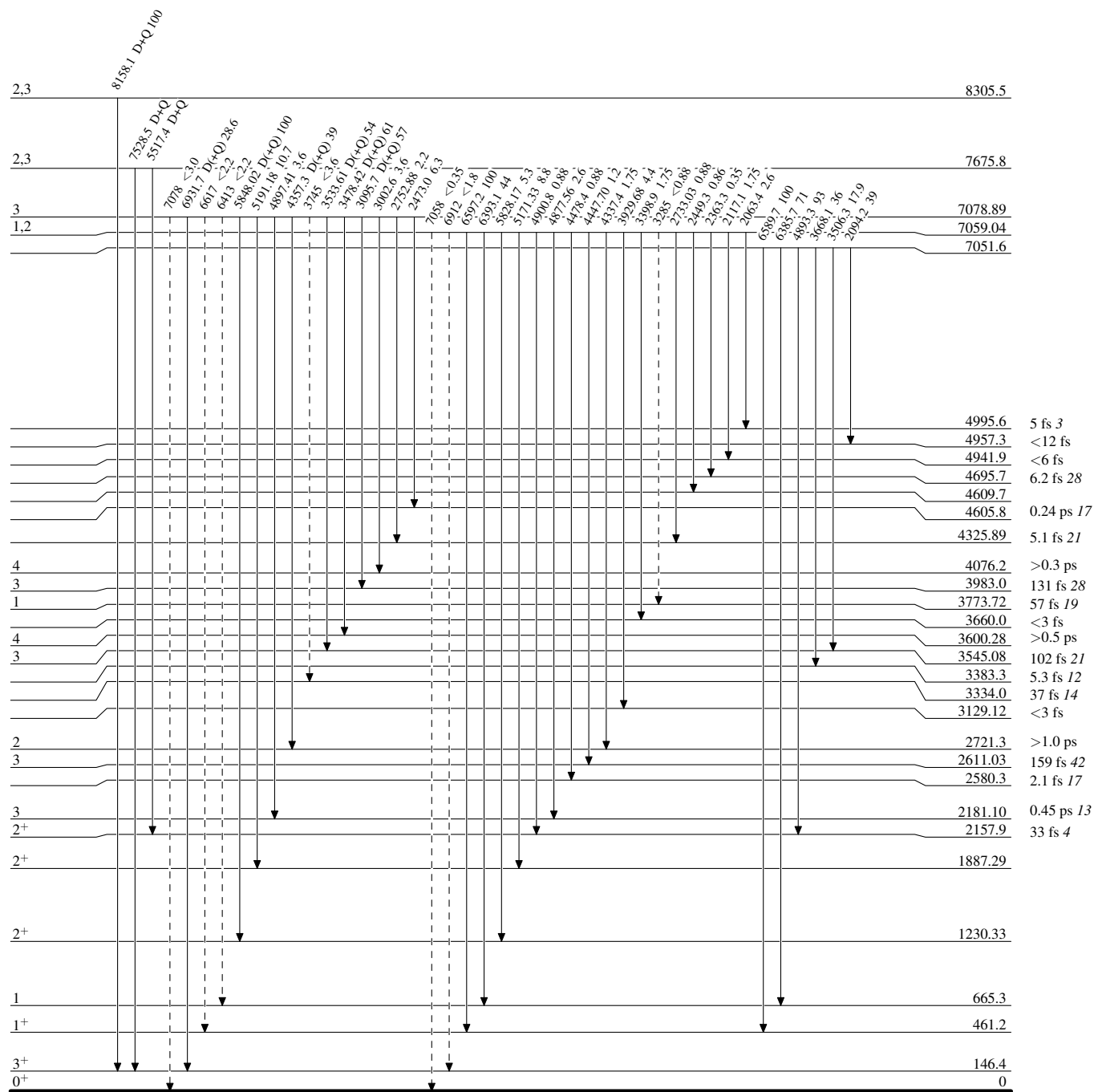
& From $\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ data given under comments.

^a Placement of transition in the level scheme is uncertain.

Legend

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)

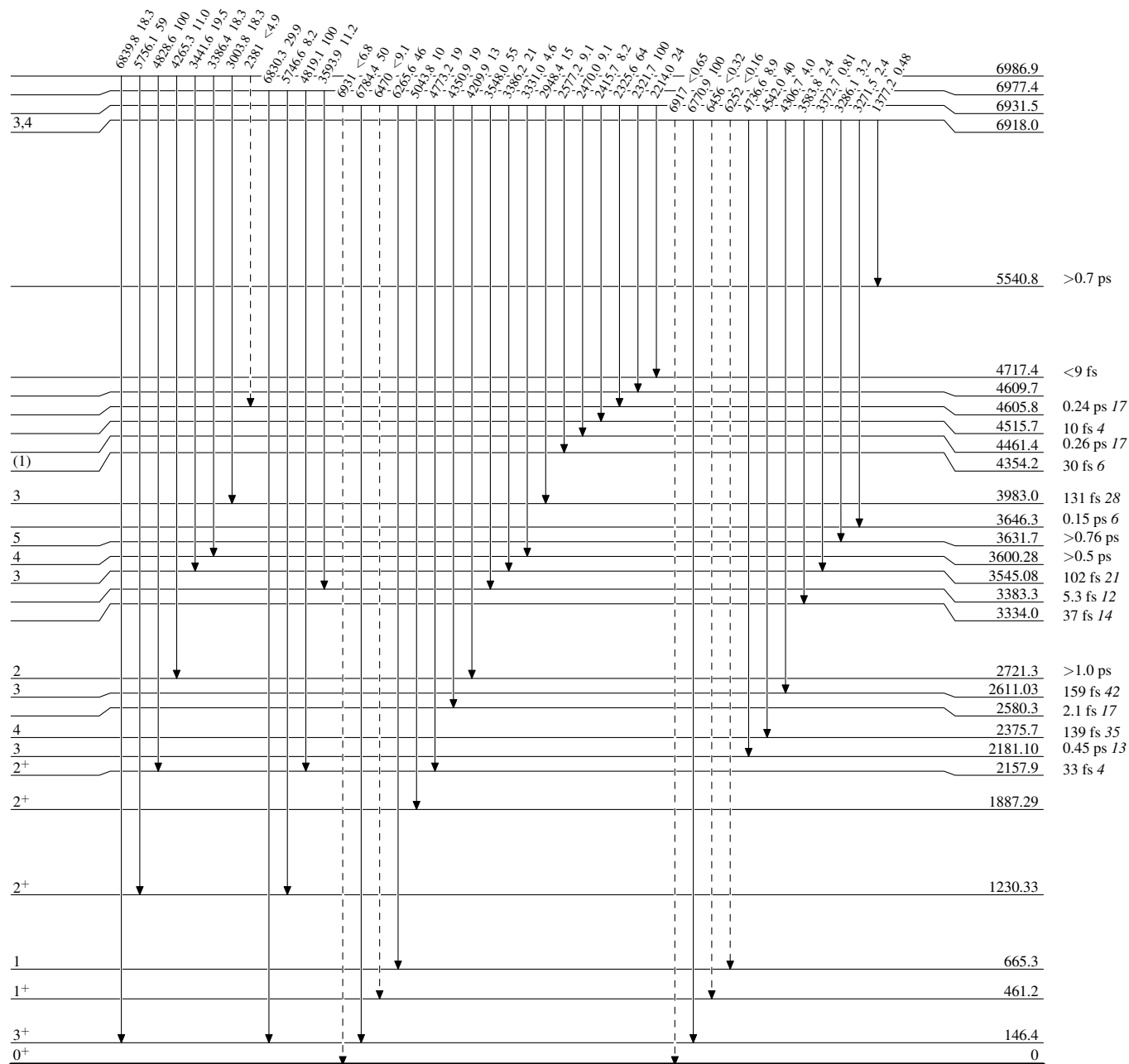
 ${}^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

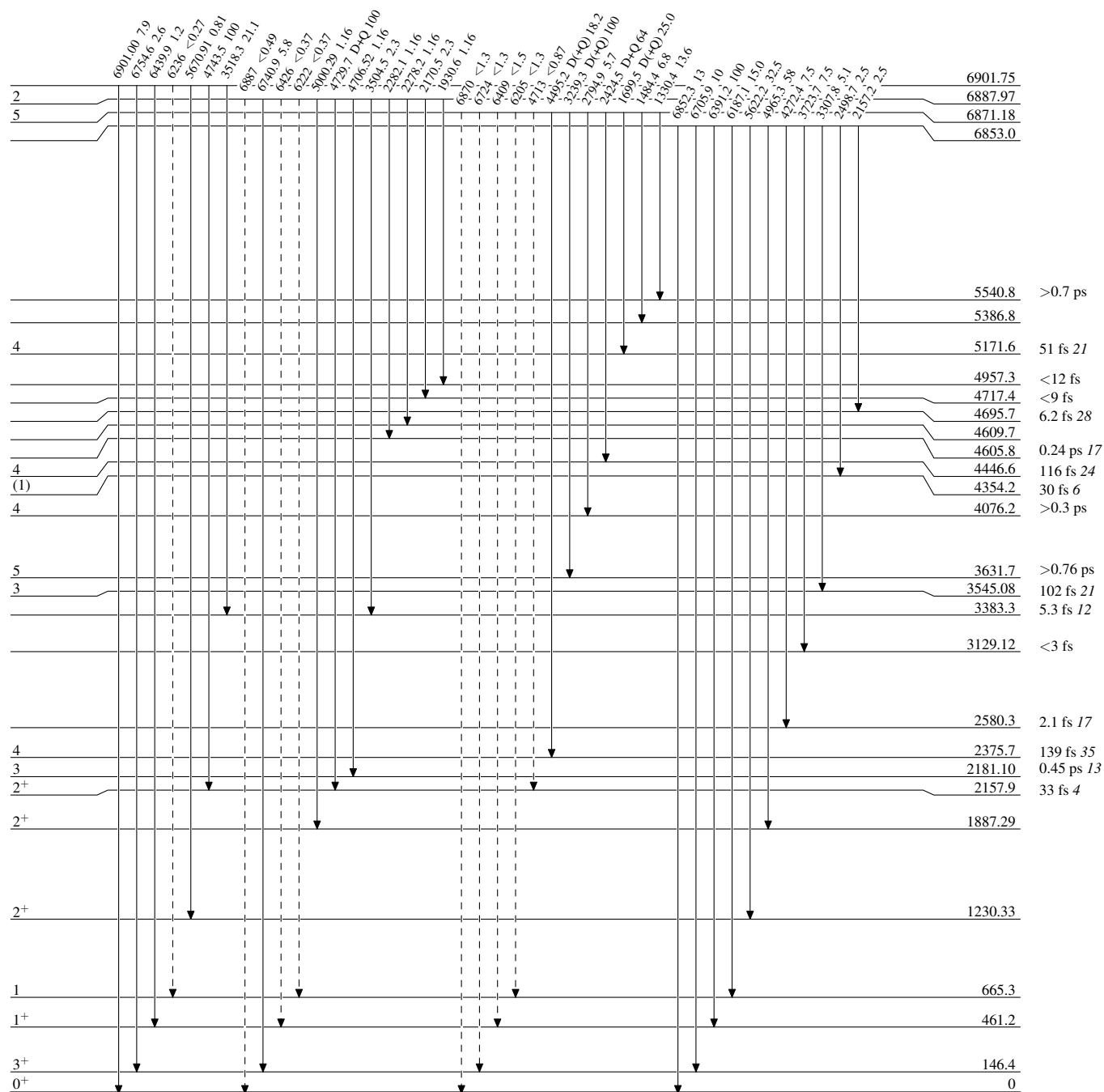
Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

Legend

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)

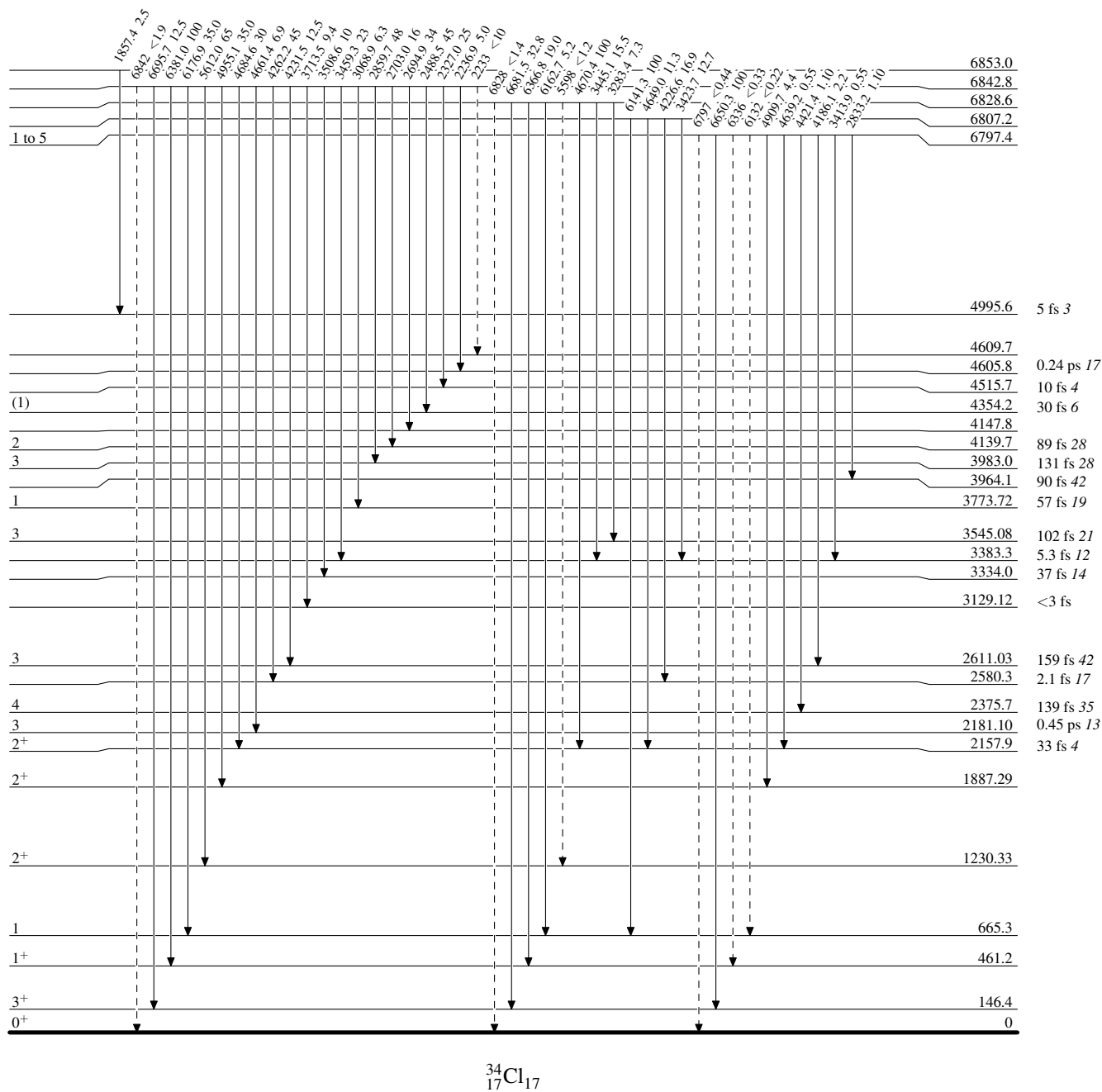
 ${}^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

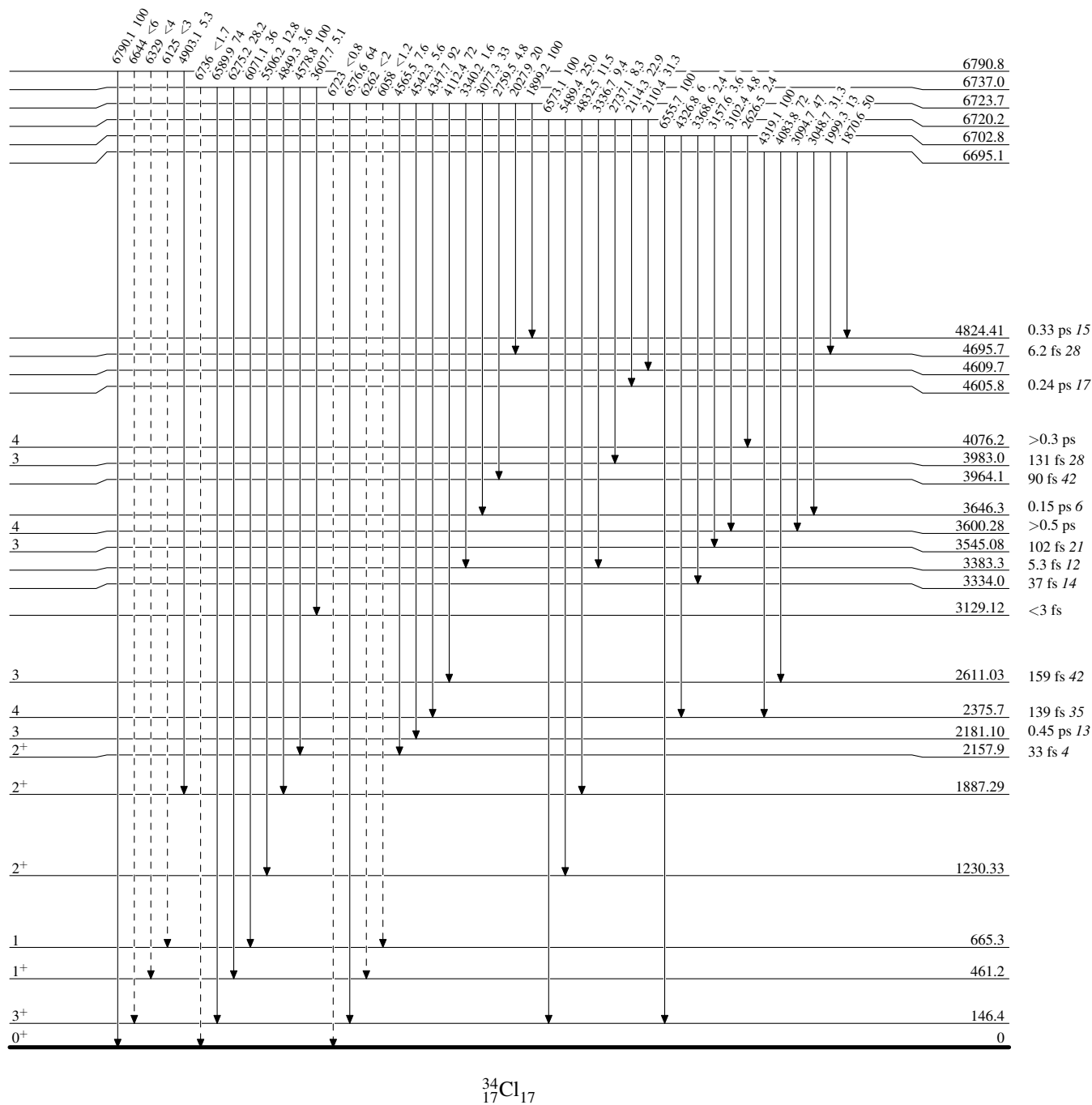
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

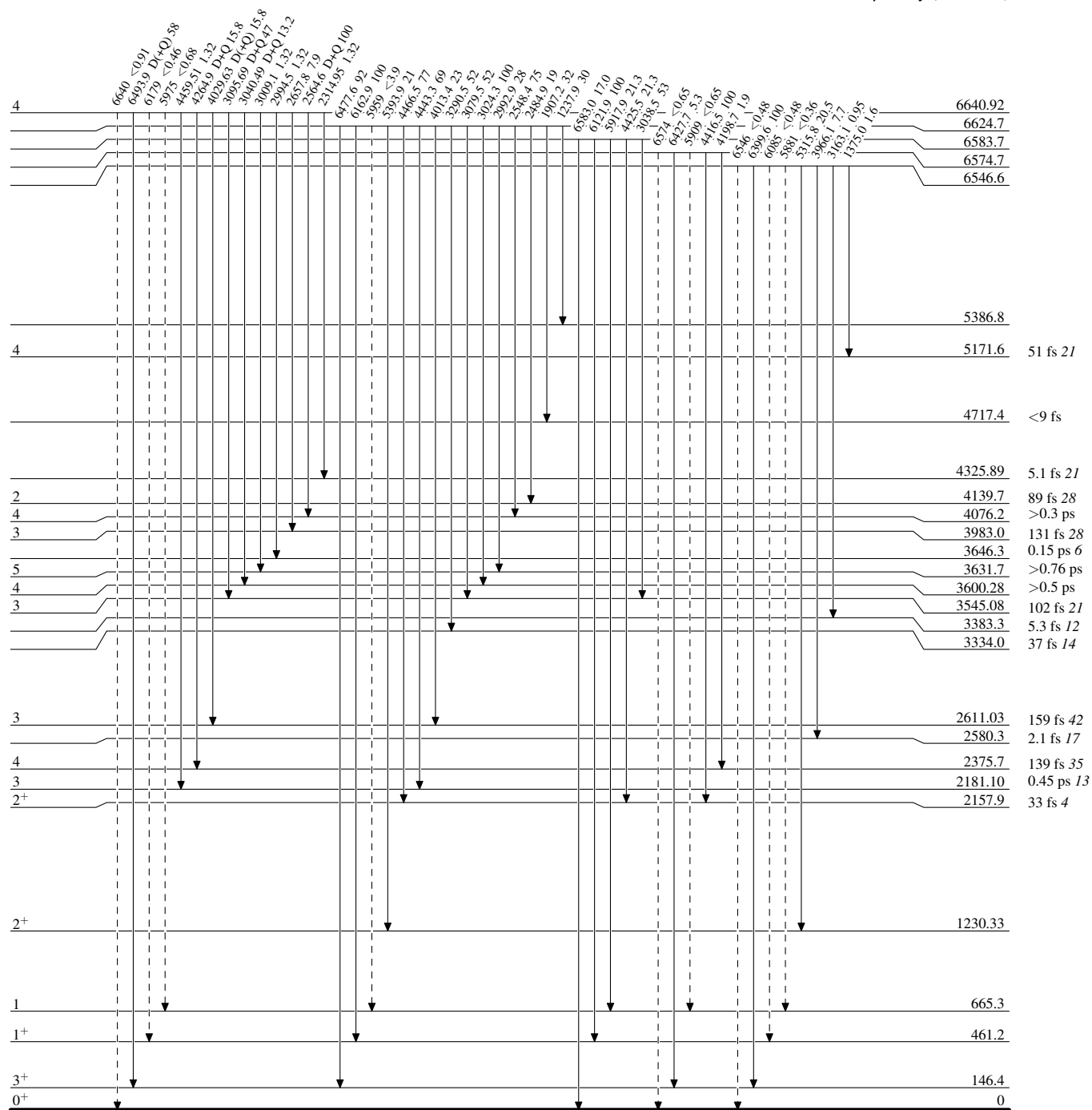
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

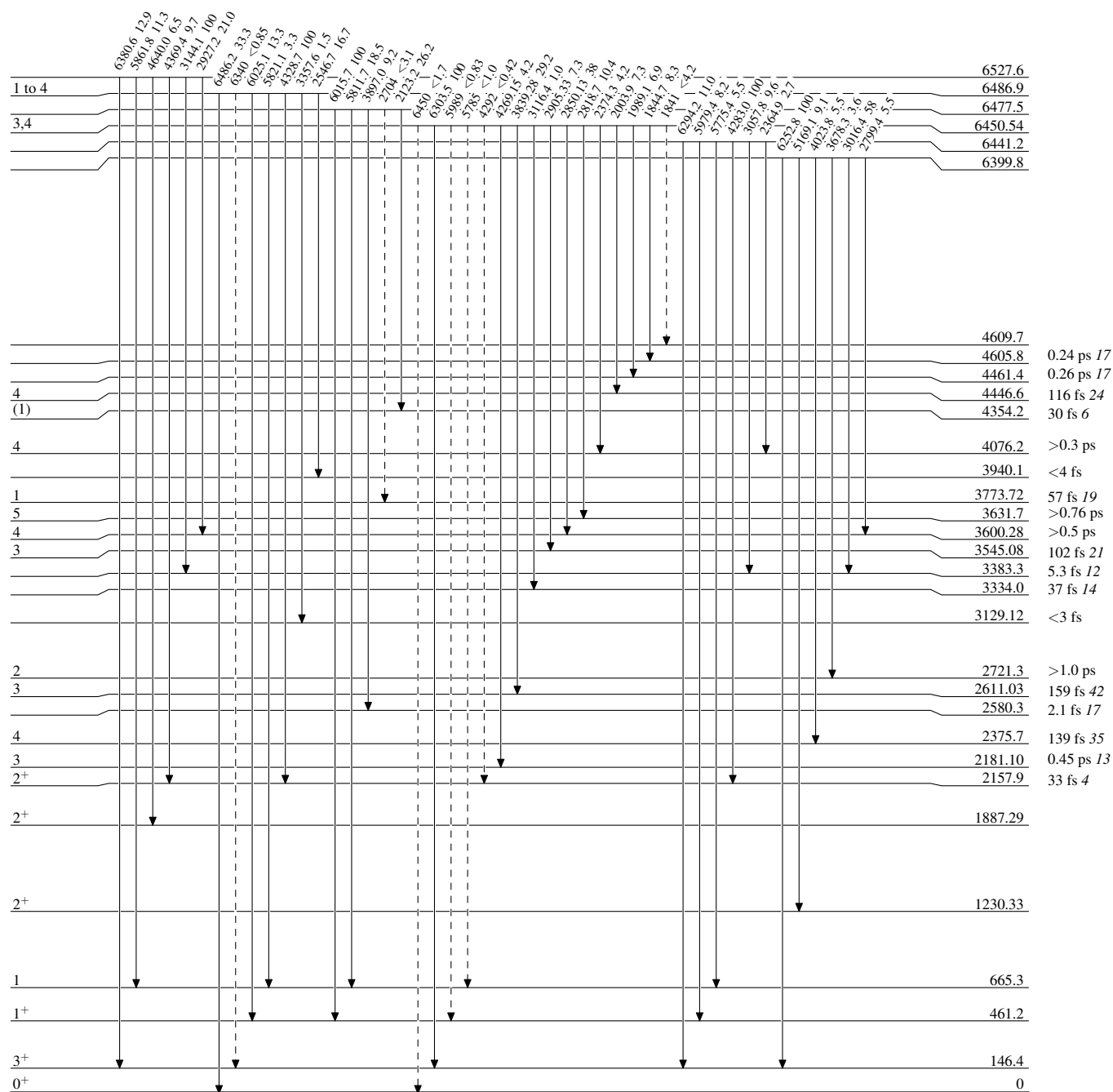
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

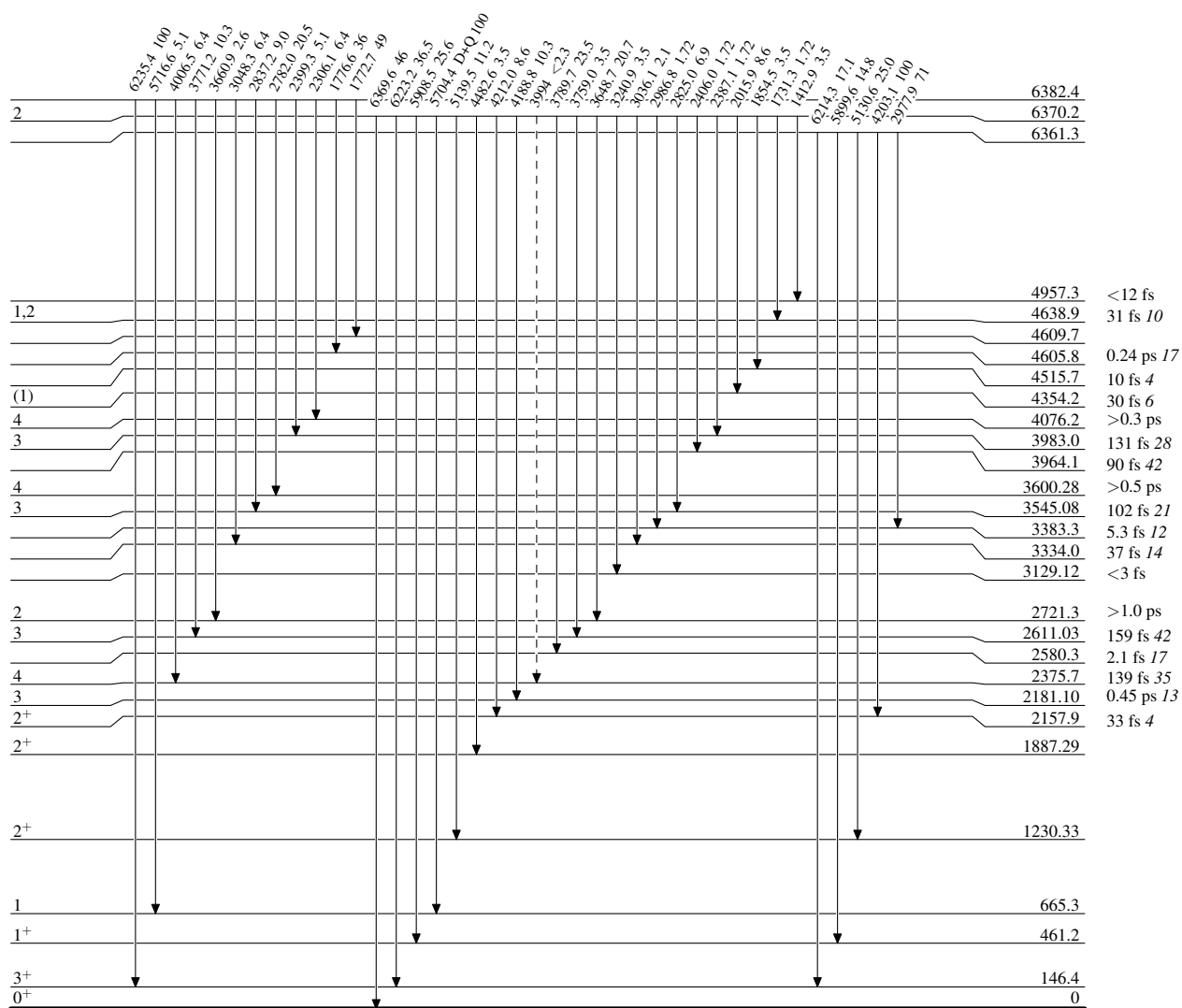
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

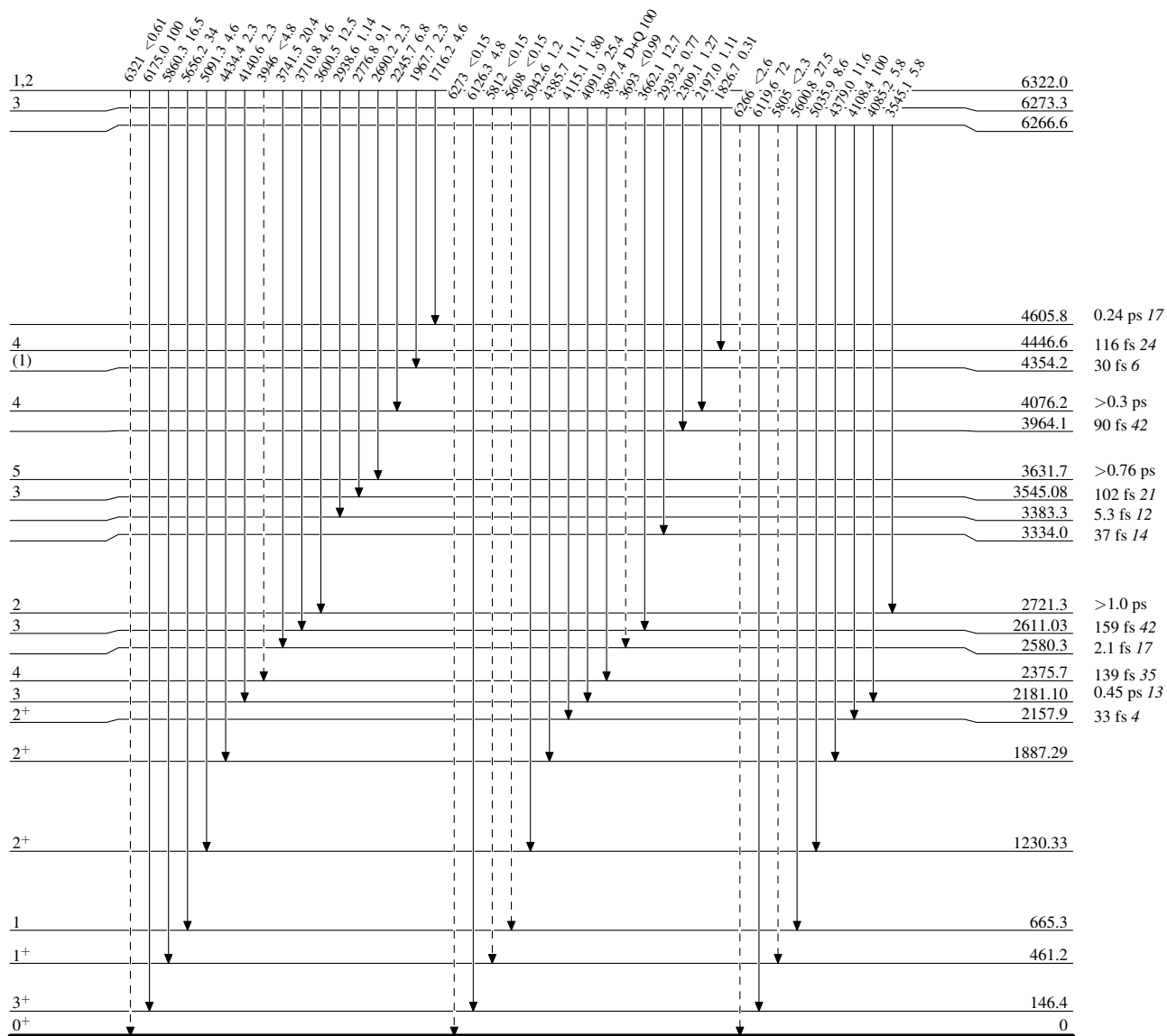
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

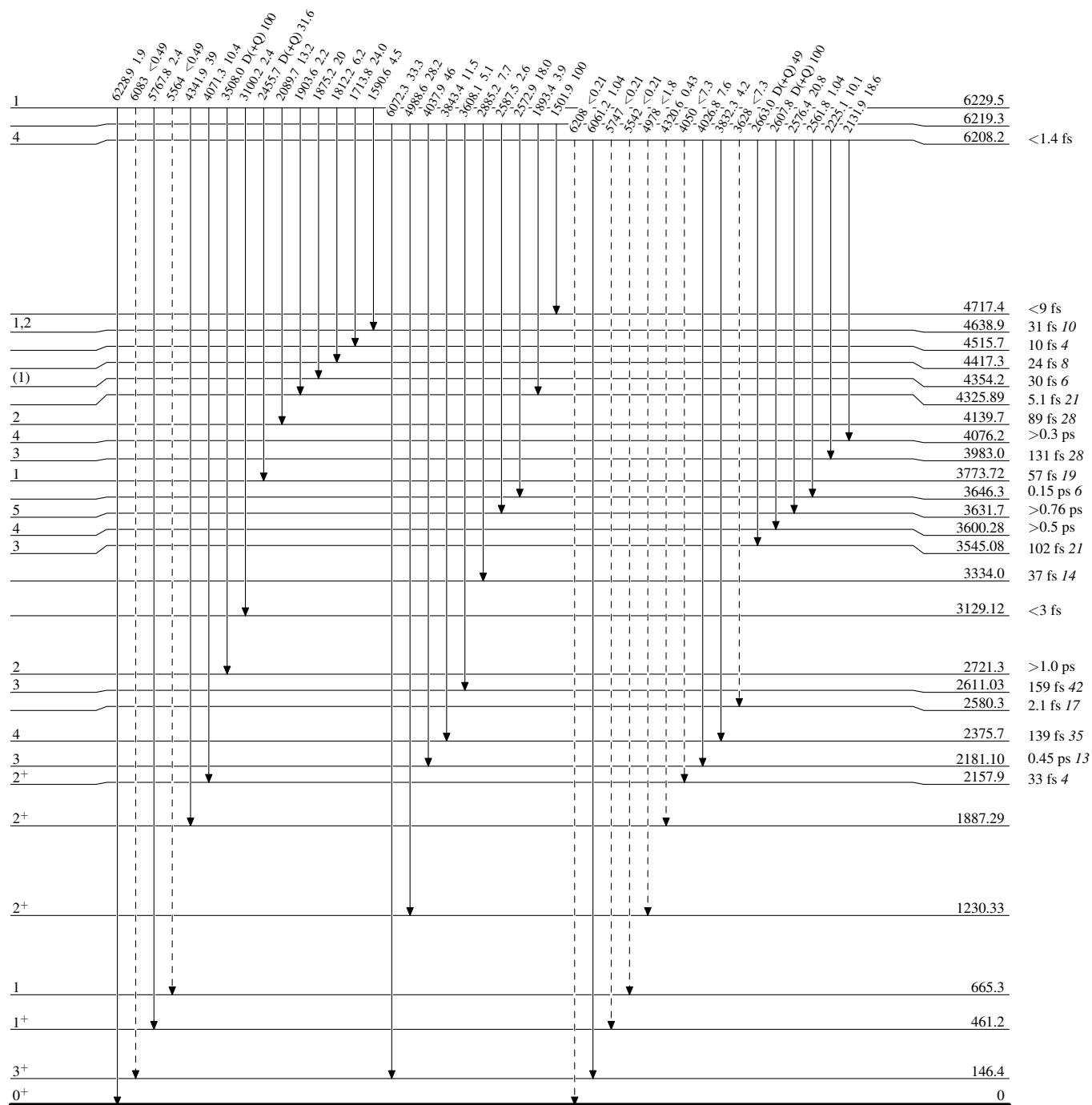
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

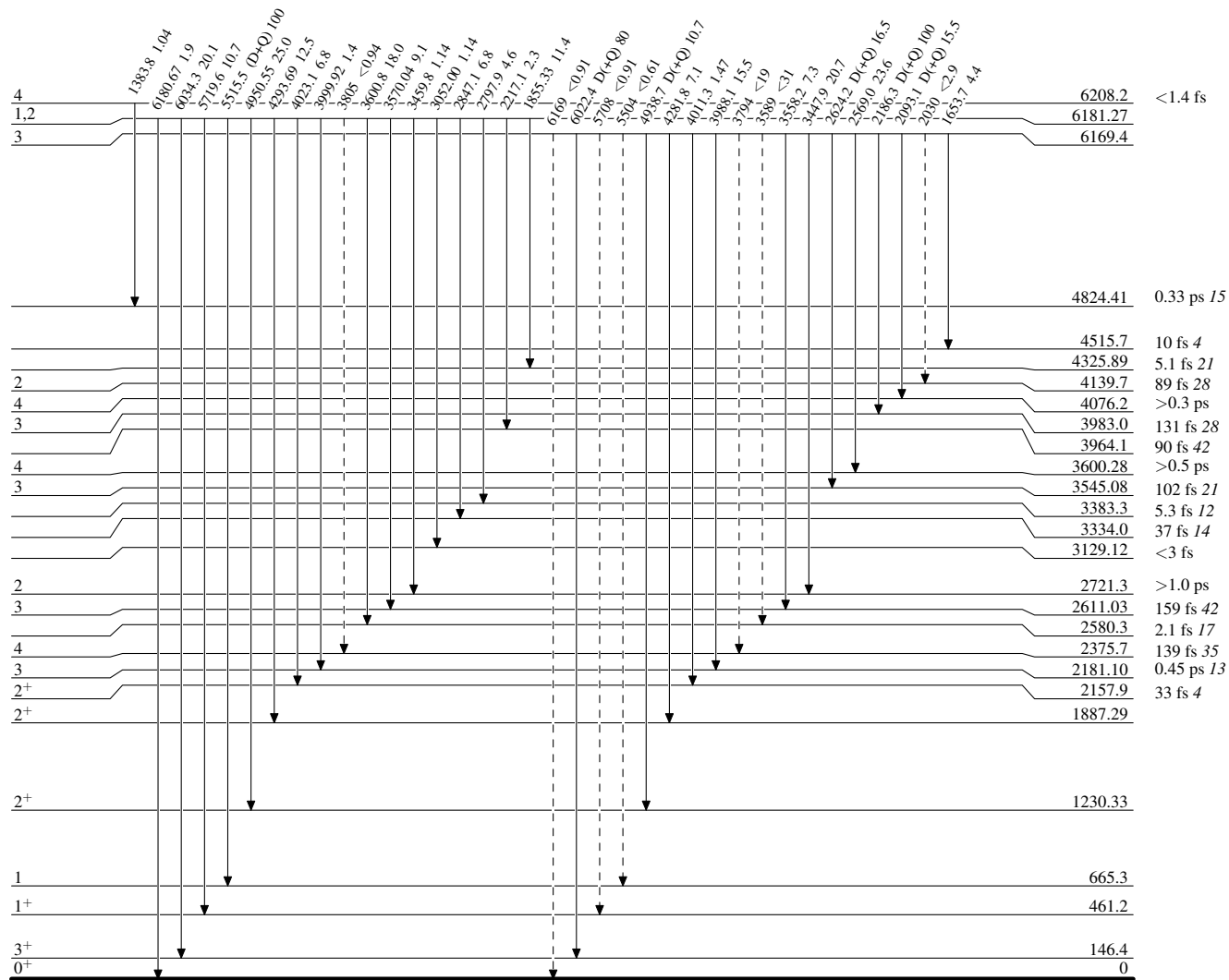
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

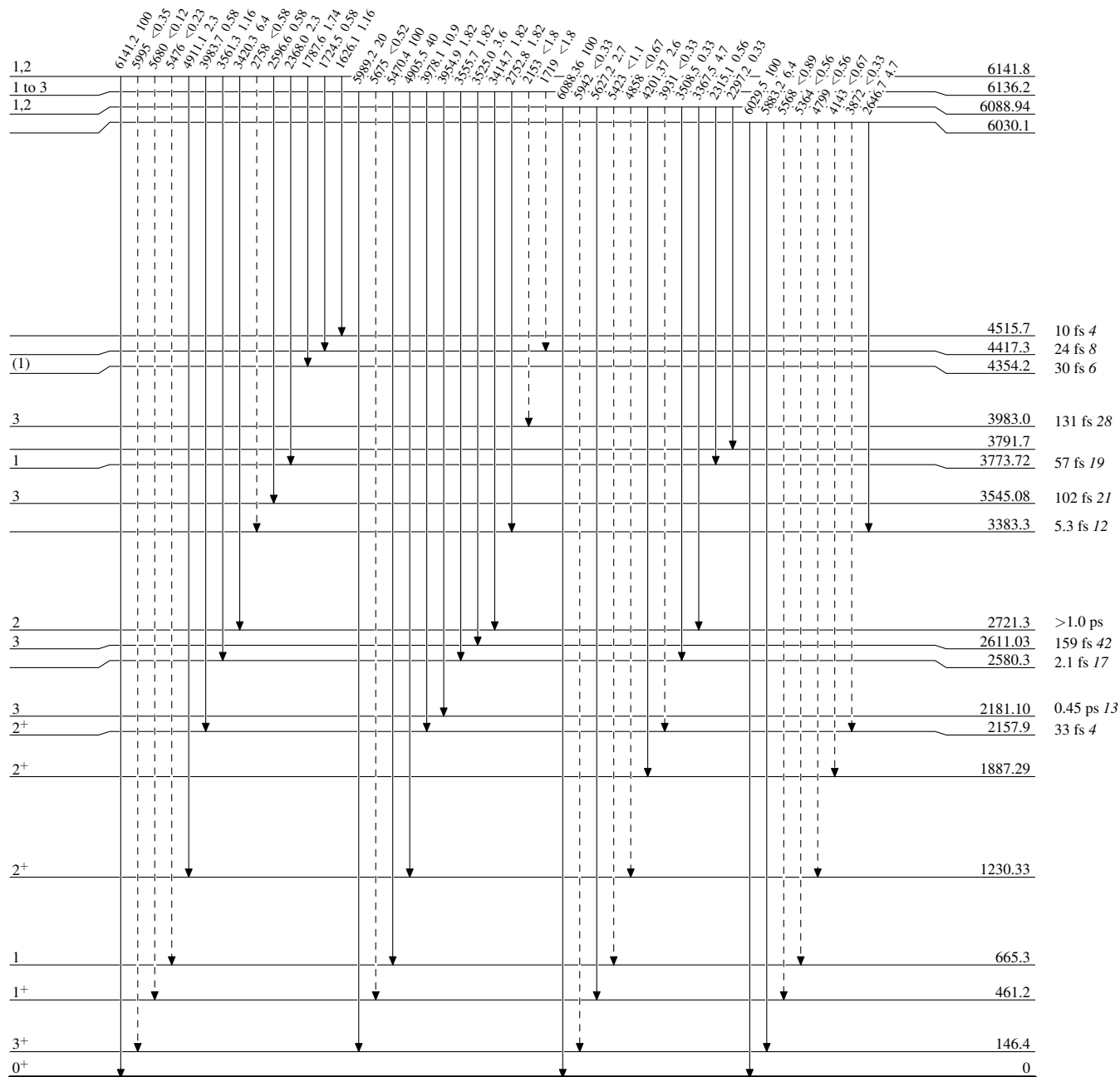
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

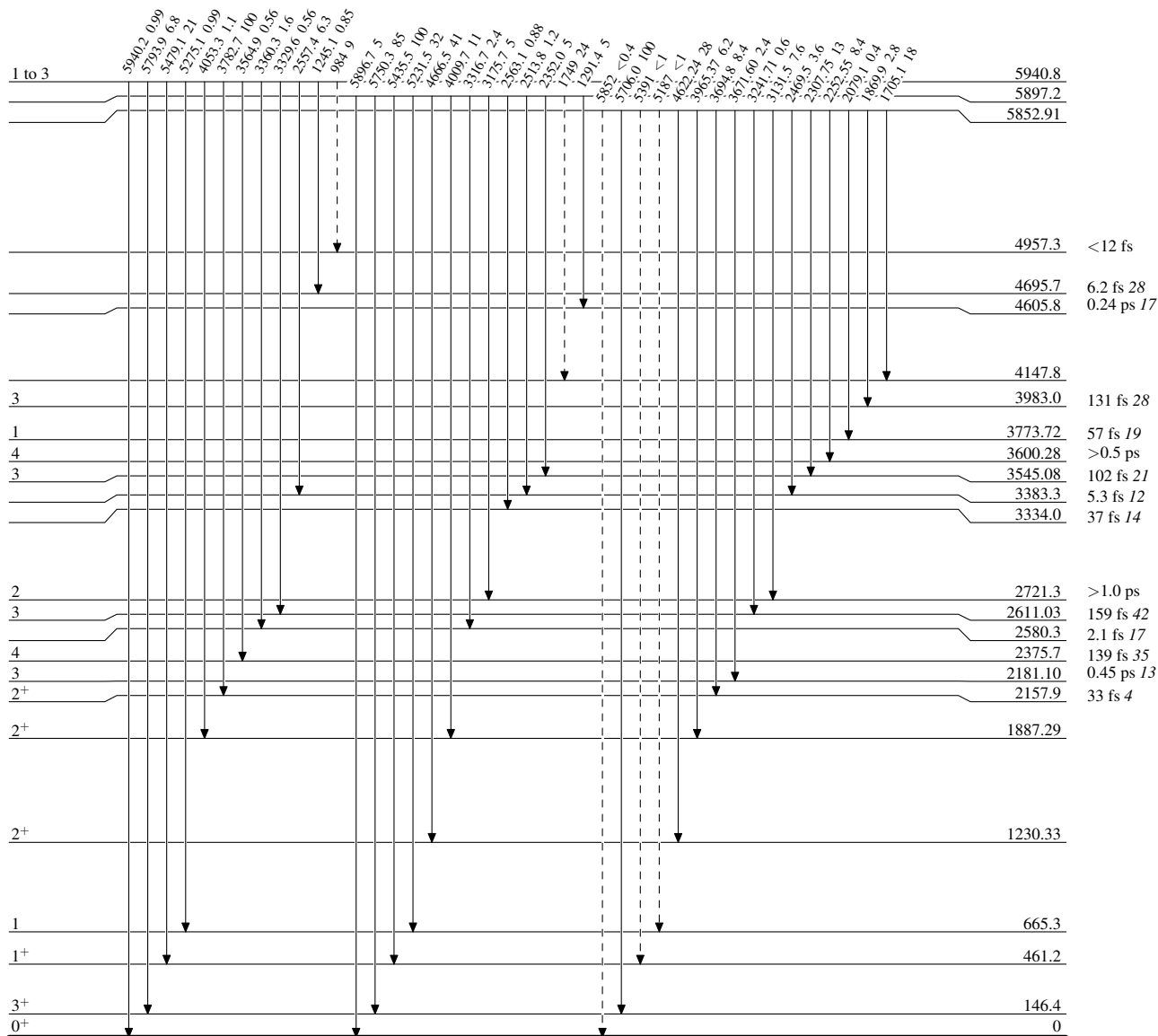
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(p,\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

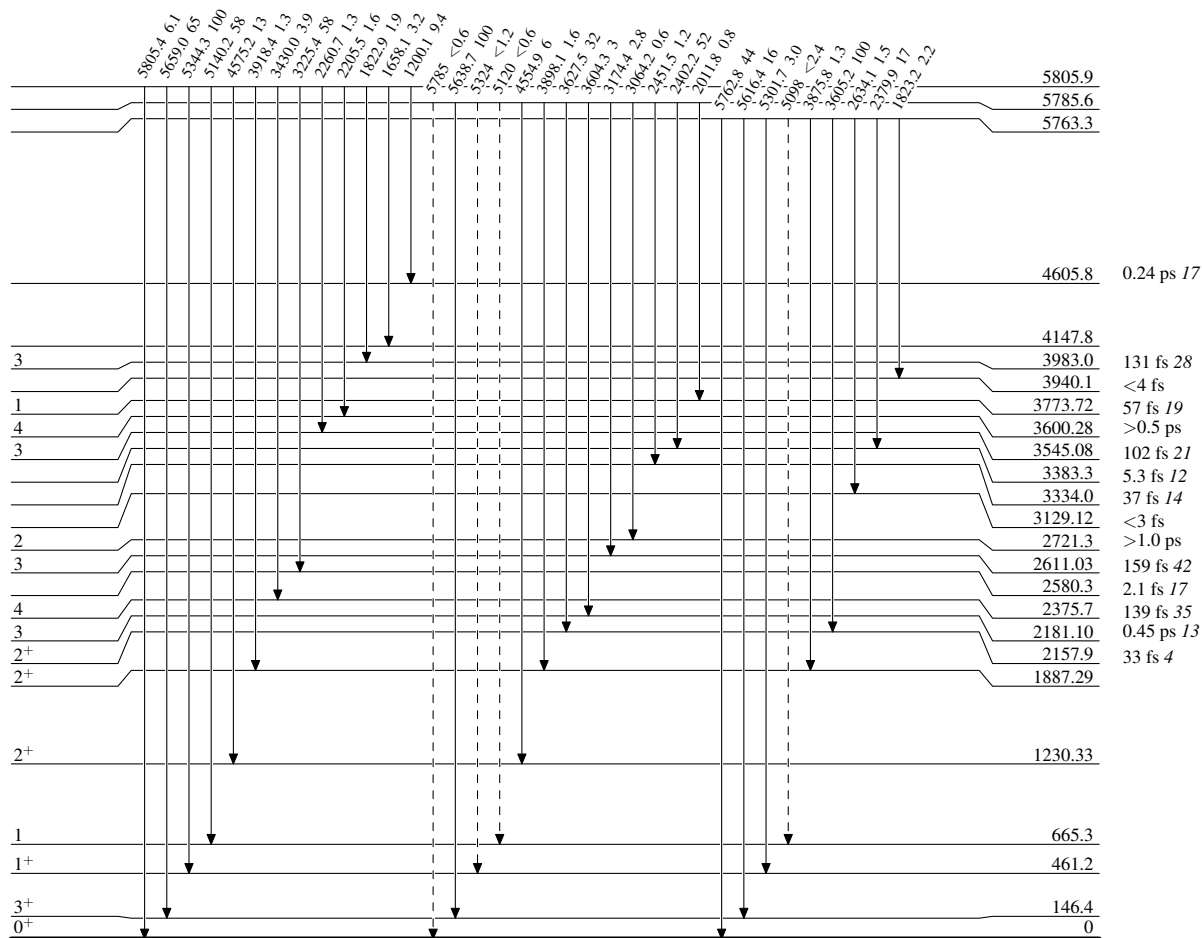
-----► γ Decay (Uncertain)

 $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

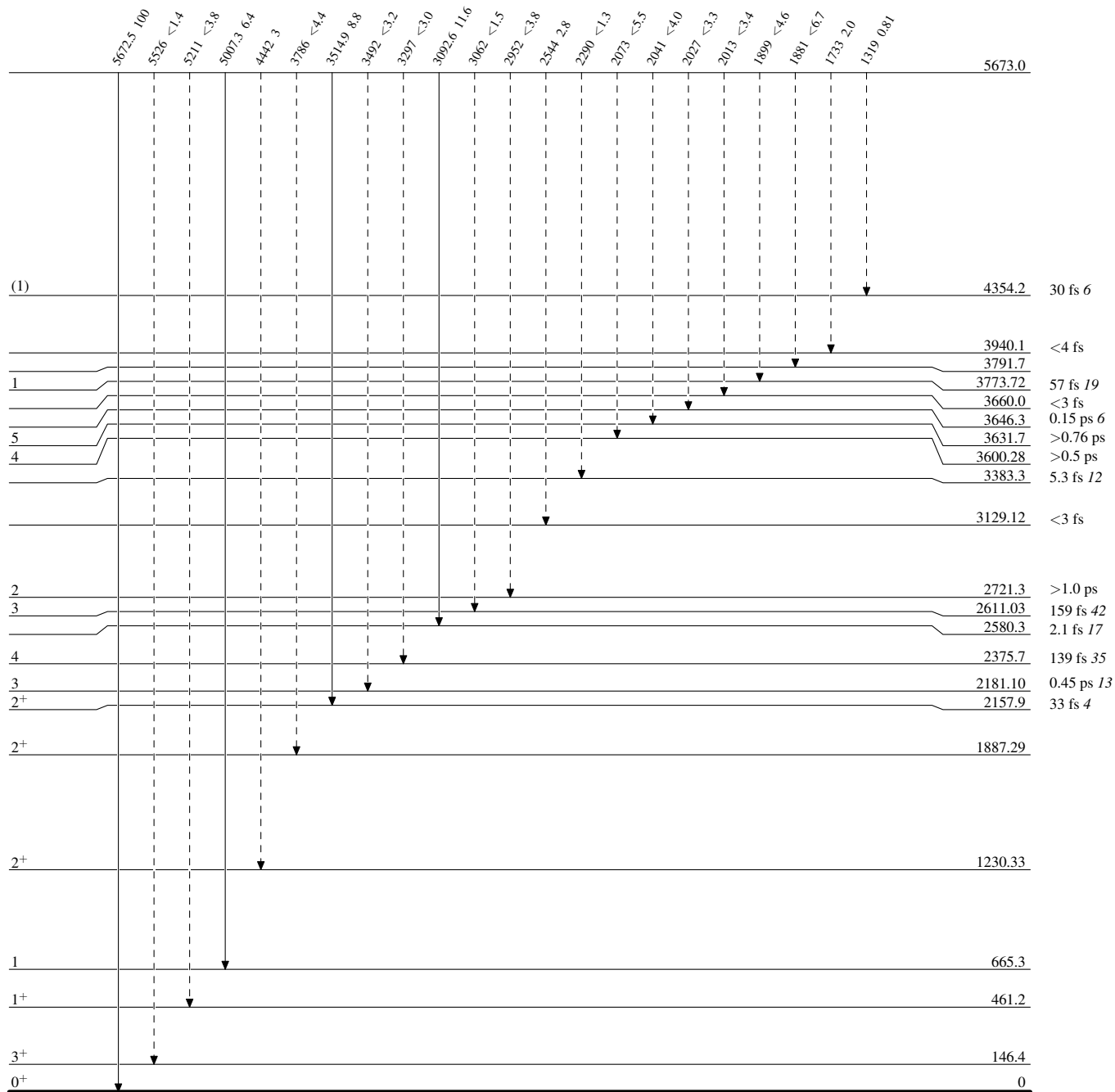
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

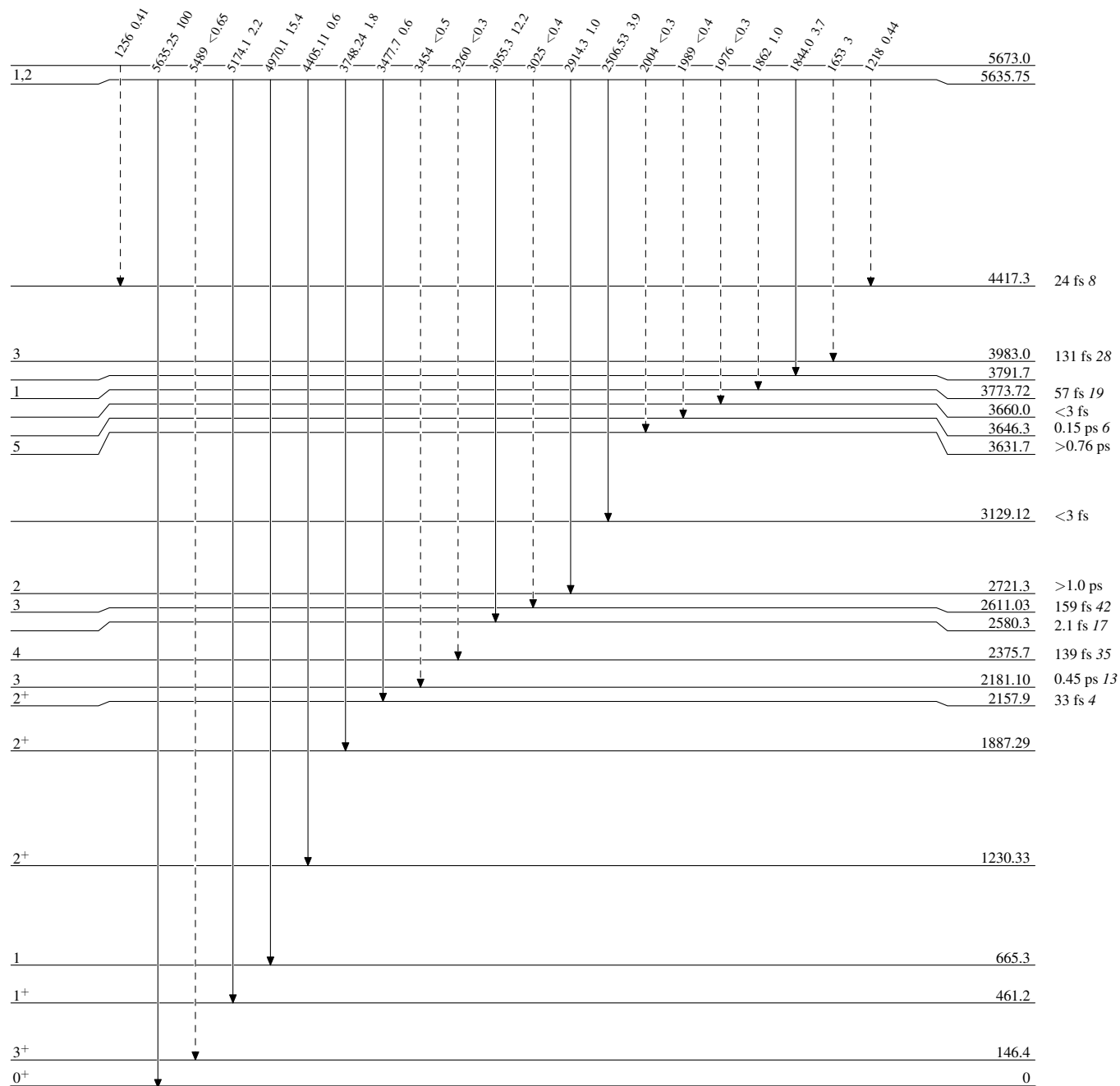
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

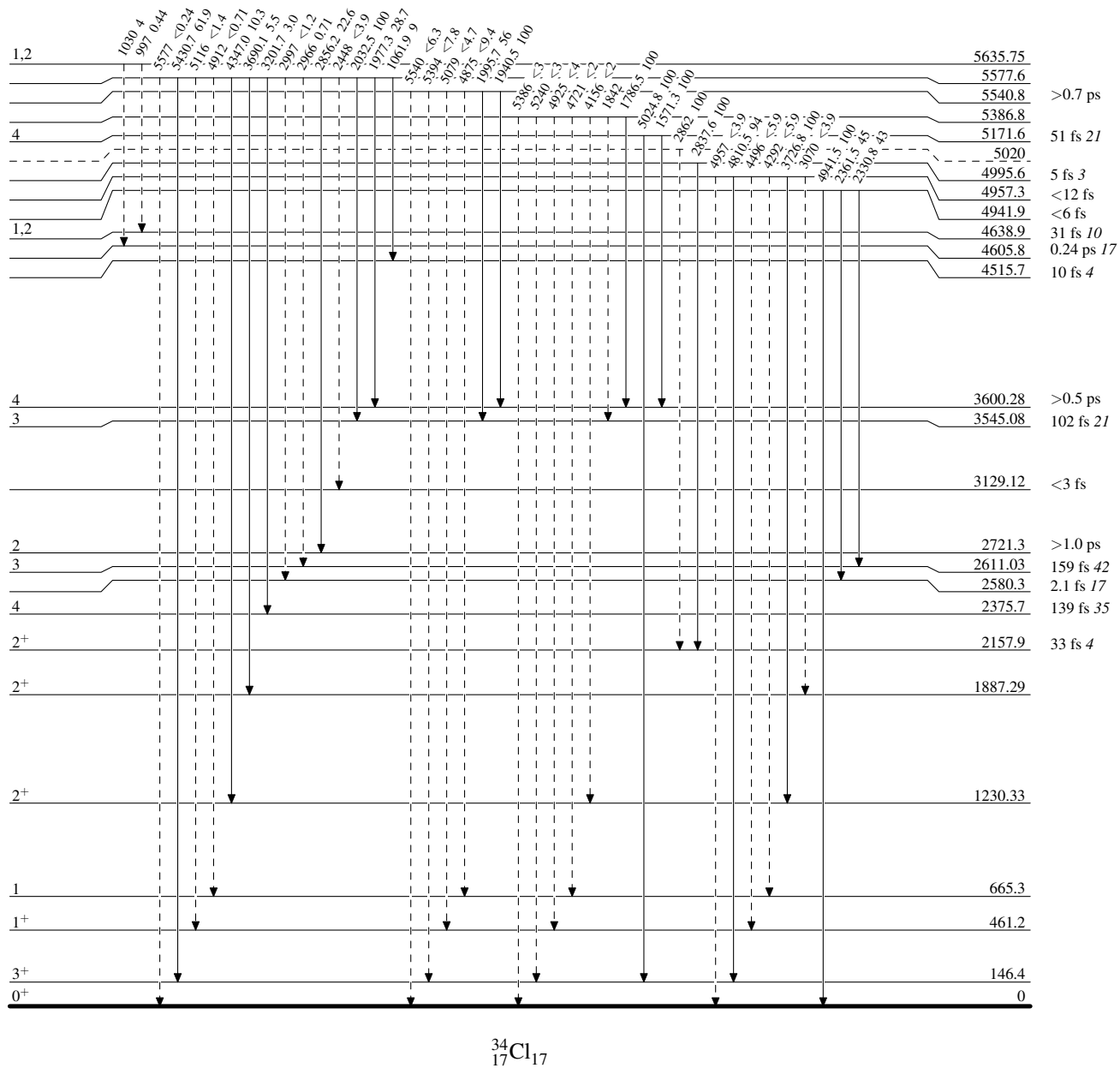
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

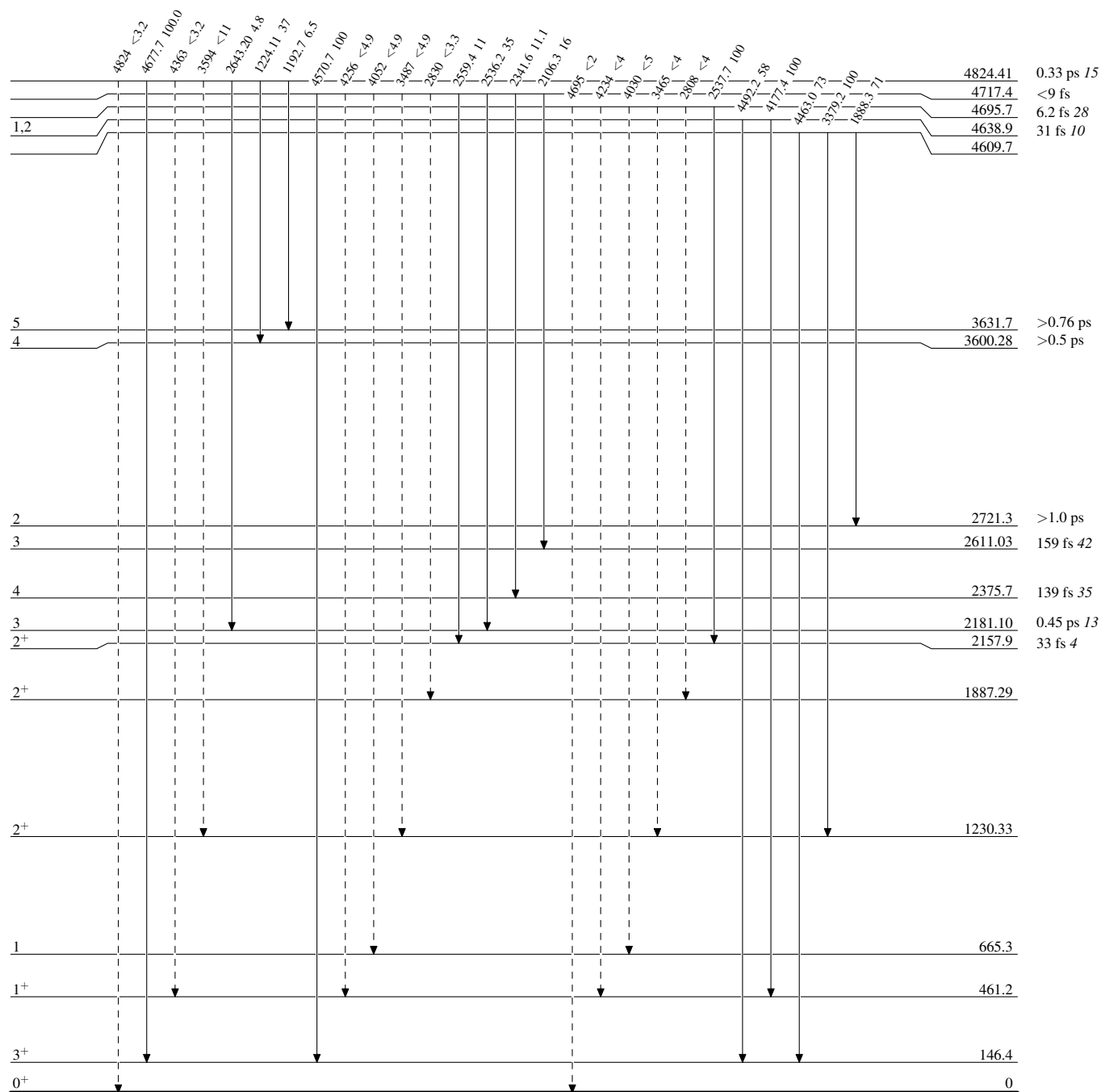
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

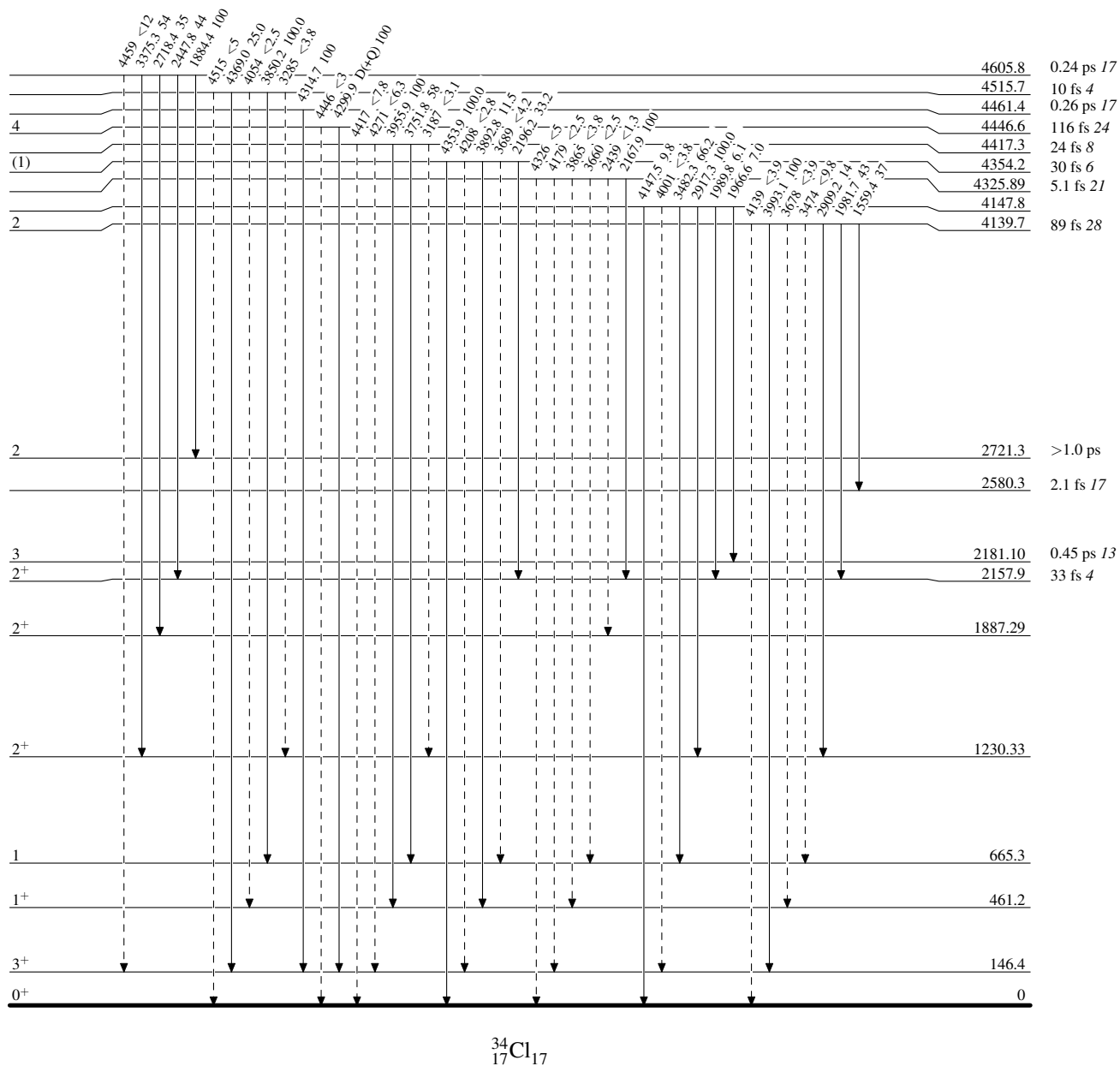
-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain) $^{34}_{17}\text{Cl}_{17}$

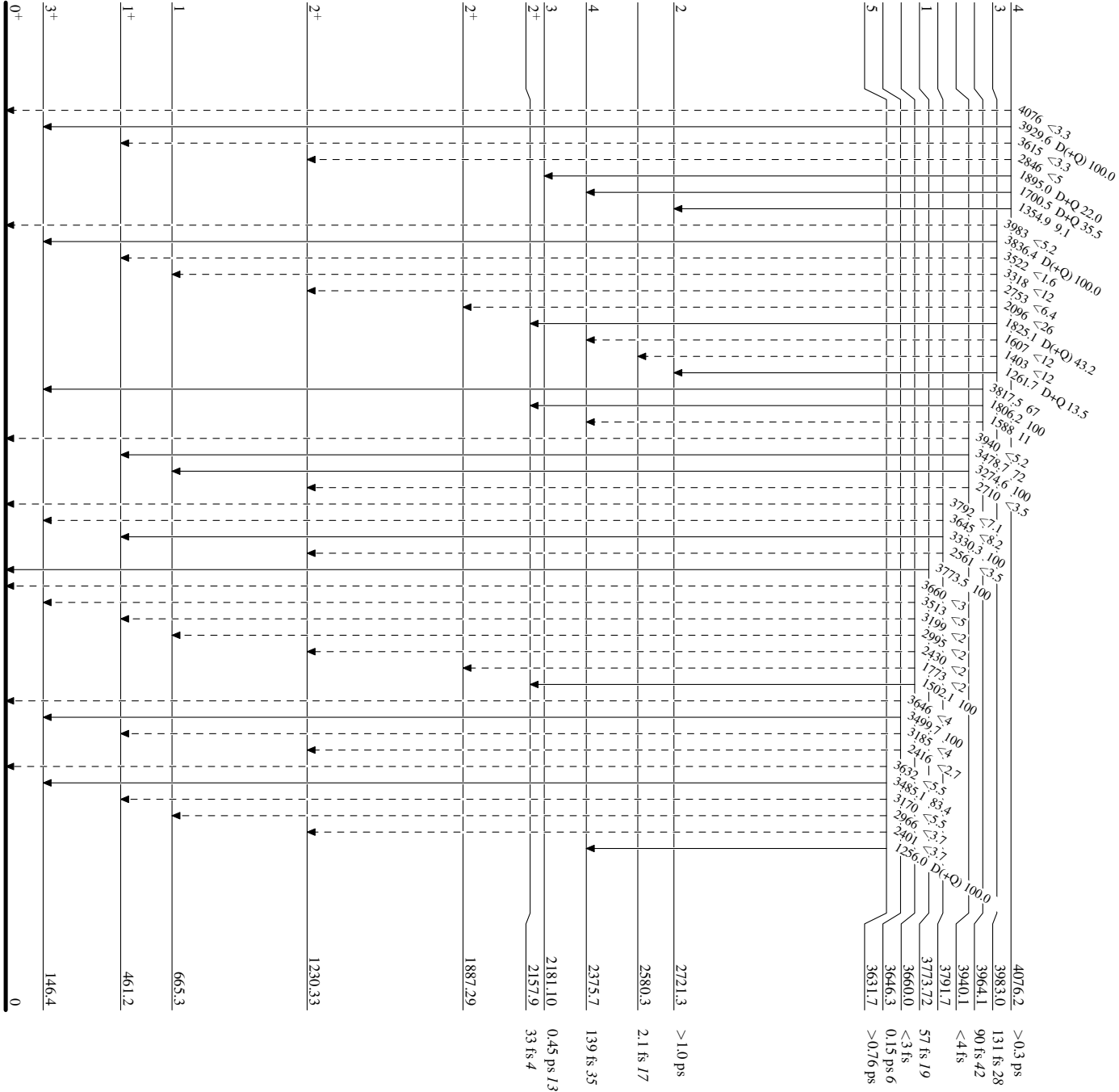
³³S(p,γ) 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

-----▶ γ Decay (Uncertain)



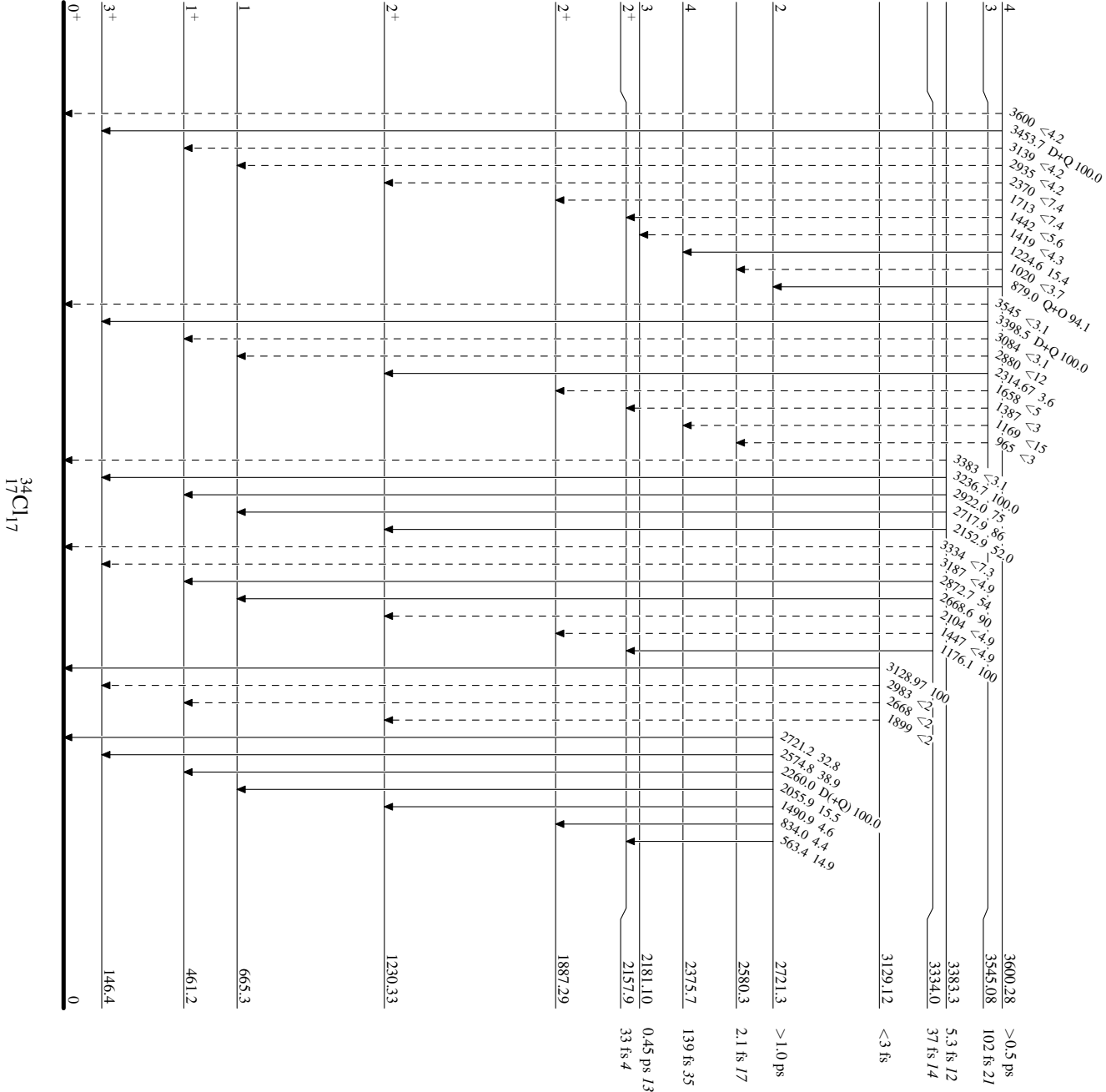
³³S(p, γ) 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)



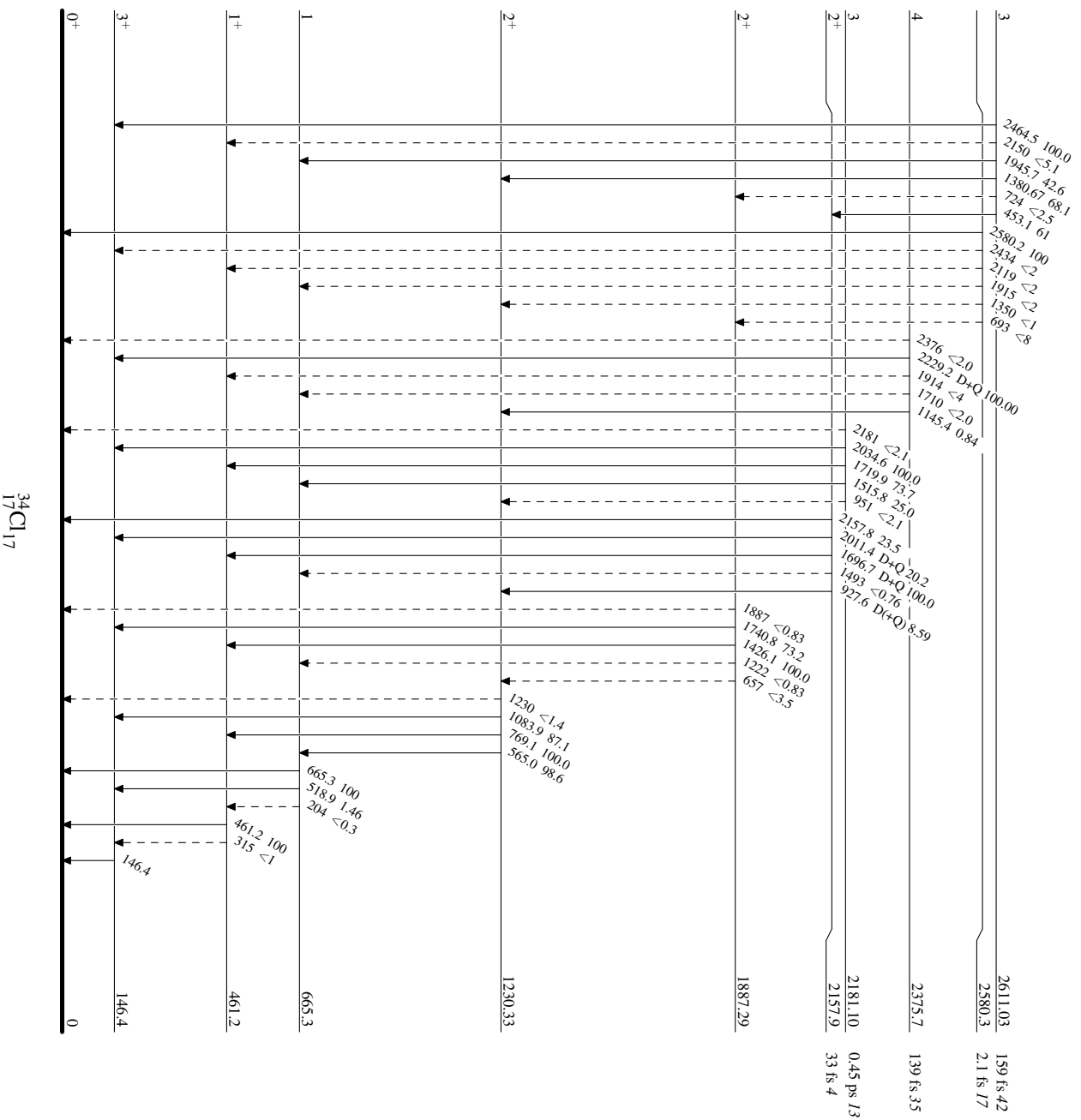
$^{33}\text{S}(\text{p},\gamma)$ 1983Wa27,1977Da02,1992Ka39

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)



$^{33}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$: resonances **1989Va15, 1964Va12**

$^{33}\text{S} + \text{p} \rightarrow ^{34}\text{Cl}$ from $J^\pi = 3/2^+$ ^{33}S ground state.

1989Va15: 1.2-3.77-MeV proton beams from the 4-MV Van de Graaff accelerator at TUNL impinged on a Ag_2S target 88.2% enriched in ^{33}S . Measured differential cross sections for $^{33}\text{S}(\text{p,p}_0)$, $^{33}\text{S}(\text{p,p}_1)$, and $^{33}\text{S}(\text{p,p}_2)$. Extracted resonance parameters for 144 levels by fitting the excitation functions with a multilevel, multichannel R-matrix model.

1964Va12: 1.2-3.2-MeV proton beams from the Utrecht 3-MV Van de Graaff accelerator impinged on CdS targets 22%-enriched in ^{33}S . γ rays were detected using a cylindrical $\text{NaI}(\text{Tl})$ detector. Measured γ -ray yields and deduced 44 levels via the $^{33}\text{S}(\text{p,p}'\gamma)$ reaction.

Other: **1977Da03** is primarily a $^{33}\text{S}(\text{p},\gamma)$ dataset but also measured $^{33}\text{S}(\text{p,p})$ for four resonances.

 ^{34}Cl Levels

E(level) [†]	J^π [‡]	$T_{1/2}$ [‡]	L [‡]	$E_p(\text{lab})$ (keV) [#]	Comments
6169	3^-	25 eV 4		1057	E(level), J^π , $T_{1/2}$, $E_p(\text{lab})$ (keV): from 1977Da03 .
6321.2 15	2^-	1.60 keV 16	1	1214.0 15	$\gamma^2=320.0$.
6369.0 15	2^-	1.60 keV 16	1	1263.3 15	$\gamma^2=240.0$.
6545.2 15	2^-	2.0 keV 2	1	1444.8 15	$E_p(\text{lab})$ (keV): other: 1445 2 (1964Va12). $\gamma^2=128.0$. ($2J+1$) $\Gamma_{p0}\Gamma_{p1}/\Gamma=0.19$ eV (1964Va12).
6632.7 20				1535 @ 2	($2J+1$) $\Gamma_{p0}\Gamma_{p1}/\Gamma=0.15$ eV (1964Va12).
6642	4^-	40 eV 6		1545	E(level), J^π , $T_{1/2}$, $E_p(\text{lab})$ (keV): from 1977Da03 .
6719.5 15	3^-	3.0 keV 3	1	1624.5 15	$\gamma^2=95.0$.
6737.5 15	2^-	2.0 keV 2	1	1643.0 15	$\gamma^2=60.0$.
6748.9 15	2^-	0.80 keV 8	1	1654.8 15	$\gamma^2=23.0$.
6778 2				1685 @ 2	($2J+1$) $\Gamma_{p0}\Gamma_{p1}/\Gamma=2.3$ eV (1964Va12).
6807.0 15	1^-	1.0 keV 1	1	1714.7 15	$E_p(\text{lab})$ (keV): other: 1713 3 (1964Va12). $\gamma^2=25.0$. ($2J+1$) $\Gamma_{p0}\Gamma_{p1}/\Gamma=0.6$ eV; $\Gamma=6$ keV 2 (1964Va12).
6842.8 15	2^-	2.0 keV 2	1	1751.5 15	$E_p(\text{lab})$ (keV): other: 1748 2 (1964Va12). $\gamma^2=42.0$. ($2J+1$) $\Gamma_{p0}\Gamma_{p1}/\Gamma=17$ eV (1964Va12).
6851.8 15	2^+	0.80 keV 8	0	1760.8 15	$\gamma^2=6.1$.
6872	5^-	26 eV 7		1782	E(level), J^π , $T_{1/2}$, $E_p(\text{lab})$ (keV): from 1977Da03 .
6890.78 30	1^-	0.85 keV 9	1	1801.0 3	$E_p(\text{lab})$ (keV): other: 1799 2 (1964Va12). $\gamma^2=15.0$. ($2J+1$) $\Gamma_{p0}\Gamma_{p1}/\Gamma=5$ eV (1964Va12).
6934.84 30	1^-	5.0 keV 5	1	1846.4 3	$E_p(\text{lab})$ (keV): other: 1845 2 (1964Va12). $\gamma^2=80.0$. ($2J+1$) $\Gamma_{p0}\Gamma_{p1}/\Gamma=21$ eV (1964Va12).
6986.85 30	1^-	0.60 keV 6	1	1900.0 3	$E_p(\text{lab})$ (keV): possible doublet 1902 2 (1964Va12). $\gamma^2=8.2$. ($2J+1$) $\Gamma_{p0}\Gamma_{p1}/\Gamma=1.0$ eV; $\Gamma=5$ keV 2 (1964Va12).
6991.31 30	1^+	0.220 keV 22	0+2	1904.6 3	$E_p(\text{lab})$ (keV): possible doublet 1902 2 (1964Va12). $\gamma^2=6.7$.
7037.79 30	2^-	8.0 keV 8	1	1952.5 3	$\gamma^2=95.0$.
7057.7 20				1973 @ 2	($2J+1$) $\Gamma_{p0}\Gamma_{p1}/\Gamma=1.4$ eV (1964Va12).
7080.4 15	$2^-, 3^-$	0.787 keV 30	1	1996.4 15	J^π : 3^- from 1977Da03 ; 2^- preferred and 3^- also possible (1989Va15). $T_{1/2}$: from 1977Da03 for 3^- ; 1.50 keV 15 for 2^- and 1.00 keV 10 for 3^- from 1989Va15 . $\gamma^2=15.6$.
7111.6 15	2^-	1.20 keV 12	1	2028.6 15	$\gamma^2=11.9$.
7124.3 15	2^-	4.0 keV 4	1	2041.6 15	$E_p(\text{lab})$ (keV): possible doublet 2044 3 (1964Va12). $\gamma^2=38.6$. ($2J+1$) $\Gamma_{p0}\Gamma_{p1}/\Gamma=11$ eV; $\Gamma=18$ keV 3 (1964Va12).
7130.5 15	2^-	15.0 keV 30	1	2048.0 15	$E_p(\text{lab})$ (keV): possible doublet 2044 3 (1964Va12). $\gamma^2=140.0$.
7196.2 15	2^-	1.60 keV 16	1	2115.7 15	$E_p(\text{lab})$ (keV): other: 2114 2 (1964Va12). $\gamma^2=13.0$. ($2J+1$) $\Gamma_{p0}\Gamma_{p1}/\Gamma=10$ eV (1964Va12).
7213.7 15	1^-	2.0 keV 2	1	2133.8 15	$E_p(\text{lab})$ (keV): other: 2131 2 (1964Va12). $\gamma^2=15.9$. ($2J+1$) $\Gamma_{p0}\Gamma_{p1}/\Gamma=7$ eV (1964Va12).
7241.2 15	1^+	0.55 keV 6	0+2	2162.1 15	$\gamma^2=16.46$.
7317.1 15	2^-	3.5 keV 4	1	2240.3 15	$\gamma^2=21.7$.

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances **1989Va15,1964Va12** (continued) ^{34}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2} [‡]	L [‡]	E _p (lab) (keV) [#]	Comments
7321.6 20				2245 [@] 2	(2J+1)Γ _{p0} Γ _{p1} /Γ=23 eV (1964Va12).
7336 2				2260 [@] 2	(2J+1)Γ _{p0} Γ _{p1} /Γ=33 eV (1964Va12).
7358.5 15	2 ⁻	20 keV 4	1	2283.0 15	γ ² =110.0.
7368.4 15	1 ⁻ ,2 ⁻ ,3 ⁻		3	2293.2 15	
7386.6 15	2 ⁻	0.70 keV 7	1	2312.0 15	γ ² =3.8.
7408.0 20				2334 [@] 2	(2J+1)Γ _{p0} Γ _{p1} /Γ=9 eV (1964Va12).
7412.0 15	1 ⁻ ,2 ⁻		1+3	2338.1 15	
7451.6 20				2379 [@] 2	(2J+1)Γ _{p0} Γ _{p1} /Γ=60 eV (1964Va12).
7457.6 15	1 ⁻	4.0 keV 4	1	2385.1 15	γ ² =19.0.
7475.3 15	2 ⁺	0.170 keV 17	0+2	2403.4 15	E _p (lab) (keV): other: 2399 2 (1964Va12). γ ² =1.81. (2J+1)Γ _{p0} Γ _{p1} /Γ=54 eV (1964Va12).
7491.3 15	2 ⁻	0.10 keV 1	3	2419.9 15	γ ² =23.0.
7493.9 15	2 ⁻	1.40 keV 14	1+3	2422.5 15	γ ² =50.3.
7510.2 15	1 ⁺	0.30 keV 3	2	2439.3 15	E _p (lab) (keV): other: 2435 3 (1964Va12). γ ² =6.4. (2J+1)Γ _{p0} Γ _{p1} /Γ=3.0 eV (1964Va12).
7525.7 15	1 ⁺ ,2 ⁺		0	2455.3 15	E _p (lab) (keV): other: 2450 4 (1964Va12). (2J+1)Γ _{p0} Γ _{p1} /Γ=6 eV (1964Va12).
7527.1 15	1 ⁺	3.0 keV 3	0	2456.8 15	γ ² =5.8.
7531.2 15	3 ⁻	40 keV 8	1	2461.0 15	E _p (lab) (keV): other: 2460 3 (1964Va12). γ ² =170.0. (2J+1)Γ _{p0} Γ _{p1} /Γ=60 eV (1964Va12).
7533.9 15	1 ⁻ ,2 ⁻ ,3 ⁻ ,4 ⁻ ,5 ⁻		3	2463.8 15	
7536.4 15	2 ⁻	3.6 keV 4	1	2466.3 15	γ ² =14.2.
7546.1 15	1 ⁺	0.150 keV 15	2	2476.3 15	γ ² =2.91.
7553.3 15	2 ⁺	2.50 keV 25	0	2483.8 15	γ ² =4.7.
7585.8 15	1 ⁺	1.40 keV 14	0+2	2517.3 15	γ ² =5.8.
7595.4 15	2 ⁺	0.70 keV 7	0+2	2527.2 15	γ ² =10.78.
7613.7 15	2 ⁺	0.30 keV 3	0	2546.0 15	E _p (lab) (keV): other: 2541 3 (1964Va12). γ ² =0.52. (2J+1)Γ _{p0} Γ _{p1} /Γ=3.1 eV (1964Va12).
7618.1 15	(1) ⁻	0.50 keV 5	3	2550.5 15	γ ² =82.0.
7648.6 15	1 ⁻	32 keV 7	1	2582.0 15	E _p (lab) (keV): other: 2578 4 (1964Va12). γ ² =110.0. (2J+1)Γ _{p0} Γ _{p1} /Γ=460 eV; Γ=39 keV 5 (1964Va12).
7652.1 15	2 ⁻	0.70 keV 7	1	2585.6 15	J ^π : 1 ⁻ also possible (1989Va15). γ ² =2.4.
7673.6 15	1 ⁻	2.0 keV 2	1	2607.7 15	E _p (lab) (keV): other: 2604 3 (1964Va12). γ ² =6.6. (2J+1)Γ _{p0} Γ _{p1} /Γ=280 eV; Γ=6 keV 2 (1964Va12).
7674.3 15	1 ⁺ ,2 ⁺ ,3 ⁺ ,4 ⁺		2	2608.5 15	
7677.9 15	3 ⁻	2.0 keV 2	1	2612.2 15	γ ² =6.6.
7684.7 15	2 ⁺ ,3 ⁺		2	2619.2 15	
7690.5 15	1 ⁺	0.30 keV 3	0+2	2625.2 15	J ^π : 2 ⁺ also possible (1989Va15). E _p (lab) (keV): other: 2625 3 (1964Va12). γ ² =1.81. (2J+1)Γ _{p0} Γ _{p1} /Γ=60 eV (1964Va12).
7696.8 15	1 ⁻	1.30 keV 13	1	2631.7 15	γ ² =4.2.
7704.6 15	1 ⁻ ,2 ⁻ ,3 ⁻ ,4 ⁻ ,5 ⁻		3	2639.7 15	
7719.3 15	1 ⁻	6.0 keV 6	1	2654.8 15	E _p (lab) (keV): other: 2652 3 (1964Va12). γ ² =18.2. (2J+1)Γ _{p0} Γ _{p1} /Γ=490 eV; Γ=7 keV 2 (1964Va12).
7761.6 15	0 ⁺ ,1 ⁺ ,2 ⁺		2	2698.4 15	
7768.5 15	1 ⁻ ,2 ⁻ ,3 ⁻		3	2705.5 15	
7787.6 15	1 ⁺ ,2 ⁺		0	2725.2 15	
7799.9 15	3 ⁺ ,4 ⁺		2	2737.9 15	
7825.3 15	3 ⁺	0.040 keV 4	2	2764.1 15	J ^π : 1 ⁺ ,2 ⁺ also possible (1989Va15). E _p (lab) (keV): possible doublet 2763 4 (1964Va12).

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p,p}),(\text{p,p}')\gamma$:resonances **1989Va15,1964Va12** (continued) ^{34}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2} [‡]	L [‡]	E _p (lab) (keV) [#]	Comments
					$\gamma^2=0.46$. (2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma=210$ eV; $\Gamma=8$ keV 2 (1964Va12).
7827.3 15	2 ⁻	10.0 keV 20	1	2766.1 15	E _p (lab) (keV): possible doublet 2763 4 (1964Va12). $\gamma^2=27.0$.
7849.6 15	3 ⁺ ,4 ⁺		2	2789.1 15	
7859.2 29				2799 @ 3	(2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma=24$ eV (1964Va12).
7883.4 29				2824 @ 3	(2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma=23$ eV (1964Va12).
7897.0 29				2838 @ 3	(2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma=24$ eV (1964Va12).
7909.5 15				2850.8 15	
7909.6 15	2 ⁻	8.0 keV 8	1	2851.0 15	$\gamma^2=19.0$.
7929.1 15	2 ⁺	1.40 keV 14	0+2	2871.1 15	E _p (lab) (keV): other: 2865 4 (1964Va12). $\gamma^2=4.0$. (2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma=9$ eV (1964Va12).
7934.2 15	3 ⁺	0.50 keV 5	2	2876.3 15	$\gamma^2=4.7$.
7944.2 15	1 ⁺ ,2 ⁺ ,3 ⁺ ,4 ⁺		2	2886.6 15	E _p (lab) (keV): other: 2883 3 (1964Va12). (2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma=3.6$ eV, (2J+1) $\Gamma_{p0}\Gamma_{p2}/\Gamma=6$ eV (1964Va12).
7953.3 29				2896 @ 3	(2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma<3.6$ eV, (2J+1) $\Gamma_{p0}\Gamma_{p2}/\Gamma=3.5$ eV (1964Va12).
7967.8 15	1 ⁺	0.20 keV 2	2	2910.9 15	$\gamma^2=1.8$.
7975.1 15	2 ⁺	3.0 keV 3	0	2918.5 15	E _p (lab) (keV): other: 2915 3 (1964Va12). $\gamma^2=3.5$. (2J+1) $\Gamma_{p0}\Gamma_{p2}/\Gamma=10$ eV (1964Va12).
7977.5 15	1 ⁺	2.70 keV 27	0+2	2920.9 15	$\gamma^2=8.5$.
7980.3 15	3 ⁺	0.30 keV 3	2	2923.8 15	$\gamma^2=2.6$.
7985.3 15	2 ⁻	19 keV 4	1	2929.0 15	E _p (lab) (keV): possible triplet 2921 4 (1964Va12) based on level widths in 1989Va15 and 1964Va12 and may contain E _p (lab)=2920.9 and 2918.5. $\gamma^2=55.0$. (2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma=600$ eV; $\Gamma=24$ keV 4 (1964Va12).
8002.6 15	1 ⁺ ,2 ⁺		0+2	2946.8 15	
8012.5 15	1 ⁺ ,2 ⁺ ,3 ⁺ ,4 ⁺		2	2957.0 15	
8020.8 15	2 ⁻	0.60 keV 6	1+3	2965.6 15	$\gamma^2=7.7$.
8027.6 15	1 ⁻	8.0 keV 8	1	2972.6 15	E _p (lab) (keV): possible doublet 2969 4 (1964Va12). $\gamma^2=35.0$. (2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma=450$ eV; $\Gamma=13$ keV 3 (1964Va12).
8043.2 15	2 ⁻	4.5 keV 5	1	2988.6 15	E _p (lab) (keV): other: 2984 4 (1964Va12). $\gamma^2=12.1$. (2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma=270$ eV, (2J+1) $\Gamma_{p0}\Gamma_{p2}/\Gamma=18$ eV; $\Gamma=6$ keV 2 (1964Va12).
8048.1 15	2 ⁺	0.20 keV 2	2	2993.7 15	J ^π : 1 ⁺ also possible (1989Va15). E _p (lab) (keV): other: 2988 3 (1964Va12). $\gamma^2=1.6$. (2J+1) $\Gamma_{p0}\Gamma_{p2}/\Gamma=10$ eV (1964Va12).
8052.5 15	1 ⁻	0.30 keV 3	3	2998.2 15	$\gamma^2=19.0$.
8059.3 15	3 ⁺	1.20 keV 12	2	3005.2 15	J ^π : 2 ⁺ ,4 ⁺ also possible (1989Va15). $\gamma^2=9.3$.
8069.9 15	1 ⁻ ,2 ⁻ ,3 ⁻ ,4 ⁻ ,5 ⁻		3	3016.2 15	
8084.6 15	2 ⁺	0.70 keV 7	0	3031.3 15	$\gamma^2=0.73$.
8087.5 15	1 ⁻	0.50 keV 5	3	3034.3 15	E _p (lab) (keV): other: 3029 3 (1964Va12). $\gamma^2=30.0$. (2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma<25$ eV; (2J+1) $\Gamma_{p0}\Gamma_{p2}/\Gamma=100$ eV (1964Va12); $\Gamma=4$ keV 2 (1964Va12).
8106.4 15	(2) ⁻	5.0 keV 5	1	3053.8 15	E _p (lab) (keV): other: 3048 4 (1964Va12). $\gamma^2=9.4$. (2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma=16$ eV (1964Va12).
8118.7 15	2 ⁺	1.10 keV 11	0+2	3066.4 15	E _p (lab) (keV): possible doublet 3070 4 (1964Va12). $\gamma^2=4.11$.
8127.4 15	0 ⁺	1.0 keV 1	2	3075.4 15	E _p (lab) (keV): possible doublet 3070 4 (1964Va12).

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances 1989Va15,1964Va12 (continued)

^{34}Cl Levels (continued)					
E(level) [†]	J ^π [‡]	T _{1/2} [‡]	L [‡]	E _p (lab) (keV) [#]	Comments
8149.2 15	2 ⁻	26 keV 5	1	3097.9 15	$\gamma^2=7.0$. (2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma=7$ eV (1964Va12). E _p (lab) (keV): other: 3093 4 (1964Va12). $\gamma^2=140.8$. (2J+1) $\Gamma_{p0}\Gamma_{p1}/\Gamma=700$ eV; $\Gamma=18$ keV 3 (1964Va12).
8155.6 15	1 ⁻	1.20 keV 12	1	3104.5 15	E _p (lab) (keV): other: 3100 3 (1964Va12). $\gamma^2=2.1$. (2J+1) $\Gamma_{p0}\Gamma_{p2}/\Gamma=50$ eV (1964Va12).
8159.0 15	1 ⁺	0.90 keV 9	0	3108.0 15	E _p (lab) (keV): possible doublet 3109 3 (1964Va12). $\gamma^2=0.88$.
8164.6 15	1 ⁻ ,2 ⁻		3	3113.8 15	E _p (lab) (keV): possible doublet 3109 3 (1964Va12). (2J+1) $\Gamma_{p0}\Gamma_{p2}/\Gamma=14$ eV (1964Va12).
8186.38 30	3 ⁺ ,4 ⁺		2+4	3136.2 3	
8192.59 30	2 ⁺	0.50 keV 5	0+2	3142.6 3	$\gamma^2=6.38$.
8203.75 30	1 ⁻	15.0 keV 30	1	3154.1 3	$\gamma^2=25.0$.
8210.54 30	(3) ⁺	0.20 keV 2	2	3161.1 3	$\gamma^2=1.2$.
8214.91 30	3 ⁻	0.30 keV 3	3	3165.6 3	$\gamma^2=14.0$.
8231.99 30	2 ⁻	10.0 keV 20	1	3183.2 3	$\gamma^2=31.9$.
8236.35 30	2 ⁺	0.95 keV 10	0+2	3187.7 3	$\gamma^2=7.47$.
8251.39 30	(2) ⁺	0.10 keV 1	2	3203.2 3	$\gamma^2=0.5$.
8254.30 30	2 ⁻ ,3 ⁻		3	3206.2 3	$\gamma^2=30.0$.
8260.22 30	4 ⁺	0.80 keV 8	2	3212.3 3	$\gamma^2=4.6$.
8265.56 30	2 ⁺ ,3 ⁺		2	3217.8 3	
8271.58? 30	0 ⁻	0.30 keV 3	1	3224.0 3	$\gamma^2=0.47$.
8276.72 30	2 ⁻	2.20 keV 22	1	3229.3 3	$\gamma^2=3.5$.
8316.60 30	4 ⁺	2.0 keV 2	2	3270.4 3	$\gamma^2=11.0$.
8317.57 30	1 ⁺ ,2 ⁺		2	3271.4 3	
8322.91 30	1 ⁻ ,2 ⁻ ,3 ⁻ ,4 ⁻ ,5 ⁻		3	3276.9 3	
8330.19 30	1 ⁻ ,2 ⁻ ,3 ⁻ ,4 ⁻ ,5 ⁻		3	3284.4 3	
8358.42 30	2 ⁺	1.50 keV 15	0	3313.5 3	$\gamma^2=1.3$.
8376.18 30	2 ⁺	0.80 keV 8	0	3331.8 3	$\gamma^2=0.66$.
8381.3 15	(2) ⁻	0.30 keV 3	3	3337.1 15	$\gamma^2=11.0$.
8386.5 15	3 ⁺ ,4 ⁺		2	3342.4 15	
8460.4 15	1 ⁺	0.80 keV 8	0+2	3418.6 15	$\gamma^2=2.11$.
8468.2 15	2 ⁻	2.80 keV 28	1	3426.6 15	$\gamma^2=65.6$.
8477.2 15	(2) ⁺	1.90 keV 19	0+2	3435.9 15	$\gamma^2=17.0$.
8485.9 15	1 ⁻	1.40 keV 14	1+3	3444.9 15	$\gamma^2=59.0$.
8494.2 15	2 ⁺	1.50 keV 15	0+2	3453.4 15	$\gamma^2=2.86$.
8494.7 15	2 ⁻	0.45 keV 5	3	3453.9 15	$\gamma^2=13.1$.
8509.3 15	2 ⁺	25 keV 5	2	3469.0 15	$\gamma^2=110.0$.
8537.6 15	3 ⁻	2.0 keV 2	1	3498.1 15	$\gamma^2=59.2$.
8543.4 15	3 ⁻	2.0 keV 2	1	3504.1 15	$\gamma^2=58.2$.
8548.1 15	1 ⁺	25 keV 5	0	3509.0 15	$\gamma^2=30.3$.
8549.2 15	2 ⁻	3.0 keV 3	1	3510.1 15	$\gamma^2=3.7$.
8563.4 15	4 ⁺	1.50 keV 15	2	3524.7 15	$\gamma^2=5.9$.
8579.4 15	1 ⁺	3.10 keV 31	0+2	3541.2 15	$\gamma^2=23.02$.
8587.1 15	4 ⁻	0.40 keV 4	1+3	3549.2 15	$\gamma^2=14.4$.
8597.9 15	1 ⁺	3.0 keV 3	0+2	3560.3 15	$\gamma^2=26.71$.
8601.8 15	2 ⁺	3.0 keV 3	0	3564.3 15	$\gamma^2=2.1$.
8603.2 15	3 ⁻	0.20 keV 2	3	3565.8 15	$\gamma^2=4.9$.
8610.5 15	3 ⁻	6.0 keV 6	1+3	3573.3 15	$\gamma^2=125.8$.
8613.4 15	4 ⁻ ,5 ⁻		3	3576.3 15	
8643.6 15	1 ⁻	0.60 keV 6	1	3607.4 15	$\gamma^2=11.34$.

Continued on next page (footnotes at end of table)

$^{33}\text{S}(\text{p,p}),(\text{p,p}'\gamma):\text{resonances}$ **1989Va15,1964Va12** (continued) ^{34}Cl Levels (continued)

E(level) [†]	J^π [‡]	$T_{1/2}$ [‡]	L [‡]	$E_p(\text{lab})$ (keV) [#]	Comments
8647.5 15	2 ⁺	10.0 keV 20	0	3611.4 15	$\gamma^2=6.9$.
8651.9 15	(1) ⁺	1.0 keV 1	2	3615.9 15	$\gamma^2=3.6$.
8655.7 15	(1) ⁺	1.80 keV 18	0+2	3619.9 15	$\gamma^2=210.69$.
8665.1 15	4 ⁻ ,5 ⁻		3	3629.5 15	
8693.3 15	1 ⁺	0.50 keV 5	2	3658.6 15	$\gamma^2=1.7$.
8697.2 15	2 ⁺	1.50 keV 15	0+2	3662.6 15	$\gamma^2=6.07$.
8700.1 15	1 ⁻	1.50 keV 15	1	3665.6 15	$\gamma^2=1.6$.
8706.4 15	1 ⁺	0.80 keV 8	0	3672.1 15	$\gamma^2=0.53$.
8714.6 15	3 ⁺	4.0 keV 4	2	3680.6 15	$\gamma^2=20.9$.
8733.2 15	3 ⁻	35 keV 7	1	3699.7 15	$\gamma^2=37.0$.
8754.6 15	1 ⁻	0.70 keV 7	3	3721.8 15	$\gamma^2=14.0$.
8756.6 15	1 ⁻	4.0 keV 4	1	3723.8 15	$\gamma^2=53.1$.
8761.4 15	3 ⁺	1.60 keV 16	0+2	3728.8 15	$\gamma^2=8.63$.
8779.4 15	3 ⁺	0.80 keV 8	0+2	3747.3 15	$\gamma^2=4.7$.
8790.6 15	2 ⁻	6.0 keV 6	1	3758.9 15	$\gamma^2=6.1$.
8794.0 15	1 ⁻	0.20 keV 2	3	3762.4 15	$\gamma^2=3.7$.

[†] E(level)= $E_{\text{c.m.}}+S(\text{p})(^{34}\text{Cl})$ where $S(\text{p})(^{34}\text{Cl})=5143.20$ 5 (2021Wa16) and $E_{\text{c.m.}}=E_p(\text{lab})\times m(^{33}\text{S})/[m_p+m(^{33}\text{S})]$. $E_p(\text{lab})$ is from 1989Va15, unless otherwise noted. 1989Va15 states that the calibrations provide accurate energies of $\Delta E=\pm 300$ eV near the calibration points and a maximum error of $\Delta E=\pm 1.5$ keV away from the calibration points. The calibration points are at $E(\text{p})=1.8804$ MeV and 3.2357 MeV. Therefore, the evaluators assign a $\Delta E=0.3$ keV for levels within 100 keV of either calibration point, and a $\Delta E=1.5$ keV for other levels.

[‡] From 1989Va15 multilevel, multichannel R-matrix analysis, unless otherwise noted. 1989Va15 states that the conventions for listing multiple J^π solutions are: if several solutions give equivalent fits, then the solutions are listed in order of increasing J, and if there is a strongly preferred solution it is listed first, with other solutions listed in parentheses. For the latter case, the evaluators adopt the strongly preferred solution and put the other solutions in comments. 1989Va15 states that typical uncertainties are 10% for laboratory widths of a few keV or smaller, and 20% for the larger widths of tens of keV. Reduced width $\gamma^2=\Gamma/2P$, where Γ is the laboratory width and P is the Coulomb penetrability (1989Va15).

[#] From 1989Va15, unless otherwise noted.

@ From 1964Va12.

$^{33}\text{S}(^3\text{He},\text{d})$ **2014Pa44,1971Er03**

$^{33}\text{S} + \text{p} \rightarrow ^{34}\text{Cl}$ from $J^\pi = 3/2^+$ ^{33}S ground state.

2014Pa44: a 25-MeV ^3He beam was provided from the 14-MV MP tandem accelerator at the Maier-Leibnitz Laboratory, Garching, Germany. Targets were $\text{Ag}^2\text{ }^{33}\text{S}$ ($20\text{ }\mu\text{g}/\text{cm}^2$, enriched to 99.9% ^{33}S) deposited upon a foil of ^{12}C ; Ag^2S ($20\text{ }\mu\text{g}/\text{cm}^2$, natural sulphur) deposited upon a foil of ^{12}C ; a $50\text{ }\mu\text{g}/\text{cm}^2$ Ag foil; a $7\text{ }\mu\text{g}/\text{cm}^2$ ^{12}C foil; and ^{24}Mg ($20\text{ }\mu\text{g}/\text{cm}^2$, enriched to 99.92% ^{24}Mg) deposited upon a foil of ^{12}C . All ^{12}C foils were enriched to 99.99% and of thickness $7\text{ }\mu\text{g}/\text{cm}^2$. Light reaction products were momentum-analyzed by the Q3D magnetic spectrograph and detected in the focal plane using a multiwire gas-filled proportional counter backed by a plastic scintillator at FWHM=12 keV. Measured $\sigma(E_d, \theta)$. Deduced levels, L-transfers, J, π , and spectroscopic factors from FRESCO-DWBA analysis of the measured $\sigma(\theta)$. Also deduced resonance strengths and the thermonuclear $^{33}\text{S}(\text{p}, \gamma)^{34}\text{Cl}$ reaction rate. Calculated $^{32}\text{S}/^{33}\text{S}$ ratios in the composition of nova ejecta via hydrodynamic nova simulations.

1971Er03: a 14-MeV ^3He beam was provided from the Argonne tandem Van de Graaff accelerator. Targets were CdS on C backing, enriched to 83% in ^{33}S . Deuterons were detected using nuclear track emulsions in the focal plane of the Enge split-pole magnetic spectrograph with an energy resolution of 17 keV. Measured $\sigma(E_d, \theta)$. Deduced levels, L-transfers, J, π , and spectroscopic factors from JULIE-DWBA analysis of the measured $\sigma(\theta)$.

 ^{34}Cl Levels

$E(\text{level})^\dagger$	L [#]	$[(2J_f+1)/(2J_i+1)]C^2S^\#$	Comments
0	2	0.25	
145.5	2	1.42	
461.0 12	0+2	0.04+0.08	
665.1 14	0+2	0.07+0.50	
1229.4 16	0	0.12	
1885 3	0	0.02	
1923.3 17	2	0.04	
2157.5 15	2	0.66	
2579.1 16	0	0.03	
2608.5 13	2	0.08	
2720.7 25	1+3	0.04+0.39	
3128.0 22	2	0.11	
3333.0 24	2	0.09	
3382.7 20	2	0.31	
3544.9 25	1	0.18	
3631.4 25	3	1.10	
3773.0 24	1	0.05	
3983.4 25	3	0.44	
4076 3	3	0.52	
4146 3	(1+3)	0.08+0.24	
4353 3	1	0.14	
4418 3	1	0.11	
4461 3	1	0.09	
4516.3 22	1	0.13	
4607.9 21	1	0.11	
4638.7 21	1+3	0.010+0.017	L: other: 1 (1971Er03). $[(2J_f+1)/(2J_i+1)]C^2S$: 0.010 2 for L=1, 0.017 3 for L=3 (2014Pa44). Other: 0.06 for L=1 (1971Er03).
4695 2			
4718 2			
4857 2	2,1+3	0.230 8	$[(2J_f+1)/(2J_i+1)]C^2S$: for L=2. $C^2S=0.0059$ 5 for L=1. $C^2S=0.034$ 3 for L=3.
4941.9 [‡]	1	0.0060 7	
4957.3 [‡]	2	0.079 5	
4979 2	1,0	0.0058 4	$[(2J_f+1)/(2J_i+1)]C^2S$: for L=1. $C^2S=0.0095$ 7 for L=0.
4994 2	2	0.056 5	
5009 2	2,3	0.070 3	$[(2J_f+1)/(2J_i+1)]C^2S$: for L=2, 0.020 1 for L=3.
5028 2	1	0.0010 1	

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$^{33}\text{S}(^3\text{He,d})$ [2014Pa44](#), [1971Er03](#) (continued) ^{34}Cl Levels (continued)

E(level) [†]	L [#]	$[(2J_f+1)/(2J_i+1)]C^2S^{\#}$	Comments
5171.6 [‡]	3,2	0.0159 7	$[(2J_f+1)/(2J_i+1)]C^2S$: for L=3, 0.057 3 for L=2.
5246 2	3	0.112 3	
5338 2	1+3,2	0.0060+0.022	$[(2J_f+1)/(2J_i+1)]C^2S$: 0.0060 5 for L=1, 0.022 2 for L=3, 0.200 8 for L=2.
5386.8 [‡]	3	0.0128 7	
5435 2	0	0.0110 6	
5453 2	0	0.0034 5	
5577 2	0	0.081 1	
5608 2	3	0.334 3	
5635.7 [‡]	1+3,2	0.0081+0.054	$[(2J_f+1)/(2J_i+1)]C^2S$: 0.0081 7 for L=1, 0.054 2 for L=3, 0.247 4 for L=2.
5672.9 [‡]	0	0.0118 8	
5784 2	1	0.0010 1	
5805.9 [‡]	1+3	0.0028+0.016	$[(2J_f+1)/(2J_i+1)]C^2S$: 0.0028 6 for L=1, 0.016 1 for L=3.
5852.8 [‡]	1+3	0.012+0.039	$[(2J_f+1)/(2J_i+1)]C^2S$: 0.012 1 for L=1, 0.039 3 for L=3.
5897.1 [‡]	1	0.0088 7	

[†] For levels below 4.64 MeV, from [1971Er03](#); for levels above 4.64 MeV, from [2014Pa44](#), unless otherwise noted.

[‡] E(level) from [2014Pa44](#) without uncertainties. [2014Pa44](#) observed these levels and used the E(level) values from the Adopted Levels of [2012Ni10](#) for calibration.

[#] From DWBA analysis of the measured $\sigma(\theta)$ in [1971Er03](#) (below 4.63 MeV) and [2014Pa44](#) (above 4.63 MeV). The spectroscopic factors C^2S given by [1971Er03](#) are defined as $[(2J_f+1)/(2J_i+1)]C^2S$, where $C^2=1/2$ is the isospin coefficient, J_f and J_i , are the final and initial spins, respectively. $J_i=3/2$ for ^{33}S ground state.

$^{34}\text{S}(\text{p},\text{n})$ 1987Fu11,1997Jo08

$^{34}\text{S}+1\text{p}\rightarrow 1\text{n}+^{34}\text{Cl}$ from $J^\pi=0^+$ ^{34}S ground state.

1987Fu11, 1987Or02: a 35-MeV proton beam was provided from the AVF-cyclotron and a beam swinger system at Tohoku University. The target was 2.64-mg/cm², 94.4%-enriched ^{34}S evaporated on 99.9%-enriched ^{12}C foils covered by thin gold.

Measured $\sigma(E_n, \theta)$ and analyzed with microscopic DWBA. Deduced level, L-transfers, J, and π .

2002Or05, 1999OrZZ, 1997Jo08: a 35-MeV proton beam was provided from the AVF-cyclotron and a beam swinger system at Tohoku University. Targets were prepared by evaporating ^{34}S onto enriched ^{12}C foils. Neutrons were detected using 10 organic liquid scintillator NE213. Measured $\sigma(E_n, \theta)$ and analyzed with microscopic and macroscopic DWBA. Deduced B(GT).

Reaction-threshold and excitation-function measurements: 1977Ba16, 1970Um01, 1969UmZZ, 1966Ma75, 1965Fr03, 1952Hi66.

 ^{34}Cl Levels

E(level) [†]	J^π [‡]	L [‡]	Comments
0	0 ⁺	0	T=1 IAS of ^{34}S g.s., observed in 1987Fu11 and 1997Jo08.
461	1 ⁺	2	$\Delta S=1$ (1987Fu11).
666	1 ⁺	0+2	$\Delta S=1$ (1987Fu11).
1230	2 ⁺	2	
1890	2 ⁺	2	
2581	1 ⁺	0	$\Delta S=1$ (1987Fu11).
3128	1 ⁺	0	$\Delta S=1$ (1987Fu11); B(GT)=1.30 7 (2002Or05).
3380	2 ⁺	2	T=1
4985	1 ⁺ , 2 ⁺	1, 2	E(level): unresolved 4971+4996 doublet. $\Delta S=1$ (1987Fu11).

[†] From 1987Fu11.

[‡] From DWBA analysis of measured $\sigma(\theta)$ using different contributions of $\Delta J(\Delta L, \Delta S)$ transitions (1987Fu11).

$^{34}\text{S}(^3\text{He},\text{t})$ [2009Pa28,2007Fu06](#)

$^{34}\text{S}+1\text{p}\rightarrow 1\text{n}+^{34}\text{Cl}$ from $J^\pi=0^+$ ^{34}S ground state.

[2009Pa28](#): a 25-MeV ^3He beam was produced from the MP tandem accelerator at MLL, Garching, Germany and impinged on $50\text{-}\mu\text{g}/\text{cm}^2$, 99.999% ^{34}S -enriched Ag_2S targets. Light reaction products were momentum-analyzed in the Q3D magnetic spectrograph at $\theta_{\text{lab}}=15^\circ$ and 25° and detected by a multiwire gas proportional counter backed by a plastic scintillator with FWHM=10 keV. Measured tritium spectra. Deduced resonances and $^{33}\text{S}(\text{p},\gamma)^{34}\text{Cl}$ reaction rate at relevant nova temperatures. Conference proceedings: [2011PaZV](#), [2010PaZT](#).

[2007Fu06](#): a 140-MeV/nucleon ^3He beam was produced from K400 Ring cyclotron at RCNP, Osaka University and impinged on a 93.47%-enriched $3.3\text{-mg}/\text{cm}^2$ ^{34}S self-supporting target. The outgoing tritons were momentum analyzed by the Grand Raiden spectrometer at 0° and detected using a focal-plane detector system with FWHM=21 keV. Measured $\sigma(E,\theta)$. Deduced levels, J, π , L-transfer, B(GT) and compared with DWBA and Shell-model calculations.

 ^{34}Cl Levels

All Gamow-Teller strengths are from [2007Fu06](#).

E(level) [†]	L [#]	Comments
0 [‡]	0	E(level): IAS of ^{34}S g.s. (2007Fu06).
458 [‡] 4		2007Fu06 reports L=0,≥1 and B(GT)=0.025 3.
666 [‡] 4	0	B(GT)=0.091 7.
2579 [‡] 4	0	B(GT)=0.285 21.
2718 [‡] 4	≥1	
3129 [‡] 4	0	B(GT)=1.37 9.
3330 [‡] 4	(0)	
4212 [‡] 4		2007Fu06 reports L=0,≥1 and B(GT)=0.030 3.
4611 [‡] 4	0	B(GT)=0.183 14.
4925 4		
4942 4	0	E(level): weighted average of 4941 4 (2007Fu06) and 4942 4 (2009Pa28). B(GT)=0.261 19.
4957 4		
4981 4		
4994 4	≥1	E(level): weighted average of 4993 4 (2007Fu06) and 4995 4 (2009Pa28).
5010 4		
5061 4		
5093 4		
5154 4		
5173 4		
5233 4		
5263 4		
5326 4		
5357 4		
5388?		
5424 4		
5444 4		
5485 4		
5545 4		
5572 4		
5606 4		
5635 4		
5671?		
5705 5		E(level): statistical uncertainty of 4 keV and a systematic uncertainty of 3 keV (2009Pa28) are added in quadrature by evaluators.
5761?		
5782 4		
5806 4		

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 $^{34}\text{S}(^3\text{He},\text{t})$ [2009Pa28](#), [2007Fu06](#) (continued)

 ^{34}Cl Levels (continued)E(level)[†]

5852 4

5868 4

5898 4

5917 4

5941 4

5980 4

[†] From [2009Pa28](#), unless otherwise noted. An overall statistical uncertainty of 2 keV and a systematic uncertainty of 3 keV ([2009Pa28](#)) are added in quadrature by evaluators, unless otherwise noted.

[‡] From [2007Fu06](#).

[#] The triton spectra with angular cuts with $\leq 0.5^\circ$, $1.0^\circ - 1.5^\circ$, and $1.5^\circ - 2.0^\circ$ are compared, and the states showing a decrease in intensity with increasing angles are assigned $L=0$ ([2007Fu06](#)).

$^{34}\text{S}(^6\text{Li}, ^6\text{He})$ **1975Wh01**

$^{34}\text{S}+3\text{p}3\text{n}\rightarrow 2\text{p}4\text{n}+^{34}\text{Cl}$ from $J^\pi=0^+$ ^{34}S ground state. ^6Li ($J^\pi=1^+$, $T=0$); ^6He ($J^\pi=0^+$, $T=1$).

1975Wh01: $E(^6\text{Li})=32$ MeV. ^6He ions were detected using two Si surface-barrier position sensitive detectors in the focal plane of the Argonne National Laboratory Enge Split Pole Spectrograph. Measured $\sigma(E(^6\text{He}),\theta)$. Deduced levels, L-transfers, J, and π from DWBA analysis of the measured $\sigma(\theta)$ assuming a one-step inelastic interaction.

 ^{34}Cl Levels

<u>E(level)[†]</u>	<u>J^π[†]</u>	<u>L[†]</u>	<u>Comments</u>
0	0 ⁺	1	J^π, L : appears to have pure L=1 angular distributions. The most likely explanation of the observed 0 ⁺ to 0 ⁺ transition is that it proceeds through two-step processes.
460	1 ⁺	2	
680	1 ⁺	0+2	21% for L=0 and 79% for L=2.
2560	1 ⁺	0+1	E(level): unresolved 2560 and 2610 doublet. 55% for L=0 and 45% for L=1.
3130	1 ⁺	0	

[†] From **1975Wh01**.

$^{35}\text{Cl}(\gamma, n)$ 1997Is02

1997Is02: a $E_{\text{max}}=32\text{-MeV}$ bremsstrahlung photon beam was produced from the Moscow State University betatron. The target was 380-g LiCl with natural isotopic composition. γ rays were detected using a 100-cm^3 Ge(Li) detector. Measured E_γ , I_γ . Deduced levels, integrated cross sections (σ_{int}), and Giant Dipole Resonance (GDR) features.

Other: 1974Ve06 measured cross sections of (γ, n) , (γ, np) , $(\gamma, 2n)$ at $E=8\text{-}30$ MeV.

 ^{34}Cl Levels

$E(\text{level})^\dagger$	Comments
0	$\sigma_{\text{int}}=3\text{-}13.4\text{ MeV}\times\text{mb.}$
150	$\sigma_{\text{int}}<9\text{ MeV}\times\text{mb.}$
460	$\sigma_{\text{int}}=12.1\text{ MeV}\times\text{mb.}$
670	$\sigma_{\text{int}}=3.6\text{-}19.5\text{ MeV}\times\text{mb.}$
1230	$\sigma_{\text{int}}=14.6\text{ MeV}\times\text{mb.}$
1890	$\sigma_{\text{int}}=2.3\text{ MeV}\times\text{mb.}$
2160	$\sigma_{\text{int}}<2.0\text{ MeV}\times\text{mb.}$
2180	$\sigma_{\text{int}}<34.7\text{ MeV}\times\text{mb.}$
2380	$\sigma_{\text{int}}<5.0\text{ MeV}\times\text{mb.}$
2610	
3130	
3330	
3380	
3550	
3630	$\sigma_{\text{int}}=21.6\text{ MeV}\times\text{mb.}$
3960	$\sigma_{\text{int}}=4.0\text{ MeV}\times\text{mb.}$
3980	$\sigma_{\text{int}}=2.0\text{ MeV}\times\text{mb.}$
4075	
4450	$\sigma_{\text{int}}=3.6\text{ MeV}\times\text{mb.}$
4460	
4606	$\sigma_{\text{int}}=17.8\text{ MeV}\times\text{mb.}$
4610	
4720	
4990	$\sigma_{\text{int}}<2.0\text{ MeV}\times\text{mb.}$
5010	

† From 1997Is02.

$^{35}\text{Cl}(\text{p,d})$ 1972Vi04

$^{35}\text{Cl} \rightarrow 1\text{n} + ^{34}\text{Cl}$ from $J^\pi = 3/2^+$ ^{35}Cl ground state.

1972Vi04: a 40-MeV proton beam was produced from the Grenoble variable-energy cyclotron. The target was chlorine gas of natural isotopic abundance. Deuterons were detected using a silicon telescope consisting of a 500- μm surface barrier ΔE detector, a 3-mm lithium-drifted silicon E detector, and another 3-mm lithium-drifted silicon E detector in anti-coincidence (FWHM=85 keV). Measured $\sigma(E_d, \theta)$. Deduced levels, J, π , L-transfers, and spectroscopic factors from DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

 ^{34}Cl Levels

Spectroscopic factor of is obtained by: $\sigma(\theta)_{\text{exp}} = 1.53[(2s_d+1)/(2s_p+1)][(2s_n+1)/(2(2L+1))] C^2 S \sigma(\theta)_{\text{DWBA}}$, where s_d , s_p , and s_n are the spins of the deuteron, proton, and neutron, respectively. 1.53 is a normalization factor related to the use of Hulthen wavefunction for the deuteron (1966Ba54). The overall uncertainty in $C^2 S$ is estimated to be about 30% (1972Vi04).

<u>E(level)[†]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>	<u>Comments</u>
0.0	2	0.45	
140 15	2	1.20	
450 15	2	0.28	
670 15	0	0.16	
1230 15	0	0.50	
1880 15	0	0.16	
2160 15	0	0.32	
2600 20			E(level): doublet (1972Vi04).
3360 20	0	0.75	
4160 20	0	0.30	E(level): possibly multiplet (1972Vi04).
5000 30	2	1.70	E(level): possibly multiplet (1972Vi04).

[†] From 1972Vi04.

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in 1972Vi04.

$^{35}\text{Cl}(\text{d},\text{t})$ [1981Bh04](#)

$^{35}\text{Cl} \rightarrow \text{n} + ^{34}\text{Cl}$ from $J^\pi=3/2^+$ ^{35}Cl ground state.

[1981Bh04](#): a 23.35-MeV deuteron beam was produced from the Argonne National Laboratory Cyclotron. Targets were prepared by evaporating lead chloride enriched in ^{35}Cl onto thin carbon films. Tritons were detected using a ΔE -E silicon detector telescope with FWHM=140 keV. Measured $\sigma(E_{\text{t}},\theta)$. Deduced levels, L-transfers, J, π , and spectroscopic factors from DWUCK-DWBA analysis.

 ^{34}Cl Levels

Spectroscopic factor $C^2\text{S}=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$, where $N=3.33$ is a normalization factor adopted from [1966Ba54](#).

<u>E(level)[†]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>
0	2	0.41
150	2	1.18
450	0+2	0.11+0.21
670	0+2	0.06+0.22
1210	0+2	0.23+0.09
1840	0+2	0.04+0.12

[†] From [1981Bh04](#), which did not give uncertainties.

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in [1981Bh04](#).

$^{35}\text{Cl}(^3\text{He},\alpha)$ 1974Va25

$^{35}\text{Cl} \rightarrow \text{In} + ^{34}\text{Cl}$ from $J^\pi=3/2^+$ ^{35}Cl ground state.

1974Va25: a 15-MeV ^3He beam was produced from the Utrecht 6-MV tandem accelerator. Targets were BaCl_2 99.3% enriched in ^{35}Cl on C backing. α particles were detected using seven position-sensitive detectors with FWHM=15 keV in the focal plane of the split-pole magnetic spectrograph. Measured $\sigma(E_\alpha, \theta)$. Deduced levels, J, π , L-transfers, and spectroscopic factors from DWBA analysis. Compared with mirror ^{34}S and discussed isospins.

 ^{34}Cl Levels

Spectroscopic factor $C^2S = \sigma(\theta)_{\text{exp}} / \sigma(\theta)_{\text{DWBA}} / N$, where $N=23.5$ is a normalization factor.

<u>E(level)[†]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>	<u>E(level)[†]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>	<u>E(level)[†]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>
0.0	2	0.44	3128	2	0.31	4206 10	0+2	0.20+0.05
146.36	2	1.5	3333	2	0.33	4321 10	0+2	0.007+0.04
461.00	0+2	0.09+0.26	3383	0+2	0.27+0.23	4608	1+3	0.007+0.04
665.55	0+2	0.03+0.31	3545.2	1	0.007	4715 10	2	0.20
1230.24	0+2	0.23+0.07	3632.5	3	0.030	4939 11	2	0.13
1887.5	0+2	0.06+0.14	3847 10			4958 11	2	0.30
2158.3	0+2	0.10+0.18	3937 10	0+2	0.003+0.07	4971 11	0+2	0.04+0.28
2180.7	2	0.07	3964 10	2	0.06	4998 12	0+2	0.08+0.56
2580.6	0	0.06	3982.1	3	0.040	5010 13	2	0.85
2611.2	2	0.23	4075 10	0+2	0.007+0.08			
2718 10	2	0.07	4136.6	0+2	0.011+0.04			

[†] E(level) values with uncertainties are measured by 1974Va25; E(level) values without uncertainties are quoted from 1973EnVA, which are also observed by 1974Va25 but without reporting their own values.

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in 1974Va25.

$^{36}\text{Ar}(\text{p}, ^3\text{He})$ 1969Br21

$^{36}\text{Ar} \rightarrow \text{p} + \text{n} + ^{34}\text{Cl}$ from $J^\pi=0^+$ ^{36}Ar ground state.

1969Br21: a 45-MeV proton beam was produced from the Berkeley 88-Inch Cyclotron. The target was 99.6%-enriched ^{36}Ar gas.

Reaction products were detected using a 155- μm phosphorus-diffused silicon ΔE transmission counter and a 3.0-mm lithium-drifted silicon E counter operated in coincidence, followed by a 0.5-mm lithium-drifted silicon E-reject counter operated in anti-coincidence with the first two counters to eliminate long-range particles. Measured $\sigma(\text{E}(^3\text{He}), \theta)$. Deduced levels, L-transfers, J, π , and isospin. Also measured $^{36}\text{Ar}(\text{p}, \text{t})^{34}\text{Ar}$ and $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$.

 ^{34}Cl Levels

Selection rules used by 1969Br21 for direct two-particle transfer reactions: $^{36}\text{Ar}(\text{p}, \text{t})^{34}\text{Ar}$ transfers particles with $T=1$, $S=0$; $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$ transfers particles with $T=0$, $S=1$. $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ transfers particles with $T=1$, $S=0$ if the transfer is direct two-particle transfer, $\sigma(\theta)$ of $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ matches $\sigma(\theta)$ of the mirror state in ^{34}Ar populated by $^{36}\text{Ar}(\text{p}, \text{t})^{34}\text{Ar}$, and absent in $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$.

E(level)	$L^\ddagger$	Comments
0.0	(0)	$T=1$
460 [†]		$T=0$
1231 [†]		$T=0$
		Weakly populated (1969Br21).
1891 [†]		$T=0$
		Weakly populated (1969Br21).
2162 [†]	(2)	$T=1$
2600 30		$T=0$
3130 30		$T=0$
3350 50	(2)	$T=1$
3940 40	(0)	$T=1$
4670 40	(3)	$T=1$
4970 40	(0)	J^π : $T=1$ component should be populated in $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ and corresponds to 0^+ . Observed in $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ and in $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$ and should be a doublet of $T=1$ and $T=0$.
5600 40		$T=0$
6160 40	(2)	J^π : $T=1$ component should be populated in $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ and corresponds to 2^+ . Observed in $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ and in $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$ and should be a doublet of $T=1$ and $T=0$.
7070 40		$T=0$

[†] 1969Br21 observed these levels in the $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ reaction and quoted E(level) from 1967En05 without reporting their own values (1969Br21).

[‡] From DWBA analysis of the mirror reaction $^{36}\text{Ar}(\text{p}, \text{t})^{34}\text{Ar}$ (1969Br21). The parentheses are added by evaluators.

$^{36}\text{Ar}(\text{d},\alpha),(\text{pol d},\alpha)$ 1972Ho02,1982Ta04,1969Br21

$^{36}\text{Ar} \rightarrow \text{lp} + \text{ln} + ^{34}\text{Cl}$ from $J^\pi=0^+$ ^{36}Ar ground state.

1972Ho02: a 19-MeV deuteron beam bombarded a 99.6%-enriched ^{36}Ar gas target. Outgoing α particles were momentum-analyzed by the Enge split-pole spectrograph and detected in 100- μm Ilford K0 nuclear emulsion plates. Measured $\sigma(E_\alpha, \theta)$. Deduced levels, L-transfers, L-mixings, J, π , and isospin with DWUCK-DWBA analysis and χ^2 versus L+L' analysis.

1982Ta04, 1980To13: a 16-MeV polarized deuteron beam from the TUNL Lamb-shift ion source bombarded a 99.9%-enriched ^{36}Ar target. Outgoing α particles were detected in four solid-state detectors with FWHM=100-200 keV. Measured $\sigma(E_\alpha, \theta)$, vector analyzing powers, $iT_{11}(E_\alpha, \theta)$, and tensor analyzing powers, $T_{20}(E_\alpha, \theta)$ and $T_{22}(E_\alpha, \theta)$. Deduced levels, L-transfers, J, π , and isospin with DWUCK-DWBA analysis.

1969Br21: a 45-MeV deuteron beam was produced from the Berkeley 88-Inch Cyclotron. The target was 99.6%-enriched ^{36}Ar gas. Reaction products were detected using a 155- μm phosphorus-diffused silicon ΔE transmission counter and a 3.0-mm lithium-drifted silicon E counter operated in coincidence, followed by a 0.5-mm lithium-drifted silicon E-reject counter operated in anti-coincidence with the first two counters to eliminate long-range particles. Measured $\sigma(E_\alpha, \theta)$. Deduced levels, L-transfers, J, π , and isospin. Also measured $^{36}\text{Ar}(\text{p}, \text{t})^{34}\text{Ar}$ and $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$.

Others: **2011En01** and **2012En01** measured cross section of $^{36}\text{Ar}(\text{d}, \alpha)^{34\text{m}}\text{Cl}$, $E \leq 8.4$ MeV.

 ^{34}Cl Levels

E(level) [†]	J^π [‡]	L [#]	Comments
0?			Very weakly populated at all angles due to parity selection rule and isospin selection rule of $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$ (1972Ho02).
147 20	3 ⁺	4	J^π : L-1 transfer from measured analyzing powers in 1982Ta04 . L: 4 from 1972Ho02 and 1982Ta04 .
459 20	1 ⁺	2	J^π : L-1 transfer from measured analyzing powers in 1982Ta04 . L: from 1982Ta04 ; other: L=0+2 in 1972Ho02 .
665 20	1 ⁺ , 3 ⁺	0+2, 2+4	J^π : L-transfer from measured analyzing powers in 1982Ta04 . L: from 1982Ta04 ; other: L=0+2 in 1972Ho02 .
1229 20	2 ⁺	2	
1890 20	2 ⁺	2	J^π : L-transfer from measured analyzing powers in 1982Ta04 . L: from 1982Ta04 ; other: L=2+4, 3+5 for an unresolved 1890 and 1920 doublet in 1972Ho02 .
2191 20	(3 ⁺)	(2)	J^π : L+1 transfer from measured analyzing powers in 1982Ta04 . L: from 1982Ta04 ; other: L=2+4 or pure L=4 in 1972Ho02 .
2382 20	1 ⁺	0+2	E(level): other: 3330 40 (1969Br21).
2589 20	1 ⁺	0+2	
2620 20			
2725 20	2 ⁻	1+3	
3146 20	1 ⁺	0+2	
3334 20	2 ⁻ , 3 ⁻ , 4 ⁻	3	
3377 20			
4970 40			E(level): likely determined from $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ (1969Br21). J^π : T=1 component should be populated in $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ and corresponds to L=0, $J^\pi=0^+$ (1969Br21). Observed in $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ and in $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$ and should be a doublet of T=1 and T=0 (1969Br21).
5600 40			E(level): likely determined from both $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$ and $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ data (1969Br21).
6160 40			E(level): likely determined from $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ (1969Br21). J^π : T=1 component should be populated in $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ and corresponds to L=2, $J^\pi=2^+$ (1969Br21). Observed in $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ and in $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$ and should be a doublet of T=1 and T=0 (1969Br21).
7070 40			E(level): likely determined from $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ data (1969Br21).

[†] From **1972Ho02**, unless otherwise noted.

[‡] Deduced from measured L values (**1972Ho02**), unless otherwise noted.

[#] From DWUCK-DWBA analysis of the measured $\sigma(\theta)$ and χ^2 versus L+L' analysis in **1972Ho02**, unless otherwise noted.

$^{40}\text{Ca}(\mu^-, \nu\alpha 2n\gamma)$ **2006Me08**

2006Me08: Muons were generated from the decay of 90-MeV/c π^- beam in a 6-m, 1.2-T superconducting solenoid and were selected by a bending magnet at the M9B beamline at TRIUMF. The target was 15-g natural calcium. γ rays and x rays were detected using a 186-cm³ HPGe detector. Charged particles caused by electrons from muon decay were tagged by a plastic scintillator in front of the HPGe detector. Measured E_γ , I_γ , $E(\text{x ray})$, $I(\text{x ray})$. Deduced levels, muon capture yields. 34CL t ----- 34CL t Muonic Lyman series for natural Calcium 34CL t 34CL t μ x ray Energy Intensity in percent 34CL t 34CL t 2p-1s 783.659 25 83.8 10 34CL t 3p-1s 940.63 10 6.2 2 34CL t 4p-1s 995.48 10 2.0 1 34CL t 5p-1s 1020.81 10 2.0 1 34CL t 6p-1s 1034.62 10 1.8 1 34CL t 7p-1s 1042.71 20 1.4 1 34CL t (8 $^-$ ∞)p-1s 1046-1063 2.8 4 34CL t ----- 34CL t ----- 34CL t Muonic Balmer series for natural Calcium 34CL t 34CL t μ x ray Energy Intensity in percent 34CL t 34CL t 3d-2p 157.35 13 64.5 9 34CL t 4d-2p 212.03 10 8.85 20 34CL t 5d-2p 237.31 10 4.34 20 34CL t 6d-2p 251.06 10 3.29 20 34CL t 7d-2p 259.45 10 1.37 20 34CL t (8 $^-$ ∞)d-2p 261-277 1.4 3 34CL t -----.

 ^{34}Cl Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\dagger$</u>
0	0 ⁺
146.4	3 ⁺
461.0	1 ⁺
665.6	1 ⁺
1230.3	2 ⁺

[†] From Adopted Levels. Level energies are rounded-off values.

 $\gamma(^{34}\text{Cl})$

<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
461.0	0.10 5	461.0	1 ⁺	0	0 ⁺	
564.7	0.08 8	1230.3	2 ⁺	665.6	1 ⁺	
665.6	<0.1	665.6	1 ⁺	0	0 ⁺	
769.3		1230.3	2 ⁺	461.0	1 ⁺	I_γ : intensity not listed in 2006Me08 ; the peak is obscured by other transitions.
1083.9	<0.15	1230.3	2 ⁺	146.4	3 ⁺	

[†] Rounded-off values from Adopted Gammas.

[‡] Percent γ -ray yield per muon capture ([2006Me08](#)).

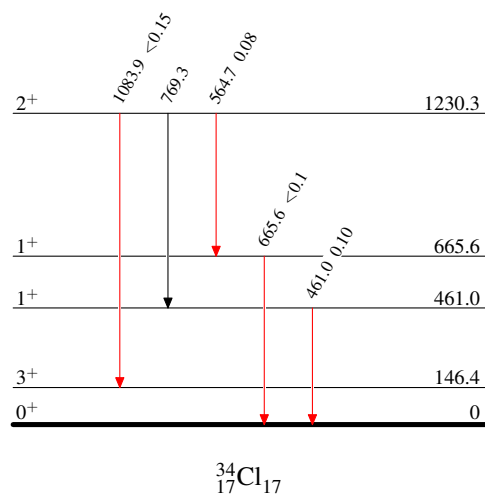
$^{40}\text{Ca}(\mu^-, \nu\alpha 2n\gamma)$ 2006Me08

Level Scheme

Intensities: percent γ -ray yield per muon capture

Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$



Adopted Levels, Gammas

$Q(\beta^-) = -16891.17$; $S(n) = 17065.34$; $S(p) = 4663.94$; $Q(\alpha) = -6743.9522$ [2021Wa16](#)

$Q(\beta^-)$: Deduced by evaluators using mass excesses of -1487.17 for ^{34}K measured by [2024Dr01](#); -18378.28980 for ^{34}Ar from [2021Wa16](#). Value from [2021Wa16](#): $Q(\beta^-) = -17160.200$ (syst).

$S(2n) = 32320.618$, $S(2p) = 6940.708$, $Q(\varepsilon) = 6061.796$, $Q(\varepsilon p) = 918.598$ ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): $^{32}\text{S}(^3\text{He}, n)$ at California Institute of Technology ([1966Mi11](#)).

^{34}Ar production:

[2012Zh06](#): ^9Be , $^{181}\text{Ta}(^{40}\text{Ar}, X)$ at $E(^{40}\text{Ar}) = 57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion- ablation, and HIPSE models. Studied target dependence of fragment cross sections.

^{34}Ar radius measurement: [2002Oz03](#).

^{34}Ar mass measurements: [2017Se09](#), [2014Sh14](#), [2002He23](#).

Theoretical calculations: [2024Xa01](#), [2022Zo01](#), [2022Ya19](#), [2021Ma33](#), [2006Or01](#), [2005Ob01](#), [2008Ba08](#) (pygmy E1 resonances), [2005Pa26](#), [2005Sa59](#), [2005Te07](#), [2004Ge02](#), [2001Ot04](#), [1999Ng03](#), [1998La02](#), [1997Ma77](#), [1997Pa38](#), [1995He18](#), [1987Sa19](#), [1977Sh13](#), [1977Sz03](#), [1975Sh10](#).

 ^{34}Ar LevelsCross Reference (XREF) Flags

A	^{35}Ca εp decay (25.7 ms)	E	$^4\text{He}(^{30}\text{S}, \alpha)$: resonances	I	$^{32}\text{S}(^3\text{He}, n\gamma)$
B	$^1\text{H}(^{34}\text{Ar}, p')$	F	$^9\text{Be}(^{37}\text{Ca}, ^{34}\text{Ar}\gamma)$	J	$^{36}\text{Ar}(p, t)$
C	$^1\text{H}(^{35}\text{Cl}, 2n)$	G	$^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$	K	Coulomb excitation
D	$^3\text{He}(^{32}\text{S}, n\gamma)$	H	$^{32}\text{S}(^3\text{He}, n)$		

E(level)	J^π	$T_{1/2}$	XREF	Comments
0.0	0^+	0.84646 s ³⁵	ABCD FGHIJK	$\% \varepsilon + \% \beta^+ = 100$ J^π : $L=0$ from 0^+ in (p,t) and ($^3\text{He}, n$). $T_{1/2}$: from $\beta(t)$ using two component fit to the decay of ^{34}Ar and ^{34}Cl (2020Ia01). Others: 0.8438 s ⁴ (2006Ia05), 0.8445 s ³⁴ (1974Ha26 , 1972Ha58), 0.74 s ¹¹ (1971Mo27), 0.85 s ¹⁰ (1967Ba36), 1.2 s ³ (1966Mi11). The two most precise results (2006Ia05 and 2020Ia01) differ by seven σ and 2020Ia01 has a comment on 2006Ia05 : at that time many of the specialized features had not yet been optimally employed, although they only recognized the impact of these deficiencies quite recently. Before subsequent half-life measurements were made of other similar cases they had rectified the deficiencies and optimally applied all the features that they use here to revisit the ^{34}Ar half-life. 2020Ia01 offers the new result as a replacement for their earlier 2006Ia05 , which they withdraw. Evaluated rms nuclear charge radius $R = 3.3654$ fm ⁴⁰ (2013An02). XREF: H(2103) J^π : $L=2$ from 0^+ in (p,t) and ($^3\text{He}, n$). $T_{1/2}$: from lifetime $\tau = 285$ fs ⁷⁰ : unweighted average of 460 fs ⁶⁰ from $^3\text{He}(^{32}\text{S}, n\gamma)$ (1985A118 , DSAM), 330 fs ⁸⁰ from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1974Be18 , DSAM), 200 fs ⁶⁰ from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1974Gr19 , DSAM), and 150 fs ⁵⁰ from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1972Ca22 , DSAM). J^π : $L=2$ from 0^+ in (p,t). $T_{1/2}$: from lifetime $\tau = 197$ fs ⁵⁰ : weighted average of 280 fs ⁵⁰ from $^3\text{He}(^{32}\text{S}, n\gamma)$ (1985A118 , DSAM), 215 fs ⁷⁰ from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1974Be18 , DSAM), 130 fs ⁶⁰ from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1974Gr19 , DSAM), and 120 fs ⁶⁵ from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1972Ca22 , DSAM). XREF: H(3900) J^π : $L=0$ from 0^+ in (p,t). Spin=0 from $n\gamma(\theta)$ in $^{32}\text{S}(^3\text{He}, n\gamma)$. $T_{1/2}$: from lifetime $\tau > 600$ fs from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1974Gr19 , DSAM). Other: > 270 fs from $^3\text{He}(^{32}\text{S}, n\gamma)$ (1985A118 , DSAM).
2091.3 ⁴	2^+	0.20 ps ⁵	ABD FGHIJK	
3288.5 ⁵	2^+	0.14 ps ³	A D FGHIJ	
3875.9	0^+	> 0.42 ps	A D GHIJ	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{34}Ar Levels (continued)

E(level)	J^π	$T_{1/2}$	XREF	Comments
4020.6 13	$(2^+)^{\dagger}$		GH J	XREF: H(4050)J(4035) J^π : 4019.8 γ to 0^+ , g.s.
4129.7 14	$(1,2)^+{\dagger}$	9×10^1 fs +12-9	D GHI	XREF: H(4150) J^π : spin=2 is favored from $n\gamma(\theta)$ in $^{32}\text{S}(^3\text{He},n\gamma)$, which did not exclude spin=1, 3, or 4 (1974Gr19); spin=4 ruled out by 841.2 γ , M1+E2, to 2^+ , 3288.5 level; 1^+ assigned in $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$ based on mirror nucleus ^{34}S . $T_{1/2}$: from lifetime $\tau=130$ fs +170-130 from $^3\text{He}(^{32}\text{S},n\gamma)$ (1985A118, DSAM). Others: <350 fs from $^{32}\text{S}(^3\text{He},n\gamma)$ (1974Be18, DSAM); >400 fs from $^{32}\text{S}(^3\text{He},n\gamma)$ (1974Gr19, DSAM).
4515.8 12	$3^{-\dagger}$	0.20 ps 6	B GHIJ	J^π : L=3 from 0^+ in $^{32}\text{S}(^3\text{He},n)$; spin=3 from $n\gamma(\theta)$ in $^{32}\text{S}(^3\text{He},n\gamma)$. $T_{1/2}$: from lifetime $\tau=290$ fs 80: weighted average of 390 fs 180 from $^{32}\text{S}(^3\text{He},n\gamma)$ (1974Be18, DSAM), 260 fs 80 from $^{32}\text{S}(^3\text{He},n\gamma)$ (1974Gr19, DSAM), and 300 fs 80 from $^{32}\text{S}(^3\text{He},n\gamma)$ (1972Ca22, DSAM).
4643.8 9	$(4^+)^{\dagger}$		G IJ	XREF: I(4631) J^π : 2552.4 γ , Q, $\Delta J=2$ to 2^+ , 2091.3 level in ($^{24}\text{Mg}, ^{34}\text{Ar}\gamma$).
4851.3 8	$(3^+)^{\dagger}$		G	
4865 4			Ij	E(level): from $^{32}\text{S}(^3\text{He},n\gamma)$.
4881.0 12	$(2^+)^{\dagger}$		G j	J^π : 4881.9 γ to 0^+ , g.s.
4961.8 17	$0^+{\dagger}$		GHIJ	XREF: H(4950)I(4967) J^π : L=0 from 0^+ in (p,t) and ($^3\text{He},n$).
4966.6 11	$2^{(-)\dagger}$		G	J^π : 2875.2 γ , D, $\Delta J=0$ to 2^+ , 2091.3 level from $\gamma(\theta)$ in ($^{24}\text{Mg}, ^{34}\text{Ar}\gamma$).
5060.3 12	$(1^+)^{\dagger}$		G	
5262 15			IJ	XREF: I(5255?) E(level): from $^{36}\text{Ar}(p,t)$. Other: 5255 4 (tentative) from $^{32}\text{S}(^3\text{He},n\gamma)$.
5316 13	(5^-)		H J	E(level): weighted average of 5310 30 from $^{32}\text{S}(^3\text{He},n)$ and 5317 13 from $^{36}\text{Ar}(p,t)$. J^π : L=(5) from 0^+ in ($^3\text{He},n$).
5542 4			IJ	E(level): from $^{32}\text{S}(^3\text{He},n\gamma)$. Other: 5535 18 from $^{36}\text{Ar}(p,t)$.
5629 5	2^+		H J	E(level): weighted average of 5620 30 from $^{32}\text{S}(^3\text{He},n)$ and 5629.6 45 from $^{36}\text{Ar}(p,t)$. J^π : L=2 from 0^+ in ($^3\text{He},n$).
5929 20	0^+		H J	XREF: H(5930) E(level): weighted average of 5930 50 from $^{32}\text{S}(^3\text{He},n)$ and 5929 20 from $^{36}\text{Ar}(p,t)$. J^π : L=0 from 0^+ in ($^3\text{He},n$).
6050 4	2^+		J	J^π : L=2 from 0^+ in (p,t).
6521 15			H J	XREF: H(6470) E(level): weighted average of 6470 30 from $^{32}\text{S}(^3\text{He},n)$ and 6525 9 from $^{36}\text{Ar}(p,t)$.
6641 14			J	
6723 22			J	
6807 4			H J	XREF: H(6820) E(level): from $^{36}\text{Ar}(p,t)$. Other: 6820 40 from $^{32}\text{S}(^3\text{He},n)$.
6917 5			J	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{34}Ar Levels (continued)

E(level)	J^π	XREF	Comments
6.99×10 ³ 5		H	
7072 8		J	
7276 2		h J	XREF: h(7300)
7340 11	2 ⁺	h J	XREF: h(7300)J(7340)
			J^π : L=2 from 0 ⁺ in (p,t).
7502 16	(2 ⁺)	J	J^π : L=(2) from 0 ⁺ in (p,t).
7921 18		J	
8166 3		J	
8660 9		J	
8746 8		J	
8899 8		J	
9004 4		J	
9148 22		J	
9226 18		J	
9549 16		J	
1.109×10 ⁴ 9	(2 ⁺ ,4 ⁺)	E	J^π : from R-Matrix analysis in $^4\text{He}(^{30}\text{S},\alpha)$:resonances.
1.152×10 ⁴ 9	(2 ⁺)	E	J^π : from R-Matrix analysis in $^4\text{He}(^{30}\text{S},\alpha)$:resonances.
1.208×10 ⁴ 10	(2 ⁺)	E	J^π : from R-Matrix analysis in $^4\text{He}(^{30}\text{S},\alpha)$:resonances.

[†] J^π assignments supported by comparison with mirror nucleus ^{34}S in $^{12}\text{C}(^{24}\text{Mg},^{34}\text{Ar})$.

 $\gamma(^{34}\text{Ar})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.	δ	Comments
2091.3	2 ⁺	2091.1 4	100	0.0	0 ⁺	[E2]		B(E2)(W.u.)=11 +4-2 E _γ : weighted average of 2091.3 5 from $^{12}\text{C}(^{24}\text{Mg},^{34}\text{Ar})$, 2090.9 4 from $^{32}\text{S}(^3\text{He},n\gamma)$, and 2091 2 from Coulomb excitation. Other: 2098 4 from $(^{37}\text{Ca},^{34}\text{Ar})$.
3288.5	2 ⁺	1197.0 5	100 4	2091.3	2 ⁺	M1+E2	+0.12 5	B(M1)(W.u.)=0.079 +22-14; B(E2)(W.u.)=3.1 +34-21 E _γ : weighted average of 1197.5 4 from $^{12}\text{C}(^{24}\text{Mg},^{34}\text{Ar})$ and 1196.5 4 from $^{32}\text{S}(^3\text{He},n\gamma)$. Other: 1193 2 from $(^{37}\text{Ca},^{34}\text{Ar})$. I _γ : other: 100 6 from $^{32}\text{S}(^3\text{He},n\gamma)$. Mult.,δ: D+Q from $n\gamma(\theta)$ in $(^3\text{He},n\gamma)$; E1+M2 ruled out by RUL.
		3288.9 10	13.9 16	0.0	0 ⁺	[E2]		B(E2)(W.u.)=0.20 +6-4 E _γ : weighted average of 3289.1 10 from $^{12}\text{C}(^{24}\text{Mg},^{34}\text{Ar})$ and 3286 4 from $^{32}\text{S}(^3\text{He},n\gamma)$. I _γ : weighted average of 14.0 16 from $^{12}\text{C}(^{24}\text{Mg},^{34}\text{Ar})$ and 12 6 from $^{32}\text{S}(^3\text{He},n\gamma)$. I _γ : from $^{32}\text{S}(^3\text{He},n\gamma)$.
3875.9	0 ⁺	587.4	<43	3288.5	2 ⁺	[E2]		E _γ : weighted average of 1784.8 8 from $^{12}\text{C}(^{24}\text{Mg},^{34}\text{Ar})$ and 1782 3 from $^{32}\text{S}(^3\text{He},n\gamma)$. I _γ : from $^{32}\text{S}(^3\text{He},n\gamma)$.
		1784.6 8	100	2091.3	2 ⁺	[E2]		
4020.6	(2 ⁺)	1930.4 23	65 12	2091.3	2 ⁺			
		4019.8 15	100 12	0.0	0 ⁺			

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Ar})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.	δ	Comments
4129.7	(1,2) ⁺	841.2 13	100 6	3288.5	2 ⁺	M1+E2	+0.52 15	E_γ : unweighted average of 842.5 7 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$ and 839.9 9 from $^{32}\text{S}(^3\text{He}, n\gamma)$. I_γ : from $^{32}\text{S}(^3\text{He}, n\gamma)$. Other: 100 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$. Mult., δ : from $n\gamma(\theta)$ in ($^3\text{He}, n\gamma$); E1+M2, E2+M3, M2+E3 ruled out by RUL. I_γ : from $^{32}\text{S}(^3\text{He}, n\gamma)$. I_γ : from $^{32}\text{S}(^3\text{He}, n\gamma)$.
4515.8	3 ⁻	2038.3 4129.4 1227.2 12	11 6 <11 100 5	2091.3 2 ⁺ 0.0 0 ⁺ 3288.5 2 ⁺		[E1]		B(E1)(W.u.)=0.0015 +7-4 E_γ : unweighted average of 1228.4 5 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$ and 1226.0 6 from $^{32}\text{S}(^3\text{He}, n\gamma)$. I_γ : from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$. Other: 100 11 from $^{32}\text{S}(^3\text{He}, n\gamma)$. Mult.: D, $\Delta J=1$ from $\gamma(\theta)$ in ($^{24}\text{Mg}, ^{34}\text{Ar}\gamma$); $\Delta\pi$ =yes from level scheme.
		2424.7 22	12.9 32	2091.3 2 ⁺		[E1]		B(E1)(W.u.)= 2.6×10^{-5} +13-8 I_γ : weighted average of 14.6 32 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$ and 11.1 33 from $^{32}\text{S}(^3\text{He}, n\gamma)$. Mult.: Q, $\Delta J=2$ from $\gamma(\theta)$ in ($^{24}\text{Mg}, ^{34}\text{Ar}\gamma$).
4643.8	(4 ⁺)	2552.4 8	100	2091.3 2 ⁺		Q		
4851.3	(3 ⁺)	1562.8 7 2759.9 12	49.2 34 100 8	3288.5 2 ⁺ 2091.3 2 ⁺				
4865		2774	100	2091.3 2 ⁺				
4881.0	(2 ⁺)	1592.5 17 2788.9 19	32 5 100 12	3288.5 2 ⁺ 2091.3 2 ⁺				
		4881.9 24	82 14	0.0 0 ⁺				
4961.8	0 ⁺	832.1 9	100	4129.7 (1,2) ⁺				
4966.6	2 ⁽⁻⁾	2875.2 10	100	2091.3 2 ⁺		D		Mult.: D, $\Delta J=0$ from $\gamma(\theta)$ in ($^{24}\text{Mg}, ^{34}\text{Ar}\gamma$).
5060.3	(1 ⁺)	1771.8 11 5059.9 [‡]	100 8 <28	3288.5 2 ⁺ 0.0 0 ⁺				
5262		3171	100	2091.3 2 ⁺				
5542		898 1026	54 16 100 16	4643.8 (4 ⁺) 4515.8 3 ⁻				I_γ : tentative, from $^{32}\text{S}(^3\text{He}, n\gamma)$. I_γ : tentative, from $^{32}\text{S}(^3\text{He}, n\gamma)$.

[†] From $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$ unless otherwise noted. E_γ values without uncertainties are deduced from level-energy differences.

[‡] Placement of transition in the level scheme is uncertain.

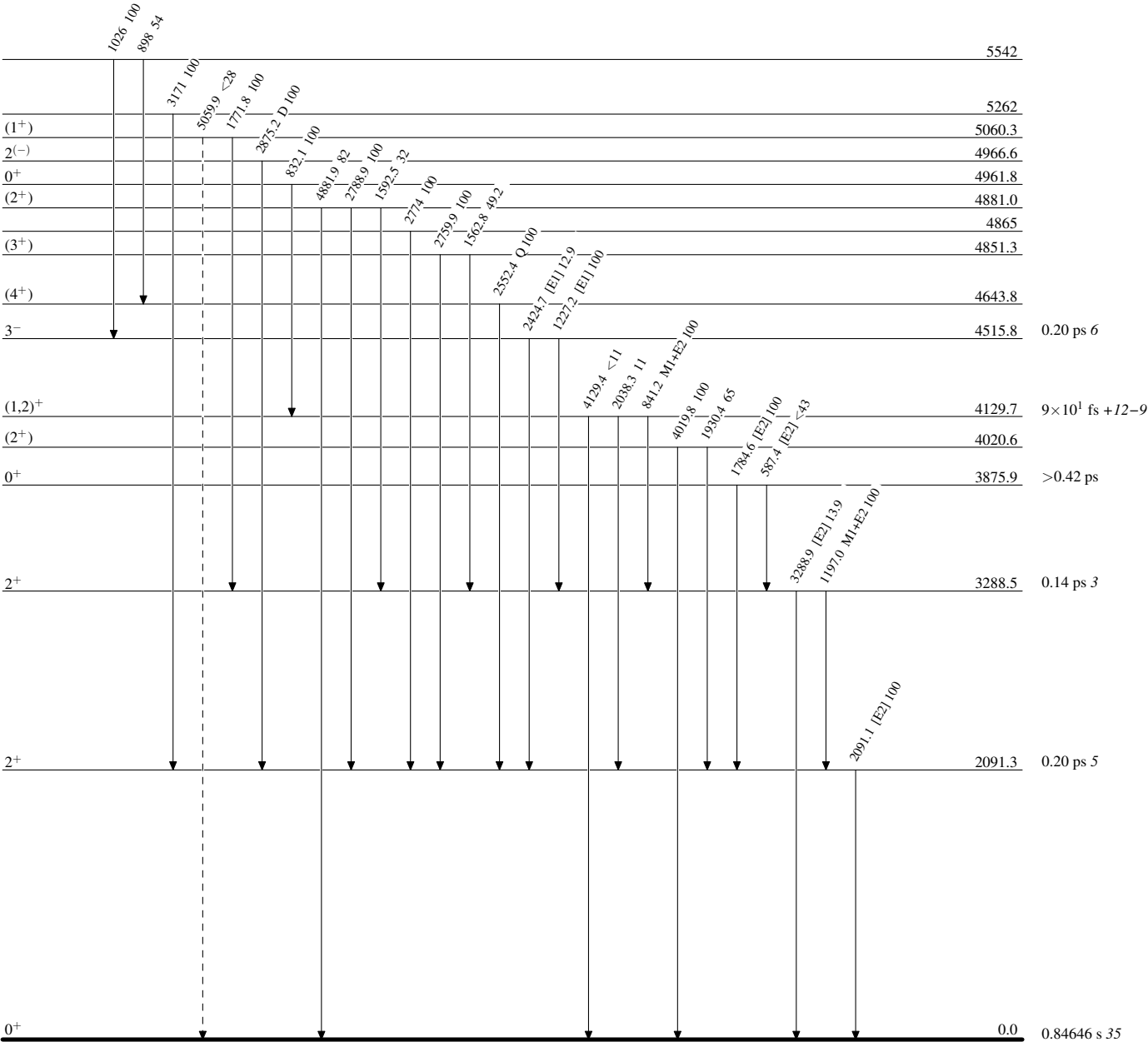
Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)



³⁴Ar₁₆

^{35}Ca εp decay (25.7 ms) 1999Tr04,1985Ay01

Parent: ^{35}Ca : $E=0$; $J^\pi=1/2^+$; $T_{1/2}=25.7$ ms 2; $Q(\varepsilon\text{p})=1.587\times 10^4$ 11; % εp decay=95.8 3

^{35}Ca - J^π , $T_{1/2}$: From the Adopted Levels of ^{35}Ca (2026Su18).

^{35}Ca - $Q(\varepsilon\text{p})$: Deduced by evaluators from mass excesses of 4777 105 for ^{35}Ca measured by 2023La09 and -18378.289 80 for ^{34}Ar from 2021Wa16. $Q(\varepsilon\text{p})$ from 2021Wa16: $Q(\varepsilon\text{p})=16280$ 200 (syst).

^{35}Ca -% εp decay: 95.8 3 derived from the renormalization of % $\Sigma\text{I}(1\text{p})$ +% $\Sigma\text{I}(2\text{p})=100.6$ in 1999Tr04 to 100. The original decay branching ratios in 1999Tr04: % $\Sigma\text{I}(1\text{p})=96.4$ 18 and % $\Sigma\text{I}(2\text{p})=4.2$ 3.

1999Tr04, 1998Le45: A secondary ^{35}Ca beam at 0.3 ions/s and 98% purity was produced via the projectile fragmentation of a 95-MeV/nucleon $^{40}\text{Ca}^{20+}$ primary beam impinging on a rotating $^{\text{nat}}\text{Ni}$ target, selected using ΔE -ToF by the GANIL LISE3 spectrometer, and implanted into a 500- μm -thick silicon detector sandwiched between two 500- μm -thick silicon detectors for detecting β^+ particles. 3.5×10^4 ^{35}Ca ions were stopped at a depth of 300 μm with FWHM=70 μm (setting 1) and 2.5×10^4 ^{35}Ca ions were stopped at a depth of 450 μm (setting 2). ε + β^+ -delayed protons were detected by the implantation detector. γ rays were detected by three Ge detectors and two NaI detectors. Measured E_{p} , I_{p} , $E\gamma$, $I\gamma$, $E_{2\text{p}}$, $I_{2\text{p}}$, βp -coin, $\text{p}\gamma$ -coin. Built the decay scheme consisting of 1p-emitting states in ^{35}K , a 2p-emitting state ($T=5/2$ IAS) in ^{35}K , 1p daughter states in ^{34}Ar , and a 2p daughter state in ^{33}Cl . Deduced decay branching ratios, B(GT) and B(F), and parent ^{35}Ca $T_{1/2}$ from implant-decay correlations.

1985Ay01: ^{35}Ca isotope discovery. ^{35}Ca was produced by bombarding a natural calcium target using a 135-MeV ^3He beam from the 88-inch Cyclotron at Lawrence Berkeley Laboratory. Recoiling products were slowed down, transported, and collected on a slowly rotating catcher wheel. The ε + β^+ -delayed protons were detected using Si detector telescopes. Measured E_{p} , I_{p} , pp -coin. Built the decay scheme consisting of a 2p-emitting state ($T=5/2$ IAS) in ^{35}K , sequential 2p intermediate states in ^{34}Ar , and 2p daughter states in ^{33}Cl . Deduced ^{35}Ca $T_{1/2}$ and ^{35}Ca mass using the known members of $A=35$, $T=5/2$ sextuplets IMME.

Theoretical studies involving ^{35}Ca decay: 2003Sm02, 1991De26, 1990Br26.

 ^{34}Ar Levels

$E(\text{level})^\dagger$	$J^\pi^\dagger$
0	0^+
2091.3 4	2^+
3288.5 5	2^+
3875.9	0^+

† From the Adopted Levels.

Delayed Protons (^{34}Ar)

Particle normalization: from the renormalization of % $\Sigma\text{I}(1\text{p})$ +% $\Sigma\text{I}(2\text{p})=100.6$ in 1999Tr04 to 100.

$E(\text{p})^\dagger$	$E(^{34}\text{Ar})$	$I(\text{p})^{\dagger a}$	$E(^{35}\text{K})^\ddagger$	Comments
1427 5	0	48.5 13	1553	
2278 [#]	2091.3	5.4 9	4520 [@]	$E(\text{p})$: unresolved $E(\text{p})_{\text{lab}}=1909$ -2647 from 1999Tr04. $I(\text{p})$: 64.0% 93 of % $I(\text{p})_{\text{tot}}=8.4$ 6 from $\text{p}\gamma$ coincidences in 1999Tr04.
2278 [#]	3288.5	1.0 4	5716 [@]	$E(\text{p})$: unresolved $E(\text{p})_{\text{lab}}=1909$ -2647 from 1999Tr04. $I(\text{p})$: 12.1% 49 of % $I(\text{p})_{\text{tot}}=8.4$ 6 from $\text{p}\gamma$ coincidences in 1999Tr04.
2278 [#]	3875.9	2.0 7	6302 [@]	$E(\text{p})$: unresolved $E(\text{p})_{\text{lab}}=1909$ -2647 from 1999Tr04. $I(\text{p})$: 23.9% 79 of % $I(\text{p})_{\text{tot}}=8.4$ 6 from $\text{p}\gamma$ coincidences in 1999Tr04.
2727 13	2091.3	6.0 5	4982	
3224 [#]	2091.3	2.2 3	5493 [@]	$E(\text{p})$: unresolved $E(\text{p})_{\text{lab}}=2947$ -3500 from 1999Tr04.
3592 25	0	3.0 3	3781	
3822 36	0	3.8 3	4018	
4041 71	2091.3	2.9 3	6335	
4570 48	0	2.9 3	4788	
4754 38	0	4.2 4	4977	
5018 71	0	3.9 3	5249	

Continued on next page (footnotes at end of table)

^{35}Ca ε p decay (25.7 ms) **1999Tr04,1985Ay01** (continued)

Delayed Protons (continued)

$E(p)^{\dagger}$	$E(^{34}\text{Ar})$	$I(p)^{\ddagger a}$	$E(^{35}\text{K})^{\ddagger}$	Comments
5294 48	0	0.72 18	5533	
5466 48	0	0.61 15	5710	
5616 37	0	1.43 17	5865	
5834 60	0	1.40 19	6089	
6316 [#]	0	1.09 17	6585 [@]	E(p): unresolved $E(p)_{\text{lab}}=5983\text{--}6649$ from 1999Tr04 .
6783 22	2091.3	3.8 2	9168 ^{&}	$E(^{35}\text{K})$: $E(p1)_{\text{lab}}=6783\text{ }22$ corresponds to $E(^{35}\text{K})=9157\text{ }23$.
7509 [#]	0	1.1 2	7813 [@]	E(p): unresolved $E(p)_{\text{lab}}=7131\text{--}7887$ from 1999Tr04 .
8802 89	0	0.41 6	9168 ^{&}	$E(^{35}\text{K})$: $E(p0)_{\text{lab}}=8802\text{ }89$ corresponds to $E(^{35}\text{K})=9144\text{ }92$.

[†] From **1999Tr04**. $E(p)$ is in lab frame. $I(p)$ is obtained from the number of observed proton events and the total number of implants, with simulated proton-detection efficiencies.

[‡] Evaluators deduced $E(\text{level})(^{35}\text{K})=E(p)_{\text{lab}}\times[m(p)+m(^{34}\text{Ar})]/m(^{34}\text{Ar})+S(p)(^{35}\text{K})+E(\text{level})(^{34}\text{Ar})$, with $S(p)(^{35}\text{K})=83.6\text{ }5$ (**2021Wa16**), $E(\text{level})(^{34}\text{Ar})$ from the Adopted Levels of ^{34}Ar , and $E(p)_{\text{lab}}$ from **1999Tr04**. For a ^{35}K proton-emitting level with multiple proton branches, evaluators take the weighted average for $E(\text{level})(^{35}\text{K})$ values deduced from each proton branch. **1999Tr04** used $S(p)(^{35}\text{K})=78\text{ }20$ from **1993Au07**, which causes a small difference between the original $E(\text{level})(^{35}\text{K})$ in **1999Tr04** and the deduced $E(\text{level})(^{35}\text{K})$ here.

[#] Unresolved protons.

[@] Unresolved proton-emitting levels in ^{35}K corresponding to a group of unresolved protons populating one daughter state in ^{34}Ar .

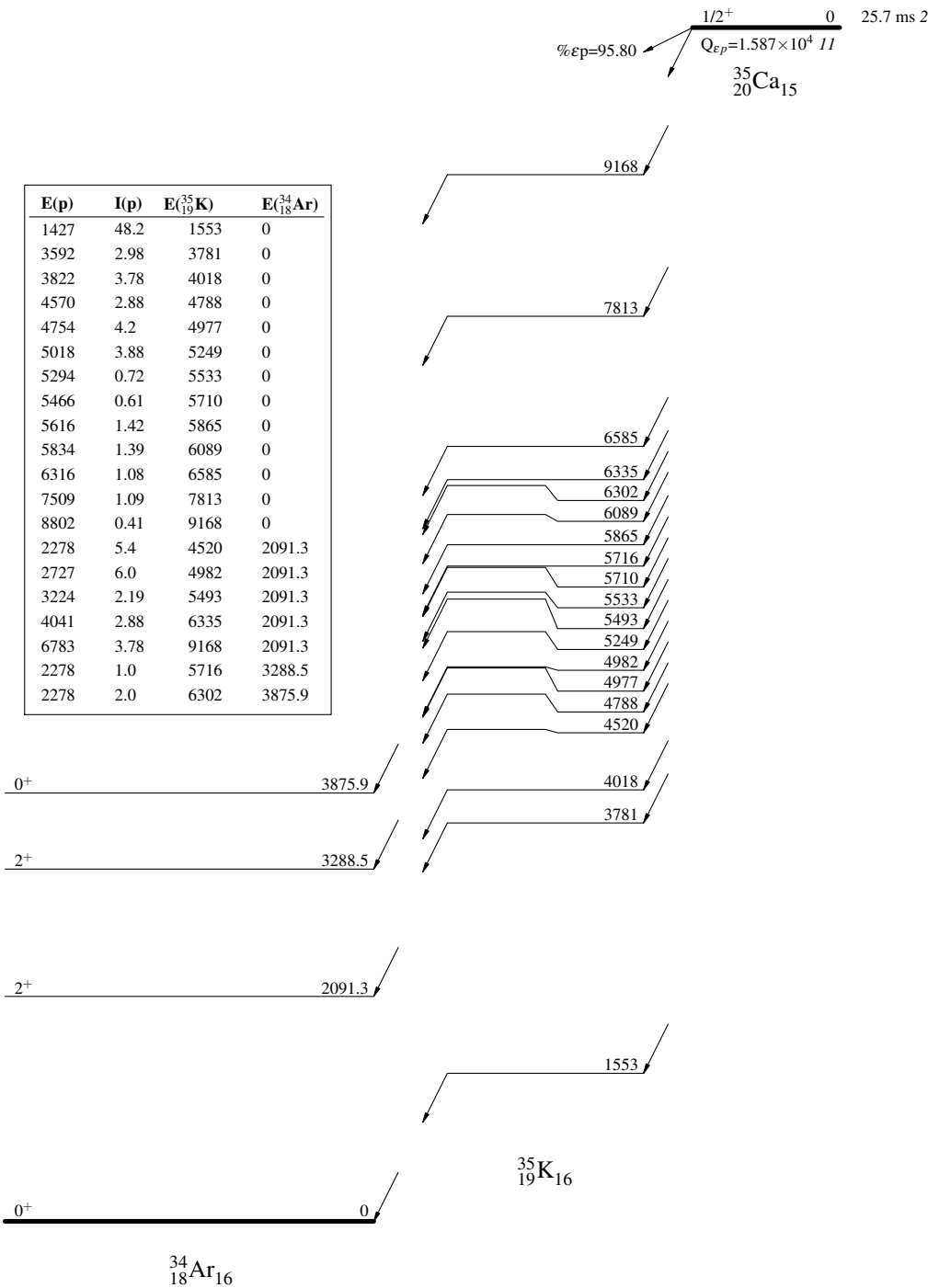
[&] $T=5/2$ isobaric analog state in ^{35}K , which also decays by $E(2p0)_{\text{lab}}=4305\text{ }26$, proton line 8 in **1999Tr04**, corresponding to $E(^{35}\text{K})=9186\text{ }27$ with $S(2p)(^{35}\text{K})=4747.5\text{ }6$ (**2021Wa16**) and adding a $+7\text{--}keV$ correction for the difference in the recoil effect between $1p$ and $2p$ emissions (**1999Tr04**). $E(^{35}\text{K})=9168\text{ }23$ is obtained from a weighted average of the three $E(^{35}\text{K})$ values of $9144\text{ }92$ ($p0$), $9157\text{ }23$ ($p1$), and $9186\text{ }27$ ($2p0$). Other: $E(2p0)_{\text{lab}}=4089\text{ }30$, $E(2p0)_{\text{c.m.}}=4311\text{ }40$ (**1985Ay01**), corresponding to $E(^{35}\text{K})=9059\text{ }41$ with $S(2p)(^{35}\text{K})=4747.5\text{ }6$ (**2021Wa16**). **1985Ay01** observed $E(2p1)_{\text{lab}}=3287\text{ }30$ and proposed both $2p0$ and $2p1$ proceed via a sequential decay mechanism with the first proton $E(p)_{\text{lab}}=2213\text{ }keV$, corresponding to an intermediate state in ^{34}Ar at $6807\text{ }keV$. However, $2p1$ has been ruled out in **1999Tr04** due to the nonobservation of expected py coincidences. **1999Tr04** also states that the observed ratio $I(2p0)/I(1p)=0.98\text{ }9$ agrees with the calculated branching ratio $I(2p)/I(1p)=1$ for the IAS (**1991De26**).

^a For absolute intensity per 100 decays, multiply by 0.994.

^{35}Ca ϵp decay (25.7 ms) 1999Tr04,1985Ay01

Decay Scheme

I(p) Intensities: I(p) per 100 parent decays



$^1\text{H}(^{35}\text{Cl}, 2\text{n})$ **2020Ia01**

2020Ia01: a 30-MeV/nucleon ^{35}Cl beam was produced from the K500 cyclotron at Texas A&M University and impinged on a liquid-nitrogen-cooled hydrogen gas target. The fully stripped reaction products recoiled into the Momentum Achromat Recoil Spectrometer (MARS), where they were spatially separated according to their charge-to-mass ratio. A 99.5% pure ^{34}Ar beam was implanted into aluminized Mylar tape and moved to a well-shielded location, where a 4π proportional gas counter was used for detecting positrons. A total of 1.7×10^8 decay positrons were recorded. Measured $E\beta$, $\beta(t)$. Deduced $T_{1/2}$ of ^{34}Ar .

2020Ia02: similar ^{34}Ar production setup at MARS. A 99.6% pure ^{34}Ar beam was implanted into aluminized Mylar tape and moved to the counting location, where a 1-mm-thick BC-404 scintillator was used to detect positrons and a HPGe detector was used to detect γ rays. Measured $E\beta$, $E\gamma$, $\beta(t)$, $\gamma(t)$, $\beta\gamma$ coin. A total of 2.6710×10^8 20 decay positrons were recorded.

 ^{34}Ar Levels

<u>E(level)</u>	<u>J^π</u>	<u>$T_{1/2}$</u>	<u>Comments</u>
0.0	0^+	0.84646 s 35	$T_{1/2}$: from $\beta(t)$ using two component fit to the decay of ^{34}Ar and ^{34}Cl (2020Ia01). The $T_{1/2}$ of ^{34}Cl was fixed to the world average of 1526.55 ms 44. The $T_{1/2}$ of ^{34}Ar with only statistical uncertainty is 0.84646 s 19.

$^1\text{H}(^{34}\text{Ar},\text{p}')$ [2001Kh17](#)

[2001Kh17](#): a 47-MeV/nucleon ^{34}Ar beam was produced via the projectile fragmentation of a 95-MeV/nucleon ^{36}Ar primary beam impinging on a ^{12}C target at GANIL. The secondary target was CH_2 . The elastic and inelastic protons were detected using the MUST Si-Si(Li)-CsI telescope and heavy nuclei were analyzed by the SPEC spectrometer. Measured $\sigma(E_p, \theta)$. Deduced levels, J, π , and deformation parameters from Coupled-Channel Born Approximation (CCBA) using the ECIS code and TAMURA-DWBA analyses of the measured $\sigma(\theta)$. Also see [2000KhZX](#), [2001BI17](#), [2003La13](#) for more discussions on the same data.

 ^{34}Ar Levels

E(level)	J π [†]	Comments
0	0 ⁺	Ratio of quadrupole matrix elements $M_n/M_p/(N/Z)=1.35$ 28 from phenomenological analysis (2001Kh17) and 0.99 17 from the QRPA transition densities normalized to the data in 2001Kh17 .
1.9×10^3 2	2 ⁺	$\beta_2=0.27$ 2 (2001Kh17)
4.5×10^3 2	3 ⁻	$\beta_3=0.39$ 3 (2001Kh17)

[†] From CCBA and DWBA analyses of measured $\sigma(\theta)$ in [2001Kh17](#).

³He(³²S,n γ) 1985A118

1985A118: inverse ³He(³²S,n γ) reaction with 80-MeV ³²S beam on 35-keV ³He-impinged Au foils. Neutrons were detected using an NE213 scintillator and γ rays were detected using 3 Ge(Li) detectors. Measured n γ coin. Deduced lifetimes for 4 levels using the Doppler-Shift Attenuation Method (DSAM).

³⁴Ar Levels

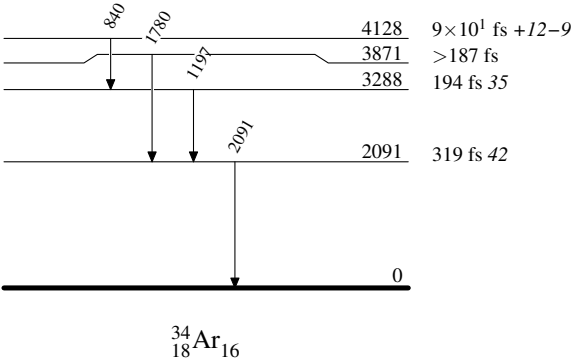
<u>E(level)</u>	<u>T_{1/2}</u>	<u>Comments</u>
0		
2091	319 fs 42	T _{1/2} : lifetime τ =460 fs 60 based on 490 fs 65 (lineshape) and 410 fs 80 (centroid-shift).
3288	194 fs 35	T _{1/2} : lifetime τ =280 fs 50 based on 255 fs 50 (lineshape) and 385 fs 95 (centroid-shift).
3871	>187 fs	T _{1/2} : lifetime τ >270 fs (lineshape). >200 fs (centroid-shift).
4128	9 $\times$ 10 ¹ fs +12-9	T _{1/2} : lifetime τ =130 fs +170-130 (centroid-shift). <310 fs (lineshape).

γ (³⁴Ar)

<u>E_{γ}</u>	<u>E_i(level)</u>	<u>E_f</u>
840	4128	3288
1197	3288	2091
1780	3871	2091
2091	2091	0

³He(³²S,n γ) 1985A118

Level Scheme



³⁴Ar₁₆

$^4\text{He}(^{30}\text{S},\alpha)$:resonances **2018Ka11**

$^{30}\text{S}+\alpha$ resonant elastic scattering. $J^\pi=0^+$ for ^{30}S g.s. and ^4He g.s.

2018Ka11: 48.4-MeV ^{20}S was produced via the $^3\text{He}(^{28}\text{Si},^{30}\text{S})\text{n}$ reaction, separated by the CRIB separator at RIKEN, passed through two PPACs, and entered an active target filled with 90% $^{\text{nat}}\text{He}$ + 10% CO_2 gas at 194.2 Torr 5. Scattered α particles were detected using the active target and 500- μm -thick SSDs. Measured E_α , I_α , and $^{30}\text{S}+\alpha$ excitation functions at $E_{\text{c.m.}}=2.0\text{--}5.5$ MeV. Deduced 3 resonances, J , π , L_α , Γ_α , and Γ_p from multichannel R-Matrix analysis.

 ^{34}Ar Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\dagger$</u>	<u>$L^\dagger$</u>	<u>Comments</u>
11092 85	($2^+, 4^+$)	(2,4)	$E_{\text{c.m.}}=4353$ keV. For $J^\pi=2^+$, $\Gamma_\alpha=20$ keV $+80-17$, $\Gamma_p=25$ keV $+500-20$. $\omega\gamma(\alpha,p)=100$ keV. For $J^\pi=4^+$, $\Gamma_\alpha=0.5$ keV $+14-4$, $\Gamma_p=0.3$ keV $+35-3$.
11518 89	(2^+)	(2)	$E_{\text{c.m.}}=4779$ keV. $\Gamma_\alpha=100$ keV $+120-60$, $\Gamma_p=210$ keV $+600-170$. $\omega\gamma(\alpha,p)=500$ keV.
12079 95	(2^+)	(2)	$E_{\text{c.m.}}=5340$ keV. $\Gamma_\alpha=260$ keV $+400-120$, $\Gamma_p=340$ keV $+550-200$. $\omega\gamma(\alpha,p)=1300$ keV.

† From R-Matrix analysis of the measured $^{30}\text{S}+\alpha$ elastic scattering excitation function in **2018Ka11**. The parentheses on L_α are added by evaluators. The parentheses on J^π of the 11518 level are added by evaluators. The selection rules dictate a unique resonance J^π for each L_α value, namely that $J = L_\alpha$ and the parity is always natural for populated states in ^{34}Ar .

$^9\text{Be}(^{37}\text{Ca}, ^{34}\text{Ar}\gamma)$ 2007DoZV

$^{37}\text{Ca} \rightarrow 2\text{p}1\text{n} + ^{34}\text{Ar}$ on ^9Be target.

2007DoZV: a 420-MeV/nucleon ^{40}Ca primary beam was provided from the heavy ion synchrotron SIS at GSI and impinged on a 4-mg/cm² ^9Be target. ^{37}Ca was selected by FRS and was incident on a 700-mg/cm² secondary ^9Be target at 195.7 MeV/u. ^{34}Ar was produced via a secondary 2p1n removal reaction. γ rays were detected using the RISING setup consisting of 15 Cluster HPGe detectors, 8 MINIBALL HPGe detectors, and 8 Hector BaF₂ detectors. Fragments were detected using the calorimeter telescope CATE consisting of position sensitive Si ΔE detectors and CsI(Tl) E_{res} detectors. Measured E_γ , I_γ .

 ^{34}Ar Levels

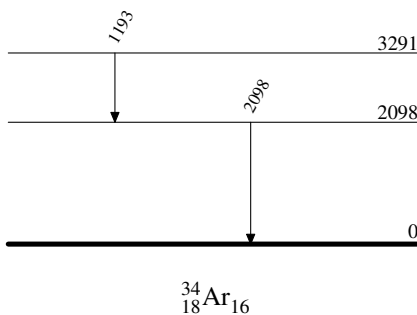
$E(\text{level})$

0
2098 4
3291 5

 $\gamma(^{34}\text{Ar})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>	<u>Comments</u>
1193 2	3291	2098	E_γ : 1193 2 from Cluster detectors and 1193 2 from MINIBALL detectors. Possible contributions from a 1184 γ from ^{35}Ar .
2098 4	2098	0	E_γ : weighted average of 2093 4 from Cluster detectors and 2100 3 from MINIBALL detectors.

[†] From 2007DoZV with only statistical uncertainties from a Gaussian fit on linear background.

 $^9\text{Be}(^{37}\text{Ca}, ^{34}\text{Ar}\gamma)$ 2007DoZVLevel Scheme

$^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$ [2020Ke03](#), [2021Ke04](#)

$^{12}\text{C}+^{24}\text{Mg}\rightarrow^{36}\text{Ar}\rightarrow 2n+^{34}\text{Ar}$ fusion evaporation.

[2020Ke03](#), [2021Ke04](#): a 95-MeV ^{24}Mg beam was produced from the ATLAS accelerator at ANL and bombarded a $\approx 200\text{-}\mu\text{g}/\text{cm}^2$

^{12}C target. Prompt γ rays were detected using the GREINA tracking array consisting of 12 modules. Fusion-evaporation residues were separated and identified by the Fragment Mass Analyzer (FMA) with a focal-plane position-sensitive parallel-grid avalanche counter (PGAC) and an ionization chamber. Measured E_γ , I_γ , $\gamma\gamma$ -coin, recoil- $\gamma(\theta)$. Deduced levels, J , π . Compared with mirror nucleus ^{34}S and shell model calculations. Calculated astrophysical $^{33}\text{Cl}(p,\gamma)^{34}\text{Ar}$ reaction rate and discussed its implications for the identification of nova presolar grains from sulfur isotopic abundances.

 ^{34}Ar Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$E(\text{level})^\dagger$	$J^\pi^\ddagger$
0.0	0^+	4020.6 13	2^+	4851.9 8	3^+
2091.5 4	2^+	4131.6 9	1^+	4881.4 12	2^+
3289.1 5	2^+	4517.5 7	3^-	4963.7 13	0^+
3876.4 9	0^+	4644.1 9	4^+	4966.9 11	2^-
				5061.8 8	1^+

† From a least-squares fit to γ -ray energies.

‡ As given in [2020Ke03](#) and [2021Ke04](#) based on known assignments of low-lying levels, comparison with mirror nucleus ^{34}S , characteristic γ -decay patterns, and $\gamma(\theta)$ analysis. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

 $\gamma(^{34}\text{Ar})$

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. †	Comments
832.1 9	1.6 2	4963.7	0^+	4131.6	1^+		
842.5 7	2.4 2	4131.6	1^+	3289.1	2^+		
1197.5 4	37.9 16	3289.1	2^+	2091.5	2^+		
1228.4 5	15.8 7	4517.5	3^-	3289.1	2^+	D	Mult.: $\Delta J=1$, dipole. $A_2=-0.35$ 10, $A_4=+0.03$ 12.
1562.8 7	5.8 4	4851.9	3^+	3289.1	2^+		
1592.5 17	1.4 2	4881.4	2^+	3289.1	2^+		
1771.8 11	5.0 4	5061.8	1^+	3289.1	2^+		
1784.8 8	5.9 4	3876.4	0^+	2091.5	2^+		
1930.4 23	4.8 9	4020.6	2^+	2091.5	2^+		
2091.3 5	100 5	2091.5	2^+	0.0	0^+		
2424.7 22	2.3 5	4517.5	3^-	2091.5	2^+		
2552.4 8	25.1 16	4644.1	4^+	2091.5	2^+	Q	Mult.: $\Delta J=2$, quadrupole. $A_2=+0.20$ 8, $A_4=-0.01$ 11.
2759.9 12	11.8 9	4851.9	3^+	2091.5	2^+		
2788.9 19	4.4 5	4881.4	2^+	2091.5	2^+		
2875.2 10	11.6 9	4966.9	2^-	2091.5	2^+	D	Mult.: $\Delta J=0$, dipole. $A_2=+0.9$ 3, $A_4=+0.4$ 5.
3289.1 10	5.3 6	3289.1	2^+	0.0	0^+		
4019.8 15	7.4 9	4020.6	2^+	0.0	0^+		
4881.9 24	3.6 6	4881.4	2^+	0.0	0^+		
5062 ‡	<1.4	5061.8	1^+	0.0	0^+		

† From [2020Ke03](#) and [2021Ke04](#). The authors state that dipole transitions, $\Delta J=1$, have negative A_2 values while quadrupole transitions, $\Delta J=2$, have positive A_2 values.

‡ Placement of transition in the level scheme is uncertain.

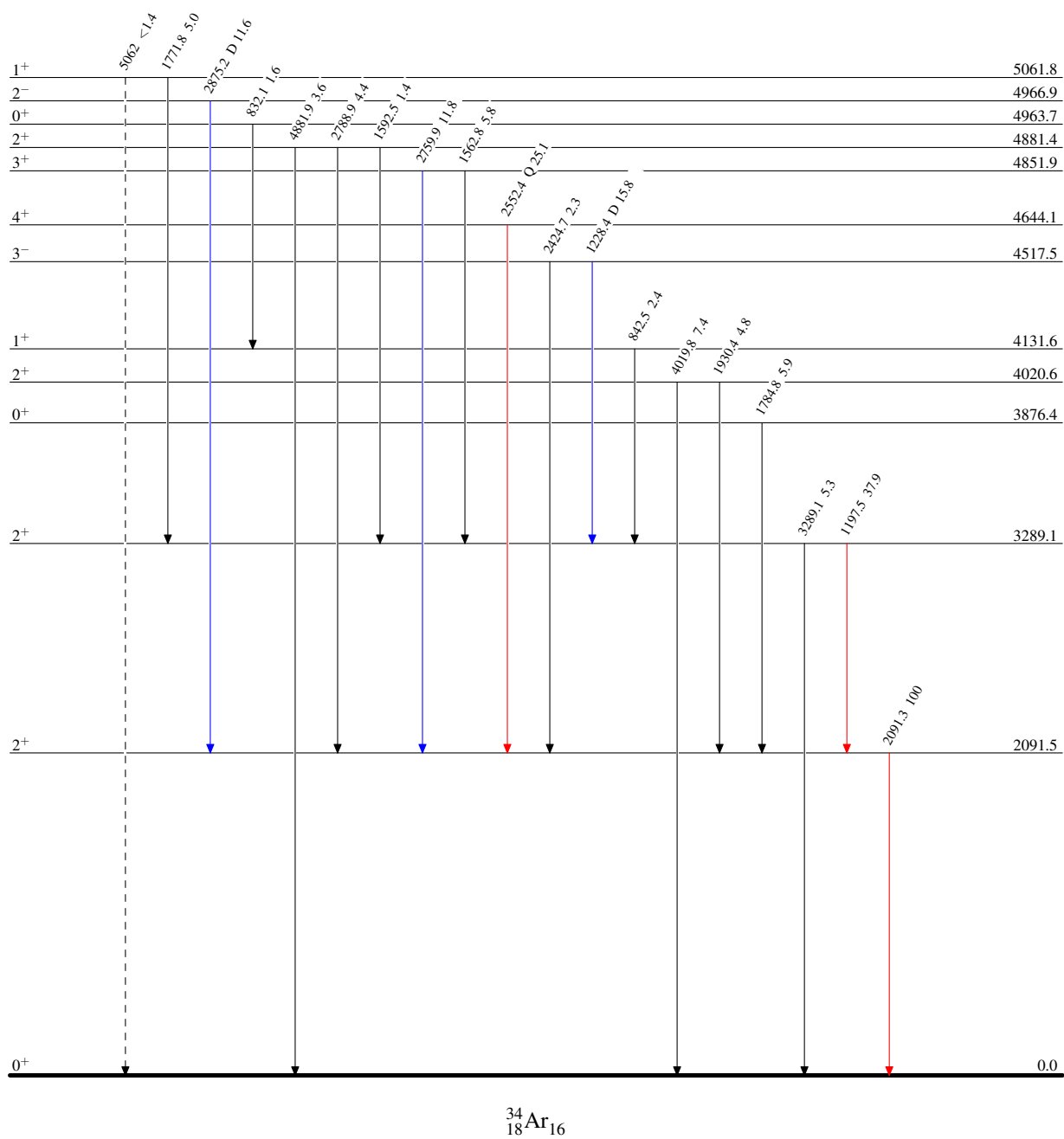
$^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$ 2020Ke03,2021Ke04

Legend

Level Scheme

Intensities: Relative I_γ

- ▶ $I_\gamma < 2\% \times I_\gamma^{\max}$
- ▶ $I_\gamma < 10\% \times I_\gamma^{\max}$
- ▶ $I_\gamma > 10\% \times I_\gamma^{\max}$
- - - -▶ γ Decay (Uncertain)

 $^{34}_{18}\text{Ar}_{16}$

$^{32}\text{S}(^3\text{He},n)$ 1986A115

$^{32}\text{S}+2p \rightarrow ^{34}\text{Ar}$ from $J^\pi=0^+$ ^{32}S ground state.

1986A115: a 25.4-MeV ^3He beam was produced from the time-of-flight facility at the University of Colorado. The target was natural sulfur evaporated onto a 2-mg/cm² gold foil. Neutrons were detected using three NE224 liquid scintillators with FWHM=350 keV. Measured $\sigma(E_n, \theta)$. Deduced levels, L-transfers, J, and π from local, zero-range DWUCK-DWBA analysis. Comparisons with shell model and analog states in ^{34}S .

1972Bb01: 14-MeV ^3He beam from the 7 MV Van de Graaff accelerator at Frankfurt University impinged on natural ^{32}S targets. Measured $\sigma(E_n, \theta)$. Deduced ground state, J, and π from zero-range DWUCK-DWBA analysis.

1968Ha09: 4.5-6.2-MeV ^3He beams from the de Graaff accelerator at Hahn-Meitner-Institut für Kernforschung Berlin and impinged on FeS targets. Measured neutron spectra using a time-of-flight spectrometer. Deduced 5 levels.

1967Mi02: 10-12-MeV ^3He beams from the ONR-CIT tandem accelerator impinged on natural sulfur targets. Neutron energies were measured by means of (n, α) reactions induced by neutrons in the active volume of a silicon semiconductor detector. Deduced Q value and one level.

1967Mc03: 4.9-5.6-MeV ^3He beams from the SUNI 6 MeV Van de Graaff accelerator at Southern Universities Nuclear Institute, South Africa impinged on natural CdS targets. Neutrons were detected using a NE213 liquid scintillator. Measured $\sigma(E_n, \theta)$ and deduced g.s. and 2 levels.

1984Be05: 4.6-6.5 MeV ^3He beams from the University of Kentucky Van de Graaff Accelerator impinged on CdS and ZnS targets. Neutron angular distributions and excitation curves were measured with a neutron time-of-flight facility.

 ^{34}Ar Levels

ε : enhancement factor = $(d\sigma/d\omega)_{\text{exp}}/[213(d\sigma/d\omega)_{\text{DWBA}}]$.

E(level) [†]	L [†]	Comments
0	0	$\varepsilon=1.4$.
2103 30	2	$\sigma(\text{exp})/\sigma(\text{theory})=1.84$ or 1.68 (1972Bb01). E(level): weighted average of 2090 30 (1986A115), 2100 30 (1968Ha09), 2190 40 (1967Mc03), and 2058 35 (1967Mi02). $\varepsilon=3.9$.
3295 30	(2)	E(level): weighted average of 3290 30 (1986A115) and 3300 30 (1968Ha09). $\varepsilon=0.32$.
3900 30	(0)	E(level): weighted average of 3900 70 (1986A115) and 3900 30 (1968Ha09). Other: 3590 60 (1967Mc03), a broad peak, possibly unresolved 3295+3900. $\varepsilon=6.1$.
4050 [‡] 30		
4150 [‡] 30		
4510 30	3	$\varepsilon=2.9$.
4950 50	0	$\varepsilon=25$.
5310 30	(5)	$\varepsilon=0.38$.
5620 30	2	$\varepsilon=18$.
5930 50	0	
6470 30		
6820 40		
6990 50		
7300 30		

[†] From **1986A115**, unless otherwise noted.

[‡] From **1968Ha09**.

$^{32}\text{S}(^3\text{He}, n\gamma)$ 1974Gr19, 1974Be18, 1972Ca22

1974Be18: an 8-MeV ^3He beam was provided from the ANU EN tandem accelerator and impinged on cooled sulphur targets.

Neutrons were detected using an NE213 liquid scintillator and γ rays were detected using a Ge(Li) detector and a NaI(Tl) detector. Measured E_γ , I_γ , $n\gamma$ -coin, $n\gamma(\theta)$. Deduced levels, J, γ -ray branching ratios; deduced lifetimes for 4 levels using the oppler-Shift Attenuation Method (DSAM).

1974Gr19: 8-12-MeV ^3He beams were provided from the 7-MV Van de Graaff accelerator at the University of Freiburg and impinged on natural ZnS targets. Neutrons were detected using an annular NE213 liquid scintillator and γ rays were detected using a Ge(Li) detector and 5 NaI(Tl) detectors. Measured E_γ , I_γ , $n\gamma$ -coin, and $n\gamma\gamma$ -coin. Deduced 10 levels, J, γ -ray branching ratios, and mixing ratios; deduced lifetimes for 5 levels using DSAM.

1972Ca22: an 8.5-MeV ^3He beam was provided from the Oxford University tandem Van de Graaff generator and impinged on Ag_2S targets. Neutrons were detected using a NE102 liquid scintillator and γ rays were detected using a Ge(Li) detector. Measured E_γ , I_γ , E_n , I_n , and $n\gamma(\theta)$. Deduced lifetimes for 3 levels using DSAM.

 ^{34}Ar Levels

E(level) [†]	J ^π	T _{1/2}	Comments
0.0	0 ⁺		
2091.0 4	2 ⁺	139 fs 35	J ^π : from the Adopted Levels. Spin=2 confirmed by $n\gamma(\theta)$ in 1974Gr19. T _{1/2} : from lifetime $\tau=200$ fs 50: weighted average of 330 fs 80 (1974Be18, DSAM), 200 fs 60 (1974Gr19, DSAM), and 150 fs 50 (1972Ca22, DSAM).
3287.5 6	2 ⁺	104 fs 42	J ^π : from the Adopted Levels. Spin=2 confirmed by $n\gamma(\theta)$ in 1974Gr19. T _{1/2} : from lifetime $\tau=150$ fs 60: weighted average of 215 fs 70 (1974Be18, DSAM), 130 fs 60 (1974Gr19, DSAM), and 120 fs 65 (1972Ca22, DSAM).
3873 3	0 ⁺	>416 fs	J ^π : from the Adopted Levels. Spin=0 confirmed by $n\gamma(\theta)$ in 1974Gr19. T _{1/2} : from lifetime $\tau>600$ fs (1974Gr19, DSAM).
4127.4 11	(2) ⁺	<243 fs	J ^π : spin=2 is favored from $n\gamma(\theta)$, which did not exclude spin=1, 3, or 4 (1974Gr19); spin=4 ruled out by 839.9 γ , M1+E2, to 2 ⁺ , 3287.5 level. T _{1/2} : from lifetime $\tau<350$ fs (1974Be18, DSAM). Other: $\tau>400$ fs (1974Gr19, DSAM).
4513.5 8	3	201 fs 55	J ^π : spin=3 from $n\gamma(\theta)$ (1974Gr19). T _{1/2} : from lifetime $\tau=290$ fs 80: weighted average of 390 fs 180 (1974Be18, DSAM), 260 fs 80 (1974Gr19, DSAM), and 300 fs 80 (1972Ca22, DSAM).
4631 [‡] 4			
4865 [‡] 4			
4967 [‡] 4			
5255 [‡] 4			
5542 [‡] 4			

[†] For E(level) below 4.6 MeV, from a least-squares fit to γ -ray energies.

[‡] From 1974Gr19 based on γ data without reporting γ energies.

 $\gamma(^{34}\text{Ar})$

E _i (level)	J _i ^π	E _γ [†]	I _γ [‡]	E _f	J _f ^π	Mult.	δ	Comments
2091.0	2 ⁺	2090.9 4	100	0.0	0 ⁺			E _γ : weighted average of 2091.1 3 (1974Be18) and 2090.0 7 (1972Ca22). A ₂ =+0.48 9 (1974Be18).
3287.5	2 ⁺	1196.5 4	89 5	2091.0	2 ⁺	M1+E2	+0.12 5	E _γ : weighted average of 1196.6 4 (1974Be18) and 1195.8 14 (1972Ca22). I _γ : weighted average of 92 5 (1974Be18), 90 7 (1972Ca22), and 85 5 (1974Gr19). Mult.,δ: D+Q from $n\gamma(\theta)$ (1974Gr19); M2 ruled out by RUL.
		3286 4	11 5	0.0	0 ⁺			A ₂ =+0.56 10 (1974Be18). E _γ : from 1974Be18.

Continued on next page (footnotes at end of table)

$^{32}\text{S}(^3\text{He}, n\gamma)$ **1974Gr19, 1974Be18, 1972Ca22 (continued)** $\gamma(^{34}\text{Ar})$ (continued)

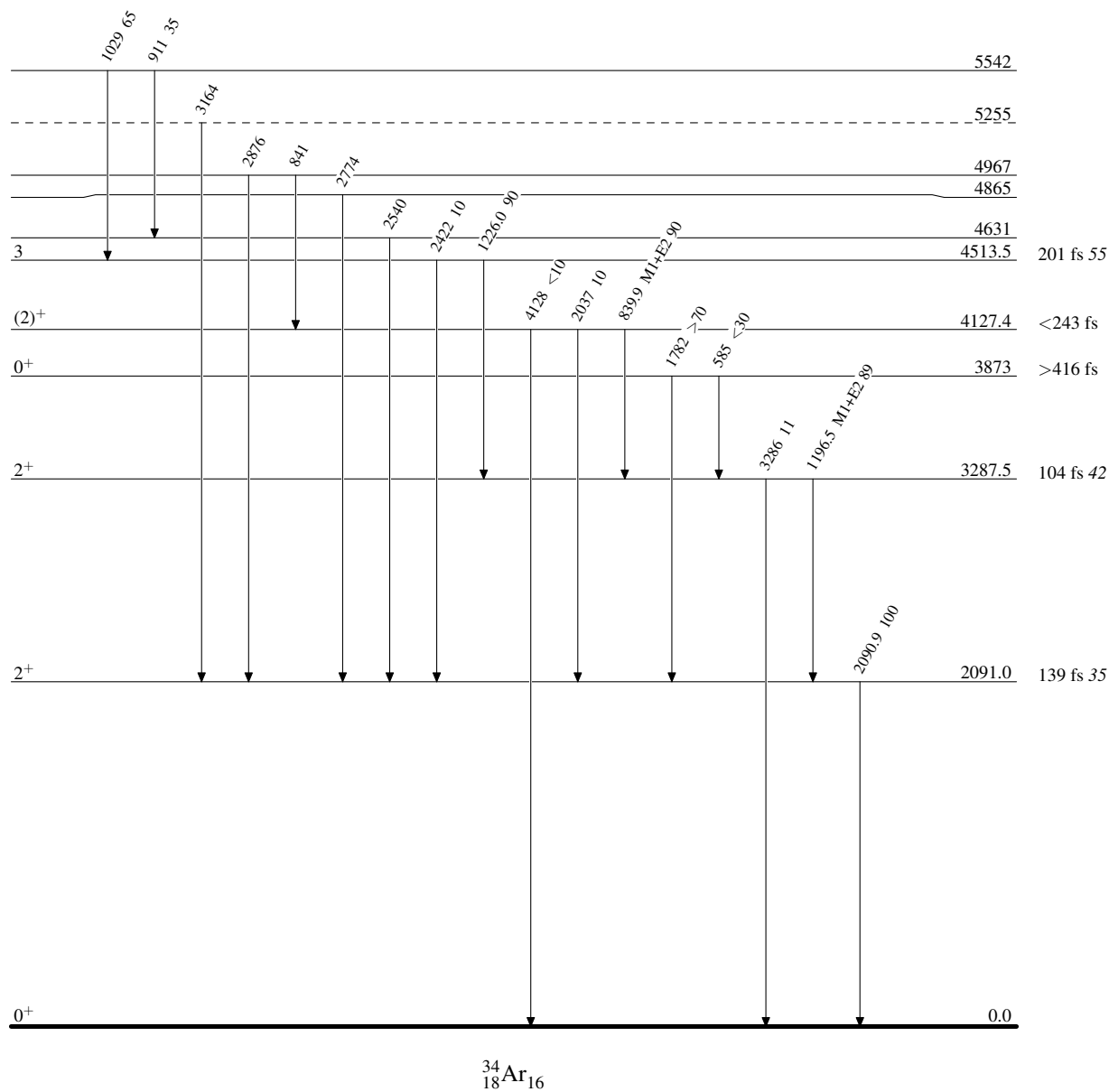
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.	δ	Comments
								I_γ : weighted average of 8 5 (1974Be18), 10 7 (1972Ca22), and 15 5 (1974Gr19).
3873	0^+	585	<30	3287.5	2^+			I_γ : from 1974Be18.
		1782 3	>70	2091.0	2^+			E_γ, I_γ : from 1974Be18; other: $I_\gamma=100$ (1974Gr19).
4127.4	$(2)^+$	839.9 9	90 5	3287.5	2^+	M1+E2	+0.52 15	E_γ : weighted average of 840.1 9 (1974Be18) and 839.1 18 (1972Ca22).
								I_γ : others: >50 (1974Be18), 100 (1972Ca22).
								Mult., δ : from $n\gamma(\theta)$ in 1974Gr19; E1+M2, E2+M3, M2+E3 ruled out by RUL.
		2037	10 5	2091.0	2^+			I_γ : other: <50 (1974Be18).
		4128	<10	0.0	0^+			
4513.5	3	1226.0 6	90 10	3287.5	2^+			E_γ : weighted average of 1225.5 6 (1974Be18) and 1226.7 7 (1972Ca22).
								I_γ : others: >90 (1974Be18), 100 (1972Ca22).
								$A_2=-0.38$ 24 (1974Be18).
		2422	10 3	2091.0	2^+			I_γ : others: <10 (1974Be18), <20 (1972Ca22).
4631		2540		2091.0	2^+			I_γ : >50 (1974Gr19).
4865		2774		2091.0	2^+			I_γ : >50 (1974Gr19).
4967		841		4127.4	$(2)^+$			I_γ : <20 (1974Gr19).
		2876		2091.0	2^+			I_γ : >50 (1974Gr19).
5255?		3164		2091.0	2^+			I_γ : >50 (1974Gr19).
5542		911	35 10	4631				I_γ : tentative (1974Gr19).
		1029	65 10	4513.5	3			I_γ : tentative (1974Gr19).

 † E_γ values without uncertainties are from level-energy differences. ‡ From 1974Gr19, unless otherwise noted.

$^{32}\text{S}(^3\text{He},n\gamma)$ 1974Gr19,1974Be18,1972Ca22

Level Scheme

Intensities: % photon branching from each level



$^{36}\text{Ar}(\text{p,t})$ 2018Lo11,1972Pa02,1969Br21

$^{36}\text{Ar} \rightarrow 2\text{n} + ^{34}\text{Ar}$ from $J^\pi=0^+$ ^{36}Ar ground state.

2018Lo11: a 100-MeV proton beam was produced from the RCNP WS beam line at Osaka University. The target was 0.76-mg/cm² enriched ^{36}Ar gas ($\geq 99\%$). Reaction products were momentum-analyzed by the Grand Raiden spectrograph and detected using the focal-plane detector system. Measured $\sigma(E_t, \theta)$. Deduced levels. Calculated astrophysical $^{30}\text{S}(\alpha, \text{p})^{33}\text{Cl}$ reaction rate. Also see **2009ObZY**.

1972Pa02: 40-45-MeV proton beams were produced from the MSU sector-focused cyclotron. The target was $>99\%$ -enriched ^{36}Ar gas. Reaction products were detected using a detector telescope made up of two silicon surface-barrier transmission mounted counters. Measured $\sigma(E_t, \theta)$. Deduced levels, L-transfers, J, and π from JULIE-DWBA analysis.

1969Br21: a 45-MeV proton beam was produced from the Berkeley 88-Inch Cyclotron. The target was 99.6%-enriched ^{36}Ar gas. Reaction products were detected using a 155- μm phosphorus-diffused silicon ΔE transmission counter and a 3.0-mm lithium-drifted silicon E counter operated in coincidence, followed by a 0.5-mm lithium-drifted silicon E-reject counter operated in anti-coincidence with the first two counters to eliminate long-range particles. Measured $\sigma(E_t, \theta)$. Deduced levels, L-transfers, J, and π from DWUCK-DWBA analysis. Also measured $^{36}\text{Ar}(\text{p}, ^3\text{He})^{34}\text{Cl}$ and $^{36}\text{Ar}(\text{d}, \alpha)^{34}\text{Cl}$.

1974Ha02: a 40.2-MeV proton beam was produced from the MSU cyclotron. Measured Q value. Tritons for the g.s. to g.s. transition measured using split-pole magnetic spectrograph, FWHM=10-25 keV. Measures Q value= -19513 ± 3 . Deduced mass excess of ^{34}Ar .

 ^{34}Ar Levels

E(level) [†]	L [‡]	Comments
0	0	
2091.6 24	2	E(level): weighted average of 2094 11 (1972Pa02) and 2091.5 24 (2018Lo11). Other: 2097 20 (1969Br21). L: 2 in 1972Pa02 .
3290.9 18	2	E(level): weighted average of 3303 25 (1969Br21), 3288 14 (1972Pa02), and 3290.9 18 (2018Lo11). L: 2 in 1972Pa02 .
3879.1 33	0	E(level): weighted average of 3899 25 (1969Br21), 3879 15 (1972Pa02), and 3878.8 33 (2018Lo11). L: 0 in 1972Pa02 for unresolved 3879+4050 and 3879 is more strongly excited.
4035 16		E(level): unweighted average of 4050 14 (1972Pa02) and 4019.1 43 (2018Lo11).
4515.1 23	(3)	E(level): weighted average of 4522 14 (1972Pa02) and 4514.9 23 (2018Lo11). Other: 4560 40 (1969Br21). L: not fitted particularly well by either L=2 or L=3 calculated distributions. However, it is distinctly different from the other experimental L=2 distributions. There is also a 3 ⁻ state at approximately the same excitation energy in the mirror nucleus ^{34}S (1969Br21).
4641.5 21		E(level): weighted average of 4651 14 (1972Pa02) and 4641.3 21 (2018Lo11).
4875.3 38		E(level): weighted average of 4867 14 (1972Pa02) and 4875.9 38 (2018Lo11).
4967.9 27	0	E(level): weighted average of 4970 40 (1969Br21), 4985 14 (1972Pa02), and 4967.2 27 (2018Lo11).
5262 15		
5317 13		E(level): weighted average of 5340 40 (1969Br21), 5307 13 (1972Pa02), and 5330 17 (2018Lo11).
5535 18		
5629.6 45		
5929 20		E(level): unweighted average of 5909 12 (1972Pa02) and 5948.2 66 (2018Lo11).
6050.1 36	2	E(level): weighted average of 6074 11 (1972Pa02), and 6049.2 20 (2018Lo11). Other: 6100 40 (1969Br21). L: other: the total angular distribution of unresolved 5909+6074 exhibits some L=2 character, and therefore possibly one or both states are 2 ⁺ , but no definite assignment can be made (1972Pa02). Based on the measured $\sigma(E_t, \theta)$ in 1972Pa02 , the 6074 contribution dominates.
6525 9		
6641 14		
6723 22		
6807 4		E(level): weighted average of 6794 11 (1972Pa02) and 6808 3 (2018Lo11). Other: 6860 40 (1969Br21).
6917 5		
7072 8		
7276 2		
7340 11	2	E(level): unweighted average of 7340 50 (1969Br21), 7322 6 (1972Pa02), and 7358 4 (2018Lo11).
7502 16	(2)	E(level): unweighted average of 7530 50 (1969Br21), 7499 4 (1972Pa02), and 7476 2 (2018Lo11).
7921 18		E(level): unweighted average of 7950 50 (1969Br21), 7925 5 (1972Pa02), and 7889 2 (2018Lo11).
8166 3		
8660 9		

Continued on next page (footnotes at end of table)

$^{36}\text{Ar}(\text{p,t})$ [2018Lo11](#), [1972Pa02](#), [1969Br21](#) (continued)

^{34}Ar Levels (continued)

E(level)[†]

8746 8
8899 8
9004 4
9148 22
9226 18
9549 16

[†] From [2018Lo11](#), unless otherwise noted.

[‡] From [1969Br21](#), unless otherwise noted.

Coulomb excitation 2023Dr01

2023Dr01: An intermediate Coulomb excitation experiment was performed to measure the $0^+ \rightarrow 2^+$ B(E2) value of $^{36,38}\text{Ca}$. ^{34}Ar beam was a contaminant produced from the fragmentation of a 140 MeV/nucleon ^{40}Ca primary beam delivered by the Coupled Cyclotron Facility at the National Superconducting Cyclotron Laboratory (NSCL) impinging upon an 800-mg/cm²-thick ^9Be production target and separated using a 300-mg/cm² Al wedge degrader in the A1900 fragment separator. Secondary beam was scattered off a 257-mg/cm²-thick Au foil located at the target position of the S800 spectrograph. The deexciting γ rays were detected using GRETTINA surrounding the S800 spectrograph. The GRETTINA comprised a total of 48 HPGe crystals; 32 of them at 90° and 16 of them at 58° with respect to the beam direction. The inelastic excitations of the projectile and target nuclei were quantified with the detection of prompt γ rays in GRETTINA. The scattered projectiles were identified using S800 focal plane detection system together with tof measurement. Measured: E_γ , tof, cross section and B(E2). Comparison with the shell model calculations.

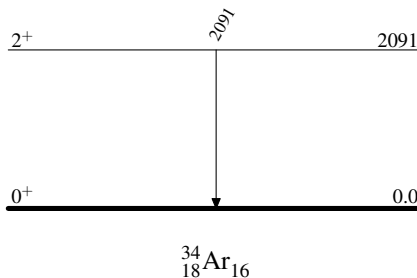
 ^{34}Ar Levels

<u>E(level)</u>	<u>J^π</u> [†]	<u>Comments</u>
0.0	0 ⁺	
2091 2	2 ⁺	B(E2) $\uparrow$ =0.0293 16 σ =52.1 mb 29 measured from laboratory scattering angles $\theta=0^\circ$ to $\theta_{\text{max}}=55^\circ$ mrad at 64.3 MeV/nucleon mid-target energy.

[†] From the Adopted Levels.

 $\gamma(^{34}\text{Ar})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
2091 2	2091	2 ⁺	0.0	0 ⁺

Coulomb excitation 2023Dr01Level Scheme

Adopted Levels

$Q(\beta^-) = -16110$ syst; $S(n) = 16830$ syst; $S(p) = -608$ 17; $Q(\alpha) = -8587$ 29 [2021Wa16](#)

$S(p), Q(\alpha)$: Deduced by evaluators using mass excesses of -1487 17 for ^{34}K measured by [2024Dr01](#); -9384.3 4 for ^{33}Ar and 4675 24 for ^{30}Cl from [2021Wa16](#). Values from [2021Wa16](#): $S(p) = -880$ 200 (syst), $Q(\alpha) = -8320$ 200 (syst).

$\Delta Q(\beta^-) = 360$, $\Delta S(n) = 280$ (syst, [2021Wa16](#)).

$S(2p) = 2730$ 17, $Q(\varepsilon) = 16891$ 17, $Q(\varepsilon p) = 12227$ 17, from mass excesses of -1487 17 for ^{34}K measured by [2024Dr01](#); -13334.7 6 for ^{32}Cl , -18378.289 80 for ^{34}Ar , and -21003.3 4 for ^{33}Cl from [2021Wa16](#). Values from [2021Wa16](#): $S(2p) = 2460$ 200 (syst),

$Q(\varepsilon) = 17160$ 200 (syst), $Q(\varepsilon p) = 12490$ 200 (syst).

$S(2n) = 39350$ 450 (syst) ([2021Wa16](#)).

Isotope discovery: $\text{Be}(^{37}\text{Ca}, X) \rightarrow p + ^{33}\text{Ar}$ at NSCL, MSU ([2024Dr01](#)).

$S(2p)$ is positive and $S(p)$ is negative. ^{34}K is a ground state one-proton emitter.

Mass and separation energy calculations: [2021Ma33](#), [2022Zo01](#), [2019Zo02](#), [2013Ti01](#).

Other structure calculations: [2023Bo17](#), [2018Ka12](#).

 ^{34}K LevelsCross Reference (XREF) Flags

A $^9\text{Be}(^{37}\text{Ca}, X)$

E(level)	J^π [†]	$T_{1/2}$	XREF	Comments
0	(1 ⁺)	<40 ns	A	$\%p = ?$ $T_{1/2}$: upper limit estimated from expected cross sections of a ^{40}Ca beam fragmentation on a Ni target at LISE, GANIL (1996PoZZ , 1986La17).
401 25	(2 ⁺)		A	
≈1240			A	
≈1810			A	

[†] From shell-model calculations with the USDC Hamiltonian ([2024Dr01](#)).

$^9\text{Be}(^{37}\text{Ca},\text{X})$ 2024Dr01

Isotope discovery of ^{34}K .

2024Dr01: a 72-MeV/nucleon ^{37}Ca secondary beam was produced via the fragmentation of a 140-MeV/nucleon $^{40}\text{Ca}^{20+}$ primary beam impinging on a 0.5-mm-thick Be target. Experimental setup includes the CAESium-iodide scintillator ARray (CAESAR) for detecting γ rays, a DSSD-CsI(Tl) $\Delta\text{E-E}$ Ring Telescope for detecting protons, a Scintillating-Fiber Array (SFA) and the S800 spectrograph for detecting heavy residuals. Measured E_p , E_γ , and $E(\text{residual})$. Deduced levels, mass excess from total decay-energy spectra using the invariant-mass method. Comparisons with shell-model calculations with the USDC Hamiltonian.

 ^{34}K Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	Comments
0	(1 ⁺)	Total decay energy $E_T=608$ 17. Mass excess= -1487 keV 17, deduced by 2024Dr01 from their measured total decay energy and mass of ^{33}Ar from 2021Wa16 .
401 25 ≈ 1240 ≈ 1810	(2 ⁺)	$E(\text{level})$: from total decay energy $E_T=1009$ 18. $E(\text{level})$: from total decay energy $E_T\approx 1850$. $E(\text{level})$: from total decay energy $E_T\approx 2420$.

[†] From the measured total decay-energy spectrum of $p+^{33}\text{Ar}$. The E_T region above 1.36 MeV contains some unresolved highly excited states in ^{34}K that proton decay to γ -decaying excited states in ^{33}Ar .

[‡] Assigned by **2024Dr01** based on shell model. Parentheses are added by evaluators.

Adopted Levels: not observed

$S(n)=24210$ syst; $S(p)=-60$ syst; $Q(\alpha)=-9610$ syst [2021Wa16](#)

$\Delta S(n)=500$, $\Delta S(p)=360$, $\Delta Q(\alpha)=350$ (syst, [2021Wa16](#)).

$S(2p)=-2510$ 300, $Q(\varepsilon)=16110$ 360, $Q(\varepsilon p)=16990$ 300 (syst, [2021Wa16](#)).

Isotope discovery: none.

$S(2p)$ is negative and $S(p)$ may be consistent with zero. ^{34}Ca is a candidate for ground-state two-proton emitter.

Theoretical calculations:

[2026Li03](#): $S(p)=401$ and $S(2p)=-1987$ from an average of Finite-Range Droplet Model (FRDM12), Koura-Tachibana-Uno-Yamada (KTUY) model, Weizsacker-Skyrme (WS4) model, and Duflo-Zuker (DZ31) model combined with the radial basis function mirror symmetry (RBFms) approach.

[2024Li59](#): $S(2p)=-1044$, $T_{1/2}=177.8$ ns. The $Q(2p)$ decay energy was determined using an empirical linear relationship. The $T_{1/2}$ was calculated by inputting the refined decay energy into the new Geiger-Nuttall formula.

[2024Ji17](#): $T_{1/2}=8.1$ fs from an extension of the Bayrak model with and without the deformation term. Used $Q(2p)=2510$ from [2021Wa16](#).

[2023Me08](#): $T_{1/2}=64.6$ ps. The WKB approximation is applied to a Yukawa-plus-exponential model potential. Used $Q(2p)=1470$ from [2017Wa10](#) and a spectroscopic factor derived from a specific cubic-polynomial relation.

[2022Zo01](#): $S(p)=259$ 96, $S(2p)=-2132$ 151, mass excess=14497 98. Mass relations of mirror nuclei.

[2022Sa41](#): $T_{1/2}=2.2$ fs from the Coulomb and proximity potential model for deformed nuclei (CPPMDN) incorporating ground-state deformation (β_2) and orientation effect. Used $Q(2p)=2510$ from [2021Wa16](#).

[2022Ro15](#): $T_{1/2}=0.35$ fs from the generalized liquid drop model (GLDM). $T_{1/2}=3.0$ as from a semi-empirical analytic expression based on a modified version of the Geiger-Nuttall law. Both calculations used $Q(2p)=2510$ from [2021Wa16](#).

[2022Pa24](#): $T_{1/2}=0.31$ ns from a two-potential approach; the nuclear potential is obtained using the Skyrme-Hartree-Fock approach and the Skyrme effective interaction of SLy8. Used $Q(2p)=1470$ from [2017Wa10](#).

[2021Xi06](#): $T_{1/2}=3.5$ fs from the Unified Fission Model (UFM), $T_{1/2}=1.7$ fs from the Generalized Liquid Drop Model (GLDM), $T_{1/2}=27.5$ fs from the Effective Liquid Drop Model (ELDM). These models share a similar physical mechanism by treating the decay as a ^2He cluster preformed on the nuclear surface that subsequently penetrates the Coulomb barrier. All used $Q(2p)=2510$ from [2021Wa16](#).

[2021Li27](#): $T_{1/2}=79.4$ ps from the Gamow-like model. Used $Q(2p)=1470$ from [2017Wa10](#).

[2021Li16](#): $T_{1/2}=117$ ps from a two-parameter empirical formula based on the Geiger-Nuttall law. Used $Q(2p)=1470$ from [2017Wa10](#).

[2020Ma35](#): $S(p)=203$ 151, $S(2p)=-2152$ 118, $T_{1/2}=3.6$ ps +51–20. Mass relations of mirror nuclei with local correlations and a four-parameter empirical formula based on the Effective Liquid Drop Model (ELDM).

[2020Cu01](#): $T_{1/2}=19.5$ ps from a generalized liquid drop model (GLDM). Used $Q(2p)=1470$ from [2017Wa10](#).

[2019Sr02](#): $T_{1/2}=2.2$ ns. A four-parameter empirical formula was fitted using the theoretical results of the Effective Liquid Drop Model (ELDM). Used $Q(2p)=1470$ from [2017Wa10](#).

Other decay half-life calculations: [2017Sa70](#), [2004Pf02](#), [2003Gr04](#), [2005Pf01](#), [2003Gr24](#), [2001Gr29](#).

Other structure calculations: [2005Wa15](#), [2005Zh28](#), [2002Zh09](#), [2001Yo12](#), [1998Co30](#), [1997Co19](#), [1997Pa38](#), [1996Sh20](#), [1995La03](#).

Giant-resonance calculations: [2021Li26](#), [2001Sa57](#), [1997Ha57](#), [2003Ma71](#), [2001Sa77](#), [1999Sa22](#), [1998Ha09](#), [1998Ha39](#), [1998Sa28](#), [1997Ha55](#).

Experimental search:

A two-hour search conducted on March 27, 2012, during the [2015Su01](#) experiment at the Radioactive Ion Beam Line in Lanzhou (RIBLL), did not observe any events at ~ 200 ns that could be attributed to ^{34}Ca in the ΔE -ToF spectrum.

 ^{34}Ca Levels

E(level)	J^π	$T_{1/2}$	Comments
0?	0^+	<35 ns	$\%p=?$; $\%2p=?$ $T_{1/2}$: upper limit estimated from expected cross sections of a ^{40}Ca beam fragmentation on a Ni target at LISE, GANIL (1996PoZZ). The authors of 1986La17 provided additional data from their experiment in a private communication 1996PoZZ.

REFERENCES FOR A=34

- 1926As01 F.W.Aston - Nature(London) 117, 893 (1926).
The isotopes of sulphur.
- 1934Fr01 O.R.Frisch - Nature 133, 721 (1934).
Induced Radioactivity of Sodium and Phosphorus.
- 1945Zu01 W.Zunti, E.Bleuler - Helv.Phys.Acta 18, 263 (1945).
Über zwei aktivitäten S^{37} und P^{34} , die durch schnelle Neutronen in Chlor induziert werden.
- 1946Bl01 E.Bleuler, W.Zunti - Helv.Phys.Acta 19, 137 (1946).
Über die Isotope S^{37} und P^{34} , die bei Bestrahlung von Chlor mit Schnellen Neutronen Entstehen.
- 1949Da16 P.W.Davison - Phys.Rev. 75, 757 (1949).
Energy Levels in Sulfur Nuclei.
- 1951Ru24 L.Ruby, J.R.Richardson - Phys.Rev. 83, 698 (1951).
Radiations from F^{18} , Mo^{93} , and Cl^{34} .
- 1952Hi66 N.M.Hintz, N.E.Ramsey - Phys.Rev. 88, 19 (1952).
Excitation Functions to 100 MeV.
- 1953St42 P.Stahelin, P.Preiswerk - Nuovo Cim. 10, 1219 (1953).
Die Lage der ersten Anregungsstufe in Kernen mit gerader Protonen-und gerader Neutronenzahl.
- 1954En19 P.M.Endt, J.C.Kluyver - Rev.Mod.Phys. 26, 95 (1954).
Energy Levels of Light Nuclei (Z = 11 to Z = 20).
- 1954Kl36 R.M.Kline, D.J.Zaffarano - Phys.Rev. 96, 1620 (1954).
Decay Characteristics of Some Short-Lived Nuclides of Low Atomic Number.
- 1956Gr07 D.Green, J.R.Richardson - Phys.Rev. 101, 776 (1956).
Radiations from P^{30} , Cl^{34} , and K^{38} .
- 1956Ha44 H.E.Handler, J.R.Richardson - Phys.Rev. 102, 833 (1956).
Gamma-Gamma Angular Correlation in the Decay of Chlorine-34.
- 1956Mo93 H.Morinaga, E.Bleuler - Phys.Rev. 103, 1423 (1956).
Weak Branchings in the Decay of P^{34} , Cl^{34} , and P^{30} .
- 1956Qu04 A.R.Quinton, W.T.Doyle - Phys.Rev. 101, 669 (1956).
(α,n) Type Reactions in Light Nuclei.
- 1957En01 P.M.Endt, C.M.Braams - Rev.Mod.Phys. 29, 683 (1957).
Energy Levels of Light Nuclei (Z = 11 to Z = 20). II.
- 1958Mi85 M.V.Mihailovic, B.Povh - Nuclear Phys. 7, 296 (1958).
The Half-Lives of Some Short-Lived Isotopes.
- 1958Sc31 R.S.Scanlan, R.W.Fink - Nuclear Phys. 9, 334 (1958).
Activation Cross-Sections for Reactions of Chlorine and Copper with 14.8 MeV Neutrons.
- 1958Sh62 G.G.Shute, P.S.Fisher - Phil.Mag. 3, 726 (1958).
Polarization Measurements of Gamma Radiation from Sulphur-34.
- 1959Ku79 J.Kuperus, P.J.M.Smulders, P.M.Endt - Physica 25, 600 (1959).
Energy measurements of proton resonances in light nuclei.
- 1960Ja12 J.Janecke - Z.Naturforsch. 15a, 593 (1960).
The Half Lives of Some Positron Emitters with Superpermitted Transitions.
- 1961Ja22 J.Janecke, H.Jung - Z.Phys. 165, 94 (1961).
Übererlaubter Positronenzerfall von K^{38m} , Sc^{42} , und Ti^{43} .
- 1962Cu07 B.Cujec - Phys.Rev. 128, 2303 (1962).
Investigation of d-s Shell Nuclei with (d,He^3) Reactions.
- 1962En05 P.M.Endt, C.van der Leun - Nucl.Phys. 34, 1 (1962).
Energy Levels of Light Nuclei. III Z = 11 to Z = 20.
- 1962Li07 A.J.Lieber, F.H.Schmidt, J.B.Gerhart - Phys.Rev. 126, 1496 (1962).
Proton Angular Distributions for (α,p) Reactions on C^{12} , Al^{27} , and P^{31} .
- 1963Br05 M.W.Brenner - Phys.Rev. 129, 765 (1963).
Proton Groups from the $S^{33}(d,p)S^{34}$ Reaction.
- 1964Gl04 P.W.M.Glaudemans, L.Eriksson, J.A.R.Werkhoven - Nucl.Phys. 55, 559(1964).
Gamma-Ray Branchings in ^{34}Cl from the $^{33}S(p,\gamma)^{34}Cl$ Reaction.
- 1964Va08 C.van der Leun, G.Wiechers - Nucl.Phys. 52, 104 (1964).
The Reaction $^{30}Si(\alpha,\gamma)^{34}S$.
- 1964Va12 C.van der Leun, P.M.Endt - Nucl.Phys. 53, 540 (1964).
The Reaction $^{33}S(p,p'\gamma)^{33}S$.
- 1965Eb01 T.G.Ebrey, P.R.Gray - Nucl.Phys. 61, 479 (1965).
Precision Half-Life Measurements of Fourteen Positron-Emitting Nuclei.
- 1965Fr03 J.M.Freeman, J.H.Montague, G.Murray, R.E.White, W.E.Burcham - Nucl.Phys. 65, 113 (1965).
The Accurate Determination of Some (p,n) Thresholds in the Proton Energy Range 5 to 10 MeV.
- 1965Fr06 J.M.Freeman, J.H.Montague, G.Murray, R.E.White, W.E.Burcham - Nucl.Phys. 69, 433 (1965).
The Half-Lives of Al^{26m} , Cl^{34} , Co^{54} and Cu^{58} .

REFERENCES FOR A=34(CONTINUED)

- 1965Mc07 W.R.McMurray, I.J.Van Heerden, G.Wiechers - Nucl.Phys. 73, 113 (1965).
The Reaction $^{30}\text{Si}(\alpha,\gamma)^{34}\text{S}$.
- 1966Ba54 R.H.Bassel - Phys.Rev. 149, 791 (1966).
Normalization of and Finite-Range Effects in ($^3\text{He},d$) and (t,d) Reactions.
- 1966Ki10 D.-H.Kim - J.Phys.Soc.Japan 21, 2445 (1966).
Two-Particle States Excited by the (α,d) Reactions on s - d Shell Nuclei.
- 1966Ma75 J.B.Marion - Phys.Letters 21, 61 (1966).
New list of recommended accelerator calibration energies.
- 1966Mi11 R.G.Miller, R.W.Kavanagh - Phys.Letters 22, 461 (1966).
Decay of ^{34}Ar .
- 1966Vo03 W.Von Witsch, P.Von Brentano, T.Mayer-Kuckuk, A.Richter - Nucl.Phys. 80, 394(1966).
The Statistical Character of the Reaction $^{37}\text{Cl}(p,\alpha)^{34}\text{S}$ at 21-22 MeV Excitation Energy of the Compound Nucleus.
- 1967Ba36 P.H.Barker, N.Drysdale, W.R.Phillips - Proc.Phys.Soc.(London) 91, 587 (1967).
The Beta Decays of ^{22}Mg , ^{30}S and ^{34}Ar .
- 1967En05 P.M.Endt, C.Van der Leun - Nucl.Phys. A105, 1 (1967); Erratum Nucl.Phys. A115, 697 (1968).
Energy Levels of $Z=11$ -21 Nuclei (IV).
- 1967Fa10 M.Fazio, S.Micheletti, M.Pignatelli, L.Zetta - Nuovo Cimento 51B, 459 (1967).
Analysis of S and Cl Wave Functions by Direct Reactions.
- 1967Mc03 W.R.McMurray, P.Van Der Merwe, I.J.Van Heerden - Nucl.Phys. A92, 401(1967).
A Study of ($^3\text{He},n$) Reactions on ^{24}Mg , ^{28}Si and ^{32}S .
- 1967Mi02 R.G.Miller, R.W.Kavanagh - Nucl.Phys. A94, 261 (1967).
Masses and Excitation Energies of Some Proton-Rich Nuclei.
- 1967Wi01 G.Wiechers, W.R.McMurray, I.J.Van Heerden - Nucl.Phys. A92, 175(1967).
Properties of Excited States in ^{34}S from Studies of the (α,γ) and (α,n) Reactions on ^{30}Si .
- 1968Ha09 M.Hagen, K.H.Maier, R.Michaelsen - Phys.Letters 26B, 432 (1968).
The Level Schemes of ^{26}Si , ^{30}S , ^{34}A and ^{42}Ti Studied with the ($^3\text{He},n$) Reaction.
- 1968Mi02 D.Miljanic, G.Paic, B.Antolkovic, P.Tomas - Nucl.Phys. A106, 401 (1968).
(n,d) Reactions on ^{14}N , ^{35}Cl , ^{39}K , ^{40}Ca and ^{75}As at 14.4 MeV.
- 1968Mo22 C.E.Moss - Nucl.Phys. A121, 285 (1968).
Excitation Energies of Levels in ^{34}S .
- 1968Wi20 B.H.Wildenthal, E.Newman - Phys.Rev. 175, 1431 (1968).
Study of Levels in ^{22}Ne , ^{26}Mg , and ^{34}S by Proton Pickup.
- 1969An35 A.E.Antropov, P.P.Zarubin, P.D.Ioannu, B.N.Orlov et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 33, 1721 (1969);
Bull.Acad.Sci.USSR, Phys.Ser. 33, 1576 (1970).
Elastic and Inelastic Proton Scattering on ^{33}S , ^{34}S and ^{35}Cl .
- 1969Br21 H.Brunnader, J.C.Hardy, J.Cerny - Nucl.Phys. A137, 487 (1969).
 ^{34}Ar and $T=1$ States in ^{34}Cl from Two-Nucleon Pick-up Reactions.
- 1969Gr03 H.Grawe, K.P.Lieb - Nucl.Phys. A127, 13 (1969).
 $E2$ Transitions in Doubly Even sd Shell Nuclei.
- 1969Gr29 H.D.Grabner, G.I.Harris - Phys.Rev. 188, 1685 (1969).
Properties of Cl^{34} Levels from the $S^{33}(p,\gamma)\text{Cl}^{34}$ Reaction.
- 1969Pu03 N.G.Puttaswamy, J.L.Yntema - Phys.Rev. 177, 1624(1969).
(d,He^3) Reaction on the Chlorine Isotopes at 23.35 MeV.
- 1969UmZZ C.J.Umbarger, H.S.Plendl, J.W.Nelson - Bull.Am.Phys.Soc. 14, No.4, 567, EE13 (1969).
Thresholds for the Reactions $^{34}\text{S}(p,n)^{34}\text{Cl}$ and $^{31}\text{P}(\alpha,n)^{34}\text{Cl}$.
- 1970An22 A.E.Antropov, V.P.Gusev, P.P.Zarubin, B.N.Orlov, A.I.Sorokin - Izv.Akad.Nauk SSSR, Ser.Fiz. 34, 1707 (1970);
Bull.Acad.Sci.USSR, Phys.Ser. 34, 1517 (1971).
Statistical Analysis of the (p,α) Reaction on Chlorine Isotopes.
- 1970Br10 F.Brandolini, I.Filosofo, C.Signorini, M.Morando - Nucl.Phys. A149, 401 (1970).
Spins and Mean Lives of ^{34}Cl Levels.
- 1970Br11 F.Brandolini, R.G.R.Engmann, C.Signorini - Nucl.Phys. A149, 411 (1970).
Gamma-Ray Transition Strengths in ^{34}Cl .
- 1970Br18 F.Brandolini, C.Signorini - Nuovo Cimento 67A, 247 (1970).
The Mean Lives of the First Two Excited States in ^{34}S .
- 1970Cu02 W.M.Currie, L.G.Earwaker, J.Martin, A.K.Sen Gupta - J.Phys. A3, 73 (1970).
Lifetimes in Some Even Nuclei in the s - d Shell.
- 1970Gr11 M.W.Greene, P.R.Alderson, D.C.Bailey, J.L.Durell et al. - Nucl.Phys. A148, 351 (1970).
Lifetimes and Decay Modes of Energy Levels in ^{34}S .
- 1970Mo09 C.E.Moss, R.V.Poore, N.R.Roberson, D.R.Tilley - Nucl.Phys. A144, 577 (1970).
Spins of Levels in ^{32}P and ^{34}S .
- 1970Ra17 C.E.Ragan, R.V.Poore, N.R.Roberson, G.E.Mitchell, D.R.Tilley - Phys.Rev. C1, 2012 (1970).

REFERENCES FOR A=34(CONTINUED)

- 1970Sy01 *Mean Lifetimes of Low-Lying Levels of ^{34}S .*
D.H.Sykes - Nucl.Phys. A149, 418 (1970).
Levels of ^{34}Cl below 3 MeV.
- 1970Th10 J.M.Thirion, G.Chourauqui, T.Muller, M.Port - J.Phys.(Paris) 31, 743 (1970).
Spin des Niveaux a 0,46 et 0,67 MeV du ^{34}Cl .
- 1970Um01 C.J.Umbarger, K.W.Kemper, J.W.Nelson, H.S.Plendl - Phys.Rev. C2, 1378 (1970).
Excitation Functions for the Reactions $^{34}\text{S}(p,n)^{34}\text{Cl}$ and $^{31}\text{P}(\alpha,n)^{34}\text{Cl}$.
- 1971AlZB REPT 1971 Annual Univ Notre Dame, P113, L. Alexander, 11/13/72, NDP.
- 1971Ar32 A.G.Artukh, V.V.Avdeichikov, G.F.Gridnev, V.L.Mikheev et al. - Nucl.Phys. A176, 284 (1971).
New Isotopes $^{29,30}\text{Mg}$, $^{31,32,33}\text{Al}$, $^{33,34,35,36}\text{Si}$, $^{35,36,37,38}\text{P}$, $^{39,40}\text{S}$ and $^{41,42}\text{Cl}$ Produced in Bombardment of a ^{232}Th Target with 290 MeV ^{40}Ar Ions.
- 1971Av10 V.D.Avchukhov, S.S.Vasilev, I.S.Dneprovskii, N.N.Kolesnikov et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 35, 838 (1971);
Bull.Acad.Sci.USSR, Phys.Ser. 35, 772 (1972).
The γ Rays of ^{29}Al , ^{34}Cl , and ^{51}Ti .
- 1971Ba98 V.Barci, F.Brandolini, C.Signorini - Lett.Nuovo Cim. 2, 863 (1971).
Mean Lives of Three ^{34}Cl Levels with the Recoil-Distance Method.
- 1971De27 P.M.DeLuca, J.C.Lawson, E.D.Berners, P.R.Chagnon - Nucl.Phys. A173, 307 (1971).
Angular Correlation Studies in $^{32}\text{S}(^3\text{He},p\gamma)^{34}\text{Cl}$.
- 1971Er03 J.R.Erskine, D.J.Crozier, J.P.Schiffer, W.P.Alford - Phys.Rev. C3, 1976 (1971).
Energy Levels in Cl^{34} from the $\text{S}^{33}(\text{He}^3, d)\text{Cl}^{34}$ Reaction: A Study of the $(d_{3/2})^2$ and the $d_{3/2}f_{7/2}$ Interactions.
- 1971Gr26 M.W.Greene, J.A.Kuehner, A.A.Pilt - Phys.Lett. 35B, 560 (1971).
The $J^\pi = 5^-$ Level in ^{34}S at 5.689 MeV.
- 1971Hy02 A.K.Hyder, G.I.Harris - Phys.Rev. C4, 2046 (1971).
Properties of Levels in Cl^{34} .
- 1971Mo27 C.E.Moss, C.Detraz, C.S.Zaidins - Nucl.Phys. A174, 408 (1971).
Beta Decay of ^{26}Si , ^{30}S and ^{34}Ar .
- 1971Mu03 P.J.Mulhern, G.D.Jones, I.G.Main, B.T.McCrone et al. - Nucl.Phys. A162, 259 (1971).
Angular Correlation Studies in ^{34}S .
- 1971Sn01 K.A.Snover, J.M.McDonald, D.B.Fossan, E.K.Warburton - Phys.Rev. C4, 398 (1971).
Gamma-Ray Decay of the Low-Lying Levels of Cl^{34} .
- 1971So01 P.Sona, G.Schrieder, W.Kutschera - Nucl.Phys. A161, 283 (1971).
Lifetimes of the 4.263 and 4.688 MeV Levels in ^{34}S .
- 1971Va21 J.G.Van der Baan, B.R.Sikora - Nucl.Phys. A173, 456 (1971).
Investigation of the $^{33}\text{S}(d,p)^{34}\text{S}$ Reaction.
- 1971Wa04 T.E.Ward, P.K.Kuroda - J.Inorg.Nucl.Chem. 33, 609 (1971).
Decay of ^{34}P and ^{34m}Cl .
- 1972AlZT L.A.Alexander, D.S.Longo, E.D.Berners, P.R.Chagnon - Bull.Am.Phys.Soc. 17, No.4, 485, DF11 (1972).
Lifetime Measurements in ^{34}Cl .
- 1972Bb01 R.Bass, U.Friedland, B.Hubert, H.Nann, A.Reiter - Nucl.Phys. A198, 449 (1972).
Study of $(^3\text{He}, n)$ Reactions on Nuclei in the 2s-1d Shell.
- 1972Bo02 G.J.Bock, E.A.Samworth, J.W.Olness, E.K.Warburton - Phys.Rev. C5, 284 (1972).
Accurate Excitation Energies for Some Levels of ^7Be , ^{27}Al , ^{34}S , and ^{37}Cl .
- 1972Ca22 J.M.G.Caraca, R.D.Gill, A.J.Cox, H.J.Rose - Nucl.Phys. A193, 1 (1972).
Gamma Decay of Low-Lying States in ^{30}S and ^{34}Ar .
- 1972Cr08 D.J.Crozier - Nucl.Phys. A198, 209 (1972).
Energy Levels of ^{34}S from the $^{33}\text{S}(d,p)^{34}\text{S}$ Reaction.
- 1972Gr15 M.W.Greene, J.A.Kuehner, G.C.Ball, C.Broude, J.S.Forster - Nucl.Phys. A188, 83 (1972).
Lifetime of the 5.69 MeV 5^- Levels in ^{34}S .
- 1972Ha28 Y.Hahn - Phys.Rev. C5, 1788 (1972).
Generalized Variational Bounds and Shape Resonances.
- 1972Ha58 J.C.Hardy, H.Schmeing, J.S.Geiger, R.L.Graham, I.S.Towner - Phys.Rev.Lett. 29, 1027 (1972).
Mirror Asymmetry of Fermi β Decay.
- 1972Ha82 J.C.Hardy, D.E.Alburger - Phys.Lett. 42B, 341 (1972).
Fermi β -Decay: The Half-Lives of $^{26}\text{Al-m}$, ^{34}Cl , $^{38}\text{K-m}$ and ^{42}Sc .
- 1972Ho02 R.N.Horoshko, M.H.Shapiro - Nucl.Phys. A180, 37 (1972).
The $^{36}\text{Ar}(d, \alpha)^{34}\text{Cl}$ Reaction at 19 MeV.
- 1972Jo10 G.D.Jones, E.M.Jayasinghe, P.J.Mulhern, I.G.Main, P.J.Twin - J.Phys.(London) A5, 1073 (1972).
Spin Assignments for Some Higher Excited States in ^{34}S .
- 1972Na23 H.Nann, L.Armbruster, B.H.Wildenthal - Nucl.Phys. A198, 11 (1972).
Investigation of the $^{32}\text{S}(^3\text{He}, p)^{34}\text{Cl}$ Reaction.
- 1972Pa02 R.A.Paddock - Phys.Rev. C5, 485 (1972).

REFERENCES FOR A=34(CONTINUED)

- (*p,t*) Reaction on Even-Even $N = Z$ Nuclei in the $2s\ 1d$ Shell.
1972PI06 A.V.Plavko, H.Kamitsubo, J.Gosset, R.M.Lombard et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 36, 625 (1972);
Bull.Acad.Sci.USSR, Phys.Ser. 36, 566 (1973).
Elastic and Inelastic Scattering of 24.5 MeV Polarized Protons by ^{32}S and ^{34}S .
1972Vi04 B.Vignon, J.P.Longequeue, I.S.Towner - Nucl.Phys. A189, 513 (1972).
Low-Lying Positive-Parity States of ^{34}Cl and ^{36}Cl from (*p,d*) Reactions And Shell-Model Interpretations.
1973An13 A.Anttila, S.Penttinen - Phys.Scr. 8, 32 (1973).
Investigation of Nuclear Levels in ^{34}Cl .
1973An24 A.E.Antropov, S.I.Vasilev, P.P.Zarubin, I.N.Kukhtina, B.N.Orlov - Izv.Akad.Nauk SSSR, Ser.Fiz. 37, 1873 (1973);
Bull.Acad.Sci.USSR, Phys.Ser. 37, No.9, 67 (1974).
Elastic and Inelastic Scattering of 24.2 MeV α Particles from ^{34}S .
1973Ca15 T.P.G.Carola, B.Sikora - Nucl.Phys. A206, 427 (1973).
Lifetimes of ^{34}Cl Levels.
1973Cr04 D.J.Crozier, H.T.Fortune, R.Middleton, S.Hinds - Phys.Lett. 46B, 189 (1973).
The $(1f_{7/2})^2$ States in ^{34}S .
1973EnVA P.M.Endt, C.Van der Leun - Nucl.Phys. A214, 1 (1973); Erratum Nucl.Phys. A248, 153 (1975).
Energy Levels of $A = 21$ -44 Nuclei (V).
1973Go33 D.R.Goosman, C.N.Davids, D.E.Alburger - Phys.Rev. C8, 1324 (1973).
Accurate Masses and β -Decay Schemes for ^{34}P and ^{33}Si .
1973Go42 J.Gosset, H.Kamitsubo, R.Lombard, B.Mayer et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 37, 1721 (1973);
Bull.Acad.Sci.USSR, Phys.Ser. 37, No.8, 136 (1974).
The (*p,p'*) and (*p,d*) Reaction with Polarized Protons on the Isotopes of Sulfur and Certain Other Light and Intermediate Nuclei.
1973La03 W.A.Lanford, B.H.Wildenthal - Phys.Rev. C7, 668 (1973).
Calculations of Allowed Beta Decay in the (*0d,1s*) Shell.
1973Ry01 J.S.Ryder, G.J.Clark, J.E.Draper, J.M.Freeman et al. - Phys.Lett. 43B, 30 (1973).
The Superallowed Fermi Decay of ^{34}Cl and the Vector Weak Interaction.
1974Al05 C.Alderliesten, P.G.A.M.Aerts, H.M.J.Van Bijlert, C.van der Leun - Nucl.Phys. A220, 284 (1974).
Investigation of ^{38}Ar Levels by $^{37}\text{Cl} + p$ Reactions (I). Yield Curves, Excitation Energies and Branching Ratios.
1974Be18 R.A.I.Bell, J.V.Thompson, I.G.Graham, L.E.Carlson - Nucl.Phys. A222, 477 (1974).
Energies, Decay Schemes and Lifetimes of Levels in ^{34}Ar Measured Using The $^{32}\text{S}(^3\text{He},n\gamma)^{34}\text{Ar}$ Reaction.
1974Cl09 C.F.Clement, S.M.Perez - J.Phys.(London) A7, 1464 (1974).
The Absolute Normalization of Spectroscopic Factors for Proton Transfer Reactions on $^{35}, ^{37}\text{Cl}$.
1974CIZK REPT CONF-740218, Paper 70.
1974Gr06 H.Grawe, R.Konig - Z.Phys. 266, 41 (1974).
Electromagnetic Transition Rates and Vibrational Modes in ^{34}S .
1974Gr15 H.Grawe, J.E.Cairns, M.W.Greene, J.A.Kuehner - Can.J.Phys. 52, 950 (1974).
An Investigation of Odd Parity States in ^{32}S .
1974Gr19 H.Grawe, U.Lohle, R.Konig - Z.Phys. 268, 419 (1974).
The Gamma-Decay of ^{34}Ar .
1974Gu26 V.B.Gubin, Z.F.Kalacheva, E.A.Romanovskii, E.I.Sirotnin et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 38, 2574 (1974);
Bull.Acad.Sci.USSR, Phys.Ser. 38, No.12, 97 (1974).
Elastic Scattering of 5-6 MeV Protons by ^{34}S .
1974Ha02 J.C.Hardy, H.Schmeing, W.Benenson, G.M.Crawley et al. - Phys.Rev. C9, 252 (1974).
Fermi Beta Decay: The Masses of ^{22}Mg , ^{26}Si , ^{30}S , and ^{34}Ar .
1974Ha26 J.C.Hardy, H.Schmeing, J.S.Geiger, R.L.Graham - Nucl.Phys. A223, 157 (1974).
The Decay of ^{34}Ar : A Mirror Comparison of Superallowed β -Decay.
1974OI02 A.Olin, O.Hausser, T.K.Alexander, A.J.Ferguson, W.Witthuhn - Nucl.Phys. A221, 555 (1974).
Static Moments of the First Excited States of $^{32}, ^{34}\text{S}$ and $^{204}, ^{206}\text{Pb}$.
1974Va25 J.J.M.Van Gasteren, B.Sikora, A.Van der Steld - Nucl.Phys. A231, 411 (1974).
Investigation of the Reaction $^{35}\text{Cl}(\tau,\alpha)^{34}\text{Cl}$ at $E = 15$ MeV.
1974Ve06 A.Veyssiere, H.Beil, R.Bergere, P.Carlos et al. - Nucl.Phys. A227, 513 (1974).
A Study of the Photoneutron Contribution to the Giant Dipole Resonance Of *s-d* Shell Nuclei.
1975Al08 G.D.Alkhozov, S.L.Belostotsky, O.A.Domchenkov, Yu.V.Dotsenko et al. - Phys.Lett. 57B, 47 (1975).
Nuclear Sizes of $^{40}, ^{48}\text{Ca}$ and $^{32}, ^{32}\text{S}$ Isotopes Determined from 1 GeV Proton Elastic Scattering.
1975Al28 G.D.Alkhozov, S.L.Belostotskii, A.A.Vorobev, V.T.Grachev et al. - Yad.Fiz. 22, 902 (1975); Sov.J.Nucl.Phys. 22, 469 (1975).
Scattering of 1-GeV Protons on ^{28}Si , ^{32}S , and ^{34}S .
1975Au06 G.F.Auchampaugh, J.Halperin, R.L.Macklin, W.M.Howard - Phys.Rev. C12, 1126 (1975).
Kilovolt $^{33}\text{S}(n,\alpha_0)$ and $^{33}\text{S}(n,\gamma)$ Cross Sections: Importance in the Nucleosynthesis of the Rare Nucleus ^{36}S .
1975Ba29 V.P.Badovsky, A.A.Vorobyev, V.G.Egoshin, V.V.Lylo et al. - Ukr.Fiz.Zh. 20, 501 (1975).
Study of the Reactions $^{50}\text{Cr}(p,p)^{50}\text{Cr}$, $^{34}\text{S}(d,d)^{34}\text{S}$ and $^{34}\text{S}(d,p)^{35}\text{S}$ with Polarized Protons and Deuterons.

REFERENCES FOR A=34(CONTINUED)

- 1975Cr07 D.J.Crozier, H.T.Fortune - Phys.Lett. 57B, 228 (1975).
Configuration Dependent $L = 0$ Shapes in $^{32}\text{S}(t,p)^{34}\text{S}$.
- 1975De16 G.Deconninck, G.Debras - Radiochem.Radioanal.Lett. 20, 175 (1975).
Analysis of Chlorine by Prompt Gamma Rays Induced by Protons.
- 1975DeZS JOUR PHCAA 31 No3 31 DF1.
- 1975Ha24 J.C.Hardy, H.Schmeing, J.S.Geiger, R.L.Graham, I.S.Towner - Izv.Akad.Nauk SSSR, Ser.Fiz. 39, 51 (1975);
Bull.Acad.Sci.USSR, Phys.Ser. 39, No.1, 42 (1975).
Superallowed Nuclear β -Decay.
- 1975He25 J.A.J.Hermans, G.A.P.Engelbertink, M.A.van Driel, H.H.Eggenhuisen, D.Bucurescu - Nucl.Phys. A255, 221 (1975).
Precision Lifetime Measurements with the Coincident High-Velocity DSA Method in Inverse Reactions.
- 1975Ke11 J.Keinonen, M.Riihonen, A.Anttila - Phys.Scr. 12, 280 (1975).
Strengths of Analogue Resonances in (p,γ) Reactions on Sulphur Isotopes.
- 1975Sh10 R.Sherr, I.Talmi - Phys.Lett. 56B, 212 (1975).
Systematic Effects of Charge Independence Breaking on Energies of Analog States.
- 1975So11 K.H.Souw, J.C.Adloff, D.Disdier, P.Chevallier - Phys.Rev. C12, 1103 (1975).
 $E0$ Decay of the First Excited 0^+ States in ^{34}S and ^{38}Ar .
- 1975Va02 M.A.Van Driel, H.Klijnman, G.A.P.Engelbertink, H.H.Eggenhuisen, J.A.J.Hermans - Nucl.Phys. A240, 98 (1975).
Allowed Electron-Capture Branches in the Decay of ^{34m}Cl .
- 1975Wh01 W.R.Wharton, P.T.Debevec - Phys.Rev. C11, 1963 (1975).
Study of the $(^6\text{Li}, ^6\text{He})$ Reactions.
- 1976Bi14 O.Bilaniuk, T.Fortune - Izv.Akad.Nauk SSSR, Ser.Fiz. 40, 2253 (1976); Bull.Acad.Sci.USSR, Phys.Ser. 40, No.10, 197 (1976).
Investigation of the States of ^{34}Cl Excited in the $^{32}\text{S}(^3\text{He},p)^{34}\text{Cl}$ Reaction.
- 1976Co11 B.E.Cooke, R.L.Kozub, W.McLatchie, T.K.Alexander et al. - J.Phys.(London) G2, 131 (1976).
Lifetimes of Levels in $^{36}, ^{37}\text{Cl}$ and ^{34}S Excited by the ^{35}Cl Bombardment Of ^3H .
- 1976De24 R.M.del Vecchio, R.T.Kouzes, R.Sherr - Nucl.Phys. A265, 220 (1976).
Experimental Study of $(f_{7/2})^{7+}$ States in s - d Shell Nuclei.
- 1976Me03 J.C.Merdinger, N.Schulz, M.Toulemonde - J.Phys.(Paris) 37, 1 (1976).
Recoil-Distance Lifetime Measurement in ^{37}Cl .
- 1976Wi08 D.H.Wilkinson, D.E.Alburger - Phys.Rev. C13, 2517 (1976).
Superallowed Fermi β Decay: Lifetimes of ^{34}Cl , $^{38}\text{K-m}$, ^{42}Sc , ^{46}V , and ^{50}Mn .
- 1977Aj01 F.Ajzenberg-Selove, E.R.Flynn, S.Orbesen, J.W.Sunier - Phys.Rev. C15, 1 (1977).
States of ^{34}P .
- 1977Ba16 P.H.Barker, R.E.White, H.Naylor, N.S.Wyatt - Nucl.Phys. A279, 199 (1977).
The Threshold Energy for the $^{34}\text{S}(p,n)^{34}\text{Cl}$ Reaction and the Q -Value of the ^{34}Cl Superallowed Beta Decay.
- 1977BeYD U.E.P.Berg, K.Wienhard - Proc.Intern.Conf.Nucl.Structure, Tokyo, Vol.1, p.216 (1977).
Nuclear Resonance Scattering from ^{34}S .
- 1977Bi04 O.M.Bilaniuk, D.J.Crozier, H.T.Fortune, R.Middleton - Phys.Rev. C16, 55 (1977).
 ^{34}Cl via the $^{32}\text{S}(^3\text{He},p)$ Reaction.
- 1977Bu11 G.W.Butler, D.G.Perry, L.P.Remsberg, A.M.Poskanzer et al. - Phys.Rev.Lett. 38, 1380 (1977).
Observation of the New Nuclides ^{27}Ne , ^{31}Mg , ^{32}Mg , ^{34}Al , and ^{39}P .
- 1977Da02 D.Dassie, F.Leccia, P.Mennrath - Nucl.Phys. A276, 260 (1977).
Etude des Etats Excites du Noyau ^{34}Cl (I). Courbe d'Excitation de la Reaction $^{33}\text{S}(p,\gamma)^{34}\text{Cl}$, Energies, Rapports d'Embranchement et Vies Moyennes des Niveaux du ^{34}Cl .
- 1977Da03 D.Dassie, F.Leccia, P.Mennrath, J.C.Sens - Nucl.Phys. A276, 279 (1977).
Etude des Etats Excites du Noyau ^{34}Cl (I). Spins, Parites, Melanges Multipolaires et Theorie.
- 1977GrZH H.Grawe, K.Kandler - Proc.Intern.Conf.Nuclear Structure, Tokyo, Vol.1, p.217 (1977).
Odd Parity High Spin States in ^{34}S .
- 1977He12 J.A.J.Hermans, G.A.P.Engelbertink, L.P.Ekstrom, H.H.Eggenhuisen, M.A.Van Driel - Nucl.Phys. A284, 307 (1977).
High-Velocity DSA Measurements in the 1-20 ps Lifetime Region: Comparison between DSA and RD.
- 1977Na05 A.M.Nathan, D.E.Alburger - Phys. Rev. C15, 1448 (1977).
Mass and Beta Decay of ^{34}Si .
- 1977Na10 H.Nann, W.S.Chien, A.Saha, B.H.Wildenthal - Phys.Rev. C15, 1959 (1977).
High-Spin States of $(fp)^2$ Character in $^{34}, ^{35}, ^{36}\text{Cl}$ and $^{37}, ^{39}\text{Ar}$.
- 1977Pa29 D.C.Palmer, D.R.Maxson, J.R.Bading - J.Phys.(London) G3, 1363 (1977).
 ^{35}Cl , ^{37}Cl and $^{36}\text{Ar}(n,d)$ Reactions at 14.1 MeV.
- 1977Pe17 F.Pellegrini, P.Guazzoni, D.Sinclair, E.Garman - Phys.Rev. C16, 1878 (1977).
Observation of the $^{30}\text{Si}(^7\text{Li}, ^3\text{He})^{34}\text{P}$ Reaction at 24 MeV.
- 1977Sc36 D.Schwalm, E.K.Warburton, J.W.Olness - Nucl.Phys. A293, 425 (1977).
Coulomb Excitation of sd Shell Nuclei: A Self-Contained Set of $B(E2)$ Values and Lifetime Measurements.
- 1977Sh13 R.Sherr - Phys.Rev. C16, 1159 (1977).
Charge-Dependent and A -Dependent Effects in Isotope Shifts of Coulomb Displacement Energies.

REFERENCES FOR A=34(CONTINUED)

- 1977Sz03 L.Szybisz, J.Davidson, V.Silbergleit - Nucl.Phys. A282, 435 (1977).
An Upper Limit for the Induced Scalar Coupling Constant $f(s)$.
- 1978Ab02 R.Abegg, S.K.Datta - Nucl.Phys. A295, 309 (1978).
Elastic and Inelastic Scattering of 11.8 MeV Polarized Deuterons from ^{34}Si .
- 1978Ba30 Y.A.Babenko, K.I.Zherebtsova, V.N.Kuzmin, Y.A.Nemilov et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 42, 205 (1978);
Bull.Acad.Sci.USSR, Phys.Ser. 42, No.1, 178 (1978).
The (d,α) Reaction on ^{36}S and ^{34}S Nuclei.
- 1978Ba39 P.Baumann, A.M.Bergdolt, G.Bergdolt, A.Huck et al. - Phys.Rev. C18, 247 (1978).
High-Spin States in ^{34}Cl .
- 1978Ba61 P.Baumann, A.M.Bergdolt, G.Bergdolt, A.Huck, G.Walter - Phys.Rev. C18, 2470 (1978).
Structure of Negative Parity High-Spin States in ^{34}Cl .
- 1978Ba77 P.Baumann, A.M.Bergdolt, G.Bergdolt, A.Huck, G.Walter - Fizika, Suppl. 10, 21 (1978).
Lifetime Measurements of Negative Parity States in ^{34}Cl .
- 1978Be46 U.E.P.Berg, K.Bangert, G.Junghans, R.Stock, K.Wienhard - Nucl.Phys. A306, 178 (1978).
Lifetimes of Highly Excited States in ^{34}S from Nuclear Resonance Fluorescence.
- 1978BeXU U.E.P.Berg, G.M.Crawley, B.Zwieglinski - Mich.State Univ.Cyclotron Lab.Ann.Rept., 1977, p.28 (1978).
Inelastic proton scattering ^{34}S .
- 1978Cr01 D.J.Crozier, H.T.Fortune, R.Middleton, S.Hinds - Phys.Rev. C17, 455 (1978).
 ^{34}S by the $^{32}\text{S}(t,p)^{34}\text{S}$ Reaction.
- 1978En02 P.M.Endt, C.van der Leun - Nucl.Phys. A310, 1 (1978).
Energy Levels of $A = 21-44$.
- 1978HaYH J.C.Hardy - AECL-6366, 29 (1978).
The Decays of ^{35}K and ^{36}K .
- 1978Ja10 R.Jahn, D.P.Stahel, G.J.Wozniak, R.J.de Meijer, J.Cerny - Phys.Rev. C18, 9 (1978).
Survey of the $(\alpha,^2\text{He})$ Reaction on $1p$ - and $2s$ $1d$ -Shell Nuclei.
- 1979AnZZ G.Andersson, E.Hagberg, B.Jonson, S.Mattsson, G.Nyman - NEANDC(OR)-152L, p.7 (1979).
On-Line Studies of Very Unstable Nuclei.
- 1979Ba54 R.Bass, J.Idzko, H.Pelz, K.Stelzer et al. - Nucl.Instrum.Methods 163, 377 (1979).
Symmetrical Four-Crystal Compton Polarimeter for Gamma Rays: Design and Application.
- 1979Ca15 L.C.Carraz, G.T.Ewan, E.Hagberg, J.C.Hardy et al. - Phys.Lett. B 85, 212 (1979).
A precise determination of the excitation energy of the lowest $T=3/2$ state in ^{35}Ar .
- 1979DaZV W.G.Davies, T.K.Alexander, G.C.Ball, J.S.Forster, I.V.Mitchell - Bull.Am.Phys.Soc. 24, No.4, 613, DM9 (1979).
High Velocity DSAM Measurements of the First Excited 2^+ Levels of ^{28}Si , ^{32}S and ^{34}S .
- 1979De02 C.Detraz, D.Guillemaud, G.Huber, R.Klapisch et al. - Phys.Rev. C19, 164 (1979).
Beta Decay of $^{27-32}\text{Na}$ and their Descendants.
- 1979Fo17 H.T.Fortune, L.Bland, R.Middleton, W.Chung, B.H.Wildenthal - Phys.Lett. 87B, 29 (1979).
Systematics of Ground-State (t,p) Cross Sections in the $2s$ - $1d$ Shell.
- 1979HaZT E.Hagberg, J.C.Hardy, L.C.Carraz, P.G.Hansen et al. - Bull.Am.Phys.Soc. 24, No.4, 613, DM11 (1979).
A Precise Determination of the Excitation Energy of the Lowest $T = 3/2$ State in ^{35}Ar .
- 1979HaZY E.Hagberg, J.C.Hardy - AECL-6452, p.14 (1979).
The decays of ^{35}K and ^{36}K .
- 1979Ku10 E.Kuhlmann, J.R.Calarco, V.K.C.Cheng, D.G.Mavis et al. - Phys.Rev. C20, 5 (1979).
 $E1$ and $E2$ Strength in ^{32}S and ^{34}S Observed in α -Capture Reactions.
- 1979Ra37 H.L.Ravn - Phys.Rep. 54, 201 (1979).
Experiments with intense secondary beams of radioactive ions.
- 1979We10 G.D.Westfall, T.J.M.Symons, D.E.Greiner, H.H.Heckman et al. - Phys.Rev.Lett. 43, 1859 (1979).
Production of Neutron-Rich Nuclides by Fragmentation of 212 MeV/amu ^{48}Ca .
- 1979Za01 P.C.Zalm, A.Holthuisen, J.A.G.De Raedt, G.van Middelkoop - Nucl.Phys. A315, 133 (1979).
Transient Field g -Factor Measurements on the 2^+_1 States of ^{32}S and ^{34}S .
- 1980Ba40 G.C.Ball, O.Hausser, T.K.Alexander, W.G.Davies et al. - Nucl.Phys. A349, 271 (1980).
Electric Quadrupole Moments of the First Excited States of ^{28}Si , ^{32}S and ^{34}S .
- 1980Ew02 G.T.Ewan, E.Hagberg, J.C.Hardy, B.Jonson et al. - Nucl.Phys. A343, 109 (1980).
The Decay of ^{35}K .
- 1980Fa07 E.Fabrizi, S.Micheletti, M.Pignanelli, F.G.Resmini et al. - Phys.Rev. C21, 844 (1980).
Proton Elastic Scattering on Light Nuclei. II. Nuclear Structure Effects.
- 1980PrZT V.S.Prokopenko, V.D.Sklyarenko, V.K.Chernievsky, A.V.Shustov - Program and Thesis, Proc.30th
Ann.Conf.Nucl.Spectrosc.Struct.At.Nuclei, Leningrad, p.318 (1980).
Investigation of spin phenomena in inelastic with energy of 6.9 MeV proton scattering on $^{32,24}\text{S}$.
- 1980To13 S.A.Tonsfeldt, T.B.Clegg, E.J.Ludwig, Y.Tagishi, J.F.Wilkerson - Phys.Rev.Lett. 45, 2008 (1980).
J-Dependent Effects in Tensor Analyzing Powers for the (d,α) Reaction.
- 1980Wi13 H.S.Wilson, R.W.Kavanagh, F.M.Mann - Phys.Rev. C22, 1696 (1980).

REFERENCES FOR A=34(CONTINUED)

- 1981Bh04 C.M.Bhat, N.G.Puttaswamy, J.L.Yntema - J.Phys.(London) G7, 1529 (1981).
The (d,t) Reaction on Chlorine Isotopes at 23 MeV
- 1981De37 R.J.de Meijer, L.W.Put, J.C.Vermeulen - Phys.Lett. 107B, 14 (1981).
Two-Nucleon Cluster Spectroscopic Factors: A search for the bound-state parameters.
- 1981Sp07 R.H.Spear - Phys.Rep. 73, 369 (1981).
Static Quadrupole Moments of the First 2^+ States in the $2s$ - $1d$ Shell: A review of experiment and theory.
- 1981To19 F.Touchard, J.Biderman, M.de Saint Simon, C.Thibault et al. - Nucl.Instrum.Methods 186, 329 (1981).
Production of ionic and atomic beams of alkaline elements.
- 1982Be32 A.M.Bernstein, R.A.Miskimen, B.Quinn, S.A.Wood et al. - Phys.Rev.Lett. 49, 451 (1982).
Determination of Relative Signs of Neutron and Proton Transition Matrix Elements: Strong cancellation observed for the $^{34}\text{S}(0^+ \rightarrow 2^+)$ transition.
- 1982De29 R.J.De Meijer, L.W.Put, J.Akkermans, J.C.Vermeulen, C.R.Bingham - Nucl.Phys. A386, 200 (1982).
Microscopic and Macroscopic DWBA Analysis of the Reactions $^{28}\text{Si}(\alpha,d)^{30}\text{P}$, $^{32}\text{S}(d,\alpha)^{30}\text{P}$ and $^{32}\text{S}(\alpha,d)^{34}\text{Cl}$ Studied at $E(\alpha) = 50$ MeV and $E(d) = 40$ MeV.
- 1982Gr07 A.Grutter - Int.J.Appl.Radiat.Isotop. 33, 456 (1982).
Decay Data of $^{81,82\text{m},83,84\text{m}}\text{Rb}$ and $^{83,85\text{m},87\text{m}}\text{Sr}$.
- 1982IvZX S.O.Ivanova, E.V.Korbetsky, P.P.Zarubin, O.M.Knyazkov et al. - Program and Theses, Proc.32nd Ann.Conf.Nucl.Spectrosc.Struct.At.Nuclei, Kiev, p.312 (1982).
- 1982MiZQ R.A.Miskimen, A.M.Bernstein, B.Quinn, S.A.Wood et al. - Bull.Am.Phys.Soc. 27, No.7, 730, ED9 (1982).
Proton Scattering at 650 and 800 MeV on ^{30}Si , ^{34}S and ^{42}Ca .
- 1982QaZX S.M.Qaim - Neutron Induced Reactions: Proceedings of the Europhysics Topical Conference, June 21-25, 1982 Smolenice, Editor P.Oblozinsky, Inst.of Physics, Slovak p.291 (1982) Physics and applications Vol.10.
Complex particle emission in fast neutron induced reactions.
- 1982Ta04 Y.Tagishi, T.B.Clegg, E.J.Ludwig, S.A.Tonsfeldt, J.F.Wilkerson - Nucl.Phys. A377, 1 (1982).
Study of (d, α) Reactions on ^{32}S , ^{36}Ar and ^{38}Ar with Vector- and Tensor-Polarized Deuterons.
- 1982Va02 C.J.Van Der Poel, G.A.P.Engelbertink, H.J.M.Aarts, D.E.C.Scherpenzeel et al. - Nucl.Phys. A373, 81 (1982).
High-Spin States in ^{34}Cl .
- 1983Ch50 M.Chenzha, M.Duma, C.Hategan, M.Tanase et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 47, 2143 (1983).
Experimental Study of Threshold Anomalies in the Elastic Scattering of Protons by Nuclei with A = 30.
- 1983Iv01 S.P.Ivanova, E.V.Korbetsky, P.P.Zarubin, O.M.Knyazkov et al. - Izv.Akad.Nauk SSSR, Ser.Fiz. 47, 127 (1983).
Analysis of 6 MeV Proton Scattering on ^{34}S .
- 1983Ko22 V.T.Koslowsky, E.Hagberg, J.C.Hardy, R.E.Azuma et al. - Nucl.Phys. A405, 29 (1983).
The Half-Lives of $^{26\text{m}}\text{Al}$, ^{34}Cl , $^{38\text{m}}\text{K}$: Precision measurements with isotope-separated samples.
- 1983Kr06 Th.Kruger, H.Appel, W.-G.Thies, H.Behrens - Phys.Lett. 121B, 303 (1983).
Spectrum Shape Studies of the β -Transitions $^{34\text{m}}\text{Cl}(3^+ \rightarrow 2^+)^{34}\text{S}$ and $^{34}\text{Cl}(0^+ \rightarrow 0^+)^{34}\text{S}$.
- 1983La12 M.Langevin, C.Detraz, D.Guillemaud-Mueller, A.C.Mueller et al. - Phys.Lett. 125B, 116 (1983).
 ^{35}Na : A new neutron-rich Sodium isotope.
- 1983LaZU M.Langevin, C.Detraz, D.Guillemaud-Mueller, A.C.Mueller et al. - IPNO-DRE-83-18 (1983).
 β -Delayed Neutrons from Very Neutron-Rich Sodium and Magnesium Isotopes.
- 1983Mi25 R.A.Miskimen, A.M.Bernstein, B.Quinn, S.A.Wood et al. - Phys.Lett. 131B, 26 (1983).
DWIA Predictions of (p,p') Data using Electromagnetically Constrained Densities.
- 1983Ra04 S.Raman, E.T.Jurney, D.A.Outlaw, I.S.Towner - Phys.Rev. C27, 1188 (1983).
 ^{34}Cl Superallowed β Decay.
- 1983Wa27 F.B.Waanders, J.P.Reinecke, H.N.Jacobs, J.J.A.Smit et al. - Nucl.Phys. A411, 81 (1983).
The Excited States of ^{34}Cl .
- 1984AlZV R.Alarcon, D.Wang, J.Rapaport, R.T.Kouzes et al. - Bull.Am.Phys.Soc. 29, No.7, 1055, ED5 (1984).
Nucleon Scattering from ^{34}S .
- 1984Be05 R.E.Benenson, M.T.McEllistrem, J.L.Weil, W.R.Coker, T.Tamura - Phys.Rev. C29, 709 (1984).
Interference between Reaction Mechanisms in $^{32}\text{S}(^3\text{He},n)^{34}\text{Ar}$.
- 1984Be26 U.E.P.Berg, K.Ackermann, K.Bangert, C.Blasing et al. - Phys.Lett. 140B, 191 (1984).
Bound State M1 Transitions in sd-Shell Nuclei.
- 1984Gu19 D.Guillemaud-Mueller, C.Detraz, M.Langevin, F.Naulin et al. - Nucl.Phys. A426, 37 (1984).
 β -Decay Schemes of very Neutron-Rich Sodium Isotopes and Their Descendants.
- 1984Ke01 J.Keinonen, A.Luukkainen, M.Bister - Nucl.Phys. A412, 101 (1984).
Isoscalar E2 Strengths in the A = 34 Mass Triplet.
- 1984La03 M.Langevin, C.Detraz, D.Guillemaud-Mueller, A.C.Mueller et al. - Nucl.Phys. A414, 151 (1984).
 β -Delayed Neutrons from very Neutron-Rich Sodium and Magnesium Isotopes.
- 1984Ma49 W.A.Mayer, W.Henning, R.Holzwarth, H.J.Korner et al. - Z.Phys. A319, 287 (1984).
Mass Excess and Excited States of Neutron-Rich Silicon, Phosphorus and Sulphur Isotopes.
- 1984Sa11 A.Saha, K.K.Seth, M.Artuso, B.Harris et al. - Phys.Rev.Lett. 52, 1876 (1984).
Determination of the Relative Sign of Neutron and Proton Transition Matrix Elements for the $^{34}\text{S}(0 \rightarrow 2^+)$ Transition.

REFERENCES FOR A=34(CONTINUED)

- 1984So13 R.J.Sobie, T.E.Drake, K.L.Erdman, R.R.Johnson et al. - Phys.Rev. C30, 1612 (1984).
Elastic and Inelastic Scattering of 50 MeV Pions from ^{12}C , ^{32}S , and ^{34}S .
- 1985A103 R.Alarcon, J.Rapaport, R.T.Kouzes, W.H.Moore, B.A.Brown - Phys.Rev. C31, 697 (1985).
Nucleon Scattering from ^{34}S and the Relative Sign of Neutron and Proton Transition Matrix Elements for the $(0 \rightarrow 2_2^+)$ Transition.
- 1985A118 T.K.Alexander, G.C.Ball, D.Horn, J.S.Forster et al. - Nucl.Phys. A444, 285 (1985).
Lifetime Measurements of Levels in ^{34}Ar : Isoscalar and isovector matrix elements for E2 analogue transitions.
- 1985Ay01 J.Aysto, D.M.Moltz, X.J.Xu, J.E.Reiff, J.Cerny - Phys.Rev.Lett. 55, 1384 (1985).
Observation of the First $T(z) = -5/2$ Nuclide, ^{35}Ca via Its β -Delayed Two-Proton Emission.
- 1985Br29 B.A.Brown, B.H.Wildenthal - At.Data Nucl.Data Tables 33, 347 (1985).
Experimental and Theoretical Gamow-Teller Beta-Decay Observables for the sd-Shell Nuclei.
- 1985Dr06 P.V.Drumm, L.K.Fifield, R.A.Bark, M.A.C.Hotchkis et al. - Nucl.Phys. A441, 95 (1985).
Ground-State Mass Excesses and Low-Lying Levels of the Neutron-Rich Nuclei ^{36}P , ^{35}P and ^{34}P .
- 1985Fi03 L.K.Fifield, C.L.Woods, R.A.Bark, P.V.Drumm, M.A.C.Hotchkis - Nucl.Phys. A440, 531 (1985).
Masses and Level Schemes of ^{33}Si and ^{34}Si .
- 1985Gy02 W.Gyles, B.M.Barnett, R.Tacik, K.L.Erdman et al. - Nucl.Phys. A439, 598 (1985).
Neutron Density Difference Measurements for ^{26}Mg , ^{24}Mg and $^{36,34,32}\text{S}$ using Low-Energy Pions.
- 1985Ki07 A.Z.Kiss, E.Koltay, B.Nyako, E.Somorjai et al. - J.Radioanal.Nucl.Chem. 89, 123 (1985).
Measurements of Relative Thick Target Yields for PIGE Analysis on Light Elements in the Proton Energy Interval 2.4-4.2 MeV.
- 1985La16 R.Lappalainen, J.Keinonen, A.Luukkainen - Nucl.Phys. A441, 1 (1985).
Short Lifetimes in ^{34}Cl .
- 1985Ra15 S.Raman, R.F.Carlton, J.C.Wells, E.T.Jurney, J.E.Lynn - Phys.Rev. C32, 18 (1985).
Thermal Neutron Capture Gamma Rays from Sulfur Isotopes: Experiment and theory.
- 1985SaZS N.Sakamoto, S.Matsuki, T.Ohsawa, H.Okamura, T.Komatuzaki - RCNP (Osaka), Ann.Rept., 1984, p.32 (1985).
Elastic and Inelastic Scattering of Polarized Protons from ^{34}S at 66 MeV.
- 1985We07 R.O.Weber, C.I.W.Tingwell, L.W.Mitchell, M.E.Sevior, D.G.Sargood - Nucl.Phys. A439, 176 (1985).
Cross Sections and Thermonuclear Reaction Rates of Proton-Induced Reactions on ^{37}Cl .
- 1985Wo06 U.Worsdorfer, H.J.Emrich, H.Miska, D.Rychel et al. - Nucl.Phys. A438, 711 (1985).
Determination of the Transition Charge Densities of the Low-Lying 2^+ States in ^{34}S .
- 1986A115 W.P.Alford, P.Craig, D.A.Lind, R.S.Raymond et al. - Nucl.Phys. A457, 317 (1986).
Structure of ^{22}Mg , ^{26}Si , ^{34}Ar and ^{38}Ca via the $(^3\text{He}, n)$ Reaction.
- 1986A121 R.Alarcon, J.Rapaport - Nucl.Phys. A458, 502 (1986).
Neutron Elastic and Inelastic Scattering from ^{28}Si , ^{32}S and ^{34}S .
- 1986Du07 J.P.Dufour, R.Del Moral, A.Fleury, F.Hubert et al. - Z.Phys. A324, 487 (1986).
Beta Decay of ^{17}C , ^{19}N , ^{22}O , ^{24}F , ^{26}Ne , ^{32}Al , ^{34}Al , $^{35} - ^{36}\text{Si}$, $^{36} - ^{37} - ^{38}\text{P}$, ^{40}S .
- 1986La17 M.Langevin, A.C.Mueller, D.Guillemaud-Mueller, M.G.Saint-Laurent et al. - Nucl.Phys. A455, 149 (1986).
Mapping of the Proton Drip-Line up to $Z = 20$: Observation of the $T(z) = -(5/2)$ series ^{23}Si , ^{27}S , ^{31}Ar and ^{35}Ca .
- 1987BaZI D.Bazin, R.Anne, D.Guerreau, D.Guillemaud-Mueller et al. - Contrib.Proc. 5th Int.Conf.Nuclei Far from Stability, Rosseau Lake, Canada, K7 (1987).
The Production Mechanisms of Nuclei Far from Stability at Ganil Energies and their Application to New β -Neutron Emitters.
- 1987Co23 G.P.Coddens, M.Salah, J.A.Harvey, N.W.Hill, N.M.Larson - Nucl.Phys. A469, 480 (1987).
Resonance Structure of $^{33}\text{S} + n$ from Transmission Measurements.
- 1987DuZU J.P.Dufour, R.Del Moral, F.Hubert, D.Jean et al. - Contrib.Proc. 5th Int.Conf.Nuclei Far from Stability, Rosseau Lake, Canada, D1 (1987).
Spectroscopic Measurements with a New Method: The projectile-fragments isotopic separation.
- 1987Fu11 K.Furukawa, M.Kabasawa, T.Kawamura, Y.Takahashi et al. - Phys.Rev. C36, 1686 (1987).
 $^{34}\text{S}(p, n)^{34}\text{Cl}$ Reaction at 35 MeV and Its Microscopic Distorted-Wave Born Approximation Analysis: Stringent test of the shell model.
- 1987Mo07 C.L.Morris, S.J.Seestrom-Morris, D.Dehnhard, C.L.Blilie et al. - Phys.Rev. C35, 1388 (1987).
Neutron and Proton Matrix Elements for 2_1^+ Transitions in $T = 1$ Nuclei from Pion Inelastic Scattering.
- 1987Nu01 J.Nurzynski, T.Kihm, K.T.Knopfle, G.Mairle, H.Clement - Nucl.Phys. A465, 365 (1987).
Reorientation Effects for 52 MeV Vector Polarized Deuterons.
- 1987Or02 H.Orihara, S.Nishihara, K.Furukawa, M.Kabasawa et al. - Nucl.Instrum.Methods Phys.Res. A257, 189 (1987).
Status of the CYRIC Neutron Tof Facilities Upgrade.
- 1987Sa19 L.Satpathy - J.Phys.(London) G13, 761 (1987).
Infinite Nuclear Matter Based Model for Masses of Atomic Nuclei.
- 1987TrZZ A.J.Trudel - Diss.Abst.Int. 48B, 1400 (1987).
Spin and Parity Assignments in ^{34}P Using the Polarized (d, α) and $(d, \alpha\gamma)$ Reactions.
- 1987Wa20 C.Wagemans, H.Weigmann, R.Barthelemy - Nucl.Phys. A469, 497 (1987).
Measurement and Resonance Analysis of the $^{33}\text{S}(n, \alpha)$ Cross Section.
- 1987Zu03 J.D.Zumbro, H.T.Fortune, M.Burlein, C.L.Morris et al. - Phys.Rev. C36, 1479 (1987).

REFERENCES FOR A=34(CONTINUED)

- 1988A114 *Double Charge Exchange to the Double Isobaric Analog State at $T(\pi)$ 292 MeV.*
G.D.Alkhazov - Yad.Fiz. 47, 920 (1988).
- 1988BaYZ *Inelastic Scattering of 1-GeV Protons and Nuclear Transition Densities.*
D.Bazin, R.Anne, D.Guerreau, D.Guillemaud-Mueller et al. - Proc. 5th Int.Conf.Nuclei Far from Stability, Rosseau Lake, Canada 1987, Ed., I.S.Towner, p.722 (1988), AIP Conf.Proc. 164 (1987).
On the Production Mechanisms of Nuclei Far from Stability at GANIL Energies and Their Application to New β -Delayed Neutron Emitters.
- 1988DuZT J.P.Dufour, R.Del Moral, F.Hubert, D.Jean et al. - Proc. 5th Int.Conf.Nuclei Far from Stability, Rosseau Lake, Canada 1987, Ed., I.S.Towner, p.344 (1988), AIP Conf.Proc. 164 (1987).
Spectroscopic Measurements with a New Method: The projectile-fragments isotopic separation.
- 1988Mc05 L.C.McIntyre, J.A.Leavitt, B.Dezfouly-Arjomandy, J.Oder - Nucl.Instrum.Methods Phys.Res. B35, 446 (1988).
Depth Profiling of Phosphorus Using Resonances in the $^{31}\text{P}(\alpha,p)^{34}\text{S}$ Reaction.
- 1988Mu08 A.C.Mueller, D.Bazin, W.D.Schmidt-Ott, R.Anne et al. - Z.Phys. A330, 63 (1988).
 β -Delayed Neutron Emission of ^{15}B , ^{18}C , ^{19}C , ^{20}N , ^{34}Al and ^{39}P .
- 1988MuZY A.C.Mueller, D.Bazin, W.D.Schmidt-Ott, R.Anne et al. - Univ.Paris, Inst.Phys.Nucl., 1987 Ann.Rept., p.E7 (1988).
New Light Beta-Delayed Neutron Emitters.
- 1988Za04 P.P.Zarubin, V.P.Gusev, A.E.Antropov - Izv.Akad.Nauk SSSR, Ser.Fiz. 52, 923 (1988); Bull.Acad.Sci.USSR, Phys.Ser. 52, No.5, 86 (1988).
Compound-Nucleus Proton Elastic Scattering Cross Sections at $E(p)$ 6 MeV.
- 1989Ba50 P.Baumann, A.Huck, G.Klotz, A.Knipper et al. - Phys.Lett. 228B, 458 (1989).
 ^{34}Si : A new doubly magic nucleus (Question).
- 1989Le16 M.Lewitowicz, Yu.E.Penionzhkevich, A.G.Artukh, A.M.Kalinin et al. - Nucl.Phys. A496, 477 (1989).
 β -Delayed Neutron Emission of the Isotopes ^{20}C , $^{40,41,42}\text{P}$, $^{43,44}\text{S}$.
- 1989MuZU A.C.Mueller, M.Lewitowicz, R.Anne, A.G.Artukh et al. - Univ.Paris, Inst.Phys.Nucl., 1988 Ann.Rept., p.25 (1989).
Etude de l'Emission de Neutrons Retardés par des Isotopes Legers tres Riches en Neutrons.
- 1989Va15 J.R.Vanhoy, E.G.Bilpuch, C.R.Westerfeldt, G.E.Mitchell - Phys.Rev. C40, 1959 (1989).
Proton Resonances in ^{34}Cl from $E(x) = 6.3$ to 8.8 MeV.
- 1990Br26 B.A.Brown - Phys.Rev.Lett. 65, 2753 (1990).
Isospin-Forbidden β -Delayed Proton Emission.
- 1990En08 P.M.Endt - Nucl.Phys. A521, 1 (1990); Errata and Addenda Nucl.Phys. A529, 763 (1991); Errata Nucl.Phys. A564, 609 (1993).
Energy Levels of $A = 21$ -44 Nuclei (VII).
- 1990SeZR H.Seifert - Thesis, Univ of Maryland (1990).
Energy Dependence of the Effective Interaction for Nucleon-Nucleus Scattering.
- 1991Bi07 R.Bilger, B.M.Barnett, H.Clement, S.Krell et al. - Phys.Lett. 269B, 247 (1991).
Measurement of Pion Induced Double Charge Exchange on ^{34}S and ^{56}Fe at $T(\pi) = 50$ MeV.
- 1991De26 C.Detraz - Z.Phys. A340, 227 (1991).
The Branching Ratio of β -Delayed Two-Proton Emission.
- 1991Ke09 J.J.Kelly, M.A.Khandaker, P.Boberg, A.E.Feldman et al. - Phys.Rev. C44, 1963 (1991).
Neutron and Proton Transition Densities from $^{32,34}\text{S}(p,p')$ at $E(p) = 318$ MeV. I. Isoscalar Densities for ^{32}S .
- 1991Kh06 M.A.Khandaker, J.J.Kelly, P.Boberg, A.E.Feldman et al. - Phys.Rev. C44, 1978 (1991).
Neutron and Proton Transition Densities from $^{32,34}\text{S}(p,p')$ at $E(p) = 318$ MeV. II. Neutron Densities for ^{34}S .
- 1991Kr06 S.Krell, B.M.Barnett, H.Clement, J.Jaki et al. - Europhys.Lett. 15, 11 (1991).
Precision Measurements of Low-Energy Pion Scattering on Sulphur Isotopes.
- 1991Or01 N.A.Orr, W.Mittig, L.K.Fifield, M.Lewitowicz et al. - Phys.Lett. 258B, 29 (1991).
New Mass Measurements of Neutron-Rich Nuclei Near $N = 20$.
- 1991Ru05 N.Runyon, P.Misaelides, G.Wolf, K.Bethge - J.Radioanal.Nucl.Chem. 153, 273 (1991).
Determination of Sulfur in Near-Surface Layers of Materials by Means of the $^{32}\text{S}(^3\text{He},p)^{34\text{m}}\text{Cl}$ Nuclear Reaction.
- 1991Zh24 X.G.Zhou, X.L.Tu, J.M.Wouters, D.J.Vieira et al. - Phys.Lett. 260B, 285 (1991).
Direct Mass Measurements of the Neutron-Rich Isotopes of Fluorine Through Chlorine.
- 1992Ka39 A.S.Kachan, A.N.Vodin, B.A.Nemashkalo, R.P.Slabospitsky - Yad.Fiz. 55, 2609 (1992); Sov.J.Nucl.Phys. 55, 1456 (1992).
Position of the M1 Resonance in the Odd-Odd Nucleus ^{34}Cl .
- 1993Au07 G.Audi, A.H.Wapstra - Nucl.Phys. A565, 66 (1993).
The 1993 Atomic Mass Evaluation. (II). Nuclear-Reaction and Separation Energies.
- 1994Br19 A.Bredbacka, M.Brenner, K.-M.Kallman, P.Manngard et al. - Nucl.Phys. A574, 397 (1994).
Low-Energy Elastic-Scattering of Alpha Particles from ^{34}S , ^{50}Cr and ^{62}Ni .
- 1994Fo04 B.Fornal, R.H.Mayer, I.G.Bearden, Ph.Benet et al. - Phys.Rev. C49, 2413 (1994).
Gamma Ray Studies of Neutron-Rich sdf Shell Nuclei Produced in Heavy Ion Collisions.
- 1994Li20 S.Lin, S.A.Brindhaban, P.H.Barker - Phys.Rev. C49, 3098 (1994).
Q Value for the ^{34}Cl Superalloyed Beta Decay.
- 1994Pe08 M.Petratis, J.P.Connelly, H.Crannell, L.W.Fagg et al. - Phys.Rev. C49, 3000 (1994).
Low-Multipolarity Magnetic Transitions in ^{30}Si , ^{32}S , and ^{34}S Studied by 180° Electron Scattering.

REFERENCES FOR A=34(CONTINUED)

- 1995He18 H.Herndl, J.Gorres, M.Wiescher, B.A.Brown, L.Van Wormer - Phys.Rev. C52, 1078 (1995).
Proton Capture Reaction Rates in the rp Process.
- 1995La03 G.A.Lalazissis, C.P.Panos - Phys.Rev. C51, 1247 (1995).
Isospin Dependence of the Oscillator Spacing.
- 1995ReZZ P.L.Reeder, Y.Kim, W.K.Hensley, H.S.Miley et al. - Proc.Intern.Conf on Exotic Nuclei and Atomic Masses, Arles, France, June 19-23, 1995, p.587 (1995).
Beta Decay Half-Lives and Delayed Particle Emission from TOFI Measurements.
- 1996PoZZ F.Pougheon - Priv.Comm. (1996).
- 1996Sh20 Y.-S.Shen, Z.Ren - Z.Phys. A356, 133 (1996).
Skyrme-Hartree-Fock Approach to Spherical Nuclei with Density-Dependent Pairing Correlations.
- 1997Co19 B.J.Cole - Phys.Rev. C56, 1866 (1997).
Systematics of Proton and Diproton Separation Energies for Light Nuclei.
- 1997Ha11 D.Habs, O.Kester, G.Bollen, L.Liljeby et al. - Nucl.Phys. A616, 29c (1997).
The REX-ISOLDE Project.
- 1997Ha55 I.Hamamoto, H.Sagawa, X.Z.Zhang - Nucl.Phys. A626, 669 (1997).
Quadrupole Strength Function and Core Polarization in Drip Line Nuclei.
- 1997Ha57 I.Hamamoto, H.Sagawa, X.Z.Zhang - Phys.Rev. C56, 3121 (1997).
Giant Monopole Resonances in Nuclei Near Stable and Drip Lines.
- 1997Is02 B.S.Ishkhanov, I.M.Kapitonov, I.A.Tutyn - Yad.Fiz. 60, No 3, 389 (1997); Phys.Atomic Nuclei 60, 319 (1997).
Partial Photoneuclear Channels and Giant Dipole Resonance in $^{35,37}\text{Cl}$ Nuclei.
- 1997Jo08 G.C.Jon, H.Orihara, T.Niizeki, M.Oura et al. - Phys.Rev. C56, 900 (1997).
Analog Transitions in sd- and f-Shell Nuclei and the Isovector Part of Optical Potentials Studied by the (p,n) Reaction at 35 MeV.
- 1997Ma77 Z.Ma, H.Toki, B.Chen, V.G.Nguyen - Prog.Theor.Phys.(Kyoto) 98, 917 (1997).
The Giant Dipole Resonance in Ar-Isotopes in the Relativistic RPA.
- 1997Pa38 S.K.Patra, R.K.Gupta, W.Greiner - Int.J.Mod.Phys. E6, 641 (1997).
Relativistic Mean-Field Theory and the Structural Properties of Ne,Mg,Si,S,Ar and Ca Nuclei from Proton- to Neutron-Drip Lines.
- 1998Co30 B.J.Cole - Phys.Rev. C58, 2831 (1998).
Proton and Two-Proton Drip Lines in the sd Shell.
- 1998En04 P.M.Endt - Nucl.Phys. A633, 1 (1998).
Supplement to Energy Levels of A = 21-44 Nuclei (VII).
- 1998Ha09 I.Hamamoto, H.Sagawa, X.Z.Zhang - Phys.Rev. C57, R1064 (1998).
Isoscalar and Isovector Dipole Mode in Drip Line Nuclei in Comparison with β -Stable Nuclei.
- 1998Ha39 I.Hamamoto, H.Sagawa, X.Z.Zhang - J.Phys.(London) G24, 1417 (1998).
Collective Modes of Nuclei Far from β -Stability Line.
- 1998Ib01 R.W.Ibbotson, T.Glasmacher, B.A.Brown, L.Chen et al. - Phys.Rev.Lett. 80, 2081 (1998).
Quadrupole Collectivity in $^{32,34,36,38}\text{Si}$ and the N = 20 Shell Closure.
- 1998La02 G.A.Lalazissis, A.R.Farhan, M.M.Sharma - Nucl.Phys. A628, 221 (1998).
Light Nuclei Near Neutron and Proton Drip Lines in Relativistic Mean-Field Theory.
- 1998Le45 M.Lewitowicz, W.Trinder, M.Bhattacharya, E.G.Adelberger et al. - Nuovo Cim. 111A, 835 (1998).
Beta-Decay of Light Nuclei Close to the Proton Drip-Line: ^{40}Ti and ^{35}Ca .
- 1998Sa28 H.Sagawa, I.Hamamoto, X.Z.Zhang - J.Phys.(London) G24, 1445 (1998).
Giant Monopole States in Nuclei Near Drip Lines.
- 1998Sc19 M.Schafer, W.-D.Schmidt-Ott, T.Dorfler, T.Hild et al. - Phys.Rev. C57, 2205 (1998).
Polarization in Fragmentation, g Factor of ^{35}K .
- 1999Ai02 N.Aissaoui, N.Added, N.Carlin, G.M.Crawley et al. - Phys.Rev. C60, 034614 (1999).
Strong Absorption Radii from Reaction Cross Section Measurements for Neutron-Rich Nuclei.
- 1999Ma63 F.Marechal, T.Suomijarvi, Y.Blumenfeld, A.Azhari et al. - Phys.Rev. C60, 034615 (1999).
Proton Scattering by Short Lived Sulfur Isotopes.
- 1999Ng03 V.G.Nguyen, Z.Ma, H.Toki, B.Chen - Nucl.Phys. A649, 37c (1999).
Relativistic Calculations of Giant Resonances with Non-Linear Models.
- 1999OrZZ H.Orihara, H.Suzuki, A.Terakawa, G.C.Jon, H.Ohnuma - Cyclotron Rad.Center, Tohoku Univ., Ann.Rept., 1998, p.13 (1999).
Comparison of (p,n) Cross-Section with Gamow-Teller β -Decay Rates Among $T_z = -1, 0$ and 1 Isobar Triplet.
- 1999Pr09 B.V.Pritychenko, T.Glasmacher, P.D.Cottle, M.Fauerbach et al. - Phys.Lett. 461B, 322 (1999); Erratum Phys.Lett. 467B, 309 (1999).
Role of Intruder Configurations in $^{26,28}\text{Ne}$ and $^{30,32}\text{Mg}$.
- 1999Sa22 H.Sagawa, I.Hamamoto, X.Z.Zhang - Nucl.Phys. A649, 319c (1999).
A Microscopic Study of Giant Resonances in Nuclei Near Drip Lines.
- 1999ShZX Y.Shimbara, H.Fujita, Y.Fujita, A.D.Bacher et al. - Osaka Univ.Lab.Nucl.Studies, Ann.Rept., 1998, p.86 (1999).
A High Resolution (p,p') Experiment at Zero Degrees and $E_p = 160$ MeV.
- 1999Tr04 W.Trinder, J.C.Angelique, R.Anne, J.Aysto et al. - Phys.Lett. 459B, 67 (1999).
 β -Decay of ^{35}Ca .

REFERENCES FOR A=34(CONTINUED)

- 1999YoZW K.Yoneda, H.Sakurai, N.Aoi, N.Fukuda et al. - RIKEN-98, p.78 (1999).
The First Measurements of Half Lives and Neutron Emission Probabilities of Very Neutron Rich Nuclides of Mg, Al, Si, and P.
- 2000KhZX E.Khan, N.Alamanos, F.Auger, E.Bauge et al. - Univ.Paris, Inst.Phys.Nucl., 1998-1999 Ann.Rept., p.31 (2000).
Proton Scattering from the Unstable Proton Rich Isotopes ^{30}S and ^{34}Ar .
- 2000Pr09 B.V.Pritychenko, T.Glasmacher, B.A.Brown, P.D.Cottle et al. - Phys.Rev. C62, 051601 (2000).
Single-Particle Structure Along the Boundary of the 'Island of Inversion': Radioactive beam spectroscopy of ^{33}Si and ^{34}P .
- 2000PrZX B.V.Pritychenko - Thesis, Michigan State University (2000).
Intermediate-Energy Coulomb Excitation of the Neutron-Rich Radioactive Isotopes $^{26,28}\text{Ne}$, $^{28-31}\text{Na}$, $^{30-34}\text{Mg}$, $^{34,35}\text{Al}$, ^{33}Si and ^{34}P .
- 2001BI17 Y.Blumenfeld, E.Khan, F.Marechal, T.Suomijarvi - Pramana 57, 493 (2001).
Proton Scattering from Unstable Nuclei.
- 2001Gr29 L.V.Grigorenko, R.C.Johnson, I.G.Mukha, I.J.Thompson, M.V.Zhukov - Phys.Rev. C64, 054002 (2001).
Two-Proton Radioactivity and Three-Body Decay: General problems and theoretical approach.
- 2001Iw07 H.Iwasaki, T.Motobayashi, H.Sakurai, K.Yoneda et al. - Phys.Lett. 522B, 227 (2001).
Large Collectivity of ^{34}Mg .
- 2001Kh17 E.Khan, T.Suomijarvi, Y.Blumenfeld, V.G.Nguyen et al. - Nucl.Phys. A694, 103 (2001).
Proton Scattering from the Unstable Nuclei ^{30}S and ^{34}Ar : Structural evolution along the sulfur and argon isotopic chains.
- 2001Nu01 S.Nummela, P.Baumann, E.Caurier, P.Dessagne et al. - Phys.Rev. C63, 044316 (2001).
Spectroscopy of $^{34,35}\text{Si}$ by β Decay: sd-fp Shell gap and single-particle states.
- 2001Ot04 T.Otsuka, R.Fujimoto, Y.Utsuno, B.A.Brown et al. - Phys.Rev.Lett. 87, 082502 (2001).
Magic Numbers in Exotic Nuclei and Spin-Isospin Properties of the NN Interaction.
- 2001Pi10 S.Pirrone, G.Politi, G.Lanzalone, S.Aiello et al. - Phys.Rev. C64, 024610 (2001).
Fusion and Competing Processes in the $^{32}\text{S} + ^{12}\text{C}$ Reaction at $E(^{32}\text{S}) = 19.5 \text{ MeV/nucleon}$.
- 2001Pr08 B.V.Pritychenko, T.Glasmacher, B.A.Brown, P.D.Cottle et al. - Phys.Rev. C63, 047308 (2001).
Shape Coexistence on the Boundary of the 'Island of Inversion': Exotic beam spectroscopy of ^{34}Al .
- 2001Sa21 W.Satula, R.Wyss - Phys.Rev.Lett. 86, 4488 (2001).
Rotations in Isospace: A doorway to the understanding of neutron-proton superfluidity in $N = Z$ Nuclei.
- 2001Sa57 H.Sagawa, H.Esbensen - Nucl.Phys. A693, 448 (2001).
Giant Resonances in Exotic Nuclei.
- 2001Sa72 F.Sarazin, H.Savajols, W.Mittig, F.Nowacki et al. - Hyperfine Interactions 132, 147 (2001).
Shape Coexistence and the $N = 28$ Shell Closure Far from Stability.
- 2001Sa77 H.Sagawa - Prog.Theor.Phys.(Kyoto), Suppl. 142, 1 (2001).
Giant Multipole States in Stable and Unstable Nuclei.
- 2001Yo03 K.Yoneda, H.Sakurai, T.Gomi, T.Motobayashi et al. - Phys.Lett. 499B, 233 (2001).
Deformation of ^{34}Mg Studied via In-Beam γ -Ray Spectroscopy using Radioactive-Ion Projectile Fragmentation.
- 2001Yo12 M.Yokoyama - Prog.Theor.Phys.(Kyoto), Suppl. 142, 325 (2001).
Excitation Modes of Nuclei Far from Stability.
- 2002AsZY M.Asai, T.Ishii, A.Makishima, M.Ogawa, M.Matsuda - Japan Atomic Energy Res.Inst.Tandem VDG Ann.Rept., 2001, p.23 (2002); JAERI-Review 2002-029 (2002).
Nanosecond Isomers in $^{32,33}\text{Si}$ and ^{34}P .
- 2002DeZU N.A.Demekhina, G.S.Karapetyan, S.M.Lukyanov, Yu.Eh.Penionzhkevich et al. - Program and Thesis, Proc.52nd Ann.Conf.Nucl.Spectrosc.Struct.At.Nuclei, Moscow, p.198 (2002).
Non-Full Fusion Reactions under Interaction of ^7Li Ions with $^{\text{nat}}\text{Cu}$ at 35 MeV/Nucleon Energy.
- 2002En02 J.Enders, A.Bauer, D.Bazin, A.Bonaccorso et al. - Phys.Rev. C65, 034318 (2002).
Single-Neutron Knockout from $^{34,35}\text{Si}$ and ^{37}S .
- 2002GI01 T.Glasmacher, B.V.Pritychenko, P.D.Cottle, R.W.Ibbotson et al. - Eur.Phys.J. A 13, 59 (2002).
In-Beam Gamma-Ray Spectroscopy with Exotic Beams at the NSCL.
- 2002He23 F.Herfurth, A.Kellerbauer, F.Ames, G.Audi et al. - Eur.Phys.J. A 15, 17 (2002).
Accurate mass measurements of very short-lived nuclei: Prerequisites for high-accuracy investigations of superallowed β -decays.
- 2002Lu09 S.M.Lukyanov, Yu.E.Penionzhkevich, R.Astabyan, S.Lobastov et al. - J.Phys.(London) G28, L41 (2002).
Experimental Evidence for the Particle Stability of ^{34}Ne and ^{37}Na .
- 2002Lu19 Yu.S.Lutostansky, S.M.Lukyanov, Yu.E.Penionzhkevich, M.V.Zverev - Part. and Nucl., Lett. 115, 86 (2002).
Neutron Drip Line in the Region of O-Mg Isotopes.
- 2002LuZT S.M.Lukyanov, Yu.E.Penionzhkevich - JINR-E7-2002-237 (2002).
Peculiarities of O-Mg Isotopes at the Neutron Drip Line.
- 2002Mi44 W.Mittig, H.Savajols, D.Baiborodin, J.M.Casandjian et al. - Eur.Phys.J. A 15, 157 (2002).
Shape coexistence and the $N = 20$ shell closure far from stability by inelastic scattering.
- 2002Mi48 W.Mittig, H.Savajols, D.Baiborodin, J.M.Casandjian et al. - Prog.Theor.Phys.(Kyoto), Suppl. 146, 16 (2002).
Surprises at the $N = 20$ Shell Closure Far from Stability by Inelastic Scattering.
- 2002Mo35 T.Motobayashi - Eur.Phys.J. A 15, 99 (2002).
Spectroscopy of nuclei around $N = 20$ with direct reactions.

REFERENCES FOR A=34(CONTINUED)

- 2002No11 M.Notani, H.Sakurai, N.Aoi, Y.Yanagisawa et al. - Phys.Lett. 542B, 49 (2002).
New Neutron-Rich Isotopes, ^{34}Ne , ^{37}Na and ^{43}Si , Produced by Fragmentation of a 64 A MeV ^{48}Ca Beam.
- 2002NoZY M.Notani, H.Sakurai, N.Aoi, Y.Yanagisawa et al. - CNS-REP-45, p.12 (2002).
New neutron-rich isotopes in the neon-to-silicon region, produced by fragmentation of a 64A MeV ^{48}Ca beam.
- 2002Nu02 S.Nummela, P.Baumann, E.Caurier, S.Courtin et al. - Nucl.Phys. A701, 410c (2002).
Study of the Neutron-Rich Nuclei with $N = 21$, ^{35}Si and ^{33}Mg , by Beta Decay of ^{35}Al and ^{33}Na .
- 2002Or05 H.Orihara, A.Terakawa, K.Itoh, H.Suzuki et al. - Phys.Lett. 539B, 40 (2002).
Gamow-Teller Matrix Elements in the Light Nuclei and (p,n) Cross Sections at $E_p = 35$ MeV.
- 2002Oz03 A.Ozawa, T.Baumann, L.Chulkov, D.Cortina et al. - Nucl.Phys. A709, 60 (2002); Erratum Nucl.Phys. A727, 465 (2003).
Measurements of the Interaction Cross Sections for Ar and Cl Isotopes.
- 2002Sa11 H.Sakurai - Eur.Phys.J. A 13, 49 (2002).
Spectroscopy on Very Neutron-Rich Nuclei Beyond $N = 20$.
- 2002Yo04 K.Yoneda, H.Sakurai, T.Gomi, T.Motobayashi et al. - Nucl.Phys. A701, 437c (2002).
Gamma-Ray Spectroscopy of ^{34}Mg via RI Beam Fragmentation.
- 2002Zh09 S.-Q.Zhang, J.Meng, S.-G.Zhou, J.-Y.Zeng - Chin.Phys.Lett. 19, 312 (2002).
Giant Neutron Halo in Exotic Calcium Nuclei.
- 2003Gr04 L.V.Grigorenko, I.G.Mukha, M.V.Zhukov - Nucl.Phys. A714, 425 (2003).
Prospective candidates for the two-proton decay studies. (II) Exploratory studies of ^{30}Ar , ^{34}Ca , and ^{45}Fe .
- 2003Gr22 S.Grevy, J.Mrazek, J.C.Angelique, P.Baumann et al. - Nucl.Phys. A722, 424c (2003).
Beta-decay studies at the $N = 28$ shell closure.
- 2003Gr24 L.V.Grigorenko, M.V.Zhukov - Phys.Rev. C 68, 054005 (2003).
Two-proton radioactivity and three-body decay. II. Exploratory studies of lifetimes and correlations.
- 2003Iw02 N.Iwasa, T.Motobayashi, H.Sakurai, H.Akiyoshi et al. - Phys.Rev. C 67, 064315 (2003).
In-beam γ spectroscopy of ^{34}Si with deuteron inelastic scattering using reverse kinematics.
- 2003La13 V.Lapoux, N.Alamamos, E.Khan - Yad.Fiz. 66, 1546 (2003); Phys.Atomic Nuclei 66, 1501 (2003).
Structure Studies of Exotic Nuclei Using (p,p') Reactions.
- 2003Lu11 S.M.Lukyanov, Yu.E.Penionzhkevich, D.Guillemaud-Mueller, M.Lewitowicz et al. - Yad.Fiz. 66, 1687 (2003); Phys.Atomic Nuclei 66, 1639 (2003).
Exploring the Neutron Drip Line at the Ne-Mg Region.
- 2003Ma71 Z.Ma, L.-G.Cao, Nguyen Van Giai, P.Ring - Nucl.Phys. A722, 491c (2003).
Giant resonances of stable and exotic nuclei in relativistic RPA.
- 2003Sm02 N.A.Smirnova, C.Volpe - Nucl.Phys. A714, 441 (2003).
On the asymmetry of Gamow-Teller β -decay rates in mirror nuclei in relation with second-class currents.
- 2003Ya20 H.Yashima, Y.Uwamino, H.Iwase, H.Sugita et al. - Radiochim.Acta 91, 689 (2003).
Measurement and calculation of radioactivities of spallation products by high-energy heavy ions.
- 2004De41 N.A.Demekhina, G.S.Karapetyan, S.M.Lukyanov, Yu.E.Penionzhkevich et al. - Yad.Fiz. 67, 1443 (2004); Phys.Atomic Nuclei 67, 1417 (2004).
Production of Heavy Nuclei in the Interaction of ^7Li Ions with Copper Nuclei at an Energy of 35 MeV per Nucleon.
- 2004Ge02 L.S.Geng, H.Toki, A.Ozawa, J.Meng - Nucl.Phys. A730, 80 (2004).
Proton and neutron skins of light nuclei within the relativistic mean field theory.
- 2004Gr20 S.Grevy, J.C.Angelique, P.Baumann, C.Borcea et al. - Phys.Lett. B 594, 252 (2004).
Beta-decay half-lives at the $N=28$ shell closure.
- 2004Lu19 S.M.Lukyanov, Yu.E.Penionzhkevich - Yad.Fiz. 67, 1654 (2004); Phys.Atomic Nuclei 67, 1627 (2004).
The Neutron Drip Line in the Region of $N = 20$ and $N = 28$ Closures.
- 2004Pf02 M.Pfutzner - Nucl.Phys. A738, 101 (2004).
Two-proton radioactivity : the status and perspectives.
- 2005Ch66 J.A.Church, C.M.Campbell, D.-C.Dinca, J.Enders et al. - Phys.Rev. C 72, 054320 (2005).
Measurement of E2 transition strengths in $^{32,34}\text{Mg}$.
- 2005Ko28 A.Kohama, K.Iida, K.Oyamatsu - Phys.Rev. C 72, 024602 (2005).
Reaction cross section described by a black sphere approximation of nuclei.
- 2005Ma03 P.Mason, N.Marginean, S.M.Lenzi, M.Ionescu-Bujor et al. - Phys.Rev. C 71, 014316 (2005).
High spin structure of ^{34}S and the proton-neutron coupling of intruder states.
- 2005Ob01 A.Obertelli, S.Peru, J.-P.Delaroche, A.Gillibert et al. - Phys.Rev. C 71, 024304 (2005).
 $N = 16$ subshell closure from stability to the neutron drip line.
- 2005OI02 J.Ollier, R.Chapman, X.Liang, M.Labiche et al. - Phys.Rev. C 71, 034316 (2005).
Intruder configurations in neutron-rich ^{34}P .
- 2005OI04 J.Ollier, A.Hodsdon, R.Chapman, X.Liang et al. - J.Phys.(London) G31, S1935 (2005).
Intruder configurations in neutron-rich P and S isotopes.
- 2005Pa26 N.Paar, D.Vretenar, P.Ring - Phys.Rev.Lett. 94, 182501 (2005).
Proton Electric Pygmy Dipole Resonance.
- 2005Pf01 M.Pfutzner - Acta Phys.Pol. B36, 1399 (2005).
The current status of 2p emission studies.

REFERENCES FOR A=34(CONTINUED)

- 2005Sa59 H.Sagawa, X.R.Zhou, X.Z.Zhang - Phys.Rev. C 72, 054311 (2005).
Evolution of deformations in medium-mass nuclei.
- 2005Te07 E.Teran, C.W.Johnson - Eur.Phys.J. A 25, Supplement 1, 673 (2005).
A statistical spectroscopy approach for calculating nuclear level densities.
- 2005Ti11 C.Timis, J.C.Angelique, A.Buta, N.L.Achouri et al. - J.Phys.(London) G31, S1965 (2005).
Spectroscopy around $N = 20$ shell closure: β -n decay study of ^{35}Al .
- 2005Wa15 Z.Wang, Z.Ren - Phys.Rev. C 71, 054323 (2005).
Systematic study of charge form factors of elastic electron-nucleus scattering with the relativistic eikonal approximation.
- 2005Zh28 C.-L.Zhang, H.-Q.Zhang, X.-Z.Zhang - Chin.Phys.Lett. 22, 1877 (2005).
Dynamical Structure of Nuclear Excitation in Continuum.
- 2006AnZW J.C.Angelique, C.Timis, S.Pietri, N.L.Achouri et al. - Proc.Frontiers in Nuclear Structure, Astrophysics, and Reactions, Isle of Kos, Greece, 12-17 Sept. 2005, S.V.Harissopulos, P.Demetriou, R.Julin, Eds., p. 134 (2006); AIP Conf.Proc. 831 (2006).
Spectroscopy Near the $N=20$ Shell Closure: β -n Decay Studies of ^{33}Mg and ^{35}Al .
- 2006EI03 D.Elekes, Zs.Dombradi, A.Saito, N.Aoi et al. - Phys.Rev. C 73, 044314 (2006).
Proton inelastic scattering studies at the borders of the "island of inversion": The $^{30,31}\text{Na}$ and $^{33,34}\text{Mg}$ case.
- 2006FuZX T.Fukui, S.Ota, S.Shimoura, N.Aoi et al. - CNS-REP-69, p.19 (2006).
High Resolution In-beam Gamma Spectroscopy of $N \sim 20$ Neutron-rich Nuclei.
- 2006Ia05 V.E.Iacob, J.C.Hardy, J.F.Brinkley, C.A.Gagliardi et al. - Phys.Rev. C 74, 055502 (2006).
Precise half-life measurements for the superallowed β^+ emitters ^{34}Ar and ^{34}Cl .
- 2006Kh08 A.Khouaja, A.C.C.Villari, M.Benjelloun, D.Hirata et al. - Nucl.Phys. A780, 1 (2006).
Reaction cross-section and reduced strong absorption radius measurements of neutron-rich nuclei in the vicinity of closed shells $N = 20$ and $N = 28$.
- 2006Kr07 Krishichayan, A.Chakraborty, S.Mukhopadhyay, S.Ray et al. - Eur.Phys.J. A 29, 151 (2006).
Spin-parity measurements in the neutron-rich $N \sim 20$ ^{34}P and ^{36}S nuclei.
- 2006Me04 T.J.Mertzimekis, P.F.Mantica, A.D.Davies, S.N.Liddick, B.E.Tomlin - Phys.Rev. C 73, 024318 (2006).
Ground state magnetic dipole moment of ^{35}K .
- 2006Me08 D.F.Measday, T.J.Stocki - Phys.Rev. C 73, 045501 (2006).
 γ rays from muon capture in natural Ca, Fe, and Ni.
- 2006Or01 J.N.Orce, V.Velazquez - Nucl.Phys. A764, 205 (2006).
Charge-independent trend of isoscalar matrix elements along the $N \sim Z$ line.
- 2006Tu03 K.Turzo, P.Himpe, D.L.Balabanski, G.Belier et al. - Phys.Rev. C 73, 044313 (2006).
Spin polarization of ^{34}Al fragments produced by nucleon pickup at intermediate energies.
- 2007ChZX H.D.Choi, R.B.Firestone, R.M.Lindstrom, G.L.Molnar et al. - IAEA-STI/PUB/1263 (2007).
Database of Prompt Gamma Rays from Slow Neutron Capture for Elemental Analysis.
- 2007DoZV P.Doornenbal - Thesis, University of Cologne, Germany (2007).
In-beam γ -ray spectroscopy of two-step fragmentation reactions at relativistic energies – The case of ^{36}Ca .
- 2007Fu06 Y.Fujita, R.Neveling, H.Fujita, T.Adachi et al. - Phys.Rev. C 75, 057305 (2007).
Gamow-Teller strengths in $A = 34$ isobars: Comparison of the mirror transitions $T_z = +1 \rightarrow 0$ and $T_z = -1 \rightarrow 0$.
- 2007GeZX M.Gelin - Thesis, Universite de Caen (2007).
 γ -Spectroscopy of Neutron-Rich Nuclei around $N=20$.
- 2007Ju03 B.Jurado, H.Savjols, W.Mittig, N.A.Orr et al. - Phys.Lett. B 649, 43 (2007).
Mass measurements of neutron-rich nuclei near the $N = 20$ and 28 shell closures.
- 2007Na31 P.Napolitani, K.-H.Schmidt, L.Tassan-Got, P.Armbruster et al. - Phys.Rev. C 76, 064609 (2007).
Measurement of the complete nuclide production and kinetic energies of the system ^{136}Xe +hydrogen at 1 GeV per nucleon.
- 2007No13 M.Notani, H.Sakurai, N.Aoi, H.Iwasaki et al. - Phys.Rev. C 76, 044605 (2007).
Projectile fragmentation reactions and production of nuclei near the neutron drip line.
- 2008Ba08 C.Barbieri, E.Caurier, K.Langanke, G.Martinez-Pinedo - Phys.Rev. C 77, 024304 (2008).
Pygmy dipole response of proton-rich argon nuclei in random-phase approximation and no-core shell model.
- 2008Hi01 P.Himpe, G.Neyens, D.L.Balabanski, G.Belier et al. - Phys.Lett. B 658, 203 (2008).
 g factor of the exotic $N = 21$ isotope ^{34}Al : probing the $N = 20$ and $N = 28$ shell gaps at the border of the "island of inversion".
- 2008Ki07 T.Kibedi, T.W.Burrows, M.B.Trzhaskovskaya, P.M.Davidson, C.W.Nestor - Nucl.Instrum.Methods Phys.Res. A589, 202 (2008).
Evaluation of theoretical conversion coefficients using BrIcc .
- 2008Na03 K.Nagatsu, T.Fukumura, M.Takei, F.Szelecsenyi et al. - Nucl.Instrum.Methods Phys.Res. B266, 709 (2008).
Measurement of thick target yields of the $^{nat}\text{S}(\alpha, x)^{34m}\text{Cl}$ nuclear reaction and estimation of its excitation function up to 70 MeV.
- 2008ReZZ P.L.Reeder - Priv.Com. (2008).
- 2008Ti05 Yu.E.Titarenko, V.F.Batyayev, A.Yu.Titarenko, M.A.Butko et al. - Phys.Rev. C 78, 034615 (2008).
Cross sections for nuclide production in a ^{56}Fe target irradiated by 300, 500, 750, 1000, 1500, and 2600 MeV protons compared with data on a hydrogen target irradiated by 300, 500, 750, 1000, and 1500 MeV/nucleon ^{56}Fe ions.
- 2008Wi09 M.Wiedeking, E.Rodriguez-Vieitez, P.Fallon, M.P.Carpenter et al. - Phys.Rev. C 78, 037302 (2008).

REFERENCES FOR A=34(CONTINUED)

- Intruder excitations in ^{35}P .*
 2009Be26 P.C.Bender, C.R.Hoffman, M.Wiedeking, J.M.Allmond et al. - Phys.Rev. C 80, 014302 (2009).
Approaching the "island of inversion": ^{34}P .
 2009Ch43 R.Chakrabarti, S.Mukhopadhyay, Krishichayan, A.Chakraborty et al. - Phys.Rev. C 80, 034326 (2009).
Experimental study of nuclei in the vicinity of the "island of inversion" through the fusion-evaporation reaction.
 2009Er07 T.Eronen, V.-V.Elomaa, J.Hakala, J.C.Hardy et al. - Phys.Rev.Lett. 103, 252501 (2009).
 Q_{EC} Values of the Superaligned β Emitters ^{34}Cl and $^{38}\text{K}^{\text{m}}$.
 2009KiZW T.Kin, M.Oshima, K.Furutaka, M.Koizumi et al. - Proc.13th Intern.Symposium on Capture Gamma-Ray Spectroscopy and Related Topics, Cologne, Germany, 25-29 Aug.2008, J.Jolie, A.Zilges, N.Warr, A.Blazhev, Eds., p.575 (2009); AIP Conf.Proc. 1090 (2009).
Identification of Nuclear Levels of ^{34}S for Determination of the Neutron Capture Cross Section.
 2009Kw02 A.A.Kwiatkowski, B.R.Barquest, G.Bollen, C.M.Campbell et al. - Phys.Rev. C 80, 051302 (2009).
Precision test of the isobaric multiplet mass equation for the $A=32$, $T=2$ quintet.
 2009ObZY S.O'Brien, T.Adachi, G.P.A.Berg, M.Couder et al. - Proc.13th Intern.Symposium on Capture Gamma-Ray Spectroscopy and Related Topics, Cologne, Germany, 25-29 Aug.2008, J.Jolie, A.Zilges, N.Warr, A.Blazhev, Eds., p.288 (2009); AIP Conf.Proc. 1090 (2009).
Exploring the α -process with Grand Raiden.
 2009Pa28 A.Parikh, T.Faestermann, R.Hertenberger, R.Krucken et al. - Phys.Rev. C 80, 015802 (2009).
New ^{34}Cl proton-threshold states and the thermonuclear $^{33}\text{S}(p,\gamma)^{34}\text{Cl}$ rate in ONe novae.
 2010Do03 P.Doornenbal, P.Reiter, H.Grawe, T.Saito et al. - Nucl.Instrum.Methods Phys.Res. A613, 218 (2010).
Lifetime effects for high-resolution gamma-ray spectroscopy at relativistic energies and their implications for the RISING spectrometer.
 2010Gh02 S.S.Ghugre - Pramana 75, 13 (2010).
Nuclei in the vicinity of 'island of inversion' through the fusion reaction.
 2010PaZT A.Parikh, T.Faestermann, R.Krucken, V.Bildstein et al. - Proc.Intern.Symposium on Nuclei in the Cosmos XI, July 19-23 2010, Heidelberg, Germany, p.52 (2010).
The $^{33}\text{S}(p,\gamma)^{34}\text{Cl}$ reaction in classical nova explosions.
 2010Ro23 C.Rodriguez-Tajes, H.Alvarez-Pol, T.Aumann, E.Benjamin et al. - Phys.Rev. C 82, 024305 (2010); Publishers Note Phys.Rev. C 82, 029910 (2010).
One-neutron knockout from light neutron-rich nuclei at relativistic energies.
 2010Wa12 Z.M.Wang, R.Chapman, X.Liang, F.Haas et al. - Phys.Rev. C 81, 054305 (2010).
 γ -ray spectroscopy of neutron-rich ^{40}S .
 2010Wa20 Z.M.Wang, R.Chapman, X.Liang, F.Haas et al. - Phys.Rev. C 81, 064301 (2010).
Intruder negative-parity states of neutron-rich ^{33}Si .
 2010WaZT Z.Wang - Thesis, Univ.of West Scotland (2010).
 γ -ray Spectroscopy of Neutron-rich Silicon and Sulphur Nuclei using Grazing Reactions.
 2010Ze03 R.G.T.Zegers, R.Meharchand, Y.Shimbara, S.M.Austin et al. - Phys.Rev.Lett. 104, 212504 (2010).
 $^{34}\text{P}(^7\text{Li}, ^7\text{Be}+\gamma)$ Reaction at 100A MeV in Inverse Kinematics.
 2011BeZZ P.C.Bender, S.L.Tabor, L.Hamilton, C.R.Hoffman et al. - Priv.Comm. (2011).
Notes on ^{34}P .
 2011En01 J.W.Engle, T.E.Barnhart, O.T.DeJesus, R.J.Nickles - Appl.Radiat.Isot. 69, 75 (2011).
Production of $^{34\text{m}}\text{Cl}$ and ^{38}Cl via the (d,α) reaction on ^{36}Ar and $^{\text{nat}}\text{Ar}$ gas at 8.4 MeV.
 2011Fr04 B.M.Freeman, C.Wrede, B.G.Delbridge, A.Garca et al. - Phys.Rev. C 83, 048801 (2011).
Branches of $^{33}\text{S}(p,\gamma)^{34}\text{Cl}$ at oxygen-neon nova temperatures.
 2011Ka01 R.Kanungo, A.Prochazka, W.Horiuchi, C.Nociforo et al. - Phys.Rev. C 83, 021302 (2011).
Matter radii of $^{32-35}\text{Mg}$.
 2011MaZP P.J.R.Mason, T.Alharbi, P.H.Regan, Zs.Podolyak et al. - Priv.Comm. (2011).
Precision Measurement of the Half-life of the $I^{\pi} = 4^{-}$ Intruder State in ^{34}P : M2 Transition Strengths Approaching the Island of Inversion.
 2011No06 M.Norrby, T.Lonnroth - Int.J.Mod.Phys. E20, 1042 (2011).
Evidence of rotational behavior of alpha-clusters in ^{32}S , ^{34}S and ^{40}Ca .
 2011No07 M.Norrby, T.Lonnroth, V.Z.Goldberg, G.V.Rogachev et al. - Eur.Phys.J. A 47, 73 (2011).
Highly excited alpha-cluster states in ^{34}S .
 2011Pa43 S.Paschalis, P.Fallon, A.O.Macchiavelli, M.Petri et al. - J.Phys.:Conf.Ser. 312, 092050 (2011).
The deformed 0^{+} state in ^{34}Si .
 2011PaZV A.Parikh, Th.Faestermann, R.Krucken, V.Bildstein et al. - Proc.Frontiers in Nuclear Structure, Astrophysics, and Reactions, Rhodes, Greece, 23-27 August 2010, P.Demetriou, R.Julin, S.Harissopulos Eds., p.188 (2011); AIP Conf.Proc. 1377 (2011).
Improving the $^{33}\text{S}(p,\gamma)^{34}\text{Cl}$ Reaction Rate for Models of Classical Nova Explosions.
 2011Ti03 Yu.E.Titarenko, V.F.Batyaev, A.Yu.Titarenko, M.A.Butko et al. - Phys.Atomic Nuclei 74, 523 (2011); Yad.Fiz. 74, 548 (2011).
Measurement and simulation of the cross sections for nuclide production in ^{56}Fe and $^{\text{nat}}\text{Cr}$ targets irradiated with 0.04- to 2.6-GeV protons.

REFERENCES FOR A=34(CONTINUED)

- 2011Ti04 Yu.E.Titarenko, V.F.Batyaev, A.Yu.Titarenko, M.A.Butko et al. - Phys.Atomic Nuclei 74, 537 (2011); Yad.Fiz. 74, 561 (2011).
Measurement and simulation of the cross sections for nuclide production in ^{93}Nb and $^{\text{nat}}\text{Ni}$ targets irradiated with 0.04- to 2.6-GeV protons.
- 2012Be11 P.C.Bender, S.L.Tabor, V.Tripathi, C.R.Hoffman et al. - Phys.Rev. C 85, 044305 (2012).
Multi-intruder structures in ^{34}P .
- 2012BoZT M.Bouhelal, F.Haas, E.Caurier, F.Nowacki - Proc.Intern.Conf.on Nuclear Structure and Dynamics,12, Opatija, Croatia, 9-13 July, 2012, T.Niksic, M.Milin, D.Vretenar, S.Szilner, Eds., p.38 (2012); AIP Conf.Proc.1491 (2012).
0 and $1\hbar\omega$ shell model description of phosphorus isotopes with $A = 30$ to 35.
- 2012En01 J.W.Engle, G.W.Severin, T.E.Barnhart, L.D.Knutson, R.J.Nickles - Appl.Radiat.Isot. 70, 355 (2012).
Cross sections of the $^{36}\text{Ar}(d,\alpha)^{34\text{m}}\text{Cl}$, $^{40}\text{Ar}(d,\alpha)^{38}\text{Cl}$, and $^{40}\text{Ar}(d,p)^{41}\text{Ar}$ nuclear reactions below 8.4 MeV.
- 2012Kw02 E.Kwan, D.J.Morrissey, D.A.Davies, M.Steiner et al. - Phys.Rev. C 86, 014612 (2012).
Systematic studies of light neutron-rich nuclei produced via the fragmentation of ^{40}Ar .
- 2012Ma21 P.J.R.Mason, T.Alharbi, P.H.Regan, N.Marginean et al. - Phys.Rev. C 85, 064303 (2012).
Half-life of the $1^{\pi} = 4^{-}$ intruder state in ^{34}P : $M2$ transition strengths approaching the island of inversion.
- 2012Ni10 N.Nica, B.Singh - Nucl.Data Sheets 113, 1563 (2012).
Nuclear Data Sheets for $A = 34$.
- 2012No05 C.Nociforo, A.Prochazka, R.Kanungo, T.Aumann et al. - Phys.Rev. C 85, 044312 (2012).
One-neutron removal reactions on Al isotopes around the $N=20$ shell closure.
- 2012Ro25 F.Rotaru, F.Negoita, S.Grevy, J.Mrazek et al. - Phys.Rev.Lett. 109, 092503 (2012).
Unveiling the Intruder Deformed 0_2^{+} State in ^{34}Si .
- 2012SrZZ P.C.Srivastava, J.G.Hirsch, M.J.Ermamatov, V.K.B.Kota - Proc.Beauty in Physics: Theory and Experiment, in Honor of Francesco Iachello: Hacienda Cocoyoc, Mexico, 14-8 May 2012, R.Bijker, Eds. p.731 (2011);AIP 1488 (2012).
Large-scale shell-model calculations for $^{32-39}\text{P}$ isotopes.
- 2012Th10 M.Thoennessen - At.Data Nucl.Data Tables 98, 933 (2012).
Discovery of the Isotopes with $11 \leq Z \leq 19$.
- 2012Zh06 X.H.Zhang, Z.Y.Sun, R.F.Chen, Z.Q.Chen et al. - Phys.Rev. C 85, 024621 (2012).
Projectile fragmentation reactions of ^{40}Ar at 57 MeV/nucleon.
- 2013An02 I.Angeli, K.P.Marinova - At.Data Nucl.Data Tables 99, 69 (2013).
Table of experimental nuclear ground state charge radii: An update.
- 2013Ch49 A.Chaudhuri, C.Andreoioiu, T.Brunner, U.Chowdhury et al. - Phys.Rev. C 88, 054317 (2013).
Evidence for the extinction of the $N=20$ neutron-shell closure for ^{32}Mg from direct mass measurements.
- 2013De37 A.Deppman, E.Andrade, V.Guimaraes, G.S.Karapetyan et al. - Phys.Rev. C 88, 064609 (2013).
Supersymmetric fission of heavy nuclei induced by intermediate-energy protons.
- 2013Do22 P.Doornenbal, H.Scheit, S.Takeuchi, N.Aoi et al. - Phys.Rev.Lett. 111, 212502 (2013).
In-Beam γ -Ray Spectroscopy of $^{34,36,38}\text{Mg}$: Merging the $N=20$ and $N=28$ Shell Quenching.
- 2013Fa10 J.Fallis, A.Parikh, P.F.Bertone, S.Bishop et al. - Phys.Rev. C 88, 045801 (2013).
Constraining nova observables: Direct measurements of resonance strengths in $^{33}\text{S}(p,\gamma)^{34}\text{Cl}$.
- 2013Li39 H.Li, Z.Ren - J.Phys.(London) G40, 105110 (2013).
Shell model calculations for the β^{-} -decays of $Z = 9-13$ nuclei.
- 2013Na17 H.Nakada, K.Sugiura, J.Margueron - Phys.Rev. C 87, 067305 (2013).
Tensor-force effects on single-particle levels and proton bubble structure around the Z or $N=20$ magic number.
- 2013SeZV Y.M.Sereda, S.M.Lukyanov, A.G.Artyukit, A.N.Vorontsov et al. - JINR-P7-2013-95 (2013).
A Study of Fragmentation of ^{20}Ne and ^{40}Ar Ions at the COMBAS Set-up.
- 2013StZY K.Steiger - Fakultat fur Physik der Technischen Universitat Munchen (2013).
Decay spectroscopy of neutron-rich nuclei around ^{37}Al .
- 2013Su18 Z.-X.Su, C.Qi - Phys.Lett. B 724, 247 (2013).
Shell evolution and its indication on the isospin dependence of the spin-orbit splitting.
- 2013Ti01 J.Tian, N.Wang, C.Li, J.Li - Phys.Rev. C 87, 014313 (2013).
Improved Kelson-Garvey mass relations for proton-rich nuclei.
- 2014Bi03 A.Bisoi, M.S.Sarkar, S.Sarkar, S.Ray et al. - Phys.Rev. C 89, 024303 (2014).
High spin spectroscopy in ^{34}Cl .
- 2014Ca21 E.Caurier, F.Nowacki, A.Poves - Phys.Rev. C 90, 014302 (2014).
Merging of the islands of inversion at $N = 20$ and $N = 28$.
- 2014ChZZ S.Chakraborty - Int.Nuclear Physics Conf. 2013,(IUPAP),Firenze,Italy, June 2-7,2013, S.Lunardi, P.G.Bizzeti, W.S.Kabana, C.Bucci,et al.Eds.p.02019 (2014); EPJ web of Conf.v.66, (2014).
Ground-state configuration of neutron-rich Aluminum isotopes through Coulomb Breakup.
- 2014Do05 P.Doornenbal, H.Scheit, S.Takeuchi, Y.Utsuno et al. - Prog.Theor.Exp.Phys. 2014, 053D01 (2014).
Rotational level structure of sodium isotopes inside the "island of inversion".
- 2014Mi09 S.Michimasa, Y.Yanagisawa, K.Inafuku, N.Aoi et al. - Phys.Rev. C 89, 054307 (2014).
Quadrupole collectivity in island-of-inversion nuclei $^{28,30}\text{Ne}$ and $^{34,36}\text{Mg}$.
- 2014Pa44 A.Parikh, K.Wimmer, T.Faestermann, R.Hertenberger et al. - Phys.Lett. B 737, 314 (2014).

REFERENCES FOR A=34(CONTINUED)

- 2014Pr09 *Isotopic $^{32}\text{S}/^{33}\text{S}$ ratio as a diagnostic of presolar grains from novae.*
B.Pritychenko, E.Betak, B.Singh, J.Totans - Nucl.Data Sheets 120, 291 (2014).
Nuclear Science References Database.
- 2014Sh14 P.Shuai, H.S.Xu, X.L.Tu, Y.H.Zhang et al. - Phys.Lett. B 735, 327 (2014).
Charge and frequency resolved isochronous mass spectrometry and the mass of ^{51}Co .
- 2014Wa03 Z.-J.Wang, Z.-Z.Ren, T.-K.Dong - Chin.Phys.C 38, 024102 (2014).
Probe the $2s_{1/2}$ and $1d_{3/2}$ state level inversion with electron-nucleus scattering.
- 2015Ch56 R.Chapman, A.Hodsdon, M.Bouhelal, F.Haas et al. - Phys.Rev. C 92, 044308 (2015).
Spectroscopy of neutron-rich $^{34,35,36,37,38}\text{P}$ populated in binary grazing reactions.
- 2015Gr15 M.Grasso, M.Anguiano - Phys.Rev. C 92, 054316 (2015).
Neutron $2p$ and $1f$ spin-orbit splittings in ^{40}Ca , ^{36}S and ^{34}Si $N=20$ isotones: Tensor-induced and pure spin-orbit effects.
- 2015Kw01 A.A.Kwiatkowski, C.Andreoiu, J.C.Bale, A.Chaudhuri et al. - Phys.Rev. C 92, 061301 (2015).
Observation of a crossover of S_{2n} in the island of inversion from precision mass spectrometry.
- 2015Mo17 S.Momota, I.Tanihata, A.Ozawa, M.Notani et al. - Phys.Rev. C 92, 024608 (2015).
Velocity-dependent transverse momentum distribution of fragments produced from $^{40}\text{Ar}+^9\text{Be}$ at 95 MeV/nucleon.
- 2015Pu01 G.Puddu - Eur.Phys.J. A 51, 14 (2015).
Description of nuclei around $N=20$ starting from the Argonne V18 interaction.
- 2015RoZU F.Rotaru, R.Lica, F.Negoita, S.Grevy et al. - Proc.of the Conf.on Advances in Radioactive Isotope Science (ARIS2014), Tokyo, Japan, June 1-6, 2014, p.020006 (2015); JPS Conf. Proc. 6, (2015).
Properties of Intruder States in ^{34}Al and ^{34}Si .
- 2015St17 A.Staszczak, C.-Y.Wong - Phys.Scr. 90, 114006 (2015).
Toroidal high-spin isomers in light nuclei with $N \neq Z$.
- 2015Su01 L.-J.Sun, C.-J.Lin, X.-X.Xu, J.-S.Wang et al. - Chin.Phys.Lett. 32, 012301 (2015).
Experimental Study of Beta-Delayed Proton Emission of $^{36,37}\text{Ca}$.
- 2015Vl01 G.N.Vlaskin, Yu.S.Khomyakov, V.I.Bulanenko - At.Energ. 117, 357 (2015).
Neutron Yield of the Reaction (α, n) on Thick Targets Comprised of Light Elements.
- 2015Xu06 R.L.Xu, C.Wu, W.-L.Qian, Z.Z.Ren - Eur.Phys.J. A 51, 20 (2015).
Dynamics of kaonic nuclei in an improved quark mass density-dependent model.
- 2016Eb02 J.-P.Ebran, A.Mutschler, E.Khan, D.Vretenar - Phys.Rev. C 94, 024304 (2016).
Spin-orbit interaction in relativistic nuclear structure models.
- 2016Ka36 G.S.Karapetyan, A.Deppman, V.Guimaraes, A.Balabekyan, N.A.Demekhina - Phys.Rev. C 94, 024618 (2016).
Interaction of ^{11}B with ^{209}Bi and ^{181}Ta targets at intermediate energies.
- 2016Li27 J.Li, W.-H.Long, J.-L.Song, Q.Zhao - Phys.Rev. C 93, 054312 (2016).
Pseudospin-orbit splitting and its consequences for the central depression in nuclear density.
- 2016Si02 J.Simonis, K.Hebeler, J.D.Holt, J.Menendez, A.Schwenk - Phys.Rev. C 93, 011302 (2016).
Exploring sd -shell nuclei from two- and three-nucleon interactions with realistic saturation properties.
- 2017Ay03 Y.Ayyad, J.Lee, A.Tamii, J.A.Lay et al. - Phys.Rev. C 96, 021303 (2017).
Investigating neutron-proton pairing in sd -shell nuclei via $(p, ^3\text{He})$ and $(^3\text{He}, p)$ transfer reactions.
- 2017Ch36 S.Chakraborty, U.Datta, T.Aumann, S.Beceiro Novo et al. - Phys.Rev. C 96, 034301 (2017).
Ground-state configuration of neutron-rich ^{35}Al via Coulomb breakup.
- 2017De15 V.De Donno, G.Co, M.Anguiano, A.M.Lallena - Phys.Rev. C 95, 054329 (2017).
Pairing in spherical nuclei: Quasiparticle random-phase approximation calculations with the Gogny interaction.
- 2017Du03 T.Duguet, V.Soma, S.Lecluse, C.Barbieri, P.Navratil - Phys.Rev. C 95, 034319 (2017).
Ab initio calculation of the potential bubble nucleus ^{34}Si .
- 2017Ha23 R.Han, X.Q.Li, W.G.Jiang, Z.H.Li et al. - Phys.Lett. B 772, 529 (2017).
Northern boundary of the "island of inversion" and triaxiality in ^{34}Si .
- 2017Ha34 R.Han, X.Li, W.Jiang, Zh.Li et al. - Sci.China: Phys.Mech.Astron. 60, 042021 (2017).
 β -decay study of neutron-rich nucleus ^{34}Al .
- 2017Ka11 K.Karakatsanis, G.A.Lalazissis, P.Ring, E.Litvinova - Phys.Rev. C 95, 034318 (2017).
Spin-orbit splittings of neutron states in $N=20$ isotones from covariant density functionals and their extensions.
- 2017Li03 R.Lica - Phys.Rev. C 95, 021301 (2017).
Identification of the crossing point at $N=21$ between normal and intruder configurations.
- 2017LiZZ R.Lica, A.Negret - Priv. Comm. (2017).
re: ^{34}Mg decay to ^{34}Al .
- 2017Sa70 G.Saxena, M.Kumawat, M.Kaushik, S.K.Jain, M.Aggarwal - Phys.Lett. B 775, 126 (2017).
Two-proton radioactivity with $2p$ halo in light mass nuclei $A=18-34$.
- 2017Se09 N.Severijs, B.Blank - J.Phys.(London) G44, 074002 (2017).
Weak interaction physics at ISOLDE.
- 2017Tr02 V.Tripathi, R.S.Lubna, B.Abromeit, H.L.Crawford et al. - Phys.Rev. C 95, 024308 (2017).
 β decay of $^{38,40}\text{Si}$ ($T_z = +5, +6$) to low-lying core excited states in odd-odd $^{38-40}\text{P}$ isotopes.
- 2017Ts01 N.Tsunoda, T.Otsuka, N.Shimizu, M.Hjorth-Jensen et al. - Phys.Rev. C 95, 021304 (2017).
Exotic neutron-rich medium-mass nuclei with realistic nuclear forces.

REFERENCES FOR A=34(CONTINUED)

- 2017Wa10 M.Wang, G.Audi, F.G.Kondev, W.J.Huang et al. - Chin.Phys.C 41, 030003 (2017).
The AME2016 atomic mass evaluation (II). Tables, graphs and references.
- 2017Wu11 F.Wu, C.L.Bai, J.M.Yao, H.Q.Zhang, X.Z.Zhang - Eur.Phys.J. A 53, 183 (2017).
Existence problem of proton semi-bubble structure in the 2_1^+ state of ^{34}Si .
- 2018Co03 G.Co, M.Anguiano, V.De Donno, A.M.Lallena - Phys.Rev. C 97, 034313 (2018).
Matter distribution and spin-orbit force in spherical nuclei.
- 2018Ka11 D.Kahl, H.Yamaguchi, S.Kubono, A.A.Chen et al. - Phys.Rev. C 97, 015802 (2018).
First measurement of $^{30}\text{S} + \alpha$ resonant elastic scattering for the $^{30}\text{S}(\alpha, p)$ reaction rate.
- 2018Ka12 S.Karampagia, R.A.Senkov, V.Zelevinsky - At.Data Nucl.Data Tables 120, 1 (2018).
Level density of the sd-nuclei-Statistical shell-model predictions.
- 2018Lo11 A.M.Long, T.Adachi, M.Beard, G.P.A.Berg et al. - Phys.Rev. C 97, 054613 (2018).
 α -unbound levels in ^{34}Ar from $^{36}\text{Ar}(p,t)^{34}\text{Ar}$ reaction measurements and implications for the astrophysical $^{30}\text{S}(\alpha, p)^{33}\text{Cl}$ reaction rate.
- 2018Lu08 R.S.Lubna, V.Tripathi, S.L.Tabor, P.-L.Tai et al. - Phys.Rev. C 97, 044312 (2018).
Intruder configurations of excited states in the neutron-rich isotopes ^{33}P and ^{34}P .
- 2018MuZY S.Mughabghab - Atlas of Neutron Resonances, 6th ed. Vol.1, Resonance Properties and Thermal Cross Sections Z = 1-60, Elsevier, Amsterdam (2018).
- 2018Sa54 A.Saastamoinen, G.J.Lotay, A.Kankainen, B.T.Roeder et al. - J.Phys.:Conf.Ser. 940, 012004 (2018).
Study of excited states of ^{35}Ar through β -decay of ^{35}K for nucleosynthesis in novae and X-ray bursts.
- 2018Va09 J.J.Valiente-Dobon, A.Poves, A.Gadea, B.Fernandez-Dominguez - Phys.Rev. C 98, 011302 (2018).
Broken mirror symmetry in ^{36}S and ^{36}Ca .
- 2018Xu05 Z.Y.Xu, H.Heylen, K.Asahi, F.Boulay et al. - Phys.Lett. B 782, 619 (2018).
Nuclear moments of the low-lying isomeric I^+ state of ^{34}Al : Investigation on the neutron $1p1h$ excitation across $N=20$ in the island of inversion.
- 2018Yo06 S.Yoshida, Y.Utsuno, N.Shimizu, T.Otsuka - Phys.Rev. C 97, 054321 (2018); Erratum Phys.Rev. C 109, 029904 (2024).
Systematic shell-model study of β -decay properties and Gamow-Teller strength distributions in $A = 40$ neutron-rich nuclei.
- 2019Ah07 D.S.Ahn, N.Fukuda, H.Geissel, N.Inabe et al. - Phys.Rev.Lett. 123, 212501 (2019).
Location of the Neutron Dripline at Fluorine and Neon.
- 2019As04 P.Ascher, N.Althubiti, D.Atanasov, K.Blaum et al. - Phys.Rev. C 100, 014304 (2019).
Mass measurements of neutron-rich isotopes near $N=20$ by in-trap decay with the ISOLTRAP spectrometer.
- 2019ChZU R.Chyzh - Thesis, Texas A and M University (2019).
Measurement of β -delayed Protons from ^{35}K Relevant to the $^{34}\text{Cl}^{\text{g.m.}}(p, \gamma)^{35}\text{Ar}$ Reaction.
- 2019Gr08 L.Grocutt, R.Chapman, M.Bouhelal, F.Haas et al. - Phys.Rev. C 100, 064308 (2019).
Lifetime measurements of $N = 20$ phosphorus isotopes using the AGATA γ -ray tracking spectrometer.
- 2019Li41 R.Lica - Phys.Rev. C 100, 034306 (2019).
Normal and intruder configurations in ^{34}Si populated in the β^- decay of ^{34}Mg and ^{34}Al .
- 2019Mo01 P.Moller, M.R.Mumpower, T.Kawano, W.D.Myers - At.Data Nucl.Data Tables 125, 1 (2019).
Nuclear properties for astrophysical and radioactive-ion-beam applications (II).
- 2019Sa02 G.Saxena, M.Kumawat, M.Kaushik, S.K.Jain, M.Aggarwal - Phys.Lett. B 788, 1 (2019).
Bubble structure in magic nuclei.
- 2019Sa23 A.Saxena, A.Kumar, V.Kumar, P.C.Srivastava, T.Suzuki - Hyperfine Interactions 240, 37 (2019).
Ab initio description of collectivity for sd shell nuclei.
- 2019Sr02 I.Sreeja, M.Balasubramaniam - Eur.Phys.J. A 55, 33 (2019).
An empirical formula for the half-lives of exotic two-proton emission.
- 2019Wa38 Ch.Wang, X.Li, R.Han, Zh.Li et al. - Nucl.Instrum.Methods Phys.Res. A940, 83 (2019).
A β -delayed neutron detection system working with the continuous beam mode.
- 2019Zo02 Y.Y.Zong, M.Q.Lin, M.Bao, Y.M.Zhao, A.Arima - Phys.Rev. C 100, 054315 (2019).
Mass relations of corresponding mirror nuclei.
- 2020Cu01 J.P.Cui, Y.H.Gao, Y.Z.Wang, J.Z.Gu - Phys.Rev. C 101, 014301 (2020), Erratum Phys.Rev. C 104, 029902 (2021).
Two-proton radioactivity within a generalized liquid drop model.
- 2020Fo04 J.M.R.Fox, C.W.Johnson, R.N.Perez - Phys.Rev. C 101, 054308 (2020).
Uncertainty quantification of an empirical shell-model interaction using principal component analysis.
- 2020Ia01 V.E.Iacob, J.C.Hardy, H.I.Park, M.Bencomo et al. - Phys.Rev. C 101, 015504 (2020).
New precise half-life measurement for the superallowed β^+ emitter ^{34}Ar .
- 2020Ia02 V.E.Iacob, J.C.Hardy, H.I.Park, M.Bencomo et al. - Phys.Rev. C 101, 045501 (2020).
Precise β branching-ratio measurement for the $0^+ \rightarrow 0^+$ superallowed decay of ^{34}Ar .
- 2020Ke03 A.R.L.Kennington, G.Lotay, D.T.Doherty, D.Seweryniak et al. - Phys.Rev.Lett. 124, 252702 (2020).
Search for Nova Presolar Grains: γ -Ray Spectroscopy of ^{34}Ar and its Relevance for the Astrophysical $^{33}\text{Cl}(p, \gamma)$ Reaction.
- 2020Ma25 A.Magilligan, B.A.Brown - Phys.Rev. C 101, 064312 (2020).
New isospin-breaking “USD” Hamiltonians for the sd shell.
- 2020Ma35 C.Ma, Y.Y.Zong, Y.M.Zhao, A.Arima - Phys.Rev. C 102, 024330 (2020).
Mass relations of mirror nuclei with local correlations.
- 2020Mi15 T.Miyagi, S.R.Stroberg, J.D.Holt, N.Shimizu - Phys.Rev. C 102, 034320 (2020).

REFERENCES FOR A=34(CONTINUED)

- 2020Pa40 *Ab initio multishell valence-space Hamiltonians and the island of inversion.*
H.I.Park, J.C.Hardy, M.Bencomo, V.E.Iacob et al. - Phys.Rev. C 102, 045502 (2020).
Intensity of a weak 519-keV γ ray following β decay of the superallowed emitter ^{34}Ar determined via the $^{33}\text{S}(p, \gamma)^{34}\text{Cl}$ reaction.
- 2020StZV N.Stone - INDC(NDS)-0816 (2020).
Table of Recommended Nuclear Magnetic Dipole Moments: Part II, Short-Lived States.
- 2021Bh12 A.Bhattacharyya, U.Datta, A.Rahaman, S.Chakraborty et al. - Phys.Rev. C 104, 045801 (2021).
Neutron capture cross sections of light neutron-rich nuclei relevant for r-process nucleosynthesis.
- 2021Ka07 H.Kasuya, K.Yoshida - Prog.Theor.Exp.Phys. 2021, 013D01 (2021).
Hartree-Fock-Bogoliubov theory for odd-mass nuclei with a time-odd constraint and application to deformed halo nuclei.
- 2021Ke04 A.R.L.Kennington, G.Lotay, D.T.Doherty, D.Seweryniak et al. - Phys.Rev. C 103, 035805 (2021).
Level structure of the $T_z = -1$ nucleus ^{34}Ar and its relevance for nucleosynthesis in ONe novae.
- 2021Ko07 F.G.Kondev, M.Wang, W.J.Huang, S.Naimi, G.Audi - Chin.Phys.C 45, 030001 (2021).
The NUBASE2020 evaluation of nuclear physics properties.
- 2021Li16 H.-M.Liu, Y.-T.Zou, X.Pan, J.-L.Chen et al. - Chin.Phys.C 45, 024108 (2021).
New Geiger-Nuttall law for two-proton radioactivity.
- 2021Li26 L.Liu, S.Liu, S.-S.Zhang, L.-G.Cao - Chin.Phys.C 45, 044105 (2021).
Isovector giant dipole resonances in proton-rich Ar and Ca isotopes.
- 2021Li27 L.Liu, S.Liu, S.-S.Zhang, L.-G.Cao - Chin.Phys.C 45, 044110 (2021).
Systematic study of two-proton radioactivity within a Gamow-like model.
- 2021Ma33 C.Ma, Y.Y.Zong, S.Q.Zhang, J.Li et al. - Phys.Rev. C 103, 054326 (2021).
Mass relations of mirror nuclei in terms of Coulomb energies based on relativistic continuum Hartree-Bogoliubov calculations.
- 2021Mi17 F.Minato, T.Marketin, N.Paar - Phys.Rev. C 104, 044321 (2021).
 β -delayed neutron-emission and fission calculations within relativistic quasiparticle random-phase approximation and a statistical model.
- 2021Sh34 V.R.Sharma, E.F.Aguilera, T.L.Belyaeva, J.C.Morales-Rivera et al. - Phys.Rev. C 104, 044621 (2021).
Fusion and proton transfer in the $^{10}\text{B} + ^{27}\text{Al}$ system at sub-barrier energies.
- 2021StZZ N.J.Stone - INDC(NDS)-0833 (2021).
Table of Nuclear Electric Quadrupole Moments.
- 2021Vl03 G.N.Vlaskin, Yu.S.Khomyakov - At.Energ. 130, 104 (2021).
(α, n) Neutron Spectra on Thick Light Targets.
- 2021Wa16 M.Wang, W.J.Huang, F.G.Kondev, G.Audi, S.Naimi - Chin.Phys.C 45, 030003 (2021).
The AME 2020 atomic mass evaluation (II). Tables, graphs and references.
- 2021Xi06 F.Xing, J.Cui, Y.Wang, J.Gu - Chin.Phys.C 45, 124105 (2021).
Two-proton radioactivity of ground and excited states within a unified fission model.
- 2022Ah02 D.S.Ahn, J.Amano, H.Baba, N.Fukuda et al. - Phys.Rev.Lett. 129, 212502 (2022).
Discovery of ^{39}Na .
- 2022Cr03 H.L.Crawford, V.Tripathi, J.M.Allmond, B.P.Crider et al. - Phys.Rev.Lett. 129, 212501 (2022).
Crossing $N = 28$ Toward the Neutron Drip Line: First Measurement of Half-Lives at FRIB.
- 2022Lu03 Y.Lu, Y.Lei, C.W.Johnson, J.Shen - Phys.Rev. C 105, 034317 (2022).
Nuclear states projected from a pair condensate.
- 2022Mo10 O.M.Molchanov, K.D.Launey, A.Mercenne, G.H.Sargsyan et al. - Phys.Rev. C 105, 034306 (2022).
Machine learning approach to pattern recognition in nuclear dynamics from the ab initio symmetry-adapted no-core shell model.
- 2022Ot01 T.Otsuka, N.Shimizu, Y.Tsunoda - Phys.Rev. C 105, 014319 (2022).
Moments and radii of exotic Na and Mg isotopes.
- 2022Pa24 X.Pan, Y.-T.Zou, B.He, X.-H.Li et al. - Int.J.Mod.Phys. E31, 2250051 (2022).
Systematic study of two-proton radioactivity half-lives using two-potential approach with different Skyrme interactions.
- 2022Ro15 G.Royer - Phys.Rev. C 106, 034605 (2022).
Calculation of two-proton radioactivity and application to ^9Be , $^6,7\text{Li}$, $^3,6\text{He}$, and $^2,3\text{H}$ emissions.
- 2022Sa41 K.P.Santhosh - Phys.Rev. C 106, 054604 (2022).
Two-proton radioactivity within a Coulomb and proximity potential model for deformed nuclei.
- 2022Sh22 O.A.Shehu, B.P.Crider, T.Ginter, C.R.Hoffman et al. - Nucl.Instrum.Methods Phys.Res. A1035, 166789 (2022).
Experimental study of the $^{34\text{m}}\text{Cl}$ beam production at intermediate energies.
- 2022St03 S.R.Stroberg, J.Henderson, G.Hackman, P.Ruotsalainen et al. - Phys.Rev. C 105, 034333 (2022).
Systematics of E2 strength in the sd shell with the valence-space in-medium similarity renormalization group.
- 2022Ya19 J.M.Yao, I.Ginnett, A.Belley, T.Miyagi et al. - Phys.Rev. C 106, 014315 (2022).
Ab initio studies of the double-Gamow-Teller transition and its correlation with neutrinoless double- β decay.
- 2022Yo04 K.Yoshida - Phys.Rev. C 105, 024313 (2022).
Cranked Skyrme-Hartree-Fock-Bogoliubov approach for a mean-field description of nuclear rotations near the drip line.
- 2022Zo01 Y.Y.Zong, C.Ma, M.Q.Lin, Y.M.Zhao - Phys.Rev. C 105, 034321 (2022).
Mass relations of mirror nuclei for both bound and unbound systems.
- 2023Ba33 H.Bashir, H.Abusara, S.Ahmad - Int.J.Mod.Phys. E32, 2350046 (2023).

REFERENCES FOR A=34(CONTINUED)

- 2023Bo17 *Shape evolution of nuclei in the region of ($A \approx 30$) using covariant density functional theory.*
I.N.Borzov, S.V.Tolokonnikov - Physics of Part.and Nuclei 54, 586 (2023).
- 2023Ch17 *Self-Consistent Calculation of Nuclear Charge Radii in K Isotopes.*
P.Choudhary, P.C.Srivastava - Nucl.Phys. A1033, 122629 (2023).
- 2023Ch34 *Nuclear structure properties of Si and P isotopes with the microscopic effective interactions.*
P.Choudhary, P.C.Srivastava - Eur.Phys.J. A 59, 97 (2023).
- 2023Dr01 *Study of S, Cl and Ar isotopes with $N \geq Z$ using microscopic effective sd -shell interactions.*
N.Dronchi, D.Weisshaar, B.A.Brown, A.Gade et al. - Phys.Rev. C 107, 034306 (2023).
- 2023La09 *Measurement of the $B(E2 \uparrow)$ strengths of ^{36}Ca and ^{38}Ca .*
L.Lalanne, O.Sorlin, A.Poves, M.Assie et al. - Phys.Rev.Lett. 131, 092501 (2023).
- 2023Lu07 *$N = 16$ Magicity Revealed at the Proton Drip Line through the Study of ^{35}Ca .*
R.S.Lubna, S.N.Liddick, T.H.Ogunbeku, A.Chester et al. - Phys.Rev. C 108, 014329 (2023).
- 2023Me08 *β decay of ^{36}Mg and ^{36}Al : Identification of a β -decaying isomer in ^{36}Al .*
P.Mehana, N.S.Rajeswari - Eur.Phys.J. A 59, 104 (2023).
- 2023Ra22 *Two-proton and one-proton emission of two-proton emitters.*
N.Ravlic, E.Yuksel, T.Niksic, N.Paar - Phys.Rev. C 108, 054305 (2023).
- 2024Ab03 *Influence of the symmetry energy on the nuclear binding energies and the neutron drip line position.*
K.Abe, H.Nakada - J.Phys.(London) G51, 035101 (2024).
- 2024Dr01 *Pairing effects on pure rotational energy of nuclei.*
N.Dronchi, R.J.Charity, L.G.Sobotka, B.A.Brown et al. - Phys.Rev. C 110, L031302 (2024).
- 2024Ha15 *Evolution of shell gaps in the neutron-poor calcium region from invariant-mass spectroscopy of $^{37,38}\text{Sc}$, ^{35}Ca , and ^{34}K .*
M.A.Hasan, A.H.Taqi, R.A.Radhi - Int.J.Mod.Phys. E33, 2450013 (2024).
- 2024Ji17 *Effect of the pairing types and pairing strength on the ground state properties of even and odd Mg isotopes.*
J.-D.Jiang, X.Liu, D.-M.Zhang, M.Li et al. - Chin.Phys.C 48, 104108 (2024).
- 2024Li59 *A simple model for two-proton radioactivity.*
T.Li, N.Wang, H.Yao, X.Zhang, M.Liu - Phys. Rev. C 110, 034318 (2024); Erratum Phys.Rev. C 113, 029901 (2026).
- 2024Lu03 *Mirror corrections in predictions of nucleon separation energies for nuclei near the proton drip line.*
R.S.Lubna, A.B.Garnsworthy, V.Tripathi, G.C.Ball et al. - Phys.Rev. C 109, 014309 (2024).
- 2024Sa36 *Cross-shell excited configurations in the structure of ^{34}Si .*
S.Sarkar - J.Phys.(London) G51, 065103 (2024).
- 2024Xa01 *A method for semi-empirical calculation of isospin mixing probability in nuclei.*
L.Xayavong, N.A.Smirmova - Phys.Rev. C 109, 014317 (2024).
- 2025Ch07 *Higher-order isospin-symmetry-breaking corrections to nuclear matrix elements of Fermi β decays.*
J.Chen, B.Singh - Nucl.Data Sheets 199, 1 (2025).
- 2025Ch19 *Nuclear Structure and Decay Data for $A=33$ Isobars.*
J.Chen - Nucl.Data Sheets 201, 1 (2025).
- 2025Ku15 *Nuclear Structure and Decay Data for $A=32$ Isobars.*
T.Kubo - Nucl.Phys. A1062, 123154 (2025).
- 2025Li35 *Neutron dripline search for fluorine, neon and sodium and the discovery of ^{39}Na at RIKEN RIBF.*
H.H.Li, J.G.Li, M.R.Xie, W.Zuo - Phys.Lett. B 867, 139609 (2025).
- 2025Li67 *Ab initio calculations for shell evolution in proton-dripline nuclei.*
Zh.J.Lian, Z.Ch.Gao - Int.J.Mod.Phys. E34, 2543006 (2025).
- 2025Ly01 *The variation after projection shell model.*
E.M.Lykiardopoulou, C.Walls, J.Bergmann, M.Brodeur et al. - Phys.Rev.Lett. 134, 052503 (2025).
- 2025StZZ *Refined Topology of the $N=20$ Island of Inversion with High Precision Mass Measurements of $^{31-33}\text{Na}$ and $^{31-35}\text{Mg}$.*
N.J.Stone - INDC(NDS)-0915 (2025).
- 2025Zh24 *Table of Recommended Nuclear Magnetic Dipole Moments.*
E.F.Zhou, C.R.Ding, J.M.Yao, B.Bally et al. - Phys.Lett. B 865, 139464 (2025).
- 2025Zh57 *Ab initio nuclear shape coexistence and emergence of island of inversion around $N=20$.*
K.Y.Zhang, S.Q.Zhang, J.Meng - Phys.Rev. C 112, 044308 (2025).
- 2026Ji03 *Triaxial relativistic Hartree-Bogoliubov theory in continuum for exotic nuclei.*
X.F.Jiang, Z.Z.Li, X.W.Sun, J.Meng - Phys.Lett. B 874, 140264 (2026).
- 2026Li03 *Dipole response in deformed halo nuclei ^{42}Mg and ^{44}Mg .*
T.Li, M.Liu, N.Wang - Chin.Phys.C 50, 024104 (2026).
- 2026Su18 *Proton separation energy predictions for proton-rich nuclei with the radial basis function approach and mirror symmetry.*
L.Sun, J.Chen - Nucl.Data Sheets 211, 1 (2026).
- Nuclear Structure and Decay Data for $A=35$ Isobars.*